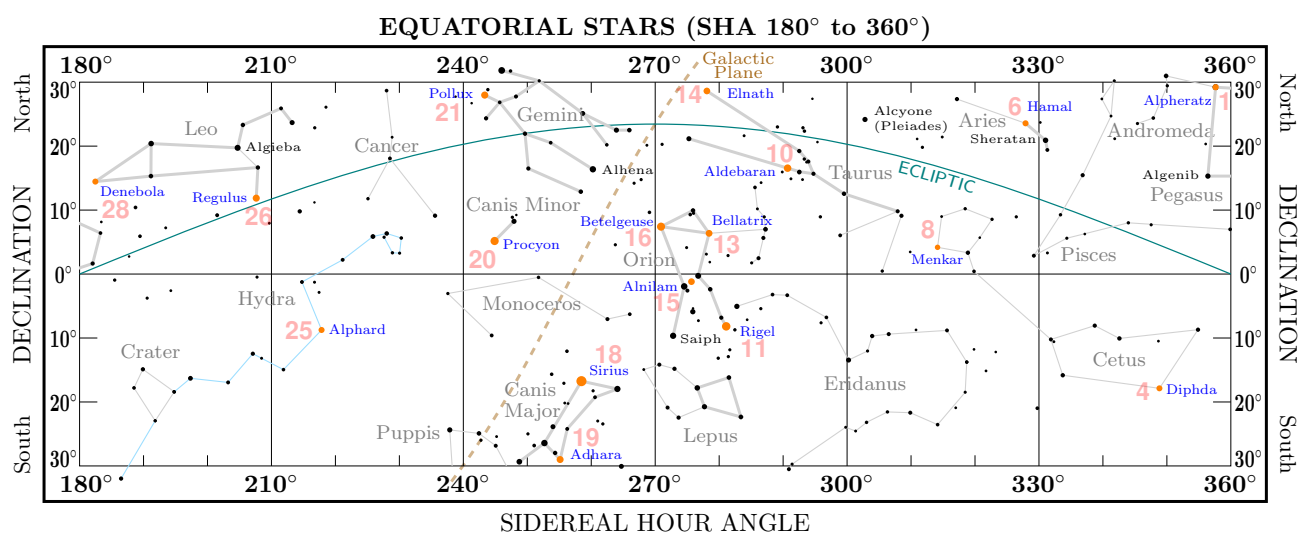
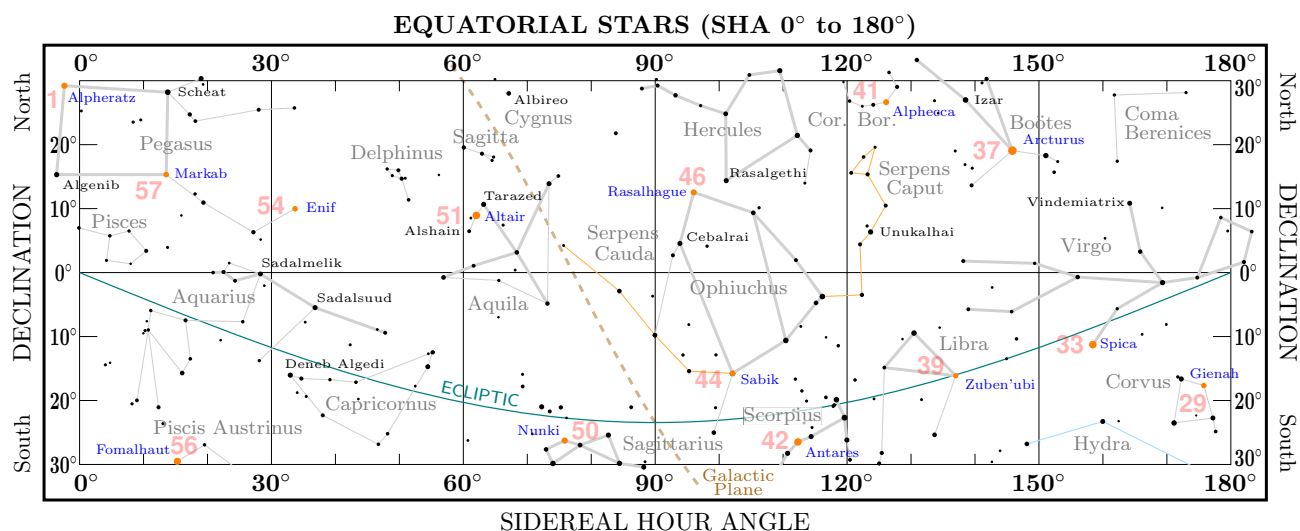




THE NAUTICAL ALMANAC

2010

Author: Andrew BAUER
Original concept from: Enno RODEGERDTS

November 13, 2024

Disclaimer: These are computer generated tables - use them at your own risk. The accuracy has been randomly checked with JPL HORIZONS System, but cannot be guaranteed. The author claims no liability for any consequences arising from use of these tables. Besides, this publication only contains the 'daily pages' of the Nautical Almanac: an official version of the Nautical Almanac is indispensable.

Information in the data page footers

Information pertaining to the IERS EOP data has been added to the odd data page footers if using MiKTeX or TeX Live (2020 or later). The International Earth Rotation Service (IERS) provides accurate data (updated weekly) on the Earth Orientation Parameters (EOP).

Earth's speed of rotation is not constant, i.e. the day length fluctuates.¹ This is due to *internal torques* caused by relative movements and mass redistribution of Earth's core, mantle, oceans, atmosphere, and cryosphere. This has an immediate impact on the GHA values of all celestial objects.

The IERS monitors and measures several parameters taking the actual speed of Earth's rotation into account. Their measured data begins on 2nd January 1973. Predictive data begins following the last day of (obtained) data and extends about 360 days into the future. (The IERS results are published with a delay of about 18-hours between the date of publication and the last available date with measured EOP.²) These Nautical Almanac daily pages take the (measured or predicted) UT1-UTC values into account providing highly accurate navigational data especially if the predictions are fairly recent.

As long as either measured or predicted data is available the footer will show:

[IERS Earth Orientation data as of dd-mmm-yyyy](#)

This indicates that IERS EOP data is in use - older dates are measured; newer dates are predictions.

If the final date of IERS prediction data is on the current data page, the footer shows:

[IERS Earth Orientation predictions end dd-mmm-yyyy](#)

Pages with dates beyond the final date of IERS prediction data have the following footer:

[No IERS EOP prediction data available](#)

Skyfield then defaults to using the ΔT and leap second files that ship with Skyfield internally.

The footers mentioned are only displayed as long as `'useIERS = True'` is set in `config.py` to enable use of IERS EOP data.

Brief historical overview

The story begins with the XEphem astronomical library, which is declared 'end of life' by its author, Elwood Charles Downey, as no further updates are planned. He generously gave permission for use of XEphem code in Ephem (also known as Pyephem), an astronomical library authored by Brandon Rhodes. Enno Rodegerdts (<https://sv-inua.net/>) created the original Nautical Almanac 'daily pages' in Pyalmanac using Python 2 and LaTeX. After contacting him I obtained permission for its future enhancement and maintenance. Pyalmanac uses Ephem.

Meanwhile Brandon Rhodes was working on a far more sophisticated astronomical library, Skyfield. This was 'state of the art' and clearly surpassed the 'Jean Meeus'-based Pyephem/Ephem. Skyfield uses NASA's NAIF (Navigation and Ancillary Information Facility) SPICE algorithms. The results agree with those from the HORIZONS System (*operated by NASA JPL (Jet Propulsion Laboratory) SSD (Solar System Dynamics) group, not by NAIF*). This in turn implies that celestial positions calculated by Skyfield agree with those generated by the United States Naval Observatory and their *Astronomical Almanac* to within 0.0005 arcseconds (half a milliarcsecond).

Pyephem was then in 'maintenance mode'. Clearly Pyalmanac needed adaptation to use Skyfield, and thus SFalmanac was born. However its performance was poor regarding the calculation of 'events' such as: sunrise, sunset, moonrise, moonset, civil twilight start/end and nautical twilight start/end. An interim (faster) solution was required.

A hybrid application, originally named Skyalmanac, was developed using Ephem to calculate 'events' and Skyfield for the rest. This was indeed much faster at the cost of poorer 'event time' data. It took a while to find a better solution: multiprocessing, which was built into SFalmanac. This now could compare to the execution times in Pyalmanac but with improved data.

New functionality was added to SFalmanac, e.g. lunar phase as a graphic; Lunar Distance tables and charts. The original Skyalmanac is deprecated and has now been replaced with the latest SFalmanac code, so Skyalmanac and SFalmanac are now identical apart from the name. Since April 2019 <http://thenauticalalmanac.com> has been publishing Celestial Navigation related material with software provided here.

¹https://en.wikipedia.org/wiki/Day_length_fluctuations

²<https://hpiers.obspm.fr/eoppc/bul/bulb/explanatory.html>

January 01, 02, 03 UT (Fri., Sat., Sun.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	100°32.5	181°58.1	S23°38.5	318°03.9	N18°45.1	131°37.2	S13°36.5	275°29.9	N00°18.6	Alpheratz	357°46.2 29°09.0
1	115°35.0	196°57.1	38.4	333°06.8	45.4	146°39.2	36.4	290°32.3	18.5	Ankaa	353°18.1 -42°15.2
2	130°37.4	211°56.1	38.3	348°09.6	45.6	161°41.2	36.2	305°34.7	18.5	Schedar	349°43.6 56°35.9
3	145°39.9	226°55.2	•• 38.2	3°12.4	•• 45.8	176°43.2	•• 36.0	320°37.1	•• 18.5	Diphda	348°58.4 -17°55.9
4	160°42.4	241°54.2	38.1	18°15.2	46.0	191°45.1	35.8	335°39.5	18.5	Achernar	335°28.3 -57°11.3
5	175°44.8	256°53.2	38.0	33°18.0	46.2	206°47.1	35.7	350°41.9	18.5	Hamal	328°03.5 23°30.8
6	190°47.3	271°52.2	S23°37.9	48°20.9	N18°46.5	221°49.1	S13°35.5	5°44.3	N00°18.5	Polaris	318°46.3 89°18.8
7	205°49.8	286°51.3	37.8	63°23.7	46.7	236°51.1	35.3	20°46.7	18.5	Acamar	315°19.9 -40°16.0
8	220°52.2	301°50.3	37.7	78°26.5	46.9	251°53.0	35.1	35°49.2	18.5	Menkar	314°17.5 4°07.8
9	235°54.7	316°49.3	•• 37.6	93°29.4	•• 47.1	266°55.0	•• 35.0	50°51.6	•• 18.4	Mirfak	308°43.7 49°54.1
10	250°57.2	331°48.4	37.5	108°32.2	47.3	281°57.0	34.8	65°54.0	18.4	Aldebaran	290°52.0 16°31.8
11	265°59.6	346°47.4	37.4	123°35.0	47.6	296°59.0	34.6	80°56.4	18.4	Rigel	281°14.1 -8°11.4
12	281°02.1	1°46.4	S23°37.3	138°37.8	N18°47.8	312°00.9	S13°34.4	95°58.8	N00°18.4	Capella	280°37.7 46°00.6
13	296°04.5	16°45.5	37.2	153°40.7	48.0	327°02.9	34.3	111°01.2	18.4	Bellatrix	278°34.4 6°21.5
14	311°07.0	31°44.5	37.1	168°43.5	48.2	342°04.9	34.1	126°03.6	18.4	Elnath	278°15.4 28°37.0
15	326°09.5	46°43.5	•• 37.0	183°46.4	•• 48.5	357°06.9	•• 33.9	141°06.0	•• 18.4	Alnilam	275°48.6 -1°11.7
16	341°11.9	61°42.6	36.9	198°49.2	48.7	12°08.8	33.7	156°08.4	18.4	Betelgeuse	271°03.7 7°24.5
17	356°14.4	76°41.6	36.8	213°52.0	48.9	27°10.8	33.6	171°10.8	18.4	Canopus	263°56.7 -52°42.1
18	11°16.9	91°40.6	S23°36.7	228°54.9	N18°49.1	42°12.8	S13°33.4	186°13.2	N00°18.4	Sirius	258°35.6 -16°43.8
19	26°19.3	106°39.7	36.6	243°57.7	49.4	57°14.7	33.2	201°15.7	18.3	Adhara	255°14.1 -28°59.2
20	41°21.8	121°38.7	36.4	259°00.6	49.6	72°16.7	33.0	216°18.1	18.3	Procyon	245°02.0 5°11.9
21	56°24.3	136°37.7	•• 36.3	274°03.4	•• 49.8	87°18.7	•• 32.9	231°20.5	•• 18.3	Pollux	243°30.4 28°00.0
22	71°26.7	151°36.8	36.2	289°06.2	50.0	102°20.7	32.7	246°22.9	18.3	Avior	234°18.6 -59°32.4
23	86°29.2	166°35.8	36.1	304°09.1	50.3	117°22.6	32.5	261°25.3	18.3	Suhail	222°54.0 -43°28.3
Mer.pass. 17:15		ν -1.0' d -0.1' m -3.92		ν 2.8' d 0.2' m -0.76		ν 2.0' d -0.2' m -2.11		ν 2.4' d -0.0' m 0.87		Miaplacidus	221°39.7 -69°45.4
Alphard 217°58.3 -8°42.2 Regulus 207°46.0 11°54.9 Dubhe 193°54.2 61°41.4 Denebola 182°36.1 14°30.7 Gienah 175°54.9 -17°35.9 Acrux 173°12.3 -63°09.1 Gacrux 172°03.9 -57°10.0 Alioth 166°22.7 55°53.9 Spica 158°34.0 -11°12.9 Alkaid 153°00.9 49°15.4 Hadar 148°51.9 -60°25.1 Menkent 148°10.8 -36°25.1 Arcturus 145°58.1 19°07.6 Rigel Kent. 139°55.6 -60°52.4 Kochab 137°20.3 74°06.5 Zuben'ubi 137°08.4 -16°05.0 Alphecca 126°13.4 26°40.6 Antares 112°29.7 -26°27.2 Atria 107°34.4 -69°02.6 Sabik 102°15.8 -15°44.3 Shaula 96°25.8 -37°06.6 Rasalhague 96°09.1 12°33.1 Eltanin 90°47.8 51°29.2 Kaus Aust. 83°47.6 -34°22.8 Vega 80°41.1 38°47.6 Nunki 76°01.8 -26°17.1 Altair 62°11.0 8°53.7 Peacock 53°23.6 -56°42.2 Deneb 49°33.7 45°19.1 Enif 33°49.9 9°55.4 Al Na'ir 27°47.1 -46°54.9 Fomalhaut 15°26.9 -29°34.2 Scheat 13°56.0 28°08.5 Markab 13°41.0 15°15.7											
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	101°31.7	181°34.8	S23°36.0	319°11.9	N18°50.5	132°24.6	S13°32.3	276°27.7	N00°18.3		
1	116°34.1	196°33.8	35.8	334°14.8	50.7	147°26.6	32.2	291°30.1	18.3		
2	131°36.6	211°32.9	35.7	349°17.7	51.0	162°28.6	32.0	306°32.5	18.3		
3	146°39.0	226°31.9	•• 35.6	4°20.5	•• 51.2	177°30.5	•• 31.8	321°34.9	•• 18.3		
4	161°41.5	241°30.9	35.5	19°23.4	51.4	192°32.5	31.6	336°37.4	18.2		
5	176°44.0	256°30.0	35.3	34°26.2	51.6	207°34.5	31.5	351°39.8	18.2		
6	191°46.4	271°29.0	S23°35.2	49°29.1	N18°51.9	222°36.4	S13°31.3	6°42.2	N00°18.2		
7	206°48.9	286°28.0	35.1	64°31.9	52.1	237°38.4	31.1	21°44.6	18.2		
8	221°51.4	301°27.1	34.9	79°34.8	52.3	252°40.4	30.9	36°47.0	18.2		
9	236°53.8	316°26.1	•• 34.8	94°37.7	•• 52.6	267°42.4	•• 30.8	51°49.4	•• 18.2		
10	251°56.3	331°25.1	34.7	109°40.5	52.8	282°44.3	30.6	66°51.8	18.2		
11	266°58.8	346°24.2	34.5	124°43.4	53.0	297°46.3	30.4	81°54.2	18.2		
12	282°01.2	1°23.2	S23°34.4	139°46.3	N18°53.3	312°48.3	S13°30.2	96°56.7	N00°18.2		
13	297°03.7	16°22.3	34.3	154°49.1	53.5	327°50.2	30.1	111°59.1	18.2		
14	312°06.1	31°21.3	34.1	169°52.0	53.7	342°52.2	29.9	127°01.5	18.2		
15	327°08.6	46°20.3	•• 34.0	184°54.9	•• 54.0	357°54.2	•• 29.7	142°03.9	•• 18.1		
16	342°11.1	61°19.4	33.8	199°57.7	54.2	12°56.2	29.5	157°06.3	18.1		
17	357°13.5	76°18.4	33.7	215°00.6	54.4	27°58.1	29.3	172°08.7	18.1		
18	12°16.0	91°17.4	S23°33.6	230°03.5	N18°54.7	43°00.1	S13°29.2	187°11.1	N00°18.1		
19	27°18.5	106°16.5	33.4	245°06.4	54.9	58°02.1	29.0	202°13.5	18.1		
20	42°20.9	121°15.5	33.3	260°09.2	55.1	73°04.0	28.8	217°16.0	18.1		
21	57°23.4	136°14.5	•• 33.1	275°12.1	•• 55.4	88°06.0	•• 28.6	232°18.4	•• 18.1		
22	72°25.9	151°13.6	33.0	290°15.0	55.6	103°08.0	28.5	247°20.8	18.1		
23	87°28.3	166°12.6	32.8	305°17.9	55.9	118°10.0	28.3	262°23.2	18.1		
Mer.pass. 17:11		ν -1.0' d -0.1' m -3.92		ν 2.9' d 0.2' m -0.78		ν 2.0' d -0.2' m -2.10		ν 2.4' d -0.0' m 0.86			
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	102°30.8	181°11.6	S23°32.7	320°20.8	N18°56.1	133°11.9	S13°28.1	277°25.6	N00°18.1		
1	117°33.3	196°10.7	32.5	335°23.7	56.3	148°13.9	27.9	292°28.0	18.1		
2	132°35.7	211°09.7	32.4	350°26.5	56.6	163°15.9	27.8	307°30.4	18.1		
3	147°38.2	226°08.7	•• 32.2	5°29.4	•• 56.8	178°17.8	•• 27.6	322°32.9	•• 18.0		
4	162°40.6	241°07.8	32.0	20°32.3	57.0	193°19.8	27.4	337°35.3	18.0		
5	177°43.1	256°06.8	31.9	35°35.2	57.3	208°21.8	27.2	352°37.7	18.0		
6	192°45.6	271°05.9	S23°31.7	50°38.1	N18°57.5	223°23.7	S13°27.0	7°40.1	N00°18.0		
7	207°48.0	286°04.9	31.6	65°41.0	57.8	238°25.7	26.9	22°42.5	18.0		
8	222°50.5	301°03.9	31.4	80°43.9	58.0	253°27.7	26.7	37°44.9	18.0		
9	237°53.0	316°03.0	•• 31.2	95°46.8	•• 58.2	268°29.6	•• 26.5	52°47.4	•• 18.0		
10	252°55.4	331°02.0	31.1	110°49.7	58.5	283°31.6	26.3	67°49.8	18.0		
11	267°57.9	346°01.0	30.9	125°52.6	58.7	298°33.6	26.2	82°52.2	18.0		
12	283°00.4	1°00.1	S23°30.7	140°55.5	N18°59.0	313°35.5	S13°26.0	97°54.6	N00°18.0		
13	298°02.8	15°59.1	30.6	155°58.4	59.2	328°37.5	25.8	112°57.0	18.0		
14	313°05.3	30°58.1	30.4	171°01.3	59.5	343°39.5	25.6	127°59.4	18.0		
15	328°07.8	45°57.2	•• 30.2	186°04.2	•• 59.7	358°41.5	•• 25.4	143°01.9	•• 18.0		
16	343°10.2	60°56.2	30.1	201°07.1	18°59.9	13°43.4	25.3	158°04.3	17.9		
17	358°12.7	75°55.3	29.9	216°10.0	19°00.2	28°45.4	25.1	173°06.7	17.9		
18	13°15.1	90°54.3	S23°29.7	231°12.9	N19°00.4	43°47.4	S13°24.9	188°09.1	N00°17.9		
19	28°17.6	105°53.3	29.5	246°15.9	00.7	58°49.3	24.7	203°11.5	17.9		
20	43°20.1	120°52.4	29.4	261°18.8	00.9	73°51.3	24.6	218°14.0	17.9		
21	58°22.5	135°51.4	•• 29.2	276°21.7	•• 01.2	88°53.3	•• 24.4	233°16.4	•• 17.9		
22	73°25.0	150°50.5	29.0	291°24.6	01.4	103°55.2	24.2	248°18.8	17.9		
23	88°27.5	165°49.5	28.8	306°27.5	01.7	118°57.2	24.0	263°21.2	17.9		
Mer.pass. 17:07		ν -1.0' d -0.2' m -3.92		ν 2.9' d 0.2' m -0.81		ν 2.0' d -0.2' m -2.10		ν 2.4' d -0.0' m 0.86			

Jan 01 Fri

Venus 81°25.6 11:53

Mars 217°31.4 02:47

Jupiter 31°04.7 15:12

Saturn 174°57.4 05:37

Jan 02 Sat

Venus 80°03.2 11:54

Mars 217°40.3 02:43

Jupiter 30°53.0 15:08

Saturn 174°56.1 05:33

Jan 03 Sun

Venus 78°40.8 11:56

Mars 217°50

h	Sun			Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	179°10.5	S23°01.6	356°04.6	3.1'	N23°30.0	-7.3'	60.9'
1	194°10.2	01.4	10°26.6	3.1'	23°22.7	-7.4'	61.0'
2	209°09.9	01.2	24°48.8	3.2'	23°15.3	-7.6'	61.0'
3	224°09.6	· · 01.0	39°11.0	3.3'	23°07.7	-7.8'	61.0'
4	239°09.3	00.8	53°33.3	3.3'	22°59.9	-7.9'	61.0'
5	254°09.0	00.6	67°55.6	3.4'	22°52.0	-8.1'	61.0'
6	269°08.7	S23°00.4	82°18.0	3.5'	N22°43.8	-8.3'	61.0'
7	284°08.4	00.2	96°40.5	3.6'	22°35.6	-8.4'	61.0'
8	299°08.1	23°00.0	111°03.1	3.6'	22°27.1	-8.6'	61.0'
9	314°07.8	22°59.8	125°25.7	3.7'	22°18.5	-8.8'	61.0'
10	329°07.5	59.6	139°48.4	3.8'	22°09.7	-8.9'	61.0'
11	344°07.2	59.4	154°11.2	3.9'	22°00.8	-9.1'	61.0'
12	359°06.9	S22°59.2	168°34.0	3.9'	N21°51.7	-9.3'	61.0'
13	14°06.6	59.0	182°57.0	4.0'	21°42.4	-9.4'	61.0'
14	29°06.3	58.8	197°20.0	4.1'	21°33.0	-9.6'	61.0'
15	44°06.0	· · 58.6	211°43.1	4.2'	21°23.4	-9.7'	61.0'
16	59°05.8	58.3	226°06.3	4.3'	21°13.7	-9.9'	61.1'
17	74°05.5	58.1	240°29.5	4.4'	21°03.8	-10.0'	61.1'
18	89°05.2	S22°57.9	254°52.9	4.4'	N20°53.8	-10.2'	61.1'
19	104°04.9	57.7	269°16.3	4.5'	20°43.6	-10.3'	61.1'
20	119°04.6	57.5	283°39.8	4.6'	20°33.3	-10.5'	61.1'
21	134°04.3	· · 57.3	298°03.5	4.7'	20°22.8	-10.6'	61.1'
22	149°04.0	57.1	312°27.2	4.8'	20°12.2	-10.7'	61.1'
23	164°03.7	56.8	326°51.0	4.9'	20°01.5	-10.9'	61.1'
SD = 16.3' $d = -0.2'$ SD = 16.6'							

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	179°03.4	S22°56.6	341°14.9	5.0'	N19°50.6	-11.0'	61.1'
1	194°03.1	56.4	355°38.8	5.1'	19°39.6	-11.2'	61.1'
2	209°02.8	56.2	10°02.9	5.2'	19°28.4	-11.3'	61.0'
3	224°02.5	· · 56.0	24°27.1	5.3'	19°17.1	-11.4'	61.0'
4	239°02.3	55.7	38°51.4	5.4'	19°05.7	-11.6'	61.0'
5	254°02.0	55.5	53°15.7	5.5'	18°54.2	-11.7'	61.0'
6	269°01.7	S22°55.3	67°40.2	5.6'	N18°42.5	-11.8'	61.0'
7	284°01.4	55.1	82°04.7	5.6'	18°30.7	-11.9'	61.0'
8	299°01.1	54.8	96°29.4	5.7'	18°18.7	-12.0'	61.0'
9	314°00.8	· · 54.6	110°54.1	5.8'	18°06.7	-12.2'	61.0'
10	329°00.5	54.4	125°19.0	5.9'	17°54.5	-12.3'	61.0'
11	344°00.2	54.2	139°43.9	6.0'	17°42.3	-12.4'	61.0'
12	358°59.9	S22°53.9	154°08.9	6.1'	N17°29.9	-12.5'	61.0'
13	13°59.6	53.7	168°34.1	6.2'	17°17.3	-12.6'	61.0'
14	28°59.3	53.5	182°59.3	6.3'	17°04.7	-12.7'	61.0'
15	43°59.1	· · 53.3	197°24.6	6.4'	16°52.0	-12.8'	61.0'
16	58°58.8	53.0	211°50.1	6.5'	16°39.2	-12.9'	61.0'
17	73°58.5	52.8	226°15.6	6.6'	16°26.2	-13.0'	60.9'
18	88°58.2	S22°52.6	240°41.2	6.7'	N16°13.2	-13.1'	60.9'
19	103°57.9	52.3	255°06.9	6.8'	16°00.0	-13.2'	60.9'
20	118°57.6	52.1	269°32.7	6.9'	15°46.8	-13.3'	60.9'
21	133°57.3	· · 51.9	283°58.7	7.0'	15°33.5	-13.4'	60.9'
22	148°57.0	51.6	298°24.7	7.1'	15°20.0	-13.5'	60.9'
23	163°56.7	51.4	312°50.8	7.2'	15°06.5	-13.6'	60.9'
SD = 16.3' $d = -0.2'$ SD = 16.7'							

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	178°56.5	S22°51.1	327°17.0	7.3'	N14°52.9	-13.7'	60.9'
1	193°56.2	50.9	341°43.3	7.4'	14°39.2	-13.8'	60.8'
2	208°55.9	50.7	356°09.7	7.5'	14°25.4	-13.9'	60.8'
3	223°55.6	· · 50.4	10°36.2	7.6'	14°11.6	-13.9'	60.8'
4	238°55.3	50.2	25°02.8	7.7'	13°57.6	-14.0'	60.8'
5	253°55.0	49.9	39°29.4	7.8'	13°43.6	-14.1'	60.8'
6	268°54.7	S22°49.7	53°56.2	7.9'	N13°29.5	-14.2'	60.8'
7	283°54.4	49.5	68°23.1	8.0'	13°15.3	-14.3'	60.7'
8	298°54.2	49.2	82°50.0	8.0'	13°01.0	-14.3'	60.7'
9	313°53.9	· · 49.0	97°17.1	8.1'	12°46.7	-14.4'	60.7'
10	328°53.6	48.7	111°44.2	8.2'	12°32.3	-14.5'	60.7'
11	343°53.3	48.5	126°11.4	8.3'	12°17.9	-14.5'	60.7'
12	358°53.0	S22°48.2	140°38.8	8.4'	N12°03.4	-14.6'	60.7'
13	13°52.7	48.0	155°06.2	8.5'	11°48.8	-14.6'	60.6'
14	28°52.4	47.7	169°33.7	8.6'	11°34.1	-14.7'	60.6'
15	43°52.2	· · 47.5	184°01.2	8.7'	11°19.4	-14.8'	60.6'
16	58°51.9	47.2	198°28.9	8.8'	11°04.7	-14.8'	60.6'
17	73°51.6	47.0	212°56.7	8.8'	10°49.8	-14.9'	60.6'
18	88°51.3	S22°46.7	227°24.5	8.9'	N10°35.0	-14.9'	60.5'
19	103°51.0	46.5	241°52.4	9.0'	10°20.0	-15.0'	60.5'
20	118°50.7	46.2	256°20.4	9.1'	10°05.1	-15.0'	60.5'
21	133°50.4	· · 46.0	270°48.5	9.2'	09°50.1	-15.1'	60.5'
22	148°50.2	45.7	285°16.7	9.2'	09°35.0	-15.1'	60.4'
23	163°49.9	45.5	299°44.9	9.3'	09°19.9	-15.1'	60.4'
SD = 16.3' $d = -0.2'$ SD = 16.6'							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:23	10:40	■	■	13:29	15:45
N 70°	08:04	09:48	■	■	14:20	16:04
68°	07:49	09:16	■	■	14:52	16:19
66°	07:37	08:52	10:26	13:42	15:16	16:31
64°	07:26	08:34	09:49	14:20	15:35	16:42
62°	07:17	08:18	09:22	14:46	15:50	16:51
60°	07:09	08:05	09:02	15:06	16:03	16:59
N 58°	07:02	07:54	08:45	15:23	16:14	17:06
56°	06:55	07:44	08:31	15:37	16:24	17:13
54°	06:50	07:35	08:19	15:49	16:33	17:19
52°	06:44	07:28	08:08	16:00	16:41	17:24
50°	06:39	07:20	07:58	16:10	16:48	17:29
45°	06:28	07:05	07:38	16:30	17:03	17:40
N 40°	06:18	06:52	07:22	16:46	17:17	17:50
35°	06:09	06:40	07:08	17:00	17:28	18:00
30°	06:00	06:30	06:56	17:12	17:38	18:08
20°	05:44	06:11	06:35	17:33	17:57	18:24
N 10°	05:28	05:55	06:17	17:51	18:13	18:40
0°	05:12	05:38	06:00	18:08	18:30	18:56
S 10°	04:53	05:20	05:43	18:25	18:48	19:15
20°	04:31	05:00	05:25	18:43	19:08	19:37
30°	04:03	04:36	05:03	19:05	19:32	20:05
35°	03:44	04:21	04:50	19:17	19:47	20:23
40°	03:22	04:03	04:36	19:32	20:05	20:46
45°	02:52	03:41	04:18	19:50	20:27	21:15
S 50°	02:09	03:12	03:56	20:11	20:55	21:59
52°	01:43	02:58	03:46	20:22	21:10	22:24
54°	01:03	02:40	03:34	20:34	21:27	23:02
56°	///	02:19	03:20	20:47	21:48	///
58°	///	01:52	03:04	21:03	22:15	///
S 60°	///	01:09	02:44	21:23	22:57	///

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	□	15:25	18:36	□	12:34	11:20
N 70°	□	16:17	18:54	□	11:40	11:00
68°	13:46	16:49	19:08	12:05	11:07	10:44
66°	14:49	17:12	19:19	11:00	10:42	10:31
64°	15:24	17:30	19:28	10:24	10:23	10:20
62°	15:49	17:45	19:36	09:58	10:07	10:10
60°	16:09	17:58	19:43	09:38	09:53	10:02
N 58°	16:25	18:08	19:49	09:21	09:42	09:55
56°	16:39	18:18	19:54	09:07	09:31	09:49
54°	16:51	18:26	19:59	08:54	09:22	09:43
52°	17:01	18:33	20:03	08:43	09:14	09:38
50°	17:11	18:40	20:07	08:33	09:07	09:33
45°	17:30	18:54	20:15	08:13	08:51	09:23
N 40°	17:46	19:05	20:22	07:56	08:39	09:14
35°	17:59	19:15	20:28	07:41	08:28	09:07
30°	18:11	19:23	20:33	07:29	08:18	09:00
20°	18:31	19:38	20:42	07:08	08:01	08:49
N 10°	18:48	19:51	20:50	06:49	07:46	08:39
0°	19:04	20:03	20:57	06:31	07:32	08:29
S 10°	19:20	20:14	21:05	06:14	07:18	08:19
20°	19:37	20:27	21:12	05:55	07:03	08:09
30°	19:56	20:41	21:21	05:33	06:46	07:57
35°	20:07	20:50	21:26	05:20	06:35	07:50
40°	20:20	20:59	21:32	05:05	06:24	07:42
45°	20:35	21:10	21:38	04:47	06:10	07:33
S 50°	20:54	21:23	21:46	04:24	05:52	07:21
52°	21:03	21:29	21:50	04:13	05:44	07:16
54°	21:12	21:36	21:54	04:01	05:35	07:10
56°	21:23	21:43	21:58	03:47	05:25	07:04
58°	21:36	21:52	22:03	03:31	05:13	06:56
S 60°	21:50	22:01	22:08	03:11	05:00	06:48

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss	hh:mm	Upper Mer. Pass hh:mm	Lower Mer. Pass hh:mm	16-18 100-93%
01	03:18	03:32	12:04	00:16	12:48	
02	03:46	04:00	12:04	01:18	13:48	
03	04:14	04:28	12:04	02:16	14:43	

January 04, 05, 06 UT (Mon., Tue., Wed.)

Aries		Venus		Mars		Jupiter		Saturn	
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	103°29.9	180°48.5	S23°28.6	321°30.4	N19°01.9	133°59.2	S13°23.8	278°23.6	N00°17.9
1	118°32.4	195°47.6	28.4	336°33.4	02.1	149°01.1	23.7	293°26.0	17.9
2	133°34.9	210°46.6	28.3	351°36.3	02.4	164°03.1	23.5	308°28.5	17.9
3	148°37.3	225°45.7	· · 28.1	6°39.2	· · 02.6	179°05.1	· · 23.3	323°30.9	· · 17.9
4	163°39.8	240°44.7	27.9	21°42.1	02.9	194°07.0	23.1	338°33.3	17.9
5	178°42.3	255°43.7	27.7	36°45.1	03.1	209°09.0	22.9	353°35.7	17.9
6	193°44.7	270°42.8	S23°27.5	51°48.0	N19°03.4	224°11.0	S13°22.8	8°38.1	N00°17.9
7	208°47.2	285°41.8	27.3	66°50.9	03.6	239°12.9	22.6	23°40.6	17.8
8	223°49.6	300°40.9	27.1	81°53.8	03.9	254°14.9	22.4	38°43.0	17.8
9	238°52.1	315°39.9	· · 26.9	96°56.8	· · 04.1	269°16.9	· · 22.2	53°45.4	· · 17.8
10	253°54.6	330°38.9	26.7	111°59.7	04.4	284°18.8	22.1	68°47.8	17.8
11	268°57.0	345°38.0	26.5	127°02.6	04.6	299°20.8	21.9	83°50.2	17.8
12	283°59.5	0°37.0	S23°26.3	142°05.6	N19°04.9	314°22.7	S13°21.7	98°52.7	N00°17.8
13	299°02.0	15°36.1	26.1	157°08.5	05.1	329°24.7	21.5	113°55.1	17.8
14	314°04.4	30°35.1	25.9	172°11.5	05.4	344°26.7	21.3	128°57.5	17.8
15	329°06.9	45°34.1	· · 25.7	187°14.4	· · 05.6	359°28.6	· · 21.2	143°59.9	· · 17.8
16	344°09.4	60°33.2	25.5	202°17.3	05.9	14°30.6	21.0	159°02.4	17.8
17	359°11.8	75°32.2	25.3	217°20.3	06.1	29°32.6	20.8	174°04.8	17.8
18	14°14.3	90°31.3	S23°25.1	232°23.2	N19°06.4	44°34.5	S13°20.6	189°07.2	N00°17.8
19	29°16.7	105°30.3	24.9	247°26.2	06.6	59°36.5	20.4	204°09.6	17.8
20	44°19.2	120°29.4	24.7	262°29.1	06.9	74°38.5	20.3	219°12.0	17.8
21	59°21.7	135°28.4	· · 24.5	277°32.1	· · 07.2	89°40.4	· · 20.1	234°14.5	· · 17.8
22	74°24.1	150°27.4	24.3	292°35.0	07.4	104°42.4	19.9	249°16.9	17.8
23	89°26.6	165°26.5	24.1	307°38.0	07.7	119°44.4	19.7	264°19.3	17.8
Mer.pass. 17:03		ν -1.0' d -0.2' m -3.92		ν 2.9' d 0.2' m -0.83		ν 2.0' d -0.2' m -2.10		ν 2.4' d -0.0' m 0.85	

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	104°29.1	180°25.5	S23°23.9	322°40.9	N19°07.9	134°46.3	S13°19.5	279°21.7	N00°17.8
1	119°31.5	195°24.6	23.7	337°43.9	08.2	149°48.3	19.4	294°24.2	17.8
2	134°34.0	210°23.6	23.4	352°46.8	08.4	164°50.3	19.2	309°26.6	17.7
3	149°36.5	225°22.7	· · 23.2	7°49.8	· · 08.7	179°52.2	· · 19.0	324°29.0	· · 17.7
4	164°38.9	240°21.7	23.0	22°52.8	08.9	194°54.2	18.8	339°31.4	17.7
5	179°41.4	255°20.7	22.8	37°55.7	09.2	209°56.1	18.6	354°33.9	17.7
6	194°43.9	270°19.8	S23°22.6	52°58.7	N19°09.5	224°58.1	S13°18.5	9°36.3	N00°17.7
7	209°46.3	285°18.8	22.4	68°01.6	09.7	240°00.1	18.3	24°38.7	17.7
8	224°48.8	300°17.9	22.1	83°04.6	10.0	255°02.0	18.1	39°41.1	17.7
9	239°51.2	315°16.9	· · 21.9	98°07.6	· · 10.2	270°04.0	· · 17.9	54°43.6	· · 17.7
10	254°53.7	330°16.0	21.7	113°10.5	10.5	285°06.0	17.7	69°46.0	17.7
11	269°56.2	345°15.0	21.5	128°13.5	10.7	300°07.9	17.6	84°48.4	17.7
12	284°58.6	0°14.1	S23°21.2	143°16.5	N19°11.0	315°09.9	S13°17.4	99°50.8	N00°17.7
13	300°01.1	15°13.1	21.0	158°19.4	11.3	330°11.8	17.2	114°53.3	17.7
14	315°03.6	30°12.2	20.8	173°22.4	11.5	345°13.8	17.0	129°55.7	17.7
15	330°06.0	45°11.2	· · 20.5	188°25.4	· · 11.8	0°15.8	· · 16.8	144°58.1	· · 17.7
16	345°08.5	60°10.2	20.3	203°28.4	12.0	15°17.7	16.7	160°00.5	17.7
17	0°11.0	75°09.3	20.1	218°31.4	12.3	30°19.7	16.5	175°03.0	17.7
18	15°13.4	90°08.3	S23°19.8	233°34.3	N19°12.6	45°21.7	S13°16.3	190°05.4	N00°17.7
19	30°15.9	105°07.4	19.6	248°37.3	12.8	60°23.6	16.1	205°07.8	17.7
20	45°18.3	120°06.4	19.4	263°40.3	13.1	75°25.6	15.9	220°10.2	17.7
21	60°20.8	135°05.5	· · 19.1	278°43.3	· · 13.4	90°27.5	· · 15.7	235°12.7	· · 17.7
22	75°23.3	150°04.5	18.9	293°46.3	13.6	105°29.5	15.6	250°15.1	17.7
23	90°25.7	165°03.6	18.6	308°49.2	13.9	120°31.5	15.4	265°17.5	17.7
Mer.pass. 16:59		ν -1.0' d -0.2' m -3.92		ν 3.0' d 0.3' m -0.86		ν 2.0' d -0.2' m -2.10		ν 2.4' d -0.0' m 0.85	

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	105°28.2	180°02.6	S23°18.4	323°52.2	N19°14.1	135°33.4	S13°15.2	280°19.9	N00°17.7
1	120°30.7	195°01.7	18.2	338°55.2	14.4	150°35.4	15.0	295°22.4	17.7
2	135°33.1	210°00.7	17.9	353°58.2	14.7	165°37.3	14.8	310°24.8	17.7
3	150°35.6	224°59.8	· · 17.7	9°01.2	· · 14.9	180°39.3	· · 14.7	325°27.2	· · 17.7
4	165°38.1	239°58.8	17.4	24°04.2	15.2	195°41.3	14.5	340°29.7	17.7
5	180°40.5	254°57.9	17.2	39°07.2	15.5	210°43.2	14.3	355°32.1	17.7
6	195°43.0	269°56.9	S23°16.9	54°10.2	N19°15.7	225°45.2	S13°14.1	10°34.5	N00°17.7
7	210°45.5	284°56.0	16.7	69°13.2	16.0	240°47.1	13.9	25°36.9	17.7
8	225°47.9	299°55.0	16.4	84°16.2	16.3	255°49.1	13.7	40°39.4	17.6
9	240°50.4	314°54.1	· · 16.2	99°19.2	· · 16.5	270°51.1	· · 13.6	55°41.8	· · 17.6
10	255°52.8	329°53.1	15.9	114°22.2	16.8	285°53.0	13.4	70°44.2	17.6
11	270°55.3	344°52.2	15.6	129°25.2	17.1	300°55.0	13.2	85°46.7	17.6
12	285°57.8	359°51.2	S23°15.4	144°28.2	N19°17.3	315°56.9	S13°13.0	100°49.1	N00°17.6
13	301°00.2	14°50.3	15.1	159°31.2	17.6	330°58.9	12.8	115°51.5	17.6
14	316°02.7	29°49.3	14.9	174°34.2	17.9	346°00.9	12.7	130°53.9	17.6
15	331°05.2	44°48.4	· · 14.6	189°37.2	· · 18.1	1°02.8	· · 12.5	145°56.4	· · 17.6
16	346°07.6	59°47.4	14.3	204°40.2	18.4	16°04.8	12.3	160°58.8	17.6
17	1°10.1	74°46.5	14.1	219°43.2	18.7	31°06.7	12.1	176°01.2	17.6
18	16°12.6	89°45.5	S23°13.8	234°46.3	N19°18.9	46°08.7	S13°11.9	191°03.7	N00°17.6
19	31°15.0	104°44.6	13.5	249°49.3	19.2	61°10.7	11.7	206°06.1	17.6
20	46°17.5	119°43.6	13.3	264°52.3	19.5	76°12.6	11.6	221°08.5	17.6
21	61°20.0	134°42.7	· · 13.0	279°55.3	· · 19.8	91°14.6	· · 11.4	236°11.0	· · 17.6
22	76°22.4	149°41.7	12.7	294°58.3	20.0	106°16.5	11.2	251°13.4	17.6
23	91°24.9	164°40.8	12.5	310°01.3	20.3	121°18.5	11.0	266°15.8	17.6
Mer.pass. 16:55		ν -1.0' d -0.2' m -3.92		ν 3.0' d 0.3' m -0.88		ν 2.0' d -0.2' m -2.09		ν 2.4' d -0.0' m 0.85	

Stars		
	SHA	Dec
Alpheratz	357°46.2	29°09.0
Ankaa	353°18.1	-42°15.2
Schedar	349°43.6	56°35.9
Diphda	348°58.4	-17°55.9
Achernar	335°28.3	-57°11.4
Hamal	328°03.5	23°30.8
Polaris	318°47.6	89°18.8
Acamar	315°19.9	-40°16.0
Menkar	314°17.5	4°07.8
Mirfak	308°43.7	49°54.1
Aldebaran	290°52.0	16°31.8
Rigel	281°14.1	-8°11.4
Capella	280°37.7	46°00.6
Bellatrix	278°34.4	6°21.5
Elnath	278°15.4	28°37.0
Alnilam	275°48.6	-1°11.7
Betelgeuse	271°03.7	7°24.5
Canopus	263°56.7	-52°42.1
Sirius	258°35.6	-16°43.8
Adhara	255°14.1	-28°59.2
Procyon	245°02.0	5°11.9
Pollux	243°30.4	28°00.0
Avior	234°18.5	-59°32.4
Suhail	222°54.0	-43°28.3
Miaplacidus	221°39.6	-69°45.4
Alphard	217°58.3	-8°42.2
Regulus	207°45.9	11°54.9
Dubhe	193°54.2	61°41.4
Denebola	182°36.1	14°30.7
Gienah	175°54.9	-17°35.9
Acrux	173°12.3	-63°09.1
Gacrux	172°03.8	-57°10.0
Alioth	166°22.7	55°53.9
Spica	158°34.0	-11°12.9
Alkaid	153°00.9	49°15.4
Hadar	148°51.9	-60°25.1
Menkent	148°10.7	-36°25.1
Arcturus	145°58.1	19°07.6
Rigil Kent.	139°55.6	-60°52.4
Kochab	137°20.2	74°06.5
Zuben'ubi	137°08.4	-16°05.0
Alphecca	126°13.3	26°40.6
Antares	112°29.7	-26°27.2
Atria	107°34.3	-69°02.6
Sabik	102°15.7	-15°44.3
Shaula	96°25.7	-37°06.6
Rasalhague	96°09.1	12°33.1
Eltanin	90°47.8	51°29.1
Kaus Aust.	83°47.5	-34°22.8
Vega	80°41.1	38°47.5
Nunki	76°01.8	-26°17.1
Altair	62°11.0	8°53.7
Peacock	53°23.6	-56°42.2
Deneb	49°33.7	45°19.1
Enif	33°49.9	9°55.3
Al Na'ir	27°47.1	-46°54.9
Fomalhaut	15°26.9	-29°34.2
Scheat	13°56.1	28°08.4
Markab	13°41.0	15°15.7

Jan 04 Mon	SHA	Mer.pass
Venus	77°18.6	11:58
Mars	218°00.5	02:33
Jupiter	30°29.2	15:02
Saturn	174°53.7	05:26

Jan 05 Tue	SHA	Mer.pass
Venus	75°56.5	11:59
Mars	218°11.9	02:29
Jupiter	30°17.3	14:59
Saturn	174°52.7	05:22

Jan 06 Wed	SHA	Mer.pass
Venus	74°34.4	12:01
Mars	218°24.0	02:24
Jupiter	30°05.2	14:56
Saturn	174°51.7	05:18

Horizontal parallax		
Venus:		0.1
Mars:		0.2

DUT1 = UT1-UTC = +0.1114 sec ΔT = TT-UT1 = +66.0726 sec

2010 January 04 to Jan. 06 UT

h Sun Moon						
Mon	GHA	Dec	GHA	ν	Dec	d HP
0	178°49.6	S22°45.2	314°13.2	9.4'	N09°04.7	-15.2' 60.4'
1	193°49.3	45.0	328°41.7	9.5'	08°49.6	-15.2' 60.4'
2	208°49.0	44.7	343°10.1	9.6'	08°34.3	-15.3' 60.3'
3	223°48.7	· · 44.4	357°38.7	9.6'	08°19.1	-15.3' 60.3'
4	238°48.5	44.2	12°07.3	9.7'	08°03.8	-15.3' 60.3'
5	253°48.2	43.9	26°36.0	9.8'	07°48.4	-15.4' 60.3'
6	268°47.9	S22°43.7	41°04.8	9.8'	N07°33.1	-15.4' 60.2'
7	283°47.6	43.4	55°33.6	9.9'	07°17.7	-15.4' 60.2'
8	298°47.3	43.1	70°02.6	10.0'	07°02.3	-15.4' 60.2'
9	313°47.0	· · 42.9	84°31.6	10.1'	06°46.9	-15.5' 60.2'
10	328°46.8	42.6	99°00.6	10.1'	06°31.4	-15.5' 60.1'
11	343°46.5	42.3	113°29.7	10.2'	06°15.9	-15.5' 60.1'
12	358°46.2	S22°42.1	127°58.9	10.3'	N06°00.4	-15.5' 60.1'
13	13°45.9	41.8	142°28.2	10.3'	05°44.9	-15.5' 60.1'
14	28°45.6	41.5	156°57.5	10.4'	05°29.4	-15.6' 60.0'
15	43°45.4	· · 41.3	171°26.9	10.4'	05°13.8	-15.6' 60.0'
16	58°45.1	41.0	185°56.3	10.5'	04°58.2	-15.6' 60.0'
17	73°44.8	40.7	200°25.8	10.6'	04°42.7	-15.6' 59.9'
18	88°44.5	S22°40.5	214°55.4	10.6'	N04°27.1	-15.6' 59.9'
19	103°44.2	40.2	229°25.0	10.7'	04°11.5	-15.6' 59.9'
20	118°43.9	39.9	243°54.7	10.7'	03°55.9	-15.6' 59.9'
21	133°43.7	· · 39.7	258°24.4	10.8'	03°40.3	-15.6' 59.8'
22	148°43.4	39.4	272°54.2	10.8'	03°24.7	-15.6' 59.8'
23	163°43.1	39.1	287°24.0	10.9'	03°09.1	-15.6' 59.8'
	SD = 16.3'	d = -0.3'			SD = 16.5'	
Tue	GHA	Dec	GHA	ν	Dec	d HP
0	178°42.8	S22°38.8	301°53.9	10.9'	N02°53.5	-15.6' 59.7'
1	193°42.5	38.6	316°23.9	11.0'	02°37.9	-15.6' 59.7'
2	208°42.3	38.3	330°53.8	11.0'	02°22.3	-15.6' 59.7'
3	223°42.0	· · 38.0	345°23.9	11.1'	02°06.7	-15.6' 59.6'
4	238°41.7	37.7	359°54.0	11.1'	01°51.1	-15.6' 59.6'
5	253°41.4	37.4	14°24.1	11.2'	01°35.5	-15.6' 59.6'
6	268°41.1	S22°37.2	28°54.3	11.2'	N01°19.9	-15.6' 59.5'
7	283°40.9	36.9	43°24.5	11.3'	01°04.3	-15.6' 59.5'
8	298°40.6	36.6	57°54.7	11.3'	00°48.8	-15.5' 59.5'
9	313°40.3	· · 36.3	72°25.0	11.3'	00°33.2	-15.5' 59.4'
10	328°40.0	36.0	86°55.4	11.4'	00°17.7	-15.5' 59.4'
11	343°39.8	35.8	101°25.8	11.4'	N00°02.2	-15.5' 59.4'
12	358°39.5	S22°35.5	115°56.2	11.5'	S00°13.3	15.5' 59.4'
13	13°39.2	35.2	130°26.6	11.5'	00°28.8	15.5' 59.3'
14	28°38.9	34.9	144°57.1	11.5'	00°44.2	15.4' 59.3'
15	43°38.6	· · 34.6	159°27.6	11.6'	00°59.6	15.4' 59.3'
16	58°38.4	34.3	173°58.2	11.6'	01°15.0	15.4' 59.2'
17	73°38.1	34.0	188°28.8	11.6'	01°30.4	15.4' 59.2'
18	88°37.8	S22°33.7	202°59.4	11.6'	S01°45.8	15.3' 59.2'
19	103°37.5	33.5	217°30.0	11.7'	02°01.1	15.3' 59.1'
20	118°37.3	33.2	232°00.7	11.7'	02°16.4	15.3' 59.1'
21	133°37.0	· · 32.9	246°31.4	11.7'	02°31.7	15.2' 59.1'
22	148°36.7	32.6	261°02.1	11.7'	02°46.9	15.2' 59.0'
23	163°36.4	32.3	275°32.9	11.8'	03°02.1	15.2' 59.0'
	SD = 16.3'	d = -0.3'			SD = 16.3'	
Wed	GHA	Dec	GHA	ν	Dec	d HP
0	178°36.2	S22°32.0	290°03.6	11.8'	S03°17.3	15.1' 59.0'
1	193°35.9	31.7	304°34.4	11.8'	03°32.4	15.1' 58.9'
2	208°35.6	31.4	319°05.2	11.8'	03°47.6	15.1' 58.9'
3	223°35.3	· · 31.1	333°36.1	11.8'	04°02.6	15.0' 58.9'
4	238°35.1	30.8	348°06.9	11.9'	04°17.6	15.0' 58.8'
5	253°34.8	30.5	2°37.8	11.9'	04°32.6	14.9' 58.8'
6	268°34.5	S22°30.2	17°08.6	11.9'	S04°47.6	14.9' 58.8'
7	283°34.2	29.9	31°39.5	11.9'	05°02.5	14.9' 58.7'
8	298°34.0	29.6	46°10.5	11.9'	05°17.4	14.8' 58.7'
9	313°33.7	· · 29.3	60°41.4	11.9'	05°32.2	14.8' 58.6'
10	328°33.4	29.0	75°12.3	11.9'	05°46.9	14.7' 58.6'
11	343°33.1	28.7	89°43.3	12.0'	06°01.7	14.7' 58.6'
12	358°32.9	S22°28.4	104°14.2	12.0'	S06°16.3	14.6' 58.5'
13	13°32.6	28.1	118°45.2	12.0'	06°31.0	14.6' 58.5'
14	28°32.3	27.8	133°16.1	12.0'	06°45.6	14.5' 58.5'
15	43°32.0	· · 27.5	147°47.1	12.0'	07°00.1	14.5' 58.4'
16	58°31.8	27.2	162°18.1	12.0'	07°14.6	14.4' 58.4'
17	73°31.5	26.9	176°49.1	12.0'	07°29.0	14.4' 58.4'
18	88°31.2	S22°26.6	191°20.1	12.0'	S07°43.4	14.3' 58.3'
19	103°31.0	26.3	205°51.1	12.0'	07°57.7	14.3' 58.3'
20	118°30.7	26.0	220°22.0	12.0'	08°11.9	14.2' 58.3'
21	133°30.4	· · 25.7	234°53.0	12.0'	08°26.1	14.1' 58.2'
22	148°30.1	25.3	249°24.0	12.0'	08°40.3	14.1' 58.2'
23	163°29.9	25.0	263°55.0	12.0'	08°54.3	14.0' 58.2'
	SD = 16.3'	d = -0.3'			SD = 16.1'	

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:20	10:30	■	■	13:41	15:52
N 70°	08:02	09:43	■	■	14:28	16:10
68°	07:47	09:12	11:30	12:41	14:59	16:24
66°	07:35	08:50	10:20	13:51	15:22	16:36
64°	07:25	08:32	09:45	14:26	15:40	16:46
62°	07:16	08:17	09:20	14:52	15:54	16:55
60°	07:08	08:04	09:00	15:11	16:07	17:03
N 58°	07:01	07:53	08:44	15:28	16:18	17:10
56°	06:55	07:43	08:30	15:41	16:28	17:16
54°	06:49	07:35	08:18	15:53	16:36	17:22
52°	06:44	07:27	08:07	16:04	16:44	17:27
50°	06:39	07:20	07:58	16:13	16:51	17:32
45°	06:28	07:05	07:38	16:33	17:06	17:43
N 40°	06:18	06:52	07:22	16:49	17:19	17:53
35°	06:09	06:41	07:09	17:02	17:30	18:02
30°	06:01	06:30	06:57	17:14	17:40	18:10
20°	05:45	06:12	06:36	17:35	17:58	18:26
N 10°	05:30	05:56	06:18	17:52	18:15	18:41
0°	05:13	05:39	06:02	18:09	18:31	18:58
S 10°	04:55	05:22	05:45	18:26	18:49	19:16
20°	04:33	05:02	05:26	18:44	19:09	19:38
30°	04:05	04:38	05:05	19:05	19:32	20:05
35°	03:47	04:23	04:53	19:18	19:47	20:23
40°	03:25	04:06	04:38	19:32	20:05	20:45
45°	02:56	03:44	04:21	19:49	20:26	21:14
S 50°	02:13	03:16	04:00	20:11	20:54	21:56
52°	01:49	03:02	03:49	20:21	21:08	22:21
54°	01:12	02:45	03:38	20:33	21:25	22:56
56°	///	02:25	03:24	20:46	21:45	///
58°	///	01:58	03:08	21:01	22:11	///
S 60°	///	01:19	02:50	21:20	22:49	///

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	21:03	23:19	· ·	10:43	10:14	09:45
N 70°	21:08	23:14	· ·	10:35	10:14	09:53
68°	21:12	23:10	· ·	10:28	10:13	10:00
66°	21:16	23:07	· ·	10:22	10:13	10:05
64°	21:19	23:04	· ·	10:17	10:13	10:10
62°	21:21	23:02	· ·	10:12	10:13	10:14
60°	21:24	23:00	· ·	10:08	10:13	10:18
N 58°	21:26	22:58	· ·	10:05	10:13	10:21
56°	21:27	22:57	· ·	10:02	10:13	10:24
54°	21:29	22:55	· ·	09:59	10:13	10:26
52°	21:30	22:54	· ·	09:57	10:13	10:29
50°	21:32	22:53	· ·	09:54	10:13	10:31
45°	21:34	22:50	· ·	09:49	10:13	10:36
N 40°	21:37	22:48	23:58	09:45	10:13	10:40
35°	21:39	22:46	23:52	09:41	10:13	10:43
30°	21:40	22:45	23:48	09:38	10:13	10:46
20°	21:43	22:42	23:40	09:32	10:12	10:51
N 10°	21:46	22:40	23:32	09:27	10:12	10:56
0°	21:49	22:38	23:26	09:22	10:12	11:01
S 10°	21:51	22:36	23:19	09:17	10:12	11:05
20°	21:54	22:33	23:12	09:12	10:12	11:10
30°	21:57	22:31	23:04	09:06	10:11	11:15
35°	21:59	22:29	23:00	09:02	10:11	11:18
40°	22:01	22:28	22:55	08:58	10:11	11:22
45°	22:03	22:26	22:49	08:53	10:11	11:26
S 50°	22:06	22:24	22:42	08:48	10:11	11:31
52°	22:07	22:23	22:39	08:45	10:10	11:33
54°	22:08	22:22	22:35	08:42	10:10	11:36
56°	22:10	22:21	22:31	08:39	10:10	11:38
58°	22:12	22:19	22:27	08:35	10:10	11:41
S 60°	22:13	22:18	22:22	08:31	10:10	11:45

Day	Sun		Mer. Pass	Moon		Age 19-21 86-66%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
04	04:42	04:55	12:05	03:10	15:35	
05	05:09	05:22	12:05	04:00	16:25	
06	05:35	05:49	12:06	04:49	17:13	

January 07, 08, 09 UT (Thu., Fri., Sat.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	106°27.3	179°39.9	S23°12.2	325°04.4	N19°20.6	136°20.4	S13°10.8	281°18.2	N00°17.6	Alpheratz	357°46.3	29°09.0											
1	121°29.8	194°38.9	11.9	340°07.4	20.8	151°22.4	10.6	296°20.7	17.6	Ankaa	353°18.1	-42°15.2											
2	136°32.3	209°38.0	11.6	355°10.4	21.1	166°24.4	10.5	311°23.1	17.6	Schedar	349°43.6	56°35.9											
3	151°34.7	224°37.0	•• 11.4	10°13.4	•• 21.4	181°26.3	•• 10.3	326°25.5	•• 17.6	Diphda	348°58.4	-17°55.9											
4	166°37.2	239°36.1	11.1	25°16.5	21.7	196°28.3	10.1	341°28.0	17.6	Achernar	335°28.4	-57°11.4											
5	181°39.7	254°35.1	10.8	40°19.5	21.9	211°30.2	09.9	356°30.4	17.6	Hamal	328°03.5	23°30.8											
6	196°42.1	269°34.2	S23°10.5	55°22.5	N19°22.2	226°32.2	S13°09.7	11°32.8	N00°17.6	Polaris	318°48.7	89°18.9											
7	211°44.6	284°33.2	10.3	70°25.6	22.5	241°34.2	09.6	26°35.3	17.6	Acamar	315°19.9	-40°16.0											
8	226°47.1	299°32.3	10.0	85°28.6	22.7	256°36.1	09.4	41°37.7	17.6	Menkar	314°17.5	4°07.8											
9	241°49.5	314°31.3	•• 09.7	100°31.6	•• 23.0	271°38.1	•• 09.2	56°40.1	•• 17.6	Mirfak	308°43.7	49°54.1											
10	256°52.0	329°30.4	09.4	115°34.7	23.3	286°40.0	09.0	71°42.6	17.6	Aldebaran	290°52.0	16°31.8											
11	271°54.4	344°29.5	09.1	130°37.7	23.6	301°42.0	08.8	86°45.0	17.6	Rigel	281°14.2	-8°11.4											
12	286°56.9	359°28.5	S23°08.8	145°40.7	N19°23.8	316°43.9	S13°08.6	101°47.4	N00°17.6	Capella	280°37.7	46°00.6											
13	301°59.4	14°27.6	08.5	160°43.8	24.1	331°45.9	08.5	116°49.9	17.6	Bellatrix	278°34.4	6°21.5											
14	317°01.8	29°26.6	08.2	175°46.8	24.4	346°47.9	08.3	131°52.3	17.6	Elnath	278°15.4	28°37.0											
15	332°04.3	44°25.7	•• 08.0	190°49.9	•• 24.7	1°49.8	•• 08.1	146°54.7	•• 17.6	Alnilam	275°48.6	-1°11.8											
16	347°06.8	59°24.7	07.7	205°52.9	24.9	16°51.8	07.9	161°57.2	17.6	Betelgeuse	271°03.7	7°24.5											
17	2°09.2	74°23.8	07.4	220°55.9	25.2	31°53.7	07.7	176°59.6	17.6	Canopus	263°56.7	-52°42.1											
18	17°11.7	89°22.9	S23°07.1	235°59.0	N19°25.5	46°55.7	S13°07.5	192°02.0	N00°17.6	Sirius	258°35.6	-16°43.9											
19	32°14.2	104°21.9	06.8	251°02.0	25.8	61°57.6	07.3	207°04.5	17.6	Adhara	255°14.1	-28°59.2											
20	47°16.6	119°21.0	06.5	266°05.1	26.1	76°59.6	07.2	222°06.9	17.6	Procyon	245°02.0	5°11.9											
21	62°19.1	134°20.0	•• 06.2	281°08.1	•• 26.3	92°01.5	•• 07.0	237°09.3	•• 17.6	Pollux	243°30.4	28°00.0											
22	77°21.6	149°19.1	05.9	296°11.2	26.6	107°03.5	06.8	252°11.8	17.6	Avior	234°18.5	-59°32.5											
23	92°24.0	164°18.2	05.6	311°14.2	26.9	122°05.5	06.6	267°14.2	17.6	Suhail	222°54.0	-43°28.4											
Mer.pass. 16:51				ν -0.9' d -0.3' m -3.92				ν 3.0' d 0.3' m -0.90				ν 2.0' d -0.2' m -2.09				ν 2.4' d -0.0' m 0.84							

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	107°26.5	179°17.2	S23°05.3	326°17.3	N19°27.2	137°07.4	S13°06.4	282°16.7	N00°17.6										
1	122°28.9	194°16.3	05.0	341°20.4	27.4	152°09.4	06.2	297°19.1	17.6										
2	137°31.4	209°15.3	04.7	356°23.4	27.7	167°11.3	06.1	312°21.5	17.6										
3	152°33.9	224°14.4	•• 04.4	11°26.5	•• 28.0	182°13.3	•• 05.9	327°24.0	•• 17.6										
4	167°36.3	239°13.5	04.1	26°29.5	28.3	197°15.2	05.7	342°26.4	17.6										
5	182°38.8	254°12.5	03.7	41°32.6	28.6	212°17.2	05.5	357°28.8	17.6										
6	197°41.3	269°11.6	S23°03.4	56°35.7	N19°28.8	227°19.1	S13°05.3	12°31.3	N00°17.6										
7	212°43.7	284°10.6	03.1	71°38.7	29.1	242°21.1	05.1	27°33.7	17.6										
8	227°46.2	299°09.7	02.8	86°41.8	29.4	257°23.0	05.0	42°36.1	17.6										
9	242°48.7	314°08.8	•• 02.5	101°44.9	•• 29.7	272°25.0	•• 04.8	57°38.6	•• 17.6										
10	257°51.1	329°07.8	02.2	116°47.9	30.0	287°27.0	04.6	72°41.0	17.6										
11	272°53.6	344°06.9	01.9	131°51.0	30.3	302°28.9	04.4	87°43.5	17.6										
12	287°56.1	359°05.9	S23°01.6	146°54.1	N19°30.5	317°30.9	S13°04.2	102°45.9	N00°17.6										
13	302°58.5	14°05.0	01.2	161°57.1	30.8	332°32.8	04.0	117°48.3	17.6										
14	318°01.0	29°04.1	00.9	177°00.2	31.1	347°34.8	03.8	132°50.8	17.6										
15	333°03.4	44°03.1	•• 00.6	192°03.3	•• 31.4	2°36.7	•• 03.7	147°53.2	•• 17.6										
16	348°05.9	59°02.2	23°00.3	207°06.4	31.7	17°38.7	03.5	162°55.6	17.6										
17	3°08.4	74°01.3	22°59.9	222°09.5	32.0	32°40.6	03.3	177°58.1	17.6										
18	18°10.8	89°00.3	S22°59.6	237°12.5	N19°32.2	47°42.6	S13°03.1	193°00.5	N00°17.6										
19	33°13.3	103°59.4	59.3	252°15.6	32.5	62°44.5	02.9	208°03.0	17.6										
20	48°15.8	118°58.5	59.0	267°18.7	32.8	77°46.5	02.7	223°05.4	17.7										
21	63°18.2	133°57.5	•• 58.6	282°21.8	•• 33.1	92°48.4	•• 02.5	238°07.8	•• 17.7										
22	78°20.7	148°56.6	58.3	297°24.9	33.4	107°50.4	02.4	253°10.3	17.7										
23	93°23.2	163°55.7	58.0	312°28.0	33.7	122°52.4	02.2	268°12.7	17.7										
Mer.pass. 16:47				ν -0.9' d -0.3' m -3.92				ν 3.1' d 0.3' m -0.93				ν 2.0' d -0.2' m -2.09				ν 2.4' d 0.0' m 0.84			

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	108°25.6	178°54.7	S22°57.6	327°31.0	N19°33.9	137°54.3	S13°02.0	283°15.2	N00°17.7										
1	123°28.1	193°53.8	57.3	342°34.1	34.2	152°56.3	01.8	298°17.6	17.7										
2	138°30.5	208°52.9	57.0	357°37.2	34.5	167°58.2	01.6	313°20.0	17.7										
3	153°33.0	223°51.9	•• 56.6	12°40.3	•• 34.8	183°00.2	•• 01.4	328°22.5	•• 17.7										
4	168°35.5	238°51.0	56.3	27°43.4	35.1	198°02.1	01.2	343°24.9	17.7										
5	183°37.9	253°50.1	56.0	42°46.5	35.4	213°04.1	01.1	358°27.4	17.7										
6	198°40.4	268°49.1	S22°55.6	57°49.6	N19°35.7	228°06.0	S13°00.9	13°29.8	N00°17.7										
7	213°42.9	283°48.2	55.3	72°52.7	36.0	243°08.0	00.7	28°32.2	17.7										
8	228°45.3	298°47.3	54.9	87°55.8	36.2	258°09.9	00.5	43°34.7	17.7										
9	243°47.8	313°46.3	•• 54.6	102°58.9	•• 36.5	273°11.9	•• 00.3	58°37.1	•• 17.7										
10	258°50.3	328°45.4	54.3	118°02.0	36.8	288°13.8	13°00.1	73°39.6	17.7										
11	273°52.7	343°44.5	53.9	133°05.1	37.1	303°15.8	12°59.9	88°42.0	17.7										
12	288°55.2	358°43.5	S22°53.6	148°08.2	N19°37.4	318°17.7	S12°59.8	103°44.5	N00°17.7										
13	303°57.7	13°42.6	53.2	163°11.3	37.7	333°19.7	59.6	118°46.9	17.7										
14	319°00.1	28°41.7	52.9	178°14.4	38.0	348°21.6	59.4	133°49.3	17.7										
15	334°02.6	43°40.8	•• 52.5	193°17.5	•• 38.3	3°23.6	•• 59.2	148°51.8	•• 17.7										
16	349°05.0	58°39.8	52.2	208°20.6	38.6	18°25.5	59.0	163°54.2	17.7										
17	4°07.5	73°38.9	51.8	223°23.7	38.8	33°27.5	58.8	178°56.7	17.7										
18	19°10.0	88°38.0	S22°51.5	238°26.9	N19°39.1	48°29.4	S12°58.6	193°59.1	N00°17.7										
19	34°12.4	103°37.0	51.1	253°30.0	39.4	63°31.4	58.5	209°01.5	17.7										
20	49°14.9	118°36.1	50.7	268°33.1	39.7	78°33.3	58.3	224°04.0	17.7										
21	64°17.4	133°35.2	•• 50.4	283°36.2	•• 40.0	93°35.3	•• 58.1	239°06.4	•• 17.7										
22	79°19.8	148°34.3	50.0	298°39.3	40.3	108°37.2	57.9	254°08.9	17.7										
23	94°22.3	163°33.3	49.7	313°42.4	40.6	123°39.2	57.7	269°11.3	17.7										
Mer.pass. 16:44				ν -0.9' d -0.3' m -3.92				ν 3.1' d 0.3' m -0.95				ν 2.0' d -0.2' m -2.08				ν 2.4' d 0.0' m 0.83			

Jan 07 Thu	SHA	Mer.pass
Venus	73°12.5	12:02
Mars	218°37.0	02:19
Jupiter	29°53.1	14:53
Saturn	174°50.9	05:14

Jan 08 Fri	SHA	Mer.pass
Venus	71°50.7	12:04
Mars	218°50.8	02:14
Jupiter	29°40.9	14:50
Saturn	174°50.2	05:10

Jan 09 Sat	SHA	Mer.pass
Venus	70°29.1	12:05
Mars	219°05.4	02:09
Jupiter	29°28.7	14:46
Saturn	174°49.5	05:06

Horizontal parallax		
Venus:	0.1	
Mars:	0.2	

h	Sun			Moon			
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	178°29.6	S22°24.7	278°26.0	12.0'	S09°08.4	14.0'	58.1'
1	193°29.3	24.4	292°57.0	12.0'	09°22.3	13.9'	58.1'
2	208°29.1	24.1	307°27.9	12.0'	09°36.2	13.8'	58.1'
3	223°28.8	· · 23.8	321°58.9	12.0'	09°50.0	13.8'	58.0'
4	238°28.5	23.5	336°29.9	12.0'	10°03.8	13.7'	58.0'
5	253°28.2	23.1	351°00.8	11.9'	10°17.5	13.6'	58.0'
6	268°28.0	S22°22.8	5°31.8	11.9'	S10°31.1	13.6'	57.9'
7	283°27.7	22.5	20°02.7	11.9'	10°44.7	13.5'	57.9'
8	298°27.4	22.2	34°33.6	11.9'	10°58.2	13.4'	57.9'
9	313°27.2	· · 21.9	49°04.6	11.9'	11°11.6	13.4'	57.8'
10	328°26.9	21.6	63°35.5	11.9'	11°25.0	13.3'	57.8'
11	343°26.6	21.2	78°06.3	11.9'	11°38.3	13.2'	57.8'
12	358°26.4	S22°20.9	92°37.2	11.9'	S11°51.5	13.1'	57.7'
13	13°26.1	20.6	107°08.1	11.8'	12°04.6	13.1'	57.7'
14	28°25.8	20.3	121°38.9	11.8'	12°17.7	13.0'	57.7'
15	43°25.6	· · 19.9	136°09.8	11.8'	12°30.7	12.9'	57.6'
16	58°25.3	19.6	150°40.6	11.8'	12°43.6	12.8'	57.6'
17	73°25.0	19.3	165°11.4	11.8'	12°56.4	12.8'	57.6'
18	88°24.7	S22°19.0	179°42.1	11.8'	S13°09.2	12.7'	57.5'
19	103°24.5	18.6	194°12.9	11.7'	13°21.9	12.6'	57.5'
20	118°24.2	18.3	208°43.6	11.7'	13°34.5	12.5'	57.5'
21	133°23.9	· · 18.0	223°14.4	11.7'	13°47.0	12.4'	57.4'
22	148°23.7	17.7	237°45.0	11.7'	13°59.4	12.4'	57.4'
23	163°23.4	17.3	252°15.7	11.7'	14°11.8	12.3'	57.4'
SD = 16.3'			SD = 15.9'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	178°23.1	S22°17.0	266°46.4	11.6'	S14°24.0	12.2'	57.3'
1	193°22.9	16.7	281°17.0	11.6'	14°36.2	12.1'	57.3'
2	208°22.6	16.3	295°47.6	11.6'	14°48.3	12.0'	57.3'
3	223°22.4	· · 16.0	310°18.2	11.6'	15°00.3	11.9'	57.2'
4	238°22.1	15.7	324°48.7	11.5'	15°12.3	11.8'	57.2'
5	253°21.8	15.3	339°19.3	11.5'	15°24.1	11.8'	57.2'
6	268°21.6	S22°15.0	353°49.8	11.5'	S15°35.9	11.7'	57.2'
7	283°21.3	14.7	8°20.2	11.4'	15°47.5	11.6'	57.1'
8	298°21.0	14.3	22°50.7	11.4'	15°59.1	11.5'	57.1'
9	313°20.8	· · 14.0	37°21.1	11.4'	16°10.6	11.4'	57.1'
10	328°20.5	13.7	51°51.5	11.4'	16°22.0	11.3'	57.0'
11	343°20.2	13.3	66°21.9	11.3'	16°33.3	11.2'	57.0'
12	358°20.0	S22°13.0	80°52.2	11.3'	S16°44.5	11.1'	57.0'
13	13°19.7	12.6	95°22.5	11.3'	16°55.6	11.0'	56.9'
14	28°19.4	12.3	109°52.8	11.2'	17°06.6	10.9'	56.9'
15	43°19.2	· · 11.9	124°23.0	11.2'	17°17.5	10.8'	56.9'
16	58°18.9	11.6	138°53.3	11.2'	17°28.3	10.7'	56.8'
17	73°18.7	11.3	153°23.4	11.2'	17°39.0	10.6'	56.8'
18	88°18.4	S22°10.9	167°53.6	11.1'	S17°49.7	10.5'	56.8'
19	103°18.1	10.6	182°23.7	11.1'	18°00.2	10.4'	56.8'
20	118°17.9	10.2	196°53.8	11.1'	18°10.6	10.3'	56.7'
21	133°17.6	· · 09.9	211°23.9	11.0'	18°20.9	10.2'	56.7'
22	148°17.3	09.5	225°53.9	11.0'	18°31.1	10.1'	56.7'
23	163°17.1	09.2	240°23.9	11.0'	18°41.3	10.0'	56.6'
SD = 16.3'			SD = 15.6'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	178°16.8	S22°08.8	254°53.8	10.9'	S18°51.3	9.9'	56.6'
1	193°16.6	08.5	269°23.8	10.9'	19°01.2	9.8'	56.6'
2	208°16.3	08.1	283°53.7	10.9'	19°11.0	9.7'	56.6'
3	223°16.0	· · 07.8	298°23.5	10.8'	19°20.7	9.6'	56.5'
4	238°15.8	07.4	312°53.3	10.8'	19°30.3	9.5'	56.5'
5	253°15.5	07.1	327°23.1	10.8'	19°39.8	9.4'	56.5'
6	268°15.3	S22°06.7	341°52.9	10.7'	S19°49.2	9.3'	56.4'
7	283°15.0	06.4	356°22.6	10.7'	19°58.4	9.2'	56.4'
8	298°14.7	06.0	10°52.3	10.7'	20°07.6	9.1'	56.4'
9	313°14.5	· · 05.7	25°22.0	10.6'	20°16.7	8.9'	56.4'
10	328°14.2	05.3	39°51.6	10.6'	20°25.6	8.8'	56.3'
11	343°14.0	05.0	54°21.2	10.6'	20°34.5	8.7'	56.3'
12	358°13.7	S22°04.6	68°50.7	10.5'	S20°43.2	8.6'	56.3'
13	13°13.5	04.2	83°20.3	10.5'	20°51.8	8.5'	56.3'
14	28°13.2	03.9	97°49.8	10.5'	21°00.3	8.4'	56.2'
15	43°12.9	· · 03.5	112°19.2	10.4'	21°08.7	8.3'	56.2'
16	58°12.7	03.2	126°48.6	10.4'	21°16.9	8.2'	56.2'
17	73°12.4	02.8	141°18.0	10.4'	21°25.1	8.0'	56.1'
18	88°12.2	S22°02.4	155°47.4	10.3'	S21°33.1	7.9'	56.1'
19	103°11.9	02.1	170°16.7	10.3'	21°41.1	7.8'	56.1'
20	118°11.7	01.7	184°46.0	10.3'	21°48.9	7.7'	56.1'
21	133°11.4	· · 01.3	199°15.3	10.2'	21°56.6	7.6'	56.0'
22	148°11.1	01.0	213°44.5	10.2'	22°04.2	7.5'	56.0'
23	163°10.9	00.6	228°13.7	10.2'	22°11.6	7.3'	56.0'
SD = 16.3'			SD = 15.4'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:15	10:19	■	■	13:55	15:59
N 70°	07:58	09:36	■	■	14:37	16:16
68°	07:44	09:08	11:10	13:04	15:06	16:30
66°	07:33	08:46	10:13	14:01	15:28	16:41
64°	07:23	08:29	09:40	14:34	15:45	16:51
62°	07:14	08:14	09:16	14:58	16:00	17:00
60°	07:07	08:02	08:57	15:17	16:12	17:07
N 58°	07:00	07:51	08:41	15:32	16:22	17:14
56°	06:54	07:42	08:28	15:46	16:32	17:20
54°	06:48	07:34	08:16	15:57	16:40	17:25
52°	06:43	07:26	08:06	16:08	16:48	17:31
50°	06:38	07:19	07:57	16:17	16:54	17:35
45°	06:28	07:04	07:38	16:36	17:09	17:46
N 40°	06:18	06:52	07:22	16:52	17:22	17:56
35°	06:09	06:41	07:09	17:05	17:33	18:04
30°	06:01	06:31	06:57	17:17	17:43	18:12
20°	05:46	06:13	06:37	17:36	18:00	18:28
N 10°	05:31	05:57	06:19	17:54	18:17	18:43
0°	05:15	05:41	06:03	18:10	18:33	18:59
S 10°	04:57	05:23	05:46	18:27	18:50	19:17
20°	04:35	05:04	05:28	18:45	19:09	19:38
30°	04:08	04:41	05:08	19:06	19:33	20:05
35°	03:50	04:26	04:55	19:18	19:47	20:23
40°	03:28	04:09	04:41	19:32	20:04	20:44
45°	03:00	03:48	04:24	19:49	20:25	21:13
S 50°	02:19	03:20	04:03	20:09	20:52	21:53
52°	01:55	03:06	03:53	20:19	21:06	22:16
54°	01:22	02:50	03:42	20:31	21:22	22:49
56°	///	02:30	03:29	20:44	21:42	///
58°	///	02:05	03:14	20:59	22:06	///
S 60°	///	01:29	02:55	21:17	22:41	///

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	01:36	04:12	■	09:11	08:17	■
N 70°	01:19	03:34	■	09:30	08:57	■
68°	01:07	03:08	05:24	09:44	09:25	08:51
66°	00:56	02:48	04:45	09:56	09:46	09:32
64°	00:48	02:32	04:18	10:06	10:03	10:00
62°	00:41	02:19	03:57	10:15	10:17	10:21
60°	00:34	02:07	03:40	10:23	10:29	10:39
N 58°	00:29	01:58	03:26	10:29	10:40	10:54
56°	00:24	01:50	03:14	10:35	10:49	11:06
54°	00:19	01:42	03:03	10:41	10:57	11:17
52°	00:16	01:36	02:54	10:45	11:04	11:27
50°	00:12	01:30	02:46	10:50	11:11	11:36
45°	00:04	01:17	02:28	10:59	11:25	11:55
N 40°	· · ·	01:06	02:14	11:07	11:37	12:10
35°	· · ·	00:57	02:02	11:14	11:47	12:23
30°	· · ·	00:50	01:51	11:20	11:56	12:35
20°	· · ·	00:36	01:33	11:31	12:11	12:54
N 10°	· · ·	00:25	01:18	11:40	12:25	13:11
0°	· · ·	00:14	01:03	11:49	12:37	13:27
S 10°	· · ·	00:03	00:49	11:57	12:50	13:43
20°	23:52	· · ·	00:33	12:07	13:04	14:00
30°	23:39	· · ·	00:16	12:17	13:19	14:20
35°	23:31	· · ·	00:06	12:24	13:28	14:32
40°	23:23	23:54	· · ·	12:31	13:39	14:45
45°	23:13	23:41	· · ·	12:39	13:51	15:01
S 50°	23:02	23:25	23:53	12:49	14:06	15:21
52°	22:56	23:17	23:43	12:54	14:13	15:30
54°	22:50	23:08	23:32	12:59	14:21	15:40
56°	22:44	22:59	23:20	13:05	14:30	15:52
58°	22:36	22:48	23:05	13:11	14:40	16:06
S 60°	22:28	22:36	22:48	13:18	14:51	16:22

Day	Sun		Mer.	Moon		Age
	Eqn.of Time	12h	Pass	Upper	Lower	22-24
	00h	mm:ss	hh:mm	hh:mm	hh:mm	55-34%
07	06:02	06:15	12:06	05:37	18:01	
08	06:27	06:40	12:07	06:26	18:50	
09	06:53	07:05	12:07	07:15	19:40	

January 10, 11, 12 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	109°24.8	178°32.4	S22°49.3	328°45.6	N19°40.9	138°41.1	S12°57.5	284°13.8	N00°17.7	Alpheratz	357°46.3	29°09.0
1	124°27.2	193°31.5	48.9	343°48.7	41.2	153°43.1	57.3	299°16.2	17.7	Ankaa	353°18.1	-42°15.2
2	139°29.7	208°30.5	48.6	358°51.8	41.5	168°45.0	57.1	314°18.7	17.8	Schedar	349°43.6	56°35.9
3	154°32.2	223°29.6	•• 48.2	13°54.9	•• 41.8	183°47.0	•• 57.0	329°21.1	•• 17.8	Diphda	348°58.4	-17°55.9
4	169°34.6	238°28.7	47.9	28°58.1	42.1	198°48.9	56.8	344°23.5	17.8	Achernar	335°28.4	-57°11.4
5	184°37.1	253°27.8	47.5	44°01.2	42.4	213°50.9	56.6	359°26.0	17.8	Hamal	328°03.5	23°30.8
6	199°39.5	268°26.8	S22°47.1	59°04.3	N19°42.6	228°52.8	S12°56.4	14°28.4	N00°17.8	Polaris	318°49.7	89°18.9
7	214°42.0	283°25.9	46.7	74°07.4	42.9	243°54.8	56.2	29°30.9	17.8	Acamar	315°20.0	-40°16.0
8	229°44.5	298°25.0	46.4	89°10.6	43.2	258°56.7	56.0	44°33.3	17.8	Menkar	314°17.5	4°07.8
9	244°46.9	313°24.1	•• 46.0	104°13.7	•• 43.5	273°58.7	•• 55.8	59°35.8	•• 17.8	Mirfak	308°43.7	49°54.1
10	259°49.4	328°23.2	45.6	119°16.8	43.8	289°00.6	55.6	74°38.2	17.8	Aldebaran	290°52.0	16°31.8
11	274°51.9	343°22.2	45.3	134°20.3	44.1	304°02.6	55.5	89°40.7	17.8	Rigel	281°14.2	-8°11.4
12	289°54.3	358°21.3	S22°44.9	149°23.1	N19°44.4	319°04.5	S12°55.3	104°43.1	N00°17.8	Capella	280°37.7	46°00.6
13	304°56.8	13°20.4	44.5	164°26.3	44.7	334°06.5	55.1	119°45.6	17.8	Bellatrix	278°34.4	6°21.5
14	319°59.3	28°19.5	44.1	179°29.4	45.0	349°08.4	54.9	134°48.0	17.8	Elnath	278°15.4	28°37.0
15	335°01.7	43°18.5	•• 43.7	194°32.5	•• 45.3	4°10.4	•• 54.7	149°50.4	•• 17.8	Alnilam	275°48.6	-1°11.8
16	350°04.2	58°17.6	43.4	209°35.7	45.6	19°12.3	54.5	164°52.9	17.8	Betelgeuse	271°03.7	7°24.5
17	5°06.7	73°16.7	43.0	224°38.8	45.9	34°14.3	54.3	179°55.3	17.8	Canopus	263°56.7	-52°42.1
18	20°09.1	88°15.8	S22°42.6	239°42.0	N19°46.2	49°16.2	S12°54.1	194°57.8	N00°17.8	Sirius	258°35.6	-16°43.9
19	35°11.6	103°14.9	42.2	254°45.1	46.5	64°18.2	54.0	210°00.2	17.8	Adhara	255°14.1	-28°59.2
20	50°14.0	118°13.9	41.8	269°48.3	46.8	79°20.1	53.8	225°02.7	17.8	Procyon	245°02.0	5°11.9
21	65°16.5	133°13.0	•• 41.4	284°51.4	•• 47.1	94°22.1	•• 53.6	240°05.1	•• 17.9	Pollux	243°30.4	28°00.0
22	80°19.0	148°12.1	41.1	299°54.6	47.4	109°24.0	53.4	255°07.6	17.9	Avior	234°18.5	-59°32.5
23	95°21.4	163°11.2	40.7	314°57.7	47.7	124°26.0	53.2	270°10.0	17.9	Suhail	222°54.0	-43°28.4
Mer.pass. 16:40		ν -0.9' d -0.4' m -3.92		ν 3.1' d 0.3' m -0.98		ν 2.0' d -0.2' m -2.08		ν 2.4' d 0.0' m 0.83				

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	110°23.9	178°10.3	S22°40.3	330°00.9	N19°48.0	139°27.9	S12°53.0	285°12.5	N00°17.9	Alphard	217°58.3	-8°42.2
1	125°26.4	193°09.3	39.9	345°04.0	48.3	154°29.9	52.8	300°14.9	17.9	Regulus	207°45.9	11°54.9
2	140°28.8	208°08.4	39.5	0°07.2	48.6	169°31.8	52.6	315°17.4	17.9	Dubhe	193°54.1	61°41.4
3	155°31.3	223°07.5	•• 39.1	15°10.3	•• 48.9	184°33.8	•• 52.4	330°19.8	•• 17.9	Denebola	182°36.0	14°30.7
4	170°33.8	238°06.6	38.7	30°13.5	49.2	199°35.7	52.3	345°22.3	17.9	Gienah	175°54.8	-17°35.9
5	185°36.2	253°05.7	38.3	45°16.6	49.5	214°37.6	52.1	0°24.7	17.9	Acrux	173°12.2	-63°09.1
6	200°38.7	268°04.7	S22°37.9	60°19.8	N19°49.8	229°39.6	S12°51.9	15°27.2	N00°17.9	Gacrux	172°03.8	-57°10.0
7	215°41.2	283°03.8	37.5	75°23.0	50.1	244°41.5	51.7	30°29.6	17.9	Alioth	166°22.6	55°53.9
8	230°43.6	298°02.9	37.1	90°26.1	50.4	259°43.5	51.5	45°32.1	17.9	Spica	158°33.9	-11°12.9
9	245°46.1	313°02.0	•• 36.7	105°29.3	•• 50.7	274°45.4	•• 51.3	60°34.5	•• 17.9	Alkaid	153°00.8	49°15.4
10	260°48.5	328°01.1	36.3	120°32.4	51.0	289°47.4	51.1	75°37.0	17.9	Hadar	148°51.8	-60°25.1
11	275°51.0	343°00.2	35.9	135°35.6	51.3	304°49.3	50.9	90°39.4	17.9	Menkent	148°10.7	-36°25.1
12	290°53.5	357°59.3	S22°35.5	150°38.8	N19°51.6	319°51.3	S12°50.8	105°41.9	N00°17.9	Arcturus	145°58.1	19°07.6
13	305°55.9	12°58.3	35.1	165°42.0	51.9	334°53.2	50.6	120°44.3	18.0	Rigil Kent.	139°55.5	-60°52.4
14	320°58.4	27°57.4	34.7	180°45.1	52.2	349°55.2	50.4	135°46.8	18.0	Kochab	137°20.1	74°06.4
15	336°00.9	42°56.5	•• 34.3	195°48.3	•• 52.5	4°57.1	•• 50.2	150°49.2	•• 18.0	Zuben'ubi	137°08.4	-16°05.0
16	351°03.3	57°55.6	33.9	210°51.5	52.8	19°59.1	50.0	165°51.7	18.0	Alphecca	126°13.3	26°40.6
17	6°05.8	72°54.7	33.5	225°54.6	53.1	35°01.0	49.8	180°54.1	18.0	Antares	112°29.6	-26°27.2
18	21°08.3	87°53.8	S22°33.0	240°57.8	N19°53.4	50°03.0	S12°49.6	195°56.6	N00°18.0	Atria	107°34.3	-69°02.6
19	36°10.7	102°52.9	32.6	256°01.0	53.7	65°04.9	49.4	210°59.0	18.0	Sabik	102°15.7	-15°44.3
20	51°13.2	117°51.9	32.2	271°04.2	54.0	80°06.8	49.2	226°01.5	18.0	Shaula	96°25.7	-37°06.6
21	66°15.6	132°51.0	•• 31.8	286°07.4	•• 54.3	95°08.8	•• 49.0	241°03.9	•• 18.0	Rasalhague	96°09.1	12°33.1
22	81°18.1	147°50.1	31.4	301°10.5	54.6	110°10.7	48.9	256°06.4	18.0	Eltanin	90°47.8	51°29.1
23	96°20.6	162°49.2	31.0	316°13.7	54.9	125°12.7	48.7	271°08.8	18.0	Kaus Aust.	83°47.5	-34°22.8
Mer.pass. 16:36		ν -0.9' d -0.4' m -3.92		ν 3.2' d 0.3' m -1.00		ν 1.9' d -0.2' m -2.08		ν 2.4' d 0.0' m 0.82		Vega	80°41.1	38°47.5

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	111°23.0	177°48.3	S22°30.5	331°16.9	N19°55.2	140°14.6	S12°48.5	286°11.3	N00°18.0	Nunki	76°01.8	-26°17.1
1	126°25.5	192°47.4	30.1	346°20.1	55.5	155°16.6	48.3	301°13.7	18.0	Altair	62°11.0	8°53.7
2	141°28.0	207°46.5	29.7	1°23.3	55.8	170°18.5	48.1	316°16.2	18.1	Peacock	53°23.6	-56°42.2
3	156°30.4	222°45.6	•• 29.3	16°26.5	•• 56.1	185°20.5	•• 47.9	331°18.6	•• 18.1	Deneb	49°33.7	45°19.1
4	171°32.9	237°44.7	28.9	31°29.6	56.4	200°22.4	47.7	346°21.1	18.1	Enif	33°49.9	9°55.3
5	186°35.4	252°43.8	28.4	46°32.8	56.7	215°24.3	47.5	1°23.5	18.1	Al Na'ir	27°47.1	-46°54.9
6	201°37.8	267°42.8	S22°28.0	61°36.0	N19°57.0	230°26.3	S12°47.3	16°26.0	N00°18.1	Fomalhaut	15°26.9	-29°34.2
7	216°40.3	282°41.9	27.6	76°39.2	57.3	245°28.2	47.2	31°28.5	18.1	Scheat	13°56.1	28°08.4
8	231°42.8	297°41.0	27.1	91°42.4	57.6	260°30.2	47.0	46°30.9	18.1	Markab	13°41.0	15°15.7
9	246°45.2	312°40.1	•• 26.7	106°45.6	•• 57.9	275°32.1	•• 46.8	61°33.4	•• 18.1			
10	261°47.7	327°39.2	26.3	121°48.8	58.3	290°34.1	46.6	76°35.8	18.1	Jan 10 Sun	SHA	Mer.pass
11	276°50.1	342°38.3	25.8	136°52.0	58.6	305°36.0	46.4	91°38.3	18.1	Venus	69°07.6	12:07
12	291°52.6	357°37.4	S22°25.4	151°55.2	N19°58.9	320°38.0	S12°46.2	106°40.7	N00°18.1	Mars	219°20.8	02:05
13	306°55.1	12°36.5	25.0	166°58.4	59.2	335°39.9	46.0	121°43.2	18.1	Jupiter	29°16.4	14:43
14	321°57.5	27°35.6	24.5	182°01.6	59.5	350°41.8	45.8	136°45.6	18.2	Saturn	174°49.0	05:02
15	337°00.0	42°34.7	•• 24.1	197°04.8	19°59.8	5°43.8	•• 45.6	151°48.1	•• 18.2			
16	352°02.5	57°33.8	23.7	212°08.0	20°00.1	20°45.7	45.4	166°50.5	18.2	Jan 11 Mon	SHA	Mer.pass
17	7°04.9	72°32.9	23.2	227°11.2	00.4	35°47.7	45.2	181°53.0	18.2	Venus	67°46.4	12:08
18	22°07.4	87°32.0	S22°22.8	242°14.4	N20°00.7	50°49.6	S12°45.1	196°55.5	N00°18.2	Mars	219°37.0	02:00
19	37°09.9	102°31.1	22.3	257°17.6	01.0	65°51.6	44.9	211°57.9	18.2	Jupiter	29°04.0	14:40
20	52°12.3	117°30.2	21.9	272°20.8	01.3	80°53.5	44.7	227°00.4	18.2	Saturn	174°48.6	04:58
21	67°14.8	132°29.3	•• 21.5	287°24.0	•• 01.6	95°55.4</						

Sun							Moon						
h	GHA	Dec	GHA	ν	Dec	d	HP						
0	178°10.6	S22°00.2	242°42.9	10.1'	S22°18.9	7.2'	56.0'						
1	193°10.4	21°59.9	257°12.0	10.1'	22°26.2	7.1'	55.9'						
2	208°10.1	59.5	271°41.1	10.1'	22°33.3	7.0'	55.9'						
3	223°09.9	· · 59.1	286°10.2	10.0'	22°40.2	6.9'	55.9'						
4	238°09.6	58.8	300°39.3	10.0'	22°47.1	6.7'	55.9'						
5	253°09.4	58.4	315°08.3	10.0'	22°53.8	6.6'	55.8'						
6	268°09.1	S21°58.0	329°37.3	10.0'	S23°00.5	6.5'	55.8'						
7	283°08.8	57.7	344°06.2	9.9'	23°06.9	6.4'	55.8'						
8	298°08.6	57.3	358°35.2	9.9'	23°13.3	6.2'	55.8'						
9	313°08.3	· · 56.9	13°04.1	9.9'	23°19.6	6.1'	55.7'						
10	328°08.1	56.5	27°32.9	9.9'	23°25.7	6.0'	55.7'						
11	343°07.8	56.2	42°01.8	9.8'	23°31.7	5.9'	55.7'						
12	358°07.6	S21°55.8	56°30.6	9.8'	S23°37.6	5.7'	55.7'						
13	13°07.3	55.4	70°59.4	9.8'	23°43.3	5.6'	55.7'						
14	28°07.1	55.0	85°28.2	9.8'	23°48.9	5.5'	55.6'						
15	43°06.8	· · 54.7	99°57.0	9.7'	23°54.4	5.4'	55.6'						
16	58°06.6	54.3	114°25.7	9.7'	23°59.8	5.2'	55.6'						
17	73°06.3	53.9	128°54.4	9.7'	24°05.0	5.1'	55.6'						
18	88°06.1	S21°53.5	143°23.1	9.7'	S24°10.1	5.0'	55.5'						
19	103°05.8	53.1	157°51.8	9.7'	24°15.1	4.9'	55.5'						
20	118°05.6	52.7	172°20.4	9.6'	24°20.0	4.7'	55.5'						
21	133°05.3	· · 52.4	186°49.1	9.6'	24°24.7	4.6'	55.5'						
22	148°05.1	52.0	201°17.7	9.6'	24°29.3	4.5'	55.5'						
23	163°04.8	51.6	215°46.3	9.6'	24°33.8	4.3'	55.4'						
SD = 16.3' d = -0.4'							SD = 15.3'						
Mon	GHA	Dec	GHA	ν	Dec	d	HP						
0	178°04.6	S21°51.2	230°14.8	9.6'	S24°38.1	4.2'	55.4'						
1	193°04.3	50.8	244°43.4	9.5'	24°42.4	4.1'	55.4'						
2	208°04.1	50.4	259°12.0	9.5'	24°46.4	4.0'	55.4'						
3	223°03.8	· · 50.1	273°40.5	9.5'	24°50.4	3.8'	55.3'						
4	238°03.6	49.7	288°09.0	9.5'	24°54.2	3.7'	55.3'						
5	253°03.3	49.3	302°37.5	9.5'	24°57.9	3.6'	55.3'						
6	268°03.1	S21°48.9	317°06.0	9.5'	S25°01.5	3.4'	55.3'						
7	283°02.8	48.5	331°34.5	9.5'	25°04.9	3.3'	55.3'						
8	298°02.6	48.1	346°03.0	9.5'	25°08.2	3.2'	55.2'						
9	313°02.3	· · 47.7	0°31.4	9.5'	25°11.4	3.0'	55.2'						
10	328°02.1	47.3	14°59.9	9.5'	25°14.4	2.9'	55.2'						
11	343°01.8	46.9	29°28.3	9.4'	25°17.3	2.8'	55.2'						
12	358°01.6	S21°46.5	43°56.8	9.4'	S25°20.1	2.6'	55.2'						
13	13°01.3	46.1	58°25.2	9.4'	25°22.8	2.5'	55.2'						
14	28°01.1	45.8	72°53.7	9.4'	25°25.3	2.4'	55.1'						
15	43°00.9	· · 45.4	87°22.1	9.4'	25°27.6	2.2'	55.1'						
16	58°00.6	45.0	101°50.5	9.4'	25°29.9	2.1'	55.1'						
17	73°00.4	44.6	116°19.0	9.4'	25°32.0	2.0'	55.1'						
18	88°00.1	S21°44.2	130°47.4	9.4'	S25°34.0	1.9'	55.1'						
19	102°59.9	43.8	145°15.8	9.4'	25°35.8	1.7'	55.0'						
20	117°59.6	43.4	159°44.3	9.4'	25°37.6	1.6'	55.0'						
21	132°59.4	· · 43.0	174°12.7	9.4'	25°39.1	1.5'	55.0'						
22	147°59.1	42.6	188°41.2	9.4'	25°40.6	1.3'	55.0'						
23	162°58.9	42.2	203°09.6	9.5'	25°41.9	1.2'	55.0'						
SD = 16.3' d = -0.4'							SD = 15.1'						
Tue	GHA	Dec	GHA	ν	Dec	d	HP						
0	177°58.6	S21°41.8	217°38.1	9.5'	S25°43.1	1.1'	55.0'						
1	192°58.4	41.4	232°06.5	9.5'	25°44.2	0.9'	54.9'						
2	207°58.2	41.0	246°35.0	9.5'	25°45.1	0.8'	54.9'						
3	222°57.9	· · 40.6	261°03.5	9.5'	25°45.9	0.7'	54.9'						
4	237°57.7	40.1	275°32.0	9.5'	25°46.6	0.5'	54.9'						
5	252°57.4	39.7	290°00.5	9.5'	25°47.1	0.4'	54.9'						
6	267°57.2	S21°39.3	304°29.0	9.5'	S25°47.5	0.3'	54.8'						
7	282°57.0	38.9	318°57.5	9.5'	25°47.8	0.1'	54.8'						
8	297°56.7	38.5	333°26.1	9.6'	25°47.9	0.0'	54.8'						
9	312°56.5	· · 38.1	347°54.6	9.6'	25°47.9	-0.1'	54.8'						
10	327°56.2	37.7	2°23.2	9.6'	25°47.8	-0.3'	54.8'						
11	342°56.0	37.3	16°51.8	9.6'	25°47.5	-0.4'	54.8'						
12	357°55.7	S21°36.9	31°20.5	9.6'	S25°47.1	-0.5'	54.8'						
13	12°55.5	36.5	45°49.1	9.7'	25°46.6	-0.6'	54.7'						
14	27°55.3	36.1	60°17.8	9.7'	25°46.0	-0.8'	54.7'						
15	42°55.0	· · 35.6	74°46.4	9.7'	25°45.2	-0.9'	54.7'						
16	57°54.8	35.2	89°15.2	9.7'	25°44.3	-1.0'	54.7'						
17	72°54.5	34.8	103°43.9	9.8'	25°43.2	-1.2'	54.7'						
18	87°54.3	S21°34.4	118°12.7	9.8'	S25°42.1	-1.3'	54.7'						
19	102°54.1	34.0	132°41.4	9.8'	25°40.8	-1.4'	54.6'						
20	117°53.8	33.6	147°10.3	9.8'	25°39.4	-1.5'	54.6'						
21	132°53.6	· · 33.1	161°39.1	9.9'	25°37.8	-1.7'	54.6'						
22	147°53.4	32.7	176°08.0	9.9'	25°36.1	-1.8'	54.6'						
23	162°53.1	32.3	190°36.9	9.9'	25°34.3	-1.9'	54.6'						
SD = 16.3' d = -0.4'							SD = 15.0'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:10	10:08	■	■	14:09	16:07
N 70°	07:53	09:29	■	■	14:47	16:23
68°	07:40	09:02	10:53	13:23	15:14	16:36
66°	07:29	08:42	10:05	14:11	15:35	16:47
64°	07:20	08:25	09:34	14:42	15:51	16:56
62°	07:12	08:11	09:12	15:05	16:05	17:05
60°	07:05	07:59	08:53	15:23	16:17	17:12
N 58°	06:58	07:49	08:38	15:38	16:27	17:18
56°	06:52	07:40	08:25	15:51	16:36	17:24
54°	06:47	07:32	08:14	16:02	16:44	17:29
52°	06:42	07:25	08:04	16:12	16:51	17:34
50°	06:37	07:18	07:55	16:21	16:58	17:39
45°	06:27	07:04	07:37	16:39	17:13	17:49
N 40°	06:18	06:51	07:21	16:55	17:25	17:58
35°	06:09	06:41	07:08	17:08	17:35	18:07
30°	06:02	06:31	06:57	17:19	17:45	18:14
20°	05:47	06:14	06:37	17:38	18:02	18:29
N 10°	05:32	05:58	06:20	17:56	18:18	18:44
0°	05:16	05:42	06:04	18:12	18:34	19:00
S 10°	04:58	05:25	05:48	18:28	18:51	19:17
20°	04:37	05:06	05:30	18:45	19:09	19:38
30°	04:10	04:43	05:10	19:05	19:32	20:05
35°	03:53	04:29	04:58	19:17	19:46	20:22
40°	03:32	04:12	04:44	19:31	20:03	20:43
45°	03:04	03:52	04:28	19:47	20:24	21:11
S 50°	02:25	03:25	04:07	20:08	20:50	21:49
52°	02:03	03:11	03:58	20:17	21:03	22:11
54°	01:32	02:56	03:47	20:28	21:19	22:41
56°	00:31	02:37	03:34	20:41	21:37	23:36
58°	////	02:13	03:19	20:55	22:01	////
S 60°	////	01:40	03:02	21:13	22:33	////

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	■	■	■	■	■	■
66°	07:02	■	■	09:01	■	■
64°	06:06	07:56	09:27	09:57	09:55	10:13
62°	05:34	07:03	08:10	10:30	10:49	11:30
60°	05:09	06:30	07:34	10:55	11:21	12:06
N 58°	04:50	06:07	07:08	11:14	11:46	12:32
56°	04:34	05:47	06:48	11:31	12:05	12:52
54°	04:21	05:31	06:31	11:45	12:21	13:09
52°	04:09	05:17	06:16	11:57	12:35	13:24
50°	03:58	05:05	06:04	12:08	12:47	13:37
45°	03:36	04:40	05:38	12:30	13:13	14:03
N 40°	03:19	04:21	05:17	12:49	13:33	14:23
35°	03:04	04:04	05:00	13:04	13:50	14:41
30°	02:51	03:50	04:45	13:17	14:04	14:56
20°	02:30	03:26	04:20	13:40	14:29	15:21
N 10°	02:11	03:05	03:58	14:00	14:50	15:42
0°	01:54	02:45	03:38	14:18	15:10	16:03
S 10°	01:36	02:26	03:17	14:37	15:30	16:23
20°	01:18	02:05	02:56	14:57	15:52	16:44
30°	00:57	01:41	02:31	15:20	16:17	17:09
35°	00:44	01:27	02:16	15:33	16:31	17:24
40°	00:30	01:11	01:59	15:49	16:48	17:41
45°	00:13	00:52	01:38	16:08	17:08	18:02
S 50°		00:28	01:13	16:31	17:34	18:27
52°		00:17	01:00	16:42	17:46	18:40
54°		00:04	00:46	16:55	18:01	18:54
56°	23:49		00:29	17:10	18:17	19:10
58°	23:31		00:09	17:27	18:37	19:30
S 60°	23:09	23:44		17:48	19:02	19:55

January 13, 14, 15 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	112°22.2	177°26.5	S22°20.1	332°33.7	N20°02.6	141°01.3	S12°43.9	287°10.2	N00°18.3	Alpheratz	357°46.3	29°09.0
1	127°24.6	192°25.6	19.7	347°36.9	02.9	156°03.2	43.7	302°12.7	18.3	Ankaa	353°18.2	-42°15.2
2	142°27.1	207°24.7	19.2	2°40.1	03.2	171°05.2	43.5	317°15.1	18.3	Schedar	349°43.7	56°35.9
3	157°29.6	222°23.8	•• 18.8	17°43.3	•• 03.5	186°07.1	•• 43.3	332°17.6	•• 18.3	Diphda	348°58.4	-17°55.9
4	172°32.0	237°22.9	18.3	32°46.5	03.8	201°09.0	43.1	347°20.0	18.3	Achernar	335°28.4	-57°11.4
5	187°34.5	252°22.0	17.9	47°49.8	04.1	216°11.0	43.0	2°22.5	18.3	Hamal	328°03.5	23°30.8
6	202°37.0	267°21.1	S22°17.4	62°53.0	N20°04.4	231°12.9	S12°42.8	17°24.9	N00°18.3	Polaris	318°50.7	89°18.9
7	217°39.4	282°20.2	17.0	77°56.2	04.7	246°14.9	42.6	32°27.4	18.3	Acamar	315°20.0	-40°16.0
8	232°41.9	297°19.3	16.5	92°59.4	05.0	261°16.8	42.4	47°29.9	18.3	Menkar	314°17.5	4°07.8
9	247°44.4	312°18.4	•• 16.0	108°02.6	•• 05.3	276°18.8	•• 42.2	62°32.3	•• 18.3	Mirfak	308°43.7	49°54.1
10	262°46.8	327°17.5	15.6	123°05.9	05.7	291°20.7	42.0	77°34.8	18.4	Aldebaran	290°52.0	16°31.8
11	277°49.3	342°16.7	15.1	138°09.1	06.0	306°22.6	41.8	92°37.2	18.4	Rigel	281°14.2	-8°11.5
12	292°51.8	357°15.8	S22°14.7	153°12.3	N20°06.3	321°24.6	S12°41.6	107°39.7	N00°18.4	Capella	280°37.7	46°00.6
13	307°54.2	12°14.9	14.2	168°15.6	06.6	336°26.5	41.4	122°42.1	18.4	Bellatrix	278°34.4	6°21.5
14	322°56.7	27°14.0	13.7	183°18.8	06.9	351°28.5	41.2	137°44.6	18.4	Elnath	278°15.4	28°37.0
15	337°59.1	42°13.1	•• 13.3	198°22.0	•• 07.2	6°30.4	•• 41.0	152°47.1	•• 18.4	Alnilam	275°48.6	-1°11.8
16	353°01.6	57°12.2	12.8	213°25.2	07.5	21°32.3	40.9	167°49.5	18.4	Betelgeuse	271°03.7	7°24.5
17	8°04.1	72°11.3	12.3	228°28.5	07.8	36°34.3	40.7	182°52.0	18.4	Canopus	263°56.7	-52°42.2
18	23°06.5	87°10.4	S22°11.9	243°31.7	N20°08.1	51°36.2	S12°40.5	197°54.4	N00°18.4	Sirius	258°35.6	-16°43.9
19	38°09.0	102°09.5	11.4	258°34.9	08.5	66°38.2	40.3	212°56.9	18.4	Adhara	255°14.1	-28°59.2
20	53°11.5	117°08.6	10.9	273°38.2	08.8	81°40.1	40.1	227°59.4	18.5	Procyon	245°02.0	5°11.9
21	68°13.9	132°07.7	•• 10.4	288°41.4	•• 09.1	96°42.1	•• 39.9	243°01.8	•• 18.5	Pollux	243°30.4	28°00.0
22	83°16.4	147°06.8	10.0	303°44.7	09.4	111°44.0	39.7	258°04.3	18.5	Avior	234°18.5	-59°32.5
23	98°18.9	162°05.9	09.5	318°47.9	09.7	126°45.9	39.5	273°06.7	18.5	Suhail	222°53.9	-43°28.4
Mer.pass. 16:28		ν -0.9' d -0.4' m -3.92		ν 3.2' d 0.3' m -1.05		ν 1.9' d -0.2' m -2.07		ν 2.5' d 0.0' m 0.81		Miaplacidus	221°39.6	-69°45.4
										Alphard	217°58.2	-8°42.2
										Regulus	207°45.9	11°54.9
										Dubhe	193°54.1	61°41.4
										Denebola	182°36.0	14°30.7
										Gienah	175°54.8	-17°35.9
										Acrux	173°12.2	-63°09.1
										Gacrux	172°03.7	-57°10.0
										Alioth	166°22.6	55°53.9
										Spica	158°33.9	-11°12.9
										Alkaid	153°00.8	49°15.4
										Hadar	148°51.7	-60°25.1
										Menkent	148°10.7	-36°25.1
										Arcturus	145°58.0	19°07.5
										Rigil Kent.	139°55.4	-60°52.4
										Kochab	137°20.0	74°06.4
										Zuben'ubi	137°08.3	-16°05.1
										Alphecca	126°13.3	26°40.6
										Antares	112°29.6	-26°27.2
										Atria	107°34.2	-69°02.6
										Sabik	102°15.7	-15°44.3
										Shaula	96°25.7	-37°06.6
										Rasalhague	96°09.1	12°33.1
										Eltanin	90°47.8	51°29.1
										Kaus Aust.	83°47.5	-34°22.8
										Vega	80°41.1	38°47.5
										Nunki	76°01.8	-26°17.1
										Altair	62°11.0	8°53.7
										Peacock	53°23.6	-56°42.2
										Deneb	49°33.7	45°19.1
										Enif	33°49.9	9°55.3
										Al Na'ir	27°47.1	-46°54.8
										Fomalhaut	15°26.9	-29°34.2
										Scheat	13°56.1	28°08.4
										Markab	13°41.0	15°15.7
										Jan 13 Wed SHA Mer.pass		
										Venus	65°04.4	12:11
										Mars	220°11.5	01:49
										Jupiter	28°39.1	14:34
										Saturn	174°48.0	04:51
										Jan 14 Thu SHA Mer.pass		
										Venus	63°43.7	12:12
										Mars	220°29.8	01:44
										Jupiter	28°26.6	14:31
										Saturn	174°47.9	04:47
										Jan 15 Fri SHA Mer.pass		
										Venus	62°23.2	12:14
										Mars	220°48.8	01:39
										Jupiter	28°14.0	14:28
										Saturn	174°47.9	04:43
										Horizontal parallax Venus: 0.1		
										Mars:	0.2	
Mer.pass. 16:20		ν -0.9' d -0.5' m -3.92		ν 3.3' d 0.3' m -1.09		ν 1.9' d -0.2' m -2.07		ν 2.5' d 0.0' m 0.80				

DUT1 = UT1-UTC = +0.1039 sec ΔT = TT-UT1 = +66.0801 sec

2010 January 13 to Jan. 15 UT

h	Sun			Moon			
Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	177°52.9	S21°31.9	205°05.8	10.0'	S25°32.4	-2.1'	54.6'
1	192°52.6	31.5	219°34.8	10.0'	25°30.4	-2.2'	54.6'
2	207°52.4	31.0	234°03.8	10.0'	25°28.2	-2.3'	54.5'
3	222°52.2	· · 30.6	248°32.8	10.1'	25°25.9	-2.4'	54.5'
4	237°51.9	30.2	263°01.9	10.1'	25°23.4	-2.6'	54.5'
5	252°51.7	29.8	277°31.0	10.2'	25°20.9	-2.7'	54.5'
6	267°51.5	S21°29.4	292°00.2	10.2'	S25°18.2	-2.8'	54.5'
7	282°51.2	28.9	306°29.4	10.2'	25°15.4	-2.9'	54.5'
8	297°51.0	28.5	320°58.6	10.3'	25°12.5	-3.0'	54.5'
9	312°50.8	· · 28.1	335°27.9	10.3'	25°09.4	-3.2'	54.5'
10	327°50.5	27.7	349°57.2	10.4'	25°06.3	-3.3'	54.4'
11	342°50.3	27.2	4°26.5	10.4'	25°03.0	-3.4'	54.4'
12	357°50.1	S21°26.8	18°55.9	10.4'	S24°59.5	-3.5'	54.4'
13	12°49.8	26.4	33°25.4	10.5'	24°56.0	-3.7'	54.4'
14	27°49.6	25.9	47°54.9	10.5'	24°52.4	-3.8'	54.4'
15	42°49.3	· · 25.5	62°24.4	10.6'	24°48.6	-3.9'	54.4'
16	57°49.1	25.1	76°54.0	10.6'	24°44.7	-4.0'	54.4'
17	72°48.9	24.6	91°23.6	10.7'	24°40.7	-4.1'	54.4'
18	87°48.7	S21°24.2	105°53.3	10.7'	S24°36.6	-4.2'	54.4'
19	102°48.4	23.8	120°23.0	10.8'	24°32.3	-4.4'	54.3'
20	117°48.2	23.3	134°52.8	10.8'	24°28.0	-4.5'	54.3'
21	132°48.0	· · 22.9	149°22.6	10.9'	24°23.5	-4.6'	54.3'
22	147°47.7	22.5	163°52.5	10.9'	24°18.9	-4.7'	54.3'
23	162°47.5	22.0	178°22.4	11.0'	24°14.2	-4.8'	54.3'
SD = 16.3'			SD = 14.9'				
$d = -0.4'$							

Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	177°47.3	S21°21.6	192°52.4	11.0'	S24°09.4	-4.9'	54.3'
1	192°47.0	21.2	207°22.4	11.1'	24°04.5	-5.0'	54.3'
2	207°46.8	20.7	221°52.5	11.1'	23°59.4	-5.2'	54.3'
3	222°46.6	· · 20.3	236°22.6	11.2'	23°54.3	-5.3'	54.3'
4	237°46.3	19.8	250°52.8	11.2'	23°49.0	-5.4'	54.2'
5	252°46.1	19.4	265°23.0	11.3'	23°43.6	-5.5'	54.2'
6	267°45.9	S21°19.0	279°53.3	11.4'	S23°38.1	-5.6'	54.2'
7	282°45.7	18.5	294°23.7	11.4'	23°32.6	-5.7'	54.2'
8	297°45.4	18.1	308°54.1	11.5'	23°26.9	-5.8'	54.2'
9	312°45.2	· · 17.6	323°24.5	11.5'	23°21.0	-5.9'	54.2'
10	327°45.0	17.2	337°55.1	11.6'	23°15.1	-6.0'	54.2'
11	342°44.7	16.7	352°25.6	11.6'	23°09.1	-6.1'	54.2'
12	357°44.5	S21°16.3	6°56.3	11.7'	S23°03.0	-6.2'	54.2'
13	12°44.3	15.8	21°27.0	11.8'	22°56.8	-6.3'	54.2'
14	27°44.1	15.4	35°57.7	11.8'	22°50.4	-6.4'	54.2'
15	42°43.8	· · 15.0	50°28.5	11.9'	22°44.0	-6.5'	54.1'
16	57°43.6	14.5	64°59.4	11.9'	22°37.5	-6.6'	54.1'
17	72°43.4	14.1	79°30.3	12.0'	22°30.8	-6.7'	54.1'
18	87°43.2	S21°13.6	94°01.3	12.1'	S22°24.1	-6.8'	54.1'
19	102°42.9	13.2	108°32.4	12.1'	22°17.3	-6.9'	54.1'
20	117°42.7	12.7	123°03.5	12.2'	22°10.3	-7.0'	54.1'
21	132°42.5	· · 12.3	137°34.7	12.2'	22°03.3	-7.1'	54.1'
22	147°42.3	11.8	152°05.9	12.3'	21°56.2	-7.2'	54.1'
23	162°42.0	11.3	166°37.2	12.4'	21°48.9	-7.3'	54.1'
SD = 16.3'			SD = 14.8'				
$d = -0.4'$							

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	177°41.8	S21°10.9	181°08.5	12.4'	S21°41.6	-7.4'	54.1'
1	192°41.6	10.4	195°40.0	12.5'	21°34.2	-7.5'	54.1'
2	207°41.4	10.0	210°11.4	12.5'	21°26.7	-7.6'	54.1'
3	222°41.1	· · 09.5	224°43.0	12.6'	21°19.1	-7.7'	54.0'
4	237°40.9	09.1	239°14.6	12.7'	21°11.4	-7.8'	54.0'
5	252°40.7	08.6	253°46.2	12.7'	21°03.6	-7.9'	54.0'
6	267°40.5	S21°08.2	268°17.9	12.8'	S20°55.8	-8.0'	54.0'
7	282°40.2	07.7	282°49.7	12.8'	20°47.8	-8.1'	54.0'
8	297°40.0	07.2	297°21.6	12.9'	20°39.7	-8.1'	54.0'
9	312°39.8	· · 06.8	311°53.5	13.0'	20°31.6	-8.2'	54.0'
10	327°39.6	06.3	326°25.4	13.0'	20°23.4	-8.3'	54.0'
11	342°39.4	05.9	340°57.5	13.1'	20°15.1	-8.4'	54.0'
12	357°39.1	S21°05.4	355°29.6	13.2'	S20°06.6	-8.5'	54.0'
13	12°38.9	04.9	10°01.7	13.2'	19°58.2	-8.6'	54.0'
14	27°38.7	04.5	24°33.9	13.3'	19°49.6	-8.7'	54.0'
15	42°38.5	· · 04.0	39°06.2	13.3'	19°40.9	-8.7'	54.0'
16	57°38.3	03.5	53°38.5	13.4'	19°32.2	-8.8'	54.0'
17	72°38.0	03.1	68°10.9	13.5'	19°23.4	-8.9'	54.0'
18	87°37.8	S21°02.6	82°43.4	13.5'	S19°14.5	-9.0'	54.0'
19	102°37.6	02.1	97°15.9	13.6'	19°05.5	-9.1'	54.0'
20	117°37.4	01.7	111°48.5	13.6'	18°56.5	-9.1'	54.0'
21	132°37.2	· · 01.2	126°21.1	13.7'	18°47.3	-9.2'	53.9'
22	147°36.9	00.7	140°53.8	13.8'	18°38.1	-9.3'	53.9'
23	162°36.7	00.3	155°26.5	13.8'	18°28.8	-9.4'	53.9'
SD = 16.3'			SD = 14.7'				
$d = -0.5'$							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:03	09:56	■	■	14:23	16:16
N 70°	07:48	09:21	■	■	14:58	16:31
68°	07:36	08:56	10:38	13:41	15:23	16:43
66°	07:25	08:36	09:56	14:22	15:42	16:53
64°	07:17	08:21	09:28	14:51	15:58	17:02
62°	07:09	08:07	09:07	15:12	16:11	17:10
60°	07:02	07:56	08:49	15:29	16:22	17:17
N 58°	06:56	07:46	08:35	15:44	16:32	17:23
56°	06:50	07:38	08:22	15:56	16:41	17:28
54°	06:45	07:30	08:12	16:07	16:49	17:33
52°	06:40	07:23	08:02	16:16	16:56	17:38
50°	06:36	07:16	07:54	16:25	17:02	17:42
45°	06:26	07:02	07:35	16:43	17:16	17:52
N 40°	06:17	06:51	07:20	16:58	17:28	18:01
35°	06:09	06:40	07:08	17:10	17:38	18:09
30°	06:02	06:31	06:57	17:21	17:47	18:17
20°	05:47	06:14	06:38	17:40	18:04	18:31
N 10°	05:33	05:59	06:21	17:57	18:20	18:45
0°	05:17	05:43	06:05	18:13	18:35	19:01
S 10°	05:00	05:27	05:50	18:28	18:51	19:18
20°	04:40	05:08	05:32	18:45	19:10	19:38
30°	04:13	04:46	05:13	19:05	19:32	20:04
35°	03:57	04:32	05:01	19:17	19:46	20:21
40°	03:36	04:16	04:47	19:30	20:02	20:41
45°	03:09	03:56	04:31	19:46	20:22	21:08
S 50°	02:31	03:30	04:12	20:06	20:47	21:45
52°	02:10	03:17	04:02	20:15	21:00	22:06
54°	01:43	03:02	03:52	20:26	21:15	22:33
56°	00:56	02:44	03:39	20:38	21:33	23:16
58°	////	02:22	03:25	20:51	21:54	////
S 60°	////	01:52	03:09	21:08	22:24	////

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	■	■	10:43	■	■	14:08
66°	■	10:30	10:02	■	12:41	14:47
64°	09:39	09:37	09:34	11:49	13:34	15:15
62°	08:48	09:05	09:13	12:40	14:06	15:35
60°	08:16	08:41	08:55	13:11	14:29	15:52
N 58°	07:53	08:22	08:40	13:34	14:48	16:07
56°	07:33	08:05	08:28	13:53	15:04	16:19
54°	07:17	07:52	08:17	14:09	15:17	16:29
52°	07:04	07:40	08:07	14:23	15:29	16:39
50°	06:51	07:29	07:58	14:35	15:39	16:47
45°	06:26	07:07	07:39	15:00	16:01	17:04
N 40°	06:06	06:48	07:24	15:19	16:18	17:19
35°	05:50	06:33	07:11	15:36	16:33	17:31
30°	05:35	06:20	07:00	15:50	16:46	17:41
20°	05:10	05:57	06:40	16:14	17:07	17:59
N 10°	04:49	05:38	06:23	16:35	17:26	18:15
0°	04:29	05:20	06:07	16:54	17:43	18:30
S 10°	04:09	05:01	05:51	17:13	18:00	18:44
20°	03:48	04:41	05:34	17:34	18:19	18:59
30°	03:23	04:19	05:15	17:57	18:40	19:17
35°	03:09	04:05	05:03	18:11	18:52	19:27
40°	02:52	03:50	04:50	18:27	19:06	19:39
45°	02:32	03:31	04:34	18:46	19:23	19:52
S 50°	02:06	03:08	04:14	19:10	19:43	20:09
52°	01:54	02:57	04:05	19:21	19:53	20:17
54°	01:40	02:44	03:55	19:34	20:04	20:25
56°	01:24	02:30	03:43	19:49	20:16	20:35
58°	01:04	02:13	03:29	20:06	20:30	20:46
S 60°	00:39	01:52	03:13	20:28	20:47	20:58

Day	Sun		Mer. Pass hh:mm	Moon		Age 28-0 5-0%
	Eqn.of Time 00 ^h	12 ^h		Mer.Pass.	Lower	
	mm:ss	mm:ss		Upper hh:mm	hh:mm	
13	08:28	08:40	12:09	10:42	23:07	
14	08:51	09:02	12:09	11:31	23:55	
15	09:13	09:23	12:09	12:19	—:—	

January 16, 17, 18 UT (Sat., Sun., Mon.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	115°19.6	176°22.6	S21°44.8	336°28.1	N20°25.2	143°20.9	S12°30.0	290°07.5	N00°19.1					Alpheratz	357°46.3	29°09.0	
1	130°22.1	191°21.8	44.3	351°31.4	25.5	158°22.8	29.8	305°10.0	19.2					Ankaa	353°18.2	-42°15.2	
2	145°24.5	206°20.9	43.8	6°34.7	25.8	173°24.8	29.6	320°12.5	19.2					Schedar	349°43.7	56°35.9	
3	160°27.0	221°20.0	•• 43.2	21°38.0	•• 26.1	188°26.7	•• 29.5	335°14.9	•• 19.2					Diphda	348°58.4	-17°55.9	
4	175°29.5	236°19.2	42.7	36°41.3	26.5	203°28.6	29.3	350°17.4	19.2					Achernar	335°28.4	-57°11.4	
5	190°31.9	251°18.3	42.1	51°44.6	26.8	218°30.6	29.1	5°19.9	19.2					Hamal	328°03.6	23°30.8	
6	205°34.4	266°17.4	S21°41.6	66°47.9	N20°27.1	233°32.5	S12°28.9	20°22.3	N00°19.2					Polaris	318°52.0	89°18.9	
7	220°36.9	281°16.5	41.1	81°51.2	27.4	248°34.5	28.7	35°24.8	19.3					Acamar	315°20.0	-40°16.0	
8	235°39.3	296°15.7	40.5	96°54.5	27.7	263°36.4	28.5	50°27.3	19.3					Menkar	314°17.5	4°07.8	
9	250°41.8	311°14.8	•• 40.0	111°57.8	•• 28.1	278°38.3	•• 28.3	65°29.8	•• 19.3					Mirfak	308°43.8	49°54.1	
10	265°44.2	326°13.9	39.4	127°01.1	28.4	293°40.3	28.1	80°32.2	19.3					Aldebaran	290°52.0	16°31.8	
11	280°46.7	341°13.1	38.9	142°04.5	28.7	308°42.2	27.9	95°34.7	19.3					Rigel	281°14.2	-8°11.5	
12	295°49.2	356°12.2	S21°38.4	157°07.8	N20°29.0	323°44.1	S12°27.7	110°37.2	N00°19.3					Capella	280°37.7	46°00.6	
13	310°51.6	11°11.3	37.8	172°11.1	29.3	338°46.1	27.5	125°39.6	19.4					Bellatrix	278°34.4	6°21.5	
14	325°54.1	26°10.5	37.3	187°14.4	29.7	353°48.0	27.3	140°42.1	19.4					Elnath	278°15.4	28°37.0	
15	340°56.6	41°09.6	•• 36.7	202°17.7	•• 30.0	8°49.9	•• 27.1	155°44.6	•• 19.4					Alnilam	275°48.6	-1°11.8	
16	355°59.0	56°08.7	36.2	217°21.0	30.3	23°51.9	26.9	170°47.1	19.4					Betelgeuse	271°03.7	7°24.5	
17	11°01.5	71°07.9	35.6	232°24.3	30.6	38°53.8	26.7	185°49.5	19.4					Canopus	263°56.7	-52°42.2	
18	26°04.0	86°07.0	S21°35.1	247°27.6	N20°30.9	53°55.7	S12°26.5	200°52.0	N00°19.4					Sirius	258°35.6	-16°43.9	
19	41°06.4	101°06.1	34.5	262°31.0	31.3	68°57.7	26.3	215°54.5	19.4					Adhara	255°14.1	-28°59.2	
20	56°08.9	116°05.3	34.0	277°34.3	31.6	83°59.6	26.1	230°56.9	19.5					Procyon	245°02.0	5°11.8	
21	71°11.4	131°04.4	•• 33.4	292°37.6	•• 31.9	99°01.5	•• 25.9	245°59.4	•• 19.5					Pollux	243°30.3	28°00.0	
22	86°13.8	146°03.5	32.8	307°40.9	32.2	114°03.5	25.7	261°01.9	19.5					Avior	234°18.5	-59°32.5	
23	101°16.3	161°02.7	32.3	322°44.2	32.6	129°05.4	25.5	276°04.4	19.5					Suhail	222°53.9	-43°28.4	
Mer.pass. 16:16			ν -0.9' d -0.5' m -3.92			ν 3.3' d 0.3' m -1.11			ν 1.9' d -0.2' m -2.07			ν 2.5' d 0.0' m 0.80					

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	116°18.7	176°01.8	S21°31.7	337°47.6	N20°32.9	144°07.3	S12°25.4	291°06.8	N00°19.5					
1	131°21.2	191°01.0	31.2	352°50.9	33.2	159°09.3	25.2	306°09.3	19.5					
2	146°23.7	206°00.1	30.6	7°54.2	33.5	174°11.2	25.0	321°11.8	19.6					
3	161°26.1	220°59.2	•• 30.0	22°57.5	•• 33.8	189°13.1	•• 24.8	336°14.3	•• 19.6					
4	176°28.6	235°58.4	29.5	38°00.9	34.2	204°15.1	24.6	351°16.7	19.6					
5	191°31.1	250°57.5	28.9	53°04.2	34.5	219°17.0	24.4	6°19.2	19.6					
6	206°33.5	265°56.7	S21°28.4	68°07.5	N20°34.8	234°18.9	S12°24.2	21°21.7	N00°19.6					
7	221°36.0	280°55.8	27.8	83°10.8	35.1	249°20.9	24.0	36°24.2	19.7					
8	236°38.5	295°54.9	27.2	98°14.2	35.4	264°22.8	23.8	51°26.6	19.7					
9	251°40.9	310°54.1	•• 26.7	113°17.5	•• 35.8	279°24.7	•• 23.6	66°29.1	•• 19.7					
10	266°43.4	325°53.2	26.1	128°20.8	36.1	294°26.7	23.4	81°31.6	19.7					
11	281°45.8	340°52.4	25.5	143°24.2	36.4	309°28.6	23.2	96°34.1	19.7					
12	296°48.3	355°51.5	S21°24.9	158°27.5	N20°36.7	324°30.5	S12°23.0	111°36.5	N00°19.7					
13	311°50.8	10°50.7	24.4	173°30.8	37.1	339°32.5	22.8	126°39.0	19.8					
14	326°53.2	25°49.8	23.8	188°34.2	37.4	354°34.4	22.6	141°41.5	19.8					
15	341°55.7	40°48.9	•• 23.2	203°37.5	•• 37.7	9°36.3	•• 22.4	156°44.0	•• 19.8					
16	356°58.2	55°48.1	22.6	218°40.8	38.0	24°38.3	22.2	171°46.4	19.8					
17	12°00.6	70°47.2	22.1	233°44.2	38.3	39°40.2	22.0	186°48.9	19.8					
18	27°03.1	85°46.4	S21°21.5	248°47.5	N20°38.7	54°42.1	S12°21.8	201°51.4	N00°19.8					
19	42°05.6	100°45.5	20.9	263°50.9	39.0	69°44.1	21.6	216°53.9	19.9					
20	57°08.0	115°44.7	20.3	278°54.2	39.3	84°46.0	21.4	231°56.3	19.9					
21	72°10.5	130°43.8	•• 19.7	293°57.5	•• 39.6	99°47.9	•• 21.2	246°58.8	•• 19.9					
22	87°13.0	145°43.0	19.2	309°00.9	40.0	114°49.9	21.0	262°01.3	19.9					
23	102°15.4	160°42.1	18.6	324°04.2	40.3	129°51.8	20.8	277°03.8	19.9					
Mer.pass. 16:12			ν -0.9' d -0.6' m -3.92			ν 3.3' d 0.3' m -1.14			ν 1.9' d -0.2' m -2.07			ν 2.5' d 0.0' m 0.79		

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	117°17.9	175°41.3	S21°18.0	339°07.6	N20°40.6	144°53.7	S12°20.6	292°06.3	N00°20.0					
1	132°20.3	190°40.4	17.4	354°10.9	40.9	159°55.7	20.4	307°08.7	20.0					
2	147°22.8	205°39.6	16.8	9°14.3	41.2	174°57.6	20.2	322°11.2	20.0					
3	162°25.3	220°38.7	•• 16.2	24°17.6	•• 41.6	189°59.5	•• 20.0	337°13.7	•• 20.0					
4	177°27.7	235°37.9	15.6	39°21.0	41.9	205°01.5	19.9	352°16.2	20.0					
5	192°30.2	250°37.0	15.1	54°24.3	42.2	220°03.4	19.7	7°18.6	20.1					
6	207°32.7	265°36.2	S21°14.5	69°27.7	N20°42.5	235°05.3	S12°19.5	22°21.1	N00°20.1					
7	222°35.1	280°35.3	13.9	84°31.0	42.9	250°07.2	19.3	37°23.6	20.1					
8	237°37.6	295°34.5	13.3	99°34.4	43.2	265°09.2	19.1	52°26.1	20.1					
9	252°40.1	310°33.6	•• 12.7	114°37.7	•• 43.5	280°11.1	•• 18.9	67°28.6	•• 20.1					
10	267°42.5	325°32.8	12.1	129°41.1	43.8	295°13.0	18.7	82°31.0	20.1					
11	282°45.0	340°31.9	11.5	144°44.4	44.2	310°15.0	18.5	97°33.5	20.2					
12	297°47.5	355°31.1	S21°10.9	159°47.8	N20°44.5	325°16.9	S12°18.3	112°36.0	N00°20.2					
13	312°49.9	10°30.2	10.3	174°51.2	44.8	340°18.8	18.1	127°38.5	20.2					
14	327°52.4	25°29.4	09.7	189°54.5	45.1	355°20.8	17.9	142°41.0	20.2					
15	342°54.8	40°28.5	•• 09.1	204°57.9	•• 45.4	10°22.7	•• 17.7	157°43.4	•• 20.2					
16	357°57.3	55°27.7	08.5	220°01.2	45.8	25°24.6	17.5	172°45.9	20.3					
17	12°59.8	70°26.9	07.9	235°04.6	46.1	40°26.6	17.3	187°48.4	20.3					
18	28°02.2	85°26.0	S21°07.3	250°08.0	N20°46.4	55°28.5	S12°17.1	202°50.9	N00°20.3					
19	43°04.7	100°25.2	06.7	265°11.3	46.7	70°30.4	16.9	217°53.4	20.3					
20	58°07.2	115°24.3	06.1	280°14.7	47.1	85°32.3	16.7	232°55.8	20.3					
21	73°09.6	130°23.5	•• 05.5	295°18.1	•• 47.4	100°34.3	•• 16.5	247°58.3	•• 20.4					
22	88°12.1	145°22.6	04.8	310°21.4	47.7	115°36.2	16.3	263°00.8	20.4					
23	103°14.6	160°21.8	04.2	325°24.8	48.0	130°38.1	16.1	278°03.3	20.4					
Mer.pass. 16:08			ν -0.9' d -0.6' m -3.92			ν 3.3' d 0.3' m -1.16			ν 1.9' d -0.2' m -2.06			ν 2.5' d 0.0' m 0.79		

Jan 16 Sat			SHA	Mer.pass
Venus			61°03.0	12:15
Mars			221°08.5	01:34
Jupiter			28°01.3	14:25
Saturn			174°47.9	04:39

Jan 17 Sun			SHA	Mer.pass
Venus			59°43.1	12:17
Mars			221°28.8	01:29
Jupiter			27°48.6	14:22
Saturn			174°48.1	04:35

Jan 18 Mon			SHA	Mer.pass
Venus			58°23.4	12:18
Mars			221°49.7	01:23
Jupiter			27°35.8	14:19
Saturn			174°48.4	04:31

Horizontal parallax			
Venus:			0.1
Mars:			0.2

h	Sun			Moon			
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	177°36.5	S20°59.8	169°59.4	13.9'	S18°19.5	-9.4'	53.9'
1	192°36.3	59.3	184°32.2	13.9'	18°10.1	-9.5'	53.9'
2	207°36.1	58.8	199°05.1	14.0'	18°00.5	-9.6'	53.9'
3	222°35.9	· · 58.4	213°38.1	14.0'	17°51.0	-9.7'	53.9'
4	237°35.6	57.9	228°11.2	14.1'	17°41.3	-9.7'	53.9'
5	252°35.4	57.4	242°44.3	14.2'	17°31.6	-9.8'	53.9'
6	267°35.2	S20°56.9	257°17.4	14.2'	S17°21.8	-9.9'	53.9'
7	282°35.0	56.5	271°50.7	14.3'	17°11.9	-9.9'	53.9'
8	297°34.8	56.0	286°23.9	14.3'	17°02.0	-10.0'	53.9'
9	312°34.6	· · 55.5	300°57.3	14.4'	16°52.0	-10.1'	53.9'
10	327°34.4	55.0	315°30.6	14.4'	16°41.9	-10.1'	53.9'
11	342°34.1	54.6	330°04.1	14.5'	16°31.8	-10.2'	53.9'
12	357°33.9	S20°54.1	344°37.6	14.5'	S16°21.6	-10.3'	53.9'
13	12°33.7	53.6	359°11.1	14.6'	16°11.3	-10.3'	53.9'
14	27°33.5	53.1	13°44.7	14.7'	16°01.0	-10.4'	53.9'
15	42°33.3	· · 52.6	28°18.4	14.7'	15°50.6	-10.5'	53.9'
16	57°33.1	52.2	42°52.1	14.8'	15°40.2	-10.5'	53.9'
17	72°32.9	51.7	57°25.8	14.8'	15°29.6	-10.6'	53.9'
18	87°32.7	S20°51.2	71°59.6	14.9'	S15°19.1	-10.6'	53.9'
19	102°32.4	50.7	86°33.5	14.9'	15°08.4	-10.7'	53.9'
20	117°32.2	50.2	101°07.4	15.0'	14°57.7	-10.8'	53.9'
21	132°32.0	· · 49.7	115°41.4	15.0'	14°47.0	-10.8'	53.9'
22	147°31.8	49.2	130°15.4	15.1'	14°36.2	-10.9'	53.9'
23	162°31.6	48.8	144°49.4	15.1'	14°25.3	-10.9'	53.9'
SD = 16.3'			SD = 14.7'				

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	177°31.4	S20°48.3	159°23.6	15.2'	S14°14.4	-11.0'	53.9'
1	192°31.2	47.8	173°57.7	15.2'	14°03.4	-11.0'	53.9'
2	207°31.0	47.3	188°31.9	15.2'	13°52.4	-11.1'	53.9'
3	222°30.8	· · 46.8	203°06.2	15.3'	13°41.3	-11.1'	53.9'
4	237°30.6	46.3	217°40.5	15.3'	13°30.2	-11.2'	53.9'
5	252°30.3	45.8	232°14.8	15.4'	13°19.0	-11.2'	53.9'
6	267°30.1	S20°45.3	246°49.2	15.4'	S13°07.7	-11.3'	53.9'
7	282°29.9	44.8	261°23.6	15.5'	12°56.4	-11.3'	53.9'
8	297°29.7	44.3	275°58.1	15.5'	12°45.1	-11.4'	53.9'
9	312°29.5	· · 43.9	290°32.6	15.6'	12°33.7	-11.4'	53.9'
10	327°29.3	43.4	305°07.2	15.6'	12°22.3	-11.5'	53.9'
11	342°29.1	42.9	319°41.7	15.6'	12°10.8	-11.5'	53.9'
12	357°28.9	S20°42.4	334°16.4	15.7'	S11°59.3	-11.6'	53.9'
13	12°28.7	41.9	348°51.1	15.7'	11°47.7	-11.6'	53.9'
14	27°28.5	41.4	3°25.8	15.8'	11°36.1	-11.7'	53.9'
15	42°28.3	· · 40.9	18°00.5	15.8'	11°24.4	-11.7'	53.9'
16	57°28.1	40.4	32°35.3	15.8'	11°12.7	-11.7'	53.9'
17	72°27.9	39.9	47°10.2	15.9'	11°01.0	-11.8'	53.9'
18	87°27.7	S20°39.4	61°45.0	15.9'	S10°49.2	-11.8'	53.9'
19	102°27.5	38.9	76°19.9	15.9'	10°37.3	-11.9'	53.9'
20	117°27.3	38.4	90°54.9	16.0'	10°25.5	-11.9'	53.9'
21	132°27.1	· · 37.9	105°29.8	16.0'	10°13.6	-12.0'	53.9'
22	147°26.9	37.4	120°04.8	16.0'	10°01.6	-12.0'	53.9'
23	162°26.7	36.9	134°39.9	16.1'	09°49.6	-12.0'	53.9'
SD = 16.3'			SD = 14.7'				

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	177°26.5	S20°36.4	149°14.9	16.1'	S09°37.6	-12.1'	53.9'
1	192°26.2	35.9	163°50.0	16.1'	09°25.5	-12.1'	53.9'
2	207°26.0	35.4	178°25.2	16.2'	09°13.4	-12.1'	53.9'
3	222°25.8	· · 34.9	193°00.3	16.2'	09°01.3	-12.2'	53.9'
4	237°25.6	34.3	207°35.5	16.2'	08°49.1	-12.2'	53.9'
5	252°25.4	33.8	222°10.7	16.2'	08°36.9	-12.2'	53.9'
6	267°25.2	S20°33.3	236°46.0	16.3'	S08°24.7	-12.3'	53.9'
7	282°25.0	32.8	251°21.2	16.3'	08°12.4	-12.3'	54.0'
8	297°24.8	32.3	265°56.5	16.3'	08°00.1	-12.3'	54.0'
9	312°24.6	· · 31.8	280°31.8	16.3'	07°47.8	-12.4'	54.0'
10	327°24.4	31.3	295°07.2	16.4'	07°35.4	-12.4'	54.0'
11	342°24.2	30.8	309°42.5	16.4'	07°23.0	-12.4'	54.0'
12	357°24.0	S20°30.3	324°17.9	16.4'	S07°10.6	-12.4'	54.0'
13	12°23.8	29.8	338°53.3	16.4'	06°58.2	-12.5'	54.0'
14	27°23.7	29.2	353°28.7	16.4'	06°45.7	-12.5'	54.0'
15	42°23.5	· · 28.7	8°04.2	16.5'	06°33.2	-12.5'	54.0'
16	57°23.3	28.2	22°39.6	16.5'	06°20.7	-12.6'	54.0'
17	72°23.1	27.7	37°15.1	16.5'	06°08.1	-12.6'	54.0'
18	87°22.9	S20°27.2	51°50.6	16.5'	S05°55.5	-12.6'	54.0'
19	102°22.7	26.7	66°26.1	16.5'	05°42.9	-12.6'	54.0'
20	117°22.5	26.1	81°01.6	16.5'	05°30.3	-12.6'	54.0'
21	132°22.3	· · 25.6	95°37.2	16.6'	05°17.7	-12.7'	54.0'
22	147°22.1	25.1	110°12.7	16.6'	05°05.0	-12.7'	54.1'
23	162°21.9	24.6	124°48.3	16.6'	04°52.3	-12.7'	54.1'
SD = 16.2'			SD = 14.7'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:56	09:44	■	■	14:37	16:25
N 70°	07:42	09:12	11:44	12:37	15:09	16:39
68°	07:31	08:49	10:24	13:57	15:32	16:51
66°	07:21	08:31	09:47	14:34	15:50	17:00
64°	07:13	08:16	09:21	15:00	16:05	17:08
62°	07:05	08:03	09:01	15:20	16:18	17:16
60°	06:59	07:53	08:45	15:36	16:28	17:22
N 58°	06:53	07:43	08:31	15:50	16:38	17:28
56°	06:48	07:35	08:19	16:02	16:46	17:33
54°	06:43	07:27	08:09	16:12	16:53	17:38
52°	06:38	07:21	07:59	16:21	17:00	17:42
50°	06:34	07:14	07:51	16:29	17:06	17:46
45°	06:25	07:01	07:34	16:47	17:20	17:56
N 40°	06:16	06:50	07:19	17:01	17:31	18:04
35°	06:09	06:40	07:07	17:13	17:41	18:12
30°	06:01	06:31	06:56	17:24	17:50	18:19
20°	05:47	06:14	06:38	17:42	18:06	18:33
N 10°	05:33	05:59	06:22	17:59	18:21	18:47
0°	05:19	05:44	06:06	18:14	18:36	19:02
S 10°	05:02	05:28	05:51	18:29	18:52	19:18
20°	04:42	05:10	05:34	18:46	19:09	19:38
30°	04:16	04:49	05:15	19:05	19:31	20:03
35°	04:00	04:35	05:04	19:16	19:45	20:20
40°	03:40	04:19	04:51	19:29	20:00	20:39
45°	03:14	04:00	04:35	19:44	20:19	21:05
S 50°	02:38	03:35	04:16	20:03	20:44	21:40
52°	02:18	03:23	04:07	20:12	20:56	22:00
54°	01:53	03:08	03:57	20:22	21:11	22:25
56°	01:14	02:51	03:45	20:34	21:27	23:01
58°	////	02:30	03:32	20:47	21:48	////
S 60°	////	02:03	03:16	21:03	22:14	////

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	11:25	10:30	09:58	15:01	17:25	19:23
N 70°	10:41	10:09	09:47	15:44	17:44	19:31
68°	10:11	09:52	09:38	16:12	17:59	19:38
66°	09:49	09:39	09:30	16:33	18:11	19:44
64°	09:31	09:27	09:24	16:50	18:21	19:49
62°	09:16	09:18	09:18	17:04	18:30	19:53
60°	09:04	09:09	09:14	17:16	18:37	19:57
N 58°	08:53	09:02	09:09	17:26	18:43	20:00
56°	08:44	08:56	09:05	17:34	18:49	20:03
54°	08:35	08:50	09:02	17:42	18:54	20:05
52°	08:28	08:45	08:59	17:49	18:58	20:07
50°	08:21	08:40	08:56	17:55	19:03	20:09
45°	08:06	08:29	08:50	18:08	19:11	20:14
N 40°	07:54	08:21	08:44	18:19	19:19	20:18
35°	07:44	08:13	08:40	18:28	19:25	20:21
30°	07:35	08:07	08:36	18:37	19:30	20:24
20°	07:19	07:55	08:29	18:50	19:40	20:28
N 10°	07:06	07:45	08:23	19:03	19:48	20:33
0°	06:53	07:36	08:17	19:14	19:56	20:37
S 10°	06:40	07:26	08:11	19:25	20:04	20:40
20°	06:26	07:16	08:05	19:37	20:12	20:45
30°	06:10	07:05	07:58	19:50	20:21	20:49
35°	06:01	06:58	07:54	19:58	20:26	20:52
40°	05:50	06:50	07:49	20:07	20:32	20:55
45°	05:38	06:41	07:44	20:17	20:39	20:58
S 50°	05:22	06:30	07:37	20:30	20:47	21:02
52°	05:15	06:25	07:34	20:35	20:51	21:04
54°	05:07	06:19	07:31	20:42	20:55	21:06
56°	04:58	06:13	07:27	20:49	20:59	21:09
58°	04:48	06:06	07:23	20:56	21:04	21:11
S 60°	04:36	05:58	07:18	21:05	21:10	21:14

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time	12 ^h	Pass	Mer. Pass.	Lower	1-3
	00 ^h	mm:ss	hh:mm	Upper	hh:mm	0-6%
16	09:34	09:44	12:10	13:03	00:41	
17	09:54	10:04	12:10	13:46	01:25	
18	10:14	10:24	12:10	14:27	02:07	

January 19, 20, 21 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	118°17.0	175°21.0	S21°03.6	340°28.2	N20°48.4	145°40.1	S12°15.9	293°05.8	N00°20.4	Alpheratz	357°46.3	29°09.0
1	133°19.5	190°20.1	03.0	355°31.5	48.7	160°42.0	15.7	308°08.3	20.4	Ankaa	353°18.2	-42°15.2
2	148°21.9	205°19.3	02.4	10°34.9	49.0	175°43.9	15.5	323°10.7	20.5	Schedar	349°43.7	56°35.9
3	163°24.4	220°18.4	•• 01.8	25°38.3	•• 49.3	190°45.9	•• 15.3	338°13.2	•• 20.5	Diphda	348°58.4	-17°55.9
4	178°26.9	235°17.6	01.2	40°41.6	49.6	205°47.8	15.1	353°15.7	20.5	Achernar	335°28.5	-57°11.4
5	193°29.3	250°16.8	21°00.5	55°45.0	50.0	220°49.7	14.9	8°18.2	20.5	Hamal	328°03.6	23°30.8
6	208°31.8	265°15.9	S20°59.9	70°48.4	N20°50.3	235°51.6	S12°14.7	23°20.7	N00°20.6	Polaris	318°53.5	89°18.9
7	223°34.3	280°15.1	59.3	85°51.8	50.6	250°53.6	14.5	38°23.1	20.6	Acamar	315°20.0	-40°16.0
8	238°36.7	295°14.3	58.7	100°55.1	50.9	265°55.5	14.3	53°25.6	20.6	Menkar	314°17.5	4°07.8
9	253°39.2	310°13.4	•• 58.1	115°58.5	•• 51.3	280°57.4	•• 14.1	68°28.1	•• 20.6	Mirfak	308°43.8	49°54.1
10	268°41.7	325°12.6	57.4	131°01.9	51.6	295°59.4	13.9	83°30.6	20.6	Aldebaran	290°52.0	16°31.8
11	283°44.1	340°11.8	56.8	146°05.3	51.9	311°01.3	13.7	98°33.1	20.7	Rigel	281°14.2	-8°11.5
12	298°46.6	355°10.9	S20°56.2	161°08.6	N20°52.2	326°03.2	S12°13.5	113°35.6	N00°20.7	Capella	280°37.7	46°00.6
13	313°49.1	10°10.1	55.6	176°12.0	52.6	341°05.1	13.3	128°38.0	20.7	Bellatrix	278°34.4	6°21.5
14	328°51.5	25°09.3	55.0	191°15.4	52.9	356°07.1	13.1	143°40.5	20.7	Elnath	278°15.4	28°37.0
15	343°54.0	40°08.4	•• 54.3	206°18.8	•• 53.2	11°09.0	•• 12.9	158°43.0	•• 20.7	Alnilam	275°48.6	-1°11.8
16	358°56.4	55°07.6	53.7	221°22.2	53.5	26°10.9	12.7	173°45.5	20.8	Betelgeuse	271°03.7	7°24.5
17	13°58.9	70°06.8	53.1	236°25.6	53.8	41°12.9	12.5	188°48.0	20.8	Canopus	263°56.7	-52°42.2
18	29°01.4	85°05.9	S20°52.4	251°28.9	N20°54.2	56°14.8	S12°12.3	203°50.5	N00°20.8	Sirius	258°35.6	-16°43.9
19	44°03.8	100°05.1	51.8	266°32.3	54.5	71°16.7	12.1	218°53.0	20.8	Adhara	255°14.1	-28°59.3
20	59°06.3	115°04.3	51.2	281°35.7	54.8	86°18.6	11.9	233°55.4	20.9	Procyon	245°02.0	5°11.8
21	74°08.8	130°03.4	•• 50.5	296°39.1	•• 55.1	101°20.6	•• 11.7	248°57.9	•• 20.9	Pollux	243°30.3	28°00.0
22	89°11.2	145°02.6	49.9	311°42.5	55.5	116°22.5	11.5	264°00.4	20.9	Avior	234°18.5	-59°32.5
23	104°13.7	160°01.8	49.3	326°45.9	55.8	131°24.4	11.3	279°02.9	20.9	Suhail	222°53.9	-43°28.4
Mer.pass. 16:04		ν -0.8' d -0.6' m -3.92		ν 3.4' d 0.3' m -1.18		ν 1.9' d -0.2' m -2.06		ν 2.5' d 0.0' m 0.78				
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	119°16.2	175°00.9	S20°48.6	341°49.3	N20°56.1	146°26.4	S12°11.1	294°05.4	N00°20.9	Alphard	217°58.2	-8°42.3
1	134°18.6	190°00.1	48.0	356°52.6	56.4	161°28.3	10.9	309°07.9	21.0	Regulus	207°45.8	11°54.9
2	149°21.1	204°59.3	47.3	11°56.0	56.8	176°30.2	10.7	324°10.4	21.0	Dubhe	193°54.0	61°41.4
3	164°23.5	219°58.5	•• 46.7	26°59.4	•• 57.1	191°32.1	•• 10.5	339°12.8	•• 21.0	Denebola	182°36.0	14°30.7
4	179°26.0	234°57.6	46.1	42°02.8	57.4	206°34.1	10.3	354°15.3	21.0	Gienah	175°54.7	-17°35.9
5	194°28.5	249°56.8	45.4	57°06.2	57.7	221°36.0	10.1	9°17.8	21.1	Acrux	173°12.1	-63°09.1
6	209°30.9	264°56.0	S20°44.8	72°09.6	N20°58.0	236°37.9	S12°09.9	24°20.3	N00°21.1	Gacrux	172°03.6	-57°10.1
7	224°33.4	279°55.2	44.1	87°13.0	58.4	251°39.8	09.7	39°22.8	21.1	Alioth	166°22.5	55°53.9
8	239°35.9	294°54.3	43.5	102°16.4	58.7	266°41.8	09.5	54°25.3	21.1	Spica	158°33.8	-11°12.9
9	254°38.3	309°53.5	•• 42.8	117°19.8	•• 59.0	281°43.7	•• 09.3	69°27.8	•• 21.1	Alkaid	153°00.7	49°15.4
10	269°40.8	324°52.7	42.2	132°23.2	59.3	296°45.6	09.1	84°30.2	21.2	Hadar	148°51.6	-60°25.1
11	284°43.3	339°51.9	41.5	147°26.6	20°59.7	311°47.6	08.9	99°32.7	21.2	Menkent	148°10.6	-36°25.1
12	299°45.7	354°51.0	S20°40.9	162°30.0	N21°00.0	326°49.5	S12°08.7	114°35.2	N00°21.2	Arcturus	145°58.0	19°07.5
13	314°48.2	9°50.2	40.2	177°33.4	00.3	341°51.4	08.5	129°37.7	21.2	Rigil Kent.	139°55.4	-60°52.4
14	329°50.7	24°49.4	39.6	192°36.8	00.6	356°53.3	08.3	144°40.2	21.3	Kochab	137°19.9	74°06.4
15	344°53.1	39°48.6	•• 38.9	207°40.2	•• 01.0	11°55.3	•• 08.1	159°42.7	•• 21.3	Zuben'ubi	137°08.3	-16°05.1
16	359°55.6	54°47.8	38.3	222°43.6	01.3	26°57.2	07.9	174°45.2	21.3	Alphecca	126°13.2	26°40.6
17	14°58.0	69°46.9	37.6	237°47.0	01.6	41°59.1	07.7	189°47.7	21.3	Antares	112°29.6	-26°27.2
18	30°00.5	84°46.1	S20°37.0	252°50.4	N21°01.9	57°01.0	S12°07.5	204°50.2	N00°21.4	Atria	107°34.1	-69°02.6
19	45°03.0	99°45.3	36.3	267°53.8	02.2	72°03.0	07.3	219°52.6	21.4	Sabik	102°15.7	-15°44.3
20	60°05.4	114°44.5	35.7	282°57.2	02.6	87°04.9	07.1	234°55.1	21.4	Shaula	96°25.6	-37°06.6
21	75°07.9	129°43.7	•• 35.0	298°00.6	•• 02.9	102°06.8	•• 06.9	249°57.6	•• 21.4	Rasalhague	96°09.0	12°33.0
22	90°10.4	144°42.8	34.3	313°04.0	03.2	117°08.7	06.7	265°00.1	21.4	Eltanin	90°47.8	51°29.1
23	105°12.8	159°42.0	33.7	328°07.4	03.5	132°10.7	06.5	280°02.6	21.5	Kaus Aust.	83°47.5	-34°22.8
Mer.pass. 16:00		ν -0.8' d -0.6' m -3.92		ν 3.4' d 0.3' m -1.20		ν 1.9' d -0.2' m -2.06		ν 2.5' d 0.0' m 0.78		Vega	80°41.1	38°47.5
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	120°15.3	174°41.2	S20°33.0	343°10.8	N21°03.9	147°12.6	S12°06.3	295°05.1	N00°21.5	Nunki	76°01.7	-26°17.1
1	135°17.8	189°40.4	32.3	358°14.2	04.2	162°14.5	06.1	310°07.6	21.5	Altair	62°11.0	8°53.7
2	150°20.2	204°39.6	31.7	13°17.6	04.5	177°16.4	05.9	325°10.1	21.5	Peacock	53°23.6	-56°42.2
3	165°22.7	219°38.8	•• 31.0	28°21.1	•• 04.8	192°18.4	•• 05.7	340°12.6	•• 21.6	Deneb	49°33.7	45°19.0
4	180°25.2	234°37.9	30.4	43°24.5	05.1	207°20.3	05.5	355°15.1	21.6	Enif	33°49.9	9°55.3
5	195°27.6	249°37.1	29.7	58°27.9	05.5	222°22.2	05.3	10°17.5	21.6	Al Na'ir	27°47.1	-46°54.8
6	210°30.1	264°36.3	S20°29.0	73°31.3	N21°05.8	237°24.2	S12°05.1	25°20.0	N00°21.6	Fomalhaut	15°26.9	-29°34.2
7	225°32.5	279°35.5	28.3	88°34.7	06.1	252°26.1	04.9	40°22.5	21.7	Scheat	13°56.1	28°08.4
8	240°35.0	294°34.7	27.7	103°38.1	06.4	267°28.0	04.7	55°25.0	21.7	Markab	13°41.1	15°15.7
9	255°37.5	309°33.9	•• 27.0	118°41.5	•• 06.8	282°29.9	•• 04.5	70°27.5	•• 21.7			
10	270°39.9	324°33.1	26.3	133°44.9	07.1	297°31.9	04.3	85°30.0	21.7	Jan 19 Tue	SHA	Mer.pass
11	285°42.4	339°32.3	25.7	148°48.4	07.4	312°33.8	04.1	100°32.5	21.8	Venus	57°03.9	12:19
12	300°44.9	354°31.4	S20°25.0	163°51.8	N21°07.7	327°35.7	S12°03.9	115°35.0	N00°21.8	Mars	222°11.1	01:18
13	315°47.3	9°30.6	24.3	178°55.2	08.0	342°37.6	03.7	130°37.5	21.8	Jupiter	27°23.0	14:15
14	330°49.8	24°29.8	23.6	193°58.6	08.4	357°39.6	03.5	145°40.0	21.8	Saturn	174°48.7	04:27
15	345°52.3	39°29.0	•• 22.9	209°02.0	•• 08.7	12°41.5	•• 03.3	160°42.5	•• 21.9			
16	0°54.7	54°28.2	22.3	224°05.5	09.0	27°43.4	03.1	175°45.0	21.9	Jan 20 Wed	SHA	Mer.pass
17	15°57.2	69°27.4	21.6	239°08.9	09.3	42°45.3	02.9	190°47.5	21.9	Venus	55°44.8	12:21
18	30°59.6	84°26.6	S20°20.9	254°12.3	N21°09.6	57°47.3	S12°02.7	205°49.9	N00°21.9	Mars	222°33.1	01:12
19	46°02.1	99°25.8	20.2	269°15.7	10.0	72°49.2	02.5	220°52.4	22.0	Jupiter	26°57.3	14:09
20	61°04.6	114°25.0	19.5	284°19.1	10.3	87°51.1	02.3	235°54.9	22.0	Saturn	174°49.8	04:19
21	76°07.0	129°24.2	•• 18.9									

DUT1 = UT1-UTC = +0.1004 sec $\Delta T = TT-UT1 = +66.0836$ sec

2010 January 19 to Jan. 21 UT

h Sun Moon						
Tue	GHA	Dec	GHA	ν	Dec	d HP
0	177°21.7	S20°24.1	139°23.9	16.6'	S04°39.6	-12.7' 54.1'
1	192°21.5	23.6	153°59.4	16.6'	04°26.9	-12.7' 54.1'
2	207°21.3	23.0	168°35.0	16.6'	04°14.1	-12.8' 54.1'
3	222°21.1	· · 22.5	183°10.6	16.6'	04°01.4	-12.8' 54.1'
4	237°20.9	22.0	197°46.2	16.6'	03°48.6	-12.8' 54.1'
5	252°20.7	21.5	212°21.9	16.6'	03°35.8	-12.8' 54.1'
6	267°20.5	S20°20.9	226°57.5	16.6'	S03°22.9	-12.8' 54.1'
7	282°20.3	20.4	241°33.1	16.6'	03°10.1	-12.9' 54.1'
8	297°20.1	19.9	256°08.7	16.6'	02°57.3	-12.9' 54.1'
9	312°20.0	· · 19.4	270°44.4	16.6'	02°44.4	-12.9' 54.2'
10	327°19.8	18.8	285°20.0	16.6'	02°31.5	-12.9' 54.2'
11	342°19.6	18.3	299°55.6	16.6'	02°18.6	-12.9' 54.2'
12	357°19.4	S20°17.8	314°31.3	16.6'	S02°05.7	-12.9' 54.2'
13	12°19.2	17.3	329°06.9	16.6'	01°52.8	-12.9' 54.2'
14	27°19.0	16.7	343°42.5	16.6'	01°39.9	-12.9' 54.2'
15	42°18.8	· · 16.2	358°18.1	16.6'	01°26.9	-12.9' 54.2'
16	57°18.6	15.7	12°53.8	16.6'	01°14.0	-13.0' 54.2'
17	72°18.4	15.1	27°29.4	16.6'	01°01.0	-13.0' 54.3'
18	87°18.2	S20°14.6	42°05.0	16.6'	S00°48.1	-13.0' 54.3'
19	102°18.1	14.1	56°40.6	16.6'	00°35.1	-13.0' 54.3'
20	117°17.9	13.5	71°16.2	16.6'	00°22.1	-13.0' 54.3'
21	132°17.7	· · 13.0	85°51.8	16.6'	S00°09.1	-13.0' 54.3'
22	147°17.5	12.5	100°27.3	16.6'	N00°03.9	13.0' 54.3'
23	162°17.3	11.9	115°02.9	16.6'	00°16.9	13.0' 54.3'
SD = 16.2' $d = -0.5'$ SD = 14.7'						
Wed	GHA	Dec	GHA	ν	Dec	d HP
0	177°17.1	S20°11.4	129°38.5	16.5'	N00°29.9	13.0' 54.3'
1	192°16.9	10.9	144°14.0	16.5'	00°42.9	13.0' 54.4'
2	207°16.7	10.3	158°49.5	16.5'	00°55.9	13.0' 54.4'
3	222°16.6	· · 09.8	173°25.0	16.5'	01°08.9	13.0' 54.4'
4	237°16.4	09.2	188°00.5	16.5'	01°21.9	13.0' 54.4'
5	252°16.2	08.7	202°36.0	16.5'	01°34.9	13.0' 54.4'
6	267°16.0	S20°08.2	217°11.5	16.4'	N01°47.9	13.0' 54.4'
7	282°15.8	07.6	231°46.9	16.4'	02°00.9	13.0' 54.4'
8	297°15.6	07.1	246°22.3	16.4'	02°14.0	13.0' 54.5'
9	312°15.4	· · 06.5	260°57.7	16.4'	02°27.0	13.0' 54.5'
10	327°15.3	06.0	275°33.1	16.3'	02°40.0	13.0' 54.5'
11	342°15.1	05.5	290°08.4	16.3'	02°53.0	13.0' 54.5'
12	357°14.9	S20°04.9	304°43.7	16.3'	N03°06.0	13.0' 54.5'
13	12°14.7	04.4	319°19.0	16.3'	03°19.0	13.0' 54.5'
14	27°14.5	03.8	333°54.3	16.2'	03°32.0	13.0' 54.6'
15	42°14.3	· · 03.3	348°29.6	16.2'	03°44.9	13.0' 54.6'
16	57°14.2	02.7	3°04.8	16.2'	03°57.9	13.0' 54.6'
17	72°14.0	02.2	17°40.0	16.2'	04°10.9	13.0' 54.6'
18	87°13.8	S20°01.6	32°15.1	16.1'	N04°23.9	13.0' 54.6'
19	102°13.6	01.1	46°50.2	16.1'	04°36.8	12.9' 54.7'
20	117°13.4	00.5	61°25.3	16.1'	04°49.7	12.9' 54.7'
21	132°13.3	20°00.0	76°00.4	16.0'	05°02.7	12.9' 54.7'
22	147°13.1	19°59.4	90°35.4	16.0'	05°15.6	12.9' 54.7'
23	162°12.9	58.9	105°10.4	15.9'	05°28.5	12.9' 54.7'
SD = 16.2' $d = -0.5'$ SD = 14.8'						
Thu	GHA	Dec	GHA	ν	Dec	d HP
0	177°12.7	S19°58.3	119°45.3	15.9'	N05°41.4	12.9' 54.7'
1	192°12.5	57.8	134°20.2	15.9'	05°54.3	12.9' 54.8'
2	207°12.4	57.2	148°55.1	15.8'	06°07.2	12.9' 54.8'
3	222°12.2	· · 56.7	163°29.9	15.8'	06°20.0	12.8' 54.8'
4	237°12.0	56.1	178°04.7	15.7'	06°32.9	12.8' 54.8'
5	252°11.8	55.6	192°39.5	15.7'	06°45.7	12.8' 54.9'
6	267°11.7	S19°55.0	207°14.2	15.7'	N06°58.5	12.8' 54.9'
7	282°11.5	54.5	221°48.8	15.6'	07°11.3	12.8' 54.9'
8	297°11.3	53.9	236°23.4	15.6'	07°24.0	12.7' 54.9'
9	312°11.1	· · 53.3	250°58.0	15.5'	07°36.8	12.7' 54.9'
10	327°11.0	52.8	265°32.5	15.5'	07°49.5	12.7' 55.0'
11	342°10.8	52.2	280°06.9	15.4'	08°02.2	12.7' 55.0'
12	357°10.6	S19°51.7	294°41.4	15.4'	N08°14.9	12.7' 55.0'
13	12°10.4	51.1	309°15.7	15.3'	08°27.6	12.6' 55.0'
14	27°10.3	50.5	323°50.0	15.3'	08°40.2	12.6' 55.1'
15	42°10.1	· · 50.0	338°24.3	15.2'	08°52.8	12.6' 55.1'
16	57°09.9	49.4	352°58.5	15.1'	09°05.4	12.6' 55.1'
17	72°09.7	48.9	7°32.6	15.1'	09°18.0	12.5' 55.1'
18	87°09.6	S19°48.3	22°06.7	15.0'	N09°30.5	12.5' 55.1'
19	102°09.4	47.7	36°40.7	15.0'	09°43.0	12.5' 55.2'
20	117°09.2	47.2	51°14.7	14.9'	09°55.5	12.4' 55.2'
21	132°09.0	· · 46.6	65°48.6	14.8'	10°07.9	12.4' 55.2'
22	147°08.9	46.0	80°22.4	14.8'	10°20.4	12.4' 55.2'
23	162°08.7	45.5	94°56.2	14.7'	10°32.7	12.4' 55.3'
SD = 16.2' $d = -0.6'$ SD = 14.9'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:48	09:31	■	■	14:52	16:35
N 70°	07:35	09:03	11:06	13:17	15:20	16:48
68°	07:25	08:41	10:10	14:13	15:42	16:58
66°	07:16	08:24	09:38	14:45	15:59	17:07
64°	07:08	08:10	09:14	15:09	16:13	17:15
62°	07:01	07:58	08:55	15:28	16:24	17:22
60°	06:55	07:48	08:39	15:43	16:34	17:27
N 58°	06:50	07:39	08:26	15:56	16:43	17:33
56°	06:45	07:31	08:15	16:08	16:51	17:38
54°	06:40	07:24	08:05	16:17	16:58	17:42
52°	06:36	07:18	07:56	16:26	17:05	17:46
50°	06:32	07:12	07:48	16:34	17:10	17:50
45°	06:23	06:59	07:32	16:51	17:23	17:59
N 40°	06:15	06:48	07:18	17:05	17:34	18:07
35°	06:08	06:39	07:06	17:16	17:44	18:14
30°	06:01	06:30	06:56	17:27	17:52	18:21
20°	05:47	06:14	06:38	17:44	18:08	18:35
N 10°	05:34	06:00	06:22	18:00	18:22	18:48
0°	05:20	05:45	06:07	18:15	18:37	19:02
S 10°	05:04	05:30	05:52	18:29	18:52	19:18
20°	04:44	05:13	05:36	18:45	19:09	19:37
30°	04:19	04:51	05:18	19:04	19:30	20:02
35°	04:04	04:38	05:07	19:15	19:43	20:18
40°	03:44	04:23	04:54	19:27	19:58	20:37
45°	03:19	04:04	04:39	19:42	20:17	21:01
S 50°	02:45	03:40	04:21	20:00	20:41	21:35
52°	02:27	03:29	04:12	20:09	20:52	21:53
54°	02:03	03:15	04:03	20:18	21:06	22:16
56°	01:30	02:59	03:51	20:29	21:21	22:48
58°	////	02:39	03:39	20:42	21:41	////
S 60°	////	02:14	03:24	20:57	22:05	////

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	09:32	09:07	08:41	21:15	23:07	· · ·
N 70°	09:28	09:10	08:51	21:15	22:59	· · ·
68°	09:25	09:13	09:00	21:15	22:53	· · ·
66°	09:22	09:15	09:07	21:15	22:47	· · ·
64°	09:20	09:17	09:13	21:15	22:43	· · ·
62°	09:18	09:18	09:19	21:15	22:39	· · ·
60°	09:17	09:20	09:23	21:16	22:35	23:58
N 58°	09:15	09:21	09:28	21:16	22:32	23:51
56°	09:14	09:22	09:31	21:16	22:30	23:46
54°	09:13	09:23	09:35	21:16	22:27	23:41
52°	09:12	09:24	09:38	21:16	22:25	23:36
50°	09:11	09:25	09:41	21:16	22:23	23:32
45°	09:09	09:27	09:47	21:16	22:19	23:24
N 40°	09:07	09:29	09:52	21:16	22:15	23:16
35°	09:05	09:30	09:56	21:16	22:12	23:10
30°	09:04	09:32	10:00	21:16	22:10	23:05
20°	09:01	09:34	10:07	21:16	22:05	22:55
N 10°	08:59	09:36	10:13	21:17	22:01	22:47
0°	08:57	09:38	10:19	21:17	21:57	22:39
S 10°	08:55	09:39	10:25	21:17	21:53	22:32
20°	08:53	09:41	10:31	21:17	21:49	22:23
30°	08:51	09:44	10:38	21:17	21:45	22:14
35°	08:49	09:45	10:42	21:17	21:42	22:09
40°	08:48	09:47	10:47	21:17	21:39	22:03
45°	08:46	09:49	10:52	21:17	21:36	21:56
S 50°	08:44	09:51	10:59	21:17	21:32	21:47
52°	08:43	09:52	11:02	21:17	21:30	21:44
54°	08:42	09:53	11:05	21:17	21:28	21:39
56°	08:40	09:54	11:09	21:17	21:25	21:35
58°	08:39	09:55	11:13	21:17	21:23	21:30
S 60°	08:38	09:57	11:18	21:17	21:20	21:24

Day	Sun		Mer. Pass	Moon		Age 4-6 12-27%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
19	10:33	10:42	12:11	15:07	02:47	
20	10:52	11:00	12:11	15:47	03:27	
21	11:09	11:18	12:11	16:29	04:08	

January 22, 23, 24 UT (Fri., Sat., Sun.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Fri	GHA	Dec	GHA	Dec		GHA	Dec		GHA	Dec		GHA	Dec		SHA	Dec	
0	121°14.4	174°21.8	S20°16.8	344°32.8	N21°11.6	147°58.8	S12°01.5	296°04.9	N00°22.1			296°04.9	N00°22.1		Alpheratz	357°46.3	29°09.0
1	136°16.9	189°21.0	16.1	359°36.3	11.9	163°00.7	01.3	311°07.4	22.1			311°07.4	22.1		Ankaa	353°18.2	-42°15.2
2	151°19.4	204°20.2	15.4	14°39.7	12.2	178°02.6	01.1	326°09.9	22.1			326°09.9	22.1		Schedar	349°43.7	56°35.9
3	166°21.8	219°19.3	•• 14.7	29°43.1	•• 12.5	193°04.6	•• 00.9	341°12.4	•• 22.2			341°12.4	•• 22.2		Diphda	348°58.4	-17°55.9
4	181°24.3	234°18.5	14.0	44°46.5	12.9	208°06.5	00.7	356°14.9	22.2			356°14.9	22.2		Achernar	335°28.5	-57°11.4
5	196°26.8	249°17.7	13.3	59°50.0	13.2	223°08.4	00.5	11°17.4	22.2			11°17.4	22.2		Hamal	328°03.6	23°30.8
6	211°29.2	264°16.9	S20°12.6	74°53.4	N21°13.5	238°10.3	S12°00.3	26°19.9	N00°22.2			26°19.9	N00°22.2		Polaris	318°54.8	89°18.9
7	226°31.7	279°16.1	11.9	89°56.8	13.8	253°12.3	12°00.1	41°22.4	22.3			41°22.4	22.3		Acamar	315°20.0	-40°16.0
8	241°34.1	294°15.3	11.3	105°00.3	14.1	268°14.2	11°59.9	56°24.9	22.3			56°24.9	22.3		Menkar	314°17.6	4°07.8
9	256°36.6	309°14.5	•• 10.6	120°03.7	•• 14.5	283°16.1	•• 59.7	71°27.4	•• 22.3			71°27.4	•• 22.3		Mirfak	308°43.8	49°54.1
10	271°39.1	324°13.7	09.9	135°07.1	14.8	298°18.0	59.5	86°29.9	22.3			86°29.9	22.3		Aldebaran	290°52.0	16°31.8
11	286°41.5	339°12.9	09.2	150°10.6	15.1	313°20.0	59.3	101°32.4	22.4			101°32.4	22.4		Rigel	281°14.2	-8°11.5
12	301°44.0	354°12.1	S20°08.5	165°14.0	N21°15.4	328°21.9	S11°59.1	116°34.8	N00°22.4			116°34.8	N00°22.4		Capella	280°37.8	46°00.6
13	316°46.5	9°11.3	07.8	180°17.4	15.7	343°23.8	58.9	131°37.3	22.4			131°37.3	22.4		Bellatrix	278°34.4	6°21.5
14	331°48.9	24°10.5	07.1	195°20.9	16.1	358°25.7	58.7	146°39.8	22.5			146°39.8	22.5		Elnath	278°15.5	28°37.0
15	346°51.4	39°09.8	•• 06.4	210°24.3	•• 16.4	13°27.7	•• 58.5	161°42.3	•• 22.5			161°42.3	•• 22.5		Alnilam	275°48.6	-1°11.8
16	1°53.9	54°09.0	05.6	225°27.7	16.7	28°29.6	58.3	176°44.8	22.5			176°44.8	22.5		Betelgeuse	271°03.7	7°24.5
17	16°56.3	69°08.2	04.9	240°31.2	17.0	43°31.5	58.1	191°47.3	22.5			191°47.3	22.5		Canopus	263°56.8	-52°42.2
18	31°58.8	84°07.4	S20°04.2	255°34.6	N21°17.3	58°33.4	S11°57.9	206°49.8	N00°22.6			206°49.8	N00°22.6		Sirius	258°35.6	-16°43.9
19	47°01.3	99°06.6	03.5	270°38.0	17.6	73°35.3	57.7	221°52.3	22.6			221°52.3	22.6		Adhara	255°14.1	-28°59.3
20	62°03.7	114°05.8	02.8	285°41.5	18.0	88°37.3	57.5	236°54.8	22.6			236°54.8	22.6		Procyon	245°02.0	5°11.8
21	77°06.2	129°05.0	•• 02.1	300°44.9	•• 18.3	103°39.2	•• 57.3	251°57.3	•• 22.6			251°57.3	•• 22.6		Pollux	243°30.3	28°00.0
22	92°08.6	144°04.2	01.4	315°48.4	18.6	118°41.1	57.1	266°59.8	22.7			266°59.8	22.7		Avior	234°18.5	-59°32.6
23	107°11.1	159°03.4	00.7	330°51.8	18.9	133°43.0	56.9	282°02.3	22.7			282°02.3	22.7		Suhail	222°53.9	-43°28.4
Mer.pass. 15:52			ν -0.8' d -0.7' m -3.92	ν 3.4' d 0.3' m -1.24	ν 1.9' d -0.2' m -2.06	ν 2.5' d 0.0' m 0.77											

Sat	GHA	Dec	GHA	Dec		GHA	Dec		GHA	Dec	
0	122°13.6	174°02.6	S20°00.0	345°55.2	N21°19.2	148°45.0	S11°56.7	297°04.8	N00°22.7		
1	137°16.0	189°01.8	19°59.3	0°58.7	19.6	163°46.9	56.5	312°07.3	22.8		
2	152°18.5	204°01.0	58.5	16°02.1	19.9	178°48.8	56.3	327°09.8	22.8		
3	167°21.0	219°00.2	•• 57.8	31°05.6	•• 20.2	193°50.7	•• 56.1	342°12.3	•• 22.8		
4	182°23.4	233°59.4	57.1	46°09.0	20.5	208°52.7	55.9	357°14.8	22.8		
5	197°25.9	248°58.7	56.4	61°12.5	20.8	223°54.6	55.7	12°17.3	22.9		
6	212°28.4	263°57.9	S19°55.7	76°15.9	N21°21.1	238°56.5	S11°55.5	27°19.8	N00°22.9		
7	227°30.8	278°57.1	55.0	91°19.3	21.5	253°58.4	55.3	42°22.3	22.9		
8	242°33.3	293°56.3	54.2	106°22.8	21.8	269°00.3	55.1	57°24.8	22.9		
9	257°35.7	308°55.5	•• 53.5	121°26.2	•• 22.1	284°02.3	•• 54.8	72°27.3	•• 23.0		
10	272°38.2	323°54.7	52.8	136°29.7	22.4	299°04.2	54.6	87°29.8	23.0		
11	287°40.7	338°53.9	52.1	151°33.1	22.7	314°06.1	54.4	102°32.3	23.0		
12	302°43.1	353°53.1	S19°51.3	166°36.6	N21°23.1	329°08.0	S11°54.2	117°34.8	N00°23.1		
13	317°45.6	8°52.4	50.6	181°40.0	23.4	344°09.9	54.0	132°37.3	23.1		
14	332°48.1	23°51.6	49.9	196°43.5	23.7	359°11.9	53.8	147°39.8	23.1		
15	347°50.5	38°50.8	•• 49.2	211°46.9	•• 24.0	14°13.8	•• 53.6	162°42.3	•• 23.1		
16	2°53.0	53°50.0	48.4	226°50.4	24.3	29°15.7	53.4	177°44.8	23.2		
17	17°55.5	68°49.2	47.7	241°53.8	24.6	44°17.6	53.2	192°47.3	23.2		
18	32°57.9	83°48.4	S19°47.0	256°57.3	N21°25.0	59°19.6	S11°53.0	207°49.8	N00°23.2		
19	48°00.4	98°47.7	46.2	272°00.7	25.3	74°21.5	52.8	222°52.3	23.3		
20	63°02.9	113°46.9	45.5	287°04.2	25.6	89°23.4	52.6	237°54.8	23.3		
21	78°05.3	128°46.1	•• 44.8	302°07.6	•• 25.9	104°25.3	•• 52.4	252°57.3	•• 23.3		
22	93°07.8	143°45.3	44.0	317°11.1	26.2	119°27.2	52.2	267°59.8	23.3		
23	108°10.2	158°44.5	43.3	332°14.5	26.5	134°29.2	52.0	283°02.3	23.4		
Mer.pass. 15:48			ν -0.8' d -0.7' m -3.92	ν 3.4' d 0.3' m -1.26	ν 1.9' d -0.2' m -2.05	ν 2.5' d 0.0' m 0.76					

Sun	GHA	Dec	GHA	Dec		GHA	Dec		GHA	Dec	
0	123°12.7	173°43.7	S19°42.6	347°18.0	N21°26.8	149°31.1	S11°51.8	298°04.8	N00°23.4		
1	138°15.2	188°43.0	41.8	2°21.5	27.2	164°33.0	51.6	313°07.3	23.4		
2	153°17.6	203°42.2	41.1	17°24.9	27.5	179°34.9	51.4	328°09.8	23.5		
3	168°20.1	218°41.4	•• 40.4	32°28.4	•• 27.8	194°36.8	•• 51.2	343°12.3	•• 23.5		
4	183°22.6	233°40.6	39.6	47°31.8	28.1	209°38.8	51.0	358°14.8	23.5		
5	198°25.0	248°39.9	38.9	62°35.3	28.4	224°40.7	50.8	13°17.3	23.5		
6	213°27.5	263°39.1	S19°38.1	77°38.7	N21°28.7	239°42.6	S11°50.6	28°19.8	N00°23.6		
7	228°30.0	278°38.3	37.4	92°42.2	29.0	254°44.5	50.4	43°22.3	23.6		
8	243°32.4	293°37.5	36.6	107°45.7	29.4	269°46.4	50.2	58°24.8	23.6		
9	258°34.9	308°36.8	•• 35.9	122°49.1	•• 29.7	284°48.4	•• 50.0	73°27.4	•• 23.7		
10	273°37.4	323°36.0	35.2	137°52.6	30.0	299°50.3	49.8	88°29.9	23.7		
11	288°39.8	338°35.2	34.4	152°56.0	30.3	314°52.2	49.6	103°32.4	23.7		
12	303°42.3	353°34.4	S19°33.7	167°59.5	N21°30.6	329°54.1	S11°49.4	118°34.9	N00°23.8		
13	318°44.7	8°33.7	32.9	183°03.0	30.9	344°56.0	49.2	133°37.4	23.8		
14	333°47.2	23°32.9	32.2	198°06.4	31.2	359°58.0	48.9	148°39.9	23.8		
15	348°49.7	38°32.1	•• 31.4	213°09.9	•• 31.6	14°59.9	•• 48.7	163°42.4	•• 23.8		
16	3°52.1	53°31.3	30.7	228°13.3	31.9	30°01.8	48.5	178°44.9	23.9		
17	18°54.6	68°30.6	29.9	243°16.8	32.2	45°03.7	48.3	193°47.4	23.9		
18	33°57.1	83°29.8	S19°29.1	258°20.3	N21°32.5	60°05.6	S11°48.1	208°49.9	N00°23.9		
19	48°59.5	98°29.0	28.4	273°23.7	32.8	75°07.6	47.9	223°52.4	24.0		
20	64°02.0	113°28.3	27.6	288°27.2	33.1	90°09.5	47.7	238°54.9	24.0		
21	79°04.5	128°27.5	•• 26.9	303°30.7	•• 33.4	105°11.4	•• 47.5	253°57.4	•• 24.0		
22	94°06.9	143°26.7	26.1	318°34.1	33.7	120°13.3	47.3	268°59.9	24.1		
23	109°09.4	158°26.0	25.4	333°37.6	34.1	135°15.2	47.1	284°02.4	24.1		
Mer.pass. 15:45			ν -0.8' d -0.7' m -3.92	ν 3.5' d 0.3' m -1.28	ν 1.9' d -0.2' m -2.05	ν 2.5' d 0.0' m 0.76					

Stars		
Jan 22 Fri	SHA	Mer.pass
Venus	53°07.3	12:23
Mars	223°18.4	01:02
Jupiter	26°44.4	14:06
Saturn	174°50.5	04:15
Jan 23 Sat	SHA	Mer.pass
Venus	51°49.0	12:24
Mars	223°41.7	00:56
Jupiter	26°31.4	14:03
Saturn	174°51.2	04:11
Jan 24 Sun	SHA	Mer.pass
Venus	50°31.0	12:26
Mars	224°05.3	00:51
Jupiter	26°18.4	14:00
Saturn	174°52.1	04:07
Horizontal parallax		
Venus:	0.1	
Mars:	0.2	

h	Sun				Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP	
0	177°08.5	S19°44.9	109°30.0	14.7'	N10°45.1	12.3'	55.3'	
1	192°08.4	44.3	124°03.6	14.6'	10°57.4	12.3'	55.3'	
2	207°08.2	43.8	138°37.2	14.5'	11°09.7	12.3'	55.4'	
3	222°08.0	· · 43.2	153°10.7	14.5'	11°22.0	12.2'	55.4'	
4	237°07.9	42.6	167°44.2	14.4'	11°34.2	12.2'	55.4'	
5	252°07.7	42.1	182°17.6	14.3'	11°46.4	12.1'	55.4'	
6	267°07.5	S19°41.5	196°50.9	14.2'	N11°58.5	12.1'	55.5'	
7	282°07.3	40.9	211°24.1	14.2'	12°10.6	12.1'	55.5'	
8	297°07.2	40.3	225°57.3	14.1'	12°22.7	12.0'	55.5'	
9	312°07.0	· · 39.8	240°30.4	14.0'	12°34.7	12.0'	55.5'	
10	327°06.8	39.2	255°03.4	13.9'	12°46.7	11.9'	55.6'	
11	342°06.7	38.6	269°36.3	13.9'	12°58.6	11.9'	55.6'	
12	357°06.5	S19°38.0	284°09.2	13.8'	N13°10.5	11.8'	55.6'	
13	12°06.3	37.5	298°42.0	13.7'	13°22.3	11.8'	55.7'	
14	27°06.2	36.9	313°14.7	13.6'	13°34.1	11.8'	55.7'	
15	42°06.0	· · 36.3	327°47.3	13.6'	13°45.9	11.7'	55.7'	
16	57°05.8	35.7	342°19.9	13.5'	13°57.6	11.7'	55.7'	
17	72°05.7	35.2	356°52.4	13.4'	14°09.3	11.6'	55.8'	
18	87°05.5	S19°34.6	11°24.8	13.3'	N14°20.9	11.6'	55.8'	
19	102°05.3	34.0	25°57.1	13.2'	14°32.4	11.5'	55.8'	
20	117°05.2	33.4	40°29.3	13.1'	14°44.0	11.5'	55.9'	
21	132°05.0	· · 32.8	55°01.4	13.0'	14°55.4	11.4'	55.9'	
22	147°04.9	32.3	69°33.5	13.0'	15°06.8	11.3'	55.9'	
23	162°04.7	31.7	84°05.4	12.9'	15°18.1	11.3'	56.0'	
SD = 16.2' $d = -0.6'$ SD = 15.1'								

Sat	GHA	Dec	GHA	ν	Dec	d	HP	
0	177°04.5	S19°31.1	98°37.3	12.8'	N15°29.4	11.2'	56.0'	
1	192°04.4	30.5	113°09.1	12.7'	15°40.7	11.2'	56.0'	
2	207°04.2	29.9	127°40.7	12.6'	15°51.8	11.1'	56.1'	
3	222°04.0	· · 29.4	142°12.3	12.5'	16°02.9	11.0'	56.1'	
4	237°03.9	28.8	156°43.8	12.4'	16°14.0	11.0'	56.1'	
5	252°03.7	28.2	171°15.2	12.3'	16°25.0	10.9'	56.2'	
6	267°03.6	S19°27.6	185°46.6	12.2'	N16°35.9	10.9'	56.2'	
7	282°03.4	27.0	200°17.8	12.1'	16°46.8	10.8'	56.2'	
8	297°03.2	26.4	214°48.9	12.0'	16°57.6	10.7'	56.3'	
9	312°03.1	· · 25.8	229°19.9	11.9'	17°08.3	10.7'	56.3'	
10	327°02.9	25.2	243°50.9	11.8'	17°18.9	10.6'	56.3'	
11	342°02.8	24.7	258°21.7	11.7'	17°29.5	10.5'	56.4'	
12	357°02.6	S19°24.1	272°52.4	11.6'	N17°40.0	10.4'	56.4'	
13	12°02.4	23.5	287°23.0	11.5'	17°50.5	10.4'	56.4'	
14	27°02.3	22.9	301°53.6	11.4'	18°00.9	10.3'	56.5'	
15	42°02.1	· · 22.3	316°24.0	11.3'	18°11.2	10.2'	56.5'	
16	57°02.0	21.7	330°54.3	11.2'	18°21.4	10.1'	56.5'	
17	72°01.8	21.1	345°24.5	11.1'	18°31.5	10.1'	56.6'	
18	87°01.7	S19°20.5	359°54.6	11.0'	N18°41.6	10.0'	56.6'	
19	102°01.5	19.9	14°24.6	10.9'	18°51.6	9.9'	56.6'	
20	117°01.4	19.3	28°54.5	10.8'	19°01.5	9.8'	56.7'	
21	132°01.2	· · 18.7	43°24.3	10.7'	19°11.3	9.7'	56.7'	
22	147°01.0	18.1	57°54.0	10.6'	19°21.0	9.7'	56.7'	
23	162°00.9	17.5	72°23.6	10.5'	19°30.7	9.6'	56.8'	
SD = 16.2' $d = -0.6'$ SD = 15.3'								

Sun	GHA	Dec	GHA	ν	Dec	d	HP	
0	177°00.7	S19°16.9	86°53.1	10.4'	N19°40.3	9.5'	56.8'	
1	192°00.6	16.4	101°22.4	10.3'	19°49.7	9.4'	56.9'	
2	207°00.4	15.8	115°51.7	10.1'	19°59.1	9.3'	56.9'	
3	222°00.3	· · 15.2	130°20.8	10.0'	20°08.4	9.2'	56.9'	
4	237°00.1	14.6	144°49.9	9.9'	20°17.6	9.1'	57.0'	
5	252°00.0	14.0	159°18.8	9.8'	20°26.7	9.0'	57.0'	
6	266°59.8	S19°13.4	173°47.6	9.7'	N20°35.7	8.9'	57.0'	
7	281°59.7	12.8	188°16.3	9.6'	20°44.7	8.8'	57.1'	
8	296°59.5	12.2	202°44.9	9.5'	20°53.5	8.7'	57.1'	
9	311°59.4	· · 11.5	217°13.4	9.4'	21°02.2	8.6'	57.2'	
10	326°59.2	10.9	231°41.7	9.3'	21°10.8	8.5'	57.2'	
11	341°59.1	10.3	246°10.0	9.1'	21°19.3	8.4'	57.2'	
12	356°58.9	S19°09.7	260°38.1	9.0'	N21°27.7	8.3'	57.3'	
13	11°58.8	09.1	275°06.2	8.9'	21°36.0	8.2'	57.3'	
14	26°58.6	08.5	289°34.1	8.8'	21°44.2	8.1'	57.3'	
15	41°58.5	· · 07.9	304°01.9	8.7'	21°52.3	8.0'	57.4'	
16	56°58.3	07.3	318°29.6	8.6'	22°00.3	7.9'	57.4'	
17	71°58.2	06.7	332°57.1	8.5'	22°08.2	7.8'	57.5'	
18	86°58.0	S19°06.1	347°24.6	8.3'	N22°15.9	7.6'	57.5'	
19	101°57.9	05.5	1°51.9	8.2'	22°23.6	7.5'	57.5'	
20	116°57.7	04.9	16°19.1	8.1'	22°31.1	7.4'	57.6'	
21	131°57.6	· · 04.3	30°46.3	8.0'	22°38.5	7.3'	57.6'	
22	146°57.4	03.7	45°13.3	7.9'	22°45.8	7.2'	57.7'	
23	161°57.3	03.0	59°40.2	7.8'	22°53.0	7.0'	57.7'	
SD = 16.2' $d = -0.6'$ SD = 15.5'								

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:39	09:18	■	■	15:06	16:45
N 70°	07:28	08:53	10:41	13:43	15:32	16:57
68°	07:18	08:33	09:57	14:28	15:52	17:07
66°	07:10	08:17	09:28	14:57	16:07	17:15
64°	07:03	08:04	09:06	15:19	16:20	17:22
62°	06:57	07:53	08:48	15:36	16:31	17:28
60°	06:51	07:44	08:34	15:51	16:41	17:33
N 58°	06:46	07:35	08:21	16:03	16:49	17:38
56°	06:42	07:28	08:11	16:14	16:57	17:43
54°	06:37	07:21	08:01	16:23	17:03	17:47
52°	06:33	07:15	07:53	16:31	17:09	17:51
50°	06:30	07:09	07:45	16:39	17:15	17:55
45°	06:21	06:57	07:29	16:55	17:27	18:03
N 40°	06:14	06:47	07:16	17:08	17:37	18:10
35°	06:07	06:38	07:05	17:19	17:46	18:17
30°	06:00	06:29	06:55	17:29	17:55	18:24
20°	05:47	06:14	06:38	17:46	18:10	18:36
N 10°	05:35	06:00	06:22	18:01	18:23	18:49
0°	05:21	05:46	06:08	18:15	18:37	19:03
S 10°	05:05	05:31	05:54	18:30	18:52	19:18
20°	04:47	05:15	05:38	18:45	19:09	19:37
30°	04:23	04:54	05:20	19:03	19:29	20:01
35°	04:07	04:42	05:10	19:13	19:41	20:16
40°	03:49	04:27	04:58	19:25	19:56	20:34
45°	03:25	04:09	04:44	19:39	20:14	20:58
S 50°	02:52	03:46	04:26	19:57	20:37	21:30
52°	02:35	03:35	04:18	20:05	20:48	21:47
54°	02:13	03:22	04:08	20:14	21:00	22:08
56°	01:44	03:07	03:58	20:25	21:15	22:36
58°	00:54	02:48	03:46	20:37	21:33	23:21
S 60°	////	02:25	03:32	20:50	21:56	////

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	08:08	07:10	□	01:09	03:42	□
N 70°	08:29	07:56	□	00:50	02:58	□
68°	08:45	08:26	07:50	00:35	02:29	04:47
66°	08:59	08:49	08:35	00:23	02:07	04:03
64°	09:10	09:07	09:05	00:13	01:50	03:34
62°	09:20	09:22	09:28	00:05	01:36	03:12
60°	09:28	09:35	09:46	· · ·	01:24	02:55
N 58°	09:35	09:46	10:01	· · ·	01:14	02:40
56°	09:42	09:55	10:14	· · ·	01:05	02:27
54°	09:48	10:04	10:26	· · ·	00:57	02:17
52°	09:53	10:12	10:36	· · ·	00:50	02:07
50°	09:58	10:18	10:45	· · ·	00:44	01:58
45°	10:08	10:33	11:04	· · ·	00:31	01:40
N 40°	10:17	10:46	11:20	· · ·	00:20	01:25
35°	10:24	10:56	11:34	· · ·	00:10	01:13
30°	10:31	11:05	11:45	· · ·	00:02	01:02
20°	10:42	11:21	12:05	23:48	· · ·	00:43
N 10°	10:53	11:35	12:23	23:35	· · ·	00:27
0°	11:02	11:49	12:39	23:24	· · ·	00:12
S 10°	11:12	12:02	12:56	23:12	23:57	· · ·
20°	11:22	12:16	13:14	23:00	23:41	· · ·
30°	11:34	12:33	13:35	22:46	23:23	· · ·
35°	11:41	12:42	13:47	22:38	23:13	23:53
40°	11:49	12:54	14:01	22:29	23:01	23:39
45°	11:58	13:07	14:17	22:19	22:46	23:21
S 50°	12:09	13:23	14:38	22:06	22:29	22:59
52°	12:15	13:30	14:48	22:00	22:21	22:49
54°	12:20	13:39	14:59	21:54	22:12	22:38
56°	12:27	13:48	15:12	21:46	22:02	22:24
58°	12:34	13:59	15:27	21:38	21:50	22:09
S 60°	12:42	14:11	15:44	21:29	21:37	21:51

Day	Sun			Moon		Age 7-9 36-56%
	Eqn. of Time		Mer.	Mer.Pass.		
	00 ^h mm:ss	12 ^h mm:ss	Pass hh:mm	Upper hh:mm	Lower hh:mm	
22	11:26	11:34	12:12	17:13	04:51	
23	11:42	11:50	12:12	18:00	05:36	
24	11:57	12:04	12:12	18:52	06:26	

January 25, 26, 27 UT (Mon., Tue., Wed.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	124°11.8	173°25.2	S19°24.6	348°41.1	N21°34.4	150°17.2	S11°46.9	299°04.9	N00°24.1	Alpheratz	357°46.3	29°08.9
1	139°14.3	188°24.4	23.8	3°44.5	34.7	165°19.1	46.7	314°07.4	24.1	Ankaa	353°18.2	-42°15.2
2	154°16.8	203°23.7	23.1	18°48.0	35.0	180°21.0	46.5	329°09.9	24.2	Schedar	349°43.8	56°35.9
3	169°19.2	218°22.9	•• 22.3	33°51.5	•• 35.3	195°22.9	•• 46.3	344°12.5	•• 24.2	Diphda	348°58.4	-17°55.9
4	184°21.7	233°22.1	21.6	48°54.9	35.6	210°24.8	46.1	359°15.0	24.2	Achernar	335°28.5	-57°11.4
5	199°24.2	248°21.4	20.8	63°58.4	35.9	225°26.8	45.9	14°17.5	24.3	Hamal	328°03.6	23°30.8
6	214°26.6	263°20.6	S19°20.0	79°01.9	N21°36.2	240°28.7	S11°45.7	29°20.0	N00°24.3	Polaris	318°55.9	89°18.9
7	229°29.1	278°19.8	19.3	94°05.3	36.5	255°30.6	45.5	44°22.5	24.3	Acamar	315°20.0	-40°16.0
8	244°31.6	293°19.1	18.5	109°08.8	36.9	270°32.5	45.3	59°25.0	24.4	Menkar	314°17.6	4°07.8
9	259°34.0	308°18.3	•• 17.7	124°12.3	•• 37.2	285°34.4	•• 45.1	74°27.5	•• 24.4	Mirfak	308°43.8	49°54.1
10	274°36.5	323°17.6	16.9	139°15.8	37.5	300°36.4	44.9	89°30.0	24.4	Aldebaran	290°52.0	16°31.8
11	289°39.0	338°16.8	16.2	154°19.2	37.8	315°38.3	44.7	104°32.5	24.5	Rigel	281°14.2	-8°11.5
12	304°41.4	353°16.0	S19°15.4	169°22.7	N21°38.1	330°40.2	S11°44.4	119°35.0	N00°24.5	Capella	280°37.8	46°00.6
13	319°43.9	8°15.3	14.6	184°26.2	38.4	345°42.1	44.2	134°37.5	24.5	Bellatrix	278°34.4	6°21.5
14	334°46.3	23°14.5	13.9	199°29.6	38.7	0°44.0	44.0	149°40.0	24.6	Elnath	278°15.5	28°37.0
15	349°48.8	38°13.8	•• 13.1	214°33.1	•• 39.0	15°45.9	•• 43.8	164°42.5	•• 24.6	Alnilam	275°48.6	-1°11.8
16	4°51.3	53°13.0	12.3	229°36.6	39.3	30°47.9	43.6	179°45.1	24.6	Betelgeuse	271°03.7	7°24.5
17	19°53.7	68°12.2	11.5	244°40.1	39.6	45°49.8	43.4	194°47.6	24.6	Canopus	263°56.8	-52°42.2
18	34°56.2	83°11.5	S19°10.8	259°43.5	N21°40.0	60°51.7	S11°43.2	209°50.1	N00°24.7	Sirius	258°35.6	-16°43.9
19	49°58.7	98°10.7	10.0	274°47.0	40.3	75°53.6	43.0	224°52.6	24.7	Adhara	255°14.1	-28°59.3
20	65°01.1	113°10.0	09.2	289°50.5	40.6	90°55.5	42.8	239°55.1	24.7	Procyon	245°02.0	5°11.8
21	80°03.6	128°09.2	•• 08.4	304°54.0	•• 40.9	105°57.5	•• 42.6	254°57.6	•• 24.8	Pollux	243°30.3	28°00.0
22	95°06.1	143°08.5	07.6	319°57.4	41.2	120°59.4	42.4	270°00.1	24.8	Avior	234°18.5	-59°32.6
23	110°08.5	158°07.7	06.8	335°00.9	41.5	136°01.3	42.2	285°02.6	24.8	Suhail	222°53.9	-43°28.5
Mer.pass. 15:41		ν -0.8' d -0.8' m -3.92		ν 3.5' d 0.3' m -1.29		ν 1.9' d -0.2' m -2.05		ν 2.5' d 0.0' m 0.75		Miaplacidus	221°39.5	-69°45.5
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	125°11.0	173°07.0	S19°06.1	350°04.4	N21°41.8	151°03.2	S11°42.0	300°05.1	N00°24.9	Alphard	217°58.2	-8°42.3
1	140°13.5	188°06.2	05.3	5°07.9	42.1	166°05.1	41.8	315°07.6	24.9	Regulus	207°45.8	11°54.9
2	155°15.9	203°05.5	04.5	20°11.3	42.4	181°07.0	41.6	330°10.2	24.9	Dubhe	193°54.0	61°41.5
3	170°18.4	218°04.7	•• 03.7	35°14.8	•• 42.7	196°09.0	•• 41.4	345°12.7	•• 25.0	Denebola	182°35.9	14°30.7
4	185°20.8	233°03.9	02.9	50°18.3	43.0	211°10.9	41.2	0°15.2	25.0	Gienah	175°54.7	-17°36.0
5	200°23.3	248°03.2	02.1	65°21.8	43.3	226°12.8	41.0	15°17.7	25.0	Acrux	173°12.0	-63°09.2
6	215°25.8	263°02.4	S19°01.3	80°25.2	N21°43.6	241°14.7	S11°40.8	30°20.2	N00°25.1	Gacrux	172°03.6	-57°10.1
7	230°28.2	278°01.7	19°00.6	95°28.7	43.9	256°16.6	40.5	45°22.7	25.1	Alioth	166°22.5	55°53.9
8	245°30.7	293°00.9	18°59.8	110°32.2	44.3	271°18.5	40.3	60°25.2	25.1	Spica	158°33.8	-11°13.0
9	260°33.2	308°00.2	•• 59.0	125°35.7	•• 44.6	286°20.5	•• 40.1	75°27.7	•• 25.2	Alkaid	153°00.6	49°15.4
10	275°35.6	322°59.4	58.2	140°39.2	44.9	301°22.4	39.9	90°30.2	25.2	Hadar	148°51.6	-60°25.1
11	290°38.1	337°58.7	57.4	155°42.6	45.2	316°24.3	39.7	105°32.8	25.2	Menkent	148°10.5	-36°25.1
12	305°40.6	352°58.0	S18°56.6	170°46.1	N21°45.5	331°26.2	S11°39.5	120°35.3	N00°25.3	Arcturus	145°57.9	19°07.5
13	320°43.0	7°57.2	55.8	185°49.6	45.8	346°28.1	39.3	135°37.8	25.3	Rigil Kent.	139°55.3	-60°52.4
14	335°45.5	22°56.5	55.0	200°53.1	46.1	1°30.0	39.1	150°40.3	25.3	Kochab	137°19.8	74°06.4
15	350°48.0	37°55.7	•• 54.2	215°56.6	•• 46.4	16°32.0	•• 38.9	165°42.8	•• 25.4	Zuben'ubi	137°08.2	-16°05.1
16	5°50.4	52°55.0	53.4	231°00.1	46.7	31°33.9	38.7	180°45.3	25.4	Alphecca	126°13.2	26°40.6
17	20°52.9	67°54.2	52.6	246°03.5	47.0	46°35.8	38.5	195°47.8	25.4	Antares	112°29.5	-26°27.3
18	35°55.3	82°53.5	S18°51.8	261°07.0	N21°47.3	61°37.7	S11°38.3	210°50.3	N00°25.5	Atria	107°34.0	-69°02.6
19	50°57.8	97°52.7	51.0	276°10.5	47.6	76°39.6	38.1	225°52.9	25.5	Sabik	102°15.6	-15°44.3
20	66°00.3	112°52.0	50.2	291°14.0	47.9	91°41.6	37.9	240°55.4	25.5	Shaula	96°25.6	-37°06.6
21	81°02.7	127°51.3	•• 49.4	306°17.5	•• 48.2	106°43.5	•• 37.7	255°57.9	•• 25.6	Rasalhague	96°09.0	12°33.0
22	96°05.2	142°50.5	48.6	321°20.9	48.5	121°45.4	37.5	271°00.4	25.6	Eltanin	90°47.7	51°29.0
23	111°07.7	157°49.8	47.8	336°24.4	48.8	136°47.3	37.2	286°02.9	25.6	Kaus Aust.	83°47.4	-34°22.8
Mer.pass. 15:37		ν -0.8' d -0.8' m -3.92		ν 3.5' d 0.3' m -1.31		ν 1.9' d -0.2' m -2.05		ν 2.5' d 0.0' m 0.75		Vega	80°41.0	38°47.4
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	126°10.1	172°49.0	S18°47.0	351°27.9	N21°49.1	151°49.2	S11°37.0	301°05.4	N00°25.7	Nunki	76°01.7	-26°17.0
1	141°12.6	187°48.3	46.2	6°31.4	49.4	166°51.1	36.8	316°07.9	25.7	Altair	62°11.0	8°53.7
2	156°15.1	202°47.5	45.4	21°34.9	49.7	181°53.0	36.6	331°10.5	25.7	Peacock	53°23.6	-56°42.2
3	171°17.5	217°46.8	•• 44.6	36°38.4	•• 50.0	196°55.0	•• 36.4	346°13.0	•• 25.8	Deneb	49°33.7	45°19.0
4	186°20.0	232°46.1	43.7	51°41.9	50.3	211°56.9	36.2	1°15.5	25.8	Enif	33°49.9	9°55.3
5	201°22.5	247°45.3	42.9	66°45.3	50.6	226°58.8	36.0	16°18.0	25.8	Al Na'ir	27°47.1	-46°54.8
6	216°24.9	262°44.6	S18°42.1	81°48.8	N21°50.9	242°00.7	S11°35.8	31°20.5	N00°25.9	Fomalhaut	15°26.9	-29°34.2
7	231°27.4	277°43.9	41.3	96°52.3	51.2	257°02.6	35.6	46°23.0	25.9	Scheat	13°56.1	28°08.4
8	246°29.8	292°43.1	40.5	111°55.8	51.5	272°04.5	35.4	61°25.6	25.9	Markab	13°41.1	15°15.7
9	261°32.3	307°42.4	•• 39.7	126°59.3	•• 51.8	287°06.5	•• 35.2	76°28.1	•• 26.0			
10	276°34.8	322°41.6	38.9	142°02.8	52.1	302°08.4	35.0	91°30.6	26.0			
11	291°37.2	337°40.9	38.1	157°06.3	52.4	317°10.3	34.8	106°33.1	26.0			
12	306°39.7	352°40.2	S18°37.2	172°09.7	N21°52.7	332°12.2	S11°34.6	121°35.6	N00°26.1			
13	321°42.2	7°39.4	36.4	187°13.2	53.0	347°14.1	34.4	136°38.1	26.1			
14	336°44.6	22°38.7	35.6	202°16.7	53.3	2°16.0	34.1	151°40.6	26.2			
15	351°47.1	37°38.0	•• 34.8	217°20.2	•• 53.6	17°18.0	•• 33.9	166°43.2	•• 26.2			
16	6°49.6	52°37.2	34.0	232°23.7	53.9	32°19.9	33.7	181°45.7	26.2			
17	21°52.0	67°36.5	33.1	247°27.2	54.2	47°21.8	33.5	196°48.2	26.3			
18	36°54.5	82°35.8	S18°32.3	262°30.7	N21°54.5	62°23.7	S11°33.3	211°50.7	N00°26.3			
19	51°57.0	97°35.1	31.5	277°34.2	54.8	77°25.6	33.1	226°53.2	26.3			
20	66°59.4	112°34.3	30.7	292°37.7	55.1	92°27.5	32.9	241°55.8	26.4			
21	82°01.9	127°33.6	•• 29.8	307°41.1	•• 55.4	107°29.4	•• 32.7	256°58.3	•• 26.4			
22	97°04.3	142°32.9	29.0	322°44.6	55.7	122°31.4	32.5	272°00.8	26.4			
23	112°06.8	157°32.1	28.2	337°48.1	56.0	137°33.3	32.3	287°03.3	26.5			
Mer.pass. 15:33		ν -0.7' d -0.8' m -3.92		ν 3.5' d 0.3' m -1.32		ν 1.9' d -0.2' m -2.05		ν 2.5' d 0.0' m 0.74				

Jan 25 Mon			SHA	Mer.pass
Venus			49°13.4	12:27
Mars			224°29.2	00:45
Jupiter			26°05.3	13:57
Saturn			174°53.1	04:03
Jan 26 Tue			SHA	Mer.pass
Venus			47°56.0	12:28
Mars			224°53.4	00:40
Jupiter			25°52.2	13:54
Saturn				

Sun				Moon			
h	GHA	Dec		GHA	ν	Dec	HP
Mon							
0	176°57.1	S19°02.4		74°06.9	7.7'	N23°00.0	6.9'
1	191°57.0	01.8		88°33.6	7.5'	23°07.0	6.8'
2	206°56.8	01.2		103°00.1	7.4'	23°13.7	6.7'
3	221°56.7	· · 00.6		117°26.6	7.3'	23°20.4	6.5'
4	236°56.6	19°00.0		131°52.9	7.2'	23°27.0	6.4'
5	251°56.4	18°59.4		146°19.1	7.1'	23°33.4	6.3'
6	266°56.3	S18°58.8		160°45.2	7.0'	N23°39.7	6.1'
7	281°56.1	58.1		175°11.2	6.9'	23°45.8	6.0'
8	296°56.0	57.5		189°37.1	6.8'	23°51.8	5.9'
9	311°55.8	· · 56.9		204°02.9	6.7'	23°57.7	5.7'
10	326°55.7	56.3		218°28.5	6.6'	24°03.4	5.6'
11	341°55.6	55.7		232°54.1	6.4'	24°09.0	5.5'
12	356°55.4	S18°55.1		247°19.5	6.3'	N24°14.5	5.3'
13	11°55.3	54.4		261°44.9	6.2'	24°19.8	5.2'
14	26°55.1	53.8		276°10.1	6.1'	24°25.0	5.0'
15	41°55.0	· · 53.2		290°35.2	6.0'	24°30.0	4.9'
16	56°54.8	52.6		305°00.3	5.9'	24°34.9	4.7'
17	71°54.7	51.9		319°25.2	5.8'	24°39.7	4.6'
18	86°54.6	S18°51.3		333°50.0	5.7'	N24°44.3	4.4'
19	101°54.4	50.7		348°14.7	5.6'	24°48.7	4.3'
20	116°54.3	50.1		2°39.4	5.5'	24°53.0	4.1'
21	131°54.2	· · 49.5		17°03.9	5.4'	24°57.1	4.0'
22	146°54.0	48.8		31°28.3	5.3'	25°01.1	3.8'
23	161°53.9	48.2		45°52.7	5.2'	25°04.9	3.7'
	SD = 16.2'	d = -0.6'				SD = 15.7'	

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	176°53.7	S18°47.6	60°16.9	5.1'	N25°08.6	3.5'	58.7'
1	191°53.6	47.0	74°41.0	5.1'	25°12.1	3.3'	58.8'
2	206°53.5	46.3	89°05.1	5.0'	25°15.5	3.2'	58.8'
3	221°53.3	· · 45.7	103°29.1	4.9'	25°18.6	3.0'	58.8'
4	236°53.2	45.1	117°52.9	4.8'	25°21.7	2.9'	58.9'
5	251°53.1	44.4	132°16.7	4.7'	25°24.5	2.7'	58.9'
6	266°52.9	S18°43.8	146°40.4	4.6'	N25°27.2	2.5'	59.0'
7	281°52.8	43.2	161°04.0	4.5'	25°29.8	2.4'	59.0'
8	296°52.7	42.6	175°27.6	4.5'	25°32.1	2.2'	59.0'
9	311°52.5	· · 41.9	189°51.0	4.4'	25°34.3	2.0'	59.1'
10	326°52.4	41.3	204°14.4	4.3'	25°36.3	1.9'	59.1'
11	341°52.2	40.7	218°37.7	4.2'	25°38.2	1.7'	59.2'
12	356°52.1	S18°40.0	233°01.0	4.2'	N25°39.9	1.5'	59.2'
13	11°52.0	39.4	247°24.1	4.1'	25°41.4	1.3'	59.2'
14	26°51.9	38.8	261°47.2	4.0'	25°42.7	1.2'	59.3'
15	41°51.7	· · 38.1	276°10.2	3.9'	25°43.9	1.0'	59.3'
16	56°51.6	37.5	290°33.2	3.9'	25°44.9	0.8'	59.4'
17	71°51.5	36.8	304°56.0	3.8'	25°45.7	0.6'	59.4'
18	86°51.3	S18°36.2	319°18.9	3.8'	N25°46.3	0.5'	59.4'
19	101°51.2	35.6	333°41.6	3.7'	25°46.8	0.3'	59.5'
20	116°51.1	34.9	348°04.3	3.6'	25°47.0	0.1'	59.5'
21	131°50.9	· · 34.3	2°27.0	3.6'	25°47.1	-0.1'	59.5'
22	146°50.8	33.7	16°49.6	3.5'	25°47.1	-0.3'	59.6'
23	161°50.7	33.0	31°12.1	3.5'	25°46.8	-0.5'	59.6'
	SD = 16.2'	d = -0.6'			SD = 16.0'		

Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	176°50.5	S18°32.4	45°34.6	3.4'	N25°46.3	-0.6'	59.7'
1	191°50.4	31.7	59°57.0	3.4'	25°45.7	-0.8'	59.7'
2	206°50.3	31.1	74°19.4	3.4'	25°44.9	-1.0'	59.7'
3	221°50.2	· · 30.5	88°41.8	3.3'	25°43.9	-1.2'	59.8'
4	236°50.0	29.8	103°04.1	3.3'	25°42.7	-1.4'	59.8'
5	251°49.9	29.2	117°26.4	3.2'	25°41.3	-1.6'	59.8'
6	266°49.8	S18°28.5	131°48.6	3.2'	N25°39.8	-1.7'	59.9'
7	281°49.7	27.9	146°10.8	3.2'	25°38.0	-1.9'	59.9'
8	296°49.5	27.2	160°33.0	3.2'	25°36.1	-2.1'	60.0'
9	311°49.4	· · 26.6	174°55.2	3.1'	25°34.0	-2.3'	60.0'
10	326°49.3	25.9	189°17.3	3.1'	25°31.7	-2.5'	60.0'
11	341°49.2	25.3	203°39.4	3.1'	25°29.2	-2.7'	60.1'
12	356°49.0	S18°24.7	218°01.5	3.1'	N25°26.5	-2.9'	60.1'
13	11°48.9	24.0	232°23.5	3.0'	25°23.7	-3.1'	60.1'
14	26°48.8	23.4	246°45.6	3.0'	25°20.6	-3.2'	60.2'
15	41°48.7	· · 22.7	261°07.6	3.0'	25°17.4	-3.4'	60.2'
16	56°48.5	22.1	275°29.6	3.0'	25°13.9	-3.6'	60.2'
17	71°48.4	21.4	289°51.6	3.0'	25°10.3	-3.8'	60.3'
18	86°48.3	S18°20.8	304°13.6	3.0'	N25°06.5	-4.0'	60.3'
19	101°48.2	20.1	318°35.6	3.0'	25°02.5	-4.2'	60.3'
20	116°48.0	19.5	332°57.7	3.0'	24°58.4	-4.4'	60.4'
21	131°47.9	· · 18.8	347°19.7	3.0'	24°54.0	-4.5'	60.4'
22	146°47.8	18.2	1°41.7	3.0'	24°49.4	-4.7'	60.4'
23	161°47.7	17.5	16°03.7	3.0'	24°44.7	-4.9'	60.5'
	SD = 16.2'	d = -0.6'			SD = 16.3'		

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:30	09:05	11:41	12:45	15:21	16:56
N 70°	07:20	08:42	10:21	14:05	15:44	17:07
68°	07:11	08:24	09:44	14:42	16:02	17:15
66°	07:04	08:10	09:18	15:08	16:16	17:22
64°	06:57	07:58	08:57	15:28	16:28	17:29
62°	06:52	07:47	08:41	15:45	16:39	17:34
60°	06:47	07:38	08:28	15:58	16:47	17:39
N 58°	06:42	07:31	08:16	16:10	16:55	17:44
56°	06:38	07:24	08:06	16:20	17:02	17:48
54°	06:34	07:17	07:57	16:29	17:08	17:52
52°	06:30	07:11	07:49	16:37	17:14	17:55
50°	06:27	07:06	07:42	16:44	17:19	17:59
45°	06:19	06:55	07:26	16:59	17:31	18:07
N 40°	06:12	06:45	07:14	17:12	17:41	18:13
35°	06:05	06:36	07:03	17:22	17:49	18:20
30°	05:59	06:28	06:53	17:32	17:57	18:26
20°	05:47	06:14	06:37	17:48	18:11	18:38
N 10°	05:35	06:00	06:23	18:03	18:25	18:50
0°	05:22	05:47	06:09	18:16	18:38	19:03
S 10°	05:07	05:33	05:55	18:30	18:52	19:18
20°	04:49	05:17	05:40	18:44	19:08	19:36
30°	04:26	04:57	05:23	19:02	19:28	19:59
35°	04:11	04:45	05:13	19:11	19:39	20:13
40°	03:53	04:31	05:02	19:23	19:53	20:31
45°	03:30	04:14	04:48	19:36	20:10	20:54
S 50°	02:59	03:52	04:31	19:53	20:32	21:24
52°	02:43	03:41	04:23	20:01	20:43	21:40
54°	02:23	03:29	04:14	20:10	20:55	21:59
56°	01:58	03:14	04:04	20:19	21:09	22:24
58°	01:19	02:57	03:53	20:31	21:25	23:00
S 60°	////	02:36	03:40	20:44	21:46	////

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	□	□	□	□	□	□
N 70°	□	□	□	□	□	□
68°	□	□	□	□	□	□
66°	07:58	□	□	06:33	□	□
64°	09:04	09:13	10:18	05:27	07:22	08:27
62°	09:40	10:09	11:15	04:52	06:26	07:31
60°	10:05	10:42	11:48	04:27	05:53	06:58
N 58°	10:26	11:07	12:12	04:07	05:28	06:33
56°	10:43	11:27	12:32	03:51	05:09	06:13
54°	10:57	11:43	12:48	03:37	04:52	05:57
52°	11:10	11:57	13:02	03:24	04:38	05:42
50°	11:21	12:10	13:15	03:14	04:26	05:30
45°	11:44	12:36	13:40	02:51	04:01	05:04
N 40°	12:03	12:56	14:00	02:33	03:40	04:44
35°	12:19	13:13	14:17	02:18	03:24	04:27
30°	12:32	13:28	14:32	02:05	03:09	04:12
20°	12:56	13:53	14:57	01:43	02:44	03:47
N 10°	13:16	14:15	15:18	01:23	02:23	03:25
0°	13:35	14:35	15:38	01:05	02:03	03:05
S 10°	13:54	14:55	15:58	00:47	01:43	02:44
20°	14:15	15:17	16:19	00:28	01:22	02:22
30°	14:38	15:43	16:44	00:06	00:57	01:57
35°	14:53	15:58	16:58	· · ·	00:43	01:42
40°	15:09	16:15	17:15	· · ·	00:26	01:25
45°	15:28	16:36	17:35	· · ·	00:06	01:04
S 50°	15:53	17:02	18:00	23:41	· · ·	00:37
52°	16:05	17:15	18:12	23:29	· · ·	00:25
54°	16:18	17:30	18:26	23:15	· · ·	00:10
56°	16:34	17:47	18:42	22:59	23:53	· · ·
58°	16:53	18:07	19:01	22:40	23:32	· · ·
S 60°	17:16	18:34	19:25	22:17	23:06	· · ·

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Upper hh:mm	Lower hh:mm	
25	12:11	12:18	12:12	19:49	07:20	
26	12:25	12:32	12:13	20:50	08:19	
27	12:38	12:44	12:13	21:53	09:21	

January 28, 29, 30 UT (Thu., Fri., Sat.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	127°09.3	172°31.4	S18°27.4	352°51.6	N21°56.3	152°35.2	S11°32.1	302°05.8	N00°26.5	Alpheratz	357°46.3	29°08.9
1	142°11.7	187°30.7	26.5	7°55.1	56.6	167°37.1	31.9	317°08.3	26.5	Ankaa	353°18.2	-42°15.2
2	157°14.2	202°30.0	25.7	22°58.6	56.9	182°39.0	31.7	332°10.9	26.6	Schedar	349°43.8	56°35.9
3	172°16.7	217°29.2	· · 24.9	38°02.1	· · 57.2	197°40.9	· · 31.5	347°13.4	· · 26.6	Diphda	348°58.5	-17°55.9
4	187°19.1	232°28.5	24.0	53°05.6	57.5	212°42.8	31.2	2°15.9	26.6	Achernar	335°28.5	-57°11.3
5	202°21.6	247°27.8	23.2	68°09.1	57.8	227°44.8	31.0	17°18.4	26.7	Hamal	328°03.6	23°30.8
6	217°24.1	262°27.1	S18°22.4	83°12.6	N21°58.1	242°46.7	S11°30.8	32°20.9	N00°26.7	Polaris	318°57.0	89°18.9
7	232°26.5	277°26.3	21.5	98°16.1	58.4	257°48.6	30.6	47°23.5	26.8	Acamar	315°20.1	-40°16.0
8	247°29.0	292°25.6	20.7	113°19.5	58.7	272°50.5	30.4	62°26.0	26.8	Menkar	314°17.6	4°07.8
9	262°31.5	307°24.9	· · 19.9	128°23.0	· · 59.0	287°52.4	· · 30.2	77°28.5	· · 26.8	Mirfak	308°43.8	49°54.1
10	277°33.9	322°24.2	19.0	143°26.5	59.3	302°54.3	30.0	92°31.0	26.9	Aldebaran	290°52.0	16°31.8
11	292°36.4	337°23.4	18.2	158°30.0	59.6	317°56.3	29.8	107°33.5	26.9	Rigel	281°14.2	-8°11.5
12	307°38.8	352°22.7	S18°17.3	173°33.5	N21°59.8	332°58.2	S11°29.6	122°36.1	N00°26.9	Capella	280°37.8	46°00.6
13	322°41.3	7°22.0	16.5	188°37.0	22°00.1	348°00.1	29.4	137°38.6	27.0	Bellatrix	278°34.4	6°21.5
14	337°43.8	22°21.3	15.7	203°40.5	00.4	3°02.0	29.2	152°41.1	27.0	Elnath	278°15.5	28°37.0
15	352°46.2	37°20.6	· · 14.8	218°44.0	· · 00.7	18°03.9	· · 29.0	167°43.6	· · 27.0	Alnilam	275°48.6	-1°11.8
16	7°48.7	52°19.8	14.0	233°47.5	01.0	33°05.8	28.8	182°46.1	27.1	Betelgeuse	271°03.7	7°24.5
17	22°51.2	67°19.1	13.1	248°51.0	01.3	48°07.7	28.5	197°48.7	27.1	Canopus	263°56.8	-52°42.2
18	37°53.6	82°18.4	S18°12.3	263°54.5	N22°01.6	63°09.6	S11°28.3	212°51.2	N00°27.2	Sirius	258°35.6	-16°43.9
19	52°56.1	97°17.7	11.4	278°58.0	01.9	78°11.6	28.1	227°53.7	27.2	Adhara	255°14.1	-28°59.3
20	67°58.6	112°17.0	10.6	294°01.5	02.2	93°13.5	27.9	242°56.2	27.2	Procyon	245°02.0	5°11.8
21	83°01.0	127°16.2	· · 09.7	309°05.0	· · 02.5	108°15.4	· · 27.7	257°58.7	· · 27.3	Pollux	243°30.3	28°00.0
22	98°03.5	142°15.5	08.9	324°08.4	02.8	123°17.3	27.5	273°01.3	27.3	Avior	234°18.5	-59°32.6
23	113°05.9	157°14.8	08.1	339°11.9	03.1	138°19.2	27.3	288°03.8	27.3	Suhail	222°53.9	-43°28.5
Mer.pass. 15:29		ν -0.7' d -0.8' m -3.92		ν 3.5' d 0.3' m -1.33		ν 1.9' d -0.2' m -2.05		ν 2.5' d 0.0' m 0.74		Miaplacidus	221°39.5	-69°45.5
Alphard		217°58.2		-8°42.3		Regulus		207°45.8		11°54.9		
Dubhe		193°53.9		61°41.5		Denebola		182°35.9		14°30.7		
Gienah		175°54.7		-17°36.0		Acrux		173°12.0		-63°09.2		
Gacrux		172°03.5		-57°10.1		Alioth		166°22.4		55°53.9		
Spica		158°33.8		-11°13.0		Alkaid		153°00.6		49°15.4		
Hadar		148°51.5		-60°25.1		Menkent		148°10.5		-36°25.1		
Arcturus		145°57.9		19°07.5		Rigel Kent.		139°55.2		-60°52.4		
Kochab		137°19.7		74°06.4		Zuben'ubi		137°08.2		-16°05.1		
Alphecca		126°13.2		26°40.5		Antares		112°29.5		-26°27.3		
Atria		107°33.9		-69°02.6		Sabik		102°15.6		-15°44.3		
Shaula		96°25.6		-37°06.6		Rasalhague		96°09.0		12°33.0		
Eltanin		90°47.7		51°29.0		Kaus Aust.		83°47.4		-34°22.8		
Vega		80°41.0		38°47.4		Nunki		76°01.7		-26°17.0		
Altair		62°11.0		8°53.6		Peacock		53°23.6		-56°42.1		
Deneb		49°33.7		45°19.0		Enif		33°49.9		9°55.3		
Al Na'ir		27°47.1		-46°54.8		Fomalhaut		15°26.9		-29°34.2		
Scheat		13°56.1		28°08.4		Markab		13°41.1		15°15.7		
Jan 28 Thu		SHA		Mer.pass		Venus		45°22.1		12:31		
Mars		225°42.3		00:28		Jupiter		25°25.9		13:48		
Saturn		174°56.6		03:51		Jan 29 Fri		SHA		Mer.pass		
Venus		44°05.7		12:32		Mars		226°07.0		00:23		
Jupiter		25°12.7		13:45		Saturn		174°57.9		03:47		
Jan 30 Sat		SHA		Mer.pass		Venus		42°49.6		12:33		
Mars		226°31.8		00:17		Jupiter		24°59.5		13:42		
Saturn		174°59.3		03:43		Horizontal parallax		Venus:		0.1		
Mars:				0.2		Mer.pass. 15:21		ν -0.7' d -0.9' m -3.92		ν 3.5' d 0.3' m -1.34		
ν 1.9' d -0.2' m -2.04		ν 2.5' d 0.0' m 0.73										

h	Sun			Moon			
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	176°47.6	S18°16.8	30°25.7	3.0'	N24°39.8	-5.1'	60.5'
1	191°47.4	16.2	44°47.7	3.0'	24°34.7	-5.3'	60.5'
2	206°47.3	15.5	59°09.8	3.1'	24°29.4	-5.5'	60.5'
3	221°47.2	· · 14.9	73°31.8	3.1'	24°23.9	-5.7'	60.6'
4	236°47.1	14.2	87°53.9	3.1'	24°18.3	-5.8'	60.6'
5	251°47.0	13.6	102°16.0	3.1'	24°12.4	-6.0'	60.6'
6	266°46.8	S18°12.9	116°38.2	3.1'	N24°06.4	-6.2'	60.7'
7	281°46.7	12.3	131°00.3	3.2'	24°00.2	-6.4'	60.7'
8	296°46.6	11.6	145°22.5	3.2'	23°53.8	-6.6'	60.7'
9	311°46.5	· · 10.9	159°44.7	3.2'	23°47.2	-6.8'	60.7'
10	326°46.4	10.3	174°06.9	3.3'	23°40.5	-6.9'	60.8'
11	341°46.3	09.6	188°29.2	3.3'	23°33.5	-7.1'	60.8'
12	356°46.1	S18°09.0	202°51.5	3.3'	N23°26.4	-7.3'	60.8'
13	11°46.0	08.3	217°13.8	3.4'	23°19.1	-7.5'	60.8'
14	26°45.9	07.6	231°36.2	3.4'	23°11.7	-7.6'	60.9'
15	41°45.8	· · 07.0	245°58.6	3.5'	23°04.0	-7.8'	60.9'
16	56°45.7	06.3	260°21.1	3.5'	22°56.2	-8.0'	60.9'
17	71°45.6	05.6	274°43.6	3.6'	22°48.2	-8.2'	60.9'
18	86°45.5	S18°05.0	289°06.2	3.6'	N22°40.1	-8.3'	61.0'
19	101°45.3	04.3	303°28.8	3.7'	22°31.8	-8.5'	61.0'
20	116°45.2	03.7	317°51.4	3.7'	22°23.3	-8.7'	61.0'
21	131°45.1	· · 03.0	332°14.1	3.8'	22°14.6	-8.8'	61.0'
22	146°45.0	02.3	346°36.9	3.8'	22°05.8	-9.0'	61.0'
23	161°44.9	01.7	0°59.7	3.9'	21°56.8	-9.2'	61.1'
SD = 16.2'			SD = 16.5'				
$d = -0.7'$							

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	176°44.8	S18°01.0	15°22.6	3.9'	N21°47.6	-9.3'	61.1'
1	191°44.7	18°00.3	29°45.5	4.0'	21°38.3	-9.5'	61.1'
2	206°44.6	17°59.6	44°08.5	4.1'	21°28.8	-9.6'	61.1'
3	221°44.4	· · 59.0	58°31.5	4.1'	21°19.1	-9.8'	61.1'
4	236°44.3	58.3	72°54.7	4.2'	21°09.3	-10.0'	61.2'
5	251°44.2	57.6	87°17.8	4.2'	20°59.4	-10.1'	61.2'
6	266°44.1	S17°57.0	101°41.1	4.3'	N20°49.2	-10.3'	61.2'
7	281°44.0	56.3	116°04.4	4.4'	20°39.0	-10.4'	61.2'
8	296°43.9	55.6	130°27.8	4.4'	20°28.5	-10.6'	61.2'
9	311°43.8	· · 55.0	144°51.2	4.5'	20°18.0	-10.7'	61.2'
10	326°43.7	54.3	159°14.7	4.6'	20°07.2	-10.9'	61.3'
11	341°43.6	53.6	173°38.3	4.7'	19°56.4	-11.0'	61.3'
12	356°43.5	S17°52.9	188°02.0	4.7'	N19°45.3	-11.2'	61.3'
13	11°43.4	52.3	202°25.7	4.8'	19°34.2	-11.3'	61.3'
14	26°43.2	51.6	216°49.5	4.9'	19°22.9	-11.4'	61.3'
15	41°43.1	· · 50.9	231°13.4	5.0'	19°11.4	-11.6'	61.3'
16	56°43.0	50.2	245°37.4	5.0'	18°59.8	-11.7'	61.3'
17	71°42.9	49.6	260°01.4	5.1'	18°48.1	-11.9'	61.3'
18	86°42.8	S17°48.9	274°25.5	5.2'	N18°36.3	-12.0'	61.3'
19	101°42.7	48.2	288°49.7	5.3'	18°24.3	-12.1'	61.4'
20	116°42.6	47.5	303°13.9	5.3'	18°12.2	-12.2'	61.4'
21	131°42.5	· · 46.8	317°38.3	5.4'	17°59.9	-12.4'	61.4'
22	146°42.4	46.2	332°02.7	5.5'	17°47.5	-12.5'	61.4'
23	161°42.3	45.5	346°27.2	5.6'	17°35.0	-12.6'	61.4'
SD = 16.2'			SD = 16.7'				
$d = -0.7'$							

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	176°42.2	S17°44.8	0°51.8	5.7'	N17°22.4	-12.7'	61.4'
1	191°42.1	44.1	15°16.5	5.7'	17°09.7	-12.9'	61.4'
2	206°42.0	43.4	29°41.2	5.8'	16°56.8	-13.0'	61.4'
3	221°41.9	· · 42.8	44°06.0	5.9'	16°43.8	-13.1'	61.4'
4	236°41.8	42.1	58°30.9	6.0'	16°30.7	-13.2'	61.4'
5	251°41.7	41.4	72°55.9	6.1'	16°17.5	-13.3'	61.4'
6	266°41.6	S17°40.7	87°21.0	6.1'	N16°04.2	-13.4'	61.4'
7	281°41.5	40.0	101°46.1	6.2'	15°50.8	-13.5'	61.4'
8	296°41.4	39.3	116°11.3	6.3'	15°37.3	-13.6'	61.4'
9	311°41.3	· · 38.7	130°36.6	6.4'	15°23.6	-13.7'	61.4'
10	326°41.2	38.0	145°02.0	6.5'	15°09.9	-13.8'	61.4'
11	341°41.1	37.3	159°27.5	6.6'	14°56.1	-13.9'	61.4'
12	356°41.0	S17°36.6	173°53.1	6.6'	N14°42.1	-14.0'	61.4'
13	11°40.9	35.9	188°18.7	6.7'	14°28.1	-14.1'	61.4'
14	26°40.8	35.2	202°44.4	6.8'	14°14.0	-14.2'	61.4'
15	41°40.7	· · 34.5	217°10.2	6.9'	13°59.7	-14.3'	61.4'
16	56°40.6	33.8	231°36.1	7.0'	13°45.4	-14.4'	61.4'
17	71°40.5	33.1	246°02.0	7.0'	13°31.0	-14.5'	61.4'
18	86°40.4	S17°32.5	260°28.1	7.1'	N13°16.6	-14.6'	61.4'
19	101°40.3	31.8	274°54.2	7.2'	13°02.0	-14.6'	61.4'
20	116°40.2	31.1	289°20.4	7.3'	12°47.3	-14.7'	61.4'
21	131°40.1	· · 30.4	303°46.7	7.4'	12°32.6	-14.8'	61.4'
22	146°40.0	29.7	318°13.0	7.4'	12°17.8	-14.9'	61.4'
23	161°39.9	29.0	332°39.4	7.5'	12°02.9	-14.9'	61.4'
SD = 16.2'			SD = 16.7'				
$d = -0.7'$							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:20	08:52	10:56	13:31	15:35	17:07
N 70°	07:11	08:31	10:03	14:24	15:56	17:16
68°	07:03	08:15	09:31	14:57	16:12	17:24
66°	06:57	08:02	09:07	15:20	16:25	17:30
64°	06:51	07:51	08:49	15:38	16:37	17:36
62°	06:46	07:41	08:34	15:53	16:46	17:41
60°	06:42	07:33	08:21	16:06	16:54	17:46
N 58°	06:37	07:25	08:10	16:17	17:02	17:50
56°	06:34	07:19	08:01	16:26	17:08	17:53
54°	06:30	07:13	07:52	16:35	17:14	17:57
52°	06:27	07:08	07:45	16:42	17:19	18:00
50°	06:24	07:03	07:38	16:49	17:24	18:03
45°	06:16	06:52	07:23	17:03	17:35	18:10
N 40°	06:10	06:43	07:11	17:15	17:44	18:17
35°	06:04	06:34	07:01	17:25	17:52	18:23
30°	05:58	06:27	06:52	17:34	18:00	18:28
20°	05:47	06:13	06:36	17:50	18:13	18:40
N 10°	05:35	06:01	06:23	18:04	18:26	18:51
0°	05:23	05:48	06:10	18:17	18:38	19:04
S 10°	05:08	05:34	05:56	18:30	18:52	19:18
20°	04:51	05:19	05:42	18:44	19:07	19:35
30°	04:29	05:00	05:26	19:00	19:26	19:57
35°	04:15	04:48	05:16	19:10	19:37	20:11
40°	03:58	04:35	05:05	19:20	19:51	20:28
45°	03:36	04:18	04:52	19:33	20:07	20:49
S 50°	03:07	03:58	04:36	19:49	20:27	21:18
52°	02:51	03:47	04:29	19:56	20:37	21:33
54°	02:33	03:36	04:20	20:05	20:49	21:51
56°	02:10	03:22	04:11	20:14	21:02	22:13
58°	01:38	03:07	04:00	20:24	21:17	22:44
S 60°	////	02:47	03:48	20:37	21:36	23:42

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	□	□	15:16	□	□	10:02
N 70°	□	12:21	15:46	□	10:51	09:30
68°	□	13:31	16:07	□	09:39	09:07
66°	11:24	14:08	16:24	09:36	09:02	08:48
64°	12:23	14:33	16:38	08:36	08:35	08:33
62°	12:57	14:53	16:49	08:01	08:14	08:20
60°	13:22	15:09	16:59	07:36	07:57	08:09
N 58°	13:41	15:23	17:07	07:16	07:43	08:00
56°	13:57	15:35	17:15	06:59	07:31	07:52
54°	14:11	15:45	17:21	06:45	07:20	07:44
52°	14:23	15:54	17:27	06:33	07:10	07:37
50°	14:34	16:02	17:32	06:22	07:01	07:31
45°	14:56	16:19	17:44	05:59	06:43	07:18
N 40°	15:14	16:33	17:53	05:40	06:27	07:07
35°	15:29	16:45	18:01	05:24	06:15	06:58
30°	15:42	16:55	18:08	05:11	06:03	06:49
20°	16:04	17:13	18:20	04:47	05:44	06:35
N 10°	16:23	17:28	18:31	04:27	05:27	06:22
0°	16:41	17:42	18:40	04:08	05:10	06:10
S 10°	16:59	17:57	18:50	03:49	04:54	05:58
20°	17:18	18:12	19:01	03:28	04:37	05:45
30°	17:40	18:29	19:12	03:04	04:17	05:30
35°	17:52	18:39	19:19	02:50	04:05	05:21
40°	18:07	18:50	19:27	02:34	03:51	05:11
45°	18:24	19:04	19:36	02:15	03:35	04:59
S 50°	18:46	19:20	19:46	01:50	03:15	04:45
52°	18:56	19:27	19:51	01:38	03:05	04:38
54°	19:07	19:36	19:57	01:25	02:54	04:30
56°	19:20	19:45	20:03	01:09	02:42	04:22
58°	19:35	19:56	20:09	00:50	02:27	04:12
S 60°	19:53	20:08	20:17	00:27	02:10	04:01

Day	Sun		Mer. Pass hh:mm	Moon		Age 13-15 92-100%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
28	12:50	12:55	12:13	22:56	10:25	
29	13:01	13:06	12:13	23:56	11:27	
30	13:11	13:16	12:13	—:—	12:25	

January 31, 01, 02 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn	
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	130°06.7	171°40.4	S17°25.4	357°03.2	N22°17.0	154°52.9	S11°17.0	305°07.6	N00°29.2
1	145°09.2	186°39.8	24.5	12°06.7	17.2	169°54.8	16.8	320°10.1	29.3
2	160°11.6	201°39.1	23.6	27°10.2	17.5	184°56.7	16.6	335°12.6	29.3
3	175°14.1	216°38.4	•• 22.7	42°13.7	•• 17.8	199°58.7	•• 16.4	350°15.2	•• 29.4
4	190°16.5	231°37.7	21.8	57°17.2	18.1	215°00.6	16.2	5°17.7	29.4
5	205°19.0	246°37.0	20.9	72°20.7	18.4	230°02.5	16.0	20°20.2	29.4
6	220°21.5	261°36.3	S17°20.0	87°24.2	N22°18.6	245°04.4	S11°15.8	35°22.8	N00°29.5
7	235°23.9	276°35.6	19.1	102°27.7	18.9	260°06.3	15.6	50°25.3	29.5
8	250°26.4	291°35.0	18.2	117°31.2	19.2	275°08.2	15.4	65°27.8	29.6
9	265°28.9	306°34.3	•• 17.3	132°34.7	•• 19.4	290°10.1	•• 15.2	80°30.3	•• 29.6
10	280°31.3	321°33.6	16.4	147°38.2	19.7	305°12.0	14.9	95°32.9	29.6
11	295°33.8	336°32.9	15.5	162°41.7	20.0	320°13.9	14.7	110°35.4	29.7
12	310°36.3	351°32.2	S17°14.6	177°45.2	N22°20.3	335°15.8	S11°14.5	125°37.9	N00°29.7
13	325°38.7	6°31.5	13.7	192°48.7	20.5	350°17.8	14.3	140°40.5	29.8
14	340°41.2	21°30.9	12.8	207°52.2	20.8	5°19.7	14.1	155°43.0	29.8
15	355°43.7	36°30.2	•• 11.9	222°55.7	•• 21.1	20°21.6	•• 13.9	170°45.5	•• 29.9
16	10°46.1	51°29.5	11.0	237°59.2	21.4	35°23.5	13.7	185°48.1	29.9
17	25°48.6	66°28.8	10.1	253°02.7	21.6	50°25.4	13.5	200°50.6	29.9
18	40°51.0	81°28.1	S17°09.2	268°06.2	N22°21.9	65°27.3	S11°13.3	215°53.1	N00°30.0
19	55°53.5	96°27.5	08.3	283°09.7	22.2	80°29.2	13.0	230°55.7	30.0
20	70°56.0	111°26.8	07.3	298°13.2	22.4	95°31.1	12.8	245°58.2	30.1
21	85°58.4	126°26.1	•• 06.4	313°16.6	•• 22.7	110°33.0	•• 12.6	261°00.7	•• 30.1
22	101°00.9	141°25.4	05.5	328°20.1	23.0	125°34.9	12.4	276°03.3	30.1
23	116°03.4	156°24.8	04.6	343°23.6	23.3	140°36.9	12.2	291°05.8	30.2
Mer.pass. 15:17		ν -0.7' d -0.9' m -3.92		ν 3.5' d 0.3' m -1.33		ν 1.9' d -0.2' m -2.04		ν 2.5' d 0.0' m 0.72	

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	131°05.8	171°24.1	S17°03.7	358°27.1	N22°23.5	155°38.8	S11°12.0	306°08.3	N00°30.2
1	146°08.3	186°23.4	02.8	13°30.6	23.8	170°40.7	11.8	321°10.9	30.3
2	161°10.8	201°22.7	01.9	28°34.1	24.1	185°42.6	11.6	336°13.4	30.3
3	176°13.2	216°22.1	•• 01.0	43°37.6	•• 24.3	200°44.5	•• 11.4	351°15.9	•• 30.4
4	191°15.7	231°21.4	17°00.0	58°41.1	24.6	215°46.4	11.1	6°18.5	30.4
5	206°18.2	246°20.7	16°59.1	73°44.6	24.9	230°48.3	10.9	21°21.0	30.4
6	221°20.6	261°20.0	S16°58.2	88°48.1	N22°25.1	245°50.2	S11°10.7	36°23.5	N00°30.5
7	236°23.1	276°19.4	57.3	103°51.6	25.4	260°52.1	10.5	51°26.1	30.5
8	251°25.5	291°18.7	56.4	118°55.1	25.7	275°54.0	10.3	66°28.6	30.6
9	266°28.0	306°18.0	•• 55.5	133°58.6	•• 25.9	290°56.0	•• 10.1	81°31.2	•• 30.6
10	281°30.5	321°17.4	54.5	149°02.1	26.2	305°57.9	09.9	96°33.7	30.6
11	296°32.9	336°16.7	53.6	164°05.6	26.5	320°59.8	09.7	111°36.2	30.7
12	311°35.4	351°16.0	S16°52.7	179°09.1	N22°26.7	336°01.7	S11°09.5	126°38.8	N00°30.7
13	326°37.9	6°15.3	51.8	194°12.5	27.0	351°03.6	09.2	141°41.3	30.8
14	341°40.3	21°14.7	50.8	209°16.0	27.3	6°05.5	09.0	156°43.8	30.8
15	356°42.8	36°14.0	•• 49.9	224°19.5	•• 27.5	21°07.4	•• 08.8	171°46.4	•• 30.9
16	11°45.3	51°13.3	49.0	239°23.0	27.8	36°09.3	08.6	186°48.9	30.9
17	26°47.7	66°12.7	48.1	254°26.5	28.0	51°11.2	08.4	201°51.4	30.9
18	41°50.2	81°12.0	S16°47.1	269°30.0	N22°28.3	66°13.1	S11°08.2	216°54.0	N00°31.0
19	56°52.6	96°11.3	46.2	284°33.5	28.6	81°15.0	08.0	231°56.5	31.0
20	71°55.1	111°10.7	45.3	299°37.0	28.8	96°17.0	07.8	246°59.0	31.1
21	86°57.6	126°10.0	•• 44.4	314°40.5	•• 29.1	111°18.9	•• 07.6	262°01.6	•• 31.1
22	102°00.0	141°09.4	43.4	329°44.0	29.4	126°20.8	07.3	277°04.1	31.2
23	117°02.5	156°08.7	42.5	344°47.5	29.6	141°22.7	07.1	292°06.7	31.2
Mer.pass. 15:13		ν -0.7' d -0.9' m -3.92		ν 3.5' d 0.3' m -1.32		ν 1.9' d -0.2' m -2.04		ν 2.5' d 0.0' m 0.72	

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	132°05.0	171°08.0	S16°41.6	359°51.0	N22°29.9	156°24.6	S11°06.9	307°09.2	N00°31.3
1	147°07.4	186°07.4	40.6	14°54.4	30.1	171°26.5	06.7	322°11.7	31.3
2	162°09.9	201°06.7	39.7	29°57.9	30.4	186°28.4	06.5	337°14.3	31.3
3	177°12.4	216°06.0	•• 38.8	45°01.4	•• 30.7	201°30.3	•• 06.3	352°16.8	•• 31.4
4	192°14.8	231°05.4	37.8	60°04.9	30.9	216°32.2	06.1	7°19.4	31.4
5	207°17.3	246°04.7	36.9	75°08.4	31.2	231°34.1	05.9	22°21.9	31.5
6	222°19.8	261°04.1	S16°36.0	90°11.9	N22°31.4	246°36.0	S11°05.6	37°24.4	N00°31.5
7	237°22.2	276°03.4	35.0	105°15.4	31.7	261°38.0	05.4	52°27.0	31.6
8	252°24.7	291°02.8	34.1	120°18.9	32.0	276°39.9	05.2	67°29.5	31.6
9	267°27.1	306°02.1	•• 33.1	135°22.4	•• 32.2	291°41.8	•• 05.0	82°32.0	•• 31.6
10	282°29.6	321°01.4	32.2	150°25.9	32.5	306°43.7	04.8	97°34.6	31.7
11	297°32.1	336°00.8	31.3	165°29.3	32.7	321°45.6	04.6	112°37.1	31.7
12	312°34.5	351°00.1	S16°30.3	180°32.8	N22°33.0	336°47.5	S11°04.4	127°39.7	N00°31.8
13	327°37.0	5°59.5	29.4	195°36.3	33.2	351°49.4	04.2	142°42.2	31.8
14	342°39.5	20°58.8	28.4	210°39.8	33.5	6°51.3	04.0	157°44.7	31.9
15	357°41.9	35°58.2	•• 27.5	225°43.3	•• 33.7	21°53.2	•• 03.7	172°47.3	•• 31.9
16	12°44.4	50°57.5	26.5	240°46.8	34.0	36°55.1	03.5	187°49.8	32.0
17	27°46.9	65°56.9	25.6	255°50.3	34.3	51°57.0	03.3	202°52.4	32.0
18	42°49.3	80°56.2	S16°24.7	270°53.8	N22°34.5	66°58.9	S11°03.1	217°54.9	N00°32.0
19	57°51.8	95°55.5	23.7	285°57.2	34.8	82°00.8	02.9	232°57.4	32.1
20	72°54.2	110°54.9	22.8	301°00.7	35.0	97°02.8	02.7	248°00.0	32.1
21	87°56.7	125°54.2	•• 21.8	316°04.2	•• 35.3	112°04.7	•• 02.5	263°02.5	•• 32.2
22	102°59.2	140°53.6	20.9	331°07.7	35.5	127°06.6	02.3	278°05.1	32.2
23	118°01.6	155°52.9	19.9	346°11.2	35.8	142°08.5	02.0	293°07.6	32.3
Mer.pass. 15:09		ν -0.7' d -0.9' m -3.92		ν 3.5' d 0.3' m -1.30		ν 1.9' d -0.2' m -2.04		ν 2.5' d 0.0' m 0.71	

Stars		
	SHA	Dec
Alpheratz	357°46.3	29°08.9
Ankaa	353°18.2	-42°15.2
Schedar	349°43.8	56°35.9
Diphda	348°58.5	-17°55.9
Achernar	335°28.6	-57°11.3
Hamal	328°03.6	23°30.8
Polaris	318°58.5	89°18.9
Acamar	315°20.1	-40°16.0
Menkar	314°17.6	4°07.8
Mirfak	308°43.8	49°54.1
Aldebaran	290°52.0	16°31.8
Rigel	281°14.2	-8°11.5
Capella	280°37.8	46°00.6
Bellatrix	278°34.4	6°21.5
Elnath	278°15.5	28°37.0
Alnilam	275°48.6	-1°11.8
Betelgeuse	271°03.7	7°24.5
Canopus	263°56.8	-52°42.2
Sirius	258°35.6	-16°43.9
Adhara	255°14.1	-28°59.3
Procyon	245°02.0	5°11.8
Pollux	243°30.3	28°00.0
Avior	234°18.5	-59°32.6
Suhail	222°53.9	-43°28.5
Miaplacidus	221°39.5	-69°45.6
Alphard	217°58.2	-8°42.3
Regulus	207°45.8	11°54.9
Dubhe	193°53.9	61°41.5
Denebola	182°35.9	14°30.7
Gienah	175°54.7	-17°36.0
Acrux	173°11.9	-63°09.2
Gacrux	172°03.5	-57°10.1
Alioth	166°22.4	55°53.9
Spica	158°33.7	-11°13.0
Alkaid	153°00.6	49°15.4
Hadar	148°51.5	-60°25.2
Menkent	148°10.5	-36°25.2
Arcturus	145°57.9	19°07.5
Rigil Kent.	139°55.2	-60°52.4
Kochab	137°19.7	74°06.4
Zuben'ubi	137°08.2	-16°05.1
Alphecca	126°13.1	26°40.5
Antares	112°29.5	-26°27.3
Atria	107°33.9	-69°02.6
Sabik	102°15.6	-15°44.3
Shaula	96°25.5	-37°06.6
Rasalhague	96°09.0	12°33.0
Eltanin	90°47.7	51°29.0
Kaus Aust.	83°47.4	-34°22.8
Vega	80°41.0	38°47.4
Nunki	76°01.7	-26°17.0
Altair	62°11.0	8°53.6
Peacock	53°23.6	-56°42.1
Deneb	49°33.7	45°19.0
Enif	33°49.9	9°55.3
Al Na'ir	27°47.1	-46°54.8
Fomalhaut	15°26.9	-29°34.2
Scheat	13°56.1	28°08.4
Markab	13°41.1	15°15.6

Jan 31 Sun	SHA	Mer.pass
Venus	41°33.7	12:34
Mars	226°56.5	00:12
Jupiter	24°46.2	13:39
Saturn	175°00.9	03:39

Feb 01 Mon	SHA	Mer.pass
Venus	40°18.2	12:35
Mars	227°21.3	00:06
Jupiter	24°32.9	13:36
Saturn	175°02.5	03:35

Feb 02 Tue	SHA	Mer.pass
Venus	39°03.1	12:36
Mars	227°46.0	00:01
Jupiter	24°19.6	13:33
Saturn	175°04.2	03:31

Horizontal parallax	
Venus:	0.1
Mars:	0.2

DUT1 = UT1-UTC = +0.0896 sec $\Delta T = TT-UT1 = +66.0944$ sec

2010 January 31 to Feb. 02 UT

h	Sun			Moon			
Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	176°39.8	S17°28.3	347°06.0	7.6'	N11°48.0	-15.0'	61.3'
1	191°39.7	27.6	1°32.5	7.7'	11°33.0	-15.1'	61.3'
2	206°39.6	26.9	15°59.2	7.7'	11°17.9	-15.2'	61.3'
3	221°39.5	· · 26.2	30°25.9	7.8'	11°02.7	-15.2'	61.3'
4	236°39.5	25.5	44°52.7	7.9'	10°47.5	-15.3'	61.3'
5	251°39.4	24.8	59°19.6	8.0'	10°32.3	-15.3'	61.3'
6	266°39.3	S17°24.1	73°46.6	8.0'	N10°16.9	-15.4'	61.3'
7	281°39.2	23.4	88°13.6	8.1'	10°01.5	-15.4'	61.3'
8	296°39.1	22.7	102°40.7	8.2'	09°46.1	-15.5'	61.3'
9	311°39.0	· · 22.0	117°07.9	8.2'	09°30.6	-15.6'	61.2'
10	326°38.9	21.3	131°35.2	8.3'	09°15.0	-15.6'	61.2'
11	341°38.8	20.6	146°02.5	8.4'	08°59.4	-15.6'	61.2'
12	356°38.7	S17°19.9	160°29.9	8.5'	N08°43.8	-15.7'	61.2'
13	11°38.6	19.2	174°57.3	8.5'	08°28.1	-15.7'	61.2'
14	26°38.5	18.5	189°24.8	8.6'	08°12.3	-15.8'	61.2'
15	41°38.4	· · 17.8	203°52.4	8.7'	07°56.6	-15.8'	61.1'
16	56°38.4	17.1	218°20.1	8.7'	07°40.8	-15.9'	61.1'
17	71°38.3	16.4	232°47.8	8.8'	07°24.9	-15.9'	61.1'
18	86°38.2	S17°15.7	247°15.6	8.8'	N07°09.0	-15.9'	61.1'
19	101°38.1	15.0	261°43.4	8.9'	06°53.1	-15.9'	61.1'
20	116°38.0	14.3	276°11.4	9.0'	06°37.2	-16.0'	61.1'
21	131°37.9	· · 13.6	290°39.3	9.0'	06°21.2	-16.0'	61.0'
22	146°37.8	12.9	305°07.4	9.1'	06°05.2	-16.0'	61.0'
23	161°37.7	12.2	319°35.5	9.2'	05°49.1	-16.0'	61.0'
SD = 16.2' $d = -0.7'$			SD = 16.7'				

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	176°37.7	S17°11.5	334°03.6	9.2'	N05°33.1	-16.1'	61.0'
1	191°37.6	10.8	348°31.8	9.3'	05°17.0	-16.1'	60.9'
2	206°37.5	10.1	3°00.1	9.3'	05°00.9	-16.1'	60.9'
3	221°37.4	· · 09.4	17°28.4	9.4'	04°44.8	-16.1'	60.9'
4	236°37.3	08.7	31°56.8	9.4'	04°28.7	-16.1'	60.9'
5	251°37.2	08.0	46°25.2	9.5'	04°12.6	-16.1'	60.9'
6	266°37.1	S17°07.2	60°53.7	9.5'	N03°56.5	-16.1'	60.8'
7	281°37.1	06.5	75°22.2	9.6'	03°40.3	-16.2'	60.8'
8	296°37.0	05.8	89°50.8	9.6'	03°24.1	-16.2'	60.8'
9	311°36.9	· · 05.1	104°19.5	9.7'	03°08.0	-16.2'	60.8'
10	326°36.8	04.4	118°48.2	9.7'	02°51.8	-16.2'	60.7'
11	341°36.7	03.7	133°16.9	9.8'	02°35.7	-16.2'	60.7'
12	356°36.7	S17°03.0	147°45.7	9.8'	N02°19.5	-16.2'	60.7'
13	11°36.6	02.3	162°14.5	9.9'	02°03.3	-16.2'	60.6'
14	26°36.5	01.5	176°43.4	9.9'	01°47.2	-16.2'	60.6'
15	41°36.4	· · 00.8	191°12.3	10.0'	01°31.0	-16.1'	60.6'
16	56°36.3	17°00.1	205°41.2	10.0'	01°14.9	-16.1'	60.6'
17	71°36.2	16°59.4	220°10.2	10.0'	00°58.7	-16.1'	60.5'
18	86°36.2	S16°58.7	234°39.3	10.1'	N00°42.6	-16.1'	60.5'
19	101°36.1	58.0	249°08.4	10.1'	00°26.5	-16.1'	60.5'
20	116°36.0	57.3	263°37.5	10.2'	N00°10.4	-16.1'	60.4'
21	131°35.9	· · 56.5	278°06.6	10.2'	S00°05.7	16.1'	60.4'
22	146°35.8	55.8	292°35.8	10.2'	00°21.7	16.0'	60.4'
23	161°35.8	55.1	307°05.0	10.3'	00°37.8	16.0'	60.3'
SD = 16.2' $d = -0.7'$			SD = 16.6'				

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	176°35.7	S16°54.4	321°34.3	10.3'	S00°53.8	16.0'	60.3'
1	191°35.6	53.7	336°03.6	10.3'	01°09.8	16.0'	60.3'
2	206°35.5	52.9	350°32.9	10.4'	01°25.7	15.9'	60.3'
3	221°35.5	· · 52.2	5°02.3	10.4'	01°41.7	15.9'	60.2'
4	236°35.4	51.5	19°31.6	10.4'	01°57.6	15.9'	60.2'
5	251°35.3	50.8	34°01.0	10.4'	02°13.5	15.9'	60.2'
6	266°35.2	S16°50.1	48°30.5	10.5'	S02°29.3	15.8'	60.1'
7	281°35.2	49.3	62°59.9	10.5'	02°45.2	15.8'	60.1'
8	296°35.1	48.6	77°29.4	10.5'	03°00.9	15.8'	60.0'
9	311°35.0	· · 47.9	91°59.0	10.5'	03°16.7	15.7'	60.0'
10	326°34.9	47.2	106°28.5	10.6'	03°32.4	15.7'	60.0'
11	341°34.9	46.4	120°58.0	10.6'	03°48.1	15.6'	59.9'
12	356°34.8	S16°45.7	135°27.6	10.6'	S04°03.7	15.6'	59.9'
13	11°34.7	45.0	149°57.2	10.6'	04°19.3	15.5'	59.9'
14	26°34.6	44.3	164°26.8	10.6'	04°34.8	15.5'	59.8'
15	41°34.6	· · 43.5	178°56.5	10.7'	04°50.4	15.5'	59.8'
16	56°34.5	42.8	193°26.1	10.7'	05°05.8	15.4'	59.8'
17	71°34.4	42.1	207°55.8	10.7'	05°21.2	15.4'	59.7'
18	86°34.3	S16°41.3	222°25.5	10.7'	S05°36.6	15.3'	59.7'
19	101°34.3	40.6	236°55.2	10.7'	05°51.9	15.3'	59.7'
20	116°34.2	39.9	251°24.9	10.7'	06°07.1	15.2'	59.6'
21	131°34.1	· · 39.2	265°54.6	10.7'	06°22.3	15.1'	59.6'
22	146°34.1	38.4	280°24.4	10.7'	06°37.5	15.1'	59.5'
23	161°34.0	37.7	294°54.1	10.8'	06°52.6	15.0'	59.5'
SD = 16.2' $d = -0.7'$			SD = 16.4'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:10	08:39	10:28	14:00	15:50	17:19
N 70°	07:02	08:20	09:46	14:42	16:08	17:27
68°	06:55	08:06	09:18	15:10	16:23	17:33
66°	06:50	07:53	08:57	15:31	16:35	17:39
64°	06:45	07:43	08:40	15:48	16:45	17:44
62°	06:40	07:34	08:26	16:02	16:54	17:48
60°	06:36	07:27	08:14	16:14	17:01	17:52
N 58°	06:32	07:20	08:04	16:24	17:08	17:56
56°	06:29	07:14	07:55	16:33	17:14	17:59
54°	06:26	07:09	07:47	16:41	17:19	18:02
52°	06:23	07:04	07:40	16:48	17:24	18:05
50°	06:20	06:59	07:34	16:54	17:29	18:08
45°	06:14	06:49	07:20	17:08	17:39	18:14
N 40°	06:08	06:40	07:09	17:19	17:47	18:20
35°	06:02	06:32	06:59	17:29	17:55	18:25
30°	05:57	06:25	06:50	17:37	18:02	18:31
20°	05:46	06:13	06:36	17:52	18:15	18:41
N 10°	05:35	06:01	06:22	18:05	18:27	18:52
0°	05:23	05:48	06:10	18:17	18:39	19:04
S 10°	05:10	05:35	05:58	18:29	18:52	19:17
20°	04:53	05:21	05:44	18:43	19:06	19:34
30°	04:32	05:03	05:28	18:58	19:24	19:55
35°	04:18	04:52	05:19	19:07	19:35	20:08
40°	04:02	04:39	05:09	19:18	19:48	20:24
45°	03:41	04:23	04:57	19:30	20:03	20:45
S 50°	03:14	04:03	04:41	19:45	20:22	21:12
52°	03:00	03:54	04:34	19:52	20:32	21:25
54°	02:43	03:43	04:26	19:59	20:43	21:42
56°	02:22	03:30	04:18	20:08	20:55	22:02
58°	01:54	03:16	04:08	20:18	21:09	22:29
S 60°	01:09	02:58	03:56	20:29	21:27	23:10

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	18:03	20:28	22:49	09:13	08:40	08:10
N 70°	18:14	20:28	22:38	08:59	08:36	08:15
68°	18:22	20:28	22:29	08:48	08:32	08:18
66°	18:30	20:28	22:22	08:38	08:29	08:21
64°	18:36	20:28	22:16	08:30	08:27	08:24
62°	18:41	20:27	22:11	08:23	08:25	08:26
60°	18:45	20:27	22:06	08:17	08:23	08:28
N 58°	18:49	20:27	22:03	08:12	08:21	08:30
56°	18:53	20:27	21:59	08:07	08:20	08:31
54°	18:56	20:27	21:56	08:03	08:18	08:33
52°	18:59	20:27	21:53	07:59	08:17	08:34
50°	19:01	20:27	21:51	07:55	08:16	08:35
45°	19:07	20:27	21:45	07:48	08:14	08:38
N 40°	19:11	20:27	21:41	07:41	08:11	08:40
35°	19:15	20:27	21:37	07:36	08:10	08:42
30°	19:19	20:27	21:33	07:30	08:08	08:43
20°	19:25	20:27	21:27	07:22	08:05	08:46
N 10°	19:30	20:27	21:22	07:14	08:02	08:49
0°	19:35	20:27	21:18	07:07	08:00	08:51
S 10°	19:40	20:27	21:13	06:59	07:57	08:53
20°	19:45	20:27	21:08	06:51	07:55	08:56
30°	19:51	20:27	21:02	06:42	07:52	08:59
35°	19:55	20:27	20:59	06:37	07:50	09:00
40°	19:58	20:27	20:56	06:31	07:48	09:02
45°	20:03	20:28	20:51	06:23	07:45	09:05
S 50°	20:08	20:28	20:47	06:15	07:42	09:07
52°	20:11	20:28	20:44	06:11	07:41	09:08
54°	20:13	20:28	20:42	06:06	07:39	09:10
56°	20:16	20:28	20:39	06:01	07:38	09:11
58°	20:20	20:28	20:36	05:56	07:36	09:13
S 60°	20:23	20:28	20:33	05:49	07:34	09:14

Day	Sun		Mer. Pass	Moon		Age
	Eqn.of Time	12 ^h		Mer.Pass.		
	00 ^h	mm:ss	hh:mm	Upper	Lower	16-18
				hh:mm	hh:mm	99-89%
31	13:21	13:25	12:13	00:54	13:21	
01	13:29	13:33	12:14	01:48	14:14	
02	13:37	13:41	12:14	02:39	15:04	

February 03, 04, 05 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	133°04.1	170°52.3	S16°19.0	1°14.7	N22°36.0	157°10.4	S11°01.8	308°10.1	N00°32.3	Alpheratz	357°46.3	29°08.9
1	148°06.6	185°51.6	18.0	16°18.2	36.3	172°12.3	01.6	323°12.7	32.4	Ankaa	353°18.2	-42°15.2
2	163°09.0	200°51.0	17.1	31°21.6	36.5	187°14.2	01.4	338°15.2	32.4	Schedar	349°43.8	56°35.9
3	178°11.5	215°50.3	•• 16.1	46°25.1	•• 36.8	202°16.1	•• 01.2	353°17.8	•• 32.4	Diphda	348°58.5	-17°55.9
4	193°14.0	230°49.7	15.1	61°28.6	37.0	217°18.0	01.0	8°20.3	32.5	Achernar	335°28.6	-57°11.3
5	208°16.4	245°49.1	14.2	76°32.1	37.3	232°19.9	00.8	23°22.9	32.5	Hamal	328°03.6	23°30.8
6	223°18.9	260°48.4	S16°13.2	91°35.6	N22°37.5	247°21.8	S11°00.5	38°25.4	N00°32.6	Polaris	318°59.9	89°18.9
7	238°21.4	275°47.8	12.3	106°39.1	37.8	262°23.7	00.3	53°27.9	32.6	Acamar	315°20.1	-40°16.0
8	253°23.8	290°47.1	11.3	121°42.5	38.0	277°25.6	11°00.1	68°30.5	32.7	Menkar	314°17.6	4°07.8
9	268°26.3	305°46.5	•• 10.4	136°46.0	•• 38.3	292°27.6	10°59.9	83°33.0	•• 32.7	Mirfak	308°43.9	49°54.1
10	283°28.7	320°45.8	09.4	151°49.5	38.5	307°29.5	59.7	98°35.6	32.8	Aldebaran	290°52.0	16°31.8
11	298°31.2	335°45.2	08.4	166°53.0	38.8	322°31.4	59.5	113°38.1	32.8	Rigel	281°14.2	-8°11.5
12	313°33.7	350°44.5	S16°07.5	181°56.5	N22°39.0	337°33.3	S10°59.3	128°40.7	N00°32.9	Capella	280°37.8	46°00.6
13	328°36.1	5°43.9	06.5	196°59.9	39.3	352°35.2	59.1	143°43.2	32.9	Bellatrix	278°34.4	6°21.5
14	343°38.6	20°43.3	05.6	212°03.4	39.5	7°37.1	58.8	158°45.7	33.0	Elnath	278°15.5	28°37.0
15	358°41.1	35°42.6	•• 04.6	227°06.9	•• 39.8	22°39.0	•• 58.6	173°48.3	•• 33.0	Alnilam	275°48.6	-1°11.8
16	13°43.5	50°42.0	03.6	242°10.4	40.0	37°40.9	58.4	188°50.8	33.0	Betelgeuse	271°03.7	7°24.5
17	28°46.0	65°41.3	02.7	257°13.9	40.3	52°42.8	58.2	203°53.4	33.1	Canopus	263°56.8	-52°42.3
18	43°48.5	80°40.7	S16°01.7	272°17.3	N22°40.5	67°44.7	S10°58.0	218°55.9	N00°33.1	Sirius	258°35.6	-16°43.9
19	58°50.9	95°40.1	16°00.7	287°20.8	40.8	82°46.6	57.8	233°58.5	33.2	Adhara	255°14.1	-28°59.3
20	73°53.4	110°39.4	15°59.8	302°24.3	41.0	97°48.5	57.6	249°01.0	33.2	Procyon	245°02.0	5°11.8
21	88°55.9	125°38.8	•• 58.8	317°27.8	•• 41.2	112°50.4	•• 57.4	264°03.6	•• 33.3	Pollux	243°30.3	28°00.0
22	103°58.3	140°38.1	57.8	332°31.3	41.5	127°52.3	57.1	279°06.1	33.3	Avior	234°18.5	-59°32.6
23	119°00.8	155°37.5	56.9	347°34.7	41.7	142°54.2	56.9	294°08.6	33.4	Suhail	222°53.9	-43°28.5
Mer.pass. 15:05		ν -0.6' d -1.0' m -3.92		ν 3.5' d 0.3' m -1.28		ν 1.9' d -0.2' m -2.04		ν 2.5' d 0.0' m 0.71				

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	134°03.2	170°36.9	S15°55.9	2°38.2	N22°42.0	157°56.2	S10°56.7	309°11.2	N00°33.4
1	149°05.7	185°36.2	54.9	17°41.7	42.2	172°58.1	56.5	324°13.7	33.5
2	164°08.2	200°35.6	54.0	32°45.2	42.5	188°00.0	56.3	339°16.3	33.5
3	179°10.6	215°35.0	•• 53.0	47°48.6	•• 42.7	203°01.9	•• 56.1	354°18.8	•• 33.6
4	194°13.1	230°34.3	52.0	62°52.1	42.9	218°03.8	55.9	9°21.4	33.6
5	209°15.6	245°33.7	51.0	77°55.6	43.2	233°05.7	55.6	24°23.9	33.6
6	224°18.0	260°33.1	S15°50.1	92°59.1	N22°43.4	248°07.6	S10°55.4	39°26.5	N00°33.7
7	239°20.5	275°32.4	49.1	108°02.5	43.7	263°09.5	55.2	54°29.0	33.7
8	254°23.0	290°31.8	48.1	123°06.0	43.9	278°11.4	55.0	69°31.6	33.8
9	269°25.4	305°31.2	•• 47.1	138°09.5	•• 44.1	293°13.3	•• 54.8	84°34.1	•• 33.8
10	284°27.9	320°30.5	46.2	153°13.0	44.4	308°15.2	54.6	99°36.7	33.9
11	299°30.3	335°29.9	45.2	168°16.4	44.6	323°17.1	54.4	114°39.2	33.9
12	314°32.8	350°29.3	S15°44.2	183°19.9	N22°44.9	338°19.0	S10°54.1	129°41.7	N00°34.0
13	329°35.3	5°28.6	43.2	198°23.4	45.1	353°20.9	53.9	144°44.3	34.0
14	344°37.7	20°28.0	42.2	213°26.9	45.3	8°22.8	53.7	159°46.8	34.1
15	359°40.2	35°27.4	•• 41.3	228°30.3	•• 45.6	23°24.7	•• 53.5	174°49.4	•• 34.1
16	14°42.7	50°26.7	40.3	243°33.8	45.8	38°26.7	53.3	189°51.9	34.2
17	29°45.1	65°26.1	39.3	258°37.3	46.1	53°28.6	53.1	204°54.5	34.2
18	44°47.6	80°25.5	S15°38.3	273°40.7	N22°46.3	68°30.5	S10°52.9	219°57.0	N00°34.3
19	59°50.1	95°24.9	37.3	288°44.2	46.5	83°32.4	52.6	234°59.6	34.3
20	74°52.5	110°24.2	36.3	303°47.7	46.8	98°34.3	52.4	250°02.1	34.4
21	89°55.0	125°23.6	•• 35.3	318°51.1	•• 47.0	113°36.2	•• 52.2	265°04.7	•• 34.4
22	104°57.5	140°23.0	34.4	333°54.6	47.2	128°38.1	52.0	280°07.2	34.5
23	119°59.9	155°22.4	33.4	348°58.1	47.5	143°40.0	51.8	295°09.8	34.5
Mer.pass. 15:01		ν -0.6' d -1.0' m -3.92		ν 3.5' d 0.2' m -1.26		ν 1.9' d -0.2' m -2.04		ν 2.5' d 0.0' m 0.70	

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	135°02.4	170°21.7	S15°32.4	4°01.6	N22°47.7	158°41.9	S10°51.6	310°12.3	N00°34.5
1	150°04.8	185°21.1	31.4	19°05.0	47.9	173°43.8	51.4	325°14.9	34.6
2	165°07.3	200°20.5	30.4	34°08.5	48.2	188°45.7	51.1	340°17.4	34.6
3	180°09.8	215°19.9	•• 29.4	49°12.0	•• 48.4	203°47.6	•• 50.9	355°20.0	•• 34.7
4	195°12.2	230°19.2	28.4	64°15.4	48.6	218°49.5	50.7	10°22.5	34.7
5	210°14.7	245°18.6	27.4	79°18.9	48.9	233°51.4	50.5	25°25.1	34.8
6	225°17.2	260°18.0	S15°26.4	94°22.3	N22°49.1	248°53.3	S10°50.3	40°27.6	N00°34.8
7	240°19.6	275°17.4	25.5	109°25.8	49.3	263°55.2	50.1	55°30.2	34.9
8	255°22.1	290°16.8	24.5	124°29.3	49.6	278°57.1	49.9	70°32.7	34.9
9	270°24.6	305°16.1	•• 23.5	139°32.7	•• 49.8	293°59.0	•• 49.6	85°35.3	•• 35.0
10	285°27.0	320°15.5	22.5	154°36.2	50.0	309°01.0	49.4	100°37.8	35.0
11	300°29.5	335°14.9	21.5	169°39.7	50.2	324°02.9	49.2	115°40.4	35.1
12	315°32.0	350°14.3	S15°20.5	184°43.1	N22°50.5	339°04.8	S10°49.0	130°42.9	N00°35.1
13	330°34.4	5°13.7	19.5	199°46.6	50.7	354°06.7	48.8	145°45.5	35.2
14	345°36.9	20°13.1	18.5	214°50.0	50.9	9°08.6	48.6	160°48.0	35.2
15	0°39.3	35°12.4	•• 17.5	229°53.5	•• 51.2	24°10.5	•• 48.4	175°50.6	•• 35.3
16	15°41.8	50°11.8	16.5	244°57.0	51.4	39°12.4	48.1	190°53.1	35.3
17	30°44.3	65°11.2	15.5	260°00.4	51.6	54°14.3	47.9	205°55.7	35.4
18	45°46.7	80°10.6	S15°14.5	275°03.9	N22°51.8	69°16.2	S10°47.7	220°58.2	N00°35.4
19	60°49.2	95°10.0	13.5	290°07.3	52.1	84°18.1	47.5	236°00.8	35.5
20	75°51.7	110°09.4	12.5	305°10.8	52.3	99°20.0	47.3	251°03.3	35.5
21	90°54.1	125°08.8	•• 11.5	320°14.3	•• 52.5	114°21.9	•• 47.1	266°05.9	•• 35.6
22	105°56.6	140°08.1	10.5	335°17.7	52.7	129°23.8	46.9	281°08.4	35.6
23	120°59.1	155°07.5	09.5	350°21.2	53.0	144°25.7	46.6	296°11.0	35.7
Mer.pass. 14:57		ν -0.6' d -1.0' m -3.92		ν 3.5' d 0.2' m -1.24		ν 1.9' d -0.2' m -2.04		ν 2.5' d 0.0' m 0.70	

Feb 03 Wed	SHA	Mer.pass
Venus	37°48.2	12:37
Mars	228°10.6	23:49
Jupiter	24°06.3	13:30
Saturn	175°06.0	03:27

Feb 04 Thu	SHA	Mer.pass
Venus	36°33.6	12:38
Mars	228°35.0	23:44
Jupiter	23°52.9	13:27
Saturn	175°07.9	03:23

Feb 05 Fri	SHA	Mer.pass
Venus	35°19.4	12:39
Mars	228°59.2	23:38
Jupiter	23°39.5	13:24
Saturn	175°09.9	03:19

Horizontal parallax		
Venus:	0.1	
Mars:	0.2	

DUT1 = UT1-UTC = +0.0841 sec $\Delta T = TT-UT1 = +66.0999$ sec

2010 February 03 to Feb. 05 UT

h Sun Moon						
Wed	GHA	Dec	GHA	ν	Dec	d HP
0	176°33.9	S16°37.0	309°23.9	10.8'	S07°07.6	15.0' 59.5'
1	191°33.9	36.2	323°53.7	10.8'	07°22.6	14.9' 59.4'
2	206°33.8	35.5	338°23.4	10.8'	07°37.5	14.9' 59.4'
3	221°33.7	· · 34.8	352°53.2	10.8'	07°52.4	14.8' 59.4'
4	236°33.6	34.0	7°23.0	10.8'	08°07.2	14.7' 59.3'
5	251°33.6	33.3	21°52.8	10.8'	08°21.9	14.7' 59.3'
6	266°33.5	S16°32.6	36°22.6	10.8'	S08°36.6	14.6' 59.2'
7	281°33.4	31.8	50°52.4	10.8'	08°51.1	14.5' 59.2'
8	296°33.4	31.1	65°22.2	10.8'	09°05.7	14.5' 59.2'
9	311°33.3	· · 30.4	79°52.0	10.8'	09°20.1	14.4' 59.1'
10	326°33.2	29.6	94°21.8	10.8'	09°34.5	14.3' 59.1'
11	341°33.2	28.9	108°51.6	10.8'	09°48.9	14.2' 59.0'
12	356°33.1	S16°28.2	123°21.4	10.8'	S10°03.1	14.2' 59.0'
13	11°33.1	27.4	137°51.2	10.8'	10°17.3	14.1' 59.0'
14	26°33.0	26.7	152°21.0	10.8'	10°31.4	14.0' 58.9'
15	41°32.9	· · 25.9	166°50.8	10.8'	10°45.4	13.9' 58.9'
16	56°32.9	25.2	181°20.5	10.8'	10°59.3	13.9' 58.8'
17	71°32.8	24.5	195°50.3	10.8'	11°13.2	13.8' 58.8'
18	86°32.7	S16°23.7	210°20.1	10.8'	S11°27.0	13.7' 58.8'
19	101°32.7	23.0	224°49.9	10.8'	11°40.7	13.6' 58.7'
20	116°32.6	22.2	239°19.6	10.8'	11°54.3	13.5' 58.7'
21	131°32.5	· · 21.5	253°49.4	10.7'	12°07.9	13.5' 58.6'
22	146°32.5	20.8	268°19.1	10.7'	12°21.4	13.4' 58.6'
23	161°32.4	20.0	282°48.9	10.7'	12°34.7	13.3' 58.6'
SD = 16.2' $d = -0.7'$ SD = 16.2'						
Thu	GHA	Dec	GHA	ν	Dec	d HP
0	176°32.4	S16°19.3	297°18.6	10.7'	S12°48.0	13.2' 58.5'
1	191°32.3	18.5	311°48.3	10.7'	13°01.2	13.1' 58.5'
2	206°32.2	17.8	326°18.0	10.7'	13°14.4	13.0' 58.4'
3	221°32.2	· · 17.0	340°47.7	10.7'	13°27.4	12.9' 58.4'
4	236°32.1	16.3	355°17.4	10.7'	13°40.3	12.9' 58.4'
5	251°32.1	15.5	9°47.0	10.7'	13°53.2	12.8' 58.3'
6	266°32.0	S16°14.8	24°16.7	10.6'	S14°06.0	12.7' 58.3'
7	281°31.9	14.0	38°46.3	10.6'	14°18.6	12.6' 58.2'
8	296°31.9	13.3	53°16.0	10.6'	14°31.2	12.5' 58.2'
9	311°31.8	· · 12.6	67°45.6	10.6'	14°43.7	12.4' 58.2'
10	326°31.8	11.8	82°15.2	10.6'	14°56.1	12.3' 58.1'
11	341°31.7	11.1	96°44.7	10.6'	15°08.4	12.2' 58.1'
12	356°31.6	S16°10.3	111°14.3	10.5'	S15°20.6	12.1' 58.0'
13	11°31.6	09.6	125°43.8	10.5'	15°32.6	12.0' 58.0'
14	26°31.5	08.8	140°13.3	10.5'	15°44.6	11.9' 58.0'
15	41°31.5	· · 08.1	154°42.9	10.5'	15°56.5	11.8' 57.9'
16	56°31.4	07.3	169°12.3	10.5'	16°08.3	11.7' 57.9'
17	71°31.4	06.6	183°41.8	10.4'	16°20.0	11.6' 57.9'
18	86°31.3	S16°05.8	198°11.3	10.4'	S16°31.6	11.5' 57.8'
19	101°31.3	05.0	212°40.7	10.4'	16°43.1	11.4' 57.8'
20	116°31.2	04.3	227°10.1	10.4'	16°54.5	11.3' 57.7'
21	131°31.1	· · 03.5	241°39.5	10.4'	17°05.8	11.2' 57.7'
22	146°31.1	02.8	256°08.9	10.3'	17°17.0	11.1' 57.7'
23	161°31.0	02.0	270°38.2	10.3'	17°28.1	11.0' 57.6'
SD = 16.2' $d = -0.7'$ SD = 16.0'						
Fri	GHA	Dec	GHA	ν	Dec	d HP
0	176°31.0	S16°01.3	285°07.5	10.3'	S17°39.0	10.9' 57.6'
1	191°30.9	16°00.5	299°36.8	10.3'	17°49.9	10.8' 57.5'
2	206°30.9	15°59.8	314°06.1	10.3'	18°00.6	10.6' 57.5'
3	221°30.8	· · 59.0	328°35.4	10.2'	18°11.3	10.5' 57.5'
4	236°30.8	58.3	343°04.6	10.2'	18°21.8	10.4' 57.4'
5	251°30.7	57.5	357°33.8	10.2'	18°32.2	10.3' 57.4'
6	266°30.7	S15°56.7	12°03.0	10.2'	S18°42.5	10.2' 57.3'
7	281°30.6	56.0	26°32.2	10.2'	18°52.7	10.1' 57.3'
8	296°30.6	55.2	41°01.3	10.1'	19°02.8	10.0' 57.3'
9	311°30.5	· · 54.5	55°30.5	10.1'	19°12.8	9.9' 57.2'
10	326°30.5	53.7	69°59.6	10.1'	19°22.7	9.7' 57.2'
11	341°30.4	52.9	84°28.7	10.1'	19°32.4	9.6' 57.2'
12	356°30.4	S15°52.2	98°57.7	10.0'	S19°42.0	9.5' 57.1'
13	11°30.3	51.4	113°26.8	10.0'	19°51.6	9.4' 57.1'
14	26°30.3	50.7	127°55.8	10.0'	20°01.0	9.3' 57.1'
15	41°30.2	· · 49.9	142°24.8	10.0'	20°10.2	9.2' 57.0'
16	56°30.2	49.1	156°53.7	9.9'	20°19.4	9.0' 57.0'
17	71°30.1	48.4	171°22.7	9.9'	20°28.5	8.9' 56.9'
18	86°30.1	S15°47.6	185°51.6	9.9'	S20°37.4	8.8' 56.9'
19	101°30.0	46.8	200°20.5	9.9'	20°46.2	8.7' 56.9'
20	116°30.0	46.1	214°49.4	9.9'	20°54.9	8.6' 56.8'
21	131°30.0	· · 45.3	229°18.2	9.8'	21°03.5	8.4' 56.8'
22	146°29.9	44.5	243°47.1	9.8'	21°11.9	8.3' 56.8'
23	161°29.9	43.8	258°15.9	9.8'	21°20.2	8.2' 56.7'
SD = 16.2' $d = -0.8'$ SD = 15.7'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	06:59	08:25	10:05	14:25	16:04	17:30
N 70°	06:52	08:09	09:30	14:59	16:20	17:37
68°	06:47	07:56	09:05	15:24	16:33	17:42
66°	06:42	07:45	08:46	15:43	16:44	17:47
64°	06:38	07:35	08:31	15:58	16:53	17:51
62°	06:34	07:27	08:18	16:11	17:01	17:55
60°	06:30	07:20	08:07	16:22	17:08	17:59
N 58°	06:27	07:14	07:58	16:31	17:14	18:02
56°	06:24	07:09	07:49	16:39	17:20	18:05
54°	06:21	07:04	07:42	16:47	17:25	18:07
52°	06:19	06:59	07:35	16:53	17:29	18:10
50°	06:16	06:55	07:29	16:59	17:34	18:12
45°	06:10	06:45	07:16	17:12	17:43	18:18
N 40°	06:05	06:37	07:06	17:22	17:51	18:23
35°	06:00	06:30	06:57	17:32	17:58	18:28
30°	05:55	06:24	06:49	17:40	18:05	18:33
20°	05:45	06:12	06:35	17:53	18:16	18:43
N 10°	05:35	06:00	06:22	18:06	18:28	18:53
0°	05:24	05:49	06:10	18:17	18:39	19:04
S 10°	05:11	05:37	05:59	18:29	18:51	19:17
20°	04:55	05:23	05:46	18:42	19:05	19:32
30°	04:35	05:05	05:31	18:56	19:22	19:52
35°	04:22	04:55	05:22	19:05	19:32	20:05
40°	04:06	04:43	05:13	19:15	19:44	20:20
45°	03:47	04:28	05:01	19:26	19:59	20:40
S 50°	03:21	04:09	04:47	19:40	20:17	21:05
52°	03:08	04:00	04:40	19:47	20:26	21:18
54°	02:52	03:50	04:33	19:54	20:36	21:33
56°	02:33	03:38	04:24	20:02	20:48	21:52
58°	02:09	03:25	04:15	20:11	21:01	22:15
S 60°	01:34	03:08	04:04	20:22	21:17	22:48

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	· · ·	01:19	■	07:39	06:55	■
N 70°	· · ·	00:52	03:31	07:52	07:24	06:31
68°	· · ·	00:32	02:44	08:03	07:46	07:19
66°	· · ·	00:16	02:14	08:13	08:03	07:51
64°	· · ·	00:04	01:52	08:20	08:17	08:14
62°	23:53	· · ·	01:34	08:27	08:29	08:33
60°	23:44	· · ·	01:20	08:33	08:39	08:48
N 58°	23:36	· · ·	01:07	08:38	08:48	09:01
56°	23:29	· · ·	00:56	08:43	08:56	09:13
54°	23:23	· · ·	00:47	08:47	09:03	09:23
52°	23:17	· · ·	00:39	08:51	09:09	09:32
50°	23:12	· · ·	00:31	08:54	09:15	09:40
45°	23:01	· · ·	00:15	09:02	09:28	09:57
N 40°	22:52	· · ·	00:02	09:08	09:38	10:11
35°	22:45	23:51	· · ·	09:14	09:47	10:23
30°	22:38	23:42	· · ·	09:18	09:55	10:33
20°	22:27	23:25	· · ·	09:27	10:08	10:52
N 10°	22:17	23:11	· · ·	09:34	10:20	11:07
0°	22:08	22:58	23:49	09:41	10:31	11:22
S 10°	21:58	22:45	23:33	09:48	10:43	11:37
20°	21:49	22:31	23:15	09:56	10:55	11:53
30°	21:38	22:15	22:55	10:04	11:09	12:12
35°	21:32	22:06	22:44	10:09	11:17	12:23
40°	21:24	21:56	22:30	10:15	11:26	12:35
45°	21:16	21:43	22:15	10:22	11:37	12:50
S 50°	21:06	21:29	21:56	10:30	11:50	13:08
52°	21:02	21:22	21:47	10:33	11:56	13:16
54°	20:57	21:14	21:36	10:37	12:03	13:26
56°	20:51	21:06	21:25	10:42	12:11	13:37
58°	20:45	20:57	21:12	10:47	12:19	13:49
S 60°	20:38	20:46	20:57	10:53	12:29	14:04

Day	Sun		Mer. Pass	Moon		Age 19-21 81-61%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
03	13:44	13:48	12:14	03:29	15:54	
04	13:51	13:53	12:14	04:19	16:45	
05	13:56	13:58	12:14	05:10	17:36	

February 06, 07, 08 UT (Sat., Sun., Mon.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	136°01.5	170°06.9	S15°08.5	5°24.6	N22°53.2	159°27.6	S10°46.4	311°13.5	N00°35.7	Alpheratz	357°46.3	29°08.9											
1	151°04.0	185°06.3	07.4	20°28.1	53.4	174°29.5	46.2	326°16.1	35.8	Ankaa	353°18.3	-42°15.2											
2	166°06.4	200°05.7	06.4	35°31.5	53.6	189°31.4	46.0	341°18.6	35.8	Schedar	349°43.8	56°35.8											
3	181°08.9	215°05.1	•• 05.4	50°35.0	•• 53.9	204°33.3	•• 45.8	356°21.2	•• 35.9	Diphda	348°58.5	-17°55.9											
4	196°11.4	230°04.5	04.4	65°38.4	54.1	219°35.2	45.6	11°23.8	35.9	Achernar	335°28.6	-57°11.3											
5	211°13.8	245°03.9	03.4	80°41.9	54.3	234°37.1	45.3	26°26.3	36.0	Hamal	328°03.6	23°30.7											
6	226°16.3	260°03.3	S15°02.4	95°45.4	N22°54.5	249°39.0	S10°45.1	41°28.9	N00°36.0	Polaris	319°01.1	89°18.9											
7	241°18.8	275°02.7	01.4	110°48.8	54.8	264°40.9	44.9	56°31.4	36.1	Acamar	315°20.1	-40°16.0											
8	256°21.2	290°02.0	15°00.4	125°52.3	55.0	279°42.9	44.7	71°34.0	36.1	Menkar	314°17.6	4°07.8											
9	271°23.7	305°01.4	14°59.4	140°55.7	•• 55.2	294°44.8	•• 44.5	86°36.5	•• 36.2	Mirfak	308°43.9	49°54.1											
10	286°26.2	320°00.8	58.4	155°59.2	55.4	309°46.7	44.3	101°39.1	36.2	Aldebaran	290°52.0	16°31.8											
11	301°28.6	335°00.2	57.3	171°02.6	55.6	324°48.6	44.1	116°41.6	36.3	Rigel	281°14.2	-8°11.5											
12	316°31.1	349°59.6	S14°56.3	186°06.1	N22°55.9	339°50.5	S10°43.8	131°44.2	N00°36.3	Capella	280°37.8	46°00.6											
13	331°33.6	4°59.0	55.3	201°09.5	56.1	354°52.4	43.6	146°46.7	36.4	Bellatrix	278°34.4	6°21.5											
14	346°36.0	19°58.4	54.3	216°13.0	56.3	9°54.3	43.4	161°49.3	36.4	Elnath	278°15.5	28°37.0											
15	1°38.5	34°57.8	•• 53.3	231°16.4	•• 56.5	24°56.2	•• 43.2	176°51.8	•• 36.5	Alnilam	275°48.6	-1°11.8											
16	16°40.9	49°57.2	52.3	246°19.9	56.7	39°58.1	43.0	191°54.4	36.5	Betelgeuse	271°03.7	7°24.5											
17	31°43.4	64°56.6	51.2	261°23.3	56.9	55°00.0	42.8	206°57.0	36.6	Canopus	263°56.8	-52°42.3											
18	46°45.9	79°56.0	S14°50.2	276°26.7	N22°57.2	70°01.9	S10°42.5	221°59.5	N00°36.6	Sirius	258°35.6	-16°44.0											
19	61°48.3	94°55.4	49.2	291°30.2	57.4	85°03.8	42.3	237°02.1	36.7	Adhara	255°14.1	-28°59.3											
20	76°50.8	109°54.8	48.2	306°33.6	57.6	100°05.7	42.1	252°04.6	36.7	Procyon	245°02.0	5°11.8											
21	91°53.3	124°54.2	•• 47.2	321°37.1	•• 57.8	115°07.6	•• 41.9	267°07.2	•• 36.8	Pollux	243°30.3	28°00.0											
22	106°55.7	139°53.6	46.1	336°40.5	58.0	130°09.5	41.7	282°09.7	36.8	Avior	234°18.5	-59°32.7											
23	121°58.2	154°53.0	45.1	351°44.0	58.2	145°11.4	41.5	297°12.3	36.9	Suhail	222°53.9	-43°28.5											
Mer.pass. 14:53				ν -0.6' d -1.0' m -3.92				ν 3.5' d 0.2' m -1.22				ν 1.9' d -0.2' m -2.03				ν 2.6' d 0.0' m 0.69							

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	137°00.7	169°52.4	S14°44.1	6°47.4	N22°58.5	160°13.3	S10°41.3	312°14.8	N00°36.9	Denebola	182°35.9	14°30.7							
1	152°03.1	184°51.8	43.1	21°50.8	58.7	175°15.2	41.0	327°17.4	37.0	Gienah	175°54.6	-17°36.0							
2	167°05.6	199°51.2	42.1	36°54.3	58.9	190°17.1	40.8	342°20.0	37.0	Acrux	173°11.9	-63°09.2							
3	182°08.1	214°50.6	•• 41.0	51°57.7	•• 59.1	205°19.0	•• 40.6	357°22.5	•• 37.1	Gacrux	172°03.4	-57°10.1							
4	197°10.5	229°50.0	40.0	67°01.2	59.3	220°20.9	40.4	12°25.1	37.1	Alioth	166°22.3	55°53.9							
5	212°13.0	244°49.4	39.0	82°04.6	59.5	235°22.8	40.2	27°27.6	37.2	Spica	158°33.7	-11°13.0							
6	227°15.4	259°48.8	S14°38.0	97°08.0	N22°59.7	250°24.7	S10°40.0	42°30.2	N00°37.2	Alkaid	153°00.5	49°15.4							
7	242°17.9	274°48.2	36.9	112°11.5	22°59.9	265°26.6	39.7	57°32.7	37.3	Hadar	148°51.4	-60°25.2							
8	257°20.4	289°47.6	35.9	127°14.9	23°00.2	280°28.5	39.5	72°35.3	37.3	Menkent	148°10.4	-36°25.2							
9	272°22.8	304°47.0	•• 34.9	142°18.4	•• 00.4	295°30.4	•• 39.3	87°37.9	•• 37.4	Arcturus	145°57.8	19°07.5							
10	287°25.3	319°46.4	33.8	157°21.8	00.6	310°32.3	39.1	102°40.4	37.4	Rigel Kent.	139°55.1	-60°52.4							
11	302°27.8	334°45.8	32.8	172°25.2	00.8	325°34.3	38.9	117°43.0	37.5	Kochab	137°19.5	74°06.4							
12	317°30.2	349°45.2	S14°31.8	187°28.7	N23°01.0	340°36.2	S10°38.7	132°45.5	N00°37.5	Zuben'ubi	137°08.1	-16°05.1							
13	332°32.7	4°44.7	30.7	202°32.1	01.2	355°38.1	38.4	147°48.1	37.6	Alphecca	126°13.1	26°40.5							
14	347°35.2	19°44.1	29.7	217°35.5	01.4	10°40.0	38.2	162°50.7	37.6	Antares	112°29.4	-26°27.3							
15	2°37.6	34°43.5	•• 28.7	232°39.0	•• 01.6	25°41.9	•• 38.0	177°53.2	•• 37.7	Atria	107°33.8	-69°02.5							
16	17°40.1	49°42.9	27.6	247°42.4	01.8	40°43.8	37.8	192°55.8	37.7	Sabik	102°15.5	-15°44.3							
17	32°42.6	64°42.3	26.6	262°45.8	02.0	55°45.7	37.6	207°58.3	37.8	Shaula	96°25.5	-37°06.6							
18	47°45.0	79°41.7	S14°25.6	277°49.3	N23°02.2	70°47.6	S10°37.4	223°00.9	N00°37.8	Rasalhague	96°08.9	12°33.0							
19	62°47.5	94°41.1	24.5	292°52.7	02.4	85°49.5	37.1	238°03.4	37.9	Eltanin	90°47.6	51°29.0							
20	77°49.9	109°40.5	23.5	307°56.1	02.7	100°51.4	36.9	253°06.0	38.0	Kaus Aust.	83°47.3	-34°22.7							
21	92°52.4	124°39.9	•• 22.5	322°59.5	•• 02.9	115°53.3	•• 36.7	268°08.6	•• 38.0	Vega	80°41.0	38°47.4							
22	107°54.9	139°39.3	21.4	338°03.0	03.1	130°55.2	36.5	283°11.1	38.1	Nunki	76°01.6	-26°17.0							
23	122°57.3	154°38.8	20.4	353°06.4	03.3	145°57.1	36.3	298°13.7	38.1	Altair	62°10.9	8°53.6							
Mer.pass. 14:50				ν -0.6' d -1.0' m -3.92				ν 3.4' d 0.2' m -1.20				ν 1.9' d -0.2' m -2.03				ν 2.6' d 0.1' m 0.69			

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	137°59.8	169°38.2	S14°19.3	8°09.8	N23°03.5	160°59.0	S10°36.1	313°16.2	N00°38.2	Feb 06 Sat	SHA	Mer.pass					
1	153°02.3	184°37.6	18.3	23°13.3	03.7	176°00.9	35.8	328°18.8	38.2	Venus	34°05.4	12:40					
2	168°04.7	199°37.0	17.3	38°16.7	03.9	191°02.8	35.6	343°21.4	38.3	Mars	229°23.1	23:33					
3	183°07.2	214°36.4	•• 16.2	53°20.1	•• 04.1	206°04.7	•• 35.4	358°23.9	•• 38.3	Jupiter	23°26.1	13:20					
4	198°09.7	229°35.8	15.2	68°23.5	04.3	221°06.6	35.2	13°26.5	38.4	Saturn	175°12.0	03:15					
5	213°12.1	244°35.2	14.1	83°27.0	04.5	236°08.5	35.0	28°29.0	38.4								
6	228°14.6	259°34.7	S14°13.1	98°30.4	N23°04.7	251°10.4	S10°34.8	43°31.6	N00°38.5								
7	243°17.0	274°34.1	12.1	113°33.8	04.9	266°12.3	34.5	58°34.2	38.5	Feb 07 Sun	SHA	Mer.pass					
8	258°19.5	289°33.5	11.0	128°37.2	05.1	281°14.2	34.3	73°36.7	38.6	Venus	32°51.7	12:41					
9	273°22.0	304°32.9	•• 10.0	143°40.6	•• 05.3	296°16.1	•• 34.1	88°39.3	•• 38.6	Mars	229°46.7	23:27					
10	288°24.4	319°32.3	08.9	158°44.1	05.5	311°18.0	33.9	103°41.8	38.7	Jupiter	23°12.7	13:17					
11	303°26.9	334°31.7	07.9	173°47.5	05.7	326°19.9	33.7	118°44.4	38.7	Saturn	175°14.2	03:10					
12	318°29.4	349°31.2	S14°06.8	188°50.9	N23°05.9	341°21.8	S10°33.5	133°47.0	N00°38.8								
13	333°31.8	4°30.6	05.8	203°54.3	06.1	356°23.7	33.2	148°49.5	38.8	Feb 08 Mon	SHA	Mer.pass					
14	348°34.3	19°30.0	04.7	218°57.7	06.3	11°25.6	33.0	163°52.1	38.9	Venus	31°38.4	12:42					
15	3°36.8	34°29.4	•• 03.7	234°01.1	•• 06.5	26°27.5	•• 32.8	178°54.7	•• 39.0	Mars	230°10.0	23:22					
16	18°39.2	49°28.8	02.6	249°04.6	06.7	41°29.4	32.6	193°57.2	39.0	Jupiter	22°59.2	13:14					
17	33°41.7																

DUT1 = UT1-UTC = +0.0799 sec $\Delta T = TT-UT1 = +66.1041$ sec

2010 February 06 to Feb. 08 UT

Sun				Moon			
h	GHA	Dec	ν	Dec	d	HP	
0	176°29.8	S15°43.0	272°44.7	9.8'	S21°28.4	8.1'	56.7'
1	191°29.8	42.2	287°13.5	9.8'	21°36.5	8.0'	56.7'
2	206°29.7	41.5	301°42.2	9.7'	21°44.5	7.8'	56.6'
3	221°29.7	40.7	316°11.0	9.7'	21°52.3	7.7'	56.6'
4	236°29.6	39.9	330°39.7	9.7'	22°00.0	7.6'	56.6'
5	251°29.6	39.2	345°08.4	9.7'	22°07.6	7.5'	56.5'
6	266°29.6	S15°38.4	359°37.0	9.7'	S22°15.1	7.3'	56.5'
7	281°29.5	37.6	14°05.7	9.6'	22°22.4	7.2'	56.4'
8	296°29.5	36.9	28°34.3	9.6'	22°29.6	7.1'	56.4'
9	311°29.4	36.1	43°02.9	9.6'	22°36.7	7.0'	56.4'
10	326°29.4	35.3	57°31.5	9.6'	22°43.7	6.8'	56.3'
11	341°29.3	34.5	72°00.1	9.6'	22°50.5	6.7'	56.3'
12	356°29.3	S15°33.8	86°28.7	9.5'	S22°57.2	6.6'	56.3'
13	11°29.3	33.0	100°57.2	9.5'	23°03.8	6.4'	56.3'
14	26°29.2	32.2	115°25.7	9.5'	23°10.2	6.3'	56.2'
15	41°29.2	31.5	129°54.3	9.5'	23°16.5	6.2'	56.2'
16	56°29.1	30.7	144°22.8	9.5'	23°22.7	6.1'	56.2'
17	71°29.1	29.9	158°51.2	9.5'	23°28.8	5.9'	56.1'
18	86°29.1	S15°29.1	173°19.7	9.5'	S23°34.7	5.8'	56.1'
19	101°29.0	28.4	187°48.2	9.4'	23°40.5	5.7'	56.1'
20	116°29.0	27.6	202°16.6	9.4'	23°46.2	5.5'	56.0'
21	131°29.0	26.8	216°45.0	9.4'	23°51.7	5.4'	56.0'
22	146°28.9	26.0	231°13.5	9.4'	23°57.1	5.3'	56.0'
23	161°28.9	25.3	245°41.9	9.4'	24°02.4	5.2'	55.9'
SD = 16.2'				SD = 15.5'			

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	176°28.8	S15°24.5	260°10.3	9.4'	S24°07.6	5.0'	55.9'
1	191°28.8	23.7	274°38.7	9.4'	24°12.6	4.9'	55.9'
2	206°28.8	22.9	289°07.0	9.4'	24°17.5	4.8'	55.8'
3	221°28.7	22.1	303°35.4	9.4'	24°22.2	4.6'	55.8'
4	236°28.7	21.4	318°03.8	9.4'	24°26.9	4.5'	55.8'
5	251°28.7	20.6	332°32.1	9.4'	24°31.4	4.4'	55.8'
6	266°28.6	S15°19.8	347°00.5	9.3'	S24°35.7	4.2'	55.7'
7	281°28.6	19.0	1°28.8	9.3'	24°39.9	4.1'	55.7'
8	296°28.6	18.2	15°57.2	9.3'	24°44.0	4.0'	55.7'
9	311°28.5	17.5	30°25.5	9.3'	24°48.0	3.8'	55.6'
10	326°28.5	16.7	44°53.8	9.3'	24°51.8	3.7'	55.6'
11	341°28.5	15.9	59°22.2	9.3'	24°55.5	3.6'	55.6'
12	356°28.4	S15°15.1	73°50.5	9.3'	S24°59.1	3.4'	55.6'
13	11°28.4	14.3	88°18.8	9.3'	25°02.5	3.3'	55.5'
14	26°28.4	13.5	102°47.2	9.3'	25°05.8	3.2'	55.5'
15	41°28.3	12.8	117°15.5	9.3'	25°09.0	3.0'	55.5'
16	56°28.3	12.0	131°43.8	9.3'	25°12.0	2.9'	55.5'
17	71°28.3	11.2	146°12.2	9.3'	25°14.9	2.8'	55.4'
18	86°28.2	S15°10.4	160°40.5	9.3'	S25°17.7	2.6'	55.4'
19	101°28.2	09.6	175°08.9	9.4'	25°20.3	2.5'	55.4'
20	116°28.2	08.8	189°37.2	9.4'	25°22.8	2.4'	55.4'
21	131°28.1	08.0	204°05.6	9.4'	25°25.2	2.2'	55.3'
22	146°28.1	07.3	218°33.9	9.4'	25°27.4	2.1'	55.3'
23	161°28.1	06.5	233°02.3	9.4'	25°29.5	2.0'	55.3'
SD = 16.2'				SD = 15.2'			

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	176°28.1	S15°05.7	247°30.7	9.4'	S25°31.5	1.8'	55.3'
1	191°28.0	04.9	261°59.1	9.4'	25°33.3	1.7'	55.2'
2	206°28.0	04.1	276°27.5	9.4'	25°35.0	1.6'	55.2'
3	221°28.0	03.3	290°55.9	9.4'	25°36.6	1.4'	55.2'
4	236°27.9	02.5	305°24.3	9.4'	25°38.0	1.3'	55.2'
5	251°27.9	01.7	319°52.8	9.5'	25°39.3	1.2'	55.1'
6	266°27.9	S15°00.9	334°21.2	9.5'	S25°40.5	1.0'	55.1'
7	281°27.9	15°00.1	348°49.7	9.5'	25°41.6	0.9'	55.1'
8	296°27.8	14°59.3	3°18.2	9.5'	25°42.5	0.8'	55.1'
9	311°27.8	58.6	17°46.7	9.5'	25°43.2	0.6'	55.0'
10	326°27.8	57.8	32°15.2	9.5'	25°43.9	0.5'	55.0'
11	341°27.8	57.0	46°43.8	9.6'	25°44.4	0.4'	55.0'
12	356°27.7	S14°56.2	61°12.3	9.6'	S25°44.8	0.3'	55.0'
13	11°27.7	55.4	75°40.9	9.6'	25°45.0	0.1'	55.0'
14	26°27.7	54.6	90°09.5	9.6'	25°45.2	-0.0'	54.9'
15	41°27.7	53.8	104°38.2	9.7'	25°45.2	-0.1'	54.9'
16	56°27.6	53.0	119°06.8	9.7'	25°45.0	-0.3'	54.9'
17	71°27.6	52.2	133°35.5	9.7'	25°44.7	-0.4'	54.9'
18	86°27.6	S14°51.4	148°04.2	9.7'	S25°44.3	-0.5'	54.8'
19	101°27.6	50.6	162°32.9	9.8'	25°43.8	-0.7'	54.8'
20	116°27.6	49.8	177°01.7	9.8'	25°43.2	-0.8'	54.8'
21	131°27.5	49.0	191°30.5	9.8'	25°42.4	-0.9'	54.8'
22	146°27.5	48.2	205°59.3	9.8'	25°41.5	-1.0'	54.8'
23	161°27.5	47.4	220°28.1	9.9'	25°40.4	-1.2'	54.7'
SD = 16.2'				SD = 15.1'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	06:48	08:12	09:44	14:46	16:18	17:42
N 70°	06:42	07:57	09:14	15:15	16:32	17:48
68°	06:38	07:45	08:52	15:37	16:44	17:52
66°	06:34	07:36	08:35	15:54	16:54	17:56
64°	06:30	07:27	08:21	16:08	17:02	17:59
62°	06:27	07:20	08:10	16:20	17:09	18:02
60°	06:24	07:14	08:00	16:30	17:15	18:05
N 58°	06:21	07:08	07:51	16:38	17:21	18:08
56°	06:19	07:03	07:43	16:46	17:26	18:10
54°	06:17	06:59	07:36	16:53	17:30	18:13
52°	06:14	06:54	07:30	16:59	17:35	18:15
50°	06:12	06:50	07:25	17:04	17:38	18:17
45°	06:07	06:42	07:13	17:16	17:47	18:22
N 40°	06:02	06:34	07:03	17:26	17:54	18:26
35°	05:58	06:28	06:54	17:35	18:01	18:31
30°	05:53	06:22	06:46	17:42	18:07	18:35
20°	05:44	06:11	06:33	17:55	18:18	18:44
N 10°	05:35	06:00	06:22	18:07	18:28	18:53
0°	05:24	05:49	06:11	18:18	18:39	19:04
S 10°	05:12	05:38	06:00	18:29	18:50	19:16
20°	04:57	05:24	05:47	18:40	19:04	19:31
30°	04:38	05:08	05:34	18:54	19:19	19:50
35°	04:26	04:58	05:26	19:02	19:29	20:02
40°	04:11	04:47	05:16	19:11	19:41	20:16
45°	03:52	04:33	05:05	19:22	19:54	20:35
S 50°	03:28	04:15	04:52	19:35	20:12	20:59
52°	03:16	04:07	04:46	19:41	20:20	21:11
54°	03:01	03:57	04:39	19:48	20:29	21:25
56°	02:44	03:46	04:31	19:56	20:40	21:41
58°	02:23	03:34	04:23	20:04	20:52	22:02
S 60°	01:53	03:19	04:13	20:14	21:07	22:30

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	■	■	■	■	■	■
66°	04:23	■	■	07:29	■	■
64°	03:42	05:32	07:14	08:11	08:10	08:17
62°	03:14	04:48	06:04	08:40	08:55	09:28
60°	02:52	04:18	05:28	09:02	09:25	10:03
N 58°	02:35	03:56	05:03	09:20	09:47	10:29
56°	02:20	03:37	04:43	09:35	10:06	10:49
54°	02:08	03:22	04:26	09:48	10:21	11:06
52°	01:57	03:09	04:11	09:59	10:35	11:20
50°	01:47	02:57	03:59	10:09	10:47	11:33
45°	01:26	02:33	03:33	10:31	11:11	11:59
N 40°	01:10	02:14	03:12	10:48	11:31	12:20
35°	00:56	01:58	02:55	11:03	11:47	12:37
30°	00:44	01:44	02:40	11:16	12:02	12:52
20°	00:23	01:20	02:15	11:37	12:26	13:17
N 10°	00:06	01:00	01:54	11:56	12:47	13:39
0°		00:41	01:34	12:14	13:06	13:59
S 10°		00:22	01:14	12:32	13:26	14:19
20°		00:02	00:52	12:51	13:47	14:41
30°	23:39		00:27	13:13	14:11	15:06
35°	23:26		00:13	13:26	14:26	15:21
40°	23:10	23:56		13:41	14:42	15:38
45°	22:52	23:36		13:59	15:02	15:58
S 50°	22:29	23:11		14:21	15:27	16:24
52°	22:18	22:58	23:49	14:32	15:39	16:36
54°	22:05	22:44	23:35	14:44	15:53	16:51
56°	21:51	22:28	23:18	14:58	16:09	17:07
58°	21:35	22:09	22:58	15:14	16:28	17:27
S 60°	21:15	21:45	22:33	15:33	16:52	17:52

Day	Sun		Mer. Pass	Moon		Age 22-24 50-30%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Upper hh:mm	Lower hh:mm	
06	14:01	14:03	12:14	06:02	18:28	
07	14:05	14:06	12:14	06:54	19:20	
08	14:08	14:09	12:14	07:46	20:12	

February 09, 10, 11 UT (Tue., Wed., Thu.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	138°58.9	169°24.2	S13°54.2	9°31.9	N23°08.2	161°44.6	S10°30.9	314°17.7	N00°39.4	Alpheratz	357°46.3	29°08.9											
1	154°01.4	184°23.7	53.1	24°35.3	08.4	176°46.5	30.6	329°20.3	39.5	Ankaa	353°18.3	-42°15.2											
2	169°03.9	199°23.1	52.1	39°38.7	08.6	191°48.5	30.4	344°22.8	39.5	Schedar	349°43.8	56°35.8											
3	184°06.3	214°22.5	•• 51.0	54°42.1	•• 08.8	206°50.4	•• 30.2	359°25.4	•• 39.6	Diphda	348°58.5	-17°55.9											
4	199°08.8	229°21.9	50.0	69°45.5	09.0	221°52.3	30.0	14°28.0	39.6	Achernar	335°28.6	-57°11.3											
5	214°11.3	244°21.4	48.9	84°48.9	09.2	236°54.2	29.8	29°30.5	39.7	Hamal	328°03.6	23°30.7											
6	229°13.7	259°20.8	S13°47.9	99°52.3	N23°09.4	251°56.1	S10°29.6	44°33.1	N00°39.8	Polaris	319°02.3	89°18.9											
7	244°16.2	274°20.2	46.8	114°55.7	09.6	266°58.0	29.3	59°35.7	39.8	Acamar	315°20.1	-40°16.0											
8	259°18.7	289°19.6	45.7	129°59.1	09.8	281°59.9	29.1	74°38.2	39.9	Menkar	314°17.6	4°07.8											
9	274°21.1	304°19.1	•• 44.7	145°02.5	•• 10.0	297°01.8	•• 28.9	89°40.8	•• 39.9	Mirfak	308°43.9	49°54.1											
10	289°23.6	319°18.5	43.6	160°05.9	10.1	312°03.7	28.7	104°43.4	40.0	Aldebaran	290°52.1	16°31.8											
11	304°26.0	334°17.9	42.6	175°09.3	10.3	327°05.6	28.5	119°45.9	40.0	Rigel	281°14.2	-8°11.5											
12	319°28.5	349°17.4	S13°41.5	190°12.7	N23°10.5	342°07.5	S10°28.2	134°48.5	N00°40.1	Capella	280°37.8	46°00.6											
13	334°31.0	4°16.8	40.4	205°16.1	10.7	357°09.4	28.0	149°51.1	40.1	Bellatrix	278°34.4	6°21.5											
14	349°33.4	19°16.2	39.4	220°19.5	10.9	12°11.3	27.8	164°53.6	40.2	Elnath	278°15.5	28°37.0											
15	4°35.9	34°15.7	•• 38.3	235°22.9	•• 11.1	27°13.2	•• 27.6	179°56.2	•• 40.2	Alnilam	275°48.6	-1°11.8											
16	19°38.4	49°15.1	37.2	250°26.3	11.3	42°15.1	27.4	194°58.8	40.3	Betelgeuse	271°03.7	7°24.5											
17	34°40.8	64°14.5	36.2	265°29.7	11.5	57°17.0	27.2	210°01.3	40.4	Canopus	263°56.8	-52°42.3											
18	49°43.3	79°14.0	S13°35.1	280°33.1	N23°11.6	72°18.9	S10°26.9	225°03.9	N00°40.4	Sirius	258°35.6	-16°44.0											
19	64°45.8	94°13.4	34.0	295°36.5	11.8	87°20.8	26.7	240°06.5	40.5	Adhara	255°14.1	-28°59.3											
20	79°48.2	109°12.8	33.0	310°39.9	12.0	102°22.7	26.5	255°09.0	40.5	Procyon	245°02.0	5°11.8											
21	94°50.7	124°12.3	•• 31.9	325°43.3	•• 12.2	117°24.6	•• 26.3	270°11.6	•• 40.6	Pollux	243°30.3	28°00.0											
22	109°53.2	139°11.7	30.8	340°46.7	12.4	132°26.5	26.1	285°14.2	40.6	Avior	234°18.5	-59°32.7											
23	124°55.6	154°11.1	29.8	355°50.1	12.6	147°28.4	25.9	300°16.7	40.7	Suhail	222°53.9	-43°28.6											
Mer.pass. 14:42		ν -0.6' d -1.1' m -3.92		ν 3.4' d 0.2' m -1.15		ν 1.9' d -0.2' m -2.03		ν 2.6' d 0.1' m 0.68															

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	139°58.1	169°10.6	S13°28.7	10°53.5	N23°12.7	162°30.3	S10°25.6	315°19.3	N00°40.7
1	155°00.5	184°10.0	27.6	25°56.8	12.9	177°32.2	25.4	330°21.9	40.8
2	170°03.0	199°09.5	26.5	41°00.2	13.1	192°34.1	25.2	345°24.4	40.8
3	185°05.5	214°08.9	•• 25.5	56°03.6	•• 13.3	207°36.0	•• 25.0	0°27.0	•• 40.9
4	200°07.9	229°08.3	24.4	71°07.0	13.5	222°37.9	24.8	15°29.6	41.0
5	215°10.4	244°07.8	23.3	86°10.4	13.7	237°39.8	24.5	30°32.1	41.0
6	230°12.9	259°07.2	S13°22.2	101°13.8	N23°13.8	252°41.7	S10°24.3	45°34.7	N00°41.1
7	245°15.3	274°06.7	21.2	116°17.2	14.0	267°43.6	24.1	60°37.3	41.1
8	260°17.8	289°06.1	20.1	131°20.5	14.2	282°45.5	23.9	75°39.8	41.2
9	275°20.3	304°05.5	•• 19.0	146°23.9	•• 14.4	297°47.4	•• 23.7	90°42.4	•• 41.2
10	290°22.7	319°05.0	17.9	161°27.3	14.6	312°49.3	23.5	105°45.0	41.3
11	305°25.2	334°04.4	16.9	176°30.7	14.7	327°51.2	23.2	120°47.5	41.3
12	320°27.7	349°03.9	S13°15.8	191°34.1	N23°14.9	342°53.1	S10°23.0	135°50.1	N00°41.4
13	335°30.1	4°03.3	14.7	206°37.5	15.1	357°55.0	22.8	150°52.7	41.5
14	350°32.6	19°02.7	13.6	221°40.8	15.3	12°56.9	22.6	165°55.2	41.5
15	5°35.0	34°02.2	•• 12.5	236°44.2	•• 15.4	27°58.8	•• 22.4	180°57.8	•• 41.6
16	20°37.5	49°01.6	11.5	251°47.6	15.6	43°00.7	22.1	196°00.4	41.6
17	35°40.0	64°01.1	10.4	266°51.0	15.8	58°02.6	21.9	211°02.9	41.7
18	50°42.4	79°00.5	S13°09.3	281°54.3	N23°16.0	73°04.5	S10°21.7	226°05.5	N00°41.7
19	65°44.9	94°00.0	08.2	296°57.7	16.1	88°06.4	21.5	241°08.1	41.8
20	80°47.4	108°59.4	07.1	312°01.1	16.3	103°08.3	21.3	256°10.7	41.9
21	95°49.8	123°58.9	•• 06.0	327°04.5	•• 16.5	118°10.2	•• 21.1	271°13.2	•• 41.9
22	110°52.3	138°58.3	05.0	342°07.8	16.7	133°12.1	20.8	286°15.8	42.0
23	125°54.8	153°57.8	03.9	357°11.2	16.8	148°14.0	20.6	301°18.4	42.0
Mer.pass. 14:38		ν -0.6' d -1.1' m -3.92		ν 3.4' d 0.2' m -1.12		ν 1.9' d -0.2' m -2.03		ν 2.6' d 0.1' m 0.67	

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	140°57.2	168°57.2	S13°02.8	12°14.6	N23°17.0	163°15.9	S10°20.4	316°20.9	N00°42.1
1	155°59.7	183°56.7	01.7	27°17.9	17.2	178°17.8	20.2	331°23.5	42.1
2	171°02.1	198°56.1	13°00.6	42°21.3	17.3	193°19.7	20.0	346°26.1	42.2
3	186°04.6	213°55.6	12°59.5	57°24.7	•• 17.5	208°21.6	•• 19.7	1°28.6	•• 42.2
4	201°07.1	228°55.0	58.4	72°28.0	17.7	223°23.5	19.5	16°31.2	42.3
5	216°09.5	243°54.5	57.4	87°31.4	17.9	238°25.4	19.3	31°33.8	42.4
6	231°12.0	258°53.9	S12°56.3	102°34.8	N23°18.0	253°27.3	S10°19.1	46°36.4	N00°42.4
7	246°14.5	273°53.4	55.2	117°38.1	18.2	268°29.2	18.9	61°38.9	42.5
8	261°16.9	288°52.8	54.1	132°41.5	18.4	283°31.1	18.6	76°41.5	42.5
9	276°19.4	303°52.3	•• 53.0	147°44.9	•• 18.5	298°33.0	•• 18.4	91°44.1	•• 42.6
10	291°21.9	318°51.7	51.9	162°48.2	18.7	313°34.9	18.2	106°46.6	42.6
11	306°24.3	333°51.2	50.8	177°51.6	18.9	328°36.8	18.0	121°49.2	42.7
12	321°26.8	348°50.6	S12°49.7	192°54.9	N23°19.0	343°38.7	S10°17.8	136°51.8	N00°42.8
13	336°29.3	3°50.1	48.6	207°58.3	19.2	358°40.6	17.6	151°54.4	42.8
14	351°31.7	18°49.5	47.5	223°01.7	19.4	13°42.5	•• 17.3	166°56.9	42.9
15	6°34.2	33°49.0	•• 46.4	238°05.0	•• 19.5	28°44.4	•• 17.1	181°59.5	•• 42.9
16	21°36.6	48°48.4	45.3	253°08.4	19.7	43°46.3	16.9	197°02.1	43.0
17	36°39.1	63°47.9	44.2	268°11.7	19.9	58°48.2	16.7	212°04.6	43.0
18	51°41.6	78°47.4	S12°43.1	283°15.1	N23°20.0	73°50.1	S10°16.5	227°07.2	N00°43.1
19	66°44.0	93°46.8	42.1	298°18.4	20.2	88°52.0	16.2	242°09.8	43.2
20	81°46.5	108°46.3	41.0	313°21.8	20.4	103°53.9	16.0	257°12.4	43.2
21	96°49.0	123°45.7	•• 39.9	328°25.1	•• 20.5	118°55.8	•• 15.8	272°14.9	•• 43.3
22	111°51.4	138°45.2	38.8	343°28.5	20.7	133°57.7	15.6	287°17.5	43.3
23	126°53.9	153°44.7	37.7	358°31.8	20.8	148°59.6	15.4	302°20.1	43.4
Mer.pass. 14:34		ν -0.6' d -1.1' m -3.92		ν 3.4' d 0.2' m -1.10		ν 1.9' d -0.2' m -2.03		ν 2.6' d 0.1' m 0.67	

Feb 09 Tue			SHA	Mer.pass
Venus			30°25.3	12:43
Mars			230°32.9	23:17
Jupiter			22°45.7	13:11
Saturn			175°18.8	03:02

Feb 10 Wed			SHA	Mer.pass
Venus			29°12.5	12:44
Mars			230°55.4	23:11
Jupiter			22°32.2	13:08
Saturn			175°21.2	02:58

Feb 11 Thu			SHA	Mer.pass
Venus			28°00.0	12:45
Mars			231°17.3	23:06
Jupiter			22°18.7	13:05
Saturn			175°23.7	02:54

Horizontal parallax			
Venus:			0.1
Mars:			0.2

DUT1 = UT1-UTC = +0.0775 sec ΔT = TT-UT1 = +66.1065 sec

2010 February 09 to Feb. 11 UT

h	Sun			Moon			
Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	176°27.5	S14°46.6	234°57.0	9.9'	S25°39.3	-1.3'	54.7'
1	191°27.5	45.8	249°25.9	9.9'	25°38.0	-1.4'	54.7'
2	206°27.4	45.0	263°54.9	10.0'	25°36.5	-1.5'	54.7'
3	221°27.4	· · 44.2	278°23.8	10.0'	25°35.0	-1.7'	54.7'
4	236°27.4	43.4	292°52.8	10.0'	25°33.3	-1.8'	54.7'
5	251°27.4	42.6	307°21.9	10.1'	25°31.5	-1.9'	54.6'
6	266°27.4	S14°41.8	321°51.0	10.1'	S25°29.6	-2.0'	54.6'
7	281°27.3	41.0	336°20.1	10.2'	25°27.6	-2.2'	54.6'
8	296°27.3	40.2	350°49.2	10.2'	25°25.4	-2.3'	54.6'
9	311°27.3	· · 39.4	5°18.4	10.2'	25°23.1	-2.4'	54.6'
10	326°27.3	38.6	19°47.7	10.3'	25°20.7	-2.5'	54.6'
11	341°27.3	37.8	34°16.9	10.3'	25°18.1	-2.7'	54.5'
12	356°27.3	S14°37.0	48°46.2	10.4'	S25°15.5	-2.8'	54.5'
13	11°27.2	36.2	63°15.6	10.4'	25°12.7	-2.9'	54.5'
14	26°27.2	35.4	77°45.0	10.4'	25°09.8	-3.0'	54.5'
15	41°27.2	· · 34.6	92°14.4	10.5'	25°06.7	-3.1'	54.5'
16	56°27.2	33.8	106°43.9	10.5'	25°03.6	-3.3'	54.5'
17	71°27.2	33.0	121°13.4	10.6'	25°00.3	-3.4'	54.4'
18	86°27.2	S14°32.2	135°43.0	10.6'	S24°56.9	-3.5'	54.4'
19	101°27.1	31.4	150°12.6	10.7'	24°53.4	-3.6'	54.4'
20	116°27.1	30.5	164°42.3	10.7'	24°49.8	-3.7'	54.4'
21	131°27.1	· · 29.7	179°12.0	10.8'	24°46.1	-3.9'	54.4'
22	146°27.1	28.9	193°41.7	10.8'	24°42.2	-4.0'	54.4'
23	161°27.1	28.1	208°11.5	10.9'	24°38.2	-4.1'	54.4'
	SD = 16.2'	$d = -0.8'$	SD = 14.9'				

Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	176°27.1	S14°27.3	222°41.4	10.9'	S24°34.1	-4.2'	54.3'
1	191°27.1	26.5	237°11.3	11.0'	24°29.9	-4.3'	54.3'
2	206°27.1	25.7	251°41.3	11.0'	24°25.6	-4.4'	54.3'
3	221°27.0	· · 24.9	266°11.3	11.1'	24°21.2	-4.5'	54.3'
4	236°27.0	24.1	280°41.3	11.1'	24°16.7	-4.7'	54.3'
5	251°27.0	23.3	295°11.4	11.2'	24°12.0	-4.8'	54.3'
6	266°27.0	S14°22.4	309°41.6	11.2'	S24°07.2	-4.9'	54.3'
7	281°27.0	21.6	324°11.8	11.3'	24°02.3	-5.0'	54.3'
8	296°27.0	20.8	338°42.0	11.3'	23°57.4	-5.1'	54.2'
9	311°27.0	· · 20.0	353°12.3	11.4'	23°52.3	-5.2'	54.2'
10	326°27.0	19.2	7°42.7	11.4'	23°47.0	-5.3'	54.2'
11	341°27.0	18.4	22°13.1	11.5'	23°41.7	-5.4'	54.2'
12	356°27.0	S14°17.6	36°43.6	11.5'	S23°36.3	-5.5'	54.2'
13	11°26.9	16.7	51°14.1	11.6'	23°30.8	-5.6'	54.2'
14	26°26.9	15.9	65°44.7	11.6'	23°25.1	-5.7'	54.2'
15	41°26.9	· · 15.1	80°15.3	11.7'	23°19.4	-5.9'	54.2'
16	56°26.9	14.3	94°46.0	11.7'	23°13.5	-6.0'	54.2'
17	71°26.9	13.5	109°16.8	11.8'	23°07.6	-6.1'	54.1'
18	86°26.9	S14°12.7	123°47.6	11.9'	S23°01.5	-6.2'	54.1'
19	101°26.9	11.9	138°18.5	11.9'	22°55.4	-6.3'	54.1'
20	116°26.9	11.0	152°49.4	12.0'	22°49.1	-6.4'	54.1'
21	131°26.9	· · 10.2	167°20.3	12.0'	22°42.7	-6.5'	54.1'
22	146°26.9	09.4	181°51.4	12.1'	22°36.3	-6.6'	54.1'
23	161°26.9	08.6	196°22.5	12.1'	22°29.7	-6.7'	54.1'
	SD = 16.2'	$d = -0.8'$	SD = 14.8'				

Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	176°26.9	S14°07.8	210°53.6	12.2'	S22°23.0	-6.8'	54.1'
1	191°26.9	06.9	225°24.8	12.3'	22°16.3	-6.9'	54.1'
2	206°26.9	06.1	239°56.1	12.3'	22°09.4	-7.0'	54.1'
3	221°26.9	· · 05.3	254°27.4	12.4'	22°02.4	-7.1'	54.1'
4	236°26.9	04.5	268°58.8	12.4'	21°55.4	-7.2'	54.0'
5	251°26.9	03.7	283°30.2	12.5'	21°48.2	-7.2'	54.0'
6	266°26.9	S14°02.8	298°01.7	12.6'	S21°41.0	-7.3'	54.0'
7	281°26.9	02.0	312°33.3	12.6'	21°33.7	-7.4'	54.0'
8	296°26.9	01.2	327°04.9	12.7'	21°26.2	-7.5'	54.0'
9	311°26.9	14°00.4	341°36.5	12.7'	21°18.7	-7.6'	54.0'
10	326°26.8	13°59.5	356°08.3	12.8'	21°11.1	-7.7'	54.0'
11	341°26.8	58.7	10°40.0	12.8'	21°03.4	-7.8'	54.0'
12	356°26.8	S13°57.9	25°11.9	12.9'	S20°55.6	-7.9'	54.0'
13	11°26.8	57.1	39°43.8	13.0'	20°47.7	-8.0'	54.0'
14	26°26.8	56.2	54°15.7	13.0'	20°39.7	-8.1'	54.0'
15	41°26.8	· · 55.4	68°47.8	13.1'	20°31.7	-8.2'	54.0'
16	56°26.8	54.6	83°19.8	13.1'	20°23.5	-8.2'	54.0'
17	71°26.8	53.8	97°52.0	13.2'	20°15.3	-8.3'	54.0'
18	86°26.9	S13°52.9	112°24.2	13.2'	S20°06.9	-8.4'	54.0'
19	101°26.9	52.1	126°56.4	13.3'	19°58.5	-8.5'	53.9'
20	116°26.9	51.3	141°28.7	13.4'	19°50.1	-8.6'	53.9'
21	131°26.9	· · 50.5	156°01.1	13.4'	19°41.5	-8.7'	53.9'
22	146°26.9	49.6	170°33.5	13.5'	19°32.8	-8.7'	53.9'
23	161°26.9	48.8	185°06.0	13.5'	19°24.1	-8.8'	53.9'
	SD = 16.2'	$d = -0.8'$	SD = 14.7'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	06:36	07:58	09:24	15:06	16:32	17:55
N 70°	06:32	07:45	08:59	15:31	16:44	17:58
68°	06:28	07:35	08:40	15:50	16:55	18:02
66°	06:25	07:26	08:24	16:05	17:03	18:05
64°	06:22	07:19	08:12	16:18	17:11	18:07
62°	06:20	07:13	08:01	16:28	17:17	18:10
60°	06:18	07:07	07:52	16:38	17:23	18:12
N 58°	06:15	07:02	07:44	16:45	17:28	18:14
56°	06:13	06:57	07:37	16:52	17:32	18:16
54°	06:11	06:53	07:30	16:59	17:36	18:18
52°	06:09	06:49	07:25	17:04	17:40	18:20
50°	06:08	06:46	07:20	17:09	17:43	18:22
45°	06:03	06:38	07:09	17:20	17:51	18:26
N 40°	05:59	06:31	06:59	17:30	17:58	18:30
35°	05:55	06:25	06:51	17:38	18:04	18:34
30°	05:51	06:20	06:44	17:45	18:09	18:38
20°	05:43	06:09	06:32	17:57	18:19	18:45
N 10°	05:34	05:59	06:21	18:07	18:29	18:54
0°	05:25	05:49	06:11	18:18	18:39	19:04
S 10°	05:13	05:39	06:00	18:28	18:50	19:15
20°	04:59	05:26	05:49	18:39	19:02	19:29
30°	04:41	05:11	05:36	18:52	19:17	19:47
35°	04:29	05:02	05:29	18:59	19:26	19:58
40°	04:15	04:51	05:20	19:08	19:37	20:12
45°	03:58	04:38	05:10	19:18	19:50	20:29
S 50°	03:35	04:21	04:57	19:30	20:06	20:52
52°	03:24	04:13	04:52	19:36	20:14	21:03
54°	03:10	04:04	04:45	19:42	20:23	21:16
56°	02:55	03:54	04:38	19:49	20:32	21:31
58°	02:35	03:43	04:30	19:57	20:44	21:50
S 60°	02:10	03:29	04:21	20:06	20:57	22:14

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	■	■	09:16	■	■	11:28
66°	■	09:11	08:21	■	09:53	12:22
64°	07:49	07:50	07:48	09:30	11:13	12:55
62°	06:51	07:13	07:23	10:28	11:50	13:19
60°	06:17	06:47	07:04	11:02	12:16	13:38
N 58°	05:53	06:26	06:47	11:26	12:36	13:53
56°	05:33	06:09	06:34	11:46	12:53	14:07
54°	05:16	05:54	06:22	12:02	13:07	14:18
52°	05:02	05:41	06:11	12:16	13:20	14:28
50°	04:50	05:30	06:02	12:28	13:31	14:37
45°	04:24	05:07	05:42	12:54	13:53	14:56
N 40°	04:04	04:48	05:26	13:14	14:12	15:12
35°	03:47	04:32	05:12	13:31	14:27	15:24
30°	03:32	04:19	05:00	13:45	14:40	15:36
20°	03:07	03:55	04:39	14:10	15:03	15:55
N 10°	02:46	03:35	04:21	14:31	15:22	16:12
0°	02:26	03:16	04:05	14:50	15:40	16:27
S 10°	02:06	02:57	03:48	15:10	15:58	16:43
20°	01:44	02:37	03:30	15:31	16:17	16:59
30°	01:19	02:13	03:09	15:55	16:39	17:18
35°	01:04	02:00	02:57	16:09	16:52	17:29
40°	00:47	01:44	02:43	16:26	17:07	17:41
45°	00:27	01:24	02:26	16:45	17:24	17:56
S 50°	00:01	01:00	02:05	17:10	17:46	18:14
52°	· · ·	00:49	01:55	17:21	17:56	18:22
54°	· · ·	00:36	01:44	17:35	18:07	18:31
56°	· · ·	00:20	01:31	17:50	18:21	18:42
58°	· · ·	00:02	01:16	18:09	18:36	18:54
S 60°	23:40	· · ·	00:59	18:31	18:54	19:07

Day	Sun		Mer. Pass	Moon		Age 25-27 22-8%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
09	14:10	14:11	12:14	08:38	21:03	
10	14:12	14:12	12:14	09:28	21:52	
11	14:12	14:13	12:14	10:16	22:39	

February 12, 13, 14 UT (Fri., Sat., Sun.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	141°56.4	168°44.1	S12°36.6	13°35.2	N23°21.0	164°01.5	S10°15.1	317°22.7	N00°43.4	Alpheratz	357°46.4	29°08.9											
1	156°58.8	183°43.6	35.5	28°38.5	21.2	179°03.4	14.9	332°25.2	43.5	Ankaa	353°18.3	-42°15.2											
2	172°01.3	198°43.0	34.4	43°41.9	21.3	194°05.3	14.7	347°27.8	43.6	Schedar	349°43.9	56°35.8											
3	187°03.8	213°42.5	•• 33.3	58°45.2	•• 21.5	209°07.2	•• 14.5	2°30.4	•• 43.6	Diphda	348°58.5	-17°55.9											
4	202°06.2	228°42.0	32.2	73°48.5	21.6	224°09.1	14.3	17°33.0	43.7	Achernar	335°28.7	-57°11.3											
5	217°08.7	243°41.4	31.0	88°51.9	21.8	239°11.0	14.0	32°35.5	43.7	Hamal	328°03.7	23°30.7											
6	232°11.1	258°40.9	S12°29.9	103°55.2	N23°22.0	254°12.9	S10°13.8	47°38.1	N00°43.8	Polaris	319°03.7	89°18.9											
7	247°13.6	273°40.3	28.8	118°58.6	22.1	269°14.8	13.6	62°40.7	43.9	Acamar	315°20.1	-40°16.0											
8	262°16.1	288°39.8	27.7	134°01.9	22.3	284°16.7	13.4	77°43.3	43.9	Menkar	314°17.6	4°07.8											
9	277°18.5	303°39.3	•• 26.6	149°05.3	•• 22.4	299°18.6	•• 13.2	92°45.8	•• 44.0	Mirfak	308°43.9	49°54.1											
10	292°21.0	318°38.7	25.5	164°08.6	22.6	314°20.5	12.9	107°48.4	44.0	Aldebaran	290°52.1	16°31.8											
11	307°23.5	333°38.2	24.4	179°11.9	22.7	329°22.4	12.7	122°51.0	44.1	Rigel	281°14.2	-8°11.5											
12	322°25.9	348°37.7	S12°23.3	194°15.3	N23°22.9	344°24.3	S10°12.5	137°53.6	N00°44.1	Capella	280°37.8	46°00.6											
13	337°28.4	3°37.1	22.2	209°18.6	23.1	359°26.2	12.3	152°56.1	44.2	Bellatrix	278°34.4	6°21.5											
14	352°30.9	18°36.6	21.1	224°21.9	23.2	14°28.1	12.1	167°58.7	44.3	Elnath	278°15.5	28°37.0											
15	7°33.3	33°36.1	•• 20.0	239°25.3	•• 23.4	29°30.0	•• 11.8	183°01.3	•• 44.3	Alnilam	275°48.6	-1°11.8											
16	22°35.8	48°35.5	18.9	254°28.6	23.5	44°31.9	11.6	198°03.9	44.4	Betelgeuse	271°03.7	7°24.5											
17	37°38.2	63°35.0	17.8	269°31.9	23.7	59°33.8	11.4	213°06.4	44.4	Canopus	263°56.9	-52°42.3											
18	52°40.7	78°34.5	S12°16.7	284°35.3	N23°23.8	74°35.7	S10°11.2	228°09.0	N00°44.5	Sirius	258°35.6	-16°44.0											
19	67°43.2	93°33.9	15.5	299°38.6	24.0	89°37.6	11.0	243°11.6	44.6	Adhara	255°14.1	-28°59.4											
20	82°45.6	108°33.4	14.4	314°41.9	24.1	104°39.5	10.8	258°14.2	44.6	Procyon	245°02.0	5°11.8											
21	97°48.1	123°32.9	•• 13.3	329°45.2	•• 24.3	119°41.4	•• 10.5	273°16.7	•• 44.7	Pollux	243°30.3	28°00.0											
22	112°50.6	138°32.3	12.2	344°48.6	24.4	134°43.3	10.3	288°19.3	44.7	Avior	234°18.5	-59°32.7											
23	127°53.0	153°31.8	11.1	359°51.9	24.6	149°45.2	10.1	303°21.9	44.8	Suhail	222°53.9	-43°28.6											
Mer.pass. 14:30		ν -0.5' d -1.1' m -3.92		ν 3.3' d 0.2' m -1.07		ν 1.9' d -0.2' m -2.03		ν 2.6' d 0.1' m 0.66		Miaplacidus			221°39.5	-69°45.6									
										Alphard			217°58.1	-8°42.3									
										Regulus			207°45.7	11°54.8									
										Dubhe			193°53.8	61°41.5									
										Denebola			182°35.8	14°30.7									
										Gienah			175°54.6	-17°36.0									
										Acrux			173°11.8	-63°09.3									
										Gacrux			172°03.4	-57°10.2									
										Alioth			166°22.3	55°53.9									
										Spica			158°33.7	-11°13.0									
										Alkaid			153°00.5	49°15.4									
										Hadar			148°51.3	-60°25.2									
										Menkent			148°10.4	-36°25.2									
										Arcturus			145°57.8	19°07.5									
										Rigel Kent.			139°55.0	-60°52.4									
										Kochab			137°19.4	74°06.4									
										Zuben'ubi			137°08.1	-16°05.1									
										Alphecca			126°13.0	26°40.5									
										Antares			112°29.3	-26°27.3									
										Atria			107°33.6	-69°02.5									
										Sabik			102°15.5	-15°44.3									
										Shaula			96°25.4	-37°06.6									
										Rasalhague			96°08.9	12°33.0									
										Eltanin			90°47.6	51°29.0									
										Kaus Aust.			83°47.3	-34°22.7									
										Vega			80°40.9	38°47.4									
										Nunki			76°01.6	-26°17.0									
										Altair			62°10.9	8°53.6									
										Peacock			53°23.5	-56°42.1									
										Deneb			49°33.7	45°18.9									
										Enif			33°49.8	9°55.3									
										Al Na'ir			27°47.1	-46°54.7									
										Fomalhaut			15°26.9	-29°34.2									
										Scheat			13°56.1	28°08.3									
										Markab			13°41.1	15°15.6									

Feb 12 Fri	SHA	Mer.pass
Venus	26°47.8	12:46
Mars	231°38.8	23:01
Jupiter	22°05.2	13:02
Saturn	175°26.3	02:50

Feb 13 Sat	SHA	Mer.pass
Venus	25°35.8	12:46
Mars	231°59.7	22:55
Jupiter	21°51.6	12:59
Saturn	175°29.0	02:46

Feb 14 Sun	SHA	Mer.pass
Venus	24°24.1	12:47
Mars	232°20.0	22:50
Jupiter	21°38.0	12:56
Saturn	175°31.7	02:42

Horizontal parallax		
Venus:	0.1	
Mars:	0.2	

h	Sun			Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	176°26.9	S13°48.0	199°38.5	13.6'	S19°15.3	-8.9'	53.9'
1	191°26.9	47.1	214°11.1	13.6'	19°06.4	-9.0'	53.9'
2	206°26.9	46.3	228°43.7	13.7'	18°57.4	-9.1'	53.9'
3	221°26.9	· · 45.5	243°16.4	13.8'	18°48.3	-9.1'	53.9'
4	236°26.9	44.7	257°49.2	13.8'	18°39.2	-9.2'	53.9'
5	251°26.9	43.8	272°22.0	13.9'	18°30.0	-9.3'	53.9'
6	266°26.9	S13°43.0	286°54.9	13.9'	S18°20.7	-9.4'	53.9'
7	281°26.9	42.2	301°27.8	14.0'	18°11.3	-9.4'	53.9'
8	296°26.9	41.3	316°00.8	14.0'	18°01.9	-9.5'	53.9'
9	311°26.9	· · 40.5	330°33.8	14.1'	17°52.4	-9.6'	53.9'
10	326°26.9	39.7	345°06.9	14.1'	17°42.8	-9.7'	53.9'
11	341°26.9	38.8	359°40.0	14.2'	17°33.2	-9.7'	53.9'
12	356°26.9	S13°38.0	14°13.2	14.2'	S17°23.4	-9.8'	53.9'
13	11°26.9	37.2	28°46.5	14.3'	17°13.6	-9.9'	53.9'
14	26°26.9	36.3	43°19.8	14.3'	17°03.8	-9.9'	53.9'
15	41°27.0	· · 35.5	57°53.1	14.4'	16°53.8	-10.0'	53.9'
16	56°27.0	34.7	72°26.5	14.5'	16°43.8	-10.1'	53.9'
17	71°27.0	33.8	87°00.0	14.5'	16°33.8	-10.1'	53.9'
18	86°27.0	S13°33.0	101°33.5	14.6'	S16°23.6	-10.2'	53.9'
19	101°27.0	32.1	116°07.0	14.6'	16°13.4	-10.3'	53.9'
20	116°27.0	31.3	130°40.6	14.7'	16°03.2	-10.3'	53.9'
21	131°27.0	· · 30.5	145°14.3	14.7'	15°52.8	-10.4'	53.9'
22	146°27.0	29.6	159°48.0	14.7'	15°42.4	-10.5'	53.9'
23	161°27.0	28.8	174°21.7	14.8'	15°32.0	-10.5'	53.9'
SD = 16.2' $d = -0.8'$ SD = 14.7'							

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	176°27.0	S13°28.0	188°55.5	14.8'	S15°21.5	-10.6'	53.9'
1	191°27.1	27.1	203°29.4	14.9'	15°10.9	-10.6'	53.9'
2	206°27.1	26.3	218°03.2	14.9'	15°00.3	-10.7'	53.9'
3	221°27.1	· · 25.4	232°37.2	15.0'	14°49.6	-10.8'	53.9'
4	236°27.1	24.6	247°11.2	15.0'	14°38.8	-10.8'	53.9'
5	251°27.1	23.8	261°45.2	15.1'	14°28.0	-10.9'	53.9'
6	266°27.1	S13°22.9	276°19.3	15.1'	S14°17.1	-10.9'	53.9'
7	281°27.1	22.1	290°53.4	15.2'	14°06.2	-11.0'	53.9'
8	296°27.1	21.2	305°27.6	15.2'	13°55.2	-11.0'	53.9'
9	311°27.2	· · 20.4	320°01.8	15.2'	13°44.2	-11.1'	53.9'
10	326°27.2	19.6	334°36.0	15.3'	13°33.1	-11.1'	53.9'
11	341°27.2	18.7	349°10.3	15.3'	13°21.9	-11.2'	53.9'
12	356°27.2	S13°17.9	3°44.6	15.4'	S13°10.7	-11.2'	53.9'
13	11°27.2	17.0	18°19.0	15.4'	12°59.5	-11.3'	53.9'
14	26°27.2	16.2	32°53.4	15.5'	12°48.2	-11.4'	53.9'
15	41°27.2	· · 15.3	47°27.9	15.5'	12°36.8	-11.4'	53.9'
16	56°27.3	14.5	62°02.3	15.5'	12°25.4	-11.4'	53.9'
17	71°27.3	13.7	76°36.9	15.6'	12°14.0	-11.5'	53.9'
18	86°27.3	S13°12.8	91°11.4	15.6'	S12°02.5	-11.5'	53.9'
19	101°27.3	12.0	105°46.0	15.6'	11°50.9	-11.6'	53.9'
20	116°27.3	11.1	120°20.7	15.7'	11°39.3	-11.6'	53.9'
21	131°27.3	· · 10.3	134°55.4	15.7'	11°27.7	-11.7'	53.9'
22	146°27.4	09.4	149°30.1	15.7'	11°16.0	-11.7'	53.9'
23	161°27.4	08.6	164°04.8	15.8'	11°04.3	-11.8'	53.9'
SD = 16.2' $d = -0.8'$ SD = 14.7'							

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	176°27.4	S13°07.7	178°39.6	15.8'	S10°52.5	-11.8'	53.9'
1	191°27.4	06.9	193°14.4	15.8'	10°40.7	-11.9'	53.9'
2	206°27.4	06.0	207°49.2	15.9'	10°28.9	-11.9'	53.9'
3	221°27.5	· · 05.2	222°24.1	15.9'	10°17.0	-11.9'	53.9'
4	236°27.5	04.3	236°59.0	15.9'	10°05.0	-12.0'	53.9'
5	251°27.5	03.5	251°34.0	16.0'	09°53.1	-12.0'	53.9'
6	266°27.5	S13°02.6	266°08.9	16.0'	S09°41.1	-12.0'	53.9'
7	281°27.5	01.8	280°43.9	16.0'	09°29.0	-12.1'	53.9'
8	296°27.6	00.9	295°18.9	16.0'	09°16.9	-12.1'	53.9'
9	311°27.6	13°00.1	309°54.0	16.1'	09°04.8	-12.2'	53.9'
10	326°27.6	12°59.2	324°29.1	16.1'	08°52.7	-12.2'	53.9'
11	341°27.6	58.4	339°04.2	16.1'	08°40.5	-12.2'	53.9'
12	356°27.7	S12°57.5	353°39.3	16.1'	S08°28.2	-12.3'	54.0'
13	11°27.7	56.7	8°14.4	16.2'	08°16.0	-12.3'	54.0'
14	26°27.7	55.8	22°49.6	16.2'	08°03.7	-12.3'	54.0'
15	41°27.7	· · 55.0	37°24.8	16.2'	07°51.4	-12.4'	54.0'
16	56°27.7	54.1	52°00.0	16.2'	07°39.0	-12.4'	54.0'
17	71°27.8	53.3	66°35.3	16.3'	07°26.6	-12.4'	54.0'
18	86°27.8	S12°52.4	81°10.5	16.3'	S07°14.2	-12.4'	54.0'
19	101°27.8	51.6	95°45.8	16.3'	07°01.7	-12.5'	54.0'
20	116°27.8	50.7	110°21.1	16.3'	06°49.3	-12.5'	54.0'
21	131°27.9	· · 49.8	124°56.4	16.3'	06°36.8	-12.5'	54.0'
22	146°27.9	49.0	139°31.7	16.3'	06°24.2	-12.6'	54.0'
23	161°27.9	48.1	154°07.1	16.4'	06°11.7	-12.6'	54.0'
SD = 16.2' $d = -0.8'$ SD = 14.7'							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	06:23	07:44	09:06	15:24	16:46	18:07
N 70°	06:21	07:33	08:44	15:46	16:57	18:09
68°	06:18	07:24	08:27	16:03	17:06	18:12
66°	06:16	07:17	08:13	16:16	17:13	18:14
64°	06:14	07:10	08:02	16:28	17:19	18:16
62°	06:12	07:05	07:52	16:37	17:25	18:17
60°	06:11	07:00	07:44	16:45	17:30	18:19
N 58°	06:09	06:55	07:37	16:53	17:34	18:20
56°	06:07	06:51	07:30	16:59	17:38	18:22
54°	06:06	06:47	07:24	17:05	17:42	18:23
52°	06:04	06:44	07:19	17:10	17:45	18:25
50°	06:03	06:41	07:14	17:15	17:48	18:26
45°	05:59	06:34	07:04	17:25	17:55	18:30
N 40°	05:56	06:28	06:56	17:33	18:01	18:33
35°	05:53	06:22	06:48	17:41	18:07	18:36
30°	05:49	06:17	06:42	17:47	18:12	18:40
20°	05:42	06:08	06:30	17:58	18:21	18:47
N 10°	05:34	05:59	06:20	18:08	18:30	18:55
0°	05:25	05:50	06:11	18:18	18:39	19:03
S 10°	05:14	05:39	06:01	18:27	18:49	19:14
20°	05:01	05:28	05:51	18:37	19:00	19:27
30°	04:44	05:14	05:39	18:49	19:14	19:44
35°	04:33	05:05	05:32	18:56	19:23	19:55
40°	04:20	04:55	05:24	19:04	19:33	20:08
45°	04:03	04:42	05:14	19:13	19:45	20:24
S 50°	03:42	04:27	05:03	19:25	20:00	20:45
52°	03:31	04:20	04:57	19:30	20:07	20:55
54°	03:19	04:11	04:51	19:36	20:15	21:07
56°	03:05	04:02	04:45	19:42	20:25	21:21
58°	02:47	03:51	04:38	19:49	20:35	21:38
S 60°	02:25	03:39	04:29	19:57	20:47	21:59

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	10:20	08:57	08:20	12:00	14:54	16:57
N 70°	09:09	08:31	08:07	13:10	15:18	17:09
68°	08:32	08:11	07:55	13:46	15:37	17:18
66°	08:05	07:55	07:46	14:12	15:52	17:26
64°	07:45	07:41	07:38	14:32	16:04	17:33
62°	07:28	07:30	07:31	14:47	16:14	17:38
60°	07:14	07:20	07:25	15:01	16:23	17:43
N 58°	07:02	07:12	07:20	15:12	16:30	17:47
56°	06:51	07:05	07:15	15:22	16:37	17:51
54°	06:42	06:58	07:11	15:30	16:43	17:54
52°	06:34	06:52	07:07	15:38	16:48	17:57
50°	06:26	06:47	07:04	15:45	16:53	18:00
45°	06:11	06:35	06:56	16:00	17:03	18:06
N 40°	05:57	06:25	06:50	16:12	17:12	18:11
35°	05:46	06:17	06:44	16:22	17:19	18:15
30°	05:36	06:09	06:39	16:31	17:25	18:19
20°	05:19	05:56	06:31	16:46	17:36	18:25
N 10°	05:05	05:45	06:23	17:00	17:46	18:31
0°	04:51	05:34	06:16	17:12	17:55	18:36
S 10°	04:37	05:24	06:09	17:24	18:04	18:41
20°	04:22	05:12	06:02	17:37	18:13	18:47
30°	04:04	04:59	05:53	17:52	18:24	18:53
35°	03:54	04:51	05:48	18:01	18:30	18:56
40°	03:43	04:43	05:42	18:11	18:37	19:00
45°	03:29	04:33	05:36	18:22	18:45	19:05
S 50°	03:12	04:20	05:28	18:36	18:54	19:10
52°	03:04	04:14	05:24	18:42	18:59	19:13
54°	02:56	04:08	05:20	18:49	19:03	19:16
56°	02:46	04:01	05:15	18:57	19:09	19:19
58°	02:34	03:53	05:10	19:06	19:15	19:22
S 60°	02:21	03:43	05:04	19:16	19:21	19:26

Day	Sun		Mer. Pass	Moon		Age 28-0 4-0%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
12	14:13	14:12	12:14	11:01	23:23	
13	14:12	14:11	12:14	11:45	—:—	
14	14:10	14:09	12:14	12:26	00:06	

February 15, 16, 17 UT (Mon., Tue., Wed.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	144°53.8	168°06.4	S11°15.9	17°33.5	N23°31.4	166°18.2	S09°59.3	320°28.3	N00°47.7	Alpheratz	357°46.4	29°08.9
1	159°56.2	183°05.9	14.7	32°36.8	31.5	181°20.1	59.1	335°30.9	47.8	Ankaa	353°18.3	-42°15.2
2	174°58.7	198°05.4	13.6	47°40.0	31.7	196°22.0	58.8	350°33.5	47.9	Schedar	349°43.9	56°35.8
3	190°01.2	213°04.9	•• 12.4	62°43.3	•• 31.8	211°23.9	•• 58.6	5°36.1	•• 47.9	Diphda	348°58.5	-17°55.9
4	205°03.6	228°04.4	11.3	77°46.6	31.9	226°25.8	58.4	20°38.7	48.0	Achernar	335°28.7	-57°11.3
5	220°06.1	243°03.9	10.2	92°49.8	32.0	241°27.7	58.2	35°41.2	48.0	Hamal	328°03.7	23°30.7
6	235°08.6	258°03.4	S11°09.0	107°53.1	N23°32.2	256°29.6	S09°58.0	50°43.8	N00°48.1	Polaris	319°05.1	89°18.9
7	250°11.0	273°02.9	07.9	122°56.4	32.3	271°31.5	57.7	65°46.4	48.2	Acamar	315°20.2	-40°16.0
8	265°13.5	288°02.4	06.7	137°59.6	32.4	286°33.4	57.5	80°49.0	48.2	Menkar	314°17.6	4°07.8
9	280°16.0	303°01.9	•• 05.6	153°02.9	•• 32.5	301°35.3	•• 57.3	95°51.6	•• 48.3	Mirfak	308°43.9	49°54.1
10	295°18.4	318°01.4	04.4	168°06.1	32.7	316°37.2	57.1	110°54.2	48.4	Aldebaran	290°52.1	16°31.8
11	310°20.9	333°00.9	03.3	183°09.4	32.8	331°39.1	56.9	125°56.7	48.4	Rigel	281°14.2	-8°11.5
12	325°23.3	348°00.4	S11°02.2	198°12.7	N23°32.9	346°41.0	S09°56.6	140°59.3	N00°48.5	Capella	280°37.8	46°00.6
13	340°25.8	2°59.9	11°01.0	213°15.9	33.0	1°42.9	56.4	156°01.9	48.5	Bellatrix	278°34.5	6°21.5
14	355°28.3	17°59.4	10°59.9	228°19.2	33.1	16°44.8	56.2	171°04.5	48.6	Elnath	278°15.5	28°37.0
15	10°30.7	32°58.9	•• 58.7	243°22.4	•• 33.3	31°46.7	•• 56.0	186°07.1	•• 48.7	Alnilam	275°48.7	-1°11.8
16	25°33.2	47°58.4	57.6	258°25.7	33.4	46°48.6	55.8	201°09.7	48.7	Betelgeuse	271°03.7	7°24.5
17	40°35.7	62°57.9	56.4	273°28.9	33.5	61°50.5	55.5	216°12.3	48.8	Canopus	263°56.9	-52°42.3
18	55°38.1	77°57.4	S10°55.3	288°32.2	N23°33.6	76°52.4	S09°55.3	231°14.8	N00°48.9	Sirius	258°35.6	-16°44.0
19	70°40.6	92°56.9	54.1	303°35.4	33.8	91°54.3	55.1	246°17.4	48.9	Adhara	255°14.2	-28°59.4
20	85°43.1	107°56.4	53.0	318°38.7	33.9	106°56.2	54.9	261°20.0	49.0	Procyon	245°02.0	5°11.8
21	100°45.5	122°55.9	•• 51.8	333°41.9	•• 34.0	121°58.1	•• 54.6	276°22.6	•• 49.0	Pollux	243°30.3	28°00.0
22	115°48.0	137°55.4	50.7	348°45.2	34.1	137°00.0	54.4	291°25.2	49.1	Avior	234°18.5	-59°32.7
23	130°50.4	152°54.9	49.5	3°48.4	34.2	152°01.9	54.2	306°27.8	49.2	Suhail	222°53.9	-43°28.6
Mer.pass. 14:18		ν -0.5' d -1.1' m -3.92		ν 3.3' d 0.1' m -0.99		ν 1.9' d -0.2' m -2.03		ν 2.6' d 0.1' m 0.65				

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	145°52.9	167°54.4	S10°48.4	18°51.7	N23°34.3	167°03.8	S09°54.0	321°30.4	N00°49.2	Alphard	217°58.1	-8°42.3
1	160°55.4	182°53.9	47.2	33°54.9	34.5	182°05.7	53.8	336°32.9	49.3	Regulus	207°45.7	11°54.8
2	175°57.8	197°53.4	46.1	48°58.1	34.6	197°07.6	53.5	351°35.5	49.4	Dubhe	193°53.8	61°41.5
3	191°00.3	212°52.9	•• 44.9	64°01.4	•• 34.7	212°09.5	•• 53.3	6°38.1	•• 49.4	Denebola	182°35.8	14°30.7
4	206°02.8	227°52.4	43.7	79°04.6	34.8	227°11.4	53.1	21°40.7	49.5	Gienah	175°54.6	-17°36.1
5	221°05.2	242°51.9	42.6	94°07.9	34.9	242°13.3	52.9	36°43.3	49.5	Acrux	173°11.8	-63°09.3
6	236°07.7	257°51.4	S10°41.4	109°11.1	N23°35.0	257°15.2	S09°52.7	51°45.9	N00°49.6	Gacrux	172°03.4	-57°10.2
7	251°10.2	272°50.9	40.3	124°14.3	35.1	272°17.1	52.4	66°48.5	49.7	Alioth	166°22.2	55°53.9
8	266°12.6	287°50.4	39.1	139°17.6	35.3	287°19.0	52.2	81°51.1	49.7	Spica	158°33.6	-11°13.0
9	281°15.1	302°49.9	•• 38.0	154°20.8	•• 35.4	302°20.9	•• 52.0	96°53.6	•• 49.8	Alkaid	153°00.4	49°15.4
10	296°17.6	317°49.4	36.8	169°24.0	35.5	317°22.8	51.8	111°56.2	49.9	Hadar	148°51.3	-60°25.2
11	311°20.0	332°48.9	35.7	184°27.3	35.6	332°24.7	51.5	126°58.8	49.9	Menkent	148°10.3	-36°25.2
12	326°22.5	347°48.4	S10°34.5	199°30.5	N23°35.7	347°26.6	S09°51.3	142°01.4	N00°50.0	Arcturus	145°57.8	19°07.5
13	341°24.9	2°47.9	33.3	214°33.7	35.8	2°28.5	51.1	157°04.0	50.1	Rigel Kent.	139°55.0	-60°52.5
14	356°27.4	17°47.5	32.2	229°36.9	35.9	17°30.4	50.9	172°06.6	50.1	Kochab	137°19.3	74°06.4
15	11°29.9	32°47.0	•• 31.0	244°40.2	•• 36.0	32°32.3	•• 50.7	187°09.2	•• 50.2	Zuben'ubi	137°08.0	-16°05.1
16	26°32.3	47°46.5	29.9	259°43.4	36.2	47°34.2	50.4	202°11.8	50.2	Alphecca	126°13.0	26°40.5
17	41°34.8	62°46.0	28.7	274°46.6	36.3	62°36.1	50.2	217°14.4	50.3	Antares	112°29.3	-26°27.3
18	56°37.3	77°45.5	S10°27.5	289°49.8	N23°36.4	77°38.0	S09°50.0	232°16.9	N00°50.4	Atria	107°33.6	-69°02.5
19	71°39.7	92°45.0	26.4	304°53.1	36.5	92°39.9	49.8	247°19.5	50.4	Sabik	102°15.4	-15°44.3
20	86°42.2	107°44.5	25.2	319°56.3	36.6	107°41.8	49.5	262°22.1	50.5	Shaula	96°25.4	-37°06.6
21	101°44.7	122°44.0	•• 24.1	334°59.5	•• 36.7	122°43.7	•• 49.3	277°24.7	•• 50.6	Rasalhague	96°08.8	12°33.0
22	116°47.1	137°43.5	22.9	350°02.7	36.8	137°45.6	49.1	292°27.3	50.6	Eltanin	90°47.6	51°28.9
23	131°49.6	152°43.0	21.7	5°05.9	36.9	152°47.5	48.9	307°29.9	50.7	Kaus Aust.	83°47.3	-34°22.7
Mer.pass. 14:14		ν -0.5' d -1.2' m -3.92		ν 3.2' d 0.1' m -0.97		ν 1.9' d -0.2' m -2.03		ν 2.6' d 0.1' m 0.65				

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	146°52.0	167°42.6	S10°20.6	20°09.1	N23°37.0	167°49.3	S09°48.7	322°32.5	N00°50.8	Nunki	76°01.6	-26°17.0
1	161°54.5	182°42.1	19.4	35°12.4	37.1	182°51.2	48.4	337°35.1	50.8	Altair	62°10.9	8°53.6
2	176°57.0	197°41.6	18.2	50°15.6	37.2	197°53.1	48.2	352°37.7	50.9	Peacock	53°23.5	-56°42.1
3	191°59.4	212°41.1	•• 17.1	65°18.8	•• 37.3	212°55.0	•• 48.0	7°40.2	•• 50.9	Deneb	49°33.6	45°18.9
4	207°01.9	227°40.6	15.9	80°22.0	37.4	227°56.9	47.8	22°42.8	51.0	Enif	33°49.8	9°55.3
5	222°04.4	242°40.1	14.7	95°25.2	37.5	242°58.8	47.5	37°45.4	51.1	Al Na'ir	27°47.1	-46°54.7
6	237°06.8	257°39.6	S10°13.6	110°28.4	N23°37.6	258°00.7	S09°47.3	52°48.0	N00°51.1	Fomalhaut	15°26.9	-29°34.2
7	252°09.3	272°39.2	12.4	125°31.6	37.7	273°02.6	47.1	67°50.6	51.2	Scheat	13°56.1	28°08.3
8	267°11.8	287°38.7	11.2	140°34.8	37.9	288°04.5	46.9	82°53.2	51.3	Markab	13°41.1	25°15.6
9	282°14.2	302°38.2	•• 10.1	155°38.0	•• 38.0	303°06.4	•• 46.7	97°55.8	•• 51.3			
10	297°16.7	317°37.7	08.9	170°41.2	38.1	318°08.3	46.4	112°58.4	51.4			
11	312°19.2	332°37.2	07.7	185°44.4	38.2	333°10.2	46.2	128°01.0	51.5			
12	327°21.6	347°36.7	S10°06.6	200°47.6	N23°38.3	348°12.1	S09°46.0	143°03.6	N00°51.5			
13	342°24.1	2°36.3	05.4	215°50.8	38.4	3°14.0	45.8	158°06.2	51.6			
14	357°26.5	17°35.8	04.2	230°54.0	38.5	18°15.9	45.5	173°08.7	51.6			
15	12°29.0	32°35.3	•• 03.1	245°57.2	•• 38.6	33°17.8	•• 45.3	188°11.3	•• 51.7			
16	27°31.5	47°34.8	01.9	261°00.4	38.7	48°19.7	45.1	203°13.9	51.8			
17	42°33.9	62°34.3	10°00.7	276°03.6	38.8	63°21.6	44.9	218°16.5	51.8			
18	57°36.4	77°33.9	S09°59.5	291°06.8	N23°38.9	78°23.5	S09°44.7	233°19.1	N00°51.9			
19	72°38.9	92°33.4	58.4	306°10.0	39.0	93°25.4	44.4	248°21.7	52.0			
20	87°41.3	107°32.9	57.2	321°13.2	39.0	108°27.3	44.2	263°24.3	52.0			
21	102°43.8	122°32.4	•• 56.0	336°16.4	•• 39.1	123°29.2	•• 44.0	278°26.9	•• 52.1			
22	117°46.3	137°31.9	54.8	351°19.5	39.2	138°31.1	43.8	293°29.5	52.2			
23	132°48.7	152°31.5	53.7	6°22.7	39.3	153°33.0	43.5	308°32.1	52.2			
Mer.pass. 14:10		ν -0.5' d -1.2' m -3.92		ν 3.2' d 0.1' m -0.94		ν 1.9' d -0.2' m -2.03		ν 2.6' d 0.1' m 0.64				

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DUT1 = UT1-UTC = +0.0724 sec ΔT = TT-UT1 = +66.1116 sec

2010 February 15 to Feb. 17 UT

h Sun Moon						
Mon	GHA	Dec	GHA	ν	Dec	d HP
0	176°28.0	S12°47.3	168°42.5	16.4'	S05°59.1	-12.6' 54.0'
1	191°28.0	46.4	183°17.8	16.4'	05°46.5	-12.6' 54.0'
2	206°28.0	45.6	197°53.2	16.4'	05°33.9	-12.7' 54.0'
3	221°28.0	· · 44.7	212°28.6	16.4'	05°21.2	-12.7' 54.0'
4	236°28.1	43.9	227°04.1	16.4'	05°08.5	-12.7' 54.1'
5	251°28.1	43.0	241°39.5	16.4'	04°55.8	-12.7' 54.1'
6	266°28.1	S12°42.1	256°14.9	16.5'	S04°43.1	-12.7' 54.1'
7	281°28.1	41.3	270°50.4	16.5'	04°30.4	-12.8' 54.1'
8	296°28.2	40.4	285°25.8	16.5'	04°17.6	-12.8' 54.1'
9	311°28.2	· · 39.6	300°01.3	16.5'	04°04.9	-12.8' 54.1'
10	326°28.2	38.7	314°36.8	16.5'	03°52.1	-12.8' 54.1'
11	341°28.3	37.8	329°12.3	16.5'	03°39.3	-12.8' 54.1'
12	356°28.3	S12°37.0	343°47.7	16.5'	S03°26.4	-12.8' 54.1'
13	11°28.3	36.1	358°23.2	16.5'	03°13.6	-12.9' 54.1'
14	26°28.4	35.3	12°58.7	16.5'	03°00.7	-12.9' 54.1'
15	41°28.4	· · 34.4	27°34.2	16.5'	02°47.9	-12.9' 54.1'
16	56°28.4	33.5	42°09.7	16.5'	02°35.0	-12.9' 54.1'
17	71°28.4	32.7	56°45.2	16.5'	02°22.1	-12.9' 54.2'
18	86°28.5	S12°31.8	71°20.8	16.5'	S02°09.2	-12.9' 54.2'
19	101°28.5	30.9	85°56.3	16.5'	01°56.2	-12.9' 54.2'
20	116°28.5	30.1	100°31.8	16.5'	01°43.3	-12.9' 54.2'
21	131°28.6	· · 29.2	115°07.3	16.5'	01°30.4	-13.0' 54.2'
22	146°28.6	28.4	129°42.8	16.5'	01°17.4	-13.0' 54.2'
23	161°28.6	27.5	144°18.2	16.5'	01°04.4	-13.0' 54.2'
SD = 16.2' d = -0.9' SD = 14.7'						
Tue	GHA	Dec	GHA	ν	Dec	d HP
0	176°28.7	S12°26.6	158°53.7	16.5'	S00°51.5	-13.0' 54.2'
1	191°28.7	25.8	173°29.2	16.5'	00°38.5	-13.0' 54.2'
2	206°28.7	24.9	188°04.7	16.5'	00°25.5	-13.0' 54.2'
3	221°28.8	· · 24.0	202°40.2	16.5'	S00°12.5	-13.0' 54.3'
4	236°28.8	23.2	217°15.6	16.5'	N00°00.5	13.0' 54.3'
5	251°28.9	22.3	231°51.1	16.4'	00°13.5	13.0' 54.3'
6	266°28.9	S12°21.4	246°26.5	16.4'	N00°26.5	13.0' 54.3'
7	281°28.9	20.6	261°02.0	16.4'	00°39.5	13.0' 54.3'
8	296°29.0	19.7	275°37.4	16.4'	00°52.5	13.0' 54.3'
9	311°29.0	· · 18.8	290°12.8	16.4'	01°05.5	13.0' 54.3'
10	326°29.0	18.0	304°48.2	16.4'	01°18.5	13.0' 54.3'
11	341°29.1	17.1	319°23.6	16.4'	01°31.5	13.0' 54.3'
12	356°29.1	S12°16.2	333°58.9	16.3'	N01°44.5	13.0' 54.4'
13	11°29.2	15.4	348°34.3	16.3'	01°57.5	13.0' 54.4'
14	26°29.2	14.5	3°09.6	16.3'	02°10.5	13.0' 54.4'
15	41°29.2	· · 13.6	17°44.9	16.3'	02°23.5	13.0' 54.4'
16	56°29.3	12.7	32°20.2	16.3'	02°36.5	13.0' 54.4'
17	71°29.3	11.9	46°55.5	16.3'	02°49.5	13.0' 54.4'
18	86°29.3	S12°11.0	61°30.7	16.2'	N03°02.4	13.0' 54.4'
19	101°29.4	10.1	76°05.9	16.2'	03°15.4	13.0' 54.4'
20	116°29.4	09.3	90°41.1	16.2'	03°28.4	13.0' 54.5'
21	131°29.5	· · 08.4	105°16.3	16.2'	03°41.3	13.0' 54.5'
22	146°29.5	07.5	119°51.5	16.1'	03°54.3	12.9' 54.5'
23	161°29.5	06.6	134°26.6	16.1'	04°07.2	12.9' 54.5'
SD = 16.2' d = -0.9' SD = 14.8'						
Wed	GHA	Dec	GHA	ν	Dec	d HP
0	176°29.6	S12°05.8	149°01.7	16.1'	N04°20.2	12.9' 54.5'
1	191°29.6	04.9	163°36.8	16.0'	04°33.1	12.9' 54.5'
2	206°29.7	04.0	178°11.8	16.0'	04°46.0	12.9' 54.5'
3	221°29.7	· · 03.2	192°46.9	16.0'	04°58.9	12.9' 54.6'
4	236°29.8	02.3	207°21.9	16.0'	05°11.8	12.9' 54.6'
5	251°29.8	01.4	221°56.8	15.9'	05°24.6	12.9' 54.6'
6	266°29.8	S12°00.5	236°31.7	15.9'	N05°37.5	12.8' 54.6'
7	281°29.9	11°59.7	251°06.6	15.9'	05°50.3	12.8' 54.6'
8	296°29.9	58.8	265°41.5	15.8'	06°03.2	12.8' 54.6'
9	311°30.0	· · 57.9	280°16.3	15.8'	06°16.0	12.8' 54.6'
10	326°30.0	57.0	294°51.1	15.7'	06°28.7	12.8' 54.7'
11	341°30.1	56.2	309°25.8	15.7'	06°41.5	12.7' 54.7'
12	356°30.1	S11°55.3	324°00.6	15.7'	N06°54.3	12.7' 54.7'
13	11°30.2	54.4	338°35.2	15.6'	07°07.0	12.7' 54.7'
14	26°30.2	53.5	353°09.9	15.6'	07°19.7	12.7' 54.7'
15	41°30.2	· · 52.6	7°44.5	15.6'	07°32.4	12.7' 54.7'
16	56°30.3	51.8	22°19.0	15.5'	07°45.0	12.6' 54.8'
17	71°30.3	50.9	36°53.5	15.5'	07°57.7	12.6' 54.8'
18	86°30.4	S11°50.0	51°28.0	15.4'	N08°10.3	12.6' 54.8'
19	101°30.4	49.1	66°02.4	15.4'	08°22.8	12.6' 54.8'
20	116°30.5	48.3	80°36.8	15.3'	08°35.4	12.5' 54.8'
21	131°30.5	· · 47.4	95°11.1	15.3'	08°47.9	12.5' 54.8'
22	146°30.6	46.5	109°45.4	15.2'	09°00.4	12.5' 54.9'
23	161°30.6	45.6	124°19.6	15.2'	09°12.9	12.4' 54.9'
SD = 16.2' d = -0.9' SD = 14.9'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	06:11	07:30	08:48	15:41	17:00	18:20
N 70°	06:09	07:21	08:29	16:00	17:09	18:21
68°	06:08	07:13	08:14	16:15	17:16	18:22
66°	06:07	07:07	08:02	16:27	17:23	18:23
64°	06:06	07:01	07:52	16:37	17:28	18:24
62°	06:05	06:56	07:43	16:46	17:33	18:25
60°	06:03	06:52	07:36	16:53	17:37	18:26
N 58°	06:02	06:48	07:29	17:00	17:41	18:27
56°	06:01	06:45	07:23	17:06	17:44	18:28
54°	06:00	06:41	07:18	17:11	17:48	18:29
52°	05:59	06:39	07:13	17:16	17:50	18:30
50°	05:58	06:36	07:09	17:20	17:53	18:31
45°	05:55	06:30	07:00	17:29	17:59	18:34
N 40°	05:53	06:24	06:52	17:37	18:05	18:36
35°	05:50	06:19	06:45	17:43	18:09	18:39
30°	05:47	06:15	06:39	17:49	18:14	18:42
20°	05:40	06:06	06:29	18:00	18:22	18:48
N 10°	05:33	05:58	06:19	18:09	18:30	18:55
0°	05:25	05:50	06:11	18:17	18:39	19:03
S 10°	05:15	05:40	06:02	18:26	18:48	19:13
20°	05:03	05:29	05:52	18:36	18:58	19:25
30°	04:47	05:16	05:41	18:47	19:11	19:41
35°	04:36	05:08	05:35	18:53	19:19	19:51
40°	04:24	04:59	05:27	19:00	19:29	20:03
45°	04:09	04:47	05:18	19:09	19:40	20:18
S 50°	03:49	04:33	05:08	19:19	19:54	20:38
52°	03:39	04:26	05:03	19:24	20:01	20:48
54°	03:27	04:18	04:58	19:29	20:08	20:59
56°	03:14	04:10	04:52	19:35	20:17	21:11
58°	02:58	04:00	04:45	19:42	20:26	21:27
S 60°	02:39	03:48	04:37	19:49	20:38	21:46

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	07:53	07:28	07:03	18:51	20:42	22:39
N 70°	07:47	07:29	07:11	18:53	20:37	22:25
68°	07:42	07:30	07:18	18:56	20:33	22:13
66°	07:38	07:31	07:23	18:58	20:29	22:04
64°	07:35	07:31	07:28	19:00	20:27	21:56
62°	07:31	07:32	07:32	19:01	20:24	21:49
60°	07:29	07:32	07:36	19:02	20:22	21:43
N 58°	07:26	07:32	07:39	19:03	20:20	21:38
56°	07:24	07:33	07:42	19:04	20:18	21:33
54°	07:22	07:33	07:44	19:05	20:17	21:29
52°	07:21	07:33	07:47	19:06	20:15	21:26
50°	07:19	07:34	07:49	19:07	20:14	21:22
45°	07:15	07:34	07:53	19:08	20:11	21:15
N 40°	07:12	07:35	07:57	19:10	20:09	21:09
35°	07:10	07:35	08:01	19:11	20:07	21:04
30°	07:08	07:35	08:04	19:12	20:05	20:59
20°	07:04	07:36	08:09	19:13	20:02	20:51
N 10°	07:00	07:37	08:14	19:15	19:59	20:45
0°	06:57	07:37	08:18	19:16	19:57	20:38
S 10°	06:54	07:38	08:23	19:18	19:54	20:32
20°	06:50	07:38	08:27	19:19	19:52	20:25
30°	06:46	07:39	08:33	19:21	19:49	20:17
35°	06:44	07:40	08:36	19:22	19:47	20:13
40°	06:41	07:40	08:40	19:23	19:45	20:08
45°	06:38	07:41	08:44	19:24	19:43	20:02
S 50°	06:35	07:41	08:49	19:25	19:40	19:55
52°	06:33	07:42	08:51	19:26	19:39	19:52
54°	06:31	07:42	08:54	19:27	19:37	19:49
56°	06:29	07:42	08:57	19:27	19:36	19:45
58°	06:27	07:43	09:00	19:28	19:34	19:41
S 60°	06:24	07:43	09:04	19:29	19:32	19:36

Day	Sun		Mer. Pass	Moon		Age 1-3 1-8%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
15	14:08	14:07	12:14	13:07	00:46	
16	14:05	14:04	12:14	13:47	01:27	
17	14:02	14:00	12:14	14:28	02:07	

February 18, 19, 20 UT (Thu., Fri., Sat.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	147°51.2	167°31.0	S09°52.5	21°25.9	N23°39.4	168°34.9	S09°43.3	323°34.7	N00°52.3	Alpheratz	357°46.4	29°08.9											
1	162°53.6	182°30.5	51.3	36°29.1	39.5	183°36.8	43.1	338°37.3	52.4	Ankaa	353°18.3	-42°15.2											
2	177°56.1	197°30.0	50.1	51°32.3	39.6	198°38.7	42.9	353°39.8	52.4	Schedar	349°43.9	56°35.8											
3	192°58.6	212°29.6	· · 49.0	66°35.5	· · 39.7	213°40.6	· · 42.7	8°42.4	· · 52.5	Diphda	348°58.5	-17°55.9											
4	208°01.0	227°29.1	47.8	81°38.6	39.8	228°42.5	42.4	23°45.0	52.6	Achernar	335°28.7	-57°11.3											
5	223°03.5	242°28.6	46.6	96°41.8	39.9	243°44.4	42.2	38°47.6	52.6	Hamal	328°03.7	23°30.7											
6	238°06.0	257°28.1	S09°45.4	111°45.0	N23°40.0	258°46.3	S09°42.0	53°50.2	N00°52.7	Polaris	319°06.5	89°18.9											
7	253°08.4	272°27.7	44.3	126°48.2	40.1	273°48.2	41.8	68°52.8	52.8	Acamar	315°20.2	-40°16.0											
8	268°10.9	287°27.2	43.1	141°51.3	40.2	288°50.1	41.5	83°55.4	52.8	Menkar	314°17.6	4°07.8											
9	283°13.4	302°26.7	· · 41.9	156°54.5	· · 40.3	303°52.0	· · 41.3	98°58.0	· · 52.9	Mirfak	308°44.0	49°54.1											
10	298°15.8	317°26.2	40.7	171°57.7	40.4	318°53.9	41.1	114°00.6	52.9	Aldebaran	290°52.1	16°31.8											
11	313°18.3	332°25.8	39.5	187°00.9	40.5	333°55.8	40.9	129°03.2	53.0	Rigel	281°14.3	-8°11.5											
12	328°20.8	347°25.3	S09°38.4	202°04.0	N23°40.5	348°57.7	S09°40.7	144°05.8	N00°53.1	Capella	280°37.9	46°00.6											
13	343°23.2	2°24.8	37.2	217°07.2	40.6	3°59.6	40.4	159°08.4	53.1	Bellatrix	278°34.5	6°21.5											
14	358°25.7	17°24.3	36.0	232°10.4	40.7	19°01.4	40.2	174°11.0	53.2	Elnath	278°15.5	28°37.0											
15	13°28.1	32°23.9	· · 34.8	247°13.5	· · 40.8	34°03.3	· · 40.0	189°13.6	· · 53.3	Alnilam	275°48.7	-1°11.8											
16	28°30.6	47°23.4	33.6	262°16.7	40.9	49°05.2	39.8	204°16.2	53.3	Betelgeuse	271°03.7	7°24.5											
17	43°33.1	62°22.9	32.5	277°19.9	41.0	64°07.1	39.5	219°18.8	53.4	Canopus	263°56.9	-52°42.3											
18	58°35.5	77°22.5	S09°31.3	292°23.0	N23°41.1	79°09.0	S09°39.3	234°21.4	N00°53.5	Sirius	258°35.6	-16°44.0											
19	73°38.0	92°22.0	30.1	307°26.2	41.2	94°10.9	39.1	249°23.9	53.5	Adhara	255°14.2	-28°59.4											
20	88°40.5	107°21.5	28.9	322°29.3	41.2	109°12.8	38.9	264°26.5	53.6	Procyon	245°02.0	5°11.8											
21	103°42.9	122°21.0	· · 27.7	337°32.5	· · 41.3	124°14.7	· · 38.6	279°29.1	· · 53.7	Pollux	243°30.3	28°00.0											
22	118°45.4	137°20.6	26.5	352°35.6	41.4	139°16.6	38.4	294°31.7	53.7	Avior	234°18.5	-59°32.7											
23	133°47.9	152°20.1	25.4	7°38.8	41.5	154°18.5	38.2	309°34.3	53.8	Suhail	222°53.9	-43°28.6											
Mer.pass. 14:06				ν -0.5' d -1.2' m -3.91				ν 3.2' d 0.1' m -0.91				ν 1.9' d -0.2' m -2.02				ν 2.6' d 0.1' m 0.64							

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	148°50.3	167°19.6	S09°24.2	22°42.0	N23°41.6	169°20.4	S09°38.0	324°36.9	N00°53.9										
1	163°52.8	182°19.2	23.0	37°45.1	41.7	184°22.3	37.8	339°39.5	53.9										
2	178°55.3	197°18.7	21.8	52°48.3	41.8	199°24.2	37.5	354°42.1	54.0										
3	193°57.7	212°18.2	· · 20.6	67°51.4	· · 41.8	214°26.1	· · 37.3	9°44.7	· · 54.1										
4	209°00.2	227°17.8	19.4	82°54.6	41.9	229°28.0	37.1	24°47.3	54.1										
5	224°02.6	242°17.3	18.2	97°57.7	42.0	244°29.9	36.9	39°49.9	54.2										
6	239°05.1	257°16.8	S09°17.0	113°00.9	N23°42.1	259°31.8	S09°36.6	54°52.5	N00°54.3										
7	254°07.6	272°16.4	15.9	128°04.0	42.2	274°33.7	36.4	69°55.1	54.3										
8	269°10.0	287°15.9	14.7	143°07.1	42.3	289°35.6	36.2	84°57.7	54.4										
9	284°12.5	302°15.4	· · 13.5	158°10.3	· · 42.3	304°37.5	· · 36.0	100°00.3	· · 54.5										
10	299°15.0	317°15.0	12.3	173°13.4	42.4	319°39.4	35.7	115°02.9	54.5										
11	314°17.4	332°14.5	11.1	188°16.6	42.5	334°41.3	35.5	130°05.5	54.6										
12	329°19.9	347°14.1	S09°09.9	203°19.7	N23°42.6	349°43.2	S09°35.3	145°08.1	N00°54.7										
13	344°22.4	2°13.6	08.7	218°22.8	42.6	4°45.1	35.1	160°10.7	54.7										
14	359°24.8	17°13.1	07.5	233°26.0	42.7	19°47.0	34.9	175°13.3	54.8										
15	14°27.3	32°12.7	· · 06.3	248°29.1	· · 42.8	34°48.9	· · 34.6	190°15.9	· · 54.9										
16	29°29.7	47°12.2	05.1	263°32.2	42.9	49°50.8	34.5	205°18.5	54.9										
17	44°32.2	62°11.7	04.0	278°35.4	43.0	64°52.7	34.2	220°21.1	55.0										
18	59°34.7	77°11.3	S09°02.8	293°38.5	N23°43.0	79°54.6	S09°34.0	235°23.7	N00°55.1										
19	74°37.1	92°10.8	01.6	308°41.6	43.1	94°56.5	33.7	250°26.3	55.1										
20	89°39.6	107°10.4	09°00.4	323°44.8	43.2	109°58.4	33.5	265°28.9	55.2										
21	104°42.1	122°09.9	08°59.2	338°47.9	· · 43.3	125°00.3	· · 33.3	280°31.5	· · 55.3										
22	119°44.5	137°09.4	58.0	353°51.0	43.3	140°02.2	33.1	295°34.1	55.3										
23	134°47.0	152°09.0	56.8	8°54.1	43.4	155°04.1	32.8	310°36.7	55.4										
Mer.pass. 14:02				ν -0.5' d -1.2' m -3.91				ν 3.2' d 0.1' m -0.88				ν 1.9' d -0.2' m -2.02				ν 2.6' d 0.1' m 0.63			

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	149°49.5	167°08.5	S08°55.6	23°57.2	N23°43.5	170°05.9	S09°32.6	325°39.2	N00°55.5										
1	164°51.9	182°08.1	54.4	39°00.4	43.6	185°07.8	32.4	340°41.8	55.5										
2	179°54.4	197°07.6	53.2	54°03.5	43.6	200°09.7	32.2	355°44.4	55.6										
3	194°56.9	212°07.1	· · 52.0	69°06.6	· · 43.7	215°11.6	· · 32.0	10°47.0	· · 55.7										
4	209°59.3	227°06.7	50.8	84°09.7	43.8	230°13.5	31.7	25°49.6	55.7										
5	225°01.8	242°06.2	49.6	99°12.8	43.9	245°15.4	31.5	40°52.2	55.8										
6	240°04.2	257°05.8	S08°48.4	114°15.9	N23°43.9	260°17.3	S09°31.3	55°54.8	N00°55.9										
7	255°06.7	272°05.3	47.2	129°19.1	44.0	275°19.2	31.1	70°57.4	55.9										
8	270°09.2	287°04.9	46.0	144°22.2	44.1	290°21.1	30.8	86°00.0	56.0										
9	285°11.6	302°04.4	· · 44.8	159°25.3	· · 44.1	305°23.0	· · 30.6	101°02.6	· · 56.1										
10	300°14.1	317°03.9	43.6	174°28.4	44.2	320°24.9	30.6	116°05.2	56.1										
11	315°16.6	332°03.5	42.4	189°31.5	44.3	335°26.8	30.2	131°07.8	56.2										
12	330°19.0	347°03.0	S08°41.2	204°34.6	N23°44.3	350°28.7	S09°29.9	146°10.4	N00°56.3										
13	345°21.5	2°02.6	40.0	219°37.7	44.4	5°30.6	29.7	161°13.0	56.3										
14	0°24.0	17°02.1	38.8	234°40.8	44.5	20°32.5	29.5	176°15.6	56.4										
15	15°26.4	32°01.7	· · 37.6	249°43.9	· · 44.5	35°34.4	· · 29.3	191°18.2	· · 56.5										
16	30°28.9	47°01.2	36.4	264°47.0	44.6	50°36.3	29.0	206°20.8	56.5										
17	45°31.4	62°00.8	35.2	279°50.1	44.7	65°38.2	28.8	221°23.4	56.6										
18	60°33.8	77°00.3	S08°34.0	294°53.2	N23°44.7	80°40.1	S09°28.6	236°26.0	N00°56.7										
19	75°36.3	91°59.9	32.8	309°56.3	44.8	95°42.0	28.4	251°28.6	56.7										
20	90°38.7	106°59.4	31.6	324°59.4	44.9	110°43.9	28.2	266°31.2	56.8										
21	105°41.2	121°59.0	· · 30.4	340°02.5	· · 44.9	125°45.8	· · 27.9	281°33.8	· · 56.9										
22	120°43.7	136°58.5	29.2	355°05.6	45.0	140°47.7	27.7	296°36.4	56.9										
23	135°46.1	151°58.1	28.0	10°08.7	45.1	155°49.6	27.5	311°39.0	57.0										
Mer.pass. 13:58				ν -0.5' d -1.2' m -3.91				ν 3.1' d 0.1' m -0.86				ν 1.9' d -0.2' m -2.02				ν 2.6' d 0.1' m 0.63			

Feb 18 Thu			SHA	Mer.pass
Venus			19°39.8	12:50
Mars			233°34.7	22:30
Jupiter			20°43.7	12:44
Saturn			175°43.5	02:25

Feb 19 Fri			SHA	Mer.pass
Venus			18°29.3	12:51
Mars			233°51.6	22:25
Jupiter			20°30.1	12:41
Saturn			175°46.6	02:21

Feb 20 Sat			SHA	Mer.pass
Venus			17°19.1	12:52
Mars			234°07.8	22:20
Jupiter			20°16.5	12:38
Saturn			175°49.8	02:17

Horizontal parallax			
Venus:			0.1
Mars:			0.2

DUT1 = UT1-UTC = +0.0681 sec $\Delta T = TT-UT1 = +66.1159$ sec

2010 February 18 to Feb. 20 UT

h	Sun			Moon			
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	176°30.7	S11°44.7	138°53.8	15.1'	N09°25.4	12.4'	54.9'
1	191°30.7	43.9	153°27.9	15.1'	09°37.8	12.4'	54.9'
2	206°30.8	43.0	168°02.0	15.0'	09°50.1	12.3'	54.9'
3	221°30.8	· · 42.1	182°36.0	15.0'	10°02.5	12.3'	55.0'
4	236°30.9	41.2	197°10.0	14.9'	10°14.8	12.3'	55.0'
5	251°30.9	40.3	211°43.9	14.9'	10°27.1	12.2'	55.0'
6	266°31.0	S11°39.4	226°17.8	14.8'	N10°39.3	12.2'	55.0'
7	281°31.0	38.6	240°51.6	14.8'	10°51.5	12.2'	55.0'
8	296°31.1	37.7	255°25.4	14.7'	11°03.7	12.1'	55.0'
9	311°31.1	· · 36.8	269°59.1	14.6'	11°15.9	12.1'	55.1'
10	326°31.2	35.9	284°32.7	14.6'	11°28.0	12.1'	55.1'
11	341°31.2	35.0	299°06.3	14.5'	11°40.0	12.0'	55.1'
12	356°31.3	S11°34.1	313°39.8	14.5'	N11°52.0	12.0'	55.1'
13	11°31.3	33.3	328°13.3	14.4'	12°04.0	11.9'	55.1'
14	26°31.4	32.4	342°46.7	14.3'	12°15.9	11.9'	55.2'
15	41°31.4	· · 31.5	357°20.0	14.3'	12°27.8	11.8'	55.2'
16	56°31.5	30.6	11°53.3	14.2'	12°39.7	11.8'	55.2'
17	71°31.6	29.7	26°26.5	14.1'	12°51.5	11.8'	55.2'
18	86°31.6	S11°28.8	40°59.6	14.1'	N13°03.2	11.7'	55.3'
19	101°31.7	27.9	55°32.7	14.0'	13°14.9	11.7'	55.3'
20	116°31.7	27.1	70°05.7	13.9'	13°26.6	11.6'	55.3'
21	131°31.8	· · 26.2	84°38.6	13.9'	13°38.2	11.6'	55.3'
22	146°31.8	25.3	99°11.5	13.8'	13°49.7	11.5'	55.3'
23	161°31.9	24.4	113°44.3	13.7'	14°01.2	11.5'	55.4'
SD = 16.2'			SD = 15.0'				
$d = -0.9'$							

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	176°31.9	S11°23.5	128°17.0	13.7'	N14°12.7	11.4'	55.4'
1	191°32.0	22.6	142°49.7	13.6'	14°24.1	11.3'	55.4'
2	206°32.1	21.7	157°22.2	13.5'	14°35.4	11.3'	55.4'
3	221°32.1	· · 20.8	171°54.7	13.4'	14°46.7	11.2'	55.5'
4	236°32.2	19.9	186°27.2	13.4'	14°58.0	11.2'	55.5'
5	251°32.2	19.1	200°59.5	13.3'	15°09.1	11.1'	55.5'
6	266°32.3	S11°18.2	215°31.8	13.2'	N15°20.3	11.1'	55.5'
7	281°32.3	17.3	230°04.0	13.1'	15°31.3	11.0'	55.5'
8	296°32.4	16.4	244°36.1	13.0'	15°42.3	10.9'	55.6'
9	311°32.5	· · 15.5	259°08.2	13.0'	15°53.3	10.9'	55.6'
10	326°32.5	14.6	273°40.2	12.9'	16°04.1	10.8'	55.6'
11	341°32.6	13.7	288°12.0	12.8'	16°14.9	10.7'	55.6'
12	356°32.6	S11°12.8	302°43.9	12.7'	N16°25.7	10.7'	55.7'
13	11°32.7	11.9	317°15.6	12.6'	16°36.4	10.6'	55.7'
14	26°32.8	11.0	331°47.2	12.6'	16°47.0	10.5'	55.7'
15	41°32.8	· · 10.1	346°18.8	12.5'	16°57.5	10.5'	55.7'
16	56°32.9	09.3	0°50.3	12.4'	17°08.0	10.4'	55.8'
17	71°32.9	08.4	15°21.7	12.3'	17°18.4	10.3'	55.8'
18	86°33.0	S11°07.5	29°53.0	12.2'	N17°28.7	10.3'	55.8'
19	101°33.1	06.6	44°24.2	12.1'	17°39.0	10.2'	55.8'
20	116°33.1	05.7	58°55.3	12.0'	17°49.2	10.1'	55.9'
21	131°33.2	· · 04.8	73°26.4	12.0'	17°59.3	10.0'	55.9'
22	146°33.2	03.9	87°57.3	11.9'	18°09.3	10.0'	55.9'
23	161°33.3	03.0	102°28.2	11.8'	18°19.3	9.9'	56.0'
SD = 16.2'			SD = 15.1'				
$d = -0.9'$							

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	176°33.4	S11°02.1	116°59.0	11.7'	N18°29.2	9.8'	56.0'
1	191°33.4	01.2	131°29.7	11.6'	18°39.0	9.7'	56.0'
2	206°33.5	11°00.3	146°00.3	11.5'	18°48.7	9.6'	56.0'
3	221°33.6	10°59.4	160°30.8	11.4'	18°58.4	9.6'	56.1'
4	236°33.6	58.5	175°01.2	11.3'	19°07.9	9.5'	56.1'
5	251°33.7	57.6	189°31.6	11.2'	19°17.4	9.4'	56.1'
6	266°33.8	S10°56.7	204°01.8	11.1'	N19°26.8	9.3'	56.1'
7	281°33.8	55.8	218°32.0	11.1'	19°36.1	9.2'	56.2'
8	296°33.9	54.9	233°02.0	11.0'	19°45.3	9.1'	56.2'
9	311°34.0	· · 54.0	247°32.0	10.9'	19°54.4	9.0'	56.2'
10	326°34.0	53.1	262°01.8	10.8'	20°03.5	8.9'	56.3'
11	341°34.1	52.2	276°31.6	10.7'	20°12.4	8.9'	56.3'
12	356°34.2	S10°51.3	291°01.3	10.6'	N20°21.3	8.8'	56.3'
13	11°34.2	50.4	305°30.9	10.5'	20°30.0	8.7'	56.3'
14	26°34.3	49.5	320°00.4	10.4'	20°38.7	8.6'	56.4'
15	41°34.4	· · 48.6	334°29.8	10.3'	20°47.3	8.5'	56.4'
16	56°34.4	47.7	348°59.1	10.2'	20°55.7	8.4'	56.4'
17	71°34.5	46.8	3°28.3	10.1'	21°04.1	8.3'	56.5'
18	86°34.6	S10°45.9	17°57.4	10.0'	N21°12.4	8.2'	56.5'
19	101°34.6	45.0	32°26.4	9.9'	21°20.6	8.1'	56.5'
20	116°34.7	44.1	46°55.3	9.8'	21°28.6	8.0'	56.6'
21	131°34.8	· · 43.2	61°24.1	9.7'	21°36.6	7.9'	56.6'
22	146°34.8	42.3	75°52.9	9.6'	21°44.5	7.8'	56.6'
23	161°34.9	41.4	90°21.5	9.5'	21°52.2	7.6'	56.6'
SD = 16.2'			SD = 15.3'				
$d = -0.9'$							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	05:58	07:16	08:32	15:58	17:13	18:32
N 70°	05:58	07:09	08:15	16:14	17:21	18:32
68°	05:58	07:02	08:02	16:27	17:27	18:32
66°	05:57	06:57	07:51	16:38	17:32	18:32
64°	05:57	06:52	07:42	16:47	17:37	18:32
62°	05:57	06:48	07:34	16:55	17:41	18:33
60°	05:56	06:44	07:28	17:01	17:44	18:33
N 58°	05:55	06:41	07:22	17:07	17:48	18:33
56°	05:55	06:38	07:16	17:12	17:51	18:34
54°	05:54	06:35	07:12	17:17	17:53	18:34
52°	05:54	06:33	07:07	17:21	17:56	18:35
50°	05:53	06:30	07:03	17:25	17:58	18:36
45°	05:51	06:25	06:55	17:33	18:03	18:37
N 40°	05:49	06:20	06:48	17:40	18:08	18:39
35°	05:47	06:16	06:42	17:46	18:12	18:42
30°	05:44	06:12	06:36	17:52	18:16	18:44
20°	05:39	06:05	06:27	18:01	18:23	18:49
N 10°	05:32	05:57	06:18	18:09	18:31	18:55
0°	05:25	05:49	06:10	18:17	18:38	19:03
S 10°	05:16	05:41	06:02	18:25	18:47	19:12
20°	05:04	05:31	05:54	18:34	18:56	19:23
30°	04:49	05:19	05:43	18:44	19:08	19:38
35°	04:40	05:11	05:37	18:50	19:16	19:47
40°	04:28	05:02	05:31	18:56	19:25	19:58
45°	04:14	04:52	05:23	19:04	19:35	20:13
S 50°	03:55	04:38	05:13	19:13	19:48	20:31
52°	03:46	04:32	05:09	19:18	19:54	20:40
54°	03:36	04:25	05:04	19:22	20:01	20:50
56°	03:24	04:17	04:58	19:28	20:09	21:02
58°	03:09	04:08	04:52	19:34	20:18	21:16
S 60°	02:52	03:58	04:45	19:40	20:28	21:33

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	06:34	05:51	□	· · ·	00:55	□
N 70°	06:51	06:24	05:28	· · ·	00:23	02:57
68°	07:04	06:48	06:22	· · ·	00:01	02:04
66°	07:15	07:07	06:55	23:43	· · ·	01:32
64°	07:25	07:22	07:20	23:29	· · ·	01:08
62°	07:33	07:35	07:39	23:17	· · ·	00:50
60°	07:40	07:46	07:55	23:07	· · ·	00:34
N 58°	07:46	07:56	08:09	22:58	· · ·	00:22
56°	07:52	08:04	08:20	22:51	· · ·	00:11
54°	07:57	08:11	08:31	22:44	· · ·	00:01
52°	08:01	08:18	08:40	22:38	23:52	· · ·
50°	08:05	08:24	08:48	22:32	23:44	· · ·
45°	08:14	08:38	09:06	22:21	23:28	· · ·
N 40°	08:21	08:49	09:20	22:11	23:15	· · ·
35°	08:28	08:58	09:33	22:02	23:03	· · ·
30°	08:34	09:06	09:43	21:55	22:53	23:54
20°	08:43	09:21	10:02	21:43	22:36	23:33
N 10°	08:52	09:33	10:18	21:32	22:22	23:15
0°	09:00	09:45	10:33	21:22	22:08	22:58
S 10°	09:09	09:57	10:49	21:11	21:54	22:41
20°	09:18	10:10	11:05	21:01	21:40	22:23
30°	09:28	10:25	11:25	20:48	21:23	22:03
35°	09:34	10:34	11:36	20:41	21:13	21:51
40°	09:41	10:44	11:49	20:33	21:02	21:37
45°	09:49	10:55	12:04	20:24	20:50	21:21
S 50°	09:58	11:10	12:23	20:13	20:34	21:01
52°	10:03	11:16	12:32	20:08	20:27	20:51
54°	10:08	11:24	12:42	20:02	20:19	20:41
56°	10:13	11:32	12:53	19:56	20:10	20:29
58°	10:19	11:42	13:06	19:49	19:59	20:15
S 60°	10:26	11:53	13:22	19:41	19:48	19:59

Day	Sun			Moon		
	Eqn. of Time	Mer.	Mer. Pass.		Age	
	00 ^h	Pass	Upper	Lower	4-6	
	mm:ss	mm:ss	hh:mm	hh:mm	hh:mm	13-30%
18	13:57	13:55	12:14	15:11	02:49	
19	13:52	13:49	12:14	15:57	03:33	
20	13:46	13:43	12:14	16:46	04:21	

February 21, 22, 23 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	150°48.6	166°57.6	S08°26.8	25°11.8	N23°45.1	170°51.5	S09°27.3	326°41.6	N00°57.1	Alpheratz	357°46.4	29°08.9
1	165°51.1	181°57.2	25.6	40°14.8	45.2	185°53.4	27.0	341°44.2	57.2	Ankaa	353°18.3	-42°15.2
2	180°53.5	196°56.7	24.4	55°17.9	45.3	200°55.3	26.8	356°46.8	57.2	Schedar	349°43.9	56°35.8
3	195°56.0	211°56.3	· · 23.2	70°21.0	· · 45.3	215°57.2	· · 26.6	11°49.4	· · 57.3	Diphda	348°58.5	-17°55.9
4	210°58.5	226°55.8	22.0	85°24.1	45.4	230°59.1	26.4	26°52.0	57.4	Achernar	335°28.7	-57°11.3
5	226°00.9	241°55.4	20.8	100°27.2	45.4	246°01.0	26.1	41°54.7	57.4	Hamal	328°03.7	23°30.7
6	241°03.4	256°54.9	S08°19.5	115°30.3	N23°45.5	261°02.9	S09°25.9	56°57.3	N00°57.5	Polaris	319°07.6	89°18.9
7	256°05.8	271°54.5	18.3	130°33.3	45.6	276°04.8	25.7	71°59.9	57.6	Acamar	315°20.2	-40°16.0
8	271°08.3	286°54.0	17.1	145°36.4	45.6	291°06.6	25.5	87°02.5	57.6	Menkar	314°17.7	4°07.8
9	286°10.8	301°53.6	· · 15.9	160°39.5	· · 45.7	306°08.5	· · 25.2	102°05.1	· · 57.7	Mirfak	308°44.0	49°54.1
10	301°13.2	316°53.1	14.7	175°42.6	45.8	321°10.4	25.0	117°07.7	57.8	Aldebaran	290°52.1	16°31.8
11	316°15.7	331°52.7	13.5	190°45.6	45.8	336°12.3	24.8	132°10.3	57.8	Rigel	281°14.3	-8°11.5
12	331°18.2	346°52.2	S08°12.3	205°48.7	N23°45.9	351°14.2	S09°24.6	147°12.9	N00°57.9	Capella	280°37.9	46°00.6
13	346°20.6	1°51.8	11.1	220°51.8	45.9	6°16.1	24.3	162°15.5	58.0	Bellatrix	278°34.5	6°21.5
14	1°23.1	16°51.3	09.9	235°54.9	46.0	21°18.0	24.1	177°18.1	58.0	Elnath	278°15.5	28°37.0
15	16°25.6	31°50.9	· · 08.7	250°57.9	· · 46.0	36°19.9	· · 23.9	192°20.7	· · 58.1	Alnilam	275°48.7	-1°11.8
16	31°28.0	46°50.4	07.5	266°01.0	46.1	51°21.8	23.7	207°23.3	58.2	Betelgeuse	271°03.8	7°24.5
17	46°30.5	61°50.0	06.2	281°04.1	46.2	66°23.7	23.4	222°25.9	58.2	Canopus	263°56.9	-52°42.3
18	61°33.0	76°49.6	S08°05.0	296°07.1	N23°46.2	81°25.6	S09°23.2	237°28.5	N00°58.3	Sirius	258°35.6	-16°44.0
19	76°35.4	91°49.1	03.8	311°10.2	46.3	96°27.5	23.0	252°31.1	58.4	Adhara	255°14.2	-28°59.4
20	91°37.9	106°48.7	02.6	326°13.3	46.3	111°29.4	22.8	267°33.7	58.5	Procyon	245°02.0	5°11.8
21	106°40.3	121°48.2	· · 01.4	341°16.3	· · 46.4	126°31.3	· · 22.6	282°36.3	· · 58.5	Pollux	243°30.3	28°00.0
22	121°42.8	136°47.8	08°00.2	356°19.4	46.4	141°33.2	22.3	297°38.9	58.6	Avior	234°18.6	-59°32.7
23	136°45.3	151°47.3	07°59.0	11°22.4	46.5	156°35.1	22.1	312°41.5	58.7	Suhail	222°53.9	-43°28.6
Mer.pass. 13:54		ν -0.5' d -1.2' m -3.91		ν 3.1' d 0.1' m -0.83		ν 1.9' d -0.2' m -2.02		ν 2.6' d 0.1' m 0.62				

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	151°47.7	166°46.9	S07°57.8	26°25.5	N23°46.5	171°37.0	S09°21.9	327°44.1	N00°58.7
1	166°50.2	181°46.5	56.5	41°28.5	46.6	186°38.9	21.7	342°46.7	58.8
2	181°52.7	196°46.0	55.3	56°31.6	46.6	201°40.8	21.4	357°49.3	58.9
3	196°55.1	211°45.6	· · 54.1	71°34.6	· · 46.7	216°42.7	· · 21.2	12°51.9	· · 58.9
4	211°57.6	226°45.1	52.9	86°37.7	46.7	231°44.6	21.0	27°54.5	59.0
5	227°00.1	241°44.7	51.7	101°40.7	46.8	246°46.5	20.8	42°57.1	59.1
6	242°02.5	256°44.3	S07°50.5	116°43.8	N23°46.8	261°48.4	S09°20.5	57°59.7	N00°59.1
7	257°05.0	271°43.8	49.2	131°46.8	46.9	276°50.3	20.3	73°02.3	59.2
8	272°07.5	286°43.4	48.0	146°49.9	46.9	291°52.2	20.1	88°04.9	59.3
9	287°09.9	301°42.9	· · 46.8	161°52.9	· · 47.0	306°54.1	· · 19.9	103°07.5	· · 59.4
10	302°12.4	316°42.5	45.6	176°56.0	47.0	321°56.0	19.6	118°10.1	59.4
11	317°14.8	331°42.1	44.4	191°59.0	47.1	336°57.9	19.4	133°12.8	59.5
12	332°17.3	346°41.6	S07°43.2	207°02.0	N23°47.1	351°59.8	S09°19.2	148°15.4	N00°59.6
13	347°19.8	1°41.2	41.9	222°05.1	47.2	7°01.7	19.0	163°18.0	59.6
14	2°22.2	16°40.7	40.7	237°08.1	47.2	22°03.6	18.7	178°20.6	59.7
15	17°24.7	31°40.3	· · 39.5	252°11.2	· · 47.3	37°05.4	· · 18.5	193°23.2	· · 59.8
16	32°27.2	46°39.9	38.3	267°14.2	47.3	52°07.3	18.3	208°25.8	59.8
17	47°29.6	61°39.4	37.1	282°17.2	47.4	67°09.2	18.1	223°28.4	00°59.9
18	62°32.1	76°39.0	S07°35.8	297°20.3	N23°47.4	82°11.1	S09°17.8	238°31.0	N01°00.0
19	77°34.6	91°38.6	34.6	312°23.3	47.5	97°13.0	17.6	253°33.6	00.0
20	92°37.0	106°38.1	33.4	327°26.3	47.5	112°14.9	17.4	268°36.2	00.1
21	107°39.5	121°37.7	· · 32.2	342°29.3	· · 47.6	127°16.8	· · 17.2	283°38.8	· · 00.2
22	122°42.0	136°37.3	31.0	357°32.4	47.6	142°18.7	16.9	298°41.4	00.3
23	137°44.4	151°36.8	29.7	12°35.4	47.7	157°20.6	16.7	313°44.0	00.3
Mer.pass. 13:51		ν -0.4' d -1.2' m -3.91		ν 3.1' d 0.1' m -0.80		ν 1.9' d -0.2' m -2.02		ν 2.6' d 0.1' m 0.62	

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	152°46.9	166°36.4	S07°28.5	27°38.4	N23°47.7	172°22.5	S09°16.5	328°46.6	N01°00.4
1	167°49.3	181°35.9	27.3	42°41.4	47.7	187°24.4	16.3	343°49.2	00.5
2	182°51.8	196°35.5	26.1	57°44.4	47.8	202°26.3	16.0	358°51.8	00.5
3	197°54.3	211°35.1	· · 24.8	72°47.5	· · 47.8	217°28.2	· · 15.8	13°54.4	· · 00.6
4	212°56.7	226°34.6	23.6	87°50.5	47.9	232°30.1	15.6	28°57.1	00.7
5	227°59.2	241°34.2	22.4	102°53.5	47.9	247°32.0	15.4	43°59.7	00.7
6	243°01.7	256°33.8	S07°21.2	117°56.5	N23°47.9	262°33.9	S09°15.1	59°02.3	N01°00.8
7	258°04.1	271°33.4	20.0	132°59.5	48.0	277°35.8	14.9	74°04.9	00.9
8	273°06.6	286°32.9	18.7	148°02.5	48.0	292°37.7	14.7	89°07.5	01.0
9	288°09.1	301°32.5	· · 17.5	163°05.5	· · 48.1	307°39.6	· · 14.5	104°10.1	· · 01.0
10	303°11.5	316°32.1	16.3	178°08.6	48.1	322°41.5	14.2	119°12.7	01.1
11	318°14.0	331°31.6	15.1	193°11.6	48.1	337°43.4	14.0	134°15.3	01.2
12	333°16.4	346°31.2	S07°13.8	208°14.6	N23°48.2	352°45.3	S09°13.8	149°17.9	N01°01.2
13	348°18.9	1°30.8	12.6	223°17.6	48.2	7°47.2	13.6	164°20.5	01.3
14	3°21.4	16°30.3	11.4	238°20.6	48.3	22°49.1	13.3	179°23.1	01.4
15	18°23.8	31°29.9	· · 10.1	253°23.6	· · 48.3	37°51.0	· · 13.1	194°25.7	· · 01.4
16	33°26.3	46°29.5	08.9	268°26.6	48.3	52°52.9	12.9	209°28.3	01.5
17	48°28.8	61°29.0	07.7	283°29.6	48.4	67°54.8	12.7	224°30.9	01.6
18	63°31.2	76°28.6	S07°06.5	298°32.6	N23°48.4	82°56.7	S09°12.5	239°33.6	N01°01.7
19	78°33.7	91°28.2	05.2	313°35.6	48.4	97°58.6	12.2	254°36.2	01.7
20	93°36.2	106°27.8	04.0	328°38.6	48.5	113°00.5	12.0	269°38.8	01.8
21	108°38.6	121°27.3	· · 02.8	343°41.5	· · 48.5	128°02.4	· · 11.8	284°41.4	· · 01.9
22	123°41.1	136°26.9	01.5	358°44.5	48.5	143°04.2	11.6	299°44.0	01.9
23	138°43.6	151°26.5	00.3	13°47.5	48.6	158°06.1	11.3	314°46.6	02.0
Mer.pass. 13:47		ν -0.4' d -1.2' m -3.91		ν 3.0' d 0.0' m -0.77		ν 1.9' d -0.2' m -2.02		ν 2.6' d 0.1' m 0.61	

Feb 21 Sun			SHA	Mer.pass
Venus			16°09.0	12:53
Mars			234°23.2	22:15
Jupiter			20°02.9	12:35
Saturn			175°53.0	02:13

Feb 22 Mon			SHA	Mer.pass
Venus			14°59.2	12:53
Mars			234°37.7	22:10
Jupiter			19°49.3	12:32
Saturn			175°56.4	02:09

Feb 23 Tue			SHA	Mer.pass
Venus			13°49.5	12:54
Mars			234°51.5	22:05
Jupiter			19°35.6	12:29
Saturn			175°59.7	02:05

Horizontal parallax			
Venus:			0.1
Mars:			0.2

DUT1 = UT1-UTC = +0.0639 sec $\Delta T = TT-UT1 = +66.1201$ sec

2010 February 21 to Feb. 23 UT

Sun							Moon						
h	GHA	Dec	GHA	ν	Dec	d	HP						
Sun													
0	176°35.0	S10°40.5	104°50.0	9.4'	N21°59.9	7.5'	56.7'						
1	191°35.1	39.6	119°18.4	9.3'	22°07.4	7.4'	56.7'						
2	206°35.1	38.7	133°46.8	9.2'	22°14.8	7.3'	56.7'						
3	221°35.2	· · 37.8	148°15.0	9.1'	22°22.1	7.2'	56.8'						
4	236°35.3	36.9	162°43.2	9.0'	22°29.3	7.1'	56.8'						
5	251°35.3	36.0	177°11.2	8.9'	22°36.4	7.0'	56.8'						
6	266°35.4	S10°35.1	191°39.1	8.9'	N22°43.4	6.9'	56.9'						
7	281°35.5	34.2	206°07.0	8.8'	22°50.3	6.7'	56.9'						
8	296°35.6	33.3	220°34.7	8.7'	22°57.0	6.6'	56.9'						
9	311°35.6	· · 32.4	235°02.4	8.6'	23°03.6	6.5'	57.0'						
10	326°35.7	31.5	249°30.0	8.5'	23°10.1	6.4'	57.0'						
11	341°35.8	30.6	263°57.4	8.4'	23°16.5	6.3'	57.0'						
12	356°35.8	S10°29.7	278°24.8	8.3'	N23°22.8	6.1'	57.1'						
13	11°35.9	28.8	292°52.1	8.2'	23°28.9	6.0'	57.1'						
14	26°36.0	27.9	307°19.3	8.1'	23°34.9	5.9'	57.1'						
15	41°36.1	· · 27.0	321°46.4	8.0'	23°40.8	5.8'	57.2'						
16	56°36.1	26.0	336°13.4	7.9'	23°46.6	5.6'	57.2'						
17	71°36.2	25.1	350°40.3	7.8'	23°52.2	5.5'	57.2'						
18	86°36.3	S10°24.2	5°07.1	7.7'	N23°57.7	5.4'	57.3'						
19	101°36.4	23.3	19°33.8	7.6'	24°03.1	5.2'	57.3'						
20	116°36.4	22.4	34°00.4	7.5'	24°08.3	5.1'	57.3'						
21	131°36.5	· · 21.5	48°27.0	7.4'	24°13.4	5.0'	57.4'						
22	146°36.6	20.6	62°53.4	7.4'	24°18.4	4.8'	57.4'						
23	161°36.7	19.7	77°19.8	7.3'	24°23.2	4.7'	57.4'						
SD = 16.2'							d = -0.9'	SD = 15.5'					
Mon	GHA	Dec	GHA	ν	Dec	d	HP						
0	176°36.8	S10°18.8	91°46.0	7.2'	N24°27.9	4.6'	57.5'						
1	191°36.8	17.9	106°12.2	7.1'	24°32.5	4.4'	57.5'						
2	206°36.9	17.0	120°38.3	7.0'	24°36.9	4.3'	57.5'						
3	221°37.0	· · 16.1	135°04.3	6.9'	24°41.1	4.1'	57.6'						
4	236°37.1	15.1	149°30.2	6.8'	24°45.3	4.0'	57.6'						
5	251°37.1	14.2	163°56.1	6.8'	24°49.3	3.8'	57.6'						
6	266°37.2	S10°13.3	178°21.8	6.7'	N24°53.1	3.7'	57.7'						
7	281°37.3	12.4	192°47.5	6.6'	24°56.8	3.6'	57.7'						
8	296°37.4	11.5	207°13.1	6.5'	25°00.4	3.4'	57.7'						
9	311°37.5	· · 10.6	221°38.6	6.4'	25°03.8	3.3'	57.8'						
10	326°37.5	09.7	236°04.0	6.4'	25°07.0	3.1'	57.8'						
11	341°37.6	08.8	250°29.4	6.3'	25°10.1	3.0'	57.9'						
12	356°37.7	S10°07.9	264°54.7	6.2'	N25°13.1	2.8'	57.9'						
13	11°37.8	06.9	279°19.9	6.1'	25°15.9	2.7'	57.9'						
14	26°37.9	06.0	293°45.0	6.1'	25°18.5	2.5'	58.0'						
15	41°37.9	· · 05.1	308°10.1	6.0'	25°21.0	2.3'	58.0'						
16	56°38.0	04.2	322°35.0	5.9'	25°23.4	2.2'	58.0'						
17	71°38.1	03.3	336°59.9	5.8'	25°25.6	2.0'	58.1'						
18	86°38.2	S10°02.4	351°24.8	5.8'	N25°27.6	1.9'	58.1'						
19	101°38.3	01.5	5°49.6	5.7'	25°29.5	1.7'	58.1'						
20	116°38.3	10°00.5	20°14.3	5.6'	25°31.2	1.6'	58.2'						
21	131°38.4	09°59.6	34°38.9	5.6'	25°32.7	1.4'	58.2'						
22	146°38.5	58.7	49°03.5	5.5'	25°34.1	1.2'	58.2'						
23	161°38.6	57.8	63°28.0	5.5'	25°35.4	1.1'	58.3'						
SD = 16.2'							d = -0.9'	SD = 15.7'					
Tue	GHA	Dec	GHA	ν	Dec	d	HP						
0	176°38.7	S09°56.9	77°52.5	5.4'	N25°36.4	0.9'	58.3'						
1	191°38.8	56.0	92°16.9	5.3'	25°37.4	0.7'	58.4'						
2	206°38.9	55.1	106°41.2	5.3'	25°38.1	0.6'	58.4'						
3	221°38.9	· · 54.1	121°05.5	5.2'	25°38.7	0.4'	58.4'						
4	236°39.0	53.2	135°29.7	5.2'	25°39.1	0.3'	58.5'						
5	251°39.1	52.3	149°53.9	5.1'	25°39.4	0.1'	58.5'						
6	266°39.2	S09°51.4	164°18.0	5.1'	N25°39.4	-0.1'	58.5'						
7	281°39.3	50.5	178°42.1	5.0'	25°39.4	-0.2'	58.6'						
8	296°39.4	49.6	193°06.1	5.0'	25°39.1	-0.4'	58.6'						
9	311°39.4	· · 48.6	207°30.1	4.9'	25°38.7	-0.6'	58.6'						
10	326°39.5	47.7	221°54.1	4.9'	25°38.1	-0.8'	58.7'						
11	341°39.6	46.8	236°18.0	4.9'	25°37.3	-0.9'	58.7'						
12	356°39.7	S09°45.9	250°41.8	4.8'	N25°36.4	-1.1'	58.8'						
13	11°39.8	45.0	265°05.6	4.8'	25°35.3	-1.3'	58.8'						
14	26°39.9	44.0	279°29.4	4.8'	25°34.1	-1.4'	58.8'						
15	41°40.0	· · 43.1	293°53.2	4.7'	25°32.6	-1.6'	58.9'						
16	56°40.1	42.2	308°16.9	4.7'	25°31.0	-1.8'	58.9'						
17	71°40.1	41.3	322°40.6	4.7'	25°29.2	-2.0'	58.9'						
18	86°40.2	S09°40.4	337°04.2	4.6'	N25°27.3	-2.1'	59.0'						
19	101°40.3	39.4	351°27.9	4.6'	25°25.2	-2.3'	59.0'						
20	116°40.4	38.5	5°51.5	4.6'	25°22.9	-2.5'	59.0'						
21	131°40.5	· · 37.6	20°15.1	4.6'	25°20.4	-2.6'	59.1'						
22	146°40.6	36.7	34°38.6	4.5'	25°17.7	-2.8'	59.1'						
23	161°40.7	35.8	49°02.2	4.5'	25°14.9	-3.0'	59.1'						
SD = 16.2'							d = -0.9'	SD = 15.9'					

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	05:44	07:02	08:15	16:14	17:27	18:45
N 70°	05:45	06:56	08:01	16:28	17:33	18:44
68°	05:47	06:51	07:49	16:39	17:38	18:42
66°	05:47	06:47	07:40	16:49	17:42	18:41
64°	05:48	06:43	07:32	16:56	17:46	18:41
62°	05:48	06:39	07:25	17:03	17:49	18:40
60°	05:48	06:36	07:19	17:09	17:52	18:40
N 58°	05:48	06:34	07:14	17:14	17:54	18:40
56°	05:48	06:31	07:09	17:19	17:57	18:40
54°	05:48	06:29	07:05	17:23	17:59	18:40
52°	05:48	06:27	07:01	17:27	18:01	18:40
50°	05:47	06:25	06:58	17:30	18:03	18:40
45°	05:46	06:20	06:50	17:37	18:07	18:41
N 40°	05:45	06:16	06:44	17:44	18:11	18:43
35°	05:43	06:13	06:38	17:49	18:15	18:44
30°	05:41	06:09	06:33	17:54	18:18	18:46
20°	05:37	06:03	06:25	18:02	18:25	18:50
N 10°	05:31	05:56	06:17	18:10	18:31	18:56
0°	05:25	05:49	06:10	18:17	18:38	19:02
S 10°	05:16	05:41	06:03	18:24	18:45	19:10
20°	05:06	05:32	05:55	18:32	18:54	19:21
30°	04:52	05:21	05:46	18:41	19:05	19:34
35°	04:43	05:14	05:40	18:46	19:12	19:43
40°	04:32	05:06	05:34	18:52	19:20	19:54
45°	04:19	04:56	05:27	18:59	19:30	20:07
S 50°	04:02	04:44	05:18	19:07	19:42	20:24
52°	03:53	04:38	05:14	19:11	19:47	20:32
54°	03:44	04:32	05:10	19:16	19:54	20:41
56°	03:33	04:25	05:05	19:20	20:01	20:52
58°	03:20	04:16	05:00	19:26	20:09	21:05
S 60°	03:04	04:07	04:53	19:32	20:18	21:20

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	☐	☐	☐	☐	☐	☐
66°	06:36	☐	☐	03:37	☐	☐
64°	07:19	07:23	07:55	02:55	04:45	06:16
62°	07:48	08:09	08:55	02:26	04:00	05:16
60°	08:11	08:38	09:28	02:04	03:31	04:43
N 58°	08:29	09:01	09:53	01:46	03:08	04:18
56°	08:44	09:20	10:13	01:32	02:50	03:58
54°	08:57	09:36	10:30	01:19	02:34	03:41
52°	09:09	09:49	10:44	01:08	02:21	03:27
50°	09:19	10:01	10:56	00:58	02:09	03:15
45°	09:41	10:26	11:22	00:37	01:45	02:49
N 40°	09:59	10:45	11:43	00:20	01:26	02:29
35°	10:13	11:02	12:00	00:06	01:09	02:11
30°	10:26	11:16	12:14	☐ ☐	00:55	01:57
20°	10:48	11:41	12:39	☐ ☐	00:32	01:32
N 10°	11:08	12:02	13:01	☐ ☐	00:11	01:10
0°	11:26	12:22	13:21	23:52	☐ ☐	00:50
S 10°	11:44	12:42	13:41	23:33	☐ ☐	00:30
20°	12:03	13:03	14:03	23:13	☐ ☐	00:08
30°	12:26	13:28	14:28	22:49	23:43	☐ ☐
35°	12:39	13:42	14:43	22:35	23:28	☐ ☐
40°	12:55	13:59	15:00	22:19	23:11	☐ ☐
45°	13:13	14:20	15:20	22:00	22:50	23:53
S 50°	13:36	14:45	15:46	21:36	22:24	23:27
52°	13:47	14:58	15:59	21:25	22:12	23:15
54°	14:00	15:12	16:13	21:12	21:58	23:00
56°	14:14	15:29	16:30	20:57	21:41	22:44
58°	14:31	15:49	16:50	20:40	21:21	22:24
S 60°	14:52	16:14	17:15	20:19	20:56	21:59

February 24, 25, 26 UT (Wed., Thu., Fri.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	153°46.0	166°26.1	S06°59.1	28°50.5	N23°48.6	173°08.0	S09°11.1	329°49.2	N01°02.1	Alpheratz	357°46.4	29°08.9					
1	168°48.5	181°25.6	57.9	43°53.5	48.6	188°09.9	10.9	344°51.8	02.2	Ankaa	353°18.3	-42°15.1					
2	183°50.9	196°25.2	56.6	58°56.5	48.7	203°11.8	10.7	359°54.4	02.2	Schedar	349°43.9	56°35.8					
3	198°53.4	211°24.8	•• 55.4	73°59.5	•• 48.7	218°13.7	•• 10.4	14°57.0	•• 02.3	Diphda	348°58.5	-17°55.9					
4	213°55.9	226°24.4	54.2	89°02.5	48.7	233°15.6	10.2	29°59.6	02.4	Achernar	335°28.7	-57°11.3					
5	228°58.3	241°23.9	52.9	104°05.4	48.8	248°17.5	10.0	45°02.3	02.4	Hamal	328°03.7	23°30.7					
6	244°00.8	256°23.5	S06°51.7	119°08.4	N23°48.8	263°19.4	S09°09.8	60°04.9	N01°02.5	Polaris	319°08.7	89°18.9					
7	259°03.3	271°23.1	50.5	134°11.4	48.8	278°21.3	09.5	75°07.5	02.6	Acamar	315°20.2	-40°16.0					
8	274°05.7	286°22.7	49.2	149°14.4	48.9	293°23.2	09.3	90°10.1	02.7	Menkar	314°17.7	4°07.8					
9	289°08.2	301°22.2	•• 48.0	164°17.3	•• 48.9	308°25.1	•• 09.1	105°12.7	•• 02.7	Mirfak	308°44.0	49°54.1					
10	304°10.7	316°21.8	46.8	179°20.3	48.9	323°27.0	08.9	120°15.3	02.8	Aldebaran	290°52.1	16°31.8					
11	319°13.1	331°21.4	45.5	194°23.3	49.0	338°28.9	08.6	135°17.9	02.9	Rigel	281°14.3	-8°11.5					
12	334°15.6	346°21.0	S06°44.3	209°26.3	N23°49.0	353°30.8	S09°08.4	150°20.5	N01°02.9	Capella	280°37.9	46°00.7					
13	349°18.1	1°20.5	43.1	224°29.2	49.0	8°32.7	08.2	165°23.1	03.0	Bellatrix	278°34.5	6°21.5					
14	4°20.5	16°20.1	41.8	239°32.2	49.0	23°34.6	08.0	180°25.7	03.1	Elnath	278°15.5	28°37.0					
15	19°23.0	31°19.7	•• 40.6	254°35.2	•• 49.1	38°36.5	•• 07.7	195°28.4	•• 03.2	Alnilam	275°48.7	-1°11.8					
16	34°25.4	46°19.3	39.4	269°38.1	49.1	53°38.4	07.5	210°31.0	03.2	Betelgeuse	271°03.8	7°24.5					
17	49°27.9	61°18.8	38.1	284°41.1	49.1	68°40.3	07.3	225°33.6	03.3	Canopus	263°56.9	-52°42.3					
18	64°30.4	76°18.4	S06°36.9	299°44.1	N23°49.1	83°42.2	S09°07.1	240°36.2	N01°03.4	Sirius	258°35.6	-16°44.0					
19	79°32.8	91°18.0	35.7	314°47.0	49.2	98°44.1	06.8	255°38.8	03.4	Adhara	255°14.2	-28°59.4					
20	94°35.3	106°17.6	34.4	329°50.0	49.2	113°46.0	06.6	270°41.4	03.5	Procyon	245°02.0	5°11.8					
21	109°37.8	121°17.2	•• 33.2	344°52.9	•• 49.2	128°47.9	•• 06.4	285°44.0	•• 03.6	Pollux	243°30.3	28°00.0					
22	124°40.2	136°16.7	31.9	359°55.9	49.3	143°49.8	06.2	300°46.6	03.7	Avior	234°18.6	-59°32.8					
23	139°42.7	151°16.3	30.7	14°58.9	49.3	158°51.7	05.9	315°49.2	03.7	Suhail	222°53.9	-43°28.6					
Mer.pass. 13:43			ν -0.4' d -1.2' m -3.91			ν 3.0' d 0.0' m -0.75			ν 1.9' d -0.2' m -2.02			ν 2.6' d 0.1' m 0.61					

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	154°45.2	166°15.9	S06°29.5	30°01.8	N23°49.3	173°53.6	S09°05.7	330°51.9	N01°03.8					
1	169°47.6	181°15.5	28.2	45°04.8	49.3	188°55.5	05.5	345°54.5	03.9					
2	184°50.1	196°15.1	27.0	60°07.7	49.3	203°57.4	05.3	0°57.1	03.9					
3	199°52.6	211°14.6	•• 25.8	75°10.7	•• 49.4	218°59.3	•• 05.0	15°59.7	•• 04.0					
4	214°55.0	226°14.2	24.5	90°13.6	49.4	234°01.2	04.8	31°02.3	04.1					
5	229°57.5	241°13.8	23.3	105°16.6	49.4	249°03.0	04.6	46°04.9	04.2					
6	244°59.9	256°13.4	S06°22.0	120°19.5	N23°49.4	264°04.9	S09°04.3	61°07.5	N01°04.2					
7	260°02.4	271°13.0	20.8	135°22.4	49.5	279°06.8	04.1	76°10.1	04.3					
8	275°04.9	286°12.6	19.6	150°25.4	49.5	294°08.7	03.9	91°12.7	04.4					
9	290°07.3	301°12.1	•• 18.3	165°28.3	•• 49.5	309°10.6	•• 03.7	106°15.4	•• 04.4					
10	305°09.8	316°11.7	17.1	180°31.3	49.5	324°12.5	03.4	121°18.0	04.5					
11	320°12.3	331°11.3	15.8	195°34.2	49.5	339°14.4	03.2	136°20.6	04.6					
12	335°14.7	346°10.9	S06°14.6	210°37.1	N23°49.6	354°16.3	S09°03.0	151°23.2	N01°04.7					
13	350°17.2	1°10.5	13.4	225°40.1	49.6	9°18.2	02.8	166°25.8	04.7					
14	5°19.7	16°10.1	12.1	240°43.0	49.6	24°20.1	02.5	181°28.4	04.8					
15	20°22.1	31°09.6	•• 10.9	255°45.9	•• 49.6	39°22.0	•• 02.3	196°31.0	•• 04.9					
16	35°24.6	46°09.2	09.6	270°48.9	49.6	54°23.9	02.1	211°33.6	04.9					
17	50°27.1	61°08.8	08.4	285°51.8	49.6	69°25.8	01.9	226°36.3	05.0					
18	65°29.5	76°08.4	S06°07.1	300°54.7	N23°49.7	84°27.7	S09°01.6	241°38.9	N01°05.1					
19	80°32.0	91°08.0	05.9	315°57.7	49.7	99°29.6	01.4	256°41.5	05.2					
20	95°34.4	106°07.6	04.7	331°00.6	49.7	114°31.5	01.2	271°44.1	05.2					
21	110°36.9	121°07.2	•• 03.4	346°03.5	•• 49.7	129°33.4	•• 01.0	286°46.7	•• 05.3					
22	125°39.4	136°06.7	02.2	1°06.4	49.7	144°35.3	00.7	301°49.3	05.4					
23	140°41.8	151°06.3	00.9	16°09.4	49.7	159°37.2	00.5	316°51.9	05.4					
Mer.pass. 13:39			ν -0.4' d -1.2' m -3.91			ν 3.0' d 0.0' m -0.72			ν 1.9' d -0.2' m -2.02			ν 2.6' d 0.1' m 0.61		

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	155°44.3	166°05.9	S05°59.7	31°12.3	N23°49.8	174°39.1	S09°00.3	331°54.6	N01°05.5					
1	170°46.8	181°05.5	58.4	46°15.2	49.8	189°41.0	09°00.1	346°57.2	05.6					
2	185°49.2	196°05.1	57.2	61°18.1	49.8	204°42.9	08°59.8	1°59.8	05.7					
3	200°51.7	211°04.7	•• 56.0	76°21.0	•• 49.8	219°44.8	•• 59.6	17°02.4	•• 05.7					
4	215°54.2	226°04.3	54.7	91°23.9	49.8	234°46.7	59.4	32°05.0	05.8					
5	230°56.6	241°03.9	53.5	106°26.8	49.8	249°48.6	59.2	47°07.6	05.9					
6	245°59.1	256°03.4	S05°52.2	121°29.8	N23°49.8	264°50.5	S08°58.9	62°10.2	N01°06.0					
7	261°01.5	271°03.0	51.0	136°32.7	49.9	279°52.4	58.7	77°12.9	06.0					
8	276°04.0	286°02.6	49.7	151°35.6	49.9	294°54.3	58.5	92°15.5	06.1					
9	291°06.5	301°02.2	•• 48.5	166°38.5	•• 49.9	309°56.2	•• 58.3	107°18.1	•• 06.2					
10	306°08.9	316°01.8	47.2	181°41.4	49.9	324°58.1	58.0	122°20.7	06.2					
11	321°11.4	331°01.4	46.0	196°44.3	49.9	340°00.0	57.8	137°23.3	06.3					
12	336°13.9	346°01.0	S05°44.7	211°47.2	N23°49.9	355°01.9	S08°57.6	152°25.9	N01°06.4					
13	351°16.3	1°00.6	43.5	226°50.1	49.9	10°03.7	57.4	167°28.5	06.5					
14	6°18.8	16°00.2	42.2	241°53.0	49.9	25°05.6	57.1	182°31.2	06.5					
15	21°21.3	30°59.8	•• 41.0	256°55.9	•• 49.9	40°07.5	•• 56.9	197°33.8	•• 06.6					
16	36°23.7	45°59.4	39.7	271°58.8	49.9	55°09.4	56.7	212°36.4	06.7					
17	51°26.2	60°58.9	38.5	287°01.7	49.9	70°11.3	56.5	227°39.0	06.8					
18	66°28.7	75°58.5	S05°37.2	302°04.6	N23°50.0	85°13.2	S08°56.2	242°41.6	N01°06.8					
19	81°31.1	90°58.1	36.0	317°07.5	50.0	100°15.1	56.0	257°44.2	06.9					
20	96°33.6	105°57.7	34.7	332°10.4	50.0	115°17.0	55.8	272°46.8	07.0					
21	111°36.0	120°57.3	•• 33.5	347°13.2	•• 50.0	130°18.9	•• 55.6	287°49.5	•• 07.1					
22	126°38.5	135°56.9	32.2	2°16.1	50.0	145°20.8	55.3	302°52.1	07.1					
23	141°41.0	150°56.5	31.0	17°19.0	50.0	160°22.7	55.1	317°54.7	07.2					
Mer.pass. 13:35			ν -0.4' d -1.2' m -3.91			ν 2.9' d 0.0' m -0.69			ν 1.9' d -0.2' m -2.02			ν 2.6' d 0.1' m 0.60		

Feb 24 Wed	SHA	Mer.pass
Venus	12°40.0	12:55
Mars	235°04.5	22:00
Jupiter	19°22.0	12:26
Saturn	176°03.2	02:00

Feb 25 Thu	SHA	Mer.pass
Venus	11°30.7	12:55
Mars	235°16.6	21:56
Jupiter	19°08.4	12:23
Saturn	176°06.7	01:56

Feb 26 Fri	SHA	Mer.pass
Venus	10°21.6	12:56
Mars	235°28.0	21:51
Jupiter	18°54.8	12:20
Saturn	176°10.3	01:52

Horizontal parallax		
Venus:	0.1	
Mars:	0.2	

h	Sun			Moon		
Wed	GHA	Dec	GHA	ν	Dec	d HP
0	176°40.8	S09°34.8	63°25.7	4.5'	N25°11.9	-3.2' 59.2'
1	191°40.9	33.9	77°49.2	4.5'	25°08.8	-3.3' 59.2'
2	206°41.0	33.0	92°12.7	4.5'	25°05.4	-3.5' 59.3'
3	221°41.0	· · 32.1	106°36.2	4.5'	25°01.9	-3.7' 59.3'
4	236°41.1	31.2	120°59.6	4.5'	24°58.2	-3.9' 59.3'
5	251°41.2	30.2	135°23.1	4.5'	24°54.4	-4.0' 59.4'
6	266°41.3	S09°29.3	149°46.6	4.5'	N24°50.3	-4.2' 59.4'
7	281°41.4	28.4	164°10.0	4.4'	24°46.1	-4.4' 59.4'
8	296°41.5	27.5	178°33.5	4.4'	24°41.7	-4.6' 59.5'
9	311°41.6	· · 26.5	192°56.9	4.4'	24°37.2	-4.7' 59.5'
10	326°41.7	25.6	207°20.4	4.5'	24°32.4	-4.9' 59.5'
11	341°41.8	24.7	221°43.8	4.5'	24°27.5	-5.1' 59.6'
12	356°41.9	S09°23.8	236°07.3	4.5'	N24°22.5	-5.2' 59.6'
13	11°42.0	22.8	250°30.7	4.5'	24°17.2	-5.4' 59.6'
14	26°42.1	21.9	264°54.2	4.5'	24°11.8	-5.6' 59.7'
15	41°42.2	· · 21.0	279°17.7	4.5'	24°06.2	-5.8' 59.7'
16	56°42.3	20.1	293°41.2	4.5'	24°00.5	-5.9' 59.7'
17	71°42.3	19.1	308°04.7	4.5'	23°54.5	-6.1' 59.8'
18	86°42.4	S09°18.2	322°28.2	4.5'	N23°48.4	-6.3' 59.8'
19	101°42.5	17.3	336°51.7	4.6'	23°42.1	-6.4' 59.8'
20	116°42.6	16.4	351°15.3	4.6'	23°35.7	-6.6' 59.9'
21	131°42.7	· · 15.4	5°38.9	4.6'	23°29.1	-6.8' 59.9'
22	146°42.8	14.5	20°02.5	4.6'	23°22.3	-6.9' 59.9'
23	161°42.9	13.6	34°26.1	4.6'	23°15.4	-7.1' 60.0'
SD = 16.2' $d = -0.9'$			SD = 16.1'			

Thu	GHA	Dec	GHA	ν	Dec	d HP
0	176°43.0	S09°12.7	48°49.7	4.7'	N23°08.3	-7.3' 60.0'
1	191°43.1	11.7	63°13.4	4.7'	23°01.0	-7.4' 60.0'
2	206°43.2	10.8	77°37.1	4.7'	22°53.5	-7.6' 60.0'
3	221°43.3	· · 09.9	92°00.9	4.8'	22°45.9	-7.8' 60.1'
4	236°43.4	08.9	106°24.6	4.8'	22°38.2	-7.9' 60.1'
5	251°43.5	08.0	120°48.4	4.8'	22°30.2	-8.1' 60.1'
6	266°43.6	S09°07.1	135°12.2	4.9'	N22°22.1	-8.3' 60.2'
7	281°43.7	06.2	149°36.1	4.9'	22°13.9	-8.4' 60.2'
8	296°43.8	05.2	164°00.0	4.9'	22°05.5	-8.6' 60.2'
9	311°43.9	· · 04.3	178°23.9	5.0'	21°56.9	-8.7' 60.3'
10	326°44.0	03.4	192°47.9	5.0'	21°48.1	-8.9' 60.3'
11	341°44.1	02.4	207°11.9	5.1'	21°39.2	-9.0' 60.3'
12	356°44.2	S09°01.5	221°36.0	5.1'	N21°30.2	-9.2' 60.3'
13	11°44.3	09°00.6	236°00.1	5.1'	21°21.0	-9.4' 60.4'
14	26°44.4	08°59.6	250°24.2	5.2'	21°11.6	-9.5' 60.4'
15	41°44.5	· · 58.7	264°48.4	5.2'	21°02.1	-9.7' 60.4'
16	56°44.6	57.8	279°12.7	5.3'	20°52.5	-9.8' 60.4'
17	71°44.7	56.9	293°36.9	5.3'	20°42.7	-10.0' 60.5'
18	86°44.8	S08°55.9	308°01.3	5.4'	N20°32.7	-10.1' 60.5'
19	101°44.9	55.0	322°25.6	5.4'	20°22.6	-10.3' 60.5'
20	116°45.0	54.1	336°50.1	5.5'	20°12.3	-10.4' 60.5'
21	131°45.1	· · 53.1	351°14.5	5.5'	20°01.9	-10.5' 60.6'
22	146°45.2	52.2	5°39.1	5.6'	19°51.4	-10.7' 60.6'
23	161°45.3	51.3	20°03.6	5.6'	19°40.7	-10.8' 60.6'
SD = 16.2' $d = -0.9'$			SD = 16.4'			

Fri	GHA	Dec	GHA	ν	Dec	d HP
0	176°45.4	S08°50.3	34°28.3	5.7'	N19°29.9	-11.0' 60.6'
1	191°45.5	49.4	48°53.0	5.7'	19°18.9	-11.1' 60.7'
2	206°45.6	48.5	63°17.7	5.8'	19°07.8	-11.2' 60.7'
3	221°45.7	· · 47.5	77°42.5	5.8'	18°56.6	-11.4' 60.7'
4	236°45.8	46.6	92°07.3	5.9'	18°45.2	-11.5' 60.7'
5	251°45.9	45.7	106°32.2	6.0'	18°33.7	-11.6' 60.8'
6	266°46.0	S08°44.7	120°57.2	6.0'	N18°22.0	-11.8' 60.8'
7	281°46.1	43.8	135°22.2	6.1'	18°10.3	-11.9' 60.8'
8	296°46.2	42.9	149°47.3	6.1'	17°58.4	-12.0' 60.8'
9	311°46.4	· · 41.9	164°12.4	6.2'	17°46.3	-12.2' 60.8'
10	326°46.5	41.0	178°37.6	6.2'	17°34.2	-12.3' 60.9'
11	341°46.6	40.1	193°02.9	6.3'	17°21.9	-12.4' 60.9'
12	356°46.7	S08°39.1	207°28.2	6.4'	N17°09.5	-12.5' 60.9'
13	11°46.8	38.2	221°53.5	6.4'	16°57.0	-12.6' 60.9'
14	26°46.9	37.2	236°18.9	6.5'	16°44.3	-12.8' 60.9'
15	41°47.0	· · 36.3	250°44.4	6.5'	16°31.6	-12.9' 60.9'
16	56°47.1	35.4	265°10.0	6.6'	16°18.7	-13.0' 61.0'
17	71°47.2	34.4	279°35.6	6.7'	16°05.7	-13.1' 61.0'
18	86°47.3	S08°33.5	294°01.2	6.7'	N15°52.6	-13.2' 61.0'
19	101°47.4	32.6	308°27.0	6.8'	15°39.4	-13.3' 61.0'
20	116°47.5	31.6	322°52.7	6.8'	15°26.1	-13.4' 61.0'
21	131°47.6	· · 30.7	337°18.6	6.9'	15°12.7	-13.5' 61.0'
22	146°47.7	29.8	351°44.5	7.0'	14°59.1	-13.6' 61.0'
23	161°47.8	28.8	6°10.4	7.0'	14°45.5	-13.7' 61.1'
SD = 16.1' $d = -0.9'$			SD = 16.5'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	05:30	06:48	07:59	16:29	17:40	18:59
N 70°	05:33	06:43	07:47	16:41	17:45	18:55
68°	05:35	06:39	07:37	16:51	17:48	18:53
66°	05:37	06:36	07:29	16:59	17:52	18:51
64°	05:38	06:33	07:22	17:06	17:54	18:49
62°	05:39	06:31	07:16	17:12	17:57	18:48
60°	05:40	06:28	07:11	17:17	17:59	18:47
N 58°	05:41	06:26	07:06	17:21	18:01	18:47
56°	05:41	06:24	07:02	17:25	18:03	18:46
54°	05:42	06:22	06:58	17:29	18:05	18:46
52°	05:42	06:21	06:55	17:32	18:06	18:45
50°	05:42	06:19	06:52	17:35	18:08	18:45
45°	05:42	06:16	06:45	17:42	18:11	18:45
N 40°	05:41	06:12	06:40	17:47	18:14	18:46
35°	05:40	06:09	06:35	17:52	18:17	18:47
30°	05:38	06:06	06:30	17:56	18:20	18:48
20°	05:35	06:01	06:23	18:03	18:26	18:51
N 10°	05:30	05:55	06:16	18:10	18:31	18:56
0°	05:24	05:49	06:10	18:16	18:37	19:02
S 10°	05:17	05:42	06:03	18:23	18:44	19:09
20°	05:07	05:34	05:56	18:30	18:52	19:18
30°	04:55	05:23	05:48	18:38	19:02	19:31
35°	04:46	05:17	05:43	18:42	19:08	19:39
40°	04:36	05:10	05:38	18:48	19:16	19:49
45°	04:24	05:01	05:31	18:54	19:24	20:01
S 50°	04:08	04:50	05:24	19:01	19:35	20:17
52°	04:00	04:44	05:20	19:05	19:40	20:24
54°	03:51	04:38	05:16	19:09	19:46	20:33
56°	03:41	04:32	05:12	19:13	19:53	20:43
58°	03:29	04:24	05:07	19:18	20:00	20:54
S 60°	03:15	04:16	05:01	19:23	20:08	21:08

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	□	□	11:23	□	□	09:11
N 70°	□	□	12:25	□	□	08:08
68°	□	09:51	12:59	□	08:37	07:32
66°	□	11:03	13:24	□	07:23	07:06
64°	09:35	11:39	13:43	06:43	06:46	06:46
62°	10:18	12:05	13:59	06:00	06:20	06:29
60°	10:47	12:25	14:12	05:31	05:59	06:15
N 58°	11:09	12:42	14:23	05:09	05:42	06:03
56°	11:27	12:56	14:32	04:51	05:27	05:53
54°	11:42	13:08	14:41	04:35	05:15	05:43
52°	11:55	13:18	14:48	04:22	05:04	05:35
50°	12:06	13:28	14:55	04:10	04:54	05:28
45°	12:30	13:48	15:10	03:45	04:33	05:12
N 40°	12:49	14:04	15:21	03:26	04:16	04:58
35°	13:05	14:17	15:32	03:09	04:01	04:47
30°	13:19	14:29	15:40	02:55	03:49	04:37
20°	13:43	14:49	15:55	02:31	03:27	04:20
N 10°	14:03	15:06	16:09	02:10	03:09	04:05
0°	14:22	15:22	16:21	01:50	02:51	03:50
S 10°	14:41	15:39	16:33	01:30	02:33	03:36
20°	15:01	15:56	16:46	01:09	02:14	03:21
30°	15:24	16:15	17:01	00:44	01:52	03:03
35°	15:38	16:27	17:09	00:30	01:39	02:52
40°	15:54	16:40	17:19	00:13	01:24	02:40
45°	16:13	16:55	17:30	· · ·	01:06	02:26
S 50°	16:36	17:14	17:44	· · ·	00:43	02:08
52°	16:47	17:23	17:50	· · ·	00:33	02:00
54°	17:00	17:33	17:57	· · ·	00:20	01:51
56°	17:14	17:44	18:05	· · ·	00:06	01:40
58°	17:31	17:57	18:14	23:50	· · ·	01:28
S 60°	17:51	18:12	18:24	23:30	· · ·	01:14

Day	Sun		Mer. Pass	Moon		Age 10-12 71-89%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
24	13:17	13:12	12:13	20:36	08:06	
25	13:08	13:03	12:13	21:36	09:07	
26	12:58	12:53	12:13	22:34	10:06	

February 27, 28, 01 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	156°43.4	165°56.1	S05°29.7	32°21.9	N23°50.0	175°24.6	S08°54.9	332°57.3	N01°07.3	Alpheratz	357°46.4	29°08.9
1	171°45.9	180°55.7	28.5	47°24.8	50.0	190°26.5	54.7	347°59.9	07.3	Ankaa	353°18.3	-42°15.1
2	186°48.4	195°55.3	27.2	62°27.7	50.0	205°28.4	54.4	3°02.5	07.4	Schedar	349°43.9	56°35.8
3	201°50.8	210°54.9	•• 26.0	77°30.6	•• 50.0	220°30.3	•• 54.2	18°05.2	•• 07.5	Diphda	348°58.5	-17°55.9
4	216°53.3	225°54.5	24.7	92°33.4	50.0	235°32.2	54.0	33°07.8	07.6	Achernar	335°28.8	-57°11.3
5	231°55.8	240°54.1	23.5	107°36.3	50.0	250°34.1	53.8	48°10.4	07.6	Hamal	328°03.7	23°30.7
6	246°58.2	255°53.7	S05°22.2	122°39.2	N23°50.0	265°36.0	S08°53.5	63°13.0	N01°07.7	Polaris	319°10.0	89°18.9
7	262°00.7	270°53.3	21.0	137°42.1	50.0	280°37.9	53.3	78°15.6	07.8	Acamar	315°20.2	-40°16.0
8	277°03.2	285°52.8	19.7	152°44.9	50.0	295°39.8	53.1	93°18.2	07.9	Menkar	314°17.7	4°07.8
9	292°05.6	300°52.4	•• 18.5	167°47.8	•• 50.0	310°41.7	•• 52.8	108°20.9	•• 07.9	Mirfak	308°44.0	49°54.1
10	307°08.1	315°52.0	17.2	182°50.7	50.0	325°43.6	52.6	123°23.5	08.0	Aldebaran	290°52.1	16°31.8
11	322°10.5	330°51.6	16.0	197°53.5	50.0	340°45.5	52.4	138°26.1	08.1	Rigel	281°14.3	-8°11.5
12	337°13.0	345°51.2	S05°14.7	212°56.4	N23°50.0	355°47.4	S08°52.2	153°28.7	N01°08.2	Capella	280°37.9	46°00.7
13	352°15.5	0°50.8	13.5	227°59.3	50.0	10°49.3	51.9	168°31.3	08.2	Bellatrix	278°34.5	6°21.5
14	7°17.9	15°50.4	12.2	243°02.1	50.0	25°51.2	51.7	183°33.9	08.3	Elnath	278°15.6	28°37.0
15	22°20.4	30°50.0	•• 11.0	258°05.0	•• 50.0	40°53.1	•• 51.5	198°36.6	•• 08.4	Alnilam	275°48.7	-1°11.8
16	37°22.9	45°49.6	09.7	273°07.9	50.0	55°55.0	51.3	213°39.2	08.4	Betelgeuse	271°03.8	7°24.5
17	52°25.3	60°49.2	08.4	288°10.7	50.0	70°56.9	51.0	228°41.8	08.5	Canopus	263°57.0	-52°42.3
18	67°27.8	75°48.8	S05°07.2	303°13.6	N23°50.0	85°58.8	S08°50.8	243°44.4	N01°08.6	Sirius	258°35.6	-16°44.0
19	82°30.3	90°48.4	05.9	318°16.4	50.0	101°00.7	50.6	258°47.0	08.7	Adhara	255°14.2	-28°59.4
20	97°32.7	105°48.0	04.7	333°19.3	50.0	116°02.6	50.4	273°49.6	08.7	Procyon	245°02.0	5°11.8
21	112°35.2	120°47.6	•• 03.4	348°22.2	•• 50.0	131°04.5	•• 50.1	288°52.3	•• 08.8	Pollux	243°30.3	28°00.0
22	127°37.6	135°47.2	02.2	3°25.0	50.0	146°06.4	49.9	303°54.9	08.9	Avior	234°18.6	-59°32.8
23	142°40.1	150°46.8	00.9	18°27.9	50.0	161°08.3	49.7	318°57.5	09.0	Suhail	222°53.9	-43°28.6
Mer.pass. 13:31		ν -0.4' d -1.3' m -3.91		ν 2.9' d 0.0' m -0.66		ν 1.9' d -0.2' m -2.02		ν 2.6' d 0.1' m 0.60				

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	157°42.6	165°46.4	S04°59.7	33°30.7	N23°50.0	176°10.2	S08°49.5	334°00.1	N01°09.0
1	172°45.0	180°46.0	58.4	48°33.6	50.0	191°12.0	49.2	349°02.7	09.1
2	187°47.5	195°45.6	57.1	63°36.4	50.0	206°13.9	49.0	4°05.4	09.2
3	202°50.0	210°45.2	•• 55.9	78°39.2	•• 50.0	221°15.8	•• 48.8	19°08.0	•• 09.3
4	217°52.4	225°44.8	54.6	93°42.1	50.0	236°17.7	48.6	34°10.6	09.3
5	232°54.9	240°44.4	53.4	108°44.9	50.0	251°19.6	48.3	49°13.2	09.4
6	247°57.4	255°44.0	S04°52.1	123°47.8	N23°50.0	266°21.5	S08°48.1	64°15.8	N01°09.5
7	262°59.8	270°43.6	50.9	138°50.6	50.0	281°23.4	47.9	79°18.5	09.6
8	278°02.3	285°43.2	49.6	153°53.5	50.0	296°25.3	47.7	94°21.1	09.6
9	293°04.8	300°42.8	•• 48.3	168°56.3	•• 50.0	311°27.2	•• 47.4	109°23.7	•• 09.7
10	308°07.2	315°42.4	47.1	183°59.1	50.0	326°29.1	47.2	124°26.3	09.8
11	323°09.7	330°42.0	45.8	199°02.0	50.0	341°31.0	47.0	139°28.9	09.9
12	338°12.1	345°41.6	S04°44.6	214°04.8	N23°49.9	356°32.9	S08°46.8	154°31.5	N01°09.9
13	353°14.6	0°41.2	43.3	229°07.6	49.9	11°34.8	46.5	169°34.2	10.0
14	8°17.1	15°40.8	42.0	244°10.5	49.9	26°36.7	46.3	184°36.8	10.1
15	23°19.5	30°40.4	•• 40.8	259°13.3	•• 49.9	41°38.6	•• 46.1	199°39.4	•• 10.2
16	38°22.0	45°40.0	39.5	274°16.1	49.9	56°40.5	45.8	214°42.0	10.2
17	53°24.5	60°39.6	38.3	289°18.9	49.9	71°42.4	45.6	229°44.6	10.3
18	68°26.9	75°39.2	S04°37.0	304°21.8	N23°49.9	86°44.3	S08°45.4	244°47.3	N01°10.4
19	83°29.4	90°38.8	35.7	319°24.6	49.9	101°46.2	45.2	259°49.9	10.5
20	98°31.9	105°38.4	34.5	334°27.4	49.9	116°48.1	44.9	274°52.5	10.5
21	113°34.3	120°38.1	•• 33.2	349°30.2	•• 49.9	131°50.0	•• 44.7	289°55.1	•• 10.6
22	128°36.8	135°37.7	32.0	4°33.0	49.9	146°51.9	44.5	304°57.7	10.7
23	143°39.2	150°37.3	30.7	19°35.9	49.8	161°53.8	44.3	320°00.4	10.7
Mer.pass. 13:27		ν -0.4' d -1.3' m -3.91		ν 2.8' d -0.0' m -0.64		ν 1.9' d -0.2' m -2.02		ν 2.6' d 0.1' m 0.59	

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	158°41.7	165°36.9	S04°29.4	34°38.7	N23°49.8	176°55.7	S08°44.0	335°03.0	N01°10.8
1	173°44.2	180°36.5	28.2	49°41.5	49.8	191°57.6	43.8	350°05.6	10.9
2	188°46.6	195°36.1	26.9	64°44.3	49.8	206°59.5	43.6	5°08.2	11.0
3	203°49.1	210°35.7	•• 25.7	79°47.1	•• 49.8	222°01.4	•• 43.4	20°10.8	•• 11.0
4	218°51.6	225°35.3	24.4	94°49.9	49.8	237°03.3	43.1	35°13.5	11.1
5	233°54.0	240°34.9	23.1	109°52.7	49.8	252°05.2	42.9	50°16.1	11.2
6	248°56.5	255°34.5	S04°21.9	124°55.5	N23°49.7	267°07.1	S08°42.7	65°18.7	N01°11.3
7	263°59.0	270°34.1	20.6	139°58.3	49.7	282°09.0	42.5	80°21.3	11.3
8	279°01.4	285°33.7	19.3	155°01.1	49.7	297°10.9	42.2	95°23.9	11.4
9	294°03.9	300°33.3	•• 18.1	170°03.9	•• 49.7	312°12.8	•• 42.0	110°26.6	•• 11.5
10	309°06.4	315°32.9	16.8	185°06.8	49.7	327°14.7	41.8	125°29.2	11.6
11	324°08.8	330°32.5	15.6	200°09.6	49.7	342°16.6	41.5	140°31.8	11.6
12	339°11.3	345°32.1	S04°14.3	215°12.3	N23°49.7	357°18.5	S08°41.3	155°34.4	N01°11.7
13	354°13.7	0°31.7	13.0	230°15.1	49.6	12°20.4	41.1	170°37.1	11.8
14	9°16.2	15°31.4	11.8	245°17.9	49.6	27°22.3	40.9	185°39.7	11.9
15	24°18.7	30°31.0	•• 10.5	260°20.7	•• 49.6	42°24.2	•• 40.6	200°42.3	•• 11.9
16	39°21.1	45°30.6	09.2	275°23.5	49.6	57°26.1	40.4	215°44.9	12.0
17	54°23.6	60°30.2	08.0	290°26.3	49.6	72°28.0	40.2	230°47.5	12.1
18	69°26.1	75°29.8	S04°06.7	305°29.1	N23°49.6	87°29.9	S08°40.0	245°50.2	N01°12.2
19	84°28.5	90°29.4	05.4	320°31.9	49.5	102°31.8	39.7	260°52.8	12.2
20	99°31.0	105°29.0	04.2	335°34.7	49.5	117°33.7	39.5	275°55.4	12.3
21	114°33.5	120°28.6	•• 02.9	350°37.5	•• 49.5	132°35.5	•• 39.3	290°58.0	•• 12.4
22	129°35.9	135°28.2	01.6	5°40.3	49.5	147°37.4	39.1	306°00.7	12.5
23	144°38.4	150°27.8	00.4	20°43.0	49.5	162°39.3	38.8	321°03.3	12.6
Mer.pass. 13:23		ν -0.4' d -1.3' m -3.91		ν 2.8' d -0.0' m -0.61		ν 1.9' d -0.2' m -2.02		ν 2.6' d 0.1' m 0.59	

Feb 27 Sat		SHA	Mer.pass
Venus		9°12.7	12:57
Mars		235°38.5	21:46
Jupiter		18°41.2	12:17
Saturn		176°13.9	01:48
Feb 28 Sun		SHA	Mer.pass
Venus		8°03.8	12:57
Mars		235°48.1	21:42
Jupiter		18°27.6	12:14
Saturn		176°17.5	01:44
Mar 01 Mon		SHA	Mer.pass
Venus		6°55.2	12:58
Mars		235°57.0	21:37
Jupiter		18°14.0	12:11
Saturn		176°21.3	01:40
Horizontal parallax			
Venus:			0.1
Mars:			0.2

DUT1 = UT1-UTC = +0.0569 sec $\Delta T = TT-UT1 = +66.1271$ sec

2010 February 27 to Mar. 01 UT

Sun							Moon						
h	Sat	GHA	Dec	GHA	ν	Dec	d	HP					
0		176°48.0	S08°27.9	20°36.5	7.1'	N14°31.7	-13.8'	61.1'					
1		191°48.1	26.9	35°02.5	7.1'	14°17.9	-13.9'	61.1'					
2		206°48.2	26.0	49°28.7	7.2'	14°04.0	-14.0'	61.1'					
3		221°48.3	· · 25.1	63°54.9	7.3'	13°49.9	-14.1'	61.1'					
4		236°48.4	24.1	78°21.1	7.3'	13°35.8	-14.2'	61.1'					
5		251°48.5	23.2	92°47.4	7.4'	13°21.6	-14.3'	61.1'					
6		266°48.6	S08°22.2	107°13.8	7.4'	N13°07.3	-14.4'	61.1'					
7		281°48.7	21.3	121°40.2	7.5'	12°52.9	-14.5'	61.1'					
8		296°48.8	20.4	136°06.7	7.5'	12°38.4	-14.6'	61.1'					
9		311°48.9	· · 19.4	150°33.3	7.6'	12°23.9	-14.6'	61.2'					
10		326°49.1	18.5	164°59.9	7.7'	12°09.2	-14.7'	61.2'					
11		341°49.2	17.5	179°26.5	7.7'	11°54.5	-14.8'	61.2'					
12		356°49.3	S08°16.6	193°53.3	7.8'	N11°39.7	-14.9'	61.2'					
13		11°49.4	15.7	208°20.0	7.8'	11°24.8	-15.0'	61.2'					
14		26°49.5	14.7	222°46.9	7.9'	11°09.8	-15.0'	61.2'					
15		41°49.6	· · 13.8	237°13.7	7.9'	10°54.8	-15.1'	61.2'					
16		56°49.7	12.8	251°40.7	8.0'	10°39.7	-15.2'	61.2'					
17		71°49.8	11.9	266°07.7	8.0'	10°24.6	-15.2'	61.2'					
18		86°50.0	S08°11.0	280°34.7	8.1'	N10°09.3	-15.3'	61.2'					
19		101°50.1	10.0	295°01.8	8.1'	09°54.0	-15.4'	61.2'					
20		116°50.2	09.1	309°29.0	8.2'	09°38.7	-15.4'	61.2'					
21		131°50.3	· · 08.1	323°56.2	8.3'	09°23.2	-15.5'	61.2'					
22		146°50.4	07.2	338°23.4	8.3'	09°07.8	-15.5'	61.2'					
23		161°50.5	06.2	352°50.7	8.3'	08°52.2	-15.6'	61.2'					
SD = 16.1'							SD = 16.7'						
d = -0.9'													
Sun		GHA	Dec	GHA	ν	Dec	d	HP					
0		176°50.6	S08°05.3	7°18.1	8.4'	N08°36.6	-15.6'	61.2'					
1		191°50.7	04.4	21°45.5	8.4'	08°21.0	-15.7'	61.2'					
2		206°50.9	03.4	36°12.9	8.5'	08°05.3	-15.7'	61.2'					
3		221°51.0	· · 02.5	50°40.4	8.5'	07°49.6	-15.8'	61.2'					
4		236°51.1	01.5	65°07.9	8.6'	07°33.8	-15.8'	61.2'					
5		251°51.2	00°00.6	79°35.5	8.6'	07°17.9	-15.9'	61.2'					
6		266°51.3	S07°59.6	94°03.2	8.7'	N07°02.1	-15.9'	61.2'					
7		281°51.4	58.7	108°30.8	8.7'	06°46.2	-16.0'	61.2'					
8		296°51.6	57.7	122°58.6	8.8'	06°30.2	-16.0'	61.2'					
9		311°51.7	· · 56.8	137°26.3	8.8'	06°14.2	-16.0'	61.2'					
10		326°51.8	55.9	151°54.1	8.8'	05°58.2	-16.1'	61.2'					
11		341°51.9	54.9	166°22.0	8.9'	05°42.1	-16.1'	61.1'					
12		356°52.0	S07°54.0	180°49.9	8.9'	N05°26.0	-16.1'	61.1'					
13		11°52.1	53.0	195°17.8	9.0'	05°09.9	-16.1'	61.1'					
14		26°52.3	52.1	209°45.8	9.0'	04°53.8	-16.2'	61.1'					
15		41°52.4	· · 51.1	224°13.8	9.0'	04°37.6	-16.2'	61.1'					
16		56°52.5	50.2	238°41.8	9.1'	04°21.4	-16.2'	61.1'					
17		71°52.6	49.2	253°09.9	9.1'	04°05.2	-16.2'	61.1'					
18		86°52.7	S07°48.3	267°38.0	9.1'	N03°49.0	-16.2'	61.1'					
19		101°52.9	47.3	282°06.1	9.2'	03°32.8	-16.3'	61.1'					
20		116°53.0	46.4	296°34.3	9.2'	03°16.5	-16.3'	61.1'					
21		131°53.1	· · 45.4	311°02.5	9.2'	03°00.3	-16.3'	61.0'					
22		146°53.2	44.5	325°30.8	9.3'	02°44.0	-16.3'	61.0'					
23		161°53.3	43.5	339°59.1	9.3'	02°27.7	-16.3'	61.0'					
SD = 16.1'							SD = 16.7'						
d = -0.9'													
Mon		GHA	Dec	GHA	ν	Dec	d	HP					
0		176°53.4	S07°42.6	354°27.4	9.3'	N02°11.4	-16.3'	61.0'					
1		191°53.6	41.7	8°55.7	9.4'	01°55.1	-16.3'	61.0'					
2		206°53.7	40.7	23°24.1	9.4'	01°38.8	-16.3'	61.0'					
3		221°53.8	· · 39.8	37°52.5	9.4'	01°22.5	-16.3'	61.0'					
4		236°53.9	38.8	52°20.9	9.4'	01°06.2	-16.3'	60.9'					
5		251°54.0	37.9	66°49.3	9.5'	00°49.9	-16.3'	60.9'					
6		266°54.2	S07°36.9	81°17.8	9.5'	N00°33.6	-16.3'	60.9'					
7		281°54.3	36.0	95°46.3	9.5'	00°17.3	-16.3'	60.9'					
8		296°54.4	35.0	110°14.8	9.5'	N00°01.1	-16.3'	60.9'					
9		311°54.5	· · 34.1	124°43.3	9.6'	S00°15.2	16.3'	60.8'					
10		326°54.7	33.1	139°11.9	9.6'	00°31.5	16.2'	60.8'					
11		341°54.8	32.2	153°40.5	9.6'	00°47.7	16.2'	60.8'					
12		356°54.9	S07°31.2	168°09.1	9.6'	S01°03.9	16.2'	60.8'					
13		11°55.0	30.3	182°37.7	9.6'	01°20.1	16.2'	60.8'					
14		26°55.1	29.3	197°06.3	9.7'	01°36.3	16.2'	60.7'					
15		41°55.3	· · 28.4	211°35.0	9.7'	01°52.5	16.1'	60.7'					
16		56°55.4	27.4	226°03.6	9.7'	02°08.6	16.1'	60.7'					
17		71°55.5	26.5	240°32.3	9.7'	02°24.7	16.1'	60.7'					
18		86°55.6	S07°25.5	255°01.0	9.7'	S02°40.8	16.1'	60.7'					
19		101°55.8	24.6	269°29.7	9.7'	02°56.8	16.0'	60.6'					
20		116°55.9	23.6	283°58.4	9.7'	03°12.9	16.0'	60.6'					
21		131°56.0	· · 22.7	298°27.2	9.7'	03°28.9	16.0'	60.6'					
22		146°56.1	21.7	312°55.9	9.8'	03°44.8	15.9'	60.6'					
23		161°56.3	20.7	327°24.7	9.8'	04°00.7	15.9'	60.5'					
SD = 16.1'							SD = 16.6'						
d = -0.9'													

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	05:16	06:33	07:43	16:44	17:54	19:12
N 70°	05:20	06:30	07:33	16:54	17:57	19:07
68°	05:24	06:28	07:24	17:02	17:59	19:04
66°	05:26	06:25	07:17	17:09	18:01	19:01
64°	05:29	06:23	07:11	17:15	18:03	18:58
62°	05:30	06:22	07:06	17:20	18:05	18:56
60°	05:32	06:20	07:02	17:24	18:06	18:55
N 58°	05:33	06:18	06:58	17:28	18:08	18:53
56°	05:34	06:17	06:54	17:32	18:09	18:52
54°	05:35	06:16	06:51	17:35	18:10	18:51
52°	05:36	06:14	06:48	17:38	18:12	18:51
50°	05:36	06:13	06:46	17:40	18:13	18:50
45°	05:37	06:11	06:40	17:46	18:15	18:49
N 40°	05:37	06:08	06:35	17:50	18:18	18:49
35°	05:36	06:06	06:31	17:55	18:20	18:49
30°	05:35	06:03	06:27	17:58	18:22	18:50
20°	05:33	05:59	06:21	18:05	18:27	18:52
N 10°	05:29	05:54	06:15	18:10	18:32	18:56
0°	05:24	05:48	06:09	18:16	18:37	19:01
S 10°	05:18	05:42	06:03	18:21	18:43	19:07
20°	05:09	05:35	05:57	18:27	18:50	19:16
30°	04:57	05:26	05:50	18:35	18:59	19:27
35°	04:49	05:20	05:46	18:39	19:04	19:35
40°	04:40	05:13	05:41	18:43	19:11	19:44
45°	04:29	05:05	05:35	18:49	19:19	19:55
S 50°	04:14	04:55	05:29	18:55	19:29	20:10
52°	04:07	04:50	05:26	18:58	19:33	20:17
54°	03:59	04:45	05:22	19:02	19:39	20:24
56°	03:50	04:39	05:18	19:05	19:45	20:34
58°	03:39	04:32	05:14	19:10	19:51	20:44
S 60°	03:26	04:24	05:09	19:14	19:59	20:56

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	14:47	17:21	19:45	07:47	07:09	06:39
N 70°	15:06	17:26	19:40	07:26	07:00	06:38
68°	15:21	17:31	19:36	07:09	06:52	06:38
66°	15:33	17:35	19:33	06:55	06:46	06:38
64°	15:43	17:38	19:30	06:43	06:41	06:38
62°	15:51	17:41	19:28	06:33	06:36	06:38
60°	15:59	17:43	19:25	06:25	06:32	06:37

March 02, 03, 04 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	159°40.9	165°27.4	S03°59.1	35°45.8	N23°49.4	177°41.2	S08°38.6	336°05.9	N01°12.6	Alpheratz	357°46.4	29°08.9
1	174°43.3	180°27.1	57.8	50°48.6	49.4	192°43.1	38.4	351°08.5	12.7	Ankaa	353°18.3	-42°15.1
2	189°45.8	195°26.7	56.6	65°51.4	49.4	207°45.0	38.2	6°11.1	12.8	Schedar	349°43.9	56°35.8
3	204°48.2	210°26.3	•• 55.3	80°54.1	•• 49.4	222°46.9	•• 37.9	21°13.8	•• 12.9	Diphda	348°58.5	-17°55.9
4	219°50.7	225°25.9	54.0	95°56.9	49.3	237°48.8	37.7	36°16.4	12.9	Achernar	335°28.8	-57°11.3
5	234°53.2	240°25.5	52.8	110°59.7	49.3	252°50.7	37.5	51°19.0	13.0	Hamal	328°03.7	23°30.7
6	249°55.6	255°25.1	S03°51.5	126°02.5	N23°49.3	267°52.6	S08°37.2	66°21.6	N01°13.1	Polaris	319°11.3	89°18.9
7	264°58.1	270°24.7	50.2	141°05.2	49.3	282°54.5	37.0	81°24.3	13.2	Acamar	315°20.2	-40°16.0
8	280°00.6	285°24.3	49.0	156°08.0	49.3	297°56.4	36.8	96°26.9	13.2	Menkar	314°17.7	4°07.8
9	295°03.0	300°23.9	•• 47.7	171°10.8	•• 49.2	312°58.3	•• 36.6	111°29.5	•• 13.3	Mirfak	308°44.0	49°54.1
10	310°05.5	315°23.6	46.4	186°13.5	49.2	328°00.2	36.3	126°32.1	13.4	Aldebaran	290°52.1	16°31.8
11	325°08.0	330°23.2	45.2	201°16.3	49.2	343°02.1	36.1	141°34.7	13.5	Rigel	281°14.3	-8°11.5
12	340°10.4	345°22.8	S03°43.9	216°19.1	N23°49.2	358°04.0	S08°35.9	156°37.4	N01°13.5	Capella	280°37.9	46°00.7
13	355°12.9	0°22.4	42.6	231°21.8	49.1	13°05.9	35.7	171°40.0	13.6	Bellatrix	278°34.5	6°21.5
14	10°15.3	15°22.0	41.4	246°24.6	49.1	28°07.8	35.4	186°42.6	13.7	Elnath	278°15.6	28°37.0
15	25°17.8	30°21.6	•• 40.1	261°27.4	•• 49.1	43°09.7	•• 35.2	201°45.2	•• 13.8	Alnilam	275°48.7	-1°11.8
16	40°20.3	45°21.2	38.8	276°30.1	49.1	58°11.6	35.0	216°47.9	13.8	Betelgeuse	271°03.8	7°24.5
17	55°22.7	60°20.8	37.5	291°32.9	49.0	73°13.5	34.8	231°50.5	13.9	Canopus	263°57.0	-52°42.4
18	70°25.2	75°20.5	S03°36.3	306°35.6	N23°49.0	88°15.4	S08°34.5	246°53.1	N01°14.0	Sirius	258°35.7	-16°44.0
19	85°27.7	90°20.1	35.0	321°38.4	49.0	103°17.3	34.3	261°55.7	14.1	Adhara	255°14.2	-28°59.4
20	100°30.1	105°19.7	33.7	336°41.1	49.0	118°19.2	34.1	276°58.4	14.1	Procyon	245°02.0	5°11.8
21	115°32.6	120°19.3	•• 32.5	351°43.9	•• 48.9	133°21.1	•• 33.8	292°01.0	•• 14.2	Pollux	243°30.4	28°00.0
22	130°35.1	135°18.9	31.2	6°46.6	48.9	148°23.0	33.6	307°03.6	14.3	Avior	234°18.6	-59°32.8
23	145°37.5	150°18.5	29.9	21°49.4	48.9	163°24.9	33.4	322°06.2	14.4	Suhail	222°53.9	-43°28.7
Mer.pass. 13:19		ν -0.4' d -1.3' m -3.91		ν 2.8' d -0.0' m -0.58		ν 1.9' d -0.2' m -2.02		ν 2.6' d 0.1' m 0.59		Miaplacidus	221°39.6	-69°45.8
										Alphard	217°58.1	-8°42.4
										Regulus	207°45.7	11°54.8
										Dubhe	193°53.7	61°41.6
										Denebola	182°35.7	14°30.7
										Gienah	175°54.5	-17°36.1
										Acrux	173°11.6	-63°09.4
										Gacrux	172°03.3	-57°10.3
										Alioth	166°22.1	55°53.9
										Spica	158°33.6	-11°13.1
										Alkaid	153°00.3	49°15.4
										Hadar	148°51.1	-60°25.3
										Menkent	148°10.2	-36°25.3
										Arcturus	145°57.7	19°07.5
										Rigel Kent.	139°54.8	-60°52.5
										Kochab	137°19.0	74°06.4
										Zuben'ubi	137°07.9	-16°05.2
										Alphecca	126°12.9	26°40.5
										Antares	112°29.2	-26°27.3
										Atria	107°33.3	-69°02.5
										Sabik	102°15.3	-15°44.3
										Shaula	96°25.3	-37°06.6
										Rasalhague	96°08.7	12°32.9
										Eltanin	90°47.4	51°28.9
										Kaus Aust.	83°47.1	-34°22.7
										Vega	80°40.8	38°47.3
										Nunki	76°01.4	-26°17.0
										Altair	62°10.8	8°53.6
										Peacock	53°23.3	-56°42.0
										Deneb	49°33.6	45°18.8
										Enif	33°49.8	9°55.2
										Al Na'ir	27°47.0	-46°54.7
										Fomalhaut	15°26.9	-29°34.1
										Scheat	13°56.1	28°08.3
										Markab	13°41.1	15°16.6
										Mar 02 Tue	SHA	Mer.pass
										Venus	5°46.6	12:59
										Mars	236°05.0	21:33
										Jupiter	18°00.4	12:08
										Saturn	176°25.0	01:35
										Mar 03 Wed	SHA	Mer.pass
										Venus	4°38.2	12:59
										Mars	236°12.1	21:29
										Jupiter	17°46.8	12:05
										Saturn	176°28.9	01:31
										Mar 04 Thu	SHA	Mer.pass
										Venus	3°29.8	13:00
										Mars	236°18.5	21:24
										Jupiter	17°33.2	12:02
										Saturn	176°32.7	01:27
</												

h Sun Moon						
Tue	GHA	Dec	GHA	ν	Dec	d HP
0	176°56.4	S07°19.8	341°53.4	9.8'	S04°16.6	15.8' 60.5'
1	191°56.5	18.8	356°22.2	9.8'	04°32.5	15.8' 60.5'
2	206°56.6	17.9	10°51.0	9.8'	04°48.3	15.8' 60.4'
3	221°56.8	· · 16.9	25°19.7	9.8'	05°04.0	15.7' 60.4'
4	236°56.9	16.0	39°48.5	9.8'	05°19.7	15.7' 60.4'
5	251°57.0	15.0	54°17.3	9.8'	05°35.4	15.6' 60.4'
6	266°57.1	S07°14.1	68°46.1	9.8'	S05°51.0	15.6' 60.3'
7	281°57.3	13.1	83°14.9	9.8'	06°06.6	15.5' 60.3'
8	296°57.4	12.2	97°43.7	9.8'	06°22.1	15.5' 60.3'
9	311°57.5	· · 11.2	112°12.5	9.8'	06°37.6	15.4' 60.2'
10	326°57.7	10.3	126°41.3	9.8'	06°53.0	15.4' 60.2'
11	341°57.8	09.3	141°10.1	9.8'	07°08.3	15.3' 60.2'
12	356°57.9	S07°08.3	155°38.9	9.8'	S07°23.6	15.2' 60.2'
13	11°58.0	07.4	170°07.7	9.8'	07°38.9	15.2' 60.1'
14	26°58.2	06.4	184°36.5	9.8'	07°54.1	15.1' 60.1'
15	41°58.3	· · 05.5	199°05.3	9.8'	08°09.2	15.1' 60.1'
16	56°58.4	04.5	213°34.1	9.8'	08°24.2	15.0' 60.0'
17	71°58.5	03.6	228°02.8	9.8'	08°39.2	14.9' 60.0'
18	86°58.7	S07°02.6	242°31.6	9.8'	S08°54.1	14.9' 60.0'
19	101°58.8	01.7	257°00.4	9.8'	09°09.0	14.8' 59.9'
20	116°58.9	07°00.7	271°29.1	9.8'	09°23.8	14.7' 59.9'
21	131°59.1	06°59.7	285°57.9	9.7'	09°38.5	14.6' 59.9'
22	146°59.2	58.8	300°26.6	9.7'	09°53.1	14.6' 59.8'
23	161°59.3	57.8	314°55.4	9.7'	10°07.7	14.5' 59.8'
SD = 16.1' $d = -1.0'$ SD = 16.5'						
Wed	GHA	Dec	GHA	ν	Dec	d HP
0	176°59.5	S06°56.9	329°24.1	9.7'	S10°22.2	14.4' 59.8'
1	191°59.6	55.9	343°52.8	9.7'	10°36.6	14.3' 59.7'
2	206°59.7	55.0	358°21.5	9.7'	10°50.9	14.3' 59.7'
3	221°59.8	· · 54.0	12°50.2	9.7'	11°05.2	14.2' 59.6'
4	237°00.0	53.1	27°18.9	9.7'	11°19.3	14.1' 59.6'
5	252°00.1	52.1	41°47.6	9.7'	11°33.4	14.0' 59.6'
6	267°00.2	S06°51.1	56°16.2	9.6'	S11°47.4	13.9' 59.5'
7	282°00.4	50.2	70°44.9	9.6'	12°01.4	13.8' 59.5'
8	297°00.5	49.2	85°13.5	9.6'	12°15.2	13.7' 59.5'
9	312°00.6	· · 48.3	99°42.1	9.6'	12°29.0	13.7' 59.4'
10	327°00.8	47.3	114°10.7	9.6'	12°42.6	13.6' 59.4'
11	342°00.9	46.3	128°39.3	9.6'	12°56.2	13.5' 59.4'
12	357°01.0	S06°45.4	143°07.9	9.6'	S13°09.7	13.4' 59.3'
13	12°01.2	44.4	157°36.4	9.5'	13°23.0	13.3' 59.3'
14	27°01.3	43.5	172°05.0	9.5'	13°36.3	13.2' 59.2'
15	42°01.4	· · 42.5	186°33.5	9.5'	13°49.5	13.1' 59.2'
16	57°01.6	41.5	201°02.0	9.5'	14°02.6	13.0' 59.2'
17	72°01.7	40.6	215°30.5	9.5'	14°15.6	12.9' 59.1'
18	87°01.8	S06°39.6	229°59.0	9.4'	S14°28.5	12.8' 59.1'
19	102°02.0	38.7	244°27.4	9.4'	14°41.3	12.7' 59.0'
20	117°02.1	37.7	258°55.8	9.4'	14°54.1	12.6' 59.0'
21	132°02.2	· · 36.7	273°24.2	9.4'	15°06.7	12.5' 59.0'
22	147°02.4	35.8	287°52.6	9.4'	15°19.2	12.4' 58.9'
23	162°02.5	34.8	302°21.0	9.4'	15°31.6	12.3' 58.9'
SD = 16.1' $d = -1.0'$ SD = 16.3'						
Thu	GHA	Dec	GHA	ν	Dec	d HP
0	177°02.6	S06°33.9	316°49.4	9.3'	S15°43.8	12.2' 58.9'
1	192°02.8	32.9	331°17.7	9.3'	15°56.0	12.1' 58.8'
2	207°02.9	31.9	345°46.0	9.3'	16°08.1	12.0' 58.8'
3	222°03.0	· · 31.0	0°14.3	9.3'	16°20.1	11.9' 58.7'
4	237°03.2	30.0	14°42.6	9.3'	16°31.9	11.8' 58.7'
5	252°03.3	29.1	29°10.8	9.2'	16°43.7	11.6' 58.7'
6	267°03.4	S06°28.1	43°39.0	9.2'	S16°55.3	11.5' 58.6'
7	282°03.6	27.1	58°07.3	9.2'	17°06.9	11.4' 58.6'
8	297°03.7	26.2	72°35.4	9.2'	17°18.3	11.3' 58.5'
9	312°03.9	· · 25.2	87°03.6	9.1'	17°29.6	11.2' 58.5'
10	327°04.0	24.2	101°31.8	9.1'	17°40.8	11.1' 58.5'
11	342°04.1	23.3	115°59.9	9.1'	17°51.8	11.0' 58.4'
12	357°04.3	S06°22.3	130°28.0	9.1'	S18°02.8	10.8' 58.4'
13	12°04.4	21.4	144°56.1	9.1'	18°13.6	10.7' 58.3'
14	27°04.5	20.4	159°24.1	9.0'	18°24.3	10.6' 58.3'
15	42°04.7	· · 19.4	173°52.2	9.0'	18°34.9	10.5' 58.2'
16	57°04.8	18.5	188°20.2	9.0'	18°45.4	10.4' 58.2'
17	72°05.0	17.5	202°48.2	9.0'	18°55.8	10.2' 58.2'
18	87°05.1	S06°16.5	217°16.2	9.0'	S19°06.0	10.1' 58.1'
19	102°05.2	15.6	231°44.2	8.9'	19°16.1	10.0' 58.1'
20	117°05.4	14.6	246°12.1	8.9'	19°26.1	9.9' 58.0'
21	132°05.5	· · 13.7	260°40.0	8.9'	19°36.0	9.7' 58.0'
22	147°05.6	12.7	275°07.9	8.9'	19°45.7	9.6' 58.0'
23	162°05.8	11.7	289°35.8	8.9'	19°55.3	9.5' 57.9'
SD = 16.1' $d = -1.0'$ SD = 16.1'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	05:01	06:19	07:27	16:59	18:08	19:26
N 70°	05:07	06:17	07:19	17:07	18:09	19:20
68°	05:11	06:16	07:12	17:14	18:10	19:15
66°	05:15	06:15	07:06	17:19	18:11	19:10
64°	05:19	06:13	07:01	17:24	18:12	19:07
62°	05:21	06:12	06:57	17:28	18:13	19:04
60°	05:23	06:11	06:53	17:32	18:14	19:02
N 58°	05:25	06:11	06:50	17:35	18:15	19:00
56°	05:27	06:10	06:47	17:38	18:15	18:58
54°	05:28	06:09	06:44	17:41	18:16	18:57
52°	05:29	06:08	06:42	17:43	18:17	18:56
50°	05:30	06:07	06:40	17:45	18:18	18:55
45°	05:31	06:05	06:35	17:50	18:19	18:53
N 40°	05:32	06:04	06:31	17:54	18:21	18:52
35°	05:33	06:02	06:27	17:57	18:23	18:52
30°	05:32	06:00	06:24	18:00	18:24	18:52
20°	05:31	05:56	06:18	18:06	18:28	18:53
N 10°	05:28	05:52	06:13	18:11	18:32	18:56
0°	05:24	05:48	06:09	18:15	18:36	19:00
S 10°	05:18	05:43	06:04	18:20	18:41	19:06
20°	05:10	05:36	05:58	18:25	18:47	19:13
30°	04:59	05:28	05:52	18:31	18:55	19:24
35°	04:52	05:23	05:49	18:35	19:00	19:31
40°	04:44	05:17	05:44	18:39	19:06	19:39
45°	04:33	05:10	05:40	18:43	19:13	19:49
S 50°	04:20	05:00	05:34	18:49	19:22	20:02
52°	04:13	04:56	05:31	18:52	19:26	20:09
54°	04:06	04:51	05:28	18:55	19:31	20:16
56°	03:58	04:46	05:25	18:58	19:36	20:24
58°	03:48	04:40	05:21	19:01	19:42	20:34
S 60°	03:36	04:33	05:17	19:05	19:49	20:45

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	22:11	· · ·	01:06	06:09	05:33	04:30
N 70°	21:54	· · ·	00:20	06:17	05:53	05:17
68°	21:41	23:51	· · ·	06:24	06:08	05:48
66°	21:30	23:29	· · ·	06:30	06:21	06:11
64°	21:21	23:12	· · ·	06:35	06:32	06:29
62°	21:13	22:58	· · ·	06:39	06:41	06:44
60°	21:06	22:46	· · ·	06:43	06:49	06:57
N 58°	21:01	22:36	· · ·	06:46	06:56	07:08
56°	20:55	22:27	23:56	06:49	07:02	07:18
54°	20:51	22:19	23:45	06:52	07:08	07:26
52°	20:47	22:12	23:35	06:54	07:13	07:34
50°	20:43	22:06	23:26	06:57	07:17	07:41
45°	20:35	21:53	23:08	07:02	07:27	07:56
N 40°	20:28	21:42	22:53	07:06	07:36	08:08
35°	20:23	21:32	22:40	07:09	07:43	08:19
30°	20:18	21:24	22:29	07:12	07:49	08:28
20°	20:09	21:10	22:11	07:18	08:00	08:44
N 10°	20:01	20:58	21:54	07:23	08:10	08:58
0°	19:55	20:47	21:39	07:27	08:19	09:12
S 10°	19:48	20:35	21:24	07:32	08:28	09:25
20°	19:40	20:23	21:08	07:37	08:38	09:39
30°	19:32	20:10	20:50	07:42	08:49	09:55
35°	19:27	20:02	20:40	07:46	08:56	10:05
40°	19:22	19:53	20:28	07:49	09:03	10:16
45°	19:16	19:43	20:14	07:53	09:12	10:29
S 50°	19:09	19:31	19:57	07:59	09:23	10:45
52°	19:06	19:25	19:49	08:01	09:28	10:52
54°	19:02	19:19	19:40	08:04	09:33	11:00
56°	18:58	19:12	19:30	08:06	09:39	11:10
58°	18:53	19:04	19:18	08:10	09:46	11:20
S 60°	18:48	18:55	19:05	08:13	09:53	11:32

Day	Sun		Mer. Pass hh:mm	Moon		Age 16-18 97-85%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Upper hh:mm	Lower hh:mm	
02	12:14	12:08	12:12	01:15	13:41	
03	12:02	11:56	12:12	02:07	14:33	
04	11:49	11:43	12:12	02:59	15:25	

March 05, 06, 07 UT (Fri., Sat., Sun.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Fri	GHA	Dec	GHA	Dec		GHA	Dec		GHA	Dec		GHA	Dec		SHA	Dec	
0	163°38.3	164°59.8	S02°27.5	39°02.3	N23°47.1	179°57.9	S08°22.3	339°14.9	N01°18.1			Alpheratz	357°46.4	29°08.8			
1	177°40.7	179°59.4	26.2	54°04.9	47.0	194°59.8	22.1	354°17.5	18.2			Ankaa	353°18.3	-42°15.1			
2	192°43.2	194°59.1	24.9	69°07.6	47.0	210°01.7	21.8	9°20.2	18.3			Schedar	349°44.0	56°35.7			
3	207°45.7	209°58.7	· · 23.6	84°10.3	· · 46.9	225°03.6	· · 21.6	24°22.8	· · 18.4			Diphda	348°58.5	-17°55.9			
4	222°48.1	224°58.3	22.4	99°13.0	46.9	240°05.5	21.4	39°25.4	18.4			Achernar	335°28.8	-57°11.2			
5	237°50.6	239°57.9	21.1	114°15.6	46.8	255°07.4	21.1	54°28.1	18.5			Hamal	328°03.7	23°30.7			
6	252°53.0	254°57.6	S02°19.8	129°18.3	N23°46.8	270°09.3	S08°20.9	69°30.7	N01°18.6			Polaris	319°12.3	89°18.9			
7	267°55.5	269°57.2	18.5	144°21.0	46.7	285°11.2	20.7	84°33.3	18.7			Acamar	315°20.3	-40°16.0			
8	282°58.0	284°56.8	17.2	159°23.6	46.7	300°13.1	20.5	99°35.9	18.7			Menkar	314°17.7	4°07.8			
9	298°00.4	299°56.4	· · 16.0	174°26.3	· · 46.7	315°15.0	· · 20.2	114°38.6	· · 18.8			Mirfak	308°44.1	49°54.1			
10	313°02.9	314°56.1	14.7	189°29.0	46.6	330°16.9	20.0	129°41.2	18.9			Aldebaran	290°52.2	16°31.8			
11	328°05.4	329°55.7	13.4	204°31.6	46.6	345°18.8	19.8	144°43.8	19.0			Rigel	281°14.3	-8°11.5			
12	343°07.8	344°55.3	S02°12.1	219°34.3	N23°46.5	0°20.7	S08°19.6	159°46.5	N01°19.0			Capella	280°38.0	46°00.7			
13	358°10.3	359°54.9	10.9	234°36.9	46.5	15°22.6	19.3	174°49.1	19.1			Bellatrix	278°34.5	6°21.5			
14	13°12.8	14°54.5	09.6	249°39.6	46.4	30°24.5	19.1	189°51.7	19.2			Elnath	278°15.6	28°37.0			
15	28°15.2	29°54.2	· · 08.3	264°42.3	· · 46.4	45°26.4	· · 18.9	204°54.3	· · 19.3			Alnilam	275°48.7	-1°11.8			
16	43°17.7	44°53.8	07.0	279°44.9	46.3	60°28.3	18.7	219°57.0	19.4			Betelgeuse	271°03.8	7°24.5			
17	58°20.2	59°53.4	05.7	294°47.6	46.3	75°30.2	18.4	234°59.6	19.4			Canopus	263°57.0	-52°42.4			
18	73°22.6	74°53.0	S02°04.5	309°50.2	N23°46.2	90°32.1	S08°18.2	250°02.2	N01°19.5			Sirius	258°35.7	-16°44.0			
19	88°25.1	89°52.7	03.2	324°52.9	46.2	105°34.0	18.0	265°04.9	19.6			Adhara	255°14.2	-28°59.4			
20	103°27.5	104°52.3	01.9	339°55.5	46.1	120°35.9	17.7	280°07.5	19.7			Procyon	245°02.0	5°11.8			
21	118°30.0	119°51.9	02°00.6	354°58.2	· · 46.1	135°37.8	· · 17.5	295°10.1	· · 19.7			Pollux	243°30.4	28°00.0			
22	133°32.5	134°51.5	01°59.3	10°00.8	46.0	150°39.7	17.3	310°12.7	19.8			Avior	234°18.6	-59°32.8			
23	148°34.9	149°51.2	58.1	25°03.4	46.0	165°41.6	17.1	325°15.4	19.9			Suhail	222°53.9	-43°28.7			
Mer.pass. 13:07			ν -0.4' d -1.3' m -3.91	ν 2.7' d -0.0' m -0.50	ν 1.9' d -0.2' m -2.02	ν 2.6' d 0.1' m 0.57											

Sat	GHA	Dec	GHA	Dec		GHA	Dec		GHA	Dec	
0	163°37.4	164°50.8	S01°56.8	40°06.1	N23°45.9	180°43.5	S08°16.8	340°18.0	N01°20.0		
1	178°39.9	179°50.4	55.5	55°08.7	45.8	195°45.4	16.6	355°20.6	20.1		
2	193°42.3	194°50.0	54.2	70°11.4	45.8	210°47.3	16.4	10°23.3	20.1		
3	208°44.8	209°49.7	· · 52.9	85°14.0	· · 45.7	225°49.2	· · 16.2	25°25.9	· · 20.2		
4	223°47.3	224°49.3	51.6	100°16.7	45.7	240°51.1	15.9	40°28.5	20.3		
5	238°49.7	239°48.9	50.4	115°19.3	45.6	255°53.0	15.7	55°31.2	20.4		
6	253°52.2	254°48.5	S01°49.1	130°21.9	N23°45.6	270°54.9	S08°15.5	70°33.8	N01°20.4		
7	268°54.7	269°48.2	47.8	145°24.6	45.5	285°56.8	15.2	85°36.4	20.5		
8	283°57.1	284°47.8	46.5	160°27.2	45.5	300°58.7	15.0	100°39.0	20.6		
9	298°59.6	299°47.4	· · 45.2	175°29.8	· · 45.4	316°00.6	· · 14.8	115°41.7	· · 20.7		
10	314°02.0	314°47.1	44.0	190°32.5	45.4	331°02.5	14.6	130°44.3	20.8		
11	329°04.5	329°46.7	42.7	205°35.1	45.3	346°04.4	14.3	145°46.9	20.8		
12	344°07.0	344°46.3	S01°41.4	220°37.7	N23°45.2	1°06.3	S08°14.1	160°49.6	N01°20.9		
13	359°09.4	359°45.9	40.1	235°40.3	45.2	16°08.2	13.9	175°52.2	21.0		
14	14°11.9	14°45.6	38.8	250°43.0	45.1	31°10.1	13.7	190°54.8	21.1		
15	29°14.4	29°45.2	· · 37.6	265°45.6	· · 45.1	46°12.0	· · 13.4	205°57.5	· · 21.1		
16	44°16.8	44°44.8	36.3	280°48.2	45.0	61°13.9	13.2	221°00.1	21.2		
17	59°19.3	59°44.4	35.0	295°50.8	45.0	76°15.8	13.0	236°02.7	21.3		
18	74°21.8	74°44.1	S01°33.7	310°53.4	N23°44.9	91°17.7	S08°12.7	251°05.4	N01°21.4		
19	89°24.2	89°43.7	32.4	325°56.1	44.8	106°19.6	12.5	266°08.0	21.5		
20	104°26.7	104°43.3	31.1	340°58.7	44.8	121°21.5	12.3	281°10.6	21.5		
21	119°29.1	119°43.0	· · 29.9	356°01.3	· · 44.7	136°23.4	· · 12.1	296°13.2	· · 21.6		
22	134°31.6	134°42.6	28.6	11°03.9	44.7	151°25.3	11.8	311°15.9	21.7		
23	149°34.1	149°42.2	27.3	26°06.5	44.6	166°27.2	11.6	326°18.5	21.8		
Mer.pass. 13:03			ν -0.4' d -1.3' m -3.91	ν 2.6' d -0.1' m -0.47	ν 1.9' d -0.2' m -2.02	ν 2.6' d 0.1' m 0.57					

Sun	GHA	Dec	GHA	Dec		GHA	Dec		GHA	Dec	
0	164°36.5	164°41.8	S01°26.0	41°09.1	N23°44.5	181°29.1	S08°11.4	341°21.1	N01°21.8		
1	179°39.0	179°41.5	24.7	56°11.7	44.5	196°31.0	11.2	356°23.8	21.9		
2	194°41.5	194°41.1	23.4	71°14.3	44.4	211°32.9	10.9	11°26.4	22.0		
3	209°43.9	209°40.7	· · 22.2	86°16.9	· · 44.4	226°34.8	· · 10.7	26°29.0	· · 22.1		
4	224°46.4	224°40.3	20.9	101°19.5	44.3	241°36.7	10.5	41°31.7	22.2		
5	239°48.9	239°40.0	19.6	116°22.1	44.2	256°38.6	10.2	56°34.3	22.2		
6	254°51.3	254°39.6	S01°18.3	131°24.7	N23°44.2	271°40.5	S08°10.0	71°36.9	N01°22.3		
7	269°53.8	269°39.2	17.0	146°27.3	44.1	286°42.4	09.8	86°39.6	22.4		
8	284°56.3	284°38.9	15.7	161°29.9	44.1	301°44.3	09.6	101°42.2	22.5		
9	299°58.7	299°38.5	· · 14.5	176°32.5	· · 44.0	316°46.2	· · 09.3	116°44.8	· · 22.6		
10	315°01.2	314°38.1	13.2	191°35.1	43.9	331°48.1	09.1	131°47.5	22.6		
11	330°03.6	329°37.7	11.9	206°37.7	43.9	346°50.0	08.9	146°50.1	22.7		
12	345°06.1	344°37.4	S01°10.6	221°40.3	N23°43.8	1°51.9	S08°08.7	161°52.7	N01°22.8		
13	0°08.6	359°37.0	09.3	236°42.9	43.7	16°53.8	08.4	176°55.4	22.9		
14	15°11.0	14°36.6	08.0	251°45.5	43.7	31°55.7	08.2	191°58.0	22.9		
15	30°13.5	29°36.3	· · 06.8	266°48.1	· · 43.6	46°57.6	· · 08.0	207°00.6	· · 23.0		
16	45°16.0	44°35.9	05.5	281°50.7	43.5	61°59.5	07.7	222°03.3	23.1		
17	60°18.4	59°35.5	04.2	296°53.3	43.5	77°01.4	07.5	237°05.9	23.2		
18	75°20.9	74°35.2	S01°02.9	311°55.8	N23°43.4	92°03.3	S08°07.3	252°08.5	N01°23.3		
19	90°23.4	89°34.8	01.6	326°58.4	43.4	107°05.2	07.1	267°11.1	23.3		
20	105°25.8	104°34.4	01°00.3	342°01.0	43.3	122°07.1	06.8	282°13.8	23.4		
21	120°28.3	119°34.0	00°59.1	357°03.6	· · 43.2	137°09.0	· · 06.6	297°16.4	· · 23.5		
22	135°30.8	134°33.7	57.8	12°06.2	43.2	152°10.9	06.4	312°19.0	23.6		
23	150°33.2	149°33.3	56.5	27°08.7	43.1	167°12.8	06.2	327°21.7	23.6		
Mer.pass. 12:59			ν -0.4' d -1.3' m -3.91	ν 2.6' d -0.1' m -0.45	ν 1.9' d -0.2' m -2.02	ν 2.6' d 0.1' m 0.57					

Mar 05 Fri			SHA	Mer.pass
Venus			2°21.6	13:00
Mars			236°24.0	21:20
Jupiter			17°19.7	11:59
Saturn			176°36.7	01:23

Mar 06 Sat			SHA	Mer.pass
Venus			1°13.4	13:01
Mars			236°28.7	21:16
Jupiter			17°06.1	11:56
Saturn			176°40.6	01:19

Mar 07 Sun			SHA	Mer.pass
Venus			0°05.3	13:02
Mars			236°32.6	21:12
Jupiter			16°52.6	11:53
Saturn			176°44.6	01:14

Horizontal parallax			
Venus:			0.1
Mars:			0.2

h	Sun			Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	177°05.9	S06°10.8	304°03.7	8.8'	S20°04.8	9.4'	57.9'
1	192°06.1	09.8	318°31.5	8.8'	20°14.2	9.2'	57.8'
2	207°06.2	08.8	332°59.3	8.8'	20°23.4	9.1'	57.8'
3	222°06.3	· · 07.9	347°27.2	8.8'	20°32.6	9.0'	57.8'
4	237°06.5	06.9	1°54.9	8.8'	20°41.5	8.9'	57.7'
5	252°06.6	05.9	16°22.7	8.8'	20°50.4	8.7'	57.7'
6	267°06.8	S06°05.0	30°50.5	8.7'	S20°59.1	8.6'	57.6'
7	282°06.9	04.0	45°18.2	8.7'	21°07.7	8.5'	57.6'
8	297°07.0	03.0	59°45.9	8.7'	21°16.2	8.3'	57.6'
9	312°07.2	· · 02.1	74°13.7	8.7'	21°24.5	8.2'	57.5'
10	327°07.3	01.1	88°41.3	8.7'	21°32.7	8.1'	57.5'
11	342°07.5	06°00.1	103°09.0	8.7'	21°40.8	7.9'	57.4'
12	357°07.6	S05°59.2	117°36.7	8.7'	S21°48.7	7.8'	57.4'
13	12°07.8	58.2	132°04.3	8.6'	21°56.5	7.7'	57.4'
14	27°07.9	57.2	146°32.0	8.6'	22°04.2	7.5'	57.3'
15	42°08.0	· · 56.3	160°59.6	8.6'	22°11.7	7.4'	57.3'
16	57°08.2	55.3	175°27.2	8.6'	22°19.1	7.3'	57.2'
17	72°08.3	54.3	189°54.8	8.6'	22°26.4	7.1'	57.2'
18	87°08.5	S05°53.4	204°22.4	8.6'	S22°33.5	7.0'	57.2'
19	102°08.6	52.4	218°50.0	8.6'	22°40.5	6.9'	57.1'
20	117°08.7	51.4	233°17.6	8.6'	22°47.4	6.7'	57.1'
21	132°08.9	· · 50.5	247°45.1	8.6'	22°54.1	6.6'	57.1'
22	147°09.0	49.5	262°12.7	8.5'	23°00.7	6.5'	57.0'
23	162°09.2	48.5	276°40.2	8.5'	23°07.2	6.3'	57.0'
SD = 16.1' d = -1.0'							
SD = 15.8'							

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	177°09.3	S05°47.6	291°07.8	8.5'	S23°13.5	6.2'	56.9'
1	192°09.5	46.6	305°35.3	8.5'	23°19.7	6.0'	56.9'
2	207°09.6	45.6	320°02.8	8.5'	23°25.7	5.9'	56.9'
3	222°09.8	· · 44.7	334°30.4	8.5'	23°31.6	5.8'	56.8'
4	237°09.9	43.7	348°57.9	8.5'	23°37.4	5.6'	56.8'
5	252°10.0	42.7	3°25.4	8.5'	23°43.0	5.5'	56.8'
6	267°10.2	S05°41.8	17°52.9	8.5'	S23°48.5	5.3'	56.7'
7	282°10.3	40.8	32°20.4	8.5'	23°53.8	5.2'	56.7'
8	297°10.5	39.8	46°48.0	8.5'	23°59.0	5.1'	56.6'
9	312°10.6	· · 38.8	61°15.5	8.5'	24°04.1	4.9'	56.6'
10	327°10.8	37.9	75°43.0	8.5'	24°09.0	4.8'	56.6'
11	342°10.9	36.9	90°10.5	8.5'	24°13.8	4.7'	56.5'
12	357°11.1	S05°35.9	104°38.0	8.5'	S24°18.5	4.5'	56.5'
13	12°11.2	35.0	119°05.6	8.5'	24°23.0	4.4'	56.5'
14	27°11.3	34.0	133°33.1	8.5'	24°27.4	4.2'	56.4'
15	42°11.5	· · 33.0	148°00.6	8.5'	24°31.6	4.1'	56.4'
16	57°11.6	32.1	162°28.2	8.5'	24°35.7	4.0'	56.4'
17	72°11.8	31.1	176°55.7	8.6'	24°39.6	3.8'	56.3'
18	87°11.9	S05°30.1	191°23.3	8.6'	S24°43.5	3.7'	56.3'
19	102°12.1	29.1	205°50.8	8.6'	24°47.1	3.5'	56.2'
20	117°12.2	28.2	220°18.4	8.6'	24°50.7	3.4'	56.2'
21	132°12.4	· · 27.2	234°46.0	8.6'	24°54.1	3.3'	56.2'
22	147°12.5	26.2	249°13.6	8.6'	24°57.3	3.1'	56.1'
23	162°12.7	25.3	263°41.2	8.6'	25°00.5	3.0'	56.1'
SD = 16.1' d = -1.0'							
SD = 15.5'							

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	177°12.8	S05°24.3	278°08.8	8.6'	S25°03.4	2.8'	56.1'
1	192°13.0	23.3	292°36.5	8.7'	25°06.3	2.7'	56.0'
2	207°13.1	22.3	307°04.1	8.7'	25°09.0	2.6'	56.0'
3	222°13.3	· · 21.4	321°31.8	8.7'	25°11.5	2.4'	56.0'
4	237°13.4	20.4	335°59.5	8.7'	25°14.0	2.3'	55.9'
5	252°13.6	19.4	350°27.2	8.7'	25°16.3	2.1'	55.9'
6	267°13.7	S05°18.5	4°54.9	8.7'	S25°18.4	2.0'	55.9'
7	282°13.9	17.5	19°22.7	8.8'	25°20.4	1.9'	55.8'
8	297°14.0	16.5	33°50.5	8.8'	25°22.3	1.7'	55.8'
9	312°14.2	· · 15.5	48°18.3	8.8'	25°24.0	1.6'	55.8'
10	327°14.3	14.6	62°46.1	8.8'	25°25.6	1.5'	55.8'
11	342°14.5	13.6	77°13.9	8.9'	25°27.1	1.3'	55.7'
12	357°14.6	S05°12.6	91°41.8	8.9'	S25°28.4	1.2'	55.7'
13	12°14.8	11.6	106°09.7	8.9'	25°29.6	1.1'	55.7'
14	27°14.9	10.7	120°37.6	8.9'	25°30.6	0.9'	55.6'
15	42°15.1	· · 09.7	135°05.5	9.0'	25°31.6	0.8'	55.6'
16	57°15.2	08.7	149°33.5	9.0'	25°32.3	0.6'	55.6'
17	72°15.4	07.8	164°01.5	9.0'	25°33.0	0.5'	55.5'
18	87°15.5	S05°06.8	178°29.6	9.1'	S25°33.5	0.4'	55.5'
19	102°15.7	05.8	192°57.6	9.1'	25°33.9	0.2'	55.5'
20	117°15.8	04.8	207°25.7	9.1'	25°34.1	0.1'	55.5'
21	132°16.0	· · 03.9	221°53.9	9.2'	25°34.2	-0.0'	55.4'
22	147°16.1	02.9	236°22.0	9.2'	25°34.2	-0.2'	55.4'
23	162°16.3	01.9	250°50.2	9.2'	25°34.0	-0.3'	55.4'
SD = 16.1' d = -1.0'							
SD = 15.3'							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	04:45	06:04	07:11	17:13	18:21	19:41
N 70°	04:53	06:04	07:05	17:20	18:21	19:32
68°	04:59	06:04	06:59	17:25	18:21	19:26
66°	05:04	06:03	06:55	17:30	18:21	19:21
64°	05:08	06:03	06:51	17:33	18:21	19:16
62°	05:12	06:03	06:47	17:37	18:21	19:12
60°	05:15	06:03	06:44	17:40	18:21	19:09
N 58°	05:17	06:02	06:42	17:42	18:21	19:07
56°	05:19	06:02	06:39	17:44	18:22	19:05
54°	05:21	06:02	06:37	17:46	18:22	19:03
52°	05:23	06:01	06:35	17:48	18:22	19:01
50°	05:24	06:01	06:33	17:50	18:22	19:00
45°	05:26	06:00	06:29	17:54	18:23	18:57
N 40°	05:28	05:59	06:26	17:57	18:24	18:55
35°	05:29	05:58	06:23	18:00	18:25	18:54
30°	05:29	05:57	06:21	18:02	18:26	18:54
20°	05:28	05:54	06:16	18:07	18:29	18:54
N 10°	05:26	05:51	06:12	18:11	18:32	18:56
0°	05:23	05:47	06:08	18:15	18:35	18:59
S 10°	05:18	05:43	06:04	18:18	18:40	19:04
20°	05:11	05:37	05:59	18:23	18:45	19:11
30°	05:02	05:30	05:54	18:28	18:52	19:20
35°	04:55	05:26	05:51	18:31	18:56	19:26
40°	04:48	05:20	05:48	18:34	19:01	19:34
45°	04:38	05:14	05:44	18:38	19:08	19:43
S 50°	04:26	05:06	05:39	18:43	19:16	19:55
52°	04:20	05:02	05:36	18:45	19:19	20:01
54°	04:13	04:58	05:34	18:47	19:24	20:08
56°	04:05	04:53	05:31	18:50	19:28	20:15
58°	03:57	04:47	05:28	18:53	19:34	20:24
S 60°	03:46	04:41	05:25	18:56	19:40	20:34

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	02:22	■	■	05:08	■	■
66°	01:35	04:10	■	05:56	05:13	■
64°	01:05	02:57	04:44	06:27	06:27	06:33
62°	00:42	02:21	03:47	06:51	07:03	07:29
60°	00:24	01:56	03:15	07:09	07:29	08:02
N 58°	00:09	01:36	02:50	07:25	07:50	08:27
56°	· · ·	01:19	02:31	07:38	08:07	08:46
54°	· · ·	01:05	02:14	07:50	08:21	09:03
52°	· · ·	00:52	02:01	08:00	08:34	09:17
50°	· · ·	00:41	01:48	08:10	08:45	09:29
45°	· · ·	00:19	01:23	08:29	09:08	09:54
N 40°	· · ·	00:01	01:03	08:45	09:27	10:15
35°	23:45	· · ·	00:46	08:59	09:43	10:32
30°	23:32	· · ·	00:32	09:10	09:56	10:46
20°	23:10	· · ·	00:07	09:31	10:20	11:11
N 10°	22:51	23:46	· · ·	09:48	10:40	11:32
0°	22:33	23:27	· · ·	10:05	10:59	11:52
S 10°	22:15	23:07	· · ·	10:21	11:18	12:12
20°	21:56	22:46	23:38	10:39	11:38	12:34
30°	21:34	22:22	23:14	11:00	12:01	12:59
35°	21:21	22:08	22:59	11:12	12:15	13:13
40°	21:07	21:52	22:42	11:26	12:31	13:30
45°	20:50	21:32	22:22	11:42	12:50	13:50
S 50°	20:28	21:08	21:56	12:03	13:14	14:16
52°	20:18	20:56	21:44	12:12	13:26	14:28
54°	20:07	20:43	21:30	12:23	13:39	14:42
56°	19:54	20:27	21:13	12:36	13:54	14:59
58°	19:39	20:09	20:54	12:50	14:12	15:18
S 60°	19:21	19:47	20:29	13:08	14:34	15:43

Day	Sun		Mer.	Moon		Age
	Eqn.of Time	12h	Pass	Mer.Pass.	Lower	19-21
	mm:ss	mm:ss	hh:mm	hh:mm	hh:mm	76-57%
05	11:36	11:30	12:11	03:52	16:19	
06	11:23	11:16	12:11	04:46	17:13	
07	11:09	11:02	12:11	05:40	18:06	

Aries		Venus		Mars		Jupiter		Saturn	
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	165° 35.7	164° 32.9	S00° 55.2	42° 11.3	N23° 43.0	182° 14.7	S08° 05.9	342° 24.3	N01° 23.7
1	180° 38.1	179° 32.6	53.9	57° 13.9	43.0	197° 16.6	05.7	357° 26.9	23.8
2	195° 40.6	194° 32.2	52.6	72° 16.5	42.9	212° 18.5	05.5	12° 29.6	23.9
3	210° 43.1	209° 31.8	· · 51.3	87° 19.0	· · 42.8	227° 20.4	· · 05.2	27° 32.2	· · 24.0
4	225° 45.5	224° 31.5	50.1	102° 21.6	42.7	242° 22.3	05.0	42° 34.8	24.0
5	240° 48.0	239° 31.1	48.8	117° 24.2	42.7	257° 24.2	04.8	57° 37.5	24.1
6	255° 50.5	254° 30.7	S00° 47.5	132° 26.7	N23° 42.6	272° 26.1	S08° 04.6	72° 40.1	N01° 24.2
7	270° 52.9	269° 30.3	46.2	147° 29.3	42.5	287° 28.0	04.3	87° 42.7	24.3
8	285° 55.4	284° 30.0	44.9	162° 31.9	42.5	302° 29.9	04.1	102° 45.4	24.4
9	300° 57.9	299° 29.6	· · 43.6	177° 34.4	· · 42.4	317° 31.8	· · 03.9	117° 48.0	· · 24.4
10	316° 00.3	314° 29.2	42.3	192° 37.0	42.3	332° 33.7	03.7	132° 50.6	24.5
11	331° 02.8	329° 28.9	41.1	207° 39.6	42.3	347° 35.6	03.4	147° 53.3	24.6
12	346° 05.3	344° 28.5	S00° 39.8	222° 42.1	N23° 42.2	2° 37.5	S08° 03.2	162° 55.9	N01° 24.7
13	1° 07.7	359° 28.1	38.5	237° 44.7	42.1	17° 39.4	03.0	177° 58.5	24.8
14	16° 10.2	14° 27.8	37.2	252° 47.2	42.0	32° 41.3	02.7	193° 05.2	24.8
15	31° 12.6	29° 27.4	· · 35.9	267° 49.8	· · 42.0	47° 43.2	· · 02.5	208° 03.8	· · 24.9
16	46° 15.1	44° 27.0	34.6	282° 52.3	41.9	62° 45.1	02.3	223° 06.4	25.0
17	61° 17.6	59° 26.7	33.4	297° 54.9	41.8	77° 47.0	02.1	238° 09.1	25.1
18	76° 20.0	74° 26.3	S00° 32.1	312° 57.4	N23° 41.8	92° 48.9	S08° 01.8	253° 11.7	N01° 25.1
19	91° 22.5	89° 25.9	30.8	328° 00.0	41.7	107° 50.8	01.6	268° 14.3	25.2
20	106° 25.0	104° 25.6	29.5	343° 02.5	41.6	122° 52.7	01.4	283° 17.0	25.3
21	121° 27.4	119° 25.2	· · 28.2	358° 05.1	· · 41.5	137° 54.6	· · 01.2	298° 19.6	· · 25.4
22	136° 29.9	134° 24.8	26.9	13° 07.6	41.5	152° 56.5	00.9	313° 22.3	25.5
23	151° 32.4	149° 24.5	25.6	28° 10.2	41.4	167° 58.4	00.7	328° 24.9	25.5
Mer.pass. 12:55		ν -0.4' d -1.3' m -3.91		ν 2.6' d -0.1' m -0.42		ν 1.9' d -0.2' m -2.02		ν 2.6' d 0.1' m 0.56	

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	166°34.8	164°24.1	S00°24.4	43°12.7	N23°41.3	183°00.3	S08°00.5	343°27.5	N01°25.6
1	181°37.3	179°23.7	23.1	58°15.3	41.2	198°02.2	00.2	358°30.2	25.7
2	196°39.7	194°23.3	21.8	73°17.8	41.2	213°04.1	08°00.0	13°32.8	25.8
3	211°42.2	209°23.0	· · 20.5	88°20.3	· · 41.1	228°06.0	07°59.8	28°35.4	· · 25.9
4	226°44.7	224°22.6	19.2	103°22.9	41.0	243°07.9	59.6	43°38.1	25.9
5	241°47.1	239°22.2	17.9	118°25.4	40.9	258°09.8	59.3	58°40.7	26.0
6	256°49.6	254°21.9	S00°16.6	133°27.9	N23°40.9	273°11.7	S07°59.1	73°43.3	N01°26.1
7	271°52.1	269°21.5	15.3	148°30.5	40.8	288°13.6	58.9	88°46.0	26.2
8	286°54.5	284°21.1	14.1	163°33.0	40.7	303°15.5	58.7	103°48.6	26.3
9	301°57.0	299°20.8	· · 12.8	178°35.5	· · 40.6	318°17.4	· · 58.4	118°51.2	· · 26.3
10	316°59.5	314°20.4	11.5	193°38.1	40.6	333°19.3	58.2	133°53.9	26.4
11	332°01.9	329°20.0	10.2	208°40.6	40.5	348°21.2	58.0	148°56.5	26.5
12	347°04.4	344°19.7	S00°08.9	223°43.1	N23°40.4	3°23.1	S07°57.7	163°59.1	N01°26.6
13	2°06.9	359°19.3	07.6	238°45.7	40.3	18°25.0	57.5	179°01.8	26.6
14	17°09.3	14°18.9	06.3	253°48.2	40.2	33°26.9	57.3	194°04.4	26.7
15	32°11.8	29°18.6	· · 05.1	268°50.7	· · 40.2	48°28.8	· · 57.1	209°07.0	· · 26.8
16	47°14.2	44°18.2	03.8	283°53.2	40.1	63°30.7	56.8	224°09.7	26.9
17	62°16.7	59°17.8	02.5	298°55.7	40.0	78°32.6	56.6	239°12.3	27.0
18	77°19.2	74°17.5	S00°01.2	313°58.3	N23°39.9	93°34.5	S07°56.4	254°14.9	N01°27.0
19	92°21.6	89°17.1	N00°00.1	329°00.8	39.9	108°36.4	56.2	269°17.6	27.1
20	107°24.1	104°16.7	01.4	344°03.3	39.8	123°38.3	55.9	284°20.2	27.2
21	122°26.6	119°16.4	· · 02.7	359°05.8	· · 39.7	138°40.2	· · 55.7	299°22.9	· · 27.3
22	137°29.0	134°16.0	03.9	14°08.3	39.6	153°42.1	55.5	314°25.5	27.4
23	152°31.5	149°15.6	05.2	29°10.8	39.5	168°44.0	55.2	329°28.1	27.4
Mer.pass. 12:52	ν -0.4' d -1.3' m -3.91			ν 2.5' d -0.1' m -0.39		ν 1.9' d -0.2' m -2.03		ν 2.6' d 0.1' m 0.56	

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	167°34.0	164°15.3	N00°06.5	44°13.3	N23°39.4	183°45.9	S07°55.0	344°30.8	N01°27.5
1	182°36.4	179°14.9	07.8	59°15.8	39.4	198°47.8	54.8	359°33.4	27.6
2	197°38.9	194°14.5	09.1	74°18.3	39.3	213°49.7	54.6	14°36.0	27.7
3	212°41.4	209°14.2	· · 10.4	89°20.8	· · 39.2	228°51.6	· · 54.3	29°38.7	· · 27.8
4	227°43.8	224°13.8	11.7	104°23.4	39.1	243°53.5	54.1	44°41.3	27.8
5	242°46.3	239°13.4	13.0	119°25.9	39.0	258°55.5	53.9	59°43.9	27.9
6	257°48.7	254°13.1	N00°14.2	134°28.4	N23°39.0	273°57.4	S07°53.7	74°46.6	N01°28.0
7	272°51.2	269°12.7	15.5	149°30.9	38.9	288°59.3	53.4	89°49.2	28.1
8	287°53.7	284°12.3	16.8	164°33.4	38.8	304°01.2	53.2	104°51.8	28.2
9	302°56.1	299°12.0	· · 18.1	179°35.9	· · 38.7	319°03.1	· · 53.0	119°54.5	· · 28.2
10	317°58.6	314°11.6	19.4	194°38.3	38.6	334°05.0	52.7	134°57.1	28.3
11	333°01.1	329°11.2	20.7	209°40.8	38.5	349°06.9	52.5	149°59.8	28.4
12	348°03.5	344°10.9	N00°22.0	224°43.3	N23°38.5	4°08.8	S07°52.3	165°02.4	N01°28.5
13	3°06.0	359°10.5	23.2	239°45.8	38.4	19°10.7	52.1	180°05.0	28.5
14	18°08.5	14°10.1	24.5	254°48.3	38.3	34°12.6	51.8	195°07.7	28.6
15	33°10.9	29°09.8	· · 25.8	269°50.8	· · 38.2	49°14.5	· · 51.6	210°10.3	· · 28.7
16	48°13.4	44°09.4	27.1	284°53.3	38.1	64°16.4	51.4	225°12.9	28.8
17	63°15.9	59°09.0	28.4	299°55.8	38.0	79°18.3	51.2	240°15.6	28.9
18	78°18.3	74°08.7	N00°29.7	314°58.3	N23°37.9	94°20.2	S07°50.9	255°18.2	N01°28.9
19	93°20.8	89°08.3	31.0	330°00.7	37.9	109°22.1	50.7	270°20.8	29.0
20	108°23.2	104°07.9	32.3	345°03.2	37.8	124°24.0	50.5	285°23.5	29.1
21	123°25.7	119°07.6	· · 33.5	0°05.7	· · 37.7	139°25.9	· · 50.2	300°26.1	· · 29.2
22	138°28.2	134°07.2	34.8	15°08.2	37.6	154°27.8	50.0	315°28.8	29.3
23	153°30.6	149°06.9	36.1	30°10.7	37.5	169°29.7	49.8	330°31.4	29.3
Mer.pass. 12:48	ν -0.4' d 1.3' m -3.90			ν 2.5' d -0.1' m -0.37		ν 1.9' d -0.2' m -2.03		ν 2.6' d 0.1' m 0.56	

	Stars	
	SHA	Dec
Alpheratz	357° 46.4	29° 08.8
Ankaa	353° 18.3	-42° 15.1
Schedar	349° 44.0	-56° 35.7
Diphda	348° 58.5	-17° 55.9
Achernar	335° 28.8	-57° 11.2
Hamal	328° 03.7	23° 30.7
Polaris	319° 13.2	89° 18.8
Acamar	315° 20.3	-40° 16.6
Menkar	314° 17.7	4° 07.8
Mirfak	308° 44.1	49° 54.0
Aldebaran	290° 52.2	16° 31.8
Rigel	281° 14.3	-8° 11.5
Capella	280° 38.0	46° 00.7
Bellatrix	278° 34.5	6° 21.5
Elnath	278° 15.6	28° 37.0
Anilam	275° 48.7	-1° 11.8
Betelgeuse	271° 03.8	7° 24.5
Canopus	263° 57.0	-52° 42.4
Sirius	258° 35.7	-16° 40.4
Adhara	255° 14.2	-28° 59.4
Procyon	245° 02.0	5° 11.8
Pollux	243° 30.4	28° 00.0
Avior	234° 18.6	-59° 32.8
Suhail	222° 53.9	-43° 28.7
Miaplacidus	221° 39.6	-69° 45.8
Alphard	217° 58.1	-8° 42.4
Regulus	207° 45.7	11° 54.8
Dubhe	193° 53.7	61° 41.6
Denebola	182° 35.7	14° 30.7
Giennah	175° 54.5	-17° 36.1
Acrux	173° 11.6	-63° 09.4
Gacrux	172° 03.2	-57° 10.3
Alioth	166° 22.1	55° 54.0
Spica	158° 33.5	-11° 13.1
Alkaid	157° 00.3	49° 15.4
Hadar	148° 51.0	-60° 25.3
Menkent	148° 10.2	-36° 25.3
Arcturus	145° 57.6	19° 07.5
Rigil Kent.	139° 54.7	-60° 52.5
Kochab	137° 18.9	74° 06.4
Zuben'ubi	137° 07.9	-16° 05.2
Alphecca	126° 12.8	26° 40.5
Antares	112° 29.1	-26° 27.3
Atria	107° 33.2	-69° 02.5
Sabik	102° 15.3	-15° 44.3
Shaula	96° 25.2	-37° 06.6
Rasalhague	96° 08.7	12° 32.9
Eltanin	90° 47.3	51° 28.9
Kaus Aust.	83° 47.1	-34° 22.7
Vega	80° 40.7	38° 47.3
Nunki	76° 01.4	-26° 17.0
Altair	62° 10.8	8° 53.6
Peacock	53° 23.3	-56° 42.0
Deneb	49° 33.5	45° 18.8
Enif	33° 49.8	9° 55.2
Al Na'ir	27° 47.0	-46° 54.6
Fomalhaut	15° 26.9	-29° 34.1
Scheat	13° 56.1	28° 08.3
Markab	13° 41.1	15° 15.5

Mar 08 Mon	SHA	Mer.pass
Venus	358° 57.3	13:02
Mars	236° 35.6	21:08
Jupiter	16° 39.0	11:50
Saturn	176° 48.6	01:10

Mar 09 Tue	SHA	Mer.pass
Venus	357° 49.3	13:03
Mars	236° 37.9	21:04
Jupiter	16° 25.5	11:46
Saturn	176° 52.7	01:06

Mar 10 Wed	SHA	Mer.pass
Venus	356° 41.3	13:03
Mars	236° 39.4	21:00
Jupiter	16° 12.0	11:43
Saturn	176° 56.8	01:02

Horizontal parallax	
Venus:	0.1
Mars:	0.2

DUT1 = UT1-UTC = +0.0433 sec ΔT = TT-UT1 = +66.1407 sec

2010 March 08 to Mar. 10 UT

h							
Sun				Moon			
Mon	GHA	Dec		GHA	ν	Dec	HP
0	177°16.4	S05°00.9		265°18.5	9.3'	S25°33.7	-0.4'
1	192°16.6	05°00.0		279°46.8	9.3'	25°33.3	-0.6'
2	207°16.7	04°59.0		294°15.1	9.4'	25°32.8	-0.7'
3	222°16.9	· · 58.0		308°43.4	9.4'	25°32.1	-0.8'
4	237°17.0	57.0		323°11.8	9.4'	25°31.3	-0.9'
5	252°17.2	56.1		337°40.2	9.5'	25°30.3	-1.1'
6	267°17.3	S04°55.1		352°08.7	9.5'	S25°29.2	-1.2'
7	282°17.5	54.1		6°37.2	9.6'	25°28.0	-1.3'
8	297°17.6	53.1		21°05.8	9.6'	25°26.7	-1.5'
9	312°17.8	· · 52.2		35°34.4	9.6'	25°25.2	-1.6'
10	327°17.9	51.2		50°03.0	9.7'	25°23.6	-1.7'
11	342°18.1	50.2		64°31.7	9.7'	25°21.9	-1.8'
12	357°18.2	S04°49.2		79°00.5	9.8'	S25°20.1	-2.0'
13	12°18.4	48.3		93°29.3	9.8'	25°18.1	-2.1'
14	27°18.5	47.3		107°58.1	9.9'	25°16.0	-2.2'
15	42°18.7	· · 46.3		122°27.0	9.9'	25°13.8	-2.3'
16	57°18.9	45.3		136°55.9	10.0'	25°11.4	-2.5'
17	72°19.0	44.4		151°24.9	10.0'	25°09.0	-2.6'
18	87°19.2	S04°43.4		165°53.9	10.1'	S25°06.4	-2.7'
19	102°19.3	42.4		180°23.0	10.1'	25°03.7	-2.8'
20	117°19.5	41.4		194°52.1	10.2'	25°00.8	-3.0'
21	132°19.6	· · 40.4		209°21.3	10.2'	24°57.9	-3.1'
22	147°19.8	39.5		223°50.5	10.3'	24°54.8	-3.2'
23	162°19.9	38.5		238°19.8	10.3'	24°51.6	-3.3'
SD = 16.1'			d = -1.0'	SD = 15.1'			
Tue	GHA	Dec		GHA	ν	Dec	HP
0	177°20.1	S04°37.5		252°49.1	10.4'	S24°48.3	-3.4'
1	192°20.3	36.5		267°18.5	10.4'	24°44.8	-3.6'
2	207°20.4	35.6		281°48.0	10.5'	24°41.3	-3.7'
3	222°20.6	· · 34.6		296°17.5	10.6'	24°37.6	-3.8'
4	237°20.7	33.6		310°47.0	10.6'	24°33.8	-3.9'
5	252°20.9	32.6		325°16.6	10.7'	24°29.9	-4.0'
6	267°21.0	S04°31.7		339°46.3	10.7'	S24°25.9	-4.1'
7	282°21.2	30.7		354°16.0	10.8'	24°21.7	-4.3'
8	297°21.3	29.7		8°45.8	10.8'	24°17.5	-4.4'
9	312°21.5	· · 28.7		23°15.7	10.9'	24°13.1	-4.5'
10	327°21.7	27.7		37°45.6	11.0'	24°08.7	-4.6'
11	342°21.8	26.8		52°15.5	11.0'	24°04.1	-4.7'
12	357°22.0	S04°25.8		66°45.5	11.1'	S23°59.4	-4.8'
13	12°22.1	24.8		81°15.6	11.1'	23°54.6	-4.9'
14	27°22.3	23.8		95°45.7	11.2'	23°49.6	-5.0'
15	42°22.4	· · 22.9		110°15.9	11.3'	23°44.6	-5.1'
16	57°22.6	21.9		124°46.2	11.3'	23°39.5	-5.2'
17	72°22.8	20.9		139°16.5	11.4'	23°34.2	-5.4'
18	87°22.9	S04°19.9		153°46.9	11.4'	S23°28.9	-5.5'
19	102°23.1	18.9		168°17.3	11.5'	23°23.4	-5.6'
20	117°23.2	18.0		182°47.8	11.6'	23°17.8	-5.7'
21	132°23.4	· · 17.0		197°18.3	11.6'	23°12.2	-5.8'
22	147°23.5	16.0		211°49.0	11.7'	23°06.4	-5.9'
23	162°23.7	15.0		226°19.6	11.7'	23°00.5	-6.0'
SD = 16.1'			d = -1.0'	SD = 14.9'			
Wed	GHA	Dec		GHA	ν	Dec	HP
0	177°23.9	S04°14.0		240°50.4	11.8'	S22°54.5	-6.1'
1	192°24.0	13.1		255°21.2	11.9'	22°48.5	-6.2'
2	207°24.2	12.1		269°52.0	11.9'	22°42.3	-6.3'
3	222°24.3	· · 11.1		284°22.9	12.0'	22°36.0	-6.4'
4	237°24.5	10.1		298°53.9	12.0'	22°29.6	-6.5'
5	252°24.7	09.1		313°25.0	12.1'	22°23.1	-6.6'
6	267°24.8	S04°08.2		327°56.1	12.2'	S22°16.5	-6.7'
7	282°25.0	07.2		342°27.2	12.2'	22°09.9	-6.8'
8	297°25.1	06.2		356°58.5	12.3'	22°03.1	-6.9'
9	312°25.3	· · 05.2		11°29.8	12.4'	21°56.2	-7.0'
10	327°25.5	04.2		26°01.1	12.4'	21°49.2	-7.1'
11	342°25.6	03.3		40°32.5	12.5'	21°42.2	-7.2'
12	357°25.8	S04°02.3		55°04.0	12.5'	S21°35.0	-7.2'
13	12°25.9	01.3		69°35.5	12.6'	21°27.8	-7.3'
14	27°26.1	04°00.3		84°07.1	12.7'	21°20.4	-7.4'
15	42°26.3	03°59.3		98°38.8	12.7'	21°13.0	-7.5'
16	57°26.4	58.4		113°10.5	12.8'	21°05.5	-7.6'
17	72°26.6	57.4		127°42.3	12.8'	20°57.9	-7.7'
18	87°26.7	S03°56.4		142°14.1	12.9'	S20°50.2	-7.8'
19	102°26.9	55.4		156°46.0	13.0'	20°42.4	-7.9'
20	117°27.1	54.4		171°18.0	13.0'	20°34.5	-8.0'
21	132°27.2	· · 53.5		185°50.0	13.1'	20°26.6	-8.0'
22	147°27.4	52.5		200°22.1	13.1'	20°18.5	-8.1'
23	162°27.6	51.5		214°54.2	13.2'	20°10.4	-8.2'
SD = 16.1'			d = -1.0'	SD = 14.8'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	04:29	05:49	06:56	17:27	18:35	19:56
N 70°	04:39	05:50	06:51	17:32	18:33	19:45
68°	04:46	05:51	06:47	17:36	18:32	19:37
66°	04:53	05:52	06:43	17:40	18:31	19:31
64°	04:58	05:53	06:40	17:42	18:30	19:25
62°	05:02	05:54	06:38	17:45	18:29	19:21
60°	05:06	05:54	06:35	17:47	18:28	19:17
N 58°	05:09	05:54	06:33	17:49	18:28	19:14
56°	05:11	05:55	06:32	17:51	18:28	19:11
54°	05:14	05:55	06:30	17:52	18:27	19:09
52°	05:16	05:55	06:28	17:54	18:27	19:06
50°	05:17	05:55	06:27	17:55	18:27	19:05
45°	05:21	05:55	06:24	17:58	18:27	19:01
N 40°	05:23	05:54	06:21	18:00	18:27	18:59
35°	05:25	05:54	06:19	18:02	18:28	18:57
30°	05:26	05:53	06:17	18:04	18:28	18:56
20°	05:26	05:52	06:14	18:08	18:30	18:55
N 10°	05:25	05:49	06:10	18:11	18:32	18:56
0°	05:22	05:47	06:07	18:14	18:35	18:59
S 10°	05:18	05:43	06:04	18:17	18:38	19:02
20°	05:13	05:38	06:00	18:20	18:42	19:08
30°	05:04	05:32	05:56	18:24	18:48	19:17
35°	04:58	05:28	05:54	18:27	18:52	19:22
40°	04:51	05:24	05:51	18:29	18:57	19:29
45°	04:43	05:18	05:48	18:33	19:02	19:37
S 50°	04:31	05:11	05:44	18:36	19:09	19:48
52°	04:26	05:07	05:42	18:38	19:12	19:54
54°	04:20	05:04	05:40	18:40	19:16	20:00
56°	04:13	05:00	05:38	18:42	19:20	20:06
58°	04:05	04:55	05:35	18:45	19:25	20:14
S 60°	03:56	04:49	05:32	18:47	19:30	20:23

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	■	■	■	■	■	■
66°	■	■	06:38	■	■	09:57
64°	05:47	05:58	05:58	07:19	08:55	10:36
62°	04:46	05:17	05:31	08:20	09:36	11:03
60°	04:12	04:49	05:10	08:54	10:04	11:24
N 58°	03:47	04:27	04:52	09:19	10:26	11:41
56°	03:28	04:09	04:37	09:39	10:43	11:55
54°	03:11	03:54	04:25	09:55	10:58	12:07
52°	02:57	03:40	04:13	10:10	11:11	12:18
50°	02:44	03:29	04:03	10:22	11:23	12:28
45°	02:19	03:05	03:42	10:47	11:46	12:48
N 40°	01:58	02:45	03:25	11:08	12:05	13:04
35°	01:41	02:29	03:11	11:25	12:21	13:18
30°	01:26	02:15	02:59	11:39	12:34	13:30
20°	01:02	01:52	02:37	12:04	12:57	13:50
N 10°	00:40	01:31	02:18	12:25	13:17	14:08
0°	00:20	01:12	02:01	12:45	13:36	14:24
S 10°	00:00	00:52	01:43	13:05	13:54	14:40
20°	· ·	00:32	01:25	13:26	14:14	14:57
30°	· ·	00:08	01:03	13:51	14:37	15:17
35°	23:54	· ·	00:50	14:05	14:50	15:29
40°	23:37	· ·	00:36	14:22	15:05	15:42
45°	23:18	· ·	00:18	14:41	15:23	15:57
S 50°	22:53	23:56	· ·	15:06	15:46	16:16
52°	22:41	23:46	· ·	15:18	15:56	16:25
54°	22:28	23:34	· ·	15:32	16:09	16:35
56°	22:12	23:20	· ·	15:48	16:22	16:47
58°	21:53	23:05	· ·	16:07	16:39	17:00
S 60°	21:30	22:46	· ·	16:30	16:58	17:15

Day	Sun		Mer. Pass	Moon		Age 22-24 47-28%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
08	10:54	10:47	12:11	06:33	18:58	
09	10:40	10:32	12:11	07:24	19:48	
10	10:25	10:17	12:10	08:13	20:36	

March 11, 12, 13 UT (Thu., Fri., Sat.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	168°33.1	164°06.5	N00°37.4	45°13.1	N23°37.4	184°31.6	S07°49.6	345°34.0	N01°29.4					Alpheratz	357°46.4 29°08.8		
1	183°35.6	179°06.1	38.7	60°15.6	37.3	199°33.5	49.3	0°36.7	29.5					Ankaa	353°18.3 -42°15.1		
2	198°38.0	194°05.8	40.0	75°18.1	37.2	214°35.4	49.1	15°39.3	29.6					Schedar	349°44.0 56°35.7		
3	213°40.5	209°05.4	· · 41.3	90°20.6	· · 37.1	229°37.3	· · 48.9	30°41.9	· · 29.7					Diphda	348°58.5 -17°55.9		
4	228°43.0	224°05.0	42.5	105°23.0	37.1	244°39.2	48.7	45°44.6	29.7					Achernar	335°28.8 -57°11.2		
5	243°45.4	239°04.7	43.8	120°25.5	37.0	259°41.1	48.4	60°47.2	29.8					Hamal	328°03.7 23°30.7		
6	258°47.9	254°04.3	N00°45.1	135°28.0	N23°36.9	274°43.0	S07°48.2	75°49.9	N01°29.9					Polaris	319°14.3 89°18.8		
7	273°50.3	269°03.9	46.4	150°30.4	36.8	289°44.9	48.0	90°52.5	30.0					Acamar	315°20.3 -40°16.0		
8	288°52.8	284°03.6	47.7	165°32.9	36.7	304°46.8	47.7	105°55.1	30.1					Menkar	314°17.7 4°07.8		
9	303°55.3	299°03.2	· · 49.0	180°35.4	· · 36.6	319°48.7	· · 47.5	120°57.8	· · 30.1					Mirfak	308°44.1 49°54.0		
10	318°57.7	314°02.8	50.3	195°37.8	36.5	334°50.6	47.3	136°00.4	30.2					Aldebaran	290°52.2 16°31.8		
11	334°00.2	329°02.5	51.6	210°40.3	36.4	349°52.5	47.1	151°03.0	30.3					Rigel	281°14.3 -8°11.5		
12	349°02.7	344°02.1	N00°52.8	225°42.8	N23°36.3	4°54.4	S07°46.8	166°05.7	N01°30.4					Capella	280°38.0 46°00.7		
13	4°05.1	359°01.7	54.1	240°45.2	36.2	19°56.3	46.6	181°08.3	30.5					Bellatrix	278°34.5 6°21.5		
14	19°07.6	14°01.4	55.4	255°47.7	36.1	34°58.2	46.4	196°11.0	30.5					Elnath	278°15.6 28°37.0		
15	34°10.1	29°01.0	· · 56.7	270°50.1	· · 36.1	50°00.1	· · 46.2	211°13.6	· · 30.6					Alnilam	275°48.7 -1°11.8		
16	49°12.5	44°00.6	58.0	285°52.6	36.0	65°02.0	45.9	226°16.2	30.7					Betelgeuse	271°03.8 7°24.5		
17	64°15.0	59°00.3	00°59.3	300°55.0	35.9	80°03.9	45.7	241°18.9	30.8					Canopus	263°57.1 -52°42.4		
18	79°17.5	73°59.9	N01°00.6	315°57.5	N23°35.8	95°05.8	S07°45.5	256°21.5	N01°30.9					Sirius	258°35.7 -16°44.0		
19	94°19.9	88°59.5	01.8	330°59.9	35.7	110°07.7	45.2	271°24.1	30.9					Adhara	255°14.3 -28°59.4		
20	109°22.4	103°59.2	03.1	346°02.4	35.6	125°09.6	45.0	286°26.8	31.0					Procyon	245°02.0 5°11.8		
21	124°24.8	118°58.8	· · 04.4	1°04.8	· · 35.5	140°11.5	· · 44.8	301°29.4	· · 31.1					Pollux	243°30.4 28°00.1		
22	139°27.3	133°58.4	05.7	16°07.3	35.4	155°13.4	44.6	316°32.1	31.2					Avior	234°18.7 -59°32.8		
23	154°29.8	148°58.1	07.0	31°09.7	35.3	170°15.3	44.3	331°34.7	31.3					Suhail	222°53.9 -43°28.7		
Mer.pass. 12:44		ν -0.4' d 1.3' m -3.90			ν 2.5' d -0.1' m -0.34			ν 1.9' d -0.2' m -2.03			ν 2.6' d 0.1' m 0.55						

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	169°32.2	163°57.7	N01°08.3	46°12.2	N23°35.2	185°17.3	S07°44.1	346°37.3	N01°31.3			
1	184°34.7	178°57.4	09.6	61°14.6	35.1	200°19.2	43.9	1°40.0	31.4			
2	199°37.2	193°57.0	10.9	76°17.1	35.0	215°21.1	43.7	16°42.6	31.5			
3	214°39.6	208°56.6	· · 12.1	91°19.5	· · 34.9	230°23.0	· · 43.4	31°45.2	· · 31.6			
4	229°42.1	223°56.3	13.4	106°21.9	34.8	245°24.9	43.2	46°47.9	31.7			
5	244°44.6	238°55.9	14.7	121°24.4	34.7	260°26.8	43.0	61°50.5	31.7			
6	259°47.0	253°55.5	N01°16.0	136°26.8	N23°34.6	275°28.7	S07°42.7	76°53.2	N01°31.8			
7	274°49.5	268°55.2	17.3	151°29.3	34.5	290°30.6	42.5	91°55.8	31.9			
8	289°51.9	283°54.8	18.6	166°31.7	34.4	305°32.5	42.3	106°58.4	32.0			
9	304°54.4	298°54.4	· · 19.9	181°34.1	· · 34.3	320°34.4	· · 42.1	122°01.1	· · 32.1			
10	319°56.9	313°54.1	21.1	196°36.5	34.2	335°36.3	41.8	137°03.7	32.1			
11	334°59.3	328°53.7	22.4	211°39.0	34.2	350°38.2	41.6	152°06.4	32.2			
12	350°01.8	343°53.3	N01°23.7	226°41.4	N23°34.1	5°40.1	S07°41.4	167°09.0	N01°32.3			
13	5°04.3	358°53.0	25.0	241°43.8	34.0	20°42.0	41.2	182°11.6	32.4			
14	20°06.7	13°52.6	26.3	256°46.3	33.9	35°43.9	40.9	197°14.3	32.5			
15	35°09.2	28°52.2	· · 27.6	271°48.7	· · 33.8	50°45.8	· · 40.7	212°16.9	· · 32.5			
16	50°11.7	43°51.9	28.9	286°51.1	33.7	65°47.7	40.5	227°19.5	32.6			
17	65°14.1	58°51.5	30.1	301°53.5	33.6	80°49.6	40.2	242°22.2	32.7			
18	80°16.6	73°51.1	N01°31.4	316°55.9	N23°33.5	95°51.5	S07°40.0	257°24.8	N01°32.8			
19	95°19.1	88°50.8	32.7	331°58.4	33.4	110°53.4	39.8	272°27.5	32.9			
20	110°21.5	103°50.4	34.0	347°00.8	33.3	125°55.3	39.6	287°30.1	32.9			
21	125°24.0	118°50.1	· · 35.3	2°03.2	· · 33.2	140°57.2	· · 39.3	302°32.7	· · 33.0			
22	140°26.4	133°49.7	36.6	17°05.6	33.1	155°59.1	39.1	317°35.4	33.1			
23	155°28.9	148°49.3	37.9	32°08.0	33.0	171°01.0	38.9	332°38.0	33.2			
Mer.pass. 12:40		ν -0.4' d 1.3' m -3.90			ν 2.4' d -0.1' m -0.31			ν 1.9' d -0.2' m -2.03			ν 2.6' d 0.1' m 0.55	

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	170°31.4	163°49.0	N01°39.1	47°10.4	N23°32.9	186°02.9	S07°38.7	347°40.7	N01°33.3			
1	185°33.8	178°48.6	40.4	62°12.9	32.8	201°04.8	38.4	2°43.3	33.3			
2	200°36.3	193°48.2	41.7	77°15.3	32.7	216°06.7	38.2	17°45.9	33.4			
3	215°38.8	208°47.9	· · 43.0	92°17.7	· · 32.5	231°08.6	· · 38.0	32°48.6	· · 33.5			
4	230°41.2	223°47.5	44.3	107°20.1	32.4	246°10.5	37.7	47°51.2	33.6			
5	245°43.7	238°47.1	45.6	122°22.5	32.3	261°12.5	37.5	62°53.9	33.7			
6	260°46.2	253°46.8	N01°46.9	137°24.9	N23°32.2	276°14.4	S07°37.3	77°56.5	N01°33.7			
7	275°48.6	268°46.4	48.1	152°27.3	32.1	291°16.3	37.1	92°59.1	33.8			
8	290°51.1	283°46.0	49.4	167°29.7	32.0	306°18.2	36.8	108°01.8	33.9			
9	305°53.6	298°45.7	· · 50.7	182°32.1	· · 31.9	321°20.1	· · 36.6	123°04.4	· · 34.0			
10	320°56.0	313°45.3	52.0	197°34.5	31.8	336°22.0	36.4	138°07.1	34.1			
11	335°58.5	328°44.9	53.3	212°36.9	31.7	351°23.9	36.2	153°09.7	34.1			
12	351°00.9	343°44.6	N01°54.6	227°39.3	N23°31.6	6°25.8	S07°35.9	168°12.3	N01°34.2			
13	6°03.4	358°44.2	55.8	242°41.7	31.5	21°27.7	35.7	183°15.0	34.3			
14	21°05.9	13°43.8	57.1	257°44.1	31.4	36°29.6	35.5	198°17.6	34.4			
15	36°08.3	28°43.5	· · 58.4	272°46.5	· · 31.3	51°31.5	· · 35.2	213°20.3	· · 34.5			
16	51°10.8	43°43.1	01°59.7	287°48.9	31.2	66°33.4	35.0	228°22.9	34.5			
17	66°13.3	58°42.7	02°01.0	302°51.2	31.1	81°35.3	34.8	243°25.5	34.6			
18	81°15.7	73°42.4	N02°02.3	317°53.6	N23°31.0	96°37.2	S07°34.6	258°28.2	N01°34.7			
19	96°18.2	88°42.0	03.6	332°56.0	30.9	111°39.1	34.3	273°30.8	34.8			
20	111°20.7	103°41.7	04.8	347°58.4	30.8	126°41.0	34.1	288°33.4	34.9			
21	126°23.1	118°41.3	· · 06.1	3°00.8	· · 30.7	141°42.9	· · 33.9	303°36.1	· · 34.9			
22	141°25.6	133°40.9	07.4	18°03.2	30.6	156°44.8	33.7	318°38.7	35.0			
23	156°28.0	148°40.6	08.7	33°05.6	30.4	171°46.7	33.4	333°41.4	35.1			
Mer.pass. 12:36		ν -0.4' d 1.3' m -3.90			ν 2.4' d -0.1' m -0.29			ν 1.9' d -0.2' m -2.03			ν 2.6' d 0.1' m 0.54	

Mar 11 Thu	SHA	Mer.pass
Venus	355°33.4	13:04
Mars	236°40.0	20:56
Jupiter	15°58.5	11:40
Saturn	177°00.9	00:58

Mar 12 Fri	SHA	Mer.pass
Venus	354°25.5	13:04
Mars	236°39.9	20:52
Jupiter	15°45.0	11:37
Saturn	177°05.1	00:53

Mar 13 Sat	SHA	Mer.pass
Venus	353°17.6	13:05
Mars	236°39.1	20:48
Jupiter	15°31.6	11:34
Saturn	177°09.3	00:49

Horizontal parallax		
Venus:		0.1
Mars:		0.2

DUT1 = UT1-UTC = +0.0406 sec $\Delta T = TT-UT1 = +66.1434$ sec

2010 March 11 to Mar. 13 UT

h	Sun			Moon			
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	177°27.7	S03°50.5	229°26.4	13.3'	S20°02.2	-8.3'	54.1'
1	192°27.9	49.5	243°58.7	13.3'	19°53.9	-8.4'	54.1'
2	207°28.0	48.5	258°31.0	13.4'	19°45.5	-8.5'	54.1'
3	222°28.2	· · 47.6	273°03.4	13.4'	19°37.0	-8.5'	54.1'
4	237°28.4	46.6	287°35.8	13.5'	19°28.5	-8.6'	54.0'
5	252°28.5	45.6	302°08.3	13.6'	19°19.8	-8.7'	54.0'
6	267°28.7	S03°44.6	316°40.9	13.6'	S19°11.1	-8.8'	54.0'
7	282°28.9	43.6	331°13.5	13.7'	19°02.4	-8.9'	54.0'
8	297°29.0	42.7	345°46.1	13.7'	18°53.5	-8.9'	54.0'
9	312°29.2	· · 41.7	0°18.9	13.8'	18°44.5	-9.0'	54.0'
10	327°29.3	40.7	14°51.6	13.8'	18°35.5	-9.1'	54.0'
11	342°29.5	39.7	29°24.5	13.9'	18°26.4	-9.2'	54.0'
12	357°29.7	S03°38.7	43°57.4	13.9'	S18°17.3	-9.2'	54.0'
13	12°29.8	37.7	58°30.3	14.0'	18°08.0	-9.3'	54.0'
14	27°30.0	36.8	73°03.3	14.1'	17°58.7	-9.4'	54.0'
15	42°30.2	· · 35.8	87°36.4	14.1'	17°49.3	-9.5'	54.0'
16	57°30.3	34.8	102°09.5	14.2'	17°39.8	-9.5'	54.0'
17	72°30.5	33.8	116°42.6	14.2'	17°30.3	-9.6'	54.0'
18	87°30.7	S03°32.8	131°15.8	14.3'	S17°20.7	-9.7'	54.0'
19	102°30.8	31.8	145°49.1	14.3'	17°11.0	-9.7'	54.0'
20	117°31.0	30.9	160°22.4	14.4'	17°01.3	-9.8'	54.0'
21	132°31.2	· · 29.9	174°55.8	14.4'	16°51.5	-9.9'	54.0'
22	147°31.3	28.9	189°29.2	14.5'	16°41.6	-9.9'	54.0'
23	162°31.5	27.9	204°02.7	14.5'	16°31.6	-10.0'	54.0'
SD = 16.1'			SD = 14.7'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	177°31.6	S03°26.9	218°36.2	14.6'	S16°21.6	-10.1'	54.0'
1	192°31.8	25.9	233°09.8	14.6'	16°11.5	-10.1'	54.0'
2	207°32.0	25.0	247°43.4	14.7'	16°01.4	-10.2'	54.0'
3	222°32.1	· · 24.0	262°17.0	14.7'	15°51.2	-10.3'	53.9'
4	237°32.3	23.0	276°50.8	14.8'	15°40.9	-10.3'	53.9'
5	252°32.5	22.0	291°24.5	14.8'	15°30.6	-10.4'	53.9'
6	267°32.6	S03°21.0	305°58.3	14.9'	S15°20.2	-10.5'	53.9'
7	282°32.8	20.0	320°32.2	14.9'	15°09.7	-10.5'	53.9'
8	297°33.0	19.1	335°06.1	14.9'	14°59.2	-10.6'	53.9'
9	312°33.1	· · 18.1	349°40.0	15.0'	14°48.6	-10.6'	53.9'
10	327°33.3	17.1	4°14.0	15.0'	14°38.0	-10.7'	53.9'
11	342°33.5	16.1	18°48.0	15.1'	14°27.3	-10.8'	53.9'
12	357°33.6	S03°15.1	33°22.1	15.1'	S14°16.5	-10.8'	53.9'
13	12°33.8	14.1	47°56.2	15.2'	14°05.7	-10.9'	53.9'
14	27°34.0	13.1	62°30.4	15.2'	13°54.9	-10.9'	53.9'
15	42°34.1	· · 12.2	77°04.6	15.2'	13°43.9	-11.0'	53.9'
16	57°34.3	11.2	91°38.8	15.3'	13°33.0	-11.0'	53.9'
17	72°34.5	10.2	106°13.1	15.3'	13°21.9	-11.1'	53.9'
18	87°34.6	S03°09.2	120°47.4	15.4'	S13°10.8	-11.1'	53.9'
19	102°34.8	08.2	135°21.8	15.4'	12°59.7	-11.2'	54.0'
20	117°35.0	07.2	149°56.1	15.4'	12°48.5	-11.2'	54.0'
21	132°35.1	· · 06.3	164°30.6	15.5'	12°37.3	-11.3'	54.0'
22	147°35.3	05.3	179°05.0	15.5'	12°26.0	-11.3'	54.0'
23	162°35.5	04.3	193°39.5	15.5'	12°14.7	-11.4'	54.0'
SD = 16.1'			SD = 14.7'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	177°35.6	S03°03.3	208°14.1	15.6'	S12°03.3	-11.4'	54.0'
1	192°35.8	02.3	222°48.6	15.6'	11°51.8	-11.5'	54.0'
2	207°36.0	01.3	237°23.2	15.6'	11°40.4	-11.5'	54.0'
3	222°36.2	03°00.3	251°57.9	15.7'	11°28.8	-11.6'	54.0'
4	237°36.3	02°59.4	266°32.6	15.7'	11°17.3	-11.6'	54.0'
5	252°36.5	58.4	281°07.3	15.7'	11°05.6	-11.7'	54.0'
6	267°36.7	S02°57.4	295°42.0	15.8'	S10°54.0	-11.7'	54.0'
7	282°36.8	56.4	310°16.7	15.8'	10°42.3	-11.8'	54.0'
8	297°37.0	55.4	324°51.5	15.8'	10°30.5	-11.8'	54.0'
9	312°37.2	· · 54.4	339°26.4	15.8'	10°18.7	-11.8'	54.0'
10	327°37.3	53.4	354°01.2	15.9'	10°06.9	-11.9'	54.0'
11	342°37.5	52.5	8°36.1	15.9'	09°55.0	-11.9'	54.0'
12	357°37.7	S02°51.5	23°11.0	15.9'	S09°43.1	-12.0'	54.0'
13	12°37.8	50.5	37°45.9	16.0'	09°31.1	-12.0'	54.0'
14	27°38.0	49.5	52°20.9	16.0'	09°19.1	-12.0'	54.0'
15	42°38.2	· · 48.5	66°55.8	16.0'	09°07.1	-12.1'	54.0'
16	57°38.4	47.5	81°30.8	16.0'	08°55.0	-12.1'	54.0'
17	72°38.5	46.5	96°05.8	16.0'	08°42.9	-12.1'	54.0'
18	87°38.7	S02°45.6	110°40.9	16.1'	S08°30.8	-12.2'	54.0'
19	102°38.9	44.6	125°15.9	16.1'	08°18.6	-12.2'	54.0'
20	117°39.0	43.6	139°51.0	16.1'	08°06.4	-12.2'	54.1'
21	132°39.2	· · 42.6	154°26.1	16.1'	07°54.2	-12.3'	54.1'
22	147°39.4	41.6	169°01.3	16.1'	07°41.9	-12.3'	54.1'
23	162°39.5	40.6	183°36.4	16.2'	07°29.6	-12.3'	54.1'
SD = 16.1'			SD = 14.7'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	04:12	05:33	06:40	17:41	18:49	20:11
N 70°	04:24	05:37	06:37	17:45	18:45	19:59
68°	04:33	05:39	06:34	17:47	18:43	19:49
66°	04:41	05:41	06:32	17:49	18:40	19:41
64°	04:47	05:43	06:30	17:51	18:39	19:35
62°	04:52	05:44	06:28	17:53	18:37	19:29
60°	04:57	05:45	06:26	17:54	18:36	19:25
N 58°	05:00	05:46	06:25	17:56	18:35	19:21
56°	05:04	05:47	06:24	17:57	18:34	19:17
54°	05:06	05:47	06:23	17:58	18:33	19:14
52°	05:09	05:48	06:22	17:59	18:33	19:12
50°	05:11	05:48	06:21	18:00	18:32	19:10
45°	05:15	05:49	06:19	18:02	18:31	19:05
N 40°	05:18	05:50	06:17	18:03	18:30	19:02
35°	05:21	05:50	06:15	18:05	18:30	19:00
30°	05:22	05:50	06:14	18:06	18:30	18:58
20°	05:24	05:49	06:11	18:09	18:31	18:56
N 10°	05:23	05:48	06:09	18:11	18:32	18:56
0°	05:22	05:46	06:06	18:13	18:34	18:58
S 10°	05:19	05:43	06:04	18:15	18:36	19:01
20°	05:14	05:39	06:01	18:18	18:40	19:06
30°	05:06	05:34	05:58	18:21	18:45	19:13
35°	05:01	05:31	05:56	18:23	18:48	19:18
40°	04:55	05:27	05:54	18:25	18:52	19:24
45°	04:47	05:22	05:52	18:27	18:56	19:31
S 50°	04:37	05:16	05:49	18:30	19:02	19:41
52°	04:32	05:13	05:47	18:31	19:05	19:46
54°	04:27	05:10	05:46	18:33	19:08	19:51
56°	04:20	05:06	05:44	18:34	19:12	19:57
58°	04:13	05:02	05:42	18:36	19:16	20:04
S 60°	04:05	04:57	05:40	18:38	19:20	20:13

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	■	07:25	06:43	■	12:20	14:30
N 70°	07:42	06:52	06:26	10:31	12:52	14:45
68°	06:52	06:28	06:12	11:20	13:14	14:57
66°	06:20	06:09	06:01	11:50	13:32	15:07
64°	05:57	05:54	05:51	12:13	13:46	15:15
62°	05:38	05:41	05:43	12:31	13:58	15:22
60°	05:22	05:30	05:36	12:46	14:08	15:28
N 58°	05:09	05:21	05:30	12:59	14:17	15:34
56°	04:58	05:12	05:24	13:10	14:24	15:38
54°	04:47	05:05	05:19	13:19	14:31	15:43
52°	04:39	04:58	05:14	13:27	14:37	15:46
50°	04:30	04:52	05:10	13:35	14:43	15:50
45°	04:13	04:39	05:01	13:51	14:54	15:57
N 40°	03:59	04:28	04:54	14:04	15:04	16:03
35°	03:47	04:19	04:47	14:15	15:12	16:09
30°	03:37	04:11	04:42	14:25	15:20	16:13
20°	03:18	03:56	04:32	14:42	15:32	16:21
N 10°	03:03	03:44	04:23	14:56	15:43	16:28
0°	02:48	03:32	04:15	15:09	15:53	16:35
S 10°	02:33	03:20	04:06	15:23	16:03	16:41
20°	02:17	03:08	03:57	15:37	16:13	16:48
30°	01:58	02:53	03:47	15:53	16:26	16:55
35°	01:48	02:45	03:41	16:02	16:32	17:00
40°	01:35	02:35	03:35	16:13	16:40	17:05
45°	01:21	02:24	03:27	16:26	16:49	17:11
S 50°	01:03	02:10	03:17	16:41	17:00	17:17
52°	00:54	02:04	03:13	16:48	17:05	17:20
54°	00:45	01:56	03:08	16:55	17:11	17:24
56°	00:34	01:48	03:03	17:04	17:17	17:28
58°	00:21	01:39	02:57	17:14	17:24	17:32
S 60°	00:07	01:29	02:50	17:25	17:32	17:37

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time	12 ^h		Mer. Pass.		
	00 ^h	mm:ss	hh:mm	Upper	Lower	25-27
				hh:mm	hh:mm	20-7%
11	10:09	10:01	12:10	08:59	21:21	
12	09:53	09:45	12:10	09:43	22:04	
13	09:37	09:29	12:09	10:25	22:45	

March 14, 15, 16 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	171°30.5	163°40.2	N02°10.0	48°07.9	N23°30.3	186°48.6	S07°33.2	348°44.0	N01°35.2	Alpheratz	357°46.4	29°08.8
1	186°33.0	178°39.8	11.3	63°10.3	30.2	201°50.5	33.0	3°46.6	35.3	Ankaa	353°18.3	-42°15.1
2	201°35.4	193°39.5	12.5	78°12.7	30.1	216°52.4	32.8	18°49.3	35.3	Schedar	349°44.0	56°35.7
3	216°37.9	208°39.1	•• 13.8	93°15.1	•• 30.0	231°54.3	•• 32.5	33°51.9	•• 35.4	Diphda	348°58.5	-17°55.9
4	231°40.4	223°38.7	15.1	108°17.4	29.9	246°56.3	32.3	48°54.6	35.5	Achernar	335°28.8	-57°11.2
5	246°42.8	238°38.4	16.4	123°19.8	29.8	261°58.2	32.1	63°57.2	35.6	Hamal	328°03.7	23°30.7
6	261°45.3	253°38.0	N02°17.7	138°22.2	N23°29.7	277°00.1	S07°31.8	78°59.9	N01°35.7	Polaris	319°15.4	89°18.8
7	276°47.8	268°37.6	19.0	153°24.6	29.6	292°02.0	31.6	94°02.5	35.7	Acamar	315°20.3	-40°16.0
8	291°50.2	283°37.3	20.3	168°26.9	29.5	307°03.9	31.4	109°05.1	35.8	Menkar	314°17.7	4°07.8
9	306°52.7	298°36.9	•• 21.5	183°29.3	•• 29.4	322°05.8	•• 31.2	124°07.8	•• 35.9	Mirfak	308°44.1	49°54.0
10	321°55.2	313°36.5	22.8	198°31.7	29.2	337°07.7	30.9	139°10.4	36.0	Aldebaran	290°52.2	16°31.8
11	336°57.6	328°36.2	24.1	213°34.0	29.1	352°09.6	30.7	154°13.1	36.1	Rigel	281°14.4	-8°11.5
12	352°00.1	343°35.8	N02°25.4	228°36.4	N23°29.0	7°11.5	S07°30.5	169°15.7	N01°36.1	Capella	280°38.0	46°00.7
13	7°02.5	358°35.4	26.7	243°38.8	28.9	22°13.4	30.3	184°18.3	36.2	Bellatrix	278°34.6	6°21.5
14	22°05.0	13°35.1	28.0	258°41.1	28.8	37°15.3	30.0	199°21.0	36.3	Elnath	278°15.6	28°37.0
15	37°07.5	28°34.7	•• 29.2	273°43.5	•• 28.7	52°17.2	•• 29.8	214°23.6	•• 36.4	Alnilam	275°48.8	-1°11.8
16	52°09.9	43°34.3	30.5	288°45.9	28.6	67°19.1	29.6	229°26.3	36.5	Betelgeuse	271°03.8	7°24.5
17	67°12.4	58°34.0	31.8	303°48.2	28.5	82°21.0	29.3	244°28.9	36.5	Canopus	263°57.1	-52°42.4
18	82°14.9	73°33.6	N02°33.1	318°50.6	N23°28.4	97°22.9	S07°29.1	259°31.5	N01°36.6	Sirius	258°35.7	-16°44.0
19	97°17.3	88°33.2	34.4	333°52.9	28.2	112°24.8	28.9	274°34.2	36.7	Adhara	255°14.3	-28°59.4
20	112°19.8	103°32.9	35.6	348°55.3	28.1	127°26.7	28.7	289°36.8	36.8	Procyon	245°02.0	5°11.8
21	127°22.3	118°32.5	•• 36.9	3°57.6	•• 28.0	142°28.6	•• 28.4	304°39.5	•• 36.9	Pollux	243°30.4	28°00.1
22	142°24.7	133°32.1	38.2	19°00.0	27.9	157°30.5	28.2	319°42.1	36.9	Avior	234°18.7	-59°32.8
23	157°27.2	148°31.8	39.5	34°02.3	27.8	172°32.4	28.0	334°44.7	37.0	Suhail	222°53.9	-43°28.7
Mer.pass. 12:32		ν -0.4' d 1.3' m -3.90		ν 2.4' d -0.1' m -0.26		ν 1.9' d -0.2' m -2.03		ν 2.6' d 0.1' m 0.54				

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	172°29.6	163°31.4	N02°40.8	49°04.7	N23°27.7	187°34.4	S07°27.8	349°47.4	N01°37.1
1	187°32.1	178°31.0	42.1	64°07.0	27.6	202°36.3	27.5	4°50.0	37.2
2	202°34.6	193°30.7	43.3	79°09.4	27.4	217°38.2	27.3	19°52.7	37.3
3	217°37.0	208°30.3	•• 44.6	94°11.7	•• 27.3	232°40.1	•• 27.1	34°55.3	•• 37.4
4	232°39.5	223°29.9	45.9	109°14.1	27.2	247°42.0	26.8	49°57.9	37.4
5	247°42.0	238°29.6	47.2	124°16.4	27.1	262°43.9	26.6	65°00.6	37.5
6	262°44.4	253°29.2	N02°48.5	139°18.8	N23°27.0	277°45.8	S07°26.4	80°03.2	N01°37.6
7	277°46.9	268°28.8	49.8	154°21.1	26.9	292°47.7	26.2	95°05.9	37.7
8	292°49.4	283°28.5	51.0	169°23.4	26.7	307°49.6	25.9	110°08.5	37.8
9	307°51.8	298°28.1	•• 52.3	184°25.8	•• 26.6	322°51.5	•• 25.7	125°11.2	•• 37.8
10	322°54.3	313°27.7	53.6	199°28.1	26.5	337°53.4	25.5	140°13.8	37.9
11	337°56.8	328°27.4	54.9	214°30.4	26.4	352°55.3	25.3	155°16.4	38.0
12	352°59.2	343°27.0	N02°56.2	229°32.8	N23°26.3	7°57.2	S07°25.0	170°19.1	N01°38.1
13	8°01.7	358°26.6	57.4	244°35.1	26.2	22°59.1	24.8	185°21.7	38.2
14	23°04.1	13°26.3	02°58.7	259°37.4	26.0	38°01.0	24.6	200°24.4	38.2
15	38°06.6	28°25.9	03°00.0	274°39.8	•• 25.9	53°02.9	•• 24.4	215°27.0	•• 38.3
16	53°09.1	43°25.5	01.3	289°42.1	25.8	68°04.8	24.1	230°29.6	38.4
17	68°11.5	58°25.2	02.6	304°44.4	25.7	83°06.8	23.9	245°32.3	38.5
18	83°14.0	73°24.8	N03°03.8	319°46.8	N23°25.6	98°08.7	S07°23.7	260°34.9	N01°38.6
19	98°16.5	88°24.4	05.1	334°49.1	25.5	113°10.6	23.4	275°37.6	38.6
20	113°18.9	103°24.1	06.4	349°51.4	25.3	128°12.5	23.2	290°40.2	38.7
21	128°21.4	118°23.7	•• 07.7	4°53.7	•• 25.2	143°14.4	•• 23.0	305°42.8	•• 38.8
22	143°23.9	133°23.3	09.0	19°56.0	25.1	158°16.3	22.8	320°45.5	38.9
23	158°26.3	148°23.0	10.2	34°58.4	25.0	173°18.2	22.5	335°48.1	39.0
Mer.pass. 12:28		ν -0.4' d 1.3' m -3.90		ν 2.3' d -0.1' m -0.24		ν 1.9' d -0.2' m -2.03		ν 2.6' d 0.1' m 0.54	

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	173°28.8	163°22.6	N03°11.5	50°00.7	N23°24.9	188°20.1	S07°22.3	350°50.8	N01°39.0
1	188°31.2	178°22.2	12.8	65°03.0	24.7	203°22.0	22.1	5°53.4	39.1
2	203°33.7	193°21.9	14.1	80°05.3	24.6	218°23.9	21.9	20°56.1	39.2
3	218°36.2	208°21.5	•• 15.4	95°07.6	•• 24.5	233°25.8	•• 21.6	35°58.7	•• 39.3
4	233°38.6	223°21.1	16.6	110°09.9	24.4	248°27.7	21.4	51°01.3	39.4
5	248°41.1	238°20.8	17.9	125°12.3	24.2	263°29.6	21.2	66°04.0	39.4
6	263°43.6	253°20.4	N03°19.2	140°14.6	N23°24.1	278°31.5	S07°21.0	81°06.6	N01°39.5
7	278°46.0	268°20.0	20.5	155°16.9	24.0	293°33.4	20.7	96°09.3	39.6
8	293°48.5	283°19.7	21.8	170°19.2	23.9	308°35.3	20.5	111°11.9	39.7
9	308°51.0	298°19.3	•• 23.0	185°21.5	•• 23.8	323°37.3	•• 20.3	126°14.6	•• 39.8
10	323°53.4	313°18.9	24.3	200°23.8	23.6	338°39.2	20.0	141°17.2	39.8
11	338°55.9	328°18.6	25.6	215°26.1	23.5	353°41.1	19.8	156°19.8	39.9
12	353°58.4	343°18.2	N03°26.9	230°28.4	N23°23.4	8°43.0	S07°19.6	171°22.5	N01°40.0
13	9°00.8	358°17.8	28.2	245°30.7	23.3	23°44.9	19.4	186°25.1	40.1
14	24°03.3	13°17.5	29.4	260°33.0	23.1	38°46.8	19.1	201°27.8	40.2
15	39°05.7	28°17.1	•• 30.7	275°35.3	•• 23.0	53°48.7	•• 18.9	216°30.4	•• 40.2
16	54°08.2	43°16.7	32.0	290°37.6	22.9	68°50.6	18.7	231°33.0	40.3
17	69°10.7	58°16.3	33.3	305°39.9	22.8	83°52.5	18.5	246°35.7	40.4
18	84°13.1	73°16.0	N03°34.5	320°42.2	N23°22.6	98°54.4	S07°18.2	261°38.3	N01°40.5
19	99°15.6	88°15.6	35.8	335°44.5	22.5	113°56.3	18.0	276°41.0	40.6
20	114°18.1	103°15.2	37.1	350°46.8	22.4	128°58.2	17.8	291°43.6	40.6
21	129°20.5	118°14.9	•• 38.4	5°49.1	•• 22.3	144°00.1	•• 17.5	306°46.3	•• 40.7
22	144°23.0	133°14.5	39.7	20°51.4	22.1	159°02.0	17.3	321°48.9	40.8
23	159°25.5	148°14.1	40.9	35°53.7	22.0	174°04.0	17.1	336°51.5	40.9
Mer.pass. 12:24		ν -0.4' d 1.3' m -3.90		ν 2.3' d -0.1' m -0.21		ν 1.9' d -0.2' m -2.03		ν 2.6' d 0.1' m 0.53	

Mar 14 Sun	SHA	Mer.pass
Venus	352°09.7	13:06
Mars	236°37.4	20:44
Jupiter	15°18.1	11:31
Saturn	177°13.5	00:45

Mar 15 Mon	SHA	Mer.pass
Venus	351°01.8	13:06
Mars	236°35.0	20:40
Jupiter	15°04.7	11:28
Saturn	177°17.7	00:41

Mar 16 Tue	SHA	Mer.pass
Venus	349°53.8	13:07
Mars	236°31.9	20:37
Jupiter	14°51.3	11:25
Saturn	177°22.0	00:37

Horizontal parallax	
Venus:	0.1
Mars:	0.2

DUT1 = UT1-UTC = +0.0374 sec $\Delta T = TT-UT1 = +66.1466$ sec

2010 March 14 to Mar. 16 UT

h	Sun			Moon			
Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	177°39.7	S02°39.6	198°11.5	16.2'	S07°17.2	-12.4'	54.1'
1	192°39.9	38.7	212°46.7	16.2'	07°04.9	-12.4'	54.1'
2	207°40.1	37.7	227°21.9	16.2'	06°52.5	-12.4'	54.1'
3	222°40.2	· · 36.7	241°57.1	16.2'	06°40.0	-12.5'	54.1'
4	237°40.4	35.7	256°32.3	16.2'	06°27.6	-12.5'	54.1'
5	252°40.6	34.7	271°07.6	16.2'	06°15.1	-12.5'	54.1'
6	267°40.7	S02°33.7	285°42.8	16.3'	S06°02.6	-12.5'	54.1'
7	282°40.9	32.7	300°18.0	16.3'	05°50.0	-12.6'	54.1'
8	297°41.1	31.8	314°53.3	16.3'	05°37.4	-12.6'	54.1'
9	312°41.3	· · 30.8	329°28.6	16.3'	05°24.9	-12.6'	54.1'
10	327°41.4	29.8	344°03.9	16.3'	05°12.2	-12.6'	54.2'
11	342°41.6	28.8	358°39.2	16.3'	04°59.6	-12.7'	54.2'
12	357°41.8	S02°27.8	13°14.5	16.3'	S04°46.9	-12.7'	54.2'
13	12°41.9	26.8	27°49.8	16.3'	04°34.2	-12.7'	54.2'
14	27°42.1	25.8	42°25.1	16.3'	04°21.5	-12.7'	54.2'
15	42°42.3	· · 24.8	57°00.4	16.3'	04°08.8	-12.7'	54.2'
16	57°42.5	23.9	71°35.7	16.3'	03°56.1	-12.8'	54.2'
17	72°42.6	22.9	86°11.0	16.3'	03°43.3	-12.8'	54.2'
18	87°42.8	S02°21.9	100°46.3	16.3'	S03°30.5	-12.8'	54.2'
19	102°43.0	20.9	115°21.7	16.3'	03°17.7	-12.8'	54.2'
20	117°43.2	19.9	129°57.0	16.3'	03°04.9	-12.8'	54.3'
21	132°43.3	· · 18.9	144°32.3	16.3'	02°52.0	-12.9'	54.3'
22	147°43.5	17.9	159°07.6	16.3'	02°39.2	-12.9'	54.3'
23	162°43.7	16.9	173°43.0	16.3'	02°26.3	-12.9'	54.3'
SD = 16.1' $d = -1.0'$			SD = 14.7'				

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	177°43.9	S02°16.0	188°18.3	16.3'	S02°13.4	-12.9'	54.3'
1	192°44.0	15.0	202°53.6	16.3'	02°00.5	-12.9'	54.3'
2	207°44.2	14.0	217°28.9	16.3'	01°47.6	-12.9'	54.3'
3	222°44.4	· · 13.0	232°04.2	16.3'	01°34.7	-12.9'	54.3'
4	237°44.5	12.0	246°39.5	16.3'	01°21.8	-12.9'	54.3'
5	252°44.7	11.0	261°14.8	16.3'	01°08.8	-13.0'	54.3'
6	267°44.9	S02°10.0	275°50.1	16.3'	S00°55.9	-13.0'	54.4'
7	282°45.1	09.0	290°25.4	16.3'	00°42.9	-13.0'	54.4'
8	297°45.2	08.1	305°00.6	16.3'	00°30.0	-13.0'	54.4'
9	312°45.4	· · 07.1	319°35.9	16.2'	00°17.0	-13.0'	54.4'
10	327°45.6	06.1	334°11.1	16.2'	S00°04.0	-13.0'	54.4'
11	342°45.8	05.1	348°46.4	16.2'	N00°09.0	13.0'	54.4'
12	357°45.9	S02°04.1	3°21.6	16.2'	N00°22.0	13.0'	54.4'
13	12°46.1	03.1	17°56.8	16.2'	00°35.0	13.0'	54.4'
14	27°46.3	02.1	32°32.0	16.2'	00°48.0	13.0'	54.5'
15	42°46.5	· · 01.1	47°07.2	16.2'	01°01.0	13.0'	54.5'
16	57°46.6	02°00.2	61°42.3	16.1'	01°14.0	13.0'	54.5'
17	72°46.8	01°59.2	76°17.5	16.1'	01°27.0	13.0'	54.5'
18	87°47.0	S01°58.2	90°52.6	16.1'	N01°40.0	13.0'	54.5'
19	102°47.2	57.2	105°27.7	16.1'	01°53.0	13.0'	54.5'
20	117°47.3	56.2	120°02.8	16.1'	02°06.0	13.0'	54.5'
21	132°47.5	· · 55.2	134°37.9	16.0'	02°19.0	13.0'	54.5'
22	147°47.7	54.2	149°12.9	16.0'	02°32.0	13.0'	54.6'
23	162°47.9	53.2	163°47.9	16.0'	02°45.0	13.0'	54.6'
SD = 16.1' $d = -1.0'$			SD = 14.8'				

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	177°48.0	S01°52.2	178°22.9	16.0'	N02°58.0	13.0'	54.6'
1	192°48.2	51.3	192°57.9	15.9'	03°10.9	13.0'	54.6'
2	207°48.4	50.3	207°32.8	15.9'	03°23.9	13.0'	54.6'
3	222°48.6	· · 49.3	222°07.8	15.9'	03°36.9	13.0'	54.6'
4	237°48.7	48.3	236°42.7	15.9'	03°49.8	13.0'	54.6'
5	252°48.9	47.3	251°17.5	15.8'	04°02.8	12.9'	54.6'
6	267°49.1	S01°46.3	265°52.4	15.8'	N04°15.7	12.9'	54.7'
7	282°49.3	45.3	280°27.2	15.8'	04°28.7	12.9'	54.7'
8	297°49.5	44.3	295°01.9	15.7'	04°41.6	12.9'	54.7'
9	312°49.6	· · 43.4	309°36.7	15.7'	04°54.5	12.9'	54.7'
10	327°49.8	42.4	324°11.4	15.7'	05°07.4	12.9'	54.7'
11	342°50.0	41.4	338°46.1	15.6'	05°20.3	12.9'	54.7'
12	357°50.2	S01°40.4	353°20.7	15.6'	N05°33.2	12.9'	54.7'
13	12°50.3	39.4	7°55.3	15.6'	05°46.0	12.8'	54.8'
14	27°50.5	38.4	22°29.9	15.5'	05°58.8	12.8'	54.8'
15	42°50.7	· · 37.4	37°04.5	15.5'	06°11.7	12.8'	54.8'
16	57°50.9	36.4	51°39.0	15.5'	06°24.5	12.8'	54.8'
17	72°51.0	35.4	66°13.4	15.4'	06°37.3	12.8'	54.8'
18	87°51.2	S01°34.5	80°47.9	15.4'	N06°50.0	12.7'	54.8'
19	102°51.4	33.5	95°22.2	15.3'	07°02.8	12.7'	54.9'
20	117°51.6	32.5	109°56.6	15.3'	07°15.5	12.7'	54.9'
21	132°51.8	· · 31.5	124°30.9	15.3'	07°28.2	12.7'	54.9'
22	147°51.9	30.5	139°05.2	15.2'	07°40.8	12.7'	54.9'
23	162°52.1	29.5	153°39.4	15.2'	07°53.5	12.6'	54.9'
SD = 16.1' $d = -1.0'$			SD = 14.9'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	03:54	05:18	06:25	17:55	19:03	20:28
N 70°	04:08	05:23	06:23	17:57	18:58	20:13
68°	04:19	05:26	06:22	17:58	18:54	20:01
66°	04:28	05:29	06:20	17:59	18:50	19:52
64°	04:36	05:32	06:19	18:00	18:48	19:44
62°	04:42	05:34	06:18	18:01	18:45	19:38
60°	04:47	05:36	06:17	18:02	18:43	19:33
N 58°	04:52	05:38	06:17	18:03	18:42	19:28
56°	04:55	05:39	06:16	18:03	18:40	19:24
54°	04:59	05:40	06:15	18:04	18:39	19:20
52°	05:02	05:41	06:15	18:04	18:38	19:17
50°	05:04	05:42	06:14	18:05	18:37	19:15
45°	05:10	05:44	06:13	18:06	18:35	19:09
N 40°	05:14	05:45	06:12	18:07	18:34	19:05
35°	05:16	05:46	06:11	18:07	18:33	19:02
30°	05:19	05:46	06:10	18:08	18:32	19:00
20°	05:21	05:47	06:09	18:10	18:32	18:57
N 10°	05:22	05:46	06:07	18:11	18:32	18:56
0°	05:21	05:45	06:06	18:12	18:33	18:57
S 10°	05:19	05:43	06:04	18:14	18:35	18:59
20°	05:15	05:40	06:02	18:15	18:37	19:03
30°	05:08	05:36	06:00	18:17	18:41	19:09
35°	05:04	05:34	05:59	18:18	18:44	19:13
40°	04:58	05:30	05:57	18:20	18:47	19:19
45°	04:51	05:26	05:56	18:21	18:51	19:25
S 50°	04:42	05:21	05:53	18:23	18:56	19:34
52°	04:38	05:19	05:52	18:24	18:58	19:38
54°	04:33	05:16	05:51	18:25	19:01	19:43
56°	04:27	05:13	05:50	18:26	19:04	19:49
58°	04:21	05:09	05:49	18:28	19:07	19:55
S 60°	04:14	05:05	05:47	18:29	19:11	20:02

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	06:14	05:49	05:25	16:25	18:17	20:12
N 70°	06:06	05:48	05:31	16:31	18:15	20:01
68°	05:59	05:47	05:35	16:36	18:13	19:52
66°	05:53	05:46	05:39	16:40	18:11	19:45
64°	05:48	05:45	05:42	16:43	18:10	19:39
62°	05:44	05:44	05:45	16:46	18:09	19:33
60°	05:40	05:44	05:48	16:48	18:08	19:29
N 58°	05:37	05:43	05:50	16:50	18:07	19:25
56°	05:34	05:43	05:52	16:52	18:06	19:21
54°	05:31	05:42	05:54	16:54	18:05	19:18
52°	05:29	05:42	05:55	16:55	18:05	19:15
50°	05:26	05:41	05:57	16:57	18:04	19:12
45°	05:21	05:41	06:00	17:00	18:03	19:07
N 40°	05:17	05:40	06:03	17:02	18:02	19:02
35°	05:14	05:39	06:05	17:05	18:01	18:58
30°	05:11	05:39	06:07	17:06	18:00	18:54
20°	05:05	05:38	06:11	17:10	17:58	18:48
N 10°	05:00	05:37	06:14	17:13	17:57	18:43
0°	04:56	05:36	06:17	17:15	17:56	18:37
S 10°	04:51	05:36	06:21	17:18	17:55	18:32
20°	04:46	05:35	06:24	17:21	17:54	18:27
30°	04:41	05:34	06:28	17:24	17:52	18:21
35°	04:37	05:34	06:30	17:26	17:51	18:17
40°	04:34	05:33	06:33	17:28	17:50	18:13
45°	04:30	05:32	06:36	17:30	17:49	18:09
S 50°	04:24	05:32	06:39	17:33	17:48	18:03
52°	04:22	05:31	06:41	17:34	17:47	18:01
54°	04:20	05:31	06:43	17:35	17:47	17:58
56°	04:17	05:30	06:45	17:37	17:46	17:55
58°	04:13	05:30	06:47	17:39	17:45	17:52
S 60°	04:10	05:29	06:50	17:41	17:44	17:48

Day	Sun		Mer.	Moon		Age
	Eqn.of Time	12 ^h	Pass	Mer.Pass.	Lower	28-1
	00 ^h	mm:ss	hh:mm	Upper	hh:mm	3-0%
14	09:21	09:13	12:09	11:06	23:26	
15	09:05	08:56	12:09	11:46	—:—	
16	08:48	08:39	12:09	12:27	00:07	

March 17, 18, 19 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	174°27.9	163°13.8	N03°42.2	50°56.0	N23°21.9	189°05.9	S07°16.9	351°54.2	N01°41.0	Alpheratz	357°46.4	29°08.8
1	189°30.4	178°13.4	43.5	65°58.2	21.8	204°07.8	16.6	6°56.8	41.1	Ankaa	353°18.3	-42°15.1
2	204°32.9	193°13.0	44.8	81°00.5	21.6	219°09.7	16.4	21°59.5	41.1	Schedar	349°44.0	56°35.7
3	219°35.3	208°12.7	· · 46.0	96°02.8	· · 21.5	234°11.6	· · 16.2	37°02.1	· · 41.2	Diphda	348°58.5	-17°55.9
4	234°37.8	223°12.3	47.3	111°05.1	21.4	249°13.5	16.0	52°04.8	41.3	Achernar	335°28.9	-57°11.2
5	249°40.2	238°11.9	48.6	126°07.4	21.3	264°15.4	15.7	67°07.4	41.4	Hamal	328°03.8	23°30.7
6	264°42.7	253°11.6	N03°49.9	141°09.7	N23°21.1	279°17.3	S07°15.5	82°10.0	N01°41.5	Polaris	319°16.4	89°18.8
7	279°45.2	268°11.2	51.1	156°11.9	21.0	294°19.2	15.3	97°12.7	41.5	Acamar	315°20.3	-40°16.0
8	294°47.6	283°10.8	52.4	171°14.2	20.9	309°21.1	15.1	112°15.3	41.6	Menkar	314°17.7	4°07.8
9	309°50.1	298°10.4	· · 53.7	186°16.5	· · 20.7	324°23.0	· · 14.8	127°18.0	· · 41.7	Mirfak	308°44.1	49°54.0
10	324°52.6	313°10.1	55.0	201°18.8	20.6	339°24.9	14.6	142°20.6	41.8	Aldebaran	290°52.2	16°31.8
11	339°55.0	328°09.7	56.2	216°21.1	20.5	354°26.8	14.4	157°23.3	41.9	Rigel	281°14.4	-8°11.5
12	354°57.5	343°09.3	N03°57.5	231°23.3	N23°20.3	9°28.8	S07°14.2	172°25.9	N01°41.9	Capella	280°38.0	46°00.6
13	10°00.0	358°09.0	03°58.8	246°25.6	20.2	24°30.7	13.9	187°28.5	42.0	Bellatrix	278°34.6	6°21.5
14	25°02.4	13°08.6	04°00.1	261°27.9	20.1	39°32.6	13.7	202°31.2	42.1	Elnath	278°15.7	28°37.0
15	40°04.9	28°08.2	· · 01.3	276°30.1	· · 20.0	54°34.5	· · 13.5	217°33.8	· · 42.2	Alnilam	275°48.8	-1°11.8
16	55°07.3	43°07.9	02.6	291°32.4	19.8	69°36.4	13.2	232°36.5	42.3	Betelgeuse	271°03.8	7°24.5
17	70°09.8	58°07.5	03.9	306°34.7	19.7	84°38.3	13.0	247°39.1	42.3	Canopus	263°57.1	-52°42.4
18	85°12.3	73°07.1	N04°05.2	321°36.9	N23°19.6	99°40.2	S07°12.8	262°41.7	N01°42.4	Sirius	258°35.7	-16°44.0
19	100°14.7	88°06.7	06.4	336°39.2	19.4	114°42.1	12.6	277°44.4	42.5	Adhara	255°14.3	-28°59.4
20	115°17.2	103°06.4	07.7	351°41.5	19.3	129°44.0	12.3	292°47.0	42.6	Procyon	245°02.1	5°11.8
21	130°19.7	118°06.0	· · 09.0	6°43.7	· · 19.2	144°45.9	· · 12.1	307°49.7	· · 42.7	Pollux	243°30.4	28°00.1
22	145°22.1	133°05.6	10.3	21°46.0	19.0	159°47.8	11.9	322°52.3	42.7	Avior	234°18.7	-59°32.8
23	160°24.6	148°05.3	11.5	36°48.3	18.9	174°49.7	11.7	337°55.0	42.8	Suhail	222°53.9	-43°28.7
Mer.pass. 12:20		ν -0.4' d 1.3' m -3.90		ν 2.3' d -0.1' m -0.19		ν 1.9' d -0.2' m -2.03		ν 2.6' d 0.1' m 0.53				

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	175°27.1	163°04.9	N04°12.8	51°50.5	N23°18.8	189°51.7	S07°11.4	352°57.6	N01°42.9
1	190°29.5	178°04.5	14.1	66°52.8	18.6	204°53.6	11.2	8°00.2	43.0
2	205°32.0	193°04.1	15.4	81°55.0	18.5	219°55.5	11.0	23°02.9	43.1
3	220°34.5	208°03.8	· · 16.6	96°57.3	· · 18.4	234°57.4	· · 10.8	38°05.5	· · 43.1
4	235°36.9	223°03.4	17.9	111°59.5	18.2	249°59.3	10.5	53°08.2	43.2
5	250°39.4	238°03.0	19.2	127°01.8	18.1	265°01.2	10.3	68°10.8	43.3
6	265°41.8	253°02.7	N04°20.5	142°04.0	N23°18.0	280°03.1	S07°10.1	83°13.5	N01°43.4
7	280°44.3	268°02.3	21.7	157°06.3	17.8	295°05.0	09.8	98°16.1	43.5
8	295°46.8	283°01.9	23.0	172°08.5	17.7	310°06.9	09.6	113°18.7	43.5
9	310°49.2	298°01.5	· · 24.3	187°10.8	· · 17.6	325°08.8	· · 09.4	128°21.4	· · 43.6
10	325°51.7	313°01.2	25.5	202°13.0	17.4	340°10.7	09.2	143°24.0	43.7
11	340°54.2	328°00.8	26.8	217°15.3	17.3	355°12.6	08.9	158°26.7	43.8
12	355°56.6	343°00.4	N04°28.1	232°17.5	N23°17.2	10°14.6	S07°08.7	173°29.3	N01°43.9
13	10°59.1	358°00.0	29.4	247°19.8	17.0	25°16.5	08.5	188°32.0	43.9
14	26°01.6	12°59.7	30.6	262°22.0	16.9	40°18.4	08.3	203°34.6	44.0
15	41°04.0	27°59.3	· · 31.9	277°24.3	· · 16.8	55°20.3	· · 08.0	218°37.3	· · 44.1
16	56°06.5	42°58.9	33.2	292°26.5	16.6	70°22.2	07.8	233°39.9	44.2
17	71°08.9	57°58.6	34.4	307°28.7	16.5	85°24.1	07.6	248°42.5	44.3
18	86°11.4	72°58.2	N04°35.7	322°31.0	N23°16.3	100°26.0	S07°07.4	263°45.2	N01°44.4
19	101°13.9	87°57.8	37.0	337°33.2	16.2	115°27.9	07.1	278°47.8	44.4
20	116°16.3	102°57.4	38.3	352°35.4	16.1	130°29.8	06.9	293°50.5	44.5
21	131°18.8	117°57.1	· · 39.5	7°37.7	· · 15.9	145°31.7	· · 06.7	308°53.1	· · 44.6
22	146°21.3	132°56.7	40.8	22°39.9	15.8	160°33.6	06.5	323°55.8	44.7
23	161°23.7	147°56.3	42.1	37°42.1	15.7	175°35.6	06.2	338°58.4	44.8
Mer.pass. 12:16		ν -0.4' d 1.3' m -3.90		ν 2.3' d -0.1' m -0.16		ν 1.9' d -0.2' m -2.03		ν 2.6' d 0.1' m 0.53	

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	176°26.2	162°55.9	N04°43.3	52°44.4	N23°15.5	190°37.5	S07°06.0	354°01.0	N01°44.8
1	191°28.7	177°55.6	44.6	67°46.6	15.4	205°39.4	05.8	9°03.7	44.9
2	206°31.1	192°55.2	45.9	82°48.8	15.2	220°41.3	05.5	24°06.3	45.0
3	221°33.6	207°54.8	· · 47.1	97°51.0	· · 15.1	235°43.2	· · 05.3	39°09.0	· · 45.1
4	236°36.1	222°54.5	48.4	112°53.3	15.0	250°45.1	05.1	54°11.6	45.2
5	251°38.5	237°54.1	49.7	127°55.5	14.8	265°47.0	04.9	69°14.3	45.2
6	266°41.0	252°53.7	N04°50.9	142°57.7	N23°14.7	280°48.9	S07°04.6	84°16.9	N01°45.3
7	281°43.4	267°53.3	52.2	157°59.9	14.5	295°50.8	04.4	99°19.5	45.4
8	296°45.9	282°53.0	53.5	173°02.2	14.4	310°52.7	04.2	114°22.2	45.5
9	311°48.4	297°52.6	· · 54.8	188°04.4	· · 14.3	325°54.7	· · 04.0	129°24.8	· · 45.6
10	326°50.8	312°52.2	56.0	203°06.6	14.1	340°56.6	03.7	144°27.5	45.6
11	341°53.3	327°51.8	57.3	218°08.8	14.0	355°58.5	03.5	159°30.1	45.7
12	356°55.8	342°51.5	N04°58.6	233°11.0	N23°13.8	11°00.4	S07°03.3	174°32.8	N01°45.8
13	11°58.2	357°51.1	04°59.8	248°13.2	13.7	26°02.3	03.1	189°35.4	45.9
14	27°00.7	12°50.7	05°01.1	263°15.5	13.6	41°04.2	02.8	204°38.0	46.0
15	42°03.2	27°50.3	· · 02.4	278°17.7	· · 13.4	56°06.1	· · 02.6	219°40.7	· · 46.0
16	57°05.6	42°50.0	03.6	293°19.9	13.3	71°08.0	02.4	234°43.3	46.1
17	72°08.1	57°49.6	04.9	308°22.1	13.1	86°09.9	02.2	249°46.0	46.2
18	87°10.6	72°49.2	N05°06.2	323°24.3	N23°13.0	101°11.8	S07°01.9	264°48.6	N01°46.3
19	102°13.0	87°48.8	07.4	338°26.5	12.8	116°13.8	01.7	279°51.3	46.4
20	117°15.5	102°48.4	08.7	353°28.7	12.7	131°15.7	01.5	294°53.9	46.4
21	132°17.9	117°48.1	· · 10.0	8°30.9	· · 12.6	146°17.6	· · 01.3	309°56.6	· · 46.5
22	147°20.4	132°47.7	11.2	23°33.1	12.4	161°19.5	01.0	324°59.2	46.6
23	162°22.9	147°47.3	12.5	38°35.3	12.3	176°21.4	00.8	340°01.8	46.7
Mer.pass. 12:12		ν -0.4' d 1.3' m -3.90		ν 2.2' d -0.1' m -0.14		ν 1.9' d -0.2' m -2.03		ν 2.6' d 0.1' m 0.52	

Mar 17 Wed		SHA	Mer.pass
Venus	348°45.8		13:07
Mars	236°28.0		20:33
Jupiter	14°37.9		11:22
Saturn	177°26.3		00:32

Mar 18 Thu		SHA	Mer.pass
Venus	347°37.8		13:08
Mars	236°23.5		20:30
Jupiter	14°24.6		11:19
Saturn	177°30.5		00:28

Mar 19 Fri		SHA	Mer.pass
Venus	346°29.7		13:09
Mars	236°18.2		20:26
Jupiter	14°11.3		11:16
Saturn	177°34.8		00:24

Horizontal parallax		
Venus:	0.1	
Mars:	0.2	

DUT1 = UT1-UTC = +0.0338 sec $\Delta T = TT-UT1 = +66.1502$ sec

2010 March 17 to Mar. 19 UT

Sun				Moon			
Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	177°52.3	S01°28.5	168°13.5	15.1'	N08°06.1	12.6'	54.9'
1	192°52.5	27.5	182°47.7	15.1'	08°18.7	12.6'	54.9'
2	207°52.6	26.5	197°21.8	15.0'	08°31.3	12.5'	55.0'
3	222°52.8	· · 25.6	211°55.8	15.0'	08°43.8	12.5'	55.0'
4	237°53.0	24.6	226°29.8	14.9'	08°56.4	12.5'	55.0'
5	252°53.2	23.6	241°03.7	14.9'	09°08.8	12.5'	55.0'
6	267°53.4	S01°22.6	255°37.6	14.8'	N09°21.3	12.4'	55.0'
7	282°53.5	21.6	270°11.5	14.8'	09°33.7	12.4'	55.0'
8	297°53.7	20.6	284°45.2	14.7'	09°46.1	12.4'	55.1'
9	312°53.9	· · 19.6	299°19.0	14.7'	09°58.5	12.3'	55.1'
10	327°54.1	18.6	313°52.7	14.6'	10°10.8	12.3'	55.1'
11	342°54.3	17.6	328°26.3	14.6'	10°23.1	12.3'	55.1'
12	357°54.4	S01°16.7	342°59.9	14.5'	N10°35.3	12.2'	55.1'
13	12°54.6	15.7	357°33.4	14.5'	10°47.5	12.2'	55.1'
14	27°54.8	14.7	12°06.9	14.4'	10°59.7	12.1'	55.2'
15	42°55.0	· · 13.7	26°40.3	14.4'	11°11.9	12.1'	55.2'
16	57°55.2	12.7	41°13.6	14.3'	11°23.9	12.1'	55.2'
17	72°55.3	11.7	55°46.9	14.2'	11°36.0	12.0'	55.2'
18	87°55.5	S01°10.7	70°20.2	14.2'	N11°48.0	12.0'	55.2'
19	102°55.7	09.7	84°53.3	14.1'	12°00.0	11.9'	55.2'
20	117°55.9	08.8	99°26.4	14.1'	12°11.9	11.9'	55.3'
21	132°56.0	· · 07.8	113°59.5	14.0'	12°23.8	11.8'	55.3'
22	147°56.2	06.8	128°32.5	13.9'	12°35.6	11.8'	55.3'
23	162°56.4	05.8	143°05.4	13.9'	12°47.4	11.7'	55.3'
	SD = 16.1'	d = -1.0'	SD = 15.0'				
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	177°56.6	S01°04.8	157°38.3	13.8'	N12°59.1	11.7'	55.3'
1	192°56.8	03.8	172°11.1	13.7'	13°10.8	11.6'	55.4'
2	207°56.9	02.8	186°43.8	13.7'	13°22.5	11.6'	55.4'
3	222°57.1	· · 01.8	201°16.5	13.6'	13°34.1	11.5'	55.4'
4	237°57.3	01°00.8	215°49.1	13.5'	13°45.6	11.5'	55.4'
5	252°57.5	00°59.8	230°21.6	13.5'	13°57.1	11.4'	55.4'
6	267°57.7	S00°58.9	244°54.1	13.4'	N14°08.5	11.4'	55.4'
7	282°57.9	57.9	259°26.5	13.3'	14°19.9	11.3'	55.5'
8	297°58.0	56.9	273°58.8	13.3'	14°31.2	11.3'	55.5'
9	312°58.2	· · 55.9	288°31.1	13.2'	14°42.4	11.2'	55.5'
10	327°58.4	54.9	303°03.3	13.1'	14°53.6	11.1'	55.5'
11	342°58.6	53.9	317°35.4	13.0'	15°04.8	11.1'	55.5'
12	357°58.8	S00°52.9	332°07.5	13.0'	N15°15.8	11.0'	55.6'
13	12°58.9	51.9	346°39.4	12.9'	15°26.8	11.0'	55.6'
14	27°59.1	50.9	1°11.3	12.8'	15°37.8	10.9'	55.6'
15	42°59.3	· · 50.0	15°43.2	12.8'	15°48.7	10.8'	55.6'
16	57°59.5	49.0	30°14.9	12.7'	15°59.5	10.8'	55.6'
17	72°59.7	48.0	44°46.6	12.6'	16°10.3	10.7'	55.7'
18	87°59.8	S00°47.0	59°18.2	12.5'	N16°20.9	10.6'	55.7'
19	103°00.0	46.0	73°49.7	12.5'	16°31.6	10.5'	55.7'
20	118°00.2	45.0	88°21.2	12.4'	16°42.1	10.5'	55.7'
21	133°00.4	· · 44.0	102°52.6	12.3'	16°52.6	10.4'	55.7'
22	148°00.6	43.0	117°23.9	12.2'	17°03.0	10.3'	55.8'
23	163°00.8	42.0	131°55.1	12.1'	17°13.3	10.3'	55.8'
	SD = 16.1'	d = -1.0'	SD = 15.1'				
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	178°00.9	S00°41.1	146°26.2	12.1'	N17°23.6	10.2'	55.8'
1	193°01.1	40.1	160°57.3	12.0'	17°33.8	10.1'	55.8'
2	208°01.3	39.1	175°28.3	11.9'	17°43.9	10.0'	55.8'
3	223°01.5	· · 38.1	189°59.2	11.8'	17°53.9	10.0'	55.9'
4	238°01.7	37.1	204°30.0	11.7'	18°03.9	9.9'	55.9'
5	253°01.8	36.1	219°00.7	11.7'	18°13.7	9.8'	55.9'
6	268°02.0	S00°35.1	233°31.4	11.6'	N18°23.5	9.7'	55.9'
7	283°02.2	34.1	248°02.0	11.5'	18°33.2	9.6'	55.9'
8	298°02.4	33.1	262°32.5	11.4'	18°42.9	9.5'	56.0'
9	313°02.6	· · 32.2	277°02.9	11.3'	18°52.4	9.5'	56.0'
10	328°02.8	31.2	291°33.2	11.3'	19°01.9	9.4'	56.0'
11	343°02.9	30.2	306°03.5	11.2'	19°11.2	9.3'	56.0'
12	358°03.1	S00°29.2	320°33.6	11.1'	N19°20.5	9.2'	56.0'
13	13°03.3	28.2	335°03.7	11.0'	19°29.7	9.1'	56.1'
14	28°03.5	27.2	349°33.7	10.9'	19°38.8	9.0'	56.1'
15	43°03.7	· · 26.2	4°03.6	10.8'	19°47.8	8.9'	56.1'
16	58°03.9	25.2	18°33.5	10.8'	19°56.7	8.8'	56.1'
17	73°04.0	24.3	33°03.2	10.7'	20°05.5	8.7'	56.2'
18	88°04.2	S00°23.3	47°32.9	10.6'	N20°14.3	8.6'	56.2'
19	103°04.4	22.3	62°02.5	10.5'	20°22.9	8.5'	56.2'
20	118°04.6	21.3	76°32.0	10.4'	20°31.4	8.4'	56.2'
21	133°04.8	· · 20.3	91°01.4	10.3'	20°39.9	8.3'	56.2'
22	148°05.0	19.3	105°30.7	10.2'	20°48.2	8.2'	56.3'
23	163°05.1	18.3	120°00.0	10.2'	20°56.4	8.1'	56.3'
	SD = 16.1'	d = -1.0'	SD = 15.2'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	03:36	05:02	06:09	18:09	19:17	20:45
N 70°	03:52	05:08	06:09	18:09	19:10	20:28
68°	04:05	05:14	06:09	18:09	19:05	20:14
66°	04:16	05:18	06:09	18:09	19:00	20:03
64°	04:24	05:21	06:09	18:09	18:57	19:54
62°	04:31	05:24	06:08	18:09	18:54	19:47
60°	04:37	05:27	06:08	18:09	18:51	19:41
N 58°	04:43	05:29	06:08	18:09	18:48	19:35
56°	04:47	05:31	06:08	18:09	18:46	19:30
54°	04:51	05:33	06:08	18:09	18:45	19:26
52°	04:55	05:34	06:08	18:09	18:43	19:23
50°	04:58	05:35	06:08	18:09	18:42	19:20
45°	05:04	05:38	06:07	18:10	18:39	19:13
N 40°	05:09	05:40	06:07	18:10	18:37	19:08
35°	05:12	05:42	06:07	18:10	18:35	19:05
30°	05:15	05:43	06:07	18:10	18:34	19:02
20°	05:19	05:44	06:06	18:10	18:32	18:58
N 10°	05:20	05:44	06:05	18:11	18:32	18:56
0°	05:20	05:44	06:05	18:11	18:32	18:56
S 10°	05:19	05:43	06:04	18:12	18:33	18:57
20°	05:16	05:41	06:03	18:13	18:35	19:00
30°	05:10	05:38	06:02	18:14	18:38	19:05
35°	05:06	05:36	06:01	18:14	18:40	19:09
40°	05:02	05:33	06:01	18:15	18:42	19:14
45°	04:56	05:30	06:00	18:16	18:45	19:20
S 50°	04:48	05:26	05:58	18:17	18:49	19:27
52°	04:44	05:24	05:58	18:17	18:51	19:31
54°	04:39	05:22	05:57	18:18	18:53	19:35
56°	04:34	05:19	05:56	18:19	18:56	19:40
58°	04:29	05:16	05:56	18:19	18:59	19:46
S 60°	04:22	05:13	05:55	18:20	19:02	19:52

March 20, 21, 22 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn	
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	177°25.3	162°46.9	N05°13.7	53°37.5	N23°12.1	191°23.3	S07°00.6	355°04.5	N01°46.8
1	192°27.8	177°46.6	15.0	68°39.7	12.0	206°25.2	00.3	10°07.1	46.8
2	207°30.3	192°46.2	16.3	83°41.9	11.8	221°27.1	07°00.1	25°09.8	46.9
3	222°32.7	207°45.8	•• 17.5	98°44.1	•• 11.7	236°29.0	06°59.9	40°12.4	•• 47.0
4	237°35.2	222°45.4	18.8	113°46.3	11.6	251°31.0	59.7	55°15.1	47.1
5	252°37.7	237°45.1	20.1	128°48.5	11.4	266°32.9	59.4	70°17.7	47.2
6	267°40.1	252°44.7	N05°21.3	143°50.7	N23°11.3	281°34.8	S06°59.2	85°20.3	N01°47.2
7	282°42.6	267°44.3	22.6	158°52.9	11.1	296°36.7	59.0	100°23.0	47.3
8	297°45.0	282°43.9	23.9	173°55.1	11.0	311°38.6	58.8	115°25.6	47.4
9	312°47.5	297°43.5	•• 25.1	188°57.3	•• 10.8	326°40.5	•• 58.5	130°28.3	•• 47.5
10	327°50.0	312°43.2	26.4	203°59.5	10.7	341°42.4	58.3	145°30.9	47.6
11	342°52.4	327°42.8	27.7	219°01.7	10.5	356°44.3	58.1	160°33.6	47.6
12	357°54.9	342°42.4	N05°28.9	234°03.8	N23°10.4	11°46.2	S06°57.9	175°36.2	N01°47.7
13	12°57.4	357°42.0	30.2	249°06.0	10.2	26°48.2	57.6	190°38.9	47.8
14	27°59.8	12°41.7	31.4	264°08.2	10.1	41°50.1	57.4	205°41.5	47.9
15	43°02.3	27°41.3	•• 32.7	279°10.4	•• 09.9	56°52.0	•• 57.2	220°44.1	•• 48.0
16	58°04.8	42°40.9	34.0	294°12.6	09.8	71°53.9	57.0	235°46.8	48.0
17	73°07.2	57°40.5	35.2	309°14.8	09.6	86°55.8	56.7	250°49.4	48.1
18	88°09.7	72°40.1	N05°36.5	324°16.9	N23°09.5	101°57.7	S06°56.5	265°52.1	N01°48.2
19	103°12.2	87°39.8	37.8	339°19.1	09.3	116°59.6	56.3	280°54.7	48.3
20	118°14.6	102°39.4	39.0	354°21.3	09.2	132°01.5	56.1	295°57.4	48.4
21	133°17.1	117°39.0	•• 40.3	9°23.5	•• 09.1	147°03.4	•• 55.8	311°00.0	•• 48.4
22	148°19.5	132°38.6	41.5	24°25.6	08.9	162°05.4	55.6	326°02.6	48.5
23	163°22.0	147°38.2	42.8	39°27.8	08.8	177°07.3	55.4	341°05.3	48.6
Mer.pass. 12:08		ν -0.4' d 1.3' m -3.90		ν 2.2' d -0.1' m -0.11		ν 1.9' d -0.2' m -2.03		ν 2.6' d 0.1' m 0.52	

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	178°24.5	162°37.9	N05°44.1	54°30.0	N23°08.6	192°09.2	S06°55.2	356°07.9	N01°48.7
1	193°26.9	177°37.5	45.3	69°32.2	08.5	207°11.1	54.9	11°10.6	48.8
2	208°29.4	192°37.1	46.6	84°34.3	08.3	222°13.0	54.7	26°13.2	48.8
3	223°31.9	207°36.7	· · 47.8	99°36.5	· · 08.2	237°14.9	· · 54.5	41°15.9	· · 48.9
4	238°34.3	222°36.3	49.1	114°38.7	08.0	252°16.8	54.2	56°18.5	49.0
5	253°36.8	237°36.0	50.4	129°40.8	07.8	267°18.7	54.0	71°21.2	49.1
6	268°39.3	252°35.6	N05°51.6	144°43.0	N23°07.7	282°20.6	S06°53.8	86°23.8	N01°49.2
7	283°41.7	267°35.2	52.9	159°45.2	07.5	297°22.6	53.6	101°26.4	49.2
8	298°44.2	282°34.8	54.1	174°47.3	07.4	312°24.5	53.3	116°29.1	49.3
9	313°46.7	297°34.4	· · 55.4	189°49.5	· · 07.2	327°26.4	· · 53.1	131°31.7	· · 49.4
10	328°49.1	312°34.1	56.6	204°51.7	07.1	342°28.3	52.9	146°34.4	49.5
11	343°51.6	327°33.7	57.9	219°53.8	06.9	357°30.2	52.7	161°37.0	49.6
12	358°54.0	342°33.3	N05°59.2	234°56.0	N23°06.8	12°32.1	S06°52.4	176°39.7	N01°49.6
13	13°56.5	357°32.9	06°00.4	249°58.1	06.6	27°34.0	52.2	191°42.3	49.7
14	28°59.0	12°32.5	01.7	265°00.3	06.5	42°35.9	52.0	206°45.0	49.8
15	44°01.4	27°32.2	· · 02.9	280°02.5	· · 06.3	57°37.9	· · 51.8	221°47.6	· · 49.9
16	59°03.9	42°31.8	04.2	295°04.6	06.2	72°39.8	51.5	236°50.2	50.0
17	74°06.4	57°31.4	05.4	310°06.8	06.0	87°41.7	51.3	251°52.9	50.0
18	89°08.8	72°31.0	N06°06.7	325°08.9	N23°05.9	102°43.6	S06°51.1	266°55.5	N01°50.1
19	104°11.3	87°30.6	08.0	340°11.1	05.7	117°45.5	50.9	281°58.2	50.2
20	119°13.8	102°30.2	09.2	355°13.2	05.6	132°47.4	50.6	297°00.8	50.3
21	134°16.2	117°29.9	· · 10.5	10°15.4	· · 05.4	147°49.3	· · 50.4	312°03.5	· · 50.4
22	149°18.7	132°29.5	11.7	25°17.5	05.2	162°51.2	50.2	327°06.1	50.4
23	164°21.1	147°29.1	13.0	40°19.7	05.1	177°53.2	50.0	342°08.7	50.5
Mer.pass. 12:04		ν -0.4' d 1.3' m -3.90		ν 2.2' d -0.1' m -0.09		ν 1.9' d -0.2' m -2.04		ν 2.6' d 0.1' m 0.52	

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	179°23.6	162°28.7	N06°14.2	55°21.8	N23°04.9	192°55.1	S06°49.7	357°11.4	N01°50.6
1	194°26.1	177°28.3	15.5	70°23.9	04.8	207°57.0	49.5	12°14.0	50.7
2	209°28.5	192°27.9	16.7	85°26.1	04.6	222°58.9	49.3	27°16.7	50.8
3	224°31.0	207°27.6	•• 18.0	100°28.2	•• 04.5	238°00.8	•• 49.1	42°19.3	•• 50.8
4	239°33.5	222°27.2	19.2	115°30.4	04.3	253°02.7	48.8	57°22.0	50.9
5	254°35.9	237°26.8	20.5	130°32.5	04.2	268°04.6	48.6	72°24.6	51.0
6	269°38.4	252°26.4	N06°21.8	145°34.7	N23°04.0	283°06.6	S06°48.4	87°27.3	N01°51.1
7	284°40.9	267°26.0	23.0	160°36.8	03.8	298°08.5	48.2	102°29.9	51.2
8	299°43.3	282°25.6	24.3	175°38.9	03.7	313°10.4	47.9	117°32.5	51.2
9	314°45.8	297°25.2	•• 25.5	190°41.1	•• 03.5	328°12.3	•• 47.7	132°35.2	•• 51.3
10	329°48.3	312°24.9	26.8	205°43.2	03.4	343°14.2	47.5	147°37.8	51.4
11	344°50.7	327°24.5	28.0	220°45.3	03.2	358°16.1	47.3	162°40.5	51.5
12	359°53.2	342°24.1	N06°29.3	235°47.5	N23°03.1	13°18.0	S06°47.0	177°43.1	N01°51.6
13	14°55.6	357°23.7	30.5	250°49.6	02.9	28°20.0	46.8	192°45.8	51.6
14	29°58.1	12°23.3	31.8	265°51.7	02.7	43°21.9	46.6	207°48.4	51.7
15	45°00.6	27°22.9	•• 33.0	280°53.9	•• 02.6	58°23.8	•• 46.4	222°51.1	•• 51.8
16	60°03.0	42°22.5	34.3	295°56.0	02.4	73°25.7	46.1	237°53.7	51.9
17	75°05.5	57°22.2	35.5	310°58.1	02.3	88°27.6	45.9	252°56.3	52.0
18	90°08.0	72°21.8	N06°36.8	326°00.2	N23°02.1	103°29.5	S06°45.7	267°59.0	N01°52.0
19	105°10.4	87°21.4	38.0	341°02.4	01.9	118°31.4	45.5	283°01.6	52.1
20	120°12.9	102°21.0	39.3	356°04.5	01.8	133°33.3	45.2	298°04.3	52.2
21	135°15.4	117°20.6	•• 40.5	11°06.6	•• 01.6	148°35.3	•• 45.0	313°06.9	•• 52.3
22	150°17.8	132°20.2	41.8	26°08.7	01.5	163°37.2	44.8	328°09.6	52.4
23	165°20.3	147°19.8	43.0	41°10.8	01.3	178°39.1	44.6	343°12.2	52.4
Mer.pass. 12:00		ν -0.4' d 1.3' m -3.90		ν 2.1' d -0.2' m -0.06		ν 1.9' d -0.2' m -2.04		ν 2.6' d 0.1' m 0.51	

	SHA	Dec
Alpheratz	357° 46.4	29° 08.8
Ankaa	353° 18.3	-42° 15.0
Schedar	349° 44.0	56° 35.7
Diphda	348° 58.5	-17° 55.9
Achernar	335° 28.9	-57° 11.2
Hamal	328° 03.8	23° 30.7
Polaris	319° 17.2	89° 18.8
Acamar	315° 20.3	-40° 16.0
Menkar	314° 17.7	4° 07.8
Mirfak	308° 44.1	49° 54.4
Aldebaran	290° 52.2	16° 31.8
Rigel	281° 14.4	-8° 11.5
Capella	280° 38.0	46° 00.6
Bellatrix	278° 34.6	6° 21.5
Elnath	278° 15.7	28° 37.0
Alnilam	275° 48.8	-1° 11.8
Betelgeuse	271° 03.9	7° 24.5
Canopus	263° 57.2	-52° 42.4
Sirius	258° 35.7	-16° 40.4
Adhara	255° 14.3	-28° 59.4
Procyon	245° 02.1	5° 11.8
Pollux	243° 30.4	28° 00.1
Avior	234° 18.7	-59° 32.9
Shuail	222° 54.0	-43° 28.7
Miaplacidus	221° 39.7	-69° 45.8
Alphard	217° 58.2	-8° 42.4
Regulus	207° 45.7	11° 54.8
Dubhe	193° 53.7	61° 41.7
Denebola	182° 35.7	14° 30.7
Gienah	175° 54.4	-17° 36.1
Acrux	173° 11.5	-63° 09.5
Gacrux	172° 03.2	-57° 10.4
Alioth	166° 22.0	55° 54.0
Spica	158° 33.5	-11° 13.1
Alkaid	153° 00.2	49° 15.4
Hadar	148° 50.9	-60° 25.4
Menkent	148° 10.1	-36° 25.3
Arcturus	145° 57.6	19° 07.5
Rigil Kent.	139° 54.6	-60° 52.6
Kochab	137° 18.8	74° 06.5
Zuben 'ubi	137° 07.8	-16° 05.2
Alphecca	126° 12.8	26° 40.5
Antares	112° 29.0	-26° 27.3
Atria	107° 33.0	-69° 02.6
Sabik	102° 15.2	-15° 44.3
Shaula	96° 25.1	-37° 06.6
Rasalhague	96° 08.6	12° 32.9
Eltanin	90° 47.2	51° 28.9
Kaus Aust.	83° 47.0	-34° 22.7
Vega	80° 40.6	38° 47.3
Nunki	76° 01.3	-26° 17.0
Altair	62° 10.7	8° 53.6
Peacock	53° 23.2	-56° 41.9
Deneb	49° 33.5	45° 18.8
Enif	33° 49.7	9° 55.2
Al Na'ir	27° 47.0	-46° 54.6
Fomalhaut	15° 26.9	-29° 34.1
Scheat	13° 56.1	28° 08.2
Markab	13° 41.0	15° 15.6

Mar 20 Sat	SHA	Mer.pass
Venus	345°21.6	13:09
Mars	236°12.2	20:23
Jupiter	13°58.0	11:13
Saturn	177°39.2	00:20

Mar 21 Sun	SHA	Mer.pass
Venus	344°13.4	13:10
Mars	236°05.5	20:19
Jupiter	13°44.7	11:10
Saturn	177°43.5	00:15

Mar 22 Mon	SHA	Mer.pass
Venus	343°05.1	13:10
Mars	235°58.2	20:16
Jupiter	13°31.5	11:07
Saturn	177°47.8	00:11

Horizontal parallax	
Venus:	0.1
Mars:	0.2

DUT1 = UT1-UTC = +0.0306 sec ΔT = TT-UT1 = +66.1534 sec

2010 March 20 to Mar. 22 UT

Sun				Moon					
Sat	GHA	Dec		GHA	ν	Dec	d	HP	
0	178°05.3	S00°17.3		134°29.1	10.1'	N21°04.6	8.0'	56.3'	
1	193°05.5	16.3		148°58.2	10.0'	21°12.6	7.9'	56.3'	
2	208°05.7	15.4		163°27.2	9.9'	21°20.5	7.8'	56.4'	
3	223°05.9	· · 14.4		177°56.1	9.8'	21°28.4	7.7'	56.4'	
4	238°06.1	13.4		192°24.9	9.7'	21°36.1	7.6'	56.4'	
5	253°06.2	12.4		206°53.6	9.7'	21°43.7	7.5'	56.4'	
6	268°06.4	S00°11.4		221°22.3	9.6'	N21°51.2	7.4'	56.5'	
7	283°06.6	10.4		235°50.9	9.5'	21°58.6	7.3'	56.5'	
8	298°06.8	09.4		250°19.4	9.4'	22°05.8	7.2'	56.5'	
9	313°07.0	· · 08.4		264°47.8	9.3'	22°13.0	7.1'	56.5'	
10	328°07.2	07.4		279°16.1	9.2'	22°20.1	6.9'	56.5'	
11	343°07.3	06.5		293°44.3	9.2'	22°27.0	6.8'	56.6'	
12	358°07.5	S00°05.5		308°12.5	9.1'	N22°33.8	6.7'	56.6'	
13	13°07.7	04.5		322°40.5	9.0'	22°40.5	6.6'	56.6'	
14	28°07.9	03.5		337°08.5	8.9'	22°47.1	6.5'	56.6'	
15	43°08.1	· · 02.5		351°36.4	8.8'	22°53.6	6.3'	56.7'	
16	58°08.3	01.5		6°04.3	8.7'	22°59.9	6.2'	56.7'	
17	73°08.4	S00°00.5		20°32.0	8.7'	23°06.2	6.1'	56.7'	
18	88°08.6	N00°00.5		34°59.7	8.6'	N23°12.3	6.0'	56.7'	
19	103°08.8	01.4		49°27.3	8.5'	23°18.2	5.9'	56.8'	
20	118°09.0	02.4		63°54.8	8.4'	23°24.1	5.7'	56.8'	
21	133°09.2	· · 03.4		78°22.2	8.4'	23°29.8	5.6'	56.8'	
22	148°09.4	04.4		92°49.6	8.3'	23°35.4	5.5'	56.8'	
23	163°09.6	05.4		107°16.8	8.2'	23°40.9	5.3'	56.9'	
SD = 16.1' $d = -1.0'$				SD = 15.4'					

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	178°09.7	N00°06.4	121°44.0	8.1'	N23°46.3	5.2'	56.9'
1	193°09.9	07.4	136°11.2	8.0'	23°51.5	5.1'	56.9'
2	208°10.1	08.4	150°38.2	8.0'	23°56.6	5.0'	56.9'
3	223°10.3	· · 09.4	165°05.2	7.9'	24°01.5	4.8'	57.0'
4	238°10.5	10.3	179°32.1	7.8'	24°06.3	4.7'	57.0'
5	253°10.7	11.3	193°58.9	7.8'	24°11.0	4.6'	57.0'
6	268°10.9	N00°12.3	208°25.7	7.7'	N24°15.6	4.4'	57.0'
7	283°11.0	13.3	222°52.3	7.6'	24°20.0	4.3'	57.1'
8	298°11.2	14.3	237°18.9	7.5'	24°24.3	4.1'	57.1'
9	313°11.4	· · 15.3	251°45.5	7.5'	24°28.4	4.0'	57.1'
10	328°11.6	16.3	266°12.0	7.4'	24°32.4	3.9'	57.1'
11	343°11.8	17.3	280°38.4	7.3'	24°36.3	3.7'	57.2'
12	358°12.0	N00°18.2	295°04.7	7.3'	N24°40.0	3.6'	57.2'
13	13°12.2	19.2	309°31.0	7.2'	24°43.6	3.4'	57.2'
14	28°12.3	20.2	323°57.2	7.1'	24°47.1	3.3'	57.2'
15	43°12.5	· · 21.2	338°23.3	7.1'	24°50.4	3.2'	57.3'
16	58°12.7	22.2	352°49.4	7.0'	24°53.5	3.0'	57.3'
17	73°12.9	23.2	7°15.4	7.0'	24°56.5	2.9'	57.3'
18	88°13.1	N00°24.2	21°41.4	6.9'	N24°59.4	2.7'	57.3'
19	103°13.3	25.2	36°07.3	6.8'	25°02.1	2.6'	57.4'
20	118°13.5	26.1	50°33.2	6.8'	25°04.7	2.4'	57.4'
21	133°13.6	· · 27.1	64°59.0	6.7'	25°07.1	2.3'	57.4'
22	148°13.8	28.1	79°24.7	6.7'	25°09.4	2.1'	57.4'
23	163°14.0	29.1	93°50.4	6.6'	25°11.5	2.0'	57.5'
SD = 16.1'		$d = 1.0'$	SD = 15.5'				

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	178°14.2	N00°30.1	108°16.0	6.6'	N25°13.5	1.8'	57.5'
1	193°14.4	31.1	122°41.6	6.5'	25°15.3	1.7'	57.5'
2	208°14.6	32.1	137°07.1	6.5'	25°17.0	1.5'	57.5'
3	223°14.8	· · 33.1	151°32.6	6.4'	25°18.5	1.4'	57.6'
4	238°14.9	34.0	165°58.0	6.4'	25°19.9	1.2'	57.6'
5	253°15.1	35.0	180°23.4	6.3'	25°21.1	1.1'	57.6'
6	268°15.3	N00°36.0	194°48.7	6.3'	N25°22.1	0.9'	57.7'
7	283°15.5	37.0	209°14.0	6.3'	25°23.0	0.7'	57.7'
8	298°15.7	38.0	223°39.3	6.2'	25°23.8	0.6'	57.7'
9	313°15.9	· · 39.0	238°04.5	6.2'	25°24.4	0.4'	57.7'
10	328°16.1	40.0	252°29.7	6.1'	25°24.8	0.3'	57.8'
11	343°16.3	40.9	266°54.8	6.1'	25°25.1	0.1'	57.8'
12	358°16.4	N00°41.9	281°19.9	6.1'	N25°25.2	-0.0'	57.8'
13	13°16.6	42.9	295°45.0	6.0'	25°25.2	-0.2'	57.8'
14	28°16.8	43.9	310°10.0	6.0'	25°25.0	-0.3'	57.9'
15	43°17.0	· · 44.9	324°35.0	6.0'	25°24.6	-0.5'	57.9'
16	58°17.2	45.9	339°00.0	5.9'	25°24.1	-0.7'	57.9'
17	73°17.4	46.9	353°25.0	5.9'	25°23.5	-0.8'	57.9'
18	88°17.6	N00°47.9	7°49.9	5.9'	N25°22.6	-1.0'	58.0'
19	103°17.7	48.8	22°14.8	5.9'	25°21.7	-1.1'	58.0'
20	118°17.9	49.8	36°39.7	5.9'	25°20.5	-1.3'	58.0'
21	133°18.1	· · 50.8	51°04.5	5.8'	25°19.2	-1.5'	58.1'
22	148°18.3	51.8	65°29.3	5.8'	25°17.7	-1.6'	58.1'
23	163°18.5	52.8	79°54.2	5.8'	25°16.1	-1.8'	58.1'
	SD = 16.0'	$d = 1.0'$			SD = 15.7'		

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	03:16	04:46	05:54	18:23	19:32	21:04
N 70°	03:35	04:54	05:55	18:21	19:23	20:43
68°	03:50	05:01	05:56	18:20	19:16	20:27
66°	04:03	05:06	05:57	18:19	19:11	20:15
64°	04:12	05:11	05:58	18:18	19:06	20:04
62°	04:21	05:14	05:59	18:17	19:02	19:56
60°	04:28	05:18	05:59	18:17	18:58	19:49
N 58°	04:34	05:21	06:00	18:16	18:55	19:42
56°	04:39	05:23	06:00	18:15	18:53	19:37
54°	04:43	05:25	06:00	18:15	18:50	19:32
52°	04:47	05:27	06:01	18:15	18:48	19:28
50°	04:51	05:29	06:01	18:14	18:47	19:25
45°	04:58	05:32	06:02	18:13	18:43	19:17
N 40°	05:04	05:35	06:02	18:13	18:40	19:12
35°	05:08	05:37	06:03	18:12	18:38	19:07
30°	05:11	05:39	06:03	18:12	18:36	19:04
20°	05:16	05:41	06:03	18:11	18:33	18:59
N 10°	05:18	05:43	06:04	18:11	18:32	18:56
0°	05:19	05:43	06:04	18:10	18:31	18:55
S 10°	05:19	05:43	06:04	18:10	18:31	18:56
20°	05:16	05:42	06:04	18:10	18:32	18:58
30°	05:12	05:40	06:04	18:10	18:34	19:02
35°	05:09	05:39	06:04	18:10	18:35	19:05
40°	05:05	05:37	06:04	18:10	18:37	19:09
45°	05:00	05:34	06:03	18:10	18:39	19:14
S 50°	04:53	05:31	06:03	18:10	18:43	19:20
52°	04:49	05:29	06:03	18:10	18:44	19:24
54°	04:46	05:27	06:03	18:11	18:46	19:27
56°	04:41	05:25	06:03	18:11	18:48	19:32
58°	04:36	05:23	06:02	18:11	18:50	19:37
S 60°	04:30	05:20	06:02	18:11	18:52	19:42

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	04:02	☐	☐	02:03	☐	☐
66°	05:01	04:08	☐	01:05	03:49	☐
64°	05:35	05:39	06:00	00:31	02:18	03:55
62°	06:00	06:17	06:53	00:07	01:41	03:02
60°	06:20	06:44	07:25		01:14	02:31
N 58°	06:37	07:05	07:49		00:54	02:07
56°	06:51	07:22	08:08		00:37	01:47
54°	07:03	07:37	08:24		00:22	01:31
52°	07:13	07:50	08:38		00:10	01:17
50°	07:23	08:01	08:51	23:59		01:05
45°	07:43	08:24	09:16	23:36		00:40
N 40°	08:00	08:43	09:36	23:17		00:20
35°	08:14	08:59	09:53	23:02		00:03
30°	08:26	09:13	10:07	22:48	23:49	
20°	08:47	09:37	10:32	22:26	23:24	
N 10°	09:05	09:57	10:53	22:06	23:03	
0°	09:22	10:16	11:13	21:48	22:43	23:41
S 10°	09:39	10:36	11:33	21:29	22:24	23:22
20°	09:58	10:56	11:55	21:10	22:02	23:00
30°	10:19	11:20	12:20	20:47	21:38	22:35
35°	10:32	11:34	12:35	20:34	21:24	22:21
40°	10:46	11:51	12:52	20:19	21:07	22:04
45°	11:04	12:10	13:12	20:01	20:47	21:44
S 50°	11:25	12:35	13:38	19:39	20:22	21:18
52°	11:35	12:47	13:50	19:28	20:10	21:06
54°	11:47	13:01	14:04	19:16	19:56	20:51
56°	12:00	13:16	14:21	19:02	19:40	20:35
58°	12:16	13:35	14:41	18:46	19:21	20:15
S 60°	12:35	13:58	15:05	18:27	18:58	19:50

March 23, 24, 25 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	180°22.8	162°19.5	N06°44.3	56°13.0	N23°01.1	193°41.0	S06°44.3	358°14.8	N01°52.5	Alpheratz	357°46.4	29°08.8
1	195°25.2	177°19.1	45.5	71°15.1	01.0	208°42.9	44.1	13°17.5	52.6	Ankaa	353°18.3	-42°15.0
2	210°27.7	192°18.7	46.8	86°17.2	00.8	223°44.8	43.9	28°20.1	52.7	Schedar	349°44.0	56°35.7
3	225°30.1	207°18.3	· · 48.0	101°19.3	· · 00.7	238°46.7	· · 43.7	43°22.8	· · 52.8	Diphda	348°58.5	-17°55.9
4	240°32.6	222°17.9	49.3	116°21.4	00.5	253°48.7	43.4	58°25.4	52.8	Achernar	335°28.9	-57°11.2
5	255°35.1	237°17.5	50.5	131°23.5	00.3	268°50.6	43.2	73°28.1	52.9	Hamal	328°03.8	23°30.7
6	270°37.5	252°17.1	N06°51.8	146°25.6	N23°00.2	283°52.5	S06°43.0	88°30.7	N01°53.0	Polaris	319°17.8	89°18.8
7	285°40.0	267°16.7	53.0	161°27.8	23°00.0	298°54.4	42.8	103°33.4	53.1	Acamar	315°20.4	-40°16.0
8	300°42.5	282°16.3	54.3	176°29.9	22°59.8	313°56.3	42.5	118°36.0	53.2	Menkar	314°17.7	4°07.8
9	315°44.9	297°16.0	· · 55.5	191°32.0	· · 59.7	328°58.2	· · 42.3	133°38.6	· · 53.2	Mirfak	308°44.1	49°54.0
10	330°47.4	312°15.6	56.7	206°34.1	59.5	344°00.2	42.1	148°41.3	53.3	Aldebaran	290°52.2	16°31.8
11	345°49.9	327°15.2	58.0	221°36.2	59.4	359°02.1	41.9	163°43.9	53.4	Rigel	281°14.4	-8°11.5
12	0°52.3	342°14.8	N06°59.2	236°38.3	N22°59.2	14°04.0	S06°41.6	178°46.6	N01°53.5	Capella	280°38.1	46°00.6
13	15°54.8	357°14.4	07°00.5	251°40.4	59.0	29°05.9	41.4	193°49.2	53.6	Bellatrix	278°34.6	6°21.5
14	30°57.3	12°14.0	01.7	266°42.5	58.9	44°07.8	41.2	208°51.9	53.6	Elnath	278°15.7	28°37.0
15	45°59.7	27°13.6	· · 03.0	281°44.6	· · 58.7	59°09.7	· · 41.0	223°54.5	· · 53.7	Alnilam	275°48.8	-1°11.8
16	61°02.2	42°13.2	04.2	296°46.7	58.5	74°11.6	40.7	238°57.2	53.8	Betelgeuse	271°03.9	7°24.5
17	76°04.6	57°12.8	05.5	311°48.8	58.4	89°13.6	40.5	253°59.8	53.9	Canopus	263°57.2	-52°42.4
18	91°07.1	72°12.4	N07°06.7	326°50.9	N22°58.2	104°15.5	S06°40.3	269°02.4	N01°53.9	Sirius	258°35.8	-16°44.0
19	106°09.6	87°12.1	07.9	341°53.0	58.0	119°17.4	40.1	284°05.1	54.0	Adhara	255°14.3	-28°59.4
20	121°12.0	102°11.7	09.2	356°55.1	57.9	134°19.3	39.8	299°07.7	54.1	Procyon	245°02.1	5°11.8
21	136°14.5	117°11.3	· · 10.4	11°57.2	· · 57.7	149°21.2	· · 39.6	314°10.4	· · 54.2	Pollux	243°30.4	28°00.1
22	151°17.0	132°10.9	11.7	26°59.3	57.5	164°23.1	39.4	329°13.0	54.3	Avior	234°18.8	-59°32.9
23	166°19.4	147°10.5	12.9	42°01.4	57.4	179°25.0	39.2	344°15.7	54.3	Suhail	222°54.0	-43°28.7
Mer.pass. 11:57		ν -0.4' d 1.2' m -3.90		ν 2.1' d -0.2' m -0.04		ν 1.9' d -0.2' m -2.04		ν 2.6' d 0.1' m 0.52				

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	181°21.9	162°10.1	N07°14.2	57°03.5	N22°57.2	194°27.0	S06°38.9	359°18.3	N01°54.4
1	196°24.4	177°09.7	15.4	72°05.6	57.1	209°28.9	38.7	14°20.9	54.5
2	211°26.8	192°09.3	16.6	87°07.7	56.9	224°30.8	38.5	29°23.6	54.6
3	226°29.3	207°08.9	· · 17.9	102°09.7	· · 56.7	239°32.7	· · 38.3	44°26.2	· · 54.7
4	241°31.7	222°08.5	19.1	117°11.8	56.6	254°34.6	38.0	59°28.9	54.7
5	256°34.2	237°08.1	20.4	132°13.9	56.4	269°36.5	37.8	74°31.5	54.8
6	271°36.7	252°07.7	N07°21.6	147°16.0	N22°56.2	284°38.5	S06°37.6	89°34.2	N01°54.9
7	286°39.1	267°07.3	22.8	162°18.1	56.0	299°40.4	37.4	104°36.8	55.0
8	301°41.6	282°07.0	24.1	177°20.2	55.9	314°42.3	37.1	119°39.5	55.1
9	316°44.1	297°06.6	· · 25.3	192°22.3	· · 55.7	329°44.2	· · 36.9	134°42.1	· · 55.1
10	331°46.5	312°06.2	26.6	207°24.3	55.5	344°46.1	36.7	149°44.7	55.2
11	346°49.0	327°05.8	27.8	222°26.4	55.4	359°48.0	36.5	164°47.4	55.3
12	1°51.5	342°05.4	N07°29.0	237°28.5	N22°55.2	14°50.0	S06°36.2	179°50.0	N01°55.4
13	16°53.9	357°05.0	30.3	252°30.6	55.0	29°51.9	36.0	194°52.7	55.5
14	31°56.4	12°04.6	31.5	267°32.7	54.9	44°53.8	35.8	209°55.3	55.5
15	46°58.9	27°04.2	· · 32.8	282°34.7	· · 54.7	59°55.7	· · 35.6	224°58.0	· · 55.6
16	62°01.3	42°03.8	34.0	297°36.8	54.5	74°57.6	35.3	240°00.6	55.7
17	77°03.8	57°03.4	35.2	312°38.9	54.4	89°59.5	35.1	255°03.3	55.8
18	92°06.2	72°03.0	N07°36.5	327°41.0	N22°54.2	105°01.5	S06°34.9	270°05.9	N01°55.9
19	107°08.7	87°02.6	37.7	342°43.0	54.0	120°03.4	34.7	285°08.5	55.9
20	122°11.2	102°02.2	38.9	357°45.1	53.9	135°05.3	34.4	300°11.2	56.0
21	137°13.6	117°01.8	· · 40.2	12°47.2	· · 53.7	150°07.2	· · 34.2	315°13.8	· · 56.1
22	152°16.1	132°01.4	41.4	27°49.2	53.5	165°09.1	34.0	330°16.5	56.2
23	167°18.6	147°01.0	42.7	42°51.3	53.3	180°11.0	33.8	345°19.1	56.2
Mer.pass. 11:53		ν -0.4' d 1.2' m -3.90		ν 2.1' d -0.2' m -0.02		ν 1.9' d -0.2' m -2.04		ν 2.6' d 0.1' m 0.52	

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	182°21.0	162°00.6	N07°43.9	57°53.4	N22°53.2	195°13.0	S06°33.5	0°21.8	N01°56.3
1	197°23.5	177°00.2	45.1	72°55.4	53.0	210°14.9	33.3	15°24.4	56.4
2	212°26.0	191°59.8	46.4	87°57.5	52.8	225°16.8	33.1	30°27.0	56.5
3	227°28.4	206°59.4	· · 47.6	102°59.6	· · 52.7	240°18.7	· · 32.9	45°29.7	· · 56.6
4	242°30.9	221°59.0	48.8	118°01.6	52.5	255°20.6	32.6	60°32.3	56.6
5	257°33.4	236°58.6	50.1	133°03.7	52.3	270°22.5	32.4	75°35.0	56.7
6	272°35.8	251°58.2	N07°51.3	148°05.8	N22°52.1	285°24.5	S06°32.2	90°37.6	N01°56.8
7	287°38.3	266°57.8	52.5	163°07.8	52.0	300°26.4	32.0	105°40.3	56.9
8	302°40.7	281°57.4	53.8	178°09.9	51.8	315°28.3	31.7	120°42.9	57.0
9	317°43.2	296°57.1	· · 55.0	193°11.9	· · 51.6	330°30.2	· · 31.5	135°45.6	· · 57.0
10	332°45.7	311°56.7	56.2	208°14.0	51.4	345°32.1	31.3	150°48.2	57.1
11	347°48.1	326°56.3	57.5	223°16.0	51.3	0°34.0	31.1	165°50.8	57.2
12	2°50.6	341°55.9	N07°58.7	238°18.1	N22°51.1	15°36.0	S06°30.9	180°53.5	N01°57.3
13	17°53.1	356°55.5	07°59.9	253°20.1	50.9	30°37.9	30.6	195°56.1	57.4
14	32°55.5	11°55.1	08°01.1	268°22.2	50.7	45°39.8	30.4	210°58.8	57.4
15	47°58.0	26°54.7	· · 02.4	283°24.2	· · 50.6	60°41.7	· · 30.2	226°01.4	· · 57.5
16	63°00.5	41°54.3	03.6	298°26.3	50.4	75°43.6	30.0	241°04.1	57.6
17	78°02.9	56°53.9	04.8	313°28.3	50.2	90°45.6	29.7	256°06.7	57.7
18	93°05.4	71°53.5	N08°06.1	328°30.4	N22°50.0	105°47.5	S06°29.5	271°09.3	N01°57.7
19	108°07.9	86°53.1	07.3	343°32.4	49.9	120°49.4	29.3	286°12.0	57.8
20	123°10.3	101°52.6	08.5	358°34.5	49.7	135°51.3	29.1	301°14.6	57.9
21	138°12.8	116°52.2	· · 09.8	13°36.5	· · 49.5	150°53.2	· · 28.8	316°17.3	· · 58.0
22	153°15.2	131°51.8	11.0	28°38.6	49.3	165°55.1	28.6	331°19.9	58.1
23	168°17.7	146°51.4	12.2	43°40.6	49.2	180°57.1	28.4	346°22.6	58.1
Mer.pass. 11:49		ν -0.4' d 1.2' m -3.90		ν 2.1' d -0.2' m 0.01		ν 1.9' d -0.2' m -2.04		ν 2.6' d 0.1' m 0.52	

Mar 23 Tue	SHA	Mer.pass
Venus	341°56.7	13:11
Mars	235°50.2	20:12
Jupiter	13°18.3	11:04
Saturn	177°52.1	00:07

Mar 24 Wed	SHA	Mer.pass
Venus	340°48.2	13:12
Mars	235°41.6	20:09
Jupiter	13°05.1	11:01
Saturn	177°56.4	00:03

Mar 25 Thu	SHA	Mer.pass
Venus	339°39.6	13:12
Mars	235°32.3	20:06
Jupiter	12°51.9	10:58
Saturn	178°00.7	23:54

Horizontal parallax	
Venus:	0.1
Mars:	0.2

DUT1 = UT1-UTC = +0.0287 sec $\Delta T = TT-UT1 = +66.1553$ sec

2010 March 23 to Mar. 25 UT

h	Sun			Moon			
Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	178°18.7	N00°53.8	94°18.9	5.8'	N25°14.3	-2.0'	58.1'
1	193°18.9	54.8	108°43.7	5.8'	25°12.4	-2.1'	58.2'
2	208°19.1	55.7	123°08.5	5.8'	25°10.3	-2.3'	58.2'
3	223°19.2	· · 56.7	137°33.2	5.7'	25°08.0	-2.4'	58.2'
4	238°19.4	57.7	151°58.0	5.7'	25°05.5	-2.6'	58.3'
5	253°19.6	58.7	166°22.7	5.7'	25°03.0	-2.8'	58.3'
6	268°19.8	N00°59.7	180°47.5	5.7'	N25°00.2	-2.9'	58.3'
7	283°20.0	01°00.7	195°12.2	5.7'	24°57.3	-3.1'	58.3'
8	298°20.2	01.7	209°36.9	5.7'	24°54.2	-3.2'	58.4'
9	313°20.4	· · 02.6	224°01.6	5.7'	24°51.0	-3.4'	58.4'
10	328°20.6	03.6	238°26.3	5.7'	24°47.6	-3.6'	58.4'
11	343°20.7	04.6	252°51.0	5.7'	24°44.0	-3.7'	58.4'
12	358°20.9	N01°05.6	267°15.7	5.7'	N24°40.3	-3.9'	58.5'
13	13°21.1	06.6	281°40.4	5.7'	24°36.4	-4.0'	58.5'
14	28°21.3	07.6	296°05.2	5.7'	24°32.3	-4.2'	58.5'
15	43°21.5	· · 08.6	310°29.9	5.7'	24°28.1	-4.4'	58.6'
16	58°21.7	09.5	324°54.6	5.7'	24°23.8	-4.5'	58.6'
17	73°21.9	10.5	339°19.4	5.8'	24°19.2	-4.7'	58.6'
18	88°22.1	N01°11.5	353°44.1	5.8'	N24°14.5	-4.8'	58.6'
19	103°22.3	12.5	8°08.9	5.8'	24°09.7	-5.0'	58.7'
20	118°22.4	13.5	22°33.7	5.8'	24°04.7	-5.2'	58.7'
21	133°22.6	· · 14.5	36°58.5	5.8'	23°59.5	-5.3'	58.7'
22	148°22.8	15.5	51°23.3	5.8'	23°54.2	-5.5'	58.7'
23	163°23.0	16.4	65°48.1	5.8'	23°48.7	-5.6'	58.8'
	SD = 16.0'	$d = 1.0'$			SD = 15.9'		

Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	178°23.2	N01°17.4	80°12.9	5.9'	N23°43.1	-5.8'	58.8'
1	193°23.4	18.4	94°37.8	5.9'	23°37.3	-5.9'	58.8'
2	208°23.6	19.4	109°02.7	5.9'	23°31.4	-6.1'	58.8'
3	223°23.8	· · 20.4	123°27.6	5.9'	23°25.3	-6.3'	58.9'
4	238°23.9	21.4	137°52.5	6.0'	23°19.0	-6.4'	58.9'
5	253°24.1	22.3	152°17.5	6.0'	23°12.6	-6.6'	58.9'
6	268°24.3	N01°23.3	166°42.4	6.0'	N23°06.0	-6.7'	59.0'
7	283°24.5	24.3	181°07.4	6.0'	22°59.3	-6.9'	59.0'
8	298°24.7	25.3	195°32.5	6.1'	22°52.4	-7.0'	59.0'
9	313°24.9	· · 26.3	209°57.5	6.1'	22°45.4	-7.2'	59.0'
10	328°25.1	27.3	224°22.6	6.1'	22°38.2	-7.3'	59.1'
11	343°25.3	28.3	238°47.8	6.2'	22°30.9	-7.5'	59.1'
12	358°25.5	N01°29.2	253°12.9	6.2'	N22°23.4	-7.6'	59.1'
13	13°25.6	30.2	267°38.1	6.2'	22°15.8	-7.8'	59.1'
14	28°25.8	31.2	282°03.3	6.3'	22°08.0	-7.9'	59.2'
15	43°26.0	· · 32.2	296°28.6	6.3'	22°00.1	-8.1'	59.2'
16	58°26.2	33.2	310°53.9	6.3'	21°52.0	-8.2'	59.2'
17	73°26.4	34.2	325°19.2	6.4'	21°43.8	-8.4'	59.2'
18	88°26.6	N01°35.1	339°44.6	6.4'	N21°35.4	-8.5'	59.3'
19	103°26.8	36.1	354°10.0	6.5'	21°26.9	-8.7'	59.3'
20	118°27.0	37.1	8°35.5	6.5'	21°18.3	-8.8'	59.3'
21	133°27.2	· · 38.1	23°01.0	6.5'	21°09.5	-8.9'	59.3'
22	148°27.3	39.1	37°26.5	6.6'	21°00.5	-9.1'	59.4'
23	163°27.5	40.1	51°52.1	6.6'	20°51.5	-9.2'	59.4'
	SD = 16.0'	$d = 1.0'$			SD = 16.0'		

Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	178°27.7	N01°41.0	66°17.7	6.7'	N20°42.2	-9.4'	59.4'
1	193°27.9	42.0	80°43.4	6.7'	20°32.9	-9.5'	59.4'
2	208°28.1	43.0	95°09.1	6.7'	20°23.4	-9.6'	59.5'
3	223°28.3	· · 44.0	109°34.8	6.8'	20°13.8	-9.8'	59.5'
4	238°28.5	45.0	124°00.6	6.8'	20°04.0	-9.9'	59.5'
5	253°28.7	46.0	138°26.4	6.9'	19°54.1	-10.0'	59.5'
6	268°28.9	N01°46.9	152°52.3	6.9'	N19°44.1	-10.2'	59.6'
7	283°29.0	47.9	167°18.3	7.0'	19°33.9	-10.3'	59.6'
8	298°29.2	48.9	181°44.2	7.0'	19°23.6	-10.4'	59.6'
9	313°29.4	· · 49.9	196°10.3	7.1'	19°13.2	-10.5'	59.6'
10	328°29.6	50.9	210°36.3	7.1'	19°02.7	-10.7'	59.7'
11	343°29.8	51.8	225°02.4	7.2'	18°52.0	-10.8'	59.7'
12	358°30.0	N01°52.8	239°28.6	7.2'	N18°41.2	-10.9'	59.7'
13	13°30.2	53.8	253°54.8	7.3'	18°30.3	-11.1'	59.7'
14	28°30.4	54.8	268°21.1	7.3'	18°19.2	-11.2'	59.7'
15	43°30.6	· · 55.8	282°47.4	7.4'	18°08.0	-11.3'	59.8'
16	58°30.7	56.8	297°13.8	7.4'	17°56.8	-11.4'	59.8'
17	73°30.9	57.7	311°40.2	7.5'	17°45.3	-11.5'	59.8'
18	88°31.1	N01°58.7	326°06.6	7.5'	N17°33.8	-11.6'	59.8'
19	103°31.3	01°59.7	340°33.1	7.6'	17°22.2	-11.8'	59.8'
20	118°31.5	02°00.7	354°59.7	7.6'	17°10.4	-11.9'	59.9'
21	133°31.7	· · 01.7	9°26.3	7.7'	16°58.5	-12.0'	59.9'
22	148°31.9	02.7	23°52.9	7.7'	16°46.5	-12.1'	59.9'
23	163°32.1	03.6	38°19.7	7.8'	16°34.4	-12.2'	59.9'
	SD = 16.0'	$d = 1.0'$			SD = 16.2'		

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	02:54	04:29	05:39	18:37	19:47	21:24
N 70°	03:17	04:39	05:41	18:34	19:36	21:00
68°	03:35	04:47	05:44	18:31	19:28	20:41
66°	03:49	04:54	05:46	18:29	19:21	20:27
64°	04:00	05:00	05:47	18:27	19:15	20:15
62°	04:10	05:04	05:49	18:25	19:10	20:05
60°	04:18	05:08	05:50	18:24	19:06	19:57
N 58°	04:24	05:12	05:51	18:23	19:02	19:50
56°	04:30	05:15	05:52	18:22	18:59	19:44
54°	04:35	05:18	05:53	18:21	18:56	19:39
52°	04:40	05:20	05:54	18:20	18:54	19:34
50°	04:44	05:22	05:55	18:19	18:51	19:30
45°	04:52	05:27	05:56	18:17	18:47	19:21
N 40°	04:59	05:30	05:57	18:16	18:43	19:15
35°	05:04	05:33	05:58	18:15	18:40	19:10
30°	05:08	05:35	05:59	18:14	18:38	19:06
20°	05:13	05:39	06:01	18:12	18:34	19:00
N 10°	05:17	05:41	06:02	18:11	18:32	18:56
0°	05:18	05:42	06:03	18:10	18:30	18:54
S 10°	05:19	05:43	06:04	18:08	18:29	18:54
20°	05:17	05:43	06:05	18:07	18:29	18:55
30°	05:14	05:42	06:06	18:06	18:30	18:58
35°	05:12	05:41	06:06	18:06	18:31	19:00
40°	05:08	05:40	06:07	18:05	18:32	19:04
45°	05:04	05:38	06:07	18:05	18:34	19:08
S 50°	04:58	05:36	06:08	18:04	18:36	19:14
52°	04:55	05:34	06:08	18:04	18:37	19:16
54°	04:52	05:33	06:08	18:03	18:38	19:20
56°	04:48	05:32	06:09	18:03	18:40	19:23
58°	04:43	05:30	06:09	18:02	18:41	19:28
S 60°	04:38	05:28	06:09	18:02	18:43	19:32

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	□	□	□	□	□	□
N 70°	□	□	09:10	□	□	06:53
68°	□	□	10:07	□	□	05:55
66°	□	08:19	10:40	□	05:42	05:21
64°	07:14	09:06	11:04	04:43	04:54	04:56
62°	08:01	09:36	11:23	03:56	04:23	04:36
60°	08:31	09:59	11:39	03:26	04:00	04:19
N 58°	08:54	10:18	11:52	03:03	03:41	04:06
56°	09:13	10:33	12:03	02:44	03:25	03:54
54°	09:28	10:46	12:13	02:28	03:12	03:43
52°	09:42	10:58	12:22	02:15	03:00	03:34
50°	09:53	11:08	12:30	02:03	02:49	03:25
45°	10:18	11:29	12:46	01:38	02:27	03:07
N 40°	10:38	11:47	13:00	01:18	02:09	02:53
35°	10:54	12:01	13:12	01:01	01:54	02:40
30°	11:08	12:13	13:22	00:47	01:41	02:29
20°	11:32	12:35	13:39	00:22	01:18	02:10
N 10°	11:53	12:53	13:54	00:01	00:58	01:54
0°	12:12	13:11	14:08	· · ·	00:40	01:38
S 10°	12:31	13:28	14:21	· · ·	00:22	01:22
20°	12:52	13:46	14:36	· · ·	00:02	01:05
30°	13:16	14:07	14:53	23:39	· · ·	00:46
35°	13:30	14:19	15:03	23:25	· · ·	00:34
40°	13:46	14:33	15:14	23:09	· · ·	00:21
45°	14:05	14:50	15:27	22:51	· · ·	00:05
S 50°	14:30	15:11	15:43	22:27	23:46	· · ·
52°	14:41	15:20	15:50	22:16	23:37	· · ·
54°	14:54	15:31	15:58	22:03	23:26	· · ·
56°	15:10	15:44	16:07	21:48	23:14	· · ·
58°	15:27	15:58	16:18	21:30	23:01	· · ·
S 60°	15:49	16:15	16:29	21:09	22:45	· · ·

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time	12 ^h		Mer. Pass.		
	00 ^h	mm:ss	hh:mm	Upper	Lower	8-10
		mm:ss		hh:mm	hh:mm	45-67%
23	06:45	06:36	12:07	18:26	05:57	
24	06:27	06:18	12:06	19:24	06:55	
25	06:09	06:00		20:21	07:53	

March 26, 27, 28 UT (Fri., Sat., Sun.)

Aries			Venus			Mars			Jupiter			Saturn			Stars					
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec					
0	183°20.2	161°51.0	N08°13.4	58°42.7	N22°49.0	195°59.0	S06°28.2	1°25.2	N01°58.2			1°25.2	N01°58.2	Alpheratz	357°46.4	29°08.8				
1	198°22.6	176°50.6	14.7	73°44.7	48.8	211°00.9	27.9	16°27.9	58.3			16°27.9	58.3	Ankaa	353°18.3	-42°15.0				
2	213°25.1	191°50.2	15.9	88°46.7	48.6	226°02.8	27.7	31°30.5	58.4			31°30.5	58.4	Schedar	349°44.0	56°35.7				
3	228°27.6	206°49.8	· · 17.1	103°48.8	· · 48.5	241°04.7	· · 27.5	46°33.1	· · 58.5			46°33.1	· · 58.5	Diphda	348°58.5	-17°55.9				
4	243°30.0	221°49.4	18.3	118°50.8	48.3	256°06.7	27.3	61°35.8	58.5			61°35.8	58.5	Achernar	335°28.9	-57°11.1				
5	258°32.5	236°49.0	19.6	133°52.9	48.1	271°08.6	27.0	76°38.4	58.6			76°38.4	58.6	Hamal	328°03.8	23°30.7				
6	273°35.0	251°48.6	N08°20.8	148°54.9	N22°47.9	286°10.5	S06°26.8	91°41.1	N01°58.7			91°41.1	N01°58.7	Polaris	319°18.6	89°18.8				
7	288°37.4	266°48.2	22.0	163°56.9	47.7	301°12.4	26.6	106°43.7	58.8			106°43.7	58.8	Acamar	315°20.4	-40°16.0				
8	303°39.9	281°47.8	23.2	178°59.0	47.6	316°14.3	26.4	121°46.4	58.8			121°46.4	58.8	Menkar	314°17.8	4°07.8				
9	318°42.3	296°47.4	· · 24.5	194°01.0	· · 47.4	331°16.2	· · 26.1	136°49.0	· · 58.9			136°49.0	· · 58.9	Mirfak	308°44.2	49°54.0				
10	333°44.8	311°47.0	25.7	209°03.0	47.2	346°18.2	25.9	151°51.6	59.0			151°51.6	59.0	Aldebaran	290°52.2	16°31.8				
11	348°47.3	326°46.6	26.9	224°05.1	47.0	1°20.1	25.7	166°54.3	59.1			166°54.3	59.1	Rigel	281°14.4	-8°11.5				
12	3°49.7	341°46.2	N08°28.1	239°07.1	N22°46.9	16°22.0	S06°25.5	181°56.9	N01°59.2			181°56.9	N01°59.2	Capella	280°38.1	46°00.6				
13	18°52.2	356°45.8	29.4	254°09.1	46.7	31°23.9	25.3	196°59.6	59.2			196°59.6	59.2	Bellatrix	278°34.6	6°21.5				
14	33°54.7	11°45.4	30.6	269°11.1	46.5	46°25.8	25.0	212°02.2	59.3			212°02.2	59.3	Elnath	278°15.7	28°37.0				
15	48°57.1	26°45.0	· · 31.8	284°13.2	· · 46.3	61°27.8	· · 24.8	227°04.9	· · 59.4			227°04.9	· · 59.4	Alnilam	275°48.8	-1°11.8				
16	63°59.6	41°44.6	33.0	299°15.2	46.1	76°29.7	24.6	242°07.5	59.5			242°07.5	59.5	Betelgeuse	271°03.9	7°24.5				
17	79°02.1	56°44.2	34.3	314°17.2	46.0	91°31.6	24.4	257°10.1	59.6			257°10.1	59.6	Canopus	263°57.2	-52°42.4				
18	94°04.5	71°43.8	N08°35.5	329°19.2	N22°45.8	106°33.5	S06°24.1	272°12.8	N01°59.6			272°12.8	N01°59.6	Sirius	258°35.8	-16°44.0				
19	109°07.0	86°43.4	36.7	344°21.3	45.6	121°35.4	23.9	287°15.4	59.7			287°15.4	59.7	Adhara	255°14.3	-28°59.4				
20	124°09.5	101°43.0	37.9	359°23.3	45.4	136°37.4	23.7	302°18.1	59.8			302°18.1	59.8	Procyon	245°02.1	5°11.8				
21	139°11.9	116°42.5	· · 39.1	14°25.3	· · 45.2	151°39.3	· · 23.5	317°20.7	· · 59.9			317°20.7	· · 59.9	Pollux	243°30.4	28°00.1				
22	154°14.4	131°42.1	40.4	29°27.3	45.1	166°41.2	23.2	332°23.4	01°59.9			332°23.4	01°59.9	Avior	234°18.8	-59°32.9				
23	169°16.8	146°41.7	41.6	44°29.3	44.9	181°43.1	23.0	347°26.0	02°00.0			347°26.0	02°00.0	Suhail	222°54.0	-43°28.8				
Mer.pass. 11:45			ν -0.4' d 1.2' m -3.90			ν 2.0' d -0.2' m 0.03			ν 1.9' d -0.2' m -2.04			ν 2.6' d 0.1' m 0.53			Miaplacidus			221°39.8	-69°45.9	
															Alphard			217°58.2	-8°42.4	
															Regulus			207°45.7	11°54.8	
															Dubhe			193°53.7	61°41.7	
															Denebola			182°35.7	14°30.7	
															Gienah			175°54.4	-17°36.2	
															Acrux			173°11.5	-63°09.5	
															Gacrux			172°03.1	-57°10.4	
															Alioth			166°22.0	55°54.0	
															Spica			158°33.4	-11°13.1	
															Alkaid			153°00.2	49°15.4	
															Hadar			148°50.9	-60°25.4	
															Menkent			148°10.1	-36°25.3	
															Arcturus			145°57.5	19°07.5	
															Rigel Kent.			139°54.5	-60°52.6	
															Kochab			137°18.7	74°06.5	
															Zuben'ubi			137°07.8	-16°05.2	
															Alphecca			126°12.7	26°40.5	
															Antares			112°29.0	-26°27.3	
															Atria			107°32.8	-69°02.6	
															Sabik			102°15.1	-15°44.3	
															Shaula			96°25.0	-37°06.6	
															Rasalhague			96°08.6	12°32.9	
															Eltanin			90°47.2	51°28.9	
															Kaus Aust.			83°46.9	-34°22.7	
															Vega			80°40.6	38°47.3	
															Nunki			76°01.3	-26°17.0	
															Altair			62°10.6	8°53.6	
															Peacock			53°23.1	-56°41.9	
															Deneb			49°33.4	45°18.8	
															Enif			33°49.7	9°55.2	
															Al Na'ir			27°46.9	-46°54.6	
															Fomalhaut			15°26.8	-29°34.0	
															Scheat			13°56.1	28°08.2	
															Markab			13°41.0	15°15.6	
															Mar 26 Fri			SHA	Mer.pass	
															Venus			338°30.9	13:13	
															Mars			235°22.5	20:02	
															Jupiter			12°38.8	10:55	
															Saturn			178°05.0	23:50	
															Mar 27 Sat			SHA	Mer.pass	
															Venus			337°22.0	13:14	
															Mars			235°12.0	19:59	
															Jupiter			12°25.7	10:52	
															Saturn			178°09.3	23:46	
															Mar 28 Sun			SHA	Mer.pass	
															Venus			336°13.0	13:14	
															Mars			235°01.0	19:56	
															Jupiter			12°12.7	10:49	
															Saturn			178°13.6	23:42	
															Horizontal parallax					
															Venus:				0.1	
															Mars:				0.1	
Mer.pass. 11:37			ν -0.4' d 1.2' m -3.90			ν 2.0' d -0.2' m 0.08			ν 1.9' d -0.2' m -2.04			ν 2.6' d 0.1' m 0.54								

h	Sun			Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	178°32.3	N02°04.6	52°46.4	7.8'	N16°22.2	-12.3'	60.0'
1	193°32.5	05.6	67°13.2	7.9'	16°09.9	-12.4'	60.0'
2	208°32.6	06.6	81°40.1	7.9'	15°57.5	-12.5'	60.0'
3	223°32.8	· · 07.6	96°07.0	7.9'	15°44.9	-12.6'	60.0'
4	238°33.0	08.5	110°33.9	8.0'	15°32.3	-12.7'	60.0'
5	253°33.2	09.5	125°00.9	8.0'	15°19.5	-12.8'	60.0'
6	268°33.4	N02°10.5	139°28.0	8.1'	N15°06.7	-12.9'	60.1'
7	283°33.6	11.5	153°55.0	8.1'	14°53.8	-13.0'	60.1'
8	298°33.8	12.5	168°22.2	8.2'	14°40.7	-13.1'	60.1'
9	313°34.0	· · 13.4	182°49.4	8.2'	14°27.6	-13.2'	60.1'
10	328°34.2	14.4	197°16.6	8.3'	14°14.3	-13.3'	60.1'
11	343°34.3	15.4	211°43.9	8.3'	14°01.0	-13.4'	60.1'
12	358°34.5	N02°16.4	226°11.2	8.4'	N13°47.6	-13.5'	60.2'
13	13°34.7	17.4	240°38.6	8.4'	13°34.1	-13.6'	60.2'
14	28°34.9	18.3	255°06.0	8.5'	13°20.5	-13.7'	60.2'
15	43°35.1	· · 19.3	269°33.5	8.5'	13°06.8	-13.8'	60.2'
16	58°35.3	20.3	284°01.0	8.6'	12°53.0	-13.9'	60.2'
17	73°35.5	21.3	298°28.6	8.6'	12°39.1	-13.9'	60.2'
18	88°35.7	N02°22.3	312°56.2	8.6'	N12°25.2	-14.0'	60.3'
19	103°35.9	23.2	327°23.8	8.7'	12°11.2	-14.1'	60.3'
20	118°36.1	24.2	341°51.5	8.7'	11°57.1	-14.2'	60.3'
21	133°36.2	· · 25.2	356°19.2	8.8'	11°42.9	-14.3'	60.3'
22	148°36.4	26.2	10°47.0	8.8'	11°28.6	-14.3'	60.3'
23	163°36.6	27.2	25°14.8	8.9'	11°14.3	-14.4'	60.3'
SD = 16.0'			SD = 16.4'				
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	178°36.8	N02°28.1	39°42.7	8.9'	N10°59.9	-14.5'	60.3'
1	193°37.0	29.1	54°10.6	8.9'	10°45.4	-14.5'	60.3'
2	208°37.2	30.1	68°38.5	9.0'	10°30.9	-14.6'	60.4'
3	223°37.4	· · 31.1	83°06.5	9.0'	10°16.2	-14.7'	60.4'
4	238°37.6	32.1	97°34.5	9.0'	10°01.6	-14.7'	60.4'
5	253°37.8	33.0	112°02.5	9.1'	09°46.8	-14.8'	60.4'
6	268°37.9	N02°34.0	126°30.6	9.1'	N09°32.0	-14.9'	60.4'
7	283°38.1	35.0	140°58.7	9.2'	09°17.1	-14.9'	60.4'
8	298°38.3	36.0	155°26.9	9.2'	09°02.2	-15.0'	60.4'
9	313°38.5	· · 37.0	169°55.1	9.2'	08°47.2	-15.0'	60.4'
10	328°38.7	37.9	184°23.3	9.3'	08°32.2	-15.1'	60.4'
11	343°38.9	38.9	198°51.6	9.3'	08°17.1	-15.2'	60.4'
12	358°39.1	N02°39.9	213°19.9	9.3'	N08°01.9	-15.2'	60.5'
13	13°39.3	40.9	227°48.2	9.4'	07°46.7	-15.3'	60.5'
14	28°39.5	41.8	242°16.6	9.4'	07°31.4	-15.3'	60.5'
15	43°39.7	· · 42.8	256°44.9	9.4'	07°16.1	-15.4'	60.5'
16	58°39.8	43.8	271°13.4	9.4'	07°00.8	-15.4'	60.5'
17	73°40.0	44.8	285°41.8	9.5'	06°45.4	-15.4'	60.5'
18	88°40.2	N02°45.8	300°10.3	9.5'	N06°29.9	-15.5'	60.5'
19	103°40.4	46.7	314°38.8	9.5'	06°14.5	-15.5'	60.5'
20	118°40.6	47.7	329°07.3	9.6'	05°59.0	-15.6'	60.5'
21	133°40.8	· · 48.7	343°35.8	9.6'	05°43.4	-15.6'	60.5'
22	148°41.0	49.7	358°04.4	9.6'	05°27.8	-15.6'	60.5'
23	163°41.2	50.6	12°33.0	9.6'	05°12.2	-15.7'	60.5'
SD = 16.0'			SD = 16.5'				
Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	178°41.4	N02°51.6	27°01.6	9.6'	N04°56.5	-15.7'	60.5'
1	193°41.6	52.6	41°30.3	9.7'	04°40.8	-15.7'	60.5'
2	208°41.7	53.6	55°58.9	9.7'	04°25.1	-15.7'	60.5'
3	223°41.9	· · 54.5	70°27.6	9.7'	04°09.4	-15.8'	60.5'
4	238°42.1	55.5	84°56.3	9.7'	03°53.6	-15.8'	60.5'
5	253°42.3	56.5	99°25.1	9.7'	03°37.8	-15.8'	60.5'
6	268°42.5	N02°57.5	113°53.8	9.8'	N03°22.0	-15.8'	60.5'
7	283°42.7	58.5	128°22.6	9.8'	03°06.2	-15.8'	60.5'
8	298°42.9	02°59.4	142°51.3	9.8'	02°50.3	-15.9'	60.5'
9	313°43.1	03°00.4	157°20.1	9.8'	02°34.5	-15.9'	60.5'
10	328°43.3	01.4	171°48.9	9.8'	02°18.6	-15.9'	60.5'
11	343°43.4	02.4	186°17.7	9.8'	02°02.7	-15.9'	60.5'
12	358°43.6	N03°03.3	200°46.6	9.8'	N01°46.8	-15.9'	60.5'
13	13°43.8	04.3	215°15.4	9.8'	01°30.9	-15.9'	60.5'
14	28°44.0	05.3	229°44.3	9.9'	01°15.0	-15.9'	60.5'
15	43°44.2	· · 06.3	244°13.1	9.9'	00°59.0	-15.9'	60.5'
16	58°44.4	07.2	258°42.0	9.9'	00°43.1	-15.9'	60.5'
17	73°44.6	08.2	273°10.9	9.9'	00°27.2	-15.9'	60.5'
18	88°44.8	N03°09.2	287°39.7	9.9'	N00°11.3	-15.9'	60.5'
19	103°45.0	10.2	302°08.6	9.9'	S00°04.7	15.9'	60.5'
20	118°45.2	11.1	316°37.5	9.9'	00°20.6	15.9'	60.5'
21	133°45.3	· · 12.1	331°06.4	9.9'	00°36.5	15.9'	60.5'
22	148°45.5	13.1	345°35.3	9.9'	00°52.4	15.9'	60.4'
23	163°45.7	14.1	0°04.2	9.9'	01°08.3	15.9'	60.4'
SD = 16.0'			SD = 16.5'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	02:30	04:12	05:23	18:51	20:02	21:48
N 70°	02:58	04:24	05:27	18:46	19:50	21:17
68°	03:19	04:34	05:31	18:42	19:40	20:56
66°	03:35	04:42	05:34	18:39	19:31	20:39
64°	03:48	04:48	05:37	18:36	19:24	20:26
62°	03:58	04:54	05:39	18:33	19:19	20:15
60°	04:07	04:59	05:41	18:31	19:14	20:06
N 58°	04:15	05:03	05:43	18:29	19:09	19:58
56°	04:22	05:07	05:44	18:28	19:05	19:51
54°	04:27	05:10	05:46	18:26	19:02	19:45
52°	04:32	05:13	05:47	18:25	18:59	19:40
50°	04:37	05:16	05:48	18:24	18:56	19:35
45°	04:46	05:21	05:50	18:21	18:51	19:26
N 40°	04:53	05:25	05:53	18:19	18:46	19:18
35°	04:59	05:29	05:54	18:17	18:43	19:12
30°	05:04	05:32	05:56	18:16	18:40	19:08
20°	05:10	05:36	05:58	18:13	18:35	19:01
N 10°	05:15	05:39	06:00	18:11	18:32	18:56
0°	05:17	05:41	06:02	18:09	18:29	18:53
S 10°	05:19	05:43	06:04	18:07	18:28	18:52
20°	05:18	05:44	06:06	18:05	18:27	18:52
30°	05:16	05:44	06:08	18:03	18:27	18:54
35°	05:14	05:43	06:09	18:02	18:27	18:56
40°	05:11	05:43	06:10	18:00	18:27	18:59
45°	05:08	05:42	06:11	17:59	18:28	19:02
S 50°	05:03	05:40	06:13	17:57	18:29	19:07
52°	05:00	05:40	06:13	17:57	18:30	19:09
54°	04:57	05:39	06:14	17:56	18:31	19:12
56°	04:54	05:38	06:15	17:55	18:32	19:15
58°	04:50	05:37	06:16	17:54	18:33	19:19
S 60°	04:46	05:35	06:17	17:53	18:34	19:23

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	11:40	14:19	16:42	06:22	05:37	05:05
N 70°	12:09	14:30	16:43	05:51	05:22	05:01
68°	12:30	14:40	16:43	05:28	05:11	04:57
66°	12:47	14:47	16:44	05:10	05:01	04:53
64°	13:01	14:53	16:44	04:55	04:53	04:51
62°	13:12	14:59	16:44	04:42	04:46	04:48
60°	13:21	15:04	16:44	04:32	04:40	04:46
N 58°	13:30	15:08	16:45	04:22	04:34	04:44
56°	13:37	15:11	16:45	04:14	04:30	04:43
54°	13:44	15:15	16:45	04:07	04:25	04:41
52°	13:49	15:18	16:45	04:00	04:21	04:40
50°	13:55	15:20	16:45	03:54	04:18	04:38
45°	14:06	15:26	16:46	03:41	04:10	04:36
N 40°	14:15	15:31	16:46	03:30	04:03	04:33
35°	14:23	15:35	16:46	03:21	03:57	04:31
30°	14:30	15:39	16:46	03:13	03:52	04:29
20°	14:42	15:45	16:47	02:58	03:43	04:26
N 10°	14:53	15:50	16:47	02:46	03:35	04:23
0°	15:03	15:56	16:48	02:34	03:28	04:20
S 10°	15:12	16:01	16:48	02:22	03:20	04:17
20°	15:23	16:06	16:48	02:09	03:12	04:14
30°	15:34	16:12	16:49	01:54	03:03	04:11
35°	15:41	16:16	16:49	01:46	02:57	04:09
40°	15:49	16:20	16:49	01:36	02:51	04:07
45°	15:58	16:25	16:50	01:24	02:44	04:04
S 50°	16:08	16:30	16:50	01:10	02:35	04:01
52°	16:13	16:33	16:51	01:03	02:31	03:59
54°	16:19	16:36	16:51	00:56	02:27	03:57
56°	16:25	16:39	16:51	00:47	02:22	03:56
58°	16:31	16:42	16:51	00:38	02:16	03:54
S 60°	16:39	16:46	16:52	00:27	02:10	03:51

Day	Sun		Mer. Pass hh:mm	Moon		Age 11-13 78-94%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Upper hh:mm	Lower hh:mm	
26	05:51	05:42	12:06	21:15	08:48	
27	05:33	05:24	12:05	22:08	09:42	
28	05:15	05:05	12:05	23:00	10:34	

March 29, 30, 31 UT (Mon., Tue., Wed.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	186°17.6	161°21.5	N09°40.9	61°07.0	N22°35.7	198°17.3	S06°12.1	4°35.5	N02°03.8					Alpheratz	357°46.4	29°08.8	
1	201°20.0	176°21.0	42.1	76°09.0	35.5	213°19.2	11.9	19°38.1	03.9					Ankaa	353°18.3	-42°15.0	
2	216°22.5	191°20.6	43.3	91°11.0	35.3	228°21.1	11.6	34°40.8	04.0					Schedar	349°44.0	56°35.6	
3	231°25.0	206°20.2	· · 44.5	106°12.9	· · 35.1	243°23.0	· · 11.4	49°43.4	· · 04.1					Diphda	348°58.5	-17°55.8	
4	246°27.4	221°19.8	45.7	121°14.9	35.0	258°25.0	11.2	64°46.1	04.1					Achernar	335°28.9	-57°11.1	
5	261°29.9	236°19.4	46.9	136°16.9	34.8	273°26.9	11.0	79°48.7	04.2					Hamal	328°03.8	23°30.7	
6	276°32.4	251°18.9	N09°48.1	151°18.8	N22°34.6	288°28.8	S06°10.8	94°51.3	N02°04.3					Polaris	319°19.4	89°18.8	
7	291°34.8	266°18.5	49.3	166°20.8	34.4	303°30.7	10.5	109°54.0	04.4					Acamar	315°20.4	-40°15.9	
8	306°37.3	281°18.1	50.5	181°22.8	34.2	318°32.7	10.3	124°56.6	04.5					Menkar	314°17.8	4°07.8	
9	321°39.8	296°17.7	· · 51.7	196°24.7	· · 34.0	333°34.6	· · 10.1	139°59.3	· · 04.5					Mirfak	308°44.2	49°54.0	
10	336°42.2	311°17.2	52.9	211°26.7	33.8	348°36.5	09.9	155°01.9	04.6					Aldebaran	290°52.3	16°31.8	
11	351°44.7	326°16.8	54.1	226°28.6	33.6	3°38.4	09.6	170°04.6	04.7					Rigel	281°14.4	-8°11.5	
12	6°47.2	341°16.4	N09°55.3	241°30.6	N22°33.4	18°40.3	S06°09.4	185°07.2	N02°04.8					Capella	280°38.1	46°00.6	
13	21°49.6	356°16.0	56.5	256°32.6	33.2	33°42.3	09.2	200°09.8	04.8					Bellatrix	278°34.6	6°21.5	
14	36°52.1	11°15.5	57.7	271°34.5	33.0	48°44.2	09.0	215°12.5	04.9					Elnath	278°15.7	28°37.0	
15	51°54.5	26°15.1	09°58.9	286°36.5	· · 32.8	63°46.1	· · 08.7	230°15.1	· · 05.0					Alnilam	275°48.8	-1°11.8	
16	66°57.0	41°14.7	10°00.1	301°38.4	32.6	78°48.0	08.5	245°17.8	05.1					Betelgeuse	271°03.9	7°24.5	
17	81°59.5	56°14.3	01.3	316°40.4	32.4	93°50.0	08.3	260°20.4	05.1					Canopus	263°57.2	-52°42.4	
18	97°01.9	71°13.8	N10°02.5	331°42.3	N22°32.2	108°51.9	S06°08.1	275°23.1	N02°05.2					Sirius	258°35.8	-16°44.0	
19	112°04.4	86°13.4	03.7	346°44.3	32.0	123°53.8	07.9	290°25.7	05.3					Adhara	255°14.4	-28°59.4	
20	127°06.9	101°13.0	04.9	1°46.2	31.8	138°55.7	07.6	305°28.3	05.4					Procyon	245°02.1	5°11.8	
21	142°09.3	116°12.6	· · 06.1	16°48.2	· · 31.7	153°57.7	· · 07.4	320°31.0	· · 05.5					Pollux	243°30.5	28°00.1	
22	157°11.8	131°12.1	07.3	31°50.1	31.5	168°59.6	07.2	335°33.6	05.5					Avior	234°18.8	-59°32.9	
23	172°14.3	146°11.7	08.4	46°52.1	31.3	184°01.5	07.0	350°36.3	05.6					Suhail	222°54.0	-43°28.8	
Mer.pass. 11:33		ν -0.4' d 1.2' m -3.90			ν 2.0' d -0.2' m 0.10			ν 1.9' d -0.2' m -2.05			ν 2.6' d 0.1' m 0.54						

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec				
0	187°16.7	161°11.3	N10°09.6	61°54.0	N22°31.1	199°03.4	S06°06.7	5°38.9	N02°05.7				
1	202°19.2	176°10.9	10.8	76°56.0	30.9	214°05.4	06.5	20°41.5	05.8				
2	217°21.6	191°10.4	12.0	91°57.9	30.7	229°07.3	06.3	35°44.2	05.8				
3	232°24.1	206°10.0	· · 13.2	106°59.9	· · 30.5	244°09.2	· · 06.1	50°46.8	· · 05.9				
4	247°26.6	221°09.6	14.4	122°01.8	30.3	259°11.1	05.9	65°49.5	06.0				
5	262°29.0	236°09.1	15.6	137°03.8	30.1	274°13.1	05.6	80°52.1	06.1				
6	277°31.5	251°08.7	N10°16.8	152°05.7	N22°29.9	289°15.0	S06°05.4	95°54.8	N02°06.1				
7	292°34.0	266°08.3	18.0	167°07.6	29.7	304°16.9	05.2	110°57.4	06.2				
8	307°36.4	281°07.9	19.2	182°09.6	29.5	319°18.8	05.0	126°00.0	06.3				
9	322°38.9	296°07.4	· · 20.4	197°11.5	· · 29.3	334°20.8	· · 04.8	141°02.7	· · 06.4				
10	337°41.4	311°07.0	21.5	212°13.5	29.1	349°22.7	04.5	156°05.3	06.5				
11	352°43.8	326°06.6	22.7	227°15.4	28.9	4°24.6	04.3	171°08.0	06.5				
12	7°46.3	341°06.1	N10°23.9	242°17.3	N22°28.7	19°26.5	S06°04.1	186°10.6	N02°06.6				
13	22°48.8	356°05.7	25.1	257°19.3	28.5	34°28.5	03.9	201°13.2	06.7				
14	37°51.2	11°05.3	26.3	272°21.2	28.3	49°30.4	03.6	216°15.9	06.8				
15	52°53.7	26°04.8	· · 27.5	287°23.1	· · 28.1	64°32.3	· · 03.4	231°18.5	· · 06.8				
16	67°56.1	41°04.4	28.7	302°25.1	27.9	79°34.2	03.2	246°21.2	06.9				
17	82°58.6	56°04.0	29.8	317°27.0	27.7	94°36.2	03.0	261°23.8	07.0				
18	98°01.1	71°03.6	N10°31.0	332°28.9	N22°27.5	109°38.1	S06°02.8	276°26.4	N02°07.1				
19	113°03.5	86°03.1	32.2	347°30.9	27.3	124°40.0	02.5	291°29.1	07.1				
20	128°06.0	101°02.7	33.4	2°32.8	27.1	139°41.9	02.3	306°31.7	07.2				
21	143°08.5	116°02.3	· · 34.6	17°34.7	· · 26.9	154°43.9	· · 02.1	321°34.4	· · 07.3				
22	158°10.9	131°01.8	35.8	32°36.6	26.7	169°45.8	01.9	336°37.0	07.4				
23	173°13.4	146°01.4	36.9	47°38.6	26.5	184°47.7	01.6	351°39.6	07.4				
Mer.pass. 11:29		ν -0.4' d 1.2' m -3.89			ν 1.9' d -0.2' m 0.12			ν 1.9' d -0.2' m -2.05			ν 2.6' d 0.1' m 0.55		

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec				
0	188°15.9	161°01.0	N10°38.1	62°40.5	N22°26.3	199°49.6	S06°01.4	6°42.3	N02°07.5				
1	203°18.3	176°00.5	39.3	77°42.4	26.1	214°51.6	01.2	21°44.9	07.6				
2	218°20.8	191°00.1	40.5	92°44.3	25.9	229°53.5	01.0	36°47.6	07.7				
3	233°23.2	205°59.6	· · 41.7	107°46.3	· · 25.7	244°55.4	· · 00.8	51°50.2	· · 07.8				
4	248°25.7	220°59.2	42.9	122°48.2	25.5	259°57.3	00.5	66°52.9	07.8				
5	263°28.2	235°58.8	44.0	137°50.1	25.3	274°59.3	00.3	81°55.5	07.9				
6	278°30.6	250°58.3	N10°45.2	152°52.0	N22°25.1	290°01.2	S06°00.1	96°58.1	N02°08.0				
7	293°33.1	265°57.9	46.4	167°53.9	24.9	305°03.1	05°59.9	112°00.8	08.1				
8	308°35.6	280°57.5	47.6	182°55.9	24.7	320°05.0	59.7	127°03.4	08.1				
9	323°38.0	295°57.0	· · 48.7	197°57.8	· · 24.5	335°07.0	· · 59.4	142°06.1	· · 08.2				
10	338°40.5	310°56.6	49.9	212°59.7	24.3	350°08.9	59.2	157°08.7	08.3				
11	353°43.0	325°56.2	51.1	228°01.6	24.1	5°10.8	59.0	172°11.3	08.4				
12	8°45.4	340°55.7	N10°52.3	243°03.5	N22°23.9	20°12.8	S05°58.8	187°14.0	N02°08.4				
13	23°47.9	355°55.3	53.5	258°05.4	23.6	35°14.7	58.5	202°16.6	08.5				
14	38°50.4	10°54.8	54.6	273°07.4	23.4	50°16.6	58.3	217°19.3	08.6				
15	53°52.8	25°54.4	· · 55.8	288°09.3	· · 23.2	65°18.5	· · 58.1	232°21.9	· · 08.7				
16	68°55.3	40°54.0	57.0	303°11.2	23.0	80°20.5	57.9	247°24.5	08.7				
17	83°57.7	55°53.5	58.2	318°13.1	22.8	95°22.4	57.7	262°27.2	08.8				
18	99°00.2	70°53.1	N10°59.3	333°15.0	N22°22.6	110°24.3	S05°57.4	277°29.8	N02°08.9				
19	114°02.7	85°52.6	11°00.5	348°16.9	22.4	125°26.2	57.2	292°32.5	09.0				
20	129°05.1	100°52.2	01.7	3°18.8	22.2	140°28.2	57.0	307°35.1	09.0				
21	144°07.6	115°51.8	· · 02.9	18°20.7	· · 22.0	155°30.1	· · 56.8	322°37.7	· · 09.1				
22	159°10.1	130°51.3	04.0	33°22.6	21.8	170°32.0	56.6	337°40.4	09.2				
23	174°12.5	145°50.9	05.2	48°24.5	21.6	185°34.0	56.3	352°43.0	09.3				
Mer.pass. 11:25		ν -0.4' d 1.2' m -3.89			ν 1.9' d -0.2' m 0.14			ν 1.9' d -0.2' m -2.05			ν 2.6' d 0.1' m 0.55		

Mar 29 Mon	SHA	Mer.pass
Venus	335°03.9	13:15
Mars	234°49.4	19:53
Jupiter	11°59.7	10:45
Saturn	178°17.9	23:37

Mar 30 Tue	SHA	Mer.pass
Venus	333°54.6	13:16
Mars	234°37.3	19:50
Jupiter	11°46.7	10:42
Saturn	178°22.2	23:33

Mar 31 Wed	SHA	Mer.pass
Venus	332°45.1	13:16
Mars	234°24.6	19:47
Jupiter	11°33.8	10:39
Saturn	178°26.4	23:29

Horizontal parallax		
Venus:	0.1	
Mars:	0.1	

h	Sun			Moon			
Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	178°45.9	N03°15.0	14°33.1	9.9'	S01°24.2	15.9'	60.4'
1	193°46.1	16.0	29°02.0	9.9'	01°40.1	15.9'	60.4'
2	208°46.3	17.0	43°30.9	9.9'	01°56.0	15.9'	60.4'
3	223°46.5	· · 18.0	57°59.8	9.9'	02°11.8	15.8'	60.4'
4	238°46.7	18.9	72°28.7	9.9'	02°27.7	15.8'	60.4'
5	253°46.9	19.9	86°57.6	9.9'	02°43.5	15.8'	60.4'
6	268°47.0	N03°20.9	101°26.5	9.9'	S02°59.3	15.8'	60.4'
7	283°47.2	21.9	115°55.4	9.9'	03°15.0	15.7'	60.4'
8	298°47.4	22.8	130°24.3	9.9'	03°30.8	15.7'	60.3'
9	313°47.6	· · 23.8	144°53.2	9.9'	03°46.5	15.7'	60.3'
10	328°47.8	24.8	159°22.1	9.9'	04°02.2	15.7'	60.3'
11	343°48.0	25.7	173°50.9	9.9'	04°17.8	15.6'	60.3'
12	358°48.2	N03°26.7	188°19.8	9.8'	S04°33.5	15.6'	60.3'
13	13°48.4	27.7	202°48.6	9.8'	04°49.1	15.6'	60.3'
14	28°48.6	28.7	217°17.5	9.8'	05°04.6	15.5'	60.3'
15	43°48.8	· · 29.6	231°46.3	9.8'	05°20.1	15.5'	60.3'
16	58°48.9	30.6	246°15.1	9.8'	05°35.6	15.4'	60.2'
17	73°49.1	31.6	260°43.9	9.8'	05°51.1	15.4'	60.2'
18	88°49.3	N03°32.6	275°12.7	9.8'	S06°06.5	15.4'	60.2'
19	103°49.5	33.5	289°41.5	9.8'	06°21.8	15.3'	60.2'
20	118°49.7	34.5	304°10.2	9.7'	06°37.1	15.3'	60.2'
21	133°49.9	· · 35.5	318°39.0	9.7'	06°52.4	15.2'	60.2'
22	148°50.1	36.4	333°07.7	9.7'	07°07.6	15.2'	60.1'
23	163°50.3	37.4	347°36.4	9.7'	07°22.8	15.1'	60.1'
SD = 16.0'			SD = 16.5'				

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	178°50.5	N03°38.4	2°05.1	9.7'	S07°37.9	15.1'	60.1'
1	193°50.6	39.4	16°33.8	9.7'	07°53.0	15.0'	60.1'
2	208°50.8	40.3	31°02.4	9.6'	08°08.0	14.9'	60.1'
3	223°51.0	· · 41.3	45°31.1	9.6'	08°22.9	14.9'	60.0'
4	238°51.2	42.3	59°59.7	9.6'	08°37.8	14.8'	60.0'
5	253°51.4	43.2	74°28.3	9.6'	08°52.6	14.8'	60.0'
6	268°51.6	N03°44.2	88°56.8	9.5'	S09°07.4	14.7'	60.0'
7	283°51.8	45.2	103°25.4	9.5'	09°22.1	14.6'	60.0'
8	298°52.0	46.2	117°53.9	9.5'	09°36.7	14.6'	59.9'
9	313°52.2	· · 47.1	132°22.4	9.5'	09°51.3	14.5'	59.9'
10	328°52.3	48.1	146°50.9	9.5'	10°05.8	14.4'	59.9'
11	343°52.5	49.1	161°19.3	9.4'	10°20.3	14.4'	59.9'
12	358°52.7	N03°50.0	175°47.8	9.4'	S10°34.6	14.3'	59.8'
13	13°52.9	51.0	190°16.2	9.4'	10°48.9	14.2'	59.8'
14	28°53.1	52.0	204°44.5	9.4'	11°03.1	14.1'	59.8'
15	43°53.3	· · 53.0	219°12.9	9.3'	11°17.3	14.1'	59.8'
16	58°53.5	53.9	233°41.2	9.3'	11°31.3	14.0'	59.7'
17	73°53.7	54.9	248°09.5	9.3'	11°45.3	13.9'	59.7'
18	88°53.8	N03°55.9	262°37.8	9.2'	S11°59.2	13.8'	59.7'
19	103°54.0	56.8	277°06.0	9.2'	12°13.0	13.7'	59.7'
20	118°54.2	57.8	291°34.2	9.2'	12°26.8	13.7'	59.6'
21	133°54.4	· · 58.8	306°02.4	9.2'	12°40.4	13.6'	59.6'
22	148°54.6	03°59.7	320°30.6	9.1'	12°54.0	13.5'	59.6'
23	163°54.8	04°00.7	334°58.7	9.1'	13°07.5	13.4'	59.6'
SD = 16.0'			SD = 16.4'				

Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	178°55.0	N04°01.7	349°26.8	9.1'	S13°20.8	13.3'	59.5'
1	193°55.2	02.6	3°54.9	9.0'	13°34.1	13.2'	59.5'
2	208°55.4	03.6	18°22.9	9.0'	13°47.4	13.1'	59.5'
3	223°55.5	· · 04.6	32°50.9	9.0'	14°00.5	13.0'	59.4'
4	238°55.7	05.6	47°18.9	8.9'	14°13.5	12.9'	59.4'
5	253°55.9	06.5	61°46.8	8.9'	14°26.4	12.8'	59.4'
6	268°56.1	N04°07.5	76°14.7	8.9'	S14°39.2	12.7'	59.4'
7	283°56.3	08.5	90°42.6	8.8'	14°52.0	12.6'	59.3'
8	298°56.5	09.4	105°10.5	8.8'	15°04.6	12.5'	59.3'
9	313°56.7	· · 10.4	119°38.3	8.8'	15°17.1	12.4'	59.3'
10	328°56.9	11.4	134°06.1	8.8'	15°29.5	12.3'	59.2'
11	343°57.0	12.3	148°33.8	8.7'	15°41.8	12.2'	59.2'
12	358°57.2	N04°13.3	163°01.5	8.7'	S15°54.1	12.1'	59.2'
13	13°57.4	14.3	177°29.2	8.7'	16°06.2	12.0'	59.1'
14	28°57.6	15.2	191°56.9	8.6'	16°18.2	11.9'	59.1'
15	43°57.8	· · 16.2	206°24.5	8.6'	16°30.0	11.8'	59.1'
16	58°58.0	17.2	220°52.1	8.6'	16°41.8	11.7'	59.0'
17	73°58.2	18.1	235°19.7	8.5'	16°53.5	11.6'	59.0'
18	88°58.4	N04°19.1	249°47.2	8.5'	S17°05.0	11.4'	59.0'
19	103°58.5	20.1	264°14.7	8.5'	17°16.5	11.3'	58.9'
20	118°58.7	21.0	278°42.2	8.4'	17°27.8	11.2'	58.9'
21	133°58.9	· · 22.0	293°09.6	8.4'	17°39.0	11.1'	58.9'
22	148°59.1	23.0	307°37.0	8.4'	17°50.1	11.0'	58.8'
23	163°59.3	23.9	322°04.4	8.3'	18°01.1	10.9'	58.8'
SD = 16.0'			SD = 16.2'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	02:02	03:54	05:07	19:05	20:19	22:16
N 70°	02:37	04:09	05:13	18:58	20:03	21:37
68°	03:02	04:20	05:18	18:53	19:52	21:11
66°	03:20	04:29	05:23	18:48	19:42	20:52
64°	03:35	04:37	05:26	18:45	19:34	20:37
62°	03:47	04:44	05:29	18:41	19:27	20:25
60°	03:57	04:49	05:32	18:39	19:21	20:14
N 58°	04:05	04:54	05:34	18:36	19:16	20:05
56°	04:13	04:59	05:36	18:34	19:12	19:58
54°	04:19	05:02	05:38	18:32	19:08	19:51
52°	04:25	05:06	05:40	18:30	19:04	19:46
50°	04:30	05:09	05:42	18:28	19:01	19:40
45°	04:40	05:15	05:45	18:25	18:55	19:30
N 40°	04:48	05:20	05:48	18:22	18:49	19:21
35°	04:55	05:25	05:50	18:20	18:45	19:15
30°	05:00	05:28	05:52	18:17	18:41	19:10
20°	05:08	05:33	05:56	18:14	18:36	19:02
N 10°	05:13	05:38	05:59	18:11	18:32	18:56
0°	05:16	05:41	06:01	18:08	18:28	18:53
S 10°	05:18	05:43	06:04	18:05	18:26	18:50
20°	05:19	05:44	06:06	18:02	18:24	18:50
30°	05:18	05:45	06:09	17:59	18:23	18:51
35°	05:16	05:46	06:11	17:57	18:23	18:52
40°	05:14	05:46	06:13	17:56	18:23	18:54
45°	05:11	05:46	06:15	17:53	18:23	18:57
S 50°	05:07	05:45	06:17	17:51	18:23	19:00
52°	05:05	05:45	06:18	17:50	18:23	19:02
54°	05:03	05:44	06:20	17:48	18:24	19:05
56°	05:00	05:44	06:21	17:47	18:24	19:07
58°	04:57	05:43	06:22	17:45	18:24	19:10
S 60°	04:54	05:42	06:24	17:44	18:25	19:14

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	19:04	21:38	■	04:37	04:06	03:24
N 70°	18:54	21:12	23:56	04:40	04:19	03:52
68°	18:46	20:53	23:10	04:43	04:30	04:13
66°	18:39	20:37	22:40	04:46	04:38	04:30
64°	18:34	20:25	22:18	04:48	04:46	04:44
62°	18:29	20:14	22:00	04:50	04:52	04:55
60°	18:25	20:05	21:46	04:52	04:58	05:06
N 58°	18:21	19:58	21:33	04:53	05:03	05:14
56°	18:18	19:51	21:23	04:55	05:07	05:22
54°	18:15	19:45	21:13	04:56	05:11	05:29
52°	18:12	19:39	21:05	04:57	05:15	05:35
50°	18:10	19:34	20:57	04:58	05:18	05:41
45°	18:05	19:24	20:42	05:00	05:26	05:53
N 40°	18:01	19:15	20:29	05:02	05:32	06:04
35°	17:57	19:08	20:18	05:04	05:37	06:12
30°	17:54	19:01	20:08	05:05	05:42	06:20
20°	17:48	18:50	19:52	05:08	05:50	06:34
N 10°	17:44	18:40	19:38	05:10	05:57	06:45
0°	17:39	18:31	19:25	05:12	06:04	06:56
S 10°	17:35	18:22	19:11	05:14	06:10	07:08
20°	17:30	18:13	18:58	05:16	06:18	07:19
30°	17:25	18:02	18:42	05:19	06:26	07:33
35°	17:22	17:56	18:33	05:20	06:31	07:41
40°	17:19	17:49	18:22	05:22	06:36	07:50
45°	17:15	17:41	18:10	05:23	06:42	08:01
S 50°	17:10	17:31	17:56	05:26	06:50	08:14
52°	17:08	17:27	17:49	05:27	06:54	08:20
54°	17:06	17:22	17:42	05:28	06:58	08:27
56°	17:03	17:17	17:33	05:29	07:02	08:34
58°	17:01	17:11	17:24	05:30	07:07	08:43
S 60°	16:57	17:04	17:13	05:32	07:12	08:53

Day	Sun		Mer. Pass	Moon		Age 14-16 98-99%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Upper Mer. Pass. hh:mm	Lower Mer. Pass. hh:mm	
29	04:56	04:47	12:05	23:51	11:25	
30	04:38	04:29	12:04	—:—	12:17	
31	04:20	04:11	12:04	00:44	13:10	

April 01, 02, 03 UT (Thu., Fri., Sat.)

Aries		Venus		Mars		Jupiter		Saturn		Stars					
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec				
0	189°15.0	160°50.4	N11°06.4	63°26.4	N22°21.4	200°35.9	S05°56.1	7°45.7	N02°09.3	Alpheratz	357°46.3	29°08.8			
1	204°17.5	175°50.0	07.5	78°28.3	21.2	215°37.8	55.9	22°48.3	09.4	Ankaa	353°18.3	-42°15.0			
2	219°19.9	190°49.6	08.7	93°30.2	21.0	230°39.7	55.7	37°50.9	09.5	Schedar	349°44.0	56°35.6			
3	234°22.4	205°49.1	· · 09.9	108°32.1	· · 20.8	245°41.7	· · 55.4	52°53.6	· · 09.6	Diphda	348°58.5	-17°55.8			
4	249°24.9	220°48.7	11.1	123°34.0	20.6	260°43.6	55.2	67°56.2	09.6	Achernar	335°28.9	-57°11.1			
5	264°27.3	235°48.2	12.2	138°35.9	20.4	275°45.5	55.0	82°58.9	09.7	Hamal	328°03.8	23°30.7			
6	279°29.8	250°47.8	N11°13.4	153°37.8	N22°20.1	290°47.4	S05°54.8	98°01.5	N02°09.8	Polaris	319°19.9	89°18.8			
7	294°32.2	265°47.3	14.6	168°39.7	19.9	305°49.4	54.6	113°04.1	09.9	Acamar	315°20.4	-40°15.9			
8	309°34.7	280°46.9	15.7	183°41.6	19.7	320°51.3	54.3	128°06.8	10.0	Menkar	314°17.8	4°07.8			
9	324°37.2	295°46.4	· · 16.9	198°43.5	· · 19.5	335°53.2	· · 54.1	143°09.4	· · 10.0	Mirfak	308°44.2	49°54.0			
10	339°39.6	310°46.0	18.1	213°45.4	19.3	350°55.2	53.9	158°12.1	10.1	Aldebaran	290°52.3	16°31.8			
11	354°42.1	325°45.6	19.2	228°47.3	19.1	5°57.1	53.7	173°14.7	10.2	Rigel	281°14.4	-8°11.5			
12	9°44.6	340°45.1	N11°20.4	243°49.2	N22°18.9	20°59.0	S05°53.5	188°17.3	N02°10.3	Capella	280°38.1	46°00.6			
13	24°47.0	355°44.7	21.6	258°51.1	18.7	36°00.9	53.2	203°20.0	10.3	Bellatrix	278°34.6	6°21.5			
14	39°49.5	10°44.2	22.7	273°53.0	18.5	51°02.9	53.0	218°22.6	10.4	Elnath	278°15.7	28°37.0			
15	54°52.0	25°43.8	· · 23.9	288°54.9	· · 18.3	66°04.8	· · 52.8	233°25.3	· · 10.5	Alnilam	275°48.8	-1°11.8			
16	69°54.4	40°43.3	25.1	303°56.8	18.1	81°06.7	52.6	248°27.9	10.6	Betelgeuse	271°03.9	7°24.5			
17	84°56.9	55°42.9	26.2	318°58.7	17.9	96°08.7	52.4	263°30.5	10.6	Canopus	263°57.3	-52°42.4			
18	99°59.3	70°42.4	N11°27.4	334°00.6	N22°17.6	111°10.6	S05°52.1	278°33.2	N02°10.7	Sirius	258°35.8	-16°44.0			
19	115°01.8	85°42.0	28.5	349°02.4	17.4	126°12.5	51.9	293°35.8	10.8	Adhara	255°14.4	-28°59.4			
20	130°04.3	100°41.5	29.7	4°04.3	17.2	141°14.4	51.7	308°38.5	10.9	Procyon	245°02.1	5°11.8			
21	145°06.7	115°41.1	· · 30.9	19°06.2	· · 17.0	156°16.4	· · 51.5	323°41.1	· · 10.9	Pollux	243°30.5	28°00.1			
22	160°09.2	130°40.6	32.0	34°08.1	16.8	171°18.3	51.3	338°43.7	11.0	Avior	234°18.9	-59°32.9			
23	175°11.7	145°40.2	33.2	49°10.0	16.6	186°20.2	51.0	353°46.4	11.1	Suhail	222°54.0	-43°28.8			
Mer.pass. 11:21		ν -0.4' d 1.2' m -3.89		ν 1.9' d -0.2' m 0.16		ν 1.9' d -0.2' m -2.05		ν 2.6' d 0.1' m 0.56		Miaplacidus	221°39.9	-69°45.9			
Alphard		217°58.2		-8°42.4		Regulus		207°45.7		11°54.8					
Dubhe		193°53.7		61°41.7		Denebola		182°35.7		14°30.7					
Gienah		175°54.4		-17°36.2		Acrux		173°11.5		-63°09.5					
Gacrux		172°03.1		-57°10.4		Alioth		166°22.0		55°54.1					
Spica		158°33.4		-11°13.1		Alkaid		153°00.1		49°15.5					
Hadar		148°50.8		-60°25.4		Menkent		148°10.0		-36°25.4					
Arcturus		145°57.5		19°07.5		Rigel Kent.		139°54.5		-60°52.6					
Kochab		137°18.6		74°06.5		Zuben'ubi		137°07.8		-16°05.2					
Alphecca		126°12.7		26°40.5		Antares		112°29.0		-26°27.3					
Atria		107°32.7		-69°02.6		Sabik		102°15.1		-15°44.3					
Shaula		96°25.0		-37°06.6		Rasalhague		96°08.5		12°32.9					
Eltanin		90°47.1		51°28.9		Kaus Aust.		83°46.9		-34°22.7					
Vega		80°40.5		38°47.3		Nunki		76°01.2		-26°17.0					
Altair		62°10.6		8°53.6		Peacock		53°23.0		-56°41.9					
Deneb		49°33.4		45°18.8		Enif		33°49.7		9°55.2					
Al Na'ir		27°46.9		-46°54.5		Fomalhaut		15°26.8		-29°34.0					
Scheat		13°56.0		28°08.2		Markab		13°41.0		15°15.6					
Apr 01 Thu		SHA		Mer.pass											
Venus		331°35.4		13:17											
Mars		234°11.4		19:44											
Jupiter		11°20.9		10:36											
Saturn		178°30.7		23:25											
Apr 02 Fri		SHA		Mer.pass											
Venus		330°25.6		13:18											
Mars		233°57.7		19:41											
Jupiter		11°08.0		10:33											
Saturn		178°34.9		23:21											
Apr 03 Sat		SHA		Mer.pass											
Venus		329°15.6		13:18											
Mars		233°43.5		19:38											
Jupiter		10°55.2		10:30											
Saturn		178°39.1		23:16											
Horizontal parallax															
Venus:										0.1					
Mars:										0.1					

Sun				Moon			
h	GHA	Dec		GHA	ν	Dec	d
Thu							HP
0	178°59.5	N04°24.9		336°31.7	8.3'	S18°12.0	10.7'
1	193°59.7	25.9		350°59.1	8.3'	18°22.7	10.6'
2	208°59.9	26.8		5°26.3	8.3'	18°33.3	10.5'
3	224°00.0	· · 27.8		19°53.6	8.2'	18°43.8	10.4'
4	239°00.2	28.8		34°20.8	8.2'	18°54.2	10.2'
5	254°00.4	29.7		48°48.0	8.2'	19°04.4	10.1'
6	269°00.6	N04°30.7		63°15.2	8.1'	S19°14.5	10.0'
7	284°00.8	31.6		77°42.3	8.1'	19°24.5	9.9'
8	299°01.0	32.6		92°09.5	8.1'	19°34.4	9.7'
9	314°01.2	· · 33.6		106°36.6	8.1'	19°44.2	9.6'
10	329°01.3	34.5		121°03.6	8.0'	19°53.8	9.5'
11	344°01.5	35.5		135°30.7	8.0'	20°03.2	9.4'
12	359°01.7	N04°36.5		149°57.7	8.0'	S20°12.6	9.2'
13	14°01.9	37.4		164°24.6	8.0'	20°21.8	9.1'
14	29°02.1	38.4		178°51.6	7.9'	20°30.9	9.0'
15	44°02.3	· · 39.4		193°18.6	7.9'	20°39.9	8.8'
16	59°02.5	40.3		207°45.5	7.9'	20°48.7	8.7'
17	74°02.7	41.3		222°12.4	7.9'	20°57.4	8.6'
18	89°02.8	N04°42.3		236°39.2	7.9'	S21°05.9	8.4'
19	104°03.0	43.2		251°06.1	7.8'	21°14.4	8.3'
20	119°03.2	44.2		265°32.9	7.8'	21°22.7	8.1'
21	134°03.4	· · 45.1		279°59.7	7.8'	21°30.8	8.0'
22	149°03.6	46.1		294°26.5	7.8'	21°38.8	7.9'
23	164°03.8	47.1		308°53.3	7.8'	21°46.7	7.7'
SD = 16.0'				SD = 16.0'			
d = 1.0'							
Fri	GHA	Dec		GHA	ν	Dec	d
0	179°04.0	N04°48.0		323°20.1	7.7'	S21°54.4	7.6'
1	194°04.1	49.0		337°46.8	7.7'	22°02.0	7.5'
2	209°04.3	50.0		352°13.5	7.7'	22°09.5	7.3'
3	224°04.5	· · 50.9		6°40.2	7.7'	22°16.8	7.2'
4	239°04.7	51.9		21°06.9	7.7'	22°24.0	7.0'
5	254°04.9	52.8		35°33.6	7.7'	22°31.0	6.9'
6	269°05.1	N04°53.8		50°00.3	7.7'	S22°37.9	6.8'
7	284°05.3	54.8		64°27.0	7.7'	22°44.6	6.6'
8	299°05.4	55.7		78°53.6	7.6'	22°51.3	6.5'
9	314°05.6	· · 56.7		93°20.3	7.6'	22°57.7	6.3'
10	329°05.8	57.6		107°46.9	7.6'	23°04.0	6.2'
11	344°06.0	58.6		122°13.5	7.6'	23°10.2	6.0'
12	359°06.2	N04°59.6		136°40.1	7.6'	S23°16.3	5.9'
13	14°06.4	05°00.5		151°06.8	7.6'	23°22.1	5.7'
14	29°06.5	01.5		165°33.4	7.6'	23°27.9	5.6'
15	44°06.7	· · 02.4		180°00.0	7.6'	23°33.5	5.5'
16	59°06.9	03.4		194°26.6	7.6'	23°39.0	5.3'
17	74°07.1	04.4		208°53.2	7.6'	23°44.3	5.2'
18	89°07.3	N05°05.3		223°19.8	7.6'	S23°49.4	5.0'
19	104°07.5	06.3		237°46.4	7.6'	23°54.5	4.9'
20	119°07.7	07.2		252°13.0	7.6'	23°59.3	4.7'
21	134°07.8	· · 08.2		266°39.7	7.6'	24°04.1	4.6'
22	149°08.0	09.2		281°06.3	7.6'	24°08.6	4.4'
23	164°08.2	10.1		295°32.9	7.6'	24°13.1	4.3'
SD = 16.0'				SD = 15.8'			
d = 1.0'							
Sat	GHA	Dec		GHA	ν	Dec	d
0	179°08.4	N05°11.1		309°59.6	7.6'	S24°17.4	4.1'
1	194°08.6	12.0		324°26.2	7.7'	24°21.5	4.0'
2	209°08.8	13.0		338°52.9	7.7'	24°25.5	3.9'
3	224°08.9	· · 14.0		353°19.5	7.7'	24°29.4	3.7'
4	239°09.1	14.9		7°46.2	7.7'	24°33.1	3.6'
5	254°09.3	15.9		22°12.9	7.7'	24°36.7	3.4'
6	269°09.5	N05°16.8		36°39.6	7.7'	S24°40.1	3.3'
7	284°09.7	17.8		51°06.3	7.7'	24°43.3	3.1'
8	299°09.9	18.7		65°33.1	7.8'	24°46.5	3.0'
9	314°10.1	· · 19.7		79°59.8	7.8'	24°49.5	2.8'
10	329°10.2	20.7		94°26.6	7.8'	24°52.3	2.7'
11	344°10.4	21.6		108°53.4	7.8'	24°55.0	2.5'
12	359°10.6	N05°22.6		123°20.2	7.8'	S24°57.5	2.4'
13	14°10.8	23.5		137°47.0	7.9'	24°59.9	2.3'
14	29°11.0	24.5		152°13.9	7.9'	25°02.2	2.1'
15	44°11.2	· · 25.4		166°40.8	7.9'	25°04.3	2.0'
16	59°11.3	26.4		181°07.7	7.9'	25°06.3	1.8'
17	74°11.5	27.4		195°34.7	8.0'	25°08.1	1.7'
18	89°11.7	N05°28.3		210°01.6	8.0'	S25°09.8	1.5'
19	104°11.9	29.3		224°28.6	8.0'	25°11.3	1.4'
20	119°12.1	30.2		238°55.7	8.1'	25°12.7	1.3'
21	134°12.3	· · 31.2		253°22.7	8.1'	25°13.9	1.1'
22	149°12.4	32.1		267°49.8	8.1'	25°15.1	1.0'
23	164°12.6	33.1		282°17.0	8.2'	25°16.0	0.8'
SD = 16.0'				SD = 15.6'			
d = 1.0'							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	01:27	03:36	04:51	19:19	20:36	22:53
N 70°	02:14	03:53	04:59	19:10	20:18	22:00
68°	02:43	04:06	05:06	19:04	20:04	21:29
66°	03:05	04:17	05:11	18:58	19:53	21:06
64°	03:22	04:26	05:16	18:53	19:44	20:49
62°	03:35	04:33	05:19	18:49	19:36	20:35
60°	03:46	04:40	05:23	18:46	19:29	20:23
N 58°	03:56	04:46	05:26	18:43	19:23	20:14
56°	04:04	04:50	05:29	18:40	19:18	20:05
54°	04:11	04:55	05:31	18:37	19:14	19:58
52°	04:17	04:59	05:33	18:35	19:10	19:51
50°	04:23	05:02	05:35	18:33	19:06	19:46
45°	04:34	05:10	05:39	18:29	18:59	19:34
N 40°	04:43	05:16	05:43	18:25	18:52	19:25
35°	04:50	05:20	05:46	18:22	18:47	19:18
30°	04:56	05:24	05:48	18:19	18:43	19:12
20°	05:05	05:31	05:53	18:14	18:37	19:03
N 10°	05:11	05:36	05:57	18:11	18:32	18:56
0°	05:16	05:40	06:00	18:07	18:28	18:52
S 10°	05:18	05:43	06:04	18:03	18:24	18:49
20°	05:20	05:45	06:07	18:00	18:22	18:47
30°	05:19	05:47	06:11	17:56	18:19	18:47
35°	05:19	05:48	06:13	17:53	18:19	18:48
40°	05:17	05:49	06:16	17:51	18:18	18:49
45°	05:15	05:49	06:19	17:48	18:17	18:51
S 50°	05:12	05:50	06:22	17:44	18:17	18:54
52°	05:11	05:50	06:23	17:43	18:16	18:55
54°	05:09	05:50	06:25	17:41	18:16	18:57
56°	05:07	05:50	06:27	17:39	18:16	18:59
58°	05:04	05:50	06:29	17:37	18:16	19:02
S 60°	05:01	05:50	06:31	17:35	18:16	19:04

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	■	■	■	■	■	■
N 70°	■	■	■	03:01	■	■
68°	■	■	■	03:48	■	■
66°	· ·	00:53	■	04:19	04:01	■
64°	· ·	00:12	02:03	04:42	04:42	04:48
62°	23:44	· ·	01:19	05:01	05:11	05:32
60°	23:23	· ·	00:50	05:16	05:33	06:01
N 58°	23:05	· ·	00:28	05:29	05:51	06:23
56°	22:50	· ·	00:10	05:41	06:06	06:42
54°	22:38	23:55	· ·	05:51	06:19	06:57
52°	22:27	23:41	· ·	06:00	06:31	07:10
50°	22:17	23:30	· ·	06:08	06:41	07:22
45°	21:56	23:06	· ·	06:25	07:02	07:47
N 40°	21:40	22:47	23:47	06:39	07:20	08:06
35°	21:26	22:31	23:30	06:51	07:34	08:22
30°	21:14	22:17	23:15	07:02	07:47	08:37
20°	20:53	21:53	22:51	07:20	08:09	09:01
N 10°	20:36	21:33	22:29	07:36	08:28	09:22
0°	20:19	21:14	22:10	07:50	08:46	09:41
S 10°	20:03	20:56	21:50	08:05	09:03	10:00
20°	19:45	20:36	21:29	08:21	09:22	10:21
30°	19:25	20:13	21:04	08:40	09:44	10:45
35°	19:14	19:59	20:50	08:51	09:57	11:00
40°	19:00	19:44	20:33	09:03	10:12	11:16
45°	18:45	19:25	20:13	09:18	10:30	11:36
S 50°	18:25	19:02	19:48	09:36	10:52	12:00
52°	18:16	18:51	19:36	09:44	11:03	12:12
54°	18:06	18:39	19:23	09:54	11:15	12:26
56°	17:55	18:25	19:07	10:05	11:29	12:42
58°	17:42	18:08	18:48	10:17	11:45	13:00
S 60°	17:27	17:49	18:25	10:31	12:05	13:24

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Upper Mer. Pass. hh:mm	Lower Mer. Pass. hh:mm	
01	04:02	03:53	12:04	01:37	14:05	
02	03:44	03:35	12:04	02:32	15:00	
03	03:26	03:18	12:03	03:28	15:55	

April 04, 05, 06 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	192°12.4	160°17.8	N12°29.5	65°41.2	N22°06.0	202°54.9	S05°40.3	10°55.6	N02°14.7	Alpheratz	357°46.3	29°08.8
1	207°14.9	175°17.3	30.6	80°43.1	05.8	217°56.8	40.0	25°58.3	14.8	Ankaa	353°18.3	-42°15.0
2	222°17.3	190°16.8	31.8	95°44.9	05.6	232°58.7	39.8	41°00.9	14.9	Schedar	349°43.9	56°35.6
3	237°19.8	205°16.3	•• 32.9	110°46.8	•• 05.3	248°00.7	•• 39.6	56°03.5	•• 14.9	Diphda	348°58.5	-17°55.8
4	252°22.3	220°15.9	34.0	125°48.6	05.1	263°02.6	39.4	71°06.2	15.0	Achernar	335°28.9	-57°11.1
5	267°24.7	235°15.4	35.2	140°50.4	04.9	278°04.5	39.2	86°08.8	15.1	Hamal	328°03.8	23°30.7
6	282°27.2	250°14.9	N12°36.3	155°52.3	N22°04.7	293°06.5	S05°38.9	101°11.5	N02°15.2	Polaris	319°20.3	89°18.7
7	297°29.7	265°14.5	37.4	170°54.1	04.5	308°08.4	38.7	116°14.1	15.2	Acamar	315°20.4	-40°15.9
8	312°32.1	280°14.0	38.6	185°56.0	04.2	323°10.3	38.5	131°16.7	15.3	Menkar	314°17.8	4°07.8
9	327°34.6	295°13.5	•• 39.7	200°57.8	•• 04.0	338°12.2	•• 38.3	146°19.4	•• 15.4	Mirfak	308°44.2	49°54.0
10	342°37.1	310°13.1	40.8	215°59.6	03.8	353°14.2	38.1	161°22.0	15.5	Aldebaran	290°52.3	16°31.8
11	357°39.5	325°12.6	42.0	231°01.5	03.6	8°16.1	37.8	176°24.6	15.5	Rigel	281°14.4	-8°11.5
12	12°42.0	340°12.1	N12°43.1	246°03.3	N22°03.3	23°18.1	S05°37.6	191°27.3	N02°15.6	Capella	280°38.1	46°00.6
13	27°44.4	355°11.7	44.2	261°05.1	03.1	38°20.0	37.4	206°29.9	15.7	Bellatrix	278°34.6	6°21.5
14	42°46.9	10°11.2	45.3	276°07.0	02.9	53°21.9	37.2	221°32.5	15.7	Elnath	278°15.7	28°37.0
15	57°49.4	25°10.7	•• 46.5	291°08.8	•• 02.7	68°23.9	•• 37.0	236°35.2	•• 15.8	Alnilam	275°48.9	-1°11.8
16	72°51.8	40°10.2	47.6	306°10.6	02.4	83°25.8	36.7	251°37.8	15.9	Betelgeuse	271°03.9	7°24.5
17	87°54.3	55°09.8	48.7	321°12.4	02.2	98°27.7	36.5	266°40.5	16.0	Canopus	263°57.3	-52°42.4
18	102°56.8	70°09.3	N12°49.9	336°14.3	N22°02.0	113°29.7	S05°36.3	281°43.1	N02°16.0	Sirius	258°35.8	-16°44.0
19	117°59.2	85°08.8	51.0	351°16.1	01.8	128°31.6	36.1	296°45.7	16.1	Adhara	255°14.4	-28°59.4
20	133°01.7	100°08.3	52.1	6°17.9	01.6	143°33.5	35.9	311°48.4	16.2	Procyon	245°02.1	5°11.8
21	148°04.2	115°07.9	•• 53.2	21°19.7	•• 01.3	158°35.5	•• 35.7	326°51.0	•• 16.3	Pollux	243°30.5	28°00.1
22	163°06.6	130°07.4	54.4	36°21.6	01.1	173°37.4	35.4	341°53.6	16.3	Avior	234°18.9	-59°32.9
23	178°09.1	145°06.9	55.5	51°23.4	00.9	188°39.3	35.2	356°56.3	16.4	Suhail	222°54.0	-43°28.8
Mer.pass. 11:09		ν -0.5' d 1.1' m -3.89		ν 1.8' d -0.2' m 0.23		ν 1.9' d -0.2' m -2.06		ν 2.6' d 0.1' m 0.57				

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	193°11.6	160°06.4	N12°56.6	66°25.2	N22°00.7	203°41.3	S05°35.0	11°58.9	N02°16.5
1	208°14.0	175°06.0	57.7	81°27.0	00.4	218°43.2	34.8	27°01.5	16.6
2	223°16.5	190°05.5	12°58.9	96°28.9	00.2	233°45.1	34.6	42°04.2	16.6
3	238°18.9	205°05.0	13°00.0	111°30.7	22°00.0	248°47.1	•• 34.3	57°06.8	•• 16.7
4	253°21.4	220°04.5	01.1	126°32.5	21°59.7	263°49.0	34.1	72°09.5	16.8
5	268°23.9	235°04.1	02.2	141°34.3	59.5	278°50.9	33.9	87°12.1	16.8
6	283°26.3	250°03.6	N13°03.3	156°36.1	N21°59.3	293°52.9	S05°33.7	102°14.7	N02°16.9
7	298°28.8	265°03.1	04.5	171°37.9	59.1	308°54.8	33.5	117°17.4	17.0
8	313°31.3	280°02.6	05.6	186°39.8	58.8	323°56.7	33.2	132°20.0	17.1
9	328°33.7	295°02.2	•• 06.7	201°41.6	•• 58.6	338°58.7	•• 33.0	147°22.6	•• 17.1
10	343°36.2	310°01.7	07.8	216°43.4	58.4	354°00.6	32.8	162°25.3	17.2
11	358°38.7	325°01.2	08.9	231°45.2	58.2	9°02.5	32.6	177°27.9	17.3
12	13°41.1	340°00.7	N13°10.1	246°47.0	N21°57.9	24°04.5	S05°32.4	192°30.5	N02°17.3
13	28°43.6	355°00.2	11.2	261°48.8	57.7	39°06.4	32.2	207°33.2	17.4
14	43°46.1	9°59.8	12.3	276°50.6	57.5	54°08.4	31.9	222°35.8	17.5
15	58°48.5	24°59.3	•• 13.4	291°52.5	•• 57.2	69°10.3	•• 31.7	237°38.4	•• 17.6
16	73°51.0	39°58.8	14.5	306°54.3	57.0	84°12.2	31.5	252°41.1	17.6
17	88°53.4	54°58.3	15.6	321°56.1	56.8	99°14.2	31.3	267°43.7	17.7
18	103°55.9	69°57.8	N13°16.7	336°57.9	N21°56.6	114°16.1	S05°31.1	282°46.3	N02°17.8
19	118°58.4	84°57.3	17.9	351°59.7	56.3	129°18.0	30.8	297°49.0	17.9
20	134°00.8	99°56.9	19.0	7°01.5	56.1	144°20.0	30.6	312°51.6	17.9
21	149°03.3	114°56.4	•• 20.1	22°03.3	•• 55.9	159°21.9	•• 30.4	327°54.3	•• 18.0
22	164°05.8	129°55.9	21.2	37°05.1	55.6	174°23.8	30.2	342°56.9	18.1
23	179°08.2	144°55.4	22.3	52°06.9	55.4	189°25.8	30.0	357°59.5	18.1
Mer.pass. 11:05		ν -0.5' d 1.1' m -3.89		ν 1.8' d -0.2' m 0.25		ν 1.9' d -0.2' m -2.06		ν 2.6' d 0.1' m 0.57	

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	194°10.7	159°54.9	N13°23.4	67°08.7	N21°55.2	204°27.7	S05°29.8	13°02.2	N02°18.2
1	209°13.2	174°54.4	24.5	82°10.5	55.0	219°29.6	29.5	28°04.8	18.3
2	224°15.6	189°54.0	25.6	97°12.3	54.7	234°31.6	29.3	43°07.4	18.4
3	239°18.1	204°53.5	•• 26.8	112°14.1	•• 54.5	249°33.5	•• 29.1	58°10.1	•• 18.4
4	254°20.5	219°53.0	27.9	127°15.9	54.3	264°35.5	28.9	73°12.7	18.5
5	269°23.0	234°52.5	29.0	142°17.7	54.0	279°37.4	28.7	88°15.3	18.6
6	284°25.5	249°52.0	N13°30.1	157°19.5	N21°53.8	294°39.3	S05°28.4	103°18.0	N02°18.6
7	299°27.9	264°51.5	31.2	172°21.3	53.6	309°41.3	28.2	118°20.6	18.7
8	314°30.4	279°51.0	32.3	187°23.1	53.3	324°43.2	28.0	133°23.2	18.8
9	329°32.9	294°50.5	•• 33.4	202°24.9	•• 53.1	339°45.1	•• 27.8	148°25.9	•• 18.9
10	344°35.3	309°50.1	34.5	217°26.7	52.9	354°47.1	27.6	163°28.5	18.9
11	359°37.8	324°49.6	35.6	232°28.5	52.6	9°49.0	27.4	178°31.1	19.0
12	14°40.3	339°49.1	N13°36.7	247°30.3	N21°52.4	24°51.0	S05°27.1	193°33.8	N02°19.1
13	29°42.7	354°48.6	37.8	262°32.1	52.2	39°52.9	26.9	208°36.4	19.1
14	44°45.2	9°48.1	38.9	277°33.9	51.9	54°54.8	26.7	223°39.0	19.2
15	59°47.7	24°47.6	•• 40.0	292°35.7	•• 51.7	69°56.8	•• 26.5	238°41.7	•• 19.3
16	74°50.1	39°47.1	41.1	307°37.5	51.5	84°58.7	26.3	253°44.3	19.4
17	89°52.6	54°46.6	42.2	322°39.3	51.2	100°00.6	26.0	268°46.9	19.4
18	104°55.0	69°46.1	N13°43.3	337°41.0	N21°51.0	115°02.6	S05°25.8	283°49.6	N02°19.5
19	119°57.5	84°45.6	44.4	352°42.8	50.8	130°04.5	25.6	298°52.2	19.6
20	135°00.0	99°45.2	45.5	7°44.6	50.5	145°06.5	25.4	313°54.8	19.6
21	150°02.4	114°44.7	•• 46.6	22°46.4	•• 50.3	160°08.4	•• 25.2	328°57.5	•• 19.7
22	165°04.9	129°44.2	47.7	37°48.2	50.1	175°10.3	25.0	344°00.1	19.8
23	180°07.4	144°43.7	48.8	52°50.0	49.8	190°12.3	24.7	359°02.7	19.9
Mer.pass. 11:01		ν -0.5' d 1.1' m -3.89		ν 1.8' d -0.2' m 0.27		ν 1.9' d -0.2' m -2.06		ν 2.6' d 0.1' m 0.58	

Apr 04 Sun	SHA	Mer.pass
Venus	328°05.3	13:19
Mars	233°28.8	19:35
Jupiter	10°42.4	10:27
Saturn	178°43.2	23:12

Apr 05 Mon	SHA	Mer.pass
Venus	326°54.9	13:20
Mars	233°13.7	19:32
Jupiter	10°29.7	10:24
Saturn	178°47.4	23:08

Apr 06 Tue	SHA	Mer.pass
Venus	325°44.2	13:21
Mars	232°58.0	19:29
Jupiter	10°17.0	10:21
Saturn	178°51.5	23:04

Horizontal parallax	
Venus:	0.1
Mars:	0.1

h	Sun			Moon			
Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	179°12.8	N05°34.0	296°44.1	8.2'	S25°16.8	0.7'	56.2'
1	194°13.0	35.0	311°11.3	8.2'	25°17.5	0.5'	56.2'
2	209°13.2	36.0	325°38.6	8.3'	25°18.1	0.4'	56.1'
3	224°13.3	· · 36.9	340°05.8	8.3'	25°18.5	0.3'	56.1'
4	239°13.5	37.9	354°33.2	8.4'	25°18.7	0.1'	56.1'
5	254°13.7	38.8	9°00.5	8.4'	25°18.9	-0.0'	56.0'
6	269°13.9	N05°39.8	23°27.9	8.4'	S25°18.9	-0.2'	56.0'
7	284°14.1	40.7	37°55.4	8.5'	25°18.7	-0.3'	56.0'
8	299°14.3	41.7	52°22.9	8.5'	25°18.4	-0.4'	56.0'
9	314°14.4	· · 42.6	66°50.4	8.6'	25°18.0	-0.6'	55.9'
10	329°14.6	43.6	81°18.0	8.6'	25°17.4	-0.7'	55.9'
11	344°14.8	44.5	95°45.6	8.7'	25°16.7	-0.8'	55.9'
12	359°15.0	N05°45.5	110°13.3	8.7'	S25°15.9	-1.0'	55.8'
13	14°15.2	46.4	124°41.0	8.8'	25°14.9	-1.1'	55.8'
14	29°15.3	47.4	139°08.8	8.8'	25°13.8	-1.2'	55.8'
15	44°15.5	· · 48.3	153°36.6	8.9'	25°12.6	-1.4'	55.7'
16	59°15.7	49.3	168°04.5	8.9'	25°11.2	-1.5'	55.7'
17	74°15.9	50.2	182°32.4	9.0'	25°09.7	-1.6'	55.7'
18	89°16.1	N05°51.2	197°00.4	9.0'	S25°08.1	-1.8'	55.6'
19	104°16.3	52.2	211°28.4	9.1'	25°06.3	-1.9'	55.6'
20	119°16.4	53.1	225°56.5	9.1'	25°04.4	-2.0'	55.6'
21	134°16.6	· · 54.1	240°24.7	9.2'	25°02.4	-2.2'	55.6'
22	149°16.8	55.0	254°52.9	9.3'	25°00.3	-2.3'	55.5'
23	164°17.0	56.0	269°21.1	9.3'	24°58.0	-2.4'	55.5'
SD = 16.0'			SD = 15.3'				
$d = 1.0'$							

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	179°17.2	N05°56.9	283°49.4	9.4'	S24°55.6	-2.5'	55.5'
1	194°17.3	57.9	298°17.8	9.4'	24°53.0	-2.7'	55.4'
2	209°17.5	58.8	312°46.3	9.5'	24°50.4	-2.8'	55.4'
3	224°17.7	05°59.8	327°14.8	9.6'	24°47.6	-2.9'	55.4'
4	239°17.9	06°00.7	341°43.3	9.6'	24°44.7	-3.0'	55.4'
5	254°18.1	01.7	356°11.9	9.7'	24°41.6	-3.2'	55.3'
6	269°18.2	N06°02.6	10°40.6	9.7'	S24°38.5	-3.3'	55.3'
7	284°18.4	03.6	25°09.4	9.8'	24°35.2	-3.4'	55.3'
8	299°18.6	04.5	39°38.2	9.9'	24°31.8	-3.5'	55.3'
9	314°18.8	· · 05.5	54°07.0	9.9'	24°28.3	-3.6'	55.2'
10	329°19.0	06.4	68°36.0	10.0'	24°24.6	-3.8'	55.2'
11	344°19.1	07.4	83°05.0	10.1'	24°20.9	-3.9'	55.2'
12	359°19.3	N06°08.3	97°34.0	10.1'	S24°17.0	-4.0'	55.2'
13	14°19.5	09.3	112°03.2	10.2'	24°13.0	-4.1'	55.1'
14	29°19.7	10.2	126°32.4	10.3'	24°08.8	-4.2'	55.1'
15	44°19.9	· · 11.1	141°01.6	10.3'	24°04.6	-4.3'	55.1'
16	59°20.0	12.1	155°30.9	10.4'	24°00.3	-4.5'	55.1'
17	74°20.2	13.0	170°00.3	10.5'	23°55.8	-4.6'	55.0'
18	89°20.4	N06°14.0	184°29.8	10.5'	S23°51.2	-4.7'	55.0'
19	104°20.6	14.9	198°59.3	10.6'	23°46.5	-4.8'	55.0'
20	119°20.8	15.9	213°28.9	10.7'	23°41.7	-4.9'	55.0'
21	134°20.9	· · 16.8	227°58.6	10.7'	23°36.8	-5.0'	54.9'
22	149°21.1	17.8	242°28.3	10.8'	23°31.8	-5.1'	54.9'
23	164°21.3	18.7	256°58.1	10.9'	23°26.7	-5.2'	54.9'
SD = 16.0'			SD = 15.1'				
$d = 1.0'$							

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	179°21.5	N06°19.7	271°28.0	10.9'	S23°21.4	-5.4'	54.9'
1	194°21.6	20.6	285°58.0	11.0'	23°16.1	-5.5'	54.9'
2	209°21.8	21.6	300°28.0	11.1'	23°10.6	-5.6'	54.8'
3	224°22.0	· · 22.5	314°58.1	11.2'	23°05.0	-5.7'	54.8'
4	239°22.2	23.5	329°28.2	11.2'	22°59.4	-5.8'	54.8'
5	254°22.4	24.4	343°58.5	11.3'	22°53.6	-5.9'	54.8'
6	269°22.5	N06°25.3	358°28.7	11.4'	S22°47.7	-6.0'	54.8'
7	284°22.7	26.3	12°59.1	11.4'	22°41.7	-6.1'	54.7'
8	299°22.9	27.2	27°29.5	11.5'	22°35.7	-6.2'	54.7'
9	314°23.1	· · 28.2	42°00.1	11.6'	22°29.5	-6.3'	54.7'
10	329°23.3	29.1	56°30.6	11.6'	22°23.2	-6.4'	54.7'
11	344°23.4	30.1	71°01.3	11.7'	22°16.8	-6.5'	54.7'
12	359°23.6	N06°31.0	85°32.0	11.8'	S22°10.3	-6.6'	54.6'
13	14°23.8	32.0	100°02.8	11.9'	22°03.7	-6.7'	54.6'
14	29°24.0	32.9	114°33.6	11.9'	21°57.1	-6.8'	54.6'
15	44°24.1	· · 33.8	129°04.6	12.0'	21°50.3	-6.9'	54.6'
16	59°24.3	34.8	143°35.6	12.1'	21°43.4	-7.0'	54.6'
17	74°24.5	35.7	158°06.6	12.1'	21°36.4	-7.1'	54.6'
18	89°24.7	N06°36.7	172°37.8	12.2'	S21°29.4	-7.2'	54.5'
19	104°24.8	37.6	187°09.0	12.3'	21°22.2	-7.2'	54.5'
20	119°25.0	38.6	201°40.2	12.3'	21°15.0	-7.3'	54.5'
21	134°25.2	· · 39.5	216°11.6	12.4'	21°07.7	-7.4'	54.5'
22	149°25.4	40.4	230°43.0	12.5'	21°00.2	-7.5'	54.5'
23	164°25.6	41.4	245°14.5	12.5'	20°52.7	-7.6'	54.5'
SD = 16.0'			SD = 15.0'				
$d = 0.9'$							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	00:20	03:17	04:35	19:33	20:54	////
N 70°	01:47	03:36	04:45	19:23	20:33	22:27
68°	02:23	03:52	04:53	19:15	20:17	21:47
66°	02:49	04:04	04:59	19:08	20:04	21:21
64°	03:08	04:14	05:05	19:02	19:53	21:01
62°	03:23	04:23	05:10	18:57	19:45	20:45
60°	03:35	04:30	05:14	18:53	19:37	20:33
N 58°	03:46	04:37	05:17	18:49	19:30	20:22
56°	03:55	04:42	05:21	18:46	19:25	20:12
54°	04:03	04:47	05:24	18:43	19:20	20:04
52°	04:10	04:52	05:26	18:40	19:15	19:57
50°	04:16	04:56	05:29	18:38	19:11	19:51
45°	04:28	05:04	05:34	18:33	19:02	19:38
N 40°	04:38	05:11	05:38	18:28	18:56	19:28
35°	04:46	05:16	05:42	18:24	18:50	19:20
30°	04:52	05:21	05:45	18:21	18:45	19:14
20°	05:02	05:28	05:50	18:15	18:38	19:03
N 10°	05:09	05:34	05:55	18:10	18:32	18:56
0°	05:15	05:39	05:59	18:06	18:27	18:51
S 10°	05:18	05:43	06:04	18:02	18:23	18:47
20°	05:20	05:46	06:08	17:57	18:19	18:45
30°	05:21	05:49	06:13	17:52	18:16	18:44
35°	05:21	05:50	06:16	17:49	18:15	18:44
40°	05:20	05:52	06:19	17:46	18:13	18:44
45°	05:19	05:53	06:22	17:42	18:12	18:46
S 50°	05:17	05:54	06:27	17:38	18:10	18:48
52°	05:16	05:55	06:28	17:36	18:10	18:49
54°	05:14	05:55	06:31	17:34	18:09	18:50
56°	05:13	05:56	06:33	17:31	18:09	18:52
58°	05:11	05:56	06:35	17:29	18:08	18:53
S 60°	05:08	05:57	06:38	17:26	18:07	18:55

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	■	■	■	■	■	■
66°	■	■	04:54	■	■	07:28
64°	03:29	04:00	04:06	05:16	06:36	08:15
62°	02:32	03:15	03:35	06:13	07:21	08:45
60°	01:59	02:45	03:12	06:46	07:51	09:08
N 58°	01:35	02:22	02:53	07:11	08:13	09:26
56°	01:15	02:04	02:37	07:30	08:31	09:42
54°	00:59	01:48	02:24	07:46	08:47	09:55
52°	00:45	01:34	02:12	08:01	09:00	10:06
50°	00:32	01:23	02:02	08:13	09:12	10:16
45°	00:07	00:58	01:40	08:38	09:36	10:38
N 40°	· ·	00:38	01:22	08:58	09:55	10:55
35°	· ·	00:22	01:07	09:15	10:11	11:09
30°	· ·	00:08	00:54	09:30	10:25	11:21
20°	23:44	· ·	00:32	09:54	10:49	11:43
N 10°	23:23	· ·	00:12	10:16	11:09	12:01
0°	23:03	23:54	· ·	10:36	11:28	12:18
S 10°	22:44	23:36	· ·	10:55	11:47	12:35
20°	22:23	23:17	· ·	11:17	12:07	12:53
30°	21:58	22:54	23:51	11:41	12:31	13:14
35°	21:44	22:41	23:39	11:56	12:44	13:26
40°	21:28	22:26	23:26	12:12	13:00	13:40
45°	21:08	22:08	23:10	12:32	13:19	13:56
S 50°	20:43	21:45	22:51	12:57	13:42	14:16
52°	20:31	21:34	22:42	13:09	13:53	14:26
54°	20:18	21:22	22:32	13:23	14:05	14:36
56°	20:02	21:08	22:20	13:39	14:20	14:48
58°	19:43	20:51	22:07	13:58	14:37	15:02
S 60°	19:20	20:31	21:51	14:21	14:57	15:19

Day	Sun		Mer. Pass hh:mm	Moon		Age 20-22 73-54%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
04	03:09	03:00	12:03	04:23	16:49	
05	02:51	02:43	12:03	05:16	17:41	
06	02:34	02:26	12:02	06:06	18:30	

April 07, 08, 09 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	195°09.8	159°43.2	N13°49.9	67°51.8	N21°49.6	205°14.2	S05°24.5	14°05.4	N02°19.9	Alpheratz	357°46.3	29°08.8
1	210°12.3	174°42.7	51.0	82°53.5	49.4	220°16.1	24.3	29°08.0	20.0	Ankaa	353°18.3	-42°15.0
2	225°14.8	189°42.2	52.1	97°55.3	49.1	235°18.1	24.1	44°10.6	20.1	Schedar	349°43.9	56°35.6
3	240°17.2	204°41.7	· · 53.2	112°57.1	· · 48.9	250°20.0	· · 23.9	59°13.3	· · 20.1	Diphda	348°58.5	-17°55.8
4	255°19.7	219°41.2	54.3	127°58.9	48.7	265°22.0	23.7	74°15.9	20.2	Achernar	335°28.9	-57°11.1
5	270°22.2	234°40.7	55.4	143°00.7	48.4	280°23.9	23.4	89°18.5	20.3	Hamal	328°03.8	23°30.7
6	285°24.6	249°40.2	N13°56.5	158°02.5	N21°48.2	295°25.8	S05°23.2	104°21.2	N02°20.4	Polaris	319°20.7	89°18.7
7	300°27.1	264°39.7	57.6	173°04.2	47.9	310°27.8	23.0	119°23.8	20.4	Acamar	315°20.4	-40°15.9
8	315°29.5	279°39.2	58.7	188°06.0	47.7	325°29.7	22.8	134°26.4	20.5	Menkar	314°17.8	4°07.8
9	330°32.0	294°38.7	13°59.8	203°07.8	· · 47.5	340°31.7	· · 22.6	149°29.1	· · 20.6	Mirfak	308°44.2	49°54.0
10	345°34.5	309°38.2	14°00.9	218°09.6	47.2	355°33.6	22.4	164°31.7	20.6	Aldebaran	290°52.3	16°31.8
11	0°36.9	324°37.7	01.9	233°11.3	47.0	10°35.5	22.1	179°34.3	20.7	Rigel	281°14.5	-8°11.5
12	15°39.4	339°37.2	N14°03.0	248°13.1	N21°46.8	25°37.5	S05°21.9	194°37.0	N02°20.8	Capella	280°38.1	46°00.6
13	30°41.9	354°36.7	04.1	263°14.9	46.5	40°39.4	21.7	209°39.6	20.9	Bellatrix	278°34.7	6°21.5
14	45°44.3	9°36.2	05.2	278°16.7	46.3	55°41.3	21.5	224°42.2	20.9	Elnath	278°15.7	28°37.0
15	60°46.8	24°35.7	· · 06.3	293°18.4	· · 46.1	70°43.3	· · 21.3	239°44.9	· · 21.0	Alnilam	275°48.9	-1°11.8
16	75°49.3	39°35.2	07.4	308°20.2	45.8	85°45.2	21.1	254°47.5	21.1	Betelgeuse	271°03.9	7°24.5
17	90°51.7	54°34.7	08.5	323°22.0	45.6	100°47.2	20.8	269°50.1	21.1	Canopus	263°57.3	-52°42.4
18	105°54.2	69°34.2	N14°09.6	338°23.8	N21°45.3	115°49.1	S05°20.6	284°52.8	N02°21.2	Sirius	258°35.8	-16°44.0
19	120°56.6	84°33.7	10.7	353°25.5	45.1	130°51.0	20.4	299°55.4	21.3	Adhara	255°14.4	-28°59.4
20	135°59.1	99°33.2	11.7	8°27.3	44.9	145°53.0	20.2	314°58.0	21.4	Procyon	245°02.1	5°11.8
21	151°01.6	114°32.7	· · 12.8	23°29.1	· · 44.6	160°54.9	· · 20.0	330°00.7	· · 21.4	Pollux	243°30.5	28°00.1
22	166°04.0	129°32.2	13.9	38°30.8	44.4	175°56.9	19.7	345°03.3	21.5	Avior	234°18.9	-59°32.9
23	181°06.5	144°31.7	15.0	53°32.6	44.1	190°58.8	19.5	0°05.9	21.6	Suhail	222°54.0	-43°28.8
Mer.pass. 10:58		ν -0.5' d 1.1' m -3.89		ν 1.8' d -0.2' m 0.29		ν 1.9' d -0.2' m -2.06		ν 2.6' d 0.1' m 0.58		Miaplacidus	221°40.0	-69°45.9
Alphard	217°58.2	-8°42.4	Regulus	207°45.7	11°54.9	Dubhe	193°53.8	61°41.7	Denebola	182°35.7	14°30.7	
Gienah	175°54.4	-17°36.2	Acrux	173°11.5	-63°09.6	Gacrux	172°03.1	-57°10.5	Alioth	166°22.0	55°54.1	
Spica	158°33.4	-11°13.1	Alkaid	153°00.1	49°15.5	Hadar	148°50.8	-60°25.4	Menkent	148°10.0	-36°25.4	
Arcturus	145°57.5	19°07.5	Rigel Kent.	139°54.4	-60°52.7	Kochab	137°18.5	74°06.5	Zuben'ubi	137°07.7	-16°05.2	
Alphecca	126°12.7	26°40.5	Antares	112°28.9	-26°27.3	Atria	107°32.6	-69°02.6	Sabik	102°15.0	-15°44.3	
Shaula	96°24.9	-37°06.6	Rasalhague	96°08.5	12°32.9	Eltanin	90°47.0	51°28.9	Kaus Aust.	83°46.8	-34°22.7	
Vega	80°40.5	38°47.3	Nunki	76°01.1	-26°17.0	Altair	62°10.6	8°53.6	Peacock	53°22.9	-56°41.9	
Deneb	49°33.3	45°18.8	Enif	33°49.6	9°55.2	Al Na'ir	27°46.8	-46°54.5	Fomalhaut	15°26.8	-29°34.0	
Scheat	13°56.0	28°08.2	Markab	13°41.0	15°15.6							
Apr 07 Wed		SHA		Mer.pass								
Venus		324°33.3		13:22								
Mars		232°41.9		19:26								
Jupiter		10°04.4		10:18								
Saturn		178°55.5		23:00								
Apr 08 Thu		SHA		Mer.pass								
Venus		323°22.2		13:22								
Mars		232°25.4		19:23								
Jupiter		9°51.8		10:15								
Saturn		178°59.6		22:55								
Apr 09 Fri		SHA		Mer.pass								
Venus		322°10.9		13:23								
Mars		232°08.4		19:21								
Jupiter		9°39.2		10:12								
Saturn		179°03.6		22:51								
Horizontal parallax												
Venus: 0.1												
Mars: 0.1												

DUT1 = UT1-UTC = +0.0102 sec ΔT = TT-UT1 = +66.1738 sec

2010 April 07 to Apr. 09 UT

h Sun Moon						
Wed	GHA	Dec	GHA	ν	Dec	d HP
0	179°25.7	N06°42.3	259°46.0	12.6'	S20°45.1	-7.7' 54.4'
1	194°25.9	43.3	274°17.6	12.7'	20°37.4	-7.8' 54.4'
2	209°26.1	44.2	288°49.3	12.8'	20°29.6	-7.9' 54.4'
3	224°26.3	· · 45.2	303°21.1	12.8'	20°21.8	-8.0' 54.4'
4	239°26.4	46.1	317°52.9	12.9'	20°13.8	-8.0' 54.4'
5	254°26.6	47.0	332°24.8	13.0'	20°05.8	-8.1' 54.4'
6	269°26.8	N06°48.0	346°56.7	13.0'	S19°57.7	-8.2' 54.4'
7	284°27.0	48.9	1°28.7	13.1'	19°49.5	-8.3' 54.3'
8	299°27.1	49.9	16°00.8	13.1'	19°41.2	-8.4' 54.3'
9	314°27.3	· · 50.8	30°33.0	13.2'	19°32.8	-8.4' 54.3'
10	329°27.5	51.7	45°05.2	13.3'	19°24.4	-8.5' 54.3'
11	344°27.7	52.7	59°37.5	13.3'	19°15.8	-8.6' 54.3'
12	359°27.8	N06°53.6	74°09.8	13.4'	S19°07.2	-8.7' 54.3'
13	14°28.0	54.6	88°42.2	13.5'	18°58.6	-8.8' 54.3'
14	29°28.2	55.5	103°14.7	13.5'	18°49.8	-8.8' 54.3'
15	44°28.4	· · 56.4	117°47.2	13.6'	18°41.0	-8.9' 54.3'
16	59°28.5	57.4	132°19.8	13.7'	18°32.1	-9.0' 54.2'
17	74°28.7	58.3	146°52.5	13.7'	18°23.1	-9.1' 54.2'
18	89°28.9	N06°59.3	161°25.2	13.8'	S18°14.0	-9.1' 54.2'
19	104°29.1	07°00.2	175°58.0	13.8'	18°04.9	-9.2' 54.2'
20	119°29.2	01.1	190°30.8	13.9'	17°55.7	-9.3' 54.2'
21	134°29.4	· · 02.1	205°03.7	14.0'	17°46.4	-9.4' 54.2'
22	149°29.6	03.0	219°36.7	14.0'	17°37.0	-9.4' 54.2'
23	164°29.8	03.9	234°09.7	14.1'	17°27.6	-9.5' 54.2'
SD = 16.0' d = 0.9' SD = 14.8'						
Thu	GHA	Dec	GHA	ν	Dec	d HP
0	179°29.9	N07°04.9	248°42.7	14.1'	S17°18.1	-9.6' 54.2'
1	194°30.1	05.8	263°15.9	14.2'	17°08.6	-9.6' 54.2'
2	209°30.3	06.8	277°49.1	14.2'	16°58.9	-9.7' 54.2'
3	224°30.5	· · 07.7	292°22.3	14.3'	16°49.2	-9.8' 54.2'
4	239°30.6	08.6	306°55.6	14.4'	16°39.5	-9.8' 54.2'
5	254°30.8	09.6	321°29.0	14.4'	16°29.6	-9.9' 54.1'
6	269°31.0	N07°10.5	336°02.4	14.5'	S16°19.7	-10.0' 54.1'
7	284°31.1	11.4	350°35.8	14.5'	16°09.8	-10.0' 54.1'
8	299°31.3	12.4	5°09.3	14.6'	15°59.8	-10.1' 54.1'
9	314°31.5	· · 13.3	19°42.9	14.6'	15°49.7	-10.1' 54.1'
10	329°31.7	14.2	34°16.5	14.7'	15°39.5	-10.2' 54.1'
11	344°31.8	15.2	48°50.2	14.7'	15°29.3	-10.3' 54.1'
12	359°32.0	N07°16.1	63°23.9	14.8'	S15°19.1	-10.3' 54.1'
13	14°32.2	17.0	77°57.7	14.8'	15°08.7	-10.4' 54.1'
14	29°32.4	18.0	92°31.5	14.9'	14°58.3	-10.4' 54.1'
15	44°32.5	· · 18.9	107°05.4	14.9'	14°47.9	-10.5' 54.1'
16	59°32.7	19.8	121°39.3	15.0'	14°37.4	-10.6' 54.1'
17	74°32.9	20.8	136°13.2	15.0'	14°26.8	-10.6' 54.1'
18	89°33.0	N07°21.7	150°47.2	15.1'	S14°16.2	-10.7' 54.1'
19	104°33.2	22.6	165°21.3	15.1'	14°05.5	-10.7' 54.1'
20	119°33.4	23.6	179°55.4	15.1'	13°54.8	-10.8' 54.1'
21	134°33.6	· · 24.5	194°29.5	15.2'	13°44.0	-10.8' 54.1'
22	149°33.7	25.4	209°03.7	15.2'	13°33.2	-10.9' 54.1'
23	164°33.9	26.4	223°37.9	15.3'	13°22.3	-10.9' 54.1'
SD = 16.0' d = 0.9' SD = 14.8'						
Fri	GHA	Dec	GHA	ν	Dec	d HP
0	179°34.1	N07°27.3	238°12.2	15.3'	S13°11.3	-11.0' 54.1'
1	194°34.2	28.2	252°46.5	15.3'	13°00.3	-11.0' 54.1'
2	209°34.4	29.2	267°20.9	15.4'	12°49.3	-11.1' 54.1'
3	224°34.6	· · 30.1	281°55.3	15.4'	12°38.2	-11.1' 54.1'
4	239°34.8	31.0	296°29.7	15.5'	12°27.1	-11.2' 54.1'
5	254°34.9	32.0	311°04.1	15.5'	12°15.9	-11.2' 54.1'
6	269°35.1	N07°32.9	325°38.6	15.5'	S12°04.6	-11.3' 54.1'
7	284°35.3	33.8	340°13.2	15.6'	11°53.3	-11.3' 54.1'
8	299°35.4	34.8	354°47.7	15.6'	11°42.0	-11.4' 54.1'
9	314°35.6	· · 35.7	9°22.4	15.6'	11°30.6	-11.4' 54.1'
10	329°35.8	36.6	23°57.0	15.7'	11°19.2	-11.5' 54.1'
11	344°35.9	37.5	38°31.7	15.7'	11°07.7	-11.5' 54.1'
12	359°36.1	N07°38.5	53°06.4	15.7'	S10°56.2	-11.6' 54.1'
13	14°36.3	39.4	67°41.1	15.8'	10°44.6	-11.6' 54.1'
14	29°36.5	40.3	82°15.9	15.8'	10°33.0	-11.6' 54.1'
15	44°36.6	· · 41.3	96°50.7	15.8'	10°21.4	-11.7' 54.1'
16	59°36.8	42.2	111°25.5	15.9'	10°09.7	-11.7' 54.1'
17	74°37.0	43.1	126°00.3	15.9'	09°58.0	-11.8' 54.1'
18	89°37.1	N07°44.0	140°35.2	15.9'	S09°46.2	-11.8' 54.1'
19	104°37.3	45.0	155°10.1	15.9'	09°34.4	-11.8' 54.1'
20	119°37.5	45.9	169°45.0	16.0'	09°22.5	-11.9' 54.1'
21	134°37.6	· · 46.8	184°20.0	16.0'	09°10.7	-11.9' 54.1'
22	149°37.8	47.8	198°55.0	16.0'	08°58.7	-12.0' 54.1'
23	164°38.0	48.7	213°30.0	16.0'	08°46.8	-12.0' 54.1'
SD = 16.0' d = 0.9' SD = 14.7'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	////	02:56	04:19	19:48	21:13	////
N 70°	01:11	03:19	04:31	19:36	20:49	23:07
68°	02:01	03:37	04:40	19:26	20:30	22:09
66°	02:31	03:51	04:48	19:18	20:15	21:37
64°	02:53	04:03	04:54	19:11	20:03	21:14
62°	03:10	04:12	05:00	19:06	19:53	20:57
60°	03:24	04:21	05:05	19:01	19:45	20:42
N 58°	03:36	04:28	05:09	18:56	19:38	20:30
56°	03:46	04:34	05:13	18:52	19:31	20:20
54°	03:54	04:40	05:16	18:49	19:26	20:11
52°	04:02	04:44	05:19	18:45	19:21	20:04
50°	04:08	04:49	05:22	18:43	19:16	19:57
48°	04:22	04:58	05:28	18:36	19:06	19:43
N 40°	04:33	05:06	05:33	18:31	18:59	19:32
38°	04:42	05:12	05:38	18:27	18:52	19:23
36°	04:49	05:17	05:41	18:23	18:47	19:16
34°	05:00	05:26	05:48	18:16	18:38	19:04
N 10°	05:08	05:32	05:53	18:10	18:32	18:56
0°	05:14	05:38	05:59	18:05	18:26	18:50
S 10°	05:18	05:42	06:04	18:00	18:21	18:46
20°	05:21	05:47	06:09	17:55	18:17	18:42
30°	05:23	05:51	06:15	17:49	18:13	18:40
38°	05:23	05:53	06:18	17:45	18:11	18:40
40°	05:23	05:55	06:22	17:41	18:08	18:40
45°	05:23	05:57	06:26	17:37	18:06	18:40
S 50°	05:21	05:59	06:31	17:32	18:04	18:41
52°	05:21	06:00	06:34	17:29	18:03	18:42
54°	05:20	06:01	06:36	17:27	18:02	18:43
56°	05:19	06:02	06:39	17:24	18:01	18:44
58°	05:17	06:03	06:42	17:21	18:00	18:45
S 60°	05:15	06:04	06:46	17:17	17:59	18:47

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	■	05:56	05:05	■	09:40	12:00
N 70°	06:39	05:13	04:44	07:23	10:22	12:19
68°	05:12	04:44	04:27	08:49	10:49	12:34
66°	04:33	04:22	04:14	09:27	11:10	12:47
64°	04:06	04:05	04:03	09:53	11:27	12:57
62°	03:45	03:50	03:53	10:13	11:40	13:05
60°	03:28	03:38	03:45	10:30	11:52	13:12
N 58°	03:13	03:27	03:37	10:44	12:02	13:19
56°	03:01	03:18	03:31	10:56	12:10	13:24
54°	02:50	03:10	03:25	11:06	12:18	13:29
52°	02:40	03:02	03:20	11:15	12:25	13:34
50°	02:32	02:56	03:15	11:23	12:31	13:38
45°	02:13	02:41	03:05	11:41	12:44	13:47
N 40°	01:58	02:29	02:56	11:55	12:55	13:54
38°	01:46	02:19	02:49	12:07	13:04	14:01
36°	01:34	02:10	02:42	12:17	13:12	14:06
34°	01:15	01:55	02:31	12:35	13:26	14:16
N 10°	00:58	01:41	02:21	12:51	13:38	14:24
0°	00:43	01:28	02:11	13:05	13:49	14:31
S 10°	00:27	01:15	02:02	13:19	14:00	14:39
20°	00:10	01:02	01:52	13:34	14:12	14:47
30°	· · ·	00:46	01:40	13:52	14:26	14:56
38°	· · ·	00:37	01:33	14:02	14:33	15:02
40°	· · ·	00:26	01:26	14:13	14:42	15:08
45°	· · ·	00:14	01:17	14:27	14:52	15:14
S 50°	23:59	· · ·	01:06	14:43	15:05	15:23
52°	23:51	· · ·	01:01	14:51	15:10	15:26
54°	23:44	· · ·	00:55	14:59	15:16	15:31
56°	23:35	· · ·	00:49	15:09	15:23	15:35
58°	23:25	· · ·	00:42	15:19	15:31	15:40
S 60°	23:13	· · ·	00:34	15:32	15:40	15:46

Day	Sun		Mer. Pass	Moon		Age 23-25 44-26%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
07	02:17	02:09	12:02	06:54	19:17	
08	02:00	01:52	12:02	07:39	20:00	
09	01:44	01:36	12:02	08:21	20:42	

April 10, 11, 12 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	198°07.2	159°06.5	N15°07.3	69°58.3	N21°32.2	207°34.0	S05°09.0	17°14.8	N02°25.0	Alpheratz	357°46.3	29°08.8
1	213°09.7	174°06.0	08.4	85°00.0	31.9	222°35.9	08.7	32°17.4	25.0	Ankaa	353°18.3	-42°14.9
2	228°12.2	189°05.5	09.4	100°01.7	31.7	237°37.9	08.5	47°20.1	25.1	Schedar	349°43.9	56°35.6
3	243°14.6	204°05.0	•• 10.5	115°03.5	•• 31.4	252°39.8	•• 08.3	62°22.7	•• 25.2	Diphda	348°58.5	-17°55.8
4	258°17.1	219°04.4	11.5	130°05.2	31.2	267°41.8	08.1	77°25.3	25.2	Achernar	335°28.9	-57°11.1
5	273°19.6	234°03.9	12.6	145°06.9	30.9	282°43.7	07.9	92°27.9	25.3	Hamal	328°03.8	23°30.7
6	288°22.0	249°03.4	N15°13.6	160°08.7	N21°30.7	297°45.6	S05°07.7	107°30.6	N02°25.4	Polaris	319°21.3	89°18.7
7	303°24.5	264°02.9	14.7	175°10.4	30.4	312°47.6	07.4	122°33.2	25.4	Acamar	315°20.4	-40°15.9
8	318°27.0	279°02.3	15.7	190°12.1	30.2	327°49.5	07.2	137°35.8	25.5	Menkar	314°17.8	4°07.8
9	333°29.4	294°01.8	•• 16.8	205°13.8	•• 29.9	342°51.5	•• 07.0	152°38.5	•• 25.6	Mirfak	308°44.2	49°54.0
10	348°31.9	309°01.3	17.8	220°15.6	29.7	357°53.4	06.8	167°41.1	25.7	Aldebaran	290°52.3	16°31.8
11	3°34.4	324°00.7	18.9	235°17.3	29.4	12°55.4	06.6	182°43.7	25.7	Rigel	281°14.5	-8°11.5
12	18°36.8	339°00.2	N15°19.9	250°19.0	N21°29.2	27°57.3	S05°06.4	197°46.3	N02°25.8	Capella	280°38.2	46°00.6
13	33°39.3	353°59.7	21.0	265°20.7	28.9	42°59.3	06.2	212°49.0	25.9	Bellatrix	278°34.7	6°21.5
14	48°41.7	8°59.2	22.0	280°22.4	28.7	58°01.2	05.9	227°51.6	25.9	Elnath	278°15.8	28°37.0
15	63°44.2	23°58.6	•• 23.1	295°24.2	•• 28.4	73°03.1	•• 05.7	242°54.2	•• 26.0	Alnilam	275°48.9	-1°11.8
16	78°46.7	38°58.1	24.1	310°25.9	28.2	88°05.1	05.5	257°56.8	26.1	Betelgeuse	271°03.9	7°24.5
17	93°49.1	53°57.6	25.1	325°27.6	27.9	103°07.0	05.3	272°59.5	26.1	Canopus	263°57.3	-52°42.4
18	108°51.6	68°57.0	N15°26.2	340°29.3	N21°27.6	118°09.0	S05°05.1	288°02.1	N02°26.2	Sirius	258°35.8	-16°44.0
19	123°54.1	83°56.5	27.2	355°31.0	27.4	133°10.9	04.9	303°04.7	26.3	Adhara	255°14.4	-28°59.4
20	138°56.5	98°56.0	28.3	10°32.8	27.1	148°12.9	04.7	318°07.4	26.3	Procyon	245°02.1	5°11.8
21	153°59.0	113°55.4	•• 29.3	25°34.5	•• 26.9	163°14.8	•• 04.4	333°10.0	•• 26.4	Pollux	243°30.5	28°00.1
22	169°01.5	128°54.9	30.3	40°36.2	26.6	178°16.8	04.2	348°12.6	26.5	Avior	234°18.9	-59°32.9
23	184°03.9	143°54.4	31.4	55°37.9	26.4	193°18.7	04.0	3°15.2	26.5	Suhail	222°54.1	-43°28.8
Mer.pass. 10:46		ν -0.5' d 1.1' m -3.89		ν 1.7' d -0.2' m 0.35		ν 1.9' d -0.2' m -2.07		ν 2.6' d 0.1' m 0.60		Miaplacidus	221°40.0	-69°45.9
										Alphard	217°58.2	-8°42.4
										Regulus	207°45.8	11°54.9
										Dubhe	193°53.8	61°41.7
										Denebola	182°35.7	14°30.7
										Gienah	175°54.4	-17°36.2
										Acrux	173°11.5	-63°09.6
										Gacrux	172°03.1	-57°10.5
										Alioth	166°22.0	55°54.1
										Spica	158°33.4	-11°13.1
										Alkaid	153°00.1	49°15.5
										Hadar	148°50.8	-60°25.5
										Menkent	148°10.0	-36°25.4
										Arcturus	145°57.5	19°07.5
										Rigel Kent.	139°54.4	-60°52.7
										Kochab	137°18.5	74°06.6
										Zuben'ubi	137°07.7	-16°05.2
										Alphecca	126°12.6	26°40.5
										Antares	112°28.9	-26°27.3
										Atria	107°32.6	-69°02.6
										Sabik	102°15.0	-15°44.3
										Shaula	96°24.9	-37°06.6
										Rasalhague	96°08.5	12°32.9
										Eltanin	90°47.0	51°28.9
										Kaus Aust.	83°46.8	-34°22.7
										Vega	80°40.5	38°47.3
										Nunki	76°01.1	-26°17.0
										Altair	62°10.5	8°53.6
										Peacock	53°22.9	-56°41.9
										Deneb	49°33.3	45°18.8
										Enif	33°49.6	9°55.2
										Al Na'ir	27°46.8	-46°54.5
										Fomalhaut	15°26.8	-29°34.0
										Scheat	13°56.0	28°08.2
										Markab	13°40.9	15°15.6
										Apr 10 Sat SHA Mer.pass		
										Venus	320°59.3	13:24
										Mars	231°51.0	19:18
										Jupiter	9°26.7	10:08
										Saturn	179°07.6	22:47
										Apr 11 Sun SHA Mer.pass		
										Venus	319°47.4	13:25
										Mars	231°33.2	19:15
										Jupiter	9°14.3	10:05
										Saturn	179°11.5	22:43
										Apr 12 Mon SHA Mer.pass		
										Venus	318°35.4	13:26
										Mars	231°15.0	19:12
										Jupiter	9°01.9	10:02
										Saturn	179°15.4	22:39
										Horizontal parallax		
										Venus:	0.1	
										Mars:	0.1	

h	Sun			Moon			
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	179°38.1	N07°49.6	228°05.0	16.0'	S08°34.8	-12.0'	54.1'
1	194°38.3	50.5	242°40.0	16.1'	08°22.8	-12.1'	54.1'
2	209°38.5	51.5	257°15.1	16.1'	08°10.7	-12.1'	54.1'
3	224°38.6	· · 52.4	271°50.2	16.1'	07°58.6	-12.1'	54.2'
4	239°38.8	53.3	286°25.3	16.1'	07°46.5	-12.2'	54.2'
5	254°39.0	54.2	301°00.4	16.1'	07°34.3	-12.2'	54.2'
6	269°39.2	N07°55.2	315°35.5	16.1'	S07°22.1	-12.2'	54.2'
7	284°39.3	56.1	330°10.7	16.2'	07°09.9	-12.3'	54.2'
8	299°39.5	57.0	344°45.8	16.2'	06°57.6	-12.3'	54.2'
9	314°39.7	· · 57.9	359°21.0	16.2'	06°45.3	-12.3'	54.2'
10	329°39.8	58.9	13°56.2	16.2'	06°33.0	-12.3'	54.2'
11	344°40.0	07°59.8	28°31.4	16.2'	06°20.6	-12.4'	54.2'
12	359°40.2	N08°00.7	43°06.6	16.2'	S06°08.3	-12.4'	54.2'
13	14°40.3	01.6	57°41.9	16.2'	05°55.9	-12.4'	54.2'
14	29°40.5	02.6	72°17.1	16.2'	05°43.4	-12.5'	54.2'
15	44°40.7	· · 03.5	86°52.3	16.3'	05°31.0	-12.5'	54.2'
16	59°40.8	04.4	101°27.6	16.3'	05°18.5	-12.5'	54.3'
17	74°41.0	05.3	116°02.8	16.3'	05°06.0	-12.5'	54.3'
18	89°41.2	N08°06.2	130°38.1	16.3'	S04°53.5	-12.6'	54.3'
19	104°41.3	07.2	145°13.4	16.3'	04°40.9	-12.6'	54.3'
20	119°41.5	08.1	159°48.6	16.3'	04°28.3	-12.6'	54.3'
21	134°41.7	· · 09.0	174°23.9	16.3'	04°15.7	-12.6'	54.3'
22	149°41.8	09.9	188°59.2	16.3'	04°03.1	-12.6'	54.3'
23	164°42.0	10.9	203°34.5	16.3'	03°50.5	-12.7'	54.3'
SD = 16.0' d = 0.9'			SD = 14.8'				

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	179°42.1	N08°11.8	218°09.8	16.3'	S03°37.8	-12.7'	54.3'
1	194°42.3	12.7	232°45.0	16.3'	03°25.1	-12.7'	54.3'
2	209°42.5	13.6	247°20.3	16.3'	03°12.4	-12.7'	54.4'
3	224°42.6	· · 14.5	261°55.6	16.3'	02°59.7	-12.7'	54.4'
4	239°42.8	15.5	276°30.9	16.3'	02°47.0	-12.8'	54.4'
5	254°43.0	16.4	291°06.1	16.3'	02°34.2	-12.8'	54.4'
6	269°43.1	N08°17.3	305°41.4	16.3'	S02°21.4	-12.8'	54.4'
7	284°43.3	18.2	320°16.6	16.2'	02°08.7	-12.8'	54.4'
8	299°43.5	19.1	334°51.9	16.2'	01°55.9	-12.8'	54.4'
9	314°43.6	· · 20.1	349°27.1	16.2'	01°43.0	-12.8'	54.4'
10	329°43.8	21.0	4°02.4	16.2'	01°30.2	-12.8'	54.4'
11	344°44.0	21.9	18°37.6	16.2'	01°17.4	-12.8'	54.5'
12	359°44.1	N08°22.8	33°12.8	16.2'	S01°04.5	-12.9'	54.5'
13	14°44.3	23.7	47°48.0	16.2'	00°51.7	-12.9'	54.5'
14	29°44.4	24.6	62°23.2	16.2'	00°38.8	-12.9'	54.5'
15	44°44.6	· · 25.6	76°58.4	16.2'	00°25.9	-12.9'	54.5'
16	59°44.8	26.5	91°33.5	16.1'	00°13.0	-12.9'	54.5'
17	74°44.9	27.4	106°08.7	16.1'	S00°00.2	-12.9'	54.5'
18	89°45.1	N08°28.3	120°43.8	16.1'	N00°12.8	12.9'	54.6'
19	104°45.3	29.2	135°18.9	16.1'	00°25.7	12.9'	54.6'
20	119°45.4	30.1	149°54.0	16.1'	00°38.6	12.9'	54.6'
21	134°45.6	· · 31.1	164°29.1	16.1'	00°51.5	12.9'	54.6'
22	149°45.8	32.0	179°04.2	16.0'	01°04.4	12.9'	54.6'
23	164°45.9	32.9	193°39.2	16.0'	01°17.3	12.9'	54.6'
SD = 16.0' d = 0.9'			SD = 14.8'				

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	179°46.1	N08°33.8	208°14.2	16.0'	N01°30.3	12.9'	54.6'
1	194°46.2	34.7	222°49.2	16.0'	01°43.2	12.9'	54.7'
2	209°46.4	35.6	237°24.2	15.9'	01°56.1	12.9'	54.7'
3	224°46.6	· · 36.6	251°59.1	15.9'	02°09.1	12.9'	54.7'
4	239°46.7	37.5	266°34.0	15.9'	02°22.0	12.9'	54.7'
5	254°46.9	38.4	281°08.9	15.9'	02°34.9	12.9'	54.7'
6	269°47.0	N08°39.3	295°43.8	15.8'	N02°47.9	12.9'	54.7'
7	284°47.2	40.2	310°18.6	15.8'	03°00.8	12.9'	54.7'
8	299°47.4	41.1	324°53.4	15.8'	03°13.7	12.9'	54.8'
9	314°47.5	· · 42.0	339°28.2	15.7'	03°26.6	12.9'	54.8'
10	329°47.7	42.9	354°02.9	15.7'	03°39.5	12.9'	54.8'
11	344°47.9	43.9	8°37.7	15.7'	03°52.5	12.9'	54.8'
12	359°48.0	N08°44.8	23°12.3	15.6'	N04°05.4	12.9'	54.8'
13	14°48.2	45.7	37°47.0	15.6'	04°18.2	12.9'	54.8'
14	29°48.3	46.6	52°21.6	15.6'	04°31.1	12.9'	54.9'
15	44°48.5	· · 47.5	66°56.2	15.5'	04°44.0	12.9'	54.9'
16	59°48.7	48.4	81°30.7	15.5'	04°56.9	12.9'	54.9'
17	74°48.8	49.3	96°05.2	15.5'	05°09.7	12.8'	54.9'
18	89°49.0	N08°50.2	110°39.7	15.4'	N05°22.6	12.8'	54.9'
19	104°49.1	51.2	125°14.1	15.4'	05°35.4	12.8'	54.9'
20	119°49.3	52.1	139°48.5	15.3'	05°48.2	12.8'	55.0'
21	134°49.5	· · 53.0	154°22.9	15.3'	06°01.0	12.8'	55.0'
22	149°49.6	53.9	168°57.2	15.3'	06°13.8	12.8'	55.0'
23	164°49.8	54.8	183°31.4	15.2'	06°26.6	12.8'	55.0'
SD = 16.0' d = 0.9'			SD = 14.9'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	////	02:34	04:03	20:03	21:35	////
N 70°	////	03:01	04:16	19:49	21:05	////
68°	01:34	03:22	04:27	19:38	20:44	22:37
66°	02:12	03:38	04:36	19:28	20:27	21:55
64°	02:38	03:51	04:44	19:20	20:14	21:29
62°	02:57	04:02	04:50	19:14	20:03	21:08
60°	03:13	04:11	04:56	19:08	19:53	20:52
N 58°	03:25	04:19	05:01	19:03	19:45	20:39
56°	03:36	04:26	05:05	18:58	19:38	20:28
54°	03:46	04:32	05:09	18:54	19:32	20:18
52°	03:54	04:37	05:13	18:51	19:26	20:10
50°	04:01	04:42	05:16	18:47	19:21	20:02
45°	04:16	04:53	05:23	18:40	19:10	19:47
N 40°	04:28	05:01	05:29	18:34	19:02	19:35
35°	04:37	05:08	05:34	18:29	18:55	19:26
30°	04:45	05:14	05:38	18:25	18:49	19:18
20°	04:57	05:23	05:45	18:17	18:39	19:06
N 10°	05:06	05:31	05:52	18:10	18:32	18:56
0°	05:13	05:37	05:58	18:04	18:25	18:50
S 10°	05:18	05:42	06:04	17:58	18:20	18:44
20°	05:22	05:48	06:10	17:52	18:14	18:40
30°	05:25	05:52	06:17	17:45	18:09	18:37
35°	05:26	05:55	06:20	17:41	18:07	18:36
40°	05:26	05:58	06:25	17:37	18:04	18:35
45°	05:26	06:00	06:30	17:32	18:01	18:35
S 50°	05:26	06:03	06:36	17:25	17:58	18:35
52°	05:25	06:05	06:39	17:23	17:57	18:36
54°	05:25	06:06	06:42	17:20	17:55	18:36
56°	05:24	06:07	06:45	17:16	17:54	18:37
58°	05:23	06:09	06:49	17:12	17:52	18:37
S 60°	05:22	06:11	06:53	17:08	17:50	18:38

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	04:34	04:09	03:46	13:58	15:49	17:42
N 70°	04:23	04:06	03:49	14:06	15:50	17:34
68°	04:14	04:03	03:51	14:13	15:50	17:28
66°	04:07	04:00	03:53	14:19	15:50	17:23
64°	04:00	03:58	03:55	14:24	15:51	17:19
62°	03:55	03:56	03:57	14:28	15:51	17:15
60°	03:50	03:54	03:58	14:32	15:51	17:11
N 58°	03:45	03:52	03:59	14:35	15:51	17:09
56°	03:42	03:51	04:00	14:38	15:51	17:06
54°	03:38	03:50	04:01	14:40	15:52	17:04
52°	03:35	03:49	04:02	14:43	15:52	17:02
50°	03:32	03:48	04:03	14:45	15:52	17:00
45°	03:26	03:45	04:05	14:49	15:52	16:56
N 40°	03:20	03:44	04:06	14:53	15:52	16:52
35°	03:16	03:42	04:08	14:56	15:52	16:49
30°	03:12	03:40	04:09	14:59	15:53	16:47
20°	03:05	03:38	04:11	15:04	15:53	16:42
N 10°	02:59	03:36	04:13	15:08	15:53	16:38
0°	02:53	03:34	04:15	15:12	15:53	16:35
S 10°	02:47	03:32	04:16	15:16	15:53	16:31
20°	02:41	03:29	04:18	15:21	15:54	16:27
30°	02:34	03:27	04:20	15:25	15:54	16:22
35°	02:29	03:25	04:22	15:28	15:54	16:20
40°	02:25	03:24	04:23	15:31	15:54	16:17
45°	02:19	03:22	04:25	15:35	15:54	16:14
S 50°	02:13	03:19	04:27	15:39	15:54	16:10
52°	02:10	03:18	04:28	15:41	15:54	16:08
54°	02:06	03:17	04:29	15:43	15:54	16:06
56°	02:03	03:16	04:30	15:45	15:54	16:04
58°	01:58	03:14	04:31	15:48	15:54	16:01
S 60°	01:54	03:13	04:33	15:50	15:55	15:59

Day	Sun		Mer. Pass hh:mm	Moon		Age 26-28 19-6%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
10	01:27	01:19	12:01	09:03	21:23	
11	01:11	01:04	12:01	09:43	22:04	
12	00:56	00:48	12:01	10:24	22:45	

April 13, 14, 15 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	201°04.7	158°27.7	N16°21.4	72°01.1	N21°13.7	209°54.2	S04°53.5	20°23.9	N02°29.8	Alpheratz	357°46.3	29°08.8
1	216°07.1	173°27.1	22.4	87°02.8	13.5	224°56.1	53.3	35°26.5	29.9	Ankaa	353°18.3	-42°14.9
2	231°09.6	188°26.6	23.4	102°04.4	13.2	239°58.1	53.1	50°29.1	29.9	Schedar	349°43.9	56°35.6
3	246°12.0	203°26.0	•• 24.4	117°06.1	•• 12.9	255°00.0	•• 52.9	65°31.7	•• 30.0	Diphda	348°58.5	-17°55.8
4	261°14.5	218°25.4	25.4	132°07.8	12.7	270°02.0	52.7	80°34.4	30.1	Achernar	335°28.9	-57°11.0
5	276°17.0	233°24.9	26.4	147°09.5	12.4	285°03.9	52.5	95°37.0	30.1	Hamal	328°03.8	23°30.7
6	291°19.4	248°24.3	N16°27.4	162°11.2	N21°12.1	300°05.9	S04°52.3	110°39.6	N02°30.2	Polaris	319°21.7	89°18.7
7	306°21.9	263°23.8	28.4	177°12.8	11.9	315°07.8	52.1	125°42.2	30.3	Acamar	315°20.4	-40°15.9
8	321°24.4	278°23.2	29.4	192°14.5	11.6	330°09.8	51.8	140°44.8	30.3	Menkar	314°17.8	4°07.8
9	336°26.8	293°22.7	•• 30.4	207°16.2	•• 11.3	345°11.7	•• 51.6	155°47.5	•• 30.4	Mirfak	308°44.2	49°54.0
10	351°29.3	308°22.1	31.4	222°17.9	11.1	0°13.7	51.4	170°50.1	30.4	Aldebaran	290°52.3	16°31.8
11	6°31.8	323°21.5	32.4	237°19.5	10.8	15°15.7	51.2	185°52.7	30.5	Rigel	281°14.5	-8°11.5
12	21°34.2	338°21.0	N16°33.4	252°21.2	N21°10.5	30°17.6	S04°51.0	200°55.3	N02°30.6	Capella	280°38.2	46°00.6
13	36°36.7	353°20.4	34.4	267°22.9	10.3	45°19.6	50.8	215°58.0	30.6	Bellatrix	278°34.7	6°21.5
14	51°39.2	8°19.9	35.4	282°24.6	10.0	60°21.5	50.6	231°00.6	30.7	Elnath	278°15.8	28°37.0
15	66°41.6	23°19.3	•• 36.4	297°26.2	•• 09.7	75°23.5	•• 50.4	246°03.2	•• 30.8	Alnilam	275°48.9	-1°11.8
16	81°44.1	38°18.7	37.4	312°27.9	09.5	90°25.4	50.2	261°05.8	30.8	Betelgeuse	271°04.0	7°24.5
17	96°46.5	53°18.2	38.4	327°29.6	09.2	105°27.4	49.9	276°08.4	30.9	Canopus	263°57.4	-52°42.4
18	111°49.0	68°17.6	N16°39.3	342°31.2	N21°08.9	120°29.3	S04°49.7	291°11.1	N02°31.0	Sirius	258°35.9	-16°44.0
19	126°51.5	83°17.0	40.3	357°32.9	08.7	135°31.3	49.5	306°13.7	31.0	Adhara	255°14.4	-28°59.4
20	141°53.9	98°16.5	41.3	12°34.6	08.4	150°33.2	49.3	321°16.3	31.1	Procyon	245°02.2	5°11.8
21	156°56.4	113°15.9	•• 42.3	27°36.2	•• 08.1	165°35.2	•• 49.1	336°18.9	•• 31.2	Pollux	243°30.5	28°00.1
22	171°58.9	128°15.3	43.3	42°37.9	07.9	180°37.1	48.9	351°21.6	31.2	Avior	234°19.0	-59°32.9
23	187°01.3	143°14.8	44.3	57°39.6	07.6	195°39.1	48.7	6°24.2	31.3	Suhail	222°54.1	-43°28.8
Mer.pass. 10:34		ν -0.6' d 1.0' m -3.89		ν 1.7' d -0.3' m 0.41		ν 2.0' d -0.2' m -2.07		ν 2.6' d 0.1' m 0.61				

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	202°03.8	158°14.2	N16°45.3	72°41.2	N21°07.3	210°41.0	S04°48.5	21°26.8	N02°31.4
1	217°06.3	173°13.6	46.3	87°42.9	07.1	225°43.0	48.2	36°29.4	31.4
2	232°08.7	188°13.1	47.2	102°44.6	06.8	240°44.9	48.0	51°32.0	31.5
3	247°11.2	203°12.5	•• 48.2	117°46.2	•• 06.5	255°46.9	•• 47.8	66°34.7	•• 31.5
4	262°13.6	218°11.9	49.2	132°47.9	06.3	270°48.8	47.6	81°37.3	31.6
5	277°16.1	233°11.4	50.2	147°49.6	06.0	285°50.8	47.4	96°39.9	31.7
6	292°18.6	248°10.8	N16°51.2	162°51.2	N21°05.7	300°52.8	S04°47.2	111°42.5	N02°31.7
7	307°21.0	263°10.2	52.2	177°52.9	05.5	315°54.7	47.0	126°45.1	31.8
8	322°23.5	278°09.7	53.1	192°54.5	05.2	330°56.7	46.8	141°47.8	31.9
9	337°26.0	293°09.1	•• 54.1	207°56.2	•• 04.9	345°58.6	•• 46.6	156°50.4	•• 31.9
10	352°28.4	308°08.5	55.1	222°57.9	04.6	1°00.6	46.3	171°53.0	32.0
11	7°30.9	323°08.0	56.1	237°59.5	04.4	16°02.5	46.1	186°55.6	32.1
12	22°33.4	338°07.4	N16°57.1	253°01.2	N21°04.1	31°04.5	S04°45.9	201°58.2	N02°32.1
13	37°35.8	353°06.8	56.8	268°02.8	03.8	46°06.4	45.7	217°00.9	32.2
14	52°38.3	8°06.2	16°59.0	283°04.5	03.6	61°08.4	45.5	232°03.5	32.2
15	67°40.8	23°05.7	17°00.0	298°06.1	•• 03.3	76°10.3	•• 45.3	247°06.1	•• 32.3
16	82°43.2	38°05.1	01.0	313°07.8	03.0	91°12.3	45.1	262°08.7	32.4
17	97°45.7	53°04.5	01.9	328°09.4	02.8	106°14.2	44.9	277°11.3	32.4
18	112°48.1	68°03.9	N17°02.9	343°11.1	N21°02.5	121°16.2	S04°44.7	292°14.0	N02°32.5
19	127°50.6	83°03.4	03.9	358°12.8	02.2	136°18.2	44.4	307°16.6	32.6
20	142°53.1	98°02.8	04.9	13°14.4	01.9	151°20.1	44.2	322°19.2	32.6
21	157°55.5	113°02.2	•• 05.8	28°16.1	•• 01.7	166°22.1	•• 44.0	337°21.8	•• 32.7
22	172°58.0	128°01.6	06.8	43°17.7	01.4	181°24.0	43.8	352°24.4	32.8
23	188°00.5	143°01.1	07.8	58°19.4	01.1	196°26.0	43.6	7°27.1	32.8
Mer.pass. 10:30		ν -0.6' d 1.0' m -3.89		ν 1.7' d -0.3' m 0.43		ν 2.0' d -0.2' m -2.08		ν 2.6' d 0.1' m 0.62	

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	203°02.9	158°00.5	N17°08.7	73°21.0	N21°00.8	211°27.9	S04°43.4	22°29.7	N02°32.9
1	218°05.4	172°59.9	09.7	88°22.7	00.6	226°29.9	43.2	37°32.3	32.9
2	233°07.9	187°59.3	10.7	103°24.3	00.3	241°31.8	43.0	52°34.9	33.0
3	248°10.3	202°58.7	•• 11.6	118°26.0	21°00.0	256°33.8	•• 42.8	67°37.5	•• 33.1
4	263°12.8	217°58.2	12.6	133°27.6	20°59.8	271°35.8	42.5	82°40.2	33.1
5	278°15.3	232°57.6	13.6	148°29.3	59.5	286°37.7	42.3	97°42.8	33.2
6	293°17.7	247°57.0	N17°14.5	163°30.9	N20°59.2	301°39.7	S04°42.1	112°45.4	N02°33.3
7	308°20.2	262°56.4	15.5	178°32.5	58.9	316°41.6	41.9	127°48.0	33.3
8	323°22.6	277°55.9	16.5	193°34.2	58.7	331°43.6	41.7	142°50.6	33.4
9	338°25.1	292°55.3	•• 17.4	208°35.8	•• 58.4	346°45.5	•• 41.5	157°53.2	•• 33.5
10	353°27.6	307°54.7	18.4	223°37.5	58.1	1°47.5	41.3	172°55.9	33.5
11	8°30.0	322°54.1	19.3	238°39.1	57.8	16°49.4	41.1	187°58.5	33.6
12	23°32.5	337°53.5	N17°20.3	253°40.8	N20°57.6	31°51.4	S04°40.9	203°01.1	N02°33.6
13	38°35.0	352°52.9	21.3	268°42.4	57.3	46°53.4	40.6	218°03.7	33.7
14	53°37.4	7°52.4	22.2	283°44.0	57.0	61°55.3	40.4	233°06.3	33.8
15	68°39.9	22°51.8	•• 23.2	298°45.7	•• 56.7	76°57.3	•• 40.2	248°09.0	•• 33.8
16	83°42.4	37°51.2	24.1	313°47.3	56.5	91°59.2	40.0	263°11.6	33.9
17	98°44.8	52°50.6	25.1	328°49.0	56.2	107°01.2	39.8	278°14.2	34.0
18	113°47.3	67°50.0	N17°26.0	343°50.6	N20°55.9	122°03.1	S04°39.6	293°16.8	N02°34.0
19	128°49.7	82°49.4	27.0	358°52.2	55.6	137°05.1	39.4	308°19.4	34.1
20	143°52.2	97°48.8	28.0	13°53.9	55.4	152°07.1	39.2	323°22.0	34.1
21	158°54.7	112°48.3	•• 28.9	28°55.5	•• 55.1	167°09.0	•• 39.0	338°24.7	•• 34.2
22	173°57.1	127°47.7	29.9	43°57.2	54.8	182°11.0	38.8	353°27.3	34.3
23	188°59.6	142°47.1	30.8	58°58.8	54.5	197°12.9	38.5	8°29.9	34.3
Mer.pass. 10:26		ν -0.6' d 1.0' m -3.89		ν 1.6' d -0.3' m 0.45		ν 2.0' d -0.2' m -2.08		ν 2.6' d 0.1' m 0.62	

Apr 13 Tue	SHA	Mer.pass
Venus	317°23.0	13:27
Mars	230°56.4	19:10
Jupiter	8°49.5	09:59
Saturn	179°19.2	22:34
Apr 14 Wed	SHA	Mer.pass
Venus	316°10.4	13:28
Mars	230°37.4	19:07
Jupiter	8°37.2	09:56
Saturn	179°23.0	22:30
Apr 15 Thu	SHA	Mer.pass
Venus	314°57.6	13:28
Mars	230°18.1	19:05
Jupiter	8°25.0	09:53
Saturn	179°26.7	22:26
Horizontal parallax		
Venus:	0.1	
Mars:	0.1	

h	Sun			Moon			
Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	179°49.9	N08°55.7	198°05.6	15.2'	N06°39.3	12.7'	55.0'
1	194°50.1	56.6	212°39.8	15.1'	06°52.0	12.7'	55.1'
2	209°50.2	57.5	227°13.9	15.1'	07°04.8	12.7'	55.1'
3	224°50.4	· · 58.4	241°48.0	15.0'	07°17.4	12.7'	55.1'
4	239°50.6	08°59.3	256°22.1	15.0'	07°30.1	12.7'	55.1'
5	254°50.7	09°00.2	270°56.0	14.9'	07°42.8	12.6'	55.1'
6	269°50.9	N09°01.1	285°30.0	14.9'	N07°55.4	12.6'	55.1'
7	284°51.0	02.1	300°03.9	14.8'	08°08.0	12.6'	55.2'
8	299°51.2	03.0	314°37.7	14.8'	08°20.6	12.6'	55.2'
9	314°51.4	· · 03.9	329°11.5	14.7'	08°33.1	12.5'	55.2'
10	329°51.5	04.8	343°45.2	14.7'	08°45.7	12.5'	55.2'
11	344°51.7	05.7	358°18.9	14.6'	08°58.2	12.5'	55.2'
12	359°51.8	N09°06.6	12°52.5	14.6'	N09°10.6	12.4'	55.3'
13	14°52.0	07.5	27°26.1	14.5'	09°23.1	12.4'	55.3'
14	29°52.1	08.4	41°59.6	14.5'	09°35.5	12.4'	55.3'
15	44°52.3	· · 09.3	56°33.1	14.4'	09°47.9	12.3'	55.3'
16	59°52.5	10.2	71°06.5	14.3'	10°00.2	12.3'	55.3'
17	74°52.6	11.1	85°39.8	14.3'	10°12.5	12.3'	55.4'
18	89°52.8	N09°12.0	100°13.1	14.2'	N10°24.8	12.2'	55.4'
19	104°52.9	12.9	114°46.3	14.2'	10°37.0	12.2'	55.4'
20	119°53.1	13.8	129°19.5	14.1'	10°49.3	12.2'	55.4'
21	134°53.2	· · 14.7	143°52.6	14.0'	11°01.4	12.1'	55.4'
22	149°53.4	15.6	158°25.6	14.0'	11°13.6	12.1'	55.5'
23	164°53.5	16.5	172°58.6	13.9'	11°25.6	12.0'	55.5'
SD = 15.9'			SD = 15.0'				

Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	179°53.7	N09°17.4	187°31.5	13.8'	N11°37.7	12.0'	55.5'
1	194°53.9	18.3	202°04.4	13.8'	11°49.7	12.0'	55.5'
2	209°54.0	19.2	216°37.2	13.7'	12°01.7	11.9'	55.5'
3	224°54.2	· · 20.1	231°09.9	13.7'	12°13.6	11.9'	55.6'
4	239°54.3	21.0	245°42.5	13.6'	12°25.5	11.8'	55.6'
5	254°54.5	22.0	260°15.1	13.5'	12°37.3	11.8'	55.6'
6	269°54.6	N09°22.9	274°47.6	13.4'	N12°49.1	11.7'	55.6'
7	284°54.8	23.8	289°20.1	13.4'	13°00.8	11.7'	55.6'
8	299°54.9	24.7	303°52.4	13.3'	13°12.5	11.6'	55.7'
9	314°55.1	· · 25.6	318°24.7	13.2'	13°24.1	11.6'	55.7'
10	329°55.2	26.5	332°57.0	13.2'	13°35.7	11.5'	55.7'
11	344°55.4	27.4	347°29.1	13.1'	13°47.2	11.5'	55.7'
12	359°55.6	N09°28.3	2°01.2	13.0'	N13°58.7	11.4'	55.7'
13	14°55.7	29.2	16°33.2	12.9'	14°10.1	11.4'	55.8'
14	29°55.9	30.1	31°05.2	12.9'	14°21.5	11.3'	55.8'
15	44°56.0	· · 31.0	45°37.1	12.8'	14°32.8	11.2'	55.8'
16	59°56.2	31.8	60°08.9	12.7'	14°44.1	11.2'	55.8'
17	74°56.3	32.7	74°40.6	12.6'	14°55.2	11.1'	55.8'
18	89°56.5	N09°33.6	89°12.2	12.6'	N15°06.4	11.1'	55.9'
19	104°56.6	34.5	103°43.8	12.5'	15°17.4	11.0'	55.9'
20	119°56.8	35.4	118°15.3	12.4'	15°28.4	10.9'	55.9'
21	134°56.9	· · 36.3	132°46.7	12.3'	15°39.4	10.9'	55.9'
22	149°57.1	37.2	147°18.0	12.3'	15°50.3	10.8'	55.9'
23	164°57.2	38.1	161°49.3	12.2'	16°01.1	10.7'	56.0'
SD = 15.9'			SD = 15.1'				

Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	179°57.4	N09°39.0	176°20.4	12.1'	N16°11.8	10.7'	56.0'
1	194°57.5	39.9	190°51.5	12.0'	16°22.5	10.6'	56.0'
2	209°57.7	40.8	205°22.6	11.9'	16°33.1	10.5'	56.0'
3	224°57.8	· · 41.7	219°53.5	11.9'	16°43.6	10.5'	56.0'
4	239°58.0	42.6	234°24.3	11.8'	16°54.1	10.4'	56.1'
5	254°58.1	43.5	248°55.1	11.7'	17°04.4	10.3'	56.1'
6	269°58.3	N09°44.4	263°25.8	11.6'	N17°14.7	10.2'	56.1'
7	284°58.4	45.3	277°56.4	11.5'	17°25.0	10.2'	56.1'
8	299°58.6	46.2	292°26.9	11.4'	17°35.1	10.1'	56.2'
9	314°58.8	· · 47.1	306°57.4	11.4'	17°45.2	10.0'	56.2'
10	329°58.9	48.0	321°27.7	11.3'	17°55.2	9.9'	56.2'
11	344°59.1	48.9	335°58.0	11.2'	18°05.1	9.8'	56.2'
12	359°59.2	N09°49.8	350°28.2	11.1'	N18°14.9	9.7'	56.2'
13	14°59.4	50.7	4°58.3	11.0'	18°24.7	9.7'	56.3'
14	29°59.5	51.5	19°28.3	10.9'	18°34.3	9.6'	56.3'
15	44°59.7	· · 52.4	33°58.3	10.9'	18°43.9	9.5'	56.3'
16	59°59.8	53.3	48°28.1	10.8'	18°53.4	9.4'	56.3'
17	75°00.0	54.2	62°57.9	10.7'	19°02.8	9.3'	56.3'
18	90°00.1	N09°55.1	77°27.6	10.6'	N19°12.1	9.2'	56.4'
19	105°00.2	56.0	91°57.2	10.5'	19°21.3	9.1'	56.4'
20	120°00.4	56.9	106°26.7	10.4'	19°30.5	9.0'	56.4'
21	135°00.5	· · 57.8	120°56.2	10.3'	19°39.5	8.9'	56.4'
22	150°00.7	58.7	135°25.5	10.3'	19°48.4	8.8'	56.4'
23	165°00.8	59.6	149°54.8	10.2'	19°57.3	8.7'	56.5'
SD = 15.9'			SD = 15.3'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	////	02:08	03:46	20:19	22:00	////
N 70°	////	02:42	04:02	20:02	21:23	////
68°	00:59	03:06	04:14	19:49	20:58	23:18
66°	01:51	03:24	04:25	19:38	20:40	22:16
64°	02:21	03:39	04:33	19:29	20:24	21:44
62°	02:43	03:51	04:41	19:22	20:12	21:21
60°	03:01	04:01	04:47	19:15	20:01	21:03
N 58°	03:15	04:10	04:53	19:10	19:52	20:48
56°	03:27	04:18	04:58	19:04	19:45	20:36
54°	03:37	04:24	05:02	19:00	19:38	20:25
52°	03:46	04:30	05:06	18:56	19:32	20:16
50°	03:54	04:36	05:10	18:52	19:26	20:08
45°	04:10	04:47	05:18	18:44	19:15	19:52
N 40°	04:23	04:56	05:24	18:37	19:05	19:39
35°	04:33	05:04	05:30	18:31	18:58	19:28
30°	04:41	05:10	05:35	18:26	18:51	19:20
20°	04:54	05:21	05:43	18:18	18:40	19:07
N 10°	05:04	05:29	05:50	18:10	18:32	18:57
0°	05:12	05:36	05:57	18:04	18:25	18:49
S 10°	05:18	05:42	06:04	17:57	18:18	18:43
20°	05:23	05:48	06:11	17:50	18:12	18:38
30°	05:26	05:54	06:18	17:42	18:06	18:34
35°	05:28	05:57	06:23	17:37	18:03	18:32
40°	05:29	06:00	06:28	17:32	18:00	18:31
45°	05:30	06:04	06:34	17:26	17:56	18:30
S 50°	05:30	06:08	06:40	17:19	17:52	18:29
52°	05:30	06:09	06:44	17:16	17:50	18:29
54°	05:30	06:11	06:47	17:13	17:48	18:29
56°	05:30	06:13	06:51	17:09	17:46	18:29
58°	05:30	06:15	06:55	17:04	17:44	18:30
S 60°	05:29	06:17	07:00	17:00	17:42	18:30

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	03:21	02:50	01:58	19:44	22:12	☐
N 70°	03:31	03:11	02:41	19:25	21:31	☐
68°	03:40	03:27	03:10	19:11	21:03	23:15
66°	03:47	03:40	03:32	18:59	20:42	22:35
64°	03:53	03:51	03:50	18:50	20:26	22:08
62°	03:58	04:00	04:04	18:41	20:12	21:47
60°	04:03	04:08	04:17	18:34	20:01	21:30
N 58°	04:07	04:16	04:27	18:28	19:51	21:15
56°	04:10	04:22	04:37	18:23	19:42	21:03
54°	04:14	04:28	04:45	18:18	19:34	20:53
52°	04:17	04:33	04:52	18:13	19:28	20:43
50°	04:19	04:37	04:59	18:10	19:21	20:35
45°	04:25	04:48	05:14	18:01	19:08	20:17
N 40°	04:30	04:56	05:26	17:54	18:57	20:03
35°	04:35	05:04	05:36	17:48	18:48	19:51
30°	04:38	05:10	05:45	17:42	18:40	19:40
20°	04:45	05:21	06:01	17:33	18:26	19:22
N 10°	04:51	05:31	06:15	17:25	18:14	19:06
0°	04:57	05:41	06:28	17:17	18:03	18:51
S 10°	05:02	05:50	06:41	17:10	17:52	18:37
20°	05:08	06:00	06:55	17:02	17:40	18:21
30°	05:15	06:12	07:11	16:53	17:26	18:03
35°	05:19	06:19	07:20	16:48	17:18	17:53
40°	05:24	06:27	07:31	16:42	17:09	17:41
45°	05:29	06:36	07:44	16:35	16:59	17:27
S 50°	05:36	06:47	08:00	16:27	16:46	17:10
52°	05:39	06:52	08:07	16:23	16:40	17:02
54°	05:42	06:58	08:15	16:19	16:34	16:54
56°	05:46	07:04	08:25	16:14	16:27	16:44
58°	05:50	07:11	08:35	16:09	16:19	16:33
S 60°	05:54	07:19	08:47	16:04	16:10	16:20

Day	Sun		Mer. Pass hh:mm	Moon		Age 29-1 3-0%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
13	00:40	00:33	12:01	11:07	23:29	
14	00:25	00:18	12:00	11:52	—:—	
15	00:10	00:03	12:00	12:39	00:15	

April 16, 17, 18 UT (Fri., Sat., Sun.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	204°02.1	157°46.5	N17°31.8	74°00.4	N20°54.2	212°14.9	S04°38.3	23°32.5	N02°34.4	Alpheratz	357°46.3	29°08.8					
1	219°04.5	172°45.9	32.7	89°02.1	54.0	227°16.8	38.1	38°35.1	34.5	Ankaa	353°18.3	-42°14.9					
2	234°07.0	187°45.3	33.7	104°03.7	53.7	242°18.8	37.9	53°37.7	34.5	Schedar	349°43.9	56°35.6					
3	249°09.5	202°44.7	•• 34.6	119°05.3	•• 53.4	257°20.8	•• 37.7	68°40.4	•• 34.6	Diphda	348°58.5	-17°55.8					
4	264°11.9	217°44.1	35.6	134°07.0	53.1	272°22.7	37.5	83°43.0	34.6	Achernar	335°28.9	-57°11.0					
5	279°14.4	232°43.5	36.5	149°08.6	52.9	287°24.7	37.3	98°45.6	34.7	Hamal	328°03.8	23°30.6					
6	294°16.9	247°43.0	N17°37.4	164°10.2	N20°52.6	302°26.6	S04°37.1	113°48.2	N02°34.8	Polaris	319°21.8	89°18.7					
7	309°19.3	262°42.4	38.4	179°11.9	52.3	317°28.6	36.9	128°50.8	34.8	Acamar	315°20.4	-40°15.9					
8	324°21.8	277°41.8	39.3	194°13.5	52.0	332°30.5	36.7	143°53.4	34.9	Menkar	314°17.8	4°07.8					
9	339°24.2	292°41.2	•• 40.3	209°15.1	•• 51.7	347°32.5	•• 36.4	158°56.1	•• 34.9	Mirfak	308°44.2	49°53.9					
10	354°26.7	307°40.6	41.2	224°16.7	51.5	2°34.5	36.2	173°58.7	35.0	Aldebaran	290°52.3	16°31.8					
11	9°29.2	322°40.0	42.2	239°18.4	51.2	17°36.4	36.0	189°01.3	35.1	Rigel	281°14.5	-8°11.5					
12	24°31.6	337°39.4	N17°43.1	254°20.0	N20°50.9	32°38.4	S04°35.8	204°03.9	N02°35.1	Capella	280°38.2	46°00.6					
13	39°34.1	352°38.8	44.0	269°21.6	50.6	47°40.3	35.6	219°06.5	35.2	Bellatrix	278°34.7	6°21.5					
14	54°36.6	7°38.2	45.0	284°23.3	50.3	62°42.3	35.4	234°09.1	35.3	Elnath	278°15.8	28°37.0					
15	69°39.0	22°37.6	•• 45.9	299°24.9	•• 50.1	77°44.3	•• 35.2	249°11.8	•• 35.3	Alnilam	275°48.9	-1°11.8					
16	84°41.5	37°37.0	46.9	314°26.5	49.8	92°46.2	35.0	264°14.4	35.4	Betelgeuse	271°04.0	7°24.5					
17	99°44.0	52°36.4	47.8	329°28.1	49.5	107°48.2	34.8	279°17.0	35.4	Canopus	263°57.4	-52°42.4					
18	114°46.4	67°35.8	N17°48.7	344°29.8	N20°49.2	122°50.1	S04°34.6	294°19.6	N02°35.5	Sirius	258°35.9	-16°44.0					
19	129°48.9	82°35.2	49.7	359°31.4	48.9	137°52.1	34.4	309°22.2	35.6	Adhara	255°14.5	-28°59.4					
20	144°51.4	97°34.6	50.6	14°33.0	48.7	152°54.1	34.1	324°24.8	35.6	Procyon	245°02.2	5°11.8					
21	159°53.8	112°34.0	•• 51.5	29°34.6	•• 48.4	167°56.0	•• 33.9	339°27.4	•• 35.7	Pollux	243°30.6	28°00.1					
22	174°56.3	127°33.4	52.5	44°36.2	48.1	182°58.0	33.7	354°30.1	35.7	Avior	234°19.0	-59°32.9					
23	189°58.7	142°32.8	53.4	59°37.9	47.8	197°59.9	33.5	9°32.7	35.8	Suhail	222°54.1	-43°28.8					
Mer.pass. 10:22			ν -0.6' d 0.9' m -3.89			ν 1.6' d -0.3' m 0.47			ν 2.0' d -0.2' m -2.08			ν 2.6' d 0.1' m 0.63					

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	205°01.2	157°32.2	N17°54.3	74°39.5	N20°47.5	213°01.9	S04°33.3	24°35.3	N02°35.9					
1	220°03.7	172°31.6	55.3	89°41.1	47.2	228°03.9	33.1	39°37.9	35.9					
2	235°06.1	187°31.0	56.2	104°42.7	47.0	243°05.8	32.9	54°40.5	36.0					
3	250°08.6	202°30.4	•• 57.1	119°44.3	•• 46.7	258°07.8	•• 32.7	69°43.1	•• 36.0					
4	265°11.1	217°29.8	58.1	134°46.0	46.4	273°09.7	32.5	84°45.7	36.1					
5	280°13.5	232°29.2	59.0	149°47.6	46.1	288°11.7	32.3	99°48.4	36.2					
6	295°16.0	247°28.6	N17°59.9	164°49.2	N20°45.8	303°13.7	S04°32.1	114°51.0	N02°36.2					
7	310°18.5	262°28.0	18°00.8	179°50.8	45.5	318°15.6	31.8	129°53.6	36.3					
8	325°20.9	277°27.4	01.8	194°52.4	45.3	333°17.6	31.6	144°56.2	36.3					
9	340°23.4	292°26.8	•• 02.7	209°54.0	•• 45.0	348°19.5	•• 31.4	159°58.8	•• 36.4					
10	355°25.8	307°26.2	03.6	224°55.7	44.7	3°21.5	31.2	175°01.4	36.5					
11	10°28.3	322°25.6	04.5	239°57.3	44.4	18°23.5	31.0	190°04.0	36.5					
12	25°30.8	337°25.0	N18°05.5	254°58.9	N20°44.1	33°25.4	S04°30.8	205°06.7	N02°36.6					
13	40°33.2	352°24.4	06.4	270°00.5	43.8	48°27.4	30.6	220°09.3	36.7					
14	55°35.7	7°23.8	07.3	285°02.1	43.6	63°29.3	30.4	235°11.9	36.7					
15	70°38.2	22°23.2	•• 08.2	300°03.7	•• 43.3	78°31.3	•• 30.2	250°14.5	•• 36.8					
16	85°40.6	37°22.6	09.1	315°05.3	43.0	93°33.3	30.0	265°17.1	36.8					
17	100°43.1	52°22.0	10.1	330°06.9	42.7	108°35.2	29.8	280°19.7	36.9					
18	115°45.6	67°21.4	N18°11.0	345°08.6	N20°42.4	123°37.2	S04°29.6	295°22.3	N02°37.0					
19	130°48.0	82°20.8	11.9	0°10.2	42.1	138°39.2	29.3	310°25.0	37.0					
20	145°50.5	97°20.2	12.8	15°11.8	41.8	153°41.1	29.1	325°27.6	37.1					
21	160°53.0	112°19.6	•• 13.7	30°13.4	•• 41.6	168°43.1	•• 28.9	340°30.2	•• 37.1					
22	175°55.4	127°18.9	14.6	45°15.0	41.3	183°45.0	28.7	355°32.8	37.2					
23	190°57.9	142°18.3	15.5	60°16.6	41.0	198°47.0	28.5	10°35.4	37.3					
Mer.pass. 10:18			ν -0.6' d 0.9' m -3.89			ν 1.6' d -0.3' m 0.48			ν 2.0' d -0.2' m -2.08			ν 2.6' d 0.1' m 0.63		

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	206°00.3	157°17.7	N18°16.5	75°18.2	N20°40.7	213°49.0	S04°28.3	25°38.0	N02°37.3					
1	221°02.8	172°17.1	17.4	90°19.8	40.4	228°50.9	28.1	40°40.6	37.4					
2	236°05.3	187°16.5	18.3	105°21.4	40.1	243°52.9	27.9	55°43.2	37.4					
3	251°07.7	202°15.9	•• 19.2	120°23.0	•• 39.8	258°54.9	•• 27.7	70°45.9	•• 37.5					
4	266°10.2	217°15.3	20.1	135°24.6	39.6	273°56.8	27.5	85°48.5	37.5					
5	281°12.7	232°14.7	21.0	150°26.2	39.3	288°58.8	27.3	100°51.1	37.6					
6	296°15.1	247°14.1	N18°21.9	165°27.8	N20°39.0	304°00.7	S04°27.1	115°53.7	N02°37.7					
7	311°17.6	262°13.4	22.8	180°29.4	38.7	319°02.7	26.8	130°56.3	37.7					
8	326°20.1	277°12.8	23.7	195°31.0	38.4	334°04.7	26.6	145°58.9	37.8					
9	341°22.5	292°12.2	•• 24.6	210°32.6	•• 38.1	349°06.6	•• 26.4	161°01.5	•• 37.8					
10	356°25.0	307°11.6	25.5	225°34.2	37.8	4°08.6	26.2	176°04.1	37.9					
11	11°27.5	322°11.0	26.4	240°35.8	37.5	19°10.6	26.0	191°06.7	38.0					
12	26°29.9	337°10.4	N18°27.3	255°37.4	N20°37.2	34°12.5	S04°25.8	206°09.4	N02°38.0					
13	41°32.4	352°09.7	28.2	270°39.0	37.0	49°14.5	25.6	221°12.0	38.1					
14	56°34.8	7°09.1	29.1	285°40.6	36.7	64°16.5	25.4	236°14.6	38.1					
15	71°37.3	22°08.5	•• 30.0	300°42.2	•• 36.4	79°18.4	•• 25.2	251°17.2	•• 38.2					
16	86°39.8	37°07.9	30.9	315°43.8	36.1	94°20.4	25.0	266°19.8	38.3					
17	101°42.2	52°07.3	31.8	330°45.4	35.8	109°22.3	24.8	281°22.4	38.3					
18	116°44.7	67°06.7	N18°32.7	345°47.0	N20°35.5	124°24.3	S04°24.6	296°25.0	N02°38.4					
19	131°47.2	82°06.0	33.6	0°48.6	35.2	139°26.3	24.4	311°27.6	38.4					
20	146°49.6	97°05.4	34.5	15°50.2	34.9	154°28.2	24.2	326°30.2	38.5					
21	161°52.1	112°04.8	•• 35.4	30°51.8	•• 34.6	169°30.2	•• 23.9	341°32.9	•• 38.6					
22	176°54.6	127°04.2	36.3	45°53.4	34.3	184°32.2	23.7	356°35.5	38.6					
23	191°57.0	142°03.6	37.2	60°55.0	34.1	199°34.1	23.5	11°38.1	38.7					
Mer.pass. 10:14			ν -0.6' d 0.9' m -3.89			ν 1.6' d -0.3' m 0.50			ν 2.0' d -0.2' m -2.09			ν 2.6' d 0.1' m 0.64		

Apr 16 Fri	SHA	Mer.pass
Venus	313°44.4	13:29
Mars	229°58.4	19:02
Jupiter	8°12.8	09:50
Saturn	179°30.4	22:22

Apr 17 Sat	SHA	Mer.pass
Venus	312°31.0	13:30
Mars	229°38.3	18:59
Jupiter	8°00.7	09:47
Saturn	179°34.1	22:18

Apr 18 Sun	SHA	Mer.pass
Venus	311°17.4	13:31
Mars	229°17.8	18:57
Jupiter	7°48.6	09:43
Saturn	179°37.7	22:14

Horizontal parallax">		
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h	Sun			Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	180°01.0	N10°00.5	164°23.9	10.1'	N20°06.0	8.6'	56.5'
1	195°01.1	01.3	178°53.0	10.0'	20°14.7	8.5'	56.5'
2	210°01.3	02.2	193°22.0	9.9'	20°23.2	8.4'	56.5'
3	225°01.4	· · 03.1	207°51.0	9.8'	20°31.6	8.3'	56.5'
4	240°01.6	04.0	222°19.8	9.8'	20°40.0	8.2'	56.6'
5	255°01.7	04.9	236°48.5	9.7'	20°48.2	8.1'	56.6'
6	270°01.9	N10°05.8	251°17.2	9.6'	N20°56.4	8.0'	56.6'
7	285°02.0	06.7	265°45.8	9.5'	21°04.4	7.9'	56.6'
8	300°02.2	07.6	280°14.3	9.4'	21°12.3	7.8'	56.7'
9	315°02.3	· · 08.4	294°42.7	9.3'	21°20.1	7.7'	56.7'
10	330°02.5	09.3	309°11.0	9.2'	21°27.8	7.6'	56.7'
11	345°02.6	10.2	323°39.3	9.2'	21°35.4	7.5'	56.7'
12	0°02.8	N10°11.1	338°07.5	9.1'	N21°42.9	7.4'	56.7'
13	15°02.9	12.0	352°35.5	9.0'	21°50.2	7.3'	56.8'
14	30°03.1	12.9	7°03.5	8.9'	21°57.5	7.1'	56.8'
15	45°03.2	· · 13.8	21°31.5	8.8'	22°04.6	7.0'	56.8'
16	60°03.3	14.6	35°59.3	8.8'	22°11.7	6.9'	56.8'
17	75°03.5	15.5	50°27.1	8.7'	22°18.6	6.8'	56.8'
18	90°03.6	N10°16.4	64°54.7	8.6'	N22°25.3	6.7'	56.9'
19	105°03.8	17.3	79°22.3	8.5'	22°32.0	6.5'	56.9'
20	120°03.9	18.2	93°49.9	8.4'	22°38.5	6.4'	56.9'
21	135°04.1	· · 19.1	108°17.3	8.4'	22°44.9	6.3'	56.9'
22	150°04.2	19.9	122°44.7	8.3'	22°51.2	6.2'	56.9'
23	165°04.4	20.8	137°11.9	8.2'	22°57.4	6.0'	57.0'
SD = 15.9'			SD = 15.4'				
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	180°04.5	N10°21.7	151°39.2	8.1'	N23°03.5	5.9'	57.0'
1	195°04.6	22.6	166°06.3	8.1'	23°09.4	5.8'	57.0'
2	210°04.8	23.5	180°33.4	8.0'	23°15.2	5.7'	57.0'
3	225°04.9	· · 24.4	195°00.3	7.9'	23°20.8	5.5'	57.0'
4	240°05.1	25.2	209°27.3	7.8'	23°26.3	5.4'	57.1'
5	255°05.2	26.1	223°54.1	7.8'	23°31.7	5.3'	57.1'
6	270°05.4	N10°27.0	238°20.9	7.7'	N23°37.0	5.1'	57.1'
7	285°05.5	27.9	252°47.6	7.6'	23°42.1	5.0'	57.1'
8	300°05.7	28.8	267°14.2	7.6'	23°47.1	4.9'	57.1'
9	315°05.8	· · 29.6	281°40.8	7.5'	23°52.0	4.7'	57.2'
10	330°05.9	30.5	296°07.3	7.4'	23°56.7	4.6'	57.2'
11	345°06.1	31.4	310°33.7	7.4'	24°01.3	4.4'	57.2'
12	0°06.2	N10°32.3	325°00.1	7.3'	N24°05.7	4.3'	57.2'
13	15°06.4	33.2	339°26.4	7.2'	24°10.1	4.2'	57.2'
14	30°06.5	34.0	353°52.6	7.2'	24°14.2	4.0'	57.3'
15	45°06.7	· · 34.9	8°18.8	7.1'	24°18.3	3.9'	57.3'
16	60°06.8	35.8	22°44.9	7.1'	24°22.1	3.7'	57.3'
17	75°06.9	36.7	37°11.0	7.0'	24°25.9	3.6'	57.3'
18	90°07.1	N10°37.5	51°37.0	6.9'	N24°29.5	3.5'	57.3'
19	105°07.2	38.4	66°02.9	6.9'	24°32.9	3.3'	57.4'
20	120°07.4	39.3	80°28.8	6.8'	24°36.3	3.2'	57.4'
21	135°07.5	· · 40.2	94°54.6	6.8'	24°39.4	3.0'	57.4'
22	150°07.6	41.1	109°20.4	6.7'	24°42.4	2.9'	57.4'
23	165°07.8	41.9	123°46.1	6.7'	24°45.3	2.7'	57.4'
SD = 15.9'			SD = 15.5'				
Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	180°07.9	N10°42.8	138°11.8	6.6'	N24°48.0	2.6'	57.5'
1	195°08.1	43.7	152°37.4	6.6'	24°50.6	2.4'	57.5'
2	210°08.2	44.6	167°03.0	6.5'	24°53.0	2.3'	57.5'
3	225°08.3	· · 45.4	181°28.5	6.5'	24°55.3	2.1'	57.5'
4	240°08.5	46.3	195°54.0	6.4'	24°57.4	2.0'	57.5'
5	255°08.6	47.2	210°19.5	6.4'	24°59.4	1.8'	57.6'
6	270°08.8	N10°48.0	224°44.9	6.4'	N25°01.2	1.7'	57.6'
7	285°08.9	48.9	239°10.3	6.3'	25°02.9	1.5'	57.6'
8	300°09.0	49.8	253°35.6	6.3'	25°04.4	1.4'	57.6'
9	315°09.2	· · 50.7	268°00.9	6.3'	25°05.8	1.2'	57.6'
10	330°09.3	51.5	282°26.1	6.2'	25°07.0	1.1'	57.7'
11	345°09.5	52.4	296°51.4	6.2'	25°08.0	0.9'	57.7'
12	0°09.6	N10°53.3	311°16.6	6.2'	N25°08.9	0.7'	57.7'
13	15°09.7	54.1	325°41.7	6.1'	25°09.7	0.6'	57.7'
14	30°09.9	55.0	340°06.9	6.1'	25°10.3	0.4'	57.7'
15	45°10.0	· · 55.9	354°32.0	6.1'	25°10.7	0.3'	57.8'
16	60°10.2	56.8	8°57.0	6.1'	25°11.0	0.1'	57.8'
17	75°10.3	57.6	23°22.1	6.0'	25°11.1	-0.0'	57.8'
18	90°10.4	N10°58.5	37°47.1	6.0'	N25°11.1	-0.2'	57.8'
19	105°10.6	10°59.4	52°12.1	6.0'	25°10.9	-0.4'	57.8'
20	120°10.7	11°00.2	66°37.1	6.0'	25°10.5	-0.5'	57.9'
21	135°10.8	· · 01.1	81°02.1	6.0'	25°10.0	-0.7'	57.9'
22	150°11.0	02.0	95°27.1	5.9'	25°09.3	-0.8'	57.9'
23	165°11.1	02.8	109°52.0	5.9'	25°08.5	-1.0'	57.9'
SD = 15.9'			SD = 15.7'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	////	01:39	03:28	20:35	22:30	////
N 70°	////	02:21	03:47	20:16	21:43	////
68°	////	02:49	04:01	20:01	21:14	////
66°	01:25	03:10	04:13	19:49	20:52	22:43
64°	02:03	03:27	04:23	19:39	20:35	22:01
62°	02:29	03:40	04:31	19:30	20:21	21:34
60°	02:49	03:51	04:38	19:23	20:10	21:14
N 58°	03:04	04:01	04:44	19:16	20:00	20:57
56°	03:17	04:09	04:50	19:11	19:51	20:44
54°	03:29	04:17	04:55	19:05	19:44	20:32
52°	03:38	04:23	04:59	19:01	19:37	20:23
50°	03:47	04:29	05:04	18:57	19:31	20:14
45°	04:04	04:42	05:12	18:48	19:19	19:56
N 40°	04:18	04:51	05:20	18:40	19:08	19:42
35°	04:29	05:00	05:26	18:34	19:00	19:31
30°	04:38	05:07	05:31	18:28	18:53	19:22
20°	04:52	05:18	05:41	18:19	18:41	19:08
N 10°	05:02	05:27	05:49	18:11	18:32	18:57
0°	05:11	05:35	05:56	18:03	18:24	18:48
S 10°	05:18	05:42	06:04	17:55	18:17	18:41
20°	05:23	05:49	06:11	17:47	18:10	18:36
30°	05:28	05:56	06:20	17:39	18:03	18:31
35°	05:30	06:00	06:25	17:34	17:59	18:29
40°	05:32	06:03	06:31	17:28	17:55	18:27
45°	05:33	06:07	06:37	17:21	17:51	18:25
S 50°	05:35	06:12	06:45	17:13	17:46	18:24
52°	05:35	06:14	06:49	17:10	17:44	18:23
54°	05:35	06:16	06:53	17:06	17:42	18:23
56°	05:36	06:19	06:57	17:01	17:39	18:22
58°	05:36	06:21	07:02	16:56	17:37	18:22
S 60°	05:36	06:24	07:07	16:51	17:34	18:22

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	□	□	□	□	□	□
N 70°	□	□	□	□	□	□
68°	02:41	□	□	□	□	□
66°	03:22	03:02	□	· ·	00:45	□
64°	03:50	03:54	04:10	23:54	· ·	01:34
62°	04:12	04:26	04:56	23:22	· ·	00:49
60°	04:29	04:50	05:26	22:58	· ·	00:19
N 58°	04:44	05:09	05:49	22:39	23:56	· ·
56°	04:57	05:25	06:08	22:24	23:38	· ·
54°	05:08	05:39	06:23	22:10	23:22	· ·
52°	05:18	05:51	06:37	21:59	23:09	· ·
50°	05:27	06:02	06:49	21:48	22:57	23:58
45°	05:45	06:24	07:13	21:26	22:33	23:33
N 40°	06:01	06:43	07:33	21:09	22:13	23:13
35°	06:14	06:58	07:49	20:54	21:57	22:56
30°	06:25	07:11	08:04	20:41	21:43	22:42
20°	06:45	07:34	08:28	20:20	21:19	22:18
N 10°	07:02	07:54	08:49	20:01	20:58	21:56
0°	07:18	08:12	09:09	19:44	20:39	21:37
S 10°	07:34	08:31	09:29	19:26	20:20	21:17
20°	07:52	08:51	09:50	19:07	19:59	20:56
30°	08:12	09:14	10:14	18:46	19:35	20:31
35°	08:24	09:27	10:29	18:33	19:21	20:17
40°	08:37	09:43	10:46	18:19	19:05	20:00
45°	08:54	10:02	11:06	18:02	18:46	19:40
S 50°	09:14	10:26	11:31	17:41	18:22	19:14
52°	09:23	10:37	11:43	17:31	18:10	19:02
54°	09:34	10:50	11:57	17:20	17:57	18:48
56°	09:46	11:05	12:13	17:07	17:42	18:32
58°	10:01	11:22	12:33	16:53	17:24	18:12
S 60°	10:17	11:44	12:57	16:35	17:02	17:48

Day	Sun		Mer. Pass	Moon		Age 2-4 2-13%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Upper hh:mm	Lower hh:mm	
16	00:04	00:11	12:00	13:31	01:05	
17	00:18	00:25	12:00	14:25	01:58	
18	00:32	00:38	11:59	15:23	02:54	

h	Sun			Moon			
Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	180°11.2	N11°03.7	124°17.0	5.9'	N25°07.5	-1.1'	57.9'
1	195°11.4	04.6	138°41.9	5.9'	25°06.4	-1.3'	57.9'
2	210°11.5	05.4	153°06.8	5.9'	25°05.1	-1.5'	58.0'
3	225°11.7	· · 06.3	167°31.7	5.9'	25°03.6	-1.6'	58.0'
4	240°11.8	07.2	181°56.6	5.9'	25°02.0	-1.8'	58.0'
5	255°11.9	08.0	196°21.5	5.9'	25°00.2	-1.9'	58.0'
6	270°12.1	N11°08.9	210°46.4	5.9'	N24°58.3	-2.1'	58.0'
7	285°12.2	09.8	225°11.3	5.9'	24°56.2	-2.3'	58.1'
8	300°12.3	10.6	239°36.2	5.9'	24°53.9	-2.4'	58.1'
9	315°12.5	· · 11.5	254°01.1	5.9'	24°51.5	-2.6'	58.1'
10	330°12.6	12.4	268°26.0	5.9'	24°48.9	-2.7'	58.1'
11	345°12.7	13.2	282°50.9	5.9'	24°46.2	-2.9'	58.1'
12	0°12.9	N11°14.1	297°15.8	5.9'	N24°43.3	-3.0'	58.2'
13	15°13.0	15.0	311°40.7	5.9'	24°40.3	-3.2'	58.2'
14	30°13.1	15.8	326°05.7	5.9'	24°37.1	-3.4'	58.2'
15	45°13.3	· · 16.7	340°30.6	6.0'	24°33.7	-3.5'	58.2'
16	60°13.4	17.5	354°55.6	6.0'	24°30.2	-3.7'	58.2'
17	75°13.5	18.4	9°20.5	6.0'	24°26.5	-3.8'	58.2'
18	90°13.7	N11°19.3	23°45.5	6.0'	N24°22.7	-4.0'	58.3'
19	105°13.8	20.1	38°10.5	6.0'	24°18.7	-4.1'	58.3'
20	120°13.9	21.0	52°35.6	6.0'	24°14.6	-4.3'	58.3'
21	135°14.1	· · 21.9	67°00.6	6.1'	24°10.3	-4.5'	58.3'
22	150°14.2	22.7	81°25.7	6.1'	24°05.8	-4.6'	58.3'
23	165°14.3	23.6	95°50.8	6.1'	24°01.2	-4.8'	58.3'
SD = 15.9'			SD = 15.8'				

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	180°14.5	N11°24.4	110°15.9	6.1'	N23°56.4	-4.9'	58.4'
1	195°14.6	25.3	124°41.1	6.2'	23°51.5	-5.1'	58.4'
2	210°14.7	26.2	139°06.2	6.2'	23°46.5	-5.2'	58.4'
3	225°14.9	· · 27.0	153°31.4	6.2'	23°41.3	-5.4'	58.4'
4	240°15.0	27.9	167°56.7	6.3'	23°35.9	-5.5'	58.4'
5	255°15.1	28.7	182°21.9	6.3'	23°30.4	-5.7'	58.5'
6	270°15.3	N11°29.6	196°47.2	6.3'	N23°24.7	-5.8'	58.5'
7	285°15.4	30.4	211°12.5	6.4'	23°18.9	-6.0'	58.5'
8	300°15.5	31.3	225°37.9	6.4'	23°12.9	-6.1'	58.5'
9	315°15.7	· · 32.2	240°03.3	6.4'	23°06.8	-6.3'	58.5'
10	330°15.8	33.0	254°28.8	6.5'	23°00.5	-6.4'	58.5'
11	345°15.9	33.9	268°54.2	6.5'	22°54.1	-6.6'	58.6'
12	0°16.1	N11°34.7	283°19.7	6.6'	N22°47.6	-6.7'	58.6'
13	15°16.2	35.6	297°45.3	6.6'	22°40.9	-6.8'	58.6'
14	30°16.3	36.4	312°10.9	6.6'	22°34.0	-7.0'	58.6'
15	45°16.4	· · 37.3	326°36.5	6.7'	22°27.0	-7.1'	58.6'
16	60°16.6	38.1	341°02.2	6.7'	22°19.9	-7.3'	58.6'
17	75°16.7	39.0	355°27.9	6.8'	22°12.6	-7.4'	58.7'
18	90°16.8	N11°39.9	9°53.7	6.8'	N22°05.2	-7.6'	58.7'
19	105°17.0	40.7	24°19.5	6.9'	21°57.7	-7.7'	58.7'
20	120°17.1	41.6	38°45.4	6.9'	21°50.0	-7.8'	58.7'
21	135°17.2	· · 42.4	53°11.3	7.0'	21°42.1	-8.0'	58.7'
22	150°17.3	43.3	67°37.3	7.0'	21°34.2	-8.1'	58.7'
23	165°17.5	44.1	82°03.3	7.1'	21°26.0	-8.2'	58.7'
SD = 15.9'			SD = 15.9'				

Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	180°17.6	N11°45.0	96°29.3	7.1'	N21°17.8	-8.4'	58.8'
1	195°17.7	45.8	110°55.4	7.2'	21°09.4	-8.5'	58.8'
2	210°17.9	46.7	125°21.6	7.2'	21°00.9	-8.6'	58.8'
3	225°18.0	· · 47.5	139°47.8	7.3'	20°52.3	-8.8'	58.8'
4	240°18.1	48.4	154°14.1	7.3'	20°43.5	-8.9'	58.8'
5	255°18.2	49.2	168°40.4	7.4'	20°34.6	-9.0'	58.8'
6	270°18.4	N11°50.1	183°06.7	7.4'	N20°25.6	-9.2'	58.9'
7	285°18.5	50.9	197°33.1	7.5'	20°16.4	-9.3'	58.9'
8	300°18.6	51.8	211°59.6	7.5'	20°07.1	-9.4'	58.9'
9	315°18.7	· · 52.6	226°26.1	7.6'	19°57.7	-9.5'	58.9'
10	330°18.9	53.5	240°52.7	7.6'	19°48.1	-9.7'	58.9'
11	345°19.0	54.3	255°19.4	7.7'	19°38.5	-9.8'	58.9'
12	0°19.1	N11°55.2	269°46.0	7.7'	N19°28.7	-9.9'	58.9'
13	15°19.3	56.0	284°12.8	7.8'	19°18.8	-10.0'	59.0'
14	30°19.4	56.9	298°39.6	7.9'	19°08.7	-10.2'	59.0'
15	45°19.5	· · 57.7	313°06.4	7.9'	18°58.6	-10.3'	59.0'
16	60°19.6	58.6	327°33.3	8.0'	18°48.3	-10.4'	59.0'
17	75°19.8	11°59.4	342°00.3	8.0'	18°37.9	-10.5'	59.0'
18	90°19.9	N12°00.2	356°27.3	8.1'	N18°27.4	-10.6'	59.0'
19	105°20.0	01.1	10°54.4	8.1'	18°16.8	-10.7'	59.0'
20	120°20.1	01.9	25°21.5	8.2'	18°06.1	-10.8'	59.1'
21	135°20.3	· · 02.8	39°48.7	8.2'	17°55.2	-11.0'	59.1'
22	150°20.4	03.6	54°16.0	8.3'	17°44.3	-11.1'	59.1'
23	165°20.5	04.5	68°43.3	8.4'	17°33.2	-11.2'	59.1'
SD = 15.9'			SD = 16.0'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	////	00:58	03:10	20:52	23:19	////
N 70°	////	01:58	03:31	20:30	22:06	////
68°	////	02:32	03:48	20:13	21:31	////
66°	00:50	02:56	04:01	19:59	21:06	23:25
64°	01:43	03:14	04:12	19:48	20:47	22:21
62°	02:14	03:29	04:21	19:38	20:31	21:48
60°	02:36	03:42	04:29	19:30	20:18	21:25
N 58°	02:53	03:52	04:36	19:23	20:08	21:07
56°	03:08	04:01	04:43	19:17	19:58	20:52
54°	03:20	04:09	04:48	19:11	19:50	20:40
52°	03:30	04:16	04:53	19:06	19:43	20:29
50°	03:40	04:23	04:58	19:01	19:36	20:20
45°	03:58	04:36	05:07	18:51	19:23	20:01
N 40°	04:13	04:47	05:15	18:43	19:12	19:46
35°	04:24	04:56	05:22	18:36	19:03	19:34
30°	04:34	05:03	05:28	18:30	18:55	19:24
20°	04:49	05:16	05:38	18:20	18:42	19:09
N 10°	05:01	05:26	05:47	18:11	18:32	18:57
0°	05:10	05:35	05:56	18:02	18:23	18:48
S 10°	05:18	05:42	06:04	17:54	18:15	18:40
20°	05:24	05:50	06:12	17:45	18:08	18:34
30°	05:30	05:58	06:22	17:35	18:00	18:28
35°	05:32	06:02	06:28	17:30	17:56	18:25
40°	05:35	06:06	06:34	17:24	17:51	18:23
45°	05:37	06:11	06:41	17:16	17:46	18:20
S 50°	05:39	06:17	06:50	17:07	17:41	18:18
52°	05:40	06:19	06:54	17:03	17:38	18:17
54°	05:40	06:22	06:58	16:59	17:35	18:17
56°	05:41	06:24	07:03	16:54	17:32	18:16
58°	05:42	06:28	07:08	16:49	17:29	18:15
S 60°	05:42	06:31	07:14	16:42	17:26	18:14

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	□	□	□	□	□	□
N 70°	□	□	□	□	□	□
68°	□	□	07:33	□	□	04:16
66°	□	05:52	08:14	□	03:57	03:33
64°	05:08	06:49	08:43	02:39	02:59	03:04
62°	05:55	07:22	09:04	01:51	02:25	02:42
60°	06:25	07:47	09:21	01:21	02:01	02:24
N 58°	06:48	08:06	09:36	00:58	01:41	02:09
56°	07:07	08:22	09:48	00:39	01:24	01:56
54°	07:22	08:36	09:59	00:23	01:10	01:45
52°	07:36	08:48	10:08	00:10	00:58	01:35
50°	07:48	08:59	10:17	· · ·	00:47	01:26
45°	08:12	09:21	10:35	· · ·	00:24	01:07
N 40°	08:32	09:38	10:49	· · ·	00:06	00:51
35°	08:49	09:53	11:02	23:50	· · ·	00:38
30°	09:03	10:06	11:12	23:37	· · ·	00:26
20°	09:27	10:28	11:31	23:14	· · ·	00:06
N 10°	09:48	10:47	11:47	22:54	23:49	· · ·
0°	10:07	11:05	12:02	22:35	23:32	· · ·
S 10°	10:27	11:23	12:16	22:16	23:16	· · ·
20°	10:47	11:42	12:32	21:56	22:58	· · ·
30°	11:12	12:04	12:50	21:33	22:38	23:44
35°	11:26	12:16	13:01	21:19	22:26	23:35
40°	11:42	12:31	13:13	21:03	22:12	23:24
45°	12:02	12:48	13:27	20:44	21:55	23:11
S 50°	12:26	13:10	13:44	20:20	21:35	22:55
52°	12:38	13:20	13:52	20:08	21:25	22:48
54°	12:51	13:31	14:01	19:55	21:14	22:40
56°	13:07	13:44	14:11	19:40	21:01	22:30
58°	13:25	14:00	14:22	19:22	20:47	22:20
S 60°	13:47	14:17	14:35	18:59	20:29	22:07

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time	12 ^h		Mer. Pass.	Lower	5-7
	00 ^h	mm:ss	hh:mm	Upper	hh:mm	21-41%
19	00:45	00:51	11:59	16:21	03:52	
20	00:58	01:04	11:59	17:19	04:50	
21	01:10	01:17	11:59	18:15	05:47	

April 22, 23, 24 UT (Thu., Fri., Sat.)

Aries		Venus		Mars		Jupiter		Saturn		Stars			
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	209°56.9	156°17.0	N19°40.1	77°49.8	N20°12.3	216°57.9	S04°08.5	29°48.3	N02°42.8	Alpheratz	357°46.3	29°08.8	
1	224°59.4	171°16.3	40.9	92°51.3	12.0	231°59.8	08.3	44°50.9	42.8	Ankaa	353°18.2	-42°14.9	
2	240°01.8	186°15.7	41.8	107°52.9	11.7	247°01.8	08.1	59°53.5	42.9	Schedar	349°43.9	56°35.6	
3	255°04.3	201°15.0	· · 42.6	122°54.4	· · 11.4	262°03.8	· · 07.9	74°56.1	· · 43.0	Diphda	348°58.5	-17°55.8	
4	270°06.8	216°14.4	43.4	137°56.0	11.1	277°05.8	07.7	89°58.7	43.0	Achernar	335°28.9	-57°11.0	
5	285°09.2	231°13.7	44.2	152°57.5	10.8	292°07.7	07.5	105°01.3	43.1	Hamal	328°03.8	23°30.6	
6	300°11.7	246°13.1	N19°45.1	167°59.1	N20°10.4	307°09.7	S04°07.3	120°03.9	N02°43.1	Polaris	319°21.9	89°18.6	
7	315°14.2	261°12.4	45.9	183°00.6	10.1	322°11.7	07.1	135°06.5	43.2	Acamar	315°20.4	-40°15.8	
8	330°16.6	276°11.7	46.7	198°02.2	09.8	337°13.6	06.9	150°09.2	43.2	Menkar	314°17.8	4°07.8	
9	345°19.1	291°11.1	· · 47.5	213°03.7	· · 09.5	352°15.6	· · 06.7	165°11.8	· · 43.3	Mirfak	308°44.2	49°53.9	
10	0°21.5	306°10.4	48.3	228°05.3	09.2	7°17.6	06.5	180°14.4	43.3	Aldebaran	290°52.3	16°31.8	
11	15°24.0	321°09.8	49.2	243°06.8	08.9	22°19.6	06.3	195°17.0	43.4	Rigel	281°14.5	-8°11.5	
12	30°26.5	336°09.1	N19°50.0	258°08.4	N20°08.6	37°21.5	S04°06.1	210°19.6	N02°43.4	Capella	280°38.2	46°00.6	
13	45°28.9	351°08.4	50.8	273°09.9	08.3	52°23.5	05.9	225°22.2	43.5	Bellatrix	278°34.7	6°21.5	
14	60°31.4	6°07.8	51.6	288°11.5	08.0	67°25.5	05.7	240°24.8	43.6	Elnath	278°15.8	28°37.0	
15	75°33.9	21°07.1	· · 52.4	303°13.0	· · 07.7	82°27.5	· · 05.5	255°27.4	· · 43.6	Alnilam	275°48.9	-1°11.8	
16	90°36.3	36°06.5	53.3	318°14.6	07.4	97°29.4	05.3	270°30.0	43.7	Betelgeuse	271°04.0	7°24.5	
17	105°38.8	51°05.8	54.1	333°16.1	07.1	112°31.4	05.1	285°32.6	43.7	Canopus	263°57.4	-52°42.4	
18	120°41.3	66°05.1	N19°54.9	348°17.7	N20°06.8	127°33.4	S04°04.9	300°35.2	N02°43.8	Sirius	258°35.9	-16°44.0	
19	135°43.7	81°04.5	55.7	3°19.2	06.4	142°35.4	04.7	315°37.8	43.8	Adhara	255°14.5	-28°59.4	
20	150°46.2	96°03.8	56.5	18°20.7	06.1	157°37.3	04.5	330°40.4	43.9	Procyon	245°02.2	5°11.8	
21	165°48.7	111°03.2	· · 57.3	33°22.3	· · 05.8	172°39.3	· · 04.3	345°43.0	· · 43.9	Pollux	243°30.6	28°00.1	
22	180°51.1	126°02.5	58.1	48°23.8	05.5	187°41.3	04.1	0°45.6	44.0	Avior	234°19.1	-59°32.9	
23	195°53.6	141°01.8	58.9	63°25.4	05.2	202°43.3	03.9	15°48.2	44.0	Suhail	222°54.1	-43°28.8	
Mer.pass. 09:59		ν -0.7' d 0.8' m -3.89		ν 1.6' d -0.3' m 0.57		ν 2.0' d -0.2' m -2.10		ν 2.6' d 0.1' m 0.66		Miaplacidus	221°40.2	-69°46.0	
										Alphard	217°58.3	-8°42.4	
										Regulus	207°45.8	11°54.9	
										Dubhe	193°53.8	61°41.8	
										Denebola	182°35.7	14°30.7	
										Gienah	175°54.4	-17°36.2	
										Acrux	173°11.5	-63°09.7	
										Gacrux	172°03.1	-57°10.5	
										Alioth	166°22.0	55°54.2	
										Spica	158°33.4	-11°13.1	
										Alkaid	153°00.1	49°15.6	
										Hadar	148°50.7	-60°25.5	
										Menkent	148°10.0	-36°25.4	
										Arcturus	145°57.5	19°07.5	
										Rigel Kent.	139°54.3	-60°52.7	
										Kochab	137°18.5	74°06.6	
										Zuben'ubi	137°07.7	-16°05.2	
										Alphecca	126°12.6	26°40.6	
										Antares	112°28.8	-26°27.4	
										Atria	107°32.4	-69°02.6	
										Sabik	102°14.9	-15°44.3	
										Shaula	96°24.8	-37°06.6	
										Rasalhague	96°08.4	12°33.0	
										Eltanin	90°46.9	51°29.0	
										Kaus Aust.	83°46.7	-34°22.7	
										Vega	80°40.4	38°47.3	
										Nunki	76°01.0	-26°17.0	
										Altair	62°10.4	8°53.6	
										Peacock	53°22.8	-56°41.8	
										Deneb	49°33.2	45°18.8	
										Enif	33°49.5	9°55.3	
										Al Na'ir	27°46.7	-46°54.4	
										Fomalhaut	15°26.7	-29°33.9	
										Scheat	13°55.9	28°08.2	
										Markab	13°40.9	15°15.6	
										Apr 22 Thu		SHA	Mer.pass
										Venus		306°20.1	13:35
										Mars		227°52.9	18:47
										Jupiter		7°00.9	09:31
										Saturn		179°51.4	21:57
										Apr 23 Fri		SHA	Mer.pass
										Venus		305°05.1	13:37
										Mars		227°30.9	18:44
										Jupiter		6°49.2	09:28
										Saturn		179°54.7	21:53
										Apr 24 Sat		SHA	Mer.pass
										Venus		303°49.9	13:38
										Mars		227°08.6	18:42
										Jupiter		6°37.5	09:25
										Saturn		179°57.9	21:49
										Horizontal parallax			
										Venus:		0.1	
										Mars:		0.1	
Mer.pass. 09:51		ν -0.7' d 0.8' m -3.89		ν 1.5' d -0.3' m 0.61		ν 2.0' d -0.2' m -2.10		ν 2.6' d 0.1' m 0.67					

h	Sun			Moon			
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	180°20.6	N12°05.3	83°10.6	8.4'	N17°22.0	-11.3'	59.1'
1	195°20.7	06.2	97°38.0	8.5'	17°10.8	-11.4'	59.1'
2	210°20.9	07.0	112°05.5	8.5'	16°59.4	-11.5'	59.1'
3	225°21.0	· · 07.8	126°33.0	8.6'	16°47.9	-11.6'	59.2'
4	240°21.1	08.7	141°00.6	8.6'	16°36.3	-11.7'	59.2'
5	255°21.2	09.5	155°28.2	8.7'	16°24.6	-11.8'	59.2'
6	270°21.4	N12°10.4	169°55.9	8.7'	N16°12.8	-11.9'	59.2'
7	285°21.5	11.2	184°23.6	8.8'	16°00.9	-12.0'	59.2'
8	300°21.6	12.1	198°51.4	8.8'	15°48.9	-12.1'	59.2'
9	315°21.7	· · 12.9	213°19.3	8.9'	15°36.8	-12.2'	59.2'
10	330°21.9	13.7	227°47.1	9.0'	15°24.6	-12.3'	59.2'
11	345°22.0	14.6	242°15.1	9.0'	15°12.4	-12.4'	59.3'
12	0°22.1	N12°15.4	256°43.1	9.1'	N15°00.0	-12.5'	59.3'
13	15°22.2	16.2	271°11.2	9.1'	14°47.5	-12.6'	59.3'
14	30°22.3	17.1	285°39.3	9.2'	14°35.0	-12.6'	59.3'
15	45°22.5	· · 17.9	300°07.4	9.2'	14°22.3	-12.7'	59.3'
16	60°22.6	18.8	314°35.6	9.3'	14°09.6	-12.8'	59.3'
17	75°22.7	19.6	329°03.9	9.3'	13°56.8	-12.9'	59.3'
18	90°22.8	N12°20.4	343°32.2	9.4'	N13°43.9	-13.0'	59.3'
19	105°22.9	21.3	358°00.5	9.4'	13°30.9	-13.1'	59.3'
20	120°23.1	22.1	12°29.0	9.5'	13°17.8	-13.1'	59.4'
21	135°23.2	· · 22.9	26°57.4	9.5'	13°04.7	-13.2'	59.4'
22	150°23.3	23.8	41°25.9	9.5'	12°51.5	-13.3'	59.4'
23	165°23.4	24.6	55°54.5	9.6'	12°38.1	-13.4'	59.4'
SD = 15.9'			SD = 16.1'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	180°23.5	N12°25.5	70°23.0	9.6'	N12°24.8	-13.5'	59.4'
1	195°23.7	26.3	84°51.7	9.7'	12°11.3	-13.5'	59.4'
2	210°23.8	27.1	99°20.4	9.7'	11°57.8	-13.6'	59.4'
3	225°23.9	· · 28.0	113°49.1	9.8'	11°44.2	-13.7'	59.4'
4	240°24.0	28.8	128°17.9	9.8'	11°30.5	-13.7'	59.4'
5	255°24.1	29.6	142°46.7	9.9'	11°16.8	-13.8'	59.4'
6	270°24.3	N12°30.5	157°15.5	9.9'	N11°03.0	-13.9'	59.5'
7	285°24.4	31.3	171°44.4	9.9'	10°49.1	-13.9'	59.5'
8	300°24.5	32.1	186°13.4	10.0'	10°35.2	-14.0'	59.5'
9	315°24.6	· · 33.0	200°42.3	10.0'	10°21.2	-14.1'	59.5'
10	330°24.7	33.8	215°11.4	10.1'	10°07.1	-14.1'	59.5'
11	345°24.8	34.6	229°40.4	10.1'	09°53.0	-14.2'	59.5'
12	0°25.0	N12°35.4	244°09.5	10.1'	N09°38.8	-14.2'	59.5'
13	15°25.1	36.3	258°38.6	10.2'	09°24.5	-14.3'	59.5'
14	30°25.2	37.1	273°07.8	10.2'	09°10.2	-14.4'	59.5'
15	45°25.3	· · 37.9	287°37.0	10.2'	08°55.9	-14.4'	59.5'
16	60°25.4	38.8	302°06.2	10.3'	08°41.5	-14.5'	59.5'
17	75°25.5	39.6	316°35.5	10.3'	08°27.0	-14.5'	59.5'
18	90°25.7	N12°40.4	331°04.8	10.3'	N08°12.5	-14.6'	59.6'
19	105°25.8	41.3	345°34.1	10.4'	07°58.0	-14.6'	59.6'
20	120°25.9	42.1	0°03.5	10.4'	07°43.3	-14.7'	59.6'
21	135°26.0	· · 42.9	14°32.8	10.4'	07°28.7	-14.7'	59.6'
22	150°26.1	43.7	29°02.3	10.4'	07°14.0	-14.7'	59.6'
23	165°26.2	44.6	43°31.7	10.5'	06°59.3	-14.8'	59.6'
SD = 15.9'			SD = 16.2'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	180°26.4	N12°45.4	58°01.2	10.5'	N06°44.5	-14.8'	59.6'
1	195°26.5	46.2	72°30.7	10.5'	06°29.7	-14.9'	59.6'
2	210°26.6	47.0	87°00.2	10.5'	06°14.8	-14.9'	59.6'
3	225°26.7	· · 47.9	101°29.7	10.6'	05°59.9	-14.9'	59.6'
4	240°26.8	48.7	115°59.3	10.6'	05°45.0	-15.0'	59.6'
5	255°26.9	49.5	130°28.9	10.6'	05°30.0	-15.0'	59.6'
6	270°27.0	N12°50.3	144°58.5	10.6'	N05°15.0	-15.0'	59.6'
7	285°27.2	51.2	159°28.1	10.6'	05°00.0	-15.1'	59.6'
8	300°27.3	52.0	173°57.8	10.7'	04°44.9	-15.1'	59.6'
9	315°27.4	· · 52.8	188°27.4	10.7'	04°29.8	-15.1'	59.6'
10	330°27.5	53.6	202°57.1	10.7'	04°14.7	-15.1'	59.6'
11	345°27.6	54.5	217°26.8	10.7'	03°59.6	-15.2'	59.6'
12	0°27.7	N12°55.3	231°56.5	10.7'	N03°44.4	-15.2'	59.6'
13	15°27.8	56.1	246°26.3	10.7'	03°29.2	-15.2'	59.6'
14	30°27.9	56.9	260°56.0	10.8'	03°14.0	-15.2'	59.6'
15	45°28.1	· · 57.7	275°25.8	10.8'	02°58.8	-15.2'	59.6'
16	60°28.2	58.6	289°55.5	10.8'	02°43.5	-15.3'	59.6'
17	75°28.3	12°59.4	304°25.3	10.8'	02°28.3	-15.3'	59.6'
18	90°28.4	N13°00.2	318°55.1	10.8'	N02°13.0	-15.3'	59.6'
19	105°28.5	01.0	333°24.9	10.8'	01°57.7	-15.3'	59.6'
20	120°28.6	01.8	347°54.7	10.8'	01°42.4	-15.3'	59.6'
21	135°28.7	· · 02.7	2°24.5	10.8'	01°27.1	-15.3'	59.6'
22	150°28.8	03.5	16°54.3	10.8'	01°11.8	-15.3'	59.6'
23	165°28.9	04.3	31°24.1	10.8'	00°56.4	-15.3'	59.6'
SD = 15.9'			SD = 16.3'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	////	////	02:51	21:10	////	////
N 70°	////	01:31	03:16	20:44	22:35	////
68°	////	02:13	03:35	20:25	21:49	////
66°	////	02:41	03:49	20:10	21:20	////
64°	01:20	03:02	04:02	19:57	20:58	22:45
62°	01:57	03:18	04:12	19:47	20:41	22:04
60°	02:23	03:32	04:21	19:38	20:27	21:38
N 58°	02:42	03:43	04:28	19:30	20:15	21:17
56°	02:58	03:53	04:35	19:23	20:05	21:01
54°	03:11	04:02	04:41	19:17	19:56	20:47
52°	03:23	04:10	04:47	19:11	19:48	20:36
50°	03:32	04:16	04:52	19:06	19:41	20:26
45°	03:52	04:31	05:02	18:55	19:27	20:05
N 40°	04:08	04:42	05:11	18:46	19:15	19:50
35°	04:20	04:52	05:18	18:39	19:05	19:37
30°	04:31	05:00	05:25	18:32	18:57	19:27
20°	04:47	05:14	05:36	18:21	18:43	19:10
N 10°	04:59	05:24	05:46	18:11	18:32	18:58
0°	05:09	05:34	05:55	18:02	18:23	18:47
S 10°	05:18	05:42	06:04	17:53	18:14	18:39
20°	05:25	05:51	06:13	17:43	18:06	18:32
30°	05:31	05:59	06:24	17:32	17:57	18:25
35°	05:34	06:04	06:30	17:26	17:52	18:22
40°	05:37	06:09	06:37	17:19	17:47	18:19
45°	05:40	06:15	06:45	17:11	17:41	18:16
S 50°	05:43	06:21	06:54	17:02	17:35	18:13
52°	05:44	06:24	06:59	16:57	17:32	18:12
54°	05:45	06:27	07:04	16:52	17:29	18:10
56°	05:46	06:30	07:09	16:47	17:26	18:09
58°	05:48	06:34	07:15	16:41	17:22	18:08
S 60°	05:49	06:38	07:22	16:34	17:18	18:07

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	08:52	11:37	13:58	04:54	03:59	03:27
N 70°	09:32	11:54	14:03	04:13	03:41	03:19
68°	09:59	12:06	14:06	03:44	03:26	03:12
66°	10:19	12:16	14:09	03:22	03:14	03:07
64°	10:35	12:25	14:12	03:05	03:04	03:02
62°	10:49	12:32	14:14	02:51	02:55	02:58
60°	11:00	12:39	14:16	02:38	02:48	02:55
N 58°	11:10	12:44	14:18	02:28	02:41	02:51
56°	11:18	12:49	14:19	02:18	02:35	02:49
54°	11:26	12:53	14:21	02:10	02:30	02:46
52°	11:32	12:57	14:22	02:03	02:25	02:44
50°	11:39	13:01	14:23	01:56	02:20	02:42
45°	11:51	13:09	14:25	01:41	02:11	02:37
N 40°	12:02	13:15	14:28	01:29	02:03	02:33
35°	12:11	13:20	14:29	01:19	01:56	02:29
30°	12:19	13:25	14:31	01:10	01:50	02:26
20°	12:33	13:34	14:34	00:54	01:39	02:21
N 10°	12:45	13:41	14:36	00:41	01:30	02:16
0°	12:56	13:48	14:38	00:28	01:21	02:12
S 10°	13:07	13:54	14:40	00:15	01:12	02:07
20°	13:19	14:02	14:43	00:00	01:02	02:02
30°	13:32	14:10	14:46	· · ·	00:51	01:57
35°	13:40	14:15	14:47	· · ·	00:44	01:53
40°	13:48	14:20	14:49	· · ·	00:37	01:50
45°	13:59	14:26	14:51	· · ·	00:28	01:45
S 50°	14:11	14:33	14:53	· · ·	00:18	01:40
52°	14:17	14:37	14:55	· · ·	00:13	01:38
54°	14:23	14:40	14:56	· · ·	00:07	01:35
56°	14:30	14:44	14:57	· · ·	00:01	01:32
58°	14:37	14:49	14:59	23:54	· · ·	01:29
S 60°	14:46	14:54	15:00	23:47	· · ·	01:25

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time	12 ^h		Mer. Pass.		8-10
	00 ^h	mm:ss	hh:mm	Upper	Lower	53-75%
22	01:23	01:28	11:59	19:08	06:42	
23	01:34	01:40	11:58	20:00	07:34	
24	01:45	01:51	11:58	20:50	08:25	

April 25, 26, 27 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	212°54.3	155°28.7	N20°37.4	79°40.3	N19°49.8	219°20.2	S03°54.0	32°55.4	N02°46.6	Alpheratz	357°46.2	29°08.8
1	227°56.8	170°28.0	38.2	94°41.8	49.5	234°22.2	53.8	47°58.0	46.6	Ankaa	353°18.2	-42°14.9
2	242°59.3	185°27.3	38.9	109°43.4	49.2	249°24.2	53.6	63°00.6	46.7	Schedar	349°43.8	56°35.6
3	258°01.7	200°26.6	· · 39.7	124°44.9	· · 48.9	264°26.1	· · 53.4	78°03.2	· · 46.7	Diphda	348°58.5	-17°55.8
4	273°04.2	215°26.0	40.5	139°46.4	48.5	279°28.1	53.2	93°05.8	46.8	Achernar	335°28.9	-57°11.0
5	288°06.6	230°25.3	41.2	154°47.9	48.2	294°30.1	53.0	108°08.4	46.8	Hamal	328°03.8	23°30.6
6	303°09.1	245°24.6	N20°42.0	169°49.4	N19°47.9	309°32.1	S03°52.8	123°11.0	N02°46.9	Polaris	319°22.1	89°18.6
7	318°11.6	260°23.9	42.7	184°50.9	47.6	324°34.1	52.6	138°13.6	46.9	Acamar	315°20.4	-40°15.8
8	333°14.0	275°23.2	43.5	199°52.4	47.3	339°36.1	52.4	153°16.2	47.0	Menkar	314°17.8	4°07.8
9	348°16.5	290°22.5	· · 44.2	214°54.0	· · 46.9	354°38.0	· · 52.2	168°18.8	· · 47.0	Mirfak	308°44.2	49°53.9
10	3°19.0	305°21.8	45.0	229°55.5	46.6	9°40.0	52.0	183°21.4	47.1	Aldebaran	290°52.3	16°31.8
11	18°21.4	320°21.1	45.8	244°57.0	46.3	24°42.0	51.8	198°23.9	47.1	Rigel	281°14.5	-8°11.5
12	33°23.9	335°20.4	N20°46.5	259°58.5	N19°46.0	39°44.0	S03°51.6	213°26.5	N02°47.2	Capella	280°38.2	46°00.6
13	48°26.4	350°19.7	47.3	275°00.0	45.7	54°46.0	51.4	228°29.1	47.2	Bellatrix	278°34.7	6°21.5
14	63°28.8	5°19.1	48.0	290°01.5	45.3	69°47.9	51.2	243°31.7	47.3	Elnath	278°15.8	28°37.0
15	78°31.3	20°18.4	· · 48.8	305°03.0	· · 45.0	84°49.9	· · 51.0	258°34.3	· · 47.3	Alnilam	275°48.9	-1°11.8
16	93°33.7	35°17.7	49.5	320°04.5	44.7	99°51.9	50.8	273°36.9	47.4	Betelgeuse	271°04.0	7°24.5
17	108°36.2	50°17.0	50.2	335°06.1	44.4	114°53.9	50.6	288°39.5	47.4	Canopus	263°57.5	-52°42.3
18	123°38.7	65°16.3	N20°51.0	350°07.6	N19°44.1	129°55.9	S03°50.4	303°42.1	N02°47.5	Sirius	258°35.9	-16°44.0
19	138°41.1	80°15.6	51.7	5°09.1	43.7	144°57.9	50.2	318°44.7	47.5	Adhara	255°14.5	-28°59.4
20	153°43.6	95°14.9	52.5	20°10.6	43.4	159°59.8	50.0	333°47.3	47.6	Procyon	245°02.2	5°11.8
21	168°46.1	110°14.2	· · 53.2	35°12.1	· · 43.1	175°01.8	· · 49.8	348°49.9	· · 47.6	Pollux	243°30.6	28°00.1
22	183°48.5	125°13.5	54.0	50°13.6	42.8	190°03.8	49.6	3°52.5	47.7	Avior	234°19.1	-59°32.9
23	198°51.0	140°12.8	54.7	65°15.1	42.4	205°05.8	49.4	18°55.1	47.7	Suhail	222°54.1	-43°28.8
Mer.pass. 09:47		ν -0.7' d 0.8' m -3.89		ν 1.5' d -0.3' m 0.62		ν 2.0' d -0.2' m -2.11		ν 2.6' d 0.1' m 0.67		Miaplacidus	221°40.2	-69°46.0
										Alphard	217°58.3	-8°42.4
										Regulus	207°45.8	11°54.9
										Dubhe	193°53.9	61°41.8
										Denebola	182°35.7	14°30.7
										Gienah	175°54.4	-17°36.2
										Acrux	173°11.5	-63°09.7
										Gacrux	172°03.1	-57°10.6
										Alioth	166°22.0	55°54.2
										Spica	158°33.4	-11°13.1
										Alkaid	153°00.1	49°15.6
										Hadar	148°50.7	-60°25.5
										Menkent	148°10.0	-36°25.4
										Arcturus	145°57.5	19°07.5
										Rigil Kent.	139°54.3	-60°52.8
										Kochab	137°18.4	74°06.6
										Zuben'ubi	137°07.7	-16°05.3
										Alphecca	126°12.6	26°40.6
										Antares	112°28.8	-26°27.4
										Atria	107°32.3	-69°02.7
										Sabik	102°14.9	-15°44.3
										Shaula	96°24.8	-37°06.6
										Rasalhague	96°08.4	12°33.0
										Eltanin	90°46.9	51°29.0
										Kaus Aust.	83°46.6	-34°22.7
										Vega	80°40.3	38°47.3
										Nunki	76°01.0	-26°17.0
										Altair	62°10.4	8°53.6
										Peacock	53°22.7	-56°41.8
										Deneb	49°33.1	45°18.8
										Enif	33°49.5	9°55.3
										Al Na'ir	27°46.7	-46°54.4
										Fomalhaut	15°26.7	-29°33.9
										Scheat	13°55.9	28°08.2
										Markab	13°40.9	15°15.6
Apr 25 Sun		SHA		Mer.pass								
		Venus		302°34.4		13:39						
		Mars		226°46.0		18:39						
		Jupiter		6°25.9		09:21						
		Saturn		180°01.1		21:45						
Apr 26 Mon		SHA		Mer.pass								
		Venus		301°18.6		13:40						
		Mars		226°23.2		18:37						
		Jupiter		6°14.3		09:18						
		Saturn		180°04.2		21:40						
Apr 27 Tue		SHA		Mer.pass								
		Venus		300°02.6		13:41						
		Mars		226°00.0		18:35						
		Jupiter		6°02.8		09:15						
		Saturn		180°07.2		21:36						
Horizontal parallax												
											Venus:	0.1
											Mars:	0.1

h	Sun			Moon			
Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	180°29.0	N13°05.1	45°53.9	10.8'	N00°41.1	-15.3'	59.6'
1	195°29.2	05.9	60°23.7	10.8'	00°25.8	-15.3'	59.6'
2	210°29.3	06.7	74°53.5	10.8'	N00°10.4	-15.3'	59.6'
3	225°29.4	· · 07.6	89°23.3	10.8'	S00°04.9	15.3'	59.6'
4	240°29.5	08.4	103°53.2	10.8'	00°20.3	15.3'	59.6'
5	255°29.6	09.2	118°23.0	10.8'	00°35.6	15.3'	59.6'
6	270°29.7	N13°10.0	132°52.8	10.8'	S00°50.9	15.3'	59.6'
7	285°29.8	10.8	147°22.6	10.8'	01°06.3	15.3'	59.6'
8	300°29.9	11.6	161°52.4	10.8'	01°21.6	15.3'	59.6'
9	315°30.0	· · 12.5	176°22.2	10.8'	01°36.9	15.3'	59.6'
10	330°30.1	13.3	190°52.0	10.8'	01°52.2	15.3'	59.6'
11	345°30.2	14.1	205°21.8	10.8'	02°07.5	15.3'	59.6'
12	0°30.4	N13°14.9	219°51.5	10.8'	S02°22.8	15.3'	59.6'
13	15°30.5	15.7	234°21.3	10.8'	02°38.1	15.3'	59.6'
14	30°30.6	16.5	248°51.0	10.7'	02°53.3	15.2'	59.6'
15	45°30.7	· · 17.3	263°20.8	10.7'	03°08.6	15.2'	59.6'
16	60°30.8	18.1	277°50.5	10.7'	03°23.8	15.2'	59.6'
17	75°30.9	19.0	292°20.2	10.7'	03°39.0	15.2'	59.6'
18	90°31.0	N13°19.8	306°49.9	10.7'	S03°54.1	15.1'	59.6'
19	105°31.1	20.6	321°19.6	10.7'	04°09.3	15.1'	59.6'
20	120°31.2	21.4	335°49.3	10.6'	04°24.4	15.1'	59.6'
21	135°31.3	· · 22.2	350°18.9	10.6'	04°39.5	15.1'	59.6'
22	150°31.4	23.0	4°48.6	10.6'	04°54.6	15.0'	59.6'
23	165°31.5	23.8	19°18.2	10.6'	05°09.6	15.0'	59.6'
SD = 15.9'			SD = 16.3'				

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	180°31.6	N13°24.6	33°47.8	10.6'	S05°24.6	15.0'	59.5'
1	195°31.7	25.4	48°17.3	10.5'	05°39.6	14.9'	59.5'
2	210°31.8	26.2	62°46.9	10.5'	05°54.5	14.9'	59.5'
3	225°31.9	· · 27.0	77°16.4	10.5'	06°09.4	14.9'	59.5'
4	240°32.0	27.8	91°45.9	10.5'	06°24.3	14.8'	59.5'
5	255°32.2	28.7	106°15.4	10.5'	06°39.1	14.8'	59.5'
6	270°32.3	N13°29.5	120°44.8	10.4'	S06°53.9	14.7'	59.5'
7	285°32.4	30.3	135°14.3	10.4'	07°08.7	14.7'	59.5'
8	300°32.5	31.1	149°43.7	10.4'	07°23.4	14.7'	59.5'
9	315°32.6	· · 31.9	164°13.1	10.3'	07°38.0	14.6'	59.5'
10	330°32.7	32.7	178°42.4	10.3'	07°52.7	14.6'	59.4'
11	345°32.8	33.5	193°11.7	10.3'	08°07.2	14.5'	59.4'
12	0°32.9	N13°34.3	207°41.0	10.3'	S08°21.7	14.5'	59.4'
13	15°33.0	35.1	222°10.3	10.2'	08°36.2	14.4'	59.4'
14	30°33.1	35.9	236°39.5	10.2'	08°50.6	14.4'	59.4'
15	45°33.2	· · 36.7	251°08.7	10.2'	09°05.0	14.3'	59.4'
16	60°33.3	37.5	265°37.8	10.1'	09°19.3	14.2'	59.4'
17	75°33.4	38.3	280°07.0	10.1'	09°33.5	14.2'	59.4'
18	90°33.5	N13°39.1	294°36.1	10.1'	S09°47.7	14.1'	59.4'
19	105°33.6	39.9	309°05.1	10.0'	10°01.8	14.1'	59.3'
20	120°33.7	40.7	323°34.2	10.0'	10°15.9	14.0'	59.3'
21	135°33.8	· · 41.5	338°03.1	10.0'	10°29.9	13.9'	59.3'
22	150°33.9	42.3	352°32.1	9.9'	10°43.8	13.9'	59.3'
23	165°34.0	43.1	7°01.0	9.9'	10°57.7	13.8'	59.3'
SD = 15.9'			SD = 16.2'				

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	180°34.1	N13°43.9	21°29.9	9.8'	S11°11.5	13.7'	59.3'
1	195°34.2	44.7	35°58.7	9.8'	11°25.2	13.7'	59.3'
2	210°34.3	45.5	50°27.5	9.8'	11°38.9	13.6'	59.2'
3	225°34.4	· · 46.3	64°56.3	9.7'	11°52.4	13.5'	59.2'
4	240°34.5	47.1	79°25.0	9.7'	12°06.0	13.4'	59.2'
5	255°34.6	47.9	93°53.7	9.6'	12°19.4	13.4'	59.2'
6	270°34.7	N13°48.7	108°22.4	9.6'	S12°32.8	13.3'	59.2'
7	285°34.8	49.5	122°51.0	9.6'	12°46.0	13.2'	59.2'
8	300°34.9	50.3	137°19.5	9.5'	12°59.3	13.1'	59.1'
9	315°35.0	· · 51.1	151°48.1	9.5'	13°12.4	13.0'	59.1'
10	330°35.1	51.9	166°16.5	9.4'	13°25.4	13.0'	59.1'
11	345°35.2	52.7	180°45.0	9.4'	13°38.4	12.9'	59.1'
12	0°35.3	N13°53.5	195°13.4	9.4'	S13°51.2	12.8'	59.1'
13	15°35.4	54.3	209°41.7	9.3'	14°04.0	12.7'	59.0'
14	30°35.5	55.0	224°10.0	9.3'	14°16.7	12.6'	59.0'
15	45°35.6	· · 55.8	238°38.3	9.2'	14°29.3	12.5'	59.0'
16	60°35.7	56.6	253°06.5	9.2'	14°41.9	12.4'	59.0'
17	75°35.8	57.4	267°34.7	9.1'	14°54.3	12.3'	59.0'
18	90°35.9	N13°58.2	282°02.8	9.1'	S15°06.6	12.2'	59.0'
19	105°35.9	59.0	296°30.9	9.0'	15°18.8	12.1'	58.9'
20	120°36.0	13°59.8	310°59.0	9.0'	15°31.0	12.0'	58.9'
21	135°36.1	14°00.6	325°27.0	9.0'	15°43.0	11.9'	58.9'
22	150°36.2	01.4	339°55.0	8.9'	15°55.0	11.8'	58.9'
23	165°36.3	02.2	354°22.9	8.9'	16°06.8	11.7'	58.8'
SD = 15.9'			SD = 16.2'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	////	////	02:31	21:30	////	////
N 70°	////	00:54	03:00	21:00	23:18	////
68°	////	01:52	03:21	20:38	22:10	////
66°	////	02:25	03:38	20:21	21:35	////
64°	00:47	02:49	03:51	20:07	21:10	23:23
62°	01:39	03:07	04:03	19:55	20:51	22:22
60°	02:09	03:22	04:12	19:45	20:36	21:51
N 58°	02:31	03:34	04:21	19:36	20:23	21:28
56°	02:48	03:45	04:28	19:29	20:12	21:10
54°	03:02	03:55	04:35	19:22	20:03	20:55
52°	03:15	04:03	04:41	19:16	19:54	20:43
50°	03:25	04:10	04:46	19:11	19:47	20:32
45°	03:47	04:26	04:57	18:59	19:31	20:10
N 40°	04:03	04:38	05:07	18:49	19:18	19:53
35°	04:16	04:48	05:15	18:41	19:08	19:40
30°	04:27	04:57	05:22	18:34	18:59	19:29
20°	04:44	05:11	05:34	18:22	18:45	19:11
N 10°	04:58	05:23	05:45	18:11	18:33	18:58
0°	05:08	05:33	05:54	18:01	18:22	18:47
S 10°	05:18	05:43	06:04	17:51	18:13	18:38
20°	05:26	05:52	06:14	17:41	18:04	18:30
30°	05:33	06:01	06:26	17:30	17:54	18:22
35°	05:36	06:06	06:32	17:23	17:49	18:19
40°	05:40	06:12	06:40	17:15	17:43	18:15
45°	05:44	06:18	06:48	17:07	17:37	18:11
S 50°	05:47	06:25	06:59	16:56	17:30	18:08
52°	05:49	06:28	07:04	16:51	17:26	18:06
54°	05:50	06:32	07:09	16:46	17:23	18:05
56°	05:52	06:36	07:15	16:40	17:19	18:03
58°	05:53	06:40	07:21	16:33	17:15	18:01
S 60°	05:55	06:44	07:29	16:26	17:10	17:59

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	16:14	18:36	21:25	02:59	02:32	01:58
N 70°	16:09	18:18	20:42	03:00	02:40	02:18
68°	16:04	18:05	20:13	03:00	02:47	02:33
66°	16:01	17:54	19:52	03:00	02:53	02:46
64°	15:58	17:45	19:35	03:00	02:58	02:57
62°	15:55	17:37	19:21	03:01	03:03	03:06
60°	15:53	17:31	19:09	03:01	03:07	03:14
N 58°	15:51	17:25	18:59	03:01	03:10	03:21
56°	15:49	17:20	18:50	03:01	03:13	03:27
54°	15:48	17:15	18:43	03:01	03:16	03:32
52°	15:46	17:11	18:36	03:01	03:18	03:37
50°	15:45	17:07	18:29	03:01	03:21	03:42
45°	15:42	16:59	18:16	03:01	03:26	03:52
N 40°	15:40	16:52	18:05	03:01	03:30	04:00
35°	15:38	16:47	17:56	03:01	03:34	04:07
30°	15:36	16:42	17:48	03:02	03:37	04:14
20°	15:33	16:33	17:34	03:02	03:42	04:24
N 10°	15:31	16:26	17:22	03:02	03:47	04:34
0°	15:28	16:19	17:11	03:02	03:52	04:43
S 10°	15:26	16:12	17:00	03:02	03:57	04:52
20°	15:23	16:05	16:48	03:02	04:02	05:02
30°	15:21	15:56	16:34	03:02	04:07	05:13
35°	15:19	15:52	16:27	03:02	04:11	05:20
40°	15:17	15:46	16:18	03:02	04:14	05:27
45°	15:15	15:40	16:08	03:02	04:19	05:36
S 50°	15:13	15:33	15:56	03:02	04:24	05:46
52°	15:12	15:30	15:50	03:02	04:26	05:51
54°	15:10	15:26	15:44	03:02	04:29	05:57
56°	15:09	15:22	15:37	03:02	04:32	06:03
58°	15:08	15:17	15:29	03:02	04:35	06:09
S 60°	15:06	15:12	15:21	03:02	04:39	06:17

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time	12 ^h		Mer. Pass.		
	00 ^h	mm:ss	hh:mm	Upper	Lower	11-13
				hh:mm	hh:mm	84-97%
25	01:56	02:01	11:58	21:40	09:15	
26	02:07	02:11	11:58	22:31	10:05	
27	02:16	02:21	11:58	23:23	10:57	

April 28, 29, 30 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	215°51.7	154°38.1	N21°29.8	81°28.4	N19°26.4	221°43.1	S03°39.7	36°01.9	N02°50.1	Alpheratz	357°46.2	29°08.8
1	230°54.2	169°37.4	30.5	96°29.9	26.1	236°45.1	39.5	51°04.5	50.1	Ankaa	353°18.2	-42°14.8
2	245°56.7	184°36.7	31.2	111°31.4	25.7	251°47.1	39.3	66°07.1	50.1	Schedar	349°43.8	56°35.5
3	260°59.1	199°36.0	•• 31.9	126°32.9	•• 25.4	266°49.1	•• 39.1	81°09.6	•• 50.2	Diphda	348°58.4	-17°55.8
4	276°01.6	214°35.3	32.5	141°34.4	25.1	281°51.1	38.9	96°12.2	50.2	Achernar	335°28.9	-57°10.9
5	291°04.1	229°34.5	33.2	156°35.8	24.7	296°53.1	38.7	111°14.8	50.3	Hamal	328°03.7	23°30.6
6	306°06.5	244°33.8	N21°33.9	171°37.3	N19°24.4	311°55.1	S03°38.5	126°17.4	N02°50.3	Polaris	319°21.9	89°18.6
7	321°09.0	259°33.1	34.6	186°38.8	24.1	326°57.1	38.3	141°20.0	50.4	Acamar	315°20.4	-40°15.8
8	336°11.4	274°32.4	35.3	201°40.3	23.7	341°59.1	38.1	156°22.6	50.4	Menkar	314°17.8	4°07.8
9	351°13.9	289°31.7	•• 36.0	216°41.8	•• 23.4	357°01.1	•• 37.9	171°25.2	•• 50.5	Mirfak	308°44.2	49°53.9
10	6°16.4	304°30.9	36.7	231°43.3	23.1	12°03.1	37.8	186°27.7	50.5	Aldebaran	290°52.3	16°31.8
11	21°18.8	319°30.2	37.3	246°44.7	22.7	27°05.0	37.6	201°30.3	50.6	Rigel	281°14.5	-8°11.5
12	36°21.3	334°29.5	N21°38.0	261°46.2	N19°22.4	42°07.0	S03°37.4	216°32.9	N02°50.6	Capella	280°38.2	46°00.6
13	51°23.8	349°28.8	38.7	276°47.7	22.0	57°09.0	37.2	231°35.5	50.7	Bellatrix	278°34.7	6°21.5
14	66°26.2	4°28.1	39.4	291°49.2	21.7	72°11.0	37.0	246°38.1	50.7	Elnath	278°15.8	28°37.0
15	81°28.7	19°27.3	•• 40.0	306°50.7	•• 21.4	87°13.0	•• 36.8	261°40.7	•• 50.7	Alnilam	275°48.9	-1°11.8
16	96°31.2	34°26.6	40.7	321°52.1	21.0	102°15.0	36.6	276°43.2	50.8	Betelgeuse	271°04.0	7°24.5
17	111°33.6	49°25.9	41.4	336°53.6	20.7	117°17.0	36.4	291°45.8	50.8	Canopus	263°57.5	-52°42.3
18	126°36.1	64°25.2	N21°42.1	351°55.1	N19°20.4	132°19.0	S03°36.2	306°48.4	N02°50.9	Sirius	258°35.9	-16°44.0
19	141°38.6	79°24.4	42.7	6°56.6	20.0	147°21.0	36.0	321°51.0	50.9	Adhara	255°14.5	-28°59.4
20	156°41.0	94°23.7	43.4	21°58.0	19.7	162°23.0	35.8	336°53.6	51.0	Procyon	245°02.2	5°11.8
21	171°43.5	109°23.0	•• 44.1	36°59.5	•• 19.4	177°25.0	•• 35.6	351°56.2	•• 51.0	Pollux	243°30.6	28°00.1
22	186°45.9	124°22.3	44.8	52°01.0	19.0	192°27.0	35.4	6°58.7	51.1	Avior	234°19.1	-59°32.9
23	201°48.4	139°21.5	45.4	67°02.5	18.7	207°28.9	35.2	22°01.3	51.1	Suhail	222°54.2	-43°28.8
Mer.pass. 09:35		ν -0.7' d 0.7' m -3.89		ν 1.5' d -0.3' m 0.67		ν 2.0' d -0.2' m -2.11		ν 2.6' d 0.0' m 0.69		Miaplacidus	221°40.3	-69°46.0
										Alphard	217°58.3	-8°42.4
										Regulus	207°45.8	11°54.9
										Dubhe	193°53.9	61°41.8
										Denebola	182°35.7	14°30.7
										Gienah	175°54.4	-17°36.2
										Acrux	173°11.5	-63°09.7
										Gacrux	172°03.1	-57°10.6
										Alioth	166°22.0	55°54.2
										Spica	158°33.4	-11°13.1
										Alkaid	153°00.1	49°15.6
										Hadar	148°50.7	-60°25.5
										Menkent	148°10.0	-36°25.4
										Arcturus	145°57.5	19°07.5
										Rigel Kent.	139°54.3	-60°52.8
										Kochab	137°18.4	74°06.7
										Zuben'ubi	137°07.7	-16°05.3
										Alphecca	126°12.6	26°40.6
										Antares	112°28.8	-26°27.4
										Atria	107°32.3	-69°02.7
										Sabik	102°14.9	-15°44.3
										Shaula	96°24.7	-37°06.6
										Rasalhague	96°08.3	12°33.0
										Eltanin	90°46.9	51°29.0
										Kaus Aust.	83°46.6	-34°22.7
										Vega	80°40.3	38°47.3
										Nunki	76°01.0	-26°17.0
										Altair	62°10.4	8°53.6
										Peacock	53°22.7	-56°41.8
										Deneb	49°33.1	45°18.8
										Enif	33°49.5	9°55.3
										Al Na'ir	27°46.7	-46°54.4
										Fomalhaut	15°26.7	-29°33.9
										Scheat	13°55.9	28°08.2
										Markab	13°40.8	15°15.6
										Apr 28 Wed	SHA	Mer.pass
										Venus	298°46.4	13:42
										Mars	225°36.7	18:32
										Jupiter	5°51.4	09:12
										Saturn	180°10.2	21:32
										Apr 29 Thu	SHA	Mer.pass
										Venus	297°29.9	13:43
										Mars	225°13.1	18:30
										Jupiter	5°40.1	09:09
										Saturn	180°13.0	21:28
										Apr 30 Fri	SHA	Mer.pass
										Venus	296°13.2	13:44
										Mars	224°49.2	18:28
										Jupiter	5°28.8	09:06
										Saturn	180°15.8	21:24

h	Sun				Moon			
Wed	GHA	Dec			GHA	ν	Dec	HP
0	180°36.4	N14°03.0			8°50.7	8.8'	S16°18.6	11.6' 58.8'
1	195°36.5	03.8			23°18.6	8.8'	16°30.2	11.5' 58.8'
2	210°36.6	04.5			37°46.4	8.7'	16°41.7	11.4' 58.8'
3	225°36.7	· · 05.3			52°14.1	8.7'	16°53.2	11.3' 58.8'
4	240°36.8	06.1			66°41.8	8.7'	17°04.5	11.2' 58.7'
5	255°36.9	06.9			81°09.4	8.6'	17°15.7	11.1' 58.7'
6	270°37.0	N14°07.7			95°37.1	8.6'	S17°26.8	11.0' 58.7'
7	285°37.1	08.5			110°04.6	8.5'	17°37.8	10.9' 58.7'
8	300°37.2	09.3			124°32.1	8.5'	17°48.7	10.8' 58.6'
9	315°37.3	· · 10.1			138°59.6	8.4'	17°59.5	10.7' 58.6'
10	330°37.4	10.8			153°27.1	8.4'	18°10.1	10.5' 58.6'
11	345°37.5	11.6			167°54.5	8.4'	18°20.7	10.4' 58.6'
12	0°37.5	N14°12.4			182°21.8	8.3'	S18°31.1	10.3' 58.5'
13	15°37.6	13.2			196°49.1	8.3'	18°41.4	10.2' 58.5'
14	30°37.7	14.0			211°16.4	8.2'	18°51.6	10.1' 58.5'
15	45°37.8	· · 14.8			225°43.6	8.2'	19°01.7	10.0' 58.5'
16	60°37.9	15.5			240°10.8	8.1'	19°11.6	9.8' 58.5'
17	75°38.0	16.3			254°38.0	8.1'	19°21.5	9.7' 58.4'
18	90°38.1	N14°17.1			269°05.1	8.1'	S19°31.2	9.6' 58.4'
19	105°38.2	17.9			283°32.1	8.0'	19°40.8	9.5' 58.4'
20	120°38.3	18.7			297°59.2	8.0'	19°50.2	9.3' 58.3'
21	135°38.4	· · 19.5			312°26.2	8.0'	19°59.6	9.2' 58.3'
22	150°38.5	20.2			326°53.1	7.9'	20°08.8	9.1' 58.3'
23	165°38.5	21.0			341°20.0	7.9'	20°17.8	9.0' 58.3'
SD = 15.9' d = 0.8' SD = 16.0'								
Thu	GHA	Dec			GHA	ν	Dec	HP
0	180°38.6	N14°21.8			355°46.9	7.8'	S20°26.8	8.8' 58.2'
1	195°38.7	22.6			10°13.8	7.8'	20°35.6	8.7' 58.2'
2	210°38.8	23.3			24°40.6	7.8'	20°44.3	8.6' 58.2'
3	225°38.9	· · 24.1			39°07.4	7.7'	20°52.9	8.4' 58.2'
4	240°39.0	24.9			53°34.1	7.7'	21°01.3	8.3' 58.1'
5	255°39.1	25.7			68°00.8	7.7'	21°09.6	8.2' 58.1'
6	270°39.2	N14°26.5			82°27.5	7.7'	S21°17.8	8.0' 58.1'
7	285°39.3	27.2			96°54.2	7.6'	21°25.8	7.9' 58.0'
8	300°39.3	28.0			111°20.8	7.6'	21°33.7	7.8' 58.0'
9	315°39.4	· · 28.8			125°47.4	7.6'	21°41.4	7.6' 58.0'
10	330°39.5	29.6			140°14.0	7.5'	21°49.1	7.5' 58.0'
11	345°39.6	30.3			154°40.5	7.5'	21°56.5	7.3' 57.9'
12	0°39.7	N14°31.1			169°07.0	7.5'	S22°03.9	7.2' 57.9'
13	15°39.8	31.9			183°33.5	7.5'	22°11.1	7.1' 57.9'
14	30°39.9	32.7			198°00.0	7.4'	22°18.2	6.9' 57.9'
15	45°40.0	· · 33.4			212°26.4	7.4'	22°25.1	6.8' 57.8'
16	60°40.0	34.2			226°52.8	7.4'	22°31.9	6.6' 57.8'
17	75°40.1	35.0			241°19.2	7.4'	22°38.5	6.5' 57.8'
18	90°40.2	N14°35.8			255°45.6	7.4'	S22°45.0	6.4' 57.7'
19	105°40.3	36.5			270°12.0	7.3'	22°51.4	6.2' 57.7'
20	120°40.4	37.3			284°38.3	7.3'	22°57.6	6.1' 57.7'
21	135°40.5	· · 38.1			299°04.6	7.3'	23°03.7	5.9' 57.6'
22	150°40.5	38.8			313°30.9	7.3'	23°09.6	5.8' 57.6'
23	165°40.6	39.6			327°57.2	7.3'	23°15.4	5.6' 57.6'
SD = 15.9' d = 0.8' SD = 15.9'								
Fri	GHA	Dec			GHA	ν	Dec	HP
0	180°40.7	N14°40.4			342°23.5	7.3'	S23°21.0	5.5' 57.6'
1	195°40.8	41.2			356°49.8	7.3'	23°26.5	5.4' 57.5'
2	210°40.9	41.9			11°16.1	7.3'	23°31.9	5.2' 57.5'
3	225°41.0	· · 42.7			25°42.3	7.2'	23°37.1	5.1' 57.5'
4	240°41.1	43.5			40°08.6	7.2'	23°42.1	4.9' 57.4'
5	255°41.1	44.2			54°34.8	7.2'	23°47.0	4.8' 57.4'
6	270°41.2	N14°45.0			69°01.1	7.2'	S23°51.8	4.6' 57.4'
7	285°41.3	45.8			83°27.3	7.2'	23°56.4	4.5' 57.3'
8	300°41.4	46.5			97°53.5	7.2'	24°00.9	4.3' 57.3'
9	315°41.5	· · 47.3			112°19.8	7.2'	24°05.2	4.2' 57.3'
10	330°41.5	48.1			126°46.0	7.2'	24°09.4	4.0' 57.3'
11	345°41.6	48.8			141°12.2	7.2'	24°13.4	3.9' 57.2'
12	0°41.7	N14°49.6			155°38.5	7.2'	S24°17.3	3.7' 57.2'
13	15°41.8	50.4			170°04.7	7.2'	24°21.1	3.6' 57.2'
14	30°41.9	51.1			184°31.0	7.3'	24°24.6	3.4' 57.1'
15	45°42.0	· · 51.9			198°57.2	7.3'	24°28.1	3.3' 57.1'
16	60°42.0	52.6			213°23.5	7.3'	24°31.4	3.1' 57.1'
17	75°42.1	53.4			227°49.7	7.3'	24°34.5	3.0' 57.0'
18	90°42.2	N14°54.2			242°16.0	7.3'	S24°37.5	2.8' 57.0'
19	105°42.3	54.9			256°42.3	7.3'	24°40.4	2.7' 57.0'
20	120°42.4	55.7			271°08.6	7.3'	24°43.1	2.6' 56.9'
21	135°42.4	· · 56.5			285°35.0	7.3'	24°45.6	2.4' 56.9'
22	150°42.5	57.2			300°01.3	7.4'	24°48.0	2.3' 56.9'
23	165°42.6	58.0			314°27.7	7.4'	24°50.3	2.1' 56.9'
SD = 15.9' d = 0.8' SD = 15.7'								

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	////	////	02:10	21:51	////	////
N 70°	////	////	02:43	21:16	////	////
68°	////	01:28	03:07	20:51	22:35	////
66°	////	02:08	03:26	20:31	21:51	////
64°	////	02:35	03:41	20:16	21:23	////
62°	01:18	02:56	03:53	20:03	21:02	22:44
60°	01:54	03:12	04:04	19:53	20:45	22:05
N 58°	02:19	03:26	04:13	19:43	20:31	21:39
56°	02:38	03:37	04:21	19:35	20:19	21:19
54°	02:54	03:47	04:28	19:28	20:09	21:03
52°	03:07	03:56	04:35	19:21	20:00	20:50
50°	03:18	04:04	04:40	19:15	19:52	20:38
45°	03:41	04:21	04:53	19:03	19:35	20:15
N 40°	03:58	04:34	05:03	18:52	19:22	19:57
35°	04:12	04:45	05:12	18:44	19:11	19:43
30°	04:24	04:54	05:19	18:36	19:01	19:31
20°	04:42	05:09	05:32	18:23	18:46	19:13
N 10°	04:56	05:22	05:43	18:11	18:33	18:59
0°	05:08	05:33	05:54	18:01	18:22	18:47
S 10°	05:18	05:43	06:04	17:50	18:12	18:37
20°	05:26	05:53	06:15	17:39	18:02	18:28
30°	05:35	06:03	06:28	17:27	17:51	18:20
35°	05:39	06:09	06:35	17:20	17:46	18:16
40°	05:43	06:15	06:43	17:11	17:40	18:12
45°	05:47	06:21	06:52	17:02	17:33	18:07
S 50°	05:51	06:29	07:03	16:51	17:25	18:03
52°	05:53	06:33	07:09	16:45	17:21	18:01
54°	05:55	06:37	07:14	16:40	17:17	17:59
56°	05:57	06:41	07:21	16:33	17:13	17:57
58°	05:59	06:46	07:28	16:26	17:08	17:55
S 60°	06:01	06:51	07:36	16:18	17:03	17:52

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	■	■	■	00:59	■	■
N 70°	■	■	■	01:44	■	■
68°	22:42	■	■	02:15	01:39	■
66°	21:57	· ·	00:22	02:37	02:25	01:56
64°	21:27	23:19	· ·	02:55	02:55	02:59
62°	21:05	22:45	· ·	03:10	03:18	03:34
60°	20:47	22:20	23:39	03:23	03:37	04:00
N 58°	20:32	22:00	23:16	03:34	03:52	04:20
56°	20:20	21:44	22:57	03:43	04:06	04:37
54°	20:09	21:30	22:41	03:52	04:17	04:51
52°	19:59	21:18	22:27	04:00	04:27	05:03
50°	19:50	21:07	22:15	04:06	04:36	05:14
45°	19:32	20:45	21:50	04:21	04:56	05:37
N 40°	19:17	20:27	21:31	04:33	05:12	05:56
35°	19:05	20:12	21:14	04:44	05:25	06:11
30°	18:54	19:59	21:00	04:53	05:37	06:25
20°	18:35	19:37	20:36	05:09	05:57	06:48
N 10°	18:19	19:18	20:15	05:23	06:14	07:08
0°	18:04	19:00	19:56	05:36	06:31	07:27
S 10°	17:50	18:42	19:36	05:49	06:47	07:45
20°	17:34	18:23	19:16	06:03	07:05	08:05
30°	17:16	18:02	18:52	06:19	07:25	08:29
35°	17:05	17:49	18:38	06:29	07:37	08:42
40°	16:54	17:35	18:22	06:40	07:51	08:58
45°	16:40	17:17	18:03	06:53	08:07	09:17
S 50°	16:23	16:56	17:39	07:08	08:27	09:40
52°	16:15	16:46	17:27	07:15	08:37	09:52
54°	16:06	16:35	17:14	07:24	08:48	10:04
56°	15:56	16:22	16:59	07:33	09:00	10:19
58°	15:45	16:08	16:41	07:43	09:14	10:37
S 60°	15:32	15:50	16:20	07:55	09:31	10:58

Day	Sun			Moon		
	Eqn. of Time		Mer.	Mer. Pass.	Age	
	00 ^h	12 ^h	Pass	Upper	Lower	
	mm:ss	mm:ss	hh:mm	hh:mm	hh:mm	
28	02:26	02:30	11:57	--:--	11:50	
29	02:35	02:39	11:57	00:18	12:45	
30	02:43	02:47	11:57	01:13	13:41	

May 01, 02, 03 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	218°49.2	153°45.4	N22°16.9	83°14.3	N19°02.0	224°06.7	S03°25.7	39°07.7	N02°53.2	Alpheratz	357°46.2	29°08.8
1	233°51.6	168°44.7	17.5	98°15.7	01.6	239°08.7	25.5	54°10.3	53.3	Ankaa	353°18.2	-42°14.8
2	248°54.1	183°43.9	18.2	113°17.2	01.3	254°10.7	25.3	69°12.9	53.3	Schedar	349°43.8	56°35.5
3	263°56.5	198°43.2	•• 18.8	128°18.7	•• 00.9	269°12.7	•• 25.1	84°15.4	•• 53.3	Diphda	348°58.4	-17°55.7
4	278°59.0	213°42.4	19.4	143°20.1	00.6	284°14.7	24.9	99°18.0	53.4	Achernar	335°28.9	-57°10.9
5	294°01.5	228°41.7	20.0	158°21.6	19°00.3	299°16.7	24.7	114°20.6	53.4	Hamal	328°03.7	23°30.6
6	309°03.9	243°40.9	N22°20.6	173°23.0	N18°59.9	314°18.7	S03°24.5	129°23.2	N02°53.5	Polaris	319°21.6	89°18.6
7	324°06.4	258°40.2	21.2	188°24.5	59.6	329°20.7	24.4	144°25.8	53.5	Acamar	315°20.4	-40°15.8
8	339°08.9	273°39.4	21.8	203°25.9	59.2	344°22.7	24.2	159°28.3	53.5	Menkar	314°17.8	4°07.8
9	354°11.3	288°38.7	•• 22.4	218°27.4	•• 58.9	359°24.7	•• 24.0	174°30.9	•• 53.6	Mirfak	308°44.2	49°53.9
10	9°13.8	303°37.9	23.0	233°28.8	58.5	14°26.7	23.8	189°33.5	53.6	Aldebaran	290°52.3	16°31.8
11	24°16.3	318°37.2	23.6	248°30.3	58.2	29°28.7	23.6	204°36.1	53.7	Rigel	281°14.5	-8°11.5
12	39°18.7	333°36.4	N22°24.2	263°31.7	N18°57.8	44°30.7	S03°23.4	219°38.6	N02°53.7	Capella	280°38.2	46°00.6
13	54°21.2	348°35.7	24.8	278°33.2	57.5	59°32.7	23.2	234°41.2	53.8	Bellatrix	278°34.7	6°21.5
14	69°23.7	3°34.9	25.5	293°34.6	57.1	74°34.7	23.0	249°43.8	53.8	Elnath	278°15.8	28°37.0
15	84°26.1	18°34.2	•• 26.1	308°36.1	•• 56.8	89°36.7	•• 22.8	264°46.4	•• 53.8	Alnilam	275°48.9	-1°11.8
16	99°28.6	33°33.4	26.7	323°37.5	56.4	104°38.7	22.6	279°48.9	53.9	Betelgeuse	271°04.0	7°24.5
17	114°31.0	48°32.7	27.3	338°39.0	56.1	119°40.7	22.4	294°51.5	53.9	Canopus	263°57.5	-52°42.3
18	129°33.5	63°31.9	N22°27.9	353°40.4	N18°55.7	134°42.7	S03°22.2	309°54.1	N02°54.0	Sirius	258°35.9	-16°44.0
19	144°36.0	78°31.2	28.4	8°41.9	55.4	149°44.7	22.0	324°56.6	54.0	Adhara	255°14.5	-28°59.4
20	159°38.4	93°30.4	29.0	23°43.3	55.0	164°46.7	21.9	339°59.2	54.0	Procyon	245°02.2	5°11.8
21	174°40.9	108°29.7	•• 29.6	38°44.8	•• 54.7	179°48.7	•• 21.7	355°01.8	•• 54.1	Pollux	243°30.6	28°00.1
22	189°43.4	123°28.9	30.2	53°46.2	54.3	194°50.7	21.5	10°04.4	54.1	Avior	234°19.2	-59°32.9
23	204°45.8	138°28.1	30.8	68°47.7	54.0	209°52.7	21.3	25°06.9	54.2	Suhail	222°54.2	-43°28.8
Mer.pass. 09:23		ν -0.7' d 0.6' m -3.89		ν 1.5' d -0.3' m 0.72		ν 2.0' d -0.2' m -2.12		ν 2.6' d 0.0' m 0.70				

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	219°48.3	153°27.4	N22°31.4	83°49.1	N18°53.6	224°54.7	S03°21.1	40°09.5	N02°54.2
1	234°50.8	168°26.6	32.0	98°50.6	53.3	239°56.7	20.9	55°12.1	54.2
2	249°53.2	183°25.9	32.6	113°52.0	52.9	254°58.7	20.7	70°14.7	54.3
3	264°55.7	198°25.1	•• 33.2	128°53.5	•• 52.6	270°00.7	•• 20.5	85°17.2	•• 54.3
4	279°58.1	213°24.4	33.8	143°54.9	52.2	285°02.7	20.3	100°19.8	54.4
5	295°00.6	228°23.6	34.4	158°56.3	51.9	300°04.7	20.1	115°22.4	54.4
6	310°03.1	243°22.8	N22°34.9	173°57.8	N18°51.5	315°06.8	S03°19.9	130°25.0	N02°54.4
7	325°05.5	258°22.1	35.5	188°59.2	51.2	330°08.8	19.8	145°27.5	54.5
8	340°08.0	273°21.3	36.1	204°00.7	50.8	345°10.8	19.6	160°30.1	54.5
9	355°10.5	288°20.6	•• 36.7	219°02.1	•• 50.5	0°12.8	•• 19.4	175°32.7	•• 54.6
10	10°12.9	303°19.8	37.3	234°03.6	50.1	15°14.8	19.2	190°35.2	54.6
11	25°15.4	318°19.1	37.9	249°05.0	49.8	30°16.8	19.0	205°37.8	54.6
12	40°17.9	333°18.3	N22°38.4	264°06.4	N18°49.4	45°18.8	S03°18.8	220°40.4	N02°54.7
13	55°20.3	348°17.5	39.0	279°07.9	49.1	60°20.8	18.6	235°43.0	54.7
14	70°22.8	3°16.8	39.6	294°09.3	48.7	75°22.8	18.4	250°45.5	54.8
15	85°25.3	18°16.0	•• 40.2	309°10.8	•• 48.4	90°24.8	•• 18.2	265°48.1	•• 54.8
16	100°27.7	33°15.2	40.7	324°12.2	48.0	105°26.8	18.0	280°50.7	54.8
17	115°30.2	48°14.5	41.3	339°13.6	47.7	120°28.8	17.8	295°53.2	54.9
18	130°32.6	63°13.7	N22°41.9	354°15.1	N18°47.3	135°30.8	S03°17.7	310°55.8	N02°54.9
19	145°35.1	78°13.0	42.5	9°16.5	46.9	150°32.8	17.5	325°58.4	55.0
20	160°37.6	93°12.2	43.0	24°18.0	46.6	165°34.8	17.3	341°01.0	55.0
21	175°40.0	108°11.4	•• 43.6	39°19.4	•• 46.2	180°36.8	•• 17.1	356°03.5	•• 55.0
22	190°42.5	123°10.7	44.2	54°20.8	45.9	195°38.8	16.9	11°06.1	55.1
23	205°45.0	138°09.9	44.7	69°22.3	45.5	210°40.8	16.7	26°08.7	55.1
Mer.pass. 09:19		ν -0.8' d 0.6' m -3.89		ν 1.4' d -0.4' m 0.73		ν 2.0' d -0.2' m -2.13		ν 2.6' d 0.0' m 0.71	

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	220°47.4	153°09.1	N22°45.3	84°23.7	N18°45.2	225°42.8	S03°16.5	41°11.2	N02°55.1
1	235°49.9	168°08.4	45.9	99°25.2	44.8	240°44.8	16.3	56°13.8	55.2
2	250°52.4	183°07.6	46.4	114°26.6	44.5	255°46.8	16.1	71°16.4	55.2
3	265°54.8	198°06.8	•• 47.0	129°28.0	•• 44.1	270°48.8	•• 15.9	86°18.9	•• 55.3
4	280°57.3	213°06.1	47.6	144°29.5	43.8	285°50.8	15.8	101°21.5	55.3
5	295°59.8	228°05.3	48.1	159°30.9	43.4	300°52.8	15.6	116°24.1	55.3
6	311°02.2	243°04.5	N22°48.7	174°32.3	N18°43.0	315°54.9	S03°15.4	131°26.7	N02°55.4
7	326°04.7	258°03.8	49.2	189°33.8	42.7	330°56.9	15.2	146°29.2	55.4
8	341°07.1	273°03.0	49.8	204°35.2	42.3	345°58.9	15.0	161°31.8	55.5
9	356°09.6	288°02.2	•• 50.3	219°36.6	•• 42.0	1°00.9	•• 14.8	176°34.4	•• 55.5
10	11°12.1	303°01.5	50.9	234°38.1	41.6	16°02.9	14.6	191°36.9	55.5
11	26°14.5	318°00.7	51.5	249°39.5	41.3	31°04.9	14.4	206°39.5	55.6
12	41°17.0	332°59.9	N22°52.0	264°40.9	N18°40.9	46°06.9	S03°14.2	221°42.1	N02°55.6
13	56°19.5	347°59.2	52.6	279°42.4	40.6	61°08.9	14.0	236°44.6	55.6
14	71°21.9	2°58.4	53.1	294°43.8	40.2	76°10.9	13.9	251°47.2	55.7
15	86°24.4	17°57.6	•• 53.7	309°45.2	•• 39.8	91°12.9	•• 13.7	266°49.8	•• 55.7
16	101°26.9	32°56.9	54.2	324°46.7	39.5	106°14.9	13.5	281°52.3	55.8
17	116°29.3	47°56.1	54.8	339°48.1	39.1	121°16.9	13.3	296°54.9	55.8
18	131°31.8	62°55.3	N22°55.3	354°49.5	N18°38.8	136°18.9	S03°13.1	311°57.5	N02°55.8
19	146°34.3	77°54.5	55.8	9°51.0	38.4	151°20.9	12.9	327°00.0	55.9
20	161°36.7	92°53.8	56.4	24°52.4	38.1	166°22.9	12.7	342°02.6	55.9
21	176°39.2	107°53.0	•• 56.9	39°53.8	•• 37.7	181°25.0	•• 12.5	357°05.2	•• 55.9
22	191°41.6	122°52.2	57.5	54°55.3	37.3	196°27.0	12.3	12°07.7	56.0
23	206°44.1	137°51.5	58.0	69°56.7	37.0	211°29.0	12.2	27°10.3	56.0
Mer.pass. 09:15		ν -0.8' d 0.6' m -3.89		ν 1.4' d -0.4' m 0.75		ν 2.0' d -0.2' m -2.13		ν 2.6' d 0.0' m 0.71	

May 01 Sat	SHA	Mer.pass
Venus	294°56.3	13:46
Mars	224°25.1	18:25
Jupiter	5°17.6	09:02
Saturn	180°18.6	21:20

May 02 Sun	SHA	Mer.pass
Venus	293°39.1	13:47
Mars	224°00.8	18:23
Jupiter	5°06.4	08:59
Saturn	180°21.2	21:16

May 03 Mon	SHA	Mer.pass
Venus	292°21.7	13:48
Mars	223°36.3	18:21
Jupiter	4°55.4	08:56
Saturn	180°23.8	21:12

Horizontal parallax	
Venus:	0.1
Mars:	0.1

Sun				Moon			
h	Sat	GHA	Dec	GHA	ν	Dec	d HP
0	180°42.7	N14°58.7	328°54.1	7.4'	S24°52.4	2.0'	56.8'
1	195°42.7	14°59.5	343°20.5	7.4'	24°54.3	1.8'	56.8'
2	210°42.8	15°00.3	357°46.9	7.5'	24°56.2	1.7'	56.8'
3	225°42.9	· · 01.0	12°13.4	7.5'	24°57.8	1.5'	56.7'
4	240°43.0	01.8	26°39.9	7.5'	24°59.3	1.4'	56.7'
5	255°43.1	02.5	41°06.4	7.5'	25°00.7	1.2'	56.7'
6	270°43.1	N15°03.3	55°32.9	7.6'	S25°02.0	1.1'	56.6'
7	285°43.2	04.0	69°59.5	7.6'	25°03.0	0.9'	56.6'
8	300°43.3	04.8	84°26.1	7.6'	25°04.0	0.8'	56.6'
9	315°43.4	· · 05.6	98°52.7	7.7'	25°04.8	0.7'	56.5'
10	330°43.4	06.3	113°19.4	7.7'	25°05.4	0.5'	56.5'
11	345°43.5	07.1	127°46.1	7.7'	25°05.9	0.4'	56.5'
12	0°43.6	N15°07.8	142°12.8	7.8'	S25°06.3	0.2'	56.5'
13	15°43.7	08.6	156°39.6	7.8'	25°06.5	0.1'	56.4'
14	30°43.7	09.3	171°06.4	7.9'	25°06.6	-0.1'	56.4'
15	45°43.8	· · 10.1	185°33.2	7.9'	25°06.5	-0.2'	56.4'
16	60°43.9	10.8	200°00.1	7.9'	25°06.3	-0.3'	56.3'
17	75°44.0	11.6	214°27.1	8.0'	25°06.0	-0.5'	56.3'
18	90°44.0	N15°12.3	228°54.1	8.0'	S25°05.5	-0.6'	56.3'
19	105°44.1	13.1	243°21.1	8.1'	25°04.9	-0.8'	56.2'
20	120°44.2	13.8	257°48.2	8.1'	25°04.1	-0.9'	56.2'
21	135°44.3	· · 14.6	272°15.3	8.2'	25°03.2	-1.0'	56.2'
22	150°44.4	15.3	286°42.5	8.2'	25°02.1	-1.2'	56.2'
23	165°44.4	16.1	301°09.7	8.3'	25°01.0	-1.3'	56.1'
SD = 15.9'				SD = 15.5'			

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	180°44.5	N15°16.8	315°37.0	8.3'	S24°59.6	-1.5'	56.1'
1	195°44.6	17.6	330°04.3	8.4'	24°58.2	-1.6'	56.1'
2	210°44.6	18.3	344°31.7	8.4'	24°56.6	-1.7'	56.0'
3	225°44.7	· · 19.1	358°59.1	8.5'	24°54.9	-1.9'	56.0'
4	240°44.8	19.8	13°26.6	8.5'	24°53.0	-2.0'	56.0'
5	255°44.8	20.6	27°54.2	8.6'	24°51.0	-2.1'	56.0'
6	270°44.9	N15°21.3	42°21.8	8.7'	S24°48.9	-2.3'	55.9'
7	285°45.0	22.1	56°49.4	8.7'	24°46.6	-2.4'	55.9'
8	300°45.1	22.8	71°17.2	8.8'	24°44.2	-2.5'	55.9'
9	315°45.1	· · 23.6	85°44.9	8.8'	24°41.7	-2.7'	55.8'
10	330°45.2	24.3	100°12.8	8.9'	24°39.0	-2.8'	55.8'
11	345°45.3	25.1	114°40.7	9.0'	24°36.2	-2.9'	55.8'
12	0°45.3	N15°25.8	129°08.7	9.0'	S24°33.3	-3.0'	55.8'
13	15°45.4	26.5	143°36.7	9.1'	24°30.3	-3.2'	55.7'
14	30°45.5	27.3	158°04.8	9.2'	24°27.1	-3.3'	55.7'
15	45°45.5	· · 28.0	172°33.0	9.2'	24°23.8	-3.4'	55.7'
16	60°45.6	28.8	187°01.2	9.3'	24°20.4	-3.5'	55.7'
17	75°45.7	29.5	201°29.5	9.4'	24°16.9	-3.7'	55.6'
18	90°45.7	N15°30.3	215°57.9	9.4'	S24°13.2	-3.8'	55.6'
19	105°45.8	31.0	230°26.3	9.5'	24°09.4	-3.9'	55.6'
20	120°45.9	31.7	244°54.8	9.6'	24°05.5	-4.0'	55.6'
21	135°46.0	· · 32.5	259°23.4	9.6'	24°01.5	-4.2'	55.5'
22	150°46.0	33.2	273°52.0	9.7'	23°57.3	-4.3'	55.5'
23	165°46.1	34.0	288°20.7	9.8'	23°53.1	-4.4'	55.5'
	SD = 15.9'	$d = 0.7'$			SD = 15.3'		

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	180°46.2	N15°34.7	302°49.5	9.9'	S23°48.7	-4.5'	55.4'
1	195°46.2	35.4	317°18.4	9.9'	23°44.2	-4.6'	55.4'
2	210°46.3	36.2	331°47.3	10.0'	23°39.5	-4.7'	55.4'
3	225°46.4	· · 36.9	346°16.3	10.1'	23°34.8	-4.9'	55.4'
4	240°46.4	37.6	0°45.4	10.1'	23°30.0	-5.0'	55.3'
5	255°46.5	38.4	15°14.5	10.2'	23°25.0	-5.1'	55.3'
6	270°46.5	N15°39.1	29°43.8	10.3'	S23°19.9	-5.2'	55.3'
7	285°46.6	39.9	44°13.1	10.4'	23°14.7	-5.3'	55.3'
8	300°46.7	40.6	58°42.4	10.4'	23°09.4	-5.4'	55.3'
9	315°46.7	· · 41.3	73°11.9	10.5'	23°04.0	-5.5'	55.2'
10	330°46.8	42.1	87°41.4	10.6'	22°58.5	-5.6'	55.2'
11	345°46.9	42.8	102°11.0	10.7'	22°52.9	-5.7'	55.2'
12	0°46.9	N15°43.5	116°40.7	10.7'	S22°47.1	-5.8'	55.2'
13	15°47.0	44.3	131°10.4	10.8'	22°41.3	-5.9'	55.1'
14	30°47.1	45.0	145°40.2	10.9'	22°35.3	-6.1'	55.1'
15	45°47.1	· · 45.7	160°10.1	11.0'	22°29.3	-6.2'	55.1'
16	60°47.2	46.5	174°40.1	11.1'	22°23.1	-6.3'	55.1'
17	75°47.3	47.2	189°10.2	11.1'	22°16.9	-6.4'	55.0'
18	90°47.3	N15°47.9	203°40.3	11.2'	S22°10.5	-6.5'	55.0'
19	105°47.4	48.6	218°10.5	11.3'	22°04.0	-6.6'	55.0'
20	120°47.4	49.4	232°40.8	11.4'	21°57.5	-6.7'	55.0'
21	135°47.5	· · 50.1	247°11.2	11.4'	21°50.8	-6.8'	55.0'
22	150°47.6	50.8	261°41.6	11.5'	21°44.0	-6.9'	54.9'
23	165°47.6	51.6	276°12.1	11.6'	21°37.2	-7.0'	54.9'
	SD = 15.9'	$d = 0.7'$			SD = 15.1'		

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	////	////	01:45	22:17	////	////
N 70°	////	////	02:26	21:33	////	////
68°	////	00:57	02:53	21:04	23:10	////
66°	////	01:50	03:14	20:43	22:09	////
64°	////	02:21	03:31	20:26	21:36	////
62°	00:50	02:44	03:44	20:12	21:13	23:15
60°	01:38	03:02	03:56	20:00	20:54	22:21
N 58°	02:06	03:17	04:06	19:50	20:39	21:51
56°	02:28	03:30	04:14	19:41	20:26	21:29
54°	02:45	03:40	04:22	19:33	20:15	21:11
52°	02:59	03:50	04:29	19:26	20:05	20:57
50°	03:11	03:58	04:35	19:20	19:57	20:44
45°	03:36	04:16	04:48	19:07	19:39	20:20
N 40°	03:54	04:30	04:59	18:55	19:25	20:01
35°	04:09	04:41	05:08	18:46	19:13	19:46
30°	04:21	04:51	05:17	18:38	19:03	19:34
20°	04:40	05:07	05:30	18:24	18:47	19:14
N 10°	04:55	05:21	05:42	18:12	18:34	18:59
0°	05:07	05:32	05:54	18:00	18:22	18:47
S 10°	05:18	05:43	06:05	17:49	18:11	18:36
20°	05:27	05:53	06:16	17:37	18:00	18:26
30°	05:36	06:05	06:29	17:24	17:49	18:17
35°	05:41	06:11	06:37	17:16	17:43	18:13
40°	05:45	06:17	06:46	17:08	17:36	18:08
45°	05:50	06:25	06:56	16:58	17:28	18:03
S 50°	05:55	06:34	07:08	16:45	17:20	17:58
52°	05:57	06:37	07:13	16:40	17:16	17:56
54°	06:00	06:42	07:20	16:34	17:11	17:54
56°	06:02	06:46	07:27	16:27	17:07	17:51
58°	06:05	06:51	07:34	16:19	17:02	17:49
S 60°	06:07	06:57	07:43	16:10	16:56	17:46

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	■	■	■	■	■	■
66°	■	■	03:15	■	■	04:46
64°	00:59	01:56	02:11	03:17	04:15	05:50
62°	00:10	01:07	01:36	04:06	05:04	06:24
60°	· ·	00:36	01:11	04:37	05:35	06:49
N 58°	· ·	00:12	00:51	05:01	05:58	07:09
56°	23:53	· ·	00:34	05:20	06:17	07:25
54°	23:38	· ·	00:20	05:36	06:33	07:39
52°	23:24	· ·	00:07	05:50	06:46	07:51
50°	23:12	23:56	· ·	06:02	06:58	08:02
45°	22:47	23:33	· ·	06:27	07:23	08:24
N 40°	22:27	23:15	23:55	06:46	07:43	08:42
35°	22:11	22:59	23:41	07:03	07:59	08:57
30°	21:56	22:46	23:29	07:17	08:13	09:10
20°	21:32	22:23	23:09	07:42	08:37	09:32
N 10°	21:11	22:03	22:51	08:03	08:58	09:51
0°	20:51	21:44	22:35	08:23	09:17	10:09
S 10°	20:32	21:26	22:18	08:42	09:36	10:27
20°	20:11	21:06	22:00	09:03	09:57	10:46
30°	19:46	20:43	21:40	09:28	10:21	11:07
35°	19:32	20:29	21:28	09:42	10:35	11:20
40°	19:15	20:13	21:14	09:58	10:51	11:35
45°	18:56	19:55	20:57	10:18	11:10	11:52
S 50°	18:31	19:31	20:37	10:43	11:34	12:13
52°	18:19	19:20	20:27	10:55	11:45	12:23
54°	18:05	19:07	20:16	11:09	11:58	12:34
56°	17:49	18:52	20:04	11:25	12:13	12:47
58°	17:31	18:35	19:49	11:43	12:31	13:02
S 60°	17:07	18:14	19:32	12:07	12:52	13:20

Day	Sun		Mer. Pass hh:mm	Moon		Age 17-19 93-79%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
01	02:51	02:54	11:57	02:09	14:37	
02	02:58	03:01	11:57	03:04	15:31	
03	03:05	03:08	11:57	03:57	16:22	

May 04, 05, 06 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	221°46.6	152°50.7	N22°58.6	84°58.1	N18°36.6	226°31.0	S03°12.0	42°12.9	N02°56.1	Alpheratz	357°46.2	29°08.8
1	236°49.0	167°49.9	59.1	99°59.5	36.3	241°33.0	11.8	57°15.4	56.1	Ankaa	353°18.2	-42°14.8
2	251°51.5	182°49.1	22°59.6	115°01.0	35.9	256°35.0	11.6	72°18.0	56.1	Schedar	349°43.8	56°35.5
3	266°54.0	197°48.4	23°00.2	130°02.4	· · 35.5	271°37.0	· · 11.4	87°20.6	· · 56.2	Diphda	348°58.4	-17°55.7
4	281°56.4	212°47.6	00.7	145°03.8	35.2	286°39.0	11.2	102°23.1	56.2	Achernar	335°28.9	-57°10.9
5	296°58.9	227°46.8	01.2	160°05.2	34.8	301°41.0	11.0	117°25.7	56.2	Hamal	328°03.7	23°30.6
6	312°01.4	242°46.0	N23°01.8	175°06.7	N18°34.5	316°43.0	S03°10.8	132°28.3	N02°56.3	Polaris	319°21.3	89°18.6
7	327°03.8	257°45.3	02.3	190°08.1	34.1	331°45.0	10.7	147°30.8	56.3	Acamar	315°20.4	-40°15.8
8	342°06.3	272°44.5	02.8	205°09.5	33.7	346°47.0	10.5	162°33.4	56.3	Menkar	314°17.8	4°07.8
9	357°08.8	287°43.7	· · 03.4	220°10.9	· · 33.4	1°49.1	· · 10.3	177°36.0	· · 56.4	Mirfak	308°44.2	49°53.9
10	12°11.2	302°42.9	03.9	235°12.4	33.0	16°51.1	10.1	192°38.5	56.4	Aldebaran	290°52.3	16°31.8
11	27°13.7	317°42.2	04.4	250°13.8	32.7	31°53.1	09.9	207°41.1	56.5	Rigel	281°14.5	-8°11.5
12	42°16.1	332°41.4	N23°04.9	265°15.2	N18°32.3	46°55.1	S03°09.7	222°43.7	N02°56.5	Capella	280°38.2	46°00.6
13	57°18.6	347°40.6	05.5	280°16.6	31.9	61°57.1	09.5	237°46.2	56.5	Bellatrix	278°34.7	6°21.5
14	72°21.1	2°39.8	06.0	295°18.1	31.6	76°59.1	09.3	252°48.8	56.6	Elnath	278°15.8	28°37.0
15	87°23.5	17°39.1	· · 06.5	310°19.5	· · 31.2	92°01.1	· · 09.1	267°51.4	· · 56.6	Alnilam	275°48.9	-1°11.8
16	102°26.0	32°38.3	07.0	325°20.9	30.8	107°03.1	09.0	282°53.9	56.6	Betelgeuse	271°04.0	7°24.5
17	117°28.5	47°37.5	07.6	340°22.3	30.5	122°05.1	08.8	297°56.5	56.7	Canopus	263°57.5	-52°42.3
18	132°30.9	62°36.7	N23°08.1	355°23.8	N18°30.1	137°07.1	S03°08.6	312°59.1	N02°56.7	Sirius	258°35.9	-16°44.0
19	147°33.4	77°35.9	08.6	10°25.2	29.8	152°09.2	08.4	328°01.6	56.7	Adhara	255°14.5	-28°59.4
20	162°35.9	92°35.2	09.1	25°26.6	29.4	167°11.2	08.2	343°04.2	56.8	Procyon	245°02.2	5°11.8
21	177°38.3	107°34.4	· · 09.6	40°28.0	· · 29.0	182°13.2	· · 08.0	358°06.7	· · 56.8	Pollux	243°30.6	28°00.1
22	192°40.8	122°33.6	10.1	55°29.4	28.7	197°15.2	07.8	13°09.3	56.9	Avior	234°19.2	-59°32.9
23	207°43.2	137°32.8	10.7	70°30.9	28.3	212°17.2	07.6	28°11.9	56.9	Suhail	222°54.2	-43°28.8
Mer.pass. 09:11		ν -0.8' d 0.5' m -3.89		ν 1.4' d -0.4' m 0.76		ν 2.0' d -0.2' m -2.13		ν 2.6' d 0.0' m 0.72		Miaplacidus	221°40.4	-69°46.0
										Alphard	217°58.3	-8°42.4
										Regulus	207°45.8	11°54.9
										Dubhe	193°53.9	61°41.8
										Denebola	182°35.7	14°30.7
										Gienah	175°54.4	-17°36.2
										Acrux	173°11.6	-63°09.7
										Gacrux	172°03.2	-57°10.6
										Alioth	166°22.0	55°54.2
										Spica	158°33.4	-11°13.1
										Alkaid	153°00.1	49°15.6
										Hadar	148°50.7	-60°25.6
										Menkent	148°09.9	-36°25.5
										Arcturus	145°57.4	19°07.6
										Rigil Kent.	139°54.3	-60°52.8
										Kochab	137°18.4	74°06.7
										Zuben'ubi	137°07.6	-16°05.3
										Alphecca	126°12.5	26°40.6
										Antares	112°28.7	-26°27.4
										Atria	107°32.2	-69°02.7
										Sabik	102°14.9	-15°44.3
										Shaula	96°24.7	-37°06.6
										Rasalhague	96°08.3	12°33.0
										Eltanin	90°46.8	51°29.0
										Kaus Aust.	83°46.6	-34°22.7
										Vega	80°40.3	38°47.4
										Nunki	76°00.9	-26°17.0
										Altair	62°10.4	8°53.6
										Peacock	53°22.6	-56°41.8
										Deneb	49°33.1	45°18.8
										Enif	33°49.4	9°55.3
										Al Na'ir	27°46.6	-46°54.4
										Fomalhaut	15°26.6	-29°33.9
										Scheat	13°55.8	28°08.2
										Markab	13°40.8	15°15.6
										May 04 Tue	SHA	Mer.pass
										Venus	291°04.1	13:49
										Mars	223°11.5	18:18
										Jupiter	4°44.4	08:53
										Saturn	180°26.3	21:08
										May 05 Wed	SHA	Mer.pass
										Venus	289°46.3	13:51
										Mars	222°46.6	18:16
										Jupiter	4°33.5	08:50
										Saturn	180°28.7	21:03
										May 06 Thu	SHA	Mer.pass
										Venus	288°28.3	13:52
										Mars	222°21.4	18:14
										Jupiter	4°22.7	08:46
										Saturn	180°31.1	20:59

h Sun Moon						
Tue	GHA	Dec	GHA	ν	Dec	d HP
0	180°47.7	N15°52.3	290°42.7	11.7'	S21°30.2	-7.0' 54.9'
1	195°47.7	53.0	305°13.4	11.7'	21°23.2	-7.1' 54.9'
2	210°47.8	53.7	319°44.1	11.8'	21°16.1	-7.2' 54.9'
3	225°47.9	· · 54.5	334°14.9	11.9'	21°08.8	-7.3' 54.8'
4	240°47.9	55.2	348°45.8	12.0'	21°01.5	-7.4' 54.8'
5	255°48.0	55.9	3°16.8	12.0'	20°54.1	-7.5' 54.8'
6	270°48.0	N15°56.7	17°47.8	12.1'	S20°46.6	-7.6' 54.8'
7	285°48.1	57.4	32°19.0	12.2'	20°39.0	-7.7' 54.8'
8	300°48.2	58.1	46°50.2	12.3'	20°31.3	-7.8' 54.8'
9	315°48.2	· · 58.8	61°21.4	12.3'	20°23.5	-7.9' 54.7'
10	330°48.3	15°59.5	75°52.8	12.4'	20°15.6	-7.9' 54.7'
11	345°48.3	16°00.3	90°24.2	12.5'	20°07.7	-8.0' 54.7'
12	0°48.4	N16°01.0	104°55.7	12.6'	S19°59.7	-8.1' 54.7'
13	15°48.5	01.7	119°27.3	12.6'	19°51.5	-8.2' 54.7'
14	30°48.5	02.4	133°58.9	12.7'	19°43.3	-8.3' 54.6'
15	45°48.6	· · 03.2	148°30.6	12.8'	19°35.1	-8.4' 54.6'
16	60°48.6	03.9	163°02.4	12.9'	19°26.7	-8.4' 54.6'
17	75°48.7	04.6	177°34.3	12.9'	19°18.2	-8.5' 54.6'
18	90°48.7	N16°05.3	192°06.2	13.0'	S19°09.7	-8.6' 54.6'
19	105°48.8	06.0	206°38.2	13.1'	19°01.1	-8.7' 54.6'
20	120°48.9	06.8	221°10.3	13.1'	18°52.4	-8.8' 54.6'
21	135°48.9	· · 07.5	235°42.4	13.2'	18°43.7	-8.8' 54.5'
22	150°49.0	08.2	250°14.6	13.3'	18°34.8	-8.9' 54.5'
23	165°49.0	08.9	264°46.9	13.4'	18°25.9	-9.0' 54.5'
SD = 15.9' $d = 0.7'$ SD = 15.0'						
Wed	GHA	Dec	GHA	ν	Dec	d HP
0	180°49.1	N16°09.6	279°19.3	13.4'	S18°16.9	-9.1' 54.5'
1	195°49.1	10.3	293°51.7	13.5'	18°07.9	-9.1' 54.5'
2	210°49.2	11.1	308°24.2	13.6'	17°58.8	-9.2' 54.5'
3	225°49.2	· · 11.8	322°56.7	13.6'	17°49.6	-9.3' 54.5'
4	240°49.3	12.5	337°29.3	13.7'	17°40.3	-9.3' 54.4'
5	255°49.3	13.2	352°02.0	13.8'	17°30.9	-9.4' 54.4'
6	270°49.4	N16°13.9	6°34.8	13.8'	S17°21.5	-9.5' 54.4'
7	285°49.5	14.6	21°07.6	13.9'	17°12.0	-9.5' 54.4'
8	300°49.5	15.3	35°40.5	13.9'	17°02.5	-9.6' 54.4'
9	315°49.6	· · 16.1	50°13.4	14.0'	16°52.9	-9.7' 54.4'
10	330°49.6	16.8	64°46.4	14.1'	16°43.2	-9.7' 54.4'
11	345°49.7	17.5	79°19.5	14.1'	16°33.4	-9.8' 54.4'
12	0°49.7	N16°18.2	93°52.6	14.2'	S16°23.6	-9.9' 54.4'
13	15°49.8	18.9	108°25.8	14.3'	16°13.8	-9.9' 54.3'
14	30°49.8	19.6	122°59.1	14.3'	16°03.8	-10.0' 54.3'
15	45°49.9	· · 20.3	137°32.4	14.4'	15°53.8	-10.1' 54.3'
16	60°49.9	21.0	152°05.8	14.4'	15°43.8	-10.1' 54.3'
17	75°50.0	21.7	166°39.2	14.5'	15°33.6	-10.2' 54.3'
18	90°50.0	N16°22.5	181°12.7	14.5'	S15°23.4	-10.2' 54.3'
19	105°50.1	23.2	195°46.2	14.6'	15°13.2	-10.3' 54.3'
20	120°50.1	23.9	210°19.8	14.7'	15°02.9	-10.4' 54.3'
21	135°50.2	· · 24.6	224°53.5	14.7'	14°52.5	-10.4' 54.3'
22	150°50.2	25.3	239°27.2	14.8'	14°42.1	-10.5' 54.3'
23	165°50.3	26.0	254°01.0	14.8'	14°31.7	-10.5' 54.3'
SD = 15.9' $d = 0.7'$ SD = 14.9'						
Thu	GHA	Dec	GHA	ν	Dec	d HP
0	180°50.3	N16°26.7	268°34.8	14.9'	S14°21.1	-10.6' 54.3'
1	195°50.4	27.4	283°08.6	14.9'	14°10.5	-10.6' 54.2'
2	210°50.4	28.1	297°42.6	15.0'	13°59.9	-10.7' 54.2'
3	225°50.5	· · 28.8	312°16.5	15.0'	13°49.2	-10.7' 54.2'
4	240°50.5	29.5	326°50.6	15.1'	13°38.5	-10.8' 54.2'
5	255°50.6	30.2	341°24.6	15.1'	13°27.7	-10.8' 54.2'
6	270°50.6	N16°30.9	355°58.7	15.2'	S13°16.8	-10.9' 54.2'
7	285°50.7	31.6	10°32.9	15.2'	13°05.9	-10.9' 54.2'
8	300°50.7	32.3	25°07.1	15.3'	12°55.0	-11.0' 54.2'
9	315°50.8	· · 33.0	39°41.4	15.3'	12°44.0	-11.0' 54.2'
10	330°50.8	33.7	54°15.7	15.3'	12°33.0	-11.1' 54.2'
11	345°50.8	34.4	68°50.0	15.4'	12°21.9	-11.1' 54.2'
12	0°50.9	N16°35.1	83°24.4	15.4'	S12°10.7	-11.2' 54.2'
13	15°50.9	35.8	97°58.8	15.5'	11°59.6	-11.2' 54.2'
14	30°51.0	36.5	112°33.3	15.5'	11°48.3	-11.3' 54.2'
15	45°51.0	· · 37.2	127°07.8	15.6'	11°37.1	-11.3' 54.2'
16	60°51.1	37.9	141°42.4	15.6'	11°25.7	-11.4' 54.2'
17	75°51.1	38.6	156°17.0	15.6'	11°14.4	-11.4' 54.2'
18	90°51.2	N16°39.3	170°51.6	15.7'	S11°03.0	-11.4' 54.2'
19	105°51.2	40.0	185°26.3	15.7'	10°51.5	-11.5' 54.2'
20	120°51.3	40.7	200°01.0	15.7'	10°40.1	-11.5' 54.2'
21	135°51.3	· · 41.4	214°35.7	15.8'	10°28.5	-11.6' 54.2'
22	150°51.3	42.1	229°10.5	15.8'	10°17.0	-11.6' 54.2'
23	165°51.4	42.8	243°45.3	15.8'	10°05.4	-11.6' 54.2'
SD = 15.8' $d = 0.7'$ SD = 14.8'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	////	////	01:15	22:49	////	////
N 70°	////	////	02:07	21:51	////	////
68°	////	////	02:39	21:18	////	////
66°	////	01:30	03:02	20:54	22:30	////
64°	////	02:07	03:20	20:35	21:51	////
62°	////	02:32	03:35	20:20	21:24	////
60°	01:20	02:52	03:48	20:07	21:04	22:40
N 58°	01:54	03:08	03:58	19:57	20:47	22:04
56°	02:17	03:22	04:08	19:47	20:33	21:39
54°	02:36	03:33	04:16	19:39	20:21	21:20
52°	02:51	03:44	04:23	19:31	20:11	21:04
50°	03:04	03:53	04:30	19:25	20:02	20:51
45°	03:30	04:11	04:44	19:10	19:43	20:24
N 40°	03:50	04:26	04:56	18:58	19:28	20:05
35°	04:05	04:38	05:05	18:49	19:16	19:49
30°	04:18	04:48	05:14	18:40	19:05	19:36
20°	04:38	05:05	05:29	18:25	18:48	19:16
N 10°	04:54	05:19	05:41	18:12	18:34	19:00
0°	05:07	05:32	05:53	18:00	18:22	18:47
S 10°	05:18	05:43	06:05	17:48	18:10	18:35
20°	05:28	05:54	06:17	17:36	17:59	18:25
30°	05:38	06:06	06:31	17:22	17:47	18:15
35°	05:43	06:13	06:39	17:14	17:40	18:10
40°	05:48	06:20	06:49	17:04	17:33	18:05
45°	05:53	06:28	06:59	16:53	17:25	18:00
S 50°	05:59	06:38	07:12	16:40	17:15	17:54
52°	06:02	06:42	07:18	16:34	17:11	17:51
54°	06:04	06:47	07:25	16:28	17:06	17:48
56°	06:07	06:52	07:32	16:20	17:01	17:46
58°	06:10	06:57	07:41	16:12	16:55	17:43
S 60°	06:13	07:03	07:50	16:02	16:49	17:39

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	■	04:44	03:29	■	06:39	09:25
N 70°	■	03:38	03:02	■	07:44	09:50
68°	03:38	03:01	02:42	06:07	08:20	10:09
66°	02:47	02:35	02:26	06:57	08:45	10:23
64°	02:14	02:14	02:13	07:29	09:04	10:35
62°	01:51	01:58	02:02	07:52	09:20	10:46
60°	01:31	01:44	01:52	08:10	09:33	10:54
N 58°	01:16	01:32	01:44	08:26	09:44	11:02
56°	01:02	01:22	01:36	08:39	09:54	11:09
54°	00:50	01:13	01:30	08:50	10:03	11:14
52°	00:40	01:04	01:24	09:00	10:10	11:20
50°	00:31	00:57	01:18	09:09	10:17	11:25
45°	00:11	00:41	01:07	09:28	10:32	11:35
N 40°	· · ·	00:28	00:57	09:43	10:44	11:43
35°	· · ·	00:17	00:48	09:56	10:54	11:51
30°	· · ·	00:07	00:41	10:07	11:03	11:57
20°	23:51	· · ·	00:28	10:26	11:18	12:08
N 10°	23:36	· · ·	00:17	10:43	11:31	12:18
0°	23:22	· · ·	00:06	10:58	11:44	12:27
S 10°	23:08	23:56	· · ·	11:13	11:56	12:36
20°	22:53	23:44	· · ·	11:29	12:09	12:45
30°	22:36	23:31	· · ·	11:48	12:24	12:56
35°	22:26	23:23	· · ·	11:59	12:32	13:02
40°	22:15	23:15	· · ·	12:11	12:42	13:09
45°	22:01	23:05	· · ·	12:26	12:53	13:17
S 50°	21:45	22:52	23:59	12:43	13:07	13:26
52°	21:37	22:46	23:55	12:51	13:13	13:31
54°	21:28	22:40	23:51	13:01	13:20	13:36
56°	21:18	22:33	23:47	13:11	13:28	13:41
58°	21:07	22:25	23:42	13:23	13:37	13:47
S 60°	20:54	22:16	23:36	13:36	13:47	13:54

Day	Sun		Mer. Pass hh:mm	Moon		Age 20-22 70-52%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
04	03:11	03:14	11:57	04:46	17:10	
05	03:16	03:19	11:57	05:33	17:55	
06	03:21	03:24	11:57	06:17	18:38	

May 07, 08, 09 UT (Fri., Sat., Sun.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	224°44.0	151°54.2	N23°34.5	86°40.0	N18°10.3	228°55.9	S02°58.5	45°17.3	N02°58.6	Alpheratz	357°46.2	29°08.8
1	239°46.5	166°53.4		101°41.4	09.9	243°57.9	58.4	60°19.9	58.6	Ankaa	353°18.2	-42°14.8
2	254°48.9	181°52.6	35.4	116°42.8	09.5	259°00.0	58.2	75°22.4	58.6	Schedar	349°43.8	56°35.5
3	269°51.4	196°51.8	· · 35.8	131°44.2	· · 09.2	274°02.0	· · 58.0	90°25.0	· · 58.6	Diphda	348°58.4	-17°55.7
4	284°53.8	211°51.0	36.3	146°45.6	08.8	289°04.0	57.8	105°27.5	58.7	Achernar	335°28.8	-57°10.9
5	299°56.3	226°50.2	36.7	161°47.0	08.4	304°06.0	57.6	120°30.1	58.7	Hamal	328°03.7	23°30.6
6	314°58.8	241°49.4	N23°37.2	176°48.4	N18°08.1	319°08.0	S02°57.4	135°32.7	N02°58.7	Polaris	319°21.2	89°18.6
7	330°01.2	256°48.6	37.6	191°49.8	07.7	334°10.1	57.3	150°35.2	58.8	Acamar	315°20.4	-40°15.8
8	345°03.7	271°47.8	38.1	206°51.2	07.3	349°12.1	57.1	165°37.8	58.8	Menkar	314°17.8	4°07.8
9	0°06.2	286°47.0	· · 38.5	221°52.6	· · 06.9	4°14.1	· · 56.9	180°40.3	· · 58.8	Mirfak	308°44.2	49°53.9
10	15°08.6	301°46.2	39.0	236°54.0	06.6	19°16.1	56.7	195°42.9	58.9	Aldebaran	290°52.3	16°31.8
11	30°11.1	316°45.4	39.4	251°55.4	06.2	34°18.1	56.5	210°45.4	58.9	Rigel	281°14.5	-8°11.5
12	45°13.6	331°44.6	N23°39.9	266°56.8	N18°05.8	49°20.2	S02°56.3	225°48.0	N02°58.9	Capella	280°38.2	46°00.6
13	60°16.0	346°43.8	40.3	281°58.2	05.4	64°22.2	56.2	240°50.5	59.0	Bellatrix	278°34.7	6°21.5
14	75°18.5	1°43.0	40.7	296°59.6	05.1	79°24.2	56.0	255°53.1	59.0	Elnath	278°15.8	28°37.0
15	90°21.0	16°42.2	· · 41.2	312°01.0	· · 04.7	94°26.2	· · 55.8	270°55.7	· · 59.0	Alnilam	275°48.9	-1°11.8
16	105°23.4	31°41.4	41.6	327°02.4	04.3	109°28.2	55.6	285°58.2	59.1	Betelgeuse	271°04.0	7°24.5
17	120°25.9	46°40.6	42.1	342°03.8	03.9	124°30.3	55.4	301°00.8	59.1	Canopus	263°57.5	-52°42.3
18	135°28.3	61°39.8	N23°42.5	357°05.2	N18°03.6	139°32.3	S02°55.2	316°03.3	N02°59.1	Sirius	258°35.9	-16°44.0
19	150°30.8	76°39.0	42.9	12°06.6	03.2	154°34.3	55.1	331°05.9	59.2	Adhara	255°14.5	-28°59.4
20	165°33.3	91°38.2	43.4	27°08.0	02.8	169°36.3	54.9	346°08.4	59.2	Procyon	245°02.2	5°11.8
21	180°35.7	106°37.4	· · 43.8	42°09.4	· · 02.4	184°38.3	· · 54.7	1°11.0	· · 59.2	Pollux	243°30.6	28°00.1
22	195°38.2	121°36.6	44.2	57°10.8	02.1	199°40.4	54.5	16°13.5	59.2	Avior	234°19.2	-59°32.9
23	210°40.7	136°35.8	44.7	72°12.2	01.7	214°42.4	54.3	31°16.1	59.3	Suhail	222°54.2	-43°28.8
Mer.pass. 09:00		ν -0.8' d 0.5' m -3.90		ν 1.4' d -0.4' m 0.81		ν 2.0' d -0.2' m -2.15		ν 2.6' d 0.0' m 0.73				

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	225°43.1	151°35.0	N23°45.1	87°13.6	N18°01.3	229°44.4	S02°54.1	46°18.6	N02°59.3	Denebola	182°35.7	14°30.7
1	240°45.6	166°34.2	45.5	102°15.0	00.9	244°46.4	54.0	61°21.2	59.3	Gienah	175°54.4	-17°36.2
2	255°48.1	181°33.4	46.0	117°16.4	00.5	259°48.4	53.8	76°23.8	59.4	Acrux	173°11.6	-63°09.7
3	270°50.5	196°32.6	· · 46.4	132°17.8	18°00.2	274°50.5	· · 53.6	91°26.3	· · 59.4	Gacrux	172°03.2	-57°10.6
4	285°53.0	211°31.8	46.8	147°19.2	17°59.8	289°52.5	53.4	106°28.9	59.4	Alioth	166°22.0	55°54.2
5	300°55.4	226°31.0	47.2	162°20.6	59.4	304°54.5	53.2	121°31.4	59.5	Spica	158°33.4	-11°13.1
6	315°57.9	241°30.2	N23°47.7	177°22.0	N17°59.0	319°56.5	S02°53.1	136°34.0	N02°59.5	Alkaid	153°00.1	49°15.6
7	331°00.4	256°29.4	48.1	192°23.4	58.7	334°58.6	52.9	151°36.5	59.5	Hadar	148°50.7	-60°25.6
8	346°02.8	271°28.6	48.5	207°24.8	58.3	350°00.6	52.7	166°39.1	59.6	Menkent	148°09.9	-36°25.5
9	1°05.3	286°27.8	· · 48.9	222°26.1	· · 57.9	5°02.6	· · 52.5	181°41.6	· · 59.6	Arcturus	145°57.4	19°07.6
10	16°07.8	301°26.9	49.3	237°27.5	57.5	20°04.6	52.3	196°44.2	59.6	Rigel Kent.	139°54.3	-60°52.8
11	31°10.2	316°26.1	49.8	252°28.9	57.1	35°06.6	52.1	211°46.7	59.6	Kochab	137°18.4	74°06.7
12	46°12.7	331°25.3	N23°50.2	267°30.3	N17°56.8	50°08.7	S02°52.0	226°49.3	N02°59.7	Zuben'ubi	137°07.6	-16°05.3
13	61°15.2	346°24.5	50.6	282°31.7	56.4	65°10.7	51.8	241°51.8	59.7	Alphecca	126°12.5	26°40.6
14	76°17.6	1°23.7	51.0	297°33.1	56.0	80°12.7	51.6	256°54.4	59.7	Antares	112°28.7	-26°27.4
15	91°20.1	16°22.9	· · 51.4	312°34.5	· · 55.6	95°14.7	· · 51.4	271°56.9	· · 59.8	Atria	107°32.2	-69°02.7
16	106°22.6	31°22.1	51.8	327°35.9	55.3	110°16.8	51.2	286°59.5	59.8	Sabik	102°14.9	-15°44.3
17	121°25.0	46°21.3	52.2	342°37.3	54.9	125°18.8	51.1	302°02.0	59.8	Shaula	96°24.7	-37°06.6
18	136°27.5	61°20.5	N23°52.7	357°38.7	N17°54.5	140°20.8	S02°50.9	317°04.6	N02°59.8	Rasalhague	96°08.3	12°33.0
19	151°29.9	76°19.7	53.1	12°40.0	54.1	155°22.8	50.7	332°07.1	59.9	Eltanin	90°46.8	51°29.0
20	166°32.4	91°18.9	53.5	27°41.4	53.7	170°24.9	50.5	347°09.7	59.9	Kaus Aust.	83°46.5	-34°22.7
21	181°34.9	106°18.1	· · 53.9	42°42.8	· · 53.3	185°26.9	· · 50.3	2°12.2	02°59.9	Vega	80°40.2	38°47.4
22	196°37.3	121°17.3	54.3	57°44.2	53.0	200°28.9	50.1	17°14.8	03°00.0	Nunki	76°00.9	-26°17.0
23	211°39.8	136°16.5	54.7	72°45.6	52.6	215°30.9	50.0	32°17.3	00.0	Altair	62°10.3	8°53.6
Mer.pass. 08:56		ν -0.8' d 0.4' m -3.90		ν 1.4' d -0.4' m 0.82		ν 2.0' d -0.2' m -2.15		ν 2.6' d 0.0' m 0.74		Peacock	53°22.6	-56°41.8

h	Sun			Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	180°51.4	N16°43.5	258°20.1	15.9'	S09°53.7	-11.7'	54.2'
1	195°51.5	44.2	272°55.0	15.9'	09°42.0	-11.7'	54.2'
2	210°51.5	44.9	287°29.9	15.9'	09°30.3	-11.8'	54.2'
3	225°51.6	· · 45.6	302°04.8	16.0'	09°18.6	-11.8'	54.2'
4	240°51.6	46.3	316°39.7	16.0'	09°06.8	-11.8'	54.2'
5	255°51.6	47.0	331°14.7	16.0'	08°54.9	-11.9'	54.2'
6	270°51.7	N16°47.6	345°49.7	16.0'	S08°43.1	-11.9'	54.2'
7	285°51.7	48.3	0°24.8	16.1'	08°31.2	-11.9'	54.2'
8	300°51.8	49.0	14°59.8	16.1'	08°19.3	-12.0'	54.2'
9	315°51.8	· · 49.7	29°34.9	16.1'	08°07.3	-12.0'	54.2'
10	330°51.8	50.4	44°10.0	16.1'	07°55.3	-12.0'	54.2'
11	345°51.9	51.1	58°45.1	16.1'	07°43.3	-12.1'	54.2'
12	0°51.9	N16°51.8	73°20.3	16.2'	S07°31.2	-12.1'	54.2'
13	15°52.0	52.5	87°55.4	16.2'	07°19.1	-12.1'	54.2'
14	30°52.0	53.2	102°30.6	16.2'	07°07.0	-12.1'	54.2'
15	45°52.0	· · 53.8	117°05.8	16.2'	06°54.9	-12.2'	54.2'
16	60°52.1	54.5	131°41.1	16.2'	06°42.7	-12.2'	54.2'
17	75°52.1	55.2	146°16.3	16.3'	06°30.5	-12.2'	54.2'
18	90°52.2	N16°55.9	160°51.5	16.3'	S06°18.2	-12.3'	54.2'
19	105°52.2	56.6	175°26.8	16.3'	06°06.0	-12.3'	54.2'
20	120°52.2	57.3	190°02.1	16.3'	05°53.7	-12.3'	54.3'
21	135°52.3	· · 58.0	204°37.4	16.3'	05°41.4	-12.3'	54.3'
22	150°52.3	58.6	219°12.7	16.3'	05°29.1	-12.4'	54.3'
23	165°52.3	59.3	233°48.0	16.3'	05°16.7	-12.4'	54.3'
SD = 15.8'			SD = 14.8'				
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	180°52.4	N17°00.0	248°23.3	16.3'	S05°04.3	-12.4'	54.3'
1	195°52.4	00.7	262°58.7	16.3'	04°51.9	-12.4'	54.3'
2	210°52.5	01.4	277°34.0	16.3'	04°39.5	-12.4'	54.3'
3	225°52.5	· · 02.1	292°09.3	16.3'	04°27.0	-12.5'	54.3'
4	240°52.5	02.7	306°44.7	16.4'	04°14.6	-12.5'	54.3'
5	255°52.6	03.4	321°20.0	16.4'	04°02.1	-12.5'	54.3'
6	270°52.6	N17°04.1	335°55.4	16.4'	S03°49.6	-12.5'	54.3'
7	285°52.6	04.8	350°30.8	16.4'	03°37.0	-12.5'	54.3'
8	300°52.7	05.5	5°06.1	16.4'	03°24.5	-12.6'	54.4'
9	315°52.7	· · 06.1	19°41.5	16.4'	03°11.9	-12.6'	54.4'
10	330°52.7	06.8	34°16.8	16.4'	02°59.3	-12.6'	54.4'
11	345°52.8	07.5	48°52.2	16.4'	02°46.7	-12.6'	54.4'
12	0°52.8	N17°08.2	63°27.5	16.4'	S02°34.1	-12.6'	54.4'
13	15°52.8	08.8	78°02.9	16.3'	02°21.5	-12.6'	54.4'
14	30°52.9	09.5	92°38.2	16.3'	02°08.9	-12.7'	54.4'
15	45°52.9	· · 10.2	107°13.6	16.3'	01°56.2	-12.7'	54.4'
16	60°52.9	10.9	121°48.9	16.3'	01°43.5	-12.7'	54.4'
17	75°53.0	11.5	136°24.2	16.3'	01°30.8	-12.7'	54.5'
18	90°53.0	N17°12.2	150°59.6	16.3'	S01°18.1	-12.7'	54.5'
19	105°53.0	12.9	165°34.9	16.3'	01°05.4	-12.7'	54.5'
20	120°53.1	13.6	180°10.2	16.3'	00°52.7	-12.7'	54.5'
21	135°53.1	· · 14.2	194°45.5	16.3'	00°40.0	-12.7'	54.5'
22	150°53.1	14.9	209°20.7	16.3'	00°27.2	-12.7'	54.5'
23	165°53.2	15.6	223°56.0	16.2'	00°14.5	-12.8'	54.5'
SD = 15.8'			SD = 14.8'				
Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	180°53.2	N17°16.2	238°31.2	16.2'	S00°01.7	-12.8'	54.5'
1	195°53.2	16.9	253°06.5	16.2'	N00°11.0	12.8'	54.6'
2	210°53.3	17.6	267°41.7	16.2'	00°23.8	12.8'	54.6'
3	225°53.3	· · 18.3	282°16.9	16.2'	00°36.6	12.8'	54.6'
4	240°53.3	18.9	296°52.0	16.2'	00°49.4	12.8'	54.6'
5	255°53.3	19.6	311°27.2	16.1'	01°02.1	12.8'	54.6'
6	270°53.4	N17°20.3	326°02.3	16.1'	N01°14.9	12.8'	54.6'
7	285°53.4	20.9	340°37.4	16.1'	01°27.7	12.8'	54.6'
8	300°53.4	21.6	355°12.5	16.1'	01°40.5	12.8'	54.7'
9	315°53.5	· · 22.3	9°47.6	16.0'	01°53.3	12.8'	54.7'
10	330°53.5	22.9	24°22.6	16.0'	02°06.1	12.8'	54.7'
11	345°53.5	23.6	38°57.6	16.0'	02°18.9	12.8'	54.7'
12	0°53.5	N17°24.3	53°32.6	16.0'	N02°31.7	12.8'	54.7'
13	15°53.6	24.9	68°07.6	15.9'	02°44.5	12.8'	54.7'
14	30°53.6	25.6	82°42.5	15.9'	02°57.3	12.8'	54.8'
15	45°53.6	· · 26.2	97°17.4	15.9'	03°10.1	12.8'	54.8'
16	60°53.7	26.9	111°52.3	15.8'	03°22.9	12.8'	54.8'
17	75°53.7	27.6	126°27.1	15.8'	03°35.7	12.8'	54.8'
18	90°53.7	N17°28.2	141°01.9	15.8'	N03°48.5	12.8'	54.8'
19	105°53.7	28.9	155°36.7	15.7'	04°01.2	12.8'	54.8'
20	120°53.8	29.6	170°11.4	15.7'	04°14.0	12.8'	54.9'
21	135°53.8	· · 30.2	184°46.1	15.7'	04°26.8	12.8'	54.9'
22	150°53.8	30.9	199°20.7	15.6'	04°39.5	12.7'	54.9'
23	165°53.8	31.5	213°55.4	15.6'	04°52.3	12.7'	54.9'
SD = 15.8'			SD = 14.9'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	////	////	00:28	////	////	////
N 70°	////	////	01:47	22:12	////	////
68°	////	////	02:25	21:32	////	////
66°	////	01:05	02:51	21:05	22:57	////
64°	////	01:51	03:10	20:45	22:06	////
62°	////	02:21	03:27	20:28	21:36	////
60°	00:58	02:42	03:40	20:15	21:13	23:04
N 58°	01:40	03:00	03:51	20:03	20:55	22:17
56°	02:07	03:14	04:01	19:53	20:40	21:49
54°	02:27	03:27	04:10	19:44	20:28	21:28
52°	02:44	03:38	04:18	19:36	20:17	21:11
50°	02:58	03:47	04:25	19:29	20:07	20:57
45°	03:25	04:07	04:40	19:14	19:47	20:29
N 40°	03:45	04:22	04:52	19:01	19:31	20:08
35°	04:02	04:35	05:03	18:51	19:19	19:52
30°	04:15	04:46	05:12	18:42	19:08	19:38
20°	04:36	05:04	05:27	18:26	18:49	19:17
N 10°	04:53	05:18	05:41	18:13	18:35	19:00
0°	05:06	05:31	05:53	18:00	18:22	18:47
S 10°	05:18	05:43	06:05	17:47	18:09	18:35
20°	05:29	05:55	06:18	17:34	17:57	18:24
30°	05:40	06:08	06:33	17:19	17:44	18:13
35°	05:45	06:15	06:42	17:11	17:37	18:08
40°	05:50	06:23	06:52	17:01	17:30	18:02
45°	05:56	06:32	07:03	16:50	17:21	17:56
S 50°	06:03	06:42	07:17	16:36	17:11	17:49
52°	06:06	06:46	07:23	16:29	17:06	17:47
54°	06:09	06:51	07:30	16:22	17:01	17:44
56°	06:12	06:57	07:38	16:14	16:55	17:40
58°	06:15	07:03	07:47	16:05	16:49	17:37
S 60°	06:19	07:10	07:57	15:55	16:43	17:33

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	02:54	02:29	02:05	11:27	13:19	15:10
N 70°	02:40	02:22	02:06	11:39	13:22	15:05
68°	02:29	02:17	02:06	11:49	13:25	15:01
66°	02:19	02:13	02:07	11:57	13:27	14:58
64°	02:11	02:09	02:07	12:03	13:29	14:56
62°	02:04	02:06	02:07	12:09	13:31	14:54
60°	01:58	02:03	02:07	12:14	13:32	14:52
N 58°	01:53	02:00	02:07	12:18	13:34	14:50
56°	01:48	01:58	02:08	12:22	13:35	14:48
54°	01:44	01:56	02:08	12:25	13:36	14:47
52°	01:40	01:54	02:08	12:28	13:37	14:46
50°	01:36	01:53	02:08	12:31	13:38	14:45
45°	01:29	01:49	02:08	12:37	13:39	14:42
N 40°	01:22	01:46	02:09	12:42	13:41	14:40
35°	01:17	01:43	02:09	12:47	13:42	14:38
30°	01:12	01:41	02:09	12:51	13:43	14:37
20°	01:03	01:36	02:09	12:57	13:45	14:34
N 10°	00:56	01:33	02:10	13:03	13:47	14:32
0°	00:48	01:29	02:10	13:08	13:49	14:30
S 10°	00:41	01:26	02:10	13:13	13:50	14:27
20°	00:34	01:22	02:11	13:19	13:52	14:25
30°	00:25	01:18	02:11	13:25	13:54	14:22
35°	00:20	01:15	02:11	13:29	13:55	14:21
40°	00:14	01:13	02:11	13:33	13:56	14:19
45°	00:07	01:09	02:12	13:38	13:57	14:17
S 50°	· · ·	01:06	02:12	13:43	13:59	14:15
52°	· · ·	01:04	02:12	13:46	14:00	14:14
54°	· · ·	01:02	02:13	13:49	14:01	14:12
56°	· · ·	01:00	02:13	13:52	14:02	14:11
58°	· · ·	00:57	02:13	13:55	14:03	14:10
S 60°	· · ·	00:55	02:13	13:59	14:04	14:08

Day	Sun		Mer. Pass hh:mm	Moon		Age 23-25 42-25%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
07	03:26	03:28	11:57	06:58	19:19	
08	03:30	03:31	11:56	07:39	19:59	
09	03:33	03:34	11:56	08:20	20:40	

May 10, 11, 12 UT (Mon., Tue., Wed.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	227°41.4	150°56.2	N24°04.4	88°20.2	N17°43.0	231°21.6	S02°45.5	48°21.0	N03°00.7	Alpheratz	357°46.2	29°08.8
1	242°43.9	165°55.3	04.8	103°21.6	42.6	246°23.6	45.3	63°23.6	00.7	Ankaa	353°18.1	-42°14.8
2	257°46.3	180°54.5	05.2	118°22.9	42.2	261°25.7	45.1	78°26.1	00.8	Schedar	349°43.7	56°35.5
3	272°48.8	195°53.7	•• 05.5	133°24.3	•• 41.8	276°27.7	•• 44.9	93°28.7	•• 00.8	Diphda	348°58.4	-17°55.7
4	287°51.3	210°52.9	05.9	148°25.7	41.5	291°29.7	44.7	108°31.2	00.8	Achernar	335°28.8	-57°10.9
5	302°53.7	225°52.1	06.3	163°27.1	41.1	306°31.8	44.6	123°33.8	00.8	Hamal	328°03.7	23°30.6
6	317°56.2	240°51.3	N24°06.6	178°28.5	N17°40.7	321°33.8	S02°44.4	138°36.3	N03°00.9	Polaris	319°20.9	89°18.6
7	332°58.7	255°50.4	07.0	193°29.8	40.3	336°35.8	44.2	153°38.9	00.9	Acamar	315°20.4	-40°15.8
8	348°01.1	270°49.6	07.4	208°31.2	39.9	351°37.8	44.0	168°41.4	00.9	Menkar	314°17.8	4°07.8
9	3°03.6	285°48.8	•• 07.7	223°32.6	•• 39.5	6°39.9	•• 43.8	183°43.9	•• 00.9	Mirfak	308°44.2	49°53.9
10	18°06.0	300°48.0	08.1	238°34.0	39.1	21°41.9	43.7	198°46.5	01.0	Aldebaran	290°52.3	16°31.8
11	33°08.5	315°47.2	08.4	253°35.3	38.7	36°43.9	43.5	213°49.0	01.0	Rigel	281°14.5	-8°11.5
12	48°11.0	330°46.4	N24°08.8	268°36.7	N17°38.4	51°46.0	S02°43.3	228°51.6	N03°01.0	Capella	280°38.3	46°00.6
13	63°13.4	345°45.5	09.2	283°38.1	38.0	66°48.0	43.1	243°54.1	01.1	Bellatrix	278°34.7	6°21.5
14	78°15.9	0°44.7	09.5	298°39.5	37.6	81°50.0	42.9	258°56.7	01.1	Elnath	278°15.8	28°37.0
15	93°18.4	15°43.9	•• 09.9	313°40.8	•• 37.2	96°52.1	•• 42.8	273°59.2	•• 01.1	Alnilam	275°49.0	-1°11.8
16	108°20.8	30°43.1	10.2	328°42.2	36.8	111°54.1	42.6	289°01.8	01.1	Betelgeuse	271°04.0	7°24.5
17	123°23.3	45°42.3	10.6	343°43.6	36.4	126°56.1	42.4	304°04.3	01.2	Canopus	263°57.6	-52°42.3
18	138°25.8	60°41.4	N24°10.9	358°45.0	N17°36.0	141°58.1	S02°42.2	319°06.8	N03°01.2	Sirius	258°36.0	-16°44.0
19	153°28.2	75°40.6	11.3	13°46.3	35.6	157°00.2	42.1	334°09.4	01.2	Adhara	255°14.6	-28°59.4
20	168°30.7	90°39.8	11.6	28°47.7	35.3	172°02.2	41.9	349°11.9	01.2	Procyon	245°02.3	5°11.8
21	183°33.1	105°39.0	•• 12.0	43°49.1	•• 34.9	187°04.2	•• 41.7	4°14.5	•• 01.3	Pollux	243°30.6	28°00.1
22	198°35.6	120°38.2	12.3	58°50.5	34.5	202°06.3	41.5	19°17.0	01.3	Avior	234°19.2	-59°32.9
23	213°38.1	135°37.3	12.7	73°51.8	34.1	217°08.3	41.3	34°19.6	01.3	Suhail	222°54.2	-43°28.8
Mer.pass. 08:48		ν -0.8' d 0.4' m -3.90		ν 1.4' d -0.4' m 0.85		ν 2.0' d -0.2' m -2.16		ν 2.5' d 0.0' m 0.75		Miaplacidus	221°40.4	-69°46.0
										Alphard	217°58.3	-8°42.4
										Regulus	207°45.9	11°54.9
										Dubhe	193°54.0	61°41.8
										Denebola	182°35.8	14°30.7
										Gienah	175°54.4	-17°36.2
										Acrux	173°11.6	-63°09.7
										Gacrux	172°03.2	-57°10.6
										Alioth	166°22.1	55°54.2
										Spica	158°33.4	-11°13.1
										Alkaid	153°00.1	49°15.6
										Hadar	148°50.7	-60°25.6
										Menkent	148°09.9	-36°25.5
										Arcturus	145°57.4	19°07.6
										Rigil Kent.	139°54.3	-60°52.8
										Kochab	137°18.5	74°06.7
										Zuben'ubi	137°07.6	-16°05.3
										Alphecca	126°12.5	26°40.6
										Antares	112°28.7	-26°27.4
										Atria	107°32.1	-69°02.7
										Sabik	102°14.8	-15°44.3
										Shaula	96°24.7	-37°06.6
										Rasalhague	96°08.3	12°33.0
										Eltanin	90°46.8	51°29.0
										Kaus Aust.	83°46.5	-34°22.7
										Vega	80°40.2	38°47.4
										Nunki	76°00.9	-26°17.0
										Altair	62°10.3	8°53.7
										Peacock	53°22.5	-56°41.8
										Deneb	49°33.0	45°18.8
										Enif	33°49.4	9°55.3
										Al Na'ir	27°46.5	-46°54.4
										Fomalhaut	15°26.6	-29°33.9
										Scheat	13°55.8	28°08.2
										Markab	13°40.8	15°15.6
										May 10 Mon	SHA	Mer.pass
										Venus	283°14.8	13:57
										Mars	220°38.8	18:05
										Jupiter	3°40.2	08:33
										Saturn	180°39.6	20:43
										May 11 Tue	SHA	Mer.pass
										Venus	281°56.0	13:58
										Mars	220°12.7	18:03
										Jupiter	3°29.8	08:30
										Saturn	180°41.6	20:39
										May 12 Wed	SHA	Mer.pass
										Venus	280°37.1	14:00
										Mars	219°46.4	18:01
										Jupiter	3°19.5	08:27
										Saturn	180°43.4	20:35

h	Sun			Moon			
Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	180°53.9	N17°32.2	228°29.9	15.5'	N05°05.0	12.7'	54.9'
1	195°53.9	32.8	243°04.5	15.5'	05°17.7	12.7'	55.0'
2	210°53.9	33.5	257°39.0	15.5'	05°30.4	12.7'	55.0'
3	225°53.9	· · 34.2	272°13.4	15.4'	05°43.2	12.7'	55.0'
4	240°54.0	34.8	286°47.8	15.4'	05°55.8	12.7'	55.0'
5	255°54.0	35.5	301°22.2	15.3'	06°08.5	12.7'	55.0'
6	270°54.0	N17°36.1	315°56.5	15.3'	N06°21.2	12.6'	55.1'
7	285°54.0	36.8	330°30.8	15.2'	06°33.8	12.6'	55.1'
8	300°54.0	37.4	345°05.0	15.2'	06°46.5	12.6'	55.1'
9	315°54.1	· · 38.1	359°39.2	15.1'	06°59.1	12.6'	55.1'
10	330°54.1	38.7	14°13.4	15.1'	07°11.7	12.6'	55.1'
11	345°54.1	39.4	28°47.4	15.0'	07°24.3	12.6'	55.2'
12	0°54.1	N17°40.1	43°21.5	15.0'	N07°36.8	12.5'	55.2'
13	15°54.2	40.7	57°55.4	14.9'	07°49.4	12.5'	55.2'
14	30°54.2	41.4	72°29.4	14.9'	08°01.9	12.5'	55.2'
15	45°54.2	· · 42.0	87°03.2	14.8'	08°14.4	12.5'	55.3'
16	60°54.2	42.7	101°37.0	14.8'	08°26.8	12.4'	55.3'
17	75°54.2	43.3	116°10.8	14.7'	08°39.3	12.4'	55.3'
18	90°54.3	N17°44.0	130°44.5	14.6'	N08°51.7	12.4'	55.3'
19	105°54.3	44.6	145°18.1	14.6'	09°04.1	12.4'	55.3'
20	120°54.3	45.3	159°51.7	14.5'	09°16.5	12.3'	55.4'
21	135°54.3	· · 45.9	174°25.3	14.5'	09°28.8	12.3'	55.4'
22	150°54.3	46.5	188°58.7	14.4'	09°41.1	12.3'	55.4'
23	165°54.4	47.2	203°32.1	14.3'	09°53.4	12.3'	55.4'
	SD = 15.8'	$d = 0.7'$	SD = 15.0'				
Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	180°54.4	N17°47.8	218°05.5	14.3'	N10°05.7	12.2'	55.5'
1	195°54.4	48.5	232°38.7	14.2'	10°17.9	12.2'	55.5'
2	210°54.4	49.1	247°11.9	14.1'	10°30.1	12.2'	55.5'
3	225°54.4	· · 49.8	261°45.1	14.1'	10°42.2	12.1'	55.5'
4	240°54.4	50.4	276°18.2	14.0'	10°54.4	12.1'	55.5'
5	255°54.5	51.1	290°51.2	13.9'	11°06.4	12.0'	55.6'
6	270°54.5	N17°51.7	305°24.1	13.9'	N11°18.5	12.0'	55.6'
7	285°54.5	52.4	319°57.0	13.8'	11°30.5	12.0'	55.6'
8	300°54.5	53.0	334°29.8	13.7'	11°42.5	11.9'	55.6'
9	315°54.5	· · 53.6	349°02.5	13.7'	11°54.4	11.9'	55.7'
10	330°54.5	54.3	3°35.2	13.6'	12°06.3	11.8'	55.7'
11	345°54.6	54.9	18°07.8	13.5'	12°18.1	11.8'	55.7'
12	0°54.6	N17°55.6	32°40.3	13.4'	N12°29.9	11.8'	55.7'
13	15°54.6	56.2	47°12.7	13.4'	12°41.7	11.7'	55.8'
14	30°54.6	56.8	61°45.1	13.3'	12°53.4	11.7'	55.8'
15	45°54.6	· · 57.5	76°17.4	13.2'	13°05.1	11.6'	55.8'
16	60°54.6	58.1	90°49.6	13.1'	13°16.7	11.6'	55.8'
17	75°54.7	58.7	105°21.7	13.1'	13°28.3	11.5'	55.9'
18	90°54.7	N17°59.4	119°53.8	13.0'	N13°39.8	11.5'	55.9'
19	105°54.7	18°00.0	134°25.8	12.9'	13°51.3	11.4'	55.9'
20	120°54.7	00.7	148°57.6	12.8'	14°02.7	11.4'	55.9'
21	135°54.7	· · 01.3	163°29.5	12.7'	14°14.0	11.3'	56.0'
22	150°54.7	01.9	178°01.2	12.7'	14°25.4	11.3'	56.0'
23	165°54.7	02.6	192°32.9	12.6'	14°36.6	11.2'	56.0'
	SD = 15.8'	$d = 0.6'$	SD = 15.1'				
Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	180°54.7	N18°03.2	207°04.4	12.5'	N14°47.8	11.1'	56.0'
1	195°54.8	03.8	221°35.9	12.4'	14°58.9	11.1'	56.1'
2	210°54.8	04.5	236°07.3	12.3'	15°10.0	11.0'	56.1'
3	225°54.8	· · 05.1	250°38.6	12.2'	15°21.0	11.0'	56.1'
4	240°54.8	05.7	265°09.9	12.1'	15°32.0	10.9'	56.1'
5	255°54.8	06.4	279°41.0	12.1'	15°42.9	10.8'	56.2'
6	270°54.8	N18°07.0	294°12.1	12.0'	N15°53.7	10.8'	56.2'
7	285°54.8	07.6	308°43.1	11.9'	16°04.5	10.7'	56.2'
8	300°54.8	08.2	323°14.0	11.8'	16°15.2	10.6'	56.2'
9	315°54.8	· · 08.9	337°44.8	11.7'	16°25.8	10.6'	56.3'
10	330°54.9	09.5	352°15.5	11.6'	16°36.4	10.5'	56.3'
11	345°54.9	10.1	6°46.1	11.5'	16°46.9	10.4'	56.3'
12	0°54.9	N18°10.8	21°16.6	11.4'	N16°57.3	10.3'	56.3'
13	15°54.9	11.4	35°47.1	11.4'	17°07.6	10.3'	56.4'
14	30°54.9	12.0	50°17.4	11.3'	17°17.9	10.2'	56.4'
15	45°54.9	· · 12.6	64°47.7	11.2'	17°28.1	10.1'	56.4'
16	60°54.9	13.3	79°17.8	11.1'	17°38.2	10.0'	56.5'
17	75°54.9	13.9	93°47.9	11.0'	17°48.2	10.0'	56.5'
18	90°54.9	N18°14.5	108°17.9	10.9'	N17°58.2	9.9'	56.5'
19	105°54.9	15.1	122°47.8	10.8'	18°08.1	9.8'	56.5'
20	120°54.9	15.8	137°17.6	10.7'	18°17.9	9.7'	56.6'
21	135°54.9	· · 16.4	151°47.3	10.6'	18°27.6	9.6'	56.6'
22	150°55.0	17.0	166°16.9	10.5'	18°37.2	9.5'	56.6'
23	165°55.0	17.6	180°46.5	10.4'	18°46.7	9.5'	56.6'
	SD = 15.8'	$d = 0.6'$	SD = 15.3'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	□	□	□	□	□	□
N 70°	////	////	01:24	22:36	////	////
68°	////	////	02:09	21:48	////	////
66°	////	00:27	02:39	21:17	////	////
64°	////	01:35	03:01	20:55	22:23	////
62°	////	02:09	03:18	20:37	21:48	////
60°	00:24	02:33	03:32	20:22	21:23	////
N 58°	01:25	02:52	03:45	20:10	21:03	22:32
56°	01:56	03:07	03:55	19:59	20:47	22:00
54°	02:18	03:20	04:05	19:49	20:34	21:37
52°	02:36	03:32	04:13	19:41	20:22	21:18
50°	02:51	03:42	04:20	19:33	20:12	21:03
45°	03:20	04:02	04:36	19:17	19:51	20:34
N 40°	03:42	04:19	04:49	19:04	19:35	20:12
35°	03:58	04:32	05:00	18:53	19:21	19:55
30°	04:12	04:43	05:09	18:44	19:10	19:41
20°	04:35	05:02	05:26	18:27	18:51	19:19
N 10°	04:52	05:18	05:40	18:13	18:35	19:01
0°	05:06	05:31	05:53	18:00	18:22	18:47
S 10°	05:18	05:44	06:06	17:47	18:09	18:34
20°	05:30	05:56	06:20	17:33	17:56	18:23
30°	05:41	06:10	06:35	17:17	17:42	18:11
35°	05:47	06:17	06:44	17:08	17:35	18:05
40°	05:53	06:26	06:54	16:58	17:27	17:59
45°	05:59	06:35	07:06	16:46	17:17	17:53
S 50°	06:07	06:46	07:21	16:31	17:06	17:46
52°	06:10	06:51	07:28	16:24	17:02	17:42
54°	06:13	06:56	07:35	16:17	16:56	17:39
56°	06:16	07:02	07:44	16:08	16:50	17:36
58°	06:20	07:08	07:53	15:59	16:44	17:32
S 60°	06:24	07:16	08:04	15:48	16:36	17:28

Aries Venus

<https://pypi.org/project/sfalmanac/>

h	Sun			Moon			
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	180°55.0	N18°18.2	195°15.9	10.3'	N18°56.2	9.4'	56.7'
1	195°55.0	18.9	209°45.2	10.2'	19°05.6	9.3'	56.7'
2	210°55.0	19.5	224°14.5	10.1'	19°14.8	9.2'	56.7'
3	225°55.0	· · 20.1	238°43.6	10.1'	19°24.0	9.1'	56.7'
4	240°55.0	20.7	253°12.7	10.0'	19°33.1	9.0'	56.8'
5	255°55.0	21.3	267°41.6	9.9'	19°42.1	8.9'	56.8'
6	270°55.0	N18°22.0	282°10.5	9.8'	N19°51.0	8.8'	56.8'
7	285°55.0	22.6	296°39.3	9.7'	19°59.8	8.7'	56.8'
8	300°55.0	23.2	311°07.9	9.6'	20°08.5	8.6'	56.9'
9	315°55.0	· · 23.8	325°36.5	9.5'	20°17.1	8.5'	56.9'
10	330°55.0	24.4	340°05.0	9.4'	20°25.6	8.4'	56.9'
11	345°55.0	25.0	354°33.4	9.3'	20°34.0	8.3'	56.9'
12	0°55.0	N18°25.7	9°01.7	9.2'	N20°42.3	8.2'	57.0'
13	15°55.0	26.3	23°29.9	9.1'	20°50.5	8.1'	57.0'
14	30°55.0	26.9	37°58.0	9.0'	20°58.5	8.0'	57.0'
15	45°55.0	· · 27.5	52°26.0	8.9'	21°06.5	7.9'	57.0'
16	60°55.0	28.1	66°54.0	8.8'	21°14.4	7.8'	57.1'
17	75°55.0	28.7	81°21.8	8.7'	21°22.1	7.6'	57.1'
18	90°55.0	N18°29.3	95°49.6	8.7'	N21°29.8	7.5'	57.1'
19	105°55.0	29.9	110°17.2	8.6'	21°37.3	7.4'	57.1'
20	120°55.0	30.6	124°44.8	8.5'	21°44.7	7.3'	57.2'
21	135°55.0	· · 31.2	139°12.2	8.4'	21°52.0	7.2'	57.2'
22	150°55.0	31.8	153°39.6	8.3'	21°59.2	7.1'	57.2'
23	165°55.0	32.4	168°06.9	8.2'	22°06.3	6.9'	57.2'
SD = 15.8'			SD = 15.5'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	180°55.0	N18°33.0	182°34.1	8.1'	N22°13.2	6.8'	57.3'
1	195°55.0	33.6	197°01.2	8.0'	22°20.0	6.7'	57.3'
2	210°55.0	34.2	211°28.3	7.9'	22°26.7	6.6'	57.3'
3	225°55.0	· · 34.8	225°55.2	7.9'	22°33.3	6.4'	57.3'
4	240°55.0	35.4	240°22.1	7.8'	22°39.7	6.3'	57.4'
5	255°55.0	36.0	254°48.8	7.7'	22°46.0	6.2'	57.4'
6	270°55.0	N18°36.6	269°15.5	7.6'	N22°52.2	6.1'	57.4'
7	285°55.0	37.2	283°42.1	7.5'	22°58.3	5.9'	57.4'
8	300°55.0	37.8	298°08.6	7.4'	23°04.2	5.8'	57.5'
9	315°55.0	· · 38.4	312°35.1	7.4'	23°10.0	5.7'	57.5'
10	330°55.0	39.0	327°01.4	7.3'	23°15.7	5.5'	57.5'
11	345°55.0	39.6	341°27.7	7.2'	23°21.2	5.4'	57.5'
12	0°55.0	N18°40.2	355°53.9	7.1'	N23°26.7	5.3'	57.6'
13	15°55.0	40.8	10°20.0	7.0'	23°31.9	5.1'	57.6'
14	30°55.0	41.4	24°46.0	7.0'	23°37.0	5.0'	57.6'
15	45°55.0	· · 42.0	39°12.0	6.9'	23°42.0	4.9'	57.6'
16	60°55.0	42.6	53°37.9	6.8'	23°46.9	4.7'	57.7'
17	75°55.0	43.2	68°03.7	6.7'	23°51.6	4.6'	57.7'
18	90°55.0	N18°43.8	82°29.4	6.7'	N23°56.2	4.4'	57.7'
19	105°55.0	44.4	96°55.1	6.6'	24°00.6	4.3'	57.7'
20	120°55.0	45.0	111°20.7	6.5'	24°04.9	4.1'	57.7'
21	135°55.0	· · 45.6	125°46.2	6.5'	24°09.0	4.0'	57.8'
22	150°55.0	46.2	140°11.7	6.4'	24°13.0	3.8'	57.8'
23	165°55.0	46.8	154°37.1	6.3'	24°16.8	3.7'	57.8'
SD = 15.8'			SD = 15.6'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	180°55.0	N18°47.4	169°02.4	6.3'	N24°20.5	3.5'	57.8'
1	195°55.0	48.0	183°27.7	6.2'	24°24.1	3.4'	57.9'
2	210°55.0	48.6	197°52.9	6.2'	24°27.5	3.2'	57.9'
3	225°55.0	· · 49.2	212°18.1	6.1'	24°30.7	3.1'	57.9'
4	240°55.0	49.8	226°43.2	6.0'	24°33.8	2.9'	57.9'
5	255°55.0	50.4	241°08.2	6.0'	24°36.8	2.8'	57.9'
6	270°54.9	N18°51.0	255°33.2	5.9'	N24°39.6	2.6'	58.0'
7	285°54.9	51.6	269°58.1	5.9'	24°42.2	2.5'	58.0'
8	300°54.9	52.2	284°23.0	5.8'	24°44.7	2.3'	58.0'
9	315°54.9	· · 52.7	298°47.8	5.8'	24°47.0	2.2'	58.0'
10	330°54.9	53.3	313°12.6	5.7'	24°49.2	2.0'	58.0'
11	345°54.9	53.9	327°37.4	5.7'	24°51.2	1.9'	58.1'
12	0°54.9	N18°54.5	342°02.0	5.6'	N24°53.1	1.7'	58.1'
13	15°54.9	55.1	356°26.7	5.6'	24°54.8	1.5'	58.1'
14	30°54.9	55.7	10°51.3	5.6'	24°56.3	1.4'	58.1'
15	45°54.9	· · 56.3	25°15.9	5.5'	24°57.7	1.2'	58.1'
16	60°54.9	56.9	39°40.4	5.5'	24°59.0	1.1'	58.2'
17	75°54.9	57.4	54°04.9	5.5'	25°00.0	0.9'	58.2'
18	90°54.8	N18°58.0	68°29.4	5.4'	N25°00.9	0.7'	58.2'
19	105°54.8	58.6	82°53.8	5.4'	25°01.7	0.6'	58.2'
20	120°54.8	59.2	97°18.2	5.4'	25°02.3	0.4'	58.2'
21	135°54.8	18°59.8	111°42.6	5.3'	25°02.7	0.3'	58.3'
22	150°54.8	19°00.4	126°06.9	5.3'	25°02.9	0.1'	58.3'
23	165°54.8	00.9	140°31.2	5.3'	25°03.0	-0.1'	58.3'
SD = 15.8'			SD = 15.8'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	□	□	□	□	□	□
N 70°	////	////	00:55	23:09	////	////
68°	////	////	01:53	22:04	////	////
66°	////	////	02:27	21:29	////	////
64°	////	01:16	02:51	21:04	22:43	////
62°	////	01:56	03:10	20:45	22:00	////
60°	////	02:23	03:25	20:29	21:32	////
N 58°	01:08	02:43	03:38	20:16	21:11	22:50
56°	01:45	03:00	03:49	20:05	20:54	22:11
54°	02:10	03:14	03:59	19:55	20:40	21:45
52°	02:29	03:26	04:08	19:46	20:28	21:26
50°	02:45	03:37	04:16	19:38	20:17	21:09
45°	03:15	03:58	04:32	19:21	19:55	20:39
N 40°	03:38	04:15	04:46	19:07	19:38	20:16
35°	03:55	04:29	04:57	18:56	19:24	19:58
30°	04:10	04:41	05:07	18:46	19:12	19:43
20°	04:33	05:01	05:24	18:28	18:52	19:20
N 10°	04:51	05:17	05:39	18:14	18:36	19:02
0°	05:06	05:31	05:53	18:00	18:22	18:47
S 10°	05:19	05:44	06:06	17:46	18:08	18:34
20°	05:31	05:58	06:21	17:32	17:55	18:22
30°	05:43	06:12	06:37	17:15	17:41	18:10
35°	05:49	06:20	06:47	17:06	17:33	18:03
40°	05:55	06:28	06:57	16:55	17:24	17:57
45°	06:02	06:38	07:10	16:42	17:14	17:50
S 50°	06:10	06:50	07:25	16:27	17:03	17:42
52°	06:13	06:55	07:32	16:20	16:57	17:39
54°	06:17	07:00	07:40	16:12	16:52	17:35
56°	06:21	07:07	07:49	16:03	16:45	17:31
58°	06:25	07:14	07:59	15:53	16:38	17:27
S 60°	06:30	07:22	08:11	15:41	16:30	17:22

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	□	□	□	□	□	□
N 70°	00:20	□	□	□	□	□
68°	01:08	□	□	□	□	□
66°	01:39	01:26	□	22:03	□	□
64°	02:02	02:05	02:17	21:24	23:09	· ·
62°	02:20	02:32	02:57	20:57	22:30	23:43
60°	02:36	02:54	03:25	20:36	22:02	23:13
N 58°	02:49	03:11	03:46	20:19	21:41	22:49
56°	03:00	03:26	04:04	20:05	21:23	22:31
54°	03:10	03:39	04:19	19:53	21:08	22:15
52°	03:19	03:50	04:32	19:42	20:56	22:01
50°	03:27	04:00	04:44	19:32	20:44	21:49
45°	03:44	04:21	05:08	19:12	20:21	21:25
N 40°	03:58	04:38	05:27	18:56	20:02	21:05
35°	04:10	04:53	05:43	18:42	19:46	20:48
30°	04:21	05:05	05:57	18:30	19:33	20:34
20°	04:39	05:27	06:21	18:09	19:09	20:10
N 10°	04:55	05:46	06:41	17:52	18:49	19:49
0°	05:10	06:04	07:01	17:35	18:31	19:29
S 10°	05:25	06:21	07:20	17:19	18:12	19:09
20°	05:41	06:41	07:41	17:01	17:52	18:48
30°	06:00	07:03	08:05	16:41	17:29	18:24
35°	06:11	07:16	08:19	16:29	17:15	18:09
40°	06:24	07:31	08:36	16:16	17:00	17:53
45°	06:39	07:49	08:56	16:00	16:41	17:33
S 50°	06:57	08:11	09:20	15:41	16:18	17:08
52°	07:06	08:22	09:32	15:31	16:07	16:56
54°	07:15	08:34	09:46	15:21	15:55	16:42
56°	07:27	08:48	10:02	15:10	15:40	16:26
58°	07:39	09:04	10:21	14:56	15:24	16:07
S 60°	07:54	09:24	10:44	14:41	15:04	15:43

Day	Sun		Mer. Pass hh:mm	Moon		Age 29-1 1-1%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
13	03:40	03:40	11:56	11:23	23:49	
14	03:40	03:40	11:56	12:17	—:—	
15	03:40	03:40	11:56	13:15	00:46	

Aries Venus

<https://pypi.org/project/sfalmanac/>

h	Sun			Moon			
Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	180°54.8	N19°01.5	154°55.5	5.3'	N25°03.0	-0.2'	58.3'
1	195°54.8	02.1	169°19.8	5.3'	25°02.7	-0.4'	58.3'
2	210°54.8	02.7	183°44.1	5.2'	25°02.3	-0.6'	58.4'
3	225°54.7	· · 03.3	198°08.3	5.2'	25°01.8	-0.7'	58.4'
4	240°54.7	03.9	212°32.5	5.2'	25°01.1	-0.9'	58.4'
5	255°54.7	04.4	226°56.8	5.2'	25°00.2	-1.0'	58.4'
6	270°54.7	N19°05.0	241°21.0	5.2'	N24°59.1	-1.2'	58.4'
7	285°54.7	05.6	255°45.2	5.2'	24°57.9	-1.4'	58.4'
8	300°54.7	06.2	270°09.4	5.2'	24°56.6	-1.5'	58.5'
9	315°54.7	· · 06.7	284°33.5	5.2'	24°55.0	-1.7'	58.5'
10	330°54.7	07.3	298°57.7	5.2'	24°53.3	-1.9'	58.5'
11	345°54.6	07.9	313°21.9	5.2'	24°51.5	-2.0'	58.5'
12	0°54.6	N19°08.5	327°46.1	5.2'	N24°49.4	-2.2'	58.5'
13	15°54.6	09.0	342°10.3	5.2'	24°47.2	-2.4'	58.5'
14	30°54.6	09.6	356°34.5	5.2'	24°44.9	-2.5'	58.6'
15	45°54.6	· · 10.2	10°58.7	5.2'	24°42.4	-2.7'	58.6'
16	60°54.6	10.8	25°22.9	5.2'	24°39.7	-2.8'	58.6'
17	75°54.6	11.3	39°47.1	5.2'	24°36.8	-3.0'	58.6'
18	90°54.5	N19°11.9	54°11.4	5.3'	N24°33.8	-3.2'	58.6'
19	105°54.5	12.5	68°35.6	5.3'	24°30.6	-3.3'	58.6'
20	120°54.5	13.0	82°59.9	5.3'	24°27.3	-3.5'	58.6'
21	135°54.5	· · 13.6	97°24.2	5.3'	24°23.8	-3.7'	58.7'
22	150°54.5	14.2	111°48.5	5.3'	24°20.2	-3.8'	58.7'
23	165°54.5	14.8	126°12.8	5.4'	24°16.3	-4.0'	58.7'
SD = 15.8'			SD = 15.9'				

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	180°54.4	N19°15.3	140°37.1	5.4'	N24°12.4	-4.1'	58.7'
1	195°54.4	15.9	155°01.5	5.4'	24°08.2	-4.3'	58.7'
2	210°54.4	16.5	169°25.9	5.4'	24°03.9	-4.5'	58.7'
3	225°54.4	· · 17.0	183°50.4	5.5'	23°59.5	-4.6'	58.7'
4	240°54.4	17.6	198°14.8	5.5'	23°54.9	-4.8'	58.8'
5	255°54.4	18.2	212°39.3	5.5'	23°50.1	-4.9'	58.8'
6	270°54.3	N19°18.7	227°03.8	5.6'	N23°45.2	-5.1'	58.8'
7	285°54.3	19.3	241°28.4	5.6'	23°40.1	-5.2'	58.8'
8	300°54.3	19.8	255°53.0	5.6'	23°34.8	-5.4'	58.8'
9	315°54.3	· · 20.4	270°17.6	5.7'	23°29.5	-5.5'	58.8'
10	330°54.3	21.0	284°42.3	5.7'	23°23.9	-5.7'	58.8'
11	345°54.2	21.5	299°07.0	5.8'	23°18.2	-5.9'	58.8'
12	0°54.2	N19°22.1	313°31.8	5.8'	N23°12.3	-6.0'	58.8'
13	15°54.2	22.7	327°56.6	5.8'	23°06.3	-6.2'	58.9'
14	30°54.2	23.2	342°21.4	5.9'	23°00.2	-6.3'	58.9'
15	45°54.2	· · 23.8	356°46.3	5.9'	22°53.9	-6.5'	58.9'
16	60°54.1	24.3	11°11.3	6.0'	22°47.4	-6.6'	58.9'
17	75°54.1	24.9	25°36.3	6.0'	22°40.8	-6.8'	58.9'
18	90°54.1	N19°25.4	40°01.3	6.1'	N22°34.1	-6.9'	58.9'
19	105°54.1	26.0	54°26.4	6.1'	22°27.2	-7.0'	58.9'
20	120°54.1	26.6	68°51.5	6.2'	22°20.1	-7.2'	58.9'
21	135°54.0	· · 27.1	83°16.7	6.3'	22°12.9	-7.3'	58.9'
22	150°54.0	27.7	97°42.0	6.3'	22°05.6	-7.5'	59.0'
23	165°54.0	28.2	112°07.3	6.4'	21°58.1	-7.6'	59.0'
SD = 15.8'			SD = 16.0'				

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	180°54.0	N19°28.8	126°32.7	6.4'	N21°50.5	-7.8'	59.0'
1	195°53.9	29.3	140°58.1	6.5'	21°42.8	-7.9'	59.0'
2	210°53.9	29.9	155°23.6	6.5'	21°34.9	-8.0'	59.0'
3	225°53.9	· · 30.4	169°49.1	6.6'	21°26.8	-8.2'	59.0'
4	240°53.9	31.0	184°14.7	6.7'	21°18.7	-8.3'	59.0'
5	255°53.8	31.5	198°40.4	6.7'	21°10.4	-8.4'	59.0'
6	270°53.8	N19°32.1	213°06.1	6.8'	N21°01.9	-8.6'	59.0'
7	285°53.8	32.6	227°31.9	6.8'	20°53.4	-8.7'	59.0'
8	300°53.8	33.2	241°57.7	6.9'	20°44.7	-8.8'	59.0'
9	315°53.7	· · 33.7	256°23.6	7.0'	20°35.8	-9.0'	59.0'
10	330°53.7	34.3	270°49.6	7.0'	20°26.9	-9.1'	59.1'
11	345°53.7	34.8	285°15.7	7.1'	20°17.8	-9.2'	59.1'
12	0°53.7	N19°35.4	299°41.8	7.2'	N20°08.6	-9.3'	59.1'
13	15°53.6	35.9	314°07.9	7.2'	19°59.2	-9.5'	59.1'
14	30°53.6	36.5	328°34.2	7.3'	19°49.7	-9.6'	59.1'
15	45°53.6	· · 37.0	343°00.5	7.4'	19°40.1	-9.7'	59.1'
16	60°53.6	37.6	357°26.8	7.4'	19°30.4	-9.8'	59.1'
17	75°53.5	38.1	11°53.3	7.5'	19°20.6	-10.0'	59.1'
18	90°53.5	N19°38.7	26°19.8	7.6'	N19°10.6	-10.1'	59.1'
19	105°53.5	39.2	40°46.3	7.6'	19°00.6	-10.2'	59.1'
20	120°53.5	39.7	55°13.0	7.7'	18°50.4	-10.3'	59.1'
21	135°53.4	· · 40.3	69°39.7	7.8'	18°40.1	-10.4'	59.1'
22	150°53.4	40.8	84°06.5	7.8'	18°29.6	-10.5'	59.1'
23	165°53.4	41.4	98°33.3	7.9'	18°19.1	-10.6'	59.1'
SD = 15.8'			SD = 16.1'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	□	□	□	□	□	□
N 70°	□	□	□	□	□	□
68°	////	////	01:36	22:22	////	////
66°	////	////	02:15	21:41	////	////
64°	////	00:54	02:41	21:14	23:08	////
62°	////	01:43	03:02	20:53	22:13	////
60°	////	02:13	03:18	20:36	21:42	////
N 58°	00:48	02:35	03:32	20:22	21:19	23:13
56°	01:33	02:53	03:44	20:10	21:01	22:23
54°	02:01	03:08	03:54	20:00	20:46	21:54
52°	02:22	03:21	04:03	19:50	20:33	21:33
50°	02:39	03:32	04:12	19:42	20:22	21:16
45°	03:11	03:55	04:29	19:24	19:59	20:43
N 40°	03:34	04:12	04:43	19:10	19:41	20:19
35°	03:53	04:27	04:55	18:58	19:26	20:01
30°	04:08	04:39	05:06	18:48	19:14	19:46
20°	04:32	05:00	05:23	18:30	18:53	19:22
N 10°	04:50	05:16	05:39	18:14	18:37	19:03
0°	05:05	05:31	05:53	18:00	18:22	18:47
S 10°	05:19	05:45	06:07	17:46	18:08	18:34
20°	05:32	05:59	06:22	17:31	17:54	18:21
30°	05:44	06:13	06:39	17:14	17:39	18:08
35°	05:51	06:22	06:49	17:04	17:31	18:01
40°	05:58	06:31	07:00	16:52	17:22	17:55
45°	06:05	06:41	07:13	16:39	17:11	17:47
S 50°	06:14	06:53	07:29	16:23	16:59	17:39
52°	06:17	06:59	07:37	16:16	16:53	17:35
54°	06:21	07:05	07:45	16:07	16:47	17:31
56°	06:25	07:11	07:54	15:58	16:41	17:27
58°	06:30	07:19	08:05	15:47	16:33	17:22
S 60°	06:35	07:27	08:17	15:35	16:25	17:17

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	□	□	□	□	□	□
N 70°	□	□	□	□	□	□
68°	□	□	04:57	□	□	02:41
66°	□	03:14	05:51	□	02:21	01:46
64°	02:59	04:31	06:24	00:30	01:03	01:13
62°	03:46	05:07	06:47	· · ·	00:26	00:48
60°	04:17	05:34	07:06	· · ·	00:00	00:28
N 58°	04:40	05:54	07:22	23:39	· · ·	00:12
56°	04:59	06:11	07:35	23:22	23:58	· · ·
54°	05:14	06:25	07:47	23:07	23:46	· · ·
52°	05:28	06:38	07:57	22:55	23:35	· · ·
50°	05:40	06:49	08:06	22:43	23:26	23:59
45°	06:05	07:12	08:25	22:20	23:06	23:43
N 40°	06:24	07:30	08:41	22:01	22:49	23:30
35°	06:41	07:45	08:54	21:45	22:36	23:19
30°	06:55	07:59	09:05	21:32	22:23	23:09
20°	07:19	08:21	09:25	21:08	22:03	22:53
N 10°	07:40	08:41	09:42	20:48	21:44	22:38
0°	08:00	08:59	09:57	20:28	21:27	22:24
S 10°	08:20	09:18	10:13	20:09	21:10	22:10
20°	08:41	09:37	10:30	19:49	20:52	21:55
30°	09:05	10:00	10:49	19:25	20:30	21:37
35°	09:19	10:13	11:00	19:11	20:18	21:27
40°	09:36	10:28	11:12	18:55	20:03	21:15
45°	09:56	10:46	11:27	18:35	19:46	21:01
S 50°	10:20	11:08	11:46	18:11	19:24	20:44
52°	10:32	11:19	11:54	17:59	19:14	20:36
54°	10:46	11:31	12:04	17:45	19:02	20:27
56°	11:02	11:45	12:15	17:30	18:49	20:17
58°	11:20	12:01	12:27	17:11	18:33	20:06
S 60°	11:43	12:20	12:41	16:48	18:15	19:52

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time	12h		Mer. Pass.		
	00h	mm:ss	hh:mm	Upper	Lower	2-4
		mm:ss	hh:mm	hh:mm	hh:mm	5-18%
16	03:39	03:39	11:56	14:14	01:44	
17	03:38	03:37	11:56	15:13	02:44	
18	03:36	03:35	11:56	16:11	03:42	

May 19, 20, 21 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	236°33.7	147°56.5	N24°57.0	93°11.3	N16°15.4	238°43.5	S02°08.3	57°27.5	N03°05.0	Alpheratz	357°46.1	29°08.8
1	251°36.1	162°55.7	57.2	108°12.6	15.0	253°45.5	08.2	72°30.0	05.0	Ankaa	353°18.1	-42°14.7
2	266°38.6	177°54.8	57.3	123°14.0	14.6	268°47.6	08.0	87°32.5	05.0	Schedar	349°43.6	56°35.5
3	281°41.1	192°54.0	•• 57.4	138°15.3	•• 14.1	283°49.7	•• 07.8	102°35.0	•• 05.0	Diphda	348°58.3	-17°55.7
4	296°43.5	207°53.2	57.5	153°16.6	13.7	298°51.7	07.7	117°37.5	05.0	Achernar	335°28.8	-57°10.8
5	311°46.0	222°52.3	57.6	168°17.9	13.3	313°53.8	07.5	132°40.1	05.0	Hamal	328°03.7	23°30.6
6	326°48.4	237°51.5	N24°57.7	183°19.2	N16°12.9	328°55.9	S02°07.3	147°42.6	N03°05.0	Polaris	319°19.1	89°18.5
7	341°50.9	252°50.6	57.8	198°20.6	12.4	343°57.9	07.2	162°45.1	05.1	Acamar	315°20.4	-40°15.7
8	356°53.4	267°49.8	57.9	213°21.9	12.0	359°00.0	07.0	177°47.6	05.1	Menkar	314°17.7	4°07.8
9	11°55.8	282°49.0	•• 58.0	228°23.2	•• 11.6	14°02.1	•• 06.8	192°50.1	•• 05.1	Mirfak	308°44.2	49°53.8
10	26°58.3	297°48.1	58.1	243°24.5	11.2	29°04.1	06.7	207°52.6	05.1	Aldebaran	290°52.3	16°31.8
11	42°00.8	312°47.3	58.2	258°25.8	10.7	44°06.2	06.5	222°55.1	05.1	Rigel	281°14.5	-8°11.4
12	57°03.2	327°46.4	N24°58.3	273°27.1	N16°10.3	59°08.2	S02°06.4	237°57.6	N03°05.1	Capella	280°38.2	46°00.5
13	72°05.7	342°45.6	58.4	288°28.5	09.9	74°10.3	06.2	253°00.2	05.1	Bellatrix	278°34.7	6°21.5
14	87°08.2	357°44.8	58.5	303°29.8	09.4	89°12.4	06.0	268°02.7	05.1	Elnath	278°15.8	28°37.0
15	102°10.6	12°43.9	•• 58.6	318°31.1	•• 09.0	104°14.4	•• 05.9	283°05.2	•• 05.1	Alnilam	275°49.0	-1°11.8
16	117°13.1	27°43.1	58.7	333°32.4	08.6	119°16.5	05.7	298°07.7	05.2	Betelgeuse	271°04.0	7°24.5
17	132°15.5	42°42.2	58.8	348°33.7	08.2	134°18.6	05.5	313°10.2	05.2	Canopus	263°57.6	-52°42.3
18	147°18.0	57°41.4	N24°58.8	3°35.0	N16°07.7	149°20.6	S02°05.4	328°12.7	N03°05.2	Sirius	258°36.0	-16°44.0
19	162°20.5	72°40.6	58.9	18°36.3	07.3	164°22.7	05.2	343°15.2	05.2	Adhara	255°14.6	-28°59.4
20	177°22.9	87°39.7	59.0	33°37.7	06.9	179°24.8	05.1	358°17.7	05.2	Procyon	245°02.3	5°11.8
21	192°25.4	102°38.9	•• 59.1	48°39.0	•• 06.5	194°26.8	•• 04.9	13°20.2	•• 05.2	Pollux	243°30.7	28°00.1
22	207°27.9	117°38.0	59.2	63°40.3	06.0	209°28.9	04.7	28°22.7	05.2	Avior	234°19.3	-59°32.9
23	222°30.3	132°37.2	59.3	78°41.6	05.6	224°31.0	04.6	43°25.3	05.2	Suhail	222°54.3	-43°28.8
Mer.pass. 08:12		ν -0.8' d 0.1' m -3.91		ν 1.3' d -0.4' m 0.96		ν 2.1' d -0.2' m -2.19		ν 2.5' d 0.0' m 0.80		Miaplacidus	221°40.6	-69°46.0
										Alphard	217°58.4	-8°42.4
										Regulus	207°45.9	11°54.9
										Dubhe	193°54.1	61°41.8
										Denebola	182°35.8	14°30.7
										Gienah	175°54.5	-17°36.2
										Acrux	173°11.6	-63°09.8
										Gacrux	172°03.2	-57°10.6
										Alioth	166°22.1	55°54.3
										Spica	158°33.4	-11°13.1
										Alkaid	153°00.1	49°15.7
										Hadar	148°50.7	-60°25.6
										Menkent	148°09.9	-36°25.5
										Arcturus	145°57.4	19°07.6
										Rigil Kent.	139°54.3	-60°52.9
										Kochab	137°18.5	74°06.8
										Zuben'ubi	137°07.6	-16°05.3
										Alphecca	126°12.5	26°40.7
										Antares	112°28.7	-26°27.4
										Atria	107°32.0	-69°02.7
										Sabik	102°14.8	-15°44.3
										Shaula	96°24.6	-37°06.7
										Rasalhague	96°08.2	12°33.0
										Eltanin	90°46.7	51°29.1
										Kaus Aust.	83°46.5	-34°22.7
										Vega	80°40.2	38°47.4
										Nunki	76°00.8	-26°17.0
										Altair	62°10.2	8°53.7
										Peacock	53°22.4	-56°41.8
										Deneb	49°32.9	45°18.8
										Enif	33°49.3	9°55.3
										Al Na'ir	27°46.4	-46°54.3
										Fomalhaut	15°26.5	-29°33.8
										Scheat	13°55.7	28°08.2
										Markab	13°40.7	15°15.6
May 19 Wed		SHA		Mer.pass								
Venus		271°22.9		14:09								
Mars		216°37.7		17:46								
Jupiter		2°09.8		08:04								
Saturn		180°53.8		20:07								
May 20 Thu		SHA		Mer.pass								
Venus		270°03.6		14:10								
Mars		216°10.1		17:44								
Jupiter		2°00.3		08:01								
Saturn		180°55.0		20:03								
May 21 Fri		SHA		Mer.pass								
Venus		268°44.3		14:12								
Mars		215°42.4		17:42								
Jupiter		1°50.8		07:57								
Saturn		180°56.0		19:59								

h	Sun			Moon			
Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	180°53.3	N19°41.9	113°00.2	8.0'	N18°08.5	-10.8'	59.1'
1	195°53.3	42.4	127°27.2	8.0'	17°57.7	-10.9'	59.1'
2	210°53.3	43.0	141°54.3	8.1'	17°46.9	-11.0'	59.2'
3	225°53.3	· · 43.5	156°21.4	8.2'	17°35.9	-11.1'	59.2'
4	240°53.2	44.1	170°48.6	8.3'	17°24.8	-11.2'	59.2'
5	255°53.2	44.6	185°15.8	8.3'	17°13.6	-11.3'	59.2'
6	270°53.2	N19°45.1	199°43.1	8.4'	N17°02.3	-11.4'	59.2'
7	285°53.1	45.7	214°10.5	8.5'	16°51.0	-11.5'	59.2'
8	300°53.1	46.2	228°38.0	8.5'	16°39.5	-11.6'	59.2'
9	315°53.1	· · 46.7	243°05.5	8.6'	16°27.9	-11.7'	59.2'
10	330°53.1	47.3	257°33.1	8.7'	16°16.2	-11.8'	59.2'
11	345°53.0	47.8	272°00.8	8.7'	16°04.4	-11.9'	59.2'
12	0°53.0	N19°48.3	286°28.5	8.8'	N15°52.6	-12.0'	59.2'
13	15°53.0	48.9	300°56.3	8.9'	15°40.6	-12.1'	59.2'
14	30°52.9	49.4	315°24.1	8.9'	15°28.5	-12.1'	59.2'
15	45°52.9	· · 49.9	329°52.1	9.0'	15°16.4	-12.2'	59.2'
16	60°52.9	50.5	344°20.0	9.1'	15°04.2	-12.3'	59.2'
17	75°52.8	51.0	358°48.1	9.1'	14°51.8	-12.4'	59.2'
18	90°52.8	N19°51.5	13°16.2	9.2'	N14°39.4	-12.5'	59.2'
19	105°52.8	52.1	27°44.4	9.2'	14°26.9	-12.6'	59.2'
20	120°52.7	52.6	42°12.6	9.3'	14°14.4	-12.7'	59.2'
21	135°52.7	· · 53.1	56°41.0	9.4'	14°01.7	-12.7'	59.2'
22	150°52.7	53.6	71°09.3	9.4'	13°49.0	-12.8'	59.2'
23	165°52.6	54.2	85°37.8	9.5'	13°36.2	-12.9'	59.2'
SD = 15.8'			SD = 16.1'				
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	180°52.6	N19°54.7	100°06.2	9.6'	N13°23.3	-13.0'	59.2'
1	195°52.6	55.2	114°34.8	9.6'	13°10.3	-13.0'	59.2'
2	210°52.5	55.8	129°03.4	9.7'	12°57.3	-13.1'	59.2'
3	225°52.5	· · 56.3	143°32.1	9.7'	12°44.1	-13.2'	59.2'
4	240°52.5	56.8	158°00.8	9.8'	12°31.0	-13.3'	59.2'
5	255°52.4	57.3	172°29.6	9.8'	12°17.7	-13.3'	59.2'
6	270°52.4	N19°57.8	186°58.4	9.9'	N12°04.4	-13.4'	59.2'
7	285°52.4	58.4	201°27.4	10.0'	11°51.0	-13.5'	59.2'
8	300°52.3	58.9	215°56.3	10.0'	11°37.5	-13.5'	59.2'
9	315°52.3	· · 59.4	230°25.3	10.1'	11°24.0	-13.6'	59.2'
10	330°52.3	19°59.9	244°54.4	10.1'	11°10.4	-13.6'	59.2'
11	345°52.2	20°00.5	259°23.5	10.2'	10°56.8	-13.7'	59.2'
12	0°52.2	N20°01.0	273°52.7	10.2'	N10°43.1	-13.8'	59.2'
13	15°52.1	01.5	288°21.9	10.3'	10°29.3	-13.8'	59.2'
14	30°52.1	02.0	302°51.2	10.3'	10°15.5	-13.9'	59.2'
15	45°52.1	· · 02.5	317°20.5	10.4'	10°01.6	-13.9'	59.2'
16	60°52.0	03.0	331°49.9	10.4'	09°47.7	-14.0'	59.2'
17	75°52.0	03.6	346°19.3	10.5'	09°33.7	-14.0'	59.2'
18	90°52.0	N20°04.1	0°48.7	10.5'	N09°19.7	-14.1'	59.2'
19	105°51.9	04.6	15°18.3	10.6'	09°05.6	-14.1'	59.2'
20	120°51.9	05.1	29°47.8	10.6'	08°51.5	-14.2'	59.2'
21	135°51.8	· · 05.6	44°17.4	10.6'	08°37.3	-14.2'	59.2'
22	150°51.8	06.1	58°47.1	10.7'	08°23.0	-14.3'	59.2'
23	165°51.8	06.6	73°16.7	10.7'	08°08.8	-14.3'	59.2'
SD = 15.8'			SD = 16.2'				
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	180°51.7	N20°07.2	87°46.5	10.8'	N07°54.5	-14.4'	59.2'
1	195°51.7	07.7	102°16.2	10.8'	07°40.1	-14.4'	59.2'
2	210°51.6	08.2	116°46.0	10.8'	07°25.7	-14.4'	59.2'
3	225°51.6	· · 08.7	131°15.9	10.9'	07°11.3	-14.5'	59.2'
4	240°51.6	09.2	145°45.8	10.9'	06°56.8	-14.5'	59.2'
5	255°51.5	09.7	160°15.7	11.0'	06°42.3	-14.5'	59.2'
6	270°51.5	N20°10.2	174°45.6	11.0'	N06°27.8	-14.6'	59.2'
7	285°51.4	10.7	189°15.6	11.0'	06°13.2	-14.6'	59.2'
8	300°51.4	11.2	203°45.7	11.1'	05°58.6	-14.6'	59.2'
9	315°51.4	· · 11.7	218°15.7	11.1'	05°44.0	-14.7'	59.2'
10	330°51.3	12.2	232°45.8	11.1'	05°29.3	-14.7'	59.2'
11	345°51.3	12.7	247°15.9	11.1'	05°14.6	-14.7'	59.2'
12	0°51.2	N20°13.3	261°46.1	11.2'	N04°59.9	-14.7'	59.2'
13	15°51.2	13.8	276°16.2	11.2'	04°45.1	-14.8'	59.2'
14	30°51.2	14.3	290°46.4	11.2'	04°30.4	-14.8'	59.2'
15	45°51.1	· · 14.8	305°16.6	11.2'	04°15.6	-14.8'	59.2'
16	60°51.1	15.3	319°46.9	11.3'	04°00.8	-14.8'	59.2'
17	75°51.0	15.8	334°17.2	11.3'	03°46.0	-14.8'	59.2'
18	90°51.0	N20°16.3	348°47.5	11.3'	N03°31.1	-14.9'	59.1'
19	105°50.9	16.8	3°17.8	11.3'	03°16.3	-14.9'	59.1'
20	120°50.9	17.3	17°48.1	11.4'	03°01.4	-14.9'	59.1'
21	135°50.9	· · 17.8	32°18.5	11.4'	02°46.5	-14.9'	59.1'
22	150°50.8	18.3	46°48.8	11.4'	02°31.6	-14.9'	59.1'
23	165°50.8	18.8	61°19.2	11.4'	02°16.7	-14.9'	59.1'
SD = 15.8'			SD = 16.1'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	////	////	01:17	22:42	////	////
66°	////	////	02:03	21:54	////	////
64°	////	00:16	02:32	21:24	////	////
62°	////	01:30	02:54	21:01	22:28	////
60°	////	02:04	03:12	20:43	21:52	////
N 58°	00:14	02:28	03:26	20:28	21:27	////
56°	01:21	02:47	03:39	20:15	21:08	22:36
54°	01:52	03:02	03:50	20:04	20:52	22:03
52°	02:15	03:16	03:59	19:55	20:38	21:40
50°	02:33	03:27	04:08	19:46	20:27	21:22
45°	03:07	03:51	04:26	19:28	20:03	20:48
N 40°	03:31	04:10	04:41	19:13	19:44	20:23
35°	03:50	04:25	04:53	19:00	19:29	20:04
30°	04:06	04:37	05:04	18:49	19:16	19:48
20°	04:30	04:58	05:22	18:31	18:55	19:23
N 10°	04:49	05:16	05:38	18:15	18:37	19:04
0°	05:05	05:31	05:53	18:00	18:22	18:48
S 10°	05:20	05:45	06:08	17:45	18:08	18:33
20°	05:33	06:00	06:23	17:30	17:53	18:20
30°	05:46	06:15	06:41	17:12	17:38	18:07
35°	05:53	06:24	06:51	17:02	17:29	18:00
40°	06:00	06:33	07:03	16:50	17:19	17:52
45°	06:08	06:44	07:16	16:36	17:09	17:45
S 50°	06:17	06:57	07:33	16:19	16:56	17:36
52°	06:21	07:03	07:41	16:12	16:50	17:32
54°	06:25	07:09	07:50	16:03	16:44	17:28
56°	06:29	07:16	08:00	15:53	16:36	17:23
58°	06:34	07:24	08:11	15:42	16:29	17:18
S 60°	06:40	07:33	08:24	15:29	16:20	17:13

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	06:05	09:08	11:30	03:32	02:21	01:46
N 70°	07:01	09:28	11:37	02:34	01:58	01:35
68°	07:35	09:44	11:43	02:00	01:41	01:27
66°	07:59	09:56	11:48	01:35	01:27	01:20
64°	08:17	10:07	11:52	01:15	01:15	01:14
62°	08:32	10:16	11:56	00:59	01:05	01:08
60°	08:45	10:23	11:59	00:45	00:56	01:04
N 58°	08:56	10:30	12:02	00:33	00:48	01:00
56°	09:05	10:35	12:04	00:23	00:42	00:56
54°	09:14	10:41	12:07	00:14	00:35	00:53
52°	09:21	10:45	12:09	00:06	00:30	00:49
50°	09:28	10:50	12:10	· · ·	00:25	00:47
45°	09:42	10:59	12:14	· · ·	00:14	00:41
N 40°	09:54	11:06	12:18	· · ·	00:05	00:36
35°	10:04	11:13	12:21	23:57	· · ·	00:31
30°	10:12	11:18	12:23	23:50	· · ·	00:27
20°	10:27	11:28	12:27	23:38	· · ·	00:20
N 10°	10:40	11:37	12:31	23:28	· · ·	00:14
0°	10:52	11:45	12:35	23:17	· · ·	00:09
S 10°	11:04	11:53	12:38	23:07	· · ·	00:03
20°	11:17	12:01	12:42	22:56	23:57	· · ·
30°	11:32	12:11	12:46	22:44	23:49	· · ·
35°	11:40	12:16	12:49	22:37	23:45	· · ·
40°	11:50	12:22	12:52	22:28	23:40	· · ·
45°	12:01	12:30	12:55	22:18	23:35	· · ·
S 50°	12:15	12:38	12:59	22:07	23:28	· · ·
52°	12:21	12:42	13:00	22:01	23:25	· · ·
54°	12:28	12:47	13:02	21:55	23:22	· · ·
56°	12:36	12:51	13:04	21:48	23:18	· · ·
58°	12:44	12:57	13:07	21:40	23:14	· · ·
S 60°	12:54	13:03	13:09	21:31	23:09	· · ·

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss	hh:mm	Mer. Pass. Upper hh:mm	Lower hh:mm	5-7 28-50%
19	03:33	03:32	11:56	17:05	04:38	
20	03:30	03:29	11:57	17:57	05:31	
21	03:27	03:25	11:57	18:46	06:22	

May 22, 23, 24 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	239°31.1	146°56.2	N25°01.9	94°45.7	N15°44.4	241°12.5	S01°56.7	60°28.0	N03°05.7	Alpheratz	357°46.1	29°08.8
1	254°33.5	161°55.3	01.9	109°47.0	43.9	256°14.6	56.6	75°30.5	05.7	Ankaa	353°18.0	-42°14.7
2	269°36.0	176°54.5	01.9	124°48.3	43.5	271°16.7	56.4	90°33.0	05.7	Schedar	349°43.6	56°35.5
3	284°38.5	191°53.7	•• 01.9	139°49.6	•• 43.1	286°18.7	•• 56.2	105°35.5	•• 05.7	Diphda	348°58.3	-17°55.7
4	299°40.9	206°52.8	01.9	154°50.9	42.6	301°20.8	56.1	120°38.0	05.7	Achernar	335°28.8	-57°10.8
5	314°43.4	221°52.0	02.0	169°52.2	42.2	316°22.9	55.9	135°40.5	05.7	Hamal	328°03.6	23°30.6
6	329°45.9	236°51.1	N25°02.0	184°53.5	N15°41.7	331°25.0	S01°55.8	150°43.0	N03°05.7	Polaris	319°18.6	89°18.5
7	344°48.3	251°50.3	02.0	199°54.8	41.3	346°27.0	55.6	165°45.5	05.7	Acamar	315°20.4	-40°15.7
8	359°50.8	266°49.5	02.0	214°56.1	40.9	1°29.1	55.5	180°48.0	05.7	Menkar	314°17.7	4°07.8
9	14°53.3	281°48.6	•• 02.0	229°57.4	•• 40.4	16°31.2	•• 55.3	195°50.5	•• 05.7	Mirfak	308°44.2	49°53.8
10	29°55.7	296°47.8	02.0	244°58.7	40.0	31°33.3	55.1	210°53.0	05.7	Aldebaran	290°52.3	16°31.8
11	44°58.2	311°47.0	02.0	260°00.0	39.5	46°35.3	55.0	225°55.5	05.7	Rigel	281°14.5	-8°11.4
12	60°00.6	326°46.1	N25°02.0	275°01.3	N15°39.1	61°37.4	S01°54.8	240°58.0	N03°05.7	Capella	280°38.2	46°00.5
13	75°03.1	341°45.3	02.0	290°02.6	38.7	76°39.5	54.7	256°00.5	05.7	Bellatrix	278°34.7	6°21.5
14	90°05.6	356°44.5	02.0	305°03.9	38.2	91°41.6	54.5	271°03.0	05.8	Elnath	278°15.8	28°37.0
15	105°08.0	11°43.6	•• 02.1	320°05.2	•• 37.8	106°43.7	•• 54.4	286°05.5	•• 05.8	Alnilam	275°49.0	-1°11.8
16	120°10.5	26°42.8	02.1	335°06.5	37.3	121°45.7	54.2	301°08.0	05.8	Betelgeuse	271°04.0	7°24.5
17	135°13.0	41°42.0	02.1	350°07.8	36.9	136°47.8	54.0	316°10.5	05.8	Canopus	263°57.6	-52°42.3
18	150°15.4	56°41.1	N25°02.1	5°09.1	N15°36.5	151°49.9	S01°53.9	331°13.0	N03°05.8	Sirius	258°36.0	-16°44.0
19	165°17.9	71°40.3	02.1	20°10.4	36.0	166°52.0	53.7	346°15.5	05.8	Adhara	255°14.6	-28°59.4
20	180°20.4	86°39.5	02.1	35°11.7	35.6	181°54.1	53.6	1°18.0	05.8	Procyon	245°02.3	5°11.8
21	195°22.8	101°38.6	•• 02.1	50°13.0	•• 35.1	196°56.1	•• 53.4	16°20.5	•• 05.8	Pollux	243°30.7	28°00.1
22	210°25.3	116°37.8	02.0	65°14.3	34.7	211°58.2	53.3	31°23.0	05.8	Avior	234°19.3	-59°32.9
23	225°27.7	131°37.0	02.0	80°15.6	34.3	227°00.3	53.1	46°25.5	05.8	Suhail	222°54.3	-43°28.8
Mer.pass. 08:01		ν -0.8' d 0.0' m -3.91		ν 1.3' d -0.4' m 1.00		ν 2.1' d -0.2' m -2.21		ν 2.5' d 0.0' m 0.81		Miaplacidus	221°40.6	-69°46.0
										Alphard	217°58.4	-8°42.4
										Regulus	207°45.9	11°54.9
										Dubhe	193°54.1	61°41.9
										Denebola	182°35.8	14°30.7
										Gienah	175°54.5	-17°36.2
										Acrux	173°11.7	-63°09.8
										Gacrux	172°03.2	-57°10.6
										Alioth	166°22.1	55°54.3
										Spica	158°33.4	-11°13.1
										Alkaid	153°00.1	-49°15.7
										Hadar	148°50.7	-60°25.7
										Menkent	148°09.9	-36°25.5
										Arcturus	145°57.4	19°07.6
										Rigel Kent.	139°54.3	-60°52.9
										Kochab	137°18.5	74°06.8
										Zuben'ubi	137°07.6	-16°05.3
										Alphecca	126°12.5	26°40.7
										Antares	112°28.7	-26°27.4
										Atria	107°32.0	-69°02.8
										Sabik	102°14.8	-15°44.3
										Shaula	96°24.6	-37°06.7
										Rasalhague	96°08.2	12°33.1
										Eltanin	90°46.7	51°29.1
										Kaus Aust.	83°46.4	-34°22.7
										Vega	80°40.1	38°47.4
										Nunki	76°00.8	-26°17.0
										Altair	62°10.2	8°53.7
										Peacock	53°22.4	-56°41.8
										Deneb	49°32.9	45°18.8
										Enif	33°49.3	9°55.3
										Al Na'ir	27°46.4	-46°54.3
										Fomalhaut	15°26.5	-29°33.8
										Scheat	13°55.7	28°08.2
										Markab	13°40.7	15°15.6
May 22 Sat		SHA		Mer.pass								
Venus		267°25.1		14:13								
Mars		215°14.6		17:39								
Jupiter		1°41.4		07:54								
Saturn		180°57.0		19:55								
May 23 Sun		SHA		Mer.pass								
Venus		266°05.9		14:14								
Mars		214°46.7		17:37								
Jupiter		1°32.2		07:51								
Saturn		180°57.8		19:51								
May 24 Mon		SHA		Mer.pass								
Venus		264°46.8		14:16								
Mars		214°18.6		17:35								
Jupiter		1°23.0		07:47								
Saturn		180°58.6		19:47								
Horizontal parallax												
Venus: 0.1												
Mars: 0.1												

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	241°29.3	146°16.2	N25°01.5	95°47.9	N15°23.2	242°52.4	S01°49.2	62°27.9	N03°05.9
1	256°31.8	161°15.4	01.5	110°49.2	22.7	257°54.4	49.1	77°30.4	05.9
2	271°34.3	176°14.5	01.4	125°50.5	22.3	272°56.5	48.9	92°32.9	05.9
3	286°36.7	191°13.7	•• 01.4	140°51.8	•• 21.8	287°58.6	•• 48.8	107°35.4	•• 05.9
4	301°39.2	206°12.9	01.3	155°53.1	21.4	303°00.7	48.6	122°37.9	05.9
5	316°41.7	221°12.0	01.3	170°54.4	20.9	318°02.8	48.4	137°40.4	05.9
6	331°44.1	236°11.2	N25°01.3	185°55.7	N15°20.5	333°04.9	S01°48.3	152°42.9	N03°05.9
7	346°46.6	251°10.4	01.2	200°57.0	20.0	348°07.0	48.1	167°45.4	05.9
8	1°49.1	266°09.6	01.2	215°58.3	19.6	3°09.0	48.0	182°47.9	05.9
9	16°51.5	281°08.7	•• 01.1	230°59.5	•• 19.1	18°11.1	•• 47.8	197°50.4	•• 05.9
10	31°54.0	296°07.9	01.1	246°00.8	18.7	33°13.2	47.7	212°52.9	05.9
11	46°56.5	311°07.1	01.0	261°02.1	18.3	48°15.3	47.5	227°55.3	05.9
12	61°58.9	326°06.2	N25°01.0	276°03.4	N15°17.8	63°17.4	S01°47.4	242°57.8	N03°05.9
13	77°01.4	341°05.4	00.9	291°04.7	17.4	78°19.5	47.2	258°00.3	05.9
14	92°03.8	356°04.6	00.9	306°06.0	16.9	93°21.6	47.1	273°02.8	06.0
15	107°06.3	11°03.8	•• 00.8	321°07.3	•• 16.5	108°23.7	•• 46.9	288°05.3	•• 06.0
16	122°08.8	26°02.9	00.7	336°08.6	16.0	123°25.7	46.8	303°07.8	06.0
17	137°11.2	41°02.1	00.7	351°09.9	15.6	138°27.8	46.6	318°10.3	06.0
18	152°13.7	56°01.3	N25°00.6	6°11.2	N15°15.1	153°29.9	S01°46.5	333°12.8	N03°06.0
19	167°16.2	71°00.5	00.6	21°12.4	14.7	168°32.0	46.3	348°15.3	06.0
20	182°18.6	85°59.6	00.5	36°13.7	14.2	183°34.1	46.1	3°17.8	06.0
21	197°21.1	100°58.8	•• 00.4	51°15.0	•• 13.8	198°36.2	•• 46.0	18°20.3	•• 06.0
22	212°23.6	115°58.0	00.4	66°16.3	13.3	213°38.3	45.8	33°22.7	06.0
23	227°26.0	130°57.1	00.3	81°17.6	12.9	228°40.4	45.7	48°25.2	06.0
Mer.pass. 07:53		ν -0.8' d -0.0' m -3.92		ν 1.3' d -0.4' m 1.02		ν 2.1' d -0.2' m -2.22		ν 2.5' d 0.0' m 0.82	

h Sun Moon						
Sat	GHA	Dec	GHA	ν	Dec	d HP
0	180°50.7	N20°19.3	75°49.6	11.4'	N02°01.8	-14.9' 59.1'
1	195°50.7	19.8	90°20.1	11.4'	01°46.8	-14.9' 59.1'
2	210°50.6	20.3	104°50.5	11.4'	01°31.9	-14.9' 59.1'
3	225°50.6	· · 20.8	119°20.9	11.5'	01°17.0	-14.9' 59.1'
4	240°50.5	21.2	133°51.4	11.5'	01°02.0	-14.9' 59.1'
5	255°50.5	21.7	148°21.9	11.5'	00°47.1	-15.0' 59.1'
6	270°50.5	N20°22.2	162°52.3	11.5'	N00°32.1	-15.0' 59.1'
7	285°50.4	22.7	177°22.8	11.5'	00°17.2	-14.9' 59.1'
8	300°50.4	23.2	191°53.3	11.5'	N00°02.2	-14.9' 59.1'
9	315°50.3	· · 23.7	206°23.8	11.5'	S00°12.7	14.9' 59.1'
10	330°50.3	24.2	220°54.3	11.5'	00°27.7	14.9' 59.1'
11	345°50.2	24.7	235°24.8	11.5'	00°42.6	14.9' 59.0'
12	0°50.2	N20°25.2	249°55.3	11.5'	S00°57.6	14.9' 59.0'
13	15°50.1	25.7	264°25.9	11.5'	01°12.5	14.9' 59.0'
14	30°50.1	26.2	278°56.4	11.5'	01°27.4	14.9' 59.0'
15	45°50.0	· · 26.7	293°26.9	11.5'	01°42.3	14.9' 59.0'
16	60°50.0	27.1	307°57.4	11.5'	01°57.2	14.9' 59.0'
17	75°49.9	27.6	322°27.9	11.5'	02°12.1	14.9' 59.0'
18	90°49.9	N20°28.1	336°58.4	11.5'	S02°26.9	14.8' 59.0'
19	105°49.8	28.6	351°28.9	11.5'	02°41.8	14.8' 59.0'
20	120°49.8	29.1	5°59.4	11.5'	02°56.6	14.8' 59.0'
21	135°49.7	· · 29.6	20°29.9	11.5'	03°11.4	14.8' 59.0'
22	150°49.7	30.1	35°00.4	11.5'	03°26.2	14.8' 59.0'
23	165°49.6	30.5	49°30.8	11.5'	03°41.0	14.7' 59.0'
SD = 15.8' d = 0.5' SD = 16.1'						
Sun	GHA	Dec	GHA	ν	Dec	d HP
0	180°49.6	N20°31.0	64°01.3	11.4'	S03°55.7	14.7' 58.9'
1	195°49.5	31.5	78°31.7	11.4'	04°10.4	14.7' 58.9'
2	210°49.5	32.0	93°02.2	11.4'	04°25.1	14.7' 58.9'
3	225°49.4	· · 32.5	107°32.6	11.4'	04°39.8	14.6' 58.9'
4	240°49.4	32.9	122°03.0	11.4'	04°54.5	14.6' 58.9'
5	255°49.3	33.4	136°33.4	11.4'	05°09.1	14.6' 58.9'
6	270°49.3	N20°33.9	151°03.8	11.4'	S05°23.6	14.6' 58.9'
7	285°49.2	34.4	165°34.1	11.3'	05°38.2	14.5' 58.9'
8	300°49.2	34.9	180°04.5	11.3'	05°52.7	14.5' 58.9'
9	315°49.1	· · 35.3	194°34.8	11.3'	06°07.2	14.4' 58.9'
10	330°49.1	35.8	209°05.1	11.3'	06°21.6	14.4' 58.8'
11	345°49.0	36.3	223°35.4	11.3'	06°36.1	14.4' 58.8'
12	0°49.0	N20°36.8	238°05.6	11.2'	S06°50.4	14.3' 58.8'
13	15°48.9	37.2	252°35.9	11.2'	07°04.7	14.3' 58.8'
14	30°48.9	37.7	267°06.1	11.2'	07°19.0	14.2' 58.8'
15	45°48.8	· · 38.2	281°36.3	11.2'	07°33.3	14.2' 58.8'
16	60°48.8	38.7	296°06.4	11.1'	07°47.5	14.2' 58.8'
17	75°48.7	39.1	310°36.6	11.1'	08°01.6	14.1' 58.8'
18	90°48.7	N20°39.6	325°06.7	11.1'	S08°15.8	14.1' 58.8'
19	105°48.6	40.1	339°36.8	11.1'	08°29.8	14.0' 58.8'
20	120°48.6	40.5	354°06.8	11.0'	08°43.8	14.0' 58.7'
21	135°48.5	· · 41.0	8°36.8	11.0'	08°57.8	13.9' 58.7'
22	150°48.4	41.5	23°06.8	11.0'	09°11.7	13.9' 58.7'
23	165°48.4	42.0	37°36.8	10.9'	09°25.5	13.8' 58.7'
SD = 15.8' d = 0.5' SD = 16.1'						
Mon	GHA	Dec	GHA	ν	Dec	d HP
0	180°48.3	N20°42.4	52°06.7	10.9'	S09°39.4	13.7' 58.7'
1	195°48.3	42.9	66°36.6	10.9'	09°53.1	13.7' 58.7'
2	210°48.2	43.4	81°06.5	10.8'	10°06.8	13.6' 58.7'
3	225°48.2	· · 43.8	95°36.3	10.8'	10°20.4	13.6' 58.7'
4	240°48.1	44.3	110°06.1	10.8'	10°34.0	13.5' 58.6'
5	255°48.1	44.8	124°35.8	10.7'	10°47.5	13.4' 58.6'
6	270°48.0	N20°45.2	139°05.5	10.7'	S11°00.9	13.4' 58.6'
7	285°47.9	45.7	153°35.2	10.6'	11°14.3	13.3' 58.6'
8	300°47.9	46.1	168°04.9	10.6'	11°27.6	13.3' 58.6'
9	315°47.8	· · 46.6	182°34.5	10.6'	11°40.9	13.2' 58.6'
10	330°47.8	47.1	197°04.0	10.5'	11°54.1	13.1' 58.6'
11	345°47.7	47.5	211°33.6	10.5'	12°07.2	13.0' 58.5'
12	0°47.7	N20°48.0	226°03.0	10.4'	S12°20.2	13.0' 58.5'
13	15°47.6	48.5	240°32.5	10.4'	12°33.2	12.9' 58.5'
14	30°47.5	48.9	255°01.9	10.4'	12°46.1	12.8' 58.5'
15	45°47.5	· · 49.4	269°31.2	10.3'	12°58.9	12.7' 58.5'
16	60°47.4	49.8	284°00.6	10.3'	13°11.7	12.7' 58.5'
17	75°47.4	50.3	298°29.8	10.2'	13°24.3	12.6' 58.5'
18	90°47.3	N20°50.7	312°59.1	10.2'	S13°36.9	12.5' 58.4'
19	105°47.3	51.2	327°28.2	10.1'	13°49.4	12.4' 58.4'
20	120°47.2	51.7	341°57.4	10.1'	14°01.9	12.4' 58.4'
21	135°47.1	· · 52.1	356°26.5	10.0'	14°14.2	12.3' 58.4'
22	150°47.1	52.6	10°55.5	10.0'	14°26.5	12.2' 58.4'
23	165°47.0	53.0	25°24.5	10.0'	14°38.7	12.1' 58.4'
SD = 15.8' d = 0.5' SD = 16.0'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	////	////	00:55	23:07	////	////
66°	////	////	01:50	22:07	////	////
64°	////	////	02:23	21:33	////	////
62°	////	01:15	02:47	21:09	22:43	////
60°	////	01:54	03:05	20:50	22:02	////
N 58°	////	02:20	03:21	20:34	21:35	////
56°	01:08	02:41	03:34	20:21	21:14	22:50
54°	01:44	02:57	03:45	20:09	20:58	22:12
52°	02:08	03:11	03:55	19:59	20:43	21:47
50°	02:27	03:23	04:04	19:50	20:31	21:28
45°	03:03	03:48	04:23	19:31	20:06	20:52
N 40°	03:28	04:07	04:38	19:15	19:47	20:26
35°	03:48	04:23	04:51	19:03	19:31	20:06
30°	04:04	04:36	05:02	18:51	19:18	19:50
20°	04:29	04:58	05:22	18:32	18:56	19:25
N 10°	04:49	05:15	05:38	18:16	18:38	19:05
0°	05:05	05:31	05:53	18:00	18:22	18:48
S 10°	05:20	05:46	06:08	17:45	18:08	18:33
20°	05:34	06:01	06:24	17:29	17:53	18:20
30°	05:48	06:17	06:43	17:11	17:36	18:06
35°	05:55	06:26	06:53	17:00	17:27	17:58
40°	06:02	06:36	07:05	16:48	17:17	17:51
45°	06:11	06:47	07:20	16:34	17:06	17:42
S 50°	06:20	07:00	07:37	16:16	16:53	17:33
52°	06:24	07:06	07:45	16:08	16:47	17:29
54°	06:29	07:13	07:54	15:59	16:40	17:24
56°	06:33	07:20	08:04	15:49	16:33	17:20
58°	06:39	07:29	08:16	15:37	16:24	17:14
S 60°	06:44	07:38	08:30	15:23	16:15	17:09

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	13:43	15:57	18:26	01:18	00:52	00:22 23:39
N 70°	13:40	15:44	17:57	01:16	00:58	00:38
68°	13:39	15:34	17:35	01:15	01:03	00:50
66°	13:37	15:26	17:19	01:13	01:07	01:00
64°	13:36	15:19	17:05	01:12	01:10	01:08
62°	13:35	15:14	16:53	01:11	01:13	01:16
60°	13:34	15:08	16:44	01:10	01:16	01:22
N 58°	13:33	15:04	16:35	01:09	01:18	01:28
56°	13:32	15:00	16:28	01:08	01:20	01:33
54°	13:32	14:56	16:22	01:08	01:22	01:37
52°	13:31	14:53	16:16	01:07	01:24	01:42
50°	13:31	14:50	16:10	01:06	01:25	01:45
45°	13:29	14:44	15:59	01:05	01:29	01:54
N 40°	13:28	14:39	15:50	01:04	01:32	02:00
35°	13:28	14:34	15:42	01:03	01:34	02:06
30°	13:27	14:31	15:35	01:02	01:36	02:12
20°	13:26	14:24	15:23	01:01	01:40	02:21
N 10°	13:25	14:18	15:12	00:59	01:44	02:29
0°	13:24	14:13	15:03	00:58	01:47	02:36
S 10°	13:23	14:07	14:53	00:57	01:50	02:44
20°	13:22	14:02	14:43	00:55	01:53	02:52
30°	13:21	13:55	14:31	00:54	01:57	03:01
35°	13:20	13:51	14:25	00:53	02:00	03:07
40°	13:19	13:47	14:17	00:52	02:02	03:13
45°	13:19	13:43	14:08	00:50	02:05	03:20
S 50°	13:18	13:37	13:58	00:49	02:09	03:28
52°	13:17	13:34	13:53	00:48	02:10	03:32
54°	13:17	13:32	13:48	00:47	02:12	03:37
56°	13:16	13:28	13:42	00:46	02:14	03:42
58°	13:16	13:25	13:36	00:45	02:16	03:47
S 60°	13:15	13:21	13:29	00:44	02:19	03:53

Day	Sun		Mer. Pass hh:mm	Moon		Age 8-10 62-82%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
22	03:23	03:21	11:57	19:35	07:11	
23	03:18	03:16	11:57	20:24	08:00	
24	03:13	03:11	11:57	21:15	08:49	

May 25, 26, 27 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	242°28.5	145°56.3	N25°00.3	96°18.9	N15°12.4	243°42.5	S01°45.5	63°27.7	N03°06.0	Alpheratz	357°46.0	29°08.8
1	257°31.0	160°55.5	00.2	111°20.2	12.0	258°44.5	45.4	78°30.2	06.0	Ankaa	353°18.0	-42°14.7
2	272°33.4	175°54.7	00.1	126°21.4	11.5	273°46.6	45.2	93°32.7	06.0	Schedar	349°43.6	56°35.5
3	287°35.9	190°53.8	•• 00.0	141°22.7	•• 11.1	288°48.7	•• 45.1	108°35.2	•• 06.0	Diphda	348°58.3	-17°55.6
4	302°38.3	205°53.0	25°00.0	156°24.0	10.6	303°50.8	44.9	123°37.7	06.0	Achernar	335°28.7	-57°10.8
5	317°40.8	220°52.2	24°59.9	171°25.3	10.2	318°52.9	44.8	138°40.2	06.0	Hamal	328°03.6	23°30.6
6	332°43.3	235°51.4	N24°59.8	186°26.6	N15°09.7	333°55.0	S01°44.6	153°42.7	N03°06.0	Polaris	319°17.9	89°18.5
7	347°45.7	250°50.5	59.8	201°27.9	09.3	348°57.1	44.5	168°45.2	06.0	Acamar	315°20.4	-40°15.7
8	2°48.2	265°49.7	59.7	216°29.2	08.8	3°59.2	44.3	183°47.6	06.0	Menkar	314°17.7	4°07.8
9	17°50.7	280°48.9	•• 59.6	231°30.5	•• 08.4	19°01.3	•• 44.2	198°50.1	•• 06.0	Mirfak	308°44.1	49°53.8
10	32°53.1	295°48.1	59.5	246°31.7	07.9	34°03.4	44.0	213°52.6	06.0	Aldebaran	290°52.3	16°31.8
11	47°55.6	310°47.3	59.4	261°33.0	07.5	49°05.4	43.9	228°55.1	06.0	Rigel	281°14.5	-8°11.4
12	62°58.1	325°46.4	N24°59.4	276°34.3	N15°07.0	64°07.5	S01°43.7	243°57.6	N03°06.0	Capella	280°38.2	46°00.5
13	78°00.5	340°45.6	59.3	291°35.6	06.6	79°09.6	43.6	259°00.1	06.0	Bellatrix	278°34.7	6°21.5
14	93°03.0	355°44.8	59.2	306°36.9	06.1	94°11.7	43.4	274°02.6	06.0	Elnath	278°15.8	28°37.0
15	108°05.4	10°44.0	•• 59.1	321°38.2	•• 05.7	109°13.8	•• 43.3	289°05.1	•• 06.0	Alnilam	275°49.0	-1°11.8
16	123°07.9	25°43.1	59.0	336°39.4	05.2	124°15.9	43.1	304°07.5	06.0	Betelgeuse	271°04.0	7°24.5
17	138°10.4	40°42.3	58.9	351°40.7	04.7	139°18.0	43.0	319°10.0	06.0	Canopus	263°57.6	-52°42.2
18	153°12.8	55°41.5	N24°58.9	6°42.0	N15°04.3	154°20.1	S01°42.8	334°12.5	N03°06.0	Sirius	258°36.0	-16°44.0
19	168°15.3	70°40.7	58.8	21°43.3	03.8	169°22.2	42.7	349°15.0	06.0	Adhara	255°14.6	-28°59.4
20	183°17.8	85°39.9	58.7	36°44.6	03.4	184°24.3	42.5	4°17.5	06.0	Procyon	245°02.3	5°11.8
21	198°20.2	100°39.0	•• 58.6	51°45.9	•• 02.9	199°26.4	•• 42.4	19°20.0	•• 06.0	Pollux	243°30.7	28°00.1
22	213°22.7	115°38.2	58.5	66°47.1	02.5	214°28.5	42.2	34°22.5	06.0	Avior	234°19.4	-59°32.9
23	228°25.2	130°37.4	58.4	81°48.4	02.0	229°30.6	42.1	49°25.0	06.0	Suhail	222°54.3	-43°28.8
Mer.pass. 07:49		ν -0.8' d -0.1' m -3.92		ν 1.3' d -0.4' m 1.03		ν 2.1' d -0.2' m -2.22		ν 2.5' d 0.0' m 0.82				

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	243°27.6	145°36.6	N24°58.3	96°49.7	N15°01.6	244°32.7	S01°41.9	64°27.4	N03°06.0			
1	258°30.1	160°35.7	58.2	111°51.0	01.1	259°34.7	41.8	79°29.9	06.0			
2	273°32.6	175°34.9	58.1	126°52.3	00.7	274°36.8	41.6	94°32.4	06.0			
3	288°35.0	190°34.1	•• 58.0	141°53.6	15°00.2	289°38.9	•• 41.5	109°34.9	•• 06.0			
4	303°37.5	205°33.3	57.9	156°54.8	14°59.8	304°41.0	41.3	124°37.4	06.0			
5	318°39.9	220°32.5	57.8	171°56.1	59.3	319°43.1	41.2	139°39.9	06.0			
6	333°42.4	235°31.6	N24°57.7	186°57.4	N14°58.9	334°45.2	S01°41.0	154°42.4	N03°06.0			
7	348°44.9	250°30.8	57.6	201°58.7	58.4	349°47.3	40.9	169°44.8	06.0			
8	3°47.3	265°30.0	57.5	217°00.0	57.9	4°49.4	40.7	184°47.3	06.0			
9	18°49.8	280°29.2	•• 57.4	232°01.2	•• 57.5	19°51.5	•• 40.6	199°49.8	•• 06.0			
10	33°52.3	295°28.4	57.3	247°02.5	57.0	34°53.6	40.4	214°52.3	06.0			
11	48°54.7	310°27.6	57.2	262°03.8	56.6	49°55.7	40.3	229°54.8	06.0			
12	63°57.2	325°26.7	N24°57.0	277°05.1	N14°56.1	64°57.8	S01°40.1	244°57.3	N03°06.0			
13	78°59.7	340°25.9	56.9	292°06.4	55.7	79°59.9	40.0	259°59.8	06.0			
14	94°02.1	355°25.1	56.8	307°07.6	55.2	95°02.0	39.8	275°02.2	06.0			
15	109°04.6	10°24.3	•• 56.7	322°08.9	•• 54.8	110°04.1	•• 39.7	290°04.7	•• 06.0			
16	124°07.1	25°23.5	56.6	337°10.2	54.3	125°06.2	39.5	305°07.2	06.0			
17	139°09.5	40°22.6	56.5	352°11.5	53.8	140°08.3	39.4	320°09.7	06.0			
18	154°12.0	55°21.8	N24°56.4	7°12.8	N14°53.4	155°10.4	S01°39.2	335°12.2	N03°06.0			
19	169°14.4	70°21.0	56.2	22°14.0	52.9	170°12.5	39.1	350°14.7	06.0			
20	184°16.9	85°20.2	56.1	37°15.3	52.5	185°14.6	38.9	5°17.1	06.0			
21	199°19.4	100°19.4	•• 56.0	52°16.6	•• 52.0	200°16.7	•• 38.8	20°19.6	•• 06.0			
22	214°21.8	115°18.6	55.9	67°17.9	51.6	215°18.8	38.6	35°22.1	06.0			
23	229°24.3	130°17.7	55.8	82°19.1	51.1	230°20.9	38.5	50°24.6	06.0			
Mer.pass. 07:45		ν -0.8' d -0.1' m -3.92		ν 1.3' d -0.5' m 1.04		ν 2.1' d -0.2' m -2.23		ν 2.5' d -0.0' m 0.83				

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	244°26.8	145°16.9	N24°55.6	97°20.4	N14°50.6	245°23.0	S01°38.3	65°27.1	N03°06.0			
1	259°29.2	160°16.1	55.5	112°21.7	50.2	260°25.1	38.2	80°29.5	06.0			
2	274°31.7	175°15.3	55.4	127°23.0	49.7	275°27.2	38.0	95°32.0	06.0			
3	289°34.2	190°14.5	•• 55.2	142°24.3	•• 49.3	290°29.3	•• 37.9	110°34.5	•• 06.0			
4	304°36.6	205°13.7	55.1	157°25.5	48.8	305°31.3	37.7	125°37.0	06.0			
5	319°39.1	220°12.9	55.0	172°26.8	48.3	320°33.4	37.6	140°39.5	06.0			
6	334°41.6	235°12.0	N24°54.9	187°28.1	N14°47.9	335°35.5	S01°37.4	155°42.0	N03°06.0			
7	349°44.0	250°11.2	54.7	202°29.4	47.4	350°37.6	37.3	170°44.4	06.0			
8	4°46.5	265°10.4	54.6	217°30.6	47.0	5°39.7	37.1	185°46.9	06.0			
9	19°48.9	280°09.6	•• 54.4	232°31.9	•• 46.5	20°41.8	•• 37.0	200°49.4	•• 06.0			
10	34°51.4	295°08.8	54.3	247°33.2	46.1	35°43.9	36.8	215°51.9	06.0			
11	49°53.9	310°08.0	54.2	262°34.5	45.6	50°46.0	36.7	230°54.4	05.9			
12	64°56.3	325°07.2	N24°54.0	277°35.7	N14°45.1	65°48.1	S01°36.5	245°56.8	N03°05.9			
13	79°58.8	340°06.4	53.9	292°37.0	44.7	80°50.2	36.4	260°59.3	05.9			
14	95°01.3	355°05.5	53.7	307°38.3	44.2	95°52.3	36.3	276°01.8	05.9			
15	110°03.7	10°04.7	•• 53.6	322°39.6	•• 43.8	110°54.4	•• 36.1	291°04.3	•• 05.9			
16	125°06.2	25°03.9	53.5	337°40.8	43.3	125°56.6	36.0	306°06.8	05.9			
17	140°08.7	40°03.1	53.3	352°42.1	42.8	140°58.7	35.8	321°09.2	05.9			
18	155°11.1	55°02.3	N24°53.2	7°43.4	N14°42.4	156°00.8	S01°35.7	336°11.7	N03°05.9			
19	170°13.6	70°01.5	53.0	22°44.7	41.9	171°02.9	35.5	351°14.2	05.9			
20	185°16.0	85°00.7	52.9	37°45.9	41.4	186°05.0	35.4	6°16.7	05.9			
21	200°18.5	99°59.9	•• 52.7	52°47.2	•• 41.0	201°07.1	•• 35.2	21°19.2	•• 05.9			
22	215°21.0	114°59.1	52.6	67°48.5	40.5	216°09.2	35.1	36°21.6	05.9			
23	230°23.4	129°58.2	52.4	82°49.8	40.1	231°11.3	34.9	51°24.1	05.9			
Mer.pass. 07:41		ν -0.8' d -0.1' m -3.92		ν 1.3' d -0.5' m 1.06		ν 2.1' d -0.1' m -2.23		ν 2.5' d -0.0' m 0.83				

May 25 Tue		SHA	Mer.pass
Venus		263°27.8	14:17
Mars		213°50.4	17:33
Jupiter		1°14.0	07:44
Saturn		180°59.2	19:43
May 26 Wed		SHA	Mer.pass
Venus		262°08.9	14:18
Mars		213°22.1	17:31
Jupiter		1°05.0	07:41
Saturn		180°59.8	19:39
May 27 Thu		SHA	Mer.pass
Venus		260°50.2	14:20
Mars		212°53.7	17:29
Jupiter		0°56.2	07:37
Saturn		181°00.3	19:35
Horizontal parallax			
Venus:		0.1	
Mars:		0.1	

Sun							Moon						
h	GHA	Dec	GHA	ν	Dec	d	HP						
Tue													
0	180°47.0	N20°53.5	39°53.5	9.9'	S14°50.8	12.0'	58.4'						
1	195°46.9	53.9	54°22.4	9.9'	15°02.8	11.9'	58.3'						
2	210°46.8	54.4	68°51.3	9.8'	15°14.7	11.8'	58.3'						
3	225°46.8	· · 54.8	83°20.1	9.8'	15°26.6	11.7'	58.3'						
4	240°46.7	55.3	97°48.9	9.7'	15°38.3	11.7'	58.3'						
5	255°46.7	55.7	112°17.6	9.7'	15°50.0	11.6'	58.3'						
6	270°46.6	N20°56.2	126°46.2	9.6'	S16°01.5	11.5'	58.3'						
7	285°46.5	56.6	141°14.9	9.6'	16°13.0	11.4'	58.2'						
8	300°46.5	57.1	155°43.5	9.5'	16°24.3	11.3'	58.2'						
9	315°46.4	· · 57.5	170°12.0	9.5'	16°35.6	11.2'	58.2'						
10	330°46.3	58.0	184°40.5	9.4'	16°46.8	11.1'	58.2'						
11	345°46.3	58.4	199°08.9	9.4'	16°57.9	11.0'	58.2'						
12	0°46.2	N20°58.9	213°37.3	9.3'	S17°08.8	10.9'	58.2'						
13	15°46.2	59.3	228°05.6	9.3'	17°19.7	10.8'	58.1'						
14	30°46.1	20°59.8	242°33.9	9.2'	17°30.5	10.7'	58.1'						
15	45°46.0	21°00.2	257°02.1	9.2'	17°41.1	10.6'	58.1'						
16	60°46.0	00.6	271°30.3	9.1'	17°51.7	10.5'	58.1'						
17	75°45.9	01.1	285°58.5	9.1'	18°02.2	10.3'	58.1'						
18	90°45.8	N21°01.5	300°26.6	9.0'	S18°12.5	10.2'	58.0'						
19	105°45.8	02.0	314°54.6	9.0'	18°22.8	10.1'	58.0'						
20	120°45.7	02.4	329°22.6	9.0'	18°32.9	10.0'	58.0'						
21	135°45.7	· · 02.8	343°50.6	8.9'	18°42.9	9.9'	58.0'						
22	150°45.6	03.3	358°18.5	8.9'	18°52.8	9.8'	58.0'						
23	165°45.5	03.7	12°46.3	8.8'	19°02.6	9.7'	57.9'						
	SD = 15.8'	d = 0.5'			SD = 15.9'								
Wed	GHA	Dec	GHA	ν	Dec	d	HP						
0	180°45.5	N21°04.2	27°14.1	8.8'	S19°12.3	9.6'	57.9'						
1	195°45.4	04.6	41°41.9	8.7'	19°21.9	9.5'	57.9'						
2	210°45.3	05.0	56°09.6	8.7'	19°31.3	9.3'	57.9'						
3	225°45.3	· · 05.5	70°37.3	8.6'	19°40.6	9.2'	57.9'						
4	240°45.2	05.9	85°04.9	8.6'	19°49.9	9.1'	57.9'						
5	255°45.1	06.3	99°32.5	8.5'	19°59.0	9.0'	57.8'						
6	270°45.1	N21°06.8	114°00.0	8.5'	S20°07.9	8.9'	57.8'						
7	285°45.0	07.2	128°27.5	8.4'	20°16.8	8.7'	57.8'						
8	300°44.9	07.6	142°55.0	8.4'	20°25.5	8.6'	57.8'						
9	315°44.9	· · 08.1	157°22.4	8.4'	20°34.1	8.5'	57.8'						
10	330°44.8	08.5	171°49.8	8.3'	20°42.6	8.4'	57.7'						
11	345°44.7	08.9	186°17.1	8.3'	20°51.0	8.2'	57.7'						
12	0°44.7	N21°09.4	200°44.4	8.2'	S20°59.2	8.1'	57.7'						
13	15°44.6	09.8	215°11.6	8.2'	21°07.3	8.0'	57.7'						
14	30°44.5	10.2	229°38.8	8.2'	21°15.3	7.9'	57.6'						
15	45°44.5	· · 10.7	244°06.0	8.1'	21°23.2	7.7'	57.6'						
16	60°44.4	11.1	258°33.1	8.1'	21°30.9	7.6'	57.6'						
17	75°44.3	11.5	273°00.2	8.0'	21°38.5	7.5'	57.6'						
18	90°44.3	N21°11.9	287°27.2	8.0'	S21°46.0	7.3'	57.6'						
19	105°44.2	12.4	301°54.2	8.0'	21°53.3	7.2'	57.5'						
20	120°44.1	12.8	316°21.2	7.9'	22°00.5	7.1'	57.5'						
21	135°44.0	· · 13.2	330°48.2	7.9'	22°07.6	6.9'	57.5'						
22	150°44.0	13.6	345°15.1	7.9'	22°14.5	6.8'	57.5'						
23	165°43.9	14.1	359°41.9	7.8'	22°21.3	6.7'	57.5'						
	SD = 15.8'	d = 0.4'			SD = 15.8'								
Thu	GHA	Dec	GHA	ν	Dec	d	HP						
0	180°43.8	N21°14.5	14°08.8	7.8'	S22°28.0	6.5'	57.4'						
1	195°43.8	14.9	28°35.6	7.8'	22°34.5	6.4'	57.4'						
2	210°43.7	15.3	43°02.4	7.8'	22°40.9	6.3'	57.4'						
3	225°43.6	· · 15.8	57°29.2	7.7'	22°47.2	6.1'	57.4'						
4	240°43.6	16.2	71°55.9	7.7'	22°53.3	6.0'	57.3'						
5	255°43.5	16.6	86°22.6	7.7'	22°59.3	5.8'	57.3'						
6	270°43.4	N21°17.0	100°49.3	7.7'	S23°05.2	5.7'	57.3'						
7	285°43.3	17.4	115°15.9	7.6'	23°10.9	5.6'	57.3'						
8	300°43.3	17.8	129°42.6	7.6'	23°16.4	5.4'	57.3'						
9	315°43.2	· · 18.3	144°09.2	7.6'	23°21.9	5.3'	57.2'						
10	330°43.1	18.7	158°35.8	7.6'	23°27.2	5.1'	57.2'						
11	345°43.1	19.1	173°02.3	7.6'	23°32.3	5.0'	57.2'						
12	0°43.0	N21°19.5	187°28.9	7.5'	S23°37.3	4.9'	57.2'						
13	15°42.9	19.9	201°55.4	7.5'	23°42.2	4.7'	57.1'						
14	30°42.8	20.3	216°22.0	7.5'	23°46.9	4.6'	57.1'						
15	45°42.8	· · 20.8	230°48.5	7.5'	23°51.5	4.4'	57.1'						
16	60°42.7	21.2	245°15.0	7.5'	23°55.9	4.3'	57.1'						
17	75°42.6	21.6	259°41.5	7.5'	24°00.2	4.1'	57.0'						
18	90°42.5	N21°22.0	274°07.9	7.5'	S24°04.3	4.0'	57.0'						
19	105°42.5	22.4	288°34.4	7.5'	24°08.3	3.9'	57.0'						
20	120°42.4	22.8	303°00.9	7.5'	24°12.2	3.7'	57.0'						
21	135°42.3	· · 23.2	317°27.3	7.4'	24°15.9	3.6'	57.0'						
22	150°42.3	23.6	331°53.8	7.4'	24°19.5	3.4'	56.9'						
23	165°42.2	24.0	346°20.2	7.4'	24°22.9	3.3'	56.9'						
	SD = 15.8'	d = 0.4'			SD = 15.7'								

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°						
68°	////	////	00:20	////	////	////
66°	////	////	01:38	22:20	////	////
64°	////	////	02:14	21:42	////	////
62°	////	00:59	02:40	21:16	23:00	////
60°	////	01:45	03:00	20:56	22:12	////
N 58°	////	02:13	03:16	20:39	21:43	////
56°	00:53	02:35	03:30	20:26	21:21	23:06
54°	01:35	02:52	03:41	20:13	21:03	22:21
52°	02:02	03:07	03:52	20:03	20:48	21:54
50°	02:22	03:20	04:01	19:54	20:35	21:33
45°	02:59	03:45	04:21	19:34	20:10	20:56
N 40°	03:25	04:05	04:36	19:18	19:50	20:29
35°	03:46	04:21	04:50	19:05	19:34	20:09
30°	04:02	04:35	05:01	18:53	19:20	19:52
20°	04:28	04:57	05:21	18:33	18:57	19:26
N 10°	04:49	05:15	05:38	18:16	18:39	19:06
0°	05:05	05:31	05:53	18:01	18:23	18:49
S 10°	05:21	05:47	06:09	17:45	18:07	18:33
20°	05:35	06:02	06:25	17:28	17:52	18:19
30°	05:49	06:18	06:44	17:10	17:35	18:05
35°	05:57	06:28	06:55	16:59	17:26	17:57
40°	06:05	06:38	07:08	16:46	17:16	17:49
45°	06:13	06:50	07:22	16:31	17:04	17:40
S 50°	06:23	07:03	07:40	16:13	16:50	17:31
52°	06:27	07:10	07:49	16:05	16:44	17:26
54°	06:32	07:17	07:58	15:55	16:37	17:22
56°	06:37	07:24	08:09	15:44	16:29	17:16
58°	06:43	07:33	08:21	15:32	16:21	17:11
S 60°	06:49	07:43	08:36	15:18	16:11	17:05

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	■	■	■	■	■	■
N 70°	20:41	■	■	00:11 23:16	■	■
68°	19:49	■	■	00:34	00:10	■
66°	19:17	21:26	■	00:52	00:43	00:26
64°	18:53	20:44	22:29	01:07	01:07	01:09
62°	18:35	20:15	21:47	01:20	01:26	01:38
60°	18:20	19:54	21:18	01:30	01:42	02:01
N 58°	18:07	19:36	20:56	01:40	01:56	02:19
56°	17:56	19:21	20:38	01:48	02:07	02:34
54°	17:46	19:08	20:23	01:55	02:18	02:47
52°	17:38	18:57	20:10	02:02	02:27	02:59
50°	17:30	18:47	19:58	02:08	02:35	03:09
45°	17:14	18:26	19:35	02:21	02:53	03:31
N 40°	17:00	18:10	19:15	02:32	03:07	03:48
35°	16:49	17:56	19:00	02:41	03:19	04:03
30°	16:39	17:43	18:46	02:49	03:30	04:16
20°	16:22	17:23	18:22	03:03	03:49	04:38
N 10°	16:08	17:05	18:02	03:15	04:05	04:57
0°	15:54	16:48	17:43	03:27	04:20	05:14
S 10°	15:41	16:32	17:25	03:39	04:35	05:32
20°	15:27	16:14	17:05	03:51	04:51	05:51
30°	15:10	15:54	16:42	04:05	05:10	06:14
35°	15:01	15:42	16:28	04:14	05:21	06:27
40°	14:50	15:28	16:13	04:23	05:34	06:42
45°	14:38	15:13	15:54	04:35	05:49	06:59
S 50°	14:23	14:53	15:31	04:48	06:07	07:22
52°	14:16	14:44	15:21	04:55	06:16	07:32
54°	14:08	14:34	15:08	05:02	06:25	07:44
56°	13:59	14:22	14:54	05:10	06:37	07:58
58°	13:50	14:09	14:37	05:19	06:49	08:15
S 60°	13:38	13:53	14:18	05:29	07:04	08:34

May 28, 29, 30 UT (Fri., Sat., Sun.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	245°25.9	144°57.4	N24°52.3	97°51.0	N14°39.6	246°13.4	S01°34.8	66°26.6	N03°05.9	Alpheratz	357°46.0	29°08.8
1	260°28.4	159°56.6	52.1	112°52.3	39.1	261°15.5	34.6	81°29.1	05.9	Ankaa	353°18.0	-42°14.7
2	275°30.8	174°55.8	51.9	127°53.6	38.7	276°17.6	34.5	96°31.6	05.9	Schedar	349°43.5	56°35.5
3	290°33.3	189°55.0	· · 51.8	142°54.9	· · 38.2	291°19.7	· · 34.3	111°34.0	· · 05.9	Diphda	348°58.3	-17°55.6
4	305°35.8	204°54.2	51.6	157°56.1	37.8	306°21.8	34.2	126°36.5	05.9	Achernar	335°28.7	-57°10.8
5	320°38.2	219°53.4	51.5	172°57.4	37.3	321°23.9	34.1	141°39.0	05.9	Hamal	328°03.6	23°30.6
6	335°40.7	234°52.6	N24°51.3	187°58.7	N14°36.8	336°26.0	S01°33.9	156°41.5	N03°05.9	Polaris	319°16.9	89°18.5
7	350°43.2	249°51.8	51.1	202°59.9	36.4	351°28.1	33.8	171°43.9	05.9	Acamar	315°20.4	-40°15.7
8	5°45.6	264°51.0	51.0	218°01.2	35.9	6°30.2	33.6	186°46.4	05.9	Menkar	314°17.7	4°07.8
9	20°48.1	279°50.2	· · 50.8	233°02.5	· · 35.4	21°32.3	· · 33.5	201°48.9	· · 05.9	Mirfak	308°44.1	49°53.8
10	35°50.5	294°49.4	50.6	248°03.8	35.0	36°34.4	33.3	216°51.4	05.9	Aldebaran	290°52.3	16°31.8
11	50°53.0	309°48.5	50.5	263°05.0	34.5	51°36.5	33.2	231°53.8	05.9	Rigel	281°14.5	-8°11.4
12	65°55.5	324°47.7	N24°50.3	278°06.3	N14°34.0	66°38.6	S01°33.0	246°56.3	N03°05.9	Capella	280°38.2	46°00.5
13	80°57.9	339°46.9	50.1	293°07.6	33.6	81°40.7	32.9	261°58.8	05.9	Bellatrix	278°34.7	6°21.5
14	96°00.4	354°46.1	50.0	308°08.8	33.1	96°42.8	32.7	277°01.3	05.9	Elnath	278°15.8	28°37.0
15	111°02.9	9°45.3	· · 49.8	323°10.1	· · 32.7	111°44.9	· · 32.6	292°03.8	· · 05.8	Alnilam	275°49.0	-1°11.8
16	126°05.3	24°44.5	49.6	338°11.4	32.2	126°47.0	32.5	307°06.2	05.8	Betelgeuse	271°04.0	7°24.5
17	141°07.8	39°43.7	49.4	353°12.7	31.7	141°49.1	32.3	322°08.7	05.8	Canopus	263°57.6	-52°42.2
18	156°10.3	54°42.9	N24°49.3	8°13.9	N14°31.3	156°51.3	S01°32.2	337°11.2	N03°05.8	Sirius	258°36.0	-16°44.0
19	171°12.7	69°42.1	49.1	23°15.2	30.8	171°53.4	32.0	352°13.7	05.8	Adhara	255°14.6	-28°59.3
20	186°15.2	84°41.3	48.9	38°16.5	30.3	186°55.5	31.9	7°16.1	05.8	Procyon	245°02.3	5°11.8
21	201°17.7	99°40.5	· · 48.7	53°17.7	· · 29.9	201°57.6	· · 31.7	22°18.6	· · 05.8	Pollux	243°30.7	28°00.1
22	216°20.1	114°39.7	48.6	68°19.0	29.4	216°59.7	31.6	37°21.1	05.8	Avior	234°19.4	-59°32.9
23	231°22.6	129°38.9	48.4	83°20.3	28.9	232°01.8	31.4	52°23.6	05.8	Suhail	222°54.3	-43°28.8
Mer.pass. 07:37		ν -0.8' d -0.2' m -3.92		ν 1.3' d -0.5' m 1.07		ν 2.1' d -0.1' m -2.24		ν 2.5' d -0.0' m 0.84				

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	246°25.0	144°38.1	N24°48.2	98°21.5	N14°28.5	247°03.9	S01°31.3	67°26.0	N03°05.8	Alphard	217°58.4	-8°42.4
1	261°27.5	159°37.3	48.0	113°22.8	28.0	262°06.0	31.2	82°28.5	05.8	Regulus	207°45.9	11°54.9
2	276°30.0	174°36.5	47.8	128°24.1	27.5	277°08.1	31.0	97°31.0	05.8	Dubhe	193°54.1	61°41.9
3	291°32.4	189°35.7	· · 47.6	143°25.3	· · 27.1	292°10.2	· · 30.9	112°33.5	· · 05.8	Denebola	182°35.8	14°30.8
4	306°34.9	204°34.9	47.4	158°26.6	26.6	307°12.3	30.7	127°35.9	05.8	Gienah	175°54.5	-17°36.2
5	321°37.4	219°34.1	47.2	173°27.9	26.1	322°14.4	30.6	142°38.4	05.8	Acrux	173°11.7	-63°09.8
6	336°39.8	234°33.3	N24°47.1	188°29.1	N14°25.7	337°16.5	S01°30.4	157°40.9	N03°05.8	Gacrux	172°03.2	-57°10.7
7	351°42.3	249°32.5	46.9	203°30.4	25.2	352°18.6	30.3	172°43.4	05.8	Alioth	166°22.1	55°54.3
8	6°44.8	264°31.7	46.7	218°31.7	24.7	7°20.8	30.1	187°45.8	05.8	Spica	158°33.4	-11°13.1
9	21°47.2	279°30.9	· · 46.5	233°32.9	· · 24.3	22°22.9	· · 30.0	202°48.3	· · 05.8	Alkaid	153°00.2	49°15.7
10	36°49.7	294°30.1	46.3	248°34.2	23.8	37°25.0	29.9	217°50.8	05.7	Hadar	148°50.7	-60°25.7
11	51°52.2	309°29.3	46.1	263°35.5	23.3	52°27.1	29.7	232°53.2	05.7	Menkent	148°09.9	-36°25.5
12	66°54.6	324°28.5	N24°45.9	278°36.7	N14°22.9	67°29.2	S01°29.6	247°55.7	N03°05.7	Arcturus	145°57.4	19°07.6
13	81°57.1	339°27.7	45.7	293°38.0	22.4	82°31.3	29.4	262°58.2	05.7	Rigil Kent.	139°54.3	-60°52.9
14	96°59.5	354°26.9	45.5	308°39.3	21.9	97°33.4	29.3	278°00.7	05.7	Kochab	137°18.6	74°06.8
15	112°02.0	9°26.1	· · 45.3	323°40.5	· · 21.5	112°35.5	· · 29.1	293°03.1	· · 05.7	Zuben'ubi	137°07.6	-16°05.3
16	127°04.5	24°25.3	45.1	338°41.8	21.0	127°37.6	29.0	308°05.6	05.7	Alphecca	126°12.5	26°40.7
17	142°06.9	39°24.5	44.9	353°43.1	20.5	142°39.7	28.9	323°08.1	05.7	Antares	112°28.6	-26°27.4
18	157°09.4	54°23.7	N24°44.7	8°44.3	N14°20.1	157°41.9	S01°28.7	338°10.5	N03°05.7	Atria	107°32.0	-69°02.8
19	172°11.9	69°22.9	44.5	23°45.6	19.6	172°44.0	28.6	353°13.0	05.7	Sabik	102°14.8	-15°44.3
20	187°14.3	84°22.1	44.3	38°46.9	19.1	187°46.1	28.4	8°15.5	05.7	Shaula	96°24.5	-37°06.7
21	202°16.8	99°21.3	· · 44.0	53°48.1	· · 18.7	202°48.2	· · 28.3	23°18.0	· · 05.7	Rasalhague	96°08.2	12°33.1
22	217°19.3	114°20.5	43.8	68°49.4	18.2	217°50.3	28.2	38°20.4	05.7	Eltanin	90°46.7	51°29.1
23	232°21.7	129°19.7	43.6	83°50.7	17.7	232°52.4	28.0	53°22.9	05.7	Kaus Aust.	83°46.4	-34°22.7
Mer.pass. 07:33		ν -0.8' d -0.2' m -3.93		ν 1.3' d -0.5' m 1.08		ν 2.1' d -0.1' m -2.24		ν 2.5' d -0.0' m 0.84		Vega	80°40.1	38°47.5

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	247°24.2	144°18.9	N24°43.4	98°51.9	N14°17.2	247°54.5	S01°27.9	68°25.4	N03°05.7	Nunki	76°00.8	-26°16.9
1	262°26.7	159°18.1	43.2	113°53.2	16.8	262°56.6	27.7	83°27.8	05.7	Altair	62°10.2	8°53.7
2	277°29.1	174°17.3	43.0	128°54.5	16.3	277°58.8	27.6	98°30.3	05.6	Peacock	53°22.3	-56°41.8
3	292°31.6	189°16.5	· · 42.8	143°55.7	· · 15.8	293°00.9	· · 27.4	113°32.8	· · 05.6	Deneb	49°32.8	45°18.9
4	307°34.0	204°15.7	42.6	158°57.0	15.4	308°03.0	27.3	128°35.3	05.6	Enif	33°49.2	9°55.3
5	322°36.5	219°14.9	42.3	173°58.3	14.9	323°05.1	27.2	143°37.7	05.6	Al Na'ir	27°46.4	-46°54.3
6	337°39.0	234°14.1	N24°42.1	188°59.5	N14°14.4	338°07.2	S01°27.0	158°40.2	N03°05.6	Fomalhaut	15°26.4	-29°33.8
7	352°41.4	249°13.3	41.9	204°00.8	14.0	353°09.3	26.9	173°42.7	05.6	Scheat	13°55.6	28°08.3
8	7°43.9	264°12.5	41.7	219°02.0	13.5	8°11.4	26.7	188°45.1	05.6	Markab	13°40.6	15°15.6
9	22°46.4	279°11.8	· · 41.4	234°03.3	· · 13.0	23°13.5	· · 26.6	203°47.6	· · 05.6			
10	37°48.8	294°11.0	41.2	249°04.6	12.5	38°15.7	26.5	218°50.1	05.6	May 28 Fri	SHA	Mer.pass
11	52°51.3	309°10.2	41.0	264°05.8	12.1	53°17.8	26.3	233°52.5	05.6	Venus	259°31.5	14:21
12	67°53.8	324°09.4	N24°40.8	279°07.1	N14°11.6	68°19.9	S01°26.2	248°55.0	N03°05.6	Mars	212°25.1	17:27
13	82°56.2	339°08.6	40.5	294°08.4	11.1	83°22.0	26.0	263°57.5	05.6	Jupiter	0°47.5	07:34
14	97°58.7	354°07.8	40.3	309°09.6	10.7	98°24.1	25.9	278°59.9	05.6	Saturn	181°00.7	19:31
15	113°01.2	9°07.0	· · 40.1	324°10.9	· · 10.2	113°26.2	· · 25.7	294°02.4	· · 05.5			
16	128°03.6	24°06.2	39.8	339°12.1	09.7	128°28.3	25.6	309°04.9	05.5	May 29 Sat	SHA	Mer.pass
17	143°06.1	39°05.4	39.6	354°13.4	09.2	143°30.5	25.5	324°07.4	05.5	Venus	258°13.0	14:22
18	158°08.5	54°04.6	N24°39.4	9°14.7	N14°08.8	158°32.6	S01°25.3	339°09.8	N03°05.5	Mars	211°56.5	17:25
19	173°11.0	69°03.8	39.1	24°15.9	08.3	173°34.7	25.2	354°12.3	05.5	Jupiter	0°38.8	07:31
20	188°13.5	84°03.1	38.9	39°17.2	07.8	188°36.8	25.0	9°14.8	05.5	Saturn	181°01.2	19:27
21	203°15.9	99°02.3	· · 38.7	54°18.4	· · 07.3	203°38.9	· · 24.9	24°17.2				

h	Sun			Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	180°42.1	N21°24.4	0°46.7	7.4'	S24°26.2	3.1'	56.9'
1	195°42.0	24.9	15°13.1	7.4'	24°29.3	3.0'	56.9'
2	210°42.0	25.3	29°39.5	7.4'	24°32.3	2.8'	56.8'
3	225°41.9	· · 25.7	44°06.0	7.5'	24°35.2	2.7'	56.8'
4	240°41.8	26.1	58°32.4	7.5'	24°37.9	2.6'	56.8'
5	255°41.7	26.5	72°58.9	7.5'	24°40.4	2.4'	56.8'
6	270°41.7	N21°26.9	87°25.4	7.5'	S24°42.8	2.3'	56.7'
7	285°41.6	27.3	101°51.8	7.5'	24°45.1	2.1'	56.7'
8	300°41.5	27.7	116°18.3	7.5'	24°47.2	2.0'	56.7'
9	315°41.4	· · 28.1	130°44.8	7.5'	24°49.2	1.8'	56.7'
10	330°41.3	28.5	145°11.3	7.5'	24°51.0	1.7'	56.6'
11	345°41.3	28.9	159°37.8	7.5'	24°52.7	1.5'	56.6'
12	0°41.2	N21°29.3	174°04.4	7.6'	S24°54.2	1.4'	56.6'
13	15°41.1	29.7	188°30.9	7.6'	24°55.6	1.2'	56.6'
14	30°41.0	30.1	202°57.5	7.6'	24°56.9	1.1'	56.5'
15	45°41.0	· · 30.5	217°24.1	7.6'	24°58.0	1.0'	56.5'
16	60°40.9	30.9	231°50.7	7.6'	24°58.9	0.8'	56.5'
17	75°40.8	31.3	246°17.4	7.7'	24°59.8	0.7'	56.5'
18	90°40.7	N21°31.7	260°44.0	7.7'	S25°00.4	0.5'	56.4'
19	105°40.6	32.1	275°10.7	7.7'	25°01.0	0.4'	56.4'
20	120°40.6	32.5	289°37.4	7.7'	25°01.3	0.2'	56.4'
21	135°40.5	· · 32.9	304°04.2	7.8'	25°01.6	0.1'	56.4'
22	150°40.4	33.3	318°31.0	7.8'	25°01.7	-0.0'	56.3'
23	165°40.3	33.6	332°57.8	7.8'	25°01.6	-0.2'	56.3'
SD = 15.8'			SD = 15.5'				
d = 0.4'							
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	180°40.3	N21°34.0	347°24.6	7.9'	S25°01.4	-0.3'	56.3'
1	195°40.2	34.4	1°51.5	7.9'	25°01.1	-0.5'	56.3'
2	210°40.1	34.8	16°18.4	7.9'	25°00.7	-0.6'	56.2'
3	225°40.0	· · 35.2	30°45.3	8.0'	25°00.0	-0.7'	56.2'
4	240°39.9	35.6	45°12.3	8.0'	24°59.3	-0.9'	56.2'
5	255°39.9	36.0	59°39.4	8.1'	24°58.4	-1.0'	56.2'
6	270°39.8	N21°36.4	74°06.4	8.1'	S24°57.4	-1.2'	56.2'
7	285°39.7	36.8	88°33.5	8.2'	24°56.2	-1.3'	56.1'
8	300°39.6	37.2	103°00.7	8.2'	24°54.9	-1.4'	56.1'
9	315°39.5	· · 37.5	117°27.9	8.2'	24°53.5	-1.6'	56.1'
10	330°39.4	37.9	131°55.1	8.3'	24°51.9	-1.7'	56.1'
11	345°39.4	38.3	146°22.4	8.3'	24°50.2	-1.8'	56.0'
12	0°39.3	N21°38.7	160°49.8	8.4'	S24°48.3	-2.0'	56.0'
13	15°39.2	39.1	175°17.2	8.4'	24°46.4	-2.1'	56.0'
14	30°39.1	39.5	189°44.6	8.5'	24°44.2	-2.3'	56.0'
15	45°39.0	· · 39.8	204°12.1	8.5'	24°42.0	-2.4'	55.9'
16	60°39.0	40.2	218°39.7	8.6'	24°39.6	-2.5'	55.9'
17	75°38.9	40.6	233°07.3	8.7'	24°37.1	-2.6'	55.9'
18	90°38.8	N21°41.0	247°34.9	8.7'	S24°34.4	-2.8'	55.9'
19	105°38.7	41.4	262°02.6	8.8'	24°31.7	-2.9'	55.8'
20	120°38.6	41.7	276°30.4	8.8'	24°28.7	-3.0'	55.8'
21	135°38.5	· · 42.1	290°58.2	8.9'	24°25.7	-3.2'	55.8'
22	150°38.5	42.5	305°26.1	9.0'	24°22.5	-3.3'	55.8'
23	165°38.4	42.9	319°54.1	9.0'	24°19.2	-3.4'	55.7'
SD = 15.8'			SD = 15.4'				
d = 0.4'							
Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	180°38.3	N21°43.3	334°22.1	9.1'	S24°15.8	-3.5'	55.7'
1	195°38.2	43.6	348°50.2	9.1'	24°12.3	-3.7'	55.7'
2	210°38.1	44.0	3°18.3	9.2'	24°08.6	-3.8'	55.7'
3	225°38.0	· · 44.4	17°46.5	9.3'	24°04.8	-3.9'	55.7'
4	240°37.9	44.8	32°14.8	9.3'	24°00.9	-4.0'	55.6'
5	255°37.9	45.1	46°43.1	9.4'	23°56.8	-4.2'	55.6'
6	270°37.8	N21°45.5	61°11.5	9.5'	S23°52.7	-4.3'	55.6'
7	285°37.7	45.9	75°40.0	9.5'	23°48.4	-4.4'	55.6'
8	300°37.6	46.2	90°08.5	9.6'	23°44.0	-4.5'	55.5'
9	315°37.5	· · 46.6	104°37.1	9.7'	23°39.4	-4.6'	55.5'
10	330°37.4	47.0	119°05.8	9.7'	23°34.8	-4.8'	55.5'
11	345°37.4	47.4	133°34.5	9.8'	23°30.0	-4.9'	55.5'
12	0°37.3	N21°47.7	148°03.3	9.9'	S23°25.1	-5.0'	55.5'
13	15°37.2	48.1	162°32.2	10.0'	23°20.2	-5.1'	55.4'
14	30°37.1	48.5	177°01.2	10.0'	23°15.0	-5.2'	55.4'
15	45°37.0	· · 48.8	191°30.2	10.1'	23°09.8	-5.3'	55.4'
16	60°36.9	49.2	205°59.3	10.2'	23°04.5	-5.5'	55.4'
17	75°36.8	49.6	220°28.5	10.2'	22°59.0	-5.6'	55.3'
18	90°36.7	N21°49.9	234°57.7	10.3'	S22°53.5	-5.7'	55.3'
19	105°36.7	50.3	249°27.1	10.4'	22°47.8	-5.8'	55.3'
20	120°36.6	50.7	263°56.5	10.5'	22°42.0	-5.9'	55.3'
21	135°36.5	· · 51.0	278°25.9	10.5'	22°36.1	-6.0'	55.3'
22	150°36.4	51.4	292°55.5	10.6'	22°30.1	-6.1'	55.2'
23	165°36.3	51.7	307°25.1	10.7'	22°24.0	-6.2'	55.2'
SD = 15.8'			SD = 15.2'				
d = 0.4'							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°						
68°						
66°	////	////	01:26	22:33	////	////
64°	////	////	02:06	21:51	////	////
62°	////	00:40	02:33	21:23	23:23	////
60°	////	01:35	02:54	21:02	22:22	////
N 58°	////	02:07	03:11	20:45	21:50	////
56°	00:36	02:30	03:26	20:30	21:27	23:26
54°	01:27	02:48	03:38	20:18	21:08	22:30
52°	01:56	03:03	03:49	20:07	20:53	22:01
50°	02:17	03:16	03:58	19:57	20:39	21:39
45°	02:56	03:43	04:19	19:37	20:13	21:00
N 40°	03:23	04:03	04:35	19:20	19:52	20:32
35°	03:44	04:19	04:48	19:07	19:36	20:11
30°	04:01	04:33	05:00	18:55	19:22	19:54
20°	04:28	04:56	05:20	18:35	18:59	19:27
N 10°	04:48	05:15	05:38	18:17	18:40	19:07
0°	05:06	05:32	05:54	18:01	18:23	18:49
S 10°	05:21	05:47	06:10	17:45	18:08	18:34
20°	05:36	06:03	06:27	17:28	17:52	18:19
30°	05:51	06:20	06:46	17:09	17:35	18:04
35°	05:58	06:30	06:57	16:57	17:25	17:56
40°	06:07	06:40	07:10	16:44	17:14	17:48
45°	06:16	06:52	07:25	16:29	17:02	17:39
S 50°	06:26	07:07	07:44	16:11	16:48	17:29
52°	06:30	07:13	07:53	16:02	16:41	17:24
54°	06:35	07:20	08:02	15:52	16:34	17:19
56°	06:41	07:28	08:14	15:41	16:26	17:14
58°	06:46	07:37	08:26	15:28	16:17	17:08
S 60°	06:53	07:47	08:41	15:13	16:07	17:02

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°						
N 70°						
68°						
66°						
64°	23:47	· · ·	00:16	01:20	01:58	03:22
62°	22:56	23:36	23:56	02:03	02:49	04:02
60°	22:24	23:08	23:34	02:31	03:21	04:29
N 58°	22:01	22:47	23:17	02:54	03:44	04:50
56°	21:42	22:29	23:02	03:12	04:03	05:08
54°	21:26	22:14	22:49	03:27	04:19	05:22
52°	21:12	22:01	22:38	03:40	04:33	05:35
50°	21:00	21:50	22:28	03:52	04:45	05:46
45°	20:35	21:26	22:07	04:16	05:10	06:10
N 40°	20:15	21:07	21:50	04:36	05:30	06:28
35°	19:58	20:51	21:36	04:52	05:46	06:44
30°	19:44	20:37	21:23	05:06	06:01	06:58
20°	19:20	20:13	21:02	05:30	06:25	07:21
N 10°	18:59	19:53	20:43	05:51	06:46	07:40
0°	18:39	19:34	20:26	06:10	07:05	07:59
S 10°	18:19	19:14	20:08	06:29	07:25	08:17
20°	17:58	18:54	19:49	06:50	07:46	08:37
30°	17:34	18:30	19:28	07:14	08:10	09:00
35°	17:20	18:16	19:15	07:28	08:24	09:13
40°	17:04	18:00	19:00	07:45	08:40	09:28
45°	16:44	17:41	18:43	08:04	09:00	09:46
S 50°	16:19	17:17	18:21	08:29	09:24	10:08
52°	16:07	17:05	18:11	08:40	09:36	10:19
54°	15:54	16:52	17:59	08:54	09:49	10:31
56°	15:38	16:37	17:45	09:09	10:05	10:45
58°	15:20	16:18	17:30	09:28	10:23	11:01
S 60°	14:57	15:56	17:11	09:50	10:46	11:20

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Upper hh:mm	Lower hh:mm	
28	02:48	02:45	11:57	—:—	12:25	
29	02:41	02:37	11:57	00:52	13:20	
30	02:33	02:29	11:58	01:46	14:12	

May 31, 01, 02 UT (Mon., Tue., Wed.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	248°23.3	143°59.9	N24°38.0	99°22.2	N14°05.9	248°45.3	S01°24.5	69°24.6	N03°05.5	Alpheratz	357°46.0	29°08.8					
1	263°25.8	158°59.1	37.7	114°23.5	05.5	263°47.4	24.3	84°27.1	05.5	Ankaa	353°18.0	-42°14.7					
2	278°28.3	173°58.3	37.5	129°24.7	05.0	278°49.5	24.2	99°29.6	05.5	Schedar	349°43.5	56°35.5					
3	293°30.7	188°57.5	•• 37.2	144°26.0	•• 04.5	293°51.6	•• 24.1	114°32.0	•• 05.4	Diphda	348°58.2	-17°55.6					
4	308°33.2	203°56.8	37.0	159°27.3	04.0	308°53.7	23.9	129°34.5	05.4	Achernar	335°28.7	-57°10.8					
5	323°35.6	218°56.0	36.7	174°28.5	03.6	323°55.9	23.8	144°37.0	05.4	Hamal	328°03.6	23°30.7					
6	338°38.1	233°55.2	N24°36.5	189°29.8	N14°03.1	338°58.0	S01°23.6	159°39.4	N03°05.4	Polaris	319°16.0	89°18.5					
7	353°40.6	248°54.4	36.2	204°31.0	02.6	354°00.1	23.5	174°41.9	05.4	Acamar	315°20.3	-40°15.6					
8	8°43.0	263°53.6	36.0	219°32.3	02.1	9°02.2	23.4	189°44.4	05.4	Menkar	314°17.7	4°07.8					
9	23°45.5	278°52.8	•• 35.7	234°33.6	•• 01.7	24°04.3	•• 23.2	204°46.8	•• 05.4	Mirfak	308°44.1	49°53.8					
10	38°48.0	293°52.0	35.5	249°34.8	01.2	39°06.5	23.1	219°49.3	05.4	Aldebaran	290°52.3	16°31.8					
11	53°50.4	308°51.3	35.2	264°36.1	00.7	54°08.6	22.9	234°51.8	05.4	Rigel	281°14.5	-8°11.4					
12	68°52.9	323°50.5	N24°35.0	279°37.3	N14°00.2	69°10.7	S01°22.8	249°54.2	N03°05.4	Capella	280°38.2	46°00.5					
13	83°55.4	338°49.7	34.7	294°38.6	13°59.8	84°12.8	22.7	264°56.7	05.4	Bellatrix	278°34.7	6°21.5					
14	98°57.8	353°48.9	34.4	309°39.9	59.3	99°14.9	22.5	279°59.1	05.3	Elnath	278°15.8	28°37.0					
15	114°00.3	8°48.1	•• 34.2	324°41.1	•• 58.8	114°17.1	•• 22.4	295°01.6	•• 05.3	Alnilam	275°48.9	-1°11.8					
16	129°02.8	23°47.3	33.9	339°42.4	58.3	129°19.2	22.3	310°04.1	05.3	Betelgeuse	271°04.0	7°24.5					
17	144°05.2	38°46.6	33.7	354°43.6	57.9	144°21.3	22.1	325°06.5	05.3	Canopus	263°57.6	-52°42.2					
18	159°07.7	53°45.8	N24°33.4	9°44.9	N13°57.4	159°23.4	S01°22.0	340°09.0	N03°05.3	Sirius	258°36.0	-16°43.9					
19	174°10.1	68°45.0	33.1	24°46.1	56.9	174°25.5	21.8	355°11.5	05.3	Adhara	255°14.6	-28°59.3					
20	189°12.6	83°44.2	32.9	39°47.4	56.4	189°27.7	21.7	10°13.9	05.3	Procyon	245°02.3	5°11.8					
21	204°15.1	98°43.4	•• 32.6	54°48.7	•• 55.9	204°29.8	•• 21.6	25°16.4	•• 05.3	Pollux	243°30.7	28°00.1					
22	219°17.5	113°42.7	32.3	69°49.9	55.5	219°31.9	21.4	40°18.9	05.3	Avior	234°19.4	-59°32.9					
23	234°20.0	128°41.9	32.1	84°51.2	55.0	234°34.0	21.3	55°21.3	05.3	Suhail	222°54.3	-43°28.8					
Mer.pass. 07:25			ν -0.8' d -0.2' m -3.93			ν 1.3' d -0.5' m 1.10			ν 2.1' d -0.1' m -2.25			ν 2.5' d -0.0' m 0.85					

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	SHA	Dec		
0	249°22.5	143°41.1	N24°31.8	99°52.4	N13°54.5	249°36.1	S01°21.2	70°23.8	N03°05.2	Denebola	182°35.8	14°30.8											
1	264°24.9	158°40.3	31.5	114°53.7	54.0	264°38.3	21.0	85°26.2	05.2	Gienah	175°54.5	-17°36.2											
2	279°27.4	173°39.5	31.3	129°54.9	53.6	279°40.4	20.9	100°28.7	05.2	Acrux	173°11.7	-63°09.8											
3	294°29.9	188°38.8	•• 31.0	144°56.2	•• 53.1	294°42.5	•• 20.7	115°31.2	•• 05.2	Gacrux	172°03.3	-57°10.7											
4	309°32.3	203°38.0	30.7	159°57.4	52.6	309°44.6	20.6	130°33.6	05.2	Alioth	166°22.2	55°54.3											
5	324°34.8	218°37.2	30.4	174°58.7	52.1	324°46.8	20.5	145°36.1	05.2	Spica	158°33.4	-11°13.1											
6	339°37.3	233°36.4	N24°30.2	190°00.0	N13°51.6	339°48.9	S01°20.3	160°38.6	N03°05.2	Alkaid	153°00.2	49°15.7											
7	354°39.7	248°35.7	29.9	205°01.2	51.2	354°51.0	20.2	175°41.0	05.2	Hadar	148°50.7	-60°25.7											
8	9°42.2	263°34.9	29.6	220°02.5	50.7	9°53.1	20.1	190°43.5	05.2	Menkent	148°09.9	-36°25.5											
9	24°44.6	278°34.1	•• 29.3	235°03.7	•• 50.2	24°55.2	•• 19.9	205°45.9	•• 05.1	Arcturus	145°57.4	19°07.6											
10	39°47.1	293°33.3	29.0	250°05.0	49.7	39°57.4	19.8	220°48.4	05.1	Rigil Kent.	139°54.3	-60°52.9											
11	54°49.6	308°32.5	28.7	265°06.2	49.3	54°59.5	19.6	235°50.9	05.1	Kochab	137°18.6	74°06.8											
12	69°52.0	323°31.8	N24°28.5	280°07.5	N13°48.8	70°01.6	S01°19.5	250°53.3	N03°05.1	Zuben'ubi	137°07.6	-16°05.3											
13	84°54.5	338°31.0	28.2	295°08.7	48.3	85°03.7	19.4	265°55.8	05.1	Alphecca	126°12.5	26°40.7											
14	99°57.0	353°30.2	27.9	310°10.0	47.8	100°05.9	19.2	280°58.2	05.1	Antares	112°28.6	-26°27.4											
15	114°59.4	8°29.5	•• 27.6	325°11.2	•• 47.3	115°08.0	•• 19.1	296°00.7	•• 05.1	Atria	107°31.9	-69°02.8											
16	130°01.9	23°28.7	27.3	340°12.5	46.9	130°10.1	19.0	311°03.2	05.1	Sabik	102°14.7	-15°44.3											
17	145°04.4	38°27.9	27.0	355°13.7	46.4	145°12.2	18.8	326°05.6	05.1	Shaula	96°24.5	-37°06.7											
18	160°06.8	53°27.1	N24°26.7	10°15.0	N13°45.9	160°14.4	S01°18.7	341°08.1	N03°05.0	Rasalhague	96°08.2	12°33.1											
19	175°09.3	68°26.4	26.4	25°16.3	45.4	175°16.5	18.6	356°10.5	05.0	Eltanin	90°46.7	51°29.1											
20	190°11.8	83°25.6	26.1	40°17.5	44.9	190°18.6	18.4	11°13.0	05.0	Kaus Aust.	83°46.4	-34°22.7											
21	205°14.2	98°24.8	•• 25.9	55°18.8	•• 44.5	205°20.7	•• 18.3	26°15.5	•• 05.0	Vega	80°40.1	38°47.5											
22	220°16.7	113°24.0	25.6	70°20.0	44.0	220°22.9	18.1	41°17.9	05.0	Nunki	76°00.7	-26°16.9											
23	235°19.1	128°23.3	25.3	85°21.3	43.5	235°25.0	18.0	56°20.4	05.0	Altair	62°10.2	8°53.7											
Mer.pass. 07:21			ν -0.8' d -0.3' m -3.93			ν 1.3' d -0.5' m 1.11			ν 2.1' d -0.1' m -2.26			ν 2.5' d -0.0' m 0.86											

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	SHA	Dec		
0	250°21.6	143°22.5	N24°25.0	100°22.5	N13°43.0	250°27.1	S01°17.9	71°22.8	N03°05.0	Markab	13°40.6	28°15.7											
1	265°24.1	158°21.7	24.7	115°23.8	42.5	265°29.2	17.7	86°25.3	05.0														
2	280°26.5	173°21.0	24.4	130°25.0	42.0	280°31.4	17.6	101°27.8	05.0														
3	295°29.0	188°20.2	•• 24.1	145°26.3	•• 41.6	295°33.5	•• 17.5	116°30.2	•• 04.9														
4	310°31.5	203°19.4	23.8	160°27.5	41.1	310°35.6	17.3	131°32.7	04.9														
5	325°33.9	218°18.7	23.4	175°28.8	40.6	325°37.8	17.2	146°35.1	04.9														
6	340°36.4	233°17.9	N24°23.1	190°30.0	N13°40.1	340°39.9	S01°17.1	161°37.6	N03°04.9														
7	355°38.9	248°17.1	22.8	205°31.3	39.6	355°42.0	16.9	176°40.1	04.9														
8	10°41.3	263°16.4	22.5	220°32.5	39.2	10°44.1	16.8	191°42.5	04.9														
9	25°43.8	278°15.6	•• 22.2	235°33.8	•• 38.7	25°46.3	•• 16.7	206°45.0	•• 04.9														
10	40°46.2	293°14.8	21.9	250°35.0	38.2	40°48.4	16.5	221°47.4	04.9														
11	55°48.7	308°14.1	21.6	265°36.3	37.7	55°50.5	16.4	236°49.9	04.8														
12	70°51.2	323°13.3	N24°21.3	280°37.5	N13°37.2	70°52.7	S01°16.3	251°52.3	N03°04.8														
13	85°53.6	338°12.5	21.0	295°38.8	36.7	85°54.8	16.1	266°54.8	04.8														
14	100°56.1	353°11.8	20.7	310°40.0	36.3	100°56.9	16.0	281°57.3	04.8														
15	115°58.6	8°11.0	•• 20.3	325°41.3	•• 35.8	115°59.0	•• 15.9	296°59.7	•• 04.8														
16	131°01.0	23°10.2	20.0	340°42.5	35.3	131°01.2	15.7	312°02.2	04.8														
17	146°03.5	38°09.5	19.7	355°43.8	34.8	146°03.3	15.6	327°04.6	04.8														
18	161°06.0	53°08.7	N24°19.4	10°45.0	N13°34.3	161°05.4	S01°15.5	342°07.1	N03°04.7														
19	176°08.4	68°07.9	19.1	25°46.3	33.8	176°07.6	15.3	357°09.5	04.7														
20	191°10.9	83°07.2	18.7	40°47.5	33.3	191°09.7	15.2	12°12.0	04.7														
21	206°13.4	98°06.4	•• 18.4	55°48.8	•• 32.9	206°11.8	•• 15.0	27°14.5	•• 04.7														
22	221°15.8	113°05.7	18.1	70°50.0	32.4	221°14.0	14.9	42°16.9	04.7														
23	236°18.3	128°04.9	17.8	85°51.3	31.9	236°16.1	14.8	57°19.4	04.7														
Mer.pass. 07:17			ν -0.8																				

h	Sun			Moon			
Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	180°36.2	N21°52.1	321°54.8	10.8'	S22°17.8	-6.3'	55.2'
1	195°36.1	52.5	336°24.6	10.8'	22°11.5	-6.4'	55.2'
2	210°36.0	52.8	350°54.4	10.9'	22°05.1	-6.5'	55.2'
3	225°35.9	· · 53.2	5°24.3	11.0'	21°58.6	-6.6'	55.1'
4	240°35.9	53.5	19°54.3	11.1'	21°51.9	-6.7'	55.1'
5	255°35.8	53.9	34°24.4	11.2'	21°45.2	-6.8'	55.1'
6	270°35.7	N21°54.2	48°54.6	11.2'	S21°38.4	-6.9'	55.1'
7	285°35.6	54.6	63°24.8	11.3'	21°31.5	-7.0'	55.1'
8	300°35.5	55.0	77°55.1	11.4'	21°24.5	-7.1'	55.0'
9	315°35.4	· · 55.3	92°25.5	11.5'	21°17.4	-7.2'	55.0'
10	330°35.3	55.7	106°56.0	11.5'	21°10.1	-7.3'	55.0'
11	345°35.2	56.0	121°26.5	11.6'	21°02.8	-7.4'	55.0'
12	0°35.1	N21°56.4	135°57.1	11.7'	S20°55.4	-7.5'	55.0'
13	15°35.0	56.7	150°27.8	11.8'	20°48.0	-7.6'	54.9'
14	30°35.0	57.1	164°58.6	11.8'	20°40.4	-7.7'	54.9'
15	45°34.9	· · 57.4	179°29.5	11.9'	20°32.7	-7.8'	54.9'
16	60°34.8	57.8	194°00.4	12.0'	20°24.9	-7.9'	54.9'
17	75°34.7	58.1	208°31.4	12.1'	20°17.1	-7.9'	54.9'
18	90°34.6	N21°58.5	223°02.5	12.2'	S20°09.2	-8.0'	54.8'
19	105°34.5	58.8	237°33.6	12.2'	20°01.1	-8.1'	54.8'
20	120°34.4	59.2	252°04.9	12.3'	19°53.0	-8.2'	54.8'
21	135°34.3	· · 59.5	266°36.2	12.4'	19°44.8	-8.3'	54.8'
22	150°34.2	21°59.9	281°07.5	12.5'	19°36.5	-8.4'	54.8'
23	165°34.1	22°00.2	295°39.0	12.5'	19°28.2	-8.4'	54.8'
SD = 15.8'			SD = 15.1'				
$d = 0.4'$							

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	180°34.0	N22°00.6	310°10.5	12.6'	S19°19.7	-8.5'	54.7'
1	195°33.9	00.9	324°42.1	12.7'	19°11.2	-8.6'	54.7'
2	210°33.8	01.2	339°13.8	12.8'	19°02.6	-8.7'	54.7'
3	225°33.8	· · 01.6	353°45.6	12.8'	18°53.9	-8.8'	54.7'
4	240°33.7	01.9	8°17.4	12.9'	18°45.2	-8.8'	54.7'
5	255°33.6	02.3	22°49.3	13.0'	18°36.3	-8.9'	54.7'
6	270°33.5	N22°02.6	37°21.3	13.0'	S18°27.4	-9.0'	54.6'
7	285°33.4	03.0	51°53.3	13.1'	18°18.4	-9.1'	54.6'
8	300°33.3	03.3	66°25.5	13.2'	18°09.3	-9.1'	54.6'
9	315°33.2	· · 03.6	80°57.7	13.3'	18°00.2	-9.2'	54.6'
10	330°33.1	04.0	95°29.9	13.3'	17°51.0	-9.3'	54.6'
11	345°33.0	04.3	110°02.3	13.4'	17°41.7	-9.4'	54.6'
12	0°32.9	N22°04.6	124°34.7	13.5'	S17°32.4	-9.4'	54.6'
13	15°32.8	05.0	139°07.1	13.5'	17°22.9	-9.5'	54.5'
14	30°32.7	05.3	153°39.7	13.6'	17°13.4	-9.6'	54.5'
15	45°32.6	· · 05.7	168°12.3	13.7'	17°03.9	-9.6'	54.5'
16	60°32.5	06.0	182°45.0	13.7'	16°54.2	-9.7'	54.5'
17	75°32.4	06.3	197°17.7	13.8'	16°44.5	-9.8'	54.5'
18	90°32.3	N22°06.7	211°50.5	13.9'	S16°34.8	-9.8'	54.5'
19	105°32.2	07.0	226°23.4	13.9'	16°25.0	-9.9'	54.5'
20	120°32.1	07.3	240°56.4	14.0'	16°15.1	-10.0'	54.5'
21	135°32.0	· · 07.6	255°29.4	14.1'	16°05.1	-10.0'	54.4'
22	150°31.9	08.0	270°02.5	14.1'	15°55.1	-10.1'	54.4'
23	165°31.8	08.3	284°35.6	14.2'	15°45.0	-10.1'	54.4'
SD = 15.8'			SD = 14.9'				
$d = 0.3'$							

Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	180°31.7	N22°08.6	299°08.8	14.3'	S15°34.9	-10.2'	54.4'
1	195°31.6	09.0	313°42.1	14.3'	15°24.7	-10.3'	54.4'
2	210°31.6	09.3	328°15.4	14.4'	15°14.4	-10.3'	54.4'
3	225°31.5	· · 09.6	342°48.8	14.4'	15°04.1	-10.4'	54.4'
4	240°31.4	09.9	357°22.2	14.5'	14°53.8	-10.4'	54.4'
5	255°31.3	10.3	11°55.7	14.6'	14°43.3	-10.5'	54.4'
6	270°31.2	N22°10.6	26°29.3	14.6'	S14°32.9	-10.5'	54.3'
7	285°31.1	10.9	41°02.9	14.7'	14°22.3	-10.6'	54.3'
8	300°31.0	11.2	55°36.6	14.7'	14°11.7	-10.6'	54.3'
9	315°30.9	· · 11.6	70°10.4	14.8'	14°01.1	-10.7'	54.3'
10	330°30.8	11.9	84°44.2	14.9'	13°50.4	-10.7'	54.3'
11	345°30.7	12.2	99°18.0	14.9'	13°39.7	-10.8'	54.3'
12	0°30.6	N22°12.5	113°51.9	15.0'	S13°28.9	-10.8'	54.3'
13	15°30.5	12.9	128°25.9	15.0'	13°18.0	-10.9'	54.3'
14	30°30.4	13.2	142°59.9	15.1'	13°07.1	-10.9'	54.3'
15	45°30.3	· · 13.5	157°34.0	15.1'	12°56.2	-11.0'	54.3'
16	60°30.2	13.8	172°08.1	15.2'	12°45.2	-11.0'	54.3'
17	75°30.1	14.1	186°42.2	15.2'	12°34.1	-11.1'	54.3'
18	90°30.0	N22°14.4	201°16.4	15.3'	S12°23.1	-11.1'	54.3'
19	105°29.9	14.8	215°50.7	15.3'	12°11.9	-11.2'	54.2'
20	120°29.8	15.1	230°25.0	15.4'	12°00.8	-11.2'	54.2'
21	135°29.7	· · 15.4	244°59.4	15.4'	11°49.5	-11.3'	54.2'
22	150°29.6	15.7	259°33.8	15.4'	11°38.3	-11.3'	54.2'
23	165°29.5	16.0	274°08.2	15.5'	11°27.0	-11.3'	54.2'
SD = 15.8'			SD = 14.8'				
$d = 0.3'$							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	□	□	□	□	□	□
N 70°	□	□	□	□	□	□
68°	□	□	□	□	□	□
66°	////	////	01:13	22:47	////	////
64°	////	////	01:58	22:00	////	////
62°	////	////	02:27	21:30	////	////
60°	////	01:27	02:50	21:07	22:32	////
N 58°	////	02:01	03:07	20:49	21:57	////
56°	////	02:25	03:22	20:34	21:32	////
54°	01:19	02:44	03:35	20:21	21:13	22:39
52°	01:50	03:00	03:46	20:10	20:57	22:07
50°	02:13	03:13	03:56	20:00	20:43	21:44
45°	02:53	03:40	04:17	19:39	20:16	21:03
N 40°	03:21	04:01	04:33	19:23	19:55	20:35
35°	03:42	04:18	04:47	19:09	19:38	20:14
30°	04:00	04:32	04:59	18:56	19:23	19:56
20°	04:27	04:56	05:20	18:36	19:00	19:29
N 10°	04:48	05:15	05:38	18:18	18:41	19:08
0°	05:06	05:32	05:54	18:01	18:24	18:50
S 10°	05:22	05:48	06:10	17:45	18:08	18:34
20°	05:37	06:04	06:28	17:28	17:51	18:19
30°	05:52	06:22	06:48	17:08	17:34	18:03
35°	06:00	06:31	06:59	16:56	17:24	17:55
40°	06:09	06:42	07:12	16:43	17:13	17:47
45°	06:18	06:55	07:28	16:28	17:01	17:37
S 50°	06:29	07:09	07:47	16:08	16:46	17:27
52°	06:33	07:16	07:56	15:59	16:39	17:22
54°	06:38	07:24	08:06	15:49	16:32	17:17
56°	06:44	07:32	08:18	15:38	16:23	17:11
58°	06:50	07:41	08:31	15:25	16:14	17:05
S 60°	06:56	07:52	08:46	15:09	16:04	16:59

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	■	■	01:57	■	■	06:44
N 70°	■	02:12	01:23	■	04:54	07:17
68°	■	01:21	00:59	■	05:45	07:40
66°	01:03	00:49	00:40	04:22	06:16	07:58
64°	00:23	00:25	00:24	05:01	06:39	08:12
62°	· · ·	00:06	00:11	05:28	06:57	08:24
60°	23:50	· · ·	00:00	05:49	07:12	08:35
N 58°	23:37	23:50	· · ·	06:06	07:25	08:43
56°	23:25	23:42	23:54	06:20	07:36	08:51
54°	23:15	23:34	23:49	06:33	07:46	08:58
52°	23:06	23:27	23:45	06:43	07:54	09:04
50°	22:58	23:21	23:40	06:53	08:02	09:10
45°	22:40	23:08	23:31	07:13	08:18	09:22
N 40°	22:26	22:57	23:23	07:30	08:31	09:32
35°	22:14	22:47	23:17	07:43	08:42	09:40
30°	22:04	22:39	23:11	07:55	08:52	09:47
20°	21:45	22:25	23:01	08:16	09:09	10:00
N 10°	21:29	22:12	22:52	08:33	09:23	10:11
0°	21:14	22:00	22:43	08:50	09:37	10:21
S 10°	20:59	21:48	22:35	09:06	09:50	10:31
20°	20:43	21:36	22:26	09:23	10:05	10:42
30°	20:25	21:21	22:15	09:43	10:21	10:55
35°	20:14	21:12	22:09	09:55	10:30	11:02
40°	20:02	21:03	22:03	10:08	10:41	11:10
45°	19:47	20:51	21:55	10:23	10:54	11:19
S 50°	19:29	20:37	21:45	10:42	11:09	11:30
52°	19:20	20:31	21:40	10:51	11:16	11:35
54°	19:11	20:23	21:35	11:01	11:24	11:41
56°	19:00	20:15	21:30	11:13	11:32	11:47
58°	18:47	20:06	21:24	11:26	11:42	11:54
S 60°	18:33	19:56	21:17	11:41	11:53	12:02

Day	Sun			Moon		
	Eqn. of Time		Mer.	Mer. Pass.		Age
	00 ^h	12 ^h	Pass	Upper	Lower	17-19
	mm:ss	mm:ss	hh:mm	hh:mm	hh:mm	91-77%
31	02:25	02:21	11:58	02:38	15:02	
01	02:16	02:12	11:58	03:26	15:49	
02	02:07	02:02	11:58	04:11	16:32	

June 03, 04, 05 UT (Thu., Fri., Sat.)

Aries		Venus		Mars		Jupiter		Saturn		Stars											
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec										
0	251°20.7	143°04.1	N24°17.4	100°52.5	N13°31.4	251°18.2	S01°14.6	72°21.8	N03°04.7	Alpheratz	357°46.0	29°08.8									
1	266°23.2	158°03.4	17.1	115°53.8	30.9	266°20.4	14.5	87°24.3	04.7	Ankaa	353°17.9	-42°14.7									
2	281°25.7	173°02.6	16.8	130°55.0	30.4	281°22.5	14.4	102°26.7	04.6	Schedar	349°43.5	56°35.5									
3	296°28.1	188°01.8	•• 16.5	145°56.3	•• 30.0	296°24.6	•• 14.3	117°29.2	•• 04.6	Diphda	348°58.2	-17°55.6									
4	311°30.6	203°01.1	16.1	160°57.5	29.5	311°26.7	14.1	132°31.6	04.6	Achernar	335°28.7	-57°10.7									
5	326°33.1	218°00.3	15.8	175°58.8	29.0	326°28.9	14.0	147°34.1	04.6	Hamal	328°03.6	23°30.7									
6	341°35.5	232°59.6	N24°15.5	191°00.0	N13°28.5	341°31.0	S01°13.9	162°36.5	N03°04.6	Polaris	319°15.3	89°18.4									
7	356°38.0	247°58.8	15.1	206°01.2	28.0	356°33.1	13.7	177°39.0	04.6	Acamar	315°20.3	-40°15.6									
8	11°40.5	262°58.1	14.8	221°02.5	27.5	11°35.3	13.6	192°41.5	04.6	Menkar	314°17.7	4°07.8									
9	26°42.9	277°57.3	•• 14.4	236°03.7	•• 27.0	26°37.4	•• 13.5	207°43.9	•• 04.5	Mirfak	308°44.1	49°53.8									
10	41°45.4	292°56.5	14.1	251°05.0	26.5	41°39.5	13.3	222°46.4	04.5	Aldebaran	290°52.3	16°31.8									
11	56°47.9	307°55.8	13.8	266°06.2	26.1	56°41.7	13.2	237°48.8	04.5	Rigel	281°14.5	-8°11.4									
12	71°50.3	322°55.0	N24°13.4	281°07.5	N13°25.6	71°43.8	S01°13.1	252°51.3	N03°04.5	Capella	280°38.2	46°00.5									
13	86°52.8	337°54.3	13.1	296°08.7	25.1	86°45.9	12.9	267°53.7	04.5	Bellatrix	278°34.7	6°21.5									
14	101°55.2	352°53.5	12.7	311°10.0	24.6	101°48.1	12.8	282°56.2	04.5	Elnath	278°15.8	28°37.0									
15	116°57.7	7°52.8	•• 12.4	326°11.2	•• 24.1	116°50.2	•• 12.7	297°58.6	•• 04.5	Alnilam	275°48.9	-1°11.8									
16	132°00.2	22°52.0	12.1	341°12.5	23.6	131°52.4	12.5	313°01.1	04.4	Betelgeuse	271°04.0	7°24.5									
17	147°02.6	37°51.3	11.7	356°13.7	23.1	146°54.5	12.4	328°03.5	04.4	Canopus	263°57.7	-52°42.2									
18	162°05.1	52°50.5	N24°11.4	11°15.0	N13°22.6	161°56.6	S01°12.3	343°06.0	N03°04.4	Sirius	258°36.0	-16°43.9									
19	177°07.6	67°49.8	11.0	26°16.2	22.2	176°58.8	12.1	358°08.4	04.4	Adhara	255°14.6	-28°59.3									
20	192°10.0	82°49.0	10.7	41°17.4	21.7	192°00.9	12.0	13°10.9	04.4	Procyon	245°02.3	5°11.8									
21	207°12.5	97°48.2	•• 10.3	56°18.7	•• 21.2	207°03.0	•• 11.9	28°13.3	•• 04.4	Pollux	243°30.7	28°00.1									
22	222°15.0	112°47.5	10.0	71°19.9	20.7	222°05.2	11.7	43°15.8	04.3	Avior	234°19.4	-59°32.9									
23	237°17.4	127°46.7	09.6	86°21.2	20.2	237°07.3	11.6	58°18.2	04.3	Suhail	222°54.4	-43°28.8									
Mer.pass. 07:13		ν -0.8' d -0.3' m -3.94		ν 1.2' d -0.5' m 1.13		ν 2.1' d -0.1' m -2.27		ν 2.5' d -0.0' m 0.86		Miaplacidus	221°40.8	-69°46.0									
Alphard	217°58.4	-8°42.4	Regulus	207°45.9	11°54.9	Dubhe	193°54.2	61°41.9	Denebola	182°35.8	14°30.8										
Gienah	175°54.5	-17°36.2	Acrux	173°11.7	-63°09.8	Gacrux	172°03.3	-57°10.7	Alioth	166°22.2	55°54.3										
Spica	158°33.4	-11°13.1	Alkaid	153°00.2	-49°15.7	Hadar	148°50.7	-60°25.7	Menkent	148°09.9	-36°25.5										
Arcturus	145°57.4	19°07.6	Rigel Kent.	139°54.3	-60°52.9	Kochab	137°18.6	74°06.8	Zuben'ubi	137°07.6	-16°05.3										
Alphecca	126°12.5	26°40.7	Antares	112°28.6	-26°27.4	Atria	107°31.9	-69°02.8	Sabik	102°14.7	-15°44.3										
Shaula	96°24.5	-37°06.7	Rasalhague	96°08.2	12°33.1	Eltanin	90°46.6	51°29.2	Kaus Aust.	83°46.4	-34°22.7										
Vega	80°40.1	38°47.5	Nunki	76°00.7	-26°16.9	Altair	62°10.1	8°53.7	Peacock	53°22.2	-56°41.8										
Deneb	49°32.8	45°18.9	Enif	33°49.2	9°55.4	Al Na'ir	27°46.3	-46°54.3	Fomalhaut	15°26.4	-29°33.8										
Scheat	13°55.6	28°08.3	Markab	13°40.6	15°15.7																
Jun 03 Thu		SHA		Mer.pass																	
Venus		251°43.4		14:28																	
Mars		209°31.8		17:15																	
Jupiter		359°57.5		07:14																	
Saturn		181°01.1		19:07																	
Jun 04 Fri		SHA		Mer.pass																	
Venus		250°26.1		14:30																	
Mars		209°02.5		17:13																	
Jupiter		359°49.6		07:10																	
Saturn		181°00.8		19:04																	
Jun 05 Sat		SHA		Mer.pass																	
Venus		249°09.1		14:31																	
Mars		208°33.2		17:11																	
Jupiter		359°41.8		07:07																	
Saturn		181°00.5		19:00																	
Horizontal parallax																					
Venus: 0.1																					
Mars: 0.1																					

h	Sun			Moon			
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	180°29.4	N22°16.3	288°42.7	15.5'	S11°15.6	-11.4'	54.2'
1	195°29.3	16.6	303°17.3	15.6'	11°04.3	-11.4'	54.2'
2	210°29.2	17.0	317°51.8	15.6'	10°52.8	-11.5'	54.2'
3	225°29.1	· · 17.3	332°26.5	15.7'	10°41.4	-11.5'	54.2'
4	240°29.0	17.6	347°01.1	15.7'	10°29.9	-11.5'	54.2'
5	255°28.9	17.9	1°35.8	15.7'	10°18.3	-11.6'	54.2'
6	270°28.7	N22°18.2	16°10.6	15.8'	S10°06.7	-11.6'	54.2'
7	285°28.6	18.5	30°45.3	15.8'	09°55.1	-11.7'	54.2'
8	300°28.5	18.8	45°20.2	15.9'	09°43.5	-11.7'	54.2'
9	315°28.4	· · 19.1	59°55.0	15.9'	09°31.8	-11.7'	54.2'
10	330°28.3	19.4	74°29.9	15.9'	09°20.1	-11.8'	54.2'
11	345°28.2	19.7	89°04.8	16.0'	09°08.3	-11.8'	54.2'
12	0°28.1	N22°20.0	103°39.8	16.0'	S08°56.5	-11.8'	54.2'
13	15°28.0	20.3	118°14.8	16.0'	08°44.7	-11.9'	54.2'
14	30°27.9	20.6	132°49.8	16.0'	08°32.8	-11.9'	54.2'
15	45°27.8	· · 20.9	147°24.8	16.1'	08°21.0	-11.9'	54.2'
16	60°27.7	21.2	161°59.9	16.1'	08°09.0	-11.9'	54.2'
17	75°27.6	21.5	176°35.0	16.1'	07°57.1	-12.0'	54.2'
18	90°27.5	N22°21.8	191°10.1	16.2'	S07°45.1	-12.0'	54.2'
19	105°27.4	22.1	205°45.3	16.2'	07°33.1	-12.0'	54.2'
20	120°27.3	22.4	220°20.5	16.2'	07°21.1	-12.1'	54.2'
21	135°27.2	· · 22.7	234°55.7	16.2'	07°09.0	-12.1'	54.2'
22	150°27.1	23.0	249°30.9	16.3'	06°56.9	-12.1'	54.2'
23	165°27.0	23.3	264°06.2	16.3'	06°44.8	-12.1'	54.2'
SD = 15.8'			SD = 14.8'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	180°26.9	N22°23.6	278°41.5	16.3'	S06°32.7	-12.2'	54.2'
1	195°26.8	23.9	293°16.8	16.3'	06°20.5	-12.2'	54.2'
2	210°26.7	24.2	307°52.1	16.3'	06°08.3	-12.2'	54.2'
3	225°26.6	· · 24.5	322°27.4	16.4'	05°56.1	-12.2'	54.2'
4	240°26.5	24.8	337°02.8	16.4'	05°43.9	-12.3'	54.2'
5	255°26.3	25.1	351°38.2	16.4'	05°31.6	-12.3'	54.2'
6	270°26.2	N22°25.4	6°13.5	16.4'	S05°19.4	-12.3'	54.2'
7	285°26.1	25.7	20°49.0	16.4'	05°07.1	-12.3'	54.2'
8	300°26.0	26.0	35°24.4	16.4'	04°54.7	-12.3'	54.2'
9	315°25.9	· · 26.3	49°59.8	16.4'	04°42.4	-12.4'	54.2'
10	330°25.8	26.6	64°35.2	16.5'	04°30.0	-12.4'	54.2'
11	345°25.7	26.9	79°10.7	16.5'	04°17.7	-12.4'	54.2'
12	0°25.6	N22°27.1	93°46.2	16.5'	S04°05.3	-12.4'	54.2'
13	15°25.5	27.4	108°21.6	16.5'	03°52.8	-12.4'	54.2'
14	30°25.4	27.7	122°57.1	16.5'	03°40.4	-12.4'	54.2'
15	45°25.3	· · 28.0	137°32.6	16.5'	03°28.0	-12.5'	54.3'
16	60°25.2	28.3	152°08.1	16.5'	03°15.5	-12.5'	54.3'
17	75°25.1	28.6	166°43.6	16.5'	03°03.0	-12.5'	54.3'
18	90°25.0	N22°28.9	181°19.1	16.5'	S02°50.5	-12.5'	54.3'
19	105°24.8	29.1	195°54.6	16.5'	02°38.0	-12.5'	54.3'
20	120°24.7	29.4	210°30.1	16.5'	02°25.5	-12.5'	54.3'
21	135°24.6	· · 29.7	225°05.6	16.5'	02°13.0	-12.5'	54.3'
22	150°24.5	30.0	239°41.1	16.5'	02°00.4	-12.6'	54.3'
23	165°24.4	30.3	254°16.6	16.5'	01°47.9	-12.6'	54.3'
SD = 15.8'			SD = 14.8'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	180°24.3	N22°30.5	268°52.1	16.5'	S01°35.3	-12.6'	54.3'
1	195°24.2	30.8	283°27.6	16.5'	01°22.7	-12.6'	54.3'
2	210°24.1	31.1	298°03.1	16.5'	01°10.1	-12.6'	54.3'
3	225°24.0	· · 31.4	312°38.5	16.5'	00°57.6	-12.6'	54.4'
4	240°23.9	31.7	327°14.0	16.5'	00°44.9	-12.6'	54.4'
5	255°23.8	31.9	341°49.5	16.5'	00°32.3	-12.6'	54.4'
6	270°23.6	N22°32.2	356°25.0	16.4'	S00°19.7	-12.6'	54.4'
7	285°23.5	32.5	11°00.4	16.4'	S00°07.1	-12.6'	54.4'
8	300°23.4	32.8	25°35.8	16.4'	N00°05.5	12.6'	54.4'
9	315°23.3	· · 33.0	40°11.3	16.4'	00°18.2	12.6'	54.4'
10	330°23.2	33.3	54°46.7	16.4'	00°30.8	12.6'	54.4'
11	345°23.1	33.6	69°22.1	16.4'	00°43.5	12.6'	54.4'
12	0°23.0	N22°33.9	83°57.5	16.4'	N00°56.1	12.7'	54.5'
13	15°22.9	34.1	98°32.8	16.3'	01°08.8	12.7'	54.5'
14	30°22.8	34.4	113°08.2	16.3'	01°21.4	12.7'	54.5'
15	45°22.6	· · 34.7	127°43.5	16.3'	01°34.1	12.7'	54.5'
16	60°22.5	34.9	142°18.8	16.3'	01°46.7	12.7'	54.5'
17	75°22.4	35.2	156°54.1	16.3'	01°59.4	12.7'	54.5'
18	90°22.3	N22°35.5	171°29.4	16.2'	N02°12.0	12.7'	54.5'
19	105°22.2	35.7	186°04.6	16.2'	02°24.7	12.7'	54.6'
20	120°22.1	36.0	200°39.8	16.2'	02°37.3	12.6'	54.6'
21	135°22.0	· · 36.3	215°15.0	16.2'	02°50.0	12.6'	54.6'
22	150°21.9	36.5	229°50.2	16.1'	03°02.6	12.6'	54.6'
23	165°21.8	36.8	244°25.3	16.1'	03°15.3	12.6'	54.6'
SD = 15.8'			SD = 14.8'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	□	□	□	□	□	□
N 70°	□	□	□	□	□	□
68°	□	□	□	□	□	□
66°	////	////	01:00	23:01	////	////
64°	////	////	01:51	22:07	////	////
62°	////	////	02:22	21:36	////	////
60°	////	01:18	02:45	21:12	22:41	////
N 58°	////	01:55	03:04	20:54	22:03	////
56°	////	02:21	03:19	20:38	21:37	////
54°	01:11	02:40	03:32	20:25	21:17	22:48
52°	01:45	02:57	03:44	20:13	21:01	22:13
50°	02:09	03:11	03:54	20:03	20:46	21:48
45°	02:51	03:39	04:15	19:42	20:18	21:07
N 40°	03:19	04:00	04:32	19:25	19:57	20:38
35°	03:41	04:17	04:46	19:10	19:40	20:16
30°	03:59	04:32	04:59	18:58	19:25	19:58
20°	04:27	04:55	05:20	18:37	19:01	19:30
N 10°	04:48	05:15	05:38	18:19	18:42	19:09
0°	05:06	05:32	05:55	18:02	18:24	18:50
S 10°	05:22	05:49	06:11	17:45	18:08	18:34
20°	05:38	06:05	06:29	17:28	17:51	18:19
30°	05:53	06:23	06:49	17:07	17:33	18:03
35°	06:02	06:33	07:01	16:56	17:23	17:55
40°	06:10	06:44	07:14	16:42	17:12	17:46
45°	06:20	06:57	07:30	16:26	17:00	17:36
S 50°	06:31	07:12	07:50	16:07	16:44	17:25
52°	06:36	07:19	07:59	15:57	16:37	17:21
54°	06:41	07:27	08:09	15:47	16:30	17:15
56°	06:47	07:35	08:21	15:35	16:21	17:10
58°	06:53	07:45	08:35	15:21	16:12	17:03
S 60°	07:00	07:55	08:51	15:05	16:01	16:56

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	01:16	00:48	00:25	08:54	10:48	12:37
N 70°	00:58	00:39	00:23	09:10	10:54	12:36
68°	00:44	00:32	00:21	09:23	10:59	12:35
66°	00:32	00:26	00:20	09:33	11:04	12:34
64°	00:22	00:20	00:18	09:41	11:07	12:33
62°	00:14	00:16	00:17	09:49	11:11	12:32
60°	00:07	00:12	00:16	09:55	11:13	12:32
N 58°	00:00	00:08	00:15	10:00	11:16	12:31
56°	· · ·	00:05	00:15	10:05	11:18	12:31
54°	· · ·	00:02	00:14	10:10	11:20	12:30
52°	· · ·	00:00	00:13	10:13	11:22	12:30
50°	23:57	· · ·	00:13	10:17	11:23	12:29
45°	23:52	· · ·	00:12	10:25	11:27	12:29
N 40°	23:48	· · ·	00:10	10:31	11:30	12:28
35°	23:44	· · ·	00:10	10:36	11:32	12:27
30°	23:40	· · ·	00:09	10:41	11:34	12:27
20°	23:35	· · ·	00:07	10:50	11:38	12:26
N 10°	23:29	· · ·	00:06	10:57	11:41	12:25
0°	23:25	· · ·	00:05	11:03	11:44	12:24
S 10°	23:20	· · ·	00:04	11:10	11:47	12:24
20°	23:15	· · ·	00:03	11:17	11:50	12:23
30°	23:09	· · ·	00:01	11:25	11:54	12:22
35°	23:05	· · ·	00:01	11:30	11:56	12:21
40°	23:01	· · ·	00:00	11:35	11:58	12:21
45°	22:57	23:59	· · ·	11:41	12:01	12:20
S 50°	22:51	23:57	· · ·	11:48	12:04	12:19
52°	22:49	23:57	· · ·	11:51	12:05	12:19
54°	22:46	23:56	· · ·	11:55	12:07	12:19
56°	22:43	23:56	· · ·	11:59	12:09	12:18
58°	22:40	23:55	· · ·	12:03	12:11	12:18
S 60°	22:36	23:54	· · ·	12:08	12:13	12:17

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time	12 ^h		Mer. Pass.		
	00 ^h	mm:ss	hh:mm	Upper	Lower	20-22
				hh:mm	hh:mm	68-49%
03	01:57	01:53	11:58	04:53	17:14	
04	01:48	01:42	11:58	05:34	17:55	
05	01:37	01:32	11:58	06:15	18:35	

Aries		Venus		Mars		Jupiter		Saturn	
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	254°18.2	142°10.5	N23°50.9	102°22.0	N12°56.1	253°52.2	S01°05.3	75°18.2	N03°03.5
1	269°20.6	157°09.7	50.5	117°23.2	55.6	268°54.4	05.2	90°20.6	03.5
2	284°23.1	172°09.0	50.1	132°24.4	55.1	283°56.5	05.1	105°23.1	03.5
3	299°25.6	187°08.3	· · 49.6	147°25.7	· · 54.6	298°58.7	· · 04.9	120°25.5	· · 03.4
4	314°28.0	202°07.6	49.2	162°26.9	54.1	314°00.8	04.8	135°27.9	03.4
5	329°30.5	217°06.8	48.8	177°28.1	53.6	329°03.0	04.7	150°30.4	03.4
6	344°32.9	232°06.1	N23°48.4	192°29.4	N12°53.1	344°05.1	S01°04.6	165°32.8	N03°03.4
7	359°35.4	247°05.4	48.0	207°30.6	52.6	359°07.3	04.4	180°35.3	03.4
8	14°37.9	262°04.7	47.6	222°31.8	52.1	14°09.4	04.3	195°37.7	03.3
9	29°40.3	277°03.9	· · 47.1	237°33.1	· · 51.6	29°11.6	· · 04.2	210°40.2	· · 03.3
10	44°42.8	292°03.2	46.7	252°34.3	51.1	44°13.7	04.0	225°42.6	03.3
11	59°45.3	307°02.5	46.3	267°35.5	50.6	59°15.9	03.9	240°45.0	03.3
12	74°47.7	322°01.8	N23°45.9	282°36.8	N12°50.1	74°18.0	S01°03.8	255°47.5	N03°03.3
13	89°50.2	337°01.0	45.4	297°38.0	49.6	89°20.2	03.7	270°49.9	03.2
14	104°52.7	352°00.3	45.0	312°39.2	49.1	104°22.3	03.5	285°52.4	03.2
15	119°55.1	6°59.6	· · 44.6	327°40.5	· · 48.6	119°24.5	· · 03.4	300°54.8	· · 03.2
16	134°57.6	21°58.9	44.2	342°41.7	48.1	134°26.6	03.3	315°57.2	03.2
17	150°00.1	36°58.2	43.7	357°42.9	47.6	149°28.8	03.2	330°59.7	03.2
18	165°02.5	51°57.4	N23°43.3	12°44.2	N12°47.1	164°30.9	S01°03.0	346°02.1	N03°03.1
19	180°05.0	66°56.7	42.9	27°45.4	46.6	179°33.1	02.9	1°04.6	03.1
20	195°07.4	81°56.0	42.4	42°46.6	46.1	194°35.2	02.8	16°07.0	03.1
21	210°09.9	96°55.3	· · 42.0	57°47.9	· · 45.6	209°37.4	· · 02.7	31°09.4	· · 03.1
22	225°12.4	111°54.6	41.6	72°49.1	45.1	224°39.5	02.6	46°11.9	03.1
23	240°14.8	126°53.8	41.1	87°50.3	44.6	239°41.7	02.4	61°14.3	03.0
Mer.pass. 07:02		ν -0.7' d -0.4' m -3.94		ν 1.2' d -0.5' m 1.16		ν 2.1' d -0.1' m -2.29		ν 2.4' d -0.0' m 0.88	

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	255°17.3	141°53.1	N23°40.7	102°51.6	N12°44.1	254°43.8	S01°02.3	76°16.8	N03°03.0
1	270°19.8	156°52.4	40.3	117°52.8	43.6	269°46.0	02.2	91°19.2	03.0
2	285°22.2	171°51.7	39.8	132°54.0	43.1	284°48.1	02.1	106°21.6	03.0
3	300°24.7	186°51.0	· · 39.4	147°55.3	· · 42.6	299°50.3	· · 01.9	121°24.1	· · 03.0
4	315°27.2	201°50.3	38.9	162°56.5	42.1	314°52.4	01.8	136°26.5	02.9
5	330°29.6	216°49.5	38.5	177°57.7	41.6	329°54.6	01.7	151°29.0	02.9
6	345°32.1	231°48.8	N23°38.1	192°59.0	N12°41.1	344°56.7	S01°01.6	166°31.4	N03°02.9
7	0°34.5	246°48.1	37.6	208°00.2	40.6	359°58.9	01.4	181°33.8	02.9
8	15°37.0	261°47.4	37.2	223°01.4	40.1	15°01.1	01.3	196°36.3	02.9
9	30°39.5	276°46.7	· · 36.7	238°02.7	· · 39.6	30°03.2	· · 01.2	211°38.7	· · 02.8
10	45°41.9	291°46.0	36.3	253°03.9	39.1	45°05.4	01.1	226°41.1	02.8
11	60°44.4	306°45.3	35.8	268°05.1	38.6	60°07.5	00.9	241°43.6	02.8
12	75°46.9	321°44.5	N23°35.4	283°06.4	N12°38.1	75°09.7	S01°00.8	256°46.0	N03°02.8
13	90°49.3	336°43.8	34.9	298°07.6	37.6	90°11.8	00.7	271°48.5	02.7
14	105°51.8	351°43.1	34.5	313°08.8	37.1	105°14.0	00.6	286°50.9	02.7
15	120°54.3	6°42.4	· · 34.0	328°10.1	· · 36.6	120°16.1	· · 00.5	301°53.3	· · 02.7
16	135°56.7	21°41.7	33.6	343°11.3	36.1	135°18.3	00.3	316°55.8	02.7
17	150°59.2	36°41.0	33.1	358°12.5	35.6	150°20.4	00.2	331°58.2	02.7
18	166°01.7	51°40.3	N23°32.7	13°13.7	N12°35.1	165°22.6	S01°00.1	347°00.6	N03°02.6
19	181°04.1	66°39.6	32.2	28°15.0	34.6	180°24.8	01°00.0	2°03.1	02.6
20	196°06.6	81°38.9	31.7	43°16.2	34.1	195°26.9	00°59.8	17°05.5	02.6
21	211°09.0	96°38.2	· · 31.3	58°17.4	· · 33.6	210°29.1	· · 59.7	32°08.0	· · 02.6
22	226°11.5	111°37.5	30.8	73°18.7	33.1	225°31.2	59.6	47°10.4	02.6
23	241°14.0	126°36.8	30.3	88°19.9	32.5	240°33.4	59.5	62°12.8	02.5
Mer.pass. 06:58		ν -0.7' d -0.4' m -3.95		ν 1.2' d -0.5' m 1.17		ν 2.2' d -0.1' m -2.29		ν 2.4' d -0.0' m 0.88	

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	256°16.4	141°36.1	N23°29.9	103°21.1	N12°32.0	255°35.5	S00°59.4	77°15.3	N03°02.5
1	271°18.9	156°35.3	29.4	118°22.3	31.5	270°37.7	59.2	92°17.7	02.5
2	286°21.4	171°34.6	29.0	133°23.6	31.0	285°39.9	59.1	107°20.1	02.5
3	301°23.8	186°33.9	· · 28.5	148°24.8	· · 30.5	300°42.0	· · 59.0	122°22.6	· · 02.4
4	316°26.3	201°33.2	28.0	163°26.0	30.0	315°44.2	58.9	137°25.0	02.4
5	331°28.8	216°32.5	27.6	178°27.3	29.5	330°46.3	58.8	152°27.4	02.4
6	346°31.2	231°31.8	N23°27.1	193°28.5	N12°29.0	345°48.5	S00°58.6	167°29.9	N03°02.4
7	1°33.7	246°31.1	26.6	208°29.7	28.5	0°50.7	58.5	182°32.3	02.3
8	16°36.2	261°30.4	26.1	223°30.9	28.0	15°52.8	58.4	197°34.7	02.3
9	31°38.6	276°29.7	· · 25.7	238°32.2	· · 27.5	30°55.0	· · 58.3	212°37.2	· · 02.3
10	46°41.1	291°29.0	25.2	253°33.4	27.0	45°57.1	58.1	227°39.6	02.3
11	61°43.5	306°28.3	24.7	268°34.6	26.5	60°59.3	58.0	242°42.0	02.3
12	76°46.0	321°27.6	N23°24.2	283°35.9	N12°26.0	76°01.5	S00°57.9	257°44.5	N03°02.2
13	91°48.5	336°26.9	23.8	298°37.1	25.5	91°03.6	57.8	272°46.9	02.2
14	106°50.9	351°26.2	23.3	313°38.3	25.0	106°05.8	57.7	287°49.3	02.2
15	121°53.4	6°25.5	· · 22.8	328°39.5	· · 24.5	121°07.9	· · 57.5	302°51.8	· · 02.2
16	136°55.9	21°24.8	22.3	343°40.8	24.0	136°10.1	57.4	317°54.2	02.1
17	151°58.3	36°24.1	21.8	358°42.0	23.4	151°12.3	57.3	332°56.6	02.1
18	167°00.8	51°23.4	N23°21.4	13°43.2	N12°22.9	166°14.4	S00°57.2	347°59.1	N03°02.1
19	182°03.3	66°22.7	20.9	28°44.4	22.4	181°16.6	57.1	3°01.5	02.1
20	197°05.7	81°22.0	20.4	43°45.7	21.9	196°18.7	56.9	18°03.9	02.0
21	212°08.2	96°21.4	· · 19.9	58°46.9	· · 21.4	211°20.9	· · 56.8	33°06.4	· · 02.0
22	227°10.6	111°20.7	19.4	73°48.1	20.9	226°23.1	56.7	48°08.8	02.0
23	242°13.1	126°20.0	18.9	88°49.3	20.4	241°25.2	56.6	63°11.2	02.0
Mer.pass. 06:54		$\nu -0.7' d -0.5' m -3.95$		$\nu 1.2' d -0.5' m 1.17$		$\nu 2.2' d -0.1' m -2.30$		$\nu 2.4' d -0.0' m 0.89$	

Stars		
	SHA	Dec
Alpheratz	357° 45.9	29° 08.8
Ankaa	353° 17.9	-42° 14.7
Schedar	349° 43.4	-56° 35.5
Diphda	348° 58.2	-17° 55.6
Achernar	335° 28.6	-57° 10.7
Hamal	328° 03.5	23° 30.7
Polaris	319° 14.5	89° 18.4
Acamar	315° 20.3	-40° 15.6
Menkar	314° 17.7	4° 07.9
Mirfak	308° 44.1	49° 53.8
Aldebaran	290° 52.3	16° 31.8
Rigel	281° 14.5	-8° 11.4
Capella	280° 38.2	46° 00.5
Bellatrix	278° 34.7	6° 21.5
Elnath	278° 15.8	28° 37.0
Alnilam	275° 48.9	-1° 11.8
Betelgeuse	271° 04.0	7° 24.5
Canopus	263° 57.7	-52° 42.2
Sirius	258° 36.0	-16° 43.9
Adhara	255° 14.6	-28° 59.3
Procyon	245° 02.3	5° 11.8
Pollux	243° 30.7	28° 00.1
Avior	234° 19.5	-59° 32.9
Suhail	222° 54.4	-43° 28.8
Miaplacidus	221° 40.8	-69° 46.0
Alphard	217° 58.4	-8° 42.4
Regulus	207° 45.9	11° 54.9
Dubhe	193° 54.2	61° 41.9
Denebola	182° 35.8	14° 30.8
Gienah	175° 54.5	-17° 36.2
Acrux	173° 11.8	-63° 09.8
Gacrux	172° 03.3	-57° 10.7
Alioth	166° 22.2	55° 54.3
Spica	158° 33.4	-11° 13.1
Alkaid	153° 00.2	49° 15.8
Hadar	148° 50.7	-60° 25.7
Menkent	148° 09.9	-36° 25.5
Arcturus	145° 57.5	19° 07.6
Rigel Kent.	139° 54.3	-60° 52.9
Kochab	137° 18.6	74° 06.8
Zuben 'ubi	137° 07.6	-16° 05.3
Alphecca	126° 12.5	26° 40.7
Antares	112° 28.6	-26° 27.4
Atria	107° 31.9	-69° 02.8
Sabik	102° 14.7	-15° 44.3
Shaula	96° 24.5	-37° 06.7
Rasalhague	96° 08.2	12° 33.1
Eltanin	90° 46.6	51° 29.2
Kaus Aust.	83° 46.3	-34° 22.7
Vega	80° 40.1	38° 47.5
Nunki	76° 00.7	-26° 16.9
Altair	62° 10.1	8° 53.7
Peacock	53° 22.2	-56° 41.8
Deneb	49° 32.8	45° 18.9
Enif	33° 49.2	9° 55.4
Al Na'ir	27° 46.3	-46° 54.3
Fomalhaut	15° 26.3	-29° 33.8
Scheat	13° 55.6	28° 08.3
Markab	13° 40.6	15° 15.7

Jun 06 Sun	SHA	Mer.pass
Venus	247° 52.3	14:32
Mars	208° 03.8	17:09
Jupiter	359° 34.1	07:04
Saturn	181° 00.0	18:56

Jun 07 Mon	SHA	Mer.pass
Venus	246° 35.8	14:33
Mars	207° 34.3	17:07
Jupiter	359° 26.5	07:00
Saturn	180° 59.5	18:52

Jun 08 Tue	SHA	Mer.pass
Venus	245° 19.6	14:34
Mars	207° 04.7	17:05
Jupiter	359° 19.1	06:57
Saturn	180° 58.8	18:48

Horizontal parallax	
Venus:	0.1
Mars:	0.1

h	Sun			Moon			
Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	180°21.6	N22°37.1	259°00.5	16.1'	N03°27.9	12.6'	54.6'
1	195°21.5	37.3	273°35.5	16.1'	03°40.5	12.6'	54.6'
2	210°21.4	37.6	288°10.6	16.0'	03°53.2	12.6'	54.7'
3	225°21.3	· · 37.9	302°45.6	16.0'	04°05.8	12.6'	54.7'
4	240°21.2	38.1	317°20.6	16.0'	04°18.4	12.6'	54.7'
5	255°21.1	38.4	331°55.6	15.9'	04°31.0	12.6'	54.7'
6	270°21.0	N22°38.6	346°30.5	15.9'	N04°43.6	12.6'	54.7'
7	285°20.8	38.9	1°05.4	15.8'	04°56.2	12.6'	54.7'
8	300°20.7	39.1	15°40.2	15.8'	05°08.7	12.6'	54.8'
9	315°20.6	· · 39.4	30°15.0	15.8'	05°21.3	12.6'	54.8'
10	330°20.5	39.7	44°49.8	15.7'	05°33.9	12.5'	54.8'
11	345°20.4	39.9	59°24.5	15.7'	05°46.4	12.5'	54.8'
12	0°20.3	N22°40.2	73°59.2	15.6'	N05°58.9	12.5'	54.8'
13	15°20.2	40.4	88°33.9	15.6'	06°11.4	12.5'	54.9'
14	30°20.0	40.7	103°08.5	15.6'	06°23.9	12.5'	54.9'
15	45°19.9	· · 40.9	117°43.0	15.5'	06°36.4	12.5'	54.9'
16	60°19.8	41.2	132°17.6	15.5'	06°48.9	12.5'	54.9'
17	75°19.7	41.4	146°52.0	15.4'	07°01.3	12.4'	54.9'
18	90°19.6	N22°41.7	161°26.4	15.4'	N07°13.8	12.4'	55.0'
19	105°19.5	41.9	176°00.8	15.3'	07°26.2	12.4'	55.0'
20	120°19.4	42.2	190°35.1	15.3'	07°38.6	12.4'	55.0'
21	135°19.2	· · 42.4	205°09.4	15.2'	07°51.0	12.4'	55.0'
22	150°19.1	42.7	219°43.6	15.2'	08°03.3	12.3'	55.0'
23	165°19.0	42.9	234°17.8	15.1'	08°15.6	12.3'	55.1'
SD = 15.8'			SD = 14.9'				
$d = 0.3'$							

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	180°18.9	N22°43.2	248°51.9	15.1'	N08°27.9	12.3'	55.1'
1	195°18.8	43.4	263°26.0	15.0'	08°40.2	12.3'	55.1'
2	210°18.7	43.7	278°00.0	14.9'	08°52.5	12.2'	55.1'
3	225°18.5	· · 43.9	292°33.9	14.9'	09°04.7	12.2'	55.2'
4	240°18.4	44.2	307°07.8	14.8'	09°16.9	12.2'	55.2'
5	255°18.3	44.4	321°41.6	14.8'	09°29.1	12.2'	55.2'
6	270°18.2	N22°44.7	336°15.4	14.7'	N09°41.3	12.1'	55.2'
7	285°18.1	44.9	350°49.1	14.6'	09°53.4	12.1'	55.3'
8	300°18.0	45.1	5°22.7	14.6'	10°05.5	12.1'	55.3'
9	315°17.8	· · 45.4	19°56.3	14.5'	10°17.6	12.0'	55.3'
10	330°17.7	45.6	34°29.8	14.4'	10°29.6	12.0'	55.3'
11	345°17.6	45.9	49°03.3	14.4'	10°41.6	12.0'	55.4'
12	0°17.5	N22°46.1	63°36.7	14.3'	N10°53.6	11.9'	55.4'
13	15°17.4	46.3	78°10.0	14.2'	11°05.5	11.9'	55.4'
14	30°17.3	46.6	92°43.2	14.2'	11°17.4	11.9'	55.4'
15	45°17.1	· · 46.8	107°16.4	14.1'	11°29.3	11.8'	55.5'
16	60°17.0	47.0	121°49.5	14.0'	11°41.1	11.8'	55.5'
17	75°16.9	47.3	136°22.5	14.0'	11°52.9	11.8'	55.5'
18	90°16.8	N22°47.5	150°55.5	13.9'	N12°04.7	11.7'	55.5'
19	105°16.7	47.7	165°28.4	13.8'	12°16.4	11.7'	55.6'
20	120°16.5	48.0	180°01.2	13.7'	12°28.0	11.6'	55.6'
21	135°16.4	· · 48.2	194°33.9	13.7'	12°39.7	11.6'	55.6'
22	150°16.3	48.4	209°06.6	13.6'	12°51.3	11.5'	55.6'
23	165°16.2	48.7	223°39.2	13.5'	13°02.8	11.5'	55.7'
SD = 15.8'			SD = 15.0'				
$d = 0.2'$							

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	180°16.1	N22°48.9	238°11.7	13.4'	N13°14.3	11.4'	55.7'
1	195°16.0	49.1	252°44.1	13.3'	13°25.7	11.4'	55.7'
2	210°15.8	49.4	267°16.5	13.3'	13°37.1	11.4'	55.8'
3	225°15.7	· · 49.6	281°48.7	13.2'	13°48.5	11.3'	55.8'
4	240°15.6	49.8	296°20.9	13.1'	13°59.8	11.3'	55.8'
5	255°15.5	50.0	310°53.0	13.0'	14°11.1	11.2'	55.8'
6	270°15.4	N22°50.3	325°25.0	12.9'	N14°22.3	11.1'	55.9'
7	285°15.2	50.5	339°57.0	12.8'	14°33.4	11.1'	55.9'
8	300°15.1	50.7	354°28.8	12.8'	14°44.5	11.0'	55.9'
9	315°15.0	· · 50.9	9°00.6	12.7'	14°55.5	11.0'	56.0'
10	330°14.9	51.2	23°32.3	12.6'	15°06.5	10.9'	56.0'
11	345°14.8	51.4	38°03.8	12.5'	15°17.5	10.9'	56.0'
12	0°14.6	N22°51.6	52°35.3	12.4'	N15°28.3	10.8'	56.0'
13	15°14.5	51.8	67°06.8	12.3'	15°39.1	10.7'	56.1'
14	30°14.4	52.1	81°38.1	12.2'	15°49.9	10.7'	56.1'
15	45°14.3	· · 52.3	96°09.3	12.1'	16°00.6	10.6'	56.1'
16	60°14.1	52.5	110°40.4	12.0'	16°11.2	10.6'	56.2'
17	75°14.0	52.7	125°11.5	12.0'	16°21.7	10.5'	56.2'
18	90°13.9	N22°52.9	139°42.4	11.9'	N16°32.2	10.4'	56.2'
19	105°13.8	53.2	154°13.3	11.8'	16°42.7	10.4'	56.3'
20	120°13.7	53.4	168°44.1	11.7'	16°53.0	10.3'	56.3'
21	135°13.5	· · 53.6	183°14.7	11.6'	17°03.3	10.2'	56.3'
22	150°13.4	53.8	197°45.3	11.5'	17°13.5	10.1'	56.3'
23	165°13.3	54.0	212°15.8	11.4'	17°23.7	10.1'	56.4'
SD = 15.8'			SD = 15.2'				
$d = 0.2'$							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°						
68°						
66°	////	////	00:46	23:17	////	////
64°	////	////	01:45	22:15	////	////
62°	////	////	02:18	21:41	////	////
60°	////	01:10	02:42	21:17	22:50	////
N 58°	////	01:50	03:01	20:57	22:09	////
56°	////	02:17	03:17	20:41	21:42	////
54°	01:04	02:38	03:30	20:28	21:21	22:56
52°	01:41	02:54	03:42	20:16	21:04	22:18
50°	02:06	03:09	03:52	20:06	20:49	21:53
45°	02:49	03:37	04:14	19:44	20:21	21:10
N 40°	03:18	03:59	04:31	19:27	19:59	20:40
35°	03:40	04:16	04:46	19:12	19:41	20:18
30°	03:58	04:31	04:58	18:59	19:27	20:00
20°	04:26	04:55	05:20	18:38	19:02	19:31
N 10°	04:48	05:15	05:38	18:20	18:43	19:09
0°	05:07	05:33	05:55	18:02	18:25	18:51
S 10°	05:23	05:49	06:12	17:46	18:08	18:35
20°	05:39	06:06	06:30	17:28	17:51	18:19
30°	05:55	06:24	06:51	17:07	17:33	18:03
35°	06:03	06:35	07:03	16:55	17:23	17:55
40°	06:12	06:46	07:16	16:41	17:12	17:46
45°	06:22	06:59	07:32	16:25	16:59	17:36
S 50°	06:33	07:14	07:52	16:05	16:43	17:24
52°	06:38	07:21	08:02	15:56	16:36	17:19
54°	06:43	07:29	08:12	15:45	16:28	17:14
56°	06:49	07:38	08:24	15:33	16:20	17:08
58°	06:56	07:48	08:38	15:19	16:10	17:02
S 60°	07:03	07:59	08:55	15:02	15:59	16:55

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	00:02 23:37	23:05	22:03	14:29	16:32	19:11
N 70°	00:07 23:49	23:28	22:56	14:19	16:10	18:19
68°	00:10 23:59	23:46	23:29	14:12	15:54	17:48
66°	00:14	00:07	00:01 23:53	14:05	15:41	17:24
64°	00:16	00:14	00:13	14:00	15:30	17:06
62°	00:19	00:20	00:23	13:55	15:21	16:52
60°	00:21	00:26	00:32	13:51	15:13	16:39
N 58°	00:23	00:30	00:40	13:47	15:06	16:29
56°	00:24	00:35	00:47	13:44	15:00	16:19
54°	00:26	00:38	00:53	13:41	14:55	16:11
52°	00:27	00:42	00:59	13:39	14:50	16:04
50°	00:28	00:45	01:04	13:36	14:46	15:57
45°	00:31	00:52	01:15	13:31	14:36	15:43
N 40°	00:33	00:57	01:24	13:27	14:28	15:32
35°	00:35	01:02	01:32	13:24	14:22	15:22
30°	00:37	01:07	01:39	13:20	14:16	15:13
20°	00:40	01:14	01:51	13:15	14:06	14:59
N 10°	00:43	01:21	02:02	13:10	13:57	14:46
0°	00:46	01:27	02:12	13:05	13:48	14:34
S 10°	00:48	01:34	02:22	13:01	13:40	14:22
20°	00:51	01:41	02:33	12:56	13:31	14:10
30°	00:54	01:49	02:45	12:51	13:21	13:55
35°	00:56	01:53	02:53	12:48	13:16	13:47
40°	00:58	01:59	03:01	12:44	13:09	13:37
45°	01:01	02:05	03:11	12:40	13:02	13:26
S 50°	01:04	02:12	03:23	12:35	12:53	13:13
52°	01:05	02:16	03:28	12:33	12:48	13:07
54°	01:07	02:19	03:34	12:30	12:44	13:00
56°	01:09	02:23	03:41	12:28	12:39	12:52
58°	01:10	02:28	03:49	12:25	12:33	12:44
S 60°	01:13	02:33	03:58	12:22	12:27	12:34

Day	Sun			Moon		
	Eqn. of Time		Mer.	Mer.Pass.		Age
	00 ^h	12 ^h	Pass	Upper	Lower	23-25
	mm:ss	mm:ss	hh:mm	hh:mm	hh:mm	40-22%
06	01:27	01:21	11:59	06:56	19:16	
07	01:16	01:10	11:59	07:38	20:00	
08	01:04	00:59	11:59	08:23	20:47	

June 09, 10, 11 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	257° 15.6	141° 19.3	N23° 18.4	103° 50.6	N12° 19.9	256° 27.4	S00° 56.5	78° 13.7	N03° 02.0	Alpheratz	357° 45.9	29° 08.8
1	272° 18.0	156° 18.6	17.9	118° 51.8	19.4	271° 29.6	56.4	93° 16.1	01.9	Ankaa	353° 17.9	-42° 14.7
2	287° 20.5	171° 17.9	17.4	133° 53.0	18.9	286° 31.7	56.2	108° 18.5	01.9	Schedar	349° 43.4	56° 35.5
3	302° 23.0	186° 17.2	•• 17.0	148° 54.2	•• 18.4	301° 33.9	•• 56.1	123° 21.0	•• 01.9	Diphda	348° 58.2	-17° 55.6
4	317° 25.4	201° 16.5	16.5	163° 55.5	17.9	316° 36.0	56.0	138° 23.4	01.9	Achernar	335° 28.6	-57° 10.7
5	332° 27.9	216° 15.8	16.0	178° 56.7	17.4	331° 38.2	55.9	153° 25.8	01.8	Hamal	328° 03.5	23° 30.7
6	347° 30.4	231° 15.1	N23° 15.5	193° 57.9	N12° 16.8	346° 40.4	S00° 55.8	168° 28.3	N03° 01.8	Polaris	319° 13.4	89° 18.4
7	2° 32.8	246° 14.4	15.0	208° 59.1	16.3	1° 42.5	55.6	183° 30.7	01.8	Acamar	315° 20.3	-40° 15.6
8	17° 35.3	261° 13.8	14.5	224° 00.4	15.8	16° 44.7	55.5	198° 33.1	01.8	Menkar	314° 17.6	4° 07.9
9	32° 37.8	276° 13.1	•• 14.0	239° 01.6	•• 15.3	31° 46.9	•• 55.4	213° 35.6	•• 01.7	Mirfak	308° 44.0	49° 53.8
10	47° 40.2	291° 12.4	13.5	254° 02.8	14.8	46° 49.0	55.3	228° 38.0	01.7	Aldebaran	290° 52.3	16° 31.8
11	62° 42.7	306° 11.7	13.0	269° 04.0	14.3	61° 51.2	55.2	243° 40.4	01.7	Rigel	281° 14.5	-8° 11.4
12	77° 45.1	321° 11.0	N23° 12.5	284° 05.3	N12° 13.8	76° 53.4	S00° 55.0	258° 42.8	N03° 01.7	Capella	280° 38.2	46° 00.5
13	92° 47.6	336° 10.3	12.0	299° 06.5	13.3	91° 55.5	54.9	273° 45.3	01.6	Bellatrix	278° 34.7	6° 21.5
14	107° 50.1	351° 09.6	11.5	314° 07.7	12.8	106° 57.7	54.8	288° 47.7	01.6	Elnath	278° 15.8	28° 36.9
15	122° 52.5	6° 08.9	•• 11.0	329° 08.9	•• 12.3	121° 59.9	•• 54.7	303° 50.1	•• 01.6	Alnilam	275° 48.9	-1° 11.7
16	137° 55.0	21° 08.3	10.4	344° 10.2	11.8	137° 02.0	54.6	318° 52.6	01.6	Betelgeuse	271° 04.0	7° 24.5
17	152° 57.5	36° 07.6	09.9	359° 11.4	11.2	152° 04.2	54.5	333° 55.0	01.5	Canopus	263° 57.7	-52° 42.2
18	167° 59.9	51° 06.9	N23° 09.4	14° 12.6	N12° 10.7	167° 06.4	S00° 54.3	348° 57.4	N03° 01.5	Sirius	258° 36.0	-16° 43.9
19	183° 02.4	66° 06.2	08.9	29° 13.8	10.2	182° 08.5	54.2	3° 59.8	01.5	Adhara	255° 14.6	-28° 59.3
20	198° 04.9	81° 05.5	08.4	44° 15.1	09.7	197° 10.7	54.1	19° 02.3	01.5	Procyon	245° 02.3	5° 11.8
21	213° 07.3	96° 04.8	•• 07.9	59° 16.3	•• 09.2	212° 12.9	•• 54.0	34° 04.7	•• 01.4	Pollux	243° 30.7	28° 00.1
22	228° 09.8	111° 04.2	07.4	74° 17.5	08.7	227° 15.0	53.9	49° 07.1	01.4	Avior	234° 19.5	-59° 32.9
23	243° 12.3	126° 03.5	06.9	89° 18.7	08.2	242° 17.2	53.8	64° 09.6	01.4	Suhail	222° 54.4	-43° 28.8
Mer.pass. 06:50		ν -0.7' d -0.5' m -3.95		ν 1.2' d -0.5' m 1.18		ν 2.2' d -0.1' m -2.30		ν 2.4' d -0.0' m 0.89		Miaplacidus	221° 40.8	-69° 46.0
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	Alphard <th>Dec</th>	Dec	
0	258° 14.7	141° 02.8	N23° 06.3	104° 19.9	N12° 07.7	257° 19.4	S00° 53.6	79° 12.0	N03° 01.4	217° 58.4	-8° 42.4	
1	273° 17.2	156° 02.1	05.8	119° 21.2	07.2	272° 21.5	53.5	94° 14.4	01.3	Regulus	207° 45.9	11° 54.9
2	288° 19.6	171° 01.4	05.3	134° 22.4	06.6	287° 23.7	53.4	109° 16.8	01.3	Dubhe	193° 54.2	61° 41.9
3	303° 22.1	186° 00.8	•• 04.8	149° 23.6	•• 06.1	302° 25.9	•• 53.3	124° 19.3	•• 01.3	Denebola	182° 35.8	14° 30.8
4	318° 24.6	201° 00.1	04.3	164° 24.8	05.6	317° 28.0	53.2	139° 21.7	01.3	Gienah	175° 54.5	-17° 36.2
5	333° 27.0	215° 59.4	03.8	179° 26.0	05.1	332° 30.2	53.1	154° 24.1	01.2	Acrux	173° 11.8	-63° 09.8
6	348° 29.5	230° 58.7	N23° 03.2	194° 27.3	N12° 04.6	347° 32.4	S00° 52.9	169° 26.6	N03° 01.2	Gacrux	172° 03.3	-57° 10.7
7	3° 32.0	245° 58.1	02.7	209° 28.5	04.1	2° 34.5	52.8	184° 29.0	01.2	Alioth	166° 22.2	55° 54.3
8	18° 34.4	260° 57.4	02.2	224° 29.7	03.6	17° 36.7	52.7	199° 31.4	01.2	Spica	158° 33.4	-11° 13.1
9	33° 36.9	275° 56.7	•• 01.7	239° 30.9	•• 03.1	32° 38.9	•• 52.6	214° 33.8	•• 01.1	Alkaid	153° 00.2	49° 15.8
10	48° 39.4	290° 56.0	01.1	254° 32.1	02.5	47° 41.1	52.5	229° 36.3	01.1	Hadar	148° 50.7	-60° 25.7
11	63° 41.8	305° 55.4	00.6	269° 33.4	02.0	62° 43.2	52.4	244° 38.7	01.1	Menkent	148° 10.0	-36° 25.5
12	78° 44.3	320° 54.7	N23° 00.1	284° 34.6	N12° 01.5	77° 45.4	S00° 52.2	259° 41.1	N03° 01.0	Arcturus	145° 57.5	19° 07.7
13	93° 46.7	335° 54.0	22° 59.5	299° 35.8	01.0	92° 47.6	52.1	274° 43.5	01.0	Rigil Kent.	139° 54.3	-60° 52.9
14	108° 49.2	350° 53.3	59.0	314° 37.0	00.5	107° 49.7	52.0	289° 46.0	01.0	Kochab	137° 18.7	74° 06.9
15	123° 51.7	5° 52.7	•• 58.5	329° 38.2	12° 00.0	122° 51.9	•• 51.9	304° 48.4	•• 01.0	Zuben'ubi	137° 07.6	-16° 05.3
16	138° 54.1	20° 52.0	57.9	344° 39.5	11° 59.5	137° 54.1	51.8	319° 50.8	00.9	Alphecca	126° 12.5	26° 40.8
17	153° 56.6	35° 51.3	57.4	359° 40.7	58.9	152° 56.3	51.7	334° 53.2	00.9	Antares	112° 28.6	-26° 27.4
18	168° 59.1	50° 50.7	N22° 56.9	14° 41.9	N11° 58.4	167° 58.4	S00° 51.6	349° 55.7	N03° 00.9	Atria	107° 31.9	-69° 02.8
19	184° 01.5	65° 50.0	56.3	29° 43.1	57.9	183° 00.6	51.4	4° 58.1	00.9	Sabik	102° 14.7	-15° 44.3
20	199° 04.0	80° 49.3	55.8	44° 44.3	57.4	198° 02.8	51.3	20° 00.5	00.8	Shaula	96° 24.5	-37° 06.7
21	214° 06.5	95° 48.7	•• 55.3	59° 45.6	•• 56.9	213° 04.9	•• 51.2	35° 02.9	•• 00.8	Rasalhague	96° 08.1	12° 33.1
22	229° 08.9	110° 48.0	54.7	74° 46.8	56.4	228° 07.1	51.1	50° 05.4	00.8	Eltanin	90° 46.6	51° 29.2
23	244° 11.4	125° 47.3	54.2	89° 48.0	55.9	243° 09.3	51.0	65° 07.8	00.7	Kaus Aust.	83° 46.3	-34° 22.7
Mer.pass. 06:46		ν -0.7' d -0.5' m -3.96		ν 1.2' d -0.5' m 1.19		ν 2.2' d -0.1' m -2.31		ν 2.4' d -0.0' m 0.89		Vega	80° 40.1	38° 47.5
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	Venu	Dec	
0	259° 13.9	140° 46.7	N22° 53.6	104° 49.2	N11° 55.3	258° 11.5	S00° 50.9	80° 10.2	N03° 00.7	244° 03.7	14:35	
1	274° 16.3	155° 46.0	53.1	119° 50.4	54.8	273° 13.6	50.8	95° 12.6	00.7	Mars	206° 35.0	17:03
2	289° 18.8	170° 45.3	52.6	134° 51.7	54.3	288° 15.8	50.6	110° 15.1	00.7	Jupiter	359° 11.8	06:53
3	304° 21.2	185° 44.7	•• 52.0	149° 52.9	•• 53.8	303° 18.0	•• 50.5	125° 17.5	•• 00.6	Saturn	180° 58.1	18:44
4	319° 23.7	200° 44.0	51.5	164° 54.1	53.3	318° 20.2	50.4	140° 19.9	00.6			
5	334° 26.2	215° 43.3	50.9	179° 55.3	52.8	333° 22.3	50.3	155° 22.3	00.6			
6	349° 28.6	230° 42.7	N22° 50.4	194° 56.5	N11° 52.2	348° 24.5	S00° 50.2	170° 24.8	N03° 00.6			
7	4° 31.1	245° 42.0	49.8	209° 57.7	51.7	3° 26.7	50.1	185° 27.2	00.5			
8	19° 33.6	260° 41.3	49.3	224° 59.0	51.2	18° 28.9	50.0	200° 29.6	00.5			
9	34° 36.0	275° 40.7	•• 48.7	240° 00.2	•• 50.7	33° 31.0	•• 49.8	215° 32.0	•• 00.5			
10	49° 38.5	290° 40.0	48.2	255° 01.4	50.2	48° 33.2	49.7	230° 34.4	00.4			
11	64° 41.0	305° 39.4	47.6	270° 02.6	49.7	63° 35.4	49.6	245° 36.9	00.4			
12	79° 43.4	320° 38.7	N22° 47.1	285° 03.8	N11° 49.1	78° 37.6	S00° 49.5	260° 39.3	N03° 00.4			
13	94° 45.9	335° 38.0	46.5	300° 05.0	48.6	93° 39.7	49.4	275° 41.7	00.4			
14	109° 48.4	350° 37.4	45.9	315° 06.3	48.1	108° 41.9	49.3	290° 44.1	00.3			
15	124° 50.8	5° 36.7	•• 45.4	330° 07.5	•• 47.6	123° 44.1	•• 49.2	305° 46.6	•• 00.3			
16	139° 53.3	20° 36.1	44.8									

Sun							Moon						
h	GHA	Dec	GHA	ν	Dec	d	HP						
Wed													
0	180°13.2	N22°54.2	226°46.2	11.3'	N17°33.7	10.0'	56.4'						
1	195°13.1	54.4	241°16.5	11.2'	17°43.7	9.9'	56.4'						
2	210°12.9	54.7	255°46.6	11.1'	17°53.6	9.8'	56.5'						
3	225°12.8	· · 54.9	270°16.7	11.0'	18°03.5	9.8'	56.5'						
4	240°12.7	55.1	284°46.7	10.9'	18°13.3	9.7'	56.5'						
5	255°12.6	55.3	299°16.6	10.8'	18°22.9	9.6'	56.6'						
6	270°12.4	N22°55.5	313°46.4	10.7'	N18°32.5	9.5'	56.6'						
7	285°12.3	55.7	328°16.1	10.6'	18°42.1	9.4'	56.6'						
8	300°12.2	55.9	342°45.7	10.5'	18°51.5	9.4'	56.7'						
9	315°12.1	· · 56.1	357°15.2	10.4'	19°00.8	9.3'	56.7'						
10	330°11.9	56.3	11°44.6	10.3'	19°10.1	9.2'	56.7'						
11	345°11.8	56.5	26°13.8	10.2'	19°19.3	9.1'	56.7'						
12	0°11.7	N22°56.7	40°43.0	10.1'	N19°28.4	9.0'	56.8'						
13	15°11.6	56.9	55°12.1	10.0'	19°37.4	8.9'	56.8'						
14	30°11.5	57.1	69°41.1	9.9'	19°46.3	8.8'	56.8'						
15	45°11.3	· · 57.3	84°10.0	9.8'	19°55.1	8.7'	56.9'						
16	60°11.2	57.5	98°38.7	9.7'	20°03.8	8.6'	56.9'						
17	75°11.1	57.8	113°07.4	9.6'	20°12.4	8.5'	56.9'						
18	90°11.0	N22°58.0	127°36.0	9.5'	N20°20.9	8.4'	57.0'						
19	105°10.8	58.2	142°04.5	9.4'	20°29.3	8.3'	57.0'						
20	120°10.7	58.4	156°32.8	9.3'	20°37.7	8.2'	57.0'						
21	135°10.6	· · 58.6	171°01.1	9.2'	20°45.9	8.1'	57.1'						
22	150°10.5	58.7	185°29.2	9.1'	20°54.0	8.0'	57.1'						
23	165°10.3	58.9	199°57.3	9.0'	21°02.0	7.9'	57.1'						
	SD = 15.7'	d = 0.2'			SD = 15.4'								
Thu	GHA	Dec	GHA	ν	Dec	d	HP						
0	180°10.2	N22°59.1	214°25.3	8.9'	N21°09.9	7.8'	57.2'						
1	195°10.1	59.3	228°53.1	8.8'	21°17.7	7.7'	57.2'						
2	210°10.0	59.5	243°20.9	8.6'	21°25.4	7.6'	57.2'						
3	225°09.8	· · 59.7	257°48.5	8.5'	21°33.0	7.5'	57.3'						
4	240°09.7	22°59.9	272°16.1	8.4'	21°40.4	7.3'	57.3'						
5	255°09.6	23°00.1	286°43.5	8.3'	21°47.8	7.2'	57.3'						
6	270°09.5	N23°00.3	301°10.8	8.2'	N21°55.0	7.1'	57.4'						
7	285°09.3	00.5	315°38.1	8.1'	22°02.1	7.0'	57.4'						
8	300°09.2	00.7	330°05.2	8.0'	22°09.1	6.9'	57.4'						
9	315°09.1	· · 00.9	344°32.3	7.9'	22°16.0	6.8'	57.5'						
10	330°09.0	01.1	358°59.2	7.8'	22°22.8	6.6'	57.5'						
11	345°08.8	01.3	13°26.1	7.7'	22°29.4	6.5'	57.5'						
12	0°08.7	N23°01.5	27°52.8	7.6'	N22°35.9	6.4'	57.5'						
13	15°08.6	01.6	42°19.5	7.6'	22°42.3	6.3'	57.6'						
14	30°08.5	01.8	56°46.0	7.5'	22°48.6	6.1'	57.6'						
15	45°08.3	· · 02.0	71°12.5	7.4'	22°54.7	6.0'	57.6'						
16	60°08.2	02.2	85°38.8	7.3'	23°00.7	5.9'	57.7'						
17	75°08.1	02.4	100°05.1	7.2'	23°06.6	5.7'	57.7'						
18	90°07.9	N23°02.6	114°31.3	7.1'	N23°12.4	5.6'	57.7'						
19	105°07.8	02.7	128°57.4	7.0'	23°18.0	5.5'	57.8'						
20	120°07.7	02.9	143°23.3	6.9'	23°23.5	5.3'	57.8'						
21	135°07.6	· · 03.1	157°49.2	6.8'	23°28.8	5.2'	57.8'						
22	150°07.4	03.3	172°15.0	6.7'	23°34.0	5.1'	57.9'						
23	165°07.3	03.5	186°40.8	6.6'	23°39.1	4.9'	57.9'						
	SD = 15.7'	d = 0.2'			SD = 15.6'								
Fri	GHA	Dec	GHA	ν	Dec	d	HP						
0	180°07.2	N23°03.7	201°06.4	6.5'	N23°44.0	4.8'	57.9'						
1	195°07.1	03.8	215°31.9	6.5'	23°48.8	4.6'	58.0'						
2	210°06.9	04.0	229°57.4	6.4'	23°53.5	4.5'	58.0'						
3	225°06.8	· · 04.2	244°22.8	6.3'	23°58.0	4.4'	58.0'						
4	240°06.7	04.4	258°48.1	6.2'	24°02.3	4.2'	58.0'						
5	255°06.6	04.5	273°13.3	6.1'	24°06.5	4.1'	58.1'						
6	270°06.4	N23°04.7	287°38.4	6.0'	N24°10.6	3.9'	58.1'						
7	285°06.3	04.9	302°03.4	6.0'	24°14.5	3.8'	58.1'						
8	300°06.2	05.1	316°28.4	5.9'	24°18.3	3.6'	58.2'						
9	315°06.0	· · 05.2	330°53.3	5.8'	24°21.9	3.5'	58.2'						
10	330°05.9	05.4	345°18.1	5.7'	24°25.4	3.3'	58.2'						
11	345°05.8	05.6	359°42.9	5.7'	24°28.7	3.2'	58.3'						
12	0°05.7	N23°05.8	14°07.6	5.6'	N24°31.9	3.0'	58.3'						
13	15°05.5	05.9	28°32.2	5.5'	24°34.9	2.9'	58.3'						
14	30°05.4	06.1	42°56.7	5.5'	24°37.8	2.7'	58.3'						
15	45°05.3	· · 06.3	57°21.2	5.4'	24°40.5	2.5'	58.4'						
16	60°05.1	06.4	71°45.6	5.3'	24°43.0	2.4'	58.4'						
17	75°05.0	06.6	86°09.9	5.3'	24°45.4	2.2'	58.4'						
18	90°04.9	N23°06.8	100°34.2	5.2'	N24°47.6	2.1'	58.5'						
19	105°04.8	06.9	114°58.4	5.2'	24°49.7	1.9'	58.5'						
20	120°04.6	07.1	129°22.6	5.1'	24°51.6	1.7'	58.5'						
21	135°04.5	· · 07.3	143°46.7	5.1'	24°53.3	1.6'	58.5'						
22	150°04.4	07.4	158°10.7	5.0'	24°54.9	1.4'	58.6'						
23	165°04.2	07.6	172°34.7	5.0'	24°56.4	1.3'	58.6'						
	SD = 15.7'	d = 0.2'			SD = 15.8'								

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	□	□	□	□	□	□
N 70°	□	□	□	□	□	□
68°	□	□	□	□	□	□
66°	////	////	00:29	23:36	////	////
64°	////	////	01:40	22:21	////	////
62°	////	////	02:14	21:46	////	////
60°	////	01:03	02:39	21:20	22:58	////
N 58°	////	01:46	02:59	21:01	22:14	////
56°	////	02:14	03:15	20:44	21:46	////
54°	00:57	02:35	03:29	20:31	21:24	23:03
52°	01:37	02:53	03:41	20:19	21:07	22:22
50°	02:03	03:07	03:51	20:08	20:52	21:56
45°	02:47	03:36	04:13	19:46	20:23	21:12
N 40°	03:17	03:58	04:31	19:28	20:01	20:42
35°	03:40	04:16	04:46	19:13	19:43	20:20
30°	03:58	04:31	04:58	19:01	19:28	20:01
20°	04:26	04:55	05:20	18:39	19:03	19:33
N 10°	04:49	05:16	05:39	18:20	18:43	19:10
0°	05:07	05:33	05:56	18:03	18:26	18:52
S 10°	05:24	05:50	06:13	17:46	18:09	18:35
20°	05:40	06:07	06:31	17:28	17:52	18:19
30°	05:56	06:26	06:52	17:07	17:33	18:03
35°	06:04	06:36	07:04	16:55	17:23	17:54
40°	06:13	06:47	07:18	16:41	17:11	17:45
45°	06:24	07:01	07:34	16:24	16:58	17:35
S 50°	06:35	07:16	07:55	16:04	16:42	17:24
52°	06:40	07:24	08:04	15:54	16:35	17:19
54°	06:46	07:32	08:15	15:44	16:27	17:13
56°	06:52	07:40	08:27	15:31	16:18	17:07
58°	06:58	07:50	08:42	15:17	16:08	17:01
S 60°	07:05	08:02	08:58	15:00	15:57	16:53

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	□	□	□	□	□	□
N 70°	□	□	□	□	□	□
68°	22:56	□	□	20:05	□	□
66°	23:43	23:21	□	19:19	21:36	□
64°	00:13	00:14	00:21	18:49	20:36	22:12
62°	00:28	00:37	00:56	18:27	20:02	21:26
60°	00:41	00:56	01:21	18:09	19:38	20:57
N 58°	00:53	01:12	01:41	17:54	19:18	20:34
56°	01:03	01:25	01:57	17:41	19:02	20:15
54°	01:12	01:36	02:11	17:30	18:48	20:00
52°	01:19	01:47	02:24	17:20	18:36	19:46
50°	01:27	01:56	02:35	17:11	18:25	19:34
45°	01:42	02:15	02:58	16:53	18:03	19:10
N 40°	01:55	02:31	03:16	16:38	17:45	18:50
35°	02:05	02:45	03:32	16:25	17:30	18:34
30°	02:15	02:57	03:45	16:14	17:17	18:20
20°	02:31	03:17	04:08	15:55	16:55	17:56
N 10°	02:46	03:35	04:28	15:39	16:35	17:35
0°	02:59	03:51	04:47	15:24	16:17	17:15
S 10°	03:13	04:08	05:06	15:08	16:00	16:56
20°	03:28	04:26	05:26	14:52	15:41	16:35
30°	03:45	04:47	05:50	14:34	15:19	16:11
35°	03:55	04:59	06:04	14:23	15:06	15:57
40°	04:06	05:13	06:20	14:11	14:51	15:40
45°	04:19	05:30	06:39	13:56	14:34	15:21
S 50°	04:36	05:50	07:03	13:38	14:12	14:57
52°	04:44	06:00	07:14	13:30	14:02	14:45
54°	04:52	06:11	07:28	13:21	13:50	14:31
56°	05:02	06:24	07:43	13:11	13:37	14:16
58°	05:13	06:39	08:01	12:59	13:22	13:58
S 60°	05:26	06:57	08:23	12:45	13:03	13:35

June 12, 13, 14 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn	
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	260°13.0	140°30.8	N22°40.3	105°18.4	N11°42.9	259°03.7	S00°48.2	81°08.3	N03°00.0
1	275°15.5	155°30.2	39.8	120°19.6	42.4	274°05.9	48.0	96°10.8	00.0
2	290°17.9	170°29.5	39.2	135°20.8	41.9	289°08.1	47.9	111°13.2	00.0
3	305°20.4	185°28.9	· · 38.6	150°22.1	· · 41.4	304°10.2	· · 47.8	126°15.6	03°00.0
4	320°22.9	200°28.2	38.0	165°23.3	40.9	319°12.4	47.7	141°18.0	02°59.9
5	335°25.3	215°27.6	37.5	180°24.5	40.3	334°14.6	47.6	156°20.4	59.9
6	350°27.8	230°26.9	N22°36.9	195°25.7	N11°39.8	349°16.8	S00°47.5	171°22.9	N02°59.9
7	5°30.2	245°26.3	36.3	210°26.9	39.3	4°19.0	47.4	186°25.3	59.8
8	20°32.7	260°25.6	35.8	225°28.1	38.8	19°21.1	47.3	201°27.7	59.8
9	35°35.2	275°25.0	· · 35.2	240°29.3	· · 38.3	34°23.3	· · 47.2	216°30.1	· · 59.8
10	50°37.6	290°24.3	34.6	255°30.6	37.7	49°25.5	47.0	231°32.5	59.8
11	65°40.1	305°23.7	34.0	270°31.8	37.2	64°27.7	46.9	246°35.0	59.7
12	80°42.6	320°23.0	N22°33.4	285°33.0	N11°36.7	79°29.9	S00°46.8	261°37.4	N02°59.7
13	95°45.0	335°22.4	32.9	300°34.2	36.2	94°32.1	46.7	276°39.8	59.7
14	110°47.5	350°21.7	32.3	315°35.4	35.7	109°34.2	46.6	291°42.2	59.6
15	125°50.0	5°21.1	· · 31.7	330°36.6	· · 35.1	124°36.4	· · 46.5	306°44.6	· · 59.6
16	140°52.4	20°20.5	31.1	345°37.8	34.6	139°38.6	46.4	321°47.0	59.6
17	155°54.9	35°19.8	30.5	0°39.1	34.1	154°40.8	46.3	336°49.5	59.5
18	170°57.4	50°19.2	N22°29.9	15°40.3	N11°33.6	169°43.0	S00°46.2	351°51.9	N02°59.5
19	185°59.8	65°18.5	29.4	30°41.5	33.0	184°45.2	46.1	6°54.3	59.5
20	201°02.3	80°17.9	28.8	45°42.7	32.5	199°47.3	45.9	21°56.7	59.5
21	216°04.7	95°17.2	· · 28.2	60°43.9	· · 32.0	214°49.5	· · 45.8	36°59.1	· · 59.4
22	231°07.2	110°16.6	27.6	75°45.1	31.5	229°51.7	45.7	52°01.5	59.4
23	246°09.7	125°16.0	27.0	90°46.3	31.0	244°53.9	45.6	67°04.0	59.4
Mer.pass. 06:38		ν -0.7' d -0.6' m -3.96		ν 1.2' d -0.5' m 1.21		ν 2.2' d -0.1' m -2.32		ν 2.4' d -0.0' m 0.90	

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	261°12.1	140°15.3	N22°26.4	105°47.5	N11°30.4	259°56.1	S00°45.5	82°06.4	N02°59.3
1	276°14.6	155°14.7	25.8	120°48.7	29.9	274°58.3	45.4	97°08.8	59.3
2	291°17.1	170°14.1	25.2	135°50.0	29.4	290°00.4	45.3	112°11.2	59.3
3	306°19.5	185°13.4	· · 24.6	150°51.2	· · 28.9	305°02.6	· · 45.2	127°13.6	· · 59.2
4	321°22.0	200°12.8	24.0	165°52.4	28.3	320°04.8	45.1	142°16.0	59.2
5	336°24.5	215°12.1	23.4	180°53.6	27.8	335°07.0	45.0	157°18.5	59.2
6	351°26.9	230°11.5	N22°22.8	195°54.8	N11°27.3	350°09.2	S00°44.9	172°20.9	N02°59.1
7	6°29.4	245°10.9	22.2	210°56.0	26.8	5°11.4	44.7	187°23.3	59.1
8	21°31.9	260°10.2	21.6	225°57.2	26.3	20°13.6	44.6	202°25.7	59.1
9	36°34.3	275°09.6	· · 21.0	240°58.4	· · 25.7	35°15.8	· · 44.5	217°28.1	· · 59.1
10	51°36.8	290°09.0	20.4	255°59.6	25.2	50°17.9	44.4	232°30.5	59.0
11	66°39.2	305°08.3	19.8	271°00.9	24.7	65°20.1	44.3	247°33.0	59.0
12	81°41.7	320°07.7	N22°19.2	286°02.1	N11°24.2	80°22.3	S00°44.2	262°35.4	N02°59.0
13	96°44.2	335°07.1	18.6	301°03.3	23.6	95°24.5	44.1	277°37.8	58.9
14	111°46.6	350°06.5	18.0	316°04.5	23.1	110°26.7	44.0	292°40.2	58.9
15	126°49.1	5°05.8	· · 17.4	331°05.7	· · 22.6	125°28.9	· · 43.9	307°42.6	· · 58.9
16	141°51.6	20°05.2	16.8	346°06.9	22.1	140°31.1	43.8	322°45.0	58.8
17	156°54.0	35°04.6	16.2	1°08.1	21.5	155°33.3	43.7	337°47.4	58.8
18	171°56.5	50°03.9	N22°15.6	16°09.3	N11°21.0	170°35.4	S00°43.6	352°49.8	N02°58.8
19	186°59.0	65°03.3	15.0	31°10.5	20.5	185°37.6	43.5	7°52.3	58.7
20	202°01.4	80°02.7	14.4	46°11.7	20.0	200°39.8	43.3	22°54.7	58.7
21	217°03.9	95°02.1	· · 13.7	61°12.9	· · 19.4	215°42.0	· · 43.2	37°57.1	· · 58.7
22	232°06.4	110°01.4	13.1	76°14.2	18.9	230°44.2	43.1	52°59.5	58.6
23	247°08.8	125°00.8	12.5	91°15.4	18.4	245°46.4	43.0	68°01.9	58.6
Mer.pass. 06:34		ν -0.6' d -0.6' m -3.97		ν 1.2' d -0.5' m 1.22		ν 2.2' d -0.1' m -2.33		ν 2.4' d -0.0' m 0.90	

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	262°11.3	140°00.2	N22°11.9	106°16.6	N11°17.9	260°48.6	S00°42.9	83°04.3	N02°58.6
1	277°13.7	154°59.6	11.3	121°17.8	17.3	275°50.8	42.8	98°06.7	58.5
2	292°16.2	169°58.9	10.7	136°19.0	16.8	290°53.0	42.7	113°09.2	58.5
3	307°18.7	184°58.3	· · 10.0	151°20.2	· · 16.3	305°55.2	· · 42.6	128°11.6	· · 58.5
4	322°21.1	199°57.7	09.4	166°21.4	15.8	320°57.4	42.5	143°14.0	58.4
5	337°23.6	214°57.1	08.8	181°22.6	15.2	335°59.5	42.4	158°16.4	58.4
6	352°26.1	229°56.4	N22°08.2	196°23.8	N11°14.7	351°01.7	S00°42.3	173°18.8	N02°58.4
7	7°28.5	244°55.8	07.6	211°25.0	14.2	6°03.9	42.2	188°21.2	58.3
8	22°31.0	259°55.2	06.9	226°26.2	13.6	21°06.1	42.1	203°23.6	58.3
9	37°33.5	274°54.6	· · 06.3	241°27.4	· · 13.1	36°08.3	· · 42.0	218°26.0	· · 58.3
10	52°35.9	289°54.0	05.7	256°28.6	12.6	51°10.5	41.9	233°28.4	58.2
11	67°38.4	304°53.4	05.1	271°29.9	12.1	66°12.7	41.8	248°30.9	58.2
12	82°40.9	319°52.7	N22°04.4	286°31.1	N11°11.5	81°14.9	S00°41.7	263°33.3	N02°58.2
13	97°43.3	334°52.1	03.8	301°32.3	11.0	96°17.1	41.5	278°35.7	58.1
14	112°45.8	349°51.5	03.2	316°33.5	10.5	111°19.3	41.4	293°38.1	58.1
15	127°48.2	4°50.9	· · 02.5	331°34.7	· · 10.0	126°21.5	· · 41.3	308°40.5	· · 58.1
16	142°50.7	19°50.3	01.9	346°35.9	09.4	141°23.7	41.2	323°42.9	58.0
17	157°53.2	34°49.7	01.3	1°37.1	08.9	156°25.9	41.1	338°45.3	58.0
18	172°55.6	49°49.1	N22°00.6	16°38.3	N11°08.4	171°28.1	S00°41.0	353°47.7	N02°58.0
19	187°58.1	64°48.4	22°00.0	31°39.5	07.8	186°30.3	40.9	8°50.1	57.9
20	203°00.6	79°47.8	21°59.4	46°40.7	07.3	201°32.4	40.8	23°52.5	57.9
21	218°03.0	94°47.2	· · 58.7	61°41.9	· · 06.8	216°34.6	· · 40.7	38°55.0	· · 57.9
22	233°05.5	109°46.6	58.1	76°43.1	06.3	231°36.8	40.6	53°57.4	57.8
23	248°08.0	124°46.0	57.4	91°44.3	05.7	246°39.0	40.5	68°59.8	57.8
Mer.pass. 06:30		ν -0.6' d -0.6' m -3.97		ν 1.2' d -0.5' m 1.23		ν 2.2' d -0.1' m -2.34		ν 2.4' d -0.0' m 0.91	

	SHA	Dec
Alpheratz	357°45.9	29°08.8
Ankaa	353°17.9	-42°14.6
Schedar	349°43.4	56°35.5
Diphda	348°58.2	-17°55.6
Achernar	335°28.6	-57°10.7
Hamal	328°03.5	23°30.7
Polaris	319°12.1	89°18.4
Acamar	315°20.3	-40°15.6
Menkar	314°17.6	4°07.9
Mirfak	308°44.0	49°53.8
Aldebaran	290°52.2	16°31.8
Rigel	281°14.5	-8°11.4
Capella	280°38.2	46°00.5
Bellatrix	278°34.7	6°21.5
Elnath	278°15.8	28°36.9
Alnilam	275°48.9	-1°11.7
Betelgeuse	271°04.0	7°24.5
Canopus	263°57.7	-52°42.2
Sirius	258°36.0	-16°43.9
Adhara	255°14.6	-28°59.3
Procyon	245°02.3	5°11.8
Pollux	243°30.7	28°00.1
Avior	234°19.5	-59°32.9
Suhail	222°54.4	-43°28.8
Miaplacidus	221°40.9	-69°46.0
Alphard	217°58.4	-8°42.4
Regulus	207°45.9	11°54.9
Dubhe	193°54.3	61°41.9
Denebola	182°35.8	14°30.8
Gienah	175°54.5	-17°36.2
Acrux	173°11.8	-63°09.8
Gacrux	172°03.3	-57°10.7
Alioth	166°22.2	55°54.3
Spica	158°33.4	-11°13.1
Alkaid	153°00.2	49°15.8
Hadar	148°50.7	-60°25.7
Menkent	148°09.9	-36°25.5
Arcturus	145°57.5	19°07.7
Rigel Kent.	139°54.3	-60°52.9
Kochab	137°18.7	74°06.9
Zuben'ubi	137°07.6	-16°05.3
Alphecca	126°12.5	26°40.8
Antares	112°28.6	-26°27.4
Atria	107°31.9	-69°02.8
Sabik	102°14.7	-15°44.3
Shaula	96°24.5	-37°06.7
Rasalhague	96°08.1	12°33.1
Eltanin	90°46.6	51°29.2
Kaus Aust.	83°46.3	-34°22.7
Vega	80°40.0	38°47.6
Nunki	76°00.7	-26°16.9
Altair	62°10.1	8°53.8
Peacock	53°22.1	-56°41.8
Deneb	49°32.7	45°18.9
Enif	33°49.1	9°55.4
Al Na'ir	27°46.2	-46°54.3
Fomalhaut	15°26.3	-29°33.8
Scheat	13°55.5	28°08.3
Markab	13°40.5	15°15.7

Jun 12 Sat	SHA	Mer.pass
Venus	240°17.8	14:39
Mars	205°05.4	16:57
Jupiter	358°50.7	06:43
Saturn	180°55.3	18:32

Jun 13 Sun	SHA	Mer.pass
Venus	239°03.2	14:40
Mars	204°35.4	16:55
Jupiter	358°43.9	06:39
Saturn	180°54.2	18:29

Jun 14 Mon	SHA	Mer.pass
Venus	237°48.9	14:41
Mars	204°05.3	16:54
Jupiter	358°37.3	06:36
Saturn	180°53.0	18:25

Horizontal parallax	
Venus:	0.1
Mars:	0.1

h	Sun			Moon			
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	180°04.1	N23°07.8	186°58.7	4.9'	N24°57.6	1.1'	58.6'
1	195°04.0	07.9	201°22.6	4.9'	24°58.7	0.9'	58.6'
2	210°03.9	08.1	215°46.5	4.8'	24°59.6	0.8'	58.7'
3	225°03.7	· · 08.2	230°10.3	4.8'	25°00.4	0.6'	58.7'
4	240°03.6	08.4	244°34.0	4.7'	25°01.0	0.4'	58.7'
5	255°03.5	08.6	258°57.8	4.7'	25°01.4	0.3'	58.7'
6	270°03.3	N23°08.7	273°21.5	4.7'	N25°01.7	0.1'	58.8'
7	285°03.2	08.9	287°45.1	4.6'	25°01.8	-0.1'	58.8'
8	300°03.1	09.0	302°08.7	4.6'	25°01.7	-0.2'	58.8'
9	315°02.9	· · 09.2	316°32.3	4.6'	25°01.5	-0.4'	58.8'
10	330°02.8	09.4	330°55.9	4.5'	25°01.0	-0.6'	58.9'
11	345°02.7	09.5	345°19.4	4.5'	25°00.5	-0.8'	58.9'
12	0°02.6	N23°09.7	359°43.0	4.5'	N24°59.7	-0.9'	58.9'
13	15°02.4	09.8	14°06.4	4.5'	24°58.8	-1.1'	58.9'
14	30°02.3	10.0	28°29.9	4.4'	24°57.7	-1.3'	59.0'
15	45°02.2	· · 10.1	42°53.4	4.4'	24°56.4	-1.4'	59.0'
16	60°02.0	10.3	57°16.8	4.4'	24°55.0	-1.6'	59.0'
17	75°01.9	10.4	71°40.2	4.4'	24°53.4	-1.8'	59.0'
18	90°01.8	N23°10.6	86°03.6	4.4'	N24°51.6	-1.9'	59.1'
19	105°01.6	10.7	100°27.0	4.4'	24°49.7	-2.1'	59.1'
20	120°01.5	10.9	114°50.4	4.4'	24°47.6	-2.3'	59.1'
21	135°01.4	· · 11.0	129°13.8	4.4'	24°45.3	-2.5'	59.1'
22	150°01.2	11.2	143°37.2	4.4'	24°42.8	-2.6'	59.1'
23	165°01.1	11.3	158°00.5	4.4'	24°40.2	-2.8'	59.2'
SD = 15.7'			SD = 16.0'				

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	180°01.0	N23°11.5	172°23.9	4.4'	N24°37.4	-3.0'	59.2'
1	195°00.9	11.6	186°47.3	4.4'	24°34.4	-3.1'	59.2'
2	210°00.7	11.7	201°10.7	4.4'	24°31.3	-3.3'	59.2'
3	225°00.6	· · 11.9	215°34.1	4.4'	24°28.0	-3.5'	59.3'
4	240°00.5	12.0	229°57.5	4.4'	24°24.5	-3.6'	59.3'
5	255°00.3	12.2	244°20.9	4.4'	24°20.9	-3.8'	59.3'
6	270°00.2	N23°12.3	258°44.4	4.5'	N24°17.0	-4.0'	59.3'
7	285°00.1	12.5	273°07.8	4.5'	24°13.1	-4.2'	59.3'
8	299°59.9	12.6	287°31.3	4.5'	24°08.9	-4.3'	59.3'
9	314°59.8	· · 12.7	301°54.8	4.5'	24°04.6	-4.5'	59.4'
10	329°59.7	12.9	316°18.3	4.5'	24°00.1	-4.7'	59.4'
11	344°59.5	13.0	330°41.8	4.6'	23°55.4	-4.8'	59.4'
12	359°59.4	N23°13.2	345°05.4	4.6'	N23°50.6	-5.0'	59.4'
13	14°59.3	13.3	359°29.0	4.6'	23°45.6	-5.2'	59.4'
14	29°59.1	13.4	13°52.6	4.7'	23°40.5	-5.3'	59.4'
15	44°59.0	· · 13.6	28°16.3	4.7'	23°35.2	-5.5'	59.5'
16	59°58.9	13.7	42°39.9	4.7'	23°29.7	-5.6'	59.5'
17	74°58.7	13.8	57°03.7	4.8'	23°24.0	-5.8'	59.5'
18	89°58.6	N23°14.0	71°27.4	4.8'	N23°18.2	-6.0'	59.5'
19	104°58.5	14.1	85°51.2	4.8'	23°12.3	-6.1'	59.5'
20	119°58.3	14.2	100°15.1	4.9'	23°06.2	-6.3'	59.5'
21	134°58.2	· · 14.4	114°38.9	4.9'	22°59.9	-6.4'	59.6'
22	149°58.1	14.5	129°02.9	5.0'	22°53.4	-6.6'	59.6'
23	164°57.9	14.6	143°26.8	5.0'	22°46.8	-6.8'	59.6'
SD = 15.7'			SD = 16.1'				

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	179°57.8	N23°14.7	157°50.9	5.1'	N22°40.1	-6.9'	59.6'
1	194°57.7	14.9	172°14.9	5.1'	22°33.2	-7.1'	59.6'
2	209°57.6	15.0	186°39.0	5.2'	22°26.1	-7.2'	59.6'
3	224°57.4	· · 15.1	201°03.2	5.2'	22°18.9	-7.4'	59.6'
4	239°57.3	15.3	215°27.4	5.3'	22°11.5	-7.5'	59.6'
5	254°57.2	15.4	229°51.7	5.3'	22°04.0	-7.7'	59.7'
6	269°57.0	N23°15.5	244°16.0	5.4'	N21°56.3	-7.8'	59.7'
7	284°56.9	15.6	258°40.4	5.4'	21°48.5	-8.0'	59.7'
8	299°56.8	15.8	273°04.9	5.5'	21°40.5	-8.1'	59.7'
9	314°56.6	· · 15.9	287°29.4	5.6'	21°32.4	-8.3'	59.7'
10	329°56.5	16.0	301°54.0	5.6'	21°24.1	-8.4'	59.7'
11	344°56.4	16.1	316°18.6	5.7'	21°15.7	-8.5'	59.7'
12	359°56.2	N23°16.2	330°43.3	5.8'	N21°07.2	-8.7'	59.7'
13	14°56.1	16.4	345°08.0	5.8'	20°58.5	-8.8'	59.7'
14	29°56.0	16.5	359°32.9	5.9'	20°49.7	-9.0'	59.7'
15	44°55.8	· · 16.6	13°57.8	6.0'	20°40.7	-9.1'	59.7'
16	59°55.7	16.7	28°22.7	6.0'	20°31.6	-9.2'	59.8'
17	74°55.6	16.8	42°47.7	6.1'	20°22.4	-9.4'	59.8'
18	89°55.4	N23°16.9	57°12.8	6.2'	N20°13.0	-9.5'	59.8'
19	104°55.3	17.1	71°38.0	6.2'	20°03.5	-9.6'	59.8'
20	119°55.2	17.2	86°03.2	6.3'	19°53.8	-9.8'	59.8'
21	134°55.0	· · 17.3	100°28.5	6.4'	19°44.1	-9.9'	59.8'
22	149°54.9	17.4	114°53.9	6.4'	19°34.1	-10.0'	59.8'
23	164°54.7	17.5	129°19.3	6.5'	19°24.1	-10.2'	59.8'
SD = 15.7'			SD = 16.3'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	☐	☐	☐	☐	☐	☐
66°	☐	☐	☐	☐	☐	☐
64°	////	////	01:35	22:26	////	////
62°	////	////	02:12	21:49	////	////
60°	////	00:57	02:37	21:23	23:05	////
N 58°	////	01:43	02:57	21:03	22:18	////
56°	////	02:12	03:14	20:47	21:49	////
54°	00:52	02:34	03:28	20:33	21:27	23:10
52°	01:35	02:51	03:40	20:21	21:09	22:26
50°	02:02	03:06	03:51	20:10	20:54	21:59
45°	02:46	03:36	04:13	19:48	20:25	21:14
N 40°	03:16	03:58	04:31	19:30	20:02	20:44
35°	03:39	04:16	04:45	19:15	19:44	20:21
30°	03:58	04:31	04:58	19:02	19:29	20:03
20°	04:26	04:56	05:20	18:40	19:04	19:34
N 10°	04:49	05:16	05:39	18:21	18:44	19:11
0°	05:08	05:34	05:56	18:04	18:26	18:52
S 10°	05:25	05:51	06:14	17:46	18:09	18:36
20°	05:41	06:08	06:32	17:28	17:52	18:19
30°	05:57	06:27	06:53	17:07	17:33	18:03
35°	06:06	06:37	07:05	16:55	17:23	17:54
40°	06:15	06:49	07:19	16:41	17:11	17:45
45°	06:25	07:02	07:36	16:24	16:58	17:35
S 50°	06:37	07:18	07:56	16:04	16:42	17:23
52°	06:42	07:25	08:06	15:54	16:35	17:18
54°	06:47	07:34	08:17	15:43	16:26	17:13
56°	06:53	07:42	08:30	15:30	16:17	17:06
58°	07:00	07:53	08:44	15:16	16:07	17:00
S 60°	07:07	08:04	09:01	14:59	15:56	16:52

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	☐	☐	☐	☐	☐	☐
66°	☐	☐	03:12	☐	☐	00:03 23:49
64°	00:48	02:02	03:53	23:05	23:21	23:25
62°	01:34	02:44	04:21	22:23	22:52	23:07
60°	02:04	03:12	04:43	21:55	22:30	22:51
N 58°	02:27	03:34	05:00	21:33	22:12	22:38
56°	02:45	03:52	05:15	21:14	21:57	22:27
54°	03:01	04:07	05:28	20:59	21:44	22:17
52°	03:15	04:20	05:39	20:46	21:33	22:08
50°	03:26	04:32	05:49	20:34	21:22	22:00
45°	03:51	04:56	06:09	20:10	21:01	21:42
N 40°	04:11	05:15	06:26	19:51	20:43	21:28
35°	04:27	05:31	06:40	19:34	20:29	21:16
30°	04:42	05:45	06:52	19:20	20:16	21:05
20°	05:06	06:08	07:13	18:56	19:54	20:47
N 10°	05:27	06:29	07:31	18:35	19:35	20:31
0°	05:47	06:48	07:48	18:16	19:17	20:16
S 10°	06:06	07:07	08:05	17:56	18:58	20:00
20°	06:28	07:27	08:22	17:35	18:39	19:44
30°	06:52	07:50	08:43	17:11	18:17	19:25
35°	07:06	08:04	08:55	16:56	18:03	19:14
40°	07:23	08:20	09:09	16:40	17:48	19:01
45°	07:43	08:39	09:25	16:20	17:30	18:46
S 50°	08:08	09:02	09:45	15:55	17:07	18:27
52°	08:20	09:14	09:54	15:43	16:56	18:18
54°	08:34	09:26	10:04	15:29	16:43	18:08
56°	08:50	09:41	10:16	15:13	16:29	17:57
58°	09:09	09:58	10:30	14:54	16:12	17:44
S 60°	09:33	10:19	10:46	14:30	15:51	17:29

Day	Sun		Mer. Pass	Moon		Age 0-2 0-3%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
12	00:16	00:10	12:00	12:01	—:—	
13	00:04	00:02	12:00	13:02	00:32	
14	00:09	00:15	12:00	14:02	01:32	

June 15, 16, 17 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn	
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	263°10.4	139°45.4	N21°56.8	106°45.5	N11°05.2	261°41.2	S00°40.4	84°02.2	N02°57.8
1	278°12.9	154°44.8	56.2	121°46.7	04.7	276°43.4	40.3	99°04.6	57.7
2	293°15.3	169°44.2	55.5	136°47.9	04.1	291°45.6	40.2	114°07.0	57.7
3	308°17.8	184°43.6	· · 54.9	151°49.1	· · 03.6	306°47.8	· · 40.1	129°09.4	· · 57.7
4	323°20.3	199°43.0	54.2	166°50.3	03.1	321°50.0	40.0	144°11.8	57.6
5	338°22.7	214°42.3	53.6	181°51.5	02.5	336°52.2	39.9	159°14.2	57.6
6	353°25.2	229°41.7	N21°52.9	196°52.8	N11°02.0	351°54.4	S00°39.8	174°16.6	N02°57.6
7	8°27.7	244°41.1	52.3	211°54.0	01.5	6°56.6	39.7	189°19.0	57.5
8	23°30.1	259°40.5	51.7	226°55.2	01.0	21°58.8	39.6	204°21.4	57.5
9	38°32.6	274°39.9	· · 51.0	241°56.4	11°00.4	37°01.0	· · 39.5	219°23.9	· · 57.5
10	53°35.1	289°39.3	50.4	256°57.6	10°59.9	52°03.2	39.4	234°26.3	57.4
11	68°37.5	304°38.7	49.7	271°58.8	59.4	67°05.4	39.3	249°28.7	57.4
12	83°40.0	319°38.1	N21°49.0	287°00.0	N10°58.8	82°07.6	S00°39.2	264°31.1	N02°57.4
13	98°42.5	334°37.5	48.4	302°01.2	58.3	97°09.8	39.1	279°33.5	57.3
14	113°44.9	349°36.9	47.7	317°02.4	57.8	112°12.0	39.0	294°35.9	57.3
15	128°47.4	4°36.3	· · 47.1	332°03.6	· · 57.2	127°14.2	· · 38.8	309°38.3	· · 57.3
16	143°49.8	19°35.7	46.4	347°04.8	56.7	142°16.4	38.7	324°40.7	57.2
17	158°52.3	34°35.1	45.8	2°06.0	56.2	157°18.6	38.6	339°43.1	57.2
18	173°54.8	49°34.5	N21°45.1	17°07.2	N10°55.6	172°20.8	S00°38.5	354°45.5	N02°57.2
19	188°57.2	64°33.9	44.5	32°08.4	55.1	187°23.0	38.4	9°47.9	57.1
20	203°59.7	79°33.3	43.8	47°09.6	54.6	202°25.2	38.3	24°50.3	57.1
21	219°02.2	94°32.7	· · 43.1	62°10.8	· · 54.0	217°27.4	· · 38.2	39°52.7	· · 57.0
22	234°04.6	109°32.1	42.5	77°12.0	53.5	232°29.6	38.1	54°55.1	57.0
23	249°07.1	124°31.5	41.8	92°13.2	53.0	247°31.8	38.0	69°57.5	57.0
Mer.pass. 06:26		ν -0.6' d -0.6' m -3.97		ν 1.2' d -0.5' m 1.23		ν 2.2' d -0.1' m -2.34		ν 2.4' d -0.0' m 0.91	

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	264°09.6	139°31.0	N21°41.1	107°14.4	N10°52.4	262°34.0	S00°37.9	84°59.9	N02°56.9
1	279°12.0	154°30.4	40.5	122°15.6	51.9	277°36.2	37.8	100°02.3	56.9
2	294°14.5	169°29.8	39.8	137°16.8	51.4	292°38.4	37.7	115°04.8	56.9
3	309°17.0	184°29.2	· · 39.1	152°18.0	· · 50.9	307°40.6	· · 37.6	130°07.2	· · 56.8
4	324°19.4	199°28.6	38.5	167°19.2	50.3	322°42.8	37.5	145°09.6	56.8
5	339°21.9	214°28.0	37.8	182°20.4	49.8	337°45.0	37.4	160°12.0	56.8
6	354°24.3	229°27.4	N21°37.1	197°21.6	N10°49.3	352°47.2	S00°37.3	175°14.4	N02°56.7
7	9°26.8	244°26.8	36.5	212°22.8	48.7	7°49.4	37.2	190°16.8	56.7
8	24°29.3	259°26.2	35.8	227°24.0	48.2	22°51.6	37.1	205°19.2	56.7
9	39°31.7	274°25.6	· · 35.1	242°25.2	· · 47.6	37°53.9	· · 37.0	220°21.6	· · 56.6
10	54°34.2	289°25.0	34.5	257°26.4	47.1	52°56.1	36.9	235°24.0	56.6
11	69°36.7	304°24.5	33.8	272°27.6	46.6	67°58.3	36.8	250°26.4	56.5
12	84°39.1	319°23.9	N21°33.1	287°28.8	N10°46.0	83°00.5	S00°36.7	265°28.8	N02°56.5
13	99°41.6	334°23.3	32.4	302°30.0	45.5	98°02.7	36.6	280°31.2	56.5
14	114°44.1	349°22.7	31.8	317°31.2	45.0	113°04.9	36.5	295°33.6	56.4
15	129°46.5	4°22.1	· · 31.1	332°32.4	· · 44.4	128°07.1	· · 36.4	310°36.0	· · 56.4
16	144°49.0	19°21.5	30.4	347°33.6	43.9	143°09.3	36.3	325°38.4	56.4
17	159°51.5	34°20.9	29.7	2°34.8	43.4	158°11.5	36.2	340°40.8	56.3
18	174°53.9	49°20.4	N21°29.0	17°36.0	N10°42.8	173°13.7	S00°36.1	355°43.2	N02°56.3
19	189°56.4	64°19.8	28.3	32°37.2	42.3	188°15.9	36.0	10°45.6	56.3
20	204°58.8	79°19.2	27.7	47°38.4	41.8	203°18.1	35.9	25°48.0	56.2
21	220°01.3	94°18.6	· · 27.0	62°39.6	· · 41.2	218°20.3	· · 35.8	40°50.4	· · 56.2
22	235°03.8	109°18.0	26.3	77°40.8	40.7	233°22.5	35.7	55°52.8	56.1
23	250°06.2	124°17.5	25.6	92°42.0	40.2	248°24.7	35.6	70°55.2	56.1
Mer.pass. 06:22		ν -0.6' d -0.7' m -3.98		ν 1.2' d -0.5' m 1.24		ν 2.2' d -0.1' m -2.35		ν 2.4' d -0.0' m 0.92	

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	265°08.7	139°16.9	N21°24.9	107°43.2	N10°39.6	263°26.9	S00°35.5	85°57.6	N02°56.1
1	280°11.2	154°16.3	24.2	122°44.4	39.1	278°29.2	35.4	101°00.0	56.0
2	295°13.6	169°15.7	23.5	137°45.6	38.6	293°31.4	35.3	116°02.4	56.0
3	310°16.1	184°15.2	· · 22.9	152°46.8	· · 38.0	308°33.6	· · 35.2	131°04.8	· · 56.0
4	325°18.6	199°14.6	22.2	167°48.0	37.5	323°35.8	35.1	146°07.2	55.9
5	340°21.0	214°14.0	21.5	182°49.2	36.9	338°38.0	35.0	161°09.6	55.9
6	355°23.5	229°13.4	N21°20.8	197°50.4	N10°36.4	353°40.2	S00°34.9	176°12.0	N02°55.8
7	10°25.9	244°12.9	20.1	212°51.6	35.9	8°42.4	34.8	191°14.4	55.8
8	25°28.4	259°12.3	19.4	227°52.8	35.3	23°44.6	34.7	206°16.8	55.8
9	40°30.9	274°11.7	· · 18.7	242°54.0	· · 34.8	38°46.8	· · 34.6	221°19.2	· · 55.7
10	55°33.3	289°11.1	18.0	257°55.2	34.3	53°49.0	34.6	236°21.6	55.7
11	70°35.8	304°10.6	17.3	272°56.4	33.7	68°51.3	34.5	251°24.0	55.7
12	85°38.3	319°10.0	N21°16.6	287°57.6	N10°33.2	83°53.5	S00°34.4	266°26.4	N02°55.6
13	100°40.7	334°09.4	15.9	302°58.8	32.6	98°55.7	34.3	281°28.8	55.6
14	115°43.2	349°08.9	15.2	318°00.0	32.1	113°57.9	34.2	296°31.2	55.5
15	130°45.7	4°08.3	· · 14.5	333°01.2	· · 31.6	129°00.1	· · 34.1	311°33.6	· · 55.5
16	145°48.1	19°07.7	13.8	348°02.4	31.0	144°02.3	34.0	326°36.0	55.5
17	160°50.6	34°07.1	13.1	3°03.6	30.5	159°04.5	33.9	341°38.4	55.4
18	175°53.1	49°06.6	N21°12.4	18°04.8	N10°29.9	174°06.7	S00°33.8	356°40.8	N02°55.4
19	190°55.5	64°06.0	11.7	33°06.0	29.4	189°08.9	33.7	11°43.2	55.4
20	205°58.0	79°05.5	11.0	48°07.2	28.9	204°11.2	33.6	26°45.6	55.3
21	221°00.4	94°04.9	· · 10.3	63°08.4	· · 28.3	219°13.4	· · 33.5	41°48.0	· · 55.3
22	236°02.9	109°04.3	09.6	78°09.6	27.8	234°15.6	33.4	56°50.4	55.2
23	251°05.4	124°03.8	08.9	93°10.7	27.3	249°17.8	33.3	71°52.8	55.2
Mer.pass. 06:18		ν -0.6' d -0.7' m -3.98		ν 1.2' d -0.5' m 1.25		ν 2.2' d -0.1' m -2.36		ν 2.4' d -0.0' m 0.92	

Stars		
	SHA	Dec
Alpheratz	357°45.9	29°08.8
Ankaa	353°17.8	-42°14.6
Schedar	349°43.3	56°35.5
Diphda	348°58.1	-17°55.6
Achernar	335°28.6	-57°10.7
Hamal	328°03.5	23°30.7
Polaris	319°11.0	89°18.4
Acamar	315°20.3	-40°15.6
Menkar	314°17.6	4°07.9
Mirfak	308°44.0	49°53.8
Aldebaran	290°52.2	16°31.8
Rigel	281°14.5	-8°11.4
Capella	280°38.2	46°00.5
Bellatrix	278°34.7	6°21.5
Elnath	278°15.8	28°36.9
Alnilam	275°48.9	-1°11.7
Betelgeuse	271°04.0	7°24.5
Canopus	263°57.7	-52°42.1
Sirius	258°36.0	-16°43.9
Adhara	255°14.6	-28°59.3
Procyon	245°02.3	5°11.8
Pollux	243°30.7	28°00.1
Avior	234°19.5	-59°32.8
Suhail	222°54.4	-43°28.8
Miaplacidus	221°40.9	-69°45.9
Alphard	217°58.4	-8°42.4
Regulus	207°45.9	11°54.9
Dubhe	193°54.3	61°41.9
Denebola	182°35.8	14°30.8
Gienah	175°54.5	-17°36.2
Acrux	173°11.8	-63°09.8
Gacrux	172°03.3	-57°10.7
Alioth	166°22.3	55°54.3
Spica	158°33.4	-11°13.1
Alkaid	153°00.2	49°15.8
Hadar	148°50.7	-60°25.7
Menkent	148°10.0	-36°25.5
Arcturus	145°57.5	19°07.7
Rigil Kent.	139°54.3	-60°52.9
Kochab	137°18.8	74°06.9
Zuben'ubi	137°07.6	-16°05.3
Alphecca	126°12.5	26°40.8
Antares	112°28.6	-26°27.4
Atria	107°31.9	-69°02.9
Sabik	102°14.7	-15°44.3
Shaula	96°24.5	-37°06.7
Rasalhague	96°08.1	12°33.1
Eltanin	90°46.6	51°29.2
Kaus Aust.	83°46.3	-34°22.7
Vega	80°40.0	38°47.6
Nunki	76°00.6	-26°16.9
Altair	62°10.1	8°53.8
Peacock	53°22.1	-56°41.8
Deneb	49°32.7	45°19.0
Enif	33°49.1	9°55.4
Al Na'ir	27°46.2	-46°54.3
Fomalhaut	15°26.2	-29°33.7
Scheat	13°55.5	28°08.3
Markab	13°40.5	15°15.7

Jun 15 Tue	SHA	Mer.pass
Venus	236°35.0	14:42
Mars	203°35.1	16:52
Jupiter	358°30.8	06:32
Saturn	180°51.8	18:21

Jun 16 Wed	SHA	Mer.pass
Venus	235°21.4	14:43
Mars	203°04.8	16:50
Jupiter	358°24.5	06:29
Saturn	180°50.4	18:17

Jun 17 Thu	SHA	Mer.pass
Venus	234°08.2	14:43
Mars	202°34.5	16:48
Jupiter	358°18.2	06:25
Saturn	180°48.9	18:13

Horizontal parallax	
Venus:	0.1
Mars:	0.1

h	Sun			Moon			
Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	179°54.6	N23°17.6	143°44.9	6.6'	N19°14.0	-10.3'	59.8'
1	194°54.5	17.7	158°10.5	6.7'	19°03.7	-10.4'	59.8'
2	209°54.3	17.8	172°36.1	6.7'	18°53.3	-10.5'	59.8'
3	224°54.2	· · 18.0	187°01.9	6.8'	18°42.8	-10.6'	59.8'
4	239°54.1	18.1	201°27.7	6.9'	18°32.1	-10.8'	59.8'
5	254°53.9	18.2	215°53.5	7.0'	18°21.4	-10.9'	59.8'
6	269°53.8	N23°18.3	230°19.5	7.0'	N18°10.5	-11.0'	59.8'
7	284°53.7	18.4	244°45.5	7.1'	17°59.5	-11.1'	59.8'
8	299°53.5	18.5	259°11.7	7.2'	17°48.4	-11.2'	59.8'
9	314°53.4	· · 18.6	273°37.8	7.3'	17°37.2	-11.3'	59.8'
10	329°53.3	18.7	288°04.1	7.3'	17°25.8	-11.4'	59.8'
11	344°53.1	18.8	302°30.4	7.4'	17°14.4	-11.5'	59.8'
12	359°53.0	N23°18.9	316°56.9	7.5'	N17°02.9	-11.6'	59.8'
13	14°52.9	19.0	331°23.4	7.6'	16°51.2	-11.8'	59.8'
14	29°52.7	19.1	345°49.9	7.6'	16°39.5	-11.9'	59.8'
15	44°52.6	· · 19.2	0°16.6	7.7'	16°27.6	-12.0'	59.8'
16	59°52.5	19.3	14°43.3	7.8'	16°15.6	-12.1'	59.8'
17	74°52.3	19.4	29°10.1	7.9'	16°03.6	-12.2'	59.8'
18	89°52.2	N23°19.5	43°36.9	7.9'	N15°51.4	-12.2'	59.8'
19	104°52.1	19.6	58°03.9	8.0'	15°39.2	-12.3'	59.8'
20	119°51.9	19.7	72°30.9	8.1'	15°26.8	-12.4'	59.8'
21	134°51.8	· · 19.8	86°58.0	8.2'	15°14.4	-12.5'	59.8'
22	149°51.7	19.9	101°25.2	8.2'	15°01.9	-12.6'	59.8'
23	164°51.5	20.0	115°52.4	8.3'	14°49.3	-12.7'	59.8'
SD = 15.7'			SD = 16.3'				

Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	179°51.4	N23°20.1	130°19.7	8.4'	N14°36.6	-12.8'	59.8'
1	194°51.3	20.2	144°47.1	8.5'	14°23.8	-12.9'	59.8'
2	209°51.1	20.3	159°14.6	8.5'	14°10.9	-13.0'	59.8'
3	224°51.0	· · 20.4	173°42.1	8.6'	13°58.0	-13.0'	59.8'
4	239°50.8	20.5	188°09.7	8.7'	13°44.9	-13.1'	59.8'
5	254°50.7	20.6	202°37.4	8.8'	13°31.8	-13.2'	59.8'
6	269°50.6	N23°20.6	217°05.2	8.8'	N13°18.6	-13.3'	59.8'
7	284°50.4	20.7	231°33.0	8.9'	13°05.4	-13.3'	59.8'
8	299°50.3	20.8	246°00.9	9.0'	12°52.0	-13.4'	59.8'
9	314°50.2	· · 20.9	260°28.8	9.0'	12°38.6	-13.5'	59.8'
10	329°50.0	21.0	274°56.9	9.1'	12°25.1	-13.5'	59.8'
11	344°49.9	21.1	289°25.0	9.2'	12°11.6	-13.6'	59.8'
12	359°49.8	N23°21.2	303°53.1	9.2'	N11°58.0	-13.7'	59.8'
13	14°49.6	21.3	318°21.4	9.3'	11°44.3	-13.7'	59.8'
14	29°49.5	21.3	332°49.7	9.4'	11°30.6	-13.8'	59.8'
15	44°49.4	· · 21.4	347°18.0	9.4'	11°16.7	-13.9'	59.8'
16	59°49.2	21.5	1°46.5	9.5'	11°02.9	-13.9'	59.7'
17	74°49.1	21.6	16°15.0	9.6'	10°48.9	-14.0'	59.7'
18	89°49.0	N23°21.7	30°43.5	9.6'	N10°35.0	-14.0'	59.7'
19	104°48.8	21.8	45°12.1	9.7'	10°20.9	-14.1'	59.7'
20	119°48.7	21.8	59°40.8	9.7'	10°06.8	-14.1'	59.7'
21	134°48.5	· · 21.9	74°09.6	9.8'	09°52.7	-14.2'	59.7'
22	149°48.4	22.0	88°38.4	9.9'	09°38.5	-14.2'	59.7'
23	164°48.3	22.1	103°07.2	9.9'	09°24.2	-14.3'	59.7'
SD = 15.7'			SD = 16.3'				

Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	179°48.1	N23°22.2	117°36.2	10.0'	N09°09.9	-14.3'	59.7'
1	194°48.0	22.2	132°05.1	10.0'	08°55.6	-14.4'	59.7'
2	209°47.9	22.3	146°34.2	10.1'	08°41.2	-14.4'	59.7'
3	224°47.7	· · 22.4	161°03.3	10.1'	08°26.8	-14.5'	59.7'
4	239°47.6	22.5	175°32.4	10.2'	08°12.3	-14.5'	59.6'
5	254°47.5	22.5	190°01.6	10.3'	07°57.8	-14.6'	59.6'
6	269°47.3	N23°22.6	204°30.8	10.3'	N07°43.2	-14.6'	59.6'
7	284°47.2	22.7	219°00.1	10.4'	07°28.6	-14.6'	59.6'
8	299°47.1	22.7	233°29.5	10.4'	07°14.0	-14.7'	59.6'
9	314°46.9	· · 22.8	247°58.9	10.5'	06°59.4	-14.7'	59.6'
10	329°46.8	22.9	262°28.4	10.5'	06°44.7	-14.7'	59.6'
11	344°46.6	23.0	276°57.9	10.5'	06°30.0	-14.7'	59.6'
12	359°46.5	N23°23.0	291°27.4	10.6'	N06°15.2	-14.8'	59.6'
13	14°46.4	23.1	305°57.0	10.6'	06°00.4	-14.8'	59.5'
14	29°46.2	23.2	320°26.6	10.7'	05°45.6	-14.8'	59.5'
15	44°46.1	· · 23.2	334°56.3	10.7'	05°30.8	-14.8'	59.5'
16	59°46.0	23.3	349°26.0	10.8'	05°16.0	-14.9'	59.5'
17	74°45.8	23.4	3°55.8	10.8'	05°01.1	-14.9'	59.5'
18	89°45.7	N23°23.4	18°25.6	10.8'	N04°46.2	-14.9'	59.5'
19	104°45.6	23.5	32°55.4	10.9'	04°31.3	-14.9'	59.5'
20	119°45.4	23.5	47°25.3	10.9'	04°16.4	-14.9'	59.5'
21	134°45.3	· · 23.6	61°55.2	11.0'	04°01.4	-15.0'	59.4'
22	149°45.2	23.7	76°25.2	11.0'	03°46.5	-15.0'	59.4'
23	164°45.0	23.7	90°55.2	11.0'	03°31.5	-15.0'	59.4'
SD = 15.7'			SD = 16.3'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°						
68°						
66°						
64°	////	////	01:33	22:30	////	////
62°	////	////	02:10	21:52	////	////
60°	////	00:52	02:36	21:26	23:11	////
N 58°	////	01:41	02:56	21:05	22:21	////
56°	////	02:11	03:13	20:49	21:51	////
54°	00:48	02:33	03:27	20:34	21:29	23:15
52°	01:33	02:51	03:39	20:22	21:11	22:29
50°	02:00	03:06	03:50	20:11	20:56	22:01
45°	02:46	03:35	04:13	19:49	20:26	21:16
N 40°	03:16	03:58	04:31	19:31	20:04	20:45
35°	03:39	04:16	04:46	19:16	19:46	20:22
30°	03:58	04:31	04:59	19:03	19:30	20:04
20°	04:27	04:56	05:21	18:41	19:05	19:35
N 10°	04:49	05:16	05:39	18:22	18:45	19:12
0°	05:08	05:34	05:57	18:04	18:27	18:53
S 10°	05:25	05:52	06:14	17:47	18:10	18:36
20°	05:41	06:09	06:33	17:28	17:52	18:20
30°	05:58	06:28	06:54	17:07	17:34	18:04
35°	06:07	06:38	07:06	16:55	17:23	17:55
40°	06:16	06:50	07:21	16:41	17:11	17:45
45°	06:26	07:03	07:37	16:24	16:58	17:35
S 50°	06:38	07:19	07:58	16:03	16:42	17:23
52°	06:43	07:27	08:08	15:53	16:34	17:18
54°	06:49	07:35	08:19	15:42	16:26	17:12
56°	06:55	07:44	08:32	15:30	16:17	17:06
58°	07:02	07:54	08:46	15:15	16:07	17:00
S 60°	07:09	08:06	09:04	14:58	15:55	16:52

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°		06:31	09:02		00:47	00:06
N 70°	04:12	06:57	09:12	01:08	00:19	23:37
68°	04:59	07:17	09:21	00:20	23:53	23:33
66°	05:28	07:32	09:28	23:40	23:33	23:27
64°	05:51	07:45	09:34	23:26	23:25	23:24
62°	06:09	07:55	09:39	23:14	23:19	23:22
60°	06:23	08:04	09:43	23:04	23:13	23:20
N 58°	06:36	08:12	09:47	22:55	23:08	23:18
56°	06:46	08:19	09:50	22:47	23:03	23:16
54°	06:56	08:25	09:53	22:40	22:59	23:15
52°	07:04	08:31	09:56	22:34	22:55	23:14
50°	07:12	08:36	09:59	22:28	22:52	23:12
45°	07:28	08:46	10:04	22:16	22:44	23:10
N 40°	07:41	08:55	10:09	22:06	22:38	23:08
35°	07:52	09:03	10:12	21:57	22:33	23:06
30°	08:01	09:09	10:16	21:49	22:28	23:04
20°	08:18	09:21	10:22	21:35	22:20	23:01
N 10°	08:32	09:31	10:27	21:23	22:12	22:58
0°	08:46	09:40	10:32	21:12	22:05	22:56
S 10°	08:59	09:49	10:37	21:00	21:58	22:53
20°	09:13	09:59	10:42	20:48	21:50	22:50
30°	09:29	10:10	10:48	20:34	21:41	22:47
35°	09:39	10:17	10:51	20:26	21:36	22:45
40°	09:49	10:24	10:55	20:16	21:30	22:43
45°	10:02	10:33	10:59	20:05	21:24	22:40
S 50°	10:17	10:43	11:04	19:51	21:15	22:37
52°	10:24	10:47	11:07	19:45	21:11	22:36
54°	10:32	10:53	11:09	19:38	21:07	22:34
56°	10:41	10:58	11:12	19:30	21:02	22:33
58°	10:50	11:05	11:15	19:21	20:57	22:31
S 60°	11:02	11:12	11:19	19:11	20:51	22:29

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Upper hh:mm	Lower hh:mm	
15	00:22	00:28	12:00	14:59	02:31	
16	00:34	00:41	12:01	15:53	03:26	
17	00:47	00:54	12:01	16:44	04:18	

June 18, 19, 20 UT (Fri., Sat., Sun.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Fri	GHA	Dec	GHA	Dec		GHA	Dec		GHA	Dec		GHA	Dec		SHA	Dec	
0	266°07.8	139°03.2	N21°08.2	108°11.9	N10°26.7	264°20.0	S00°33.2	86°55.2	N02°55.2						Alpheratz	357°45.8	29°08.8
1	281°10.3	154°02.6	07.4	123°13.1	26.2	279°22.2	33.1	101°57.6	55.1						Ankaa	353°17.8	-42°14.6
2	296°12.8	169°02.1	06.7	138°14.3	25.6	294°24.4	33.0	117°00.0	55.1						Schedar	349°43.3	56°35.5
3	311°15.2	184°01.5	•• 06.0	153°15.5	•• 25.1	309°26.7	•• 32.9	132°02.4	•• 55.0						Diphda	348°58.1	-17°55.6
4	326°17.7	199°01.0	05.3	168°16.7	24.6	324°28.9	32.8	147°04.8	55.0						Achernar	335°28.5	-57°10.7
5	341°20.2	214°00.4	04.6	183°17.9	24.0	339°31.1	32.7	162°07.2	55.0						Hamal	328°03.5	23°30.7
6	356°22.6	228°59.8	N21°03.9	198°19.1	N10°23.5	354°33.3	S00°32.6	177°09.6	N02°54.9						Polaris	319°10.0	89°18.4
7	11°25.1	243°59.3	03.2	213°20.3	22.9	9°35.5	32.5	192°12.0	54.9						Acamar	315°20.2	-40°15.5
8	26°27.6	258°58.7	02.4	228°21.5	22.4	24°37.7	32.4	207°14.4	54.8						Menkar	314°17.6	4°07.9
9	41°30.0	273°58.2	•• 01.7	243°22.7	•• 21.9	39°40.0	•• 32.3	222°16.8	•• 54.8						Mirfak	308°44.0	49°53.8
10	56°32.5	288°57.6	01.0	258°23.9	21.3	54°42.2	32.2	237°19.2	54.8						Aldebaran	290°52.2	16°31.8
11	71°34.9	303°57.0	21°00.3	273°25.1	20.8	69°44.4	32.1	252°21.6	54.7						Rigel	281°14.5	-8°11.4
12	86°37.4	318°56.5	N20°59.6	288°26.3	N10°20.2	84°46.6	S00°32.1	267°23.9	N02°54.7						Capella	280°38.2	46°00.5
13	101°39.9	333°55.9	58.8	303°27.5	19.7	99°48.8	32.0	282°26.3	54.6						Bellatrix	278°34.7	6°21.5
14	116°42.3	348°55.4	58.1	318°28.7	19.2	114°51.0	31.9	297°28.7	54.6						Elnath	278°15.8	28°36.9
15	131°44.8	3°54.8	•• 57.4	333°29.9	•• 18.6	129°53.3	•• 31.8	312°31.1	•• 54.6						Alnilam	275°48.9	-1°11.7
16	146°47.3	18°54.3	56.7	348°31.1	18.1	144°55.5	31.7	327°33.5	54.5						Betelgeuse	271°04.0	7°24.5
17	161°49.7	33°53.7	55.9	3°32.3	17.5	159°57.7	31.6	342°35.9	54.5						Canopus	263°57.7	-52°42.1
18	176°52.2	48°53.2	N20°55.2	18°33.4	N10°17.0	174°59.9	S00°31.5	357°38.3	N02°54.4						Sirius	258°36.0	-16°43.9
19	191°54.7	63°52.6	54.5	33°34.6	16.4	190°02.1	31.4	12°40.7	54.4						Adhara	255°14.6	-28°59.3
20	206°57.1	78°52.1	53.8	48°35.8	15.9	205°04.4	31.3	27°43.1	54.4						Procyon	245°02.3	5°11.8
21	221°59.6	93°51.5	•• 53.0	63°37.0	•• 15.4	220°06.6	•• 31.2	42°45.5	•• 54.3						Pollux	243°30.7	28°00.1
22	237°02.0	108°51.0	52.3	78°38.2	14.8	235°08.8	31.1	57°47.9	54.3						Avior	234°19.5	-59°32.8
23	252°04.5	123°50.4	51.6	93°39.4	14.3	250°11.0	31.0	72°50.3	54.2						Suhail	222°54.4	-43°28.8
Mer.pass. 06:14			ν -0.6' d -0.7' m -3.98	ν 1.2' d -0.5' m 1.26	ν 2.2' d -0.1' m -2.36	ν 2.4' d -0.0' m 0.92											

Sat	GHA	Dec	GHA	Dec		GHA	Dec		GHA	Dec	
0	267°07.0	138°49.9	N20°50.8	108°40.6	N10°13.7	265°13.2	S00°30.9	87°52.7	N02°54.2		
1	282°09.4	153°49.3	50.1	123°41.8	13.2	280°15.5	30.8	102°55.1	54.2		
2	297°11.9	168°48.8	49.4	138°43.0	12.6	295°17.7	30.7	117°57.5	54.1		
3	312°14.4	183°48.3	•• 48.6	153°44.2	•• 12.1	310°19.9	•• 30.6	132°59.9	•• 54.1		
4	327°16.8	198°47.7	47.9	168°45.4	11.6	325°22.1	30.6	148°02.3	54.0		
5	342°19.3	213°47.2	47.2	183°46.6	11.0	340°24.3	30.5	163°04.6	54.0		
6	357°21.8	228°46.6	N20°46.4	198°47.8	N10°10.5	355°26.6	S00°30.4	178°07.0	N02°54.0		
7	12°24.2	243°46.1	45.7	213°49.0	09.9	10°28.8	30.3	193°09.4	53.9		
8	27°26.7	258°45.5	45.0	228°50.1	09.4	25°31.0	30.2	208°11.8	53.9		
9	42°29.2	273°45.0	•• 44.2	243°51.3	•• 08.8	40°33.2	•• 30.1	223°14.2	•• 53.8		
10	57°31.6	288°44.5	43.5	258°52.5	08.3	55°35.4	30.0	238°16.6	53.8		
11	72°34.1	303°43.9	42.7	273°53.7	07.8	70°37.7	29.9	253°19.0	53.8		
12	87°36.5	318°43.4	N20°42.0	288°54.9	N10°07.2	85°39.9	S00°29.8	268°21.4	N02°53.7		
13	102°39.0	333°42.8	41.2	303°56.1	06.7	100°42.1	29.7	283°23.8	53.7		
14	117°41.5	348°42.3	40.5	318°57.3	06.1	115°44.3	29.6	298°26.2	53.6		
15	132°43.9	3°41.8	•• 39.8	333°58.5	•• 05.6	130°46.6	•• 29.5	313°28.6	•• 53.6		
16	147°46.4	18°41.2	39.0	348°59.7	05.0	145°48.8	29.4	328°31.0	53.6		
17	162°48.9	33°40.7	38.3	4°00.9	04.5	160°51.0	29.4	343°33.3	53.5		
18	177°51.3	48°40.2	N20°37.5	19°02.1	N10°03.9	175°53.2	S00°29.3	358°35.7	N02°53.5		
19	192°53.8	63°39.6	36.8	34°03.2	03.4	190°55.5	29.2	13°38.1	53.4		
20	207°56.3	78°39.1	36.0	49°04.4	02.8	205°57.7	29.1	28°40.5	53.4		
21	222°58.7	93°38.6	•• 35.3	64°05.6	•• 02.3	220°59.9	•• 29.0	43°42.9	•• 53.3		
22	238°01.2	108°38.0	34.5	79°06.8	01.8	236°02.1	28.9	58°45.3	53.3		
23	253°03.7	123°37.5	33.8	94°08.0	01.2	251°04.4	28.8	73°47.7	53.3		
Mer.pass. 06:11			ν -0.5' d -0.7' m -3.99	ν 1.2' d -0.5' m 1.27	ν 2.2' d -0.1' m -2.37	ν 2.4' d -0.0' m 0.93					

Sun	GHA	Dec	GHA	Dec		GHA	Dec		GHA	Dec	
0	268°06.1	138°37.0	N20°33.0	109°09.2	N10°00.7	266°06.6	S00°28.7	88°50.1	N02°53.2		
1	283°08.6	153°36.4	32.2	124°10.4	10°00.1	281°08.8	28.6	103°52.5	53.2		
2	298°11.0	168°35.9	31.5	139°11.6	09°59.6	296°11.0	28.5	118°54.9	53.1		
3	313°13.5	183°35.4	•• 30.7	154°12.8	•• 59.0	311°13.3	•• 28.4	133°57.2	•• 53.1		
4	328°16.0	198°34.9	30.0	169°14.0	58.5	326°15.5	28.4	148°59.6	53.1		
5	343°18.4	213°34.3	29.2	184°15.1	57.9	341°17.7	28.3	164°02.0	53.0		
6	358°20.9	228°33.8	N20°28.5	199°16.3	N09°57.4	356°20.0	S00°28.2	179°04.4	N02°53.0		
7	13°23.4	243°33.3	27.7	214°17.5	56.8	11°22.2	28.1	194°06.8	52.9		
8	28°25.8	258°32.7	26.9	229°18.7	56.3	26°24.4	28.0	209°09.2	52.9		
9	43°28.3	273°32.2	•• 26.2	244°19.9	•• 55.7	41°26.6	•• 27.9	224°11.6	•• 52.8		
10	58°30.8	288°31.7	25.4	259°21.1	55.2	56°28.9	27.8	239°14.0	52.8		
11	73°33.2	303°31.2	24.7	274°22.3	54.7	71°31.1	27.7	254°16.4	52.8		
12	88°35.7	318°30.7	N20°23.9	289°23.5	N09°54.1	86°33.3	S00°27.6	269°18.7	N02°52.7		
13	103°38.1	333°30.1	23.1	304°24.7	53.6	101°35.6	27.6	284°21.1	52.7		
14	118°40.6	348°29.6	22.4	319°25.8	53.0	116°37.8	27.5	299°23.5	52.6		
15	133°43.1	3°29.1	•• 21.6	334°27.0	•• 52.5	131°40.0	•• 27.4	314°25.9	•• 52.6		
16	148°45.5	18°28.6	20.8	349°28.2	51.9	146°42.2	27.3	329°28.3	52.5		
17	163°48.0	33°28.1	20.1	4°29.4	51.4	161°44.5	27.2	344°30.7	52.5		
18	178°50.5	48°27.5	N20°19.3	19°30.6	N09°50.8	176°46.7	S00°27.1	359°33.1	N02°52.5		
19	193°52.9	63°27.0	18.5	34°31.8	50.3	191°48.9	27.0	14°35.5	52.4		
20	208°55.4	78°26.5	17.7	49°33.0	49.7	206°51.2	26.9	29°37.8	52.4		
21	223°57.9	93°26.0	•• 17.0	64°34.2	•• 49.2	221°53.4	•• 26.8	44°40.2	•• 52.3		
22	239°00.3	108°25.5	16.2	79°35.3	48.6	236°55.6	26.8	59°42.6	52.3		
23	254°02.8	123°25.0	15.4	94°36.5	48.1	251°57.9	26.7	74°45.0	52.2		
Mer.pass. 06:07			ν -0.5' d -0.8' m -3.99	ν 1.2' d -0.5' m 1.27	ν 2.2' d -0.1' m -2.37	ν 2.4' d -0.0' m 0.93					

Jun 18 Fri	SHA	Mer.pass
Venus	232°55.4	14:44
Mars	202°04.1	16:46
Jupiter	358°12.2	06:22
Saturn	180°47.4	18:09

Jun 19 Sat	SHA	Mer.pass
Venus	231°42.9	14:45
Mars	201°33.6	16:44
Jupiter	358°06.3	06:18
Saturn	180°45.7	18:06

Jun 20 Sun	SHA	Mer.pass
Venus	230°30.9	14:46
Mars	201°03.1	16:42
Jupiter	358°00.5	06:15
Saturn	180°44.0	18:02

Horizontal parallax		
Venus:	0.1	
Mars:	0.1	

h	Sun			Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	179°44.9	N23°23.8	105°25.2	11.1'	N03°16.5	-15.0'	59.4'
1	194°44.7	23.9	119°55.2	11.1'	03°01.5	-15.0'	59.4'
2	209°44.6	23.9	134°25.3	11.1'	02°46.5	-15.0'	59.4'
3	224°44.5	· · 24.0	148°55.4	11.1'	02°31.5	-15.0'	59.4'
4	239°44.3	24.0	163°25.6	11.2'	02°16.5	-15.0'	59.3'
5	254°44.2	24.1	177°55.8	11.2'	02°01.5	-15.0'	59.3'
6	269°44.1	N23°24.1	192°26.0	11.2'	N01°46.5	-15.0'	59.3'
7	284°43.9	24.2	206°56.2	11.3'	01°31.4	-15.0'	59.3'
8	299°43.8	24.3	221°26.4	11.3'	01°16.4	-15.0'	59.3'
9	314°43.7	· · 24.3	235°56.7	11.3'	01°01.4	-15.0'	59.3'
10	329°43.5	24.4	250°27.0	11.3'	00°46.4	-15.0'	59.3'
11	344°43.4	24.4	264°57.3	11.3'	00°31.4	-15.0'	59.2'
12	359°43.2	N23°24.5	279°27.7	11.4'	N00°16.4	-15.0'	59.2'
13	14°43.1	24.5	293°58.0	11.4'	N00°01.4	-15.0'	59.2'
14	29°43.0	24.6	308°28.4	11.4'	S00°13.6	15.0'	59.2'
15	44°42.8	· · 24.6	322°58.8	11.4'	00°28.6	15.0'	59.2'
16	59°42.7	24.7	337°29.2	11.4'	00°43.6	15.0'	59.2'
17	74°42.6	24.7	351°59.6	11.4'	00°58.5	14.9'	59.1'
18	89°42.4	N23°24.8	6°30.1	11.4'	S01°13.5	14.9'	59.1'
19	104°42.3	24.8	21°00.5	11.5'	01°28.4	14.9'	59.1'
20	119°42.2	24.8	35°31.0	11.5'	01°43.3	14.9'	59.1'
21	134°42.0	· · 24.9	50°01.4	11.5'	01°58.2	14.9'	59.1'
22	149°41.9	24.9	64°31.9	11.5'	02°13.1	14.9'	59.1'
23	164°41.7	25.0	79°02.4	11.5'	02°28.0	14.8'	59.0'
SD = 15.7'			SD = 16.2'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	179°41.6	N23°25.0	93°32.9	11.5'	S02°42.8	14.8'	59.0'
1	194°41.5	25.1	108°03.4	11.5'	02°57.6	14.8'	59.0'
2	209°41.3	25.1	122°33.9	11.5'	03°12.4	14.8'	59.0'
3	224°41.2	· · 25.1	137°04.4	11.5'	03°27.2	14.7'	59.0'
4	239°41.1	25.2	151°34.9	11.5'	03°41.9	14.7'	59.0'
5	254°40.9	25.2	166°05.4	11.5'	03°56.6	14.7'	58.9'
6	269°40.8	N23°25.3	180°35.9	11.5'	S04°11.3	14.7'	58.9'
7	284°40.7	25.3	195°06.4	11.5'	04°25.9	14.6'	58.9'
8	299°40.5	25.3	209°36.9	11.5'	04°40.6	14.6'	58.9'
9	314°40.4	· · 25.4	224°07.5	11.5'	04°55.1	14.6'	58.9'
10	329°40.3	25.4	238°38.0	11.5'	05°09.7	14.5'	58.9'
11	344°40.1	25.5	253°08.4	11.5'	05°24.2	14.5'	58.8'
12	359°40.0	N23°25.5	267°38.9	11.5'	S05°38.7	14.4'	58.8'
13	14°39.8	25.5	282°09.4	11.5'	05°53.1	14.4'	58.8'
14	29°39.7	25.6	296°39.9	11.5'	06°07.5	14.4'	58.8'
15	44°39.6	· · 25.6	311°10.4	11.5'	06°21.9	14.3'	58.8'
16	59°39.4	25.6	325°40.8	11.4'	06°36.2	14.3'	58.7'
17	74°39.3	25.6	340°11.3	11.4'	06°50.5	14.2'	58.7'
18	89°39.2	N23°25.7	354°41.7	11.4'	S07°04.7	14.2'	58.7'
19	104°39.0	25.7	9°12.1	11.4'	07°18.9	14.1'	58.7'
20	119°38.9	25.7	23°42.5	11.4'	07°33.1	14.1'	58.7'
21	134°38.8	· · 25.8	38°12.9	11.4'	07°47.1	14.0'	58.6'
22	149°38.6	25.8	52°43.3	11.4'	08°01.2	14.0'	58.6'
23	164°38.5	25.8	67°13.7	11.3'	08°15.2	13.9'	58.6'
SD = 15.7'			SD = 16.1'				

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	179°38.3	N23°25.8	81°44.0	11.3'	S08°29.1	13.9'	58.6'
1	194°38.2	25.9	96°14.3	11.3'	08°43.0	13.8'	58.6'
2	209°38.1	25.9	110°44.6	11.3'	08°56.9	13.8'	58.6'
3	224°37.9	· · 25.9	125°14.9	11.3'	09°10.6	13.7'	58.5'
4	239°37.8	25.9	139°45.2	11.2'	09°24.4	13.7'	58.5'
5	254°37.7	26.0	154°15.4	11.2'	09°38.0	13.6'	58.5'
6	269°37.5	N23°26.0	168°45.6	11.2'	S09°51.6	13.5'	58.5'
7	284°37.4	26.0	183°15.8	11.2'	10°05.2	13.5'	58.5'
8	299°37.3	26.0	197°46.0	11.1'	10°18.7	13.4'	58.4'
9	314°37.1	· · 26.0	212°16.1	11.1'	10°32.1	13.4'	58.4'
10	329°37.0	26.1	226°46.2	11.1'	10°45.5	13.3'	58.4'
11	344°36.8	26.1	241°16.3	11.1'	10°58.8	13.2'	58.4'
12	359°36.7	N23°26.1	255°46.4	11.0'	S11°12.0	13.2'	58.4'
13	14°36.6	26.1	270°16.4	11.0'	11°25.2	13.1'	58.3'
14	29°36.4	26.1	284°46.4	11.0'	11°38.3	13.0'	58.3'
15	44°36.3	· · 26.1	299°16.4	10.9'	11°51.3	13.0'	58.3'
16	59°36.2	26.2	313°46.3	10.9'	12°04.3	12.9'	58.3'
17	74°36.0	26.2	328°16.2	10.9'	12°17.2	12.8'	58.3'
18	89°35.9	N23°26.2	342°46.1	10.8'	S12°30.0	12.7'	58.2'
19	104°35.8	26.2	357°15.9	10.8'	12°42.7	12.7'	58.2'
20	119°35.6	26.2	11°45.7	10.8'	12°55.4	12.6'	58.2'
21	134°35.5	· · 26.2	26°15.5	10.7'	13°08.0	12.5'	58.2'
22	149°35.4	26.2	40°45.3	10.7'	13°20.5	12.4'	58.2'
23	164°35.2	26.2	55°15.0	10.7'	13°32.9	12.4'	58.1'
SD = 15.7'			SD = 16.0'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	☐	☐	☐	☐	☐	☐
66°	☐	☐	☐	☐	☐	☐
64°	////	////	01:31	22:32	////	////
62°	////	////	02:09	21:54	////	////
60°	////	00:50	02:36	21:27	23:14	////
N 58°	////	01:40	02:56	21:07	22:23	////
56°	////	02:10	03:13	20:50	21:53	////
54°	00:45	02:33	03:27	20:36	21:30	23:18
52°	01:32	02:51	03:39	20:23	21:12	22:31
50°	02:00	03:06	03:50	20:12	20:57	22:03
45°	02:46	03:35	04:13	19:50	20:27	21:17
N 40°	03:16	03:58	04:31	19:32	20:05	20:46
35°	03:40	04:16	04:46	19:17	19:46	20:23
30°	03:58	04:32	04:59	19:04	19:31	20:05
20°	04:27	04:57	05:21	18:42	19:06	19:35
N 10°	04:50	05:17	05:40	18:23	18:46	19:13
0°	05:09	05:35	05:58	18:05	18:28	18:54
S 10°	05:26	05:52	06:15	17:48	18:10	18:37
20°	05:42	06:10	06:34	17:29	17:53	18:21
30°	05:59	06:29	06:55	17:08	17:34	18:04
35°	06:07	06:39	07:07	16:55	17:24	17:55
40°	06:17	06:51	07:21	16:41	17:12	17:46
45°	06:27	07:04	07:38	16:24	16:58	17:35
S 50°	06:39	07:21	07:59	16:04	16:42	17:24
52°	06:44	07:28	08:09	15:54	16:35	17:18
54°	06:50	07:36	08:20	15:43	16:26	17:13
56°	06:56	07:45	08:33	15:30	16:17	17:06
58°	07:03	07:56	08:48	15:15	16:07	17:00
S 60°	07:10	08:07	09:05	14:58	15:55	16:52

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	11:17	13:30	15:51	23:11	22:43	22:07
N 70°	11:18	13:21	15:28	23:15	22:55	22:32
68°	11:18	13:13	15:11	23:18	23:05	22:51
66°	11:18	13:07	14:57	23:20	23:13	23:06
64°	11:19	13:02	14:45	23:22	23:20	23:19
62°	11:19	12:57	14:36	23:24	23:27	23:30
60°	11:19	12:53	14:27	23:26	23:32	23:39
N 58°	11:19	12:50	14:20	23:27	23:37	23:48
56°	11:19	12:47	14:14	23:28	23:41	23:55
54°	11:19	12:44	14:08	23:30	23:45	· · ·
52°	11:20	12:42	14:03	23:31	23:48	· · ·
50°	11:20	12:39	13:59	23:32	23:51	· · ·
45°	11:20	12:35	13:49	23:34	23:58	· · ·
N 40°	11:20	12:31	13:41	23:35	· · ·	00:04
35°	11:20	12:27	13:34	23:37	· · ·	00:09
30°	11:21	12:24	13:28	23:38	· · ·	00:13
20°	11:21	12:19	13:17	23:41	· · ·	00:21
N 10°	11:21	12:15	13:08	23:43	· · ·	00:27
0°	11:21	12:10	12:59	23:45	· · ·	00:34
S 10°	11:22	12:06	12:51	23:47	· · ·	00:40
20°	11:22	12:02	12:42	23:49	· · ·	00:47
30°	11:23	11:57	12:32	23:51	· · ·	00:54
35°	11:23	11:54	12:26	23:52	· · ·	00:59
40°	11:23	11:51	12:20	23:54	· · ·	01:04
45°	11:24	11:47	12:12	23:56	· · ·	01:10
S 50°	11:24	11:43	12:03	23:58	· · ·	01:17
52°	11:24	11:41	11:59	23:59	· · ·	01:20
54°	11:24	11:39	11:55	· · ·	00:00	01:24
56°	11:25	11:37	11:50	· · ·	00:01	01:28
58°	11:25	11:34	11:44	· · ·	00:02	01:33
S 60°	11:25	11:31	11:38	· · ·	00:04	01:38

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss	hh:mm	Upper hh:mm	Lower hh:mm	6-8 37-59%
18	01:00	01:07	12:01	17:33	05:09	
19	01:14	01:20	12:01	18:22	05:58	
20	01:27	01:33	12:02	19:11	06:46	

Aries		Venus		Mars		Jupiter		Saturn	
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	269°05.3	138°24.4	N20°14.7	109°37.7	N09°47.5	267°00.1	S00°26.6	89°47.4	N02°52.2
1	284°07.7	153°23.9	13.9	124°38.9	47.0	282°02.3	26.5	104°49.8	52.2
2	299°10.2	168°23.4	13.1	139°40.1	46.4	297°04.6	26.4	119°52.0	52.1
3	314°12.6	183°22.9	•• 12.3	154°41.3	•• 45.9	312°06.8	•• 26.3	134°54.5	•• 52.1
4	329°15.1	198°22.4	11.5	169°42.5	45.3	327°09.0	26.2	149°56.9	52.0
5	344°17.6	213°21.9	10.8	184°43.7	44.8	342°11.3	26.1	164°59.3	52.0
6	359°20.0	228°21.4	N20°10.0	199°44.8	N09°44.2	357°13.5	S00°26.1	180°01.7	N02°51.9
7	14°22.5	243°20.9	09.2	214°46.0	43.7	12°15.7	26.0	195°04.1	51.9
8	29°25.0	258°20.4	08.4	229°47.2	43.1	27°18.0	25.9	210°06.5	51.8
9	44°27.4	273°19.8	•• 07.6	244°48.4	•• 42.6	42°20.2	•• 25.8	225°08.9	•• 51.8
10	59°29.9	288°19.3	06.9	259°49.6	42.0	57°22.4	25.7	240°11.2	51.8
11	74°32.4	303°18.8	06.1	274°50.8	41.5	72°24.7	25.6	255°13.6	51.7
12	89°34.8	318°18.3	N20°05.3	289°52.0	N09°40.9	87°26.9	S00°25.5	270°16.0	N02°51.7
13	104°37.3	333°17.8	04.5	304°53.1	40.4	102°29.1	25.4	285°18.4	51.6
14	119°39.8	348°17.3	03.7	319°54.3	39.8	117°31.4	25.4	300°20.8	51.6
15	134°42.2	3°16.8	•• 02.9	334°55.5	•• 39.3	132°33.6	•• 25.3	315°23.2	•• 51.5
16	149°44.7	18°16.3	02.1	349°56.7	38.7	147°35.8	25.2	330°25.5	51.5
17	164°47.1	33°15.8	01.3	4°57.9	38.2	162°38.1	25.1	345°27.9	51.4
18	179°49.6	48°15.3	N20°00.6	19°59.1	N09°37.6	177°40.3	S00°25.0	0°30.3	N02°51.4
19	194°52.1	63°14.8	19°59.8	35°00.3	37.1	192°42.6	24.9	15°32.7	51.4
20	209°54.5	78°14.3	59.0	50°01.4	36.5	207°44.8	24.8	30°35.1	51.3
21	224°57.0	93°13.8	•• 58.2	65°02.6	•• 36.0	222°47.0	•• 24.8	45°37.5	•• 51.3
22	239°59.5	108°13.3	57.4	80°03.8	35.4	237°49.3	24.7	60°39.8	51.2
23	255°01.9	123°12.8	56.6	95°05.0	34.9	252°51.5	24.6	75°42.2	51.2
Mer.pass. 06:03		ν -0.5' d -0.8' m -4.00		ν 1.2' d -0.5' m 1.28		ν 2.2' d -0.1' m -2.38		ν 2.4' d -0.0' m 0.93	

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	271°03.5	138°00.6	N19°36.4	110°34.6	N09°21.0	268°47.5	S00°22.5	91°41.7	N02°50.0
1	286°06.0	153°00.1	35.6	125°35.7	20.5	283°49.8	22.4	106°44.1	50.0
2	301°08.5	167°59.6	34.8	140°36.9	19.9	298°52.0	22.3	121°46.5	49.9
3	316°10.9	182°59.1	•• 34.0	155°38.1	•• 19.4	313°54.3	•• 22.2	136°48.9	•• 49.9
4	331°13.4	197°58.7	33.2	170°39.3	18.8	328°56.5	22.2	151°51.3	49.8
5	346°15.9	212°58.2	32.3	185°40.5	18.2	343°58.8	22.1	166°53.6	49.8
6	1°18.3	227°57.7	N19°31.5	200°41.7	N09°17.7	359°01.0	S00°22.0	181°56.0	N02°49.8
7	16°20.8	242°57.2	30.7	215°42.8	17.1	14°03.3	21.9	196°58.4	49.7
8	31°23.2	257°56.8	29.9	230°44.0	16.6	29°05.5	21.8	212°00.8	49.7
9	46°25.7	272°56.3	•• 29.1	245°45.2	•• 16.0	44°07.8	•• 21.8	227°03.2	•• 49.6
10	61°28.2	287°55.8	28.2	260°46.4	15.5	59°10.0	21.7	242°05.5	49.6
11	76°30.6	302°55.3	27.4	275°47.6	14.9	74°12.3	21.6	257°07.9	49.5
12	91°33.1	317°54.9	N19°26.6	290°48.7	N09°14.3	89°14.5	S00°21.5	272°10.3	N02°49.5
13	106°35.6	332°54.4	25.8	305°49.9	13.8	104°16.8	21.4	287°12.7	49.4
14	121°38.0	347°53.9	24.9	320°51.1	13.2	119°19.0	21.3	302°15.0	49.4
15	136°40.5	2°53.4	•• 24.1	335°52.3	•• 12.7	134°21.2	•• 21.3	317°17.4	•• 49.3
16	151°43.0	17°53.0	23.3	350°53.5	12.1	149°23.5	21.2	332°19.8	49.3
17	166°45.4	32°52.5	22.4	5°54.6	11.6	164°25.7	21.1	347°22.2	49.2
18	181°47.9	47°52.0	N19°21.6	20°55.8	N09°11.0	179°28.0	S00°21.0	2°24.5	N02°49.2
19	196°50.4	62°51.6	20.8	35°57.0	10.4	194°30.2	20.9	17°26.9	49.1
20	211°52.8	77°51.1	19.9	50°58.2	09.9	209°32.5	20.9	32°29.3	49.1
21	226°55.3	92°50.6	•• 19.1	65°59.4	•• 09.3	224°34.7	•• 20.8	47°31.7	•• 49.0
22	241°57.7	107°50.2	18.3	81°00.5	08.8	239°37.0	20.7	62°34.0	49.0
23	257°00.2	122°49.7	17.4	96°01.7	08.2	254°39.2	20.6	77°36.4	48.9
Mer.pass. 05:55		ν -0.5' d -0.8' m -4.00		ν 1.2' d -0.6' m 1.29		ν 2.2' d -0.1' m -2.40		ν 2.4' d -0.0' m 0.94	

Jun 21 Mon	SHA	Mer.pass
Venus	229° 19.2	14:47
Mars	200° 32.5	16:40
Jupiter	357° 54.8	06:11
Saturn	180° 42.1	17:58

Jun 22 Tue	SHA	Mer.pass
Venus	228° 07.9	14:48
Mars	200° 01.8	16:38
Jupiter	357° 49.4	06:08
Saturn	180° 40.2	17:54

Jun 23 Wed	SHA	Mer.pass
Venus	226° 57.0	14:48
Mars	199° 31.0	16:36
Jupiter	357° 44.0	06:04
Saturn	180° 38.2	17:50

Horizontal parallax	
Venus:	0.1
Mars:	0.1

h Sun Moon						
Mon	GHA	Dec	GHA	ν	Dec	d HP
0	179°35.1	N23°26.2	69°44.6	10.6'	S13°45.3	12.3' 58.1'
1	194°34.9	26.3	84°14.3	10.6'	13°57.6	12.2' 58.1'
2	209°34.8	26.3	98°43.9	10.6'	14°09.8	12.1' 58.1'
3	224°34.7	· · 26.3	113°13.4	10.5'	14°21.9	12.0' 58.1'
4	239°34.5	26.3	127°42.9	10.5'	14°33.9	11.9' 58.0'
5	254°34.4	26.3	142°12.4	10.4'	14°45.9	11.9' 58.0'
6	269°34.3	N23°26.3	156°41.9	10.4'	S14°57.7	11.8' 58.0'
7	284°34.1	26.3	171°11.3	10.4'	15°09.5	11.7' 58.0'
8	299°34.0	26.3	185°40.6	10.3'	15°21.2	11.6' 58.0'
9	314°33.9	· · 26.3	200°09.9	10.3'	15°32.8	11.5' 57.9'
10	329°33.7	26.3	214°39.2	10.2'	15°44.3	11.4' 57.9'
11	344°33.6	26.3	229°08.5	10.2'	15°55.7	11.3' 57.9'
12	359°33.5	N23°26.3	243°37.7	10.2'	S16°07.0	11.2' 57.9'
13	14°33.3	26.3	258°06.8	10.1'	16°18.2	11.1' 57.9'
14	29°33.2	26.3	272°35.9	10.1'	16°29.4	11.0' 57.8'
15	44°33.0	· · 26.3	287°05.0	10.0'	16°40.4	10.9' 57.8'
16	59°32.9	26.3	301°34.0	10.0'	16°51.4	10.8' 57.8'
17	74°32.8	26.3	316°03.0	9.9'	17°02.2	10.7' 57.8'
18	89°32.6	N23°26.3	330°32.0	9.9'	S17°13.0	10.6' 57.8'
19	104°32.5	26.3	345°00.9	9.9'	17°23.6	10.5' 57.7'
20	119°32.4	26.3	359°29.7	9.8'	17°34.2	10.4' 57.7'
21	134°32.2	· · 26.3	13°58.6	9.8'	17°44.6	10.3' 57.7'
22	149°32.1	26.3	28°27.3	9.7'	17°54.9	10.2' 57.7'
23	164°32.0	26.2	42°56.1	9.7'	18°05.2	10.1' 57.6'
SD = 15.7' $d = 0.0'$ SD = 15.9'						

Tue	GHA	Dec	GHA	ν	Dec	d HP
0	179°31.8	N23°26.2	57°24.7	9.6'	S18°15.3	10.0' 57.6'
1	194°31.7	26.2	71°53.4	9.6'	18°25.4	9.9' 57.6'
2	209°31.6	26.2	86°22.0	9.6'	18°35.3	9.8' 57.6'
3	224°31.4	· · 26.2	100°50.5	9.5'	18°45.1	9.7' 57.6'
4	239°31.3	26.2	115°19.1	9.5'	18°54.8	9.6' 57.5'
5	254°31.2	26.2	129°47.5	9.4'	19°04.4	9.5' 57.5'
6	269°31.0	N23°26.2	144°15.9	9.4'	S19°13.9	9.4' 57.5'
7	284°30.9	26.2	158°44.3	9.3'	19°23.3	9.3' 57.5'
8	299°30.7	26.1	173°12.7	9.3'	19°32.5	9.2' 57.5'
9	314°30.6	· · 26.1	187°41.0	9.3'	19°41.7	9.0' 57.4'
10	329°30.5	26.1	202°09.2	9.2'	19°50.7	8.9' 57.4'
11	344°30.3	26.1	216°37.4	9.2'	19°59.7	8.8' 57.4'
12	359°30.2	N23°26.1	231°05.6	9.1'	S20°08.5	8.7' 57.4'
13	14°30.1	26.1	245°33.7	9.1'	20°17.2	8.6' 57.4'
14	29°29.9	26.0	260°01.8	9.0'	20°25.7	8.5' 57.3'
15	44°29.8	· · 26.0	274°29.9	9.0'	20°34.2	8.3' 57.3'
16	59°29.7	26.0	288°57.9	9.0'	20°42.5	8.2' 57.3'
17	74°29.5	26.0	303°25.8	8.9'	20°50.7	8.1' 57.3'
18	89°29.4	N23°26.0	317°53.7	8.9'	S20°58.8	8.0' 57.3'
19	104°29.3	25.9	332°21.6	8.8'	21°06.8	7.9' 57.2'
20	119°29.1	25.9	346°49.5	8.8'	21°14.7	7.7' 57.2'
21	134°29.0	· · 25.9	1°17.3	8.8'	21°22.4	7.6' 57.2'
22	149°28.9	25.9	15°45.1	8.7'	21°30.0	7.5' 57.2'
23	164°28.7	25.8	30°12.8	8.7'	21°37.5	7.4' 57.1'
SD = 15.7' $d = -0.0'$ SD = 15.7'						

Wed	GHA	Dec	GHA	ν	Dec	d HP
0	179°28.6	N23°25.8	44°40.5	8.7'	S21°44.8	7.2' 57.1'
1	194°28.5	25.8	59°08.1	8.6'	21°52.1	7.1' 57.1'
2	209°28.3	25.8	73°35.8	8.6'	21°59.2	7.0' 57.1'
3	224°28.2	· · 25.7	88°03.4	8.6'	22°06.2	6.8' 57.1'
4	239°28.1	25.7	102°30.9	8.5'	22°13.0	6.7' 57.0'
5	254°27.9	25.7	116°58.4	8.5'	22°19.7	6.6' 57.0'
6	269°27.8	N23°25.6	131°25.9	8.5'	S22°26.3	6.5' 57.0'
7	284°27.6	25.6	145°53.4	8.4'	22°32.8	6.3' 57.0'
8	299°27.5	25.6	160°20.8	8.4'	22°39.1	6.2' 57.0'
9	314°27.4	· · 25.6	174°48.2	8.4'	22°45.3	6.1' 56.9'
10	329°27.2	25.5	189°15.6	8.3'	22°51.4	5.9' 56.9'
11	344°27.1	25.5	203°42.9	8.3'	22°57.3	5.8' 56.9'
12	359°27.0	N23°25.4	218°10.2	8.3'	S23°03.1	5.7' 56.9'
13	14°26.8	25.4	232°37.5	8.3'	23°08.8	5.5' 56.9'
14	29°26.7	25.4	247°04.8	8.2'	23°14.3	5.4' 56.8'
15	44°26.6	· · 25.3	261°32.0	8.2'	23°19.7	5.3' 56.8'
16	59°26.4	25.3	275°59.2	8.2'	23°25.0	5.1' 56.8'
17	74°26.3	25.3	290°26.4	8.2'	23°30.1	5.0' 56.8'
18	89°26.2	N23°25.2	304°53.6	8.1'	S23°35.1	4.9' 56.8'
19	104°26.0	25.2	319°20.7	8.1'	23°40.0	4.7' 56.7'
20	119°25.9	25.1	333°47.8	8.1'	23°44.7	4.6' 56.7'
21	134°25.8	· · 25.1	348°14.9	8.1'	23°49.3	4.4' 56.7'
22	149°25.6	25.1	2°42.0	8.1'	23°53.7	4.3' 56.7'
23	164°25.5	25.0	17°09.1	8.1'	23°58.0	4.2' 56.6'
SD = 15.7' $d = -0.0'$ SD = 15.6'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	□	□	□	□	□	□
N 70°	□	□	□	□	□	□
68°	□	□	□	□	□	□
66°	□	□	□	□	□	□
64°	////	////	01:31	22:33	////	////
62°	////	////	02:09	21:54	////	////
60°	////	00:49	02:36	21:28	23:14	////
N 58°	////	01:41	02:57	21:07	22:23	////
56°	////	02:11	03:13	20:51	21:53	////
54°	00:45	02:33	03:28	20:36	21:31	23:18
52°	01:32	02:51	03:40	20:24	21:13	22:31
50°	02:00	03:06	03:51	20:13	20:58	22:03
45°	02:46	03:36	04:13	19:51	20:28	21:18
N 40°	03:17	03:59	04:31	19:32	20:05	20:47
35°	03:40	04:17	04:47	19:17	19:47	20:24
30°	03:59	04:32	05:00	19:04	19:32	20:05
20°	04:28	04:57	05:22	18:42	19:07	19:36
N 10°	04:51	05:18	05:41	18:23	18:46	19:13
0°	05:10	05:36	05:58	18:06	18:28	18:54
S 10°	05:27	05:53	06:16	17:48	18:11	18:37
20°	05:43	06:10	06:34	17:30	17:54	18:21
30°	05:59	06:29	06:56	17:08	17:35	18:05
35°	06:08	06:40	07:08	16:56	17:24	17:56
40°	06:18	06:52	07:22	16:42	17:12	17:46
45°	06:28	07:05	07:39	16:25	16:59	17:36
S 50°	06:40	07:21	08:00	16:04	16:43	17:24
52°	06:45	07:29	08:10	15:54	16:35	17:19
54°	06:51	07:37	08:21	15:43	16:27	17:13
56°	06:57	07:46	08:34	15:30	16:18	17:07
58°	07:04	07:56	08:48	15:16	16:08	17:00
S 60°	07:11	08:08	09:06	14:58	15:56	16:53

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	19:07	■	■	20:38	■	■
N 70°	17:54	■	■	21:52	■	■
68°	17:17	19:54	■	22:31	21:43	■
66°	16:51	18:53	21:42	22:58	22:45	21:49
64°	16:31	18:19	20:06	23:18	23:20	23:26
62°	16:15	17:54	19:28	23:35	23:45	· ·
60°	16:02	17:35	19:02	23:50	· ·	00:05
N 58°	15:50	17:19	18:41	· ·	00:02	00:22
56°	15:41	17:05	18:24	· ·	00:12	00:36
54°	15:32	16:53	18:10	00:01	00:22	00:48
52°	15:24	16:43	17:57	00:07	00:30	00:59
50°	15:17	16:34	17:46	00:12	00:38	01:08
45°	15:02	16:15	17:23	00:24	00:54	01:29
N 40°	14:50	15:59	17:05	00:34	01:07	01:45
35°	14:40	15:46	16:49	00:42	01:18	01:59
30°	14:31	15:34	16:36	00:49	01:28	02:11
20°	14:16	15:15	16:13	01:02	01:45	02:32
N 10°	14:02	14:58	15:54	01:13	02:00	02:50
0°	13:50	14:42	15:36	01:23	02:14	03:07
S 10°	13:37	14:26	15:18	01:34	02:29	03:25
20°	13:24	14:10	14:58	01:45	02:44	03:43
30°	13:10	13:51	14:36	01:58	03:01	04:04
35°	13:01	13:40	14:23	02:05	03:11	04:16
40°	12:51	13:27	14:08	02:14	03:23	04:31
45°	12:40	13:12	13:51	02:24	03:37	04:48
S 50°	12:26	12:54	13:29	02:36	03:54	05:09
52°	12:20	12:46	13:18	02:42	04:02	05:19
54°	12:13	12:36	13:07	02:48	04:11	05:30
56°	12:05	12:25	12:53	02:55	04:21	05:43
58°	11:56	12:13	12:38	03:03	04:32	05:58
S 60°	11:47	11:59	12:19	03:12	04:46	06:16

Day	Sun		Mer. Pass	Moon		Age 9-11 70-87%
	Eqn. of Time 00 ^h mm:ss	Time 12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
21	01:40	01:46	12:02	20:02	07:36	
22	01:53	01:59	12:02	20:55	08:28	
23	02:06	02:12	12:02	21:49	09:22	

June 24, 25, 26 UT (Thu., Fri., Sat.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Thu	GHA	Dec	GHA	Dec		GHA	Dec		GHA	Dec		GHA	Dec		SHA	Dec	
0	272°02.7	137°49.2	N19°16.6	111°02.9	N09°07.7	269°41.5	S00°20.5	92°38.8	N02°48.9						Alpheratz	357°45.8	29°08.9
1	287°05.1	152°48.8	15.8	126°04.1	07.1	284°43.7	20.5	107°41.2	48.8						Ankaa	353°17.8	-42°14.6
2	302°07.6	167°48.3	14.9	141°05.3	06.5	299°46.0	20.4	122°43.5	48.8						Schedar	349°43.2	56°35.5
3	317°10.1	182°47.9	•• 14.1	156°06.4	•• 06.0	314°48.3	•• 20.3	137°45.9	•• 48.8						Diphda	348°58.1	-17°55.5
4	332°12.5	197°47.4	13.3	171°07.6	05.4	329°50.5	20.2	152°48.3	48.7						Achernar	335°28.5	-57°10.6
5	347°15.0	212°46.9	12.4	186°08.8	04.9	344°52.8	20.2	167°50.7	48.7						Hamal	328°03.4	23°30.7
6	2°17.5	227°46.5	N19°11.6	201°10.0	N09°04.3	359°55.0	S00°20.1	182°53.0	N02°48.6						Polaris	319°07.4	89°18.4
7	17°19.9	242°46.0	10.7	216°11.1	03.7	14°57.3	20.0	197°55.4	48.6						Acamar	315°20.2	-40°15.5
8	32°22.4	257°45.6	09.9	231°12.3	03.2	29°59.5	19.9	212°57.8	48.5						Menkar	314°17.5	4°07.9
9	47°24.8	272°45.1	•• 09.0	246°13.5	•• 02.6	45°01.8	•• 19.8	228°00.2	•• 48.5						Mirfak	308°43.9	49°53.8
10	62°27.3	287°44.6	08.2	261°14.7	02.1	60°04.0	19.8	243°02.5	48.4						Aldebaran	290°52.2	16°31.8
11	77°29.8	302°44.2	07.4	276°15.9	01.5	75°06.3	19.7	258°04.9	48.4						Rigel	281°14.5	-8°11.3
12	92°32.2	317°43.7	N19°06.5	291°17.0	N09°00.9	90°08.5	S00°19.6	273°07.3	N02°48.3						Capella	280°38.1	46°00.4
13	107°34.7	332°43.3	05.7	306°18.2	09°00.4	105°10.8	19.5	288°09.7	48.3						Bellatrix	278°34.7	6°21.5
14	122°37.2	347°42.8	04.8	321°19.4	08°59.8	120°13.0	19.4	303°12.0	48.2						Elnath	278°15.7	28°36.9
15	137°39.6	2°42.4	•• 04.0	336°20.6	•• 59.3	135°15.3	•• 19.4	318°14.4	•• 48.2						Alnilam	275°48.9	-1°11.7
16	152°42.1	17°41.9	03.1	351°21.7	58.7	150°17.5	19.3	333°16.8	48.1						Betelgeuse	271°04.0	7°24.5
17	167°44.6	32°41.5	02.3	6°22.9	58.1	165°19.8	19.2	348°19.2	48.1						Canopus	263°57.7	-52°42.1
18	182°47.0	47°41.0	N19°01.4	21°24.1	N08°57.6	180°22.1	S00°19.1	3°21.5	N02°48.0						Sirius	258°36.0	-16°43.9
19	197°49.5	62°40.6	19°00.6	36°25.3	57.0	195°24.3	19.1	18°23.9	48.0						Adhara	255°14.6	-28°59.2
20	212°52.0	77°40.1	18°59.7	51°26.5	56.5	210°26.6	19.0	33°26.3	47.9						Procyon	245°02.3	5°11.8
21	227°54.4	92°39.7	•• 58.9	66°27.6	•• 55.9	225°28.8	•• 18.9	48°28.6	•• 47.9						Pollux	243°30.7	28°00.0
22	242°56.9	107°39.2	58.0	81°28.8	55.3	240°31.1	18.8	63°31.0	47.8						Avior	234°19.6	-59°32.8
23	257°59.3	122°38.8	57.2	96°30.0	54.8	255°33.3	18.7	78°33.4	47.8						Suhail	222°54.4	-43°28.7
Mer.pass. 05:51			ν -0.5' d -0.8' m -4.01	ν 1.2' d -0.6' m 1.30	ν 2.3' d -0.1' m -2.40	ν 2.4' d -0.0' m 0.94											
Fri	GHA	Dec	GHA	Dec		GHA	Dec		GHA	Dec		GHA	Dec				
0	273°01.8	137°38.3	N18°56.3	111°31.2	N08°54.2	270°35.6	S00°18.7	93°35.8	N02°47.7						Alphard	217°58.4	-8°42.4
1	288°04.3	152°37.9	55.4	126°32.3	53.7	285°37.9	18.6	108°38.1	47.7						Regulus	207°46.0	11°54.9
2	303°06.7	167°37.4	54.6	141°33.5	53.1	300°40.1	18.5	123°40.5	47.6						Dubhe	193°54.4	61°41.8
3	318°09.2	182°37.0	•• 53.7	156°34.7	•• 52.5	315°42.4	•• 18.4	138°42.9	•• 47.6						Denebola	182°35.9	14°30.8
4	333°11.7	197°36.5	52.9	171°35.9	52.0	330°44.6	18.4	153°45.2	47.5						Gienah	175°54.5	-17°36.2
5	348°14.1	212°36.1	52.0	186°37.0	51.4	345°46.9	18.3	168°47.6	47.5						Acrux	173°11.9	-63°09.8
6	3°16.6	227°35.7	N18°51.2	201°38.2	N08°50.9	0°49.1	S00°18.2	183°50.0	N02°47.4						Gacrux	172°03.4	-57°10.7
7	18°19.1	242°35.2	50.3	216°39.4	50.3	15°51.4	18.1	198°52.4	47.4						Alioth	166°22.3	55°54.3
8	33°21.5	257°34.8	49.4	231°40.6	49.7	30°53.7	18.1	213°54.7	47.3						Spica	158°33.4	-11°13.1
9	48°24.0	272°34.3	•• 48.6	246°41.7	•• 49.2	45°55.9	•• 18.0	228°57.1	•• 47.3						Alkaid	153°00.3	49°15.8
10	63°26.5	287°33.9	47.7	261°42.9	48.6	60°58.2	17.9	243°59.5	47.2						Hadar	148°50.8	-60°25.7
11	78°28.9	302°33.4	46.8	276°44.1	48.0	76°00.4	17.8	259°01.8	47.2						Menkent	148°10.0	-36°25.5
12	93°31.4	317°33.0	N18°46.0	291°45.3	N08°47.5	91°02.7	S00°17.8	274°04.2	N02°47.1						Arcturus	145°57.5	19°07.7
13	108°33.8	332°32.6	45.1	306°46.4	46.9	106°05.0	17.7	289°06.6	47.1						Rigel Kent.	139°54.4	-60°53.0
14	123°36.3	347°32.1	44.2	321°47.6	46.4	121°07.2	17.6	304°08.9	47.0						Kochab	137°18.9	74°06.9
15	138°38.8	2°31.7	•• 43.4	336°48.8	•• 45.8	136°09.5	•• 17.5	319°11.3	•• 47.0						Zuben'ubi	137°07.6	-16°05.2
16	153°41.2	17°31.3	42.5	351°50.0	45.2	151°11.8	17.5	334°13.7	46.9						Alphecca	126°12.5	26°40.8
17	168°43.7	32°30.8	41.6	6°51.1	44.7	166°14.0	17.4	349°16.1	46.9						Antares	112°28.6	-26°27.4
18	183°46.2	47°30.4	N18°40.8	21°52.3	N08°44.1	181°16.3	S00°17.3	4°18.4	N02°46.8						Atria	107°31.9	-69°02.9
19	198°48.6	62°30.0	39.9	36°53.5	43.5	196°18.5	17.2	19°20.8	46.8						Sabik	102°14.7	-15°44.3
20	213°51.1	77°29.5	39.0	51°54.7	43.0	211°20.8	17.2	34°23.2	46.7						Shaula	96°24.4	-37°06.7
21	228°53.6	92°29.1	•• 38.2	66°55.8	•• 42.4	226°23.1	•• 17.1	49°25.5	•• 46.7						Rasalhague	96°08.1	12°33.2
22	243°56.0	107°28.7	37.3	81°57.0	41.8	241°25.3	17.0	64°27.9	46.6						Eltanin	90°46.6	51°29.3
23	258°58.5	122°28.2	36.4	96°58.2	41.3	256°27.6	16.9	79°30.3	46.6						Kaus Aust.	83°46.3	-34°22.7
Mer.pass. 05:47			ν -0.4' d -0.9' m -4.01	ν 1.2' d -0.6' m 1.31	ν 2.3' d -0.1' m -2.41	ν 2.4' d -0.0' m 0.94											
Sat	GHA	Dec	GHA	Dec		GHA	Dec		GHA	Dec		GHA	Dec				
0	274°01.0	137°27.8	N18°35.5	111°59.4	N08°40.7	271°29.9	S00°16.9	94°32.6	N02°46.5						Jun 24 Thu	SHA	Mer.pass
1	289°03.4	152°27.4	34.7	127°00.5	40.1	286°32.1	16.8	109°35.0	46.5						Venus	225°46.6	14:49
2	304°05.9	167°26.9	33.8	142°01.7	39.6	301°34.4	16.7	124°37.4	46.4						Mars	199°00.2	16:35
3	319°08.3	182°26.5	•• 32.9	157°02.9	•• 39.0	316°36.6	•• 16.6	139°39.7	•• 46.4						Jupiter	357°38.8	06:00
4	334°10.8	197°26.1	32.0	172°04.0	38.5	331°38.9	16.6	154°42.1	46.3						Saturn	180°36.1	17:47
5	349°13.3	212°25.7	31.2	187°05.2	37.9	346°41.2	16.5	169°44.5	46.3								
6	4°15.7	227°25.2	N18°30.3	202°06.4	N08°37.3	1°43.4	S00°16.4	184°46.8	N02°46.2						Jun 25 Fri	SHA	Mer.pass
7	19°18.2	242°24.8	29.4	217°07.6	36.8	16°45.7	16.3	199°49.2	46.2						Venus	224°36.5	14:50
8	34°20.7	257°24.4	28.5	232°08.7	36.2	31°48.0	16.3	214°51.6	46.1						Mars	198°29.3	16:33
9	49°23.1	272°24.0	•• 27.6	247°09.9	•• 35.6	46°50.2	•• 16.2	229°53.9	•• 46.0						Jupiter	357°33.8	05:57
10	64°25.6	287°23.5	26.8	262°11.1	35.1	61°52.5	16.1	244°56.3	46.0						Saturn	180°33.9	17:43
11	79°28.1	302°23.1	25.9	277°12.3	34.5	76°54.8	16.1	259°58.7	45.9								
12	94°30.5	317°22.7	N18°25.0	292°13.4	N08°33.9	91°57.0	S00°16.0	275°01.0	N02°45.9						Jun 26 Sat	SHA	Mer.pass
13	109°33.0	332°22.3	24.1	307°14.6	33.4	106°59.3	15.9	290°03.4	45.8						Venus	223°26.8	14:51
14	124°35.5	347°21.8	23.2	322°15.8	32.8	122°01.6	15.8	305°05.8	45.8						Mars	197°58.4	16:31
15	139°37.9	2°21.4	•• 22.3	337°16.9	•• 32.2	137°03.8	•• 15.8	320°08.1	•• 45.7						Jupiter	357°28.9	05:53
16	154°40.4	17°21.0	21.4	352°18.1	31.7	152°06.1	15.7	335°10.5	45.7						Saturn	180°31.7	17:39
17	169°42.8	32°20.6	20.6	7°19.3	31.1	167°08.4	15.6	350°12.9	45.6								
18	184°45.3	47°20.2	N18°19.7	22°20.5	N08°30.5	182°10.6	S00°15.5	5°15.2	N02°45.6						Horizontal parallax		
19	199°47.8	62°19.8	18.8	37°21.6	30.0	197°12.9	15.5	20°17.6	45.5						Venus:	0.1	

h	Sun			Moon			
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	179°25.4	N23°25.0	31°36.2	8.0'	S24°02.2	4.0'	56.6'
1	194°25.2	24.9	46°03.2	8.0'	24°06.2	3.9'	56.6'
2	209°25.1	24.9	60°30.2	8.0'	24°10.1	3.8'	56.6'
3	224°25.0	· · 24.8	74°57.3	8.0'	24°13.9	3.6'	56.6'
4	239°24.8	24.8	89°24.3	8.0'	24°17.5	3.5'	56.5'
5	254°24.7	24.8	103°51.3	8.0'	24°20.9	3.3'	56.5'
6	269°24.6	N23°24.7	118°18.3	8.0'	S24°24.3	3.2'	56.5'
7	284°24.4	24.7	132°45.3	8.0'	24°27.5	3.1'	56.5'
8	299°24.3	24.6	147°12.2	8.0'	24°30.5	2.9'	56.5'
9	314°24.2	· · 24.6	161°39.2	8.0'	24°33.4	2.8'	56.4'
10	329°24.0	24.5	176°06.2	8.0'	24°36.2	2.6'	56.4'
11	344°23.9	24.5	190°33.2	8.0'	24°38.8	2.5'	56.4'
12	359°23.8	N23°24.4	205°00.1	8.0'	S24°41.3	2.4'	56.4'
13	14°23.6	24.4	219°27.1	8.0'	24°43.7	2.2'	56.4'
14	29°23.5	24.3	233°54.1	8.0'	24°45.9	2.1'	56.3'
15	44°23.4	· · 24.3	248°21.1	8.0'	24°48.0	1.9'	56.3'
16	59°23.2	24.2	262°48.1	8.0'	24°49.9	1.8'	56.3'
17	74°23.1	24.1	277°15.1	8.0'	24°51.7	1.6'	56.3'
18	89°23.0	N23°24.1	291°42.1	8.0'	S24°53.3	1.5'	56.3'
19	104°22.8	24.0	306°09.1	8.0'	24°54.8	1.4'	56.2'
20	119°22.7	24.0	320°36.1	8.0'	24°56.2	1.2'	56.2'
21	134°22.6	· · 23.9	335°03.1	8.0'	24°57.4	1.1'	56.2'
22	149°22.4	23.9	349°30.2	8.1'	24°58.5	0.9'	56.2'
23	164°22.3	23.8	3°57.2	8.1'	24°59.5	0.8'	56.2'
SD = 15.7'			SD = 15.4'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	179°22.2	N23°23.7	18°24.3	8.1'	S25°00.3	0.7'	56.1'
1	194°22.0	23.7	32°51.4	8.1'	25°00.9	0.5'	56.1'
2	209°21.9	23.6	47°18.5	8.1'	25°01.5	0.4'	56.1'
3	224°21.8	· · 23.6	61°45.6	8.2'	25°01.8	0.2'	56.1'
4	239°21.6	23.5	76°12.8	8.2'	25°02.1	0.1'	56.0'
5	254°21.5	23.4	90°39.9	8.2'	25°02.2	-0.0'	56.0'
6	269°21.4	N23°23.4	105°07.1	8.2'	S25°02.2	-0.2'	56.0'
7	284°21.3	23.3	119°34.4	8.2'	25°02.0	-0.3'	56.0'
8	299°21.1	23.2	134°01.6	8.3'	25°01.7	-0.4'	56.0'
9	314°21.0	· · 23.2	148°28.9	8.3'	25°01.2	-0.6'	55.9'
10	329°20.9	23.1	162°56.2	8.3'	25°00.6	-0.7'	55.9'
11	344°20.7	23.0	177°23.5	8.4'	24°59.9	-0.9'	55.9'
12	359°20.6	N23°23.0	191°50.9	8.4'	S24°59.1	-1.0'	55.9'
13	14°20.5	22.9	206°18.3	8.4'	24°58.1	-1.1'	55.9'
14	29°20.3	22.8	220°45.7	8.5'	24°56.9	-1.3'	55.8'
15	44°20.2	· · 22.7	235°13.2	8.5'	24°55.7	-1.4'	55.8'
16	59°20.1	22.7	249°40.7	8.5'	24°54.3	-1.5'	55.8'
17	74°19.9	22.6	264°08.3	8.6'	24°52.7	-1.7'	55.8'
18	89°19.8	N23°22.5	278°35.9	8.6'	S24°51.0	-1.8'	55.8'
19	104°19.7	22.5	293°03.5	8.7'	24°49.2	-1.9'	55.7'
20	119°19.5	22.4	307°31.2	8.7'	24°47.3	-2.1'	55.7'
21	134°19.4	· · 22.3	321°58.9	8.8'	24°45.2	-2.2'	55.7'
22	149°19.3	22.2	336°26.6	8.8'	24°43.0	-2.3'	55.7'
23	164°19.1	22.2	350°54.4	8.8'	24°40.7	-2.5'	55.7'
SD = 15.7'			SD = 15.3'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	179°19.0	N23°22.1	5°22.3	8.9'	S24°38.2	-2.6'	55.6'
1	194°18.9	22.0	19°50.2	8.9'	24°35.6	-2.7'	55.6'
2	209°18.8	21.9	34°18.1	9.0'	24°32.9	-2.9'	55.6'
3	224°18.6	· · 21.8	48°46.1	9.0'	24°30.1	-3.0'	55.6'
4	239°18.5	21.8	63°14.1	9.1'	24°27.1	-3.1'	55.6'
5	254°18.4	21.7	77°42.2	9.1'	24°24.0	-3.2'	55.6'
6	269°18.2	N23°21.6	92°10.4	9.2'	S24°20.7	-3.4'	55.5'
7	284°18.1	21.5	106°38.6	9.3'	24°17.4	-3.5'	55.5'
8	299°18.0	21.4	121°06.8	9.3'	24°13.9	-3.6'	55.5'
9	314°17.8	· · 21.4	135°35.1	9.4'	24°10.3	-3.7'	55.5'
10	329°17.7	21.3	150°03.5	9.4'	24°06.5	-3.9'	55.5'
11	344°17.6	21.2	164°31.9	9.5'	24°02.7	-4.0'	55.4'
12	359°17.4	N23°21.1	179°00.4	9.5'	S23°58.7	-4.1'	55.4'
13	14°17.3	21.0	193°28.9	9.6'	23°54.6	-4.2'	55.4'
14	29°17.2	20.9	207°57.5	9.7'	23°50.3	-4.3'	55.4'
15	44°17.1	· · 20.8	222°26.2	9.7'	23°46.0	-4.5'	55.4'
16	59°16.9	20.7	236°54.9	9.8'	23°41.5	-4.6'	55.3'
17	74°16.8	20.7	251°23.7	9.8'	23°36.9	-4.7'	55.3'
18	89°16.7	N23°20.6	265°52.5	9.9'	S23°32.2	-4.8'	55.3'
19	104°16.5	20.5	280°21.4	10.0'	23°27.4	-4.9'	55.3'
20	119°16.4	20.4	294°50.4	10.0'	23°22.5	-5.1'	55.3'
21	134°16.3	· · 20.3	309°19.4	10.1'	23°17.4	-5.2'	55.3'
22	149°16.1	20.2	323°48.5	10.2'	23°12.3	-5.3'	55.2'
23	164°16.0	20.1	338°17.7	10.2'	23°07.0	-5.4'	55.2'
SD = 15.7'			SD = 15.2'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	□	□	□	□	□	□
N 70°	□	□	□	□	□	□
68°	□	□	□	□	□	□
66°	□	□	□	□	□	□
64°	////	////	01:33	22:32	////	////
62°	////	////	02:11	21:54	////	////
60°	////	00:52	02:37	21:28	23:12	////
N 58°	////	01:42	02:58	21:07	22:23	////
56°	////	02:12	03:14	20:51	21:53	////
54°	00:48	02:34	03:29	20:36	21:31	23:17
52°	01:34	02:52	03:41	20:24	21:13	22:31
50°	02:02	03:07	03:52	20:13	20:58	22:03
45°	02:47	03:37	04:14	19:51	20:28	21:18
N 40°	03:18	03:59	04:32	19:33	20:06	20:47
35°	03:41	04:18	04:47	19:18	19:48	20:24
30°	04:00	04:33	05:00	19:05	19:32	20:06
20°	04:29	04:58	05:22	18:43	19:07	19:37
N 10°	04:51	05:18	05:41	18:24	18:47	19:14
0°	05:10	05:36	05:59	18:06	18:29	18:55
S 10°	05:27	05:54	06:16	17:49	18:12	18:38
20°	05:43	06:11	06:35	17:30	17:54	18:22
30°	06:00	06:30	06:56	17:09	17:36	18:05
35°	06:09	06:40	07:08	16:57	17:25	17:57
40°	06:18	06:52	07:23	16:43	17:13	17:47
45°	06:28	07:06	07:39	16:26	17:00	17:37
S 50°	06:40	07:22	08:00	16:05	16:44	17:25
52°	06:45	07:29	08:10	15:55	16:36	17:20
54°	06:51	07:37	08:21	15:44	16:28	17:14
56°	06:57	07:46	08:34	15:32	16:19	17:08
58°	07:04	07:57	08:48	15:17	16:09	17:01
S 60°	07:11	08:08	09:06	14:59	15:57	16:54

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	■	■	■	■	■	■
66°	■	■	23:24	■	■	■
64°	21:36	22:21	22:33	23:50	· ·	00:58
62°	20:46	21:35	22:01	00:04	00:40	01:43
60°	20:14	21:06	21:37	00:30	01:12	02:13
N 58°	19:51	20:43	21:18	00:51	01:35	02:35
56°	19:32	20:25	21:02	01:09	01:54	02:53
54°	19:16	20:09	20:49	01:23	02:10	03:09
52°	19:02	19:56	20:37	01:36	02:24	03:22
50°	18:50	19:44	20:26	01:47	02:36	03:34
45°	18:26	19:19	20:04	02:11	03:01	03:58
N 40°	18:06	19:00	19:46	02:30	03:21	04:17
35°	17:49	18:43	19:31	02:46	03:37	04:33
30°	17:35	18:29	19:18	02:59	03:52	04:47
20°	17:11	18:05	18:55	03:23	04:16	05:11
N 10°	16:50	17:44	18:36	03:43	04:37	05:31
0°	16:30	17:25	18:18	04:02	04:57	05:50
S 10°	16:11	17:05	17:59	04:21	05:16	06:09
20°	15:50	16:45	17:40	04:41	05:37	06:30
30°	15:26	16:20	17:17	05:05	06:02	06:53
35°	15:12	16:06	17:04	05:19	06:16	07:07
40°	14:56	15:50	16:49	05:35	06:32	07:22
45°	14:37	15:30	16:30	05:54	06:52	07:41
S 50°	14:12	15:06	16:07	06:18	07:17	08:05
52°	14:01	14:54	15:56	06:29	07:29	08:16
54°	13:47	14:40	15:44	06:42	07:42	08:28
56°	13:32	14:25	15:30	06:57	07:58	08:43
58°	13:14	14:06	15:13	07:15	08:16	09:00
S 60°	12:52	13:43	14:52	07:37	08:39	09:21

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time	12h		Mer. Pass.		
	00h	12h		Upper	Lower	12-14
	mm:ss	mm:ss	hh:mm	hh:mm	hh:mm	94-100%
24	02:19	02:25	12:02	22:44	10:16	
25	02:31	02:38	12:03	23:38	11:11	
26	02:44	02:50	12:03	—:—	12:04	

June 27, 28, 29 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn	
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	275°00.1	137°17.7	N18°14.3	112°27.5	N08°27.1	272°24.3	S00°15.1	95°29.4	N02°45.3
1	290°02.6	152°17.3	13.4	127°28.7	26.6	287°26.5	15.0	110°31.8	45.2
2	305°05.0	167°16.9	12.5	142°29.8	26.0	302°28.8	15.0	125°34.1	45.2
3	320°07.5	182°16.4	•• 11.6	157°31.0	•• 25.4	317°31.1	•• 14.9	140°36.5	•• 45.1
4	335°10.0	197°16.0	10.7	172°32.2	24.9	332°33.3	14.8	155°38.9	45.1
5	350°12.4	212°15.6	09.9	187°33.3	24.3	347°35.6	14.8	170°41.2	45.0
6	5°14.9	227°15.2	N18°09.0	202°34.5	N08°23.7	2°37.9	S00°14.7	185°43.6	N02°44.9
7	20°17.3	242°14.8	08.1	217°35.7	23.2	17°40.2	14.6	200°46.0	44.9
8	35°19.8	257°14.4	07.2	232°36.9	22.6	32°42.4	14.6	215°48.3	44.8
9	50°22.3	272°14.0	•• 06.3	247°38.0	•• 22.0	47°44.7	•• 14.5	230°50.7	•• 44.8
10	65°24.7	287°13.6	05.4	262°39.2	21.5	62°47.0	14.4	245°53.1	44.7
11	80°27.2	302°13.2	04.5	277°40.4	20.9	77°49.3	14.3	260°55.4	44.7
12	95°29.7	317°12.8	N18°03.6	292°41.5	N08°20.3	92°51.5	S00°14.3	275°57.8	N02°44.6
13	110°32.1	332°12.4	02.7	307°42.7	19.8	107°53.8	14.2	291°00.1	44.6
14	125°34.6	347°12.0	01.8	322°43.9	19.2	122°56.1	14.1	306°02.5	44.5
15	140°37.1	2°11.6	18°00.8	337°45.0	•• 18.6	137°58.3	•• 14.1	321°04.9	•• 44.5
16	155°39.5	17°11.2	17°59.9	352°46.2	18.0	153°00.6	14.0	336°07.2	44.4
17	170°42.0	32°10.8	59.0	7°47.4	17.5	168°02.9	13.9	351°09.6	44.4
18	185°44.4	47°10.4	N17°58.1	22°48.6	N08°16.9	183°05.2	S00°13.9	6°12.0	N02°44.3
19	200°46.9	62°10.0	57.2	37°49.7	16.3	198°07.4	13.8	21°14.3	44.3
20	215°49.4	77°09.6	56.3	52°50.9	15.8	213°09.7	13.7	36°16.7	44.2
21	230°51.8	92°09.2	•• 55.4	67°52.1	•• 15.2	228°12.0	•• 13.7	51°19.0	•• 44.1
22	245°54.3	107°08.8	54.5	82°53.2	14.6	243°14.3	13.6	66°21.4	44.1
23	260°56.8	122°08.4	53.6	97°54.4	14.1	258°16.5	13.5	81°23.8	44.0
Mer.pass. 05:39		ν -0.4' d -0.9' m -4.02		ν 1.2' d -0.6' m 1.32		ν 2.3' d -0.1' m -2.42		ν 2.4' d -0.1' m 0.95	

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	275°59.2	137°08.0	N17°52.7	112°55.6	N08°13.5	273°18.8	S00°13.4	96°26.1	N02°44.0
1	291°01.7	152°07.6	51.8	127°56.7	12.9	288°21.1	13.4	111°28.5	43.9
2	306°04.2	167°07.2	50.9	142°57.9	12.4	303°23.4	13.3	126°30.8	43.9
3	321°06.6	182°06.8	•• 49.9	157°59.1	•• 11.8	318°25.7	•• 13.2	141°33.2	•• 43.8
4	336°09.1	197°06.4	49.0	173°00.2	11.2	333°27.9	13.2	156°35.6	43.8
5	351°11.6	212°06.0	48.1	188°01.4	10.6	348°30.2	13.1	171°37.9	43.7
6	6°14.0	227°05.6	N17°47.2	203°02.6	N08°10.1	3°32.5	S00°13.0	186°40.3	N02°43.7
7	21°16.5	242°05.2	46.3	218°03.7	09.5	18°34.8	13.0	201°42.6	43.6
8	36°18.9	257°04.8	45.4	233°04.9	08.9	33°37.0	12.9	216°45.0	43.5
9	51°21.4	272°04.4	•• 44.5	248°06.1	•• 08.4	48°39.3	•• 12.8	231°47.4	•• 43.5
10	66°23.9	287°04.0	43.5	263°07.2	07.8	63°41.6	12.8	246°49.7	43.4
11	81°26.3	302°03.7	42.6	278°08.4	07.2	78°43.9	12.7	261°52.1	43.4
12	96°28.8	317°03.3	N17°41.7	293°09.6	N08°06.6	93°46.2	S00°12.6	276°54.4	N02°43.3
13	111°31.3	332°02.9	40.8	308°10.7	06.1	108°48.4	12.6	291°56.8	43.3
14	126°33.7	347°02.5	39.9	323°11.9	05.5	123°50.7	12.5	306°59.2	43.2
15	141°36.2	2°02.1	•• 38.9	338°13.1	•• 04.9	138°53.0	•• 12.4	322°01.5	•• 43.2
16	156°38.7	17°01.7	38.0	353°14.2	04.4	153°55.3	12.4	337°03.9	43.1
17	171°41.1	32°01.3	37.1	8°15.4	03.8	168°57.6	12.3	352°06.2	43.1
18	186°43.6	47°00.9	N17°36.2	23°16.6	N08°03.2	183°59.8	S00°12.2	7°08.6	N02°43.0
19	201°46.1	62°00.6	35.2	38°17.7	02.6	199°02.1	12.2	22°11.0	42.9
20	216°48.5	77°00.2	34.3	53°18.9	02.1	214°04.4	12.1	37°13.3	42.9
21	231°51.0	91°59.8	•• 33.4	68°20.1	•• 01.5	229°06.7	•• 12.0	52°15.7	•• 42.8
22	246°53.4	106°59.4	32.5	83°21.2	00.9	244°09.0	12.0	67°18.0	42.8
23	261°55.9	121°59.0	31.5	98°22.4	00.3	259°11.3	11.9	82°20.4	42.7
Mer.pass. 05:35		ν -0.4' d -0.9' m -4.03		ν 1.2' d -0.6' m 1.33		ν 2.3' d -0.1' m -2.43		ν 2.4' d -0.1' m 0.95	

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	276°58.4	136°58.7	N17°30.6	113°23.6	N07°59.8	274°13.5	S00°11.8	97°22.7	N02°42.7
1	292°00.8	151°58.3	29.7	128°24.7	59.2	289°15.8	11.8	112°25.1	42.6
2	307°03.3	166°57.9	28.8	143°25.9	58.6	304°18.1	11.7	127°27.5	42.6
3	322°05.8	181°57.5	•• 27.8	158°27.1	•• 58.1	319°20.4	•• 11.7	142°29.8	•• 42.5
4	337°08.2	196°57.1	26.9	173°28.2	57.5	334°22.7	11.6	157°32.2	42.4
5	352°10.7	211°56.8	26.0	188°29.4	56.9	349°25.0	11.5	172°34.5	42.4
6	7°13.2	226°56.4	N17°25.0	203°30.6	N07°56.3	4°27.2	S00°11.5	187°36.9	N02°42.3
7	22°15.6	241°56.0	24.1	218°31.7	55.8	19°29.5	11.4	202°39.2	42.3
8	37°18.1	256°55.6	23.2	233°32.9	55.2	34°31.8	11.3	217°41.6	42.2
9	52°20.6	271°55.3	•• 22.2	248°34.1	•• 54.6	49°34.1	•• 11.3	232°44.0	•• 42.2
10	67°23.0	286°54.9	21.3	263°35.2	54.0	64°36.4	11.2	247°46.3	42.1
11	82°25.5	301°54.5	20.4	278°36.4	53.5	79°38.7	11.1	262°48.7	42.1
12	97°27.9	316°54.2	N17°19.4	293°37.6	N07°52.9	94°41.0	S00°11.1	277°51.0	N02°42.0
13	112°30.4	331°53.8	18.5	308°38.7	52.3	109°43.3	11.0	292°53.4	41.9
14	127°32.9	346°53.4	17.5	323°39.9	51.7	124°45.5	10.9	307°55.7	41.9
15	142°35.3	1°53.0	•• 16.6	338°41.1	•• 51.2	139°47.8	•• 10.9	322°58.1	•• 41.8
16	157°37.8	16°52.7	15.7	353°42.2	50.6	154°50.1	10.8	338°00.4	41.8
17	172°40.3	31°52.3	14.7	8°43.4	50.0	169°52.4	10.8	353°02.8	41.7
18	187°42.7	46°51.9	N17°13.8	23°44.5	N07°49.4	184°54.7	S00°10.7	8°05.2	N02°41.7
19	202°45.2	61°51.6	12.8	38°45.7	48.9	199°57.0	10.6	23°07.5	41.6
20	217°47.7	76°51.2	11.9	53°46.9	48.3	214°59.3	10.6	38°09.9	41.5
21	232°50.1	91°50.8	•• 11.0	68°48.0	•• 47.7	230°01.6	•• 10.5	53°12.2	•• 41.5
22	247°52.6	106°50.5	10.0	83°49.2	47.1	245°03.8	10.4	68°14.6	41.4
23	262°55.0	121°50.1	09.1	98°50.4	46.6	260°06.1	10.4	83°16.9	41.4
Mer.pass. 05:31		ν -0.4' d -0.9' m -4.03		ν 1.2' d -0.6' m 1.33		ν 2.3' d -0.1' m -2.44		ν 2.4' d -0.1' m 0.96	

Stars		
	SHA	Dec
Alpheratz	357°45.7	29°08.9
Ankaa	353°17.7	-42°14.6
Schedar	349°43.2	56°35.5
Diphda	348°58.0	-17°55.5
Achernar	335°28.4	-57°10.6
Hamal	328°03.4	23°30.7
Polaris	319°06.1	89°18.4
Acamar	315°20.2	-40°15.5
Menkar	314°17.5	4°07.9
Mirfak	308°43.9	49°53.8
Aldebaran	290°52.2	16°31.8
Rigel	281°14.4	-8°11.3
Capella	280°38.1	46°00.4
Bellatrix	278°34.6	6°21.6
Elnath	278°15.7	28°36.9
Alnilam	275°48.9	-1°11.7
Betelgeuse	271°04.0	7°24.5
Canopus	263°57.7	-52°42.1
Sirius	258°36.0	-16°43.9
Adhara	255°14.6	-28°59.2
Procyon	245°02.3	5°11.8
Pollux	243°30.7	-28°00.0
Avior	234°19.6	-59°32.8
Suhail	222°54.4	-43°28.7
Miaplacidus	221°41.0	-69°45.9
Alphard	217°58.4	-8°42.4
Regulus	207°46.0	11°54.9
Dubhe	193°54.4	61°41.8
Denebola	182°35.9	14°30.8
Gienah	175°54.5	-17°36.2
Acrux	173°11.9	-63°09.8
Gacrux	172°03.4	-57°10.7
Alioth	166°22.3	55°54.3
Spica	158°33.4	-11°13.1
Alkaid	153°00.3	49°15.8
Hadar	148°50.8	-60°25.8
Menkent	148°10.0	-36°25.5
Arcturus	145°57.5	19°07.7
Rigel Kent.	139°54.4	-60°53.0
Kochab	137°18.9	74°06.9
Zuben'ubi	137°07.6	-16°05.2
Alphecca	126°12.5	26°40.8
Antares	112°28.6	-26°27.4
Atria	107°31.9	-69°02.9
Sabik	102°14.7	-15°44.3
Shaula	96°24.4	-37°06.7
Rasalhague	96°08.1	12°33.2
Eltanin	90°46.6	51°29.3
Kaus Aust.	83°46.2	-34°22.7
Vega	80°40.0	38°47.6
Nunki	76°00.6	-26°16.9
Altair	62°10.0	8°53.8
Peacock	53°22.0	-56°41.8
Deneb	49°32.6	45°19.0
Enif	33°49.0	9°55.5
Al Na'ir	27°46.0	-46°54.3
Fomalhaut	15°26.1	-29°33.7
Scheat	13°55.4	28°08.4
Markab	13°40.4	15°15.7

Jun 27 Sun	SHA	Mer.pass
Venus	222°17.6	14:51
Mars	197°27.4	16:29
Jupiter	357°24.2	05:50
Saturn	180°29.3	17:35

Jun 28 Mon	SHA	Mer.pass
Venus	221°08.7	14:52
Mars	196°56.3	16:27
Jupiter	357°19.6	05:46
Saturn	180°26.9	17:32

Jun 29 Tue	SHA	Mer.pass
Venus	220°00.3	14:52
Mars	196°25.2	16:25
Jupiter	357°15.2	05:42
Saturn	180°24.4	17:28

Horizontal parallax		
Venus:	0.1	
Mars:	0.1	

Sun							Moon						
h	GHA	Dec	GHA	ν	Dec	d	HP						
Sun													
0	179°15.9	N23°20.0	352°46.9	10.3'	S23°01.6	-5.5'	55.2'						
1	194°15.8	19.9	7°16.2	10.4'	22°56.1	-5.6'	55.2'						
2	209°15.6	19.8	21°45.6	10.4'	22°50.5	-5.7'	55.2'						
3	224°15.5	· · 19.7	36°15.0	10.5'	22°44.7	-5.8'	55.1'						
4	239°15.4	19.6	50°44.5	10.6'	22°38.9	-5.9'	55.1'						
5	254°15.2	19.5	65°14.1	10.6'	22°33.0	-6.0'	55.1'						
6	269°15.1	N23°19.4	79°43.8	10.7'	S22°26.9	-6.2'	55.1'						
7	284°15.0	19.3	94°13.5	10.8'	22°20.8	-6.3'	55.1'						
8	299°14.9	19.2	108°43.3	10.9'	22°14.5	-6.4'	55.1'						
9	314°14.7	· · 19.1	123°13.1	10.9'	22°08.1	-6.5'	55.0'						
10	329°14.6	19.0	137°43.0	11.0'	22°01.7	-6.6'	55.0'						
11	344°14.5	18.9	152°13.0	11.1'	21°55.1	-6.7'	55.0'						
12	359°14.3	N23°18.8	166°43.1	11.1'	S21°48.4	-6.8'	55.0'						
13	14°14.2	18.7	181°13.2	11.2'	21°41.7	-6.9'	55.0'						
14	29°14.1	18.6	195°43.5	11.3'	21°34.8	-7.0'	55.0'						
15	44°14.0	· · 18.5	210°13.7	11.4'	21°27.8	-7.1'	54.9'						
16	59°13.8	18.4	224°44.1	11.4'	21°20.7	-7.2'	54.9'						
17	74°13.7	18.3	239°14.5	11.5'	21°13.6	-7.3'	54.9'						
18	89°13.6	N23°18.2	253°45.0	11.6'	S21°06.3	-7.4'	54.9'						
19	104°13.4	18.1	268°15.6	11.6'	20°58.9	-7.5'	54.9'						
20	119°13.3	18.0	282°46.2	11.7'	20°51.5	-7.5'	54.9'						
21	134°13.2	· · 17.9	297°17.0	11.8'	20°43.9	-7.6'	54.8'						
22	149°13.1	17.8	311°47.7	11.9'	20°36.3	-7.7'	54.8'						
23	164°12.9	17.6	326°18.6	11.9'	20°28.6	-7.8'	54.8'						
SD = 15.7' d = -0.1'							SD = 15.1'						
Mon	GHA	Dec	GHA	ν	Dec	d	HP						
0	179°12.8	N23°17.5	340°49.5	12.0'	S20°20.8	-7.9'	54.8'						
1	194°12.7	17.4	355°20.6	12.1'	20°12.9	-8.0'	54.8'						
2	209°12.5	17.3	9°51.6	12.2'	20°04.9	-8.1'	54.8'						
3	224°12.4	· · 17.2	24°22.8	12.2'	19°56.8	-8.2'	54.7'						
4	239°12.3	17.1	38°54.0	12.3'	19°48.6	-8.3'	54.7'						
5	254°12.2	17.0	53°25.3	12.4'	19°40.3	-8.3'	54.7'						
6	269°12.0	N23°16.9	67°56.7	12.4'	S19°32.0	-8.4'	54.7'						
7	284°11.9	16.7	82°28.1	12.5'	19°23.6	-8.5'	54.7'						
8	299°11.8	16.6	96°59.7	12.6'	19°15.1	-8.6'	54.7'						
9	314°11.7	· · 16.5	111°31.3	12.7'	19°06.5	-8.7'	54.7'						
10	329°11.5	16.4	126°02.9	12.7'	18°57.8	-8.7'	54.6'						
11	344°11.4	16.3	140°34.6	12.8'	18°49.1	-8.8'	54.6'						
12	359°11.3	N23°16.1	155°06.5	12.9'	S18°40.2	-8.9'	54.6'						
13	14°11.1	16.0	169°38.3	12.9'	18°31.3	-9.0'	54.6'						
14	29°11.0	15.9	184°10.3	13.0'	18°22.4	-9.1'	54.6'						
15	44°10.9	· · 15.8	198°42.3	13.1'	18°13.3	-9.1'	54.6'						
16	59°10.8	15.7	213°14.4	13.2'	18°04.2	-9.2'	54.6'						
17	74°10.6	15.5	227°46.5	13.2'	17°55.0	-9.3'	54.5'						
18	89°10.5	N23°15.4	242°18.8	13.3'	S17°45.7	-9.4'	54.5'						
19	104°10.4	15.3	256°51.1	13.4'	17°36.3	-9.4'	54.5'						
20	119°10.3	15.2	271°23.4	13.4'	17°26.9	-9.5'	54.5'						
21	134°10.1	· · 15.0	285°55.8	13.5'	17°17.4	-9.6'	54.5'						
22	149°10.0	14.9	300°28.3	13.6'	17°07.9	-9.6'	54.5'						
23	164°09.9	14.8	315°00.9	13.6'	16°58.2	-9.7'	54.5'						
SD = 15.7' d = -0.1'							SD = 14.9'						
Tue	GHA	Dec	GHA	ν	Dec	d	HP						
0	179°09.8	N23°14.6	329°33.5	13.7'	S16°48.5	-9.8'	54.5'						
1	194°09.6	14.5	344°06.2	13.8'	16°38.8	-9.8'	54.4'						
2	209°09.5	14.4	358°39.0	13.8'	16°29.0	-9.9'	54.4'						
3	224°09.4	· · 14.3	13°11.8	13.9'	16°19.1	-10.0'	54.4'						
4	239°09.3	14.1	27°44.7	14.0'	16°09.1	-10.0'	54.4'						
5	254°09.1	14.0	42°17.7	14.0'	15°59.1	-10.1'	54.4'						
6	269°09.0	N23°13.9	56°50.7	14.1'	S15°49.0	-10.1'	54.4'						
7	284°08.9	13.7	71°23.8	14.2'	15°38.9	-10.2'	54.4'						
8	299°08.8	13.6	85°57.0	14.2'	15°28.7	-10.3'	54.4'						
9	314°08.6	· · 13.5	100°30.2	14.3'	15°18.4	-10.3'	54.4'						
10	329°08.5	13.3	115°03.5	14.3'	15°08.1	-10.4'	54.3'						
11	344°08.4	13.2	129°36.8	14.4'	14°57.7	-10.4'	54.3'						
12	359°08.3	N23°13.1	144°10.2	14.5'	S14°47.3	-10.5'	54.3'						
13	14°08.1	12.9	158°43.6	14.5'	14°36.8	-10.5'	54.3'						
14	29°08.0	12.8	173°17.2	14.6'	14°26.2	-10.6'	54.3'						
15	44°07.9	· · 12.6	187°50.7	14.6'	14°15.6	-10.7'	54.3'						
16	59°07.8	12.5	202°24.4	14.7'	14°05.0	-10.7'	54.3'						
17	74°07.6	12.4	216°58.1	14.7'	13°54.3	-10.8'	54.3'						
18	89°07.5	N23°12.2	231°31.8	14.8'	S13°43.5	-10.8'	54.3'						
19	104°07.4	12.1	246°05.6	14.9'	13°32.7	-10.9'	54.3'						
20	119°07.3	11.9	260°39.5	14.9'	13°21.8	-10.9'	54.2'						
21	134°07.1	· · 11.8	275°13.4	15.0'	13°10.9	-11.0'	54.2'						
22	149°07.0	11.6	289°47.3	15.0'	13°00.0	-11.0'	54.2'						
23	164°06.9	11.5	304°21.4	15.1'	12°49.0	-11.0'	54.2'						
SD = 15.7' d = -0.1'							SD = 14.9'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	□	□	□	□	□	□
N 70°	□	□	□	□	□	□
68°	□	□	□	□	□	□
66°	□	□	□	□	□	□
64°	////	////	01:36	22:29	////	////
62°	////	////	02:13	21:53	////	////
60°	////	00:56	02:39	21:27	23:08	////
N 58°	////	01:45	02:59	21:07	22:21	////
56°	////	02:14	03:16	20:50	21:52	////
54°	00:52	02:36	03:30	20:36	21:30	23:13
52°	01:36	02:54	03:42	20:24	21:12	22:29
50°	02:04	03:09	03:53	20:13	20:58	22:02
45°	02:49	03:38	04:16	19:51	20:28	21:18
N 40°	03:19	04:01	04:33	19:33	20:06	20:47
35°	03:42	04:19	04:48	19:18	19:48	20:24
30°	04:01	04:34	05:01	19:05	19:33	20:06
20°	04:30	04:59	05:23	18:43	19:08	19:37
N 10°	04:52	05:19	05:42	18:24	18:47	19:14
0°	05:11	05:37	06:00	18:07	18:29	18:56
S 10°	05:28	05:54	06:17	17:50	18:12	18:39
20°	05:44	06:11	06:35	17:31	17:55	18:23
30°	06:00	06:30	06:56	17:10	17:36	18:06
35°	06:09	06:41	07:09	16:58	17:26	17:58
40°	06:18	06:52	07:23	16:44	17:14	17:48
45°	06:29	07:06	07:39	16:27	17:01	17:38
S 50°	06:40	07:22	08:00	16:07	16:45	17:26
52°	06:45	07:29	08:10	15:57	16:38	17:21
54°	06:51	07:37	08:21	15:46	16:29	17:16
56°	06:57	07:46	08:33	15:33	16:20	17:09
58°	07:04	07:56	08:48	15:19	16:10	17:03
S 60°	07:11	08:08	09:05	15:01	15:59	16:55

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	■	■	00:36 23:41	■	■	03:53
N 70°	■	23:48	23:18	■	■	04:40
68°	23:46	23:17	23:01	■	03:05	05:09
66°	23:05	22:54	22:47	01:44	03:46	05:31
64°	22:36	22:36	22:35	02:34	04:13	05:49
62°	22:14	22:21	22:25	03:06	04:35	06:03
60°	21:56	22:08	22:16	03:29	04:52	06:15
N 58°	21:42	21:57	22:08	03:48	05:06	06:25
56°	21:29	21:48	22:02	04:03	05:18	06:34
54°	21:18	21:39	21:56	04:17	05:29	06:42
52°	21:08	21:31	21:50	04:28	05:38	06:49
50°	20:59	21:24	21:45	04:39	05:47	06:56
45°	20:40	21:10	21:34	05:00	06:05	07:09
N 40°	20:25	20:57	21:25	05:18	06:19	07:20
35°	20:12	20:47	21:18	05:32	06:31	07:30
30°	20:00	20:38	21:11	05:45	06:42	07:38
20°	19:41	20:22	20:59	06:06	07:00	07:52
N 10°	19:24	20:08	20:49	06:25	07:16	08:05
0°	19:08	19:55	20:39	06:42	07:31	08:16
S 10°	18:52	19:42	20:29	06:59	07:45	08:28
20°	18:34	19:27	20:19	07:18	08:01	08:40
30°	18:15	19:11	20:07	07:39	08:19	08:54
35°	18:03	19:02	20:00	07:51	08:29	09:02
40°	17:50	18:51	19:52	08:05	08:41	09:11
45°	17:34	18:38	19:42	08:22	08:54	09:21
S 50°	17:14	18:23	19:31	08:42	09:11	09:34
52°	17:05	18:15	19:26	08:52	09:19	09:40
54°	16:54	18:07	19:20	09:02	09:27	09:46
56°	16:42	17:58	19:13	09:15	09:37	09:54
58°	16:29	17:48	19:06	09:29	09:48	10:02
S 60°	16:12	17:36	18:58	09:46	10:01	10:11

June 30, 01, 02 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	277°57.5	136°49.8	N17°08.1	113°51.5	N07°46.0	275°08.4	S00°10.3	98°19.3	N02°41.3	Alpheratz	357°45.7	29°08.9
1	293°00.0	151°49.4	07.2	128°52.7	45.4	290°10.7	10.3	113°21.6	41.3	Ankaa	353°17.7	-42°14.6
2	308°02.4	166°49.0	06.2	143°53.8	44.8	305°13.0	10.2	128°24.0	41.2	Schedar	349°43.1	56°35.5
3	323°04.9	181°48.7	•• 05.3	158°55.0	•• 44.3	320°15.3	•• 10.1	143°26.3	•• 41.1	Diphda	348°58.0	-17°55.5
4	338°07.4	196°48.3	04.3	173°56.2	43.7	335°17.6	10.1	158°28.7	41.1	Achernar	335°28.4	-57°10.6
5	353°09.8	211°48.0	03.4	188°57.3	43.1	350°19.9	10.0	173°31.0	41.0	Hamal	328°03.4	23°30.7
6	8°12.3	226°47.6	N17°02.4	203°58.5	N07°42.5	5°22.2	S00°09.9	188°33.4	N02°41.0	Polaris	319°05.0	89°18.4
7	23°14.8	241°47.2	01.5	218°59.7	42.0	20°24.5	09.9	203°35.8	40.9	Acamar	315°20.2	-40°15.5
8	38°17.2	256°46.9	17°00.5	234°00.8	41.4	35°26.8	09.8	218°38.1	40.9	Menkar	314°17.5	4°07.9
9	53°19.7	271°46.5	16°59.6	249°02.0	•• 40.8	50°29.0	•• 09.8	233°40.5	•• 40.8	Mirfak	308°43.8	49°53.8
10	68°22.2	286°46.2	58.6	264°03.1	40.2	65°31.3	09.7	248°42.8	40.7	Aldebaran	290°52.2	16°31.8
11	83°24.6	301°45.8	57.7	279°04.3	39.7	80°33.6	09.6	263°45.2	40.7	Rigel	281°14.4	-8°11.3
12	98°27.1	316°45.5	N16°56.7	294°05.5	N07°39.1	95°35.9	S00°09.6	278°47.5	N02°40.6	Capella	280°38.1	46°00.4
13	113°29.5	331°45.1	55.8	309°06.6	38.5	110°38.2	09.5	293°49.9	40.6	Bellatrix	278°34.6	6°21.6
14	128°32.0	346°44.8	54.8	324°07.8	37.9	125°40.5	09.5	308°52.2	40.5	Elnath	278°15.7	28°36.9
15	143°34.5	1°44.4	•• 53.9	339°09.0	•• 37.3	140°42.8	•• 09.4	323°54.6	•• 40.5	Alnilam	275°48.9	-1°11.7
16	158°36.9	16°44.0	52.9	354°10.1	36.8	155°45.1	09.3	338°56.9	40.4	Betelgeuse	271°03.9	7°24.5
17	173°39.4	31°43.7	52.0	9°11.3	36.2	170°47.4	09.3	353°59.3	40.3	Canopus	263°57.7	-52°42.1
18	188°41.9	46°43.3	N16°51.0	24°12.4	N07°35.6	185°49.7	S00°09.2	9°01.6	N02°40.3	Sirius	258°36.0	-16°43.8
19	203°44.3	61°43.0	50.0	39°13.6	35.0	200°52.0	09.2	24°04.0	40.2	Adhara	255°14.6	-28°59.2
20	218°46.8	76°42.6	49.1	54°14.8	34.5	215°54.3	09.1	39°06.3	40.2	Procyon	245°02.3	5°11.8
21	233°49.3	91°42.3	•• 48.1	69°15.9	•• 33.9	230°56.6	•• 09.0	54°08.7	•• 40.1	Pollux	243°30.7	28°00.0
22	248°51.7	106°42.0	47.2	84°17.1	33.3	245°58.9	09.0	69°11.0	40.0	Avior	234°19.6	-59°32.8
23	263°54.2	121°41.6	46.2	99°18.2	32.7	261°01.2	08.9	84°13.4	40.0	Suhail	222°54.4	-43°28.7
Mer.pass. 05:27		ν -0.4' d -0.9' m -4.04		ν 1.2' d -0.6' m 1.34		ν 2.3' d -0.1' m -2.45		ν 2.4' d -0.1' m 0.96				

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	278°56.7	136°41.3	N16°45.2	114°19.4	N07°32.1	276°03.5	S00°08.9	99°15.7	N02°39.9	Denebola	182°35.9	14°30.8
1	293°59.1	151°40.9	44.3	129°20.6	31.6	291°05.8	08.8	114°18.1	39.9	Gienah	175°54.6	-17°36.2
2	309°01.6	166°40.6	43.3	144°21.7	31.0	306°08.1	08.7	129°20.4	39.8	Acrux	173°11.9	-63°09.8
3	324°04.0	181°40.2	•• 42.4	159°22.9	•• 30.4	321°10.4	•• 08.7	144°22.8	•• 39.8	Gacrux	172°03.4	-57°10.7
4	339°06.5	196°39.9	41.4	174°24.0	29.8	336°12.6	08.6	159°25.1	39.7	Alioth	166°22.4	55°54.4
5	354°09.0	211°39.5	40.4	189°25.2	29.3	351°14.9	08.6	174°27.5	39.6	Spica	158°33.4	-11°13.1
6	9°11.4	226°39.2	N16°39.5	204°26.4	N07°28.7	6°17.2	S00°08.5	189°29.8	N02°39.6	Alkaid	153°00.3	49°15.8
7	24°13.9	241°38.9	38.5	219°27.5	28.1	21°19.5	08.4	204°32.2	39.5	Hadar	148°50.8	-60°25.8
8	39°16.4	256°38.5	37.5	234°28.7	27.5	36°21.8	08.4	219°34.5	39.5	Menkent	148°10.0	-36°25.5
9	54°18.7	271°38.2	•• 36.6	249°29.8	•• 26.9	51°24.1	•• 08.3	234°36.9	•• 39.4	Arcturus	145°57.5	19°07.7
10	69°21.3	286°37.8	35.6	264°31.0	26.4	66°26.4	08.3	249°39.2	39.3	Rigil Kent.	139°54.4	-60°53.0
11	84°23.8	301°37.5	34.6	279°32.2	25.8	81°28.7	08.2	264°41.6	39.3	Kochab	137°19.0	74°06.9
12	99°26.2	316°37.2	N16°33.7	294°33.3	N07°25.2	96°31.0	S00°08.2	279°43.9	N02°39.2	Zuben'ubi	137°07.6	-16°05.2
13	114°28.7	331°36.8	32.7	309°34.5	24.6	111°33.3	08.1	294°46.3	39.2	Alphecca	126°12.5	26°40.8
14	129°31.1	346°36.5	31.7	324°35.6	24.0	126°35.6	08.0	309°48.6	39.1	Antares	112°28.6	-26°27.4
15	144°33.6	1°36.1	•• 30.7	339°36.8	•• 23.5	141°37.9	•• 08.0	324°51.0	•• 39.0	Atria	107°31.9	-69°02.9
16	159°36.1	16°35.8	29.8	354°38.0	22.9	156°40.2	07.9	339°53.3	39.0	Sabik	102°14.7	-15°44.3
17	174°38.5	31°35.5	28.8	9°39.1	22.3	171°42.5	07.9	354°55.7	38.9	Shaula	96°24.4	-37°06.7
18	189°41.0	46°35.1	N16°27.8	24°40.3	N07°21.7	186°44.8	S00°07.8	9°58.0	N02°38.9	Rasalhague	96°08.1	12°33.2
19	204°43.5	61°34.8	26.9	39°41.4	21.1	201°47.1	07.7	25°00.4	38.8	Eltanin	90°46.6	51°29.3
20	219°45.9	76°34.5	25.9	54°42.6	20.6	216°49.4	07.7	40°02.7	38.7	Kaus Aust.	83°46.2	-34°22.7
21	234°48.4	91°34.2	•• 24.9	69°43.8	•• 20.0	231°51.7	•• 07.6	55°05.1	•• 38.7	Vega	80°40.0	38°47.6
22	249°50.9	106°33.8	23.9	84°44.9	19.4	246°54.1	07.6	70°07.4	38.6	Nunki	76°00.6	-26°16.9
23	264°53.3	121°33.5	22.9	99°46.1	18.8	261°56.4	07.5	85°09.8	38.6	Altair	62°10.0	8°53.8
Mer.pass. 05:23		ν -0.3' d -1.0' m -4.04		ν 1.2' d -0.6' m 1.35		ν 2.3' d -0.1' m -2.45		ν 2.4' d -0.1' m 0.96		Peacock	53°22.0	-56°41.8

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	279°55.8	136°33.2	N16°22.0	114°47.2	N07°18.2	276°58.7	S00°07.5	100°12.1	N02°38.5	Jun 30 Wed				
1	294°58.3	151°32.8	21.0	129°48.4	17.7	292°01.0	07.4	115°14.5	38.4	Venus	218°52.2	Mer.pass 14:53		
2	310°00.7	166°32.5	20.0	144°49.5	17.1	307°03.3	07.4	130°16.8	38.4	Mars	195°54.0	16:23		
3	325°03.2	181°32.2	•• 19.0	159°50.7	•• 16.5	322°05.6	•• 07.3	145°19.1	•• 38.3	Jupiter	357°10.9	05:39		
4	340°05.6	196°31.8	18.1	174°51.9	15.9	337°07.9	07.2	160°21.5	38.3	Saturn	180°21.8	17:24		
5	355°08.1	211°31.5	17.1	189°53.0	15.3	352°10.2	07.2	175°23.8	38.2					
6	10°10.6	226°31.2	N16°16.1	204°54.2	N07°14.7	7°12.5	S00°07.1	190°26.2	N02°38.1	Jul 01 Thu				
7	25°13.0	241°30.9	15.1	219°55.3	14.2	22°14.8	07.1	205°28.5	38.1	Venus	217°44.6	Mer.pass 14:54		
8	40°15.5	256°30.5	14.1	234°56.5	13.6	37°17.1	07.0	220°30.9	38.0	Mars	195°22.8	16:21		
9	55°18.0	271°30.2	•• 13.1	249°57.6	•• 13.0	52°19.4	•• 07.0	235°33.2	•• 38.0	Jupiter	357°06.8	05:35		
10	70°20.4	286°29.9	12.2	264°58.8	12.4	67°21.7	06.9	250°35.6	37.9	Saturn	180°19.1	17:20		
11	85°22.9	301°29.6	11.2	280°00.0	11.8	82°24.0	06.9	265°37.9	37.8					
12	100°25.4	316°29.3	N16°10.2	295°01.1	N07°11.3	97°26.3	S00°06.8	280°40.3	N02°37.8	Jul 02 Fri				
13	115°27.8	331°28.9	09.2	310°02.3	10.7	112°28.6	06.7	295°42.6	37.7	Venus	216°37.4	Mer.pass 14:54		
14	130°30.3	346°28.6	08.2	325°03.4	10.1	127°30.9	06.7	310°45.0	37.7	Mars	194°51.4	16:20		
15	145°32.8	1°28.3	•• 07.2	340°04.6	•• 09.5	142°33.2	•• 06.6	325°47.3	•• 37.6	Jupiter	357°02.9	05:31		
16	160°35.2	16°28.0	06.2	355°05.7	08.9	157°35.5	06.6	340°49.6	37.5	Saturn	180°16.3	17:16		
17	175°37.7	31°27.7	05.3	10°06.9	08.3	172°37.9	06.5	355°52.0	37.5					
18	190°40.1	46°27.3	N16°04.3	25°08.1	N07°07.8	187°40.2	S00°06.5	10°54.3	N02°37.4	Horizontal parallax				
19	205°42.6	61°27.0	03.3	40°09.2	07.2	202°42.5	06.4	25°56.7	37.4	Venus:	0.1			
20	220°45.1	76°26.7	02.3	55°10.4	06.6	217°44.8	06.4	40°59.0	37.3	Mars:	0.1			
21	235°47.5	91°26.4	•• 01.3	70°11.5	•• 06.0	232°47.1	•• 06.3	56°01.4	•• 37.2					
22	250°50.0	106°26.1	16°00.3	85°12.7	05.4	247°49.4	06.3	71°03.7	37.2					
23	265°52.5	121°25.8	15°59.3	100°13.8	04.8	262°51.7	06.2	86°06.1	37.1					
Mer.pass. 05:19		ν -0.3' d -1.0' m -4.05		ν 1.2' d -0.6' m 1.35		ν 2.3' d -0.1' m -2.46		ν 2.3' d -0.1' m 0.96						

h Sun Moon						
Wed	GHA	Dec	GHA	ν	Dec	d HP
0	179°06.8	N23°11.4	318°55.4	15.1'	S12°37.9	-11.1' 54.2'
1	194°06.6	11.2	333°29.5	15.2'	12°26.8	-11.1' 54.2'
2	209°06.5	11.1	348°03.7	15.2'	12°15.7	-11.2' 54.2'
3	224°06.4	•• 10.9	2°37.9	15.3'	12°04.5	-11.2' 54.2'
4	239°06.3	10.8	17°12.2	15.3'	11°53.3	-11.3' 54.2'
5	254°06.1	10.6	31°46.5	15.4'	11°42.0	-11.3' 54.2'
6	269°06.0	N23°10.5	46°20.9	15.4'	S11°30.7	-11.4' 54.2'
7	284°05.9	10.3	60°55.3	15.5'	11°19.4	-11.4' 54.2'
8	299°05.8	10.2	75°29.8	15.5'	11°08.0	-11.4' 54.2'
9	314°05.7	•• 10.0	90°04.3	15.6'	10°56.5	-11.5' 54.2'
10	329°05.5	09.9	104°38.8	15.6'	10°45.1	-11.5' 54.1'
11	344°05.4	09.7	119°13.4	15.6'	10°33.5	-11.5' 54.1'
12	359°05.3	N23°09.6	133°48.1	15.7'	S10°22.0	-11.6' 54.1'
13	14°05.2	09.4	148°22.7	15.7'	10°10.4	-11.6' 54.1'
14	29°05.0	09.2	162°57.5	15.8'	09°58.8	-11.7' 54.1'
15	44°04.9	•• 09.1	177°32.2	15.8'	09°47.1	-11.7' 54.1'
16	59°04.8	08.9	192°07.0	15.8'	09°35.4	-11.7' 54.1'
17	74°04.7	08.8	206°41.9	15.9'	09°23.7	-11.8' 54.1'
18	89°04.6	N23°08.6	221°16.7	15.9'	S09°12.0	-11.8' 54.1'
19	104°04.4	08.5	235°51.7	15.9'	09°00.2	-11.8' 54.1'
20	119°04.3	08.3	250°26.6	16.0'	08°48.4	-11.9' 54.1'
21	134°04.2	•• 08.1	265°01.6	16.0'	08°36.5	-11.9' 54.1'
22	149°04.1	08.0	279°36.6	16.1'	08°24.6	-11.9' 54.1'
23	164°03.9	07.8	294°11.7	16.1'	08°12.7	-11.9' 54.1'
SD = 15.7' $d = -0.1'$ SD = 14.8'						
Thu	GHA	Dec	GHA	ν	Dec	d HP
0	179°03.8	N23°07.7	308°46.8	16.1'	S08°00.8	-12.0' 54.1'
1	194°03.7	07.5	323°21.9	16.1'	07°48.8	-12.0' 54.1'
2	209°03.6	07.3	337°57.0	16.2'	07°36.8	-12.0' 54.1'
3	224°03.5	•• 07.2	352°32.2	16.2'	07°24.8	-12.0' 54.1'
4	239°03.3	07.0	7°07.4	16.2'	07°12.8	-12.1' 54.1'
5	254°03.2	06.8	21°42.6	16.3'	07°00.7	-12.1' 54.1'
6	269°03.1	N23°06.7	36°17.9	16.3'	S06°48.6	-12.1' 54.1'
7	284°03.0	06.5	50°53.2	16.3'	06°36.5	-12.1' 54.1'
8	299°02.9	06.3	65°28.5	16.3'	06°24.3	-12.2' 54.1'
9	314°02.7	•• 06.2	80°03.8	16.4'	06°12.2	-12.2' 54.1'
10	329°02.6	06.0	94°39.2	16.4'	06°00.0	-12.2' 54.1'
11	344°02.5	05.8	109°14.6	16.4'	05°47.8	-12.2' 54.1'
12	359°02.4	N23°05.7	123°50.0	16.4'	S05°35.5	-12.2' 54.1'
13	14°02.3	05.5	138°25.4	16.4'	05°23.3	-12.3' 54.1'
14	29°02.1	05.3	153°00.8	16.5'	05°11.0	-12.3' 54.1'
15	44°02.0	•• 05.1	167°36.3	16.5'	04°58.7	-12.3' 54.1'
16	59°01.9	05.0	182°11.8	16.5'	04°46.4	-12.3' 54.1'
17	74°01.8	04.8	196°47.3	16.5'	04°34.1	-12.3' 54.1'
18	89°01.7	N23°04.6	211°22.8	16.5'	S04°21.8	-12.4' 54.1'
19	104°01.5	04.4	225°58.3	16.5'	04°09.4	-12.4' 54.1'
20	119°01.4	04.3	240°33.9	16.6'	03°57.0	-12.4' 54.1'
21	134°01.3	•• 04.1	255°09.4	16.6'	03°44.7	-12.4' 54.1'
22	149°01.2	03.9	269°45.0	16.6'	03°32.3	-12.4' 54.1'
23	164°01.1	03.7	284°20.6	16.6'	03°19.8	-12.4' 54.1'
SD = 15.7' $d = -0.2'$ SD = 14.8'						
Fri	GHA	Dec	GHA	ν	Dec	d HP
0	179°00.9	N23°03.6	298°56.2	16.6'	S03°07.4	-12.4' 54.1'
1	194°00.8	03.4	313°31.8	16.6'	02°55.0	-12.4' 54.1'
2	209°00.7	03.2	328°07.4	16.6'	02°42.5	-12.5' 54.1'
3	224°00.6	•• 03.0	342°43.0	16.6'	02°30.1	-12.5' 54.1'
4	239°00.5	02.8	357°18.6	16.6'	02°17.6	-12.5' 54.1'
5	254°00.4	02.6	11°54.2	16.6'	02°05.1	-12.5' 54.1'
6	269°00.2	N23°02.5	26°29.8	16.6'	S01°52.6	-12.5' 54.1'
7	284°00.1	02.3	41°05.5	16.6'	01°40.1	-12.5' 54.1'
8	299°00.0	02.1	55°41.1	16.6'	01°27.6	-12.5' 54.1'
9	313°59.9	•• 01.9	70°16.7	16.6'	01°15.1	-12.5' 54.1'
10	328°59.8	01.7	84°52.4	16.6'	01°02.6	-12.5' 54.1'
11	343°59.6	01.5	99°28.0	16.6'	00°50.1	-12.5' 54.2'
12	358°59.5	N23°01.4	114°03.6	16.6'	S00°37.5	-12.5' 54.2'
13	13°59.4	01.2	128°39.3	16.6'	00°25.0	-12.5' 54.2'
14	28°59.3	01.0	143°14.9	16.6'	S00°12.5	-12.5' 54.2'
15	43°59.2	•• 00.8	157°50.5	16.6'	N00°00.1	12.5' 54.2'
16	58°59.1	00.6	172°26.1	16.6'	00°12.6	12.5' 54.2'
17	73°58.9	00.4	187°01.7	16.6'	00°25.2	12.5' 54.2'
18	88°58.8	N23°00.2	201°37.3	16.6'	N00°37.7	12.6' 54.2'
19	103°58.7	23°00.0	216°12.9	16.6'	00°50.3	12.6' 54.2'
20	118°58.6	22°59.8	230°48.5	16.6'	01°02.8	12.6' 54.2'
21	133°58.5	•• 59.6	245°24.0	16.6'	01°15.4	12.5' 54.2'
22	148°58.4	59.4	259°59.6	16.5'	01°27.9	12.5' 54.2'
23	163°58.2	59.2	274°35.1	16.5'	01°40.5	12.5' 54.3'
SD = 15.7' $d = -0.2'$ SD = 14.8'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	☐	☐	☐	☐	☐	☐
66°	////	////	00:18	23:41	////	////
64°	////	////	01:41	22:25	////	////
62°	////	////	02:17	21:50	////	////
60°	////	01:03	02:42	21:25	23:03	////
N 58°	////	01:48	03:02	21:05	22:18	////
56°	////	02:17	03:18	20:49	21:50	////
54°	00:58	02:38	03:32	20:35	21:29	23:08
52°	01:40	02:56	03:44	20:23	21:11	22:27
50°	02:06	03:11	03:55	20:13	20:57	22:01
45°	02:51	03:40	04:17	19:50	20:28	21:17
N 40°	03:21	04:02	04:35	19:33	20:05	20:47
35°	03:43	04:20	04:50	19:18	19:48	20:24
30°	04:02	04:35	05:02	19:05	19:33	20:06
20°	04:30	05:00	05:24	18:43	19:08	19:37
N 10°	04:53	05:20	05:43	18:25	18:48	19:15
0°	05:12	05:38	06:00	18:08	18:30	18:56
S 10°	05:28	05:55	06:17	17:50	18:13	18:39
20°	05:44	06:12	06:36	17:32	17:56	18:23
30°	06:00	06:30	06:57	17:11	17:38	18:07
35°	06:09	06:41	07:09	16:59	17:27	17:59
40°	06:18	06:52	07:23	16:45	17:16	17:50
45°	06:28	07:06	07:39	16:29	17:02	17:39
S 50°	06:40	07:21	08:00	16:08	16:47	17:28
52°	06:45	07:29	08:09	15:59	16:39	17:23
54°	06:51	07:37	08:20	15:48	16:31	17:17
56°	06:57	07:46	08:33	15:35	16:22	17:11
58°	07:03	07:56	08:47	15:21	16:12	17:05
S 60°	07:11	08:07	09:04	15:04	16:01	16:57

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	23:10	22:45	22:23	06:19	08:17	10:07
N 70°	22:58	22:41	22:25	06:40	08:27	10:09
68°	22:48	22:37	22:26	06:56	08:35	10:10
66°	22:40	22:34	22:28	07:09	08:41	10:11
64°	22:33	22:31	22:29	07:20	08:47	10:12
62°	22:27	22:28	22:30	07:29	08:51	10:13
60°	22:22	22:26	22:31	07:36	08:55	10:13
N 58°	22:17	22:24	22:31	07:43	08:59	10:14
56°	22:13	22:23	22:32	07:49	09:02	10:14
54°	22:09	22:21	22:33	07:54	09:05	10:15
52°	22:06	22:20	22:33	07:59	09:08	10:15
50°	22:03	22:19	22:34	08:03	09:10	10:16
45°	21:56	22:16	22:35	08:13	09:15	10:17
N 40°	21:50	22:14	22:36	08:20	09:19	10:17
35°	21:46	22:12	22:37	08:27	09:23	10:18
30°	21:41	22:10	22:38	08:33	09:26	10:18
20°	21:34	22:07	22:39	08:43	09:31	10:19
N 10°	21:27	22:04	22:40	08:51	09:36	10:20
0°	21:21	22:01	22:42	08:59	09:40	10:21
S 10°	21:15	21:59	22:43	09:07	09:45	10:21
20°	21:08	21:56	22:44	09:16	09:50	10:22
30°	21:00	21:53	22:45	09:25	09:55	10:23
35°	20:56	21:51	22:46	09:31	09:58	10:23
40°	20:51	21:49	22:47	09:37	10:01	10:24
45°	20:45	21:47	22:48	09:44	10:05	10:25
S 50°	20:38	21:44	22:50	09:53	10:10	10:25
52°	20:35	21:43	22:50	09:57	10:12	10:26
54°	20:31	21:41	22:51	10:01	10:14	10:26
56°	20:27	21:40	22:52	10:06	10:17	10:26
58°	20:23	21:38	22:53	10:12	10:20	10:27
S 60°	20:18	21:36	22:54	10:18	10:23	10:27

Day	Sun		Mer. Pass	Moon		Age 18-20 89-74%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
30	03:33	03:39	12:04	02:49	15:10	
01	03:45	03:50	12:04	03:31	15:51	
02	03:56	04:02	12:04	04:11	16:31	

July 03, 04, 05 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars				
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec			
0	280°54.9	136°25.5	N15°58.3	115°15.0	N07°04.3	277°54.0	S00°06.1	101°08.4	N02°37.1	Alpheratz	357°45.7	29°08.9		
1	295°57.4	151°25.1	57.3	130°16.1	03.7	292°56.3	06.1	116°10.7	37.0	Ankaa	353°17.7	-42°14.6		
2	310°59.9	166°24.8	56.3	145°17.3	03.1	307°58.6	06.0	131°13.1	36.9	Schedar	349°43.1	56°35.5		
3	326°02.3	181°24.5	•• 55.3	160°18.5	•• 02.5	323°01.0	•• 06.0	146°15.4	•• 36.9	Diphda	348°58.0	-17°55.5		
4	341°04.8	196°24.2	54.3	175°19.6	01.9	338°03.3	05.9	161°17.8	36.8	Achernar	335°28.4	-57°10.6		
5	356°07.2	211°23.9	53.3	190°20.8	01.3	353°05.6	05.9	176°20.1	36.7	Hamal	328°03.3	23°30.7		
6	11°09.7	226°23.6	N15°52.4	205°21.9	N07°00.7	8°07.9	S00°05.8	191°22.5	N02°36.7	Polaris	319°03.8	89°18.4		
7	26°12.2	241°23.3	51.4	220°23.1	07°00.2	23°10.2	05.8	206°24.8	36.6	Acamar	315°20.1	-40°15.5		
8	41°14.6	256°23.0	50.4	235°24.2	06°59.6	38°12.5	05.7	221°27.1	36.6	Menkar	314°17.5	4°07.9		
9	56°17.1	271°22.7	•• 49.4	250°25.4	•• 59.0	53°14.8	•• 05.7	236°29.5	•• 36.5	Mirfak	308°43.8	49°53.8		
10	71°19.6	286°22.4	48.4	265°26.5	58.4	68°17.1	05.6	251°31.8	36.4	Aldebaran	290°52.1	16°31.8		
11	86°22.0	301°22.1	47.4	280°27.7	57.8	83°19.4	05.6	266°34.2	36.4	Rigel	281°14.4	-8°11.3		
12	101°24.5	316°21.8	N15°46.4	295°28.8	N06°57.2	98°21.8	S00°05.5	281°36.5	N02°36.3	Capella	280°38.1	46°00.4		
13	116°27.0	331°21.5	45.4	310°30.0	56.7	113°24.1	05.5	296°38.8	36.3	Bellatrix	278°34.6	6°21.6		
14	131°29.4	346°21.2	44.4	325°31.2	56.1	128°26.4	05.4	311°41.2	36.2	Elnath	278°15.7	28°36.9		
15	146°31.9	1°20.8	•• 43.4	340°32.3	•• 55.5	143°28.7	•• 05.4	326°43.5	•• 36.1	Alnilam	275°48.9	-1°11.7		
16	161°34.4	16°20.5	42.4	355°33.5	54.9	158°31.0	05.3	341°45.9	36.1	Betelgeuse	271°03.9	7°24.5		
17	176°36.8	31°20.2	41.3	10°34.6	54.3	173°33.3	05.3	356°48.2	36.0	Canopus	263°57.7	-52°42.1		
18	191°39.3	46°19.9	N15°40.3	25°35.8	N06°53.7	188°35.6	S00°05.2	11°50.6	N02°35.9	Sirius	258°36.0	-16°43.8		
19	206°41.7	61°19.6	39.3	40°36.9	53.1	203°38.0	05.2	26°52.9	35.9	Adhara	255°14.6	-28°59.2		
20	221°44.2	76°19.3	38.3	55°38.1	52.6	218°40.3	05.1	41°55.2	35.8	Procyon	245°02.3	5°11.9		
21	236°46.7	91°19.0	•• 37.3	70°39.2	•• 52.0	233°42.6	•• 05.1	56°57.6	•• 35.8	Pollux	243°30.7	28°00.0		
22	251°49.1	106°18.8	36.3	85°40.4	51.4	248°44.9	05.0	71°59.9	35.7	Avior	234°19.6	-59°32.8		
23	266°51.6	121°18.5	35.3	100°41.5	50.8	263°47.2	05.0	87°02.3	35.6	Suhail	222°54.5	-43°28.7		
Mer.pass. 05:15		ν -0.3' d -1.0' m -4.05		ν 1.2' d -0.6' m 1.36		ν 2.3' d -0.1' m -2.47		ν 2.3' d -0.1' m 0.97		Miaplacidus	221°41.1	-69°45.9		
												Alphard	217°58.5	-8°42.4
												Regulus	207°46.0	11°54.9
												Dubhe	193°54.4	61°41.8
												Denebola	182°35.9	14°30.8
												Gienah	175°54.6	-17°36.2
												Acrux	173°12.0	-63°09.8
												Gacrux	172°03.5	-57°10.7
												Alioth	166°22.4	55°54.4
												Spica	158°33.5	-11°13.1
												Alkaid	153°00.3	49°15.8
												Hadar	148°50.8	-60°25.8
												Menkent	148°10.0	-36°25.5
												Arcturus	145°57.5	19°07.7
												Rigel Kent.	139°54.4	-60°53.0
												Kochab	137°19.0	74°06.9
												Zuben'ubi	137°07.6	-16°05.2
												Alphecca	126°12.5	26°40.8
												Antares	112°28.6	-26°27.4
												Atria	107°31.9	-69°02.9
												Sabik	102°14.7	-15°44.3
												Shaula	96°24.4	-37°06.7
												Rasalhague	96°08.1	12°33.2
												Eltanin	90°46.6	51°29.3
												Kaus Aust.	83°46.2	-34°22.7
												Vega	80°40.0	38°47.7
												Nunki	76°00.6	-26°16.9
												Altair	62°10.0	8°53.8
												Peacock	53°21.9	-56°41.8
												Deneb	49°32.6	45°19.1
												Enif	33°49.0	9°55.5
												Al Na'ir	27°46.0	-46°54.3
												Fomalhaut	15°26.1	-29°33.7
												Scheat	13°55.3	28°08.4
												Markab	13°40.3	15°15.8
												Jul 03 Sat	SHA	Mer.pass
												Venus	215°30.5	14:55
												Mars	194°20.1	16:18
												Jupiter	356°59.1	05:28
												Saturn	180°13.5	17:13
												Jul 04 Sun	SHA	Mer.pass
												Venus	214°24.1	14:55
												Mars	193°48.6	16:16
												Jupiter	356°55.5	05:24
												Saturn	180°10.5	17:09
												Jul 05 Mon	SHA	Mer.pass
												Venus	213°18.0	14:56
												Mars	193°17.1	16:14
												Jupiter	356°52.0	05:20
												Saturn	180°07.5	17:05
												Horizontal parallax		
												Venus:	0.1	
												Mars:	0.1	
Mer.pass. 05:08		ν -0.3' d -1.0' m -4.06		ν 1.1' d -0.6' m 1.37		ν 2.3' d -0.0' m -2.48		ν 2.3' d -0.1' m 0.97						

h	Sun			Moon			
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	178°58.1	N22°59.0	289°10.7	16.5'	N01°53.0	12.5'	54.3'
1	193°58.0	58.9	303°46.2	16.5'	02°05.6	12.5'	54.3'
2	208°57.9	58.7	318°21.7	16.5'	02°18.1	12.5'	54.3'
3	223°57.8	· · 58.5	332°57.1	16.5'	02°30.6	12.5'	54.3'
4	238°57.7	58.3	347°32.6	16.4'	02°43.2	12.5'	54.3'
5	253°57.6	58.1	2°08.0	16.4'	02°55.7	12.5'	54.3'
6	268°57.4	N22°57.9	16°43.4	16.4'	N03°08.2	12.5'	54.3'
7	283°57.3	57.7	31°18.8	16.4'	03°20.7	12.5'	54.3'
8	298°57.2	57.5	45°54.2	16.3'	03°33.2	12.5'	54.4'
9	313°57.1	· · 57.3	60°29.6	16.3'	03°45.7	12.5'	54.4'
10	328°57.0	57.0	75°04.9	16.3'	03°58.2	12.5'	54.4'
11	343°56.9	56.8	89°40.2	16.3'	04°10.7	12.5'	54.4'
12	358°56.7	N22°56.6	104°15.5	16.2'	N04°23.2	12.5'	54.4'
13	13°56.6	56.4	118°50.7	16.2'	04°35.6	12.4'	54.4'
14	28°56.5	56.2	133°25.9	16.2'	04°48.1	12.4'	54.4'
15	43°56.4	· · 56.0	148°01.1	16.2'	05°00.5	12.4'	54.5'
16	58°56.3	55.8	162°36.3	16.1'	05°12.9	12.4'	54.5'
17	73°56.2	55.6	177°11.4	16.1'	05°25.3	12.4'	54.5'
18	88°56.1	N22°55.4	191°46.5	16.1'	N05°37.7	12.4'	54.5'
19	103°55.9	55.2	206°21.5	16.0'	05°50.1	12.4'	54.5'
20	118°55.8	55.0	220°56.5	16.0'	06°02.5	12.4'	54.5'
21	133°55.7	· · 54.8	235°31.5	15.9'	06°14.8	12.3'	54.5'
22	148°55.6	54.6	250°06.5	15.9'	06°27.2	12.3'	54.6'
23	163°55.5	54.3	264°41.4	15.9'	06°39.5	12.3'	54.6'
SD = 15.7'			SD = 14.8'				

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	178°55.4	N22°54.1	279°16.2	15.8'	N06°51.8	12.3'	54.6'
1	193°55.3	53.9	293°51.1	15.8'	07°04.1	12.3'	54.6'
2	208°55.2	53.7	308°25.9	15.7'	07°16.3	12.2'	54.6'
3	223°55.0	· · 53.5	323°00.6	15.7'	07°28.6	12.2'	54.6'
4	238°54.9	53.3	337°35.3	15.7'	07°40.8	12.2'	54.7'
5	253°54.8	53.1	352°10.0	15.6'	07°53.0	12.2'	54.7'
6	268°54.7	N22°52.8	6°44.6	15.6'	N08°05.2	12.2'	54.7'
7	283°54.6	52.6	21°19.1	15.5'	08°17.3	12.1'	54.7'
8	298°54.5	52.4	35°53.6	15.5'	08°29.4	12.1'	54.7'
9	313°54.4	· · 52.2	50°28.1	15.4'	08°41.5	12.1'	54.8'
10	328°54.3	52.0	65°02.5	15.4'	08°53.6	12.1'	54.8'
11	343°54.1	51.8	79°36.9	15.3'	09°05.7	12.0'	54.8'
12	358°54.0	N22°51.5	94°11.2	15.3'	N09°17.7	12.0'	54.8'
13	13°53.9	51.3	108°45.4	15.2'	09°29.7	12.0'	54.8'
14	28°53.8	51.1	123°19.6	15.1'	09°41.7	11.9'	54.9'
15	43°53.7	· · 50.9	137°53.8	15.1'	09°53.6	11.9'	54.9'
16	58°53.6	50.6	152°27.9	15.0'	10°05.5	11.9'	54.9'
17	73°53.5	50.4	167°01.9	15.0'	10°17.4	11.8'	54.9'
18	88°53.4	N22°50.2	181°35.9	14.9'	N10°29.2	11.8'	55.0'
19	103°53.3	50.0	196°09.8	14.9'	10°41.0	11.8'	55.0'
20	118°53.1	49.7	210°43.7	14.8'	10°52.8	11.7'	55.0'
21	133°53.0	· · 49.5	225°17.4	14.7'	11°04.6	11.7'	55.0'
22	148°52.9	49.3	239°51.2	14.7'	11°16.3	11.7'	55.0'
23	163°52.8	49.1	254°24.8	14.6'	11°27.9	11.6'	55.1'
SD = 15.7'			SD = 14.9'				

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	178°52.7	N22°48.8	268°58.4	14.5'	N11°39.6	11.6'	55.1'
1	193°52.6	48.6	283°32.0	14.5'	11°51.2	11.6'	55.1'
2	208°52.5	48.4	298°05.4	14.4'	12°02.7	11.5'	55.1'
3	223°52.4	· · 48.1	312°38.8	14.3'	12°14.3	11.5'	55.2'
4	238°52.3	47.9	327°12.2	14.3'	12°25.7	11.4'	55.2'
5	253°52.2	47.7	341°45.4	14.2'	12°37.1	11.4'	55.2'
6	268°52.1	N22°47.4	356°18.6	14.1'	N12°48.6	11.4'	55.2'
7	283°51.9	47.2	10°51.8	14.0'	12°59.9	11.3'	55.3'
8	298°51.8	47.0	25°24.8	14.0'	13°11.2	11.3'	55.3'
9	313°51.7	· · 46.7	39°57.8	13.9'	13°22.5	11.2'	55.3'
10	328°51.6	46.5	54°30.7	13.8'	13°33.7	11.2'	55.3'
11	343°51.5	46.3	69°03.5	13.7'	13°44.9	11.1'	55.4'
12	358°51.4	N22°46.0	83°36.2	13.7'	N13°56.0	11.1'	55.4'
13	13°51.3	45.8	98°08.9	13.6'	14°07.1	11.0'	55.4'
14	28°51.2	45.5	112°41.5	13.5'	14°18.1	11.0'	55.5'
15	43°51.1	· · 45.3	127°14.0	13.4'	14°29.1	10.9'	55.5'
16	58°51.0	45.1	141°46.4	13.3'	14°40.0	10.9'	55.5'
17	73°50.9	44.8	156°18.8	13.3'	14°50.9	10.8'	55.5'
18	88°50.8	N22°44.6	170°51.1	13.2'	N15°01.7	10.8'	55.6'
19	103°50.6	44.3	185°23.2	13.1'	15°12.4	10.7'	55.6'
20	118°50.5	44.1	199°55.3	13.0'	15°23.1	10.6'	55.6'
21	133°50.4	· · 43.9	214°27.3	12.9'	15°33.8	10.6'	55.6'
22	148°50.3	43.6	228°59.3	12.8'	15°44.4	10.5'	55.7'
23	163°50.2	43.4	243°31.1	12.8'	15°54.9	10.5'	55.7'
SD = 15.7'			SD = 15.0'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	☐	☐	☐	☐	☐	☐
66°	////	////	00:41	23:23	////	////
64°	////	////	01:47	22:20	////	////
62°	////	////	02:21	21:47	////	////
60°	////	01:11	02:45	21:22	22:56	////
N 58°	////	01:53	03:05	21:03	22:15	////
56°	////	02:20	03:21	20:47	21:47	////
54°	01:05	02:41	03:35	20:34	21:27	23:02
52°	01:44	02:59	03:46	20:22	21:10	22:24
50°	02:10	03:13	03:57	20:11	20:55	21:58
45°	02:53	03:42	04:19	19:50	20:27	21:15
N 40°	03:23	04:04	04:36	19:32	20:05	20:46
35°	03:45	04:21	04:51	19:18	19:47	20:23
30°	04:03	04:36	05:04	19:05	19:32	20:05
20°	04:32	05:01	05:25	18:44	19:08	19:37
N 10°	04:54	05:21	05:44	18:25	18:48	19:15
0°	05:12	05:38	06:01	18:08	18:30	18:57
S 10°	05:29	05:55	06:18	17:51	18:14	18:40
20°	05:45	06:12	06:36	17:33	17:57	18:24
30°	06:00	06:30	06:56	17:12	17:39	18:08
35°	06:09	06:40	07:09	17:00	17:28	18:00
40°	06:18	06:52	07:22	16:47	17:17	17:51
45°	06:28	07:05	07:39	16:30	17:04	17:41
S 50°	06:39	07:21	07:59	16:10	16:48	17:30
52°	06:44	07:28	08:08	16:01	16:41	17:25
54°	06:50	07:36	08:19	15:50	16:33	17:19
56°	06:56	07:44	08:31	15:38	16:25	17:13
58°	07:02	07:54	08:45	15:24	16:15	17:07
S 60°	07:09	08:05	09:02	15:07	16:04	17:00

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	21:59	21:32	20:49	11:56	13:51	16:06
N 70°	22:08	21:49	21:24	11:50	13:36	15:33
68°	22:15	22:04	21:49	11:45	13:23	15:09
66°	22:22	22:15	22:08	11:41	13:13	14:51
64°	22:27	22:25	22:24	11:37	13:04	14:36
62°	22:31	22:34	22:37	11:34	12:57	14:24
60°	22:35	22:41	22:49	11:31	12:51	14:14
N 58°	22:39	22:47	22:58	11:29	12:46	14:05
56°	22:42	22:53	23:07	11:27	12:41	13:57
54°	22:45	22:58	23:15	11:25	12:36	13:50
52°	22:47	23:03	23:21	11:23	12:32	13:44
50°	22:50	23:07	23:28	11:22	12:29	13:38
45°	22:55	23:16	23:41	11:18	12:21	13:26
N 40°	22:59	23:24	23:52	11:16	12:15	13:16
35°	23:03	23:31	· · ·	11:13	12:09	13:08
30°	23:06	23:37	· · ·	11:11	12:05	13:00
20°	23:12	23:47	· · ·	11:07	11:56	12:48
N 10°	23:17	23:56	· · ·	11:04	11:49	12:36
0°	23:22	· · ·	00:05	11:01	11:42	12:26
S 10°	23:27	· · ·	00:13	10:58	11:36	12:16
20°	23:32	· · ·	00:23	10:55	11:28	12:05
30°	23:39	· · ·	00:33	10:51	11:20	11:52
35°	23:42	· · ·	00:39	10:49	11:16	11:45
40°	23:46	· · ·	00:46	10:47	11:10	11:37
45°	23:51	· · ·	00:55	10:44	11:04	11:27
S 50°	23:56	· · ·	01:05	10:41	10:57	11:15
52°	23:59	· · ·	01:09	10:39	10:54	11:10
54°	· · ·	00:02	01:14	10:37	10:50	11:04
56°	· · ·	00:05	01:20	10:36	10:46	10:58
58°	· · ·	00:08	01:27	10:34	10:41	10:51
S 60°	· · ·	00:12	01:34	10:31	10:36	10:42

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss	hh:mm	Upper hh:mm	Lower hh:mm	21-23 65-46%
03	04:07	04:13	12:04	04:51	17:12	
04	04:18	04:24	12:04	05:32	17:53	
05	04:29	04:34	12:05	06:15	18:38	

July 06, 07, 08 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn	
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	283°52.3	136°04.7	N14°45.2	116°37.9	N06°21.9	280°41.1	S00°02.6	103°56.8	N02°32.5
1	298°54.8	151°04.5	44.2	131°39.0	21.3	295°43.4	02.6	118°59.1	32.4
2	313°57.3	166°04.2	43.1	146°40.2	20.8	310°45.8	02.5	134°01.4	32.4
3	328°59.7	181°03.9	•• 42.1	161°41.3	•• 20.2	325°48.1	•• 02.5	149°03.8	•• 32.3
4	344°02.2	196°03.7	41.1	176°42.5	19.6	340°50.4	02.5	164°06.1	32.2
5	359°04.7	211°03.4	40.0	191°43.6	19.0	355°52.7	02.4	179°08.4	32.2
6	14°07.1	226°03.2	N14°39.0	206°44.8	N06°18.4	10°55.1	S00°02.4	194°10.8	N02°32.1
7	29°09.6	241°02.9	38.0	221°45.9	17.8	25°57.4	02.3	209°13.1	32.0
8	44°12.1	256°02.6	36.9	236°47.1	17.2	40°59.7	02.3	224°15.4	32.0
9	59°14.5	271°02.4	•• 35.9	251°48.2	•• 16.6	56°02.1	•• 02.2	239°17.8	•• 31.9
10	74°17.0	286°02.1	34.8	266°49.3	16.0	71°04.4	02.2	254°20.1	31.8
11	89°19.4	301°01.9	33.8	281°50.5	15.4	86°06.7	02.2	269°22.4	31.8
12	104°21.9	316°01.6	N14°32.7	296°51.6	N06°14.8	101°09.1	S00°02.1	284°24.8	N02°31.7
13	119°24.4	331°01.4	31.7	311°52.8	14.2	116°11.4	02.1	299°27.1	31.6
14	134°26.8	346°01.1	30.7	326°53.9	13.6	131°13.8	02.0	314°29.4	31.6
15	149°29.3	1°00.9	•• 29.6	341°55.1	•• 13.0	146°16.1	•• 02.0	329°31.8	•• 31.5
16	164°31.8	16°00.6	28.6	356°56.2	12.5	161°18.4	01.9	344°34.1	31.4
17	179°34.2	31°00.3	27.5	11°57.4	11.9	176°20.8	01.9	359°36.4	31.4
18	194°36.7	46°00.1	N14°26.5	26°58.5	N06°11.3	191°23.1	S00°01.9	14°38.8	N02°31.3
19	209°39.2	60°59.8	25.4	41°59.7	10.7	206°25.4	01.8	29°41.1	31.2
20	224°41.6	75°59.6	24.4	57°00.8	10.1	221°27.8	01.8	44°43.4	31.2
21	239°44.1	90°59.3	•• 23.3	72°01.9	•• 09.5	236°30.1	•• 01.7	59°45.7	•• 31.1
22	254°46.6	105°59.1	22.3	87°03.1	08.9	251°32.4	01.7	74°48.1	31.0
23	269°49.0	120°58.8	21.2	102°04.2	08.3	266°34.8	01.7	89°50.4	31.0
Mer.pass. 05:04		ν -0.3' d -1.0' m -4.07		ν 1.1' d -0.6' m 1.38		ν 2.3' d -0.0' m -2.49		ν 2.3' d -0.1' m 0.97	

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	284°51.5	135°58.6	N14°20.2	117°05.4	N06°07.7	281°37.1	S00°01.6	104°52.7	N02°30.9
1	299°53.9	150°58.4	19.1	132°06.5	07.1	296°39.5	01.6	119°55.1	30.8
2	314°56.4	165°58.1	18.1	147°07.7	06.5	311°41.8	01.5	134°57.4	30.8
3	329°58.9	180°57.9	•• 17.0	162°08.8	•• 05.9	326°44.1	•• 01.5	149°59.7	•• 30.7
4	345°01.3	195°57.6	16.0	177°10.0	05.3	341°46.5	01.4	165°02.1	30.6
5	0°03.8	210°57.4	14.9	192°11.1	04.7	356°48.8	01.4	180°04.4	30.6
6	15°06.3	225°57.1	N14°13.9	207°12.2	N06°04.1	11°51.1	S00°01.4	195°06.7	N02°30.5
7	30°08.7	240°56.9	12.8	222°13.4	03.5	26°53.5	01.3	210°09.0	30.4
8	45°11.2	255°56.6	11.8	237°14.5	02.9	41°55.8	01.3	225°11.4	30.4
9	60°13.7	270°56.4	•• 10.7	252°15.7	•• 02.4	56°58.2	•• 01.2	240°13.7	•• 30.3
10	75°16.1	285°56.2	09.6	267°16.8	01.8	72°00.5	01.2	255°16.0	30.2
11	90°18.6	300°55.9	08.6	282°18.0	01.2	87°02.9	01.2	270°18.4	30.2
12	105°21.1	315°55.7	N14°07.5	297°19.1	N06°00.6	102°05.2	S00°01.1	285°20.7	N02°30.1
13	120°23.5	330°55.4	06.5	312°20.3	06°00.0	117°07.5	01.1	300°23.0	30.0
14	135°26.0	345°55.2	05.4	327°21.4	05°59.4	132°09.9	01.0	315°25.4	30.0
15	150°28.4	0°55.0	•• 04.4	342°22.5	•• 58.8	147°12.2	•• 01.0	330°27.7	•• 29.9
16	165°30.9	15°54.7	03.3	357°23.7	58.2	162°14.6	01.0	345°30.0	29.8
17	180°33.4	30°54.5	02.2	12°24.8	57.6	177°16.9	00.9	0°32.3	29.8
18	195°35.8	45°54.3	N14°01.2	27°26.0	N05°57.0	192°19.2	S00°00.9	15°34.7	N02°29.7
19	210°38.3	60°54.0	14°00.1	42°27.1	56.4	207°21.6	00.9	30°37.0	29.6
20	225°40.8	75°53.8	13°59.1	57°28.3	55.8	222°23.9	00.8	45°39.3	29.6
21	240°43.2	90°53.6	•• 58.0	72°29.4	•• 55.2	237°26.3	•• 00.8	60°41.6	•• 29.5
22	255°45.7	105°53.3	56.9	87°30.5	54.6	252°28.6	00.7	75°44.0	29.4
23	270°48.2	120°53.1	55.9	102°31.7	54.0	267°31.0	00.7	90°46.3	29.3
Mer.pass. 05:00		ν -0.2' d -1.1' m -4.08		ν 1.1' d -0.6' m 1.38		ν 2.3' d -0.0' m -2.50		ν 2.3' d -0.1' m 0.97	

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	285°50.6	135°52.9	N13°54.8	117°32.8	N05°53.4	282°33.3	S00°00.7	105°48.6	N02°29.3
1	300°53.1	150°52.6	53.8	132°34.0	52.8	297°35.7	00.6	120°51.0	29.2
2	315°55.6	165°52.4	52.7	147°35.1	52.2	312°38.0	00.6	135°53.3	29.1
3	330°58.0	180°52.2	•• 51.6	162°36.2	•• 51.6	327°40.3	•• 00.5	150°55.6	•• 29.1
4	346°00.5	195°51.9	50.6	177°37.4	51.0	342°42.7	00.5	165°57.9	29.0
5	1°02.9	210°51.7	49.5	192°38.5	50.4	357°45.0	00.5	181°00.3	28.9
6	16°05.4	225°51.5	N13°48.4	207°39.7	N05°49.8	12°47.4	S00°00.4	196°02.6	N02°28.9
7	31°07.9	240°51.3	47.4	222°40.8	49.2	27°49.7	00.4	211°04.9	28.8
8	46°10.3	255°51.0	46.3	237°42.0	48.6	42°52.1	00.4	226°07.2	28.7
9	61°12.8	270°50.8	•• 45.2	252°43.1	•• 48.0	57°54.4	•• 00.3	241°09.6	•• 28.7
10	76°15.3	285°50.6	44.2	267°44.2	47.4	72°56.8	00.3	256°11.9	28.6
11	91°17.7	300°50.4	43.1	282°45.4	46.8	87°59.1	00.2	271°14.2	28.5
12	106°20.2	315°50.1	N13°42.0	297°46.5	N05°46.2	103°01.5	S00°00.2	286°16.5	N02°28.5
13	121°22.7	330°49.9	40.9	312°47.7	45.7	118°03.8	00.2	301°18.9	28.4
14	136°25.1	345°49.7	39.9	327°48.8	45.1	133°06.2	00.1	316°21.2	28.3
15	151°27.6	0°49.5	•• 38.8	342°49.9	•• 44.5	148°08.5	•• 00.1	331°23.5	•• 28.3
16	166°30.0	15°49.2	37.7	357°51.1	43.9	163°10.9	00.1	346°25.8	28.2
17	181°32.5	30°49.0	36.7	12°52.2	43.3	178°13.2	S00°00.0	1°28.2	28.1
18	196°35.0	45°48.8	N13°35.6	27°53.4	N05°42.7	193°15.6	N00°00.0	16°30.5	N02°28.0
19	211°37.4	60°48.6	34.5	42°54.5	42.1	208°17.9	00.0	31°32.8	28.0
20	226°39.9	75°48.4	33.4	57°55.6	41.5	223°20.3	00.1	46°35.1	27.9
21	241°42.4	90°48.1	•• 32.4	72°56.8	•• 40.9	238°22.6	•• 00.1	61°37.5	•• 27.8
22	256°44.8	105°47.9	31.3	87°57.9	40.3	253°25.0	00.1	76°39.8	27.8
23	271°47.3	120°47.7	30.2	102°59.1	39.7	268°27.3	00.2	91°42.1	27.7
Mer.pass. 04:56		ν -0.2' d -1.1' m -4.08		ν 1.1' d -0.6' m 1.39		ν 2.3' d -0.0' m -2.51		ν 2.3' d -0.1' m 0.98	

Stars		
	SHA	Dec
Alpheratz	357°45.7	29°08.9
Ankaa	353°17.6	-42°14.6
Schedar	349°43.1	56°35.5
Diphda	348°58.0	-17°55.5
Achernar	335°28.3	-57°10.6
Hamal	328°03.3	23°30.7
Polaris	319°02.4	89°18.4
Acamar	315°20.1	-40°15.5
Menkar	314°17.5	4°07.9
Mirfak	308°43.8	49°53.8
Aldebaran	290°52.1	16°31.8
Rigel	281°14.4	-8°11.3
Capella	280°38.1	46°00.4
Bellatrix	278°34.6	6°21.6
Elnath	278°15.7	28°36.9
Alnilam	275°48.8	-1°11.7
Betelgeuse	271°03.9	7°24.5
Canopus	263°57.7	-52°42.0
Sirius	258°35.9	-16°43.8
Adhara	255°14.6	-28°59.2
Procyon	245°02.3	5°11.9
Pollux	243°30.7	28°00.0
Avior	234°19.6	-50°32.7
Suhail	222°54.5	-43°28.7
Miaplacidus	221°41.1	-69°45.9
Alphard	217°58.5	-8°42.3
Regulus	207°46.0	11°54.9
Dubhe	193°54.4	61°41.8
Denebola	182°35.9	14°30.8
Gienah	175°54.6	-17°36.2
Acrux	173°12.0	-63°09.8
Gacrux	172°03.5	-57°10.7
Alioth	166°22.4	55°54.4
Spica	158°33.5	-11°13.1
Alkaid	153°00.3	49°15.8
Hadar	148°50.9	-60°25.8
Menkent	148°10.0	-36°25.5
Arcturus	145°57.5	19°07.7
Rigil Kent.	139°54.4	-60°53.0
Kochab	137°19.1	74°06.9
Zuben'ubi	137°07.6	-16°05.2
Alphecca	126°12.5	26°40.8
Antares	112°28.6	-26°27.4
Atria	107°31.9	-69°02.9
Sabik	102°14.7	-15°44.3
Shaula	96°24.4	-37°06.7
Rasalhague	96°08.1	12°33.2
Eltanin	90°46.6	51°29.3
Kaus Aust.	83°46.2	-34°22.7
Vega	80°40.0	38°47.7
Nunki	76°00.6	-26°16.9
Altair	62°10.0	8°53.9
Peacock	53°21.9	-56°41.9
Deneb	49°32.6	45°19.1
Enif	33°49.0	9°55.5
Al Na'ir	27°46.0	-46°54.3
Fomalhaut	15°26.1	-29°33.7
Scheat	13°55.3	28°08.4
Markab	13°40.3	15°15.8

Jul 06 Tue	SHA	Mer.pass
Venus	212°12.4	14:56
Mars	192°45.5	16:12
Jupiter	356°48.7	05:16
Saturn	180°04.4	17:02

Jul 07 Wed	SHA	Mer.pass
Venus	211°07.1	14:56
Mars	192°13.9	16:10
Jupiter	356°45.6	05:13
Saturn	180°01.3	16:58

Jul 08 Thu	SHA	Mer.pass
Venus	210°02.2	14:57
Mars	191°42.2	16:09
Jupiter	356°42.7	05:09
Saturn	179°58.0	16:54

Horizontal parallax	
Venus:	0.1
Mars:	0.1

h Sun Moon						
Tue	GHA	Dec	GHA	ν	Dec	d HP
0	178° 50.1	N22° 43.1	258° 02.9	12.7'	N16° 05.3	10.4' 55.7'
1	193° 50.0	42.9	272° 34.5	12.6'	16° 15.7	10.3' 55.8'
2	208° 49.9	42.6	287° 06.1	12.5'	16° 26.1	10.3' 55.8'
3	223° 49.8	· · 42.4	301° 37.6	12.4'	16° 36.4	10.2' 55.8'
4	238° 49.7	42.1	316° 09.0	12.3'	16° 46.6	10.1' 55.9'
5	253° 49.6	41.9	330° 40.3	12.2'	16° 56.7	10.1' 55.9'
6	268° 49.5	N22° 41.6	345° 11.5	12.1'	N17° 06.8	10.0' 55.9'
7	283° 49.4	41.4	359° 42.6	12.0'	17° 16.8	9.9' 56.0'
8	298° 49.3	41.1	14° 13.7	11.9'	17° 26.8	9.9' 56.0'
9	313° 49.2	· · 40.9	28° 44.6	11.8'	17° 36.6	9.8' 56.0'
10	328° 49.1	40.6	43° 15.4	11.7'	17° 46.4	9.7' 56.0'
11	343° 49.0	40.4	57° 46.2	11.6'	17° 56.2	9.7' 56.1'
12	358° 48.8	N22° 40.1	72° 16.8	11.6'	N18° 05.8	9.6' 56.1'
13	13° 48.7	39.9	86° 47.4	11.5'	18° 15.4	9.5' 56.1'
14	28° 48.6	39.6	101° 17.8	11.4'	18° 24.9	9.4' 56.2'
15	43° 48.5	· · 39.4	115° 48.2	11.3'	18° 34.3	9.3' 56.2'
16	58° 48.4	39.1	130° 18.5	11.2'	18° 43.6	9.3' 56.2'
17	73° 48.3	38.8	144° 48.6	11.1'	18° 52.9	9.2' 56.3'
18	88° 48.2	N22° 38.6	159° 18.7	11.0'	N19° 02.1	9.1' 56.3'
19	103° 48.1	38.3	173° 48.6	10.9'	19° 11.2	9.0' 56.3'
20	118° 48.0	38.1	188° 18.5	10.8'	19° 20.2	8.9' 56.4'
21	133° 47.9	· · 37.8	202° 48.2	10.7'	19° 29.1	8.8' 56.4'
22	148° 47.8	37.5	217° 17.9	10.6'	19° 38.0	8.8' 56.4'
23	163° 47.7	37.3	231° 47.4	10.4'	19° 46.7	8.7' 56.5'
SD = 15.7' $d = -0.2'$ SD = 15.2'						
Wed	GHA	Dec	GHA	ν	Dec	d HP
0	178° 47.6	N22° 37.0	246° 16.9	10.3'	N19° 55.4	8.6' 56.5'
1	193° 47.5	36.8	260° 46.2	10.2'	20° 03.9	8.5' 56.5'
2	208° 47.4	36.5	275° 15.5	10.1'	20° 12.4	8.4' 56.6'
3	223° 47.3	· · 36.2	289° 44.6	10.0'	20° 20.8	8.3' 56.6'
4	238° 47.2	36.0	304° 13.7	9.9'	20° 29.1	8.2' 56.7'
5	253° 47.1	35.7	318° 42.6	9.8'	20° 37.3	8.1' 56.7'
6	268° 47.0	N22° 35.4	333° 11.4	9.7'	N20° 45.4	8.0' 56.7'
7	283° 46.9	35.2	347° 40.2	9.6'	20° 53.4	7.9' 56.8'
8	298° 46.8	34.9	2° 08.8	9.5'	21° 01.3	7.8' 56.8'
9	313° 46.7	· · 34.6	16° 37.3	9.4'	21° 09.1	7.7' 56.8'
10	328° 46.6	34.4	31° 05.7	9.3'	21° 16.8	7.6' 56.9'
11	343° 46.5	34.1	45° 34.0	9.2'	21° 24.4	7.5' 56.9'
12	358° 46.4	N22° 33.8	60° 02.2	9.1'	N21° 31.8	7.4' 56.9'
13	13° 46.3	33.5	74° 30.3	9.0'	21° 39.2	7.3' 57.0'
14	28° 46.2	33.3	88° 58.3	8.9'	21° 46.5	7.2' 57.0'
15	43° 46.1	· · 33.0	103° 26.2	8.8'	21° 53.6	7.0' 57.0'
16	58° 46.0	32.7	117° 54.0	8.7'	22° 00.7	6.9' 57.1'
17	73° 45.9	32.5	132° 21.7	8.6'	22° 07.6	6.8' 57.1'
18	88° 45.8	N22° 32.2	146° 49.2	8.5'	N22° 14.5	6.7' 57.2'
19	103° 45.7	31.9	161° 16.7	8.4'	22° 21.2	6.6' 57.2'
20	118° 45.6	31.6	175° 44.1	8.3'	22° 27.8	6.5' 57.2'
21	133° 45.5	· · 31.4	190° 11.4	8.2'	22° 34.2	6.4' 57.3'
22	148° 45.4	31.1	204° 38.5	8.1'	22° 40.6	6.2' 57.3'
23	163° 45.3	30.8	219° 05.6	8.0'	22° 46.8	6.1' 57.3'
SD = 15.7' $d = -0.3'$ SD = 15.4'						
Thu	GHA	Dec	GHA	ν	Dec	d HP
0	178° 45.2	N22° 30.5	233° 32.5	7.9'	N22° 52.9	6.0' 57.4'
1	193° 45.1	30.2	247° 59.4	7.8'	22° 58.9	5.9' 57.4'
2	208° 45.0	30.0	262° 26.2	7.7'	23° 04.8	5.7' 57.5'
3	223° 44.9	· · 29.7	276° 52.8	7.6'	23° 10.5	5.6' 57.5'
4	238° 44.8	29.4	291° 19.4	7.5'	23° 16.1	5.5' 57.5'
5	253° 44.7	29.1	305° 45.8	7.4'	23° 21.6	5.3' 57.6'
6	268° 44.6	N22° 28.8	320° 12.2	7.3'	N23° 26.9	5.2' 57.6'
7	283° 44.5	28.6	334° 38.5	7.2'	23° 32.2	5.1' 57.6'
8	298° 44.4	28.3	349° 04.6	7.1'	23° 37.2	5.0' 57.7'
9	313° 44.3	· · 28.0	3° 30.7	7.0'	23° 42.2	4.8' 57.7'
10	328° 44.2	27.7	17° 56.7	6.9'	23° 47.0	4.7' 57.8'
11	343° 44.1	27.4	32° 22.6	6.8'	23° 51.7	4.5' 57.8'
12	358° 44.0	N22° 27.1	46° 48.3	6.7'	N23° 56.2	4.4' 57.8'
13	13° 43.9	26.8	61° 14.0	6.6'	24° 00.6	4.3' 57.9'
14	28° 43.8	26.6	75° 39.6	6.5'	24° 04.9	4.1' 57.9'
15	43° 43.7	· · 26.3	90° 05.2	6.4'	24° 09.0	4.0' 57.9'
16	58° 43.6	26.0	104° 30.6	6.3'	24° 13.0	3.8' 58.0'
17	73° 43.5	25.7	118° 55.9	6.2'	24° 16.8	3.7' 58.0'
18	88° 43.4	N22° 25.4	133° 21.2	6.2'	N24° 20.5	3.5' 58.0'
19	103° 43.3	25.1	147° 46.3	6.1'	24° 24.0	3.4' 58.1'
20	118° 43.2	24.8	162° 11.4	6.0'	24° 27.4	3.2' 58.1'
21	133° 43.1	· · 24.5	176° 36.4	5.9'	24° 30.7	3.1' 58.2'
22	148° 43.0	24.2	191° 01.3	5.8'	24° 33.8	2.9' 58.2'
23	163° 42.9	23.9	205° 26.1	5.8'	24° 36.7	2.8' 58.2'
SD = 15.7' $d = -0.3'$ SD = 15.6'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	☐	☐	☐	☐	☐	☐
66°	////	////	00:57	23:08	////	////
64°	////	////	01:54	22:14	////	////
62°	////	////	02:26	21:43	////	////
60°	////	01:19	02:50	21:19	22:48	////
N 58°	////	01:58	03:08	21:01	22:10	////
56°	////	02:25	03:24	20:45	21:44	////
54°	01:13	02:45	03:37	20:32	21:24	22:55
52°	01:49	03:02	03:49	20:20	21:07	22:20
50°	02:13	03:16	03:59	20:10	20:53	21:55
45°	02:56	03:44	04:21	19:49	20:25	21:13
N 40°	03:25	04:06	04:38	19:31	20:04	20:45
35°	03:47	04:23	04:52	19:17	19:46	20:23
30°	04:05	04:38	05:05	19:05	19:32	20:05
20°	04:33	05:02	05:26	18:44	19:08	19:37
N 10°	04:55	05:21	05:44	18:25	18:48	19:15
0°	05:13	05:39	06:01	18:09	18:31	18:57
S 10°	05:29	05:55	06:18	17:52	18:15	18:41
20°	05:45	06:12	06:36	17:34	17:58	18:25
30°	06:00	06:30	06:56	17:14	17:40	18:10
35°	06:09	06:40	07:08	17:02	17:30	18:01
40°	06:18	06:51	07:22	16:48	17:19	17:52
45°	06:27	07:04	07:38	16:32	17:06	17:43
S 50°	06:38	07:19	07:57	16:13	16:51	17:32
52°	06:43	07:26	08:07	16:03	16:44	17:27
54°	06:49	07:34	08:17	15:53	16:36	17:22
56°	06:54	07:43	08:29	15:41	16:27	17:16
58°	07:01	07:52	08:43	15:27	16:18	17:10
S 60°	07:08	08:03	08:59	15:11	16:07	17:03

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	☐	☐	☐	☐	☐	☐
N 70°	20:31	☐	☐	18:05	☐	☐
68°	21:26	☐	☐	17:11	☐	☐
66°	22:00	21:46	☐	16:38	18:40	☐
64°	22:24	22:28	22:42	16:14	17:58	19:42
62°	22:44	22:57	23:23	15:56	17:30	19:01
60°	23:00	23:19	23:52	15:40	17:09	18:33
N 58°	23:14	23:37	· · ·	15:27	16:51	18:11
56°	23:25	23:52	· · ·	15:16	16:36	17:53
54°	23:36	· · ·	00:05	15:06	16:24	17:38
52°	23:45	· · ·	00:17	14:58	16:13	17:25
50°	23:53	· · ·	00:27	14:50	16:03	17:14
45°	· · ·	00:11	00:48	14:33	15:42	16:50
N 40°	· · ·	00:25	01:06	14:20	15:25	16:31
35°	00:02	00:38	01:20	14:08	15:11	16:15
30°	00:10	00:48	01:33	13:59	14:59	16:02
20°	00:25	01:07	01:55	13:41	14:38	15:38
N 10°	00:38	01:23	02:14	13:27	14:20	15:18
0°	00:50	01:39	02:32	13:13	14:04	14:59
S 10°	01:02	01:54	02:50	12:59	13:47	14:40
20°	01:15	02:11	03:09	12:44	13:29	14:20
30°	01:30	02:30	03:32	12:28	13:09	13:57
35°	01:39	02:41	03:45	12:18	12:57	13:43
40°	01:49	02:54	04:00	12:07	12:43	13:27
45°	02:01	03:09	04:18	11:54	12:27	13:09
S 50°	02:15	03:28	04:41	11:38	12:07	12:45
52°	02:22	03:37	04:52	11:31	11:57	12:34
54°	02:30	03:47	05:04	11:23	11:47	12:21
56°	02:38	03:59	05:18	11:13	11:35	12:07
58°	02:48	04:12	05:35	11:03	11:21	11:50
S 60°	02:59	04:27	05:55	10:51	11:05	11:29

Day	Sun		Mer. Pass hh:mm	Moon		Age 24-26 36-18%
	Eqn. of Time 00 ^h mm:ss	Time 12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
06	04:40	04:45	12:05	07:01	19:26	
07	04:50	04:54	12:05	07:51	20:18	
08	04:59	05:04	12:05	08:45	21:14	

July 09, 10, 11 UT (Fri., Sat., Sun.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Fri	GHA	Dec	GHA	Dec		GHA	Dec		GHA	Dec		GHA	Dec		SHA	Dec	
0	286°49.8	135°47.5	N13°29.1	118°00.2	N05°39.1	283°29.7	N00°00.2	106°44.4	N02°27.6			106°44.4	N02°27.6		Alpheratz	357°45.6	29°08.9
1	301°52.2	150°47.3	28.1	133°01.3	38.5	298°32.0	00.3	121°46.8	27.6			121°46.8			Ankaa	353°17.6	-42°14.6
2	316°54.7	165°47.1	27.0	148°02.5	37.9	313°34.4	00.3	136°49.1	27.5			136°49.1			Schedar	349°43.0	56°35.6
3	331°57.2	180°46.8	· · 25.9	163°03.6	· · 37.3	328°36.7	· · 00.3	151°51.4	· · 27.4			151°51.4	· · 27.4		Diphda	348°57.9	-17°55.5
4	346°59.6	195°46.6	24.8	178°04.7	36.7	343°39.1	00.4	166°53.7	27.3			166°53.7			Achernar	335°28.3	-57°10.6
5	2°02.1	210°46.4	23.8	193°05.9	36.1	358°41.4	00.4	181°56.1	27.3			181°56.1			Hamal	328°03.3	23°30.7
6	17°04.5	225°46.2	N13°22.7	208°07.0	N05°35.5	13°43.8	N00°00.4	196°58.4	N02°27.2			196°58.4	N02°27.2		Polaris	319°00.8	89°18.4
7	32°07.0	240°46.0	21.6	223°08.2	34.9	28°46.1	00.5	212°00.7	27.1			212°00.7			Acamar	315°20.1	-40°15.5
8	47°09.5	255°45.8	20.5	238°09.3	34.3	43°48.5	00.5	227°03.0	27.1			227°03.0			Menkar	314°17.4	4°07.9
9	62°11.9	270°45.6	· · 19.4	253°10.4	· · 33.7	58°50.9	· · 00.5	242°05.3	· · 27.0			242°05.3	· · 27.0		Mirfak	308°43.8	49°53.8
10	77°14.4	285°45.4	18.4	268°11.6	33.1	73°53.2	00.6	257°07.7	26.9			257°07.7			Aldebaran	290°52.1	16°31.8
11	92°16.9	300°45.2	17.3	283°12.7	32.5	88°55.6	00.6	272°10.0	26.9			272°10.0			Rigel	281°14.4	-8°11.3
12	107°19.3	315°45.0	N13°16.2	298°13.9	N05°31.9	103°57.9	N00°00.6	287°12.3	N02°26.8			287°12.3	N02°26.8		Capella	280°38.0	46°00.4
13	122°21.8	330°44.7	15.1	313°15.0	31.3	119°00.3	00.7	302°14.6	26.7			302°14.6			Bellatrix	278°34.6	6°21.6
14	137°24.3	345°44.5	14.0	328°16.1	30.7	134°02.6	00.7	317°17.0	26.6			317°17.0			Elnath	278°15.7	28°36.9
15	152°26.7	0°44.3	· · 12.9	343°17.3	· · 30.1	149°05.0	· · 00.7	332°19.3	· · 26.6			332°19.3	· · 26.6		Alnilam	275°48.8	-1°11.7
16	167°29.2	15°44.1	11.9	358°18.4	29.5	164°07.3	00.8	347°21.6	26.5			347°21.6			Betelgeuse	271°03.9	7°24.5
17	182°31.7	30°43.9	10.8	13°19.5	28.9	179°09.7	00.8	2°23.9	26.4			2°23.9			Canopus	263°57.6	-52°42.0
18	197°34.1	45°43.7	N13°09.7	28°20.7	N05°28.3	194°12.1	N00°00.8	17°26.2	N02°26.4			17°26.2	N02°26.4		Sirius	258°35.9	-16°43.8
19	212°36.6	60°43.5	08.6	43°21.8	27.7	209°14.4	00.9	32°28.6	26.3			32°28.6			Adhara	255°14.6	-28°59.2
20	227°39.0	75°43.3	07.5	58°23.0	27.1	224°16.8	00.9	47°30.9	26.2			47°30.9			Procyon	245°02.3	5°11.9
21	242°41.5	90°43.1	· · 06.4	73°24.1	· · 26.5	239°19.1	· · 00.9	62°33.2	· · 26.2			62°33.2	· · 26.2		Pollux	243°30.7	28°00.0
22	257°44.0	105°42.9	05.3	88°25.2	25.9	254°21.5	01.0	77°35.5	26.1			77°35.5			Avior	234°19.6	-59°32.7
23	272°46.4	120°42.7	04.3	103°26.4	25.3	269°23.9	01.0	92°37.8	26.0			92°37.8			Suhail	222°54.5	-43°28.7
Mer.pass. 04:52			ν -0.2' d -1.1' m -4.09	ν 1.1' d -0.6' m 1.39	ν 2.4' d 0.0' m -2.51	ν 2.3' d -0.1' m 0.98											

Sat	GHA	Dec	GHA	Dec		GHA	Dec		GHA	Dec	
0	287°48.9	135°42.5	N13°03.2	118°27.5	N05°24.7	284°26.2	N00°01.0	107°40.2	N02°25.9		
1	302°51.4	150°42.3	02.1	133°28.6	24.1	299°28.6	01.0	122°42.5	25.9		
2	317°53.8	165°42.1	13°01.0	148°29.8	23.5	314°30.9	01.1	137°44.8	25.8		
3	332°56.3	180°41.9	12°59.9	163°30.9	· · 22.9	329°33.3	· · 01.1	152°47.1	· · 25.7		
4	347°58.8	195°41.7	58.8	178°32.0	22.3	344°35.7	01.1	167°49.4	25.7		
5	3°01.2	210°41.5	57.7	193°33.2	21.7	359°38.0	01.2	182°51.8	25.6		
6	18°03.7	225°41.3	N12°56.6	208°34.3	N05°21.0	14°40.4	N00°01.2	197°54.1	N02°25.5		
7	33°06.2	240°41.1	55.5	223°35.5	20.4	29°42.7	01.2	212°56.4	25.4		
8	48°08.6	255°40.9	54.5	238°36.6	19.8	44°45.1	01.3	227°58.7	25.4		
9	63°11.1	270°40.7	· · 53.4	253°37.7	· · 19.2	59°47.5	· · 01.3	243°01.0	· · 25.3		
10	78°13.5	285°40.5	52.3	268°38.9	18.6	74°49.8	01.3	258°03.4	25.2		
11	93°16.0	300°40.3	51.2	283°40.0	18.0	89°52.2	01.4	273°05.7	25.2		
12	108°18.5	315°40.2	N12°50.1	298°41.1	N05°17.4	104°54.5	N00°01.4	288°08.0	N02°25.1		
13	123°20.9	330°40.0	49.0	313°42.3	16.8	119°56.9	01.4	303°10.3	25.0		
14	138°23.4	345°39.8	47.9	328°43.4	16.2	134°59.3	01.4	318°12.6	24.9		
15	153°25.9	0°39.6	· · 46.8	343°44.5	· · 15.6	150°01.6	· · 01.5	333°15.0	· · 24.9		
16	168°28.3	15°39.4	45.7	358°45.7	15.0	165°04.0	01.5	348°17.3	24.8		
17	183°30.8	30°39.2	44.6	13°46.8	14.4	180°06.4	01.5	3°19.6	24.7		
18	198°33.3	45°39.0	N12°43.5	28°47.9	N05°13.8	195°08.7	N00°01.6	18°21.9	N02°24.7		
19	213°35.7	60°38.8	42.4	43°49.1	13.2	210°11.1	01.6	33°24.2	24.6		
20	228°38.2	75°38.6	41.3	58°50.2	12.6	225°13.5	01.6	48°26.6	24.5		
21	243°40.7	90°38.5	· · 40.2	73°51.3	· · 12.0	240°15.8	· · 01.7	63°28.9	· · 24.4		
22	258°43.1	105°38.3	39.1	88°52.5	11.4	255°18.2	01.7	78°31.2	24.4		
23	273°45.6	120°38.1	38.0	103°53.6	10.8	270°20.6	01.7	93°33.5	24.3		
Mer.pass. 04:48			ν -0.2' d -1.1' m -4.09	ν 1.1' d -0.6' m 1.40	ν 2.4' d 0.0' m -2.52	ν 2.3' d -0.1' m 0.98					

Sun	GHA	Dec	GHA	Dec		GHA	Dec		GHA	Dec	
0	288°48.0	135°37.9	N12°36.9	118°54.7	N05°10.2	285°22.9	N00°01.7	108°35.8	N02°24.2		
1	303°50.5	150°37.7	35.8	133°55.9	09.6	300°25.3	01.8	123°38.1	24.1		
2	318°53.0	165°37.5	34.7	148°57.0	09.0	315°27.7	01.8	138°40.5	24.1		
3	333°55.4	180°37.3	· · 33.6	163°58.1	· · 08.4	330°30.0	· · 01.8	153°42.8	· · 24.0		
4	348°57.9	195°37.2	32.5	178°59.3	07.8	345°32.4	01.9	168°45.1	23.9		
5	4°00.4	210°37.0	31.4	194°00.4	07.2	0°34.8	01.9	183°47.4	23.9		
6	19°02.8	225°36.8	N12°30.3	209°01.5	N05°06.6	15°37.1	N00°01.9	198°49.7	N02°23.8		
7	34°05.3	240°36.6	29.2	224°02.7	06.0	30°39.5	01.9	213°52.0	23.7		
8	49°07.8	255°36.4	28.1	239°03.8	05.4	45°41.9	02.0	228°54.4	23.6		
9	64°10.2	270°36.3	· · 27.0	254°04.9	· · 04.8	60°44.2	· · 02.0	243°56.7	· · 23.6		
10	79°12.7	285°36.1	25.9	269°06.1	04.2	75°46.6	02.0	258°59.0	23.5		
11	94°15.2	300°35.9	24.8	284°07.2	03.5	90°49.0	02.0	274°01.3	23.4		
12	109°17.6	315°35.7	N12°23.7	299°08.3	N05°02.9	105°51.3	N00°02.1	289°03.6	N02°23.4		
13	124°20.1	330°35.6	22.6	314°09.5	02.3	120°53.7	02.1	304°05.9	23.3		
14	139°22.5	345°35.4	21.5	329°10.6	01.7	135°56.1	02.1	319°08.2	23.2		
15	154°25.0	0°35.2	· · 20.4	344°11.7	· · 01.1	150°58.5	· · 02.2	334°10.6	· · 23.1		
16	169°27.5	15°35.0	19.3	359°12.9	05°00.5	166°00.8	02.2	349°12.9	23.1		
17	184°29.9	30°34.9	18.2	14°14.0	04°59.9	181°03.2	02.2	4°15.2	23.0		
18	199°32.4	45°34.7	N12°17.1	29°15.1	N04°59.3	196°05.6	N00°02.2	19°17.5	N02°22.9		
19	214°34.9	60°34.5	15.9	44°16.3	58.7	211°07.9	02.3	34°19.8	22.8		
20	229°37.3	75°34.3	14.8	59°17.4	58.1	226°10.3	02.3	49°22.1	22.8		
21	244°39.8	90°34.2	· · 13.7	74°18.5	· · 57.5	241°12.7	· · 02.3	64°24.5	· · 22.7		
22	259°42.3	105°34.0	12.6	89°19.7	56.9	256°15.1	02.3	79°26.8	22.6		
23	274°44.7	120°33.8	11.5	104°20.8	56.3	271°17.4	02.4	94°29.1	22.5		
Mer.pass. 04:44			ν -0.2' d -1.1' m -4.10	ν 1.1' d -0.6' m 1.40	ν 2.4' d 0.0' m -2.53	ν 2.3' d -0.1' m 0.98					

Jul 09 Fri	SHA	Mer.pass
Venus	208°57.7	14:57
Mars	191°10.4	16:07
Jupiter	356°39.9	05:05
Saturn	179°54.7	16:50

Jul 10 Sat	SHA	Mer.pass
Venus	207°53.6	14:57
Mars	190°38.6	16:05
Jupiter	356°37.3	05:01
Saturn	179°51.3	16:47

Jul 11 Sun	SHA	Mer.pass
Venus	206°49.8	14:58
Mars	190°06.7	16:03
Jupiter	356°34.9	04:58
Saturn	179°47.8	16:43

Horizontal parallax		
Venus:	0.1	
Mars:	0.1	

h	Sun			Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	178°42.8	N22°23.6	219°50.9	5.7'	N24°39.5	2.6'	58.3'
1	193°42.8	23.3	234°15.6	5.6'	24°42.1	2.5'	58.3'
2	208°42.7	23.0	248°40.2	5.5'	24°44.6	2.3'	58.3'
3	223°42.6	· · 22.7	263°04.7	5.5'	24°46.9	2.2'	58.4'
4	238°42.5	22.5	277°29.2	5.4'	24°49.1	2.0'	58.4'
5	253°42.4	22.2	291°53.5	5.3'	24°51.1	1.9'	58.5'
6	268°42.3	N22°21.9	306°17.8	5.2'	N24°52.9	1.7'	58.5'
7	283°42.2	21.6	320°42.1	5.2'	24°54.6	1.5'	58.5'
8	298°42.1	21.3	335°06.3	5.1'	24°56.2	1.4'	58.6'
9	313°42.0	· · 21.0	349°30.4	5.0'	24°57.5	1.2'	58.6'
10	328°41.9	20.7	3°54.4	5.0'	24°58.7	1.0'	58.6'
11	343°41.8	20.3	18°18.4	4.9'	24°59.8	0.9'	58.7'
12	358°41.7	N22°20.0	32°42.3	4.9'	N25°00.7	0.7'	58.7'
13	13°41.6	19.7	47°06.2	4.8'	25°01.4	0.5'	58.7'
14	28°41.5	19.4	61°30.0	4.8'	25°01.9	0.4'	58.8'
15	43°41.4	· · 19.1	75°53.8	4.7'	25°02.3	0.2'	58.8'
16	58°41.3	18.8	90°17.5	4.7'	25°02.5	0.0'	58.8'
17	73°41.3	18.5	104°41.2	4.6'	25°02.6	-0.1'	58.9'
18	88°41.2	N22°18.2	119°04.8	4.6'	N25°02.5	-0.3'	58.9'
19	103°41.1	17.9	133°28.3	4.5'	25°02.2	-0.5'	58.9'
20	118°41.0	17.6	147°51.9	4.5'	25°01.7	-0.6'	59.0'
21	133°40.9	· · 17.3	162°15.4	4.4'	25°01.1	-0.8'	59.0'
22	148°40.8	17.0	176°38.8	4.4'	25°00.3	-1.0'	59.0'
23	163°40.7	16.7	191°02.2	4.4'	24°59.3	-1.1'	59.1'
SD = 15.7'			SD = 15.9'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	178°40.6	N22°16.4	205°25.6	4.3'	N24°58.2	-1.3'	59.1'
1	193°40.5	16.1	219°48.9	4.3'	24°56.9	-1.5'	59.1'
2	208°40.4	15.7	234°12.2	4.3'	24°55.4	-1.7'	59.2'
3	223°40.3	· · 15.4	248°35.5	4.3'	24°53.7	-1.8'	59.2'
4	238°40.3	15.1	262°58.7	4.2'	24°51.9	-2.0'	59.2'
5	253°40.2	14.8	277°22.0	4.2'	24°49.9	-2.2'	59.3'
6	268°40.1	N22°14.5	291°45.2	4.2'	N24°47.7	-2.3'	59.3'
7	283°40.0	14.2	306°08.4	4.2'	24°45.4	-2.5'	59.3'
8	298°39.9	13.9	320°31.5	4.2'	24°42.9	-2.7'	59.4'
9	313°39.8	· · 13.5	334°54.7	4.1'	24°40.2	-2.9'	59.4'
10	328°39.7	13.2	349°17.8	4.1'	24°37.3	-3.0'	59.4'
11	343°39.6	12.9	3°41.0	4.1'	24°34.3	-3.2'	59.5'
12	358°39.5	N22°12.6	18°04.1	4.1'	N24°31.1	-3.4'	59.5'
13	13°39.5	12.3	32°27.2	4.1'	24°27.7	-3.6'	59.5'
14	28°39.4	11.9	46°50.3	4.1'	24°24.1	-3.7'	59.6'
15	43°39.3	· · 11.6	61°13.4	4.1'	24°20.4	-3.9'	59.6'
16	58°39.2	11.3	75°36.5	4.1'	24°16.5	-4.1'	59.6'
17	73°39.1	11.0	89°59.6	4.1'	24°12.4	-4.3'	59.6'
18	88°39.0	N22°10.7	104°22.7	4.1'	N24°08.1	-4.4'	59.7'
19	103°38.9	10.3	118°45.8	4.1'	24°03.7	-4.6'	59.7'
20	118°38.8	10.0	133°08.9	4.1'	23°59.1	-4.8'	59.7'
21	133°38.8	· · 09.7	147°32.1	4.1'	23°54.3	-4.9'	59.8'
22	148°38.7	09.4	161°55.2	4.2'	23°49.4	-5.1'	59.8'
23	163°38.6	09.0	176°18.4	4.2'	23°44.3	-5.3'	59.8'
SD = 15.7'			SD = 16.1'				

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	178°38.5	N22°08.7	190°41.6	4.2'	N23°39.0	-5.5'	59.8'
1	193°38.4	08.4	205°04.8	4.2'	23°33.5	-5.6'	59.9'
2	208°38.3	08.0	219°28.0	4.2'	23°27.9	-5.8'	59.9'
3	223°38.2	· · 07.7	233°51.2	4.3'	23°22.1	-6.0'	59.9'
4	238°38.1	07.4	248°14.5	4.3'	23°16.2	-6.1'	59.9'
5	253°38.1	07.1	262°37.8	4.3'	23°10.1	-6.3'	60.0'
6	268°38.0	N22°06.7	277°01.1	4.3'	N23°03.8	-6.5'	60.0'
7	283°37.9	06.4	291°24.4	4.4'	22°57.3	-6.6'	60.0'
8	298°37.8	06.1	305°47.8	4.4'	22°50.7	-6.8'	60.0'
9	313°37.7	· · 05.7	320°11.2	4.4'	22°43.9	-6.9'	60.1'
10	328°37.6	05.4	334°34.6	4.5'	22°37.0	-7.1'	60.1'
11	343°37.6	05.1	348°58.1	4.5'	22°29.8	-7.3'	60.1'
12	358°37.5	N22°04.7	3°21.6	4.6'	N22°22.6	-7.4'	60.1'
13	13°37.4	04.4	17°45.2	4.6'	22°15.1	-7.6'	60.1'
14	28°37.3	04.1	32°08.8	4.6'	22°07.6	-7.8'	60.2'
15	43°37.2	· · 03.7	46°32.4	4.7'	21°59.8	-7.9'	60.2'
16	58°37.1	03.4	60°56.1	4.7'	21°51.9	-8.1'	60.2'
17	73°37.0	03.1	75°19.9	4.8'	21°43.8	-8.2'	60.2'
18	88°37.0	N22°02.7	89°43.6	4.8'	N21°35.6	-8.4'	60.2'
19	103°36.9	02.4	104°07.5	4.9'	21°27.2	-8.5'	60.3'
20	118°36.8	02.0	118°31.4	4.9'	21°18.7	-8.7'	60.3'
21	133°36.7	· · 01.7	132°55.3	5.0'	21°10.0	-8.8'	60.3'
22	148°36.6	01.4	147°19.3	5.0'	21°01.2	-9.0'	60.3'
23	163°36.6	01.0	161°43.3	5.1'	20°52.2	-9.1'	60.3'
SD = 15.7'			SD = 16.3'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°						
68°						
66°	////	////	01:12	22:55	////	////
64°	////	////	02:01	22:08	////	////
62°	////	////	02:32	21:38	////	////
60°	////	01:28	02:54	21:15	22:40	////
N 58°	////	02:04	03:13	20:57	22:05	////
56°	////	02:29	03:28	20:42	21:40	////
54°	01:21	02:49	03:41	20:29	21:21	22:47
52°	01:54	03:05	03:52	20:18	21:05	22:15
50°	02:18	03:19	04:02	20:08	20:51	21:52
45°	02:59	03:47	04:23	19:47	20:24	21:11
N 40°	03:27	04:08	04:40	19:30	20:03	20:43
35°	03:49	04:25	04:54	19:16	19:46	20:21
30°	04:06	04:39	05:06	19:04	19:31	20:04
20°	04:34	05:03	05:27	18:43	19:08	19:37
N 10°	04:55	05:22	05:45	18:26	18:48	19:15
0°	05:13	05:39	06:02	18:09	18:31	18:57
S 10°	05:29	05:56	06:18	17:53	18:15	18:41
20°	05:45	06:12	06:36	17:35	17:59	18:26
30°	06:00	06:30	06:56	17:15	17:41	18:11
35°	06:08	06:39	07:07	17:04	17:31	18:03
40°	06:17	06:50	07:21	16:50	17:20	17:54
45°	06:26	07:03	07:36	16:35	17:08	17:45
S 50°	06:37	07:18	07:56	16:15	16:53	17:34
52°	06:42	07:25	08:05	16:06	16:46	17:29
54°	06:47	07:32	08:15	15:56	16:39	17:24
56°	06:53	07:41	08:27	15:45	16:30	17:19
58°	06:59	07:50	08:40	15:31	16:21	17:12
S 60°	07:05	08:01	08:56	15:15	16:10	17:06

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°						
N 70°						
68°						22:51
66°			00:08		22:30	22:06
64°	23:31	· · ·	01:10	20:58	21:28	21:36
62°	· · ·	00:17	01:45	20:12	20:53	21:13
60°	· · ·	00:48	02:10	19:42	20:28	20:55
N 58°	00:14	01:11	02:30	19:18	20:07	20:39
56°	00:32	01:29	02:46	19:00	19:51	20:26
54°	00:47	01:45	03:00	18:44	19:36	20:15
52°	01:00	01:58	03:12	18:30	19:24	20:05
50°	01:12	02:10	03:23	18:19	19:13	19:55
45°	01:36	02:35	03:46	17:54	18:50	19:36
N 40°	01:55	02:55	04:04	17:34	18:31	19:20
35°	02:11	03:11	04:19	17:18	18:15	19:07
30°	02:25	03:25	04:32	17:03	18:02	18:55
20°	02:49	03:50	04:54	16:39	17:38	18:35
N 10°	03:10	04:10	05:13	16:18	17:18	18:17
0°	03:29	04:30	05:31	15:58	16:59	18:00
S 10°	03:49	04:50	05:49	15:38	16:40	17:44
20°	04:10	05:10	06:09	15:17	16:20	17:26
30°	04:34	05:35	06:31	14:53	15:56	17:05
35°	04:48	05:49	06:44	14:38	15:42	16:53
40°	05:05	06:05	06:59	14:22	15:26	16:38
45°	05:25	06:25	07:16	14:02	15:07	16:22
S 50°	05:50	06:50	07:38	13:37	14:43	16:01
52°	06:02	07:02	07:48	13:24	14:31	15:51
54°	06:16	07:15	08:00	13:11	14:17	15:40
56°	06:32	07:31	08:13	12:55	14:02	15:27
58°	06:51	07:49	08:29	12:36	13:44	15:12
S 60°	07:14	08:12	08:47	12:12	13:21	14:54

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time			Mer. Pass.		
	00 ^h	12 ^h	Pass	Upper	Lower	27-0
	mm:ss	mm:ss	hh:mm	hh:mm	hh:mm	10-1%
09	05:09	05:13	12:05	09:44	22:14	
10	05:18	05:22	12:05	10:45	23:15	
11	05:26	05:30	12:06	11:46	--:--	

July 12, 13, 14 UT (Mon., Tue., Wed.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	289°47.2	135°33.7	N12°10.4	119°21.9	N04°55.7	286°19.8	N00°02.4	109°31.4	N02°22.5	Alpheratz	357°45.6	29°08.9
1	304°49.7	150°33.5	09.3	134°23.0	55.1	301°22.2	02.4	124°33.7	22.4	Ankaa	353°17.6	-42°14.6
2	319°52.1	165°33.3	08.2	149°24.2	54.5	316°24.2	02.4	139°36.0	22.3	Schedar	349°43.0	56°35.6
3	334°54.6	180°33.1	•• 07.1	164°25.3	•• 53.9	331°26.9	•• 02.5	154°38.3	•• 22.3	Diphda	348°57.9	-17°55.5
4	349°57.0	195°33.0	06.0	179°26.4	53.3	346°29.3	02.5	169°40.7	22.2	Achernar	335°28.3	-57°10.6
5	4°59.5	210°32.8	04.8	194°27.6	52.6	1°31.7	02.5	184°43.0	22.1	Hamal	328°03.3	23°30.7
6	20°02.0	225°32.7	N12°03.7	209°28.7	N04°52.0	16°34.1	N00°02.5	199°45.3	N02°22.0	Polaris	318°59.4	89°18.4
7	35°04.4	240°32.5	02.6	224°29.8	51.4	31°36.4	02.6	214°47.6	22.0	Acamar	315°20.1	-40°15.4
8	50°06.9	255°32.3	01.5	239°31.0	50.8	46°38.8	02.6	229°49.9	21.9	Menkar	314°17.4	4°08.0
9	65°09.4	270°32.2	12°00.4	254°32.1	•• 50.2	61°41.2	•• 02.6	244°52.2	•• 21.8	Mirfak	308°43.7	49°53.8
10	80°11.8	285°32.0	11°59.3	269°33.2	49.6	76°43.6	02.6	259°54.5	21.7	Aldebaran	290°52.1	16°31.8
11	95°14.3	300°31.8	58.2	284°34.3	49.0	91°45.9	02.7	274°56.8	21.7	Rigel	281°14.4	-8°11.3
12	110°16.8	315°31.7	N11°57.0	299°35.5	N04°48.4	106°48.3	N00°02.7	289°59.2	N02°21.6	Capella	280°38.0	46°00.4
13	125°19.2	330°31.5	55.9	314°36.6	47.8	121°50.7	02.7	305°01.5	21.5	Bellatrix	278°34.6	6°21.6
14	140°21.7	345°31.4	54.8	329°37.7	47.2	136°53.1	02.7	320°03.8	21.4	Elnath	278°15.6	28°36.9
15	155°24.1	0°31.2	•• 53.7	344°38.9	•• 46.6	151°55.5	•• 02.8	335°06.1	•• 21.4	Alnilam	275°48.8	-1°11.7
16	170°26.6	15°31.0	52.6	359°40.0	46.0	166°57.8	02.8	350°08.4	21.3	Betelgeuse	271°03.9	7°24.6
17	185°29.1	30°30.9	51.5	14°41.1	45.4	182°00.2	02.8	5°10.7	21.2	Canopus	263°57.6	-52°42.0
18	200°31.5	45°30.7	N11°50.3	29°42.2	N04°44.8	197°02.6	N00°02.8	20°13.0	N02°21.1	Sirius	258°35.9	-16°43.8
19	215°34.0	60°30.6	49.2	44°43.4	44.1	212°05.0	02.8	35°15.3	21.1	Adhara	255°14.6	-28°59.2
20	230°36.5	75°30.4	48.1	59°44.5	43.5	227°07.3	02.9	50°17.7	21.0	Procyon	245°02.3	5°11.9
21	245°38.9	90°30.2	•• 47.0	74°45.6	•• 42.9	242°09.7	•• 02.9	65°20.0	•• 20.9	Pollux	243°30.7	28°00.0
22	260°41.4	105°30.1	45.9	89°46.8	42.3	257°12.1	02.9	80°22.3	20.8	Avior	234°19.6	-59°32.7
23	275°43.9	120°29.9	44.7	104°47.9	41.7	272°14.5	02.9	95°24.6	20.8	Suhail	222°54.5	-43°28.7
Mer.pass. 04:40		ν -0.2' d -1.1' m -4.11		ν 1.1' d -0.6' m 1.41		ν 2.4' d 0.0' m -2.54		ν 2.3' d -0.1' m 0.98				

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	290°46.3	135°29.8	N11°43.6	119°49.0	N04°41.1	287°16.9	N00°03.0	110°26.9	N02°20.7
1	305°48.8	150°29.6	42.5	134°50.1	40.5	302°19.3	03.0	125°29.2	20.6
2	320°51.3	165°29.5	41.4	149°51.3	39.9	317°21.6	03.0	140°31.5	20.5
3	335°53.7	180°29.3	•• 40.2	164°52.4	•• 39.3	332°24.0	•• 03.0	155°33.8	•• 20.5
4	350°56.2	195°29.2	39.1	179°53.5	38.7	347°26.4	03.0	170°36.1	20.4
5	5°58.6	210°29.0	38.0	194°54.7	38.1	2°28.8	03.1	185°38.5	20.3
6	21°01.1	225°28.9	N11°36.9	209°55.8	N04°37.5	17°31.2	N00°03.1	200°40.8	N02°20.2
7	36°03.6	240°28.7	35.8	224°56.9	36.8	32°33.5	03.1	215°43.1	20.2
8	51°06.0	255°28.6	34.6	239°58.0	36.2	47°35.9	03.1	230°45.4	20.1
9	66°08.5	270°28.4	•• 33.5	254°59.2	•• 35.6	62°38.3	•• 03.2	245°47.7	•• 20.0
10	81°11.0	285°28.3	32.4	270°00.3	35.0	77°40.7	03.2	260°50.0	19.9
11	96°13.4	300°28.1	31.3	285°01.4	34.4	92°43.1	03.2	275°52.3	19.9
12	111°15.9	315°28.0	N11°30.1	300°02.5	N04°33.8	107°45.5	N00°03.2	290°54.6	N02°19.8
13	126°18.4	330°27.8	29.0	315°03.7	33.2	122°47.9	03.2	305°56.9	19.7
14	141°20.8	345°27.7	27.9	330°04.8	32.6	137°50.2	03.3	320°59.2	19.6
15	156°23.3	0°27.5	•• 26.7	345°05.9	•• 32.0	152°52.6	•• 03.3	336°01.6	•• 19.6
16	171°25.8	15°27.4	25.6	0°07.1	31.4	167°55.0	03.3	351°03.9	19.5
17	186°28.2	30°27.3	24.5	15°08.2	30.7	182°57.4	03.3	6°06.2	19.4
18	201°30.7	45°27.1	N11°23.4	30°09.3	N04°30.1	197°59.8	N00°03.3	21°08.5	N02°19.3
19	216°33.1	60°27.0	22.2	45°10.4	29.5	213°02.2	03.4	36°10.8	19.3
20	231°35.6	75°26.8	21.1	60°11.6	28.9	228°04.6	03.4	51°13.1	19.2
21	246°38.1	90°26.7	•• 20.0	75°12.7	•• 28.3	243°06.9	•• 03.4	66°15.4	•• 19.1
22	261°40.5	105°26.6	18.8	90°13.8	27.7	258°09.3	03.4	81°17.7	19.0
23	276°43.0	120°26.4	17.7	105°14.9	27.1	273°11.7	03.4	96°20.0	19.0
Mer.pass. 04:36		ν -0.2' d -1.1' m -4.11		ν 1.1' d -0.6' m 1.41		ν 2.4' d 0.0' m -2.54		ν 2.3' d -0.1' m 0.99	

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	291°45.5	135°26.3	N11°16.6	120°16.1	N04°26.5	288°14.1	N00°03.5	111°22.3	N02°18.9
1	306°47.9	150°26.1	15.4	135°17.2	25.9	303°16.5	03.5	126°24.6	18.8
2	321°50.4	165°26.0	14.3	150°18.3	25.3	318°18.9	03.5	141°26.9	18.7
3	336°52.9	180°25.9	•• 13.2	165°19.4	•• 24.6	333°21.3	•• 03.5	156°29.2	•• 18.7
4	351°55.3	195°25.7	12.1	180°20.6	24.0	348°23.7	03.5	171°31.6	18.6
5	6°57.8	210°25.6	10.9	195°21.7	23.4	3°26.1	03.6	186°33.9	18.5
6	22°00.3	225°25.5	N11°09.8	210°22.8	N04°22.8	18°28.4	N00°03.6	201°36.2	N02°18.4
7	37°02.7	240°25.3	08.7	225°23.9	22.2	33°30.8	03.6	216°38.5	18.4
8	52°05.2	255°25.2	07.5	240°25.1	21.6	48°33.2	03.6	231°40.8	18.3
9	67°07.6	270°25.1	•• 06.4	255°26.2	•• 21.0	63°35.6	•• 03.6	246°43.1	•• 18.2
10	82°10.1	285°24.9	05.2	270°27.3	20.4	78°38.0	03.6	261°45.4	18.1
11	97°12.6	300°24.8	04.1	285°28.4	19.8	93°40.4	03.7	276°47.7	18.0
12	112°15.0	315°24.7	N11°03.0	300°29.6	N04°19.1	108°42.8	N00°03.7	291°50.0	N02°18.0
13	127°17.5	330°24.5	01.8	315°30.7	18.5	123°45.2	03.7	306°52.3	17.9
14	142°20.0	345°24.4	11°00.7	330°31.8	17.9	138°47.6	03.7	321°54.6	17.8
15	157°22.4	0°24.3	10°59.6	345°32.9	•• 17.3	153°50.0	•• 03.7	336°56.9	•• 17.7
16	172°24.9	15°24.1	58.4	0°34.1	16.7	168°52.4	03.7	351°59.2	17.7
17	187°27.4	30°24.0	57.3	15°35.2	16.1	183°54.8	03.8	7°01.5	17.6
18	202°29.8	45°23.9	N10°56.2	30°36.3	N04°15.5	198°57.2	N00°03.8	22°03.8	N02°17.5
19	217°32.3	60°23.8	55.0	45°37.4	14.9	213°59.5	03.8	37°06.2	17.4
20	232°34.7	75°23.6	53.9	60°38.5	14.3	229°01.9	03.8	52°08.5	17.4
21	247°37.2	90°23.5	•• 52.7	75°39.7	•• 13.6	244°04.3	•• 03.8	67°10.8	•• 17.3
22	262°39.7	105°23.4	51.6	90°40.8	13.0	259°06.7	03.8	82°13.1	17.2
23	277°42.1	120°23.3	50.5	105°41.9	12.4	274°09.1	03.9	97°15.4	17.1
Mer.pass. 04:32		ν -0.1' d -1.1' m -4.12		ν 1.1' d -0.6' m 1.42		ν 2.4' d 0.0' m -2.55		ν 2.3' d -0.1' m 0.99	

Jul 12 Mon	SHA	Mer.pass
Venus	205°46.5	14:58
Mars	189°34.7	16:01
Jupiter	356°32.6	04:54
Saturn	179°44.2	16:39

Jul 13 Tue	SHA	Mer.pass
Venus	204°43.5	14:58
Mars	189°02.7	16:00
Jupiter	356°30.5	04:50
Saturn	179°40.6	16:36

Jul 14 Wed	SHA	Mer.pass
Venus	203°40.8	14:58
Mars	188°30.6	15:58
Jupiter	356°28.6	04:46
Saturn	179°36.9	16:32

Horizontal parallax	
Venus:	0.1
Mars:	0.1

h Sun Moon						
Mon	GHA	Dec	GHA	ν	Dec	d HP
0	178°36.5	N22°00.7	176°07.4	5.2'	N20°43.1	-9.3' 60.3'
1	193°36.4	00.3	190°31.6	5.2'	20°33.8	-9.4' 60.4'
2	208°36.3	22°00.0	204°55.8	5.3'	20°24.4	-9.6' 60.4'
3	223°36.2	21°59.6	219°20.0	5.3'	20°14.9	-9.7' 60.4'
4	238°36.1	59.3	233°44.4	5.4'	20°05.1	-9.8' 60.4'
5	253°36.1	58.9	248°08.8	5.5'	19°55.3	-10.0' 60.4'
6	268°36.0	N21°58.6	262°33.2	5.5'	N19°45.3	-10.1' 60.4'
7	283°35.9	58.3	276°57.7	5.6'	19°35.2	-10.3' 60.5'
8	298°35.8	57.9	291°22.3	5.6'	19°24.9	-10.4' 60.5'
9	313°35.7	· · 57.6	305°47.0	5.7'	19°14.6	-10.5' 60.5'
10	328°35.7	57.2	320°11.7	5.8'	19°04.0	-10.7' 60.5'
11	343°35.6	56.9	334°36.4	5.8'	18°53.4	-10.8' 60.5'
12	358°35.5	N21°56.5	349°01.3	5.9'	N18°42.6	-10.9' 60.5'
13	13°35.4	56.2	3°26.2	6.0'	18°31.7	-11.0' 60.5'
14	28°35.3	55.8	17°51.2	6.0'	18°20.6	-11.2' 60.5'
15	43°35.3	· · 55.5	32°16.2	6.1'	18°09.5	-11.3' 60.5'
16	58°35.2	55.1	46°41.3	6.2'	17°58.2	-11.4' 60.6'
17	73°35.1	54.7	61°06.5	6.3'	17°46.8	-11.5' 60.6'
18	88°35.0	N21°54.4	75°31.8	6.3'	N17°35.2	-11.7' 60.6'
19	103°35.0	54.0	89°57.1	6.4'	17°23.6	-11.8' 60.6'
20	118°34.9	53.7	104°22.5	6.5'	17°11.8	-11.9' 60.6'
21	133°34.8	· · 53.3	118°48.0	6.5'	16°59.9	-12.0' 60.6'
22	148°34.7	53.0	133°13.5	6.6'	16°47.9	-12.1' 60.6'
23	163°34.6	52.6	147°39.1	6.7'	16°35.8	-12.2' 60.6'
SD = 15.7' $d = -0.3'$ SD = 16.5'						
Tue	GHA	Dec	GHA	ν	Dec	d HP
0	178°34.6	N21°52.3	162°04.8	6.8'	N16°23.6	-12.3' 60.6'
1	193°34.5	51.9	176°30.6	6.8'	16°11.3	-12.4' 60.6'
2	208°34.4	51.5	190°56.4	6.9'	15°58.8	-12.5' 60.6'
3	223°34.3	· · 51.2	205°22.3	7.0'	15°46.3	-12.6' 60.6'
4	238°34.3	50.8	219°48.3	7.0'	15°33.7	-12.7' 60.6'
5	253°34.2	50.5	234°14.3	7.1'	15°20.9	-12.8' 60.6'
6	268°34.1	N21°50.1	248°40.4	7.2'	N15°08.1	-12.9' 60.6'
7	283°34.0	49.7	263°06.6	7.3'	14°55.1	-13.0' 60.6'
8	298°34.0	49.4	277°32.9	7.3'	14°42.1	-13.1' 60.6'
9	313°33.9	· · 49.0	291°59.2	7.4'	14°29.0	-13.2' 60.6'
10	328°33.8	48.6	306°25.6	7.5'	14°15.8	-13.3' 60.6'
11	343°33.7	48.3	320°52.1	7.5'	14°02.5	-13.4' 60.6'
12	358°33.7	N21°47.9	335°18.6	7.6'	N13°49.1	-13.5' 60.6'
13	13°33.6	47.5	349°45.2	7.7'	13°35.6	-13.6' 60.6'
14	28°33.5	47.2	4°11.9	7.8'	13°22.0	-13.6' 60.6'
15	43°33.4	· · 46.8	18°38.7	7.8'	13°08.4	-13.7' 60.6'
16	58°33.4	46.4	33°05.5	7.9'	12°54.6	-13.8' 60.6'
17	73°33.3	46.1	47°32.4	8.0'	12°40.8	-13.9' 60.6'
18	88°33.2	N21°45.7	61°59.4	8.0'	N12°27.0	-14.0' 60.6'
19	103°33.1	45.3	76°26.4	8.1'	12°13.0	-14.0' 60.6'
20	118°33.1	45.0	90°53.5	8.2'	11°59.0	-14.1' 60.6'
21	133°33.0	· · 44.6	105°20.7	8.2'	11°44.9	-14.2' 60.6'
22	148°32.9	44.2	119°47.9	8.3'	11°30.7	-14.2' 60.6'
23	163°32.9	43.8	134°15.2	8.4'	11°16.5	-14.3' 60.6'
SD = 15.7' $d = -0.4'$ SD = 16.5'						
Wed	GHA	Dec	GHA	ν	Dec	d HP
0	178°32.8	N21°43.5	148°42.6	8.4'	N11°02.2	-14.4' 60.6'
1	193°32.7	43.1	163°10.1	8.5'	10°47.8	-14.4' 60.6'
2	208°32.6	42.7	177°37.6	8.6'	10°33.4	-14.5' 60.6'
3	223°32.6	· · 42.3	192°05.1	8.6'	10°18.9	-14.5' 60.6'
4	238°32.5	42.0	206°32.8	8.7'	10°04.3	-14.6' 60.6'
5	253°32.4	41.6	221°00.5	8.8'	09°49.7	-14.7' 60.6'
6	268°32.4	N21°41.2	235°28.2	8.8'	N09°35.1	-14.7' 60.6'
7	283°32.3	40.8	249°56.0	8.9'	09°20.4	-14.8' 60.6'
8	298°32.2	40.4	264°23.9	8.9'	09°05.6	-14.8' 60.5'
9	313°32.1	· · 40.1	278°51.9	9.0'	08°50.8	-14.9' 60.5'
10	328°32.1	39.7	293°19.9	9.1'	08°35.9	-14.9' 60.5'
11	343°32.0	39.3	307°47.9	9.1'	08°21.0	-14.9' 60.5'
12	358°31.9	N21°38.9	322°16.0	9.2'	N08°06.1	-15.0' 60.5'
13	13°31.9	38.5	336°44.2	9.2'	07°51.1	-15.0' 60.5'
14	28°31.8	38.2	351°12.4	9.3'	07°36.1	-15.1' 60.5'
15	43°31.7	· · 37.8	5°40.7	9.3'	07°21.0	-15.1' 60.5'
16	58°31.7	37.4	20°09.1	9.4'	07°05.9	-15.1' 60.5'
17	73°31.6	37.0	34°37.5	9.4'	06°50.8	-15.2' 60.4'
18	88°31.5	N21°36.6	49°05.9	9.5'	N06°35.6	-15.2' 60.4'
19	103°31.5	36.2	63°34.4	9.6'	06°20.4	-15.2' 60.4'
20	118°31.4	35.9	78°03.0	9.6'	06°05.2	-15.3' 60.4'
21	133°31.3	· · 35.5	92°31.6	9.6'	05°50.0	-15.3' 60.4'
22	148°31.3	35.1	107°00.2	9.7'	05°34.7	-15.3' 60.4'
23	163°31.2	34.7	121°28.9	9.7'	05°19.4	-15.3' 60.4'
SD = 15.7' $d = -0.4'$ SD = 16.5'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°						
68°						
66°	////	////	01:26	22:42	////	////
64°	////	////	02:09	22:00	////	////
62°	////	00:29	02:38	21:32	23:32	////
60°	////	01:38	03:00	21:11	22:31	////
N 58°	////	02:11	03:17	20:53	21:59	////
56°	00:27	02:35	03:32	20:39	21:35	23:35
54°	01:30	02:54	03:44	20:26	21:17	22:39
52°	02:00	03:09	03:55	20:15	21:01	22:09
50°	02:23	03:23	04:05	20:06	20:48	21:47
45°	03:02	03:49	04:26	19:45	20:21	21:08
N 40°	03:30	04:10	04:42	19:29	20:01	20:41
35°	03:51	04:27	04:56	19:15	19:44	20:20
30°	04:08	04:41	05:08	19:03	19:30	20:03
20°	04:35	05:04	05:28	18:43	19:07	19:36
N 10°	04:56	05:23	05:46	18:26	18:48	19:15
0°	05:14	05:40	06:02	18:09	18:32	18:58
S 10°	05:30	05:56	06:18	17:53	18:16	18:42
20°	05:45	06:12	06:35	17:36	18:00	18:27
30°	05:59	06:29	06:55	17:17	17:43	18:12
35°	06:07	06:39	07:06	17:05	17:33	18:04
40°	06:16	06:49	07:19	16:52	17:22	17:56
45°	06:25	07:02	07:35	16:37	17:10	17:47
S 50°	06:35	07:16	07:53	16:18	16:56	17:36
52°	06:40	07:23	08:02	16:09	16:49	17:32
54°	06:45	07:30	08:12	16:00	16:42	17:27
56°	06:50	07:38	08:24	15:48	16:34	17:22
58°	06:56	07:47	08:36	15:35	16:25	17:16
S 60°	07:03	07:57	08:52	15:20	16:14	17:09

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°		03:25	06:17	23:28	22:32	22:00
N 70°		04:05	06:33	22:46	22:14	21:52
68°	01:56	04:32	06:45	22:17	22:00	21:46
66°	02:40	04:52	06:55	21:56	21:48	21:41
64°	03:09	05:09	07:03	21:38	21:38	21:37
62°	03:31	05:22	07:10	21:23	21:29	21:33
60°	03:49	05:33	07:17	21:11	21:22	21:29
N 58°	04:04	05:43	07:22	21:00	21:15	21:26
56°	04:16	05:52	07:27	20:51	21:09	21:24
54°	04:27	05:59	07:31	20:43	21:04	21:21
52°	04:37	06:06	07:35	20:35	20:59	21:19
50°	04:45	06:12	07:38	20:28	20:55	21:17
45°	05:04	06:25	07:46	20:14	20:45	21:13
N 40°	05:18	06:36	07:52	20:02	20:37	21:09
35°	05:31	06:45	07:57	19:51	20:31	21:06
30°	05:42	06:53	08:02	19:42	20:24	21:03
20°	06:01	07:06	08:10	19:27	20:14	20:58
N 10°	06:17	07:18	08:17	19:13	20:04	20:53
0°	06:32	07:29	08:24	18:59	19:56	20:49
S 10°	06:47	07:40	08:31	18:46	19:47	20:45
20°	07:03	07:52	08:38	18:32	19:37	20:40
30°	07:21	08:06	08:46	18:16	19:26	20:34
35°	07:32	08:13	08:50	18:06	19:19	20:31
40°	07:44	08:22	08:55	17:55	19:12	20:28
45°	07:58	08:32	09:01	17:42	19:03	20:24
S 50°	08:15	08:45	09:08	17:26	18:53	20:19
52°	08:24	08:50	09:12	17:18	18:48	20:16
54°	08:33	08:57	09:15	17:10	18:43	20:14
56°	08:43	09:04	09:19	17:01	18:37	20:11
58°	08:54	09:11	09:24	16:50	18:30	20:08
S 60°	09:08	09:20	09:29	16:37	18:22	20:04

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time	12 ^h		Mer. Pass.		
	00 ^h	mm:ss	hh:mm	Upper	Lower	1-3 0-7%
12	05:34	05:38	12:06	12:46	00:16	
13	05:42	05:45	12:06	13:43	01:15	
14	05:49	05:52	12:06	14:36	02:10	

Aries Venus

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	293°43.7	135°20.3	N10°21.8	121°09.9	N03°57.1	290°09.1	N00°04.2	113°13.0	N02°15.2
1	308°46.2	150°20.2	20.7	136°11.1	56.5	305°11.5	04.2	128°15.3	15.1
2	323°48.7	165°20.1	19.5	151°12.2	55.9	320°13.9	04.2	143°17.6	15.0
3	338°51.1	180°20.0	· · 18.4	166°13.3	· · 55.2	335°16.3	· · 04.3	158°19.9	· · 14.9
4	353°53.6	195°19.9	17.2	181°14.4	54.6	350°18.7	04.3	173°22.2	14.9
5	8°56.1	210°19.8	16.1	196°15.5	54.0	5°21.1	04.3	188°24.5	14.8
6	23°58.5	225°19.7	N10°14.9	211°16.7	N03°53.4	20°23.5	N00°04.3	203°26.8	N02°14.7
7	39°01.0	240°19.6	13.8	226°17.8	52.8	35°25.9	04.3	218°29.1	14.6
8	54°03.5	255°19.5	12.6	241°18.9	52.2	50°28.3	04.3	233°31.4	14.5
9	69°05.9	270°19.4	· · 11.5	256°20.0	· · 51.5	65°30.8	· · 04.3	248°33.7	· · 14.5
10	84°08.4	285°19.3	10.3	271°21.1	50.9	80°33.2	04.3	263°36.0	14.4
11	99°10.8	300°19.2	09.2	286°22.3	50.3	95°35.6	04.3	278°38.3	14.3
12	114°13.3	315°19.1	N10°08.0	301°23.4	N03°49.7	110°38.0	N00°04.4	293°40.6	N02°14.2
13	129°15.8	330°19.0	06.8	316°24.5	49.1	125°40.4	04.4	308°42.9	14.2
14	144°18.2	345°18.9	05.7	331°25.6	48.5	140°42.8	04.4	323°45.2	14.1
15	159°20.7	0°18.8	· · 04.5	346°26.7	· · 47.9	155°45.2	· · 04.4	338°47.5	· · 14.0
16	174°23.2	15°18.7	03.4	1°27.9	47.2	170°47.6	04.4	353°49.8	13.9
17	189°25.6	30°18.6	02.2	16°29.0	46.6	185°50.0	04.4	8°52.1	13.8
18	204°28.1	45°18.5	N10°01.1	31°30.1	N03°46.0	200°52.4	N00°04.4	23°54.4	N02°13.8
19	219°30.6	60°18.4	09°59.9	46°31.2	45.4	215°54.8	04.4	38°56.7	13.7
20	234°33.0	75°18.3	58.8	61°32.3	44.8	230°57.2	04.4	53°59.0	13.6
21	249°35.5	90°18.2	· · 57.6	76°33.4	· · 44.2	245°59.6	· · 04.4	69°01.3	· · 13.5
22	264°38.0	105°18.1	56.4	91°34.6	43.5	261°02.1	04.5	84°03.6	13.4
23	279°40.4	120°18.0	55.3	106°35.7	42.9	276°04.5	04.5	99°05.9	13.4
Mer.pass. 04:24		ν -0.1' d -1.2' m -4.13		ν 1.1' d -0.6' m 1.42		ν 2.4' d 0.0' m -2.57		ν 2.3' d -0.1' m 0.99	

Jul 15 Thu	SHA	Mer.pass
Venus	202°38.5	14:59
Mars	187°58.4	15:56
Jupiter	356°26.9	04:42
Saturn	179°33.1	16:28

Jul 16 Fri	SHA	Mer.pass
Venus	201°36.6	14:59
Mars	187°26.2	15:54
Jupiter	356°25.4	04:39
Saturn	179°29.2	16:25

Jul 17 Sat	SHA	Mer.pass
Venus	200°35.0	14:59
Mars	186°53.9	15:52
Jupiter	356°24.0	04:35
Saturn	179°25.3	16:21

Horizontal parallax	
Venus:	0.2
Mars:	0.1

h	Sun			Moon			
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	178°31.1	N21°34.3	135°57.7	9.8'	N05°04.1	-15.3'	60.3'
1	193°31.1	33.9	150°26.5	9.8'	04°48.7	-15.4'	60.3'
2	208°31.0	33.5	164°55.3	9.9'	04°33.4	-15.4'	60.3'
3	223°30.9	· · 33.1	179°24.2	9.9'	04°18.0	-15.4'	60.3'
4	238°30.9	32.7	193°53.1	10.0'	04°02.6	-15.4'	60.3'
5	253°30.8	32.3	208°22.1	10.0'	03°47.2	-15.4'	60.3'
6	268°30.7	N21°32.0	222°51.1	10.0'	N03°31.8	-15.4'	60.2'
7	283°30.7	31.6	237°20.1	10.1'	03°16.4	-15.4'	60.2'
8	298°30.6	31.2	251°49.2	10.1'	03°00.9	-15.4'	60.2'
9	313°30.5	· · 30.8	266°18.3	10.2'	02°45.5	-15.4'	60.2'
10	328°30.5	30.4	280°47.5	10.2'	02°30.0	-15.4'	60.2'
11	343°30.4	30.0	295°16.7	10.2'	02°14.6	-15.5'	60.2'
12	358°30.3	N21°29.6	309°45.9	10.3'	N01°59.1	-15.5'	60.1'
13	13°30.3	29.2	324°15.2	10.3'	01°43.7	-15.4'	60.1'
14	28°30.2	28.8	338°44.5	10.3'	01°28.2	-15.4'	60.1'
15	43°30.2	· · 28.4	353°13.8	10.4'	01°12.8	-15.4'	60.1'
16	58°30.1	28.0	7°43.2	10.4'	00°57.3	-15.4'	60.1'
17	73°30.0	27.6	22°12.6	10.4'	00°41.9	-15.4'	60.0'
18	88°30.0	N21°27.2	36°42.0	10.4'	N00°26.5	-15.4'	60.0'
19	103°29.9	26.8	51°11.4	10.5'	N00°11.1	-15.4'	60.0'
20	118°29.8	26.4	65°40.9	10.5'	S00°04.3	15.4'	60.0'
21	133°29.8	· · 26.0	80°10.4	10.5'	00°19.7	15.4'	59.9'
22	148°29.7	25.6	94°39.9	10.5'	00°35.1	15.4'	59.9'
23	163°29.7	25.2	109°09.5	10.6'	00°50.5	15.3'	59.9'
	SD = 15.7'	$d = -0.4'$	SD = 16.5'				
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	178°29.6	N21°24.8	123°39.1	10.6'	S01°05.8	15.3'	59.9'
1	193°29.5	24.4	138°08.7	10.6'	01°21.2	15.3'	59.9'
2	208°29.5	24.0	152°38.3	10.6'	01°36.5	15.3'	59.8'
3	223°29.4	· · 23.6	167°07.9	10.7'	01°51.8	15.3'	59.8'
4	238°29.4	23.2	181°37.6	10.7'	02°07.0	15.2'	59.8'
5	253°29.3	22.7	196°07.2	10.7'	02°22.3	15.2'	59.8'
6	268°29.2	N21°22.3	210°36.9	10.7'	S02°37.5	15.2'	59.7'
7	283°29.2	21.9	225°06.6	10.7'	02°52.7	15.2'	59.7'
8	298°29.1	21.5	239°36.3	10.7'	03°07.8	15.1'	59.7'
9	313°29.1	· · 21.1	254°06.0	10.7'	03°22.9	15.1'	59.7'
10	328°29.0	20.7	268°35.8	10.7'	03°38.0	15.1'	59.6'
11	343°28.9	20.3	283°05.5	10.8'	03°53.1	15.0'	59.6'
12	358°28.9	N21°19.9	297°35.3	10.8'	S04°08.1	15.0'	59.6'
13	13°28.8	19.5	312°05.0	10.8'	04°23.1	15.0'	59.6'
14	28°28.8	19.0	326°34.8	10.8'	04°38.1	14.9'	59.5'
15	43°28.7	· · 18.6	341°04.6	10.8'	04°53.0	14.9'	59.5'
16	58°28.7	18.2	355°34.4	10.8'	05°07.9	14.8'	59.5'
17	73°28.6	17.8	10°04.2	10.8'	05°22.7	14.8'	59.5'
18	88°28.5	N21°17.4	24°34.0	10.8'	S05°37.5	14.7'	59.4'
19	103°28.5	17.0	39°03.8	10.8'	05°52.2	14.7'	59.4'
20	118°28.4	16.6	53°33.6	10.8'	06°06.9	14.7'	59.4'
21	133°28.4	· · 16.1	68°03.4	10.8'	06°21.6	14.6'	59.4'
22	148°28.3	15.7	82°33.2	10.8'	06°36.2	14.6'	59.3'
23	163°28.3	15.3	97°03.0	10.8'	06°50.8	14.5'	59.3'
	SD = 15.7'	$d = -0.4'$	SD = 16.3'				
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	178°28.2	N21°14.9	111°32.8	10.8'	S07°05.3	14.5'	59.3'
1	193°28.1	14.5	126°02.6	10.8'	07°19.7	14.4'	59.3'
2	208°28.1	14.0	140°32.4	10.8'	07°34.1	14.3'	59.2'
3	223°28.0	· · 13.6	155°02.1	10.8'	07°48.5	14.3'	59.2'
4	238°28.0	13.2	169°31.9	10.8'	08°02.8	14.2'	59.2'
5	253°27.9	12.8	184°01.7	10.8'	08°17.0	14.2'	59.1'
6	268°27.9	N21°12.4	198°31.5	10.8'	S08°31.2	14.1'	59.1'
7	283°27.8	11.9	213°01.2	10.7'	08°45.3	14.1'	59.1'
8	298°27.8	11.5	227°31.0	10.7'	08°59.3	14.0'	59.1'
9	313°27.7	· · 11.1	242°00.7	10.7'	09°13.3	13.9'	59.0'
10	328°27.7	10.7	256°30.4	10.7'	09°27.3	13.9'	59.0'
11	343°27.6	10.2	271°00.1	10.7'	09°41.1	13.8'	59.0'
12	358°27.6	N21°09.8	285°29.9	10.7'	S09°54.9	13.7'	58.9'
13	13°27.5	09.4	299°59.5	10.7'	10°08.7	13.7'	58.9'
14	28°27.4	08.9	314°29.2	10.7'	10°22.3	13.6'	58.9'
15	43°27.4	· · 08.5	328°58.9	10.6'	10°35.9	13.5'	58.9'
16	58°27.3	08.1	343°28.5	10.6'	10°49.5	13.5'	58.8'
17	73°27.3	07.7	357°58.1	10.6'	11°02.9	13.4'	58.8'
18	88°27.2	N21°07.2	12°27.8	10.6'	S11°16.3	13.3'	58.8'
19	103°27.2	06.8	26°57.4	10.6'	11°29.6	13.2'	58.7'
20	118°27.1	06.4	41°26.9	10.6'	11°42.9	13.2'	58.7'
21	133°27.1	· · 05.9	55°56.5	10.5'	11°56.0	13.1'	58.7'
22	148°27.0	05.5	70°26.0	10.5'	12°09.1	13.0'	58.7'
23	163°27.0	05.1	84°55.5	10.5'	12°22.1	12.9'	58.6'
	SD = 15.7'	$d = -0.4'$	SD = 16.2'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	□	□	□	□	□	□
N 70°	□	□	□	□	□	□
68°	□	□	□	□	□	□
66°	////	////	01:39	22:29	////	////
64°	////	////	02:18	21:52	////	////
62°	////	00:56	02:45	21:26	23:10	////
60°	////	01:48	03:05	21:06	22:22	///
N 58°	////	02:18	03:22	20:49	21:52	////
56°	00:51	02:40	03:36	20:35	21:30	23:15
54°	01:39	02:58	03:48	20:23	21:12	22:31
52°	02:07	03:14	03:59	20:12	20:58	22:04
50°	02:28	03:26	04:08	20:03	20:45	21:43
45°	03:06	03:52	04:28	19:43	20:19	21:05
N 40°	03:33	04:13	04:44	19:27	19:59	20:39
35°	03:53	04:29	04:58	19:14	19:43	20:18
30°	04:10	04:43	05:10	19:02	19:29	20:01
20°	04:37	05:05	05:29	18:43	19:07	19:35
N 10°	04:57	05:24	05:47	18:25	18:48	19:15
0°	05:14	05:40	06:02	18:10	18:32	18:58
S 10°	05:30	05:56	06:18	17:54	18:16	18:42
20°	05:44	06:11	06:35	17:37	18:01	18:28
30°	05:59	06:28	06:54	17:18	17:44	18:14
35°	06:06	06:38	07:05	17:07	17:35	18:06
40°	06:15	06:48	07:18	16:55	17:24	17:58
45°	06:23	07:00	07:33	16:40	17:13	17:49
S 50°	06:33	07:14	07:51	16:22	16:59	17:39
52°	06:38	07:20	08:00	16:13	16:52	17:35
54°	06:43	07:27	08:09	16:03	16:45	17:30
56°	06:48	07:35	08:20	15:53	16:37	17:25
58°	06:53	07:44	08:32	15:40	16:29	17:19
S 60°	07:00	07:54	08:47	15:26	16:19	17:13

July 18, 19, 20 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn	
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	295°42.0	135°15.8	N09°26.2	122°03.6	N03°27.5	292°04.8	N00°04.7	115°03.3	N02°11.4
1	310°44.5	150°15.8	25.0	137°04.7	26.9	307°07.2	04.7	130°05.6	11.3
2	325°46.9	165°15.7	23.9	152°05.8	26.3	322°09.7	04.7	145°07.9	11.2
3	340°49.4	180°15.6	· · 22.7	167°06.9	· · 25.6	337°12.1	· · 04.7	160°10.2	· · 11.1
4	355°51.9	195°15.5	21.5	182°08.0	25.0	352°14.5	04.7	175°12.5	11.0
5	10°54.3	210°15.5	20.4	197°09.1	24.4	7°16.9	04.7	190°14.8	10.9
6	25°56.8	225°15.4	N09°19.2	212°10.3	N03°23.8	22°19.3	N00°04.7	205°17.1	N02°10.9
7	40°59.3	240°15.3	18.0	227°11.4	23.2	37°21.8	04.7	220°19.4	10.8
8	56°01.7	255°15.2	16.9	242°12.5	22.5	52°24.2	04.7	235°21.7	10.7
9	71°04.2	270°15.2	· · 15.7	257°13.6	· · 21.9	67°26.6	· · 04.7	250°24.0	· · 10.6
10	86°06.7	285°15.1	14.5	272°14.7	21.3	82°29.0	04.7	265°26.3	10.5
11	101°09.1	300°15.0	13.4	287°15.8	20.7	97°31.4	04.7	280°28.6	10.5
12	116°11.6	315°14.9	N09°12.2	302°16.9	N03°20.1	112°33.9	N00°04.7	295°30.8	N02°10.4
13	131°14.1	330°14.9	11.0	317°18.1	19.4	127°36.3	04.7	310°33.1	10.3
14	146°16.5	345°14.8	09.8	332°19.2	18.8	142°38.7	04.7	325°35.4	10.2
15	161°19.0	0°14.7	· · 08.7	347°20.3	· · 18.2	157°41.1	· · 04.7	340°37.7	· · 10.1
16	176°21.4	15°14.7	07.5	2°21.4	17.6	172°43.6	04.7	355°40.0	10.1
17	191°23.9	30°14.6	06.3	17°22.5	17.0	187°46.0	04.8	10°42.3	10.0
18	206°26.4	45°14.5	N09°05.2	32°23.6	N03°16.4	202°48.4	N00°04.8	25°44.6	N02°09.9
19	221°28.8	60°14.4	04.0	47°24.7	15.7	217°50.8	04.8	40°46.9	09.8
20	236°31.3	75°14.4	02.8	62°25.8	15.1	232°53.3	04.8	55°49.2	09.7
21	251°33.8	90°14.3	· · 01.6	77°27.0	· · 14.5	247°55.7	· · 04.8	70°51.5	· · 09.6
22	266°36.2	105°14.2	09°00.5	92°28.1	13.9	262°58.1	04.8	85°53.8	09.6
23	281°38.7	120°14.2	08°59.3	107°29.2	13.3	278°00.5	04.8	100°56.1	09.5
Mer.pass. 04:16		ν -0.1' d -1.2' m -4.15		ν 1.1' d -0.6' m 1.43		ν 2.4' d 0.0' m -2.58		ν 2.3' d -0.1' m 0.99	

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	296°41.2	135°14.1	N08°58.1	122°30.3	N03°12.6	293°03.0	N00°04.8	115°58.4	N02°09.4
1	311°43.6	150°14.1	56.9	137°31.4	12.0	308°05.4	04.8	131°00.7	09.3
2	326°46.1	165°14.0	55.8	152°32.5	11.4	323°07.8	04.8	146°03.0	09.2
3	341°48.5	180°13.9	· · 54.6	167°33.6	· · 10.8	338°10.2	· · 04.8	161°05.2	· · 09.2
4	356°51.0	195°13.9	53.4	182°34.7	10.1	353°12.7	04.8	176°07.5	09.1
5	11°53.5	210°13.8	52.2	197°35.8	09.5	8°15.1	04.8	191°09.8	09.0
6	26°55.9	225°13.7	N08°51.1	212°37.0	N03°08.9	23°17.5	N00°04.8	206°12.1	N02°08.9
7	41°58.4	240°13.7	49.9	227°38.1	08.3	38°19.9	04.8	221°14.4	08.8
8	57°00.9	255°13.6	48.7	242°39.2	07.7	53°22.4	04.8	236°16.7	08.7
9	72°03.3	270°13.6	· · 47.5	257°40.3	· · 07.0	68°24.8	· · 04.8	251°19.0	· · 08.7
10	87°05.8	285°13.5	46.4	272°41.4	06.4	83°27.2	04.8	266°21.3	08.6
11	102°08.3	300°13.4	45.2	287°42.5	05.8	98°29.7	04.8	281°23.6	08.5
12	117°10.7	315°13.4	N08°44.0	302°43.6	N03°05.2	113°32.1	N00°04.8	296°25.9	N02°08.4
13	132°13.2	330°13.3	42.8	317°44.7	04.6	128°34.5	04.8	311°28.2	08.3
14	147°15.7	345°13.3	41.7	332°45.8	03.9	143°36.9	04.8	326°30.5	08.2
15	162°18.1	0°13.2	· · 40.5	347°47.0	· · 03.3	158°39.4	· · 04.8	341°32.8	· · 08.2
16	177°20.6	15°13.2	39.3	2°48.1	02.7	173°41.8	04.8	356°35.0	08.1
17	192°23.0	30°13.1	38.1	17°49.2	02.1	188°44.2	04.8	11°37.3	08.0
18	207°25.5	45°13.0	N08°36.9	32°50.3	N03°01.5	203°46.7	N00°04.8	26°39.6	N02°07.9
19	222°28.0	60°13.0	35.8	47°51.4	00.8	218°49.1	04.8	41°41.9	07.8
20	237°30.4	75°12.9	34.6	62°52.5	03°00.2	233°51.5	04.8	56°44.2	07.7
21	252°32.9	90°12.9	· · 33.4	77°53.6	02°59.6	248°54.0	· · 04.8	71°46.5	· · 07.7
22	267°35.4	105°12.8	32.2	92°54.7	59.0	263°56.4	04.8	86°48.8	07.6
23	282°37.8	120°12.8	31.0	107°55.8	58.3	278°58.8	04.8	101°51.1	07.5
Mer.pass. 04:13		ν -0.1' d -1.2' m -4.15		ν 1.1' d -0.6' m 1.44		ν 2.4' d 0.0' m -2.59		ν 2.3' d -0.1' m 0.99	

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	297°40.3	135°12.7	N08°29.9	122°56.9	N02°57.7	294°01.3	N00°04.8	116°53.4	N02°07.4
1	312°42.8	150°12.7	28.7	137°58.1	57.1	309°03.7	04.8	131°55.7	07.3
2	327°45.2	165°12.6	27.5	152°59.2	56.5	324°06.1	04.8	146°57.9	07.2
3	342°47.7	180°12.6	· · 26.3	168°00.3	· · 55.9	339°08.6	· · 04.8	162°00.2	· · 07.2
4	357°50.2	195°12.5	25.1	183°01.4	55.2	354°11.0	04.8	177°02.5	07.1
5	12°52.6	210°12.5	23.9	198°02.5	54.6	9°13.4	04.8	192°04.8	07.0
6	27°55.1	225°12.4	N08°22.8	213°03.6	N02°54.0	24°15.9	N00°04.8	207°07.1	N02°06.9
7	42°57.5	240°12.4	21.6	228°04.7	53.4	39°18.3	04.8	222°09.4	06.8
8	58°00.0	255°12.4	20.4	243°05.8	52.7	54°20.7	04.8	237°11.7	06.7
9	73°02.5	270°12.3	· · 19.2	258°06.9	· · 52.1	69°23.2	· · 04.8	252°14.0	· · 06.7
10	88°04.9	285°12.3	18.0	273°08.0	51.5	84°25.6	04.8	267°16.3	06.6
11	103°07.4	300°12.2	16.8	288°09.1	50.9	99°28.0	04.8	282°18.6	06.5
12	118°09.9	315°12.2	N08°15.7	303°10.2	N02°50.3	114°30.5	N00°04.8	297°20.8	N02°06.4
13	133°12.3	330°12.1	14.5	318°11.4	49.6	129°32.9	04.8	312°23.1	06.3
14	148°14.8	345°12.1	13.3	333°12.5	49.0	144°35.3	04.8	327°25.4	06.2
15	163°17.3	0°12.0	· · 12.1	348°13.6	· · 48.4	159°37.8	· · 04.8	342°27.7	· · 06.2
16	178°19.7	15°12.0	10.9	3°14.7	47.8	174°40.2	04.8	357°30.0	06.1
17	193°22.2	30°12.0	09.7	18°15.8	47.1	189°42.7	04.8	12°32.3	06.0
18	208°24.7	45°11.9	N08°08.5	33°16.9	N02°46.5	204°45.1	N00°04.8	27°34.6	N02°05.9
19	223°27.1	60°11.9	07.4	48°18.0	45.9	219°47.5	04.8	42°36.9	05.8
20	238°29.6	75°11.9	06.2	63°19.1	45.3	234°50.0	04.8	57°39.1	05.7
21	253°32.0	90°11.8	· · 05.0	78°20.2	· · 44.6	249°52.4	· · 04.8	72°41.4	· · 05.7
22	268°34.5	105°11.8	03.8	93°21.3	44.0	264°54.9	04.8	87°43.7	05.6
23	283°37.0	120°11.7	02.6	108°22.4	43.4	279°57.3	04.8	102°46.0	05.5
Mer.pass. 04:09		ν -0.0' d -1.2' m -4.16		ν 1.1' d -0.6' m 1.44		ν 2.4' d -0.0' m -2.60		ν 2.3' d -0.1' m 1.00	

Stars		
	SHA	Dec
Alpheratz	357°45.6	29°08.9
Ankaa	353°17.5	-42°14.5
Schedar	349°42.9	56°35.6
Diphda	348°57.9	-17°55.5
Achernar	335°28.2	-57°10.6
Hamal	328°03.2	23°30.7
Polaris	318°56.7	89°18.4
Acamar	315°20.0	-40°15.4
Menkar	314°17.4	4°08.0
Mirfak	308°43.7	49°53.8
Aldebaran	290°52.0	16°31.8
Rigel	281°14.3	-8°11.3
Capella	280°38.0	46°00.4
Bellatrix	278°34.5	6°21.6
Elnath	278°15.6	28°36.9
Alnilam	275°48.8	-1°11.7
Betelgeuse	271°03.9	7°24.6
Canopus	263°57.6	-52°42.0
Sirius	258°35.9	-16°43.8
Adhara	255°14.6	-28°59.1
Procyon	245°02.3	5°11.9
Pollux	243°30.6	28°00.0
Avior	234°19.6	-50°32.7
Suhail	222°54.5	-43°28.6
Miaplacidus	221°41.2	-69°45.8
Alphard	217°58.5	-8°42.3
Regulus	207°46.0	11°54.9
Dubhe	193°54.5	61°41.8
Denebola	182°35.9	14°30.8
Gienah	175°54.6	-17°36.2
Acrux	173°12.1	-63°09.8
Gacrux	172°03.6	-50°10.7
Alioth	166°22.5	55°54.3
Spica	158°33.5	-11°13.1
Alkaid	153°00.4	49°15.8
Hadar	148°50.9	-60°25.8
Menkent	148°10.1	-36°25.5
Arcturus	145°57.6	19°07.7
Rigil Kent.	139°54.5	-60°53.0
Kochab	137°19.3	74°07.0
Zuben'ubi	137°07.7	-16°05.2
Alphecca	126°12.6	26°40.9
Antares	112°28.6	-26°27.4
Atria	107°31.9	-69°03.0
Sabik	102°14.7	-15°44.3
Shaula	96°24.4	-37°06.7
Rasalhague	96°08.1	12°33.2
Eltanin	90°46.6	51°29.4
Kaus Aust.	83°46.2	-34°22.7
Vega	80°40.0	38°47.7
Nunki	76°00.5	-26°16.9
Altair	62°10.0	8°53.9
Peacock	53°21.8	-56°41.9
Deneb	49°32.5	45°19.1
Enif	33°48.9	9°55.5
Al Na'ir	27°45.9	-46°54.3
Fomalhaut	15°26.0	-29°33.7
Scheat	13°55.2	28°08.4
Markab	13°40.2	15°15.8

Jul 18 Sun	SHA	Mer.pass
Venus	199°33.8	14:59
Mars	186°21.6	15:51
Jupiter	356°22.8	04:31
Saturn	179°21.3	16:17

Jul 19 Mon	SHA	Mer.pass
Venus	198°33.0	14:59
Mars	185°49.1	15:49
Jupiter	356°21.8	04:27
Saturn	179°17.2	16:14

Jul 20 Tue	SHA	Mer.pass
Venus	197°32.4	14:59
Mars	185°16.6	15:47
Jupiter	356°21.0	04:23
Saturn	179°13.1	16:10

Horizontal parallax	
Venus:	0.2
Mars:	0.1

h Sun			Moon				
Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	178°26.9	N21°04.6	99°25.0	10.5'	S12°35.0	12.8'	58.6'
1	193°26.9	04.2	113°54.5	10.4'	12°47.9	12.8'	58.6'
2	208°26.8	03.8	128°23.9	10.4'	13°00.6	12.7'	58.5'
3	223°26.8	· · 03.3	142°53.4	10.4'	13°13.3	12.6'	58.5'
4	238°26.7	02.9	157°22.8	10.4'	13°25.9	12.5'	58.5'
5	253°26.7	02.5	171°52.1	10.4'	13°38.4	12.4'	58.5'
6	268°26.6	N21°02.0	186°21.5	10.3'	S13°50.9	12.3'	58.4'
7	283°26.6	01.6	200°50.8	10.3'	14°03.2	12.3'	58.4'
8	298°26.5	01.1	215°20.1	10.3'	14°15.5	12.2'	58.4'
9	313°26.5	· · 00.7	229°49.4	10.2'	14°27.6	12.1'	58.3'
10	328°26.5	21°00.3	244°18.6	10.2'	14°39.7	12.0'	58.3'
11	343°26.4	20°59.8	258°47.8	10.2'	14°51.7	11.9'	58.3'
12	358°26.4	N20°59.4	273°17.0	10.2'	S15°03.6	11.8'	58.3'
13	13°26.3	58.9	287°46.2	10.1'	15°15.4	11.7'	58.2'
14	28°26.3	58.5	302°15.3	10.1'	15°27.1	11.6'	58.2'
15	43°26.2	· · 58.0	316°44.4	10.1'	15°38.7	11.5'	58.2'
16	58°26.2	57.6	331°13.5	10.0'	15°50.2	11.4'	58.1'
17	73°26.1	57.2	345°42.5	10.0'	16°01.6	11.3'	58.1'
18	88°26.1	N20°56.7	0°11.6	10.0'	S16°13.0	11.2'	58.1'
19	103°26.0	56.3	14°40.5	10.0'	16°24.2	11.1'	58.0'
20	118°26.0	55.8	29°09.5	9.9'	16°35.3	11.0'	58.0'
21	133°25.9	· · 55.4	43°38.4	9.9'	16°46.3	10.9'	58.0'
22	148°25.9	54.9	58°07.3	9.9'	16°57.3	10.8'	58.0'
23	163°25.9	54.5	72°36.1	9.8'	17°08.1	10.7'	57.9'
SD = 15.7'			SD = 16.0'				
$d = -0.4'$							

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	178°25.8	N20°54.0	87°05.0	9.8'	S17°18.8	10.6'	57.9'
1	193°25.8	53.6	101°33.8	9.8'	17°29.4	10.5'	57.9'
2	208°25.7	53.1	116°02.5	9.7'	17°39.9	10.4'	57.8'
3	223°25.7	· · 52.7	130°31.2	9.7'	17°50.3	10.3'	57.8'
4	238°25.6	52.2	144°59.9	9.7'	18°00.6	10.2'	57.8'
5	253°25.6	51.8	159°28.6	9.6'	18°10.8	10.1'	57.8'
6	268°25.6	N20°51.3	173°57.2	9.6'	S18°20.9	10.0'	57.7'
7	283°25.5	50.9	188°25.8	9.6'	18°30.9	9.9'	57.7'
8	298°25.5	50.4	202°54.4	9.5'	18°40.7	9.8'	57.7'
9	313°25.4	· · 50.0	217°22.9	9.5'	18°50.5	9.6'	57.6'
10	328°25.4	49.5	231°51.4	9.5'	19°00.1	9.5'	57.6'
11	343°25.4	49.1	246°19.9	9.4'	19°09.7	9.4'	57.6'
12	358°25.3	N20°48.6	260°48.3	9.4'	S19°19.1	9.3'	57.6'
13	13°25.3	48.1	275°16.7	9.4'	19°28.4	9.2'	57.5'
14	28°25.2	47.7	289°45.1	9.3'	19°37.5	9.1'	57.5'
15	43°25.2	· · 47.2	304°13.4	9.3'	19°46.6	9.0'	57.5'
16	58°25.1	46.8	318°41.7	9.3'	19°55.6	8.8'	57.4'
17	73°25.1	46.3	333°10.0	9.2'	20°04.4	8.7'	57.4'
18	88°25.1	N20°45.8	347°38.2	9.2'	S20°13.1	8.6'	57.4'
19	103°25.0	45.4	2°06.4	9.2'	20°21.7	8.5'	57.4'
20	118°25.0	44.9	16°34.6	9.1'	20°30.2	8.4'	57.3'
21	133°25.0	· · 44.5	31°02.7	9.1'	20°38.6	8.2'	57.3'
22	148°24.9	44.0	45°30.8	9.1'	20°46.8	8.1'	57.3'
23	163°24.9	43.5	59°58.9	9.1'	20°54.9	8.0'	57.3'
SD = 15.7'			SD = 15.8'				
$d = -0.4'$							

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	178°24.8	N20°43.1	74°27.0	9.0'	S21°02.9	7.9'	57.2'
1	193°24.8	42.6	88°55.0	9.0'	21°10.8	7.7'	57.2'
2	208°24.8	42.1	103°23.0	9.0'	21°18.5	7.6'	57.2'
3	223°24.7	· · 41.7	117°51.0	8.9'	21°26.1	7.5'	57.1'
4	238°24.7	41.2	132°18.9	8.9'	21°33.6	7.4'	57.1'
5	253°24.7	40.7	146°46.8	8.9'	21°41.0	7.2'	57.1'
6	268°24.6	N20°40.3	161°14.7	8.9'	S21°48.3	7.1'	57.1'
7	283°24.6	39.8	175°42.5	8.8'	21°55.4	7.0'	57.0'
8	298°24.5	39.3	190°10.3	8.8'	22°02.4	6.9'	57.0'
9	313°24.5	· · 38.9	204°38.1	8.8'	22°09.3	6.7'	57.0'
10	328°24.5	38.4	219°05.9	8.7'	22°16.0	6.6'	57.0'
11	343°24.4	37.9	233°33.7	8.7'	22°22.6	6.5'	56.9'
12	358°24.4	N20°37.5	248°01.4	8.7'	S22°29.1	6.4'	56.9'
13	13°24.4	37.0	262°29.1	8.7'	22°35.4	6.2'	56.9'
14	28°24.3	36.5	276°56.8	8.7'	22°41.7	6.1'	56.9'
15	43°24.3	· · 36.1	291°24.4	8.6'	22°47.8	6.0'	56.8'
16	58°24.3	35.6	305°52.1	8.6'	22°53.7	5.8'	56.8'
17	73°24.2	35.1	320°19.7	8.6'	22°59.5	5.7'	56.8'
18	88°24.2	N20°34.6	334°47.3	8.6'	S23°05.2	5.6'	56.7'
19	103°24.2	34.2	349°14.8	8.6'	23°10.8	5.4'	56.7'
20	118°24.1	33.7	3°42.4	8.5'	23°16.2	5.3'	56.7'
21	133°24.1	· · 33.2	18°09.9	8.5'	23°21.5	5.2'	56.7'
22	148°24.1	32.7	32°37.4	8.5'	23°26.7	5.0'	56.6'
23	163°24.0	32.2	47°04.9	8.5'	23°31.7	4.9'	56.6'
SD = 15.7'			SD = 15.6'				
$d = -0.5'$							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	□	□	□	□	□	□
N 70°	□	□	□	□	□	□
68°	////	////	00:44	23:18	////	////
66°	////	////	01:52	22:17	////	////
64°	////	////	02:27	21:43	////	////
62°	////	01:14	02:52	21:19	22:53	////
60°	////	01:57	03:11	21:00	22:12	////
N 58°	////	02:25	03:27	20:44	21:45	////
56°	01:08	02:46	03:41	20:31	21:25	23:00
54°	01:48	03:04	03:52	20:19	21:08	22:22
52°	02:14	03:18	04:03	20:09	20:53	21:57
50°	02:33	03:31	04:12	20:00	20:41	21:38
45°	03:10	03:56	04:31	19:41	20:16	21:02
N 40°	03:36	04:15	04:47	19:25	19:57	20:36
35°	03:56	04:31	05:00	19:12	19:41	20:16
30°	04:12	04:45	05:11	19:01	19:28	20:00
20°	04:38	05:07	05:31	18:42	19:06	19:34
N 10°	04:58	05:25	05:47	18:25	18:48	19:14
0°	05:15	05:41	06:03	18:10	18:32	18:58
S 10°	05:30	05:56	06:18	17:55	18:17	18:43
20°	05:44	06:11	06:34	17:38	18:02	18:29
30°	05:58	06:27	06:53	17:20	17:46	18:15
35°	06:05	06:36	07:04	17:09	17:37	18:08
40°	06:13	06:46	07:16	16:57	17:27	18:00
45°	06:22	06:58	07:30	16:43	17:15	17:51
S 50°	06:31	07:11	07:48	16:25	17:02	17:42
52°	06:35	07:18	07:56	16:17	16:56	17:38
54°	06:40	07:24	08:06	16:07	16:49	17:33
56°	06:45	07:32	08:16	15:57	16:41	17:28
58°	06:50	07:40	08:28	15:45	16:33	17:23
S 60°	06:56	07:50	08:42	15:31	16:24	17:17

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	16:02	■	■	19:35	■	■
N 70°	15:19	■	■	20:20	■	■
68°	14:50	17:11	■	20:50	20:18	■
66°	14:28	16:28	18:42	21:13	21:02	20:39
64°	14:11	15:59	17:46	21:31	21:32	21:36
62°	13:58	15:37	17:13	21:46	21:54	22:09
60°	13:46	15:20	16:49	21:59	22:12	22:34
N 58°	13:36	15:05	16:30	22:10	22:27	22:53
56°	13:27	14:53	16:13	22:19	22:41	23:10
54°	13:19	14:42	16:00	22:28	22:52	23:24
52°	13:12	14:32	15:48	22:35	23:02	23:36
50°	13:06	14:24	15:37	22:42	23:11	23:47
45°	12:53	14:06	15:15	22:57	23:30	· ·
N 40°	12:42	13:51	14:58	23:09	23:46	· ·
35°	12:32	13:39	14:43	23:20	23:59	· ·
30°	12:24	13:28	14:30	23:29	· ·	00:11
20°	12:10	13:09	14:08	23:45	· ·	00:30
N 10°	11:58	12:54	13:49	23:58	· ·	00:48
0°	11:47	12:39	13:32	· ·	00:11	01:04
S 10°	11:36	12:24	13:14	· ·	00:25	01:20
20°	11:24	12:08	12:56	· ·	00:39	01:38
30°	11:11	11:50	12:34	· ·	00:55	01:58
35°	11:03	11:40	12:22	· ·	01:04	02:10
40°	10:54	11:28	12:08	00:05	01:15	02:23
45°	10:44	11:15	11:51	00:14	01:28	02:39
S 50°	10:32	10:58	11:30	00:24	01:43	02:59
52°	10:26	10:50	11:20	00:29	01:51	03:09
54°	10:20	10:41	11:09	00:35	01:59	03:19
56°	10:13	10:31	10:56	00:41	02:08	03:31
58°	10:05	10:20	10:42	00:48	02:18	03:46
S 60°	09:56	10:08	10:25	00:55	02:30	04:02

Day	Sun		Mer. Pass	Moon		Age 7-9
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Upper hh:mm	Lower hh:mm	
18	06:12	06:15	12:06	17:59	05:34	
19	06:17	06:19	12:06	18:51	06:25	
20	06:21	06:22	12:06	19:45	07:18	

July 21, 22, 23 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn	
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	298°39.4	135°11.7	N08°01.4	123°23.5	N02°42.8	294°59.7	N00°04.8	117°48.3	N02°05.4
1	313°41.9	150°11.7	08°00.2	138°24.6	42.2	310°02.2	04.8	132°50.6	05.3
2	328°44.4	165°11.6	07°59.0	153°25.7	41.5	325°04.6	04.8	147°52.9	05.2
3	343°46.8	180°11.6	•• 57.9	168°26.8	•• 40.9	340°07.1	•• 04.8	162°55.2	•• 05.2
4	358°49.3	195°11.6	56.7	183°28.0	40.3	355°09.5	04.7	177°57.4	05.1
5	13°51.8	210°11.5	55.5	198°29.1	39.7	10°11.9	04.7	192°59.7	05.0
6	28°54.2	225°11.5	N07°54.3	213°30.2	N02°39.0	25°14.4	N00°04.7	208°02.0	N02°04.9
7	43°56.7	240°11.5	53.1	228°31.3	38.4	40°16.8	04.7	223°04.3	04.8
8	58°59.1	255°11.4	51.9	243°32.4	37.8	55°19.3	04.7	238°06.6	04.7
9	74°01.6	270°11.4	•• 50.7	258°33.5	•• 37.2	70°21.7	•• 04.7	253°08.9	•• 04.6
10	89°04.1	285°11.4	49.5	273°34.6	36.5	85°24.2	04.7	268°11.2	04.6
11	104°06.5	300°11.3	48.3	288°35.7	35.9	100°26.6	04.7	283°13.4	04.5
12	119°09.0	315°11.3	N07°47.1	303°36.8	N02°35.3	115°29.0	N00°04.7	298°15.7	N02°04.4
13	134°11.5	330°11.3	45.9	318°37.9	34.7	130°31.5	04.7	313°18.0	04.3
14	149°13.9	345°11.3	44.8	333°39.0	34.0	145°33.9	04.7	328°20.3	04.2
15	164°16.4	0°11.2	•• 43.6	348°40.1	•• 33.4	160°36.4	•• 04.7	343°22.6	•• 04.1
16	179°18.9	15°11.2	42.4	3°41.2	32.8	175°38.8	04.7	358°24.9	04.0
17	194°21.3	30°11.2	41.2	18°42.3	32.2	190°41.3	04.7	13°27.2	04.0
18	209°23.8	45°11.1	N07°40.0	33°43.4	N02°31.5	205°43.7	N00°04.7	28°29.4	N02°03.9
19	224°26.3	60°11.1	38.8	48°44.5	30.9	220°46.2	04.7	43°31.7	03.8
20	239°28.7	75°11.1	37.6	63°45.6	30.3	235°48.6	04.7	58°34.0	03.7
21	254°31.2	90°11.1	•• 36.4	78°46.7	•• 29.7	250°51.1	•• 04.7	73°36.3	•• 03.6
22	269°33.6	105°11.1	35.2	93°47.8	29.0	265°53.5	04.7	88°38.6	03.5
23	284°36.1	120°11.0	34.0	108°48.9	28.4	280°56.0	04.6	103°40.9	03.5
Mer.pass. 04:05		ν -0.0' d -1.2' m -4.17		ν 1.1' d -0.6' m 1.44		ν 2.4' d -0.0' m -2.61		ν 2.3' d -0.1' m 1.00	

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	299°38.6	135°11.0	N07°32.8	123°50.0	N02°27.8	295°58.4	N00°04.6	118°43.2	N02°03.4
1	314°41.0	150°11.0	31.6	138°51.2	27.2	311°00.8	04.6	133°45.4	03.3
2	329°43.5	165°11.0	30.4	153°52.3	26.5	326°03.3	04.6	148°47.7	03.2
3	344°46.0	180°10.9	•• 29.2	168°53.4	•• 25.9	341°05.7	•• 04.6	163°50.0	•• 03.1
4	359°48.4	195°10.9	28.0	183°54.5	25.3	356°08.2	04.6	178°52.3	03.0
5	14°50.9	210°10.9	26.9	198°55.6	24.7	11°10.6	04.6	193°54.6	02.9
6	29°53.4	225°10.9	N07°25.7	213°56.7	N02°24.0	26°13.1	N00°04.6	208°56.9	N02°02.9
7	44°55.8	240°10.9	24.5	228°57.8	23.4	41°15.5	04.6	223°59.1	02.8
8	59°58.3	255°10.8	23.3	243°58.9	22.8	56°18.0	04.6	239°01.4	02.7
9	75°00.8	270°10.8	•• 22.1	259°00.0	•• 22.2	71°20.4	•• 04.6	254°03.7	•• 02.6
10	90°03.2	285°10.8	20.9	274°01.1	21.5	86°22.9	04.6	269°06.0	02.5
11	105°05.7	300°10.8	19.7	289°02.2	20.9	101°25.3	04.6	284°08.3	02.4
12	120°08.1	315°10.8	N07°18.5	304°03.3	N02°20.3	116°27.8	N00°04.6	299°10.6	N02°02.3
13	135°10.6	330°10.8	17.3	319°04.4	19.7	131°30.3	04.5	314°12.8	02.3
14	150°13.1	345°10.8	16.1	334°05.5	19.0	146°32.7	04.5	329°15.1	02.2
15	165°15.5	0°10.7	•• 14.9	349°06.6	•• 18.4	161°35.2	•• 04.5	344°17.4	•• 02.1
16	180°18.0	15°10.7	13.7	4°07.7	17.8	176°37.6	04.5	359°19.7	02.0
17	195°20.5	30°10.7	12.5	19°08.8	17.2	191°40.1	04.5	14°22.0	01.9
18	210°22.9	45°10.7	N07°11.3	34°09.9	N02°16.5	206°42.5	N00°04.5	29°24.3	N02°01.8
19	225°25.4	60°10.7	10.1	49°11.0	15.9	221°45.0	04.5	44°26.5	01.7
20	240°27.9	75°10.7	08.9	64°12.1	15.3	236°47.4	04.5	59°28.8	01.6
21	255°30.3	90°10.7	•• 07.7	79°13.2	•• 14.6	251°49.9	•• 04.5	74°31.1	•• 01.6
22	270°32.8	105°10.7	06.5	94°14.3	14.0	266°52.3	04.5	89°33.4	01.5
23	285°35.3	120°10.7	05.3	109°15.4	13.4	281°54.8	04.5	104°35.7	01.4
Mer.pass. 04:01		ν -0.0' d -1.2' m -4.18		ν 1.1' d -0.6' m 1.45		ν 2.4' d -0.0' m -2.62		ν 2.3' d -0.1' m 1.00	

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	300°37.7	135°10.6	N07°04.1	124°16.5	N02°12.8	296°57.2	N00°04.4	119°38.0	N02°01.3
1	315°40.2	150°10.6	02.9	139°17.6	12.1	311°59.7	04.4	134°40.2	01.2
2	330°42.6	165°10.6	01.7	154°18.7	11.5	327°02.2	04.4	149°42.5	01.1
3	345°45.1	180°10.6	07°00.5	169°19.8	•• 10.9	342°04.6	•• 04.4	164°44.8	•• 01.0
4	0°47.6	195°10.6	06°59.3	184°20.9	10.3	357°07.1	04.4	179°47.1	01.0
5	15°50.0	210°10.6	58.1	199°22.0	09.6	12°09.5	04.4	194°49.4	00.9
6	30°52.5	225°10.6	N06°56.9	214°23.1	N02°09.0	27°12.0	N00°04.4	209°51.6	N02°00.8
7	45°55.0	240°10.6	55.7	229°24.2	08.4	42°14.4	04.4	224°53.9	00.7
8	60°57.4	255°10.6	54.5	244°25.3	07.7	57°16.9	04.4	239°56.2	00.6
9	75°59.9	270°10.6	•• 53.3	259°26.4	•• 07.1	72°19.4	•• 04.4	254°58.5	•• 00.5
10	91°02.4	285°10.6	52.1	274°27.5	06.5	87°21.8	04.3	270°00.8	00.4
11	106°04.8	300°10.6	50.9	289°28.6	05.9	102°24.3	04.3	285°03.0	00.3
12	121°07.3	315°10.6	N06°49.7	304°29.7	N02°05.2	117°26.7	N00°04.3	300°05.3	N02°00.3
13	136°09.8	330°10.6	48.5	319°30.8	04.6	132°29.2	04.3	315°07.6	00.2
14	151°12.2	345°10.6	47.3	334°31.9	04.0	147°31.6	04.3	330°09.9	00.1
15	166°14.7	0°10.6	•• 46.1	349°33.0	•• 03.4	162°34.1	•• 04.3	345°12.2	02°00.0
16	181°17.1	15°10.6	44.9	4°34.1	02.7	177°36.6	04.3	0°14.4	01°59.9
17	196°19.6	30°10.6	43.7	19°35.2	02.1	192°39.0	04.3	15°16.7	59.8
18	211°22.1	45°10.6	N06°42.5	34°36.3	N02°01.5	207°41.5	N00°04.2	30°19.0	N01°59.7
19	226°24.5	60°10.6	41.2	49°37.4	00.8	222°44.0	04.2	45°21.3	59.6
20	241°27.0	75°10.6	40.0	64°38.5	02°00.2	237°46.4	04.2	60°23.6	59.6
21	256°29.5	90°10.6	•• 38.8	79°39.6	01°59.6	252°48.9	•• 04.2	75°25.8	•• 59.5
22	271°31.9	105°10.6	37.6	94°40.7	59.0	267°51.3	04.2	90°28.1	59.4
23	286°34.4	120°10.6	36.4	109°41.8	58.3	282°53.8	04.2	105°30.4	59.3
Mer.pass. 03:57		ν -0.0' d -1.2' m -4.18		ν 1.1' d -0.6' m 1.45		ν 2.5' d -0.0' m -2.62		ν 2.3' d -0.1' m 1.00	

Stars		
	SHA	Dec
Alpheratz	357°45.5	29°09.0
Ankaa	353°17.5	-42°14.5
Schedar	349°42.9	56°35.6
Diphda	348°57.8	-17°55.4
Achernar	335°28.2	-57°10.6
Hamal	328°03.2	23°30.8
Polaris	318°55.0	89°18.3
Acamar	315°20.0	-40°15.4
Menkar	314°17.3	4°08.0
Mirfak	308°43.6	49°53.8
Aldebaran	290°52.0	16°31.8
Rigel	281°14.3	-8°11.3
Capella	280°37.9	46°00.4
Bellatrix	278°34.5	6°21.6
Elnath	278°15.6	28°36.9
Alnilam	275°48.8	-1°11.6
Betelgeuse	271°03.9	7°24.6
Canopus	263°57.6	-52°42.0
Sirius	258°35.9	-16°43.8
Adhara	255°14.5	-28°59.1
Procyon	245°02.2	5°11.9
Pollux	243°30.6	28°00.0
Avior	234°19.6	-50°32.7
Suhail	222°54.5	-43°28.6
Miaplacidus	221°41.2	-69°45.8
Alphard	217°58.5	-8°42.3
Regulus	207°46.0	11°54.9
Dubhe	193°54.5	61°41.8
Denebola	182°35.9	14°30.8
Gienah	175°54.6	-17°36.2
Acrux	173°12.1	-63°09.8
Gacrux	172°03.6	-57°10.7
Alioth	166°22.5	55°54.3
Spica	158°33.5	-11°13.1
Alkaid	153°00.4	49°15.8
Hadar	148°51.0	-60°25.8
Menkent	148°10.1	-36°25.5
Arcturus	145°57.6	19°07.7
Rigil Kent.	139°54.5	-60°53.0
Kochab	137°19.4	74°07.0
Zuben'ubi	137°07.7	-16°05.2
Alphecca	126°12.6	26°40.9
Antares	112°28.6	-26°27.4
Atria	107°31.9	-69°03.0
Sabik	102°14.7	-15°44.3
Shaula	96°24.4	-37°06.7
Rasalhague	96°08.1	12°33.2
Eltanin	90°46.6	51°29.4
Kaus Aust.	83°46.2	-34°22.7
Vega	80°40.0	38°47.8
Nunki	76°00.5	-26°16.9
Altair	62°09.9	8°53.9
Peacock	53°21.8	-56°41.9
Deneb	49°32.5	45°19.2
Enif	33°48.9	9°55.5
Al Na'ir	27°45.8	-46°54.3
Fomalhaut	15°26.0	-29°33.7
Scheat	13°55.2	28°08.5
Markab	13°40.2	15°15.8

Jul 21 Wed	SHA	Mer.pass
Venus	196°32.3	14:59
Mars	184°44.1	15:45
Jupiter	356°20.3	04:19
Saturn	179°08.9	16:06

Jul 22 Thu	SHA	Mer.pass
Venus	195°32.4	14:59
Mars	184°11.5	15:44
Jupiter	356°19.8	04:15
Saturn	179°04.6	16:03

Jul 23 Fri	SHA	Mer.pass
Venus	194°32.9	14:59
Mars	183°38.8	15:42
Jupiter	356°19.5	04:11
Saturn	179°00.2	15:59

Horizontal parallax	
Venus:	0.2
Mars:	0.1

h	Sun			Moon			
Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	178°24.0	N20°31.8	61°32.4	8.5'	S23°36.6	4.8'	56.6'
1	193°24.0	31.3	75°59.9	8.5'	23°41.4	4.6'	56.6'
2	208°23.9	30.8	90°27.4	8.4'	23°46.0	4.5'	56.5'
3	223°23.9	· · 30.3	104°54.8	8.4'	23°50.5	4.4'	56.5'
4	238°23.9	29.9	119°22.3	8.4'	23°54.9	4.2'	56.5'
5	253°23.8	29.4	133°49.7	8.4'	23°59.1	4.1'	56.5'
6	268°23.8	N20°28.9	148°17.1	8.4'	S24°03.2	3.9'	56.4'
7	283°23.8	28.4	162°44.5	8.4'	24°07.1	3.8'	56.4'
8	298°23.8	27.9	177°11.9	8.4'	24°10.9	3.7'	56.4'
9	313°23.7	· · 27.4	191°39.3	8.4'	24°14.6	3.5'	56.4'
10	328°23.7	27.0	206°06.7	8.4'	24°18.2	3.4'	56.3'
11	343°23.7	26.5	220°34.1	8.4'	24°21.6	3.3'	56.3'
12	358°23.6	N20°26.0	235°01.4	8.4'	S24°24.8	3.1'	56.3'
13	13°23.6	25.5	249°28.8	8.4'	24°28.0	3.0'	56.3'
14	28°23.6	25.0	263°56.2	8.4'	24°30.9	2.9'	56.3'
15	43°23.6	· · 24.5	278°23.6	8.4'	24°33.8	2.7'	56.2'
16	58°23.5	24.0	292°50.9	8.4'	24°36.5	2.6'	56.2'
17	73°23.5	23.6	307°18.3	8.4'	24°39.1	2.4'	56.2'
18	88°23.5	N20°23.1	321°45.7	8.4'	S24°41.5	2.3'	56.2'
19	103°23.4	22.6	336°13.1	8.4'	24°43.8	2.2'	56.1'
20	118°23.4	22.1	350°40.5	8.4'	24°46.0	2.0'	56.1'
21	133°23.4	· · 21.6	5°07.9	8.4'	24°48.0	1.9'	56.1'
22	148°23.4	21.1	19°35.3	8.4'	24°49.9	1.8'	56.1'
23	163°23.3	20.6	34°02.7	8.4'	24°51.7	1.6'	56.0'
SD = 15.7'			SD = 15.4'				

Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	178°23.3	N20°20.1	48°30.1	8.4'	S24°53.3	1.5'	56.0'
1	193°23.3	19.6	62°57.5	8.4'	24°54.8	1.3'	56.0'
2	208°23.3	19.1	77°25.0	8.5'	24°56.1	1.2'	56.0'
3	223°23.2	· · 18.6	91°52.4	8.5'	24°57.3	1.1'	56.0'
4	238°23.2	18.2	106°19.9	8.5'	24°58.4	0.9'	55.9'
5	253°23.2	17.7	120°47.4	8.5'	24°59.3	0.8'	55.9'
6	268°23.2	N20°17.2	135°14.9	8.5'	S25°00.1	0.7'	55.9'
7	283°23.1	16.7	149°42.4	8.5'	25°00.8	0.5'	55.9'
8	298°23.1	16.2	164°09.9	8.6'	25°01.3	0.4'	55.8'
9	313°23.1	· · 15.7	178°37.5	8.6'	25°01.7	0.2'	55.8'
10	328°23.1	15.2	193°05.1	8.6'	25°01.9	0.1'	55.8'
11	343°23.0	14.7	207°32.7	8.6'	25°02.0	-0.0'	55.8'
12	358°23.0	N20°14.2	222°00.3	8.6'	S25°02.0	-0.2'	55.8'
13	13°23.0	13.7	236°28.0	8.7'	25°01.9	-0.3'	55.7'
14	28°23.0	13.2	250°55.6	8.7'	25°01.6	-0.4'	55.7'
15	43°22.9	· · 12.7	265°23.3	8.7'	25°01.2	-0.6'	55.7'
16	58°22.9	12.2	279°51.1	8.8'	25°00.6	-0.7'	55.7'
17	73°22.9	11.7	294°18.8	8.8'	24°59.9	-0.8'	55.7'
18	88°22.9	N20°11.2	308°46.6	8.8'	S24°59.1	-1.0'	55.6'
19	103°22.9	10.7	323°14.4	8.9'	24°58.1	-1.1'	55.6'
20	118°22.8	10.2	337°42.3	8.9'	24°57.0	-1.2'	55.6'
21	133°22.8	· · 09.7	352°10.2	8.9'	24°55.8	-1.4'	55.6'
22	148°22.8	09.2	6°38.1	9.0'	24°54.4	-1.5'	55.6'
23	163°22.8	08.6	21°06.1	9.0'	24°52.9	-1.6'	55.5'
SD = 15.7'			SD = 15.3'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	178°22.8	N20°08.1	35°34.0	9.0'	S24°51.3	-1.8'	55.5'
1	193°22.7	07.6	50°02.1	9.1'	24°49.6	-1.9'	55.5'
2	208°22.7	07.1	64°30.1	9.1'	24°47.7	-2.0'	55.5'
3	223°22.7	· · 06.6	78°58.3	9.2'	24°45.7	-2.1'	55.5'
4	238°22.7	06.1	93°26.4	9.2'	24°43.5	-2.3'	55.4'
5	253°22.7	05.6	107°54.6	9.2'	24°41.3	-2.4'	55.4'
6	268°22.6	N20°05.1	122°22.8	9.3'	S24°38.9	-2.5'	55.4'
7	283°22.6	04.6	136°51.1	9.3'	24°36.4	-2.6'	55.4'
8	298°22.6	04.1	151°19.4	9.4'	24°33.7	-2.8'	55.4'
9	313°22.6	· · 03.6	165°47.8	9.4'	24°30.9	-2.9'	55.3'
10	328°22.6	03.0	180°16.2	9.5'	24°28.0	-3.0'	55.3'
11	343°22.6	02.5	194°44.7	9.5'	24°25.0	-3.2'	55.3'
12	358°22.5	N20°02.0	209°13.2	9.6'	S24°21.9	-3.3'	55.3'
13	13°22.5	01.5	223°41.7	9.6'	24°18.6	-3.4'	55.3'
14	28°22.5	01.0	238°10.4	9.7'	24°15.2	-3.5'	55.3'
15	43°22.5	· · 00.5	252°39.0	9.7'	24°11.7	-3.6'	55.2'
16	58°22.5	20°00.0	267°07.7	9.8'	24°08.0	-3.8'	55.2'
17	73°22.5	19°59.4	281°36.5	9.8'	24°04.3	-3.9'	55.2'
18	88°22.4	N19°58.9	296°05.3	9.9'	S24°00.4	-4.0'	55.2'
19	103°22.4	58.4	310°34.2	9.9'	23°56.4	-4.1'	55.2'
20	118°22.4	57.9	325°03.1	10.0'	23°52.2	-4.2'	55.1'
21	133°22.4	· · 57.4	339°32.1	10.0'	23°48.0	-4.4'	55.1'
22	148°22.4	56.9	354°01.1	10.1'	23°43.6	-4.5'	55.1'
23	163°22.4	56.3	8°30.2	10.2'	23°39.2	-4.6'	55.1'
SD = 15.7'			SD = 15.1'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	□	□	□	□	□	□
N 70°	□	□	□	□	□	□
68°	////	////	01:13	22:53	////	////
66°	////	////	02:05	22:05	////	////
64°	////	////	02:36	21:34	////	////
62°	////	01:30	02:59	21:12	22:38	////
60°	////	02:07	03:18	20:54	22:03	////
N 58°	////	02:33	03:33	20:39	21:38	////
56°	01:22	02:53	03:46	20:26	21:19	22:47
54°	01:57	03:09	03:57	20:15	21:02	22:14
52°	02:20	03:23	04:07	20:05	20:49	21:51
50°	02:39	03:35	04:16	19:56	20:37	21:32
45°	03:14	03:59	04:34	19:38	20:13	20:58
N 40°	03:39	04:18	04:49	19:23	19:54	20:33
35°	03:59	04:33	05:02	19:11	19:39	20:14
30°	04:14	04:46	05:13	19:00	19:26	19:58
20°	04:39	05:08	05:32	18:41	19:05	19:33
N 10°	04:59	05:25	05:48	18:25	18:47	19:14
0°	05:15	05:41	06:03	18:10	18:32	18:58
S 10°	05:30	05:55	06:18	17:55	18:18	18:43
20°	05:43	06:10	06:34	17:40	18:03	18:30
30°	05:57	06:26	06:52	17:22	17:47	18:16
35°	06:04	06:35	07:02	17:11	17:39	18:09
40°	06:11	06:44	07:14	16:59	17:29	18:02
45°	06:19	06:55	07:28	16:45	17:18	17:54
S 50°	06:28	07:08	07:45	16:29	17:05	17:45
52°	06:32	07:14	07:53	16:21	16:59	17:41
54°	06:37	07:21	08:02	16:12	16:53	17:37
56°	06:41	07:28	08:12	16:02	16:45	17:32
58°	06:46	07:36	08:23	15:50	16:37	17:27
S 60°	06:52	07:45	08:36	15:37	16:29	17:22

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	■	■	■	■	■	■
66°	■	■	21:55	■	■	23:01
64°	19:23	20:24	20:43	21:52	22:43	· · ·
62°	18:36	19:35	20:07	22:39	23:32	· · ·
60°	18:06	19:04	19:41	23:09	· · ·	00:03
N 58°	17:43	18:40	19:20	23:32	· · ·	00:26
56°	17:25	18:22	19:04	23:51	· · ·	00:45
54°	17:09	18:06	18:49	· · ·	00:06	01:01
52°	16:55	17:52	18:36	· · ·	00:20	01:14
50°	16:44	17:40	18:25	· · ·	00:32	01:26
45°	16:19	17:15	18:02	00:10	00:57	01:51
N 40°	16:00	16:55	17:44	00:28	01:16	02:10
35°	15:43	16:39	17:28	00:43	01:33	02:27
30°	15:29	16:25	17:14	00:57	01:47	02:41
20°	15:05	16:00	16:51	01:19	02:11	03:05
N 10°	14:45	15:39	16:31	01:39	02:32	03:26
0°	14:26	15:20	16:12	01:58	02:52	03:45
S 10°	14:07	15:00	15:54	02:16	03:11	04:05
20°	13:46	14:39	15:33	02:36	03:32	04:25
30°	13:23	14:15	15:10	02:59	03:56	04:49
35°	13:09	14:01	14:56	03:12	04:11	05:03
40°	12:53	13:44	14:41	03:28	04:27	05:19
45°	12:34	13:24	14:22	03:46	04:47	05:38
S 50°	12:10	13:00	13:58	04:10	05:11	06:02
52°	11:59	12:48	13:47	04:21	05:23	06:14
54°	11:46	12:34	13:34	04:33	05:37	06:27
56°	11:31	12:19	13:19	04:48	05:52	06:42
58°	11:14	12:00	13:01	05:05	06:11	07:00
S 60°	10:53	11:37	12:40	05:26	06:34	07:22

Day	Sun		Mer.	Moon		Age
	Eqn. of Time	12 ^h	Pass	Mer. Pass.	Lower	10-12
	mm:ss	mm:ss	hh:mm	hh:mm	hh:mm	76-91%
21	06:24	06:25	12:06	20:39	08:12	
22	06:27	06:28	12:06	21:32	09:06	
23	06:29	06:30	12:06	22:25	09:59	

July 24, 25, 26 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars					
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec				
0	301°36.9	135°10.6	N06°35.2	124°42.9	N01°57.7	297°56.3	N00°04.2	120°32.7	N01°59.2	Alpheratz	357°45.5	29°09.0			
1	316°39.3	150°10.6	34.0	139°44.0	57.1	312°58.7	04.2	135°35.0	59.1	Ankaa	353°17.5	-42°14.5			
2	331°41.8	165°10.6	32.8	154°45.1	56.4	328°01.2	04.1	150°37.2	59.0	Schedar	349°42.8	56°35.6			
3	346°44.2	180°10.6	· · 31.6	169°46.2	· · 55.8	343°03.7	· · 04.1	165°39.5	· · 58.9	Diphda	348°57.8	-17°55.4			
4	1°46.7	195°10.6	30.4	184°47.3	55.2	358°06.1	04.1	180°41.8	58.9	Achernar	335°28.1	-57°10.6			
5	16°49.2	210°10.7	29.2	199°48.4	54.6	13°08.6	04.1	195°44.1	58.8	Hamal	328°03.1	23°30.8			
6	31°51.6	225°10.7	N06°28.0	214°49.5	N01°53.9	28°11.0	N00°04.1	210°46.4	N01°58.7	Polaris	318°53.5	89°18.4			
7	46°54.1	240°10.7	26.8	229°50.6	53.3	43°13.5	04.1	225°48.6	58.6	Acamar	315°20.0	-40°15.4			
8	61°56.6	255°10.7	25.6	244°51.7	52.7	58°16.0	04.1	240°50.9	58.5	Menkar	314°17.3	4°08.0			
9	76°59.0	270°10.7	· · 24.4	259°52.8	· · 52.0	73°18.4	· · 04.0	255°53.2	· · 58.4	Mirfak	308°43.6	49°53.8			
10	92°01.5	285°10.7	23.2	274°53.9	51.4	88°20.9	04.0	270°55.5	58.3	Aldebaran	290°52.0	16°31.8			
11	107°04.0	300°10.7	22.0	289°55.0	50.8	103°23.4	04.0	285°57.7	58.2	Rigel	281°14.3	-8°11.3			
12	122°06.4	315°10.7	N06°20.7	304°56.1	N01°50.2	118°25.8	N00°04.0	301°00.0	N01°58.2	Capella	280°37.9	46°00.4			
13	137°08.9	330°10.7	19.5	319°57.2	49.5	133°28.3	04.0	316°02.3	58.1	Bellatrix	278°34.5	6°21.6			
14	152°11.4	345°10.7	18.3	334°58.3	48.9	148°30.8	04.0	331°04.6	58.0	Elnath	278°15.5	28°36.9			
15	167°13.8	0°10.8	· · 17.1	349°59.4	· · 48.3	163°33.2	· · 04.0	346°06.9	· · 57.9	Alnilam	275°48.7	-1°11.6			
16	182°16.3	15°10.8	15.9	5°00.4	47.6	178°35.7	03.9	1°09.1	57.8	Betelgeuse	271°03.8	7°24.6			
17	197°18.7	30°10.8	14.7	20°01.5	47.0	193°38.2	03.9	16°11.4	57.7	Canopus	263°57.6	-52°41.9			
18	212°21.2	45°10.8	N06°13.5	35°02.6	N01°46.4	208°40.6	N00°03.9	31°13.7	N01°57.6	Sirius	258°35.9	-16°43.8			
19	227°23.7	60°10.8	12.3	50°03.7	45.8	223°43.1	03.9	46°16.0	57.5	Adhara	255°14.5	-28°59.1			
20	242°26.1	75°10.8	11.1	65°04.8	45.1	238°45.6	03.9	61°18.2	57.4	Procyon	245°02.2	5°11.9			
21	257°28.6	90°10.9	· · 09.9	80°05.9	· · 44.5	253°48.1	· · 03.9	76°20.5	· · 57.4	Pollux	243°30.6	28°00.0			
22	272°31.1	105°10.9	08.7	95°07.0	43.9	268°50.5	03.9	91°22.8	57.3	Avior	234°19.6	-59°32.7			
23	287°33.5	120°10.9	07.4	110°08.1	43.2	283°53.0	03.8	106°25.1	57.2	Suhail	222°54.5	-43°28.6			
Mer.pass. 03:53		ν 0.0' d -1.2' m -4.19		ν 1.1' d -0.6' m 1.46		ν 2.5' d -0.0' m -2.63		ν 2.3' d -0.1' m 1.00		Miaplacidus	221°41.2	-69°45.8			
												Alphard	217°58.4	-8°42.3	
												Regulus	207°46.0	11°54.9	
												Dubhe	193°54.5	61°41.8	
												Denebola	182°35.9	14°30.8	
												Gienah	175°54.6	-17°36.2	
												Acrux	173°12.1	-63°09.8	
												Gacrux	172°03.6	-57°10.7	
												Alioth	166°22.5	55°54.3	
												Spica	158°33.5	-11°13.1	
												Alkaid	153°00.4	-49°15.8	
												Hadar	148°51.0	-60°25.8	
												Menkent	148°10.1	-36°25.5	
												Arcturus	145°57.6	19°07.7	
												Rigel Kent.	139°54.5	-60°53.0	
												Kochab	137°19.4	74°07.0	
												Zuben'ubi	137°07.7	-16°05.2	
												Alphecca	126°12.6	26°40.9	
												Antares	112°28.6	-26°27.4	
												Atria	107°32.0	-69°03.0	
												Sabik	102°14.7	-15°44.3	
												Shaula	96°24.4	-37°06.7	
												Rasalhague	96°08.1	12°33.3	
												Eltanin	90°46.6	51°29.4	
												Kaus Aust.	83°46.2	-34°22.7	
												Vega	80°40.0	38°47.8	
												Nunki	76°00.5	-26°16.9	
												Altair	62°09.9	8°53.9	
												Peacock	53°21.8	-56°41.9	
												Deneb	49°32.5	45°19.2	
												Enif	33°48.9	9°55.6	
												Al Na'ir	27°45.8	-46°54.3	
												Fomalhaut	15°25.9	-29°33.7	
												Scheat	13°55.2	28°08.5	
												Markab	13°40.2	15°15.9	
													Jul 24 Sat	SHA	Mer.pass
												Venus	193°33.8	14:59	
												Mars	183°06.0	15:40	
												Jupiter	356°19.4	04:08	
												Saturn	178°55.8	15:55	
													Jul 25 Sun	SHA	Mer.pass
												Venus	192°34.9	14:59	
												Mars	182°33.2	15:38	
												Jupiter	356°19.5	04:04	
												Saturn	178°51.3	15:52	
													Jul 26 Mon	SHA	Mer.pass
												Venus	191°36.4	14:59	
												Mars	182°00.3	15:37	
												Jupiter	356°19.7	04:00	
												Saturn	178°46.8	15:48	
													Horizontal parallax		
												Venus:	0.2		
												Mars:	0.1		
Mer.pass. 03:45		ν 0.0' d -1.2' m -4.21		ν 1.1' d -0.6' m 1.46		ν 2.5' d -0.0' m -2.65		ν 2.3' d -0.1' m 1.00							

h	Sun			Moon			
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	178°22.4	N19°55.8	22°59.4	10.2'	S23°34.6	-4.7'	55.1'
1	193°22.4	55.3	37°28.6	10.3'	23°29.9	-4.8'	55.1'
2	208°22.3	54.8	51°57.8	10.3'	23°25.1	-4.9'	55.0'
3	223°22.3	· · 54.3	66°27.2	10.4'	23°20.1	-5.0'	55.0'
4	238°22.3	53.7	80°56.6	10.4'	23°15.1	-5.2'	55.0'
5	253°22.3	53.2	95°26.0	10.5'	23°09.9	-5.3'	55.0'
6	268°22.3	N19°52.7	109°55.5	10.6'	S23°04.6	-5.4'	55.0'
7	283°22.3	52.2	124°25.1	10.6'	22°59.3	-5.5'	55.0'
8	298°22.3	51.6	138°54.7	10.7'	22°53.8	-5.6'	54.9'
9	313°22.3	· · 51.1	153°24.4	10.8'	22°48.2	-5.7'	54.9'
10	328°22.2	50.6	167°54.2	10.8'	22°42.4	-5.8'	54.9'
11	343°22.2	50.1	182°24.0	10.9'	22°36.6	-5.9'	54.9'
12	358°22.2	N19°49.5	196°53.9	10.9'	S22°30.7	-6.0'	54.9'
13	13°22.2	49.0	211°23.8	11.0'	22°24.7	-6.1'	54.9'
14	28°22.2	48.5	225°53.8	11.1'	22°18.5	-6.2'	54.8'
15	43°22.2	· · 48.0	240°23.9	11.1'	22°12.3	-6.3'	54.8'
16	58°22.2	47.4	254°54.0	11.2'	22°06.0	-6.4'	54.8'
17	73°22.2	46.9	269°24.2	11.3'	21°59.5	-6.5'	54.8'
18	88°22.2	N19°46.4	283°54.5	11.3'	S21°53.0	-6.6'	54.8'
19	103°22.2	45.8	298°24.9	11.4'	21°46.3	-6.7'	54.8'
20	118°22.1	45.3	312°55.3	11.5'	21°39.6	-6.8'	54.8'
21	133°22.1	· · 44.8	327°25.7	11.5'	21°32.7	-6.9'	54.7'
22	148°22.1	44.2	341°56.3	11.6'	21°25.8	-7.0'	54.7'
23	163°22.1	43.7	356°26.9	11.7'	21°18.8	-7.1'	54.7'
SD = 15.7'			SD = 15.0'				

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	178°22.1	N19°43.2	10°57.5	11.7'	S21°11.6	-7.2'	54.7'
1	193°22.1	42.6	25°28.3	11.8'	21°04.4	-7.3'	54.7'
2	208°22.1	42.1	39°59.1	11.9'	20°57.1	-7.4'	54.7'
3	223°22.1	· · 41.6	54°30.0	11.9'	20°49.6	-7.5'	54.7'
4	238°22.1	41.0	69°00.9	12.0'	20°42.1	-7.6'	54.6'
5	253°22.1	40.5	83°31.9	12.1'	20°34.5	-7.7'	54.6'
6	268°22.1	N19°40.0	98°03.0	12.1'	S20°26.8	-7.8'	54.6'
7	283°22.1	39.4	112°34.1	12.2'	20°19.1	-7.9'	54.6'
8	298°22.1	38.9	127°05.3	12.3'	20°11.2	-8.0'	54.6'
9	313°22.1	· · 38.3	141°36.6	12.3'	20°03.2	-8.0'	54.6'
10	328°22.1	37.8	156°08.0	12.4'	19°55.2	-8.1'	54.6'
11	343°22.0	37.3	170°39.4	12.5'	19°47.1	-8.2'	54.5'
12	358°22.0	N19°36.7	185°10.9	12.5'	S19°38.8	-8.3'	54.5'
13	13°22.0	36.2	199°42.4	12.6'	19°30.5	-8.4'	54.5'
14	28°22.0	35.6	214°14.0	12.7'	19°22.1	-8.5'	54.5'
15	43°22.0	· · 35.1	228°45.7	12.7'	19°13.7	-8.5'	54.5'
16	58°22.0	34.6	243°17.4	12.8'	19°05.1	-8.6'	54.5'
17	73°22.0	34.0	257°49.3	12.9'	18°56.5	-8.7'	54.5'
18	88°22.0	N19°33.5	272°21.1	12.9'	S18°47.8	-8.8'	54.5'
19	103°22.0	32.9	286°53.1	13.0'	18°39.0	-8.9'	54.4'
20	118°22.0	32.4	301°25.1	13.1'	18°30.1	-8.9'	54.4'
21	133°22.0	· · 31.8	315°57.2	13.1'	18°21.2	-9.0'	54.4'
22	148°22.0	31.3	330°29.3	13.2'	18°12.2	-9.1'	54.4'
23	163°22.0	30.7	345°01.5	13.3'	18°03.1	-9.2'	54.4'
SD = 15.7'			SD = 14.9'				

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	178°22.0	N19°30.2	359°33.8	13.3'	S17°53.9	-9.2'	54.4'
1	193°22.0	29.6	14°06.1	13.4'	17°44.7	-9.3'	54.4'
2	208°22.0	29.1	28°38.5	13.5'	17°35.3	-9.4'	54.4'
3	223°22.0	· · 28.5	43°11.0	13.5'	17°26.0	-9.5'	54.4'
4	238°22.0	28.0	57°43.5	13.6'	17°16.5	-9.5'	54.3'
5	253°22.0	27.4	72°16.1	13.7'	17°07.0	-9.6'	54.3'
6	268°22.0	N19°26.9	86°48.8	13.7'	S16°57.4	-9.7'	54.3'
7	283°22.0	26.3	101°21.5	13.8'	16°47.7	-9.7'	54.3'
8	298°22.0	25.8	115°54.3	13.8'	16°38.0	-9.8'	54.3'
9	313°22.0	· · 25.2	130°27.2	13.9'	16°28.2	-9.9'	54.3'
10	328°22.0	24.7	145°00.1	14.0'	16°18.3	-9.9'	54.3'
11	343°22.0	24.1	159°33.0	14.0'	16°08.4	-10.0'	54.3'
12	358°22.0	N19°23.6	174°06.1	14.1'	S15°58.4	-10.1'	54.3'
13	13°22.0	23.0	188°39.1	14.1'	15°48.4	-10.1'	54.3'
14	28°22.0	22.5	203°12.3	14.2'	15°38.2	-10.2'	54.2'
15	43°22.0	· · 21.9	217°45.5	14.3'	15°28.1	-10.2'	54.2'
16	58°22.0	21.4	232°18.8	14.3'	15°17.8	-10.3'	54.2'
17	73°22.0	20.8	246°52.1	14.4'	15°07.5	-10.4'	54.2'
18	88°22.0	N19°20.2	261°25.5	14.4'	S14°57.2	-10.4'	54.2'
19	103°22.0	19.7	275°58.9	14.5'	14°46.8	-10.5'	54.2'
20	118°22.0	19.1	290°32.4	14.6'	14°36.3	-10.5'	54.2'
21	133°22.0	· · 18.6	305°06.0	14.6'	14°25.8	-10.6'	54.2'
22	148°22.0	18.0	319°39.6	14.7'	14°15.2	-10.6'	54.2'
23	163°22.1	17.4	334°13.2	14.7'	14°04.6	-10.7'	54.2'
SD = 15.7'			SD = 14.8'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	□	□	□	□	□	□
N 70°	□	□	□	□	□	□
68°	////	////	01:34	22:33	////	////
66°	////	////	02:17	21:53	////	////
64°	////	00:42	02:45	21:25	23:20	////
62°	////	01:45	03:07	21:04	22:25	////
60°	////	02:17	03:24	20:47	21:53	////
N 58°	00:39	02:41	03:39	20:33	21:30	23:24
56°	01:35	02:59	03:51	20:21	21:12	22:34
54°	02:05	03:15	04:02	20:10	20:57	22:05
52°	02:27	03:28	04:11	20:01	20:44	21:44
50°	02:45	03:39	04:19	19:53	20:33	21:26
45°	03:18	04:03	04:37	19:35	20:10	20:54
N 40°	03:42	04:21	04:52	19:21	19:52	20:30
35°	04:01	04:36	05:04	19:09	19:37	20:11
30°	04:17	04:48	05:15	18:58	19:24	19:56
20°	04:41	05:09	05:33	18:40	19:04	19:32
N 10°	05:00	05:26	05:48	18:24	18:47	19:13
0°	05:16	05:41	06:03	18:10	18:32	18:58
S 10°	05:29	05:55	06:17	17:56	18:18	18:44
20°	05:42	06:09	06:33	17:41	18:04	18:31
30°	05:55	06:24	06:50	17:23	17:49	18:18
35°	06:02	06:33	07:00	17:13	17:40	18:11
40°	06:09	06:42	07:11	17:02	17:31	18:04
45°	06:17	06:53	07:25	16:49	17:21	17:57
S 50°	06:25	07:05	07:41	16:32	17:08	17:48
52°	06:29	07:11	07:49	16:25	17:03	17:44
54°	06:33	07:17	07:57	16:16	16:57	17:40
56°	06:38	07:24	08:07	16:07	16:50	17:36
58°	06:42	07:31	08:18	15:56	16:42	17:31
S 60°	06:47	07:40	08:30	15:43	16:34	17:26

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	■	23:57	22:09	■	■	00:23
N 70°	■	22:18	21:40	■	■	02:01
68°	22:21	21:38	21:19	■	00:20	02:40
66°	21:22	21:10	21:02	· ·	01:18	03:07
64°	20:48	20:49	20:48	00:13	01:51	03:28
62°	20:23	20:32	20:36	00:48	02:15	03:44
60°	20:03	20:17	20:26	01:14	02:35	03:58
N 58°	19:47	20:05	20:17	01:34	02:51	04:09
56°	19:33	19:54	20:10	01:51	03:04	04:19
54°	19:21	19:45	20:03	02:05	03:16	04:28
52°	19:10	19:36	19:56	02:17	03:26	04:36
50°	19:01	19:28	19:51	02:28	03:35	04:43
45°	18:41	19:12	19:38	02:51	03:54	04:59
N 40°	18:24	18:59	19:28	03:09	04:10	05:11
35°	18:11	18:47	19:20	03:24	04:23	05:21
30°	17:58	18:37	19:12	03:37	04:34	05:31
20°	17:38	18:20	18:58	04:00	04:54	05:46
N 10°	17:20	18:05	18:47	04:19	05:11	06:00
0°	17:03	17:51	18:36	04:37	05:26	06:13
S 10°	16:46	17:36	18:25	04:55	05:42	06:25
20°	16:28	17:21	18:13	05:14	05:59	06:39
30°	16:07	17:04	17:59	05:36	06:18	06:54
35°	15:55	16:53	17:51	05:49	06:29	07:03
40°	15:41	16:42	17:42	06:04	06:41	07:13
45°	15:24	16:28	17:32	06:21	06:56	07:25
S 50°	15:03	16:11	17:19	06:43	07:14	07:39
52°	14:53	16:03	17:13	06:53	07:23	07:45
54°	14:42	15:54	17:06	07:04	07:32	07:53
56°	14:29	15:43	16:59	07:18	07:43	08:01
58°	14:14	15:32	16:50	07:33	07:55	08:10
S 60°	13:56	15:18	16:41	07:51	08:09	08:20

Day	Sun		Mer. Pass hh:mm	Moon		Age 13-15 96-100%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
24	06:31	06:31	12:07	23:15	10:50	
25	06:32	06:32	12:07	—:—	11:39	
26	06:32	06:32	12:07	00:02	12:24	

July 27, 28, 29 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	304°34.3	135°12.5	N05°07.9	126°01.6	N01°12.3	300°54.4	N00°02.9	123°16.5	N01°52.8	Alpheratz	357°45.5	29°09.0
1	319°36.7	150°12.5	06.7	141°02.7	11.7	315°56.9	02.9	138°18.8	52.7	Ankaa	353°17.4	-42°14.5
2	334°39.2	165°12.6	05.5	156°03.8	11.0	330°59.4	02.8	153°21.0	52.6	Schedar	349°42.8	56°35.6
3	349°41.7	180°12.6	•• 04.3	171°04.9	•• 10.4	346°01.9	•• 02.8	168°23.3	•• 52.5	Diphda	348°57.8	-17°55.4
4	4°44.1	195°12.7	03.1	186°06.0	09.8	1°04.4	02.8	183°25.6	52.4	Achernar	335°28.1	-57°10.6
5	19°46.6	210°12.7	01.8	201°07.1	09.1	16°06.8	02.8	198°27.8	52.3	Hamal	328°03.1	23°30.8
6	34°49.1	225°12.8	N05°00.6	216°08.2	N01°08.5	31°09.3	N00°02.7	213°30.1	N01°52.2	Polaris	318°52.2	89°18.4
7	49°51.5	240°12.8	04°59.4	231°09.3	07.9	46°11.8	02.7	228°32.4	52.2	Acamar	315°19.9	-40°15.4
8	64°54.0	255°12.9	58.2	246°10.4	07.2	61°14.3	02.7	243°34.6	52.1	Menkar	314°17.3	4°08.0
9	79°56.5	270°12.9	•• 57.0	261°11.4	•• 06.6	76°16.8	•• 02.7	258°36.9	•• 52.0	Mirfak	308°43.6	49°53.8
10	94°58.9	285°13.0	55.7	276°12.5	06.0	91°19.3	02.6	273°39.2	51.9	Aldebaran	290°52.0	16°31.9
11	110°01.4	300°13.0	54.5	291°13.6	05.3	106°21.8	02.6	288°41.5	51.8	Rigel	281°14.3	-8°11.2
12	125°03.8	315°13.1	N04°53.3	306°14.7	N01°04.7	121°24.3	N00°02.6	303°43.7	N01°51.7	Capella	280°37.9	46°00.4
13	140°06.3	330°13.1	52.1	321°15.8	04.1	136°26.7	02.6	318°46.0	51.6	Bellatrix	278°34.5	6°21.6
14	155°08.8	345°13.2	50.9	336°16.9	03.4	151°29.2	02.6	333°48.3	51.5	Elnath	278°15.5	28°36.9
15	170°11.2	0°13.2	•• 49.6	351°18.0	•• 02.8	166°31.7	•• 02.5	348°50.5	•• 51.4	Alnilam	275°48.7	-1°11.6
16	185°13.7	15°13.3	48.4	6°19.1	02.2	181°34.2	02.5	3°52.8	51.3	Betelgeuse	271°03.8	7°24.6
17	200°16.2	30°13.3	47.2	21°20.2	01.5	196°36.7	02.5	18°55.1	51.2	Canopus	263°57.6	-52°41.9
18	215°18.6	45°13.4	N04°46.0	36°21.2	N01°00.9	211°39.2	N00°02.5	33°57.3	N01°51.1	Sirius	258°35.9	-16°43.8
19	230°21.1	60°13.4	44.8	51°22.3	01°00.3	226°41.7	02.4	48°59.6	51.1	Adhara	255°14.5	-28°59.1
20	245°23.6	75°13.5	43.5	66°23.4	00°59.6	241°44.2	02.4	64°01.9	51.0	Procyon	245°02.2	5°11.9
21	260°26.0	90°13.6	•• 42.3	81°24.5	•• 59.0	256°46.7	•• 02.4	79°04.1	•• 50.9	Pollux	243°30.6	28°00.0
22	275°28.5	105°13.6	41.1	96°25.6	58.4	271°49.2	02.3	94°06.4	50.8	Avior	234°19.6	-59°32.6
23	290°31.0	120°13.7	39.9	111°26.7	57.7	286°51.7	02.3	109°08.7	50.7	Suhail	222°54.5	-43°28.6
Mer.pass. 03:41		ν 0.0' d -1.2' m -4.22		ν 1.1' d -0.6' m 1.47		ν 2.5' d -0.0' m -2.66		ν 2.3' d -0.1' m 1.00		Miaplacidus	221°41.2	-69°45.8
Alphard	217°58.4											-8°42.3
Regulus	207°46.0											11°54.9
Dubhe	193°54.5											61°41.8
Denebola	182°35.9											14°30.8
Gienah	175°54.6											-17°36.2
Acrux	173°12.2											-63°09.8
Gacrux	172°03.6											-57°10.7
Alioth	166°22.5											55°54.3
Spica	158°33.5											-11°13.1
Alkaid	153°00.5											49°15.8
Hadar	148°51.0											-60°25.8
Menkent	148°10.1											-36°25.5
Arcturus	145°57.6											19°07.7
Rigel Kent.	139°54.6											-60°53.0
Kochab	137°19.5											74°07.0
Zuben'ubi	137°07.7											-16°05.2
Alphecca	126°12.6											26°40.9
Antares	112°28.6											-26°27.4
Atria	107°32.0											-69°03.0
Sabik	102°14.7											-15°44.3
Shaula	96°24.4											-37°06.7
Rasalhague	96°08.1											12°33.3
Eltanin	90°46.7											51°29.4
Kaus Aust.	83°46.2											-34°22.8
Vega	80°40.0											38°47.8
Nunki	76°00.5											-26°16.9
Altair	62°09.9											8°53.9
Peacock	53°21.8											-56°41.9
Deneb	49°32.5											45°19.2
Enif	33°48.9											9°55.6
Al Na'ir	27°45.8											-46°54.3
Fomalhaut	15°25.9											-29°33.7
Scheat	13°55.2											28°08.5
Markab	13°40.2											15°15.9
Jul 27 Tue		SHA		Mer.pass								
Venus		190°38.2		14:59								
Mars		181°27.4		15:35								
Jupiter		356°20.1		03:56								
Saturn		178°42.2		15:45								
Jul 28 Wed		SHA		Mer.pass								
Venus		189°40.3		14:59								
Mars		180°54.3		15:33								
Jupiter		356°20.7		03:52								
Saturn		178°37.5		15:41								
Jul 29 Thu		SHA		Mer.pass								
Venus		188°42.7		14:59								
Mars		180°21.3		15:31								
Jupiter		356°21.5		03:48								
Saturn		178°32.8		15:37								
Horizontal parallax												
Venus:												0.2
Mars:												0.1

h	Sun			Moon			
Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	178°22.1	N19°16.9	348°46.9	14.8'	S13°53.9	-10.7'	54.2'
1	193°22.1	16.3	3°20.7	14.8'	13°43.2	-10.8'	54.1'
2	208°22.1	15.8	17°54.5	14.9'	13°32.4	-10.8'	54.1'
3	223°22.1	· · 15.2	32°28.4	14.9'	13°21.5	-10.9'	54.1'
4	238°22.1	14.6	47°02.3	15.0'	13°10.6	-10.9'	54.1'
5	253°22.1	14.1	61°36.3	15.0'	12°59.7	-11.0'	54.1'
6	268°22.1	N19°13.5	76°10.3	15.1'	S12°48.7	-11.0'	54.1'
7	283°22.1	12.9	90°44.4	15.1'	12°37.7	-11.1'	54.1'
8	298°22.1	12.4	105°18.6	15.2'	12°26.6	-11.1'	54.1'
9	313°22.1	· · 11.8	119°52.7	15.2'	12°15.5	-11.2'	54.1'
10	328°22.1	11.2	134°27.0	15.3'	12°04.3	-11.2'	54.1'
11	343°22.1	10.7	149°01.2	15.3'	11°53.1	-11.3'	54.1'
12	358°22.1	N19°10.1	163°35.6	15.4'	S11°41.9	-11.3'	54.1'
13	13°22.1	09.5	178°09.9	15.4'	11°30.6	-11.3'	54.1'
14	28°22.2	09.0	192°44.3	15.5'	11°19.2	-11.4'	54.1'
15	43°22.2	· · 08.4	207°18.8	15.5'	11°07.8	-11.4'	54.0'
16	58°22.2	07.8	221°53.3	15.5'	10°56.4	-11.5'	54.0'
17	73°22.2	07.3	236°27.8	15.6'	10°45.0	-11.5'	54.0'
18	88°22.2	N19°06.7	251°02.4	15.6'	S10°33.5	-11.5'	54.0'
19	103°22.2	06.1	265°37.0	15.7'	10°21.9	-11.6'	54.0'
20	118°22.2	05.6	280°11.7	15.7'	10°10.3	-11.6'	54.0'
21	133°22.2	· · 05.0	294°46.4	15.7'	09°58.7	-11.6'	54.0'
22	148°22.2	04.4	309°21.2	15.8'	09°47.1	-11.7'	54.0'
23	163°22.2	03.8	323°55.9	15.8'	09°35.4	-11.7'	54.0'
SD = 15.7' $d = -0.6'$			SD = 14.8'				
Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	178°22.3	N19°03.3	338°30.8	15.9'	S09°23.7	-11.7'	54.0'
1	193°22.3	02.7	353°05.6	15.9'	09°12.0	-11.8'	54.0'
2	208°22.3	02.1	7°40.5	15.9'	09°00.2	-11.8'	54.0'
3	223°22.3	· · 01.5	22°15.5	16.0'	08°48.4	-11.8'	54.0'
4	238°22.3	01.0	36°50.4	16.0'	08°36.5	-11.9'	54.0'
5	253°22.3	19°00.4	51°25.4	16.0'	08°24.7	-11.9'	54.0'
6	268°22.3	N18°59.8	66°00.5	16.1'	S08°12.8	-11.9'	54.0'
7	283°22.3	59.2	80°35.5	16.1'	08°00.8	-12.0'	54.0'
8	298°22.4	58.7	95°10.6	16.1'	07°48.9	-12.0'	54.0'
9	313°22.4	· · 58.1	109°45.8	16.2'	07°36.9	-12.0'	54.0'
10	328°22.4	57.5	124°20.9	16.2'	07°24.9	-12.0'	54.0'
11	343°22.4	56.9	138°56.1	16.2'	07°12.9	-12.1'	54.0'
12	358°22.4	N18°56.3	153°31.3	16.2'	S07°00.8	-12.1'	54.0'
13	13°22.4	55.8	168°06.6	16.3'	06°48.7	-12.1'	54.0'
14	28°22.4	55.2	182°41.9	16.3'	06°36.6	-12.1'	54.0'
15	43°22.5	· · 54.6	197°17.2	16.3'	06°24.5	-12.2'	54.0'
16	58°22.5	54.0	211°52.5	16.3'	06°12.3	-12.2'	54.0'
17	73°22.5	53.4	226°27.8	16.4'	06°00.1	-12.2'	54.0'
18	88°22.5	N18°52.8	241°03.2	16.4'	S05°48.0	-12.2'	54.0'
19	103°22.5	52.3	255°38.6	16.4'	05°35.7	-12.2'	54.0'
20	118°22.5	51.7	270°14.0	16.4'	05°23.5	-12.3'	54.0'
21	133°22.6	· · 51.1	284°49.4	16.5'	05°11.3	-12.3'	54.0'
22	148°22.6	50.5	299°24.9	16.5'	04°59.0	-12.3'	54.0'
23	163°22.6	49.9	314°00.4	16.5'	04°46.7	-12.3'	53.9'
SD = 15.7' $d = -0.6'$			SD = 14.7'				
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	178°22.6	N18°49.3	328°35.9	16.5'	S04°34.4	-12.3'	53.9'
1	193°22.6	48.7	343°11.4	16.5'	04°22.1	-12.3'	53.9'
2	208°22.6	48.2	357°46.9	16.5'	04°09.7	-12.3'	54.0'
3	223°22.7	· · 47.6	12°22.5	16.6'	03°57.4	-12.4'	54.0'
4	238°22.7	47.0	26°58.0	16.6'	03°45.0	-12.4'	54.0'
5	253°22.7	46.4	41°33.6	16.6'	03°32.7	-12.4'	54.0'
6	268°22.7	N18°45.8	56°09.2	16.6'	S03°20.3	-12.4'	54.0'
7	283°22.7	45.2	70°44.8	16.6'	03°07.9	-12.4'	54.0'
8	298°22.8	44.6	85°20.4	16.6'	02°55.5	-12.4'	54.0'
9	313°22.8	· · 44.0	99°56.0	16.6'	02°43.0	-12.4'	54.0'
10	328°22.8	43.4	114°31.6	16.6'	02°30.6	-12.4'	54.0'
11	343°22.8	42.8	129°07.3	16.6'	02°18.2	-12.4'	54.0'
12	358°22.8	N18°42.2	143°42.9	16.7'	S02°05.7	-12.5'	54.0'
13	13°22.9	41.7	158°18.6	16.7'	01°53.3	-12.5'	54.0'
14	28°22.9	41.1	172°54.2	16.7'	01°40.8	-12.5'	54.0'
15	43°22.9	· · 40.5	187°29.9	16.7'	01°28.4	-12.5'	54.0'
16	58°22.9	39.9	202°05.5	16.7'	01°15.9	-12.5'	54.0'
17	73°22.9	39.3	216°41.2	16.7'	01°03.4	-12.5'	54.0'
18	88°23.0	N18°38.7	231°16.9	16.7'	S00°50.9	-12.5'	54.0'
19	103°23.0	38.1	245°52.6	16.7'	00°38.4	-12.5'	54.0'
20	118°23.0	37.5	260°28.2	16.7'	00°25.9	-12.5'	54.0'
21	133°23.0	· · 36.9	275°03.9	16.7'	00°13.5	-12.5'	54.0'
22	148°23.0	36.3	289°39.6	16.7'	S00°01.0	-12.5'	54.0'
23	163°23.1	35.7	304°15.2	16.7'	N00°11.5	12.5'	54.0'
SD = 15.7' $d = -0.6'$			SD = 14.7'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°	////	////	00:33	23:22	////	////
68°	////	////	01:53	22:16	////	////
66°	////	////	02:29	21:41	////	////
64°	////	01:12	02:55	21:16	22:55	////
62°	////	01:58	03:15	20:56	22:12	////
60°	////	02:27	03:31	20:40	21:44	////
N 58°	01:05	02:48	03:45	20:27	21:22	23:02
56°	01:47	03:06	03:56	20:16	21:05	22:22
54°	02:14	03:21	04:06	20:06	20:51	21:56
52°	02:35	03:33	04:15	19:57	20:39	21:37
50°	02:51	03:44	04:23	19:49	20:28	21:20
45°	03:23	04:06	04:41	19:32	20:06	20:49
N 40°	03:46	04:24	04:55	19:18	19:49	20:26
35°	04:04	04:38	05:06	19:06	19:34	20:08
30°	04:19	04:50	05:16	18:56	19:22	19:54
20°	04:42	05:10	05:34	18:39	19:02	19:30
N 10°	05:01	05:27	05:49	18:24	18:46	19:12
0°	05:16	05:41	06:03	18:10	18:32	18:57
S 10°	05:29	05:55	06:17	17:56	18:18	18:44
20°	05:42	06:08	06:32	17:42	18:05	18:32
30°	05:54	06:23	06:48	17:25	17:50	18:19
35°	06:00	06:31	06:58	17:15	17:42	18:13
40°	06:07	06:40	07:09	17:05	17:34	18:06
45°	06:14	06:50	07:22	16:52	17:24	17:59
S 50°	06:22	07:02	07:37	16:36	17:12	17:51
52°	06:26	07:07	07:45	16:29	17:07	17:48
54°	06:29	07:13	07:53	16:21	17:01	17:44
56°	06:33	07:19	08:02	16:12	16:54	17:40
58°	06:38	07:27	08:12	16:01	16:47	17:36
S 60°	06:42	07:35	08:24	15:50	16:39	17:31

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	21:33	21:07	20:44	03:44	05:48	07:40
N 70°	21:17	21:00	20:43	04:11	06:02	07:44
68°	21:05	20:54	20:43	04:31	06:12	07:48
66°	20:55	20:49	20:43	04:47	06:21	07:51
64°	20:46	20:44	20:42	05:00	06:28	07:53
62°	20:39	20:41	20:42	05:10	06:34	07:56
60°	20:33	20:38	20:42	05:20	06:39	07:58
N 58°	20:27	20:35	20:42	05:28	06:44	07:59
56°	20:22	20:32	20:42	05:35	06:48	08:01
54°	20:17	20:30	20:41	05:41	06:52	08:02
52°	20:13	20:28	20:41	05:46	06:55	08:03
50°	20:09	20:26	20:41	05:51	06:58	08:04
45°	20:01	20:22	20:41	06:02	07:05	08:07
N 40°	19:54	20:18	20:41	06:11	07:10	08:09
35°	19:48	20:15	20:40	06:19	07:15	08:10
30°	19:43	20:12	20:40	06:26	07:19	08:12
20°	19:34	20:07	20:40	06:37	07:26	08:14
N 10°	19:26	20:03	20:40	06:47	07:33	08:17
0°	19:18	19:59	20:40	06:57	07:38	08:19
S 10°	19:11	19:55	20:39	07:06	07:44	08:21
20°	19:03	19:51	20:39	07:16	07:50	08:23
30°	18:53	19:47	20:39	07:27	07:57	08:25
35°	18:48	19:44	20:39	07:33	08:01	08:27
40°	18:42	19:41	20:39	07:41	08:05	08:28
45°	18:35	19:37	20:38	07:49	08:10	08:30
S 50°	18:26	19:33	20:38	07:59	08:17	08:32
52°	18:22	19:31	20:38	08:04	08:19	08:33
54°	18:18	19:28	20:38	08:09	08:22	08:34
56°	18:13	19:26	20:38	08:15	08:26	08:36
58°	18:08	19:23	20:38	08:21	08:30	08:37
S 60°	18:01	19:20	20:38	08:28	08:34	08:38

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss	hh:mm	Mer. Pass. Upper hh:mm	Lower hh:mm	16-18 99-92%
27	06:32	06:31	12:07	00:46	13:08	
28	06:31	06:30	12:07	01:28	13:49	
29	06:30	06:29	12:06	02:09	14:29	

July 30, 31, 01 UT (Fri., Sat., Sun.)

Aries		Venus		Mars		Jupiter		Saturn		Stars			
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	307°31.7	135°17.2	N03°39.8	127°19.8	N00°26.6	303°54.2	N00°00.9	125°59.7	N01°46.1	Alpheratz	357°45.5	29°09.0	
1	322°34.2	150°17.3		38.6	142°20.9		00.9	141°02.0	46.1	Ankaa	353°17.4	-42°14.5	
2	337°36.6	165°17.3		37.4	157°21.9		00.8	156°04.2	46.0	Schedar	349°42.8	56°35.6	
3	352°39.1	180°17.4	· ·	36.2	172°23.0	349°01.7	· ·	171°06.5	· ·	Diphda	348°57.8	-17°55.4	
4	7°41.5	195°17.5		34.9	187°24.1	4°04.2	00.8	186°08.8	45.8	Achernar	335°28.0	-57°10.6	
5	22°44.0	210°17.6		33.7	202°25.2	19°06.7	00.7	201°11.0	45.7	Hamal	328°03.1	23°30.8	
6	37°46.5	225°17.7	N03°32.5	217°26.3	N00°22.8	34°09.2	N00°00.7	216°13.3	N01°45.6	Polaris	318°50.9	89°18.4	
7	52°48.9	240°17.8		31.2	232°27.3	49°11.7	00.7	231°15.6	45.5	Acamar	315°19.9	-40°15.4	
8	67°51.4	255°17.9		30.0	247°28.4	64°14.2	00.6	246°17.8	45.4	Menkar	314°17.3	4°08.0	
9	82°53.9	270°18.0	· ·	28.8	262°29.5	79°16.8	· ·	261°20.1	· ·	Mirfak	308°43.5	49°53.8	
10	97°56.3	285°18.0		27.5	277°30.6	94°19.3	00.6	276°22.3	45.2	Aldebaran	290°52.0	16°31.9	
11	112°58.8	300°18.1		26.3	292°31.6	109°21.8	00.5	291°24.6	45.1	Rigel	281°14.3	-8°11.2	
12	128°01.3	315°18.2	N03°25.1	307°32.7	N00°19.0	124°24.3	N00°00.5	306°26.9	N01°45.0	Capella	280°37.8	46°00.4	
13	143°03.7	330°18.3		23.9	322°33.8	139°26.8	00.5	321°29.1	44.9	Bellatrix	278°34.5	6°21.6	
14	158°06.2	345°18.4		22.6	337°34.9	154°29.3	00.4	336°31.4	44.8	Elnath	278°15.5	28°36.9	
15	173°08.7	0°18.5	· ·	21.4	352°36.0	169°31.8	· ·	351°33.7	· ·	Alnilam	275°48.7	-1°11.6	
16	188°11.1	15°18.6		20.2	7°37.0	184°34.3	00.4	6°35.9	44.6	Betelgeuse	271°03.8	7°24.6	
17	203°13.6	30°18.7		18.9	22°38.1	199°36.9	00.3	21°38.2	44.5	Canopus	263°57.5	-52°41.9	
18	218°16.0	45°18.8	N03°17.7	37°39.2	N00°15.2	214°39.4	N00°00.3	36°40.4	N01°44.5	Sirius	258°35.9	-16°43.8	
19	233°18.5	60°18.9		16.5	52°40.3	229°41.9	00.3	51°42.7	44.4	Adhara	255°14.5	-28°59.1	
20	248°21.0	75°19.0		15.3	67°41.4	244°44.4	00.2	66°45.0	44.3	Procyon	245°02.2	5°11.9	
21	263°23.4	90°19.1	· ·	14.0	82°42.4	259°46.9	· ·	81°47.2	· ·	Pollux	243°30.6	28°00.0	
22	278°25.9	105°19.2		12.8	97°43.5	274°49.4	00.2	96°49.5	44.1	Avior	234°19.6	-59°32.6	
23	293°28.4	120°19.3		11.6	112°44.6	289°51.9	00.1	111°51.7	44.0	Suhail	222°54.5	-43°28.6	
Mer.pass. 03:29		ν 0.1' d -1.2' m -4.24		ν 1.1' d -0.6' m 1.48		ν 2.5' d -0.0' m -2.68		ν 2.3' d -0.1' m 1.00		Miaplacidus	221°41.2	-69°45.8	
											Alphard	217°58.4	-8°42.3
											Regulus	207°46.0	11°54.9
											Dubhe	193°54.6	61°41.7
											Denebola	182°35.9	14°30.8
											Gienah	175°54.6	-17°36.2
											Acrux	173°12.2	-63°09.8
											Gacrux	172°03.6	-57°10.7
											Alioth	166°22.6	55°54.3
											Spica	158°33.5	-11°13.1
											Alkaid	153°00.5	49°15.8
											Hadar	148°51.0	-60°25.8
											Menkent	148°10.1	-36°25.5
											Arcturus	145°57.6	19°07.7
											Rigel Kent.	139°54.6	-60°53.0
											Kochab	137°19.5	74°07.0
											Zuben'ubi	137°07.7	-16°05.2
											Alphecca	126°12.6	26°40.9
											Antares	112°28.6	-26°27.4
											Atria	107°32.0	-69°03.0
											Sabik	102°14.7	-15°44.3
											Shaula	96°24.4	-37°06.7
											Rasalhague	96°08.1	12°33.3
											Eltanin	90°46.7	51°29.5
											Kaus Aust.	83°46.2	-34°22.8
											Vega	80°40.0	38°47.8
											Nunki	76°00.5	-26°16.9
											Altair	62°09.9	8°53.9
											Peacock	53°21.8	-56°41.9
											Deneb	49°32.5	45°19.2
											Enif	33°48.8	9°55.6
											Al Na'ir	27°45.8	-46°54.3
											Fomalhaut	15°25.9	-29°33.7
											Scheat	13°55.1	28°08.5
											Markab	13°40.2	15°15.9
Jul 30 Fri		SHA		Mer.pass									
Venus		187°45.5		14:59									
Mars		179°48.1		15:30									
Jupiter		356°22.5		03:44									
Saturn		178°28.0		15:34									
Jul 31 Sat		SHA		Mer.pass									
Venus		186°48.5		14:59									
Mars		179°14.8		15:28									
Jupiter		356°23.6		03:40									
Saturn		178°23.2		15:30									
Aug 01 Sun		SHA		Mer.pass									
Venus		185°51.9		14:58									
Mars		178°41.5		15:26									
Jupiter		356°25.0		03:36									
Saturn		178°18.3		15:26									
Horizontal parallax													
Venus: 0.2													
Mars: 0.1													

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	308°30.8	135°19.3	N03°10.3	127°45.7	N00°11.3	304°54.5	N00°00.1	126°54.0	N01°43.9
1	323°33.3	150°19.4	09.1	142°46.7	10.7	319°57.0	00.0	141°56.3	43.8
2	338°35.8	165°19.5	07.9	157°47.8	10.1	334°59.5	N00°00.0	156°58.5	43.7
3	353°38.2	180°19.6	· ·	06.6	172°48.9	350°02.0	S00°00.0	172°00.8	· ·
4	8°40.7	195°19.7	05.4	187°50.0	08.8	5°04.5	00.1	187°03.1	43.5
5	23°43.1	210°19.8	04.2	202°51.1	08.1	20°07.0	00.1	202°05.3	43.4
6	38°45.6	225°19.9	N03°03.0	217°52.1	N00°07.5	35°09.6	S00°00.1	217°07.6	N01°43.3
7	53°48.1	240°20.0	01.7	232°53.2	06.9	50°12.1	00.2	232°09.8	43.2
8	68°50.5	255°20.1	03°00.5	247°54.3	06.2	65°14.6	00.2	247°12.1	43.1
9	83°53.0	270°20.2	02°59.3	262°55.4	· ·	80°17.1	· ·	262°14.4	· ·
10	98°55.5	285°20.3	58.0	277°56.4	04.9	95°19.6	00.3	277°16.6	42.9
11	113°57.9	300°20.4	56.8	292°57.5	04.3	110°22.1	00.3	292°18.9	42.8
12	129°00.4	315°20.6	N02°55.6	307°58.6	N00°03.7	125°24.7	S00°00.4	307°21.1	N01°42.7
13	144°02.9	330°20.7	54.3	322°59.7	03.0	140°27.2	00.4	322°23.4	42.6
14	159°05.3	345°20.8	53.1	338°00.7	02.4	155°29.7	00.4	337°25.6	42.6
15	174°07.8	0°20.9	· ·	51.9	353°01.8	170°32.2	· ·	352°27.9	· ·
16	189°10.3	15°21.0	50.6	8°02.9	01.1	185°34.7	00.5	7°30.2	42.4
17	204°12.7	30°21.1	49.4	23°04.0	N00°00.5	200°37.3	00.5	22°32.4	42.3
18	219°15.2	45°21.2	N02°48.2	38°05.0	S00°00.2	215°39.8	S00°00.6	37°34.7	N01°42.2
19	234°17.6	60°21.3	46.9	53°06.1	00.8	230°42.3	00.6	52°36.9	42.1
20	249°20.1	75°21.4	45.7	68°07.2	01.4	245°44.8	00.7	67°39.2	42.0
21	264°22.6	90°21.5	· ·	44.5	83°08.3	260°47.4	· ·	82°41.5	· ·
22	279°25.0	105°21.6	43.2	98°09.3	02.7	275°49.9	00.7	97°43.7	41.8
23	294°27.5	120°21.7	42.0	113°10.4	03.3	290°52.4	00.8	112°46.0	41.7
Mer.pass. 03:25		ν 0.1' d -1.2' m -4.25		ν 1.1' d -0.6' m 1.48		ν 2.5' d -0.0' m -2.69		ν 2.3' d -0.1' m 1.00	

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	309°30.0	135°21.8	N02°40.8	128°11.5	S00°04.0	305°54.9	S00°00.8	127°48.2	N01°41.6
1	324°32.4	150°21.9	39.6	143°12.6	04.6	320°57.4	00.8	142°50.5	41.5
2	339°34.9	165°22.0	38.3	158°13.6	05.3	336°00.0	00.9	157°52.8	41.4
3	354°37.4	180°22.2	· ·	37.1	173°14.7	351°02.5	· ·	172°55.0	· ·
4	9°39.8	195°22.3	35.9	188°15.8	06.5	6°05.0	01.0	187°57.3	41.2
5	24°42.3	210°22.4	34.6	203°16.9	07.2	21°07.5	01.0	202°59.5	41.1
6	39°44.7	225°22.5	N02°33.4	218°17.9	S00°07.8	36°10.1	S00°01.0	218°01.8	N01°41.0
7	54°47.2	240°22.6	32.2	233°19.0	08.5	51°12.6	01.1	233°04.0	40.9
8	69°49.7	255°22.7	30.9	248°20.1	09.1	66°15.1	01.1	248°06.3	40.8
9	84°52.1	270°22.8	· ·	29.7	263°21.1	· ·	01.2	263°08.6	· ·
10	99°54.6	285°22.9	28.5	278°22.2	10.4	96°20.2	01.2	278°10.8	40.6
11	114°57.1	300°23.1	27.2	293°23.3	11.0	111°22.7	01.2	293°13.1	40.5
12	129°59.5	315°23.2	N02°26.0	308°24.4	S00°11.7	126°25.2	S00°01.3	308°15.3	N01°40.4
13	145°02.0	330°23.3	24.8	323°25.4	12.3	141°27.7	01.3	323°17.6	40.3
14	160°04.5	345°23.4	23.5	338°26.5	12.9	156°30.3	01.4	338°19.8	40.3
15	175°06.9	0°23.5	· ·	22.3	353°27.6	· ·	01.4	353°22.1	· ·
16	190°09.4	15°23.6	21.1	8°28.7	14.2	186°35.3	01.4	8°24.4	40.1
17	205°11.9	30°23.8	19.8	23°29.7	14.9	201°37.9	01.5	23°26.6	40.0
18	220°14.3	45°23.9	N02°18.6	38°30.8	S00°15.5	216°40.4	S00°01.5	38°28.9	N01°39.9
19	235°16.8	60°24.0	17.4	53°31.9	16.1	231°42.9	01.6	53°31.1	39.8
20	250°19.2	75°24.1	16.1	68°32.9	16.8	246°45.4	01.6	68°33.4	39.7
21	265°21.7	90°24.2	· ·	14.9	83°34.0	· ·	01.6	83°35.6	· ·
22	280°24.2	105°24.4	13.7	98°35.1	18.0	276°50.5	01.7	98°37.9	39.5
23	295°26.6	120°24.5	12.4	113°36.2	18.7	291°53.0	01.7	113°40.2	39.4
Mer.pass. 03:21		ν 0.1' d -1.2' m -4.26		ν 1.1' d 0.6' m 1.48		ν 2.5' d 0.0' m -2.69		ν 2.3' d -0.1' m 1.00	

h	Sun			Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	178°23.1	N18°35.1	318°50.9	16.7'	N00°24.0	12.5'	54.0'
1	193°23.1	34.5	333°26.6	16.7'	00°36.5	12.5'	54.0'
2	208°23.1	33.9	348°02.2	16.7'	00°49.0	12.5'	54.0'
3	223°23.2	· · 33.3	2°37.9	16.6'	01°01.5	12.5'	54.0'
4	238°23.2	32.7	17°13.5	16.6'	01°14.0	12.5'	54.0'
5	253°23.2	32.1	31°49.2	16.6'	01°26.5	12.5'	54.0'
6	268°23.2	N18°31.5	46°24.8	16.6'	N01°39.0	12.5'	54.0'
7	283°23.2	30.9	61°00.4	16.6'	01°51.4	12.5'	54.1'
8	298°23.3	30.3	75°36.0	16.6'	02°03.9	12.5'	54.1'
9	313°23.3	· · 29.7	90°11.6	16.6'	02°16.4	12.5'	54.1'
10	328°23.3	29.1	104°47.2	16.6'	02°28.9	12.5'	54.1'
11	343°23.4	28.5	119°22.8	16.6'	02°41.3	12.4'	54.1'
12	358°23.4	N18°27.8	133°58.3	16.5'	N02°53.8	12.4'	54.1'
13	13°23.4	27.2	148°33.8	16.5'	03°06.2	12.4'	54.1'
14	28°23.5	26.6	163°09.4	16.5'	03°18.6	12.4'	54.1'
15	43°23.5	· · 26.0	177°44.9	16.5'	03°31.0	12.4'	54.1'
16	58°23.5	25.4	192°20.4	16.5'	03°43.5	12.4'	54.1'
17	73°23.5	24.8	206°55.8	16.4'	03°55.9	12.4'	54.1'
18	88°23.6	N18°24.2	221°31.3	16.4'	N04°08.2	12.4'	54.1'
19	103°23.6	23.6	236°06.7	16.4'	04°20.6	12.4'	54.2'
20	118°23.6	23.0	250°42.1	16.4'	04°33.0	12.4'	54.2'
21	133°23.7	· · 22.4	265°17.5	16.4'	04°45.3	12.3'	54.2'
22	148°23.7	21.8	279°52.8	16.3'	04°57.7	12.3'	54.2'
23	163°23.7	21.1	294°28.2	16.3'	05°10.0	12.3'	54.2'
SD = 15.7' $d = -0.6'$			SD = 14.7'				
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	178°23.7	N18°20.5	309°03.5	16.3'	N05°22.3	12.3'	54.2'
1	193°23.8	19.9	323°38.7	16.2'	05°34.6	12.3'	54.2'
2	208°23.8	19.3	338°14.0	16.2'	05°46.9	12.3'	54.2'
3	223°23.8	· · 18.7	352°49.2	16.2'	05°59.1	12.2'	54.2'
4	238°23.9	18.1	7°24.4	16.2'	06°11.4	12.2'	54.2'
5	253°23.9	17.5	21°59.6	16.1'	06°23.6	12.2'	54.3'
6	268°23.9	N18°16.8	36°34.7	16.1'	N06°35.8	12.2'	54.3'
7	283°24.0	16.2	51°09.8	16.1'	06°47.9	12.2'	54.3'
8	298°24.0	15.6	65°44.8	16.0'	07°00.1	12.1'	54.3'
9	313°24.0	· · 15.0	80°19.9	16.0'	07°12.2	12.1'	54.3'
10	328°24.1	14.4	94°54.9	16.0'	07°24.4	12.1'	54.3'
11	343°24.1	13.8	109°29.8	15.9'	07°36.4	12.1'	54.3'
12	358°24.1	N18°13.1	124°04.7	15.9'	N07°48.5	12.0'	54.4'
13	13°24.2	12.5	138°39.6	15.8'	08°00.5	12.0'	54.4'
14	28°24.2	11.9	153°14.5	15.8'	08°12.6	12.0'	54.4'
15	43°24.2	· · 11.3	167°49.2	15.8'	08°24.6	12.0'	54.4'
16	58°24.3	10.7	182°24.0	15.7'	08°36.5	11.9'	54.4'
17	73°24.3	10.0	196°58.7	15.7'	08°48.5	11.9'	54.4'
18	88°24.3	N18°09.4	211°33.4	15.6'	N09°00.4	11.9'	54.4'
19	103°24.4	08.8	226°08.0	15.6'	09°12.2	11.8'	54.5'
20	118°24.4	08.2	240°42.6	15.5'	09°24.1	11.8'	54.5'
21	133°24.4	· · 07.6	255°17.1	15.5'	09°35.9	11.8'	54.5'
22	148°24.5	06.9	269°51.6	15.4'	09°47.7	11.8'	54.5'
23	163°24.5	06.3	284°26.1	15.4'	09°59.4	11.7'	54.5'
SD = 15.7' $d = -0.6'$			SD = 14.8'				
Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	178°24.5	N18°05.7	299°00.4	15.3'	N10°11.2	11.7'	54.5'
1	193°24.6	05.1	313°34.8	15.3'	10°22.9	11.7'	54.6'
2	208°24.6	04.4	328°09.1	15.2'	10°34.5	11.6'	54.6'
3	223°24.6	· · 03.8	342°43.3	15.2'	10°46.1	11.6'	54.6'
4	238°24.7	03.2	357°17.5	15.1'	10°57.7	11.5'	54.6'
5	253°24.7	02.5	11°51.6	15.1'	11°09.3	11.5'	54.6'
6	268°24.8	N18°01.9	26°25.7	15.0'	N11°20.8	11.5'	54.6'
7	283°24.8	01.3	40°59.7	15.0'	11°32.2	11.4'	54.7'
8	298°24.8	00.7	55°33.7	14.9'	11°43.7	11.4'	54.7'
9	313°24.9	18°00.0	70°07.6	14.8'	11°55.1	11.4'	54.7'
10	328°24.9	17°59.4	84°41.4	14.8'	12°06.4	11.3'	54.7'
11	343°24.9	58.8	99°15.2	14.7'	12°17.7	11.3'	54.7'
12	358°25.0	N17°58.1	113°48.9	14.7'	N12°29.0	11.2'	54.8'
13	13°25.0	57.5	128°22.6	14.6'	12°40.2	11.2'	54.8'
14	28°25.1	56.9	142°56.2	14.5'	12°51.4	11.1'	54.8'
15	43°25.1	· · 56.2	157°29.8	14.5'	13°02.5	11.1'	54.8'
16	58°25.1	55.6	172°03.2	14.4'	13°13.6	11.0'	54.8'
17	73°25.2	55.0	186°36.6	14.3'	13°24.7	11.0'	54.9'
18	88°25.2	N17°54.3	201°10.0	14.3'	N13°35.7	10.9'	54.9'
19	103°25.3	53.7	215°43.3	14.2'	13°46.6	10.9'	54.9'
20	118°25.3	53.1	230°16.5	14.1'	13°57.5	10.8'	54.9'
21	133°25.3	· · 52.4	244°49.6	14.1'	14°08.4	10.8'	55.0'
22	148°25.4	51.8	259°22.7	14.0'	14°19.2	10.7'	55.0'
23	163°25.4	51.2	273°55.7	13.9'	14°29.9	10.7'	55.0'
SD = 15.8' $d = -0.6'$			SD = 14.9'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	□	□	□	□	□	□
N 70°	////	////	01:16	22:49	////	////
68°	////	////	02:09	21:59	////	////
66°	////	////	02:41	21:29	////	////
64°	////	01:33	03:04	21:06	22:35	////
62°	////	02:10	03:23	20:48	21:59	////
60°	////	02:36	03:38	20:33	21:34	////
N 58°	01:24	02:56	03:51	20:21	21:14	22:44
56°	01:59	03:13	04:02	20:10	20:58	22:11
54°	02:23	03:26	04:11	20:00	20:45	21:48
52°	02:42	03:38	04:20	19:52	20:33	21:29
50°	02:57	03:49	04:28	19:44	20:23	21:14
45°	03:27	04:10	04:44	19:28	20:02	20:45
N 40°	03:49	04:27	04:57	19:15	19:45	20:23
35°	04:07	04:41	05:08	19:04	19:32	20:05
30°	04:21	04:52	05:18	18:54	19:20	19:51
20°	04:44	05:11	05:35	18:38	19:01	19:29
N 10°	05:01	05:27	05:49	18:23	18:45	19:11
0°	05:16	05:41	06:03	18:10	18:32	18:57
S 10°	05:29	05:54	06:16	17:57	18:19	18:44
20°	05:41	06:07	06:30	17:43	18:06	18:32
30°	05:52	06:21	06:46	17:27	17:52	18:21
35°	05:58	06:29	06:55	17:18	17:44	18:15
40°	06:04	06:37	07:06	17:07	17:36	18:09
45°	06:11	06:47	07:18	16:55	17:27	18:02
S 50°	06:19	06:58	07:33	16:40	17:16	17:55
52°	06:22	07:03	07:40	16:33	17:11	17:52
54°	06:25	07:08	07:48	16:26	17:05	17:48
56°	06:29	07:14	07:56	16:17	16:59	17:45
58°	06:33	07:21	08:06	16:07	16:52	17:41
S 60°	06:37	07:29	08:17	15:56	16:45	17:36

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	20:21	19:56	19:22	09:28	11:19	13:22
N 70°	20:27	20:10	19:48	09:25	11:07	12:57
68°	20:32	20:21	20:08	09:22	10:58	12:39
66°	20:37	20:31	20:24	09:20	10:50	12:24
64°	20:40	20:39	20:37	09:18	10:44	12:12
62°	20:44	20:46	20:49	09:16	10:38	12:02
60°	20:46	20:51	20:58	09:15	10:33	11:54
N 58°	20:49	20:57	21:06	09:14	10:29	11:46
56°	20:51	21:01	21:14	09:13	10:25	11:40
54°	20:53	21:06	21:21	09:12	10:22	11:34
52°	20:55	21:10	21:26	09:11	10:19	11:28
50°	20:57	21:13	21:32	09:10	10:16	11:24
45°	21:00	21:21	21:44	09:08	10:10	11:13
N 40°	21:03	21:27	21:53	09:07	10:05	11:05
35°	21:06	21:33	22:02	09:05	10:01	10:58
30°	21:08	21:38	22:09	09:04	09:57	10:51
20°	21:13	21:46	22:22	09:02	09:50	10:40
N 10°	21:16	21:54	22:34	09:00	09:45	10:30
0°	21:20	22:01	22:44	08:59	09:39	10:21
S 10°	21:23	22:08	22:55	08:57	09:34	10:12
20°	21:27	22:16	23:07	08:55	09:28	10:03
30°	21:31	22:25	23:20	08:53	09:22	09:52
35°	21:34	22:30	23:28	08:52	09:18	09:46
40°	21:37	22:36	23:36	08:51	09:14	09:39
45°	21:40	22:43	23:47	08:49	09:09	09:31
S 50°	21:44	22:51	■	08:48	09:03	09:21
52°	21:46	22:55	· ·	08:47	09:01	09:16
54°	21:48	22:59	· ·	08:46	08:58	09:11
56°	21:50	23:04	· ·	08:45	08:55	09:06
58°	21:53	23:09	· ·	08:44	08:51	09:00
S 60°	21:56	23:15	· ·	08:43	08:47	08:53

Day	Sun		Mer. Pass hh:mm	Moon		Age 19-21 87-71%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
30	06:28	06:26	12:06	02:49	15:09	
31	06:25	06:24	12:06	03:30	15:50	
01	06:22	06:20	12:06	04:11	16:33	

Aries		Venus		Mars		Jupiter		Saturn	
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	310°29.1	135°24.6	N02°11.2	128°37.2	S00°19.3	306°55.6	S00°01.8	128°42.4	N01°39.3
1	325°31.6	150°24.7	10.0	143°38.3	20.0	321°58.1	01.8	143°44.7	39.2
2	340°34.0	165°24.8	08.7	158°39.4	20.6	337°00.6	01.9	158°46.9	39.1
3	355°36.5	180°25.0	•• 07.5	173°40.4	•• 21.2	352°03.2	•• 01.9	173°49.2	•• 39.0
4	10°39.0	195°25.1	06.3	188°41.5	21.9	7°05.7	01.9	188°51.4	38.9
5	25°41.4	210°25.2	05.0	203°42.6	22.5	22°08.2	02.0	203°53.7	38.8
6	40°43.9	225°25.3	N02°03.8	218°43.7	S00°23.2	37°10.8	S00°02.0	218°55.9	N01°38.7
7	55°46.4	240°25.5	02.6	233°44.7	23.8	52°13.3	02.1	233°58.2	38.6
8	70°48.8	255°25.6	01.3	248°45.8	24.4	67°15.8	02.1	249°00.5	38.5
9	85°51.3	270°25.7	02°00.1	263°46.9	•• 25.1	82°18.4	•• 02.2	264°02.7	•• 38.4
10	100°53.7	285°25.8	01°58.9	278°47.9	25.7	97°20.9	02.2	279°05.0	38.3
11	115°56.2	300°26.0	57.6	293°49.0	26.4	112°23.4	02.2	294°07.2	38.2
12	130°58.7	315°26.1	N01°56.4	308°50.1	S00°27.0	127°26.0	S00°02.3	309°09.5	N01°38.1
13	146°01.1	330°26.2	55.2	323°51.1	27.6	142°28.5	02.3	324°11.7	38.0
14	161°03.6	345°26.4	53.9	338°52.2	28.3	157°31.0	02.4	339°14.0	37.9
15	176°06.1	0°26.5	•• 52.7	353°53.3	•• 28.9	172°33.6	•• 02.4	354°16.2	•• 37.8
16	191°08.5	15°26.6	51.5	8°54.3	29.6	187°36.1	02.5	9°18.5	37.7
17	206°11.0	30°26.7	50.2	23°55.4	30.2	202°38.6	02.5	24°20.7	37.6
18	221°13.5	45°26.9	N01°49.0	38°56.5	S00°30.9	217°41.2	S00°02.6	39°23.0	N01°37.5
19	236°15.9	60°27.0	47.8	53°57.6	31.5	232°43.7	02.6	54°25.3	37.4
20	251°18.4	75°27.1	46.5	68°58.6	32.1	247°46.2	02.6	69°27.5	37.3
21	266°20.8	90°27.3	•• 45.3	83°59.7	•• 32.8	262°48.8	•• 02.7	84°29.8	•• 37.3
22	281°23.3	105°27.4	44.0	99°00.8	33.4	277°51.3	02.7	99°32.0	37.2
23	296°25.8	120°27.5	42.8	114°01.8	34.1	292°53.8	02.8	114°34.3	37.1
Mer.pass. 03:18		ν 0.1' d -1.2' m -4.27		ν 1.1' d 0.6' m 1.48		ν 2.5' d 0.0' m -2.70		ν 2.3' d -0.1' m 1.00	

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	311°28.2	135°27.7	N01°41.6	129°02.9	S00°34.7	307°56.4	S00°02.8	129°36.5	N01°37.0
1	326°30.7	150°27.8	40.3	144°04.0	35.3	322°58.9	02.9	144°38.8	36.9
2	341°33.2	165°27.9	39.1	159°05.0	36.0	338°01.5	02.9	159°41.0	36.8
3	356°35.6	180°28.1	· · 37.9	174°06.1	· · 36.6	353°04.0	· · 03.0	174°43.3	· · 36.7
4	11°38.1	195°28.2	36.6	189°07.2	37.3	8°06.5	03.0	189°45.5	36.6
5	26°40.6	210°28.3	35.4	204°08.2	37.9	23°09.1	03.0	204°47.8	36.5
6	41°43.0	225°28.5	N01°34.2	219°09.3	S00°38.5	38°11.6	S00°03.1	219°50.0	N01°36.4
7	56°45.5	240°28.6	32.9	234°10.4	39.2	53°14.2	03.1	234°52.3	36.3
8	71°48.0	255°28.8	31.7	249°11.4	39.8	68°16.7	03.2	249°54.6	36.2
9	86°50.4	270°28.9	· · 30.5	264°12.5	· · 40.5	83°19.2	· · 03.2	264°56.8	· · 36.1
10	101°52.9	285°29.0	29.2	279°13.6	41.1	98°21.8	03.3	279°59.1	36.0
11	116°55.3	300°29.2	28.0	294°14.6	41.7	113°24.3	03.3	295°01.3	35.9
12	131°57.8	315°29.3	N01°26.8	309°15.7	S00°42.4	128°26.9	S00°03.4	310°03.6	N01°35.8
13	147°00.3	330°29.5	25.5	324°16.8	43.0	143°29.4	03.4	325°05.8	35.7
14	162°02.7	345°29.6	24.3	339°17.8	43.7	158°31.9	03.5	340°08.1	35.6
15	177°05.2	0°29.7	· · 23.1	354°18.9	· · 44.3	173°34.5	· · 03.5	355°10.3	· · 35.5
16	192°07.7	15°29.9	21.8	9°20.0	45.0	188°37.0	03.6	10°12.6	35.4
17	207°10.1	30°30.0	20.6	24°21.0	45.6	203°39.6	03.6	25°14.8	35.3
18	222°12.6	45°30.2	N01°19.4	39°22.1	S00°46.2	218°42.1	S00°03.6	40°17.1	N01°35.2
19	237°15.1	60°30.3	18.1	54°23.2	46.9	233°44.7	03.7	55°19.3	35.1
20	252°17.5	75°30.5	16.9	69°24.2	47.5	248°47.2	03.7	70°21.6	35.0
21	267°20.0	90°30.6	· · 15.6	84°25.3	· · 48.2	263°49.8	· · 03.8	85°23.8	· · 34.9
22	282°22.5	105°30.7	14.4	99°26.3	48.8	278°52.3	03.8	100°26.1	34.8
23	297°24.9	120°30.9	13.2	114°27.4	49.4	293°54.8	03.9	115°28.3	34.7
Mer.pass. 03:14		ν 0.1' d -1.2' m -4.28		ν 1.1' d 0.6' m 1.49		ν 2.5' d 0.0' m -2.71		ν 2.3' d -0.1' m 1.00	

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	312°27.4	135°31.0	N01°11.9	129°28.5	S00°50.1	308°57.4	S00°03.9	130°30.6	N01°34.6
1	327°29.8	150°31.2	10.7	144°29.5	50.7	323°59.9	04.0	145°32.8	34.5
2	342°32.3	165°31.3	09.5	159°30.6	51.4	339°02.5	04.0	160°35.1	34.4
3	357°34.8	180°31.5	· · 08.2	174°31.7	· · 52.0	354°05.0	· · 04.1	175°37.3	· · 34.3
4	12°37.2	195°31.6	07.0	189°32.7	52.7	9°07.6	04.1	190°39.6	34.2
5	27°39.7	210°31.8	05.8	204°33.8	53.3	24°10.1	04.2	205°41.8	34.1
6	42°42.2	225°31.9	N01°04.5	219°34.9	S00°53.9	39°12.7	S00°04.2	220°44.1	N01°34.0
7	57°44.6	240°32.1	03.3	234°35.9	54.6	54°15.2	04.3	235°46.3	33.9
8	72°47.1	255°32.2	02.1	249°37.0	55.2	69°17.8	04.3	250°48.6	33.8
9	87°49.6	270°32.4	01°00.8	264°38.1	· · 55.9	84°20.3	· · 04.4	265°50.8	· · 33.7
10	102°52.0	285°32.5	00°59.6	279°39.1	56.5	99°22.9	04.4	280°53.1	33.6
11	117°54.5	300°32.7	58.4	294°40.2	57.1	114°25.4	04.5	295°55.3	33.5
12	132°56.9	315°32.8	N00°57.1	309°41.2	S00°57.8	129°28.0	S00°04.5	310°57.6	N01°33.4
13	147°59.4	330°33.0	55.9	324°42.3	58.4	144°30.5	04.6	325°59.8	33.3
14	163°01.9	345°33.1	54.6	339°43.4	59.1	159°33.1	04.6	341°02.1	33.2
15	178°04.3	0°33.3	· · 53.4	354°44.4	00°59.7	174°35.6	· · 04.7	356°04.3	· · 33.1
16	193°06.8	15°33.4	52.2	9°45.5	01°00.4	189°38.2	04.7	11°06.6	33.0
17	208°09.3	30°33.6	50.9	24°46.6	01.0	204°40.7	04.8	26°08.8	32.9
18	223°11.7	45°33.7	N00°49.7	39°47.6	S01°01.6	219°43.3	S00°04.8	41°11.1	N01°32.8
19	238°14.2	60°33.9	48.5	54°48.7	02.3	234°45.8	04.9	56°13.3	32.7
20	253°16.7	75°34.1	47.2	69°49.7	02.9	249°48.4	04.9	71°15.6	32.6
21	268°19.1	90°34.2	· · 46.0	84°50.8	· · 03.6	264°50.9	· · 05.0	86°17.8	· · 32.5
22	283°21.6	105°34.4	44.8	99°51.9	04.2	279°53.5	05.0	101°20.1	32.4
23	298°24.1	120°34.5	43.5	114°52.9	04.9	294°56.0	05.1	116°22.3	32.3
Mer.pass. 03:10	ν 0.1' d -1.2' m -4.29			ν 1.1' d 0.6' m 1.49		ν 2.5' d 0.0' m -2.72		ν 2.3' d -0.1' m 1.00	

Stars		
	SHA	Dec
Alpheratz	357° 45.5	29° 09.0
Ankaa	353° 17.4	-42° 14.5
Schedar	349° 42.7	56° 35.6
Diphda	348° 57.8	-17° 55.4
Achernar	335° 28.0	-57° 10.6
Hamal	328° 03.1	23° 30.8
Polaris	318° 49.4	89° 18.4
Acamar	315° 19.9	-40° 15.4
Menkar	314° 17.3	4° 08.0
Mirfak	308° 43.5	49° 53.8
Aldebaran	290° 51.9	16° 31.9
Rigel	281° 14.2	-8° 11.2
Capella	280° 37.8	46° 00.4
Bellatrix	278° 34.4	6° 21.6
Elnath	278° 15.5	28° 36.9
Anilam	275° 48.7	-1° 11.6
Betelgeuse	271° 03.8	7° 24.6
Canopus	263° 57.5	-52° 41.9
Sirius	258° 35.8	-16° 43.7
Adhara	255° 14.5	-28° 59.1
Procyon	245° 02.2	5° 11.9
Pollux	243° 30.6	28° 00.0
Avior	234° 19.6	-59° 32.6
Suhail	222° 54.5	-43° 28.6
Miaplacidus	221° 41.2	-69° 45.7
Alphard	217° 58.4	-8° 42.3
Regulus	207° 46.0	11° 54.9
Dubhe	193° 54.6	61° 41.7
Denebola	182° 36.0	14° 30.8
Genah	175° 54.6	-17° 36.2
Acrux	173° 12.2	-63° 09.8
Gacrux	172° 03.7	-57° 10.6
Alioth	166° 22.6	55° 54.3
Spica	158° 33.5	-11° 13.1
Alkaid	153° 00.5	49° 15.8
Hadar	148° 51.1	-60° 25.8
Menkent	148° 10.1	-36° 25.5
Arcturus	145° 57.6	19° 07.7
Rigel Kent.	139° 54.6	-60° 53.0
Kochab	137° 19.6	74° 07.0
Zuben 'ubi	137° 07.7	-16° 05.2
Alphecca	126° 12.6	26° 40.9
Antares	112° 28.6	-26° 27.4
Atria	107° 32.0	-69° 03.0
Sabik	102° 14.7	-15° 44.3
Shaula	96° 24.4	-37° 06.7
Rasalhague	96° 08.1	12° 33.3
Eltanin	90° 46.7	51° 29.5
Kaus Aust.	83° 46.2	-34° 22.8
Vega	80° 40.0	38° 47.8
Nunki	76° 00.5	-26° 16.9
Altair	62° 09.9	8° 53.9
Peacock	53° 21.8	-56° 41.9
Deneb	49° 32.5	45° 19.2
Enif	33° 48.8	9° 55.6
Al Na'ir	27° 45.8	-46° 54.3
Fomalhaut	15° 25.9	-29° 33.7
Scheat	13° 55.1	28° 08.5
Markab	13° 40.1	15° 15.9

Aug 02 Mon	SHA	Mer.pass
Venus	184°55.5	14:58
Mars	178°08.1	15:24
Jupiter	356°26.5	03:32
Saturn	178°13.3	15:23

Aug 03 Tue	SHA	Mer.pass
Venus	183°59.4	14:58
Mars	177°34.7	15:23
Jupiter	356°28.1	03:28
Saturn	178°08.3	15:19

Aug 04 Wed	SHA	Mer.pass
Venus	183°03.7	14:58
Mars	177°01.1	15:21
Jupiter	356°30.0	03:24
Saturn	178°03.2	15:16

Horizontal parallax	
Venus:	0.2
Mars:	0.1

Sun			Moon			
h	GHA	Dec	GHA	ν	Dec	d
Mon	HP					
0	178°25.5	N17°50.5	288°28.6	13.9'	N14°40.6	10.6'
1	193°25.5	49.9	303°01.5	13.8'	14°51.2	10.6'
2	208°25.6	49.3	317°34.3	13.7'	15°01.8	10.5'
3	223°25.6	· · 48.6	332°07.0	13.6'	15°12.3	10.5'
4	238°25.6	48.0	346°39.6	13.6'	15°22.8	10.4'
5	253°25.7	47.3	1°12.2	13.5'	15°33.2	10.4'
6	268°25.7	N17°46.7	15°44.6	13.4'	N15°43.6	10.3'
7	283°25.8	46.1	30°17.0	13.3'	15°53.9	10.2'
8	298°25.8	45.4	44°49.4	13.3'	16°04.1	10.2'
9	313°25.9	· · 44.8	59°21.6	13.2'	16°14.3	10.1'
10	328°25.9	44.1	73°53.8	13.1'	16°24.4	10.0'
11	343°26.0	43.5	88°25.9	13.0'	16°34.5	10.0'
12	358°26.0	N17°42.8	102°57.9	12.9'	N16°44.4	9.9'
13	13°26.0	42.2	117°29.8	12.8'	16°54.4	9.9'
14	28°26.1	41.6	132°01.7	12.8'	17°04.2	9.8'
15	43°26.1	· · 40.9	146°33.4	12.7'	17°14.0	9.7'
16	58°26.2	40.3	161°05.1	12.6'	17°23.7	9.6'
17	73°26.2	39.6	175°36.7	12.5'	17°33.4	9.6'
18	88°26.3	N17°39.0	190°08.2	12.4'	N17°42.9	9.5'
19	103°26.3	38.3	204°39.7	12.3'	17°52.5	9.4'
20	118°26.4	37.7	219°11.0	12.3'	18°01.9	9.4'
21	133°26.4	· · 37.0	233°42.2	12.2'	18°11.2	9.3'
22	148°26.5	36.4	248°13.4	12.1'	18°20.5	9.2'
23	163°26.5	35.7	262°44.5	12.0'	18°29.7	9.1'
SD = 15.8'		d = -0.6'	SD = 15.0'			

Tue	GHA	Dec	GHA	ν	Dec	d
HP						
0	178°26.6	N17°35.1	277°15.5	11.9'	N18°38.9	9.1'
1	193°26.6	34.4	291°46.4	11.8'	18°47.9	9.0'
2	208°26.7	33.8	306°17.2	11.7'	18°56.9	8.9'
3	223°26.7	· · 33.1	320°47.9	11.6'	19°05.8	8.8'
4	238°26.8	32.5	335°18.5	11.5'	19°14.6	8.7'
5	253°26.8	31.8	349°49.0	11.4'	19°23.3	8.6'
6	268°26.9	N17°31.2	4°19.5	11.3'	N19°32.0	8.6'
7	283°26.9	30.5	18°49.8	11.3'	19°40.6	8.5'
8	298°27.0	29.9	33°20.1	11.2'	19°49.0	8.4'
9	313°27.0	· · 29.2	47°50.2	11.1'	19°57.4	8.3'
10	328°27.1	28.6	62°20.3	11.0'	20°05.7	8.2'
11	343°27.1	27.9	76°50.3	10.9'	20°13.9	8.1'
12	358°27.2	N17°27.3	91°20.2	10.8'	N20°22.1	8.0'
13	13°27.2	26.6	105°49.9	10.7'	20°30.1	7.9'
14	28°27.3	25.9	120°19.6	10.6'	20°38.0	7.8'
15	43°27.3	· · 25.3	134°49.2	10.5'	20°45.9	7.7'
16	58°27.4	24.6	149°18.7	10.4'	20°53.6	7.7'
17	73°27.4	24.0	163°48.1	10.3'	21°01.3	7.6'
18	88°27.5	N17°23.3	178°17.4	10.2'	N21°08.8	7.5'
19	103°27.5	22.7	192°46.6	10.1'	21°16.3	7.4'
20	118°27.6	22.0	207°15.7	10.0'	21°23.6	7.3'
21	133°27.6	· · 21.3	221°44.8	9.9'	21°30.9	7.2'
22	148°27.7	20.7	236°13.7	9.8'	21°38.0	7.0'
23	163°27.7	20.0	250°42.5	9.7'	21°45.1	6.9'
SD = 15.8'		d = -0.6'	SD = 15.2'			

Wed	GHA	Dec	GHA	ν	Dec	d
HP						
0	178°27.8	N17°19.4	265°11.2	9.6'	N21°52.0	6.8'
1	193°27.8	18.7	279°39.9	9.5'	21°58.9	6.7'
2	208°27.9	18.0	294°08.4	9.4'	22°05.6	6.6'
3	223°28.0	· · 17.4	308°36.8	9.3'	22°12.2	6.5'
4	238°28.0	16.7	323°05.2	9.2'	22°18.7	6.4'
5	253°28.1	16.0	337°33.4	9.1'	22°25.1	6.3'
6	268°28.1	N17°15.4	352°01.5	9.0'	N22°31.4	6.2'
7	283°28.2	14.7	6°29.6	9.0'	22°37.6	6.1'
8	298°28.2	14.1	20°57.5	8.9'	22°43.7	5.9'
9	313°28.3	· · 13.4	35°25.4	8.8'	22°49.6	5.8'
10	328°28.4	12.7	49°53.2	8.7'	22°55.4	5.7'
11	343°28.4	12.1	64°20.8	8.6'	23°01.2	5.6'
12	358°28.5	N17°11.4	78°48.4	8.5'	N23°06.7	5.5'
13	13°28.5	10.7	93°15.9	8.4'	23°12.2	5.3'
14	28°28.6	10.1	107°43.2	8.3'	23°17.6	5.2'
15	43°28.6	· · 09.4	122°10.5	8.2'	23°22.8	5.1'
16	58°28.7	08.7	136°37.7	8.1'	23°27.9	5.0'
17	73°28.8	08.0	151°04.8	8.0'	23°32.9	4.8'
18	88°28.8	N17°07.4	165°31.8	7.9'	N23°37.7	4.7'
19	103°28.9	06.7	179°58.7	7.8'	23°42.4	4.6'
20	118°28.9	06.0	194°25.6	7.7'	23°47.0	4.5'
21	133°29.0	· · 05.4	208°52.3	7.6'	23°51.5	4.3'
22	148°29.1	04.7	223°18.9	7.6'	23°55.8	4.2'
23	163°29.1	04.0	237°45.5	7.5'	24°00.0	4.1'
SD = 15.8'		d = -0.7'	SD = 15.4'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	□	□	□	□	□	□
N 70°	////	////	01:42	22:24	////	////
68°	////	////	02:24	21:44	////	////
66°	////	00:51	02:53	21:17	23:10	////
64°	////	01:50	03:14	20:56	22:18	////
62°	////	02:22	03:31	20:40	21:47	////
60°	00:46	02:46	03:45	20:26	21:24	23:16
N 58°	01:40	03:04	03:57	20:14	21:06	22:29
56°	02:09	03:20	04:07	20:04	20:51	22:00
54°	02:31	03:32	04:16	19:55	20:38	21:39
52°	02:49	03:44	04:25	19:47	20:27	21:22
50°	03:03	03:54	04:32	19:40	20:18	21:08
45°	03:32	04:14	04:47	19:24	19:58	20:40
N 40°	03:53	04:30	05:00	19:12	19:42	20:19
35°	04:10	04:43	05:11	19:01	19:29	20:02
30°	04:23	04:54	05:20	18:52	19:18	19:49
20°	04:45	05:13	05:36	18:36	18:59	19:27
N 10°	05:02	05:28	05:50	18:22	18:44	19:10
0°	05:16	05:41	06:03	18:10	18:31	18:56
S 10°	05:28	05:53	06:15	17:57	18:19	18:44
20°	05:39	06:06	06:29	17:44	18:07	18:33
30°	05:50	06:19	06:44	17:29	17:54	18:22
35°	05:56	06:26	06:53	17:20	17:47	18:17
40°	06:02	06:34	07:03	17:10	17:39	18:11
45°	06:08	06:43	07:15	16:58	17:30	18:05
S 50°	06:15	06:54	07:29	16:44	17:19	17:58
52°	06:18	06:58	07:35	16:38	17:15	17:55
54°	06:21	07:04	07:43	16:30	17:09	17:52
56°	06:24	07:09	07:51	16:22	17:04	17:49
58°	06:28	07:16	08:00	16:13	16:58	17:45
S 60°	06:32	07:23	08:10	16:03	16:51	17:42

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	18:08	□	□	16:11	□	□
N 70°	19:14	□	□	15:06	□	□
68°	19:51	19:15	□	14:30	16:47	□
66°	20:17	20:07	19:44	14:05	15:56	18:09
64°	20:37	20:39	20:47	13:46	15:25	17:07
62°	20:54	21:03	21:22	13:30	15:01	16:32
60°	21:08	21:22	21:47	13:17	14:43	16:07
N 58°	21:20	21:38	22:07	13:06	14:27	15:47
56°	21:30	21:52	22:24	12:56	14:14	15:31
54°	21:39	22:04	22:38	12:48	14:03	15:17
52°	21:47	22:14	22:51	12:40	13:53	15:05
50°	21:55	22:24	23:02	12:33	13:44	14:54
45°	22:11	22:43	23:25	12:18	13:25	14:32
N 40°	22:24	23:00	23:43	12:06	13:10	14:14
35°	22:35	23:13	23:59	11:56	12:57	13:59
30°	22:45	23:25	· ·	11:47	12:45	13:45
20°	23:02	23:46	· ·	11:32	12:26	13:23
N 10°	23:16	· ·	00:04	11:18	12:10	13:04
0°	23:30	· ·	00:20	11:06	11:54	12:46
S 10°	23:45	· ·	00:37	10:54	11:38	12:28
20°	· ·	00:00	00:56	10:40	11:22	12:09
30°	· ·	00:17	01:17	10:25	11:03	11:47
35°	· ·	00:27	01:29	10:17	10:52	11:34
40°	· ·	00:39	01:43	10:07	10:39	11:19
45°	· ·	00:53	02:00	09:55	10:25	11:01
S 50°	□	01:10	02:21	09:41	10:07	10:39
52°	00:05	01:18	02:31	09:35	09:58	10:29
54°	00:12	01:27	02:42	09:28	09:49	10:17
56°	00:19	01:37	02:55	09:20	09:38	10:04
58°	00:27	01:49	03:10	09:11	09:26	09:49
S 60°	00:37	02:02	03:29	09:00	09:12	09:30

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time	12 ^h		Mer. Pass.		
	00 ^h	mm:ss	hh:mm	Upper	Lower	22-24
				hh:mm	hh:mm	62-42%
02	06:18	06:16	12:06	04:55	17:18	
03	06:14	06:11	12:06	05:42	18:07	
04	06:09	06:06	12:06	06:33	19:00	

<https://pypi.org/project/sfalmanac/>

Sun				Moon			
h	GHA	Dec		GHA	ν	Dec	d
Thu							HP
0	178°29.2	N17°03.3		252°12.0	7.4'	N24°04.1	3.9'
1	193°29.2	02.7		266°38.3	7.3'	24°08.0	3.8'
2	208°29.3	02.0		281°04.6	7.2'	24°11.8	3.7'
3	223°29.4	· · 01.3		295°30.8	7.1'	24°15.4	3.5'
4	238°29.4	00.6		309°56.9	7.0'	24°18.9	3.4'
5	253°29.5	17°00.0		324°23.0	7.0'	24°22.3	3.2'
6	268°29.5	N16°59.3		338°48.9	6.9'	N24°25.6	3.1'
7	283°29.6	58.6		353°14.8	6.8'	24°28.6	2.9'
8	298°29.7	57.9		7°40.6	6.7'	24°31.6	2.8'
9	313°29.7	· · 57.3		22°06.3	6.6'	24°34.4	2.7'
10	328°29.8	56.6		36°31.9	6.5'	24°37.1	2.5'
11	343°29.9	55.9		50°57.5	6.5'	24°39.6	2.4'
12	358°29.9	N16°55.2		65°22.9	6.4'	N24°41.9	2.2'
13	13°30.0	54.6		79°48.3	6.3'	24°44.2	2.1'
14	28°30.0	53.9		94°13.6	6.2'	24°46.2	1.9'
15	43°30.1	· · 53.2		108°38.9	6.2'	24°48.2	1.8'
16	58°30.2	52.5		123°04.1	6.1'	24°49.9	1.6'
17	73°30.2	51.8		137°29.2	6.0'	24°51.5	1.5'
18	88°30.3	N16°51.2		151°54.2	6.0'	N24°53.0	1.3'
19	103°30.4	50.5		166°19.2	5.9'	24°54.3	1.2'
20	118°30.4	49.8		180°44.1	5.8'	24°55.5	1.0'
21	133°30.5	· · 49.1		195°08.9	5.8'	24°56.5	0.8'
22	148°30.6	48.4		209°33.7	5.7'	24°57.3	0.7'
23	163°30.6	47.7		223°58.4	5.7'	24°58.0	0.5'
SD = 15.8' d = -0.7'				SD = 15.6'			
Fri	GHA	Dec		GHA	ν	Dec	d
0	178°30.7	N16°47.1		238°23.1	5.6'	N24°58.5	0.4'
1	193°30.8	46.4		252°47.6	5.5'	24°58.9	0.2'
2	208°30.8	45.7		267°12.2	5.5'	24°59.1	0.1'
3	223°30.9	· · 45.0		281°36.7	5.4'	24°59.2	-0.1'
4	238°31.0	44.3		296°01.1	5.4'	24°59.1	-0.3'
5	253°31.0	43.6		310°25.5	5.3'	24°58.8	-0.4'
6	268°31.1	N16°42.9		324°49.8	5.3'	N24°58.4	-0.6'
7	283°31.2	42.3		339°14.0	5.2'	24°57.8	-0.8'
8	298°31.2	41.6		353°38.3	5.2'	24°57.0	-0.9'
9	313°31.3	· · 40.9		8°02.4	5.1'	24°56.1	-1.1'
10	328°31.4	40.2		22°26.6	5.1'	24°55.0	-1.3'
11	343°31.5	39.5		36°50.7	5.1'	24°53.8	-1.4'
12	358°31.5	N16°38.8		51°14.7	5.0'	N24°52.3	-1.6'
13	13°31.6	38.1		65°38.7	5.0'	24°50.8	-1.7'
14	28°31.7	37.4		80°02.7	4.9'	24°49.0	-1.9'
15	43°31.7	· · 36.7		94°26.7	4.9'	24°47.1	-2.1'
16	58°31.8	36.0		108°50.6	4.9'	24°45.0	-2.3'
17	73°31.9	35.3		123°14.4	4.8'	24°42.8	-2.4'
18	88°31.9	N16°34.7		137°38.3	4.8'	N24°40.3	-2.6'
19	103°32.0	34.0		152°02.1	4.8'	24°37.7	-2.8'
20	118°32.1	33.3		166°25.9	4.8'	24°35.0	-2.9'
21	133°32.2	· · 32.6		180°49.7	4.7'	24°32.1	-3.1'
22	148°32.2	31.9		195°13.4	4.7'	24°29.0	-3.3'
23	163°32.3	31.2		209°37.1	4.7'	24°25.7	-3.4'
SD = 15.8' d = -0.7'				SD = 15.9'			
Sat	GHA	Dec		GHA	ν	Dec	d
0	178°32.4	N16°30.5		224°00.8	4.7'	N24°22.3	-3.6'
1	193°32.4	29.8		238°24.5	4.7'	24°18.7	-3.8'
2	208°32.5	29.1		252°48.2	4.7'	24°14.9	-3.9'
3	223°32.6	· · 28.4		267°11.9	4.7'	24°11.0	-4.1'
4	238°32.7	27.7		281°35.5	4.6'	24°06.9	-4.3'
5	253°32.7	27.0		295°59.2	4.6'	24°02.6	-4.4'
6	268°32.8	N16°26.3		310°22.8	4.6'	N23°58.2	-4.6'
7	283°32.9	25.6		324°46.4	4.6'	23°53.6	-4.8'
8	298°33.0	24.9		339°10.1	4.6'	23°48.8	-4.9'
9	313°33.0	· · 24.2		353°33.7	4.6'	23°43.8	-5.1'
10	328°33.1	23.5		7°57.3	4.6'	23°38.7	-5.3'
11	343°33.2	22.8		22°20.9	4.6'	23°33.4	-5.5'
12	358°33.3	N16°22.1		36°44.6	4.6'	N23°28.0	-5.6'
13	13°33.3	21.4		51°08.2	4.6'	23°22.4	-5.8'
14	28°33.4	20.7		65°31.9	4.7'	23°16.6	-5.9'
15	43°33.5	· · 20.0		79°55.5	4.7'	23°10.6	-6.1'
16	58°33.6	19.3		94°19.2	4.7'	23°04.5	-6.3'
17	73°33.6	18.6		108°42.8	4.7'	22°58.2	-6.4'
18	88°33.7	N16°17.9		123°06.5	4.7'	N22°51.8	-6.6'
19	103°33.8	17.2		137°30.2	4.7'	22°45.2	-6.8'
20	118°33.9	16.5		151°54.0	4.7'	22°38.4	-6.9'
21	133°34.0	· · 15.8		166°17.7	4.8'	22°31.5	-7.1'
22	148°34.0	15.1		180°41.5	4.8'	22°24.4	-7.3'
23	163°34.1	14.4		195°05.3	4.8'	22°17.1	-7.4'
SD = 15.8' d = -0.7'				SD = 16.1'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	////	////	00:56	23:02	////	////
N 70°	////	////	02:03	22:03	////	////
68°	////	////	02:39	21:29	////	////
66°	////	01:22	03:04	21:05	22:42	////
64°	////	02:06	03:23	20:46	22:02	////
62°	////	02:34	03:39	20:31	21:35	////
60°	01:14	02:55	03:52	20:18	21:14	22:51
N 58°	01:54	03:12	04:03	20:07	20:58	22:14
56°	02:20	03:26	04:13	19:57	20:44	21:49
54°	02:40	03:39	04:22	19:49	20:32	21:30
52°	02:56	03:49	04:29	19:41	20:21	21:14
50°	03:09	03:58	04:36	19:35	20:12	21:01
45°	03:36	04:18	04:51	19:20	19:53	20:34
N 40°	03:56	04:33	05:03	19:08	19:38	20:15
35°	04:12	04:46	05:13	18:58	19:26	19:59
30°	04:26	04:56	05:22	18:49	19:15	19:46
20°	04:46	05:14	05:37	18:34	18:58	19:25
N 10°	05:02	05:28	05:50	18:21	18:43	19:09
0°	05:16	05:41	06:02	18:09	18:31	18:56
S 10°	05:27	05:53	06:15	17:57	18:19	18:44
20°	05:38	06:04	06:27	17:45	18:08	18:34
30°	05:48	06:17	06:42	17:30	17:55	18:24
35°	05:53	06:24	06:50	17:22	17:49	18:19
40°	05:59	06:31	07:00	17:13	17:41	18:14
45°	06:04	06:39	07:11	17:02	17:33	18:08
S 50°	06:11	06:49	07:24	16:48	17:23	18:02
52°	06:13	06:54	07:30	16:42	17:19	17:59
54°	06:16	06:59	07:37	16:35	17:14	17:57
56°	06:19	07:04	07:45	16:28	17:09	17:54
58°	06:22	07:10	07:53	16:19	17:03	17:50
S 60°	06:26	07:16	08:03	16:10	16:57	17:47

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	□	□	□	□	□	□
N 70°	□	□	□	□	□	□
68°	□	□	□	□	□	□
66°	□	□	23:40	□	□	20:25
64°	21:14	22:29	· ·	18:38	19:28	19:44
62°	22:00	23:10	· ·	17:52	18:47	19:16
60°	22:30	23:38	· ·	17:23	18:19	18:54
N 58°	22:53	· ·	00:00	17:00	17:57	18:36
56°	23:11	· ·	00:18	16:41	17:39	18:21
54°	23:27	· ·	00:33	16:26	17:24	18:08
52°	23:41	· ·	00:46	16:12	17:10	17:57
50°	23:52	· ·	00:57	16:01	16:59	17:47
45°	· ·	00:17	01:21	15:36	16:35	17:25
N 40°	· ·	00:37	01:40	15:16	16:15	17:08
35°	· ·	00:53	01:56	15:00	15:59	16:53
30°	00:13	01:07	02:10	14:46	15:45	16:40
20°	00:36	01:32	02:33	14:22	15:21	16:18
N 10°	00:56	01:53	02:53	14:01	15:00	15:59
0°	01:15	02:12	03:12	13:42	14:41	15:41
S 10°	01:33	02:32	03:31	13:22	14:21	15:23
20°	01:54	02:53	03:52	13:02	14:00	15:04
30°	02:17	03:17	04:15	12:38	13:36	14:41
35°	02:31	03:32	04:29	12:23	13:22	14:28
40°	02:47	03:48	04:44	12:07	13:05	14:13
45°	03:06	04:08	05:03	11:48	12:45	13:55
S 50°	03:30	04:33	05:27	11:23	12:21	13:32
52°	03:42	04:45	05:38	11:11	12:09	13:21
54°	03:55	04:59	05:51	10:58	11:55	13:08
56°	04:10	05:15	06:05	10:43	11:39	12:54
58°	04:28	05:34	06:22	10:24	11:20	12:37
S 60°	04:51	05:58	06:43	10:02	10:56	12:17

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Upper Mer. Pass. hh:mm	Lower Mer. Pass. hh:mm	
05	06:03	06:00	12:06	07:28	19:57	
06	05:57	05:54	12:06	08:27	20:57	
07	05:51	05:47	12:06	09:27	21:57	

August 08, 09, 10 UT (Sun., Mon., Tue.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	316°23.9	135°47.3	S00°46.6	131°10.0	S01°51.8	313°03.2	S00°09.2	134°06.3	N01°25.0					Alpheratz	357°45.4	29°09.0	
1	331°26.4	150°47.5	47.8	146°11.1	52.5	328°05.7	09.2	149°08.5	24.9					Ankaa	353°17.3	-42°14.5	
2	346°28.9	165°47.7	49.1	161°12.1	53.1	343°08.3	09.3	164°10.7	24.8					Schedar	349°42.7	56°35.7	
3	1°31.3	180°47.9	· · 50.3	176°13.2	· · 53.8	358°10.9	· · 09.3	179°13.0	· · 24.7					Diphda	348°57.7	-17°55.4	
4	16°33.8	195°48.1	51.5	191°14.2	54.4	13°13.5	09.4	194°15.2	24.6					Achernar	335°27.9	-57°10.6	
5	31°36.3	210°48.3	52.8	206°15.3	55.1	28°16.0	09.5	209°17.5	24.5					Hamal	328°03.0	23°30.8	
6	46°38.7	225°48.5	S00°54.0	221°16.3	S01°55.7	43°18.6	S00°09.5	224°19.7	N01°24.4					Polaris	318°46.2	89°18.4	
7	61°41.2	240°48.7	55.2	236°17.4	56.4	58°21.2	09.6	239°22.0	24.3					Acamar	315°19.8	-40°15.4	
8	76°43.7	255°48.9	56.5	251°18.4	57.0	73°23.8	09.6	254°24.2	24.2					Menkar	314°17.2	4°08.0	
9	91°46.1	270°49.1	· · 57.7	266°19.4	· · 57.7	88°26.4	· · 09.7	269°26.4	· · 24.1					Mirfak	308°43.4	49°53.8	
10	106°48.6	285°49.3	00°58.9	281°20.5	58.3	103°28.9	09.8	284°28.7	24.0					Aldebaran	290°51.9	16°31.9	
11	121°51.0	300°49.5	01°00.2	296°21.5	58.9	118°31.5	09.8	299°30.9	23.9					Rigel	281°14.2	-8°11.2	
12	136°53.5	315°49.7	S01°01.4	311°22.6	S01°59.6	133°34.1	S00°09.9	314°33.2	N01°23.7					Capella	280°37.8	46°00.4	
13	151°56.0	330°49.9	02.6	326°23.6	02°00.2	148°36.7	10.0	329°35.4	23.6					Bellatrix	278°34.4	6°21.6	
14	166°58.4	345°50.1	03.9	341°24.7	00.9	163°39.2	10.0	344°37.6	23.5					Elnath	278°15.4	28°36.9	
15	182°00.9	0°50.3	· · 05.1	356°25.7	· · 01.5	178°41.8	· · 10.1	359°39.9	· · 23.4					Alnilam	275°48.6	-1°11.6	
16	197°03.4	15°50.5	06.3	11°26.8	02.2	193°44.4	10.1	14°42.1	23.3					Betelgeuse	271°03.7	7°24.6	
17	212°05.8	30°50.7	07.6	26°27.8	02.8	208°47.0	10.2	29°44.4	23.2					Canopus	263°57.5	-52°41.9	
18	227°08.3	45°50.9	S01°08.8	41°28.9	S02°03.5	223°49.6	S00°10.3	44°46.6	N01°23.1					Sirius	258°35.8	-16°43.7	
19	242°10.8	60°51.1	10.0	56°29.9	04.1	238°52.1	10.3	59°48.8	23.0					Adhara	255°14.5	-28°59.1	
20	257°13.2	75°51.3	11.3	71°31.0	04.8	253°54.7	10.4	74°51.1	22.9					Procyon	245°02.2	5°11.9	
21	272°15.7	90°51.5	· · 12.5	86°32.0	· · 05.4	268°57.3	· · 10.5	89°53.3	· · 22.8					Pollux	243°30.6	28°00.0	
22	287°18.2	105°51.7	13.7	101°33.1	06.0	283°59.9	10.5	104°55.6	22.7					Avior	234°19.6	-59°32.6	
23	302°20.6	120°51.9	14.9	116°34.1	06.7	299°02.5	10.6	119°57.8	22.6					Suhail	222°54.5	-43°28.6	
Mer.pass. 02:54			ν 0.2' d 1.2' m -4.32	ν 1.1' d 0.6' m 1.50	ν 2.6' d 0.1' m -2.75	ν 2.2' d -0.1' m 1.00											

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	317°23.1	135°52.1	S01°16.2	131°35.2	S02°07.3	314°05.1	S00°10.6	135°00.0	N01°22.5
1	332°25.5	150°52.3	17.4	146°36.2	08.0	329°07.6	10.7	150°02.3	22.4
2	347°28.0	165°52.6	18.6	161°37.3	08.6	344°10.2	10.8	165°04.5	22.3
3	2°30.5	180°52.8	· · 19.9	176°38.3	· · 09.3	359°12.8	· · 10.8	180°06.8	· · 22.2
4	17°32.9	195°53.0	21.1	191°39.4	09.9	14°15.4	10.9	195°09.0	22.1
5	32°35.4	210°53.2	22.3	206°40.4	10.6	29°18.0	11.0	210°11.2	22.0
6	47°37.9	225°53.4	S01°23.6	221°41.4	S02°11.2	44°20.6	S00°11.0	225°13.5	N01°21.9
7	62°40.3	240°53.6	24.8	236°42.5	11.9	59°23.1	11.1	240°15.7	21.8
8	77°42.8	255°53.8	26.0	251°43.5	12.5	74°25.7	11.2	255°18.0	21.7
9	92°45.3	270°54.0	· · 27.3	266°44.6	· · 13.1	89°28.3	· · 11.2	270°20.2	· · 21.6
10	107°47.7	285°54.2	28.5	281°45.6	13.8	104°30.9	11.3	285°22.4	21.5
11	122°50.2	300°54.4	29.7	296°46.7	14.4	119°33.5	11.3	300°24.7	21.4
12	137°52.7	315°54.6	S01°31.0	311°47.7	S02°15.1	134°36.1	S00°11.4	315°26.9	N01°21.3
13	152°55.1	330°54.9	32.2	326°48.8	15.7	149°38.7	11.5	330°29.2	21.2
14	167°57.6	345°55.1	33.4	341°49.8	16.4	164°41.2	11.5	345°31.4	21.1
15	183°00.0	0°55.3	· · 34.6	356°50.9	· · 17.0	179°43.8	· · 11.6	0°33.6	· · 21.0
16	198°02.5	15°55.5	35.9	11°51.9	17.7	194°46.4	11.7	15°35.9	20.9
17	213°05.0	30°55.7	37.1	26°52.9	18.3	209°49.0	11.7	30°38.1	20.8
18	228°07.4	45°55.9	S01°38.3	41°54.0	S02°19.0	224°51.6	S00°11.8	45°40.3	N01°20.7
19	243°09.9	60°56.1	39.6	56°55.0	19.6	239°54.2	11.9	60°42.6	20.6
20	258°12.4	75°56.4	40.8	71°56.1	20.2	254°56.8	11.9	75°44.8	20.5
21	273°14.8	90°56.6	· · 42.0	86°57.1	· · 20.9	269°59.3	· · 12.0	90°47.1	· · 20.3
22	288°17.3	105°56.8	43.3	101°58.2	21.5	285°01.9	12.1	105°49.3	20.2
23	303°19.8	120°57.0	44.5	116°59.2	22.2	300°04.5	12.1	120°51.5	20.1
Mer.pass. 02:50			ν 0.2' d 1.2' m -4.33	ν 1.0' d 0.6' m 1.50	ν 2.6' d 0.1' m -2.75	ν 2.2' d -0.1' m 1.00			

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	318°22.2	135°57.2	S01°45.7	132°00.3	S02°22.8	315°07.1	S00°12.2	135°53.8	N01°20.0
1	333°24.7	150°57.4	46.9	147°01.3	23.5	330°09.7	12.3	150°56.0	19.9
2	348°27.1	165°57.7	48.2	162°02.3	24.1	345°12.3	12.3	165°58.3	19.8
3	3°29.6	180°57.9	· · 49.4	177°03.4	· · 24.8	0°14.9	· · 12.4	181°00.5	· · 19.7
4	18°32.1	195°58.1	50.6	192°04.4	25.4	15°17.5	12.5	196°02.7	19.6
5	33°34.5	210°58.3	51.9	207°05.5	26.1	30°20.1	12.5	211°05.0	19.5
6	48°37.0	225°58.5	S01°53.1	222°06.5	S02°26.7	45°22.7	S00°12.6	226°07.2	N01°19.4
7	63°39.5	240°58.8	54.3	237°07.6	27.4	60°25.2	12.7	241°09.4	19.3
8	78°41.9	255°59.0	55.6	252°08.6	28.0	75°27.8	12.7	256°11.7	19.2
9	93°44.4	270°59.2	· · 56.8	267°09.6	· · 28.6	90°30.4	· · 12.8	271°13.9	· · 19.1
10	108°46.9	285°59.4	58.0	282°10.7	29.3	105°33.0	12.9	286°16.1	19.0
11	123°49.3	300°59.6	01°59.2	297°11.7	29.9	120°35.6	12.9	301°18.4	18.9
12	138°51.8	315°59.9	S02°00.5	312°12.8	S02°30.6	135°38.2	S00°13.0	316°20.6	N01°18.8
13	153°54.3	331°00.1	01.7	327°13.8	31.2	150°40.8	13.1	331°22.9	18.7
14	168°56.7	346°00.3	02.9	342°14.8	31.9	165°43.4	13.1	346°25.1	18.6
15	183°59.2	1°00.5	· · 04.2	357°15.9	· · 32.5	180°46.0	· · 13.2	1°27.3	· · 18.5
16	199°01.6	16°00.8	05.4	12°16.9	33.2	195°48.6	13.3	16°29.6	18.4
17	214°04.1	31°01.0	06.6	27°18.0	33.8	210°51.2	13.4	31°31.8	18.3
18	229°06.6	46°01.2	S02°07.8	42°19.0	S02°34.5	225°53.8	S00°13.4	46°34.0	N01°18.2
19	244°09.0	61°01.4	09.1	57°20.0	35.1	240°56.4	13.5	61°36.3	18.1
20	259°11.5	76°01.7	10.3	72°21.1	35.8	255°59.0	13.6	76°38.5	18.0
21	274°14.0	91°01.9	· · 11.5	87°22.1	· · 36.4	271°01.6	· · 13.6	91°40.7	· · 17.9
22	289°16.4	106°02.1	12.7	102°23.2	37.0	286°04.2	13.7	106°43.0	17.8
23	304°18.9	121°02.3	14.0	117°24.2	37.7	301°06.7	13.8	121°45.2	17.6
Mer.pass. 02:46			ν 0.2' d 1.2' m -4.34	ν 1.0' d 0.6' m 1.50	ν 2.6' d 0.1' m -2.76	ν 2.2' d -0.1' m 1.00			

Aug 08 Sun	SHA	Mer.pass
Venus	179°23.4	14:57
Mars	174°46.1	15:14
Jupiter	356°39.2	03:07
Saturn	177°42.3	15:01

Aug 09 Mon	SHA	Mer.pass
Venus	178°29.1	14:56
Mars	174°12.1	15:13
Jupiter	356°42.0	03:03
Saturn	177°37.0	14:58

Aug 10 Tue	SHA	Mer.pass
Venus	177°35.0	14:56
Mars	173°38.0	15:11
Jupiter	356°44.9	02:59
Saturn	177°31.6	14:54

Horizontal parallax	
Venus:	0.2
Mars:	0.1

Sun							Moon						
h	GHA	Dec	GHA	ν	Dec	d	HP						
0	178°34.2	N16°13.7	209°29.1	4.8'	N22°09.7	-7.6'	60.1'						
1	193°34.3	13.0	223°52.9	4.9'	22°02.1	-7.7'	60.1'						
2	208°34.3	12.2	238°16.8	4.9'	21°54.4	-7.9'	60.1'						
3	223°34.4	11.5	252°40.6	4.9'	21°46.5	-8.1'	60.1'						
4	238°34.5	10.8	267°04.6	4.9'	21°38.4	-8.2'	60.2'						
5	253°34.6	10.1	281°28.5	5.0'	21°30.2	-8.4'	60.2'						
6	268°34.7	N16°09.4	295°52.5	5.0'	N21°21.8	-8.5'	60.2'						
7	283°34.7	08.7	310°16.5	5.0'	21°13.3	-8.7'	60.3'						
8	298°34.8	08.0	324°40.5	5.1'	21°04.6	-8.8'	60.3'						
9	313°34.9	07.3	339°04.6	5.1'	20°55.8	-9.0'	60.3'						
10	328°35.0	06.6	353°28.7	5.2'	20°46.8	-9.1'	60.4'						
11	343°35.1	05.9	7°52.9	5.2'	20°37.7	-9.3'	60.4'						
12	358°35.2	N16°05.1	22°17.1	5.2'	N20°28.4	-9.4'	60.4'						
13	13°35.2	04.4	36°41.3	5.3'	20°19.0	-9.6'	60.4'						
14	28°35.3	03.7	51°05.6	5.3'	20°09.4	-9.7'	60.5'						
15	43°35.4	03.0	65°30.0	5.4'	19°59.7	-9.9'	60.5'						
16	58°35.5	02.3	79°54.3	5.4'	19°49.8	-10.0'	60.5'						
17	73°35.6	01.6	94°18.7	5.5'	19°39.8	-10.2'	60.5'						
18	88°35.6	N16°00.9	108°43.2	5.5'	N19°29.7	-10.3'	60.6'						
19	103°35.7	16°00.2	123°07.7	5.6'	19°19.4	-10.4'	60.6'						
20	118°35.8	15°59.4	137°32.2	5.6'	19°08.9	-10.6'	60.6'						
21	133°35.9	58.7	151°56.8	5.6'	18°58.4	-10.7'	60.6'						
22	148°36.0	58.0	166°21.5	5.7'	18°47.7	-10.8'	60.7'						
23	163°36.1	57.3	180°46.2	5.8'	18°36.8	-11.0'	60.7'						
SD = 15.8' $d = -0.7'$							SD = 16.4'						

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	178°36.2	N15°56.6	195°10.9	5.8'	N18°25.9	-11.1'	60.7'
1	193°36.2	55.9	209°35.7	5.9'	18°14.7	-11.2'	60.7'
2	208°36.3	55.1	224°00.6	5.9'	18°03.5	-11.4'	60.8'
3	223°36.4	54.4	238°25.5	6.0'	17°52.1	-11.5'	60.8'
4	238°36.5	53.7	252°50.5	6.0'	17°40.6	-11.6'	60.8'
5	253°36.6	53.0	267°15.5	6.1'	17°29.0	-11.7'	60.8'
6	268°36.7	N15°52.3	281°40.6	6.1'	N17°17.3	-11.9'	60.8'
7	283°36.7	51.5	296°05.7	6.2'	17°05.4	-12.0'	60.9'
8	298°36.8	50.8	310°30.8	6.2'	16°53.4	-12.1'	60.9'
9	313°36.9	50.1	324°56.1	6.3'	16°41.3	-12.2'	60.9'
10	328°37.0	49.4	339°21.4	6.3'	16°29.1	-12.3'	60.9'
11	343°37.1	48.7	353°46.7	6.4'	16°16.7	-12.5'	60.9'
12	358°37.2	N15°47.9	8°12.1	6.5'	N16°04.3	-12.6'	60.9'
13	13°37.3	47.2	22°37.6	6.5'	15°51.7	-12.7'	61.0'
14	28°37.4	46.5	37°03.1	6.6'	15°39.0	-12.8'	61.0'
15	43°37.4	45.8	51°28.7	6.6'	15°26.2	-12.9'	61.0'
16	58°37.5	45.0	65°54.3	6.7'	15°13.3	-13.0'	61.0'
17	73°37.6	44.3	80°20.0	6.7'	15°00.3	-13.1'	61.0'
18	88°37.7	N15°43.6	94°45.7	6.8'	N14°47.2	-13.2'	61.0'
19	103°37.8	42.9	109°11.5	6.9'	14°34.0	-13.3'	61.0'
20	118°37.9	42.1	123°37.4	6.9'	14°20.7	-13.4'	61.1'
21	133°38.0	41.4	138°03.3	7.0'	14°07.3	-13.5'	61.1'
22	148°38.1	40.7	152°29.3	7.0'	13°53.8	-13.6'	61.1'
23	163°38.2	40.0	166°55.3	7.1'	13°40.2	-13.7'	61.1'
	SD = 15.8'	$d = -0.7'$			SD = 16.6'		

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	178°38.3	N15°39.2	181°21.4	7.1'	N13°26.5	-13.8'	61.1'
1	193°38.3	38.5	195°47.5	7.2'	13°12.7	-13.9'	61.1'
2	208°38.4	37.8	210°13.7	7.3'	12°58.8	-14.0'	61.1'
3	223°38.5	37.0	224°40.0	7.3'	12°44.8	-14.1'	61.1'
4	238°38.6	36.3	239°06.3	7.4'	12°30.8	-14.1'	61.1'
5	253°38.7	35.6	253°32.7	7.4'	12°16.7	-14.2'	61.1'
6	268°38.8	N15°34.8	267°59.1	7.5'	N12°02.4	-14.3'	61.2'
7	283°38.9	34.1	282°25.6	7.5'	11°48.1	-14.4'	61.2'
8	298°39.0	33.4	296°52.1	7.6'	11°33.8	-14.4'	61.2'
9	313°39.1	32.7	311°18.7	7.6'	11°19.3	-14.5'	61.2'
10	328°39.2	31.9	325°45.4	7.7'	11°04.8	-14.6'	61.2'
11	343°39.3	31.2	340°12.1	7.8'	10°50.2	-14.7'	61.2'
12	358°39.4	N15°30.5	354°38.8	7.8'	N10°35.5	-14.7'	61.2'
13	13°39.5	29.7	9°05.7	7.9'	10°20.8	-14.8'	61.2'
14	28°39.5	29.0	23°32.5	7.9'	10°06.0	-14.9'	61.2'
15	43°39.6	28.3	37°59.4	8.0'	09°51.1	-14.9'	61.2'
16	58°39.7	27.5	52°26.4	8.0'	09°36.2	-15.0'	61.2'
17	73°39.8	26.8	66°53.4	8.1'	09°21.2	-15.0'	61.2'
18	88°39.9	N15°26.1	81°20.5	8.1'	N09°06.2	-15.1'	61.2'
19	103°40.0	25.3	95°47.6	8.2'	08°51.1	-15.2'	61.2'
20	118°40.1	24.6	110°14.8	8.2'	08°35.9	-15.2'	61.2'
21	133°40.2	23.8	124°42.0	8.3'	08°20.7	-15.3'	61.2'
22	148°40.3	23.1	139°09.3	8.3'	08°05.5	-15.3'	61.2'
23	163°40.4	22.4	153°36.6	8.4'	07°50.2	-15.4'	61.2'
	SD = 15.8'	$d = -0.7'$			SD = 16.7'		

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	////	////	01:34	22:29	////	////
N 70°	////	////	02:22	21:44	////	////
68°	////	00:24	02:53	21:15	23:24	////
66°	////	01:45	03:15	20:53	22:21	////
64°	////	02:20	03:33	20:36	21:47	////
62°	00:22	02:45	03:47	20:22	21:23	23:29
60°	01:34	03:04	03:59	20:10	21:04	22:32
N 58°	02:06	03:20	04:10	20:00	20:49	22:01
56°	02:29	03:33	04:19	19:51	20:36	21:39
54°	02:48	03:45	04:27	19:43	20:25	21:21
52°	03:03	03:55	04:34	19:36	20:15	21:06
50°	03:15	04:03	04:41	19:30	20:06	20:54
45°	03:41	04:22	04:54	19:16	19:49	20:29
N 40°	04:00	04:36	05:06	19:05	19:34	20:10
35°	04:15	04:48	05:15	18:55	19:22	19:55
30°	04:28	04:58	05:24	18:47	19:12	19:43
20°	04:48	05:15	05:38	18:33	18:56	19:23
N 10°	05:03	05:29	05:51	18:20	18:42	19:08
0°	05:16	05:41	06:02	18:09	18:30	18:55
S 10°	05:27	05:52	06:14	17:58	18:19	18:45
20°	05:36	06:03	06:26	17:46	18:08	18:35
30°	05:46	06:14	06:39	17:32	17:57	18:25
35°	05:51	06:21	06:47	17:24	17:51	18:21
40°	05:56	06:28	06:56	17:15	17:44	18:16
45°	06:01	06:36	07:06	17:05	17:36	18:11
S 50°	06:06	06:45	07:19	16:53	17:27	18:06
52°	06:08	06:49	07:25	16:47	17:23	18:03
54°	06:11	06:53	07:31	16:41	17:19	18:01
56°	06:14	06:58	07:38	16:33	17:14	17:58
58°	06:16	07:03	07:46	16:25	17:09	17:56
S 60°	06:19	07:09	07:55	16:16	17:03	17:53

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	☐	☐	03:09	☐	21:07	20:27
N 70°	☐	00:45	03:33	21:28	20:40	20:15
68°	☐	01:29	03:52	20:42	20:20	20:05
66°	☐ ☐	01:58	04:07	20:12	20:04	19:57
64°	00:20	02:20	04:19	19:49	19:50	19:50
62°	00:48	02:37	04:29	19:31	19:39	19:44
60°	01:09	02:52	04:37	19:15	19:29	19:38
N 58°	01:26	03:04	04:45	19:02	19:20	19:33
56°	01:41	03:14	04:51	18:51	19:13	19:29
54°	01:53	03:24	04:57	18:41	19:06	19:25
52°	02:05	03:32	05:02	18:32	19:00	19:22
50°	02:14	03:39	05:07	18:24	18:54	19:19
45°	02:35	03:55	05:18	18:07	18:42	19:12
N 40°	02:51	04:08	05:26	17:53	18:32	19:06
35°	03:05	04:19	05:33	17:41	18:23	19:01
30°	03:18	04:28	05:40	17:31	18:16	18:57
20°	03:38	04:45	05:51	17:12	18:02	18:49
N 10°	03:56	04:59	06:00	16:56	17:51	18:42
0°	04:13	05:12	06:09	16:41	17:39	18:35
S 10°	04:30	05:25	06:18	16:26	17:28	18:29
20°	04:47	05:39	06:27	16:10	17:16	18:21
30°	05:08	05:55	06:38	15:51	17:02	18:13
35°	05:20	06:05	06:44	15:40	16:54	18:08
40°	05:33	06:15	06:51	15:27	16:45	18:03
45°	05:49	06:28	07:00	15:12	16:34	17:57
S 50°	06:09	06:42	07:09	14:54	16:21	17:49
52°	06:18	06:49	07:14	14:45	16:14	17:45
54°	06:29	06:57	07:19	14:35	16:08	17:41
56°	06:41	07:06	07:24	14:24	16:00	17:37
58°	06:54	07:15	07:30	14:11	15:51	17:32
S 60°	07:10	07:26	07:37	13:56	15:41	17:26

August 11, 12, 13 UT (Wed., Thu., Fri.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	319°21.4	136°02.6	S02°15.2	132°25.2	S02°38.3	316°09.3	S00°13.8	136°47.5	N01°17.5			136°47.5	N01°17.5	Alpheratz	357°45.4	29°09.0	
1	334°23.8	151°02.8	16.4	147°26.3	39.0	331°11.9	13.9	151°49.7	17.4			151°49.7	17.4	Ankaa	353°17.3	-42°14.5	
2	349°26.3	166°03.0	17.7	162°27.3	39.6	346°14.5	14.0	166°51.9	17.3			166°51.9	17.3	Schedar	349°42.6	56°35.7	
3	4°28.8	181°03.3	•• 18.9	177°28.4	•• 40.3	1°17.1	•• 14.0	181°54.2	•• 17.2			181°54.2	•• 17.2	Diphda	348°57.7	-17°55.4	
4	19°31.2	196°03.5	20.1	192°29.4	40.9	16°19.7	14.1	196°56.4	17.1			196°56.4	17.1	Achernar	335°27.9	-57°10.6	
5	34°33.7	211°03.7	21.3	207°30.4	41.6	31°22.3	14.2	211°58.6	17.0			211°58.6	17.0	Hamal	328°03.0	23°30.8	
6	49°36.1	226°04.0	S02°22.6	222°31.5	S02°42.2	46°24.9	S00°14.3	227°00.9	N01°16.9			227°00.9	N01°16.9	Polaris	318°44.9	89°18.4	
7	64°38.6	241°04.2	23.8	237°32.5	42.9	61°27.5	14.3	242°03.1	16.8			242°03.1	16.8	Acamar	315°19.8	-40°15.4	
8	79°41.1	256°04.4	25.0	252°33.6	43.5	76°30.1	14.4	257°05.3	16.7			257°05.3	16.7	Menkar	314°17.2	4°08.0	
9	94°43.5	271°04.7	•• 26.2	267°34.6	•• 44.2	91°32.7	•• 14.5	272°07.6	•• 16.6			272°07.6	•• 16.6	Mirfak	308°43.4	49°53.8	
10	109°46.0	286°04.9	27.5	282°35.6	44.8	106°35.3	14.5	287°09.8	16.5			287°09.8	16.5	Aldebaran	290°51.9	16°31.9	
11	124°48.5	301°05.1	28.7	297°36.7	45.5	121°37.9	14.6	302°12.0	16.4			302°12.0	16.4	Rigel	281°14.2	-8°11.2	
12	139°50.9	316°05.4	S02°29.9	312°37.7	S02°46.1	136°40.5	S00°14.7	317°14.3	N01°16.3			317°14.3	N01°16.3	Capella	280°37.7	46°00.4	
13	154°53.4	331°05.6	31.1	327°38.7	46.7	151°43.1	14.8	332°16.5	16.2			332°16.5	16.2	Bellatrix	278°34.4	6°21.6	
14	169°55.9	346°05.8	32.4	342°39.8	47.4	166°45.7	14.8	347°18.7	16.1			347°18.7	16.1	Elnath	278°15.4	28°36.9	
15	184°58.3	1°06.1	•• 33.6	357°40.8	•• 48.0	181°48.3	•• 14.9	2°21.0	•• 16.0			2°21.0	•• 16.0	Alnilam	275°48.6	-1°11.6	
16	200°00.8	16°06.3	34.8	12°41.9	48.7	196°50.9	15.0	17°23.2	15.9			17°23.2	15.9	Betelgeuse	271°03.7	7°24.6	
17	215°03.2	31°06.6	36.1	27°42.9	49.3	211°53.5	15.0	32°25.4	15.8			32°25.4	15.8	Canopus	263°57.5	-52°41.9	
18	230°05.7	46°06.8	S02°37.3	42°43.9	S02°50.0	226°56.1	S00°15.1	47°27.7	N01°15.7			47°27.7	N01°15.7	Sirius	258°35.8	-16°43.7	
19	245°08.2	61°07.0	38.5	57°45.0	50.6	241°58.7	15.2	62°29.9	15.6			62°29.9	15.6	Adhara	255°14.5	-28°59.1	
20	260°10.6	76°07.3	39.7	72°46.0	51.3	257°01.3	15.3	77°32.2	15.4			77°32.2	15.4	Procyon	245°02.2	5°11.9	
21	275°13.1	91°07.5	•• 41.0	87°47.0	•• 51.9	272°03.9	•• 15.3	92°34.4	•• 15.3			92°34.4	•• 15.3	Pollux	243°30.5	28°00.0	
22	290°15.6	106°07.7	42.2	102°48.1	52.6	287°06.5	15.4	107°36.6	15.2			107°36.6	15.2	Avior	234°19.6	-59°32.6	
23	305°18.0	121°08.0	43.4	117°49.1	53.2	302°09.1	15.5	122°38.9	15.1			122°38.9	15.1	Suhail	222°54.5	-43°28.5	
Mer.pass. 02:42			ν 0.2' d 1.2' m -4.35			ν 1.0' d 0.6' m 1.50			ν 2.6' d 0.1' m -2.77			ν 2.2' d -0.1' m 1.00					
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec				
0	320°20.5	136°08.2	S02°44.6	132°50.2	S02°53.9	317°11.7	S00°15.5	137°41.1	N01°15.0			137°41.1	N01°15.0	Regulus	207°46.0	11°54.9	
1	335°23.0	151°08.5	45.8	147°51.2	54.5	332°14.4	15.6	152°43.3	14.9			152°43.3	14.9	Dubhe	193°54.6	61°41.7	
2	350°25.4	166°08.7	47.1	162°52.2	55.2	347°17.0	15.7	167°45.6	14.8			167°45.6	14.8	Denebola	182°36.0	14°30.8	
3	5°27.9	181°09.0	•• 48.3	177°53.3	•• 55.8	2°19.6	•• 15.8	182°47.8	•• 14.7			182°47.8	•• 14.7	Gienah	175°54.7	-17°36.1	
4	20°30.4	196°09.2	49.5	192°54.3	56.4	17°22.2	15.8	197°50.0	14.6			197°50.0	14.6	Acrux	173°12.3	-63°09.8	
5	35°32.8	211°09.4	50.7	207°55.3	57.1	32°24.8	15.9	212°52.3	14.5			212°52.3	14.5	Gacrux	172°03.7	-57°10.6	
6	50°35.3	226°09.7	S02°52.0	222°56.4	S02°57.7	47°27.4	S00°16.0	227°54.5	N01°14.4			227°54.5	N01°14.4	Alioth	166°22.6	55°54.3	
7	65°37.7	241°09.9	53.2	237°57.4	58.4	62°30.0	16.1	242°56.7	14.3			242°56.7	14.3	Spica	158°33.6	-11°13.1	
8	80°40.2	256°10.2	54.4	252°58.4	59.0	77°32.6	16.1	257°59.0	14.2			257°59.0	14.2	Alkaid	153°00.6	49°15.8	
9	95°42.7	271°10.4	•• 55.6	267°59.5	02°59.7	92°35.2	•• 16.2	273°01.2	•• 14.1			273°01.2	•• 14.1	Hadar	148°51.1	-60°25.8	
10	110°45.1	286°10.7	56.9	283°00.5	03°00.3	107°37.8	16.3	288°03.4	14.0			288°03.4	14.0	Menkent	148°10.1	-36°25.5	
11	125°47.6	301°10.9	58.1	298°01.5	01.0	122°40.4	16.3	303°05.7	13.9			303°05.7	13.9	Arcturus	145°57.6	19°07.7	
12	140°50.1	316°11.2	S02°59.3	313°02.6	S03°01.6	137°43.0	S00°16.4	318°07.9	N01°13.8			318°07.9	N01°13.8	Rigel Kent.	139°54.7	-60°53.0	
13	155°52.5	331°11.4	03°00.5	328°03.6	02.3	152°45.6	16.5	333°10.1	13.7			333°10.1	13.7	Kochab	137°19.8	74°07.0	
14	170°55.0	346°11.7	01.8	343°04.6	02.9	167°48.2	16.6	348°12.3	13.6			348°12.3	13.6	Zuben'ubi	137°07.7	-16°05.2	
15	185°57.5	1°11.9	•• 03.0	358°05.7	•• 03.6	182°50.8	•• 16.6	3°14.6	•• 13.4			3°14.6	•• 13.4	Alphecca	126°12.7	26°40.9	
16	200°59.9	16°12.2	04.2	13°06.7	04.2	197°53.4	16.7	18°16.8	13.3			18°16.8	13.3	Antares	112°28.7	-26°27.4	
17	216°02.4	31°12.4	05.4	28°07.7	04.9	212°56.0	16.8	33°19.0	13.2			33°19.0	13.2	Atria	107°32.1	-69°03.0	
18	231°04.9	46°12.7	S03°06.6	43°08.8	S03°05.5	227°58.7	S00°16.9	48°21.3	N01°13.1			48°21.3	N01°13.1	Sabik	102°14.7	-15°44.3	
19	246°07.3	61°12.9	07.9	58°09.8	06.2	243°01.3	16.9	63°23.5	13.0			63°23.5	13.0	Shaula	96°24.4	-37°06.8	
20	261°09.8	76°13.2	09.1	73°10.8	06.8	258°03.9	17.0	78°25.7	12.9			78°25.7	12.9	Rasalhague	96°08.1	12°33.3	
21	276°12.2	91°13.4	•• 10.3	88°11.9	•• 07.4	273°06.5	•• 17.1	93°28.0	•• 12.8			93°28.0	•• 12.8	Eltanin	90°46.7	51°29.5	
22	291°14.7	106°13.7	11.5	103°12.9	08.1	288°09.1	17.2	108°30.2	12.7			108°30.2	12.7	Kaus Aust.	83°46.2	-34°22.8	
23	306°17.2	121°13.9	12.7	118°13.9	08.7	303°11.7	17.2	123°32.4	12.6			123°32.4	12.6	Vega	80°40.0	38°47.8	
Mer.pass. 02:38			ν 0.2' d 1.2' m -4.36			ν 1.0' d 0.6' m 1.51			ν 2.6' d 0.1' m -2.78			ν 2.2' d -0.1' m 1.00					
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec				
0	321°19.6	136°14.2	S03°14.0	133°15.0	S03°09.4	318°14.3	S00°17.3	138°34.7	N01°12.5			138°34.7	N01°12.5	Aug 11 Wed	SHA	Mer.pass	
1	336°22.1	151°14.4	15.2	148°16.0	10.0	333°16.9	17.4	153°36.9	12.4			153°36.9	12.4	Venus	176°41.2	14:56	
2	351°24.6	166°14.7	16.4	163°17.0	10.7	348°19.5	17.5	168°39.1	12.3			168°39.1	12.3	Mars	173°03.9	15:09	
3	6°27.0	181°14.9	•• 17.6	178°18.1	•• 11.3	3°22.1	•• 17.5	183°41.4	•• 12.2			183°41.4	•• 12.2	Jupiter	356°48.0	02:55	
4	21°29.5	196°15.2	18.9	193°19.1	12.0	18°24.8	17.6	198°43.6</									

h	Sun			Moon			
Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	178°40.5	N15°21.6	168°04.0	8.4'	N07°34.8	-15.4'	61.2'
1	193°40.6	20.9	182°31.4	8.5'	07°19.4	-15.4'	61.2'
2	208°40.7	20.2	196°58.8	8.5'	07°04.0	-15.5'	61.2'
3	223°40.8	· · 19.4	211°26.3	8.5'	06°48.5	-15.5'	61.2'
4	238°40.9	18.7	225°53.9	8.6'	06°33.0	-15.6'	61.2'
5	253°41.0	17.9	240°21.5	8.6'	06°17.4	-15.6'	61.2'
6	268°41.1	N15°17.2	254°49.1	8.7'	N06°01.8	-15.6'	61.2'
7	283°41.2	16.4	269°16.8	8.7'	05°46.2	-15.7'	61.1'
8	298°41.3	15.7	283°44.5	8.8'	05°30.6	-15.7'	61.1'
9	313°41.4	· · 15.0	298°12.2	8.8'	05°14.9	-15.7'	61.1'
10	328°41.5	14.2	312°40.0	8.8'	04°59.2	-15.7'	61.1'
11	343°41.6	13.5	327°07.9	8.9'	04°43.4	-15.8'	61.1'
12	358°41.7	N15°12.7	341°35.7	8.9'	N04°27.7	-15.8'	61.1'
13	13°41.8	12.0	356°03.7	8.9'	04°11.9	-15.8'	61.1'
14	28°41.9	11.2	10°31.6	9.0'	03°56.1	-15.8'	61.1'
15	43°42.0	· · 10.5	24°59.6	9.0'	03°40.3	-15.8'	61.1'
16	58°42.1	09.8	39°27.6	9.1'	03°24.5	-15.8'	61.1'
17	73°42.2	09.0	53°55.7	9.1'	03°08.6	-15.9'	61.0'
18	88°42.3	N15°08.3	68°23.7	9.1'	N02°52.8	-15.9'	61.0'
19	103°42.4	07.5	82°51.9	9.1'	02°36.9	-15.9'	61.0'
20	118°42.5	06.8	97°20.0	9.2'	02°21.0	-15.9'	61.0'
21	133°42.6	· · 06.0	111°48.2	9.2'	02°05.2	-15.9'	61.0'
22	148°42.7	05.3	126°16.4	9.2'	01°49.3	-15.9'	61.0'
23	163°42.8	04.5	140°44.6	9.3'	01°33.4	-15.9'	61.0'
SD = 15.8'			SD = 16.7'				

Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	178°42.9	N15°03.8	155°12.9	9.3'	N01°17.5	-15.9'	60.9'
1	193°43.0	03.0	169°41.2	9.3'	01°01.7	-15.9'	60.9'
2	208°43.1	02.3	184°09.5	9.3'	00°45.8	-15.9'	60.9'
3	223°43.2	· · 01.5	198°37.9	9.4'	00°29.9	-15.9'	60.9'
4	238°43.3	00.8	213°06.2	9.4'	N00°14.1	-15.9'	60.9'
5	253°43.4	15°00.0	227°34.6	9.4'	S00°01.8	15.8'	60.9'
6	268°43.5	N14°59.3	242°03.0	9.4'	S00°17.6	15.8'	60.8'
7	283°43.6	58.5	256°31.5	9.5'	00°33.5	15.8'	60.8'
8	298°43.7	57.8	270°59.9	9.5'	00°49.3	15.8'	60.8'
9	313°43.8	· · 57.0	285°28.4	9.5'	01°05.1	15.8'	60.8'
10	328°43.9	56.3	299°56.9	9.5'	01°20.9	15.8'	60.8'
11	343°44.0	55.5	314°25.4	9.5'	01°36.6	15.7'	60.7'
12	358°44.1	N14°54.8	328°53.9	9.5'	S01°52.4	15.7'	60.7'
13	13°44.3	54.0	343°22.5	9.6'	02°08.1	15.7'	60.7'
14	28°44.4	53.3	357°51.0	9.6'	02°23.8	15.7'	60.7'
15	43°44.5	· · 52.5	12°19.6	9.6'	02°39.4	15.6'	60.7'
16	58°44.6	51.8	26°48.2	9.6'	02°55.1	15.6'	60.6'
17	73°44.7	51.0	41°16.8	9.6'	03°10.7	15.6'	60.6'
18	88°44.8	N14°50.2	55°45.4	9.6'	S03°26.2	15.5'	60.6'
19	103°44.9	49.5	70°14.0	9.6'	03°41.8	15.5'	60.6'
20	118°45.0	48.7	84°42.7	9.6'	03°57.3	15.5'	60.5'
21	133°45.1	· · 48.0	99°11.3	9.6'	04°12.8	15.4'	60.5'
22	148°45.2	47.2	113°40.0	9.7'	04°28.2	15.4'	60.5'
23	163°45.3	46.5	128°08.6	9.7'	04°43.6	15.3'	60.5'
SD = 15.8'			SD = 16.6'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	178°45.4	N14°45.7	142°37.3	9.7'	S04°58.9	15.3'	60.4'
1	193°45.5	44.9	157°06.0	9.7'	05°14.2	15.3'	60.4'
2	208°45.6	44.2	171°34.6	9.7'	05°29.5	15.2'	60.4'
3	223°45.8	· · 43.4	186°03.3	9.7'	05°44.7	15.2'	60.3'
4	238°45.9	42.7	200°32.0	9.7'	05°59.9	15.1'	60.3'
5	253°46.0	41.9	215°00.7	9.7'	06°15.0	15.1'	60.3'
6	268°46.1	N14°41.1	229°29.3	9.7'	S06°30.0	15.0'	60.3'
7	283°46.2	40.4	243°58.0	9.7'	06°45.0	15.0'	60.2'
8	298°46.3	39.6	258°26.7	9.7'	07°00.0	14.9'	60.2'
9	313°46.4	· · 38.9	272°55.4	9.7'	07°14.9	14.8'	60.2'
10	328°46.5	38.1	287°24.1	9.7'	07°29.7	14.8'	60.1'
11	343°46.6	37.3	301°52.7	9.7'	07°44.5	14.7'	60.1'
12	358°46.8	N14°36.6	316°21.4	9.7'	S07°59.3	14.7'	60.1'
13	13°46.9	35.8	330°50.1	9.7'	08°13.9	14.6'	60.1'
14	28°47.0	35.0	345°18.7	9.7'	08°28.5	14.5'	60.0'
15	43°47.1	· · 34.3	359°47.4	9.6'	08°43.1	14.5'	60.0'
16	58°47.2	33.5	14°16.1	9.6'	08°57.5	14.4'	60.0'
17	73°47.3	32.8	28°44.7	9.6'	09°11.9	14.3'	59.9'
18	88°47.4	N14°32.0	43°13.3	9.6'	S09°26.3	14.3'	59.9'
19	103°47.5	31.2	57°42.0	9.6'	09°40.5	14.2'	59.9'
20	118°47.7	30.5	72°10.6	9.6'	09°54.7	14.1'	59.8'
21	133°47.8	· · 29.7	86°39.2	9.6'	10°08.8	14.0'	59.8'
22	148°47.9	28.9	101°07.8	9.6'	10°22.9	14.0'	59.8'
23	163°48.0	28.2	115°36.4	9.6'	10°36.8	13.9'	59.7'
SD = 15.8'			SD = 16.5'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	////	////	02:01	22:03	////	////
N 70°	////	////	02:39	21:26	////	////
68°	////	01:15	03:06	21:01	22:46	////
66°	////	02:04	03:26	20:41	22:02	////
64°	////	02:34	03:42	20:26	21:33	////
62°	01:07	02:56	03:55	20:13	21:12	22:55
60°	01:51	03:13	04:07	20:02	20:54	22:15
N 58°	02:18	03:28	04:16	19:52	20:40	21:49
56°	02:39	03:40	04:25	19:44	20:28	21:29
54°	02:56	03:51	04:32	19:37	20:18	21:12
52°	03:10	04:00	04:39	19:30	20:09	20:59
50°	03:22	04:08	04:45	19:24	20:01	20:47
45°	03:45	04:25	04:58	19:11	19:44	20:24
N 40°	04:04	04:39	05:09	19:01	19:30	20:06
35°	04:18	04:51	05:18	18:52	19:19	19:51
30°	04:30	05:00	05:25	18:44	19:09	19:40
20°	04:49	05:16	05:39	18:31	18:54	19:21
N 10°	05:03	05:29	05:51	18:19	18:41	19:06
0°	05:15	05:40	06:02	18:08	18:30	18:55
S 10°	05:26	05:51	06:12	17:58	18:19	18:45
20°	05:35	06:01	06:24	17:47	18:09	18:35
30°	05:44	06:12	06:37	17:34	17:59	18:27
35°	05:48	06:18	06:44	17:27	17:53	18:23
40°	05:52	06:24	06:52	17:18	17:46	18:18
45°	05:57	06:31	07:02	17:09	17:39	18:14
S 50°	06:01	06:40	07:14	16:57	17:31	18:09
52°	06:04	06:43	07:19	16:52	17:27	18:07
54°	06:06	06:48	07:25	16:46	17:23	18:05
56°	06:08	06:52	07:32	16:39	17:19	18:03
58°	06:10	06:57	07:39	16:32	17:14	18:01
S 60°	06:13	07:02	07:48	16:23	17:09	17:58

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	05:46	08:09	10:31	19:58	19:30	19:00
N 70°	05:55	08:08	10:19	19:55	19:36	19:15
68°	06:02	08:06	10:09	19:52	19:40	19:27
66°	06:08	08:05	10:01	19:50	19:44	19:37
64°	06:13	08:05	09:54	19:48	19:47	19:46
62°	06:18	08:04	09:48	19:47	19:50	19:53
60°	06:21	08:03	09:43	19:46	19:52	19:59
N 58°	06:25	08:03	09:39	19:44	19:54	20:05
56°	06:28	08:02	09:35	19:43	19:56	20:10
54°	06:30	08:02	09:32	19:42	19:58	20:15
52°	06:33	08:01	09:28	19:41	20:00	20:19
50°	06:35	08:01	09:26	19:41	20:01	20:23
45°	06:40	08:00	09:19	19:39	20:04	20:31
N 40°	06:44	08:00	09:14	19:37	20:07	20:38
35°	06:47	07:59	09:10	19:36	20:09	20:43
30°	06:50	07:59	09:06	19:35	20:12	20:49
20°	06:55	07:58	09:00	19:33	20:15	20:58
N 10°	06:59	07:57	08:54	19:31	20:18	21:06
0°	07:04	07:57	08:49	19:29	20:21	21:13
S 10°	07:08	07:56	08:43	19:27	20:24	21:21
20°	07:12	07:55	08:38	19:25	20:27	21:29
30°	07:18	07:55	08:32	19:23	20:31	21:38
35°	07:20	07:54	08:28	19:21	20:33	21:43
40°	07:24	07:54	08:24	19:20	20:35	21:50
45°	07:28	07:54	08:19	19:18	20:38	21:57
S 50°	07:32	07:53	08:14	19:16	20:41	22:05
52°	07:34	07:53	08:11	19:15	20:43	22:09
54°	07:36	07:53	08:09	19:14	20:44	22:14
56°	07:39	07:52	08:06	19:13	20:46	22:18
58°	07:42	07:52	08:02	19:11	20:48	22:24
S 60°	07:45	07:52	07:58	19:10	20:51	22:30

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time	12 ^h		Mer. Pass.		
	00 ^h	mm:ss	hh:mm	Upper	Lower	1-3
				hh:mm	hh:mm	1-12%
11	05:18	05:13	12:05	13:16	00:50	
12	05:08	05:03	12:05	14:09	01:43	
13	04:58	04:53	12:05	15:01	02:35	

August 14, 15, 16 UT (Sat., Sun., Mon.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Sat	GHA	Dec	GHA	Dec		GHA	Dec		GHA	Dec		GHA	Dec		SHA	Dec	
0	322°18.8	136°20.4	S03°43.2	133°39.7	S03°24.9	319°17.1	S00°19.2	139°28.2	N01°09.9			139°28.2	N01°09.9		Alpheratz	357°45.4 29°09.1	
1	337°21.2	151°20.7	44.4	148°40.7	25.6	334°19.7	19.2	154°30.4	09.8			154°30.4	09.8		Ankaa	353°17.3 -42°14.5	
2	352°23.7	166°20.9	45.7	163°41.8	26.2	349°22.3	19.3	169°32.7	09.7			169°32.7	09.7		Schedar	349°42.6 56°35.7	
3	7°26.2	181°21.2	· · 46.9	178°42.8	· · 26.9	4°24.9	· · 19.4	184°34.9	· · 09.6			184°34.9	· · 09.6		Diphda	348°57.7 -17°55.4	
4	22°28.6	196°21.5	48.1	193°43.8	27.5	19°27.5	19.5	199°37.1	09.5			199°37.1	09.5		Achernar	335°27.9 -57°10.6	
5	37°31.1	211°21.8	49.3	208°44.8	28.2	34°30.1	19.5	214°39.3	09.4			214°39.3	09.4		Hamal	328°03.0 23°30.8	
6	52°33.6	226°22.0	S03°50.5	223°45.9	S03°28.8	49°32.8	S00°19.6	229°41.6	N01°09.3			229°41.6	N01°09.3		Polaris	318°43.4 89°18.4	
7	67°36.0	241°22.3	51.7	238°46.9	29.5	64°35.4	19.7	244°43.8	09.2			244°43.8	09.2		Acamar	315°19.8 -40°15.4	
8	82°38.5	256°22.6	53.0	253°47.9	30.1	79°38.0	19.8	259°46.0	09.1			259°46.0	09.1		Menkar	314°17.2 4°08.0	
9	97°40.9	271°22.8	· · 54.2	268°48.9	· · 30.7	94°40.6	· · 19.9	274°48.3	· · 09.0			274°48.3	· · 09.0		Mirfak	308°43.4 49°53.8	
10	112°43.4	286°23.1	55.4	283°50.0	31.4	109°43.2	19.9	289°50.5	08.9			289°50.5	08.9		Aldebaran	290°51.8 16°31.9	
11	127°45.9	301°23.4	56.6	298°51.0	32.0	124°45.9	20.0	304°52.7	08.8			304°52.7	08.8		Rigel	281°14.2 -8°11.2	
12	142°48.3	316°23.6	S03°57.8	313°52.0	S03°32.7	139°48.5	S00°20.1	319°55.0	N01°08.7			319°55.0	N01°08.7		Capella	280°37.7 46°00.4	
13	157°50.8	331°23.9	03°59.0	328°53.1	33.3	154°51.1	20.2	334°57.2	08.6			334°57.2	08.6		Bellatrix	278°34.4 6°21.6	
14	172°53.3	346°24.2	04°00.3	343°54.1	34.0	169°53.7	20.3	349°59.4	08.4			349°59.4	08.4		Elnath	278°15.4 28°36.9	
15	187°55.7	1°24.5	· · 01.5	358°55.1	· · 34.6	184°56.3	· · 20.3	5°01.6	· · 08.3			5°01.6	· · 08.3		Alnilam	275°48.6 -1°11.6	
16	202°58.2	16°24.7	02.7	13°56.1	35.3	199°59.0	20.4	20°03.9	08.2			20°03.9	08.2		Betelgeuse	271°03.7 7°24.6	
17	218°00.7	31°25.0	03.9	28°57.2	35.9	215°01.6	20.5	35°06.1	08.1			35°06.1	08.1		Canopus	263°57.4 -52°41.8	
18	233°03.1	46°25.3	S04°05.1	43°58.2	S03°36.6	230°04.2	S00°20.6	50°08.3	N01°08.0			50°08.3	N01°08.0		Sirius	258°35.8 -16°43.7	
19	248°05.6	61°25.6	06.3	58°59.2	37.2	245°06.8	20.7	65°10.6	07.9			65°10.6	07.9		Adhara	255°14.4 -28°59.0	
20	263°08.1	76°25.8	07.5	74°00.2	37.9	260°09.5	20.7	80°12.8	07.8			80°12.8	07.8		Procyon	245°02.1 5°11.9	
21	278°10.5	91°26.1	· · 08.8	89°01.3	· · 38.5	275°12.1	· · 20.8	95°15.0	· · 07.7			95°15.0	· · 07.7		Pollux	243°30.5 28°00.0	
22	293°13.0	106°26.4	10.0	104°02.3	39.2	290°14.7	20.9	110°17.2	07.6			110°17.2	07.6		Avior	234°19.6 -59°32.6	
23	308°15.4	121°26.7	11.2	119°03.3	39.8	305°17.3	21.0	125°19.5	07.5			125°19.5	07.5		Suhail	222°54.5 -43°28.5	
Mer.pass. 02:30			ν 0.3' d 1.2' m -4.38			ν 1.0' d 0.6' m 1.51			ν 2.6' d 0.1' m -2.79			ν 2.2' d -0.1' m 1.00					

Sun	GHA	Dec	GHA	Dec		GHA	Dec		GHA	Dec				
0	323°17.9	136°27.0	S04°12.4	134°04.3	S03°40.5	320°19.9	S00°21.1	140°21.7	N01°07.4					
1	338°20.4	151°27.2	13.6	149°05.4	41.1	335°22.6	21.1	155°23.9	07.3					
2	353°22.8	166°27.5	14.8	164°06.4	41.8	350°25.2	21.2	170°26.1	07.2					
3	8°25.3	181°27.8	· · 16.0	179°07.4	· · 42.4	5°27.8	· · 21.3	185°28.4	· · 07.1					
4	23°27.8	196°28.1	17.2	194°08.4	43.1	20°30.4	21.4	200°30.6	06.9					
5	38°30.2	211°28.4	18.5	209°09.5	43.7	35°33.1	21.5	215°32.8	06.8					
6	53°32.7	226°28.6	S04°19.7	224°10.5	S03°44.3	50°35.7	S00°21.5	230°35.1	N01°06.7					
7	68°35.2	241°28.9	20.9	239°11.5	45.0	65°38.3	21.6	245°37.3	06.6					
8	83°37.6	256°29.2	22.1	254°12.5	45.6	80°41.0	21.7	260°39.5	06.5					
9	98°40.1	271°29.5	· · 23.3	269°13.6	· · 46.3	95°43.6	· · 21.8	275°41.7	· · 06.4					
10	113°42.6	286°29.8	24.5	284°14.6	46.9	110°46.2	21.9	290°44.0	06.3					
11	128°45.0	301°30.1	25.7	299°15.6	47.6	125°48.8	22.0	305°46.2	06.2					
12	143°47.5	316°30.3	S04°26.9	314°16.6	S03°48.2	140°51.5	S00°22.0	320°48.4	N01°06.1					
13	158°49.9	331°30.6	28.1	329°17.6	48.9	155°54.1	22.1	335°50.6	06.0					
14	173°52.4	346°30.9	29.4	344°18.7	49.5	170°56.7	22.2	350°52.9	05.9					
15	188°54.9	1°31.2	· · 30.6	359°19.7	· · 50.2	185°59.3	· · 22.3	5°55.1	· · 05.8					
16	203°57.3	16°31.5	31.8	14°20.7	50.8	201°02.0	22.4	20°57.3	05.7					
17	218°59.8	31°31.8	33.0	29°21.7	51.5	216°04.6	22.5	35°59.5	05.5					
18	234°02.3	46°32.1	S04°34.2	44°22.8	S03°52.1	231°07.2	S00°22.5	51°01.8	N01°05.4					
19	249°04.7	61°32.4	35.4	59°23.8	52.8	246°09.9	22.6	66°04.0	05.3					
20	264°07.2	76°32.6	36.6	74°24.8	53.4	261°12.5	22.7	81°06.2	05.2					
21	279°09.7	91°32.9	· · 37.8	89°25.8	· · 54.1	276°15.1	· · 22.8	96°08.5	· · 05.1					
22	294°12.1	106°33.2	39.0	104°26.8	54.7	291°17.7	22.9	111°10.7	05.0					
23	309°14.6	121°33.5	40.2	119°27.9	55.4	306°20.4	22.9	126°12.9	04.9					
Mer.pass. 02:26			ν 0.3' d 1.2' m -4.40			ν 1.0' d 0.6' m 1.51			ν 2.6' d 0.1' m -2.80			ν 2.2' d -0.1' m 1.00		

Mon	GHA	Dec	GHA	Dec		GHA	Dec		GHA	Dec	
0	324°17.0	136°33.8	S04°41.4	134°28.9	S03°56.0	321°23.0	S00°23.0	141°15.1	N01°04.8		
1	339°19.5	151°34.1	42.7	149°29.9	56.6	336°25.6	23.1	156°17.4	04.7		
2	354°22.0	166°34.4	43.9	164°30.9	57.3	351°28.3	23.2	171°19.6	04.6		
3	9°24.4	181°34.7	· · 45.1	179°31.9	· · 57.9	6°30.9	· · 23.3	186°21.8	· · 04.5		
4	24°26.9	196°35.0	46.3	194°33.0	58.6	21°33.5	23.4	201°24.0	04.4		
5	39°29.4	211°35.3	47.5	209°34.0	59.2	36°36.2	23.5	216°26.3	04.2		
6	54°31.8	226°35.6	S04°48.7	224°35.0	S03°59.9	51°38.8	S00°23.5	231°28.5	N01°04.1		
7	69°34.3	241°35.9	49.9	239°36.0	04°00.5	66°41.4	23.6	246°30.7	04.0		
8	84°36.8	256°36.2	51.1	254°37.0	01.2	81°44.1	23.7	261°32.9	03.9		
9	99°39.2	271°36.5	· · 52.3	269°38.1	· · 01.8	96°46.7	· · 23.8	276°35.2	· · 03.8		
10	114°41.7	286°36.7	53.5	284°39.1	02.5	111°49.3	23.9	291°37.4	03.7		
11	129°44.2	301°37.0	54.7	299°40.1	03.1	126°52.0	24.0	306°39.6	03.6		
12	144°46.6	316°37.3	S04°55.9	314°41.1	S04°03.8	141°54.6	S00°24.0	321°41.8	N01°03.5		
13	159°49.1	331°37.6	57.1	329°42.1	04.4	156°57.2	24.1	336°44.1	03.4		
14	174°51.5	346°37.9	58.3	344°43.2	05.1	171°59.9	24.2	351°46.3	03.3		
15	189°54.0	1°38.2	04°59.5	359°44.2	· · 05.7	187°02.5	· · 24.3	6°48.5	· · 03.2		
16	204°56.5	16°38.5	05°00.8	14°45.2	06.4	202°05.1	24.4	21°50.7	03.1		
17	219°58.9	31°38.8	02.0	29°46.2	07.0	217°07.8	24.5	36°53.0	02.9		
18											

h	Sun			Moon			
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	178°48.1	N14°27.4	130°04.9	9.6'	S10°50.7	13.8'	59.7'
1	193°48.2	26.6	144°33.5	9.5'	11°04.6	13.7'	59.7'
2	208°48.3	25.9	159°02.0	9.5'	11°18.3	13.7'	59.6'
3	223°48.5	· · 25.1	173°30.6	9.5'	11°31.9	13.6'	59.6'
4	238°48.6	24.3	187°59.1	9.5'	11°45.5	13.5'	59.6'
5	253°48.7	23.5	202°27.6	9.5'	11°59.0	13.4'	59.5'
6	268°48.8	N14°22.8	216°56.1	9.5'	S12°12.4	13.3'	59.5'
7	283°48.9	22.0	231°24.5	9.5'	12°25.7	13.2'	59.5'
8	298°49.0	21.2	245°53.0	9.4'	12°39.0	13.1'	59.4'
9	313°49.1	· · 20.5	260°21.4	9.4'	12°52.1	13.1'	59.4'
10	328°49.3	19.7	274°49.9	9.4'	13°05.1	13.0'	59.4'
11	343°49.4	18.9	289°18.3	9.4'	13°18.1	12.9'	59.3'
12	358°49.5	N14°18.1	303°46.6	9.4'	S13°31.0	12.8'	59.3'
13	13°49.6	17.4	318°15.0	9.3'	13°43.8	12.7'	59.3'
14	28°49.7	16.6	332°43.4	9.3'	13°56.4	12.6'	59.2'
15	43°49.9	· · 15.8	347°11.7	9.3'	14°09.0	12.5'	59.2'
16	58°50.0	15.1	1°40.0	9.3'	14°21.5	12.4'	59.2'
17	73°50.1	14.3	16°08.3	9.3'	14°33.9	12.3'	59.1'
18	88°50.2	N14°13.5	30°36.5	9.2'	S14°46.2	12.2'	59.1'
19	103°50.3	12.7	45°04.8	9.2'	14°58.4	12.1'	59.1'
20	118°50.4	12.0	59°33.0	9.2'	15°10.5	12.0'	59.0'
21	133°50.6	· · 11.2	74°01.2	9.2'	15°22.5	11.9'	59.0'
22	148°50.7	10.4	88°29.4	9.2'	15°34.4	11.8'	58.9'
23	163°50.8	09.6	102°57.5	9.1'	15°46.2	11.7'	58.9'
SD = 15.8'			SD = 16.3'				

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	178°50.9	N14°08.8	117°25.7	9.1'	S15°57.9	11.6'	58.9'
1	193°51.0	08.1	131°53.8	9.1'	16°09.5	11.5'	58.8'
2	208°51.2	07.3	146°21.9	9.1'	16°20.9	11.4'	58.8'
3	223°51.3	· · 06.5	160°49.9	9.0'	16°32.3	11.3'	58.8'
4	238°51.4	05.7	175°18.0	9.0'	16°43.6	11.1'	58.7'
5	253°51.5	05.0	189°46.0	9.0'	16°54.7	11.0'	58.7'
6	268°51.7	N14°04.2	204°14.0	9.0'	S17°05.7	10.9'	58.7'
7	283°51.8	03.4	218°42.0	8.9'	17°16.7	10.8'	58.6'
8	298°51.9	02.6	233°09.9	8.9'	17°27.5	10.7'	58.6'
9	313°52.0	· · 01.8	247°37.8	8.9'	17°38.2	10.6'	58.5'
10	328°52.1	01.1	262°05.7	8.9'	17°48.8	10.5'	58.5'
11	343°52.3	14°00.3	276°33.6	8.9'	17°59.2	10.4'	58.5'
12	358°52.4	N13°59.5	291°01.5	8.8'	S18°09.6	10.2'	58.4'
13	13°52.5	58.7	305°29.3	8.8'	18°19.8	10.1'	58.4'
14	28°52.6	57.9	319°57.1	8.8'	18°29.9	10.0'	58.4'
15	43°52.8	· · 57.1	334°24.9	8.8'	18°39.9	9.9'	58.3'
16	58°52.9	56.4	348°52.7	8.7'	18°49.8	9.8'	58.3'
17	73°53.0	55.6	3°20.4	8.7'	18°59.6	9.6'	58.3'
18	88°53.1	N13°54.8	17°48.1	8.7'	S19°09.2	9.5'	58.2'
19	103°53.3	54.0	32°15.8	8.7'	19°18.7	9.4'	58.2'
20	118°53.4	53.2	46°43.5	8.6'	19°28.1	9.3'	58.1'
21	133°53.5	· · 52.4	61°11.1	8.6'	19°37.4	9.2'	58.1'
22	148°53.6	51.7	75°38.7	8.6'	19°46.6	9.0'	58.1'
23	163°53.8	50.9	90°06.4	8.6'	19°55.6	8.9'	58.0'
SD = 15.8'			SD = 16.1'				

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	178°53.9	N13°50.1	104°33.9	8.6'	S20°04.5	8.8'	58.0'
1	193°54.0	49.3	119°01.5	8.5'	20°13.3	8.7'	58.0'
2	208°54.1	48.5	133°29.0	8.5'	20°21.9	8.5'	57.9'
3	223°54.3	· · 47.7	147°56.6	8.5'	20°30.5	8.4'	57.9'
4	238°54.4	46.9	162°24.1	8.5'	20°38.9	8.3'	57.9'
5	253°54.5	46.1	176°51.5	8.5'	20°47.1	8.1'	57.8'
6	268°54.6	N13°45.4	191°19.0	8.4'	S20°55.3	8.0'	57.8'
7	283°54.8	44.6	205°46.4	8.4'	21°03.3	7.9'	57.7'
8	298°54.9	43.8	220°13.9	8.4'	21°11.2	7.8'	57.7'
9	313°55.0	· · 43.0	234°41.3	8.4'	21°18.9	7.6'	57.7'
10	328°55.2	42.2	249°08.6	8.4'	21°26.6	7.5'	57.6'
11	343°55.3	41.4	263°36.0	8.4'	21°34.0	7.4'	57.6'
12	358°55.4	N13°40.6	278°03.4	8.3'	S21°41.4	7.2'	57.6'
13	13°55.5	39.8	292°30.7	8.3'	21°48.6	7.1'	57.5'
14	28°55.7	39.0	306°58.0	8.3'	21°55.7	7.0'	57.5'
15	43°55.8	· · 38.2	321°25.3	8.3'	22°02.7	6.8'	57.5'
16	58°55.9	37.4	335°52.6	8.3'	22°09.5	6.7'	57.4'
17	73°56.1	36.7	350°19.9	8.3'	22°16.2	6.6'	57.4'
18	88°56.2	N13°35.9	4°47.2	8.3'	S22°22.8	6.4'	57.4'
19	103°56.3	35.1	19°14.4	8.2'	22°29.2	6.3'	57.3'
20	118°56.4	34.3	33°41.7	8.2'	22°35.5	6.2'	57.3'
21	133°56.6	· · 33.5	48°08.9	8.2'	22°41.6	6.0'	57.3'
22	148°56.7	32.7	62°36.1	8.2'	22°47.7	5.9'	57.2'
23	163°56.8	31.9	77°03.3	8.2'	22°53.5	5.7'	57.2'
SD = 15.8'			SD = 15.8'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	////	////	02:23	21:40	////	////
N 70°	////	////	02:56	21:09	23:35	////
68°	////	01:42	03:19	20:47	22:20	////
66°	////	02:20	03:37	20:30	21:44	////
64°	////	02:46	03:51	20:15	21:19	23:38
62°	01:31	03:06	04:03	20:04	21:00	22:32
60°	02:05	03:22	04:14	19:54	20:45	22:00
N 58°	02:29	03:36	04:23	19:45	20:32	21:37
56°	02:48	03:47	04:31	19:37	20:20	21:18
54°	03:03	03:57	04:37	19:30	20:11	21:03
52°	03:16	04:06	04:44	19:24	20:02	20:51
50°	03:28	04:13	04:49	19:19	19:55	20:40
45°	03:50	04:29	05:01	19:07	19:39	20:18
N 40°	04:07	04:42	05:11	18:57	19:26	20:01
35°	04:21	04:53	05:20	18:49	19:15	19:47
30°	04:32	05:02	05:27	18:41	19:06	19:36
20°	04:50	05:17	05:40	18:29	18:52	19:19
N 10°	05:04	05:29	05:51	18:18	18:40	19:05
0°	05:15	05:40	06:01	18:08	18:29	18:54
S 10°	05:25	05:50	06:11	17:58	18:20	18:44
20°	05:33	05:59	06:22	17:47	18:10	18:36
30°	05:41	06:09	06:34	17:36	18:00	18:28
35°	05:45	06:15	06:41	17:29	17:55	18:25
40°	05:49	06:21	06:49	17:21	17:49	18:21
45°	05:53	06:27	06:58	17:12	17:43	18:17
S 50°	05:57	06:35	07:08	17:01	17:35	18:13
52°	05:58	06:38	07:13	16:56	17:32	18:12
54°	06:00	06:42	07:19	16:51	17:28	18:10
56°	06:02	06:46	07:25	16:45	17:24	18:08
58°	06:04	06:50	07:32	16:38	17:20	18:06
S 60°	06:06	06:55	07:40	16:30	17:15	18:04

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	13:06	■	■	18:17	■	■
N 70°	12:36	15:23	■	18:48	17:53	■
68°	12:14	14:29	■	19:12	18:48	■
66°	11:57	13:57	16:05	19:30	19:21	19:06
64°	11:43	13:33	15:23	19:45	19:45	19:49
62°	11:32	13:15	14:54	19:58	20:05	20:18
60°	11:22	13:00	14:32	20:08	20:21	20:40
N 58°	11:14	12:47	14:15	20:18	20:34	20:58
56°	11:06	12:36	14:00	20:26	20:46	21:13
54°	11:00	12:26	13:47	20:33	20:57	21:26
52°	10:54	12:17	13:36	20:40	21:06	21:38
50°	10:49	12:09	13:26	20:46	21:14	21:48
45°	10:37	11:53	13:05	20:59	21:32	22:10
N 40°	10:28	11:40	12:49	21:10	21:46	22:27
35°	10:20	11:28	12:35	21:19	21:58	22:42
30°	10:13	11:18	12:22	21:27	22:09	22:55
20°	10:01	11:02	12:02	21:42	22:28	23:16
N 10°	09:50	10:47	11:44	21:54	22:44	23:35
0°	09:41	10:33	11:27	22:06	22:59	23:53
S 10°	09:31	10:20	11:11	22:17	23:14	· ·
20°	09:21	10:06	10:53	22:30	23:31	· ·
30°	09:09	09:49	10:33	22:44	23:49	· ·
35°	09:03	09:40	10:21	22:53	· ·	00:00
40°	08:55	09:29	10:07	23:02	· ·	00:13
45°	08:46	09:17	09:52	23:14	· ·	00:28
S 50°	08:36	09:01	09:32	23:27	· ·	00:46
52°	08:31	08:54	09:23	23:34	· ·	00:55
54°	08:26	08:47	09:13	23:41	· ·	01:05
56°	08:20	08:38	09:01	23:49	· ·	01:16
58°	08:14	08:28	08:48	23:58	· ·	01:29
S 60°	08:06	08:17	08:32	· ·	00:08	01:44

Day	Sun		Mer. Pass	Moon		Age 4-6 21-42%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Upper hh:mm	Lower hh:mm	
14	04:48	04:42	12:05	15:53	03:27	
15	04:36	04:30	12:05	16:46	04:19	
16	04:24	04:18	12:04	17:40	05:13	

August 17, 18, 19 UT (Tue., Wed., Thu.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	325°16.2	136°41.0	S05°10.4	134°53.3	S04°11.5	322°26.2	S00°25.1	142°08.5	N01°02.2					Alpheratz	357°45.4	29°09.1	
1	340°18.7	151°41.3	11.6	149°54.4	12.2	337°28.9	25.2	157°10.8	02.1					Ankaa	353°17.3	-42°14.5	
2	355°21.1	166°41.6	12.8	164°55.4	12.8	352°31.5	25.2	172°13.0	02.0					Schedar	349°42.6	56°35.7	
3	10°23.6	181°41.9	· · 14.0	179°56.4	· · 13.5	7°34.1	· · 25.3	187°15.2	· · 01.9					Diphda	348°57.7	-17°55.4	
4	25°26.0	196°42.2	15.2	194°57.4	14.1	22°36.8	25.4	202°17.4	01.8					Achernar	335°27.9	-57°10.6	
5	40°28.5	211°42.5	16.4	209°58.4	14.8	37°39.4	25.5	217°19.6	01.6					Hamal	328°03.0	23°30.8	
6	55°31.0	226°42.8	S05°17.6	224°59.4	S04°15.4	52°42.1	S00°25.6	232°21.9	N01°01.5					Polaris	318°41.8	89°18.4	
7	70°33.4	241°43.1	18.8	240°00.5	16.1	67°44.7	25.7	247°24.1	01.4					Acamar	315°19.8	-40°15.4	
8	85°35.9	256°43.4	20.0	255°01.5	16.7	82°47.3	25.8	262°26.3	01.3					Menkar	314°17.1	4°08.0	
9	100°38.4	271°43.7	· · 21.2	270°02.5	· · 17.4	97°50.0	· · 25.9	277°28.5	· · 01.2					Mirfak	308°43.3	49°53.8	
10	115°40.8	286°44.0	22.4	285°03.5	18.0	112°52.6	25.9	292°30.8	01.1					Aldebaran	290°51.8	16°31.9	
11	130°43.3	301°44.4	23.6	300°04.5	18.7	127°55.3	26.0	307°33.0	01.0					Rigel	281°14.1	-8°11.2	
12	145°45.8	316°44.7	S05°24.8	315°05.5	S04°19.3	142°57.9	S00°26.1	322°35.2	N01°00.9					Capella	280°37.7	46°00.4	
13	160°48.2	331°45.0	26.0	330°06.5	20.0	158°00.5	26.2	337°37.4	00.8					Bellatrix	278°34.3	6°21.6	
14	175°50.7	346°45.3	27.2	345°07.6	20.6	173°03.2	26.3	352°39.7	00.7					Elnath	278°15.4	28°36.9	
15	190°53.1	1°45.6	· · 28.4	0°08.6	· · 21.3	188°05.8	· · 26.4	7°41.9	· · 00.6					Alnilam	275°48.6	-1°11.6	
16	205°55.6	16°45.9	29.6	15°09.6	21.9	203°08.5	26.5	22°44.1	00.4					Betelgeuse	271°03.7	7°24.6	
17	220°58.1	31°46.2	30.8	30°10.6	22.6	218°11.1	26.6	37°46.3	00.3					Canopus	263°57.4	-52°41.8	
18	236°00.5	46°46.5	S05°32.0	45°11.6	S04°23.2	233°13.7	S00°26.6	52°48.5	N01°00.2					Sirius	258°35.8	-16°43.7	
19	251°03.0	61°46.9	33.2	60°12.6	23.8	248°16.4	26.7	67°50.8	00.1					Adhara	255°14.4	-28°59.0	
20	266°05.5	76°47.2	34.4	75°13.6	24.5	263°19.0	26.8	82°53.0	01°00.0					Procyon	245°02.1	5°11.9	
21	281°07.9	91°47.5	· · 35.6	90°14.7	· · 25.1	278°21.7	· · 26.9	97°55.2	00°59.9					Pollux	243°30.5	28°00.0	
22	296°10.4	106°47.8	36.8	105°15.7	25.8	293°24.3	27.0	112°57.4	59.8					Avior	234°19.5	-59°32.5	
23	311°12.9	121°48.1	38.0	120°16.7	26.4	308°27.0	27.1	127°59.7	59.7					Suhail	222°54.4	-43°28.5	
Mer.pass. 02:19			ν 0.3' d 1.2' m -4.42			ν 1.0' d 0.6' m 1.51			ν 2.6' d 0.1' m -2.81			ν 2.2' d -0.1' m 1.00					

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	326°15.3	136°48.4	S05°39.2	135°17.7	S04°27.1	323°29.6	S00°27.2	143°01.9	N00°59.6					
1	341°17.8	151°48.8	40.4	150°18.7	27.7	338°32.2	27.3	158°04.1	59.5					
2	356°20.3	166°49.1	41.6	165°19.7	28.4	353°34.9	27.3	173°06.3	59.4					
3	11°22.7	181°49.4	· · 42.8	180°20.7	· · 29.0	8°37.5	· · 27.4	188°08.5	· · 59.2					
4	26°25.2	196°49.7	44.0	195°21.8	29.7	23°40.2	27.5	203°10.8	59.1					
5	41°27.6	211°50.0	45.2	210°22.8	30.3	38°42.8	27.6	218°13.0	59.0					
6	56°30.1	226°50.4	S05°46.4	225°23.8	S04°31.0	53°45.5	S00°27.7	233°15.2	N00°58.9					
7	71°32.6	241°50.7	47.6	240°24.8	31.6	68°48.1	27.8	248°17.4	58.8					
8	86°35.0	256°51.0	48.8	255°25.8	32.3	83°50.8	27.9	263°19.7	58.7					
9	101°37.5	271°51.3	· · 50.0	270°26.8	· · 32.9	98°53.4	· · 28.0	278°21.9	· · 58.6					
10	116°40.0	286°51.7	51.2	285°27.8	33.6	113°56.1	28.1	293°24.1	58.5					
11	131°42.4	301°52.0	52.4	300°28.8	34.2	128°58.7	28.2	308°26.3	58.4					
12	146°44.9	316°52.3	S05°53.5	315°29.8	S04°34.9	144°01.3	S00°28.2	323°28.5	N00°58.3					
13	161°47.4	331°52.6	54.7	330°30.9	35.5	159°04.0	28.3	338°30.8	58.1					
14	176°49.8	346°53.0	55.9	345°31.9	36.1	174°06.6	28.4	353°33.0	58.0					
15	191°52.3	1°53.3	· · 57.1	0°32.9	· · 36.8	189°09.3	· · 28.5	8°35.2	· · 57.9					
16	206°54.8	16°53.6	58.3	15°33.9	37.4	204°11.9	28.6	23°37.4	57.8					
17	221°57.2	31°53.9	05°59.5	30°34.9	38.1	219°14.6	28.7	38°39.6	57.7					
18	236°59.7	46°54.3	S06°00.7	45°35.9	S04°38.7	234°17.2	S00°28.8	53°41.9	N00°57.6					
19	252°02.1	61°54.6	01.9	60°36.9	39.4	249°19.9	28.9	68°44.1	57.5					
20	267°04.6	76°54.9	03.1	75°37.9	40.0	264°22.5	29.0	83°46.3	57.4					
21	282°07.1	91°55.3	· · 04.3	90°38.9	· · 40.7	279°25.2	· · 29.1	98°48.5	· · 57.3					
22	297°09.5	106°55.6	05.5	105°40.0	41.3	294°27.8	29.1	113°50.7	57.2					
23	312°12.0	121°55.9	06.7	120°41.0	42.0	309°30.5	29.2	128°53.0	57.0					
Mer.pass. 02:15			ν 0.3' d 1.2' m -4.43			ν 1.0' d 0.6' m 1.51			ν 2.6' d 0.1' m -2.82			ν 2.2' d -0.1' m 1.00		

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	327°14.5	136°56.2	S06°07.9	135°42.0	S04°42.6	324°33.1	S00°29.3	143°55.2	N00°56.9
1	342°16.9	151°56.6	09.1	150°43.0	43.3	339°35.8	29.4	158°57.4	56.8
2	357°19.4	166°56.9	10.2	165°44.0	43.9	354°38.4	29.5	173°59.6	56.7
3	12°21.9	181°57.2	· · 11.4	180°45.0	· · 44.6	9°41.1	· · 29.6	189°01.8	· · 56.6
4	27°24.3	196°57.6	12.6	195°46.0	45.2	24°43.7	29.7	204°04.1	56.5
5	42°26.8	211°57.9	13.8	210°47.0	45.9	39°46.4	29.8	219°06.3	56.4
6	57°29.3	226°58.2	S06°15.0	225°48.0	S04°46.5	54°49.0	S00°29.9	234°08.5	N00°56.3
7	72°31.7	241°58.6	16.2	240°49.0	47.2	69°51.7	30.0	249°10.7	56.2
8	87°34.2	256°58.9	17.4	255°50.0	47.8	84°54.3	30.1	264°12.9	56.1
9	102°36.6	271°59.3	· · 18.6	270°51.0	· · 48.4	99°57.0	· · 30.2	279°15.2	· · 55.9
10	117°39.1	286°59.6	19.8	285°52.1	49.1	114°59.6	30.3	294°17.4	55.8
11	132°41.6	301°59.9	21.0	300°53.1	49.7	130°02.3	30.3	309°19.6	55.7
12	147°44.0	317°00.3	S06°22.1	315°54.1	S04°50.4	145°04.9	S00°30.4	324°21.8	N00°55.6
13	162°46.5	332°00.6	23.3	330°55.1	51.0	160°07.6	30.5	339°24.0	55.5
14	177°49.0	347°00.9	24.5	345°56.1	51.7	175°10.3	30.6	354°26.3	55.4
15	192°51.4	2°01.3	· · 25.7	0°57.1	· · 52.3	190°12.9	· · 30.7	9°28.5	· · 55.3
16	207°53.9	17°01.6	26.9	15°58.1	53.0	205°15.6	30.8	24°30.7	55.2
17	222°56.4	32°02.0	28.1	30°59.1	53.6	220°18.2	30.9	39°32.9	55.1
18	237°58.8	47°02.3	S06°29.3	46°00.1	S04°54.3	235°20.9	S00°31.0	54°35.1	N00°55.0
19	253°01.3	62°02.7	30.5	61°01.1	54.9	250°23.5	31.1	69°37.4	54.8
20	268°03.8	77°03.0	31.6	76°02.1	55.6	265°26.2	31.2	84°39.6	54.7
21	283°06.2	92°03.3	· · 32.8	91°03.1	· · 56.2	280°28.8	· · 31.3	99°41.8	· · 54.6
22	298°08.7	107°03.7	34.0	106					

h	Sun			Moon			
Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	178°57.0	N13°31.1	91°30.5	8.2'	S22°59.3	5.6'	57.2'
1	193°57.1	30.3	105°57.7	8.2'	23°04.9	5.5'	57.1'
2	208°57.2	29.5	120°24.9	8.2'	23°10.4	5.3'	57.1'
3	223°57.4	· · 28.7	134°52.1	8.2'	23°15.7	5.2'	57.1'
4	238°57.5	27.9	149°19.3	8.2'	23°20.9	5.1'	57.0'
5	253°57.6	27.1	163°46.5	8.2'	23°26.0	4.9'	57.0'
6	268°57.8	N13°26.3	178°13.6	8.2'	S23°30.9	4.8'	57.0'
7	283°57.9	25.5	192°40.8	8.2'	23°35.7	4.6'	56.9'
8	298°58.0	24.7	207°08.0	8.2'	23°40.3	4.5'	56.9'
9	313°58.2	· · 23.9	221°35.1	8.2'	23°44.8	4.4'	56.9'
10	328°58.3	23.1	236°02.3	8.2'	23°49.2	4.2'	56.8'
11	343°58.4	22.3	250°29.5	8.2'	23°53.4	4.1'	56.8'
12	358°58.6	N13°21.5	264°56.6	8.2'	S23°57.5	4.0'	56.8'
13	13°58.7	20.7	279°23.8	8.2'	24°01.5	3.8'	56.7'
14	28°58.8	19.9	293°51.0	8.2'	24°05.3	3.7'	56.7'
15	43°59.0	· · 19.1	308°18.1	8.2'	24°08.9	3.5'	56.7'
16	58°59.1	18.3	322°45.3	8.2'	24°12.5	3.4'	56.6'
17	73°59.2	17.5	337°12.5	8.2'	24°15.9	3.3'	56.6'
18	88°59.4	N13°16.7	351°39.7	8.2'	S24°19.1	3.1'	56.6'
19	103°59.5	15.9	6°06.9	8.2'	24°22.2	3.0'	56.5'
20	118°59.7	15.1	20°34.1	8.2'	24°25.2	2.8'	56.5'
21	133°59.8	· · 14.3	35°01.3	8.2'	24°28.1	2.7'	56.5'
22	148°59.9	13.5	49°28.6	8.2'	24°30.8	2.6'	56.4'
23	164°00.1	12.7	63°55.8	8.3'	24°33.3	2.4'	56.4'
SD = 15.8'			SD = 15.6'				

Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	179°00.2	N13°11.9	78°23.1	8.3'	S24°35.7	2.3'	56.4'
1	194°00.3	11.1	92°50.3	8.3'	24°38.0	2.1'	56.4'
2	209°00.5	10.3	107°17.6	8.3'	24°40.2	2.0'	56.3'
3	224°00.6	· · 09.5	121°44.9	8.3'	24°42.2	1.9'	56.3'
4	239°00.7	08.7	136°12.3	8.3'	24°44.1	1.7'	56.3'
5	254°00.9	07.9	150°39.6	8.4'	24°45.8	1.6'	56.2'
6	269°01.0	N13°07.1	165°07.0	8.4'	S24°47.4	1.5'	56.2'
7	284°01.2	06.3	179°34.3	8.4'	24°48.8	1.3'	56.2'
8	299°01.3	05.5	194°01.7	8.4'	24°50.1	1.2'	56.1'
9	314°01.4	· · 04.6	208°29.2	8.4'	24°51.3	1.0'	56.1'
10	329°01.6	03.8	222°56.6	8.5'	24°52.4	0.9'	56.1'
11	344°01.7	03.0	237°24.1	8.5'	24°53.3	0.8'	56.1'
12	359°01.9	N13°02.2	251°51.6	8.5'	S24°54.0	0.6'	56.0'
13	14°02.0	01.4	266°19.1	8.6'	24°54.7	0.5'	56.0'
14	29°02.1	13°00.6	280°46.6	8.6'	24°55.2	0.4'	56.0'
15	44°02.3	12°59.8	295°14.2	8.6'	24°55.5	0.2'	56.0'
16	59°02.4	59.0	309°41.8	8.6'	24°55.8	0.1'	55.9'
17	74°02.6	58.2	324°09.5	8.7'	24°55.8	-0.0'	55.9'
18	89°02.7	N12°57.4	338°37.1	8.7'	S24°55.8	-0.2'	55.9'
19	104°02.8	56.6	353°04.8	8.7'	24°55.6	-0.3'	55.8'
20	119°03.0	55.7	7°32.6	8.8'	24°55.3	-0.4'	55.8'
21	134°03.1	· · 54.9	22°00.4	8.8'	24°54.9	-0.6'	55.8'
22	149°03.3	54.1	36°28.2	8.8'	24°54.3	-0.7'	55.8'
23	164°03.4	53.3	50°56.0	8.9'	24°53.6	-0.8'	55.7'
SD = 15.8'			SD = 15.4'				

Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	179°03.6	N12°52.5	65°23.9	8.9'	S24°52.7	-1.0'	55.7'
1	194°03.7	51.7	79°51.8	9.0'	24°51.8	-1.1'	55.7'
2	209°03.8	50.9	94°19.8	9.0'	24°50.7	-1.2'	55.7'
3	224°04.0	· · 50.1	108°47.8	9.0'	24°49.4	-1.4'	55.6'
4	239°04.1	49.2	123°15.8	9.1'	24°48.0	-1.5'	55.6'
5	254°04.3	48.4	137°43.9	9.1'	24°46.6	-1.6'	55.6'
6	269°04.4	N12°47.6	152°12.0	9.2'	S24°44.9	-1.8'	55.6'
7	284°04.6	46.8	166°40.2	9.2'	24°43.2	-1.9'	55.5'
8	299°04.7	46.0	181°08.4	9.3'	24°41.3	-2.0'	55.5'
9	314°04.8	· · 45.2	195°36.6	9.3'	24°39.3	-2.1'	55.5'
10	329°05.0	44.4	210°04.9	9.3'	24°37.1	-2.3'	55.5'
11	344°05.1	43.5	224°33.3	9.4'	24°34.9	-2.4'	55.4'
12	359°05.3	N12°42.7	239°01.7	9.4'	S24°32.5	-2.5'	55.4'
13	14°05.4	41.9	253°30.1	9.5'	24°30.0	-2.6'	55.4'
14	29°05.6	41.1	267°58.6	9.5'	24°27.3	-2.8'	55.4'
15	44°05.7	· · 40.3	282°27.2	9.6'	24°24.6	-2.9'	55.3'
16	59°05.9	39.4	296°55.7	9.6'	24°21.7	-3.0'	55.3'
17	74°06.0	38.6	311°24.4	9.7'	24°18.7	-3.1'	55.3'
18	89°06.2	N12°37.8	325°53.1	9.8'	S24°15.5	-3.3'	55.3'
19	104°06.3	37.0	340°21.8	9.8'	24°12.3	-3.4'	55.3'
20	119°06.4	36.2	354°50.6	9.9'	24°08.9	-3.5'	55.2'
21	134°06.6	· · 35.4	9°19.5	9.9'	24°05.4	-3.6'	55.2'
22	149°06.7	34.5	23°48.4	10.0'	24°01.8	-3.7'	55.2'
23	164°06.9	33.7	38°17.4	10.0'	23°58.1	-3.9'	55.2'
SD = 15.8'			SD = 15.2'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	////	////	02:43	21:19	////	////
N 70°	////	01:11	03:11	20:53	22:45	////
68°	////	02:04	03:31	20:33	21:58	////
66°	////	02:36	03:47	20:18	21:28	////
64°	01:03	02:58	04:01	20:05	21:06	22:55
62°	01:50	03:16	04:12	19:54	20:49	22:13
60°	02:19	03:31	04:21	19:45	20:35	21:45
N 58°	02:40	03:43	04:29	19:37	20:23	21:25
56°	02:57	03:54	04:36	19:30	20:12	21:08
54°	03:11	04:03	04:43	19:24	20:03	20:55
52°	03:23	04:11	04:49	19:18	19:55	20:43
50°	03:33	04:18	04:54	19:13	19:48	20:33
45°	03:54	04:33	05:05	19:02	19:33	20:12
N 40°	04:11	04:45	05:14	18:53	19:22	19:56
35°	04:24	04:55	05:22	18:45	19:12	19:43
30°	04:34	05:04	05:29	18:38	19:03	19:33
20°	04:51	05:18	05:41	18:27	18:49	19:16
N 10°	05:04	05:29	05:51	18:17	18:38	19:03
0°	05:15	05:39	06:01	18:07	18:28	18:53
S 10°	05:23	05:48	06:10	17:58	18:19	18:44
20°	05:31	05:57	06:20	17:48	18:11	18:37
30°	05:38	06:06	06:31	17:37	18:02	18:30
35°	05:42	06:11	06:37	17:31	17:57	18:27
40°	05:45	06:17	06:44	17:24	17:52	18:23
45°	05:48	06:23	06:53	17:16	17:46	18:20
S 50°	05:51	06:29	07:03	17:06	17:39	18:17
52°	05:53	06:32	07:07	17:01	17:36	18:16
54°	05:54	06:36	07:13	16:56	17:33	18:14
56°	05:56	06:39	07:18	16:50	17:29	18:13
58°	05:57	06:43	07:24	16:44	17:26	18:12
S 60°	05:59	06:47	07:32	16:37	17:21	18:10

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	■	■	■	■	■	■
66°	■	■	■	■	■	■
64°	17:05	18:19	18:50	20:00	20:39	21:58
62°	16:23	17:30	18:10	20:42	21:27	22:37
60°	15:55	16:59	17:43	21:10	21:58	23:04
N 58°	15:33	16:36	17:21	21:32	22:22	23:25
56°	15:15	16:17	17:04	21:50	22:40	23:43
54°	15:00	16:01	16:49	22:06	22:56	23:57
52°	14:47	15:48	16:36	22:19	23:10	· · ·
50°	14:36	15:36	16:24	22:30	23:22	· · ·
45°	14:12	15:11	16:01	22:55	23:47	· · ·
N 40°	13:53	14:51	15:42	23:14	· · ·	00:06
35°	13:37	14:35	15:26	23:30	· · ·	00:23
30°	13:24	14:20	15:12	23:44	· · ·	00:37
20°	13:00	13:56	14:48	· · ·	00:08	01:01
N 10°	12:40	13:35	14:28	· · ·	00:28	01:22
0°	12:22	13:16	14:09	· · ·	00:48	01:42
S 10°	12:03	12:56	13:50	00:11	01:07	02:01
20°	11:43	12:35	13:29	00:30	01:28	02:22
30°	11:20	12:11	13:06	00:52	01:51	02:46
35°	11:07	11:57	12:52	01:05	02:05	03:00
40°	10:51	11:41	12:36	01:20	02:22	03:16
45°	10:33	11:21	12:17	01:38	02:41	03:35
S 50°	10:10	10:57	11:53	02:00	03:05	04:00
52°	09:59	10:45	11:41	02:11	03:17	04:11
54°	09:47	10:32	11:28	02:23	03:30	04:25
56°	09:33	10:16	11:13	02:36	03:46	04:40
58°	09:16	09:58	10:55	02:53	04:04	04:58
S 60°	08:57	09:35	10:32	03:12	04:26	05:20

Day	Sun		Mer. Pass	Moon		Age 7-9 53-73%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Upper hh:mm	Lower hh:mm	
17	04:12	04:06	12:04	18:35	06:07	
18	03:59	03:53	12:04	19:29	07:02	
19	03:46	03:39	12:04	20:21	07:55	

August 20, 21, 22 UT (Fri., Sat., Sun.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Fri	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec			
0	328°13.6	137°04.4	S06°36.4	136°06.1	S04°58.2	325°36.8	S00°31.6	144°48.5	N00°54.3				Alpheratz	357°45.3	29°09.1		
1	343°16.1	152°04.7	37.6	151°07.1	58.8	340°39.5	31.6	159°50.7	54.2				Ankaa	353°17.3	-42°14.5		
2	358°18.5	167°05.1	38.7	166°08.2	04°59.4	355°42.1	31.7	174°52.9	54.1				Schedar	349°42.6	56°35.7		
3	13°21.0	182°05.4	•• 39.9	181°09.2	05°00.1	10°44.8	•• 31.8	189°55.1	•• 54.0				Diphda	348°57.6	-17°55.4		
4	28°23.5	197°05.8	41.1	196°10.2	00.7	25°47.4	31.9	204°57.3	53.8				Achernar	335°27.8	-57°10.6		
5	43°25.9	212°06.1	42.3	211°11.2	01.4	40°50.1	32.0	219°59.5	53.7				Hamal	328°02.9	23°30.8		
6	58°28.4	227°06.5	S06°43.5	226°12.2	S05°02.0	55°52.7	S00°32.1	235°01.8	N00°53.6				Polaris	318°40.3	89°18.4		
7	73°30.9	242°06.8	44.7	241°13.2	02.7	70°55.4	32.2	250°04.0	53.5				Acamar	315°19.7	-40°15.4		
8	88°33.3	257°07.2	45.9	256°14.2	03.3	85°58.1	32.3	265°06.2	53.4				Menkar	314°17.1	4°08.1		
9	103°35.8	272°07.5	•• 47.0	271°15.2	•• 04.0	101°00.7	•• 32.4	280°08.4	•• 53.3				Mirfak	308°43.3	49°53.8		
10	118°38.2	287°07.9	48.2	286°16.2	04.6	116°03.4	32.5	295°10.6	53.2				Aldebaran	290°51.8	16°31.9		
11	133°40.7	302°08.2	49.4	301°17.2	05.3	131°06.0	32.6	310°12.9	53.1				Rigel	281°14.1	-8°11.2		
12	148°43.2	317°08.6	S06°50.6	316°18.2	S05°05.9	146°08.7	S00°32.7	325°15.1	N00°53.0				Capella	280°37.6	46°00.4		
13	163°45.6	332°08.9	51.8	331°19.2	06.6	161°11.4	32.8	340°17.3	52.8				Bellatrix	278°34.3	6°21.6		
14	178°48.1	347°09.3	52.9	346°20.2	07.2	176°14.0	32.9	355°19.5	52.7				Elnath	278°15.3	28°36.9		
15	193°50.6	2°09.6	•• 54.1	1°21.2	•• 07.9	191°16.7	•• 33.0	10°21.7	•• 52.6				Alnilam	275°48.6	-1°11.6		
16	208°53.0	17°10.0	55.3	16°22.2	08.5	206°19.3	33.1	25°23.9	52.5				Betelgeuse	271°03.7	7°24.6		
17	223°55.5	32°10.3	56.5	31°23.2	09.2	221°22.0	33.2	40°26.2	52.4				Canopus	263°57.4	-52°41.8		
18	238°58.0	47°10.7	S06°57.7	46°24.2	S05°09.8	236°24.7	S00°33.3	55°28.4	N00°52.3				Sirius	258°35.7	-16°43.7		
19	254°00.4	62°11.1	06°58.8	61°25.2	10.4	251°27.3	33.4	70°30.6	52.2				Adhara	255°14.4	-28°59.0		
20	269°02.9	77°11.4	07°00.0	76°26.2	11.1	266°30.0	33.5	85°32.8	52.1				Procyon	245°02.1	5°11.9		
21	284°05.4	92°11.8	•• 01.2	91°27.2	•• 11.7	281°32.6	•• 33.5	100°35.0	•• 52.0				Pollux	243°30.5	28°00.0		
22	299°07.8	107°12.1	02.4	106°28.2	12.4	296°35.3	33.6	115°37.2	51.8				Avior	234°19.5	-59°32.5		
23	314°10.3	122°12.5	03.6	121°29.2	13.0	311°38.0	33.7	130°39.5	51.7				Suhail	222°54.4	-43°28.5		
Mer.pass. 02:07			ν 0.3' d 1.2' m -4.45			ν 1.0' d 0.6' m 1.52			ν 2.7' d 0.1' m -2.83			ν 2.2' d -0.1' m 1.00					

Sat	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec						
0	329°12.7	137°12.9	S07°04.7	136°30.2	S05°13.7	326°40.6	S00°33.8	145°41.7	N00°51.6					
1	344°15.2	152°13.2	05.9	151°31.2	14.3	341°43.3	33.9	160°43.9	51.5					
2	359°17.7	167°13.6	07.1	166°32.2	15.0	356°46.0	34.0	175°46.1	51.4					
3	14°20.1	182°13.9	•• 08.3	181°33.2	•• 15.6	11°48.6	•• 34.1	190°48.3	•• 51.3					
4	29°22.6	197°14.3	09.4	196°34.2	16.3	26°51.3	34.2	205°50.5	51.2					
5	44°25.1	212°14.7	10.6	211°35.2	16.9	41°53.9	34.3	220°52.8	51.1					
6	59°27.5	227°15.0	S07°11.8	226°36.2	S05°17.6	56°56.6	S00°34.4	235°55.0	N00°51.0					
7	74°30.0	242°15.4	13.0	241°37.2	18.2	71°59.3	34.5	250°57.2	50.8					
8	89°32.5	257°15.8	14.2	256°38.2	18.9	87°01.9	34.6	265°59.4	50.7					
9	104°34.9	272°16.1	•• 15.3	271°39.2	•• 19.5	102°04.6	•• 34.7	281°01.6	•• 50.6					
10	119°37.4	287°16.5	16.5	286°40.2	20.1	117°07.3	34.8	296°03.8	50.5					
11	134°39.9	302°16.9	17.7	301°41.2	20.8	132°09.9	34.9	311°06.1	50.4					
12	149°42.3	317°17.2	S07°18.9	316°42.2	S05°21.4	147°12.6	S00°35.0	326°08.3	N00°50.3					
13	164°44.8	332°17.6	20.0	331°43.2	22.1	162°15.3	35.1	341°10.5	50.2					
14	179°47.2	347°18.0	21.2	346°44.2	22.7	177°17.9	35.2	356°12.7	50.1					
15	194°49.7	2°18.3	•• 22.4	1°45.2	•• 23.4	192°20.6	•• 35.3	11°14.9	•• 50.0					
16	209°52.2	17°18.7	23.6	16°46.2	24.0	207°23.3	35.4	26°17.1	49.8					
17	224°54.6	32°19.1	24.7	31°47.2	24.7	222°25.9	35.5	41°19.4	49.7					
18	239°57.1	47°19.4	S07°25.9	46°48.2	S05°25.3	237°28.6	S00°35.6	56°21.6	N00°49.6					
19	254°59.6	62°19.8	27.1	61°49.2	26.0	252°31.3	35.7	71°23.8	49.5					
20	270°02.0	77°20.2	28.2	76°50.2	26.6	267°33.9	35.8	86°26.0	49.4					
21	285°04.5	92°20.6	•• 29.4	91°51.2	•• 27.3	282°36.6	•• 35.9	101°28.2	•• 49.3					
22	300°07.0	107°20.9	30.6	106°52.2	27.9	297°39.3	36.0	116°30.4	49.2					
23	315°09.4	122°21.3	31.8	121°53.2	28.5	312°41.9	36.1	131°32.7	49.1					
Mer.pass. 02:03			ν 0.4' d 1.2' m -4.46			ν 1.0' d 0.6' m 1.52			ν 2.7' d 0.1' m -2.83			ν 2.2' d -0.1' m 1.00		

Sun	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec						
0	330°11.9	137°21.7	S07°32.9	136°54.2	S05°29.2	327°44.6	S00°36.2	146°34.9	N00°48.9					
1	345°14.4	152°22.0	34.1	151°55.2	29.8	342°47.3	36.3	161°37.1	48.8					
2	0°16.8	167°22.4	35.3	166°56.2	30.5	357°49.9	36.4	176°39.3	48.7					
3	15°19.3	182°22.8	•• 36.4	181°57.2	•• 31.1	12°52.6	•• 36.5	191°41.5	•• 48.6					
4	30°21.7	197°23.2	37.6	196°58.2	31.8	27°55.3	36.6	206°43.7	48.5					
5	45°24.2	212°23.6	38.8	211°59.2	32.4	42°58.0	36.7	221°45.9	48.4					
6	60°26.7	227°23.9	S07°39.9	227°00.2	S05°33.1	58°00.6	S00°36.8	236°48.2	N00°48.3					
7	75°29.1	242°24.3	41.1	242°01.2	33.7	73°03.3	36.9	251°50.4	48.2					
8	90°31.6	257°24.7	42.3	257°02.2	34.4	88°06.0	37.0	266°52.6	48.1					
9	105°34.1	272°25.1	•• 43.5	272°03.2	•• 35.0	103°08.6	•• 37.1	281°54.8	•• 47.9					
10	120°36.5	287°25.5	44.6	287°04.2	35.7	118°11.3	37.2	296°57.0	47.8					
11	135°39.0	302°25.8	45.8	302°05.2	36.3	133°14.0	37.3	311°59.2	47.7					
12	150°41.5	317°26.2	S07°47.0	317°06.2	S05°36.9	148°16.6	S00°37.4	327°01.4	N00°47.6					
13	165°43.9	332°26.6	48.1	332°07.2	37.6	163°19.3	37.5	342°03.7	47.5					
14	180°46.4	347°27.0	49.3	347°08.1	38.2	178°22.0	37.6	357°05.9	47.4					
15	195°48.8	2°27.4	•• 50.5	2°09.1	•• 38.9	193°24.7	•• 37.7	12°08.1	•• 47.3					
16	210°51.3	17°27.7	51.6	17°10.1	39.5	208°27.3	37.8	27°10.3	47.2					
17	225°53.8	32°28.1	52.8	32°11.1	40.2	223°30.0	37.9	42°12.5	47.0					
18	240°56.2	47°28.5	S07°53.9	47°12.1	S05°40.8	238°32.7	S00°38.0	57°14.7	N00°46.9					
19	255°58.7	62°28.9	55.1	62°13.1	41.5	253°35.4	38.1	72°16.9	46.8					
20	271°01.2	77°29.3	56.3	77°14.1	42.1	268°38.0	38.2	87°19.2	46.7					
21	286°03.6	92°29.7	•• 57.4	92°15.1	•• 42.8	283°40.7	•• 38.3	102°21.4	•• 46.6					
22	301°06.1	107°30.1	58.6	107°16.1	43.4	298°43.4	38.4	117°23.6	46.5					
23	316°08.6	122°30.5	59.8	122°17.1	44.1	313°46.0	38.5	132°25.8	46.4					
Mer.pass. 01:59			ν 0.4' d 1.2' m -4.47			ν 1.0' d 0.6' m 1.52			ν 2.7' d 0.1' m -2.84			ν 2.2' d -0.1' m 1.00		

Aug 20 Fri	SHA	Mer.pass
Venus	168°50.8	14:51
Mars	167°52.5	14:55
Jupiter	357°23.2	02:17
Saturn	176°34.8	14:19

Aug 21 Sat	SHA	Mer.pass
Venus	168°00.1	14:51
Mars	167°17.5	14:53
Jupiter	357°27.9	02:13
Saturn	176°28.9	14:15

Aug 22 Sun	SHA	Mer.pass
Venus	167°09.8	14:50
Mars	166°42.3	14:51
Jupiter	357°32.7	02:09
Saturn	176°23.0	14:12

Horizontal parallax	
Venus:	0.2
Mars:	0.1

h	Sun				Moon			
Fri	GHA	Dec		GHA	ν	Dec	d	HP
0	179°07.0	N12°32.9		52°46.4	10.1'	S23°54.2	-4.0'	55.2'
1	194°07.2	32.1		67°15.5	10.1'	23°50.2	-4.1'	55.1'
2	209°07.3	31.2		81°44.6	10.2'	23°46.2	-4.2'	55.1'
3	224°07.5	· · 30.4		96°13.8	10.3'	23°41.9	-4.3'	55.1'
4	239°07.6	29.6		110°43.1	10.3'	23°37.6	-4.4'	55.1'
5	254°07.8	28.8		125°12.4	10.4'	23°33.2	-4.5'	55.0'
6	269°07.9	N12°28.0		139°41.8	10.4'	S23°28.7	-4.7'	55.0'
7	284°08.1	27.1		154°11.2	10.5'	23°24.0	-4.8'	55.0'
8	299°08.2	26.3		168°40.7	10.6'	23°19.2	-4.9'	55.0'
9	314°08.4	· · 25.5		183°10.2	10.6'	23°14.3	-5.0'	55.0'
10	329°08.5	24.7		197°39.8	10.7'	23°09.3	-5.1'	54.9'
11	344°08.7	23.8		212°09.5	10.7'	23°04.2	-5.2'	54.9'
12	359°08.8	N12°23.0		226°39.2	10.8'	S22°59.0	-5.3'	54.9'
13	14°09.0	22.2		241°09.0	10.9'	22°53.7	-5.4'	54.9'
14	29°09.1	21.4		255°38.9	10.9'	22°48.3	-5.5'	54.9'
15	44°09.3	· · 20.5		270°08.8	11.0'	22°42.7	-5.6'	54.9'
16	59°09.4	19.7		284°38.8	11.0'	22°37.1	-5.7'	54.8'
17	74°09.6	18.9		299°08.9	11.1'	22°31.4	-5.9'	54.8'
18	89°09.7	N12°18.1		313°39.0	11.2'	S22°25.5	-6.0'	54.8'
19	104°09.9	17.2		328°09.2	11.2'	22°19.6	-6.1'	54.8'
20	119°10.0	16.4		342°39.4	11.3'	22°13.5	-6.2'	54.8'
21	134°10.2	· · 15.6		357°09.7	11.4'	22°07.3	-6.3'	54.8'
22	149°10.3	14.7		11°40.1	11.4'	22°01.1	-6.4'	54.7'
23	164°10.5	13.9		26°10.5	11.5'	21°54.7	-6.5'	54.7'
	SD = 15.8'	$d = -0.8'$				SD = 15.0'		

Sat		GHA	Dec		GHA	ν	Dec	d	HP
0		179°10.6	N12°13.1		40°41.0	11.6'	S21°48.3	-6.6'	54.7'
1		194°10.8	12.3		55°11.6	11.6'	21°41.7	-6.7'	54.7'
2		209°10.9	11.4		69°42.2	11.7'	21°35.0	-6.8'	54.7'
3		224°11.1	· · 10.6		84°12.9	11.8'	21°28.3	-6.8'	54.7'
4		239°11.3	09.8		98°43.6	11.8'	21°21.4	-6.9'	54.6'
5		254°11.4	08.9		113°14.5	11.9'	21°14.5	-7.0'	54.6'
6		269°11.6	N12°08.1		127°45.4	12.0'	S21°07.5	-7.1'	54.6'
7		284°11.7	07.3		142°16.3	12.0'	21°00.3	-7.2'	54.6'
8		299°11.9	06.4		156°47.3	12.1'	20°53.1	-7.3'	54.6'
9		314°12.0	· · 05.6		171°18.4	12.2'	20°45.8	-7.4'	54.6'
10		329°12.2	04.8		185°49.6	12.2'	20°38.4	-7.5'	54.5'
11		344°12.3	03.9		200°20.8	12.3'	20°30.9	-7.6'	54.5'
12		359°12.5	N12°03.1		214°52.1	12.3'	S20°23.3	-7.7'	54.5'
13		14°12.6	02.3		229°23.4	12.4'	20°15.6	-7.8'	54.5'
14		29°12.8	01.4		243°54.8	12.5'	20°07.8	-7.9'	54.5'
15		44°13.0	12°00.6		258°26.3	12.5'	20°00.0	-7.9'	54.5'
16		59°13.1	11°59.8		272°57.9	12.6'	19°52.1	-8.0'	54.5'
17		74°13.3	58.9		287°29.5	12.7'	19°44.0	-8.1'	54.4'
18		89°13.4	N11°58.1		302°01.1	12.7'	S19°35.9	-8.2'	54.4'
19		104°13.6	57.3		316°32.9	12.8'	19°27.7	-8.3'	54.4'
20		119°13.7	56.4		331°04.7	12.9'	19°19.5	-8.4'	54.4'
21		134°13.9	· · 55.6		345°36.5	12.9'	19°11.1	-8.4'	54.4'
22		149°14.0	54.8		0°08.5	13.0'	19°02.7	-8.5'	54.4'
23		164°14.2	53.9		14°40.5	13.1'	18°54.2	-8.6'	54.4'
		SD = 15.8'		d = -0.8'	SD = 14.9'				

Sun		GHA	Dec		GHA	ν	Dec	d	HP
0		179°14.4	N11°53.1		29°12.5	13.1'	S18°45.6	-8.7'	54.4'
1		194°14.5	52.3		43°44.7	13.2'	18°36.9	-8.7'	54.3'
2		209°14.7	51.4		58°16.8	13.2'	18°28.1	-8.8'	54.3'
3		224°14.8	· · 50.6		72°49.1	13.3'	18°19.3	-8.9'	54.3'
4		239°15.0	49.7		87°21.4	13.4'	18°10.4	-9.0'	54.3'
5		254°15.2	48.9		101°53.8	13.4'	18°01.4	-9.0'	54.3'
6		269°15.3	N11°48.1		116°26.2	13.5'	S17°52.4	-9.1'	54.3'
7		284°15.5	47.2		130°58.7	13.6'	17°43.3	-9.2'	54.3'
8		299°15.6	46.4		145°31.2	13.6'	17°34.1	-9.3'	54.3'
9		314°15.8	· · 45.5		160°03.9	13.7'	17°24.8	-9.3'	54.3'
10		329°15.9	44.7		174°36.5	13.7'	17°15.5	-9.4'	54.2'
11		344°16.1	43.9		189°09.3	13.8'	17°06.1	-9.5'	54.2'
12		359°16.3	N11°43.0		203°42.1	13.9'	S16°56.6	-9.5'	54.2'
13		14°16.4	42.2		218°14.9	13.9'	16°47.1	-9.6'	54.2'
14		29°16.6	41.3		232°47.8	14.0'	16°37.4	-9.7'	54.2'
15		44°16.7	· · 40.5		247°20.8	14.0'	16°27.8	-9.7'	54.2'
16		59°16.9	39.7		261°53.8	14.1'	16°18.0	-9.8'	54.2'
17		74°17.1	38.8		276°26.9	14.1'	16°08.2	-9.9'	54.2'
18		89°17.2	N11°38.0		291°00.1	14.2'	S15°58.3	-9.9'	54.2'
19		104°17.4	37.1		305°33.3	14.3'	15°48.4	-10.0'	54.2'
20		119°17.6	36.3		320°06.5	14.3'	15°38.4	-10.1'	54.1'
21		134°17.7	· · 35.4		334°39.8	14.4'	15°28.4	-10.1'	54.1'
22		149°17.9	34.6		349°13.2	14.4'	15°18.2	-10.2'	54.1'
23		164°18.0	33.8		3°46.6	14.5'	15°08.1	-10.2'	54.1'
		SD = 15.8'		d = -0.8'	SD = 14.8'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	////	////	03:02	21:00	23:37	////
N 70°	////	01:43	03:25	20:37	22:15	////
68°	////	02:23	03:43	20:20	21:38	////
66°	////	02:50	03:58	20:06	21:13	23:40
64°	01:31	03:10	04:10	19:55	20:53	22:29
62°	02:06	03:26	04:20	19:45	20:38	21:56
60°	02:31	03:39	04:28	19:36	20:25	21:32
N 58°	02:50	03:51	04:36	19:29	20:14	21:14
56°	03:05	04:00	04:42	19:23	20:04	20:59
54°	03:18	04:09	04:48	19:17	19:56	20:46
52°	03:30	04:16	04:53	19:12	19:49	20:35
50°	03:39	04:23	04:58	19:07	19:42	20:26
45°	03:59	04:37	05:09	18:57	19:28	20:06
N 40°	04:14	04:49	05:17	18:49	19:17	19:51
35°	04:26	04:58	05:24	18:41	19:08	19:39
30°	04:36	05:06	05:31	18:35	19:00	19:29
20°	04:52	05:19	05:42	18:24	18:47	19:14
N 10°	05:04	05:30	05:51	18:15	18:37	19:02
0°	05:14	05:39	06:00	18:07	18:28	18:52
S 10°	05:22	05:47	06:08	17:58	18:19	18:44
20°	05:29	05:55	06:18	17:49	18:11	18:37
30°	05:35	06:03	06:28	17:39	18:03	18:31
35°	05:38	06:08	06:34	17:33	17:59	18:29
40°	05:41	06:13	06:40	17:27	17:54	18:26
45°	05:44	06:18	06:48	17:19	17:49	18:23
S 50°	05:46	06:24	06:57	17:10	17:43	18:21
52°	05:47	06:27	07:01	17:06	17:41	18:20
54°	05:48	06:29	07:06	17:01	17:38	18:19
56°	05:49	06:33	07:11	16:56	17:35	18:18
58°	05:50	06:36	07:17	16:50	17:31	18:17
S 60°	05:51	06:40	07:23	16:44	17:28	18:16

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	■	■	20:38	■	■	■
N 70°	■	20:54	20:02	■	23:20	· · ·
68°	■	19:59	19:37	■	■	00:14
66°	19:39	19:25	19:17	22:54	· · ·	00:46
64°	18:59	19:01	19:01	23:34	· · ·	01:10
62°	18:31	18:41	18:47	· · ·	00:01	01:28
60°	18:09	18:25	18:36	· · ·	00:22	01:44
N 58°	17:52	18:12	18:26	· · ·	00:39	01:57
56°	17:37	18:00	18:17	· · ·	00:53	02:08
54°	17:24	17:50	18:10	· · ·	01:06	02:18
52°	17:13	17:41	18:03	00:10	01:17	02:26
50°	17:03	17:33	17:56	00:21	01:26	02:34
45°	16:42	17:15	17:43	00:45	01:47	02:50
N 40°	16:25	17:01	17:32	01:03	02:03	03:04
35°	16:10	16:48	17:22	01:19	02:17	03:15
30°	15:58	16:38	17:13	01:33	02:29	03:25
20°	15:36	16:19	16:59	01:55	02:49	03:42
N 10°	15:17	16:03	16:46	02:15	03:07	03:57
0°	15:00	15:48	16:34	02:34	03:23	04:11
S 10°	14:42	15:33	16:22	02:52	03:40	04:24
20°	14:24	15:17	16:09	03:12	03:57	04:39
30°	14:02	14:58	15:54	03:34	04:17	04:55
35°	13:49	14:47	15:45	03:48	04:29	05:05
40°	13:34	14:35	15:35	04:03	04:42	05:16
45°	13:17	14:20	15:23	04:21	04:58	05:28
S 50°	12:55	14:01	15:09	04:43	05:17	05:44
52°	12:45	13:53	15:02	04:54	05:26	05:51
54°	12:33	13:43	14:55	05:06	05:36	05:59
56°	12:19	13:32	14:47	05:20	05:48	06:08
58°	12:04	13:20	14:37	05:36	06:01	06:18
S 60°	11:45	13:05	14:27	05:55	06:16	06:29

August 23, 24, 25 UT (Mon., Tue., Wed.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Mon	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec			
0	331°11.0	137°30.8	S08°00.9	137°18.1	S05°44.7	328°48.7	S00°38.6	147°28.0	N00°46.3				Alpheratz	357°45.3	29°09.1		
1	346°13.5	152°31.2	02.1	152°19.1	45.3	343°51.4	38.7	162°30.2	46.1				Ankaa	353°17.2	-42°14.6		
2	1°16.0	167°31.6	03.3	167°20.1	46.0	358°54.1	38.8	177°32.4	46.0				Schedar	349°42.5	56°35.7		
3	16°18.4	182°32.0	· · 04.4	182°21.1	· · 46.6	13°56.7	· · 38.9	192°34.7	· · 45.9				Diphda	348°57.6	-17°55.4		
4	31°20.9	197°32.4	05.6	197°22.1	47.3	28°59.4	39.0	207°36.9	45.8				Achernar	335°27.8	-57°10.6		
5	46°23.3	212°32.8	06.7	212°23.1	47.9	44°02.1	39.1	222°39.1	45.7				Hamal	328°02.9	23°30.9		
6	61°25.8	227°33.2	S08°07.9	227°24.0	S05°48.6	59°04.8	S00°39.2	237°41.3	N00°45.6				Polaris	318°39.0	89°18.4		
7	76°28.3	242°33.6	09.1	242°25.0	49.2	74°07.4	39.3	252°43.5	45.5				Acamar	315°19.7	-40°15.4		
8	91°30.7	257°34.0	10.2	257°26.0	49.9	89°10.1	39.4	267°45.7	45.4				Menkar	314°17.1	4°08.1		
9	106°33.2	272°34.4	· · 11.4	272°27.0	· · 50.5	104°12.8	· · 39.5	282°47.9	· · 45.2				Mirfak	308°43.2	49°53.8		
10	121°35.7	287°34.8	12.5	287°28.0	51.2	119°15.5	39.6	297°50.1	45.1				Aldebaran	290°51.8	16°31.9		
11	136°38.1	302°35.2	13.7	302°29.0	51.8	134°18.2	39.7	312°52.4	45.0				Rigel	281°14.1	-8°11.2		
12	151°40.6	317°35.6	S08°14.9	317°30.0	S05°52.5	149°20.8	S00°39.8	327°54.6	N00°44.9				Capella	280°37.6	46°00.4		
13	166°43.1	332°36.0	16.0	332°31.0	53.1	164°23.5	39.9	342°56.8	44.8				Bellatrix	278°34.3	6°21.6		
14	181°45.5	347°36.4	17.2	347°32.0	53.7	179°26.2	40.0	357°59.0	44.7				Elnath	278°15.3	28°36.9		
15	196°48.0	2°36.8	· · 18.3	2°33.0	· · 54.4	194°28.9	· · 40.1	13°01.2	· · 44.6				Alnilam	275°48.5	-1°11.6		
16	211°50.4	17°37.2	19.5	17°34.0	55.0	209°31.5	40.2	28°03.4	44.5				Betelgeuse	271°03.6	7°24.6		
17	226°52.9	32°37.6	20.7	32°34.9	55.7	224°34.2	40.3	43°05.6	44.3				Canopus	263°57.4	-52°41.8		
18	241°55.4	47°38.0	S08°21.8	47°35.9	S05°56.3	239°36.9	S00°40.4	58°07.8	N00°44.2				Sirius	258°35.7	-16°43.7		
19	256°57.8	62°38.4	23.0	62°36.9	57.0	254°39.6	40.5	73°10.1	44.1				Adhara	255°14.4	-28°59.0		
20	272°00.3	77°38.8	24.1	77°37.9	57.6	269°42.3	40.6	88°12.3	44.0				Procyon	245°02.1	5°11.9		
21	287°02.8	92°39.2	· · 25.3	92°38.9	· · 58.3	284°44.9	· · 40.7	103°14.5	· · 43.9				Pollux	243°30.5	28°00.0		
22	302°05.2	107°39.6	26.4	107°39.9	58.9	299°47.6	40.8	118°16.7	43.8				Avior	234°19.5	-59°32.5		
23	317°07.7	122°40.0	27.6	122°40.9	59.6	314°50.3	40.9	133°18.9	43.7				Suhail	222°54.4	-43°28.5		
Mer.pass. 01:55			ν 0.4' d 1.2' m -4.48			ν 1.0' d 0.6' m 1.52			ν 2.7' d 0.1' m -2.85			ν 2.2' d -0.1' m 0.99					
Tue	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	332°10.2	137°40.4	S08°28.7	137°41.9	S06°00.2	329°53.0	S00°41.0	148°21.1	N00°43.6				Dubhe	193°54.6	61°41.6		
1	347°12.6	152°40.8	29.9	152°42.9	00.8	344°55.7	41.1	163°23.3	43.4				Denebola	182°36.0	14°30.8		
2	2°15.1	167°41.2	31.1	167°43.8	01.5	359°58.3	41.2	178°25.5	43.3				Gienah	175°54.7	-17°36.1		
3	17°17.6	182°41.6	· · 32.2	182°44.8	· · 02.1	15°01.0	· · 41.3	193°27.8	· · 43.2				Acrux	173°12.3	-63°09.7		
4	32°20.0	197°42.0	33.4	197°45.8	02.8	30°03.7	41.4	208°30.0	43.1				Gacrux	172°03.8	-57°10.6		
5	47°22.5	212°42.4	34.5	212°46.8	03.4	45°06.4	41.5	223°32.2	43.0				Alioth	166°22.7	55°54.3		
6	62°24.9	227°42.8	S08°35.7	227°47.8	S06°04.1	60°09.1	S00°41.7	238°34.4	N00°42.9				Spica	158°33.6	-11°13.1		
7	77°27.4	242°43.2	36.8	242°48.8	04.7	75°11.7	41.8	253°36.6	42.8				Alkaid	153°00.6	49°15.8		
8	92°29.9	257°43.7	38.0	257°49.8	05.4	90°14.4	41.9	268°38.8	42.7				Hadar	148°51.2	-60°25.7		
9	107°32.3	272°44.1	· · 39.1	272°50.8	· · 06.0	105°17.1	· · 42.0	283°41.0	· · 42.5				Menkent	148°10.2	-36°25.5		
10	122°34.8	287°44.5	40.3	287°51.8	06.6	120°19.8	42.1	298°43.2	42.4				Arcturus	145°57.7	19°07.7		
11	137°37.3	302°44.9	41.4	302°52.7	07.3	135°22.5	42.2	313°45.5	42.3				Rigel Kent.	139°54.8	-60°53.0		
12	152°39.7	317°45.3	S08°42.6	317°53.7	S06°07.9	150°25.2	S00°42.3	328°47.7	N00°42.2				Kochab	137°20.0	74°06.9		
13	167°42.2	332°45.7	43.7	332°54.7	08.6	165°27.8	42.4	343°49.9	42.1				Zuben'ubi	137°07.8	-16°05.2		
14	182°44.7	347°46.1	44.9	347°55.7	09.2	180°30.5	42.5	358°52.1	42.0				Alphecca	126°12.7	26°40.9		
15	197°47.1	2°46.5	· · 46.0	2°56.7	· · 09.9	195°33.2	· · 42.6	13°54.3	· · 41.9				Antares	112°28.7	-26°27.4		
16	212°49.6	17°47.0	47.2	17°57.7	10.5	210°35.9	42.7	28°56.5	41.7				Atria	107°32.2	-69°03.1		
17	227°52.1	32°47.4	48.3	32°58.7	11.2	225°38.6	42.8	43°58.7	41.6				Sabik	102°14.7	-15°44.3		
18	242°54.5	47°47.8	S08°49.5	47°59.6	S06°11.8	240°41.3	S00°42.9	59°00.9	N00°41.5				Shaula	96°24.5	-37°06.8		
19	257°57.0	62°48.2	50.6	63°00.6	12.5	255°43.9	43.0	74°03.1	41.4				Rasalhague	96°08.2	12°33.3		
20	272°59.4	77°48.6	51.8	78°01.6	13.1	270°46.6	43.1	89°05.4	41.3				Eltanin	90°46.8	51°29.5		
21	288°01.9	92°49.1	· · 52.9	93°02.6	· · 13.7	285°49.3	· · 43.2	104°07.6	· · 41.2				Kaus Aust.	83°46.2	-34°22.8		
22	303°04.4	107°49.5	54.1	108°03.6	14.4	300°52.0	43.3	119°09.8	41.1				Vega	80°40.1	38°47.9		
23	318°06.8	122°49.9	55.2	123°04.6	15.0	315°54.7	43.4	134°12.0	41.0				Nunki	76°00.5	-26°17.0		
Mer.pass. 01:51			ν 0.4' d 1.2' m -4.49			ν 1.0' d 0.6' m 1.52			ν 2.7' d 0.1' m -2.85			ν 2.2' d -0.1' m 0.99					
Wed	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	333°09.3	137°50.3	S08°56.4	138°05.6	S06°15.7	330°57.4	S00°43.5	149°14.2	N00°40.8				Altair	62°09.9	8°54.0		
1	348°11.8	152°50.7	57.5	153°06.5	16.3	346°00.1	43.6	164°16.4	40.7				Peacock	53°21.8	-56°42.0		
2	3°14.2	167°51.2	58.7	168°07.5	17.0	1°02.7	43.7	179°18.6	40.6				Deneb	49°32.5	45°19.3		
3	18°16.7	182°51.6	08°59.8	183°08.5	· · 17.6	16°05.4	· · 43.9	194°20.8	· · 40.5				Enif	33°48.8	9°55.7		
4	33°19.2	197°52.0	09°00.9	198°09.5	18.3	31°08.1	44.0	209°23.0	40.4				Al Na'ir	27°45.7	-46°54.4		
5	48°21.6	212°52.4	02.1	213°10.5	18.9	46°10.8	44.1	224°25.2	40.3				Fomalhaut	15°25.8	-29°33.7		
6	63°24.1	227°52.9	S09°03.2	228°11.5	S06°19.5	61°13.5	S00°44.2	239°27.5	N00°40.2				Scheat	13°55.0	28°08.6		
7	78°26.5	242°53.3	04.4	243°12.4	20.2	76°16.2	44.3	254°29.7	40.0				Markab	13°40.0	15°16.0		
8	93°29.0	257°53.7	05.5	258°13.4	20.8	91°18.9	44.4	269°31.9	39.9								
9	108°31.5	272°54.1	· · 06.7	273°14.4	· · 21.5	106°21.5	· · 44.5	284°34.1	· · 39.8				Aug 23 Mon	SHA	Mer.pass		
10	123°33.9	287°54.6	07.8	288°15.4	22.1	121°24.2	44.6	299°36.3	39.7				Venus	166°19.8	14:50		
11	138°36.4	302°55.0	08.9	303°16.4	22.8	136°26.9	44.7	314°38.5	39.6								

h Sun			Moon				
Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	179°18.2	N11°32.9	18°20.1	14.5'	S14°57.8	-10.3'	54.1'
1	194°18.4	32.1	32°53.6	14.6'	14°47.5	-10.3'	54.1'
2	209°18.5	31.2	47°27.2	14.6'	14°37.2	-10.4'	54.1'
3	224°18.7	· · 30.4	62°00.9	14.7'	14°26.8	-10.5'	54.1'
4	239°18.9	29.5	76°34.5	14.7'	14°16.3	-10.5'	54.1'
5	254°19.0	28.7	91°08.3	14.8'	14°05.8	-10.6'	54.1'
6	269°19.2	N11°27.8	105°42.1	14.8'	S13°55.2	-10.6'	54.1'
7	284°19.3	27.0	120°15.9	14.9'	13°44.6	-10.7'	54.1'
8	299°19.5	26.1	134°49.8	14.9'	13°33.9	-10.7'	54.0'
9	314°19.7	· · 25.3	149°23.7	15.0'	13°23.2	-10.8'	54.0'
10	329°19.8	24.4	163°57.7	15.0'	13°12.4	-10.8'	54.0'
11	344°20.0	23.6	178°31.8	15.1'	13°01.6	-10.9'	54.0'
12	359°20.2	N11°22.8	193°05.8	15.1'	S12°50.7	-10.9'	54.0'
13	14°20.3	21.9	207°40.0	15.2'	12°39.8	-11.0'	54.0'
14	29°20.5	21.1	222°14.1	15.2'	12°28.8	-11.0'	54.0'
15	44°20.7	· · 20.2	236°48.4	15.3'	12°17.8	-11.1'	54.0'
16	59°20.8	19.4	251°22.6	15.3'	12°06.8	-11.1'	54.0'
17	74°21.0	18.5	265°56.9	15.4'	11°55.6	-11.2'	54.0'
18	89°21.2	N11°17.7	280°31.3	15.4'	S11°44.5	-11.2'	54.0'
19	104°21.3	16.8	295°05.7	15.4'	11°33.3	-11.2'	54.0'
20	119°21.5	16.0	309°40.1	15.5'	11°22.1	-11.3'	54.0'
21	134°21.7	· · 15.1	324°14.6	15.5'	11°10.8	-11.3'	54.0'
22	149°21.8	14.3	338°49.1	15.6'	10°59.4	-11.4'	54.0'
23	164°22.0	13.4	353°23.7	15.6'	10°48.1	-11.4'	54.0'
SD = 15.8' $d = -0.8'$			SD = 14.8'				

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	179°22.2	N11°12.5	7°58.3	15.6'	S10°36.7	-11.4'	54.0'
1	194°22.3	11.7	22°32.9	15.7'	10°25.2	-11.5'	54.0'
2	209°22.5	10.8	37°07.6	15.7'	10°13.8	-11.5'	53.9'
3	224°22.7	· · 10.0	51°42.3	15.7'	10°02.2	-11.6'	53.9'
4	239°22.8	09.1	66°17.0	15.8'	09°50.7	-11.6'	53.9'
5	254°23.0	08.3	80°51.8	15.8'	09°39.1	-11.6'	53.9'
6	269°23.2	N11°07.4	95°26.6	15.9'	S09°27.5	-11.7'	53.9'
7	284°23.3	06.6	110°01.5	15.9'	09°15.8	-11.7'	53.9'
8	299°23.5	05.7	124°36.4	15.9'	09°04.1	-11.7'	53.9'
9	314°23.7	· · 04.9	139°11.3	16.0'	08°52.4	-11.8'	53.9'
10	329°23.8	04.0	153°46.3	16.0'	08°40.6	-11.8'	53.9'
11	344°24.0	03.2	168°21.2	16.0'	08°28.9	-11.8'	53.9'
12	359°24.2	N11°02.3	182°56.3	16.0'	S08°17.0	-11.8'	53.9'
13	14°24.3	01.4	197°31.3	16.1'	08°05.2	-11.9'	53.9'
14	29°24.5	11°00.6	212°06.4	16.1'	07°53.3	-11.9'	53.9'
15	44°24.7	10°59.7	226°41.5	16.1'	07°41.4	-11.9'	53.9'
16	59°24.8	58.9	241°16.6	16.2'	07°29.5	-12.0'	53.9'
17	74°25.0	58.0	255°51.8	16.2'	07°17.5	-12.0'	53.9'
18	89°25.2	N10°57.2	270°26.9	16.2'	S07°05.5	-12.0'	53.9'
19	104°25.4	56.3	285°02.2	16.2'	06°53.5	-12.0'	53.9'
20	119°25.5	55.4	299°37.4	16.3'	06°41.5	-12.1'	53.9'
21	134°25.7	· · 54.6	314°12.6	16.3'	06°29.4	-12.1'	53.9'
22	149°25.9	53.7	328°47.9	16.3'	06°17.3	-12.1'	53.9'
23	164°26.0	52.9	343°23.2	16.3'	06°05.2	-12.1'	53.9'
SD = 15.8' $d = -0.9'$			SD = 14.7'				

Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	179°26.2	N10°52.0	357°58.5	16.3'	S05°53.1	-12.1'	53.9'
1	194°26.4	51.1	12°33.9	16.4'	05°41.0	-12.2'	53.9'
2	209°26.6	50.3	27°09.3	16.4'	05°28.8	-12.2'	53.9'
3	224°26.7	· · 49.4	41°44.6	16.4'	05°16.6	-12.2'	53.9'
4	239°26.9	48.6	56°20.0	16.4'	05°04.4	-12.2'	53.9'
5	254°27.1	47.7	70°55.5	16.4'	04°52.2	-12.2'	53.9'
6	269°27.2	N10°46.8	85°30.9	16.5'	S04°39.9	-12.3'	53.9'
7	284°27.4	46.0	100°06.4	16.5'	04°27.7	-12.3'	53.9'
8	299°27.6	45.1	114°41.8	16.5'	04°15.4	-12.3'	53.9'
9	314°27.8	· · 44.3	129°17.3	16.5'	04°03.1	-12.3'	53.9'
10	329°27.9	43.4	143°52.8	16.5'	03°50.8	-12.3'	53.9'
11	344°28.1	42.5	158°28.3	16.5'	03°38.5	-12.3'	53.9'
12	359°28.3	N10°41.7	173°03.8	16.5'	S03°26.1	-12.3'	53.9'
13	14°28.5	40.8	187°39.4	16.5'	03°13.8	-12.4'	53.9'
14	29°28.6	39.9	202°14.9	16.6'	03°01.4	-12.4'	53.9'
15	44°28.8	· · 39.1	216°50.5	16.6'	02°49.1	-12.4'	53.9'
16	59°29.0	38.2	231°26.0	16.6'	02°36.7	-12.4'	53.9'
17	74°29.2	37.4	246°01.6	16.6'	02°24.3	-12.4'	53.9'
18	89°29.3	N10°36.5	260°37.2	16.6'	S02°11.9	-12.4'	53.9'
19	104°29.5	35.6	275°12.7	16.6'	01°59.5	-12.4'	53.9'
20	119°29.7	34.8	289°48.3	16.6'	01°47.1	-12.4'	53.9'
21	134°29.9	· · 33.9	304°23.9	16.6'	01°34.7	-12.4'	53.9'
22	149°30.0	33.0	318°59.5	16.6'	01°22.3	-12.4'	53.9'
23	164°30.2	32.2	333°35.1	16.6'	01°09.8	-12.4'	53.9'
SD = 15.8' $d = -0.9'$			SD = 14.7'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	////	01:13	03:19	20:42	22:40	////
N 70°	////	02:08	03:39	20:22	21:51	////
68°	////	02:40	03:55	20:07	21:20	////
66°	01:03	03:03	04:08	19:54	20:58	22:51
64°	01:52	03:21	04:19	19:44	20:41	22:08
62°	02:21	03:36	04:28	19:35	20:27	21:40
60°	02:43	03:48	04:35	19:28	20:15	21:19
N 58°	03:00	03:58	04:42	19:21	20:05	21:03
56°	03:14	04:07	04:48	19:15	19:56	20:49
54°	03:26	04:15	04:54	19:10	19:48	20:37
52°	03:36	04:22	04:58	19:05	19:42	20:27
50°	03:45	04:28	05:03	19:01	19:36	20:18
45°	04:03	04:41	05:12	18:52	19:23	20:00
N 40°	04:18	04:52	05:20	18:44	19:12	19:46
35°	04:29	05:00	05:27	18:38	19:04	19:35
30°	04:38	05:08	05:32	18:32	18:57	19:26
20°	04:53	05:20	05:42	18:22	18:45	19:11
N 10°	05:05	05:30	05:51	18:14	18:35	19:00
0°	05:14	05:38	05:59	18:06	18:27	18:51
S 10°	05:21	05:46	06:07	17:58	18:19	18:44
20°	05:27	05:53	06:15	17:50	18:12	18:38
30°	05:32	06:00	06:25	17:41	18:05	18:33
35°	05:35	06:04	06:30	17:35	18:01	18:31
40°	05:37	06:08	06:36	17:29	17:57	18:29
45°	05:39	06:13	06:43	17:23	17:52	18:27
S 50°	05:41	06:18	06:51	17:14	17:47	18:25
52°	05:41	06:20	06:55	17:11	17:45	18:24
54°	05:42	06:23	06:59	17:06	17:43	18:24
56°	05:42	06:26	07:04	17:02	17:40	18:23
58°	05:43	06:29	07:09	16:57	17:37	18:23
S 60°	05:43	06:32	07:15	16:51	17:34	18:23

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	19:56	19:28	19:05	01:10	03:21	05:15
N 70°	19:37	19:18	19:02	01:44	03:38	05:23
68°	19:22	19:10	19:00	02:09	03:52	05:28
66°	19:10	19:04	18:58	02:27	04:02	05:33
64°	19:00	18:58	18:56	02:42	04:11	05:37
62°	18:51	18:53	18:55	02:55	04:19	05:41
60°	18:43	18:49	18:54	03:05	04:25	05:44
N 58°	18:37	18:45	18:53	03:15	04:31	05:47
56°	18:31	18:42	18:52	03:23	04:36	05:49
54°	18:25	18:39	18:51	03:30	04:41	05:51
52°	18:20	18:36	18:50	03:36	04:45	05:53
50°	18:16	18:33	18:49	03:42	04:49	05:55
45°	18:07	18:28	18:47	03:54	04:57	05:59
N 40°	17:59	18:23	18:46	04:04	05:03	06:02
35°	17:52	18:19	18:45	04:13	05:09	06:05
30°	17:46	18:15	18:44	04:20	05:14	06:07
20°	17:35	18:09	18:42	04:33	05:23	06:11
N 10°	17:26	18:04	18:40	04:44	05:30	06:15
0°	17:17	17:59	18:39	04:55	05:37	06:18
S 10°	17:08	17:53	18:37	05:05	05:44	06:21
20°	16:59	17:48	18:36	05:16	05:52	06:25
30°	16:48	17:41	18:34	05:29	06:00	06:29
35°	16:42	17:38	18:33	05:36	06:05	06:31
40°	16:35	17:34	18:32	05:44	06:10	06:34
45°	16:27	17:29	18:30	05:54	06:16	06:37
S 50°	16:17	17:23	18:29	06:05	06:24	06:40
52°	16:12	17:20	18:28	06:11	06:27	06:42
54°	16:07	17:17	18:27	06:17	06:31	06:44
56°	16:01	17:14	18:26	06:23	06:35	06:45
58°	15:55	17:10	18:25	06:30	06:40	06:48
S 60°	15:47	17:06	18:24	06:38	06:45	06:50

Day	Sun		Mer. Pass	Moon		Age 13-15 97-100%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
23	02:47	02:39	12:03	23:27	11:06	
24	02:31	02:23	12:02	—:—	11:48	
25	02:15	02:07	12:02	00:08	12:29	

August 26, 27, 28 UT (Thu., Fri., Sat.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	334°08.4	138°00.6	S09°23.8	138°29.1	S06°31.1	332°01.9	S00°46.1	150°07.2	N00°38.1	Alpheratz	357°45.3	29°09.1					
1	349°10.9	153°01.1	24.9	153°30.1	31.8	347°04.6	46.2	165°09.4	38.0	Ankaa	353°17.2	-42°14.6					
2	4°13.4	168°01.5	26.0	168°31.1	32.4	2°07.3	46.3	180°11.6	37.9	Schedar	349°42.5	56°35.8					
3	19°15.8	183°01.9	•• 27.2	183°32.1	•• 33.1	17°10.0	•• 46.4	195°13.9	•• 37.8	Diphda	348°57.6	-17°55.4					
4	34°18.3	198°02.4	28.3	198°33.0	33.7	32°12.7	46.5	210°16.1	37.7	Achernar	335°27.8	-57°10.6					
5	49°20.8	213°02.8	29.4	213°34.0	34.4	47°15.3	46.6	225°18.3	37.5	Hamal	328°02.9	23°30.9					
6	64°23.2	228°03.3	S09°30.6	228°35.0	S06°35.0	62°18.0	S00°46.7	240°20.5	N00°37.4	Polaris	318°37.8	89°18.4					
7	79°25.7	243°03.7	31.7	243°36.0	35.6	77°20.7	46.9	255°22.7	37.3	Acamar	315°19.7	-40°15.4					
8	94°28.1	258°04.1	32.8	258°37.0	36.3	92°23.4	47.0	270°24.9	37.2	Menkar	314°17.1	4°08.1					
9	109°30.6	273°04.6	•• 34.0	273°37.9	•• 36.9	107°26.1	•• 47.1	285°27.1	•• 37.1	Mirfak	308°43.2	49°53.8					
10	124°33.1	288°05.0	35.1	288°38.9	37.6	122°28.8	47.2	300°29.3	37.0	Aldebaran	290°51.7	16°31.9					
11	139°35.5	303°05.5	36.2	303°39.9	38.2	137°31.5	47.3	315°31.5	36.9	Rigel	281°14.1	-8°11.2					
12	154°38.0	318°05.9	S09°37.4	318°40.9	S06°38.9	152°34.2	S00°47.4	330°33.7	N00°36.7	Capella	280°37.6	46°00.4					
13	169°40.5	333°06.4	38.5	333°41.9	39.5	167°36.9	47.5	345°35.9	36.6	Bellatrix	278°34.3	6°21.7					
14	184°42.9	348°06.8	39.6	348°42.8	40.2	182°39.6	47.6	0°38.1	36.5	Elnath	278°15.3	28°36.9					
15	199°45.4	3°07.3	•• 40.8	3°43.8	•• 40.8	197°42.3	•• 47.7	15°40.4	•• 36.4	Alnilam	275°48.5	-1°11.6					
16	214°47.9	18°07.7	41.9	18°44.8	41.4	212°45.0	47.8	30°42.6	36.3	Betelgeuse	271°03.6	7°24.6					
17	229°50.3	33°08.2	43.0	33°45.8	42.1	227°47.7	47.9	45°44.8	36.2	Canopus	263°57.3	-52°41.8					
18	244°52.8	48°08.6	S09°44.2	48°46.7	S06°42.7	242°50.4	S00°48.1	60°47.0	N00°36.1	Sirius	258°35.7	-16°43.7					
19	259°55.3	63°09.1	45.3	63°47.7	43.4	257°53.1	48.2	75°49.2	35.9	Adhara	255°14.4	-28°59.0					
20	274°57.7	78°09.5	46.4	78°48.7	44.0	272°55.8	48.3	90°51.4	35.8	Procyon	245°02.1	5°11.9					
21	290°00.2	93°10.0	•• 47.6	93°49.7	•• 44.7	287°58.5	•• 48.4	105°53.6	•• 35.7	Pollux	243°30.5	28°00.0					
22	305°02.6	108°10.4	48.7	108°50.7	45.3	303°01.2	48.5	120°55.8	35.6	Avior	234°19.5	-59°32.5					
23	320°05.1	123°10.9	49.8	123°51.6	45.9	318°03.8	48.6	135°58.0	35.5	Suhail	222°54.4	-43°28.5					
Mer.pass. 01:43			ν 0.4' d 1.1' m -4.52			ν 1.0' d 0.6' m 1.52			ν 2.7' d 0.1' m -2.86			ν 2.2' d -0.1' m 0.99					

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	335°07.6	138°11.3	S09°50.9	138°52.6	S06°46.6	333°06.5	S00°48.7	151°00.2	N00°35.4					
1	350°10.0	153°11.8	52.1	153°53.6	47.2	348°09.2	48.8	166°02.4	35.3					
2	5°12.5	168°12.2	53.2	168°54.6	47.9	3°11.9	48.9	181°04.6	35.1					
3	20°15.0	183°12.7	•• 54.3	183°55.5	•• 48.5	18°14.6	•• 49.0	196°06.8	•• 35.0					
4	35°17.4	198°13.1	55.4	198°56.5	49.2	33°17.3	49.2	211°09.1	34.9					
5	50°19.9	213°13.6	56.6	213°57.5	49.8	48°20.0	49.3	226°11.3	34.8					
6	65°22.4	228°14.1	S09°57.7	228°58.5	S06°50.4	63°22.7	S00°49.4	241°13.5	N00°34.7					
7	80°24.8	243°14.5	09°58.8	243°59.4	51.1	78°25.4	49.5	256°15.7	34.6					
8	95°27.3	258°15.0	10°00.0	259°00.4	51.7	93°28.1	49.6	271°17.9	34.5					
9	110°29.8	273°15.4	•• 01.1	274°01.4	•• 52.4	108°30.8	•• 49.7	286°20.1	•• 34.3					
10	125°32.2	288°15.9	02.2	289°02.4	53.0	123°33.5	49.8	301°22.3	34.2					
11	140°34.7	303°16.4	03.3	304°03.3	53.7	138°36.2	49.9	316°24.5	34.1					
12	155°37.1	318°16.8	S10°04.4	319°04.3	S06°54.3	153°38.9	S00°50.0	331°26.7	N00°34.0					
13	170°39.6	333°17.3	05.6	334°05.3	54.9	168°41.6	50.1	346°28.9	33.9					
14	185°42.1	348°17.8	06.7	349°06.2	55.6	183°44.3	50.3	1°31.1	33.8					
15	200°44.5	3°18.2	•• 07.8	4°07.2	•• 56.2	198°47.0	•• 50.4	16°33.3	•• 33.7					
16	215°47.0	18°18.7	08.9	19°08.2	56.9	213°49.7	50.5	31°35.5	33.5					
17	230°49.5	33°19.2	10.1	34°09.2	57.5	228°52.4	50.6	46°37.7	33.4					
18	245°51.9	48°19.6	S10°11.2	49°10.1	S06°58.2	243°55.1	S00°50.7	61°40.0	N00°33.3					
19	260°54.4	63°20.1	12.3	64°11.1	58.8	258°57.8	50.8	76°42.2	33.2					
20	275°56.9	78°20.6	13.4	79°12.1	06°59.4	274°00.5	50.9	91°44.4	33.1					
21	290°59.3	93°21.0	•• 14.5	94°13.1	07°00.1	289°03.2	•• 51.0	106°46.6	•• 33.0					
22	306°01.8	108°21.5	15.7	109°14.0	00.7	304°05.9	51.1	121°48.8	32.8					
23	321°04.2	123°22.0	16.8	124°15.0	01.4	319°08.6	51.3	136°51.0	32.7					
Mer.pass. 01:39			ν 0.5' d 1.1' m -4.53			ν 1.0' d 0.6' m 1.52			ν 2.7' d 0.1' m -2.87			ν 2.2' d -0.1' m 0.99		

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	336°06.7	138°22.4	S10°17.9	139°16.0	S07°02.0	334°11.3	S00°51.4	151°53.2	N00°32.6					
1	351°09.2	153°22.9	19.0	154°16.9	02.7	349°14.0	51.5	166°55.4	32.5					
2	6°11.6	168°23.4	20.1	169°17.9	03.3	4°16.7	51.6	181°57.6	32.4					
3	21°14.1	183°23.9	•• 21.2	184°18.9	•• 03.9	19°19.4	•• 51.7	196°59.8	•• 32.3					
4	36°16.6	198°24.3	22.4	199°19.9	04.6	34°22.1	51.8	212°02.0	32.2					
5	51°19.0	213°24.8	23.5	214°20.8	05.2	49°24.8	51.9	227°04.2	32.0					
6	66°21.5	228°25.3	S10°24.6	229°21.8	S07°05.9	64°27.5	S00°52.0	242°06.4	N00°31.9					
7	81°24.0	243°25.8	25.7	244°22.8	06.5	79°30.2	52.2	257°08.6	31.8					
8	96°26.4	258°26.2	26.8	259°23.7	07.2	94°32.9	52.3	272°10.8	31.7					
9	111°28.9	273°26.7	•• 27.9	274°24.7	•• 07.8	109°35.7	•• 52.4	287°13.0	•• 31.6					
10	126°31.4	288°27.2	29.0	289°25.7	08.4	124°38.4	52.5	302°15.2	31.5					
11	141°33.8	303°27.7	30.2	304°26.6	09.1	139°41.1	52.6	317°17.5	31.4					
12	156°36.3	318°28.2	S10°31.3	319°27.6	S07°09.7	154°43.8	S00°52.7	332°19.7	N00°31.2					
13	171°38.7	333°28.6	32.4	334°28.6	10.4	169°46.5	52.8	347°21.9	31.1					
14	186°41.2	348°29.1	33.5	349°29.5	11.0	184°49.2	52.9	2°24.1	31.0					
15	201°43.7	3°29.6	•• 34.6	4°30.5	•• 11.6	199°51.9	•• 53.1	17°26.3	•• 30.9					
16	216°46.1	18°30.1	35.7	19°31.5	12.3	214°54.6	53.2	32°28.5	30.8					
17	231°48.6	33°30.6	36.8	34°32.5	12.9	229°57.3	53.3	47°30.7	30.7					
18	246°51.1	48°31.1	S10°37.9	49°33.4	S07°13.6	245°00.0	S00°53.4	62°32.9	N00°30.5					
19	261°53.5	63°31.5	39.1	64°34.4	14.2	260°02.7	53.5	77°35.1	30.4					
20	276°56.0	78°32.0	40.2	79°35.4	14.9	275°05.4	53.6	92°37.3	30.3					
21	291°58.5	93°32.5	•• 41.3	94°36.3	•• 15.5	290°08.1	•• 53.7	107°39.5	•• 30.2					
22	307°00.9	108°33.0	42.4	109°37.3	16.1	305°10.8	53.9	122°41.7	30.1					
23	322°03.4	123°33.5	43.5	124°38.3	16.8	320°13.5	54.0	137°43.9	30.0					
Mer.pass. 01:35			ν 0.5' d 1.1' m -4.54			ν 1.0' d 0.6' m 1.52			ν 2.7' d 0.1' m -2.87			ν 2.2' d -0.1' m 0.99		

Aug 26 Thu	SHA	Mer.pass
Venus	163°52.2	14:48
Mars	164°20.7	14:45
Jupiter	357°53.5	01:52
Saturn	175°58.8	13:57

Aug 27 Fri	SHA	Mer.pass
Venus	163°03.8	14:47
Mars	163°45.0	14:44
Jupiter	357°59.0	01:47
Saturn	175°52.7	13:54

Aug 28 Sat	SHA	Mer.pass
Venus	162°15.7	14:46
Mars	163°09.3	14:42
Jupiter	358°04.6	01:43
Saturn	175°46.5	13:50

Horizontal parallax		
Venus:		

h	Sun			Moon				
Thu	GHA	Dec		GHA	ν	Dec	d	HP
0	179°30.4	N10°31.3		348°10.7	16.6'	S00°57.4	-12.4'	53.9'
1	194°30.6	30.4		2°46.3	16.6'	00°45.0	-12.4'	53.9'
2	209°30.7	29.6		17°21.9	16.6'	00°32.5	-12.4'	53.9'
3	224°30.9	· · 28.7		31°57.5	16.6'	00°20.1	-12.4'	53.9'
4	239°31.1	27.8		46°33.1	16.6'	S00°07.6	-12.4'	53.9'
5	254°31.3	27.0		61°08.7	16.6'	N00°04.8	12.4'	53.9'
6	269°31.4	N10°26.1		75°44.3	16.6'	N00°17.3	12.4'	53.9'
7	284°31.6	25.2		90°19.9	16.6'	00°29.7	12.4'	53.9'
8	299°31.8	24.4		104°55.5	16.6'	00°42.2	12.4'	53.9'
9	314°32.0	· · 23.5		119°31.1	16.6'	00°54.6	12.4'	53.9'
10	329°32.1	22.6		134°06.7	16.6'	01°07.1	12.4'	54.0'
11	344°32.3	21.7		148°42.2	16.6'	01°19.5	12.4'	54.0'
12	359°32.5	N10°20.9		163°17.8	16.6'	N01°31.9	12.4'	54.0'
13	14°32.7	20.0		177°53.4	16.5'	01°44.4	12.4'	54.0'
14	29°32.9	19.1		192°28.9	16.5'	01°56.8	12.4'	54.0'
15	44°33.0	· · 18.3		207°04.4	16.5'	02°09.2	12.4'	54.0'
16	59°33.2	17.4		221°40.0	16.5'	02°21.6	12.4'	54.0'
17	74°33.4	16.5		236°15.5	16.5'	02°34.0	12.4'	54.0'
18	89°33.6	N10°15.7		250°51.0	16.5'	N02°46.4	12.4'	54.0'
19	104°33.7	14.8		265°26.4	16.5'	02°58.8	12.4'	54.0'
20	119°33.9	13.9		280°01.9	16.5'	03°11.2	12.4'	54.0'
21	134°34.1	· · 13.0		294°37.4	16.4'	03°23.6	12.4'	54.0'
22	149°34.3	12.2		309°12.8	16.4'	03°35.9	12.4'	54.0'
23	164°34.5	11.3		323°48.2	16.4'	03°48.3	12.3'	54.0'
	SD = 15.8'	d = -0.9'		SD = 14.7'				
Fri	GHA	Dec		GHA	ν	Dec	d	HP
0	179°34.6	N10°10.4		338°23.6	16.4'	N04°00.6	12.3'	54.0'
1	194°34.8	09.5		352°59.0	16.4'	04°13.0	12.3'	54.0'
2	209°35.0	08.7		7°34.4	16.3'	04°25.3	12.3'	54.0'
3	224°35.2	· · 07.8		22°09.7	16.3'	04°37.6	12.3'	54.1'
4	239°35.4	06.9		36°45.0	16.3'	04°49.9	12.3'	54.1'
5	254°35.5	06.0		51°20.3	16.3'	05°02.1	12.3'	54.1'
6	269°35.7	N10°05.2		65°55.6	16.3'	N05°14.4	12.2'	54.1'
7	284°35.9	04.3		80°30.9	16.2'	05°26.6	12.2'	54.1'
8	299°36.1	03.4		95°06.1	16.2'	05°38.8	12.2'	54.1'
9	314°36.3	· · 02.5		109°41.3	16.2'	05°51.0	12.2'	54.1'
10	329°36.4	01.7		124°16.5	16.1'	06°03.2	12.2'	54.1'
11	344°36.6	10°00.8		138°51.6	16.1'	06°15.4	12.1'	54.1'
12	359°36.8	N09°59.9		153°26.7	16.1'	N06°27.5	12.1'	54.1'
13	14°37.0	59.0		168°01.8	16.1'	06°39.6	12.1'	54.1'
14	29°37.2	58.2		182°36.9	16.0'	06°51.7	12.1'	54.2'
15	44°37.4	· · 57.3		197°11.9	16.0'	07°03.8	12.0'	54.2'
16	59°37.5	56.4		211°46.9	16.0'	07°15.8	12.0'	54.2'
17	74°37.7	55.5		226°21.9	15.9'	07°27.8	12.0'	54.2'
18	89°37.9	N09°54.6		240°56.8	15.9'	N07°39.8	12.0'	54.2'
19	104°38.1	53.8		255°31.7	15.9'	07°51.8	11.9'	54.2'
20	119°38.3	52.9		270°06.5	15.8'	08°03.7	11.9'	54.2'
21	134°38.4	· · 52.0		284°41.4	15.8'	08°15.7	11.9'	54.2'
22	149°38.6	51.1		299°16.2	15.8'	08°27.6	11.9'	54.2'
23	164°38.8	50.3		313°50.9	15.7'	08°39.4	11.8'	54.2'
	SD = 15.8'	d = -0.9'		SD = 14.7'				
Sat	GHA	Dec		GHA	ν	Dec	d	HP
0	179°39.0	N09°49.4		328°25.6	15.7'	N08°51.2	11.8'	54.3'
1	194°39.2	48.5		343°00.3	15.6'	09°03.0	11.8'	54.3'
2	209°39.4	47.6		357°34.9	15.6'	09°14.8	11.7'	54.3'
3	224°39.5	· · 46.7		12°09.5	15.6'	09°26.6	11.7'	54.3'
4	239°39.7	45.8		26°44.1	15.5'	09°38.3	11.7'	54.3'
5	254°39.9	45.0		41°18.6	15.5'	09°49.9	11.6'	54.3'
6	269°40.1	N09°44.1		55°53.0	15.4'	N10°01.6	11.6'	54.3'
7	284°40.3	43.2		70°27.5	15.4'	10°13.2	11.6'	54.3'
8	299°40.5	42.3		85°01.8	15.3'	10°24.7	11.5'	54.4'
9	314°40.7	· · 41.4		99°36.2	15.3'	10°36.3	11.5'	54.4'
10	329°40.8	40.6		114°10.5	15.2'	10°47.8	11.5'	54.4'
11	344°41.0	39.7		128°44.7	15.2'	10°59.2	11.4'	54.4'
12	359°41.2	N09°38.8		143°18.9	15.1'	N11°10.6	11.4'	54.4'
13	14°41.4	37.9		157°53.0	15.1'	11°22.0	11.3'	54.4'
14	29°41.6	37.0		172°27.1	15.0'	11°33.3	11.3'	54.4'
15	44°41.8	· · 36.1		187°01.1	15.0'	11°44.6	11.3'	54.4'
16	59°42.0	35.3		201°35.1	14.9'	11°55.9	11.2'	54.5'
17	74°42.1	34.4		216°09.1	14.9'	12°07.1	11.2'	54.5'
18	89°42.3	N09°33.5		230°42.9	14.8'	N12°18.3	11.1'	54.5'
19	104°42.5	32.6		245°16.8	14.8'	12°29.4	11.1'	54.5'
20	119°42.7	31.7		259°50.5	14.7'	12°40.4	11.0'	54.5'
21	134°42.9	· · 30.8		274°24.2	14.7'	12°51.5	11.0'	54.5'
22	149°43.1	29.9		288°57.9	14.6'	13°02.5	10.9'	54.6'
23	164°43.3	29.1		303°31.5	14.5'	13°13.4	10.9'	54.6'
	SD = 15.8'	d = -0.9'		SD = 14.8'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	////	01:48	03:35	20:24	22:07	////
N 70°	////	02:29	03:53	20:07	21:29	////
68°	////	02:56	04:07	19:54	21:04	23:32
66°	01:33	03:16	04:18	19:43	20:44	22:23
64°	02:09	03:32	04:28	19:34	20:29	21:49
62°	02:34	03:45	04:36	19:26	20:16	21:25
60°	02:53	03:56	04:43	19:19	20:05	21:07
N 58°	03:09	04:05	04:49	19:13	19:56	20:52
56°	03:22	04:14	04:54	19:08	19:48	20:39
54°	03:33	04:21	04:59	19:03	19:41	20:29
52°	03:42	04:27	05:03	18:59	19:35	20:19
50°	03:51	04:33	05:07	18:55	19:29	20:11
45°	04:08	04:45	05:16	18:47	19:17	19:54
N 40°	04:21	04:55	05:23	18:40	19:08	19:41
35°	04:32	05:03	05:29	18:34	19:00	19:31
30°	04:40	05:10	05:34	18:29	18:53	19:22
20°	04:54	05:21	05:43	18:20	18:42	19:08
N 10°	05:05	05:30	05:51	18:12	18:33	18:58
0°	05:13	05:37	05:58	18:05	18:26	18:50
S 10°	05:19	05:44	06:05	17:58	18:19	18:44
20°	05:25	05:51	06:13	17:50	18:13	18:39
30°	05:29	05:57	06:21	17:42	18:06	18:34
35°	05:31	06:00	06:26	17:37	18:03	18:33
40°	05:33	06:04	06:31	17:32	18:00	18:31
45°	05:34	06:08	06:38	17:26	17:56	18:30
S 50°	05:35	06:12	06:45	17:19	17:52	18:29
52°	05:35	06:14	06:49	17:15	17:50	18:29
54°	05:35	06:16	06:52	17:12	17:48	18:29
56°	05:36	06:19	06:56	17:08	17:46	18:29
58°	05:36	06:21	07:01	17:03	17:43	18:29
S 60°	05:35	06:24	07:06	16:58	17:41	18:29

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	18:43	18:19	17:50	07:04	08:53	10:49
N 70°	18:47	18:30	18:11	07:03	08:45	10:31
68°	18:50	18:39	18:27	07:03	08:38	10:16
66°	18:52	18:47	18:41	07:02	08:32	10:04
64°	18:55	18:53	18:52	07:02	08:27	09:54
62°	18:56	18:58	19:01	07:02	08:23	09:46
60°	18:58	19:03	19:10	07:01	08:19	09:39
N 58°	19:00	19:08	19:17	07:01	08:16	09:32
56°	19:01	19:11	19:23	07:01	08:13	09:27
54°	19:02	19:15	19:29	07:01	08:11	09:22
52°	19:04	19:18	19:34	07:01	08:08	09:17
50°	19:05	19:21	19:39	07:01	08:06	09:13
45°	19:07	19:27	19:49	07:00	08:02	09:04
N 40°	19:09	19:32	19:58	07:00	07:58	08:57
35°	19:10	19:37	20:05	07:00	07:55	08:51
30°	19:12	19:41	20:11	06:59	07:52	08:45
20°	19:15	19:48	20:23	06:59	07:47	08:36
N 10°	19:17	19:54	20:33	06:59	07:43	08:28
0°	19:19	20:00	20:42	06:58	07:39	08:20
S 10°	19:21	20:06	20:52	06:58	07:35	08:12
20°	19:24	20:12	21:02	06:58	07:30	08:04
30°	19:26	20:19	21:14	06:57	07:25	07:55
35°	19:28	20:24	21:20	06:57	07:23	07:50
40°	19:30	20:28	21:28	06:57	07:19	07:44
45°	19:32	20:34	21:37	06:56	07:16	07:37
S 50°	19:35	20:41	21:48	06:56	07:11	07:28
52°	19:36	20:44	21:53	06:56	07:09	07:24
54°	19:37	20:47	21:59	06:55	07:07	07:20
56°	19:38	20:51	22:05	06:55	07:05	07:16
58°	19:40	20:55	22:13	06:55	07:02	07:10
S 60°	19:42	21:00	22:21	06:55	06:59	07:05

August 29, 30, 31 UT (Sun., Mon., Tue.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	337°05.8	138°34.0	S10°44.6	139°39.2	S07°17.4	335°16.2	S00°54.1	152°46.1	N00°29.9					Alpheratz	357°45.3	29°09.1	
1	352°08.3	153°34.5	45.7	154°40.2	18.1	350°18.9	54.2	167°48.3	29.7					Ankaa	353°17.2	-42°14.6	
2	7°10.8	168°35.0	46.8	169°41.2	18.7	5°21.6	54.3	182°50.5	29.6					Schedar	349°42.5	56°35.8	
3	22°13.2	183°35.5	· · 47.9	184°42.1	· · 19.3	20°24.4	· · 54.4	197°52.7	· · 29.5					Diphda	348°57.6	-17°55.4	
4	37°15.7	198°36.0	49.0	199°43.1	20.0	35°27.1	54.5	212°54.9	29.4					Achernar	335°27.7	-57°10.6	
5	52°18.2	213°36.4	50.1	214°44.1	20.6	50°29.8	54.7	227°57.1	29.3					Hamal	328°02.9	23°30.9	
6	67°20.6	228°36.9	S10°51.2	229°45.0	S07°21.3	65°32.5	S00°54.8	242°59.3	N00°29.2					Polaris	318°36.4	89°18.4	
7	82°23.1	243°37.4	52.3	244°46.0	21.9	80°35.2	54.9	258°01.5	29.0					Acamar	315°19.7	-40°15.4	
8	97°25.6	258°37.9	53.4	259°47.0	22.6	95°37.9	55.0	273°03.8	28.9					Menkar	314°17.1	4°08.1	
9	112°28.0	273°38.4	· · 54.5	274°47.9	· · 23.2	110°40.6	· · 55.1	288°06.0	· · 28.8					Mirfak	308°43.2	49°53.8	
10	127°30.5	288°38.9	55.6	289°48.9	23.8	125°43.3	55.2	303°08.2	28.7					Aldebaran	290°51.7	16°31.9	
11	142°33.0	303°39.4	56.7	304°49.8	24.5	140°46.0	55.3	318°10.4	28.6					Rigel	281°14.1	-8°11.2	
12	157°35.4	318°39.9	S10°57.9	319°50.8	S07°25.1	155°48.7	S00°55.5	333°12.6	N00°28.5					Capella	280°37.6	46°00.4	
13	172°37.9	333°40.4	10°59.0	334°51.8	25.8	170°51.4	55.6	348°14.8	28.4					Bellatrix	278°34.3	6°21.7	
14	187°40.3	348°40.9	11°00.1	349°52.7	26.4	185°54.1	55.7	3°17.0	28.2					Elnath	278°15.3	28°36.9	
15	202°42.8	3°41.4	· · 01.2	4°53.7	· · 27.0	200°56.9	· · 55.8	18°19.2	· · 28.1					Alnilam	275°48.5	-1°11.6	
16	217°45.3	18°41.9	02.3	19°54.7	27.7	215°59.6	55.9	33°21.4	28.0					Betelgeuse	271°03.6	7°24.6	
17	232°47.7	33°42.4	03.4	34°55.6	28.3	231°02.3	56.0	48°23.6	27.9					Canopus	263°57.3	-52°41.8	
18	247°50.2	48°42.9	S11°04.5	49°56.6	S07°29.0	246°05.0	S00°56.1	63°25.8	N00°27.8					Sirius	258°35.7	-16°43.7	
19	262°52.7	63°43.4	05.6	64°57.6	29.6	261°07.7	56.3	78°28.0	27.7					Adhara	255°14.4	-28°59.0	
20	277°55.1	78°43.9	06.7	79°58.5	30.2	276°10.4	56.4	93°30.2	27.5					Procyon	245°02.1	5°11.9	
21	292°57.6	93°44.5	· · 07.8	94°59.5	· · 30.9	291°13.1	· · 56.5	108°32.4	· · 27.4					Pollux	243°30.4	28°00.0	
22	308°00.1	108°45.0	08.9	110°00.4	31.5	306°15.8	56.6	123°34.6	27.3					Avior	234°19.5	-59°32.5	
23	323°02.5	123°45.5	09.9	125°01.4	32.2	321°18.5	56.7	138°36.8	27.2					Suhail	222°54.4	-43°28.5	
Mer.pass. 01:31			ν 0.5' d 1.1' m -4.55	ν 1.0' d 0.6' m 1.52	ν 2.7' d 0.1' m -2.88	ν 2.2' d -0.1' m 0.99											

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	338°05.0	138°46.0	S11°11.0	140°02.4	S07°32.8	336°21.3	S00°56.8	153°39.0	N00°27.1
1	353°07.4	153°46.5	12.1	155°03.3	33.4	351°24.0	57.0	168°41.2	27.0
2	8°09.9	168°47.0	13.2	170°04.3	34.1	6°26.7	57.1	183°43.4	26.8
3	23°12.4	183°47.5	· · 14.3	185°05.3	· · 34.7	21°29.4	· · 57.2	198°45.6	· · 26.7
4	38°14.8	198°48.0	15.4	200°06.2	35.4	36°32.1	57.3	213°47.8	26.6
5	53°17.3	213°48.5	16.5	215°07.2	36.0	51°34.8	57.4	228°50.0	26.5
6	68°19.8	228°49.0	S11°17.6	230°08.1	S07°36.7	66°37.5	S00°57.5	243°52.2	N00°26.4
7	83°22.2	243°49.6	18.7	245°09.1	37.3	81°40.2	57.6	258°54.4	26.3
8	98°24.7	258°50.1	19.8	260°10.1	37.9	96°43.0	57.8	273°56.6	26.1
9	113°27.2	273°50.6	· · 20.9	275°11.0	· · 38.6	111°45.7	· · 57.9	288°58.8	· · 26.0
10	128°29.6	288°51.1	22.0	290°12.0	39.2	126°48.4	58.0	304°01.0	25.9
11	143°32.1	303°51.6	23.1	305°12.9	39.9	141°51.1	58.1	319°03.2	25.8
12	158°34.6	318°52.1	S11°24.2	320°13.9	S07°40.5	156°53.8	S00°58.2	334°05.4	N00°25.7
13	173°37.0	333°52.7	25.3	335°14.9	41.1	171°56.5	58.3	349°07.6	25.6
14	188°39.5	348°53.2	26.4	350°15.8	41.8	186°59.2	58.5	4°09.9	25.5
15	203°41.9	3°53.7	· · 27.4	5°16.8	· · 42.4	202°02.0	· · 58.6	19°12.1	· · 25.3
16	218°44.4	18°54.2	28.5	20°17.7	43.1	217°04.7	58.7	34°14.3	25.2
17	233°46.9	33°54.7	29.6	35°18.7	43.7	232°07.4	58.8	49°16.5	25.1
18	248°49.3	48°55.3	S11°30.7	50°19.7	S07°44.3	247°10.1	S00°58.9	64°18.7	N00°25.0
19	263°51.8	63°55.8	31.8	65°20.6	45.0	262°12.8	59.0	79°20.9	24.9
20	278°54.3	78°56.3	32.9	80°21.6	45.6	277°15.5	59.2	94°23.1	24.8
21	293°56.7	93°56.8	· · 34.0	95°22.5	· · 46.2	292°18.2	· · 59.3	109°25.3	· · 24.6
22	308°59.2	108°57.4	35.1	110°23.5	46.9	307°21.0	59.4	124°27.5	24.5
23	324°01.7	123°57.9	36.1	125°24.4	47.5	322°23.7	59.5	139°29.7	24.4
Mer.pass. 01:27			ν 0.5' d 1.1' m -4.56	ν 1.0' d 0.6' m 1.52	ν 2.7' d 0.1' m -2.88	ν 2.2' d -0.1' m 0.98			

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	339°04.1	138°58.4	S11°37.2	140°25.4	S07°48.2	337°26.4	S00°59.6	154°31.9	N00°24.3
1	354°06.6	153°58.9	38.3	155°26.4	48.8	352°29.1	59.8	169°34.1	24.2
2	9°09.1	168°59.5	39.4	170°27.3	49.4	7°31.8	00°59.9	184°36.3	24.1
3	24°11.5	184°00.0	· · 40.5	185°28.3	· · 50.1	22°34.5	01°00.0	199°38.5	· · 23.9
4	39°14.0	199°00.5	41.6	200°29.2	50.7	37°37.3	00.1	214°40.7	23.8
5	54°16.4	214°01.1	42.7	215°30.2	51.4	52°40.0	00.2	229°42.9	23.7
6	69°18.9	229°01.6	S11°43.7	230°31.1	S07°52.0	67°42.7	S01°00.3	244°45.1	N00°23.6
7	84°21.4	244°02.1	44.8	245°32.1	52.6	82°45.4	00.5	259°47.3	23.5
8	99°23.8	259°02.7	45.9	260°33.1	53.3	97°48.1	00.6	274°49.5	23.4
9	114°26.3	274°03.2	· · 47.0	275°34.0	· · 53.9	112°50.9	· · 00.7	289°51.7	· · 23.2
10	129°28.8	289°03.7	48.1	290°35.0	54.6	127°53.6	00.8	304°53.9	23.1
11	144°31.2	304°04.3	49.1	305°35.9	55.2	142°56.3	00.9	319°56.1	23.0
12	159°33.7	319°04.8	S11°50.2	320°36.9	S07°55.8	157°59.0	S01°01.0	334°58.3	N00°22.9
13	174°36.2	334°05.4	51.3	335°37.8	56.5	173°01.7	01.2	350°00.5	22.8
14	189°38.6	349°05.9	52.4	350°38.8	57.1	188°04.4	01.3	5°02.7	22.7
15	204°41.1	4°06.4	· · 53.5	5°39.7	· · 57.8	203°07.2	· · 01.4	20°04.9	· · 22.5
16	219°43.5	19°07.0	54.5	20°40.7	58.4	218°09.9	01.5	35°07.1	22.4
17	234°46.0	34°07.5	55.6	35°41.7	59.0	233°12.6	01.6	50°09.3	22.3
18	249°48.5	49°08.1	S11°56.7	50°42.6	S07°59.7	248°15.3	S01°01.8	65°11.5	N00°22.2
19	264°50.9	64°08.6	57.8	65°43.6	08°00.3	263°18.0	01.9	80°13.7	22.1
20	279°53.4	79°09.1	58.8	80°44.5	00.9	278°20.8	02.0	95°15.9	22.0
21	294°55.9	94°09.7	11°59.9	95°45.5	· · 01.6	293°23.5	· · 02.1	110°18.1	· · 21.8
22	309°58.3	109°10.2	12°01.0	110°46.4	02.2	308°26.2	02.2	125°20.3	21.7
23	325°00.8	124°10.8	02.1	125°47.4	02.9	323°28.9	02.4	140°22.5	21.6
Mer.pass. 01:23			ν 0.5' d 1.1' m -4.57	ν 1.0' d 0.6' m 1.52	ν 2.7' d 0.1' m -2.89	ν 2.2' d -0.1' m 0.98			

Aug 29 Sun	SHA	Mer.pass
Venus	161°28.1	14:45
Mars	162°33.4	14:40
Jupiter	358°10.4	01:39
Saturn	175°40.3	13:47

Aug 30 Mon	SHA	Mer.pass
Venus	160°41.0	14:44
Mars	161°57.4	14:39
Jupiter	358°16.3	01:34
Saturn	175°34.0	13:43

Aug 31 Tue	SHA	Mer.pass
Venus	159°54.3	14:44
Mars	161°21.3	14:37
Jupiter	358°22.3	01:30
Saturn	175°27.7	13:40

Horizontal parallax		
Venus:	0.2	
Mars:	0.1	

Sun							Moon						
h	GHA	Dec	GHA	ν	Dec	d	HP						
0	179°43.4	N09°28.2	318°05.1	14.5'	N13°24.3	10.8'	54.6'						
1	194°43.6	27.3	332°38.5	14.4'	13°35.1	10.8'	54.6'						
2	209°43.8	26.4	347°12.0	14.4'	13°45.9	10.7'	54.6'						
3	224°44.0	· · 25.5	1°45.3	14.3'	13°56.6	10.7'	54.6'						
4	239°44.2	24.6	16°18.6	14.2'	14°07.3	10.6'	54.7'						
5	254°44.4	23.7	30°51.9	14.2'	14°17.9	10.6'	54.7'						
6	269°44.6	N09°22.8	45°25.1	14.1'	N14°28.5	10.5'	54.7'						
7	284°44.8	22.0	59°58.2	14.1'	14°39.0	10.5'	54.7'						
8	299°44.9	21.1	74°31.2	14.0'	14°49.5	10.4'	54.7'						
9	314°45.1	· · 20.2	89°04.2	13.9'	14°59.9	10.3'	54.7'						
10	329°45.3	19.3	103°37.2	13.9'	15°10.2	10.3'	54.8'						
11	344°45.5	18.4	118°10.0	13.8'	15°20.5	10.2'	54.8'						
12	359°45.7	N09°17.5	132°42.8	13.7'	N15°30.8	10.2'	54.8'						
13	14°45.9	16.6	147°15.5	13.7'	15°40.9	10.1'	54.8'						
14	29°46.1	15.7	161°48.2	13.6'	15°51.0	10.0'	54.8'						
15	44°46.3	· · 14.8	176°20.8	13.5'	16°01.1	10.0'	54.9'						
16	59°46.5	14.0	190°53.3	13.5'	16°11.1	9.9'	54.9'						
17	74°46.6	13.1	205°25.8	13.4'	16°21.0	9.9'	54.9'						
18	89°46.8	N09°12.2	219°58.2	13.3'	N16°30.9	9.8'	54.9'						
19	104°47.0	11.3	234°30.5	13.2'	16°40.6	9.7'	54.9'						
20	119°47.2	10.4	249°02.7	13.2'	16°50.4	9.7'	55.0'						
21	134°47.4	· · 09.5	263°34.9	13.1'	17°00.0	9.6'	55.0'						
22	149°47.6	08.6	278°07.0	13.0'	17°09.6	9.5'	55.0'						
23	164°47.8	07.7	292°39.0	12.9'	17°19.1	9.4'	55.0'						
SD = 15.8' $d = -0.9'$							SD = 14.9'						
Mon	GHA	Dec	GHA	ν	Dec	d	HP						
0	179°48.0	N09°06.8	307°10.9	12.9'	N17°28.6	9.4'	55.0'						
1	194°48.2	05.9	321°42.8	12.8'	17°38.0	9.3'	55.1'						
2	209°48.4	05.0	336°14.6	12.7'	17°47.3	9.2'	55.1'						
3	224°48.5	· · 04.1	350°46.3	12.6'	17°56.5	9.2'	55.1'						
4	239°48.7	03.2	5°18.0	12.6'	18°05.7	9.1'	55.1'						
5	254°48.9	02.4	19°49.6	12.5'	18°14.7	9.0'	55.2'						
6	269°49.1	N09°01.5	34°21.1	12.4'	N18°23.7	8.9'	55.2'						
7	284°49.3	09°00.6	48°52.5	12.3'	18°32.7	8.9'	55.2'						
8	299°49.5	08°59.7	63°23.8	12.3'	18°41.5	8.8'	55.2'						
9	314°49.7	· · 58.8	77°55.1	12.2'	18°50.3	8.7'	55.2'						
10	329°49.9	57.9	92°26.3	12.1'	18°59.0	8.6'	55.3'						
11	344°50.1	57.0	106°57.4	12.0'	19°07.6	8.5'	55.3'						
12	359°50.3	N08°56.1	121°28.4	11.9'	N19°16.1	8.4'	55.3'						
13	14°50.5	55.2	135°59.3	11.9'	19°24.6	8.4'	55.3'						
14	29°50.7	54.3	150°30.2	11.8'	19°32.9	8.3'	55.4'						
15	44°50.8	· · 53.4	165°01.0	11.7'	19°41.2	8.2'	55.4'						
16	59°51.0	52.5	179°31.7	11.6'	19°49.4	8.1'	55.4'						
17	74°51.2	51.6	194°02.3	11.5'	19°57.5	8.0'	55.4'						
18	89°51.4	N08°50.7	208°32.8	11.5'	N20°05.5	7.9'	55.5'						
19	104°51.6	49.8	223°03.3	11.4'	20°13.4	7.8'	55.5'						
20	119°51.8	48.9	237°33.7	11.3'	20°21.3	7.7'	55.5'						
21	134°52.0	· · 48.0	252°04.0	11.2'	20°29.0	7.7'	55.5'						
22	149°52.2	47.1	266°34.2	11.1'	20°36.7	7.6'	55.6'						
23	164°52.4	46.2	281°04.3	11.0'	20°44.2	7.5'	55.6'						
SD = 15.8' $d = -0.9'$							SD = 15.0'						
Tue	GHA	Dec	GHA	ν	Dec	d	HP						
0	179°52.6	N08°45.3	295°34.4	11.0'	N20°51.7	7.4'	55.6'						
1	194°52.8	44.4	310°04.3	10.9'	20°59.1	7.3'	55.6'						
2	209°53.0	43.5	324°34.2	10.8'	21°06.3	7.2'	55.7'						
3	224°53.2	· · 42.6	339°04.0	10.7'	21°13.5	7.1'	55.7'						
4	239°53.4	41.7	353°33.7	10.6'	21°20.6	7.0'	55.7'						
5	254°53.6	40.8	8°03.4	10.5'	21°27.6	6.9'	55.8'						
6	269°53.8	N08°39.9	22°32.9	10.5'	N21°34.4	6.8'	55.8'						
7	284°53.9	39.0	37°02.4	10.4'	21°41.2	6.7'	55.8'						
8	299°54.1	38.1	51°31.7	10.3'	21°47.9	6.6'	55.8'						
9	314°54.3	· · 37.2	66°01.0	10.2'	21°54.5	6.5'	55.9'						
10	329°54.5	36.3	80°30.3	10.1'	22°00.9	6.4'	55.9'						
11	344°54.7	35.4	94°59.4	10.0'	22°07.3	6.3'	55.9'						
12	359°54.9	N08°34.5	109°28.4	10.0'	N22°13.5	6.1'	56.0'						
13	14°55.1	33.6	123°57.4	9.9'	22°19.7	6.0'	56.0'						
14	29°55.3	32.7	138°26.3	9.8'	22°25.7	5.9'	56.0'						
15	44°55.5	· · 31.8	152°55.1	9.7'	22°31.6	5.8'	56.0'						
16	59°55.7	30.9	167°23.8	9.6'	22°37.4	5.7'	56.1'						
17	74°55.9	30.0	181°52.4	9.6'	22°43.2	5.6'	56.1'						
18	89°56.1	N08°29.1	196°21.0	9.5'	N22°48.7	5.5'	56.1'						
19	104°56.3	28.2	210°49.4	9.4'	22°54.2	5.4'	56.2'						
20	119°56.5	27.3	225°17.8	9.3'	22°59.6	5.2'	56.2'						
21	134°56.7	· · 26.4	239°46.1	9.2'	23°04.8	5.1'	56.2'						
22	149°56.9	25.5	254°14.4	9.1'	23°09.9	5.0'	56.3'						
23	164°57.1	24.6	268°42.5	9.1'	23°15.0	4.9'	56.3'						
SD = 15.8' $d = -0.9'$							SD = 15.2'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	////	02:15	03:51	20:07	21:40	////
N 70°	////	02:47	04:06	19:52	21:10	////
68°	01:07	03:10	04:18	19:41	20:47	22:43
66°	01:56	03:28	04:28	19:31	20:30	22:00
64°	02:25	03:42	04:36	19:23	20:16	21:32
62°	02:47	03:54	04:44	19:16	20:05	21:11
60°	03:04	04:04	04:50	19:10	19:55	20:55
N 58°	03:18	04:13	04:55	19:05	19:47	20:41
56°	03:30	04:20	05:00	19:00	19:40	20:30
54°	03:40	04:27	05:04	18:56	19:33	20:20
52°	03:48	04:32	05:08	18:52	19:28	20:11
50°	03:56	04:38	05:12	18:49	19:22	20:04
45°	04:12	04:49	05:19	18:41	19:12	19:48
N 40°	04:24	04:58	05:26	18:35	19:03	19:36
35°	04:34	05:05	05:31	18:30	18:56	19:26
30°	04:42	05:11	05:36	18:25	18:49	19:18
20°	04:55	05:21	05:44	18:17	18:40	19:06
N 10°	05:05	05:30	05:51	18:10	18:32	18:56
0°	05:12	05:36	05:57	18:04	18:25	18:49
S 10°	05:18	05:43	06:04	17:58	18:19	18:43
20°	05:22	05:48	06:10	17:51	18:13	18:39
30°	05:26	05:54	06:18	17:44	18:08	18:36
35°	05:27	05:57	06:22	17:40	18:05	18:35
40°	05:28	06:00	06:27	17:35	18:02	18:34
45°	05:29	06:03	06:32	17:30	17:59	18:33
S 50°	05:29	06:06	06:39	17:23	17:56	18:33
52°	05:29	06:08	06:42	17:20	17:54	18:33
54°	05:29	06:10	06:45	17:17	17:53	18:34
56°	05:28	06:11	06:49	17:13	17:51	18:34
58°	05:28	06:13	06:53	17:09	17:49	18:35
S 60°	05:27	06:15	06:58	17:05	17:47	18:35

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	17:03	□	□	13:08	□	□
N 70°	17:44	16:37	□	12:28	15:13	□
68°	18:13	17:49	□	12:01	14:02	□
66°	18:34	18:27	18:15	11:41	13:25	15:22
64°	18:52	18:53	18:59	11:25	13:00	14:38
62°	19:06	19:14	19:28	11:11	12:40	14:09
60°	19:18	19:31	19:51	11:00	12:23	13:47
N 58°	19:29	19:45	20:09	10:50	12:10	13:29
56°	19:38	19:57	20:25	10:42	11:58	13:14
54°	19:46	20:08	20:38	10:34	11:48	13:01
52°	19:53	20:18	20:49	10:27	11:39	12:50

September 01, 02, 03 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	340°03.3	139°11.3	S12°03.1	140°48.3	S08°03.5	338°31.7	S01°02.5	155°24.7	N00°21.5	Alpheratz	357°45.3	29°09.1
1	355°05.7	154°11.9	04.2	155°49.3	04.1	353°34.4	02.6	170°26.9	21.4	Ankaa	353°17.2	-42°14.6
2	10°08.2	169°12.4	05.3	170°50.2	04.8	8°37.1	02.7	185°29.1	21.3	Schedar	349°42.5	56°35.8
3	25°10.7	184°13.0	•• 06.4	185°51.2	•• 05.4	23°39.8	•• 02.8	200°31.3	•• 21.1	Diphda	348°57.6	-17°55.4
4	40°13.1	199°13.5	07.4	200°52.1	06.1	38°42.5	03.0	215°33.5	21.0	Achernar	335°27.7	-57°10.6
5	55°15.6	214°14.1	08.5	215°53.1	06.7	53°45.3	03.1	230°35.7	20.9	Hamal	328°02.9	23°30.9
6	70°18.0	229°14.6	S12°09.6	230°54.0	S08°07.3	68°48.0	S01°03.2	245°37.9	N00°20.8	Polaris	318°34.9	89°18.4
7	85°20.5	244°15.2	10.7	245°55.0	08.0	83°50.7	03.3	260°40.1	20.7	Acamar	315°19.6	-40°15.4
8	100°23.0	259°15.7	11.7	260°55.9	08.6	98°53.4	03.4	275°42.3	20.6	Menkar	314°17.0	4°08.1
9	115°25.4	274°16.3	•• 12.8	275°56.9	•• 09.2	113°56.1	•• 03.6	290°44.5	•• 20.4	Mirfak	308°43.2	49°53.8
10	130°27.9	289°16.9	13.9	290°57.8	09.9	128°58.9	03.7	305°46.7	20.3	Aldebaran	290°51.7	16°31.9
11	145°30.4	304°17.4	14.9	305°58.8	10.5	144°01.6	03.8	320°48.9	20.2	Rigel	281°14.0	-8°11.2
12	160°32.8	319°18.0	S12°16.0	320°59.7	S08°11.2	159°04.3	S01°03.9	335°51.1	N00°20.1	Capella	280°37.5	46°00.4
13	175°35.3	334°18.5	17.1	336°00.7	11.8	174°07.0	04.0	350°53.3	20.0	Bellatrix	278°34.2	6°21.7
14	190°37.8	349°19.1	18.1	351°01.6	12.4	189°09.8	04.2	5°55.5	19.9	Elnath	278°15.2	28°36.9
15	205°40.2	4°19.7	•• 19.2	6°02.6	•• 13.1	204°12.5	•• 04.3	20°57.7	•• 19.7	Alnilam	275°48.5	-1°11.6
16	220°42.7	19°20.2	20.3	21°03.5	13.7	219°15.2	04.4	35°59.9	19.6	Betelgeuse	271°03.6	7°24.6
17	235°45.2	34°20.8	21.3	36°04.5	14.3	234°17.9	04.5	51°02.1	19.5	Canopus	263°57.3	-52°41.8
18	250°47.6	49°21.3	S12°22.4	51°05.4	S08°15.0	249°20.7	S01°04.6	66°04.3	N00°19.4	Sirius	258°35.7	-16°43.7
19	265°50.1	64°21.9	23.5	66°06.4	15.6	264°23.4	04.8	81°06.5	19.3	Adhara	255°14.3	-28°59.0
20	280°52.5	79°22.5	24.5	81°07.3	16.3	279°26.1	04.9	96°08.7	19.2	Procyon	245°02.0	5°11.9
21	295°55.0	94°23.0	•• 25.6	96°08.3	•• 16.9	294°28.8	•• 05.0	111°10.9	•• 19.0	Pollux	243°30.4	28°00.0
22	310°57.5	109°23.6	26.7	111°09.2	17.5	309°31.6	05.1	126°13.1	18.9	Avior	234°19.5	-59°32.5
23	325°59.9	124°24.2	27.7	126°10.2	18.2	324°34.3	05.2	141°15.3	18.8	Suhail	222°54.4	-43°28.5
Mer.pass. 01:20		ν 0.5' d 1.1' m -4.59		ν 1.0' d 0.6' m 1.52		ν 2.7' d 0.1' m -2.89		ν 2.2' d -0.1' m 0.98		Miaplacidus	221°41.2	-69°45.6
										Alphard	217°58.4	-8°42.3
										Regulus	207°45.9	11°54.9
										Dubhe	193°54.6	61°41.6
										Denebola	182°36.0	14°30.8
										Gienah	175°54.7	-17°36.1
										Acrux	173°12.4	-63°09.7
										Gacrux	172°03.8	-57°10.5
										Alioth	166°22.8	55°54.2
										Spica	158°33.6	-11°13.0
										Alkaid	153°00.7	49°15.8
										Hadar	148°51.3	-60°25.7
										Menkent	148°10.2	-36°25.5
										Arcturus	145°57.7	19°07.7
										Rigel Kent.	139°54.9	-60°53.0
										Kochab	137°20.2	74°06.9
										Zuben'ubi	137°07.8	-16°05.2
										Alphecca	126°12.8	26°40.9
										Antares	112°28.7	-26°27.4
										Atria	107°32.4	-69°03.1
										Sabik	102°14.8	-15°44.3
										Shaula	96°24.5	-37°06.8
										Rasalhague	96°08.2	12°33.3
										Eltanin	90°46.9	51°29.6
										Kaus Aust.	83°46.3	-34°22.8
										Vega	80°40.1	38°47.9
										Nunki	76°00.6	-26°17.0
										Altair	62°10.0	8°54.0
										Peacock	53°21.8	-56°42.0
										Deneb	49°32.5	45°19.4
										Enif	33°48.8	9°55.7
										Al Na'ir	27°45.7	-46°54.4
										Fomalhaut	15°25.8	-29°33.7
										Scheat	13°55.0	28°08.6
										Markab	13°40.0	15°16.0
Sep 01 WedSHA Mer.pass Venus159°08.114:43 Mars160°45.114:36 Jupiter358°28.401:26 Saturn175°21.413:36												
Sep 02 ThuSHA Mer.pass Venus158°22.314:42 Mars160°08.714:34 Jupiter358°34.601:21 Saturn175°15.113:33												
Sep 03 FriSHA Mer.pass Venus157°37.114:41 Mars159°32.314:33 Jupiter358°40.901:17 Saturn175°08.713:29												
Horizontal parallax Venus:0.3 Mars:0.1												
Mer.pass. 01:12		ν 0.6' d 1.0' m -4.61		ν 0.9' d 0.6' m 1.52		ν 2.7' d 0.1' m -2.90		ν 2.2' d -0.1' m 0.98				

h	Sun			Moon			
Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	179°57.3	N08°23.7	283°10.6	9.0'	N23°19.8	4.8'	56.3'
1	194°57.5	22.8	297°38.6	8.9'	23°24.6	4.6'	56.4'
2	209°57.7	21.9	312°06.5	8.8'	23°29.3	4.5'	56.4'
3	224°57.9	· · 21.0	326°34.3	8.7'	23°33.8	4.4'	56.4'
4	239°58.1	20.1	341°02.0	8.7'	23°38.2	4.3'	56.5'
5	254°58.3	19.2	355°29.7	8.6'	23°42.5	4.2'	56.5'
6	269°58.5	N08°18.2	9°57.3	8.5'	N23°46.6	4.0'	56.5'
7	284°58.7	17.3	24°24.8	8.4'	23°50.6	3.9'	56.6'
8	299°58.9	16.4	38°52.3	8.4'	23°54.5	3.8'	56.6'
9	314°59.1	· · 15.5	53°19.6	8.3'	23°58.3	3.6'	56.6'
10	329°59.2	14.6	67°46.9	8.2'	24°02.0	3.5'	56.7'
11	344°59.4	13.7	82°14.1	8.1'	24°05.5	3.4'	56.7'
12	359°59.6	N08°12.8	96°41.3	8.1'	N24°08.8	3.2'	56.7'
13	14°59.8	11.9	111°08.3	8.0'	24°12.1	3.1'	56.8'
14	30°00.0	11.0	125°35.3	7.9'	24°15.2	3.0'	56.8'
15	45°00.2	· · 10.1	140°02.3	7.9'	24°18.2	2.8'	56.8'
16	60°00.4	09.2	154°29.1	7.8'	24°21.0	2.7'	56.9'
17	75°00.6	08.3	168°55.9	7.7'	24°23.8	2.6'	56.9'
18	90°00.8	N08°07.4	183°22.6	7.6'	N24°26.3	2.4'	56.9'
19	105°01.0	06.4	197°49.3	7.6'	24°28.8	2.3'	57.0'
20	120°01.2	05.5	212°15.8	7.5'	24°31.1	2.2'	57.0'
21	135°01.4	· · 04.6	226°42.4	7.5'	24°33.2	2.0'	57.0'
22	150°01.6	03.7	241°08.8	7.4'	24°35.3	1.9'	57.1'
23	165°01.8	02.8	255°35.2	7.3'	24°37.1	1.7'	57.1'
SD = 15.8'			SD = 15.4'				

Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	180°02.0	N08°01.9	270°01.5	7.3'	N24°38.9	1.6'	57.1'
1	195°02.2	01.0	284°27.8	7.2'	24°40.5	1.5'	57.2'
2	210°02.4	08°00.1	298°54.0	7.1'	24°41.9	1.3'	57.2'
3	225°02.6	07°59.2	313°20.1	7.1'	24°43.3	1.2'	57.2'
4	240°02.8	58.3	327°46.2	7.0'	24°44.4	1.0'	57.3'
5	255°03.0	57.3	342°12.2	7.0'	24°45.4	0.9'	57.3'
6	270°03.2	N07°56.4	356°38.2	6.9'	N24°46.3	0.7'	57.4'
7	285°03.4	55.5	11°04.1	6.9'	24°47.1	0.6'	57.4'
8	300°03.6	54.6	25°30.0	6.8'	24°47.6	0.4'	57.4'
9	315°03.8	· · 53.7	39°55.8	6.8'	24°48.1	0.3'	57.5'
10	330°04.0	52.8	54°21.6	6.7'	24°48.4	0.1'	57.5'
11	345°04.2	51.9	68°47.3	6.7'	24°48.5	-0.0'	57.5'
12	0°04.4	N07°51.0	83°12.9	6.6'	N24°48.5	-0.2'	57.6'
13	15°04.6	50.0	97°38.5	6.6'	24°48.3	-0.3'	57.6'
14	30°04.8	49.1	112°04.1	6.5'	24°48.0	-0.5'	57.7'
15	45°05.0	· · 48.2	126°29.6	6.5'	24°47.5	-0.6'	57.7'
16	60°05.2	47.3	140°55.1	6.4'	24°46.9	-0.8'	57.7'
17	75°05.4	46.4	155°20.5	6.4'	24°46.2	-0.9'	57.8'
18	90°05.6	N07°45.5	169°45.9	6.3'	N24°45.2	-1.1'	57.8'
19	105°05.9	44.6	184°11.2	6.3'	24°44.2	-1.2'	57.8'
20	120°06.1	43.7	198°36.6	6.3'	24°42.9	-1.4'	57.9'
21	135°06.3	· · 42.7	213°01.8	6.2'	24°41.6	-1.5'	57.9'
22	150°06.5	41.8	227°27.1	6.2'	24°40.0	-1.7'	58.0'
23	165°06.7	40.9	241°52.3	6.2'	24°38.3	-1.8'	58.0'
SD = 15.8'			SD = 15.6'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	180°06.9	N07°40.0	256°17.4	6.1'	N24°36.5	-2.0'	58.0'
1	195°07.1	39.1	270°42.6	6.1'	24°34.5	-2.2'	58.1'
2	210°07.3	38.2	285°07.7	6.1'	24°32.3	-2.3'	58.1'
3	225°07.5	· · 37.2	299°32.8	6.1'	24°30.0	-2.5'	58.1'
4	240°07.7	36.3	313°57.8	6.0'	24°27.5	-2.6'	58.2'
5	255°07.9	35.4	328°22.9	6.0'	24°24.9	-2.8'	58.2'
6	270°08.1	N07°34.5	342°47.9	6.0'	N24°22.1	-2.9'	58.3'
7	285°08.3	33.6	357°12.8	6.0'	24°19.2	-3.1'	58.3'
8	300°08.5	32.7	11°37.8	5.9'	24°16.0	-3.3'	58.3'
9	315°08.7	· · 31.7	26°02.7	5.9'	24°12.8	-3.4'	58.4'
10	330°08.9	30.8	40°27.7	5.9'	24°09.4	-3.6'	58.4'
11	345°09.1	29.9	54°52.6	5.9'	24°05.8	-3.7'	58.5'
12	0°09.3	N07°29.0	69°17.4	5.9'	N24°02.1	-3.9'	58.5'
13	15°09.5	28.1	83°42.3	5.9'	23°58.2	-4.0'	58.5'
14	30°09.7	27.2	98°07.2	5.8'	23°54.1	-4.2'	58.6'
15	45°09.9	· · 26.2	112°32.0	5.8'	23°49.9	-4.4'	58.6'
16	60°10.1	25.3	126°56.9	5.8'	23°45.5	-4.5'	58.6'
17	75°10.3	24.4	141°21.7	5.8'	23°41.0	-4.7'	58.7'
18	90°10.5	N07°23.5	155°46.5	5.8'	N23°36.3	-4.8'	58.7'
19	105°10.7	22.6	170°11.3	5.8'	23°31.5	-5.0'	58.8'
20	120°10.9	21.6	184°36.1	5.8'	23°26.5	-5.2'	58.8'
21	135°11.1	· · 20.7	199°01.0	5.8'	23°21.4	-5.3'	58.8'
22	150°11.3	19.8	213°25.8	5.8'	23°16.0	-5.5'	58.9'
23	165°11.5	18.9	227°50.6	5.8'	23°10.6	-5.6'	58.9'
SD = 15.8'			SD = 15.8'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	////	02:37	04:06	19:50	21:17	////
N 70°	////	03:04	04:19	19:38	20:51	23:18
68°	01:39	03:24	04:29	19:28	20:32	22:13
66°	02:15	03:40	04:38	19:19	20:17	21:40
64°	02:40	03:52	04:45	19:12	20:05	21:16
62°	02:59	04:03	04:51	19:06	19:54	20:58
60°	03:14	04:12	04:57	19:01	19:46	20:43
N 58°	03:26	04:20	05:02	18:56	19:38	20:31
56°	03:37	04:26	05:06	18:52	19:31	20:20
54°	03:46	04:32	05:10	18:49	19:26	20:11
52°	03:54	04:38	05:13	18:45	19:20	20:03
50°	04:02	04:43	05:16	18:42	19:16	19:56
45°	04:16	04:53	05:23	18:36	19:06	19:42
N 40°	04:28	05:01	05:28	18:30	18:58	19:31
35°	04:37	05:07	05:33	18:26	18:51	19:22
30°	04:44	05:13	05:37	18:22	18:46	19:14
20°	04:56	05:22	05:45	18:15	18:37	19:03
N 10°	05:05	05:30	05:51	18:09	18:30	18:54
0°	05:11	05:36	05:56	18:03	18:24	18:48
S 10°	05:16	05:41	06:02	17:58	18:19	18:43
20°	05:20	05:46	06:08	17:52	18:14	18:40
30°	05:23	05:50	06:14	17:45	18:10	18:37
35°	05:23	05:53	06:18	17:42	18:07	18:37
40°	05:24	05:55	06:22	17:38	18:05	18:36
45°	05:24	05:58	06:27	17:33	18:03	18:37
S 50°	05:23	06:00	06:33	17:27	18:00	18:37
52°	05:22	06:01	06:35	17:25	17:59	18:38
54°	05:22	06:03	06:38	17:22	17:58	18:39
56°	05:21	06:04	06:41	17:19	17:57	18:40
58°	05:20	06:05	06:45	17:16	17:55	18:41
S 60°	05:19	06:07	06:49	17:12	17:54	18:42

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°						
N 70°						
68°						
66°			20:45			18:44
64°	19:16	20:07	21:41	16:12	17:20	17:48
62°	19:57	20:51	22:14	15:32	16:36	17:15
60°	20:25	21:20	22:38	15:04	16:07	16:50
N 58°	20:46	21:42	22:57	14:43	15:45	16:30
56°	21:04	22:00	23:13	14:25	15:26	16:14
54°	21:19	22:15	23:27	14:10	15:11	16:00
52°	21:32	22:29	23:39	13:57	14:58	15:48
50°	21:44	22:40	23:50	13:46	14:46	15:37
45°	22:08	23:04	· · ·	13:22	14:22	15:14
N 40°	22:27	23:24	· · ·	13:03	14:02	14:56
35°	22:43	23:40	· · ·	12:48	13:46	14:40
30°	22:57	23:54	· · ·	12:34	13:32	14:27
20°	23:21	· · ·	00:18	12:11	13:08	14:04
N 10°	23:41	· · ·	00:38	11:50	12:47	13:44
0°	· · ·	00:00	00:58	11:32	12:28	13:26
S 10°	· · ·	00:20	01:17	11:13	12:08	13:07
20°	· · ·	00:40	01:38	10:53	11:47	12:47
30°	00:06	01:05	02:01	10:30	11:23	12:23
35°	00:19	01:19	02:15	10:16	11:09	12:10
40°	00:34	01:35	02:32	10:00	10:52	11:54
45°	00:53	01:55	02:51	09:42	10:33	11:35
S 50°	01:15	02:19	03:15	09:18	10:08	11:11
52°	01:26	02:31	03:27	09:07	09:56	11:00
54°	01:39	02:45	03:40	08:54	09:43	10:47
56°	01:53	03:00	03:55	08:40	09:27	10:31
58°	02:10	03:19	04:13	08:23	09:08	10:14
S 60°	02:30	03:42	04:35	08:02	08:45	09:52

Day	Sun		Mer. Pass hh:mm	Moon		Age 22-24 58-37%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
01	00:11	00:01	12:00	05:19	17:46	
02	00:08	00:18	12:00	06:14	18:43	
03	00:27	00:37	11:59	07:12	19:41	

September 04, 05, 06 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars			
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	343°00.7	139°53.1	S13°19.1	141°56.4	S08°49.3	341°48.0	S01°11.2	158°03.0	N00°13.0	Alpheratz	357°45.3	29°09.1	
1	358°03.1	154°53.7	20.2	156°57.3	50.0	356°50.8	11.4	173°05.2	12.9	Ankaa	353°17.2	-42°14.6	
2	13°05.6	169°54.3	21.2	171°58.3	50.6	11°53.5	11.5	188°07.4	12.8	Schedar	349°42.5	56°35.8	
3	28°08.1	184°55.0	· · 22.2	186°59.2	· · 51.2	26°56.2	· · 11.6	203°09.6	· · 12.7	Diphda	348°57.6	-17°55.4	
4	43°10.5	199°55.6	23.3	202°00.1	51.9	41°59.0	11.7	218°11.8	12.6	Achernar	335°27.7	-57°10.6	
5	58°13.0	214°56.2	24.3	217°01.1	52.5	57°01.7	11.9	233°14.0	12.5	Hamal	328°02.8	23°30.9	
6	73°15.5	229°56.8	S13°25.3	232°02.0	S08°53.1	72°04.5	S01°12.0	248°16.2	N00°12.3	Polaris	318°33.5	89°18.4	
7	88°17.9	244°57.4	26.4	247°02.9	53.8	87°07.2	12.1	263°18.4	12.2	Acamar	315°19.6	-40°15.4	
8	103°20.4	259°58.1	27.4	262°03.9	54.4	102°09.9	12.2	278°20.6	12.1	Menkar	314°17.0	4°08.1	
9	118°22.9	274°58.7	· · 28.4	277°04.8	· · 55.0	117°12.7	· · 12.3	293°22.8	· · 12.0	Mirfak	308°43.1	49°53.9	
10	133°25.3	289°59.3	29.5	292°05.8	55.7	132°15.4	12.5	308°25.0	11.9	Aldebaran	290°51.7	16°31.9	
11	148°27.8	304°59.9	30.5	307°06.7	56.3	147°18.1	12.6	323°27.2	11.7	Rigel	281°14.0	-8°11.2	
12	163°30.3	320°00.6	S13°31.5	322°07.6	S08°56.9	162°20.9	S01°12.7	338°29.4	N00°11.6	Capella	280°37.5	46°00.4	
13	178°32.7	335°01.2	32.6	337°08.6	57.6	177°23.6	12.8	353°31.6	11.5	Bellatrix	278°34.2	6°21.7	
14	193°35.2	350°01.8	33.6	352°09.5	58.2	192°26.3	13.0	8°33.8	11.4	Elnath	278°15.2	28°36.9	
15	208°37.6	5°02.4	· · 34.6	7°10.4	· · 58.8	207°29.1	· · 13.1	23°36.0	· · 11.3	Alnilam	275°48.5	-1°11.6	
16	223°40.1	20°03.1	35.6	22°11.4	08°59.5	222°31.8	13.2	38°38.2	11.2	Betelgeuse	271°03.6	7°24.6	
17	238°42.6	35°03.7	36.7	37°12.3	09°00.1	237°34.5	13.3	53°40.4	11.0	Canopus	263°57.3	-52°41.8	
18	253°45.0	50°04.3	S13°37.7	52°13.2	S09°00.7	252°37.3	S01°13.5	68°42.6	N00°10.9	Sirius	258°35.6	-16°43.7	
19	268°47.5	65°05.0	38.7	67°14.2	01.4	267°40.0	13.6	83°44.7	10.8	Adhara	255°14.3	-28°59.0	
20	283°50.0	80°05.6	39.8	82°15.1	02.0	282°42.8	13.7	98°46.9	10.7	Procyon	245°02.0	5°11.9	
21	298°52.4	95°06.2	· · 40.8	97°16.0	· · 02.6	297°45.5	· · 13.8	113°49.1	· · 10.6	Pollux	243°30.4	28°00.0	
22	313°54.9	110°06.9	41.8	112°17.0	03.3	312°48.2	14.0	128°51.3	10.4	Avior	234°19.4	-59°32.5	
23	328°57.4	125°07.5	42.8	127°17.9	03.9	327°51.0	14.1	143°53.5	10.3	Suhail	222°54.4	-43°28.4	
Mer.pass. 01:08		ν 0.6' d 1.0' m -4.62		ν 0.9' d 0.6' m 1.52		ν 2.7' d 0.1' m -2.90		ν 2.2' d -0.1' m 0.98					
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec				
0	343°59.8	140°08.1	S13°43.8	142°18.8	S09°04.5	342°53.7	S01°14.2	158°55.7	N00°10.2				
1	359°02.3	155°08.8	44.9	157°19.8	05.2	357°56.4	14.3	173°57.9	10.1				
2	14°04.8	170°09.4	45.9	172°20.7	05.8	12°59.2	14.5	189°00.1	10.0				
3	29°07.2	185°10.0	· · 46.9	187°21.6	· · 06.4	28°01.9	· · 14.6	204°02.3	· · 09.9				
4	44°09.7	200°10.7	47.9	202°22.6	07.1	43°04.7	14.7	219°04.5	09.7				
5	59°12.1	215°11.3	49.0	217°23.5	07.7	58°07.4	14.8	234°06.7	09.6				
6	74°14.6	230°12.0	S13°50.0	232°24.4	S09°08.3	73°10.1	S01°15.0	249°08.9	N00°09.5				
7	89°17.1	245°12.6	51.0	247°25.4	09.0	88°12.9	15.1	264°11.1	09.4				
8	104°19.5	260°13.3	52.0	262°26.3	09.6	103°15.6	15.2	279°13.3	09.3				
9	119°22.0	275°13.9	· · 53.0	277°27.2	· · 10.2	118°18.4	· · 15.4	294°15.5	· · 09.1				
10	134°24.5	290°14.6	54.0	292°28.1	10.9	133°21.1	15.5	309°17.7	09.0				
11	149°26.9	305°15.2	55.1	307°29.1	11.5	148°23.8	15.6	324°19.9	08.9				
12	164°29.4	320°15.8	S13°56.1	322°30.0	S09°12.1	163°26.6	S01°15.7	339°22.1	N00°08.8				
13	179°31.9	335°16.5	57.1	337°30.9	12.8	178°29.3	15.9	354°24.3	08.7				
14	194°34.3	350°17.2	58.1	352°31.9	13.4	193°32.1	16.0	9°26.5	08.6				
15	209°36.8	5°17.8	13°59.1	7°32.8	· · 14.0	208°34.8	· · 16.1	24°28.7	· · 08.4				
16	224°39.2	20°18.5	14°00.1	22°33.7	14.7	223°37.5	16.2	39°30.9	08.3				
17	239°41.7	35°19.1	01.1	37°34.7	15.3	238°40.3	16.4	54°33.0	08.2				
18	254°44.2	50°19.8	S14°02.2	52°35.6	S09°15.9	253°43.0	S01°16.5	69°35.2	N00°08.1				
19	269°46.6	65°20.4	03.2	67°36.5	16.6	268°45.8	16.6	84°37.4	08.0				
20	284°49.1	80°21.1	04.2	82°37.4	17.2	283°48.5	16.7	99°39.6	07.8				
21	299°51.6	95°21.7	· · 05.2	97°38.4	· · 17.8	298°51.2	· · 16.9	114°41.8	· · 07.7				
22	314°54.0	110°22.4	06.2	112°39.3	18.5	313°54.0	17.0	129°44.0	07.6				
23	329°56.5	125°23.1	07.2	127°40.2	19.1	328°56.7	17.1	144°46.2	07.5				
Mer.pass. 01:04		ν 0.6' d 1.0' m -4.63		ν 0.9' d 0.6' m 1.52		ν 2.7' d 0.1' m -2.91		ν 2.2' d -0.1' m 0.97					
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec				
0	344°59.0	140°23.7	S14°08.2	142°41.2	S09°19.7	343°59.5	S01°17.2	159°48.4	N00°07.4				
1	0°01.4	155°24.4	09.2	157°42.1	20.3	359°02.2	17.4	174°50.6	07.2				
2	15°03.9	170°25.0	10.2	172°43.0	21.0	14°04.9	17.5	189°52.8	07.1				
3	30°06.4	185°25.7	· · 11.2	187°43.9	· · 21.6	29°07.7	· · 17.6	204°55.0	· · 07.0				
4	45°08.8	200°26.4	12.3	202°44.9	22.2	44°10.4	17.8	219°57.2	06.9				
5	60°11.3	215°27.0	13.3	217°45.8	22.9	59°13.2	17.9	234°59.4	06.8				
6	75°13.7	230°27.7	S14°14.3	232°46.7	S09°23.5	74°15.9	S01°18.0	250°01.6	N00°06.7				
7	90°16.2	245°28.4	15.3	247°47.6	24.1	89°18.7	18.1	265°03.8	06.5				
8	105°18.7	260°29.1	16.3	262°48.6	24.8	104°21.4	18.3	280°06.0	06.4				
9	120°21.1	275°29.7	· · 17.3	277°49.5	· · 25.4	119°24.1	· · 18.4	295°08.2	· · 06.3				
10	135°23.6	290°30.4	18.3	292°50.4	26.0	134°26.9	18.5	310°10.4	06.2				
11	150°26.1	305°31.1	19.3	307°51.3	26.7	149°29.6	18.6	325°12.6	06.1				
12	165°28.5	320°31.7	S14°20.3	322°52.3	S09°27.3	164°32.4	S01°18.8	340°14.7	N00°05.9				
13	180°31.0	335°32.4	21.3	337°53.2	27.9	179°35.1	18.9	355°16.9	05.8				
14	195°33.5	350°33.1	22.3	352°54.1	28.5	194°37.9	19.0	10°19.1	05.7				
15	210°35.9	5°33.8	· · 23.3	7°55.0	· · 29.2	209°40.6	· · 19.1	25°21.3	· · 05.6				
16	225°38.4	20°34.5	24.3	22°56.0	29.8	224°43.3	19.3	40°23.5	05.5				
17	240°40.9	35°35.1	25.3	37°56.9	30.4	239°46.1	19.4	55°25.7	05.4				
18	255°43.3	50°35.8	S14°26.3	52°57.8	S09°31.1	254°48.8	S01°19.5	70°27.9	N00°05.2				
19	270°45.8	65°36.5	27.3	67°58.7	31.7	269°51.6	19.7	85°30.1	05.1				
20	285°48.2	80°37.2	28.3	82°59.7	32.3	284°54.3	19.8	100°32.3	05.0				
21	300°50.7	95°37.9	· · 29.3	98°00.6	· · 33.0	299°57.1	· · 19.9	115°34.5	· · 04.9				
22	315°53.2	110°38.5	30.3	113°01.5	33.6	314°59.8	20.0	130°36.7	04.8				
23	330°55.6	125°39.2	31.3	128°02.4	34.2	330°02.6	20.2	145°38.9	04.6				
Mer.pass. 01:00		ν 0.7' d 1.0' m -4.64		ν 0.9' d 0.6' m 1.52		ν 2.7' d 0.1' m -2.91		ν 2.2' d -0.1' m 0.97					
											Sep 04 Sat	SHA	Mer.pass
											Venus	156°52.4	14:40
											Mars	158°55.7	14:31
											Jupiter	358°47.4	01:13
											Saturn	175°02.3	13:26
											Sep 05 Sun	SHA	Mer.pass
											Venus	156°08.3	14:39
											Mars	158°19.0	14:30
											Jupiter	358°53.9	01:08
											Saturn	174°55.9	13:22
											Sep 06 Mon	SHA	Mer.pass
											Venus	155°24.8	14:38
											Mars	157°42.2	14:28
											Jupiter	359°00.5	01:04
											Saturn	174°49.5	13:19
											Horizontal parallax		
											Venus:	0.3	
											Mars:	0.1	

Sun				Moon			
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	180°11.8	N07°18.0	242°15.4	5.8'	N23°05.0	-5.8'	59.0'
1	195°12.0	17.0	256°40.2	5.8'	22°59.2	-5.9'	59.0'
2	210°12.2	16.1	271°05.0	5.8'	22°53.2	-6.1'	59.0'
3	225°12.4	· · 15.2	285°29.8	5.8'	22°47.1	-6.2'	59.1'
4	240°12.6	14.3	299°54.6	5.8'	22°40.9	-6.4'	59.1'
5	255°12.8	13.3	314°19.5	5.8'	22°34.5	-6.6'	59.1'
6	270°13.0	N07°12.4	328°44.3	5.9'	N22°27.9	-6.7'	59.2'
7	285°13.2	11.5	343°09.2	5.9'	22°21.2	-6.9'	59.2'
8	300°13.4	10.6	357°34.0	5.9'	22°14.4	-7.0'	59.3'
9	315°13.6	· · 09.7	11°58.9	5.9'	22°07.4	-7.2'	59.3'
10	330°13.8	08.7	26°23.8	5.9'	22°00.2	-7.3'	59.3'
11	345°14.0	07.8	40°48.7	5.9'	21°52.9	-7.5'	59.4'
12	0°14.2	N07°06.9	55°13.6	5.9'	N21°45.4	-7.6'	59.4'
13	15°14.4	06.0	69°38.6	6.0'	21°37.8	-7.8'	59.4'
14	30°14.6	05.0	84°03.5	6.0'	21°30.0	-7.9'	59.5'
15	45°14.8	· · 04.1	98°28.5	6.0'	21°22.1	-8.1'	59.5'
16	60°15.0	03.2	112°53.5	6.0'	21°14.0	-8.2'	59.5'
17	75°15.3	02.3	127°18.5	6.0'	21°05.8	-8.4'	59.6'
18	90°15.5	N07°01.3	141°43.5	6.1'	N20°57.4	-8.5'	59.6'
19	105°15.7	07°00.4	156°08.6	6.1'	20°48.9	-8.7'	59.7'
20	120°15.9	06°59.5	170°33.7	6.1'	20°40.2	-8.8'	59.7'
21	135°16.1	· · 58.6	184°58.8	6.1'	20°31.4	-8.9'	59.7'
22	150°16.3	57.6	199°23.9	6.2'	20°22.5	-9.1'	59.8'
23	165°16.5	56.7	213°49.1	6.2'	20°13.4	-9.2'	59.8'
SD = 15.9'		$d = -0.9'$		SD = 16.1'			

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	180° 16.7	N06° 55.8	228° 14.2	6.2'	N20° 04.2	-9.4'	59.8'
1	195° 16.9	54.9	242° 39.5	6.2'	19° 54.8	-9.5'	59.9'
2	210° 17.1	53.9	257° 04.7	6.3'	19° 45.3	-9.7'	59.9'
3	225° 17.3	· · 53.0	271° 30.0	6.3'	19° 35.6	-9.8'	59.9'
4	240° 17.5	52.1	285° 55.3	6.3'	19° 25.8	-9.9'	60.0'
5	255° 17.7	51.2	300° 20.6	6.4'	19° 15.9	-10.1'	60.0'
6	270° 17.9	N06° 50.2	314° 46.0	6.4'	N19° 05.9	-10.2'	60.0'
7	285° 18.2	49.3	329° 11.4	6.4'	18° 55.6	-10.3'	60.1'
8	300° 18.4	48.4	343° 36.8	6.5'	18° 45.3	-10.5'	60.1'
9	315° 18.6	· · 47.5	358° 02.3	6.5'	18° 34.8	-10.6'	60.1'
10	330° 18.8	46.5	12° 27.8	6.5'	18° 24.2	-10.7'	60.2'
11	345° 19.0	45.6	26° 53.3	6.6'	18° 13.5	-10.9'	60.2'
12	0° 19.2	N06° 44.7	41° 18.9	6.6'	N18° 02.7	-11.0'	60.2'
13	15° 19.4	43.7	55° 44.5	6.6'	17° 51.7	-11.1'	60.3'
14	30° 19.6	42.8	70° 10.1	6.7'	17° 40.6	-11.2'	60.3'
15	45° 19.8	· · 41.9	84° 35.8	6.7'	17° 29.3	-11.4'	60.3'
16	60° 20.0	41.0	99° 01.5	6.8'	17° 18.0	-11.5'	60.3'
17	75° 20.2	40.0	113° 27.3	6.8'	17° 06.5	-11.6'	60.4'
18	90° 20.4	N06° 39.1	127° 53.1	6.8'	N16° 54.9	-11.7'	60.4'
19	105° 20.7	38.2	142° 18.9	6.9'	16° 43.1	-11.8'	60.4'
20	120° 20.9	37.2	156° 44.8	6.9'	16° 31.3	-12.0'	60.5'
21	135° 21.1	· · 36.3	171° 10.7	6.9'	16° 19.3	-12.1'	60.5'
22	150° 21.3	35.4	185° 36.6	7.0'	16° 07.2	-12.2'	60.5'
23	165° 21.5	34.5	200° 02.6	7.0'	15° 55.0	-12.3'	60.5'
	SD = 15.9'	$d = -0.9'$			SD = 16.3'		

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	180°21.7	N06°33.5	214°28.6	7.1'	N15°42.7	-12.4'	60.6'
1	195°21.9	32.6	228°54.7	7.1'	15°30.3	-12.5'	60.6'
2	210°22.1	31.7	243°20.8	7.1'	15°17.8	-12.6'	60.6'
3	225°22.3	· · 30.7	257°46.9	7.2'	15°05.1	-12.8'	60.6'
4	240°22.5	29.8	272°13.1	7.2'	14°52.4	-12.9'	60.7'
5	255°22.8	28.9	286°39.3	7.3'	14°39.5	-13.0'	60.7'
6	270°23.0	N06°27.9	301°05.5	7.3'	N14°26.5	-13.1'	60.7'
7	285°23.2	27.0	315°31.8	7.3'	14°13.5	-13.2'	60.7'
8	300°23.4	26.1	329°58.2	7.4'	14°00.3	-13.3'	60.8'
9	315°23.6	· · 25.1	344°24.5	7.4'	13°47.0	-13.4'	60.8'
10	330°23.8	24.2	358°50.9	7.4'	13°33.7	-13.5'	60.8'
11	345°24.0	23.3	13°17.4	7.5'	13°20.2	-13.6'	60.8'
12	0°24.2	N06°22.4	27°43.9	7.5'	N13°06.7	-13.6'	60.9'
13	15°24.4	21.4	42°10.4	7.6'	12°53.0	-13.7'	60.9'
14	30°24.6	20.5	56°37.0	7.6'	12°39.3	-13.8'	60.9'
15	45°24.9	· · 19.6	71°03.6	7.6'	12°25.4	-13.9'	60.9'
16	60°25.1	18.6	85°30.2	7.7'	12°11.5	-14.0'	60.9'
17	75°25.3	17.7	99°56.9	7.7'	11°57.5	-14.1'	61.0'
18	90°25.5	N06°16.8	114°23.6	7.8'	N11°43.4	-14.2'	61.0'
19	105°25.7	15.8	128°50.4	7.8'	11°29.3	-14.3'	61.0'
20	120°25.9	14.9	143°17.1	7.8'	11°15.0	-14.3'	61.0'
21	135°26.1	· · 14.0	157°44.0	7.9'	11°00.7	-14.4'	61.0'
22	150°26.3	13.0	172°10.8	7.9'	10°46.3	-14.5'	61.1'
23	165°26.5	12.1	186°37.7	7.9'	10°31.8	-14.6'	61.1'
	SD = 15.9'	$d = -0.9'$			SD = 16.5'		

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	////	02:57	04:20	19:34	20:55	////
N 70°	01:15	03:20	04:31	19:24	20:34	22:32
68°	02:02	03:37	04:40	19:15	20:17	21:49
66°	02:32	03:51	04:48	19:08	20:04	21:22
64°	02:53	04:02	04:54	19:02	19:53	21:01
62°	03:10	04:12	04:59	18:57	19:44	20:45
60°	03:23	04:20	05:04	18:52	19:36	20:32
N 58°	03:35	04:27	05:08	18:48	19:29	20:21
56°	03:45	04:33	05:12	18:44	19:23	20:11
54°	03:53	04:38	05:15	18:41	19:18	20:03
52°	04:00	04:43	05:18	18:38	19:13	19:56
50°	04:07	04:47	05:21	18:36	19:09	19:49
45°	04:20	04:56	05:26	18:30	19:00	19:36
N 40°	04:31	05:04	05:31	18:25	18:53	19:26
35°	04:39	05:10	05:35	18:21	18:47	19:17
30°	04:46	05:15	05:39	18:18	18:42	19:11
20°	04:57	05:23	05:45	18:12	18:34	19:00
N 10°	05:05	05:29	05:51	18:07	18:28	18:53
0°	05:10	05:35	05:55	18:02	18:23	18:47
S 10°	05:15	05:39	06:00	17:57	18:18	18:43
20°	05:18	05:43	06:05	17:52	18:15	18:40
30°	05:19	05:47	06:11	17:47	18:11	18:39
35°	05:19	05:49	06:14	17:44	18:09	18:39
40°	05:19	05:50	06:17	17:41	18:08	18:39
45°	05:18	05:52	06:22	17:37	18:06	18:40
S 50°	05:17	05:54	06:26	17:32	18:04	18:42
52°	05:16	05:55	06:29	17:30	18:04	18:43
54°	05:15	05:56	06:31	17:27	18:03	18:44
56°	05:14	05:56	06:34	17:25	18:02	18:45
58°	05:12	05:57	06:37	17:22	18:01	18:47
S 60°	05:10	05:58	06:40	17:19	18:01	18:49

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	☐	23:45	· ·	☐	19:52	18:57
N 70°	☐	☐	00:25	☐	19:09	18:38
68°	22:22	· ·	00:53	19:11	18:40	18:24
66°	23:05	· ·	01:13	18:27	18:18	18:12
64°	23:33	· ·	01:30	17:58	18:01	18:01
62°	23:55	· ·	01:43	17:35	17:46	17:53
60°	· ·	00:13	01:55	17:17	17:34	17:45
N 58°	· ·	00:27	02:05	17:02	17:23	17:38
56°	· ·	00:40	02:13	16:49	17:14	17:32
54°	· ·	00:50	02:21	16:37	17:05	17:27
52°	· ·	01:00	02:27	16:27	16:58	17:22
50°	· ·	01:09	02:34	16:18	16:51	17:18
45°	00:12	01:27	02:47	15:59	16:36	17:08
N 40°	00:29	01:42	02:57	15:43	16:24	17:00
35°	00:44	01:54	03:07	15:30	16:14	16:53
30°	00:57	02:05	03:15	15:18	16:05	16:47
20°	01:19	02:23	03:28	14:58	15:49	16:36
N 10°	01:38	02:39	03:40	14:41	15:35	16:27
0°	01:56	02:54	03:52	14:24	15:22	16:18
S 10°	02:14	03:09	04:03	14:07	15:09	16:09
20°	02:33	03:25	04:15	13:50	14:54	15:59
30°	02:55	03:44	04:28	13:29	14:38	15:48
35°	03:08	03:54	04:36	13:17	14:28	15:41
40°	03:22	04:06	04:45	13:03	14:17	15:34
45°	03:40	04:20	04:55	12:46	14:04	15:25
S 50°	04:01	04:38	05:07	12:26	13:48	15:15
52°	04:11	04:46	05:13	12:16	13:41	15:10
54°	04:23	04:55	05:19	12:05	13:32	15:04
56°	04:36	05:05	05:26	11:52	13:23	14:58
58°	04:51	05:16	05:34	11:37	13:12	14:51
S 60°	05:09	05:30	05:43	11:20	13:00	14:43

September 07, 08, 09 UT (Tue., Wed., Thu.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	345°58.1	140°39.9	S14°32.3	143°03.4	S09°34.9	345°05.3	S01°20.3	160°41.1	N00°04.5	Alpheratz	357°45.3	29°09.2											
1	1°00.6	155°40.6	33.3	158°04.3	35.5	0°08.0	20.4	175°43.3	04.4	Ankaa	353°17.2	-42°14.6											
2	16°03.0	170°41.3	34.2	173°05.2	36.1	15°10.8	20.6	190°45.5	04.3	Schedar	349°42.4	56°35.8											
3	31°05.5	185°42.0	· · 35.2	188°06.1	· · 36.7	30°13.5	· · 20.7	205°47.7	· · 04.2	Diphda	348°57.6	-17°55.4											
4	46°08.0	200°42.7	36.2	203°07.0	37.4	45°16.3	20.8	220°49.8	04.0	Achernar	335°27.7	-57°10.6											
5	61°10.4	215°43.4	37.2	218°08.0	38.0	60°19.0	20.9	235°52.0	03.9	Hamal	328°02.8	23°30.9											
6	76°12.9	230°44.1	S14°38.2	233°08.9	S09°38.6	75°21.8	S01°21.1	250°54.2	N00°03.8	Polaris	318°32.3	89°18.4											
7	91°15.3	245°44.8	39.2	248°09.8	39.3	90°24.5	21.2	265°56.4	03.7	Acamar	315°19.6	-40°15.4											
8	106°17.8	260°45.5	40.2	263°10.7	39.9	105°27.3	21.3	280°58.6	03.6	Menkar	314°17.0	4°08.1											
9	121°20.3	275°46.2	· · 41.2	278°11.6	· · 40.5	120°30.0	· · 21.5	296°00.8	· · 03.4	Mirfak	308°43.1	49°53.9											
10	136°22.7	290°46.9	42.2	293°12.6	41.1	135°32.8	21.6	311°03.0	03.3	Aldebaran	290°51.6	16°31.9											
11	151°25.2	305°47.6	43.2	308°13.5	41.8	150°35.5	21.7	326°05.2	03.2	Rigel	281°14.0	-8°11.2											
12	166°27.7	320°48.3	S14°44.1	323°14.4	S09°42.4	165°38.2	S01°21.8	341°07.4	N00°03.1	Capella	280°37.4	46°00.4											
13	181°30.1	335°49.0	45.1	338°15.3	43.0	180°41.0	22.0	356°09.6	03.0	Bellatrix	278°34.2	6°21.7											
14	196°32.6	350°49.7	46.1	353°16.2	43.7	195°43.7	22.1	11°11.8	02.9	Elnath	278°15.2	28°36.9											
15	211°35.1	5°50.4	· · 47.1	8°17.2	· · 44.3	210°46.5	· · 22.2	26°14.0	· · 02.7	Alnilam	275°48.4	-1°11.6											
16	226°37.5	20°51.1	48.1	23°18.1	44.9	225°49.2	22.3	41°16.2	02.6	Betelgeuse	271°03.5	7°24.6											
17	241°40.0	35°51.8	49.1	38°19.0	45.5	240°52.0	22.5	56°18.4	02.5	Canopus	263°57.2	-52°41.8											
18	256°42.5	50°52.5	S14°50.1	53°19.9	S09°46.2	255°54.7	S01°22.6	71°20.5	N00°02.4	Sirius	258°35.6	-16°43.7											
19	271°44.9	65°53.2	51.0	68°20.8	46.8	270°57.5	22.7	86°22.7	02.3	Adhara	255°14.3	-28°59.0											
20	286°47.4	80°53.9	52.0	83°21.8	47.4	286°00.2	22.9	101°24.9	02.1	Procyon	245°02.0	5°11.9											
21	301°49.8	95°54.6	· · 53.0	98°22.7	· · 48.1	301°03.0	· · 23.0	116°27.1	· · 02.0	Pollux	243°30.4	28°00.0											
22	316°52.3	110°55.3	54.0	113°23.6	48.7	316°05.7	23.1	131°29.3	01.9	Avior	234°19.4	-59°32.4											
23	331°54.8	125°56.0	55.0	128°24.5	49.3	331°08.5	23.3	146°31.5	01.8	Suhail	222°54.4	-43°28.4											
Mer.pass. 00:56				ν 0.7' d 1.0' m -4.65				ν 0.9' d 0.6' m 1.52				ν 2.7' d 0.1' m -2.91				ν 2.2' d -0.1' m 0.97							

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec						
0	346°57.2	140°56.8	S14°55.9	143°25.4	S09°49.9	346°11.2	S01°23.4	161°33.7	N00°01.7	Denebola	182°36.0	14°30.8							
1	1°59.7	155°57.5	56.9	158°26.3	50.6	1°14.0	23.5	176°35.9	01.5	Gienah	175°54.7	-17°36.1							
2	17°02.2	170°58.2	57.9	173°27.3	51.2	16°16.7	23.6	191°38.1	01.4	Acrux	173°12.4	-63°09.7							
3	32°04.6	185°58.9	· · 58.9	188°28.2	· · 51.8	31°19.5	· · 23.8	206°40.3	· · 01.3	Gacrux	172°03.8	-57°10.5							
4	47°07.1	200°59.6	14°59.8	203°29.1	52.5	46°22.2	23.9	221°42.5	01.2	Alioth	166°22.8	55°54.2							
5	62°09.6	216°00.3	15°00.8	218°30.0	53.1	61°25.0	24.0	236°44.7	01.1	Spica	158°33.6	-11°13.0							
6	77°12.0	231°01.1	S15°01.8	233°30.9	S09°53.7	76°27.7	S01°24.2	251°46.9	N00°00.9	Alkaid	153°00.7	49°15.7							
7	92°14.5	246°01.8	02.8	248°31.8	54.3	91°30.5	24.3	266°49.1	00.8	Hadar	148°51.3	-60°25.7							
8	107°16.9	261°02.5	03.8	263°32.8	55.0	106°33.2	24.4	281°51.2	00.7	Menkent	148°10.3	-36°25.5							
9	122°19.4	276°03.2	· · 04.7	278°33.7	· · 55.6	121°36.0	· · 24.5	296°53.4	· · 00.6	Arcturus	145°57.7	19°07.7							
10	137°21.9	291°04.0	05.7	293°34.6	56.2	136°38.7	24.7	311°55.6	00.5	Rigel Kent.	139°54.9	-60°53.0							
11	152°24.3	306°04.7	06.7	308°35.5	56.8	151°41.5	24.8	326°57.8	00.4	Kochab	137°20.3	74°06.9							
12	167°26.8	321°05.4	S15°07.6	323°36.4	S09°57.5	166°44.2	S01°24.9	342°00.0	N00°00.2	Zuben'ubi	137°07.8	-16°05.2							
13	182°29.3	336°06.2	08.6	338°37.3	58.1	181°47.0	25.1	357°02.2	N00°00.1	Alphecca	126°12.8	26°40.9							
14	197°31.7	351°06.9	09.6	353°38.2	58.7	196°49.7	25.2	12°04.4	S00°00.0	Antares	112°28.8	-26°27.4							
15	212°34.2	6°07.6	· · 10.6	8°39.2	09°59.4	211°52.5	· · 25.3	27°06.6	· · 00.1	Atria	107°32.4	-69°03.1							
16	227°36.7	21°08.3	11.5	23°40.1	10°00.0	226°55.2	25.4	42°08.8	00.2	Sabik	102°14.8	-15°44.3							
17	242°39.1	36°09.1	12.5	38°41.0	00.6	241°58.0	25.6	57°11.0	00.4	Shaula	96°24.6	-37°06.8							
18	257°41.6	51°09.8	S15°13.5	53°41.9	S10°01.2	257°00.7	S01°25.7	72°13.2	S00°00.5	Rasalhague	96°08.2	12°33.3							
19	272°44.1	66°10.6	14.4	68°42.8	01.9	272°03.5	25.8	87°15.4	00.6	Eltanin	90°46.9	51°29.6							
20	287°46.5	81°11.3	15.4	83°43.7	02.5	287°06.2	26.0	102°17.5	00.7	Kaus Aust.	83°46.3	-34°22.8							
21	302°49.0	96°12.0	· · 16.4	98°44.6	· · 03.1	302°09.0	· · 26.1	117°19.7	· · 00.8	Vega	80°40.2	38°47.9							
22	317°51.4	111°12.8	17.3	113°45.6	03.7	317°11.7	26.2	132°21.9	01.0	Nunki	76°00.6	-26°17.0							
23	332°53.9	126°13.5	18.3	128°46.5	04.4	332°14.5	26.4	147°24.1	01.1	Altair	62°10.0	8°54.0							
Mer.pass. 00:52				ν 0.7' d 1.0' m -4.67				ν 0.9' d 0.6' m 1.52				ν 2.7' d 0.1' m -2.92				ν 2.2' d -0.1' m 0.97			

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	347°56.4	141°14.3	S15°19.2	143°47.4	S10°05.0	347°17.2	S01°26.5	162°26.3	S00°01.2	Sep 07 Tue	SHA	Mer.pass	
1	2°58.8	156°15.0	20.2	158°48.3	05.6	2°20.0	26.6	177°28.5	01.3	Venus	154°41.8	14:37	
2	18°01.3	171°15.8	21.2	173°49.2	06.2	17°22.7	26.7	192°30.7	01.4	Mars	157°05.3	14:27	
3	33°03.8	186°16.5	· · 22.1	188°50.1	· · 06.9	32°25.5	· · 26.9	207°32.9	· · 01.6	Jupiter	359°07.2	00:59	
4	48°06.2	201°17.2	23.1	203°51.0	07.5	47°28.2	27.0	222°35.1	01.7	Saturn	174°43.0	13:15	
5	63°08.7	216°18.0	24.1	218°51.9	08.1	62°31.0	27.1	237°37.3	01.8				
6	78°11.2	231°18.7	S15°25.0	233°52.8	S10°08.7	77°33.7	S01°27.3	252°39.5	S00°01.9				
7	93°13.6	246°19.5	26.0	248°53.8	09.4	92°36.5	27.4	267°41.7	02.0				
8	108°16.1	261°20.3	26.9	263°54.7	10.0	107°39.2	27.5	282°43.8	02.2				
9	123°18.6	276°21.0	· · 27.9	278°55.6	· · 10.6	122°42.0	· · 27.7	297°46.0	· · 02.3	Sep 08 Wed	SHA	Mer.pass	
10	138°21.0	291°21.8	28.8	293°56.5	11.3	137°44.7	27.8	312°48.2	02.4	Venus	153°59.5	14:36	
11	153°23.5	306°22.5	29.8	308°57.4	11.9	152°47.5	27.9	327°50.4	02.5	Mars	156°28.2	14:25	
12	168°25.9	321°23.3	S15°30.8	323°58.3	S10°12.5	167°50.2	S01°28.0	342°52.6	S00°02.6	Jupiter	359°14.0	00:55	
13	183°28.4	336°24.0	31.7	338°59.2	13.1	182°53.0	28.2	357°54.8	02.8	Saturn	174°36.5	13:12	
14	198°30.9	351°24.8	32.7	354°00.1	13.8	197°55.7	28.3	12°57.0	02.9				
15	213°33.3	6°25.6	· · 33.6	9°01.0	· · 14.4	212°58.5	· · 28.4	27°59.2	· · 03.0	Sep 09 Thu	SHA	Mer.pass	
16	228°35.8	21°26.3	34.6	24°01.9	15.0	228°01.2	28.6	43°01.4	03.1	Venus	153°17.9	14:34	
17	243°38.3	36°27.1	35.5	39°02.8	15.6	243°04.0	28.7	58°03.6	03.2	Mars	155°51.0	14:24	
18	258°40.7	51°27.8	S15°36.5	54°03.8	S10°16.3	258°06.8	S01°28.8	73°05.8	S00°03.4	Jupiter	359°20.8	00:51	
19	273°43.2	66°28.6	37.4	69°04.7	16.9	273°09.5	29.0	88°07.9	03.5	</			

h	Sun			Moon			
Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	180°26.8	N06°11.2	201°04.7	8.0'	N10°17.2	-14.6'	61.1'
1	195°27.0	10.2	215°31.6	8.0'	10°02.6	-14.7'	61.1'
2	210°27.2	09.3	229°58.6	8.0'	09°47.9	-14.8'	61.1'
3	225°27.4	· · 08.3	244°25.7	8.1'	09°33.1	-14.8'	61.1'
4	240°27.6	07.4	258°52.7	8.1'	09°18.3	-14.9'	61.1'
5	255°27.8	06.5	273°19.8	8.1'	09°03.4	-15.0'	61.2'
6	270°28.0	N06°05.5	287°47.0	8.2'	N08°48.4	-15.0'	61.2'
7	285°28.2	04.6	302°14.1	8.2'	08°33.4	-15.1'	61.2'
8	300°28.5	03.7	316°41.3	8.2'	08°18.3	-15.1'	61.2'
9	315°28.7	· · 02.7	331°08.5	8.3'	08°03.1	-15.2'	61.2'
10	330°28.9	01.8	345°35.8	8.3'	07°47.9	-15.3'	61.2'
11	345°29.1	06°00.9	0°03.1	8.3'	07°32.7	-15.3'	61.2'
12	0°29.3	N05°59.9	14°30.4	8.3'	N07°17.4	-15.4'	61.2'
13	15°29.5	59.0	28°57.7	8.4'	07°02.0	-15.4'	61.2'
14	30°29.7	58.0	43°25.1	8.4'	06°46.6	-15.5'	61.3'
15	45°29.9	· · 57.1	57°52.5	8.4'	06°31.1	-15.5'	61.3'
16	60°30.2	56.2	72°19.9	8.4'	06°15.7	-15.5'	61.3'
17	75°30.4	55.2	86°47.4	8.5'	06°00.1	-15.6'	61.3'
18	90°30.6	N05°54.3	101°14.9	8.5'	N05°44.5	-15.6'	61.3'
19	105°30.8	53.4	115°42.4	8.5'	05°28.9	-15.7'	61.3'
20	120°31.0	52.4	130°09.9	8.5'	05°13.3	-15.7'	61.3'
21	135°31.2	· · 51.5	144°37.4	8.6'	04°57.6	-15.7'	61.3'
22	150°31.4	50.5	159°05.0	8.6'	04°41.9	-15.7'	61.3'
23	165°31.6	49.6	173°32.6	8.6'	04°26.1	-15.8'	61.3'
SD = 15.9'			SD = 16.7'				

Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	180°31.9	N05°48.7	188°00.2	8.6'	N04°10.3	-15.8'	61.3'
1	195°32.1	47.7	202°27.8	8.6'	03°54.5	-15.8'	61.3'
2	210°32.3	46.8	216°55.5	8.7'	03°38.7	-15.9'	61.3'
3	225°32.5	· · 45.9	231°23.1	8.7'	03°22.9	-15.9'	61.3'
4	240°32.7	44.9	245°50.8	8.7'	03°07.0	-15.9'	61.3'
5	255°32.9	44.0	260°18.5	8.7'	02°51.1	-15.9'	61.3'
6	270°33.1	N05°43.0	274°46.2	8.7'	N02°35.2	-15.9'	61.3'
7	285°33.4	42.1	289°14.0	8.7'	02°19.3	-15.9'	61.3'
8	300°33.6	41.2	303°41.7	8.8'	02°03.3	-15.9'	61.3'
9	315°33.8	· · 40.2	318°09.5	8.8'	01°47.4	-16.0'	61.3'
10	330°34.0	39.3	332°37.3	8.8'	01°31.4	-16.0'	61.3'
11	345°34.2	38.3	347°05.0	8.8'	01°15.4	-16.0'	61.3'
12	0°34.4	N05°37.4	1°32.8	8.8'	N00°59.5	-16.0'	61.3'
13	15°34.6	36.5	16°00.7	8.8'	00°43.5	-16.0'	61.3'
14	30°34.9	35.5	30°28.5	8.8'	00°27.5	-16.0'	61.3'
15	45°35.1	· · 34.6	44°56.3	8.8'	N00°11.6	-16.0'	61.3'
16	60°35.3	33.6	59°24.1	8.8'	S00°04.4	16.0'	61.3'
17	75°35.5	32.7	73°52.0	8.9'	00°20.4	16.0'	61.3'
18	90°35.7	N05°31.7	88°19.8	8.9'	S00°36.3	16.0'	61.3'
19	105°35.9	30.8	102°47.7	8.9'	00°52.3	15.9'	61.2'
20	120°36.1	29.9	117°15.6	8.9'	01°08.2	15.9'	61.2'
21	135°36.4	· · 28.9	131°43.4	8.9'	01°24.2	15.9'	61.2'
22	150°36.6	28.0	146°11.3	8.9'	01°40.1	15.9'	61.2'
23	165°36.8	27.0	160°39.2	8.9'	01°56.0	15.9'	61.2'
SD = 15.9'			SD = 16.7'				

Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	180°37.0	N05°26.1	175°07.1	8.9'	S02°11.8	15.9'	61.2'
1	195°37.2	25.2	189°34.9	8.9'	02°27.7	15.8'	61.2'
2	210°37.4	24.2	204°02.8	8.9'	02°43.5	15.8'	61.2'
3	225°37.7	· · 23.3	218°30.7	8.9'	02°59.4	15.8'	61.2'
4	240°37.9	22.3	232°58.6	8.9'	03°15.1	15.8'	61.1'
5	255°38.1	21.4	247°26.4	8.9'	03°30.9	15.7'	61.1'
6	270°38.3	N05°20.4	261°54.3	8.9'	S03°46.6	15.7'	61.1'
7	285°38.5	19.5	276°22.2	8.9'	04°02.3	15.7'	61.1'
8	300°38.7	18.5	290°50.1	8.9'	04°18.0	15.6'	61.1'
9	315°38.9	· · 17.6	305°17.9	8.9'	04°33.6	15.6'	61.1'
10	330°39.2	16.7	319°45.8	8.9'	04°49.2	15.6'	61.0'
11	345°39.4	15.7	334°13.6	8.8'	05°04.8	15.5'	61.0'
12	0°39.6	N05°14.8	348°41.5	8.8'	S05°20.3	15.5'	61.0'
13	15°39.8	13.8	3°09.3	8.8'	05°35.8	15.4'	61.0'
14	30°40.0	12.9	17°37.1	8.8'	05°51.2	15.4'	61.0'
15	45°40.2	· · 11.9	32°05.0	8.8'	06°06.6	15.3'	61.0'
16	60°40.5	11.0	46°32.8	8.8'	06°21.9	15.3'	60.9'
17	75°40.7	10.0	61°00.6	8.8'	06°37.2	15.2'	60.9'
18	90°40.9	N05°09.1	75°28.4	8.8'	S06°52.4	15.2'	60.9'
19	105°41.1	08.2	89°56.1	8.8'	07°07.6	15.1'	60.9'
20	120°41.3	07.2	104°23.9	8.8'	07°22.7	15.1'	60.9'
21	135°41.5	· · 06.3	118°51.6	8.7'	07°37.8	15.0'	60.8'
22	150°41.8	05.3	133°19.4	8.7'	07°52.8	14.9'	60.8'
23	165°42.0	04.4	147°47.1	8.7'	08°07.8	14.9'	60.8'
SD = 15.9'			SD = 16.7'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	00:32	03:16	04:34	19:18	20:35	23:01
N 70°	01:47	03:35	04:43	19:09	20:17	22:01
68°	02:22	03:50	04:51	19:02	20:02	21:28
66°	02:47	04:02	04:57	18:56	19:51	21:05
64°	03:05	04:12	05:02	18:51	19:41	20:47
62°	03:20	04:20	05:07	18:47	19:33	20:32
60°	03:33	04:27	05:11	18:43	19:26	20:21
N 58°	03:43	04:34	05:14	18:40	19:20	20:11
56°	03:52	04:39	05:18	18:37	19:15	20:02
54°	03:59	04:44	05:20	18:34	19:10	19:54
52°	04:06	04:48	05:23	18:32	19:06	19:48
50°	04:12	04:52	05:25	18:29	19:02	19:42
45°	04:24	05:00	05:30	18:25	18:54	19:30
N 40°	04:34	05:07	05:34	18:21	18:48	19:20
35°	04:42	05:12	05:38	18:17	18:43	19:13
30°	04:48	05:17	05:41	18:14	18:38	19:07
20°	04:58	05:24	05:46	18:09	18:31	18:57
N 10°	05:05	05:29	05:50	18:05	18:26	18:51
0°	05:10	05:34	05:54	18:01	18:22	18:46
S 10°	05:13	05:37	05:58	17:57	18:18	18:43
20°	05:15	05:41	06:03	17:53	18:15	18:41
30°	05:16	05:43	06:07	17:49	18:13	18:40
35°	05:15	05:44	06:10	17:46	18:11	18:41
40°	05:14	05:46	06:13	17:43	18:10	18:42
45°	05:13	05:47	06:16	17:40	18:09	18:43
S 50°	05:10	05:48	06:20	17:36	18:09	18:46
52°	05:09	05:48	06:22	17:35	18:08	18:47
54°	05:08	05:48	06:24	17:33	18:08	18:49
56°	05:06	05:49	06:26	17:31	18:08	18:51
58°	05:04	05:49	06:28	17:28	18:08	18:53
S 60°	05:01	05:49	06:31	17:26	18:07	18:56

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	02:38	05:06	07:30	18:24	17:57	17:29
N 70°	02:54	05:10	07:23	18:17	17:58	17:39
68°	03:06	05:13	07:18	18:11	17:59	17:47
66°	03:17	05:16	07:14	18:05	18:00	17:54
64°	03:25	05:18	07:10	18:01	18:00	17:59
62°	03:32	05:20	07:07	17:57	18:01	18:04
60°	03:39	05:22	07:04	17:54	18:01	18:09
N 58°	03:44	05:23	07:02	17:51	18:01	18:12
56°	03:49	05:25	07:00	17:48	18:02	18:16
54°	03:53	05:26	06:58	17:45	18:02	18:19
52°	03:57	05:27	06:56	17:43	18:02	18:22
50°	04:01	05:28	06:55	17:41	18:03	18:24
45°	04:08	05:30	06:51	17:37	18:03	18:30
N 40°	04:15	05:32	06:48	17:33	18:04	18:35
35°	04:20	05:33	06:46	17:30	18:04	18:39
30°	04:25	05:35	06:44	17:27	18:05	18:42
20°	04:33	05:37	06:40	17:21	18:05	18:49
N 10°	04:40	05:39	06:37	17:17	18:06	18:54
0°	04:47	05:41	06:34	17:13	18:06	19:00
S 10°	04:54	05:43	06:31	17:08	18:07	19:05
20°	05:01	05:45	06:28	17:03	18:07	19:10
30°	05:09	05:47	06:25	16:58	18:08	19:17
35°	05:14	05:49	06:23	16:55	18:08	19:20
40°	05:19	05:50	06:21	16:51	18:08	19:25
45°	05:25	05:52	06:19	16:47	18:09	19:30
S 50°	05:32	05:54	06:16	16:42	18:09	19:35
52°	05:35	05:55	06:14	16:40	18:09	19:38
54°	05:39	05:56	06:13	16:37	18:09	19:41
56°	05:43	05:58	06:11	16:34	18:10	19:44
58°	05:48	05:59	06:10	16:31	18:10	19:48
S 60°	05:53	06:00	06:08	16:27	18:10	19:52

Day	Sun		Mer. Pass	Moon		Age 28-1 3-1%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
07	01:47	01:57	11:58	11:00	23:27	
08	02:07	02:18	11:58	11:54	--:-	
09	02:28	02:38	11:57	12:47	00:20	

September 10, 11, 12 UT (Fri., Sat., Sun.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Fri	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec			
0	348°55.5	141°32.5	S15°42.2	144°09.2	S10°20.0	348°23.3	S01°29.6	163°18.9	S00°04.1				Alpheratz	357°45.3	29°09.2		
1	3°58.0	156°33.2	43.1	159°10.1	20.6	3°26.0	29.7	178°21.1	04.2				Ankaa	353°17.2	-42°14.6		
2	19°00.4	171°34.0	44.1	174°11.0	21.2	18°28.8	29.9	193°23.3	04.3				Schedar	349°42.4	56°35.8		
3	34°02.9	186°34.8	•• 45.0	189°11.9	•• 21.9	33°31.5	•• 30.0	208°25.5	•• 04.4				Diphda	348°57.5	-17°55.4		
4	49°05.4	201°35.6	46.0	204°12.8	22.5	48°34.3	30.1	223°27.7	04.5				Achernar	335°27.6	-57°10.6		
5	64°07.8	216°36.4	46.9	219°13.7	23.1	63°37.0	30.3	238°29.9	04.7				Hamal	328°02.8	23°30.9		
6	79°10.3	231°37.1	S15°47.8	234°14.6	S10°23.7	78°39.8	S01°30.4	253°32.0	S00°04.8				Polaris	318°31.1	89°18.5		
7	94°12.8	246°37.9	48.8	249°15.5	24.4	93°42.6	30.5	268°34.2	04.9				Acamar	315°19.6	-40°15.4		
8	109°15.2	261°38.7	49.7	264°16.4	25.0	108°45.3	30.7	283°36.4	05.0				Menkar	314°17.0	4°08.1		
9	124°17.7	276°39.5	•• 50.7	279°17.3	•• 25.6	123°48.1	•• 30.8	298°38.6	•• 05.1				Mirfak	308°43.1	49°53.9		
10	139°20.2	291°40.3	51.6	294°18.2	26.2	138°50.8	30.9	313°40.8	05.3				Aldebaran	290°51.6	16°31.9		
11	154°22.6	306°41.1	52.6	309°19.2	26.9	153°53.6	31.1	328°43.0	05.4				Rigel	281°14.0	-8°11.2		
12	169°25.1	321°41.8	S15°53.5	324°20.1	S10°27.5	168°56.3	S01°31.2	343°45.2	S00°05.5				Capella	280°37.4	46°00.4		
13	184°27.5	336°42.6	54.4	339°21.0	28.1	183°59.1	31.3	358°47.4	05.6				Bellatrix	278°34.2	6°21.7		
14	199°30.0	351°43.4	55.4	354°21.9	28.7	199°01.8	31.5	13°49.6	05.7				Elnath	278°15.2	28°36.9		
15	214°32.5	6°44.2	•• 56.3	9°22.8	•• 29.4	214°04.6	•• 31.6	28°51.8	•• 05.9				Alnilam	275°48.4	-1°11.6		
16	229°34.9	21°45.0	57.2	24°23.7	30.0	229°07.4	31.7	43°53.9	06.0				Betelgeuse	271°03.5	7°24.6		
17	244°37.4	36°45.8	58.2	39°24.6	30.6	244°10.1	31.8	58°56.1	06.1				Canopus	263°57.2	-52°41.8		
18	259°39.9	51°46.6	S15°59.1	54°25.5	S10°31.2	259°12.9	S01°32.0	73°58.3	S00°06.2				Sirius	258°35.6	-16°43.7		
19	274°42.3	66°47.4	16°00.0	69°26.4	31.8	274°15.6	32.1	89°00.5	06.3				Adhara	255°14.3	-28°59.0		
20	289°44.8	81°48.2	01.0	84°27.3	32.5	289°18.4	32.2	104°02.7	06.5				Procyon	245°02.0	5°11.9		
21	304°47.3	96°49.0	•• 01.9	99°28.2	•• 33.1	304°21.1	•• 32.4	119°04.9	•• 06.6				Pollux	243°30.4	28°00.0		
22	319°49.7	111°49.8	02.8	114°29.1	33.7	319°23.9	32.5	134°07.1	06.7				Avior	234°19.4	-59°32.4		
23	334°52.2	126°50.6	03.8	129°30.0	34.3	334°26.7	32.6	149°09.3	06.8				Suhail	222°54.4	-43°28.4		
Mer.pass. 00:44			ν 0.8' d 0.9' m -4.69			ν 0.9' d 0.6' m 1.52			ν 2.8' d 0.1' m -2.92			ν 2.2' d 0.1' m 0.96					

Sat	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec						
0	349°54.6	141°51.4	S16°04.7	144°30.9	S10°35.0	349°29.4	S01°32.8	164°11.5	S00°06.9					
1	4°57.1	156°52.2	05.6	159°31.8	35.6	4°32.2	32.9	179°13.7	07.1					
2	19°59.6	171°53.0	06.6	174°32.7	36.2	19°34.9	33.0	194°15.8	07.2					
3	35°02.0	186°53.8	•• 07.5	189°33.6	•• 36.8	34°37.7	•• 33.2	209°18.0	•• 07.3					
4	50°04.5	201°54.6	08.4	204°34.5	37.4	49°40.4	33.3	224°20.2	07.4					
5	65°07.0	216°55.5	09.4	219°35.4	38.1	64°43.2	33.4	239°22.4	07.5					
6	80°09.4	231°56.3	S16°10.3	234°36.3	S10°38.7	79°45.9	S01°33.6	254°24.6	S00°07.7					
7	95°11.9	246°57.1	11.2	249°37.2	39.3	94°48.7	33.7	269°26.8	07.8					
8	110°14.4	261°57.9	12.1	264°38.1	39.9	109°51.5	33.8	284°29.0	07.9					
9	125°16.8	276°58.7	•• 13.1	279°39.0	•• 40.5	124°54.2	•• 34.0	299°31.2	•• 08.0					
10	140°19.3	291°59.5	14.0	294°39.9	41.2	139°57.0	34.1	314°33.4	08.1					
11	155°21.8	307°00.4	14.9	309°40.8	41.8	154°59.7	34.2	329°35.5	08.3					
12	170°24.2	322°01.2	S16°15.8	324°41.7	S10°42.4	170°02.5	S01°34.3	344°37.7	S00°08.4					
13	185°26.7	337°02.0	16.7	339°42.6	43.0	185°05.3	34.5	359°39.9	08.5					
14	200°29.1	352°02.8	17.7	354°43.5	43.7	200°08.0	34.6	14°42.1	08.6					
15	215°31.6	7°03.7	•• 18.6	9°44.4	•• 44.3	215°10.8	•• 34.7	29°44.3	•• 08.7					
16	230°34.1	22°04.5	19.5	24°45.3	44.9	230°13.5	34.9	44°46.5	08.9					
17	245°36.5	37°05.3	20.4	39°46.2	45.5	245°16.3	35.0	59°48.7	09.0					
18	260°39.0	52°06.1	S16°21.3	54°47.1	S10°46.1	260°19.0	S01°35.1	74°50.9	S00°09.1					
19	275°41.5	67°07.0	22.3	69°48.0	46.8	275°21.8	35.3	89°53.1	09.2					
20	290°43.9	82°07.8	23.2	84°48.9	47.4	290°24.6	35.4	104°55.3	09.3					
21	305°46.4	97°08.6	•• 24.1	99°49.8	•• 48.0	305°27.3	•• 35.5	119°57.4	•• 09.5					
22	320°48.9	112°09.5	25.0	114°50.7	48.6	320°30.1	35.7	134°59.6	09.6					
23	335°51.3	127°10.3	25.9	129°51.6	49.2	335°32.8	35.8	150°01.8	09.7					
Mer.pass. 00:40			ν 0.8' d 0.9' m -4.70			ν 0.9' d 0.6' m 1.52			ν 2.8' d 0.1' m -2.92			ν 2.2' d 0.1' m 0.96		

Sun	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec						
0	350°53.8	142°11.1	S16°26.8	144°52.4	S10°49.9	350°35.6	S01°35.9	165°04.0	S00°09.8					
1	5°56.2	157°12.0	27.7	159°53.3	50.5	5°38.4	36.1	180°06.2	09.9					
2	20°58.7	172°12.8	28.7	174°54.2	51.1	20°41.1	36.2	195°08.4	10.1					
3	36°01.2	187°13.7	•• 29.6	189°55.1	•• 51.7	35°43.9	•• 36.3	210°10.6	•• 10.2					
4	51°03.6	202°14.5	30.5	204°56.0	52.3	50°46.6	36.5	225°12.8	10.3					
5	66°06.1	217°15.4	31.4	219°56.9	53.0	65°49.4	36.6	240°15.0	10.4					
6	81°08.6	232°16.2	S16°32.3	234°57.8	S10°53.6	80°52.2	S01°36.7	255°17.1	S00°10.5					
7	96°11.0	247°17.0	33.2	249°58.7	54.2	95°54.9	36.9	270°19.3	10.7					
8	111°13.5	262°17.9	34.1	264°59.6	54.8	110°57.7	37.0	285°21.5	10.8					
9	126°16.0	277°18.7	•• 35.0	280°00.5	•• 55.4	126°00.4	•• 37.1	300°23.7	•• 10.9					
10	141°18.4	292°19.6	35.9	295°01.4	56.1	141°03.2	37.3	315°25.9	11.0					
11	156°20.9	307°20.5	36.8	310°02.3	56.7	156°06.0	37.4	330°28.1	11.1					
12	171°23.4	322°21.3	S16°37.7	325°03.2	S10°57.3	171°08.7	S01°37.5	345°30.3	S00°11.3					
13	186°25.8	337°22.2	38.6	340°04.1	57.9	186°11.5	37.7	0°32.5	11.4					
14	201°28.3	352°23.0	39.5	355°05.0	58.5	201°14.2	37.8	15°34.6	11.5					
15	216°30.7	7°23.9	•• 40.4	10°05.9	•• 59.1	216°17.0	•• 37.9	30°36.8	•• 11.6					
16	231°33.2	22°24.7	41.3	25°06.7	10°59.8	231°19.8	38.1	45°39.0	11.7					
17	246°35.7	37°25.6	42.2	40°07.6	11°00.4	246°22.5	38.2	60°41.2	11.9					
18	261°38.1	52°26.5	S16°43.1	55°08.5	S11°01.0	261°25.3	S01°38.3	75°43.4	S00°12.0					
19	276°40.6	67°27.3	44.0	70°09.4	01.6	276°28.0	38.5	90°45.6	12.1					
20	291°43.1	82°28.2	44.9	85°10.3	02.2	291°30.8	38.6	105°47.8	12.2					
21	306°45.5	97°29.1	•• 45.8	100°11.2	•• 02.9	306°33.6	•• 38.7	120°50.0	•• 12.3					
22	321°48.0	112°29.9	46.7	115°12.1	03.5	321°36.3	38.9	135°52.2	12.5					
23	336°50.5	127°30.8	47.6	130°13.0	04.1	336°39.1	39.0	150°54.3	12.6					
Mer.pass. 00:36			ν 0.8' d 0.9' m -4.71			ν 0.9' d 0.6' m 1.52			ν 2.8' d 0.1' m -2.93			ν 2.2' d 0.1' m 0.96		

Sep 10 Fri	SHA	Mer.pass
Venus	152°37.0	14:33
Mars	155°13.7	14:23
Jupiter	359°27.8	00:46
Saturn	174°23.4	13:05

Sep 11 Sat	SHA	Mer.pass
Venus	151°56.8	14:32
Mars	154°36.2	14:21
Jupiter	359°34.8	00:42
Saturn	174°16.8	13:01

Sep 12 Sun	SHA	Mer.pass
Venus	151°17.4	14:30
Mars	153°58.7	14:20
Jupiter	359°41.8	00:38
Saturn	174°10.2	12:58

Horizontal parallax		
Venus:	0.3	
Mars:	0.1	

h	Sun			Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	180°42.2	N05°03.4	162°14.8	8.7'	S08°22.6	14.8'	60.8'
1	195°42.4	02.5	176°42.5	8.7'	08°37.5	14.8'	60.7'
2	210°42.6	01.5	191°10.2	8.7'	08°52.2	14.7'	60.7'
3	225°42.8	05°00.6	205°37.8	8.6'	09°06.9	14.6'	60.7'
4	240°43.1	04°59.6	220°05.5	8.6'	09°21.5	14.6'	60.7'
5	255°43.3	58.7	234°33.1	8.6'	09°36.1	14.5'	60.6'
6	270°43.5	N04°57.7	249°00.7	8.6'	S09°50.6	14.4'	60.6'
7	285°43.7	56.8	263°28.3	8.6'	10°05.0	14.3'	60.6'
8	300°43.9	55.9	277°55.9	8.6'	10°19.3	14.3'	60.6'
9	315°44.1	· · 54.9	292°23.4	8.5'	10°33.6	14.2'	60.5'
10	330°44.4	54.0	306°51.0	8.5'	10°47.8	14.1'	60.5'
11	345°44.6	53.0	321°18.5	8.5'	11°01.9	14.0'	60.5'
12	0°44.8	N04°52.1	335°46.0	8.5'	S11°15.9	13.9'	60.4'
13	15°45.0	51.1	350°13.5	8.4'	11°29.8	13.9'	60.4'
14	30°45.2	50.2	4°40.9	8.4'	11°43.7	13.8'	60.4'
15	45°45.5	· · 49.2	19°08.3	8.4'	11°57.4	13.7'	60.4'
16	60°45.7	48.3	33°35.7	8.4'	12°11.1	13.6'	60.3'
17	75°45.9	47.3	48°03.1	8.4'	12°24.7	13.5'	60.3'
18	90°46.1	N04°46.4	62°30.5	8.3'	S12°38.2	13.4'	60.3'
19	105°46.3	45.4	76°57.8	8.3'	12°51.6	13.3'	60.2'
20	120°46.5	44.5	91°25.1	8.3'	13°04.9	13.2'	60.2'
21	135°46.8	· · 43.5	105°52.4	8.3'	13°18.1	13.1'	60.2'
22	150°47.0	42.6	120°19.7	8.2'	13°31.3	13.0'	60.1'
23	165°47.2	41.6	134°46.9	8.2'	13°44.3	12.9'	60.1'
SD = 15.9' d = -0.9' SD = 16.6'							

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	180°47.4	N04°40.7	149°14.1	8.2'	S13°57.2	12.8'	60.1'
1	195°47.6	39.7	163°41.3	8.2'	14°10.0	12.7'	60.0'
2	210°47.9	38.8	178°08.5	8.1'	14°22.8	12.6'	60.0'
3	225°48.1	· · 37.8	192°35.6	8.1'	14°35.4	12.5'	60.0'
4	240°48.3	36.9	207°02.8	8.1'	14°47.9	12.4'	59.9'
5	255°48.5	35.9	221°29.9	8.1'	15°00.3	12.3'	59.9'
6	270°48.7	N04°35.0	235°56.9	8.0'	S15°12.6	12.2'	59.9'
7	285°48.9	34.0	250°24.0	8.0'	15°24.8	12.1'	59.8'
8	300°49.2	33.1	264°51.0	8.0'	15°36.9	12.0'	59.8'
9	315°49.4	· · 32.1	279°18.0	8.0'	15°48.9	11.9'	59.8'
10	330°49.6	31.2	293°45.0	7.9'	16°00.8	11.8'	59.7'
11	345°49.8	30.2	308°11.9	7.9'	16°12.5	11.6'	59.7'
12	0°50.0	N04°29.3	322°38.8	7.9'	S16°24.2	11.5'	59.6'
13	15°50.3	28.3	337°05.7	7.9'	16°35.7	11.4'	59.6'
14	30°50.5	27.4	351°32.6	7.8'	16°47.1	11.3'	59.6'
15	45°50.7	· · 26.4	5°59.4	7.8'	16°58.4	11.2'	59.5'
16	60°50.9	25.5	20°26.2	7.8'	17°09.6	11.1'	59.5'
17	75°51.1	24.5	34°53.0	7.8'	17°20.6	10.9'	59.5'
18	90°51.4	N04°23.6	49°19.8	7.7'	S17°31.6	10.8'	59.4'
19	105°51.6	22.6	63°46.6	7.7'	17°42.4	10.7'	59.4'
20	120°51.8	21.7	78°13.3	7.7'	17°53.1	10.6'	59.3'
21	135°52.0	· · 20.7	92°40.0	7.7'	18°03.7	10.4'	59.3'
22	150°52.2	19.7	107°06.7	7.7'	18°14.1	10.3'	59.3'
23	165°52.5	18.8	121°33.3	7.6'	18°24.4	10.2'	59.2'
SD = 15.9' d = -0.9' SD = 16.4'							

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	180°52.7	N04°17.8	136°00.0	7.6'	S18°34.6	10.1'	59.2'
1	195°52.9	16.9	150°26.6	7.6'	18°44.7	9.9'	59.2'
2	210°53.1	15.9	164°53.2	7.6'	18°54.6	9.8'	59.1'
3	225°53.3	· · 15.0	179°19.7	7.5'	19°04.4	9.7'	59.1'
4	240°53.6	14.0	193°46.3	7.5'	19°14.1	9.6'	59.0'
5	255°53.8	13.1	208°12.8	7.5'	19°23.7	9.4'	59.0'
6	270°54.0	N04°12.1	222°39.3	7.5'	S19°33.1	9.3'	59.0'
7	285°54.2	11.2	237°05.8	7.5'	19°42.4	9.2'	58.9'
8	300°54.4	10.2	251°32.3	7.5'	19°51.5	9.0'	58.9'
9	315°54.7	· · 09.3	265°58.7	7.4'	20°00.6	8.9'	58.8'
10	330°54.9	08.3	280°25.2	7.4'	20°09.4	8.8'	58.8'
11	345°55.1	07.4	294°51.6	7.4'	20°18.2	8.6'	58.8'
12	0°55.3	N04°06.4	309°18.0	7.4'	S20°26.8	8.5'	58.7'
13	15°55.5	05.4	323°44.4	7.4'	20°35.3	8.3'	58.7'
14	30°55.8	04.5	338°10.8	7.4'	20°43.6	8.2'	58.6'
15	45°56.0	· · 03.5	352°37.1	7.3'	20°51.8	8.1'	58.6'
16	60°56.2	02.6	7°03.5	7.3'	20°59.9	7.9'	58.6'
17	75°56.4	01.6	21°29.8	7.3'	21°07.8	7.8'	58.5'
18	90°56.6	N04°00.7	35°56.1	7.3'	S21°15.6	7.7'	58.5'
19	105°56.9	03°59.7	50°22.4	7.3'	21°23.3	7.5'	58.4'
20	120°57.1	58.8	64°48.7	7.3'	21°30.8	7.4'	58.4'
21	135°57.3	· · 57.8	79°15.0	7.3'	21°38.2	7.2'	58.4'
22	150°57.5	56.8	93°41.3	7.3'	21°45.4	7.1'	58.3'
23	165°57.7	55.9	108°07.5	7.3'	21°52.5	6.9'	58.3'
SD = 15.9' d = -1.0' SD = 16.1'							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	01:26	03:33	04:48	19:02	20:17	22:17
N 70°	02:11	03:49	04:56	18:55	20:01	21:36
68°	02:40	04:02	05:02	18:49	19:48	21:09
66°	03:01	04:13	05:07	18:45	19:38	20:49
64°	03:17	04:21	05:11	18:40	19:30	20:33
62°	03:30	04:29	05:15	18:37	19:23	20:20
60°	03:41	04:35	05:18	18:34	19:17	20:10
N 58°	03:51	04:41	05:21	18:31	19:11	20:01
56°	03:59	04:45	05:23	18:29	19:07	19:53
54°	04:06	04:50	05:26	18:27	19:02	19:46
52°	04:12	04:53	05:28	18:25	18:59	19:40
50°	04:17	04:57	05:30	18:23	18:55	19:35
45°	04:29	05:04	05:34	18:19	18:49	19:24
N 40°	04:37	05:10	05:37	18:16	18:43	19:15
35°	04:44	05:14	05:40	18:13	18:38	19:08
30°	04:50	05:18	05:42	18:11	18:35	19:03
20°	04:59	05:24	05:47	18:07	18:29	18:54
N 10°	05:05	05:29	05:50	18:03	18:24	18:49
0°	05:09	05:33	05:53	18:00	18:21	18:45
S 10°	05:11	05:36	05:57	17:57	18:18	18:42
20°	05:12	05:38	06:00	17:54	18:16	18:41
30°	05:12	05:40	06:04	17:50	18:14	18:42
35°	05:11	05:40	06:06	17:48	18:14	18:43
40°	05:09	05:41	06:08	17:46	18:13	18:45
45°	05:07	05:41	06:10	17:44	18:13	18:47
S 50°	05:04	05:41	06:13	17:41	18:13	18:50
52°	05:02	05:41	06:15	17:39	18:13	18:52
54°	05:00	05:41	06:16	17:38	18:13	18:54
56°	04:58	05:41	06:18	17:36	18:13	18:57
58°	04:55	05:41	06:20	17:35	18:14	18:59
S 60°	04:52	05:41	06:22	17:33	18:14	19:03

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	09:59	13:04	■	16:54	15:45	■
N 70°	09:39	12:10	■	17:16	16:41	■
68°	09:24	11:37	14:11	17:33	17:15	16:37
66°	09:12	11:13	13:20	17:47	17:40	17:30
64°	09:02	10:55	12:48	17:59	17:59	18:02
62°	08:54	10:40	12:24	18:09	18:15	18:27
60°	08:46	10:28	12:06	18:17	18:29	18:46
N 58°	08:40	10:17	11:50	18:25	18:40	19:02
56°	08:34	10:07	11:37	18:31	18:50	19:16
54°	08:29	09:59	11:26	18:37	18:59	19:27
52°	08:25	09:52	11:16	18:43	19:07	19:38
50°	08:21	09:45	11:07	18:48	19:15	19:47
45°	08:12	09:31	10:48	18:58	19:30	20:07
N 40°	08:05	09:20	10:32	19:07	19:43	20:23
35°	07:58	09:10	10:20	19:15	19:54	20:37
30°	07:53	09:01	10:08	19:22	20:04	20:49
20°	07:44	08:47	09:49	19:33	20:20	21:09
N 10°	07:35	08:34	09:33	19:44	20:35	21:27
0°	07:28	08:22	09:17	19:53	20:48	21:44
S 10°	07:20	08:10	09:02	20:03	21:02	22:01
20°	07:12	07:58	08:46	20:14	21:17	22:19
30°	07:03	07:44	08:27	20:26	21:33	22:40
35°	06:58	07:36	08:17	20:32	21:43	22:52
40°	06:53	07:26	08:04	20:40	21:55	23:06
45°	06:46	07:16	07:50	20:50	22:08	23:22
S 50°	06:38	07:03	07:32	21:01	22:24	23:43
52°	06:34	06:57	07:24	21:06	22:32	23:53
54°	06:30	06:50	07:15	21:12	22:41	· · ·
56°	06:26	06:43	07:05	21:18	22:50	· · ·
58°	06:21	06:35	06:53	21:26	23:01	· · ·
S 60°	06:16	06:26	06:40	21:34	23:14	· · ·

Day	Sun		Mer.	Moon		Age
	Eqn. of Time	12h	Pass	Mer. Pass.	Lower	2-4
	00h		hh:mm	Upper	hh:mm	4-18%
	mm:ss	mm:ss		hh:mm	hh:mm	
10	02:49	02:59	11:57	13:41	01:14	
11	03:10	03:20	11:57	14:35	02:08	
12	03:31	03:41	11:56	15:31	03:03	

September 13, 14, 15 UT (Mon., Tue., Wed.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	351°52.9	142°31.7	S16°48.5	145°13.9	S11°04.7	351°41.8	S01°39.1	165°56.5	S00°12.7	Alpheratz	357°45.3	29°09.2
1	6°55.4	157°32.6	49.4	160°14.8	05.3	6°44.6	39.2	180°58.7	12.8	Ankaa	353°17.1	-42°14.6
2	21°57.9	172°33.4	50.3	175°15.7	05.9	21°47.4	39.4	196°00.9	12.9	Schedar	349°42.4	56°35.9
3	37°00.3	187°34.3	· · 51.2	190°16.5	· · 06.6	36°50.1	· · 39.5	211°03.1	· · 13.1	Diphda	348°57.5	-17°55.4
4	52°02.8	202°35.2	52.1	205°17.4	07.2	51°52.9	39.6	226°05.3	13.2	Achernar	335°27.6	-57°10.6
5	67°05.2	217°36.1	53.0	220°18.3	07.8	66°55.6	39.8	241°07.5	13.3	Hamal	328°02.8	23°30.9
6	82°07.7	232°37.0	S16°53.9	235°19.2	S11°08.4	81°58.4	S01°39.9	256°09.7	S00°13.4	Polaris	318°29.7	89°18.5
7	97°10.2	247°37.8	54.8	250°20.1	09.0	97°01.2	40.0	271°11.8	13.5	Acamar	315°19.5	-40°15.4
8	112°12.6	262°38.7	55.7	265°21.0	09.7	112°03.9	40.2	286°14.0	13.7	Menkar	314°17.0	4°08.1
9	127°15.1	277°39.6	· · 56.6	280°21.9	· · 10.3	127°06.7	· · 40.3	301°16.2	· · 13.8	Mirfak	308°43.0	49°53.9
10	142°17.6	292°40.5	57.4	295°22.8	10.9	142°09.5	40.4	316°18.4	13.9	Aldebaran	290°51.6	16°31.9
11	157°20.0	307°41.4	58.3	310°23.7	11.5	157°12.2	40.6	331°20.6	14.0	Rigel	281°13.9	-8°11.2
12	172°22.5	322°42.3	S16°59.2	325°24.5	S11°12.1	172°15.0	S01°40.7	346°22.8	S00°14.2	Capella	280°37.4	46°00.4
13	187°25.0	337°43.2	17°00.1	340°25.4	12.7	187°17.7	40.8	1°25.0	14.3	Bellatrix	278°34.1	6°21.7
14	202°27.4	352°44.1	01.0	355°26.3	13.4	202°20.5	41.0	16°27.2	14.4	Elnath	278°15.1	28°36.9
15	217°29.9	7°45.0	· · 01.9	10°27.2	· · 14.0	217°23.3	· · 41.1	31°29.3	· · 14.5	Alnilam	275°48.4	-1°11.6
16	232°32.4	22°45.9	02.7	25°28.1	14.6	232°26.0	41.2	46°31.5	14.6	Betelgeuse	271°03.5	7°24.6
17	247°34.8	37°46.8	03.6	40°29.0	15.2	247°28.8	41.4	61°33.7	14.8	Canopus	263°57.2	-52°41.8
18	262°37.3	52°47.7	S17°04.5	55°29.9	S11°15.8	262°31.6	S01°41.5	76°35.9	S00°14.9	Sirius	258°35.6	-16°43.7
19	277°39.7	67°48.6	05.4	70°30.8	16.4	277°34.3	41.6	91°38.1	15.0	Adhara	255°14.2	-28°59.0
20	292°42.2	82°49.5	06.3	85°31.6	17.0	292°37.1	41.8	106°40.3	15.1	Procyon	245°02.0	5°11.9
21	307°44.7	97°50.4	· · 07.2	100°32.5	· · 17.7	307°39.8	· · 41.9	121°42.5	· · 15.2	Pollux	243°30.3	27°59.9
22	322°47.1	112°51.3	08.0	115°33.4	18.3	322°42.6	42.0	136°44.7	15.4	Avior	234°19.4	-59°32.4
23	337°49.6	127°52.2	08.9	130°34.3	18.9	337°45.4	42.2	151°46.8	15.5	Suhail	222°54.3	-43°28.4
Mer.pass. 00:32		ν 0.9' d 0.9' m -4.72		ν 0.9' d 0.6' m 1.52		ν 2.8' d 0.1' m -2.93		ν 2.2' d 0.1' m 0.96		Miaplacidus	221°41.1	-69°45.5
Alphard 217°58.3 -8°42.2												
Regulus 207°45.9 11°54.9												
Dubhe 193°54.6 61°41.5												
Denebola 182°36.0 14°30.8												
Gienah 175°54.7 -17°36.1												
Acrux 173°12.4 -63°09.6												
Gacrux 172°03.8 -57°10.5												
Alioth 166°22.8 55°54.2												
Spica 158°33.6 -11°13.0												
Alkaid 153°00.7 49°15.7												
Hadar 148°51.4 -60°25.7												
Menkent 148°10.3 -36°25.5												
Arcturus 145°57.8 19°07.7												
Rigel Kent. 139°55.0 -60°52.9												
Kochab 137°20.4 74°06.9												
Zuben'ubi 137°07.8 -16°05.2												
Alphecca 126°12.8 26°40.9												
Antares 112°28.8 -26°27.4												
Atria 107°32.5 -69°03.1												
Sabik 102°14.8 -15°44.2												
Shaula 96°24.6 -37°06.8												
Rasalhague 96°08.3 12°33.3												
Eltanin 90°47.0 51°29.6												
Kaus Aust. 83°46.3 -34°22.8												
Vega 80°40.2 38°47.9												
Nunki 76°00.6 -26°17.0												
Altair 62°10.0 8°54.0												
Peacock 53°21.8 -56°42.1												
Deneb 49°32.6 45°19.4												
Enif 33°48.8 9°55.7												
Al Na'ir 27°45.7 -46°54.4												
Fomalhaut 15°25.7 -29°33.7												
Scheat 13°55.0 28°08.7												
Markab 13°40.0 15°16.0												
Sep 13 Mon SHA Mer.pass												
Venus 150°38.8 14:29												
Mars 153°21.0 14:18												
Jupiter 359°48.9 00:33												
Saturn 174°03.6 12:54												
Sep 14 Tue SHA Mer.pass												
Venus 150°01.0 14:28												
Mars 152°43.1 14:17												
Jupiter 359°56.1 00:29												
Saturn 173°57.0 12:51												
Sep 15 Wed SHA Mer.pass												
Venus 149°24.2 14:26												
Mars 152°05.1 14:15												
Jupiter 0°03.3 00:24												
Saturn 173°50.3 12:47												
Horizontal parallax												
Venus: 0.3												
Mars: 0.1												
Mer.pass. 00:25		ν 1.0' d 0.9' m -4.74		ν 0.9' d 0.6' m 1.52		ν 2.8' d 0.1' m -2.93		ν 2.2' d 0.1' m 0.95				

h		Sun			Moon		
Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	180°58.0	N03°54.9	122°33.8	7.3'	S21°59.4	6.8'	58.2'
1	195°58.2	54.0	137°00.1	7.2'	22°06.2	6.7'	58.2'
2	210°58.4	53.0	151°26.3	7.2'	22°12.9	6.5'	58.2'
3	225°58.6	· · 52.1	165°52.5	7.2'	22°19.4	6.4'	58.1'
4	240°58.8	51.1	180°18.8	7.2'	22°25.8	6.2'	58.1'
5	255°59.1	50.2	194°45.0	7.2'	22°32.0	6.1'	58.0'
6	270°59.3	N03°49.2	209°11.3	7.2'	S22°38.1	5.9'	58.0'
7	285°59.5	48.2	223°37.5	7.2'	22°44.0	5.8'	58.0'
8	300°59.7	47.3	238°03.7	7.2'	22°49.8	5.7'	57.9'
9	316°00.0	· · 46.3	252°30.0	7.2'	22°55.5	5.5'	57.9'
10	331°00.2	45.4	266°56.2	7.2'	23°01.0	5.4'	57.8'
11	346°00.4	44.4	281°22.4	7.2'	23°06.3	5.2'	57.8'
12	1°00.6	N03°43.5	295°48.7	7.2'	S23°11.6	5.1'	57.8'
13	16°00.8	42.5	310°14.9	7.3'	23°16.6	4.9'	57.7'
14	31°01.1	41.5	324°41.2	7.3'	23°21.6	4.8'	57.7'
15	46°01.3	· · 40.6	339°07.4	7.3'	23°26.3	4.6'	57.7'
16	61°01.5	39.6	353°33.7	7.3'	23°31.0	4.5'	57.6'
17	76°01.7	38.7	8°00.0	7.3'	23°35.5	4.3'	57.6'
18	91°01.9	N03°37.7	22°26.3	7.3'	S23°39.8	4.2'	57.5'
19	106°02.2	36.8	36°52.6	7.3'	23°44.0	4.0'	57.5'
20	121°02.4	35.8	51°18.9	7.3'	23°48.0	3.9'	57.5'
21	136°02.6	· · 34.8	65°45.2	7.3'	23°51.9	3.8'	57.4'
22	151°02.8	33.9	80°11.5	7.4'	23°55.7	3.6'	57.4'
23	166°03.1	32.9	94°37.9	7.4'	23°59.3	3.5'	57.3'
SD = 15.9'		$d = -1.0'$	SD = 15.9'				

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	181°03.3	N03°32.0	109°04.3	7.4'	S24°02.8	3.3'	57.3'
1	196°03.5	31.0	123°30.7	7.4'	24°06.1	3.2'	57.3'
2	211°03.7	30.0	137°57.1	7.4'	24°09.3	3.0'	57.2'
3	226°03.9	· · 29.1	152°23.5	7.4'	24°12.3	2.9'	57.2'
4	241°04.2	28.1	166°49.9	7.5'	24°15.2	2.7'	57.2'
5	256°04.4	27.2	181°16.4	7.5'	24°17.9	2.6'	57.1'
6	271°04.6	N03°26.2	195°42.9	7.5'	S24°20.5	2.4'	57.1'
7	286°04.8	25.3	210°09.4	7.5'	24°23.0	2.3'	57.0'
8	301°05.0	24.3	224°36.0	7.6'	24°25.3	2.2'	57.0'
9	316°05.3	· · 23.3	239°02.5	7.6'	24°27.4	2.0'	57.0'
10	331°05.5	22.4	253°29.1	7.6'	24°29.4	1.9'	56.9'
11	346°05.7	21.4	267°55.8	7.7'	24°31.3	1.7'	56.9'
12	1°05.9	N03°20.5	282°22.4	7.7'	S24°33.0	1.6'	56.9'
13	16°06.2	19.5	296°49.1	7.7'	24°34.6	1.4'	56.8'
14	31°06.4	18.5	311°15.8	7.8'	24°36.1	1.3'	56.8'
15	46°06.6	· · 17.6	325°42.6	7.8'	24°37.4	1.2'	56.7'
16	61°06.8	16.6	340°09.4	7.8'	24°38.5	1.0'	56.7'
17	76°07.0	15.7	354°36.2	7.9'	24°39.5	0.9'	56.7'
18	91°07.3	N03°14.7	9°03.1	7.9'	S24°40.4	0.7'	56.6'
19	106°07.5	13.7	23°30.0	7.9'	24°41.1	0.6'	56.6'
20	121°07.7	12.8	37°56.9	8.0'	24°41.7	0.5'	56.6'
21	136°07.9	· · 11.8	52°23.9	8.0'	24°42.2	0.3'	56.5'
22	151°08.2	10.9	66°50.9	8.1'	24°42.5	0.2'	56.5'
23	166°08.4	09.9	81°18.0	8.1'	24°42.7	0.0'	56.5'
SD = 15.9'		$d = -1.0'$	SD = 15.6'				

Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	181°08.6	N03°08.9	95°45.1	8.1'	S24°42.7	-0.1'	56.4'
1	196°08.8	08.0	110°12.2	8.2'	24°42.6	-0.2'	56.4'
2	211°09.1	07.0	124°39.4	8.2'	24°42.4	-0.4'	56.4'
3	226°09.3	· · 06.0	139°06.6	8.3'	24°42.0	-0.5'	56.3'
4	241°09.5	05.1	153°33.9	8.3'	24°41.5	-0.6'	56.3'
5	256°09.7	04.1	168°01.2	8.4'	24°40.8	-0.8'	56.3'
6	271°09.9	N03°03.2	182°28.6	8.4'	S24°40.1	-0.9'	56.2'
7	286°10.2	02.2	196°56.1	8.5'	24°39.1	-1.1'	56.2'
8	301°10.4	01.2	211°23.5	8.5'	24°38.1	-1.2'	56.2'
9	316°10.6	03°00.3	225°51.1	8.6'	24°36.9	-1.3'	56.1'
10	331°10.8	02°59.3	240°18.6	8.6'	24°35.6	-1.4'	56.1'
11	346°11.1	58.4	254°46.3	8.7'	24°34.1	-1.6'	56.1'
12	1°11.3	N02°57.4	269°14.0	8.7'	S24°32.6	-1.7'	56.0'
13	16°11.5	56.4	283°41.7	8.8'	24°30.9	-1.8'	56.0'
14	31°11.7	55.5	298°09.5	8.9'	24°29.0	-2.0'	56.0'
15	46°11.9	· · 54.5	312°37.3	8.9'	24°27.0	-2.1'	55.9'
16	61°12.2	53.5	327°05.3	9.0'	24°24.9	-2.2'	55.9'
17	76°12.4	52.6	341°33.2	9.0'	24°22.7	-2.4'	55.9'
18	91°12.6	N02°51.6	356°01.2	9.1'	S24°20.4	-2.5'	55.8'
19	106°12.8	50.7	10°29.3	9.1'	24°17.9	-2.6'	55.8'
20	121°13.1	49.7	24°57.5	9.2'	24°15.3	-2.7'	55.8'
21	136°13.3	· · 48.7	39°25.7	9.3'	24°12.5	-2.9'	55.8'
22	151°13.5	47.8	53°53.9	9.3'	24°09.7	-3.0'	55.7'
23	166°13.7	46.8	68°22.3	9.4'	24°06.7	-3.1'	55.7'
SD = 15.9'		$d = -1.0'$	SD = 15.4'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	01:57	03:49	05:02	18:47	19:59	21:46
N 70°	02:32	04:03	05:08	18:41	19:45	21:14
68°	02:56	04:14	05:12	18:37	19:34	20:51
66°	03:14	04:23	05:16	18:33	19:26	20:34
64°	03:28	04:31	05:20	18:30	19:18	20:20
62°	03:40	04:37	05:23	18:27	19:12	20:09
60°	03:50	04:43	05:25	18:25	19:07	19:59
N 58°	03:58	04:47	05:27	18:23	19:02	19:51
56°	04:06	04:51	05:29	18:21	18:58	19:44
54°	04:12	04:55	05:31	18:19	18:55	19:38
52°	04:17	04:58	05:33	18:18	18:52	19:32
50°	04:22	05:01	05:34	18:16	18:49	19:28
45°	04:33	05:08	05:37	18:13	18:43	19:18
N 40°	04:40	05:13	05:40	18:11	18:38	19:10
35°	04:47	05:17	05:42	18:09	18:34	19:04
30°	04:52	05:20	05:44	18:07	18:31	18:59
20°	04:59	05:25	05:47	18:04	18:26	18:52
N 10°	05:04	05:29	05:50	18:01	18:22	18:47
0°	05:08	05:32	05:52	17:59	18:20	18:44
S 10°	05:09	05:34	05:55	17:57	18:18	18:42
20°	05:10	05:35	05:57	17:54	18:16	18:42
30°	05:08	05:36	06:00	17:52	18:16	18:43
35°	05:07	05:36	06:01	17:50	18:16	18:45
40°	05:05	05:36	06:03	17:49	18:16	18:47
45°	05:02	05:36	06:05	17:47	18:16	18:51
S 50°	04:57	05:35	06:07	17:45	18:17	18:55
52°	04:55	05:34	06:08	17:44	18:18	18:57
54°	04:53	05:34	06:09	17:43	18:18	19:00
56°	04:50	05:33	06:10	17:42	18:19	19:03
58°	04:47	05:32	06:11	17:41	18:20	19:06
S 60°	04:43	05:32	06:13	17:40	18:21	19:10

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	■	■	■	■	■	■
66°	15:42	■	■	17:05	■	■
64°	14:36	16:04	16:50	18:12	18:40	19:47
62°	14:01	15:18	16:08	18:47	19:25	20:28
60°	13:35	14:49	15:40	19:13	19:55	20:56
N 58°	13:15	14:26	15:18	19:33	20:18	21:18
56°	12:59	14:08	15:00	19:50	20:36	21:35
54°	12:45	13:52	14:45	20:04	20:52	21:50
52°	12:32	13:39	14:32	20:17	21:05	22:03
50°	12:22	13:27	14:20	20:28	21:17	22:15
45°	11:59	13:03	13:56	20:51	21:42	22:39
N 40°	11:41	12:43	13:37	21:09	22:01	22:58
35°	11:26	12:27	13:21	21:25	22:17	23:13
30°	11:13	12:13	13:07	21:38	22:32	23:27
20°	10:50	11:49	12:43	22:02	22:56	23:50
N 10°	10:31	11:28	12:23	22:21	23:16	· · ·
0°	10:13	11:09	12:04	22:40	23:36	· · ·
S 10°	09:56	10:50	11:45	22:59	23:55	· · ·
20°	09:36	10:30	11:24	23:19	· · ·	00:16
30°	09:15	10:06	11:00	23:42	· · ·	00:39
35°	09:02	09:52	10:46	23:56	· · ·	00:53
40°	08:47	09:36	10:30	· · ·	00:11	01:10
45°	08:30	09:17	10:11	· · ·	00:30	01:29
S 50°	08:09	08:53	09:47	· · ·	00:54	01:53
52°	07:58	08:42	09:35	· · ·	01:05	02:05
54°	07:47	08:29	09:22	00:04	01:18	02:18
56°	07:34	08:14	09:07	00:16	01:32	02:33
58°	07:19	07:57	08:49	00:31	01:50	02:51
S 60°	07:01	07:35	08:27	00:48	02:11	03:14

Day	Sun		Mer. Pass hh:mm	Moon		Age 5-7 27-48%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
13	03:52	04:02	11:56	16:27	03:59	
14	04:13	04:24	11:56	17:22	04:55	
15	04:34	04:45	11:55	18:17	05:50	

September 16, 17, 18 UT (Thu., Fri., Sat.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	354°50.3	143°38.7	S17°50.9	146°17.4	S11°48.9	355°00.8	S01°48.7	168°34.0	S00°21.4	Alpheratz	357°45.2	29°09.2											
1	9°52.8	158°39.7	51.7	161°18.3	49.5	10°03.6	48.9	183°36.2	21.5	Ankaa	353°17.1	-42°14.6											
2	24°55.3	173°40.7	52.6	176°19.1	50.1	25°06.4	49.0	198°38.4	21.6	Schedar	349°42.4	56°35.9											
3	39°57.7	188°41.7	•• 53.4	191°20.0	•• 50.8	40°09.1	•• 49.1	213°40.6	•• 21.7	Diphda	348°57.5	-17°55.4											
4	55°00.2	203°42.7	54.2	206°20.9	51.4	55°11.9	49.3	228°42.7	21.9	Achernar	335°27.6	-57°10.6											
5	70°02.7	218°43.7	55.1	221°21.7	52.0	70°14.7	49.4	243°44.9	22.0	Hamal	328°02.8	23°30.9											
6	85°05.1	233°44.7	S17°55.9	236°22.6	S11°52.6	85°17.4	S01°49.5	258°47.1	S00°22.1	Polaris	318°28.4	89°18.5											
7	100°07.6	248°45.7	56.7	251°23.5	53.2	100°20.2	49.7	273°49.3	22.2	Acamar	315°19.5	-40°15.4											
8	115°10.1	263°46.7	57.6	266°24.4	53.8	115°23.0	49.8	288°51.5	22.4	Menkar	314°16.9	4°08.1											
9	130°12.5	278°47.7	•• 58.4	281°25.2	•• 54.4	130°25.7	•• 49.9	303°53.7	•• 22.5	Mirfak	308°43.0	49°53.9											
10	145°15.0	293°48.7	17°59.2	296°26.1	55.0	145°28.5	50.1	318°55.9	22.6	Aldebaran	290°51.6	16°31.9											
11	160°17.4	308°49.7	18°00.1	311°27.0	55.6	160°31.3	50.2	333°58.0	22.7	Rigel	281°13.9	-8°11.2											
12	175°19.9	323°50.7	S18°00.9	326°27.9	S11°56.2	175°34.0	S01°50.3	349°00.2	S00°22.8	Capella	280°37.4	46°00.4											
13	190°22.4	338°51.7	01.7	341°28.7	56.9	190°36.8	50.5	4°02.4	23.0	Bellatrix	278°34.1	6°21.7											
14	205°24.8	353°52.7	02.5	356°29.6	57.5	205°39.6	50.6	19°04.6	23.1	Elnath	278°15.1	28°37.0											
15	220°27.3	8°53.7	•• 03.4	11°30.5	•• 58.1	220°42.3	•• 50.7	34°06.8	•• 23.2	Alnilam	275°48.4	-1°11.6											
16	235°29.8	23°54.8	04.2	26°31.3	58.7	235°45.1	50.9	49°09.0	23.3	Betelgeuse	271°03.5	7°24.6											
17	250°32.2	38°55.8	05.0	41°32.2	59.3	250°47.9	51.0	64°11.1	23.4	Canopus	263°57.1	-52°41.8											
18	265°34.7	53°56.8	S18°05.8	56°33.1	S11°59.9	265°50.6	S01°51.1	79°13.3	S00°23.6	Sirius	258°35.6	-16°43.7											
19	280°37.2	68°57.8	06.7	71°33.9	12°00.5	280°53.4	51.3	94°15.5	23.7	Adhara	255°14.2	-28°59.0											
20	295°39.6	83°58.8	07.5	86°34.8	01.1	295°56.2	51.4	109°17.7	23.8	Procyon	245°01.9	5°11.9											
21	310°42.1	98°59.9	•• 08.3	101°35.7	•• 01.7	310°58.9	•• 51.5	124°19.9	•• 23.9	Pollux	243°30.3	27°59.9											
22	325°44.6	114°00.9	09.1	116°36.5	02.3	326°01.7	51.7	139°22.1	24.0	Avior	234°19.3	-59°32.4											
23	340°47.0	129°01.9	09.9	131°37.4	02.9	341°04.5	51.8	154°24.3	24.2	Suhail	222°54.3	-43°28.4											
Mer.pass. 00:21				ν 1.0' d 0.8' m -4.75				ν 0.9' d 0.6' m 1.52				ν 2.8' d 0.1' m -2.93				ν 2.2' d 0.1' m 0.95							

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	355°49.5	144°03.0	S18°10.7	146°38.3	S12°03.5	356°07.2	S01°51.9	169°26.4	S00°24.3										
1	10°51.9	159°04.0	11.6	161°39.2	04.2	11°10.0	52.1	184°28.6	24.4										
2	25°54.4	174°05.0	12.4	176°40.0	04.8	26°12.8	52.2	199°30.8	24.5										
3	40°56.9	189°06.1	•• 13.2	191°40.9	•• 05.4	41°15.5	•• 52.4	214°33.0	•• 24.6										
4	55°59.3	204°07.1	14.0	206°41.8	06.0	56°18.3	52.5	229°35.2	24.8										
5	71°01.8	219°08.1	14.8	221°42.6	06.6	71°21.1	52.6	244°37.4	24.9										
6	86°04.3	234°09.2	S18°15.6	236°43.5	S12°07.2	86°23.8	S01°52.8	259°39.6	S00°25.0										
7	101°06.7	249°10.2	16.4	251°44.4	07.8	101°26.6	52.9	274°41.7	25.1										
8	116°09.2	264°11.3	17.2	266°45.2	08.4	116°29.4	53.0	289°43.9	25.3										
9	131°11.7	279°12.3	•• 18.1	281°46.1	•• 09.0	131°32.2	•• 53.2	304°46.1	•• 25.4										
10	146°14.1	294°13.4	18.9	296°47.0	09.6	146°34.9	53.3	319°48.3	25.5										
11	161°16.6	309°14.4	19.7	311°47.8	10.2	161°37.7	53.4	334°50.5	25.6										
12	176°19.1	324°15.5	S18°20.5	326°48.7	S12°10.8	176°40.5	S01°53.6	349°52.7	S00°25.7										
13	191°21.5	339°16.5	21.3	341°49.6	11.4	191°43.2	53.7	4°54.9	25.9										
14	206°24.0	354°17.6	22.1	356°50.4	12.0	206°46.0	53.8	19°57.0	26.0										
15	221°26.4	9°18.7	•• 22.9	11°51.3	•• 12.6	221°48.8	•• 54.0	34°59.2	•• 26.1										
16	236°28.9	24°19.7	23.7	26°52.1	13.3	236°51.5	54.1	50°01.4	26.2										
17	251°31.4	39°20.8	24.5	41°53.0	13.9	251°54.3	54.2	65°03.6	26.3										
18	266°33.8	54°21.8	S18°25.3	56°53.9	S12°14.5	266°57.1	S01°54.4	80°05.8	S00°26.5										
19	281°36.8	69°22.9	26.1	71°54.7	15.1	281°56.8	54.5	95°08.0	26.6										
20	296°38.8	84°24.0	26.9	86°55.6	15.7	297°02.6	54.6	110°10.1	26.7										
21	311°41.2	99°25.1	•• 27.7	101°56.5	•• 16.3	312°05.4	•• 54.8	125°12.3	•• 26.8										
22	326°43.7	114°26.1	28.5	116°57.3	16.9	327°08.1	54.9	140°14.5	26.9										
23	341°46.2	129°27.2	29.3	131°58.2	17.5	342°10.9	55.0	155°16.7	27.1										
Mer.pass. 00:17				ν 1.0' d 0.8' m -4.76				ν 0.9' d 0.6' m 1.52				ν 2.8' d 0.1' m -2.93				ν 2.2' d 0.1' m 0.94			

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	356°48.6	144°28.3	S18°30.1	146°59.1	S12°18.1	357°13.7	S01°55.2	170°18.9	S00°27.2										
1	11°51.1	159°29.4	30.9	161°59.9	18.7	12°16.4	55.3	185°21.1	27.3										
2	26°53.5	174°30.4	31.7	177°00.8	19.3	27°19.2	55.4	200°23.3	27.4										
3	41°56.0	189°31.5	•• 32.5	192°01.6	•• 19.9	42°02.0	•• 55.6	215°25.4	•• 27.5										
4	56°58.5	204°32.6	33.2	207°02.5	20.5	57°24.7	55.7	230°27.6	27.7										
5	72°00.9	219°33.7	34.0	222°03.4	21.1	72°27.5	55.8	245°29.8	27.8										
6	87°03.4	234°34.8	S18°34.8	237°04.2	S12°21.7	87°30.3	S01°56.0	260°32.0	S00°27.9										
7	102°05.9	249°35.9	35.6	252°05.1	22.3	102°33.1	56.1	275°34.2	28.0										
8	117°08.3	264°37.0	36.4	267°05.9	22.9	117°35.8	56.2	290°36.4	28.2										
9	132°10.8	279°38.1	•• 37.2	282°06.8	•• 23.5	132°38.6	•• 56.4	305°38.5	•• 28.3										
10	147°13.3	294°39.2	38.0	297°07.7	24.1	147°41.4	56.5	320°40.7	28.4										
11	162°15.7	309°40.3	38.8	312°08.5	24.7	162°44.1	56.6	335°42.9	28.5										
12	177°18.2	324°41.4	S18°39.5	327°09.4	S12°25.3	177°46.9	S01°56.8	350°45.1	S00°28.6										
13	192°20.7	339°42.5	40.3	342°10.2	26.0	192°49.7	56.9	5°47.3	28.8										
14	207°23.1	354°43.6	41.1	357°11.1	26.6	207°52.4	57.1	20°49.5	28.9										
15	222°25.6	9°44.7	•• 41.9	12°12.0	•• 27.2	222°55.2	•• 57.2	35°51.7	•• 29.0										
16	237°28.0	24°45.8	42.7	27°12.8	27.8	237°58.0	57.3	50°53.8	29.1										
17	252°30.5	39°46.9	43.4	42°13.7	28.4	253°00.7	57.5	65°56.0	29.2										
18	267°33.0	54°48.0	S18°44.2	57°14.5	S12°29.0	268°03.5	S01°57.6	80°58.2	S00°29.4										
19	282°35.4	69°49.1	45.0	72°15.4	29.6	283°06.3	57.7	96°00.4	29.5										
20	297°37.9	84°50.2	45.8	87°16.3	30.2	298°09.0	57.9	111°02.6	29.6										
21	312°40.4	99°51.3	•• 46.6	102°17.1	•• 30.8	313°11.8	•• 58.0	126°04.8	•• 29.7										
22	327°42.8	114°52.5	47.3	117°18.0	31.4	328°14.6	58.1	141°06.9	29.8										
23	342°45.3	129°53.6	48.1	132°18.8	32.0	343°17.4	58.3	156°09.1	30.0										
Mer.pass. 00:13				ν 1.1' d 0.8' m -4.77				ν 0.9' d 0.6' m 1.52				ν 2.8' d 0.1' m -2.93				ν 2.2' d 0.1' m 0.94			

Sep 16 Thu	SHA	Mer.pass
Venus	148°48.3	14:24
Mars	151°27.0	14:14
Jupiter	0°10.5	00:20
Saturn	173°43.6	12:44

Sep 17 Fri	SHA	Mer.pass
Venus	148°13.5	14:23
Mars	150°48.8	14:13
Jupiter	0°17.8	00:15
Saturn	173°37.0	12:40

Sep 18 Sat	SHA	Mer.pass
Venus	147°39.7	14:21
Mars	150°10.4	14:11
Jupiter	0°25.1	00:11
Saturn	173°30.3	12:37

Horizontal parallax		
Venus:	0.3	
Mars:	0.1	

h	Sun			Moon			
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	181°14.0	N02°45.8	82°50.6	9.4'	S24°03.6	-3.2'	55.7'
1	196°14.2	44.9	97°19.1	9.5'	24°00.4	-3.3'	55.6'
2	211°14.4	43.9	111°47.6	9.6'	23°57.0	-3.5'	55.6'
3	226°14.6	· · 43.0	126°16.2	9.6'	23°53.5	-3.6'	55.6'
4	241°14.8	42.0	140°44.8	9.7'	23°50.0	-3.7'	55.6'
5	256°15.1	41.0	155°13.5	9.8'	23°46.2	-3.8'	55.5'
6	271°15.3	N02°40.1	169°42.3	9.8'	S23°42.4	-3.9'	55.5'
7	286°15.5	39.1	184°11.1	9.9'	23°38.5	-4.1'	55.5'
8	301°15.7	38.1	198°40.0	10.0'	23°34.4	-4.2'	55.4'
9	316°16.0	· · 37.2	213°09.0	10.0'	23°30.2	-4.3'	55.4'
10	331°16.2	36.2	227°38.0	10.1'	23°25.9	-4.4'	55.4'
11	346°16.4	35.2	242°07.1	10.2'	23°21.5	-4.5'	55.4'
12	1°16.6	N02°34.3	256°36.3	10.2'	S23°17.0	-4.6'	55.3'
13	16°16.9	33.3	271°05.6	10.3'	23°12.4	-4.7'	55.3'
14	31°17.1	32.3	285°34.9	10.4'	23°07.6	-4.9'	55.3'
15	46°17.3	· · 31.4	300°04.2	10.4'	23°02.8	-5.0'	55.3'
16	61°17.5	30.4	314°33.7	10.5'	22°57.8	-5.1'	55.2'
17	76°17.7	29.5	329°03.2	10.6'	22°52.8	-5.2'	55.2'
18	91°18.0	N02°28.5	343°32.8	10.7'	S22°47.6	-5.3'	55.2'
19	106°18.2	27.5	358°02.4	10.7'	22°42.3	-5.4'	55.2'
20	121°18.4	26.6	12°32.2	10.8'	22°36.9	-5.5'	55.1'
21	136°18.6	· · 25.6	27°02.0	10.9'	22°31.4	-5.6'	55.1'
22	151°18.9	24.6	41°31.8	10.9'	22°25.8	-5.7'	55.1'
23	166°19.1	23.7	56°01.8	11.0'	22°20.1	-5.8'	55.1'
SD = 15.9'			SD = 15.2'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	181°19.3	N02°22.7	70°31.8	11.1'	S22°14.2	-5.9'	55.0'
1	196°19.5	21.7	85°01.8	11.1'	22°08.3	-6.0'	55.0'
2	211°19.8	20.8	99°32.0	11.2'	22°02.3	-6.1'	55.0'
3	226°20.0	· · 19.8	114°02.2	11.3'	21°56.2	-6.2'	55.0'
4	241°20.2	18.8	128°32.5	11.4'	21°50.0	-6.3'	54.9'
5	256°20.4	17.9	143°02.9	11.4'	21°43.7	-6.4'	54.9'
6	271°20.7	N02°16.9	157°33.3	11.5'	S21°37.2	-6.5'	54.9'
7	286°20.9	15.9	172°03.8	11.6'	21°30.7	-6.6'	54.9'
8	301°21.1	15.0	186°34.4	11.6'	21°24.1	-6.7'	54.9'
9	316°21.3	· · 14.0	201°05.0	11.7'	21°17.4	-6.8'	54.8'
10	331°21.5	13.0	215°35.7	11.8'	21°10.6	-6.9'	54.8'
11	346°21.8	12.1	230°06.5	11.9'	21°03.7	-7.0'	54.8'
12	1°22.0	N02°11.1	244°37.4	11.9'	S20°56.7	-7.1'	54.8'
13	16°22.2	10.1	259°08.3	12.0'	20°49.7	-7.2'	54.8'
14	31°22.4	09.2	273°39.3	12.1'	20°42.5	-7.3'	54.7'
15	46°22.7	· · 08.2	288°10.3	12.1'	20°35.2	-7.3'	54.7'
16	61°22.9	07.2	302°41.5	12.2'	20°27.9	-7.4'	54.7'
17	76°23.1	06.3	317°12.7	12.3'	20°20.4	-7.5'	54.7'
18	91°23.3	N02°05.3	331°44.0	12.3'	S20°12.9	-7.6'	54.7'
19	106°23.6	04.3	346°15.3	12.4'	20°05.3	-7.7'	54.6'
20	121°23.8	03.4	0°46.7	12.5'	19°57.6	-7.8'	54.6'
21	136°24.0	· · 02.4	15°18.2	12.6'	19°49.8	-7.9'	54.6'
22	151°24.2	01.4	29°49.8	12.6'	19°42.0	-7.9'	54.6'
23	166°24.5	00.5	44°21.4	12.7'	19°34.0	-8.0'	54.6'
SD = 15.9'			SD = 15.0'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	181°24.7	N01°59.5	58°53.1	12.8'	S19°26.0	-8.1'	54.6'
1	196°24.9	58.5	73°24.8	12.8'	19°17.9	-8.2'	54.5'
2	211°25.1	57.6	87°56.6	12.9'	19°09.7	-8.3'	54.5'
3	226°25.3	· · 56.6	102°28.5	13.0'	19°01.4	-8.4'	54.5'
4	241°25.6	55.6	117°00.5	13.0'	18°53.1	-8.4'	54.5'
5	256°25.8	54.7	131°32.5	13.1'	18°44.6	-8.5'	54.5'
6	271°26.0	N01°53.7	146°04.6	13.2'	S18°36.1	-8.6'	54.5'
7	286°26.2	52.7	160°36.8	13.2'	18°27.5	-8.7'	54.4'
8	301°26.5	51.8	175°09.0	13.3'	18°18.9	-8.7'	54.4'
9	316°26.7	· · 50.8	189°41.3	13.3'	18°10.1	-8.8'	54.4'
10	331°26.9	49.8	204°13.6	13.4'	18°01.3	-8.9'	54.4'
11	346°27.1	48.9	218°46.0	13.5'	17°52.4	-9.0'	54.4'
12	1°27.4	N01°47.9	233°18.5	13.5'	S17°43.5	-9.0'	54.4'
13	16°27.6	46.9	247°51.0	13.6'	17°34.5	-9.1'	54.4'
14	31°27.8	46.0	262°23.6	13.7'	17°25.4	-9.2'	54.3'
15	46°28.0	· · 45.0	276°56.3	13.7'	17°16.2	-9.2'	54.3'
16	61°28.2	44.0	291°29.0	13.8'	17°07.0	-9.3'	54.3'
17	76°28.5	43.1	306°01.8	13.8'	16°57.6	-9.4'	54.3'
18	91°28.7	N01°42.1	320°34.7	13.9'	S16°48.3	-9.4'	54.3'
19	106°28.9	41.1	335°07.6	14.0'	16°38.8	-9.5'	54.3'
20	121°29.1	40.2	349°40.5	14.0'	16°29.3	-9.6'	54.3'
21	136°29.4	· · 39.2	4°13.6	14.1'	16°19.8	-9.6'	54.3'
22	151°29.6	38.2	18°46.7	14.1'	16°10.1	-9.7'	54.2'
23	166°29.8	37.3	33°19.8	14.2'	16°00.4	-9.8'	54.2'
SD = 15.9'			SD = 14.9'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	02:23	04:04	05:15	18:31	19:41	21:20
N 70°	02:50	04:16	05:20	18:27	19:30	20:54
68°	03:11	04:26	05:23	18:24	19:21	20:35
66°	03:26	04:33	05:26	18:21	19:13	20:20
64°	03:39	04:40	05:28	18:19	19:07	20:07
62°	03:50	04:45	05:30	18:17	19:02	19:57
60°	03:58	04:50	05:32	18:16	18:58	19:49
N 58°	04:06	04:54	05:34	18:14	18:54	19:41
56°	04:12	04:58	05:35	18:13	18:50	19:35
54°	04:18	05:01	05:36	18:12	18:47	19:30
52°	04:23	05:04	05:37	18:11	18:44	19:25
50°	04:27	05:06	05:39	18:10	18:42	19:20
45°	04:36	05:11	05:41	18:08	18:37	19:12
N 40°	04:44	05:15	05:43	18:06	18:33	19:05
35°	04:49	05:19	05:44	18:04	18:30	18:59
30°	04:54	05:22	05:46	18:03	18:27	18:55
20°	05:00	05:26	05:48	18:01	18:23	18:49
N 10°	05:04	05:29	05:50	17:59	18:20	18:45
0°	05:07	05:31	05:51	17:58	18:18	18:42
S 10°	05:08	05:32	05:53	17:56	18:17	18:42
20°	05:07	05:32	05:54	17:55	18:17	18:42
30°	05:04	05:32	05:56	17:53	18:17	18:45
35°	05:02	05:32	05:57	17:53	18:18	18:47
40°	05:00	05:31	05:58	17:52	18:19	18:50
45°	04:56	05:30	05:59	17:51	18:20	18:54
S 50°	04:51	05:28	06:00	17:50	18:22	19:00
52°	04:48	05:27	06:01	17:49	18:23	19:02
54°	04:45	05:26	06:02	17:49	18:24	19:05
56°	04:42	05:25	06:02	17:48	18:25	19:09
58°	04:38	05:24	06:03	17:47	18:26	19:13
S 60°	04:34	05:22	06:04	17:47	18:28	19:17

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	■	■	19:10	■	■	22:33
N 70°	■	■	18:23	■	■	23:18
68°	■	18:18	17:53	■	21:48	23:48
66°	17:52	17:38	17:30	20:32	22:27	· ·
64°	17:05	17:10	17:12	21:18	22:54	· ·
62°	16:35	16:49	16:57	21:48	23:14	· ·
60°	16:12	16:32	16:44	22:11	23:31	· ·
N 58°	15:54	16:17	16:33	22:29	23:45	· ·
56°	15:38	16:05	16:24	22:44	23:57	· ·
54°	15:25	15:54	16:15	22:57	· ·	00:08
52°	15:13	15:44	16:08	23:09	· ·	00:17
50°	15:02	15:35	16:01	23:19	· ·	00:25
45°	14:41	15:17	15:46	23:40	· ·	00:43
N 40°	14:23	15:01	15:34	23:57	· ·	00:57
35°	14:08	14:49	15:24	· ·	00:11	01:09
30°	13:55	14:37	15:14	· ·	00:24	01:20
20°	13:33	14:18	14:59	· ·	00:45	01:38
N 10°	13:14	14:01	14:45	00:10	01:03	01:54
0°	12:56	13:45	14:32	00:29	01:20	02:08
S 10°	12:38	13:29	14:19	00:48	01:37	02:23
20°	12:19	13:13	14:05	01:08	01:55	02:38
30°	11:56	12:53	13:49	01:31	02:16	02:55
35°	11:43	12:41	13:39	01:44	02:28	03:06
40°	11:28	12:28	13:29	02:00	02:42	03:17
45°	11:10	12:13	13:16	02:18	02:58	03:31
S 50°	10:48	11:53	13:01	02:41	03:18	03:47
52°	10:37	11:44	12:53	02:52	03:28	03:55
54°	10:25	11:34	12:45	03:04	03:38	04:04
56°	10:11	11:22	12:36	03:18	03:50	04:13
58°	09:55	11:09	12:26	03:35	04:04	04:24
S 60°	09:35	10:53	12:14	03:55	04:21	04:37

Day	Sun		Mer. Pass	Moon		Age
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss	hh:mm	Upper hh:mm	Lower hh:mm	8-10 58-76%
16	04:56	05:07	11:55	19:08	06:43	
17	05:17	05:28	11:55	19:57	07:33	
18	05:39	05:49	11:54	20:43	08:20	

September 19, 20, 21 UT (Sun., Mon., Tue.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	357°47.8	144°54.7	S18°48.9	147°19.7	S12°32.6	358°20.1	S01°58.4	171°11.3	S00°30.1					Alpheratz	357°45.2	29°09.2	
1	12°50.2	159°55.8	49.7	162°20.5	33.2	13°22.9	58.5	186°13.5	30.2					Ankaa	353°17.1	-42°14.6	
2	27°52.7	174°57.0	50.4	177°21.4	33.8	28°25.7	58.7	201°15.7	30.3					Schedar	349°42.4	56°35.9	
3	42°55.2	189°58.1	· · 51.2	192°22.3	· · 34.4	43°28.4	· · 58.8	216°17.9	· · 30.5					Diphda	348°57.5	-17°55.4	
4	57°57.6	204°59.2	52.0	207°23.1	35.0	58°31.2	58.9	231°20.0	30.6					Achernar	335°27.6	-57°10.6	
5	73°00.1	220°00.4	52.7	222°24.0	35.6	73°34.0	59.1	246°22.2	30.7					Hamal	328°02.7	23°30.9	
6	88°02.5	235°01.5	S18°53.5	237°24.8	S12°36.2	88°36.7	S01°59.2	261°24.4	S00°30.8					Polaris	318°27.4	89°18.5	
7	103°05.0	250°02.6	54.3	252°25.7	36.8	103°39.5	59.3	276°26.6	30.9					Acamar	315°19.5	-40°15.4	
8	118°07.5	265°03.8	55.0	267°26.5	37.4	118°42.3	59.5	291°28.8	31.1					Menkar	314°16.9	4°08.1	
9	133°09.9	280°04.9	· · 55.8	282°27.4	· · 38.0	133°45.0	· · 59.6	306°31.0	· · 31.2					Mirfak	308°43.0	49°53.9	
10	148°12.4	295°06.1	56.6	297°28.2	38.6	148°47.8	59.7	321°33.2	31.3					Aldebaran	290°51.6	16°31.9	
11	163°14.9	310°07.2	57.3	312°29.1	39.2	163°50.6	01°59.9	336°35.3	31.4					Rigel	281°13.9	-8°11.2	
12	178°17.3	325°08.4	S18°58.1	327°29.9	S12°39.8	178°53.4	S02°00.0	351°37.5	S00°31.5					Capella	280°37.3	46°00.4	
13	193°19.8	340°09.5	58.8	342°30.8	40.4	193°56.1	00.1	6°39.7	31.7					Bellatrix	278°34.1	6°21.7	
14	208°22.3	355°10.7	18°59.6	357°31.7	41.0	208°58.9	00.3	21°41.9	31.8					Elnath	278°15.1	28°37.0	
15	223°24.7	10°11.8	19°00.4	12°32.5	· · 41.6	224°01.7	· · 00.4	36°44.1	· · 31.9					Alnilam	275°48.4	-1°11.6	
16	238°27.2	25°13.0	01.1	27°33.4	42.2	239°04.4	00.5	51°46.3	32.0					Betelgeuse	271°03.4	7°24.6	
17	253°29.6	40°14.1	01.9	42°34.2	42.8	254°07.2	00.7	66°48.4	32.1					Canopus	263°57.1	-52°41.8	
18	268°32.1	55°15.3	S19°02.6	57°35.1	S12°43.4	269°10.0	S02°00.8	81°50.6	S00°32.3					Sirius	258°35.5	-16°43.7	
19	283°34.6	70°16.4	03.4	72°35.9	44.0	284°12.7	00.9	96°52.8	32.4					Adhara	255°14.2	-28°59.0	
20	298°37.0	85°17.6	04.1	87°36.8	44.6	299°15.5	01.1	111°55.0	32.5					Procyon	245°01.9	5°11.9	
21	313°39.5	100°18.8	· · 04.9	102°37.6	· · 45.2	314°18.3	· · 01.2	126°57.2	· · 32.6					Pollux	243°30.3	27°59.9	
22	328°42.0	115°19.9	05.6	117°38.5	45.8	329°21.1	01.3	141°59.4	32.8					Avior	234°19.3	-59°32.4	
23	343°44.4	130°21.1	06.4	132°39.3	46.4	344°23.8	01.5	157°01.5	32.9					Suhail	222°54.3	-43°28.4	
Mer.pass. 00:09			ν 1.1' d 0.8' m -4.78	ν 0.9' d 0.6' m 1.51	ν 2.8' d 0.1' m -2.93	ν 2.2' d 0.1' m 0.94											

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	358°46.9	145°22.3	S19°07.1	147°40.2	S12°47.0	359°26.6	S02°01.6	172°03.7	S00°33.0
1	13°49.4	160°23.5	07.9	162°41.0	47.6	14°29.4	01.7	187°05.9	33.1
2	28°51.8	175°24.6	08.6	177°41.9	48.2	29°32.1	01.9	202°08.1	33.2
3	43°54.3	190°25.8	· · 09.4	192°42.7	· · 48.8	44°34.9	· · 02.0	217°10.3	· · 33.4
4	58°56.8	205°27.0	10.1	207°43.6	49.4	59°37.7	02.2	232°12.5	33.5
5	73°59.2	220°28.2	10.9	222°44.4	50.0	74°40.4	02.3	247°14.6	33.6
6	89°01.7	235°29.4	S19°11.6	237°45.3	S12°50.6	89°43.2	S02°02.4	262°16.8	S00°33.7
7	104°04.1	250°30.6	12.4	252°46.1	51.2	104°46.0	02.6	277°19.0	33.8
8	119°06.6	265°31.8	13.1	267°47.0	51.8	119°48.7	02.7	292°21.2	34.0
9	134°09.1	280°32.9	· · 13.8	282°47.8	· · 52.4	134°51.5	· · 02.8	307°23.4	· · 34.1
10	149°11.5	295°34.1	14.6	297°48.7	53.0	149°54.3	03.0	322°25.6	34.2
11	164°14.0	310°35.3	15.3	312°49.5	53.6	164°57.1	03.1	337°27.7	34.3
12	179°16.5	325°36.5	S19°16.1	327°50.4	S12°54.2	179°59.8	S02°03.2	352°29.9	S00°34.4
13	194°18.9	340°37.7	16.8	342°51.2	54.8	195°02.6	03.4	7°32.1	34.6
14	209°21.4	355°38.9	17.5	357°52.1	55.4	210°05.4	03.5	22°34.3	34.7
15	224°23.9	10°40.1	· · 18.3	12°52.9	· · 56.0	225°08.1	· · 03.6	37°36.5	· · 34.8
16	239°26.3	25°41.4	19.0	27°53.8	56.6	240°10.9	03.8	52°38.7	34.9
17	254°28.8	40°42.6	19.7	42°54.6	57.2	255°13.7	03.9	67°40.8	35.1
18	269°31.3	55°43.8	S19°20.5	57°55.5	S12°57.8	270°16.4	S02°04.0	82°43.0	S00°35.2
19	284°33.7	70°45.0	21.2	72°56.3	58.4	285°19.2	04.2	97°45.2	35.3
20	299°36.2	85°46.2	21.9	87°57.1	59.0	300°22.0	04.3	112°47.4	35.4
21	314°38.6	100°47.4	· · 22.7	102°58.0	12°59.5	315°24.8	· · 04.4	127°49.6	· · 35.5
22	329°41.1	115°48.6	23.4	117°58.8	13°00.1	330°27.5	04.6	142°51.8	35.7
23	344°43.6	130°49.9	24.1	132°59.7	00.7	345°30.3	04.7	157°53.9	35.8
Mer.pass. 00:05			ν 1.2' d 0.7' m -4.79	ν 0.9' d 0.6' m 1.51	ν 2.8' d 0.1' m -2.94	ν 2.2' d 0.1' m 0.93			

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	359°46.0	145°51.1	S19°24.8	148°00.5	S13°01.3	0°33.1	S02°04.8	172°56.1	S00°35.9
1	14°48.5	160°52.3	25.6	163°01.4	01.9	15°35.8	05.0	187°58.3	36.0
2	29°51.0	175°53.5	26.3	178°02.2	02.5	30°38.6	05.1	203°00.5	36.1
3	44°53.4	190°54.8	· · 27.0	193°03.1	· · 03.1	45°41.4	· · 05.2	218°02.7	· · 36.3
4	59°55.9	205°56.0	27.7	208°03.9	03.7	60°44.1	05.4	233°04.9	36.4
5	74°58.4	220°57.2	28.5	223°04.7	04.3	75°46.9	05.5	248°07.0	36.5
6	90°00.8	235°58.5	S19°29.2	238°05.6	S13°04.9	90°49.7	S02°05.6	263°09.2	S00°36.6
7	105°03.3	250°59.7	29.9	253°06.4	05.5	105°52.5	05.8	278°11.4	36.7
8	120°05.7	266°01.0	30.6	268°07.3	06.1	120°55.2	05.9	293°13.6	36.9
9	135°08.2	281°02.2	· · 31.3	283°08.1	· · 06.7	135°58.0	· · 06.0	308°15.8	· · 37.0
10	150°10.7	296°03.5	32.0	298°09.0	07.3	151°00.8	06.2	323°18.0	37.1
11	165°13.1	311°04.7	32.8	313°09.8	07.9	166°03.5	06.3	338°20.1	37.2
12	180°15.6	326°06.0	S19°33.5	328°10.7	S13°08.5	181°06.3	S02°06.4	353°22.3	S00°37.4
13	195°18.1	341°07.2	34.2	343°11.5	09.1	196°09.1	06.6	8°24.5	37.5
14	210°20.5	356°08.5	34.9	358°12.3	09.7	211°11.8	06.7	23°26.7	37.6
15	225°23.0	11°09.7	· · 35.6	13°13.2	· · 10.3	226°14.6	· · 06.8	38°28.9	· · 37.7
16	240°25.5	26°11.0	36.3	28°14.0	10.9	241°17.4	07.0	53°31.0	37.8
17	255°27.9	41°12.2	37.0	43°14.9	11.5	256°20.2	07.1	68°33.2	38.0
18	270°30.4	56°13.5	S19°37.7	58°15.7	S13°12.0	271°22.9	S02°07.2	83°35.4	S00°38.1
19	285°32.9	71°14.8	38.4	73°16.5	12.6	286°25.7	07.4	98°37.6	38.2
20	300°35.3	86°16.1	39.1	88°17.4	13.2	301°28.5	07.5	113°39.8	38.3
21	315°37.8	101°17.3	· · 39.8	103°18.2	· · 13.8	316°31.2	· · 07.6	128°42.0	· · 38.4
22	330°40.2	116°18.6	40.5	118°19.1	14.4	331°34.0	07.8	143°44.1	38.6
23	345°42.7	131°19.9	41.2	133°19.9	15.0	346°36.8	07.9	158°46.3	38.7
Mer.pass. 23:57			ν 1.2' d 0.7' m -4.79	ν 0.8' d 0.6' m 1.51	ν 2.8' d 0.1' m -2.94	ν 2.2' d 0.1' m 0.93			

Sep 19 Sun	SHA	Mer.pass
Venus	147°06.9	14:19
Mars	149°31.9	14:10
Jupiter	0°32.4	00:07
Saturn	173°23.6	12:33

Sep 20 Mon	SHA	Mer.pass
Venus	146°35.4	14:17
Mars	148°53.3	14:09
Jupiter	0°39.7	00:02
Saturn	173°16.8	12:30

Sep 21 Tue	SHA	Mer.pass
Venus	146°05.1	14:15
Mars	148°14.5	14:07
Jupiter	0°47.0	23:53
Saturn	173°10.1	12:26

Horizontal parallax		
Venus:	0.3	
Mars:	0.1	

h	Sun			Moon			
Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	181°30.0	N01°36.3	47°53.0	14.3'	S15°50.7	-9.8'	54.2'
1	196°30.3	35.3	62°26.3	14.3'	15°40.9	-9.9'	54.2'
2	211°30.5	34.3	76°59.6	14.4'	15°31.0	-9.9'	54.2'
3	226°30.7	· · 33.4	91°32.9	14.4'	15°21.0	-10.0'	54.2'
4	241°30.9	32.4	106°06.4	14.5'	15°11.0	-10.1'	54.2'
5	256°31.2	31.4	120°39.8	14.5'	15°01.0	-10.1'	54.2'
6	271°31.4	N01°30.5	135°13.4	14.6'	S14°50.8	-10.2'	54.2'
7	286°31.6	29.5	149°47.0	14.6'	14°40.7	-10.2'	54.1'
8	301°31.8	28.5	164°20.6	14.7'	14°30.4	-10.3'	54.1'
9	316°32.0	· · 27.6	178°54.3	14.7'	14°20.2	-10.3'	54.1'
10	331°32.3	26.6	193°28.0	14.8'	14°09.8	-10.4'	54.1'
11	346°32.5	25.6	208°01.8	14.8'	13°59.4	-10.4'	54.1'
12	1°32.7	N01°24.7	222°35.7	14.9'	S13°49.0	-10.5'	54.1'
13	16°32.9	23.7	237°09.6	14.9'	13°38.5	-10.6'	54.1'
14	31°33.2	22.7	251°43.5	15.0'	13°27.9	-10.6'	54.1'
15	46°33.4	· · 21.7	266°17.5	15.0'	13°17.3	-10.7'	54.1'
16	61°33.6	20.8	280°51.5	15.1'	13°06.7	-10.7'	54.1'
17	76°33.8	19.8	295°25.6	15.1'	12°56.0	-10.8'	54.1'
18	91°34.1	N01°18.8	309°59.8	15.2'	S12°45.2	-10.8'	54.1'
19	106°34.3	17.9	324°33.9	15.2'	12°34.4	-10.8'	54.0'
20	121°34.5	16.9	339°08.1	15.3'	12°23.6	-10.9'	54.0'
21	136°34.7	· · 15.9	353°42.4	15.3'	12°12.7	-10.9'	54.0'
22	151°34.9	15.0	8°16.7	15.3'	12°01.7	-11.0'	54.0'
23	166°35.2	14.0	22°51.1	15.4'	11°50.7	-11.0'	54.0'
SD = 15.9'			SD = 14.8'				

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	181°35.4	N01°13.0	37°25.5	15.4'	S11°39.7	-11.1'	54.0'
1	196°35.6	12.1	51°59.9	15.5'	11°28.6	-11.1'	54.0'
2	211°35.8	11.1	66°34.4	15.5'	11°17.5	-11.2'	54.0'
3	226°36.1	· · 10.1	81°08.9	15.6'	11°06.4	-11.2'	54.0'
4	241°36.3	09.1	95°43.4	15.6'	10°55.2	-11.2'	54.0'
5	256°36.5	08.2	110°18.0	15.6'	10°43.9	-11.3'	54.0'
6	271°36.7	N01°07.2	124°52.6	15.7'	S10°32.7	-11.3'	54.0'
7	286°37.0	06.2	139°27.3	15.7'	10°21.3	-11.4'	54.0'
8	301°37.2	05.3	154°02.0	15.7'	10°10.0	-11.4'	54.0'
9	316°37.4	· · 04.3	168°36.7	15.8'	09°58.6	-11.4'	54.0'
10	331°37.6	03.3	183°11.5	15.8'	09°47.2	-11.5'	54.0'
11	346°37.8	02.4	197°46.3	15.8'	09°35.7	-11.5'	54.0'
12	1°38.1	N01°01.4	212°21.1	15.9'	S09°24.2	-11.5'	54.0'
13	16°38.3	01°00.4	226°56.0	15.9'	09°12.7	-11.6'	53.9'
14	31°38.5	00°59.4	241°30.9	15.9'	09°01.1	-11.6'	53.9'
15	46°38.7	· · 58.5	256°05.8	16.0'	08°49.5	-11.6'	53.9'
16	61°39.0	57.5	270°40.8	16.0'	08°37.9	-11.7'	53.9'
17	76°39.2	56.5	285°15.8	16.0'	08°26.2	-11.7'	53.9'
18	91°39.4	N00°55.6	299°50.8	16.0'	S08°14.5	-11.7'	53.9'
19	106°39.6	54.6	314°25.8	16.1'	08°02.8	-11.8'	53.9'
20	121°39.8	53.6	329°00.9	16.1'	07°51.0	-11.8'	53.9'
21	136°40.1	· · 52.6	343°36.0	16.1'	07°39.2	-11.8'	53.9'
22	151°40.3	51.7	358°11.1	16.1'	07°27.4	-11.8'	53.9'
23	166°40.5	50.7	12°46.3	16.2'	07°15.6	-11.9'	53.9'
SD = 15.9'			SD = 14.7'				

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	181°40.7	N00°49.7	27°21.5	16.2'	S07°03.7	-11.9'	53.9'
1	196°41.0	48.8	41°56.6	16.2'	06°51.8	-11.9'	53.9'
2	211°41.2	47.8	56°31.9	16.2'	06°39.9	-11.9'	53.9'
3	226°41.4	· · 46.8	71°07.1	16.3'	06°27.9	-12.0'	53.9'
4	241°41.6	45.8	85°42.4	16.3'	06°15.9	-12.0'	53.9'
5	256°41.8	44.9	100°17.6	16.3'	06°04.0	-12.0'	53.9'
6	271°42.1	N00°43.9	114°52.9	16.3'	S05°51.9	-12.0'	53.9'
7	286°42.3	42.9	129°28.2	16.3'	05°39.9	-12.1'	53.9'
8	301°42.5	42.0	144°03.6	16.3'	05°27.8	-12.1'	53.9'
9	316°42.7	· · 41.0	158°38.9	16.4'	05°15.8	-12.1'	53.9'
10	331°43.0	40.0	173°14.3	16.4'	05°03.6	-12.1'	53.9'
11	346°43.2	39.0	187°49.6	16.4'	04°51.5	-12.1'	53.9'
12	1°43.4	N00°38.1	202°25.0	16.4'	S04°39.4	-12.2'	53.9'
13	16°43.6	37.1	217°00.4	16.4'	04°27.2	-12.2'	53.9'
14	31°43.8	36.1	231°35.9	16.4'	04°15.1	-12.2'	53.9'
15	46°44.1	· · 35.2	246°11.3	16.4'	04°02.9	-12.2'	53.9'
16	61°44.3	34.2	260°46.7	16.4'	03°50.7	-12.2'	53.9'
17	76°44.5	33.2	275°22.2	16.5'	03°38.4	-12.2'	53.9'
18	91°44.7	N00°32.2	289°57.6	16.5'	S03°26.2	-12.2'	53.9'
19	106°45.0	31.3	304°33.1	16.5'	03°14.0	-12.3'	53.9'
20	121°45.2	30.3	319°08.6	16.5'	03°01.7	-12.3'	53.9'
21	136°45.4	· · 29.3	333°44.0	16.5'	02°49.4	-12.3'	53.9'
22	151°45.6	28.4	348°19.5	16.5'	02°37.1	-12.3'	53.9'
23	166°45.8	27.4	2°55.0	16.5'	02°24.8	-12.3'	53.9'
SD = 15.9'			SD = 14.7'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	02:44	04:19	05:29	18:16	19:24	20:58
N 70°	03:07	04:29	05:31	18:13	19:15	20:36
68°	03:25	04:37	05:33	18:11	19:08	20:19
66°	03:38	04:43	05:35	18:10	19:01	20:06
64°	03:49	04:49	05:37	18:09	18:56	19:55
62°	03:59	04:53	05:38	18:07	18:52	19:46
60°	04:06	04:57	05:39	18:06	18:48	19:39
N 58°	04:13	05:01	05:40	18:06	18:45	19:32
56°	04:19	05:04	05:41	18:05	18:42	19:26
54°	04:24	05:06	05:42	18:04	18:39	19:22
52°	04:28	05:09	05:42	18:04	18:37	19:17
50°	04:32	05:11	05:43	18:03	18:35	19:13
45°	04:40	05:15	05:44	18:02	18:31	19:06
N 40°	04:47	05:18	05:45	18:01	18:28	19:00
35°	04:51	05:21	05:46	18:00	18:25	18:55
30°	04:55	05:23	05:47	17:59	18:23	18:51
20°	05:01	05:26	05:48	17:58	18:20	18:46
N 10°	05:04	05:28	05:49	17:57	18:18	18:43
0°	05:06	05:30	05:50	17:57	18:17	18:41
S 10°	05:06	05:30	05:51	17:56	18:17	18:41
20°	05:04	05:30	05:52	17:56	18:18	18:43
30°	05:01	05:28	05:52	17:55	18:19	18:47
35°	04:58	05:27	05:53	17:55	18:20	18:50
40°	04:55	05:26	05:53	17:55	18:22	18:53
45°	04:50	05:24	05:53	17:54	18:24	18:58
S 50°	04:44	05:22	05:54	17:54	18:26	19:04
52°	04:41	05:20	05:54	17:54	18:28	19:07
54°	04:37	05:19	05:54	17:54	18:29	19:11
56°	04:34	05:17	05:54	17:54	18:31	19:15
58°	04:29	05:15	05:54	17:54	18:33	19:19
S 60°	04:24	05:13	05:55	17:54	18:35	19:25

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	18:18	17:48	17:24	· · ·	00:55	02:52
N 70°	17:55	17:36	17:20	· · ·	01:16	03:02
68°	17:38	17:26	17:16	· · ·	01:32	03:10
66°	17:23	17:18	17:12	00:09	01:45	03:17
64°	17:12	17:11	17:09	00:27	01:56	03:22
62°	17:01	17:05	17:07	00:41	02:05	03:27
60°	16:53	16:59	17:05	00:53	02:13	03:31
N 58°	16:45	16:55	17:03	01:03	02:19	03:35
56°	16:38	16:50	17:01	01:12	02:25	03:38
54°	16:32	16:47	16:59	01:20	02:31	03:41
52°	16:27	16:43	16:58	01:27	02:35	03:44
50°	16:22	16:40	16:56	01:33	02:40	03:46
45°	16:11	16:33	16:54	01:46	02:49	03:51
N 40°	16:02	16:27	16:51	01:58	02:57	03:55
35°	15:54	16:23	16:49	02:07	03:03	03:59
30°	15:48	16:18	16:47	02:15	03:09	04:02
20°	15:36	16:11	16:44	02:29	03:19	04:08
N 10°	15:25	16:04	16:41	02:42	03:28	04:13
0°	15:16	15:58	16:38	02:53	03:36	04:17
S 10°	15:06	15:51	16:36	03:05	03:44	04:22
20°	14:55	15:44	16:33	03:17	03:53	04:27
30°	14:43	15:37	16:29	03:31	04:02	04:32
35°	14:36	15:32	16:28	03:38	04:08	04:35
40°	14:28	15:27	16:25	03:47	04:14	04:38
45°	14:19	15:21	16:23	03:58	04:21	04:42
S 50°	14:08	15:14	16:20	04:11	04:30	04:47
52°	14:02	15:11	16:19	04:16	04:34	04:49
54°	13:57	15:07	16:17	04:23	04:38	04:52
56°	13:50	15:03	16:15	04:30	04:43	04:54
58°	13:43	14:59	16:14	04:38	04:49	04:57
S 60°	13:35	14:54	16:11	04:47	04:55	05:01

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time	12 ^h		Mer. Pass.		
	00 ^h	mm:ss	hh:mm	Upper	Lower	11-13
	mm:ss	mm:ss	mm:ss	hh:mm	hh:mm	84-95%
19	06:00	06:11	11:54	21:26	09:05	
20	06:22	06:32	11:53	22:07	09:47	
21	06:43	06:54	11:53	22:48	10:28	

September 22, 23, 24 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	0°45.2	146°21.2	S19°41.9	148°20.7	S13°15.6	1°39.6	S02°08.0	173°48.5	S00°38.8	Alpheratz	357°45.2 29°09.2
1	15°47.6	161°22.4	42.7	163°21.6	16.2	16°42.3	08.2	188°50.7	38.9	Ankaa	353°17.1 -42°14.6
2	30°50.1	176°23.7	43.3	178°22.4	16.8	31°45.1	08.3	203°52.9	39.0	Schedar	349°42.4 56°35.9
3	45°52.6	191°25.0	· · 44.0	193°23.3	· · 17.4	46°47.9	· · 08.4	218°55.1	· · 39.2	Diphda	348°57.5 -17°55.4
4	60°55.0	206°26.3	44.7	208°24.1	18.0	61°50.6	08.6	233°57.2	39.3	Achernar	335°27.6 -57°10.7
5	75°57.5	221°27.6	45.4	223°24.9	18.6	76°53.4	08.7	248°59.4	39.4	Hamal	328°02.7 23°30.9
6	91°00.0	236°28.9	S19°46.1	238°25.8	S13°19.2	91°56.2	S02°08.8	264°01.6	S00°39.5	Polaris	318°26.4 89°18.5
7	106°02.4	251°30.2	46.8	253°26.6	19.8	106°58.9	09.0	279°03.8	39.7	Acamar	315°19.5 -40°15.4
8	121°04.9	266°31.5	47.5	268°27.4	20.3	122°01.7	09.1	294°06.0	39.8	Menkar	314°16.9 4°08.1
9	136°07.3	281°32.8	· · 48.2	283°28.3	· · 20.9	137°04.5	· · 09.2	309°08.2	· · 39.9	Mirfak	308°42.9 49°53.9
10	151°09.8	296°34.1	48.9	298°29.1	21.5	152°07.3	09.4	324°10.3	40.0	Aldebaran	290°51.5 16°31.9
11	166°12.3	311°35.4	49.6	313°30.0	22.1	167°10.0	09.5	339°12.5	40.1	Rigel	281°13.9 -8°11.2
12	181°14.7	326°36.7	S19°50.3	328°30.8	S13°22.7	182°12.8	S02°09.6	354°14.7	S00°40.3	Capella	280°37.3 46°00.4
13	196°17.2	341°38.0	51.0	343°31.6	23.3	197°15.6	09.8	9°16.9	40.4	Bellatrix	278°34.1 6°21.7
14	211°19.7	356°39.3	51.7	358°32.5	23.9	212°18.3	09.9	24°19.1	40.5	Elnath	278°15.1 28°37.0
15	226°22.1	11°40.6	· · 52.3	13°33.3	· · 24.5	227°21.1	· · 10.0	39°21.2	· · 40.6	Alnilam	275°48.3 -1°11.6
16	241°24.6	26°41.9	53.0	28°34.1	25.1	242°23.9	10.2	54°23.4	40.7	Betelgeuse	271°03.4 7°24.6
17	256°27.1	41°43.2	53.7	43°35.0	25.7	257°26.6	10.3	69°25.6	40.9	Canopus	263°57.1 -52°41.8
18	271°29.5	56°44.6	S19°54.4	58°35.8	S13°26.3	272°29.4	S02°10.4	84°27.8	S00°41.0	Sirius	258°35.5 -16°43.7
19	286°32.0	71°45.9	55.1	73°36.6	26.8	287°32.2	10.6	99°30.0	41.1	Adhara	255°14.2 -28°59.0
20	301°34.5	86°47.2	55.8	88°37.5	27.4	302°35.0	10.7	114°32.2	41.2	Procyon	245°01.9 5°11.9
21	316°36.9	101°48.5	· · 56.4	103°38.3	· · 28.0	317°37.7	· · 10.8	129°34.3	· · 41.3	Pollux	243°30.3 27°59.9
22	331°39.4	116°49.9	57.1	118°39.1	28.6	332°40.5	11.0	144°36.5	41.5	Avior	234°19.3 -59°32.4
23	346°41.8	131°51.2	57.8	133°40.0	29.2	347°43.3	11.1	159°38.7	41.6	Suhail	222°54.3 -43°28.4
Mer.pass. 23:53		ν 1.3' d 0.7' m -4.80		ν 0.8' d 0.6' m 1.51		ν 2.8' d 0.1' m -2.94		ν 2.2' d 0.1' m 0.93		Miaplacidus	221°41.0 -69°45.5
Alphard 217°58.3 -8°42.2 Regulus 207°45.9 11°54.9 Dubhe 193°54.5 61°41.5 Denebola 182°36.0 14°30.7 Gienah 175°54.7 -17°36.1 Acrux 173°12.4 -63°09.6 Gacrux 172°03.8 -57°10.5 Alioth 166°22.8 55°54.1 Spica 158°33.6 -11°13.0 Alkaid 153°00.8 49°15.7 Hadar 148°51.4 -60°25.6 Menkent 148°10.3 -36°25.4 Arcturus 145°57.8 19°07.7 Rigel Kent. 139°55.0 -60°52.9 Kochab 137°20.5 74°06.8 Zuben'ubi 137°07.9 -16°05.2 Alphecca 126°12.9 26°40.9 Antares 112°28.8 -26°27.4 Atria 107°32.6 -69°03.1 Sabik 102°14.9 -15°44.2 Shaula 96°24.6 -37°06.8 Rasalhague 96°08.3 12°33.3 Eltanin 90°47.1 51°29.6 Kaus Aust. 83°46.4 -34°22.8 Vega 80°40.2 38°47.9 Nunki 76°00.7 -26°17.0 Altair 62°10.0 8°54.0 Peacock 53°21.9 -56°42.1 Deneb 49°32.6 45°19.5 Enif 33°48.8 9°55.7 Al Na'ir 27°45.7 -46°54.5 Fomalhaut 15°25.7 -29°33.7 Scheat 13°55.0 28°08.7 Markab 13°40.0 15°16.0											
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	1°44.3	146°52.5	S19°58.5	148°40.8	S13°29.8	2°46.0	S02°11.2	174°40.9	S00°41.7		
1	16°46.8	161°53.9	59.1	163°41.6	30.4	17°48.8	11.4	189°43.1	41.8		
2	31°49.2	176°55.2	19°59.8	178°42.5	31.0	32°51.6	11.5	204°45.3	42.0		
3	46°51.7	191°56.6	20°00.5	193°43.3	· · 31.6	47°54.3	· · 11.6	219°47.4	· · 42.1		
4	61°54.2	206°57.9	01.1	208°44.1	32.2	62°57.1	11.8	234°49.6	42.2		
5	76°56.6	221°59.3	01.8	223°45.0	32.7	77°59.9	11.9	249°51.8	42.3		
6	91°59.1	237°00.6	S20°02.5	238°45.8	S13°33.3	93°02.7	S02°12.0	264°54.0	S00°42.4		
7	107°01.6	252°02.0	03.2	253°46.6	33.9	108°05.4	12.2	279°56.2	42.6		
8	122°04.0	267°03.3	03.8	268°47.5	34.5	123°08.2	12.3	294°58.3	42.7		
9	137°06.5	282°04.7	· · 04.5	283°48.3	· · 35.1	138°11.0	· · 12.4	310°00.5	· · 42.8		
10	152°08.9	297°06.0	05.2	298°49.1	35.7	153°13.7	12.6	325°02.7	42.9		
11	167°11.4	312°07.4	05.8	313°50.0	36.3	168°16.5	12.7	340°04.9	43.0		
12	182°13.9	327°08.8	S20°06.5	328°50.8	S13°36.9	183°19.3	S02°12.8	355°07.1	S00°43.2		
13	197°16.3	342°10.1	07.1	343°51.6	37.4	198°22.0	13.0	10°09.3	43.3		
14	212°18.8	357°11.5	07.8	358°52.4	38.0	213°24.8	13.1	25°11.4	43.4		
15	227°21.3	12°12.9	· · 08.5	13°53.3	· · 38.6	228°27.6	· · 13.2	40°13.6	· · 43.5		
16	242°23.7	27°14.2	09.1	28°54.1	39.2	243°30.4	13.4	55°15.8	43.6		
17	257°26.2	42°15.6	09.8	43°54.9	39.8	258°33.1	13.5	70°18.0	43.8		
18	272°28.7	57°17.0	S20°10.4	58°55.8	S13°40.4	273°35.9	S02°13.6	85°20.2	S00°43.9		
19	287°31.1	72°18.4	11.1	73°56.6	41.0	288°38.7	13.8	100°22.3	44.0		
20	302°33.6	87°19.8	11.7	88°57.4	41.6	303°41.4	13.9	115°24.5	44.1		
21	317°36.1	102°21.1	· · 12.4	103°58.3	· · 42.1	318°44.2	· · 14.0	130°26.7	· · 44.3		
22	332°38.5	117°22.5	13.0	118°59.1	42.7	333°47.0	14.2	145°28.9	44.4		
23	347°41.0	132°23.9	13.7	133°59.9	43.3	348°49.7	14.3	160°31.1	44.5		
Mer.pass. 23:49		ν 1.3' d 0.7' m -4.81		ν 0.8' d 0.6' m 1.51		ν 2.8' d 0.1' m -2.94		ν 2.2' d 0.1' m 0.92			
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	2°43.4	147°25.3	S20°14.3	149°00.7	S13°43.9	3°52.5	S02°14.4	175°33.3	S00°44.6		
1	17°45.9	162°26.7	15.0	164°01.6	44.5	18°55.3	14.6	190°35.4	44.7		
2	32°48.4	177°28.1	15.6	179°02.4	45.1	33°58.0	14.7	205°37.6	44.9		
3	47°50.8	192°29.5	· · 16.3	194°03.2	· · 45.7	49°00.8	· · 14.8	220°39.8	· · 45.0		
4	62°53.3	207°30.9	16.9	209°04.0	46.2	64°03.6	15.0	235°42.0	45.1		
5	77°55.8	222°32.3	17.6	224°04.9	46.8	79°06.4	15.1	250°44.2	45.2		
6	92°58.2	237°33.7	S20°18.2	239°05.7	S13°47.4	94°09.1	S02°15.2	265°46.3	S00°45.3		
7	108°00.7	252°35.1	18.9	254°06.5	48.0	109°11.9	15.4	280°48.5	45.5		
8	123°03.2	267°36.6	19.5	269°07.3	48.6	124°14.7	15.5	295°50.7	45.6		
9	138°05.6	282°38.0	· · 20.1	284°08.2	· · 49.2	139°17.4	· · 15.6	310°52.9	· · 45.7		
10	153°08.1	297°39.4	20.8	299°09.0	49.8	154°20.2	15.8	325°55.1	45.8		
11	168°10.5	312°40.8	21.4	314°09.8	50.3	169°23.0	15.9	340°57.3	46.0		
12	183°13.0	327°42.2	S20°22.0	329°10.6	S13°50.9	184°25.7	S02°16.0	355°59.4	S00°46.1		
13	198°15.5	342°43.7	22.7	344°11.5	51.5	199°28.5	16.2	11°01.6	46.2		
14	213°17.9	357°45.1	23.3	359°12.3	52.1	214°31.3	16.3	26°03.8	46.3		
15	228°20.4	12°46.5	· · 23.9	14°13.1	· · 52.7	229°34.1	· · 16.4	41°06.0	· · 46.4		
16	243°22.9	27°48.0	24.6	29°13.9	53.3	244°36.8	16.6	56°08.2	46.6		
17	258°25.3	42°49.4	25.2	44°14.8	53.9	259°39.6	16.7	71°10.3	46.7		
18	273°27.8	57°50.8	S20°25.8	59°15.6	S13°54.4	274°42.4	S02°16.8	86°12.5	S00°46.8		
19	288°30.3	72°52.3	26.5	74°16.4	55.0	289°45.1	17.0	101°14.7	46.9		
20	303°32.7	87°53.7	27.1	89°17.2	55.6	304°47.9	17.1	116°16.9	47.0		
21	318°35.2	102°55.2	· · 27.7	104°18.1	· · 56.2	319°50.7	· · 17.2	131°19.1	· · 47.2		
22	333°37.7	117°56.6	28.3	119°18.9	56.8	334°53.4	17.4	146°21.3	47.3		
23	348°40.1	132°58.1	28.9	134°19.7	57.4	349°56.2	17.5	161°23.4	47.4		
Mer.pass. 23:45		ν 1.4' d 0.6' m -4.81		ν 0.8' d 0.6' m 1.51		ν 2.8' d 0.1' m -2.93		ν 2.2' d 0.1' m 0.92			
Sep 22 Wed SHA Mer.pass Venus 145°36.0 14:13 Mars 147°35.6 14:06 Jupiter 0°54.4 23:49 Saturn 173°03.3 12:23 Sep 23 Thu SHA Mer.pass Venus 145°08.2 14:11 Mars 146°56.5 14:04 Jupiter 1°01.7 23:45 Saturn 172°56.6 12:19 Sep 24 Fri SHA Mer.pass Venus 144°41.9 14:09 Mars 146°17.3 14:03 Jupiter 1°09.1 23:40 Saturn 172°49.8 12:16 Horizontal parallax Venus: 0.3 Mars: 0.1											

h	Sun			Moon			
Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	181°46.1	N00°26.4	17°30.5	16.5'	S02°12.5	-12.3'	53.9'
1	196°46.3	25.4	32°06.0	16.5'	02°00.2	-12.3'	53.9'
2	211°46.5	24.5	46°41.5	16.5'	01°47.9	-12.3'	53.9'
3	226°46.7	· · 23.5	61°17.0	16.5'	01°35.5	-12.3'	53.9'
4	241°47.0	22.5	75°52.5	16.5'	01°23.2	-12.3'	54.0'
5	256°47.2	21.6	90°28.0	16.5'	01°10.9	-12.4'	54.0'
6	271°47.4	N00°20.6	105°03.5	16.5'	S00°58.5	-12.4'	54.0'
7	286°47.6	19.6	119°39.0	16.5'	00°46.1	-12.4'	54.0'
8	301°47.8	18.6	134°14.5	16.5'	00°33.8	-12.4'	54.0'
9	316°48.1	· · 17.7	148°50.0	16.5'	00°21.4	-12.4'	54.0'
10	331°48.3	16.7	163°25.5	16.5'	S00°09.0	-12.4'	54.0'
11	346°48.5	15.7	178°01.0	16.5'	N00°03.3	12.4'	54.0'
12	1°48.7	N00°14.7	192°36.5	16.5'	N00°15.7	12.4'	54.0'
13	16°48.9	13.8	207°12.0	16.5'	00°28.1	12.4'	54.0'
14	31°49.2	12.8	221°47.4	16.5'	00°40.5	12.4'	54.0'
15	46°49.4	· · 11.8	236°22.9	16.5'	00°52.8	12.4'	54.0'
16	61°49.6	10.9	250°58.3	16.4'	01°05.2	12.4'	54.0'
17	76°49.8	09.9	265°33.8	16.4'	01°17.6	12.4'	54.0'
18	91°50.0	N00°08.9	280°09.2	16.4'	N01°30.0	12.4'	54.0'
19	106°50.3	07.9	294°44.6	16.4'	01°42.3	12.4'	54.0'
20	121°50.5	07.0	309°20.0	16.4'	01°54.7	12.4'	54.0'
21	136°50.7	· · 06.0	323°55.4	16.4'	02°07.1	12.4'	54.0'
22	151°50.9	05.0	338°30.8	16.4'	02°19.4	12.3'	54.0'
23	166°51.1	04.1	353°06.2	16.4'	02°31.8	12.3'	54.0'
SD = 15.9' $d = -1.0'$			SD = 14.7'				

Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	181°51.4	N00°03.1	7°41.5	16.3'	N02°44.1	12.3'	54.0'
1	196°51.6	02.1	22°16.9	16.3'	02°56.4	12.3'	54.1'
2	211°51.8	01.1	36°52.2	16.3'	03°08.8	12.3'	54.1'
3	226°52.0	N00°00.2	51°27.5	16.3'	03°21.1	12.3'	54.1'
4	241°52.3	S00°00.8	66°02.8	16.3'	03°33.4	12.3'	54.1'
5	256°52.5	01.8	80°38.0	16.2'	03°45.7	12.3'	54.1'
6	271°52.7	S00°02.8	95°13.3	16.2'	N03°58.0	12.3'	54.1'
7	286°52.9	03.7	109°48.5	16.2'	04°10.2	12.3'	54.1'
8	301°53.1	04.7	124°23.7	16.2'	04°22.5	12.2'	54.1'
9	316°53.4	· · 05.7	138°58.9	16.2'	04°34.7	12.2'	54.1'
10	331°53.6	06.6	153°34.0	16.1'	04°47.0	12.2'	54.1'
11	346°53.8	07.6	168°09.2	16.1'	04°59.2	12.2'	54.1'
12	1°54.0	S00°08.6	182°44.3	16.1'	N05°11.4	12.2'	54.1'
13	16°54.2	09.6	197°19.3	16.1'	05°23.6	12.2'	54.1'
14	31°54.5	10.5	211°54.4	16.0'	05°35.7	12.1'	54.2'
15	46°54.7	· · 11.5	226°29.4	16.0'	05°47.9	12.1'	54.2'
16	61°54.9	12.5	241°04.4	16.0'	06°00.0	12.1'	54.2'
17	76°55.1	13.5	255°39.4	15.9'	06°12.1	12.1'	54.2'
18	91°55.3	S00°14.4	270°14.3	15.9'	N06°24.2	12.1'	54.2'
19	106°55.5	15.4	284°49.2	15.9'	06°36.3	12.0'	54.2'
20	121°55.8	16.4	299°24.1	15.8'	06°48.4	12.0'	54.2'
21	136°56.0	· · 17.4	313°59.0	15.8'	07°00.4	12.0'	54.2'
22	151°56.2	18.3	328°33.8	15.8'	07°12.4	12.0'	54.2'
23	166°56.4	19.3	343°08.5	15.7'	07°24.4	12.0'	54.2'
SD = 15.9' $d = -1.0'$			SD = 14.7'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	181°56.6	S00°20.3	357°43.3	15.7'	N07°36.3	11.9'	54.2'
1	196°56.9	21.2	12°18.0	15.7'	07°48.2	11.9'	54.3'
2	211°57.1	22.2	26°52.7	15.6'	08°00.1	11.9'	54.3'
3	226°57.3	· · 23.2	41°27.3	15.6'	08°12.0	11.8'	54.3'
4	241°57.5	24.2	56°01.9	15.6'	08°23.9	11.8'	54.3'
5	256°57.7	25.1	70°36.5	15.5'	08°35.7	11.8'	54.3'
6	271°58.0	S00°26.1	85°11.0	15.5'	N08°47.5	11.8'	54.3'
7	286°58.2	27.1	99°45.5	15.4'	08°59.2	11.7'	54.3'
8	301°58.4	28.1	114°19.9	15.4'	09°10.9	11.7'	54.3'
9	316°58.6	· · 29.0	128°54.3	15.4'	09°22.6	11.7'	54.3'
10	331°58.8	30.0	143°28.6	15.3'	09°34.3	11.6'	54.3'
11	346°59.1	31.0	158°03.0	15.3'	09°45.9	11.6'	54.4'
12	1°59.3	S00°31.9	172°37.2	15.2'	N09°57.5	11.6'	54.4'
13	16°59.5	32.9	187°11.5	15.2'	10°09.1	11.5'	54.4'
14	31°59.7	33.9	201°45.6	15.1'	10°20.6	11.5'	54.4'
15	46°59.9	· · 34.9	216°19.8	15.1'	10°32.1	11.4'	54.4'
16	62°00.1	35.8	230°53.9	15.0'	10°43.5	11.4'	54.4'
17	77°00.4	36.8	245°27.9	15.0'	10°54.9	11.4'	54.4'
18	92°00.6	S00°37.8	260°01.9	14.9'	N11°06.3	11.3'	54.4'
19	107°00.8	38.8	274°35.8	14.9'	11°17.6	11.3'	54.5'
20	122°01.0	39.7	289°09.7	14.8'	11°28.9	11.2'	54.5'
21	137°01.2	· · 40.7	303°43.6	14.8'	11°40.1	11.2'	54.5'
22	152°01.5	41.7	318°17.4	14.7'	11°51.3	11.2'	54.5'
23	167°01.7	42.7	332°51.1	14.7'	12°02.5	11.1'	54.5'
SD = 15.9' $d = 1.0'$			SD = 14.8'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	03:03	04:34	05:42	18:00	19:08	20:37
N 70°	03:23	04:42	05:43	17:59	19:00	20:18
68°	03:38	04:48	05:44	17:59	18:54	20:04
66°	03:50	04:53	05:45	17:58	18:49	19:52
64°	03:59	04:58	05:45	17:58	18:45	19:43
62°	04:08	05:01	05:46	17:58	18:42	19:35
60°	04:14	05:04	05:46	17:57	18:39	19:29
N 58°	04:20	05:07	05:46	17:57	18:36	19:23
56°	04:25	05:10	05:47	17:57	18:34	19:18
54°	04:30	05:12	05:47	17:57	18:32	19:14
52°	04:34	05:14	05:47	17:56	18:30	19:10
50°	04:37	05:15	05:48	17:56	18:29	19:06
45°	04:44	05:19	05:48	17:56	18:25	19:00
N 40°	04:50	05:21	05:48	17:56	18:23	18:54
35°	04:54	05:23	05:49	17:56	18:21	18:50
30°	04:57	05:25	05:49	17:56	18:19	18:47
20°	05:01	05:27	05:49	17:56	18:17	18:43
N 10°	05:04	05:28	05:49	17:56	18:16	18:41
0°	05:05	05:29	05:49	17:56	18:16	18:40
S 10°	05:04	05:28	05:49	17:56	18:17	18:41
20°	05:01	05:27	05:49	17:56	18:18	18:44
30°	04:57	05:25	05:49	17:57	18:21	18:48
35°	04:54	05:23	05:48	17:57	18:22	18:52
40°	04:49	05:21	05:48	17:57	18:25	18:56
45°	04:44	05:18	05:48	17:58	18:27	19:02
S 50°	04:37	05:15	05:47	17:59	18:31	19:09
52°	04:33	05:13	05:47	17:59	18:33	19:13
54°	04:30	05:11	05:47	17:59	18:35	19:17
56°	04:25	05:09	05:46	18:00	18:37	19:21
58°	04:20	05:07	05:46	18:00	18:39	19:27
S 60°	04:14	05:04	05:46	18:01	18:42	19:33

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	17:03	16:41	16:15	04:41	06:29	08:22
N 70°	17:04	16:49	16:32	04:43	06:24	08:07
68°	17:06	16:56	16:45	04:45	06:19	07:56
66°	17:07	17:02	16:57	04:46	06:15	07:46
64°	17:08	17:07	17:06	04:47	06:12	07:38
62°	17:09	17:11	17:14	04:48	06:09	07:31
60°	17:10	17:15	17:21	04:49	06:06	07:25
N 58°	17:10	17:18	17:27	04:49	06:04	07:20
56°	17:11	17:21	17:33	04:50	06:02	07:15
54°	17:11	17:24	17:38	04:51	06:00	07:11
52°	17:12	17:26	17:42	04:51	05:59	07:07
50°	17:12	17:29	17:46	04:52	05:57	07:04
45°	17:13	17:33	17:55	04:53	05:54	06:57
N 40°	17:14	17:38	18:03	04:53	05:52	06:51
35°	17:15	17:41	18:09	04:54	05:49	06:45
30°	17:15	17:44	18:15	04:55	05:47	06:41
20°	17:17	17:50	18:24	04:56	05:44	06:33
N 10°	17:18	17:55	18:33	04:57	05:41	06:26
0°	17:19	17:59	18:41	04:58	05:38	06:19
S 10°	17:20	18:04	18:50	04:59	05:35	06:13
20°	17:21	18:09	18:59	04:59	05:32	06:06
30°	17:22	18:15	19:09	05:00	05:29	05:58
35°	17:23	18:18	19:15	05:01	05:27	05:54
40°	17:23	18:22	19:22	05:02	05:25	05:49
45°	17:24	18:26	19:29	05:02	05:22	05:43
S 50°	17:26	18:32	19:39	05:03	05:19	05:36
52°	17:26	18:34	19:44	05:04	05:18	05:33
54°	17:27	18:37	19:48	05:04	05:16	05:29
56°	17:27	18:40	19:54	05:05	05:15	05:25
58°	17:28	18:43	20:00	05:05	05:13	05:21
S 60°	17:29	18:47	20:07	05:06	05:11	05:16

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time	12 ^h		Mer. Pass.		
	00 ^h	mm:ss	hh:mm	Upper	Lower	14-16
				hh:mm	hh:mm	98-99%
22	07:04	07:15	11:53	23:28	11:08	
23	07:25	07:36	11:52	—	11:49	
24	07:47	07:57	11:52	00:09	12:30	

September 25, 26, 27 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	3°42.6	147°59.5	S20°29.6	149°20.5	S13°57.9	4°59.0	S02°17.6	176°25.6	S00°47.5	Alpheratz	357°45.2	29°09.2
1	18°45.0	163°01.0	30.2	164°21.3	58.5	20°01.7	17.8	191°27.8	47.6	Ankaa	353°17.1	-42°14.6
2	33°47.5	178°02.5	30.8	179°22.2	59.1	35°04.5	17.9	206°30.0	47.8	Schedar	349°42.4	56°35.9
3	48°50.0	193°03.9	· · 31.4	194°23.0	13°59.7	50°07.3	· · 18.0	221°32.2	· · 47.9	Diphda	348°57.5	-17°55.4
4	63°52.4	208°05.4	32.0	209°23.8	14°00.3	65°10.0	18.2	236°34.3	48.0	Achernar	335°27.5	-57°10.7
5	78°54.9	223°06.9	32.7	224°24.6	00.8	80°12.8	18.3	251°36.5	48.1	Hamal	328°02.7	23°30.9
6	93°57.4	238°08.3	S20°33.3	239°25.4	S14°01.4	95°15.6	S02°18.4	266°38.7	S00°48.3	Polaris	318°25.4	89°18.5
7	108°59.8	253°09.8	33.9	254°26.3	02.0	110°18.4	18.5	281°40.9	48.4	Acamar	315°19.5	-40°15.4
8	124°02.3	268°11.3	34.5	269°27.1	02.6	125°21.1	18.7	296°43.1	48.5	Menkar	314°16.9	4°08.1
9	139°04.8	283°12.8	· · 35.1	284°27.9	· · 03.2	140°23.9	· · 18.8	311°45.3	· · 48.6	Mirfak	308°42.9	49°53.9
10	154°07.2	298°14.2	35.7	299°28.7	03.8	155°26.7	18.9	326°47.4	48.7	Aldebaran	290°51.5	16°31.9
11	169°09.7	313°15.7	36.3	314°29.5	04.3	170°29.4	19.1	341°49.6	48.9	Rigel	281°13.9	-8°11.2
12	184°12.2	328°17.2	S20°36.9	329°30.4	S14°04.9	185°32.2	S02°19.2	356°51.8	S00°49.0	Capella	280°37.3	46°00.4
13	199°14.6	343°18.7	37.5	344°31.2	05.5	200°35.0	19.3	11°54.0	49.1	Bellatrix	278°34.1	6°21.7
14	214°17.1	358°20.2	38.1	359°32.0	06.1	215°37.7	19.5	26°56.2	49.2	Elnath	278°15.0	28°37.0
15	229°19.5	13°21.7	· · 38.7	14°32.8	· · 06.7	230°40.5	· · 19.6	41°58.3	· · 49.3	Alnilam	275°48.3	-1°11.6
16	244°22.0	28°23.2	39.4	29°33.6	07.2	245°43.3	19.7	57°00.5	49.5	Betelgeuse	271°03.4	7°24.6
17	259°24.5	43°24.7	40.0	44°34.4	07.8	260°46.0	19.9	72°02.7	49.6	Canopus	263°57.1	-52°41.8
18	274°26.9	58°26.2	S20°40.6	59°35.3	S14°08.4	275°48.8	S02°20.0	87°04.9	S00°49.7	Sirius	258°35.5	-16°43.7
19	289°29.4	73°27.7	41.2	74°36.1	09.0	290°51.6	20.1	102°07.1	49.8	Adhara	255°14.2	-28°59.0
20	304°31.9	88°29.2	41.7	89°36.9	09.6	305°54.3	20.3	117°09.2	49.9	Procyon	245°01.9	5°11.9
21	319°34.3	103°30.7	· · 42.3	104°37.7	· · 10.1	320°57.1	· · 20.4	132°11.4	· · 50.1	Pollux	243°30.3	27°59.9
22	334°36.8	118°32.2	42.9	119°38.5	10.7	335°59.9	20.5	147°13.6	50.2	Avior	234°19.3	-59°32.4
23	349°39.3	133°33.7	43.5	134°39.3	11.3	351°02.7	20.7	162°15.8	50.3	Suhail	222°54.3	-43°28.4
Mer.pass. 23:41		ν 1.5' d 0.6' m -4.82		ν 0.8' d 0.6' m 1.51		ν 2.8' d 0.1' m -2.93		ν 2.2' d 0.1' m 0.91		Miaplacidus	221°40.9	-69°45.5
										Alphard	217°58.3	-8°42.2
										Regulus	207°45.9	11°54.9
										Dubhe	193°54.5	61°41.5
										Denebola	182°36.0	14°30.7
										Gienah	175°54.7	-17°36.1
										Acrux	173°12.4	-63°09.6
										Gacrux	172°03.8	-57°10.4
										Alioth	166°22.8	55°54.1
										Spica	158°33.7	-11°13.0
										Alkaid	153°00.8	49°15.7
										Hadar	148°51.4	-60°25.6
										Menkent	148°10.3	-36°25.4
										Arcturus	145°57.8	19°07.7
										Rigel Kent.	139°55.0	-60°52.9
										Kochab	137°20.6	74°06.8
										Zuben'ubi	137°07.9	-16°05.2
										Alphecca	126°12.9	26°40.9
										Antares	112°28.8	-26°27.4
										Atria	107°32.7	-69°03.0
										Sabik	102°14.9	-15°44.2
										Shaula	96°24.7	-37°06.8
										Rasalhague	96°08.3	12°33.3
										Eltanin	90°47.1	51°29.6
										Kaus Aust.	83°46.4	-34°22.8
										Vega	80°40.3	38°47.9
										Nunki	76°00.7	-26°17.0
										Altair	62°10.0	8°54.0
										Peacock	53°21.9	-56°42.1
										Deneb	49°32.6	45°19.5
										Enif	33°48.8	9°55.7
										Al Na'ir	27°45.7	-46°54.5
										Fomalhaut	15°25.8	-29°33.7
										Scheat	13°55.0	28°08.7
										Markab	13°40.0	15°16.0
Sep 25 Sat		SHA		Mer.pass								
Venus		144°17.0		14:07								
Mars		145°37.9		14:02								
Jupiter		1°16.4		23:36								
Saturn		172°43.0		12:13								
Sep 26 Sun		SHA		Mer.pass								
Venus		143°53.5		14:04								
Mars		144°58.4		14:01								
Jupiter		1°23.7		23:31								
Saturn		172°36.3		12:09								
Sep 27 Mon		SHA		Mer.pass								
Venus		143°31.7		14:02								
Mars		144°18.8		13:59								
Jupiter		1°31.0		23:27								
Saturn		172°29.5		12:06								
Horizontal parallax												
Venus:						0.4						
Mars:						0.1						
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	5°40.9	149°12.6	S20°58.0	149°59.6	S14°25.7	7°11.8	S02°23.9	178°10.3	S00°53.3			
1	20°43.3	164°14.2	58.5	165°00.4	26.3	22°14.6	24.1	193°12.5	53.5			
2	35°45.8	179°15.7	59.1	180°01.3	26.9	37°17.4	24.2	208°14.7	53.6			
3	50°48.2	194°17.3	20°59.7	195°02.1	· · 27.5	52°20.1	· · 24.3	223°16.9	· · 53.7			
4	65°50.7	209°18.9	21°00.2	210°02.9	28.0	67°22.9	24.5	238°19.0	53.8			
5	80°53.2	224°20.5	00.8	225°03.7	28.6	82°25.7	24.6	253°21.2	53.9			
6	95°55.6	239°22.1	S21°01.3	240°04.5	S14°29.2	97°28.4	S02°24.7	268°23.4	S00°54.1			
7	110°58.1	254°23.7	01.9	255°05.3	29.8	112°31.2	24.8	283°25.6	54.2			
8	126°00.6	269°25.4	02.4	270°06.1	30.3	127°34.0	25.0	298°27.8	54.3			
9	141°03.0	284°27.0	· · 03.0	285°06.9	· · 30.9	142°36.8	· · 25.1	313°29.9	· · 54.4			
10	156°05.5	299°28.6	03.5	300°07.7	31.5	157°39.5	25.2	328°32.1	54.5			
11	171°08.0	314°30.2	04.1	315°08.5	32.1	172°42.3	25.4	343°34.3	54.7			
12	186°10.4	329°31.8	S21°04.6	330°09.3	S14°32.6	187°45.1	S02°25.5	358°36.5	S00°54.8			
13	201°12.9	344°33.4	05.2	345°10.1	33.2	202°47.8	25.6	13°38.7	54.9			
14	216°15.4	359°35.1	05.7	0°10.9	33.8	217°50.6	25.8	28°40.8	55.0			
15	231°17.8	14°36.7	· · 06.3	15°11.7	· · 34.4	232°53.4	· · 25.9	43°43.0	· · 55.1			
16	246°20.3	29°38.3	06.8	30°12.5	34.9	247°56.1	26.0	58°45.2	55.3			
17	261°22.7	44°40.0	07.4	45°13.3	35.5	262°58.9	26.1	73°47.4	55.4			
18	276°25.2	59°41.6	S21°07.9	60°14.1	S14°36.1	278°01.6	S02°26.3	88°49.6	S00°55.5			
19	291°27.7	74°43.2	08.4	75°15.0	36.6	293°04.4	26.4	103°51.8	55.6			
20	306°30.1	89°44.9	09.0	90°15.8	37.2	308°07.2	26.5	118°53.9	55.8			
21	321°32.6	104°46.5	· · 09.5	105°16.6	· · 37.8	323°09.9	· · 26.7	133°56.1	· · 55.9			
22	336°35.1	119°48.2	10.0	120°17.4	38.4	338°12.7	26.8	148°58.3	56.0			
23	351°37.5	134°49.8	10.6	135°18.2	38.9	353°15.5	26.9	164°00.5	56.1			
Mer.pass. 23:33		ν 1.6' d 0.6' m -4.82		ν 0.8' d 0.6' m 1.50		ν 2.8' d 0.1' m -2.93		ν 2.2' d 0.1' m 0.90				

h	Sun			Moon			
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	182°01.9	S00°43.6	347°24.8	14.6'	N12°13.6	11.1'	54.5'
1	197°02.1	44.6	1°58.5	14.6'	12°24.7	11.0'	54.5'
2	212°02.3	45.6	16°32.0	14.5'	12°35.7	11.0'	54.5'
3	227°02.5	· · 46.5	31°05.6	14.5'	12°46.7	10.9'	54.6'
4	242°02.8	47.5	45°39.1	14.4'	12°57.6	10.9'	54.6'
5	257°03.0	48.5	60°12.5	14.4'	13°08.4	10.8'	54.6'
6	272°03.2	S00°49.5	74°45.8	14.3'	N13°19.3	10.8'	54.6'
7	287°03.4	50.4	89°19.2	14.3'	13°30.0	10.7'	54.6'
8	302°03.6	51.4	103°52.4	14.2'	13°40.8	10.7'	54.6'
9	317°03.8	· · 52.4	118°25.6	14.1'	13°51.4	10.6'	54.6'
10	332°04.1	53.4	132°58.7	14.1'	14°02.0	10.6'	54.6'
11	347°04.3	54.3	147°31.8	14.0'	14°12.6	10.5'	54.7'
12	2°04.5	S00°55.3	162°04.8	14.0'	N14°23.1	10.4'	54.7'
13	17°04.7	56.3	176°37.8	13.9'	14°33.6	10.4'	54.7'
14	32°04.9	57.3	191°10.7	13.8'	14°43.9	10.3'	54.7'
15	47°05.1	· · 58.2	205°43.5	13.8'	14°54.3	10.3'	54.7'
16	62°05.4	00°59.2	220°16.3	13.7'	15°04.6	10.2'	54.7'
17	77°05.6	01°00.2	234°49.0	13.7'	15°14.8	10.2'	54.8'
18	92°05.8	S01°01.2	249°21.7	13.6'	N15°24.9	10.1'	54.8'
19	107°06.0	02.1	263°54.3	13.5'	15°35.0	10.0'	54.8'
20	122°06.2	03.1	278°26.8	13.5'	15°45.0	10.0'	54.8'
21	137°06.4	· · 04.1	292°59.3	13.4'	15°55.0	9.9'	54.8'
22	152°06.7	05.0	307°31.6	13.3'	16°04.9	9.8'	54.8'
23	167°06.9	06.0	322°04.0	13.3'	16°14.7	9.8'	54.8'
SD = 15.9'			SD = 14.9'				

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	182°07.1	S01°07.0	336°36.2	13.2'	N16°24.5	9.7'	54.9'
1	197°07.3	08.0	351°08.4	13.1'	16°34.2	9.6'	54.9'
2	212°07.5	08.9	5°40.6	13.1'	16°43.8	9.6'	54.9'
3	227°07.7	· · 09.9	20°12.6	13.0'	16°53.4	9.5'	54.9'
4	242°08.0	10.9	34°44.6	12.9'	17°02.9	9.4'	54.9'
5	257°08.2	11.9	49°16.6	12.9'	17°12.3	9.3'	54.9'
6	272°08.4	S01°12.8	63°48.4	12.8'	N17°21.7	9.3'	55.0'
7	287°08.6	13.8	78°20.2	12.7'	17°30.9	9.2'	55.0'
8	302°08.8	14.8	92°51.9	12.7'	17°40.1	9.1'	55.0'
9	317°09.0	· · 15.8	107°23.6	12.6'	17°49.3	9.1'	55.0'
10	332°09.2	16.7	121°55.2	12.5'	17°58.3	9.0'	55.0'
11	347°09.5	17.7	136°26.7	12.4'	18°07.3	8.9'	55.0'
12	2°09.7	S01°18.7	150°58.1	12.4'	N18°16.2	8.8'	55.1'
13	17°09.9	19.6	165°29.5	12.3'	18°25.0	8.7'	55.1'
14	32°10.1	20.6	180°00.8	12.2'	18°33.8	8.7'	55.1'
15	47°10.3	· · 21.6	194°32.0	12.2'	18°42.4	8.6'	55.1'
16	62°10.5	22.6	209°03.2	12.1'	18°51.0	8.5'	55.1'
17	77°10.7	23.5	223°34.3	12.0'	18°59.5	8.4'	55.2'
18	92°11.0	S01°24.5	238°05.3	11.9'	N19°07.9	8.3'	55.2'
19	107°11.2	25.5	252°36.2	11.9'	19°16.2	8.2'	55.2'
20	122°11.4	26.5	267°07.1	11.8'	19°24.4	8.2'	55.2'
21	137°11.6	· · 27.4	281°37.9	11.7'	19°32.6	8.1'	55.2'
22	152°11.8	28.4	296°08.6	11.7'	19°40.7	8.0'	55.3'
23	167°12.0	29.4	310°39.3	11.6'	19°48.6	7.9'	55.3'
SD = 15.9'			SD = 15.0'				

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	182°12.2	S01°30.4	325°09.9	11.5'	N19°56.5	7.8'	55.3'
1	197°12.5	31.3	339°40.4	11.4'	20°04.3	7.7'	55.3'
2	212°12.7	32.3	354°10.8	11.4'	20°12.0	7.6'	55.3'
3	227°12.9	· · 33.3	8°41.2	11.3'	20°19.6	7.5'	55.4'
4	242°13.1	34.2	23°11.5	11.2'	20°27.2	7.4'	55.4'
5	257°13.3	35.2	37°41.7	11.1'	20°34.6	7.3'	55.4'
6	272°13.5	S01°36.2	52°11.8	11.1'	N20°41.9	7.2'	55.4'
7	287°13.7	37.2	66°41.9	11.0'	20°49.2	7.1'	55.4'
8	302°13.9	38.1	81°11.9	10.9'	20°56.3	7.0'	55.5'
9	317°14.2	· · 39.1	95°41.8	10.9'	21°03.3	6.9'	55.5'
10	332°14.4	40.1	110°11.7	10.8'	21°10.3	6.8'	55.5'
11	347°14.6	41.1	124°41.4	10.7'	21°17.1	6.7'	55.5'
12	2°14.8	S01°42.0	139°11.2	10.6'	N21°23.8	6.6'	55.5'
13	17°15.0	43.0	153°40.8	10.6'	21°30.5	6.5'	55.6'
14	32°15.2	44.0	168°10.3	10.5'	21°37.0	6.4'	55.6'
15	47°15.4	· · 44.9	182°39.8	10.4'	21°43.4	6.3'	55.6'
16	62°15.6	45.9	197°09.3	10.3'	21°49.8	6.2'	55.6'
17	77°15.9	46.9	211°38.6	10.3'	21°56.0	6.1'	55.6'
18	92°16.1	S01°47.9	226°07.9	10.2'	N22°02.1	6.0'	55.7'
19	107°16.3	48.8	240°37.1	10.1'	22°08.1	5.9'	55.7'
20	122°16.5	49.8	255°06.2	10.1'	22°14.0	5.8'	55.7'
21	137°16.7	· · 50.8	269°35.3	10.0'	22°19.8	5.7'	55.7'
22	152°16.9	51.8	284°04.3	9.9'	22°25.5	5.6'	55.8'
23	167°17.1	52.7	298°33.2	9.9'	22°31.0	5.5'	55.8'
SD = 15.9'			SD = 15.1'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	03:21	04:48	05:56	17:45	18:52	20:17
N 70°	03:38	04:54	05:55	17:46	18:46	20:02
68°	03:50	04:59	05:55	17:46	18:42	19:49
66°	04:01	05:03	05:54	17:47	18:38	19:39
64°	04:09	05:06	05:54	17:47	18:35	19:31
62°	04:16	05:09	05:53	17:48	18:32	19:25
60°	04:22	05:12	05:53	17:48	18:30	19:19
N 58°	04:27	05:14	05:53	17:49	18:28	19:14
56°	04:32	05:16	05:53	17:49	18:26	19:10
54°	04:36	05:17	05:52	17:49	18:24	19:06
52°	04:39	05:19	05:52	17:49	18:23	19:03
50°	04:42	05:20	05:52	17:50	18:22	19:00
45°	04:48	05:22	05:52	17:50	18:20	18:54
N 40°	04:53	05:24	05:51	17:51	18:18	18:49
35°	04:56	05:26	05:51	17:51	18:17	18:46
30°	04:59	05:27	05:50	17:52	18:16	18:43
20°	05:02	05:28	05:50	17:53	18:15	18:40
N 10°	05:04	05:28	05:49	17:54	18:15	18:39
0°	05:03	05:27	05:48	17:55	18:15	18:39
S 10°	05:02	05:26	05:47	17:56	18:17	18:41
20°	04:59	05:24	05:46	17:57	18:19	18:45
30°	04:53	05:21	05:45	17:58	18:22	18:50
35°	04:49	05:19	05:44	17:59	18:25	18:54
40°	04:44	05:16	05:43	18:00	18:27	18:59
45°	04:38	05:13	05:42	18:02	18:31	19:06
S 50°	04:30	05:08	05:41	18:03	18:36	19:14
52°	04:26	05:06	05:40	18:04	18:38	19:18
54°	04:22	05:04	05:39	18:05	18:40	19:23
56°	04:17	05:01	05:38	18:06	18:43	19:28
58°	04:11	04:58	05:37	18:07	18:46	19:34
S 60°	04:04	04:55	05:36	18:08	18:50	19:41

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	15:38	□	□	10:30	□	□
N 70°	16:09	15:30	□	10:00	12:15	□
68°	16:33	16:15	15:32	09:38	11:31	13:56
66°	16:51	16:45	16:37	09:21	11:02	12:51
64°	17:06	17:08	17:13	09:07	10:40	12:16
62°	17:18	17:26	17:38	08:56	10:23	11:51
60°	17:29	17:41	17:59	08:46	10:09	11:31
N 58°	17:38	17:54	18:15	08:37	09:56	11:15
56°	17:47	18:05	18:30	08:30	09:46	11:01
54°	17:54	18:15	18:42	08:23	09:36	10:50
52°	18:01	18:23	18:53	08:17	09:28	10:39
50°	18:07	18:31	19:02	08:12	09:21	10:30
45°	18:19	18:48	19:23	08:00	09:05	10:10
N 40°	18:30	19:02	19:39	07:51	08:52	09:54
35°	18:39	19:14	19:53	07:43	08:41	09:41
30°	18:48	19:24	20:06	07:35	08:32	09:29
20°	19:02	19:42	20:27	07:23	08:15	09:09
N 10°	19:14	19:58	20:45	07:12	08:01	08:52
0°	19:26	20:12	21:02	07:02	07:48	08:36
S 10°	19:37	20:27	21:19	06:52	07:34	08:20
20°	19:50	20:43	21:38	06:42	07:20	08:03
30°	20:04	21:01	21:59	06:30	07:04	07:43
35°	20:13	21:12	22:12	06:23	06:55	07:32
40°	20:22	21:24	22:27	06:15	06:44	07:19
45°	20:34	21:39	22:44	06:06	06:32	07:03
S 50°	20:48	21:57	23:05	05:55	06:17	06:44
52°	20:54	22:05	23:16	05:50	06:10	06:35
54°	21:01	22:15	23:27	05:44	06:02	06:25
56°	21:09	22:26	23:41	05:38	05:53	06:14
58°	21:18	22:38	23:56	05:31	05:44	06:01
S 60°	21:29	22:53	· · ·	05:23	05:32	05:46

Day	Sun		Mer.	Moon		Age
	Eqn.of Time	12 ^h	Pass	Mer.Pass.	Lower	17-19
	mm:ss	mm:ss	hh:mm	hh:mm	hh:mm	97-88%
25	08:08	08:18	11:52	00:52	13:14	
26	08:28	08:39	11:51	01:37	14:00	
27	08:49	08:59	11:51	02:24	14:49	

September 28, 29, 30 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	6°40.0	149°51.5	S21°11.1	150°19.0	S14°39.5	8°18.2	S02°27.1	179°02.7	S00°56.2	Alpheratz	357°45.2 29°09.2
1	21°42.5	164°53.1	11.6	165°19.8	40.1	23°21.0	27.2	194°04.8	56.4	Ankaa	353°17.1 -42°14.6
2	36°44.9	179°54.8	12.2	180°20.6	40.7	38°23.8	27.3	209°07.0	56.5	Schedar	349°42.3 56°35.9
3	51°47.4	194°56.5	· · 12.7	195°21.4	· · 41.2	53°26.5	· · 27.4	224°09.2	· · 56.6	Diphda	348°57.5 -17°55.4
4	66°49.9	209°58.1	13.2	210°22.2	41.8	68°29.3	27.6	239°11.4	56.7	Achernar	335°27.5 -57°10.7
5	81°52.3	224°59.8	13.7	225°23.0	42.4	83°32.1	27.7	254°13.6	56.8	Hamal	328°02.7 23°30.9
6	96°54.8	240°01.5	S21°14.3	240°23.8	S14°42.9	98°34.8	S02°27.8	269°15.7	S00°57.0	Polaris	318°24.2 89°18.5
7	111°57.2	255°03.2	14.8	255°24.6	43.5	113°37.6	28.0	284°17.9	57.1	Acamar	315°19.4 -40°15.4
8	126°59.7	270°04.8	15.3	270°25.4	44.1	128°40.4	28.1	299°20.1	57.2	Menkar	314°16.9 4°08.1
9	142°02.2	285°06.5	· · 15.8	285°26.2	· · 44.6	143°43.1	· · 28.2	314°22.3	· · 57.3	Mirfak	308°42.9 49°53.9
10	157°04.6	300°08.2	16.4	300°27.0	45.2	158°45.9	28.4	329°24.5	57.4	Aldebaran	290°51.5 16°31.9
11	172°07.1	315°09.9	16.9	315°27.8	45.8	173°48.7	28.5	344°26.6	57.6	Rigel	281°13.8 -8°11.2
12	187°09.6	330°11.6	S21°17.4	330°28.6	S14°46.4	188°51.4	S02°28.6	359°28.8	S00°57.7	Capella	280°37.2 46°00.4
13	202°12.0	345°13.3	17.9	345°29.4	46.9	203°54.2	28.7	14°31.0	57.8	Bellatrix	278°34.0 6°21.7
14	217°14.5	0°15.0	18.4	0°30.2	47.5	218°57.0	28.9	29°33.2	57.9	Elnath	278°15.0 28°37.0
15	232°17.0	15°16.7	· · 18.9	15°31.0	· · 48.1	233°59.7	· · 29.0	44°35.4	· · 58.1	Alnilam	275°48.3 -1°11.6
16	247°19.4	30°18.4	19.4	30°31.8	48.6	249°02.5	29.1	59°37.5	58.2	Betelgeuse	271°03.4 7°24.6
17	262°21.9	45°20.1	19.9	45°32.6	49.2	264°05.3	29.3	74°39.7	58.3	Canopus	263°57.0 -52°41.8
18	277°24.3	60°21.8	S21°20.5	60°33.4	S14°49.8	279°08.0	S02°29.4	89°41.9	S00°58.4	Sirius	258°35.5 -16°43.7
19	292°26.8	75°23.5	21.0	75°34.2	50.3	294°10.8	29.5	104°44.1	58.5	Adhara	255°14.1 -28°59.0
20	307°29.3	90°25.2	21.5	90°35.0	50.9	309°13.6	29.6	119°46.3	58.7	Procyon	245°01.9 5°11.9
21	322°31.7	105°26.9	· · 22.0	105°35.8	· · 51.5	324°16.3	· · 29.8	134°48.4	· · 58.8	Pollux	243°30.2 27°59.9
22	337°34.2	120°28.7	22.5	120°36.6	52.1	339°19.1	29.9	149°50.6	58.9	Avior	234°19.2 -59°32.4
23	352°36.7	135°30.4	23.0	135°37.4	52.6	354°21.8	30.0	164°52.8	59.0	Suhail	222°54.2 -43°28.4
Mer.pass. 23:29		ν 1.7' d 0.5' m -4.82		ν 0.8' d 0.6' m 1.50		ν 2.8' d 0.1' m -2.93		ν 2.2' d 0.1' m 0.90		Miaplacidus	221°40.9 -69°45.5
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	7°39.1	150°32.1	S21°23.5	150°38.2	S14°53.2	9°24.6	S02°30.2	179°55.0	S00°59.1	Alphard	217°58.3 -8°42.2
1	22°41.6	165°33.8	24.0	165°38.9	53.8	24°27.4	30.3	194°57.2	59.3	Regulus	207°45.9 11°54.9
2	37°44.1	180°35.6	24.5	180°39.7	54.3	39°30.1	30.4	209°59.4	59.4	Dubhe	193°54.5 61°41.4
3	52°46.5	195°37.3	· · 25.0	195°40.5	· · 54.9	54°32.9	· · 30.5	225°01.5	· · 59.5	Denebola	182°36.0 14°30.7
4	67°49.0	210°39.1	25.5	210°41.3	55.5	69°35.7	30.7	240°03.7	59.6	Gienah	175°54.7 -17°36.1
5	82°51.5	225°40.8	26.0	225°42.1	56.0	84°38.4	30.8	255°05.9	59.7	Acrux	173°12.4 -63°09.6
6	97°53.9	240°42.5	S21°26.4	240°42.9	S14°56.6	99°41.2	S02°30.9	270°08.1	S00°59.9	Gacrux	172°03.8 -57°10.4
7	112°56.4	255°44.3	26.9	255°43.7	57.2	114°44.0	31.1	285°10.3	01°00.0	Alioth	166°22.8 55°54.1
8	127°58.8	270°46.0	27.4	270°44.5	57.7	129°46.7	31.2	300°12.4	00.1	Spica	158°33.7 -11°13.0
9	143°01.3	285°47.8	· · 27.9	285°45.3	· · 58.3	144°49.5	· · 31.3	315°14.6	· · 00.2	Alkaid	153°00.8 49°15.7
10	158°03.8	300°49.6	28.4	300°46.1	58.9	159°52.3	31.4	330°16.8	00.3	Hadar	148°51.4 -60°25.6
11	173°06.2	315°51.3	28.9	315°46.9	14°59.4	174°55.0	31.6	345°19.0	00.5	Menkent	148°10.3 -36°25.4
12	188°08.7	330°53.1	S21°29.4	330°47.7	S15°00.0	189°57.8	S02°31.7	0°21.2	S01°00.6	Arcturus	145°57.8 19°07.7
13	203°11.2	345°54.8	29.9	345°48.5	00.6	205°00.5	31.8	15°23.3	00.7	Rigel Kent.	139°55.1 -60°52.9
14	218°13.6	0°56.6	30.3	0°49.3	01.1	220°03.3	32.0	30°25.5	00.8	Kochab	137°20.6 74°06.8
15	233°16.1	15°58.4	· · 30.8	15°50.1	· · 01.7	235°06.1	· · 32.1	45°27.7	· · 01.0	Zuben'ubi	137°07.9 -16°05.2
16	248°18.6	31°00.2	31.3	30°50.9	02.3	250°08.8	32.2	60°29.9	01.1	Alphecca	126°12.9 26°40.9
17	263°21.0	46°01.9	31.8	45°51.6	02.8	265°11.6	32.3	75°32.1	01.2	Antares	112°28.9 -26°27.4
18	278°23.5	61°03.7	S21°32.2	60°52.4	S15°03.4	280°14.4	S02°32.5	90°34.2	S01°01.3	Atria	107°32.7 -69°03.0
19	293°26.1	76°05.5	32.7	75°53.2	03.9	295°17.1	32.6	105°36.4	01.4	Sabik	102°14.9 -15°44.2
20	308°28.4	91°07.3	33.2	90°54.0	04.5	310°19.9	32.7	120°38.6	01.6	Shaula	96°24.7 -37°06.8
21	323°30.9	106°09.1	· · 33.7	105°54.8	· · 05.1	325°22.7	· · 32.9	135°40.8	· · 01.7	Rasalhague	96°08.3 12°33.3
22	338°33.3	121°10.9	34.1	120°55.6	05.6	340°25.4	33.0	150°43.0	01.8	Eltanin	90°47.1 51°29.6
23	353°35.8	136°12.7	34.6	135°56.4	06.2	355°28.2	33.1	165°45.1	01.9	Kaus Aust.	83°46.4 -34°22.8
Mer.pass. 23:26		ν 1.7' d 0.5' m -4.83		ν 0.8' d 0.6' m 1.50		ν 2.8' d 0.1' m -2.93		ν 2.2' d 0.1' m 0.89		Vega	80°40.3 38°47.9
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	8°38.3	151°14.5	S21°35.1	150°57.2	S15°06.8	10°30.9	S02°33.2	180°47.3	S01°02.0	Nunki	76°00.7 -26°17.0
1	23°40.7	166°16.3	35.5	165°58.0	07.3	25°33.7	33.4	195°49.5	02.2	Altair	62°10.0 8°54.0
2	38°43.2	181°18.1	36.0	180°58.8	07.9	40°36.5	33.5	210°51.7	02.3	Peacock	53°21.9 -56°42.1
3	53°45.7	196°19.9	· · 36.5	195°59.5	· · 08.5	55°39.2	· · 33.6	225°53.9	· · 02.4	Deneb	49°32.6 45°19.5
4	68°48.1	211°21.7	36.9	211°00.3	09.0	70°42.0	33.7	240°56.0	02.5	Enif	33°48.8 9°55.7
5	83°50.6	226°23.5	37.4	226°01.1	09.6	85°44.8	33.9	255°58.2	02.6	Al Na'ir	27°45.7 -46°54.5
6	98°53.1	241°25.4	S21°37.8	241°01.9	S15°10.2	100°47.5	S02°34.0	271°00.4	S01°02.8	Fomalhaut	15°25.8 -29°33.8
7	113°55.5	256°27.2	38.3	256°02.7	10.7	115°50.3	34.1	286°02.6	02.9	Scheat	13°55.0 28°08.7
8	128°58.0	271°29.0	38.7	271°03.5	11.3	130°53.0	34.3	301°04.8	03.0	Markab	13°40.0 15°16.1
9	144°00.5	286°30.8	· · 39.2	286°04.3	· · 11.8	145°55.8	· · 34.4	316°06.9	· · 03.1		
10	159°02.9	301°32.7	39.6	301°05.1	12.4	160°58.6	34.5	331°09.1	03.2	Sep 28 Tue	SHA Mer.pass
11	174°05.4	316°34.5	40.1	316°05.9	13.0	176°01.3	34.6	346°11.3	03.4	Venus	143°11.5 13:59
12	189°07.8	331°36.4	S21°40.5	331°06.6	S15°13.5	191°04.1	S02°34.8	1°13.5	S01°03.5	Mars	143°39.0 13:58
13	204°10.3	346°38.2	41.0	346°07.4	14.1	206°06.8	34.9	16°15.7	03.6	Jupiter	1°38.3 23:22
14	219°12.8	1°40.0	41.4	1°08.2	14.7	221°09.6	35.0	31°17.8	03.7	Saturn	172°22.7 12:02
15	234°15.2	16°41.9	· · 41.9	16°09.0	· · 15.2	236°12.4	· · 35.1	46°20.0	· · 03.8	Sep 29 Wed	SHA Mer.pass
16	249°17.7	31°43.7	42.3	31°09.8	15.8	251°15.1	35.3	61°22.2	04.0	Venus	142°53.0 13:56
17	264°20.2	46°45.6	42.8	46°10.6	16.3	266°17.9	35.4	76°24.4	04.1	Mars	142°59.0 13:57
18	279°22.6	61°47.5	S21°43.2	61°11.4	S15°16.9	281°20.7	S02°35.5	91°26.6	S01°04.2	Jupiter	1°45.5 23:18
19	294°25.1	76°49.3	43.7	76°12.1	17.5	296°23.4	35.7	106°28.7	04.3	Saturn	172°15.9 11:59
20	309°27.6	91°51.2	44.1	91°12.9	18.0	311°26.2	35.8	121°30.9	04.5		
21	324°30.0	106°53.1	· · 44.5	106°13.7	· · 18.6	326°28.9	· · 35.9	136°33.1	· · 04.6	Horizontal parallax	
22	339°32.5	121°54.9	45.0	121°14.5	19.1	341°31.7	36.0	151°35.3	04.7	Venus:	0.4
23	354°35.0	136°56.8	45.4	136°15.3	19.7	356°34.5	36.2	166°37.5	04.8	Mars:	0.1
Mer.pass. 23:22		ν 1.8' d 0.5' m -4.82		ν 0.8' d 0.6' m 1.50		ν 2.8' d 0.1' m -2.93		ν 2.2' d 0.1' m 0.89			

Sun				Moon			
h	GHA	Dec		GHA	ν	Dec	HP
Tue							
0	182°17.3	S01°53.7		313°02.0	9.8'	N22°36.5	5.3'
1	197°17.6	54.7		327°30.8	9.7'	22°41.8	5.2'
2	212°17.8	55.6		341°59.5	9.6'	22°47.0	5.1'
3	227°18.0	56.6		356°28.2	9.6'	22°52.1	5.0'
4	242°18.2	57.6		10°56.7	9.5'	22°57.1	4.9'
5	257°18.4	58.6		25°25.2	9.4'	23°02.0	4.8'
6	272°18.6	S01°59.5		39°53.7	9.4'	N23°06.8	4.6'
7	287°18.8	02°00.5		54°22.1	9.3'	23°11.4	4.5'
8	302°19.0	01.5		68°50.4	9.2'	23°15.9	4.4'
9	317°19.2	02.5		83°18.6	9.2'	23°20.3	4.3'
10	332°19.5	03.4		97°46.8	9.1'	23°24.6	4.2'
11	347°19.7	04.4		112°14.9	9.1'	23°28.8	4.0'
12	2°19.9	S02°05.4		126°43.0	9.0'	N23°32.8	3.9'
13	17°20.1	06.3		141°11.0	8.9'	23°36.7	3.8'
14	32°20.3	07.3		155°38.9	8.9'	23°40.5	3.7'
15	47°20.5	08.3		170°06.7	8.8'	23°44.2	3.5'
16	62°20.7	09.3		184°34.5	8.7'	23°47.7	3.4'
17	77°20.9	10.2		199°02.3	8.7'	23°51.1	3.3'
18	92°21.1	S02°11.2		213°30.0	8.6'	N23°54.4	3.2'
19	107°21.3	12.2		227°57.6	8.6'	23°57.6	3.0'
20	122°21.6	13.2		242°25.2	8.5'	24°00.6	2.9'
21	137°21.8	14.1		256°52.7	8.5'	24°03.5	2.8'
22	152°22.0	15.1		271°20.1	8.4'	24°06.3	2.6'
23	167°22.2	16.1		285°47.5	8.3'	24°08.9	2.5'
	SD = 16.0'	d = 1.0'		SD = 15.2'			
Wed	GHA	Dec		GHA	ν	Dec	HP
0	182°22.4	S02°17.0		300°14.9	8.3'	N24°11.4	2.4'
1	197°22.6	18.0		314°42.2	8.2'	24°13.8	2.2'
2	212°22.8	19.0		329°09.4	8.2'	24°16.0	2.1'
3	227°23.0	20.0		343°36.6	8.1'	24°18.1	2.0'
4	242°23.2	20.9		358°03.8	8.1'	24°20.1	1.8'
5	257°23.4	21.9		12°30.9	8.0'	24°21.9	1.7'
6	272°23.6	S02°22.9		26°57.9	8.0'	N24°23.6	1.6'
7	287°23.8	23.8		41°24.9	7.9'	24°25.2	1.4'
8	302°24.1	24.8		55°51.8	7.9'	24°26.6	1.3'
9	317°24.3	25.8		70°18.7	7.9'	24°27.9	1.2'
10	332°24.5	26.8		84°45.6	7.8'	24°29.1	1.0'
11	347°24.7	27.7		99°12.4	7.8'	24°30.1	0.9'
12	2°24.9	S02°28.7		113°39.2	7.7'	N24°31.0	0.7'
13	17°25.1	29.7		128°05.9	7.7'	24°31.7	0.6'
14	32°25.3	30.6		142°32.6	7.7'	24°32.3	0.5'
15	47°25.5	31.6		156°59.2	7.6'	24°32.8	0.3'
16	62°25.7	32.6		171°25.9	7.6'	24°33.1	0.2'
17	77°25.9	33.6		185°52.4	7.5'	24°33.2	0.0'
18	92°26.1	S02°34.5		200°19.0	7.5'	N24°33.3	-0.1'
19	107°26.3	35.5		214°45.5	7.5'	24°33.2	-0.3'
20	122°26.5	36.5		229°12.0	7.4'	24°32.9	-0.4'
21	137°26.7	37.4		243°38.4	7.4'	24°32.5	-0.5'
22	152°27.0	38.4		258°04.8	7.4'	24°32.0	-0.7'
23	167°27.2	39.4		272°31.2	7.3'	24°31.3	-0.8'
	SD = 16.0'	d = 1.0'		SD = 15.4'			
Thu	GHA	Dec		GHA	ν	Dec	HP
0	182°27.4	S02°40.4		286°57.5	7.3'	N24°30.5	-1.0'
1	197°27.6	41.3		301°23.8	7.3'	24°29.5	-1.1'
2	212°27.8	42.3		315°50.1	7.3'	24°28.4	-1.3'
3	227°28.0	43.3		330°16.4	7.2'	24°27.1	-1.4'
4	242°28.2	44.2		344°42.7	7.2'	24°25.7	-1.6'
5	257°28.4	45.2		359°08.9	7.2'	24°24.2	-1.7'
6	272°28.6	S02°46.2		13°35.1	7.2'	N24°22.5	-1.8'
7	287°28.8	47.2		28°01.3	7.2'	24°20.6	-2.0'
8	302°29.0	48.1		42°27.4	7.1'	24°18.6	-2.1'
9	317°29.2	49.1		56°53.6	7.1'	24°16.5	-2.3'
10	332°29.4	50.1		71°19.7	7.1'	24°14.2	-2.4'
11	347°29.6	51.0		85°45.8	7.1'	24°11.8	-2.6'
12	2°29.8	S02°52.0		100°11.9	7.1'	N24°09.2	-2.7'
13	17°30.0	53.0		114°38.0	7.1'	24°06.5	-2.9'
14	32°30.2	54.0		129°04.0	7.1'	24°03.6	-3.0'
15	47°30.4	54.9		143°30.1	7.0'	24°00.6	-3.2'
16	62°30.6	55.9		157°56.2	7.0'	23°57.5	-3.3'
17	77°30.9	56.9		172°22.2	7.0'	23°54.2	-3.5'
18	92°31.1	S02°57.8		186°48.2	7.0'	N23°50.7	-3.6'
19	107°31.3	58.8		201°14.3	7.0'	23°47.1	-3.7'
20	122°31.5	02°59.8		215°40.3	7.0'	23°43.4	-3.9'
21	137°31.7	03°00.7		230°06.3	7.0'	23°39.5	-4.0'
22	152°31.9	01.7		244°32.3	7.0'	23°35.4	-4.2'
23	167°32.1	02.7		258°58.3	7.0'	23°31.2	-4.3'
	SD = 16.0'	d = 1.0'		SD = 15.6'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	03:38	05:02	06:09	17:29	18:36	19:59
N 70°	03:52	05:06	06:07	17:32	18:32	19:46
68°	04:02	05:10	06:05	17:34	18:29	19:35
66°	04:11	05:13	06:04	17:35	18:26	19:27
64°	04:19	05:15	06:02	17:37	18:24	19:20
62°	04:25	05:17	06:01	17:38	18:22	19:14
60°	04:30	05:19	06:00	17:39	18:20	19:09
N 58°	04:34	05:20	05:59	17:40	18:19	19:05
56°	04:38	05:22	05:59	17:41	18:18	19:01
54°	04:41	05:23	05:58	17:42	18:17	18:58
52°	04:44	05:24	05:57	17:42	18:16	18:55
50°	04:47	05:24	05:57	17:43	18:15	18:53
45°	04:52	05:26	05:55	17:45	18:14	18:48
N 40°	04:56	05:27	05:54	17:46	18:13	18:44
35°	04:58	05:28	05:53	17:47	18:12	18:42
30°	05:00	05:28	05:52	17:48	18:12	18:40
20°	05:03	05:28	05:50	17:50	18:12	18:38
N 10°	05:03	05:28	05:49	17:52	18:13	18:37
0°	05:02	05:26	05:47	17:54	18:14	18:38
S 10°	05:00	05:24	05:45	17:55	18:16	18:41
20°	04:56	05:21	05:43	17:58	18:20	18:45
30°	04:49	05:17	05:41	18:00	18:24	18:52
35°	04:45	05:14	05:40	18:02	18:27	18:57
40°	04:39	05:11	05:38	18:03	18:30	19:02
45°	04:32	05:07	05:36	18:05	18:35	19:10
S 50°	04:23	05:01	05:34	18:08	18:40	19:19
52°	04:18	04:59	05:33	18:09	18:43	19:24
54°	04:14	04:56	05:32	18:10	18:46	19:29
56°	04:08	04:53	05:30	18:12	18:49	19:35
58°	04:01	04:49	05:29	18:13	18:53	19:41
S 60°	03:54	04:45	05:27	18:15	18:57	19:49

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	□	□	□	□	□	□
N 70°	□	□	□	□	□	□
68°	□	□	□	□	□	□
66°	16:21	□	18:12	14:56	□	16:57
64°	17:27	18:04	19:21	13:50	15:07	15:48
62°	18:02	18:46	19:56	13:15	14:25	15:12
60°	18:28	19:14	20:22	12:50	13:57	14:46
N 58°	18:48	19:36	20:42	12:30	13:35	14:25
56°	19:05	19:54	20:59	12:14	13:17	14:08
54°	19:19	20:09	21:13	11:59	13:02	13:54
52°	19:31	20:22	21:25	11:47	12:49	13:41
50°	19:42	20:34	21:36	11:36	12:38	13:30
45°	20:05	20:58	21:59	11:14	12:14	13:07
N 40°	20:24	21:17	22:17	10:56	11:54	12:48
35°	20:40	21:33	22:33	10:40	11:38	12:33
30°	20:53	21:47	22:46	10:27	11:24	12:19
20°	21:16	22:10	23:09	10:05	11:01	11:56
N 10°	21:36	22:31	23:28	09:45	10:40	11:36
0°	21:55	22:50	23:46	09:27	10:21	11:17
S 10°	22:14	23:09	□	09:09	10:02	10:58
20°	22:34	23:30	□	08:50	09:42	10:38
30°	22:57	23:54	□	08:27	09:18	10:14
35°	23:11	□	00:08	08:14	09:04	10:00
40°	23:27	□	00:24	07:59	08:48	09:44
45°	23:46	□	00:43	07:41	08:29	09:25
S 50°	□	00:10	01:07	07:19	08:05	09:01
52°	□	00:21	01:19	07:09	07:53	08:50
54°	□	00:34	01:32	06:57	07:40	08:37
56°	□	00:49	01:47	06:43	07:25	08:22
58°	□	01:07	02:05	06:27	07:07	08:04
S 60°	00:15	01:29	02:27	06:08	06:45	07:42

Day	Sun		Mer. Pass	Moon		Age 20-22 81-62%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Upper hh:mm	Lower hh:mm	
28	09:09	09:19	11:51	03:15	15:41	
29	09:30	09:40	11:50	04:08	16:36	
30	09:49	09:59	11:50	05:04	17:32	

October 01, 02, 03 UT (Fri., Sat., Sun.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	9°37.4	151°58.7	S21°45.8	151°16.1	S15°20.3	11°37.2	S02°36.3	181°39.6	S01°04.9	Alpheratz	357°45.2	29°09.2
1	24°39.9	167°00.6	46.3	166°16.8	20.8	26°40.0	36.4	196°41.8	05.1	Ankaa	353°17.1	-42°14.7
2	39°42.3	182°02.5	46.7	181°17.6	21.4	41°42.7	36.5	211°44.0	05.2	Schedar	349°42.3	56°36.0
3	54°44.8	197°04.3	· · 47.1	196°18.4	· · 21.9	56°45.5	· · 36.7	226°46.2	· · 05.3	Diphda	348°57.5	-17°55.4
4	69°47.3	212°06.2	47.5	211°19.2	22.5	71°48.3	36.8	241°48.4	05.4	Achernar	335°27.5	-57°10.7
5	84°49.7	227°08.1	48.0	226°20.0	23.1	86°51.0	36.9	256°50.6	05.5	Hamal	328°02.7	23°31.0
6	99°52.2	242°10.0	S21°48.4	241°20.8	S15°23.6	101°53.8	S02°37.0	271°52.7	S01°05.7	Polaris	318°23.1	89°18.6
7	114°54.7	257°11.9	48.8	256°21.5	24.2	116°56.5	37.2	286°54.9	05.8	Acamar	315°19.4	-40°15.4
8	129°57.1	272°13.8	49.2	271°22.3	24.7	131°59.3	37.3	301°57.1	05.9	Menkar	314°16.8	4°08.1
9	144°59.6	287°15.7	· · 49.6	286°23.1	· · 25.3	147°02.1	· · 37.4	316°59.3	· · 06.0	Mirfak	308°42.9	49°53.9
10	160°02.1	302°17.7	50.1	301°23.9	25.9	162°04.8	37.5	332°01.5	06.1	Aldebaran	290°51.5	16°31.9
11	175°04.5	317°19.6	50.5	316°24.7	26.4	177°07.6	37.7	347°03.6	06.3	Rigel	281°13.8	-8°11.2
12	190°07.0	332°21.5	S21°50.9	331°25.4	S15°27.0	192°10.3	S02°37.8	2°05.8	S01°06.4	Capella	280°37.2	46°00.4
13	205°09.4	347°23.4	51.3	346°26.2	27.5	207°13.1	37.9	17°08.0	06.5	Bellatrix	278°34.0	6°21.7
14	220°11.9	2°25.3	51.7	1°27.0	28.1	222°15.9	38.1	32°10.2	06.6	Elnath	278°15.0	28°37.0
15	235°14.4	17°27.3	· · 52.1	16°27.8	· · 28.6	237°18.6	· · 38.2	47°12.4	· · 06.7	Alnilam	275°48.3	-1°11.6
16	250°16.8	32°29.2	52.5	31°28.6	29.2	252°21.4	38.3	62°14.5	06.9	Betelgeuse	271°03.4	7°24.6
17	265°19.3	47°31.1	52.9	46°29.3	29.8	267°24.1	38.4	77°16.7	07.0	Canopus	263°57.0	-52°41.8
18	280°21.8	62°33.1	S21°53.3	61°30.1	S15°30.3	282°26.9	S02°38.6	92°18.9	S01°07.1	Sirius	258°35.4	-16°43.7
19	295°24.2	77°35.0	53.7	76°30.9	30.9	297°29.7	38.7	107°21.1	07.2	Adhara	255°14.1	-28°59.0
20	310°26.7	92°36.9	54.1	91°31.7	31.4	312°32.4	38.8	122°23.3	07.3	Procyon	249°01.8	5°11.9
21	325°29.2	107°38.9	· · 54.5	106°32.4	· · 32.0	327°35.2	· · 38.9	137°25.4	· · 07.5	Pollux	243°30.2	27°59.9
22	340°31.6	122°40.8	54.9	121°33.2	32.5	342°37.9	39.1	152°27.6	07.6	Avior	234°19.2	-59°32.4
23	355°34.1	137°42.8	55.3	136°34.0	33.1	357°40.7	39.2	167°29.8	07.7	Suhail	222°54.2	-43°28.4
Mer.pass. 23:18		ν 1.9' d 0.4' m -4.82		ν 0.8' d 0.6' m 1.50		ν 2.8' d 0.1' m -2.93		ν 2.2' d 0.1' m 0.89				

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	10°36.6	152°44.8	S21°55.7	151°34.8	S15°33.7	12°43.5	S02°39.3	182°32.0	S01°07.8	Denebola	182°35.9	14°30.7
1	25°39.0	167°46.7	56.1	166°35.6	34.2	27°46.2	39.4	197°34.2	07.9	Gienah	175°54.7	-17°36.1
2	40°41.5	182°48.7	56.5	181°36.3	34.8	42°49.0	39.6	212°36.3	08.1	Acrux	173°12.4	-63°09.5
3	55°43.9	197°50.7	· · 56.9	196°37.1	· · 35.3	57°51.7	· · 39.7	227°38.5	· · 08.2	Gacrux	172°03.8	-57°10.4
4	70°46.4	212°52.6	57.3	211°37.9	35.9	72°54.5	39.8	242°40.7	08.3	Alioth	166°22.8	55°54.1
5	85°48.9	227°54.6	57.7	226°38.7	36.4	87°57.3	39.9	257°42.9	08.4	Spica	158°33.6	-11°13.0
6	100°51.3	242°56.6	S21°58.1	241°39.4	S15°37.0	103°00.0	S02°40.1	272°45.1	S01°08.5	Alkaid	153°00.8	49°15.6
7	115°53.8	257°58.6	58.5	256°40.2	37.5	118°02.8	40.2	287°47.2	08.7	Hadar	148°51.4	-60°25.6
8	130°56.3	273°00.5	58.8	271°41.0	38.1	133°05.5	40.3	302°49.4	08.8	Menkent	148°10.3	-36°25.4
9	145°58.7	288°02.5	· · 59.2	286°41.8	· · 38.6	148°08.3	· · 40.4	317°51.6	· · 08.9	Arcturus	145°57.8	19°07.7
10	161°01.2	303°04.5	21°59.6	301°42.5	39.2	163°11.0	40.6	332°53.8	09.0	Rigel Kent.	139°55.1	-60°52.9
11	176°03.7	318°06.5	22°00.0	316°43.3	39.8	178°13.8	40.7	347°56.0	09.2	Kochab	137°20.6	74°06.8
12	191°06.1	333°08.5	S22°00.4	331°44.1	S15°40.3	193°16.6	S02°40.8	2°58.1	S01°09.3	Zuben'ubi	137°07.9	-16°05.2
13	206°08.6	348°10.5	00.7	346°44.9	40.9	208°19.3	40.9	18°00.3	09.4	Alphecca	126°12.9	26°40.9
14	221°11.1	3°12.5	01.1	1°45.6	41.4	223°22.1	41.1	33°02.5	09.5	Antares	112°28.9	-26°27.4
15	236°13.5	18°14.5	· · 01.5	16°46.4	· · 42.0	238°24.8	· · 41.2	48°04.7	· · 09.6	Atria	107°32.7	-69°03.0
16	251°16.0	33°16.5	01.9	31°47.2	42.5	253°27.6	41.3	63°06.9	09.8	Sabik	102°14.9	-15°44.2
17	266°18.4	48°18.6	02.2	46°47.9	43.1	268°30.3	41.4	78°09.0	09.9	Shaula	96°24.7	-37°06.8
18	281°20.9	63°20.6	S22°02.6	61°48.7	S15°43.6	283°33.1	S02°41.6	93°11.2	S01°10.0	Rasalhague	96°08.4	12°33.3
19	296°23.4	78°22.6	03.0	76°49.5	44.2	298°35.9	41.7	108°13.4	10.1	Eltanin	90°47.1	51°29.6
20	311°25.8	93°24.6	03.3	91°50.3	44.7	313°38.6	41.8	123°15.6	10.2	Kaus Aust.	83°46.4	-34°22.8
21	326°28.3	108°26.7	· · 03.7	106°51.0	· · 45.3	328°41.4	· · 41.9	138°17.8	· · 10.4	Vega	80°40.3	38°47.9
22	341°30.8	123°28.7	04.0	121°51.8	45.8	343°44.1	42.0	153°19.9	10.5	Nunki	76°00.7	-26°17.0
23	356°33.2	138°30.7	04.4	136°52.6	46.4	358°46.9	42.2	168°22.1	10.6	Altair	62°10.1	8°54.0
Mer.pass. 23:14		ν 2.0' d 0.4' m -4.82		ν 0.8' d 0.6' m 1.49		ν 2.8' d 0.1' m -2.92		ν 2.2' d 0.1' m 0.89				

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	11°35.7	153°32.8	S22°04.8	151°53.3	S15°46.9	13°49.6	S02°42.3	183°24.3	S01°10.7	Fomalhaut	15°25.8	-29°33.8
1	26°38.2	168°34.8	05.1	166°54.1	47.5	28°52.4	42.4	198°26.5	10.8	Scheat	13°55.0	28°08.8
2	41°40.6	183°36.9	05.5	181°54.9	48.0	43°55.2	42.5	213°28.7	11.0	Markab	13°40.0	15°16.1
3	56°43.1	198°38.9	· · 05.8	196°55.6	· · 48.6	58°57.9	· · 42.7	228°30.8	· · 11.1	Oct 01 Fri SHA Mer.pass		
4	71°45.5	213°41.0	06.2	211°56.4	49.1	74°00.7	42.8	243°33.0	11.2	Venus	142°21.3	13:50
5	86°48.0	228°43.0	06.5	226°57.2	49.7	89°03.4	42.9	258°35.2	11.3	Mars	141°38.6	13:54
6	101°50.5	243°45.1	S22°06.9	241°58.0	S15°50.2	104°06.2	S02°43.0	273°37.4	S01°11.4	Jupiter	1°59.8	23:09
7	116°52.9	258°47.1	07.2	256°58.7	50.8	119°08.9	43.2	288°39.6	11.6	Saturn	172°02.2	11:52
8	131°55.4	273°49.2	07.6	271°59.5	51.3	134°11.7	43.3	303°41.7	11.7	Oct 02 Sat SHA Mer.pass		
9	146°57.9	288°51.3	· · 07.9	287°00.3	· · 51.9	149°14.4	· · 43.4	318°43.9	· · 11.8	Venus	142°08.2	13:47
10	162°00.3	303°53.4	08.3	302°01.0	52.4	164°17.2	43.5	333°46.1	11.9	Mars	140°58.2	13:53
11	177°02.8	318°55.4	08.6	317°01.8	53.0	179°20.0	43.7	348°48.3	12.0	Jupiter	2°06.9	23:05
12	192°05.3	333°57.5	S22°08.9	332°02.6	S15°53.5	194°22.7	S02°43.8	3°50.5	S01°12.2	Saturn	171°55.4	11:48
13	207°07.7	348°59.6	09.3	347°03.3	54.1	209°25.5	43.9	18°52.6	12.3	Oct 03 Sun SHA Mer.pass		
14	222°10.2	4°01.7	09.6	2°04.1	54.6	224°28.2	44.0	33°54.8	12.4	Venus	141°57.1	13:44
15	237°12.7	19°03.8	· · 09.9	17°04.9	· · 55.2	239°31.0	· · 44.1	48°57.0	· · 12.5	Mars	140°17.6	13:52
16	252°15.1	34°05.9	10.3	32°05.6	55.7	254°33.7	44.3	63°59.2	12.6	Jupiter	2°13.9	23:00
17	267°17.6	49°08.0	10.6	47°06.4	56.3	269°36.5	44.4	79°01.4	12.8	Saturn	171°48.6	11:45
18	282°20.0	64°10.1	S22°10.9	62°07.2	S15°56.8	284°39.2	S02°44.5	94°03.5	S01°12.9	Horizontal parallax		
19	297°22.5	79°12.2	11.3	77°07.9	57.4	299°42.0	44.6	109°05.7	13.0	Venus:	0.4	
20	312°25.0	94°14.3	11.6	92°08.7	57.9	314°44.7	44.8	124°07.9	13.1	Mars:	0.1	
21	327°27.4	109°16.4	· · 11.9	107°09.5	· · 58.5	32						

h	Sun			Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	182°32.3	S03°03.7	273°24.3	7.0'	N23°26.9	-4.5'	57.8'
1	197°32.5	04.6	287°50.3	7.0'	23°22.4	-4.6'	57.9'
2	212°32.7	05.6	302°16.4	7.0'	23°17.8	-4.8'	57.9'
3	227°32.9	· · 06.6	316°42.4	7.0'	23°13.0	-4.9'	57.9'
4	242°33.1	07.5	331°08.4	7.0'	23°08.1	-5.1'	58.0'
5	257°33.3	08.5	345°34.4	7.0'	23°03.1	-5.2'	58.0'
6	272°33.5	S03°09.5	0°00.5	7.0'	N22°57.9	-5.3'	58.0'
7	287°33.7	10.4	14°26.5	7.0'	22°52.5	-5.5'	58.0'
8	302°33.9	11.4	28°52.5	7.1'	22°47.0	-5.6'	58.1'
9	317°34.1	· · 12.4	43°18.6	7.1'	22°41.4	-5.8'	58.1'
10	332°34.3	13.4	57°44.7	7.1'	22°35.6	-5.9'	58.1'
11	347°34.5	14.3	72°10.7	7.1'	22°29.7	-6.1'	58.2'
12	2°34.7	S03°15.3	86°36.8	7.1'	N22°23.6	-6.2'	58.2'
13	17°34.9	16.3	101°02.9	7.1'	22°17.4	-6.4'	58.2'
14	32°35.1	17.2	115°29.0	7.1'	22°11.0	-6.5'	58.3'
15	47°35.3	· · 18.2	129°55.2	7.1'	22°04.5	-6.6'	58.3'
16	62°35.5	19.2	144°21.3	7.2'	21°57.9	-6.8'	58.3'
17	77°35.7	20.1	158°47.5	7.2'	21°51.1	-6.9'	58.4'
18	92°35.9	S03°21.1	173°13.6	7.2'	N21°44.2	-7.1'	58.4'
19	107°36.1	22.1	187°39.8	7.2'	21°37.2	-7.2'	58.4'
20	122°36.3	23.1	202°06.0	7.2'	21°30.0	-7.3'	58.5'
21	137°36.5	· · 24.0	216°32.3	7.2'	21°22.6	-7.5'	58.5'
22	152°36.7	25.0	230°58.5	7.3'	21°15.2	-7.6'	58.5'
23	167°36.9	26.0	245°24.8	7.3'	21°07.6	-7.7'	58.6'
SD = 16.0'			SD = 15.8'				
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	182°37.1	S03°26.9	259°51.1	7.3'	N20°59.8	-7.9'	58.6'
1	197°37.3	27.9	274°17.4	7.3'	20°51.9	-8.0'	58.6'
2	212°37.5	28.9	288°43.7	7.4'	20°43.9	-8.2'	58.7'
3	227°37.7	· · 29.8	303°10.1	7.4'	20°35.8	-8.3'	58.7'
4	242°37.9	30.8	317°36.5	7.4'	20°27.5	-8.4'	58.7'
5	257°38.1	31.8	332°02.9	7.4'	20°19.1	-8.6'	58.8'
6	272°38.3	S03°32.7	346°29.3	7.5'	N20°10.5	-8.7'	58.8'
7	287°38.5	33.7	0°55.8	7.5'	20°01.8	-8.8'	58.8'
8	302°38.7	34.7	15°22.2	7.5'	19°53.0	-8.9'	58.9'
9	317°38.9	· · 35.6	29°48.7	7.5'	19°44.1	-9.1'	58.9'
10	332°39.1	36.6	44°15.3	7.6'	19°35.0	-9.2'	58.9'
11	347°39.3	37.6	58°41.8	7.6'	19°25.8	-9.3'	59.0'
12	2°39.5	S03°38.5	73°08.4	7.6'	N19°16.5	-9.5'	59.0'
13	17°39.7	39.5	87°35.1	7.6'	19°07.0	-9.6'	59.0'
14	32°39.9	40.5	102°01.7	7.7'	18°57.4	-9.7'	59.1'
15	47°40.1	· · 41.4	116°28.4	7.7'	18°47.7	-9.8'	59.1'
16	62°40.3	42.4	130°55.1	7.7'	18°37.9	-10.0'	59.1'
17	77°40.5	43.4	145°21.8	7.8'	18°27.9	-10.1'	59.1'
18	92°40.7	S03°44.4	159°48.6	7.8'	N18°17.9	-10.2'	59.2'
19	107°40.9	45.3	174°15.4	7.8'	18°07.7	-10.3'	59.2'
20	122°41.1	46.3	188°42.2	7.9'	17°57.3	-10.4'	59.2'
21	137°41.3	· · 47.3	203°09.1	7.9'	17°46.9	-10.6'	59.3'
22	152°41.5	48.2	217°36.0	7.9'	17°36.3	-10.7'	59.3'
23	167°41.7	49.2	232°02.9	7.9'	17°25.7	-10.8'	59.3'
SD = 16.0'			SD = 16.0'				
Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	182°41.9	S03°50.2	246°29.8	8.0'	N17°14.9	-10.9'	59.4'
1	197°42.1	51.1	260°56.8	8.0'	17°04.0	-11.0'	59.4'
2	212°42.3	52.1	275°23.8	8.0'	16°52.9	-11.1'	59.4'
3	227°42.5	· · 53.1	289°50.8	8.1'	16°41.8	-11.2'	59.5'
4	242°42.7	54.0	304°17.9	8.1'	16°30.6	-11.4'	59.5'
5	257°42.8	55.0	318°45.0	8.1'	16°19.2	-11.5'	59.5'
6	272°43.0	S03°56.0	333°12.2	8.2'	N16°07.7	-11.6'	59.5'
7	287°43.2	56.9	347°39.3	8.2'	15°56.2	-11.7'	59.6'
8	302°43.4	57.9	2°06.5	8.2'	15°44.5	-11.8'	59.6'
9	317°43.6	· · 58.9	16°33.7	8.3'	15°32.7	-11.9'	59.6'
10	332°43.8	03°59.8	31°01.0	8.3'	15°20.8	-12.0'	59.7'
11	347°44.0	04°00.8	45°28.3	8.3'	15°08.8	-12.1'	59.7'
12	2°44.2	S04°01.8	59°55.6	8.3'	N14°56.7	-12.2'	59.7'
13	17°44.4	02.7	74°22.9	8.4'	14°44.5	-12.3'	59.8'
14	32°44.6	03.7	88°50.3	8.4'	14°32.2	-12.4'	59.8'
15	47°44.8	· · 04.7	103°17.7	8.4'	14°19.8	-12.5'	59.8'
16	62°45.0	05.6	117°45.2	8.5'	14°07.3	-12.6'	59.8'
17	77°45.2	06.6	132°12.6	8.5'	13°54.7	-12.7'	59.9'
18	92°45.4	S04°07.5	146°40.1	8.5'	N13°42.0	-12.8'	59.9'
19	107°45.6	08.5	161°07.7	8.6'	13°29.2	-12.9'	59.9'
20	122°45.8	09.5	175°35.2	8.6'	13°16.3	-13.0'	59.9'
21	137°46.0	· · 10.4	190°02.8	8.6'	13°03.3	-13.1'	60.0'
22	152°46.2	11.4	204°30.4	8.6'	12°50.3	-13.2'	60.0'
23	167°46.3	12.4	218°58.0	8.7'	12°37.1	-13.2'	60.0'
SD = 16.0'			SD = 16.2'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	03:53	05:15	06:23	17:14	18:21	19:42
N 70°	04:05	05:18	06:19	17:18	18:18	19:31
68°	04:14	05:20	06:16	17:21	18:16	19:22
66°	04:22	05:22	06:13	17:24	18:15	19:15
64°	04:28	05:24	06:11	17:26	18:13	19:09
62°	04:33	05:25	06:09	17:28	18:12	19:04
60°	04:37	05:26	06:07	17:30	18:11	19:00
N 58°	04:41	05:27	06:06	17:32	18:11	18:56
56°	04:44	05:28	06:05	17:33	18:10	18:53
54°	04:47	05:28	06:03	17:34	18:10	18:51
52°	04:49	05:29	06:02	17:36	18:09	18:48
50°	04:51	05:29	06:01	17:37	18:09	18:46
45°	04:56	05:30	05:59	17:39	18:08	18:42
N 40°	04:59	05:30	05:57	17:41	18:08	18:39
35°	05:01	05:30	05:55	17:43	18:08	18:37
30°	05:02	05:30	05:54	17:44	18:08	18:36
20°	05:04	05:29	05:51	17:47	18:09	18:35
N 10°	05:03	05:28	05:49	17:50	18:11	18:35
0°	05:01	05:25	05:46	17:53	18:13	18:37
S 10°	04:58	05:23	05:44	17:55	18:16	18:41
20°	04:53	05:19	05:41	17:58	18:20	18:46
30°	04:45	05:13	05:37	18:02	18:26	18:54
35°	04:40	05:10	05:35	18:04	18:29	18:59
40°	04:34	05:06	05:33	18:06	18:33	19:06
45°	04:26	05:01	05:31	18:09	18:39	19:14
S 50°	04:16	04:55	05:27	18:12	18:45	19:24
52°	04:11	04:52	05:26	18:14	18:48	19:29
54°	04:05	04:48	05:24	18:16	18:52	19:35
56°	03:59	04:45	05:22	18:18	18:55	19:41
58°	03:52	04:41	05:20	18:20	19:00	19:49
S 60°	03:43	04:36	05:18	18:22	19:05	19:57

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	□	20:00	23:37	□	19:07	17:26
N 70°	□	21:27	· ·	□	17:39	17:00
68°	19:25	22:06	· ·	17:42	16:59	16:40
66°	20:28	22:33	· ·	16:39	16:31	16:25
64°	21:03	22:53	· ·	16:03	16:09	16:11
62°	21:28	23:09	· ·	15:38	15:52	16:00
60°	21:48	23:23	· ·	15:17	15:37	15:51
N 58°	22:04	23:35	· ·	15:01	15:25	15:42
56°	22:18	23:45	· ·	14:47	15:14	15:35
54°	22:29	23:54	· ·	14:34	15:05	15:28
52°	22:40	· ·	00:02	14:23	14:56	15:22
50°	22:49	· ·	00:09	14:14	14:49	15:17
45°	23:09	· ·	00:24	13:53	14:32	15:05
N 40°	23:25	· ·	00:36	13:37	14:19	14:55
35°	23:38	· ·	00:47	13:22	14:07	14:47
30°	23:50	· ·	00:56	13:10	13:57	14:39
20°	· ·	00:10	01:12	12:49	13:39	14:26
N 10°	· ·	00:27	01:25	12:31	13:24	14:15
0°	· ·	00:43	01:38	12:13	13:09	14:04
S 10°	00:05	00:59	01:51	11:56	12:55	13:53
20°	00:24	01:16	02:05	11:37	12:39	13:42
30°	00:47	01:36	02:20	11:16	12:21	13:28
35°	01:00	01:47	02:29	11:03	12:10	13:20
40°	01:15	02:00	02:39	10:49	11:58	13:11
45°	01:33	02:15	02:51	10:31	11:44	13:01
S 50°	01:55	02:34	03:06	10:09	11:26	12:48
52°	02:06	02:43	03:12	09:59	11:18	12:42
54°	02:18	02:53	03:20	09:47	11:08	12:35
56°	02:32	03:04	03:28	09:34	10:58	12:27
58°	02:48	03:17	03:37	09:18	10:45	12:19
S 60°	03:07	03:32	03:48	08:59	10:31	12:09

Day	Sun		Mer. Pass hh:mm	Moon		Age 23-25 52-30%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Upper hh:mm	Lower hh:mm	
01	10:09	10:19	11:50	06:00	18:28	
02	10:28	10:38	11:49	06:56	19:24	
03	10:47	10:57	11:49	07:51	20:18	

October 04, 05, 06 UT (Mon., Tue., Wed.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	12°34.8	154°22.8	S22°12.9	152°11.7	S16°00.1	14°55.8	S02°45.2	184°16.6	S01°13.6	Alpheratz	357°45.2 29°09.2
1	27°37.3	169°24.9	13.2	167°12.5	00.7	29°58.5	45.4	199°18.8	13.7	Ankaa	353°17.1 -42°14.7
2	42°39.8	184°27.0	13.5	182°13.3	01.2	45°01.3	45.5	214°21.0	13.8	Schedar	349°42.3 56°36.0
3	57°42.2	199°29.2	· · 13.8	197°14.0	· · 01.8	60°04.0	· · 45.6	229°23.2	· · 14.0	Diphda	348°57.5 -17°55.4
4	72°44.7	214°31.3	14.1	212°14.8	02.3	75°06.8	45.7	244°25.3	14.1	Achernar	335°27.5 -57°10.7
5	87°47.2	229°33.4	14.4	227°15.6	02.9	90°09.5	45.9	259°27.5	14.2	Hamal	328°02.7 23°31.0
6	102°49.6	244°35.6	S22°14.7	242°16.3	S16°03.4	105°12.3	S02°46.0	274°29.7	S01°14.3	Polaris	318°22.3 89°18.6
7	117°52.1	259°37.7	15.1	257°17.1	03.9	120°15.0	46.1	289°31.9	14.4	Acamar	315°19.4 -40°15.4
8	132°54.5	274°39.9	15.4	272°17.8	04.5	135°17.8	46.2	304°34.1	14.6	Menkar	314°16.8 4°08.1
9	147°57.0	289°42.1	· · 15.7	287°18.6	· · 05.0	150°20.5	· · 46.3	319°36.2	· · 14.7	Mirfak	308°42.8 49°53.9
10	162°59.5	304°44.2	16.0	302°19.4	05.6	165°23.3	46.5	334°38.4	14.8	Aldebaran	290°51.4 16°31.9
11	178°01.9	319°46.4	16.3	317°20.1	06.1	180°26.0	46.6	349°40.6	14.9	Rigel	281°13.8 -8°11.2
12	193°04.4	334°48.5	S22°16.6	332°20.9	S16°06.7	195°28.8	S02°46.7	4°42.8	S01°15.0	Capella	280°37.2 46°00.4
13	208°06.9	349°50.7	16.9	347°21.6	07.2	210°31.5	46.8	19°45.0	15.2	Bellatrix	278°34.0 6°21.7
14	223°09.3	4°52.9	17.2	2°22.4	07.8	225°34.3	47.0	34°47.1	15.3	Elnath	278°15.0 28°37.0
15	238°11.8	19°55.1	· · 17.4	17°23.2	· · 08.3	240°37.1	· · 47.1	49°49.3	· · 15.4	Alnilam	275°48.2 -1°11.6
16	253°14.3	34°57.2	17.7	32°23.9	08.9	255°39.8	47.2	64°51.5	15.5	Betelgeuse	271°03.3 7°24.6
17	268°16.7	49°59.4	18.0	47°24.7	09.4	270°42.6	47.3	79°53.7	15.6	Canopus	263°57.0 -52°41.8
18	283°19.2	65°01.6	S22°18.3	62°25.4	S16°09.9	285°45.3	S02°47.4	94°55.9	S01°15.8	Sirius	258°35.4 -16°43.7
19	298°21.6	80°03.8	18.6	77°26.2	10.5	300°48.1	47.6	109°58.0	15.9	Adhara	255°14.1 -28°59.0
20	313°24.1	95°06.0	18.9	92°27.0	11.0	315°50.8	47.7	125°00.2	16.0	Procyon	245°01.8 5°11.9
21	328°26.6	110°08.2	· · 19.2	107°27.7	· · 11.6	330°53.6	· · 47.8	140°02.4	· · 16.1	Pollux	243°30.2 27°59.9
22	343°29.0	125°10.4	19.5	122°28.5	12.1	345°56.3	47.9	155°04.6	16.2	Avior	234°19.2 -59°32.4
23	358°31.5	140°12.6	19.7	137°29.2	12.7	0°59.1	48.0	170°06.8	16.4	Suhail	222°54.2 -43°28.4
Mer.pass. 23:06		ν 2.1' d 0.3' m -4.81		ν 0.8' d 0.5' m 1.49		ν 2.8' d 0.1' m -2.92		ν 2.2' d 0.1' m 0.89		Miaplacidus 221°40.8 -69°45.5	
Alphard 217°58.2 -8°42.2											
Regulus 207°45.8 11°54.9											
Dubhe 193°54.5 61°41.4											
Denebola 182°35.9 14°30.7											
Gienah 175°54.7 -17°36.1											
Acrux 173°12.4 -63°09.5											
Gacrux 172°03.8 -57°10.4											
Alioth 166°22.8 55°54.1											
Spica 158°33.6 -11°13.0											
Alkaid 153°00.8 49°15.6											
Hadar 148°51.4 -60°25.6											
Menkent 148°10.3 -36°25.4											
Arcturus 145°57.8 19°07.7											
Rigel Kent. 139°55.1 -60°52.9											
Kochab 137°20.7 74°06.8											
Zuben'ubi 137°07.9 -16°05.2											
Alphecca 126°12.9 26°40.9											
Antares 112°28.9 -26°27.4											
Atria 107°32.8 -69°03.0											
Sabik 102°14.9 -15°44.2											
Shaula 96°24.7 -37°06.8											
Rasalhague 96°08.4 12°33.3											
Eltanin 90°47.2 51°29.6											
Kaus Aust. 83°46.4 -34°22.8											
Vega 80°40.3 38°47.9											
Nunki 76°00.7 -26°17.0											
Altair 62°10.1 8°54.0											
Peacock 53°22.0 -56°42.1											
Deneb 49°32.7 45°19.5											
Enif 33°48.8 9°55.7											
Al Na'ir 27°45.7 -46°54.5											
Fomalhaut 15°25.8 -29°33.8											
Scheat 13°55.0 28°08.8											
Markab 13°40.0 15°16.1											
Oct 04 Mon SHA Mer.pass											
Venus 141°47.9 13:41											
Mars 139°36.9 13:51											
Jupiter 2°20.9 22:56											
Saturn 171°41.8 11:41											
Oct 05 Tue SHA Mer.pass											
Venus 141°40.9 13:37											
Mars 138°56.0 13:49											
Jupiter 2°27.8 22:52											
Saturn 171°35.0 11:38											
Oct 06 Wed SHA Mer.pass											
Venus 141°35.9 13:33											
Mars 138°15.0 13:48											
Jupiter 2°34.7 22:47											
Saturn 171°28.2 11:34											
Horizontal parallax											
Venus: 0.4											
Mars: 0.1											
Mer.pass. 22:58		ν 2.3' d 0.2' m -4.79		ν 0.7' d 0.5' m 1.49		ν 2.7' d 0.1' m -2.91		ν 2.2' d 0.1' m 0.90			

Sun							Moon						
h	GHA	Dec	GHA	ν	Dec	d	HP						
Mon													
0	182°46.5	S04°13.3	233°25.7	8.7'	N12°23.9	-13.3'	60.1'						
1	197°46.7	14.3	247°53.4	8.7'	12°10.5	-13.4'	60.1'						
2	212°46.9	15.3	262°21.1	8.7'	11°57.1	-13.5'	60.1'						
3	227°47.1	· · 16.2	276°48.9	8.8'	11°43.6	-13.6'	60.1'						
4	242°47.3	17.2	291°16.6	8.8'	11°30.1	-13.7'	60.2'						
5	257°47.5	18.2	305°44.4	8.8'	11°16.4	-13.7'	60.2'						
6	272°47.7	S04°19.1	320°12.2	8.8'	N11°02.7	-13.8'	60.2'						
7	287°47.9	20.1	334°40.1	8.9'	10°48.9	-13.9'	60.2'						
8	302°48.1	21.1	349°07.9	8.9'	10°35.0	-14.0'	60.2'						
9	317°48.3	· · 22.0	3°35.8	8.9'	10°21.0	-14.0'	60.3'						
10	332°48.5	23.0	18°03.7	8.9'	10°07.0	-14.1'	60.3'						
11	347°48.7	24.0	32°31.7	9.0'	09°52.9	-14.2'	60.3'						
12	2°48.8	S04°24.9	46°59.6	9.0'	N09°38.7	-14.2'	60.3'						
13	17°49.0	25.9	61°27.6	9.0'	09°24.5	-14.3'	60.4'						
14	32°49.2	26.8	75°55.6	9.0'	09°10.1	-14.4'	60.4'						
15	47°49.4	· · 27.8	90°23.6	9.0'	08°55.8	-14.4'	60.4'						
16	62°49.6	28.8	104°51.6	9.0'	08°41.3	-14.5'	60.4'						
17	77°49.8	29.7	119°19.7	9.1'	08°26.8	-14.6'	60.4'						
18	92°50.0	S04°30.7	133°47.7	9.1'	N08°12.3	-14.6'	60.5'						
19	107°50.2	31.7	148°15.8	9.1'	07°57.6	-14.7'	60.5'						
20	122°50.4	32.6	162°43.9	9.1'	07°43.0	-14.7'	60.5'						
21	137°50.6	· · 33.6	177°12.0	9.1'	07°28.2	-14.8'	60.5'						
22	152°50.7	34.6	191°40.1	9.1'	07°13.4	-14.8'	60.5'						
23	167°50.9	35.5	206°08.3	9.2'	06°58.6	-14.9'	60.6'						
SD = 16.0' $d = 1.0'$							SD = 16.4'						
Tue	GHA	Dec	GHA	ν	Dec	d	HP						
0	182°51.1	S04°36.5	220°36.4	9.2'	N06°43.7	-14.9'	60.6'						
1	197°51.3	37.4	235°04.6	9.2'	06°28.8	-15.0'	60.6'						
2	212°51.5	38.4	249°32.8	9.2'	06°13.8	-15.0'	60.6'						
3	227°51.7	· · 39.4	264°01.0	9.2'	05°58.7	-15.1'	60.6'						
4	242°51.9	40.3	278°29.2	9.2'	05°43.7	-15.1'	60.6'						
5	257°52.1	41.3	292°57.4	9.2'	05°28.5	-15.2'	60.7'						
6	272°52.3	S04°42.3	307°25.6	9.2'	N05°13.4	-15.2'	60.7'						
7	287°52.4	43.2	321°53.8	9.2'	04°58.2	-15.2'	60.7'						
8	302°52.6	44.2	336°22.1	9.2'	04°42.9	-15.3'	60.7'						
9	317°52.8	· · 45.1	350°50.3	9.2'	04°27.7	-15.3'	60.7'						
10	332°53.0	46.1	5°18.6	9.3'	04°12.4	-15.3'	60.7'						
11	347°53.2	47.1	19°46.8	9.3'	03°57.0	-15.4'	60.7'						
12	2°53.4	S04°48.0	34°15.1	9.3'	N03°41.6	-15.4'	60.8'						
13	17°53.6	49.0	48°43.3	9.3'	03°26.2	-15.4'	60.8'						
14	32°53.7	50.0	63°11.6	9.3'	03°10.8	-15.5'	60.8'						
15	47°53.9	· · 50.9	77°39.8	9.3'	02°55.3	-15.5'	60.8'						
16	62°54.1	51.9	92°08.1	9.3'	02°39.9	-15.5'	60.8'						
17	77°54.3	52.8	106°36.4	9.3'	02°24.4	-15.5'	60.8'						
18	92°54.5	S04°53.8	121°04.6	9.3'	N02°08.9	-15.5'	60.8'						
19	107°54.7	54.8	135°32.9	9.3'	01°53.3	-15.6'	60.8'						
20	122°54.9	55.7	150°01.1	9.3'	01°37.8	-15.6'	60.8'						
21	137°55.1	· · 56.7	164°29.4	9.2'	01°22.2	-15.6'	60.9'						
22	152°55.2	57.6	178°57.6	9.2'	01°06.6	-15.6'	60.9'						
23	167°55.4	58.6	193°25.9	9.2'	00°51.0	-15.6'	60.9'						
SD = 16.0' $d = 1.0'$							SD = 16.5'						
Wed	GHA	Dec	GHA	ν	Dec	d	HP						
0	182°55.6	S04°59.6	207°54.1	9.2'	N00°35.4	-15.6'	60.9'						
1	197°55.8	05°00.5	222°22.4	9.2'	00°19.8	-15.6'	60.9'						
2	212°56.0	01.5	236°50.6	9.2'	N00°04.2	-15.6'	60.9'						
3	227°56.2	· · 02.4	251°18.8	9.2'	S00°11.4	15.6'	60.9'						
4	242°56.3	03.4	265°47.0	9.2'	00°27.1	15.6'	60.9'						
5	257°56.5	04.4	280°15.2	9.2'	00°42.7	15.6'	60.9'						
6	272°56.7	S05°05.3	294°43.4	9.2'	S00°58.3	15.6'	60.9'						
7	287°56.9	06.3	309°11.6	9.2'	01°13.9	15.6'	60.9'						
8	302°57.1	07.2	323°39.7	9.1'	01°29.5	15.6'	60.9'						
9	317°57.3	· · 08.2	338°07.9	9.1'	01°45.1	15.6'	60.9'						
10	332°57.4	09.2	352°36.0	9.1'	02°00.7	15.6'	60.9'						
11	347°57.6	10.1	7°04.1	9.1'	02°16.3	15.6'	60.9'						
12	2°57.8	S05°11.1	21°32.2	9.1'	S02°31.9	15.6'	60.9'						
13	17°58.0	12.0	36°00.3	9.1'	02°47.5	15.5'	60.9'						
14	32°58.2	13.0	50°28.4	9.1'	03°03.0	15.5'	60.9'						
15	47°58.4	· · 14.0	64°56.4	9.0'	03°18.5	15.5'	60.9'						
16	62°58.5	14.9	79°24.5	9.0'	03°34.0	15.5'	60.9'						
17	77°58.7	15.9	93°52.5	9.0'	03°49.5	15.5'	60.9'						
18	92°58.9	S05°16.8	108°20.5	9.0'	S04°05.0	15.4'	60.9'						
19	107°59.1	17.8	122°48.5	9.0'	04°20.4	15.4'	60.9'						
20	122°59.3	18.7	137°16.4	8.9'	04°35.8	15.4'	60.9'						
21	137°59.5	· · 19.7	151°44.3	8.9'	04°51.2	15.3'	60.9'						
22	152°59.6	20.7	166°12.3	8.9'	05°06.5	15.3'	60.9'						
23	167°59.8	21.6	180°40.1	8.9'	05°21.8	15.3'	60.9'						
SD = 16.0' $d = 1.0'$							SD = 16.6'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	04:08	05:29	06:36	16:59	18:06	19:25
N 70°	04:18	05:30	06:31	17:04	18:05	19:16
68°	04:26	05:31	06:27	17:09	18:04	19:09
66°	04:32	05:32	06:23	17:12	18:03	19:03
64°	04:37	05:32	06:20	17:16	18:03	18:58
62°	04:41	05:33	06:17	17:19	18:03	18:54
60°	04:45	05:33	06:15	17:21	18:02	18:51
N 58°	04:48	05:33	06:12	17:23	18:02	18:48
56°	04:50	05:33	06:11	17:25	18:02	18:45
54°	04:53	05:34	06:09	17:27	18:02	18:43
52°	04:54	05:34	06:07	17:29	18:02	18:41
50°	04:56	05:34	06:06	17:30	18:02	18:40
45°	04:59	05:33	06:03	17:33	18:03	18:37
N 40°	05:02	05:33	06:00	17:36	18:03	18:35
35°	05:03	05:32	05:58	17:39	18:04	18:33
30°	05:04	05:32	05:56	17:41	18:05	18:32
20°	05:04	05:30	05:52	17:45	18:07	18:32
N 10°	05:03	05:27	05:48	17:48	18:09	18:34
0°	05:00	05:24	05:45	17:52	18:12	18:37
S 10°	04:56	05:21	05:42	17:55	18:16	18:41
20°	04:50	05:16	05:38	17:59	18:21	18:47
30°	04:42	05:10	05:34	18:04	18:28	18:56
35°	04:36	05:06	05:31	18:06	18:32	19:02
40°	04:29	05:01	05:28	18:09	18:37	19:09
45°	04:20	04:55	05:25	18:13	18:43	19:18
S 50°	04:09	04:48	05:21	18:17	18:50	19:30
52°	04:03	04:45	05:19	18:19	18:53	19:35
54°	03:57	04:41	05:17	18:21	18:57	19:41
56°	03:50	04:37	05:15	18:24	19:02	19:49
58°	03:42	04:32	05:12	18:26	19:07	19:57
S 60°	03:33	04:26	05:09	18:29	19:12	20:06

	Aries	Venus		Mars		Jupiter		Saturn	
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	15°32.2	157°05.3	S22°31.2	153°06.0	S16°39.0	18°13.7	S02°53.9	186°53.6	S01°22.2
1	30°34.7	172°07.7	31.4	168°06.7	39.5	33°16.5	54.0	201°55.8	22.3
2	45°37.2	187°10.1	31.6	183°07.5	40.1	48°19.2	54.1	216°58.0	22.5
3	60°39.6	202°12.5	•• 31.8	198°08.2	•• 40.6	63°21.9	•• 54.2	232°00.1	•• 22.6
4	75°42.1	217°14.9	32.0	213°09.0	41.1	78°24.7	54.4	247°02.3	22.7
5	90°44.6	232°17.3	32.2	228°09.7	41.7	93°27.4	54.5	262°04.5	22.8
6	105°47.0	247°19.7	S22°32.4	243°10.4	S16°42.2	108°30.2	S02°54.6	277°06.7	S01°22.9
7	120°49.5	262°22.1	32.5	258°11.2	42.7	123°32.9	54.7	292°08.9	23.1
8	135°52.0	277°24.5	32.7	273°11.9	43.3	138°35.7	54.8	307°11.0	23.2
9	150°54.4	292°27.0	•• 32.9	288°12.7	•• 43.8	153°38.4	•• 54.9	322°13.2	•• 23.3
10	165°56.9	307°29.4	33.1	303°13.4	44.3	168°41.1	55.1	337°15.4	23.4
11	180°59.3	322°31.8	33.2	318°14.2	44.9	183°43.9	55.2	352°17.6	23.5
12	196°01.8	337°34.2	S22°33.4	333°14.9	S16°45.4	198°46.6	S02°55.3	7°19.8	S01°23.7
13	211°04.3	352°36.7	33.6	348°15.6	45.9	213°49.4	55.4	22°22.0	23.8
14	226°06.7	7°39.1	33.7	3°16.4	46.5	228°52.1	55.5	37°24.1	23.9
15	241°09.2	22°41.6	•• 33.9	18°17.1	•• 47.0	243°54.9	•• 55.6	52°26.3	•• 24.0
16	256°11.7	37°44.0	34.0	33°17.9	47.5	258°57.6	55.8	67°28.5	24.1
17	271°14.1	52°46.5	34.2	48°18.6	48.0	274°00.3	55.9	82°30.7	24.3
18	286°16.6	67°48.9	S22°34.4	63°19.3	S16°48.6	289°03.1	S02°56.0	97°32.9	S01°24.4
19	301°19.1	82°51.4	34.5	78°20.1	49.1	304°05.8	56.1	112°35.0	24.5
20	316°21.5	97°53.9	34.7	93°20.8	49.6	319°08.6	56.2	127°37.2	24.6
21	331°24.0	112°56.3	•• 34.8	108°21.5	•• 50.2	334°11.3	•• 56.3	142°39.4	•• 24.7
22	346°26.5	127°58.8	35.0	123°22.3	50.7	349°14.1	56.4	157°41.6	24.8
23	1°28.9	143°01.3	35.1	138°23.0	51.2	4°16.8	56.6	172°43.8	25.0
Mer.pass. 22:54		ν 2.4' d 0.2' m -4.77		ν 0.7' d 0.5' m 1.48		ν 2.7' d 0.1' m -2.91		ν 2.2' d 0.1' m 0.90	

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	16°31.4	158°03.8	S22°35.3	153°23.8	S16°51.7	19°19.5	S02°56.7	187°45.9	S01°25.1
1	31°33.8	173°06.3	35.4	168°24.5	52.3	34°22.3	56.8	202°48.1	25.2
2	46°36.3	188°08.7	35.5	183°25.2	52.8	49°25.0	56.9	217°50.3	25.3
3	61°38.8	203°11.2	· · 35.7	198°26.0	· · 53.3	64°27.8	· · 57.0	232°52.5	· · 25.4
4	76°41.2	218°13.7	35.8	213°26.7	53.9	79°30.5	57.1	247°54.7	25.6
5	91°43.7	233°16.2	36.0	228°27.4	54.4	94°33.2	57.3	262°56.8	25.7
6	106°46.2	248°18.7	S22°36.1	243°28.2	S16°54.9	109°36.0	S02°57.4	277°59.0	S01°25.8
7	121°48.6	263°21.2	36.2	258°28.9	55.4	124°38.7	57.5	293°01.2	25.9
8	136°51.1	278°23.8	36.4	273°29.6	56.0	139°41.5	57.6	308°03.4	26.0
9	151°53.6	293°26.3	· · 36.5	288°30.4	· · 56.5	154°44.2	· · 57.7	323°05.6	· · 26.2
10	166°56.0	308°28.8	36.6	303°31.1	57.0	169°46.9	57.8	338°07.7	26.3
11	181°58.5	323°31.3	36.7	318°31.8	57.5	184°49.7	57.9	353°09.9	26.4
12	197°00.9	338°33.8	S22°36.9	333°32.6	S16°58.1	199°52.4	S02°58.1	8°12.1	S01°26.5
13	212°03.4	353°36.4	37.0	348°33.3	58.6	214°55.2	58.2	23°14.3	26.6
14	227°05.9	8°38.9	37.1	3°34.0	59.1	229°57.9	58.3	38°16.5	26.8
15	242°08.3	23°41.5	· · 37.2	18°34.8	16°59.6	245°00.6	· · 58.4	53°18.6	· · 26.9
16	257°10.8	38°44.0	37.3	33°35.5	17°00.2	260°03.4	58.5	68°20.8	27.0
17	272°13.3	53°46.5	37.4	48°36.2	00.7	275°06.1	58.6	83°23.0	27.1
18	287°15.7	68°49.1	S22°37.5	63°37.0	S17°01.2	290°08.9	S02°58.7	98°25.2	S01°27.2
19	302°18.2	83°51.6	37.6	78°37.7	01.7	305°11.6	58.9	113°27.4	27.3
20	317°20.7	98°54.2	37.8	93°38.4	02.3	320°14.3	59.0	128°29.6	27.5
21	332°23.1	113°56.8	· · 37.9	108°39.2	· · 02.8	335°17.1	· · 59.1	143°31.7	· · 27.6
22	347°25.6	128°59.3	38.0	123°39.9	03.3	350°19.8	59.2	158°33.9	27.7
23	2°28.1	144°01.9	38.1	138°40.6	03.8	5°22.5	59.3	173°36.1	27.8
Mer.pass. 22:50		ν 2.5' d 0.1' m -4.76		ν 0.7' d 0.5' m 1.48		ν 2.7' d 0.1' m -2.91		ν 2.2' d 0.1' m 0.90	

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	17°30.5	159°04.5	S22°38.2	153°41.4	S17°04.4	20°25.3	S02°59.4	188°38.3	S01°27.9
1	32°33.0	174°07.1	38.3	168°42.1	04.9	35°28.0	59.5	203°40.5	28.1
2	47°35.4	189°09.6	38.3	183°42.8	05.4	50°30.8	59.7	218°42.6	28.2
3	62°37.9	204°12.2	· · 38.4	198°43.5	· · 05.9	65°33.5	· · 59.8	233°44.8	· · 28.3
4	77°40.4	219°14.8	38.5	213°44.3	06.5	80°36.2	02°59.9	248°47.0	28.4
5	92°42.8	234°17.4	38.6	228°45.0	07.0	95°39.0	03°00.0	263°49.2	28.5
6	107°45.3	249°20.0	S22°38.7	243°45.7	S17°07.5	110°41.7	S03°00.1	278°51.4	S01°28.7
7	122°47.8	264°22.6	38.8	258°46.5	08.0	125°44.4	00.2	293°53.5	28.8
8	137°50.2	279°25.2	38.9	273°47.2	08.5	140°47.2	00.3	308°55.7	28.9
9	152°52.7	294°27.8	· · 38.9	288°47.9	· · 09.1	155°49.9	· · 00.4	323°57.9	· · 29.0
10	167°55.2	309°30.4	39.0	303°48.6	09.6	170°52.6	00.6	339°00.1	29.1
11	182°57.6	324°33.1	39.1	318°49.4	10.1	185°55.4	00.7	354°02.3	29.2
12	198°00.1	339°35.7	S22°39.2	333°50.1	S17°10.6	200°58.1	S03°00.8	9°04.4	S01°29.4
13	213°02.6	354°38.3	39.2	348°50.8	11.1	216°00.8	00.9	24°06.6	29.5
14	228°05.6	9°40.9	39.3	3°51.5	11.7	231°03.6	01.0	39°08.8	29.6
15	243°07.5	24°43.6	· · 39.4	18°52.3	· · 12.2	246°06.3	· · 01.1	54°11.0	· · 29.7
16	258°09.9	39°46.2	39.4	33°53.0	12.7	261°09.1	01.2	69°13.2	29.8
17	273°12.4	54°48.9	39.5	48°53.7	13.2	276°11.8	01.3	84°15.4	30.0
18	288°14.9	69°51.5	S22°39.6	63°54.4	S17°13.7	291°14.5	S03°01.5	99°17.5	S01°30.1
19	303°17.3	84°54.1	39.6	78°55.2	14.3	306°17.3	01.6	114°19.7	30.2
20	318°19.8	99°56.8	39.7	93°55.9	14.8	321°20.0	01.7	129°21.9	30.3
21	333°22.3	114°59.5	· · 39.7	108°56.6	· · 15.3	336°22.7	· · 01.8	144°24.1	· · 30.4
22	348°24.7	130°02.1	39.8	123°57.3	15.8	351°25.5	01.9	159°26.3	30.6
23	3°27.2	145°04.8	39.8	138°58.1	16.3	6°28.2	02.0	174°28.4	30.7
Mer.pass. 22:46		ν 2.6' d 0.1' m -4.74		ν 0.7' d 0.5' m 1.48		ν 2.7' d 0.1' m -2.90		ν 2.2' d 0.1' m 0.91	

	Stars	
	SHA	Dec
Alpheratz	357° 45.2	29° 09.3
Ankaa	353° 17.1	-42° 14.7
Schedar	349° 42.3	-56° 36.0
Diphda	348° 57.5	-17° 55.4
Achernar	335° 27.5	-57° 10.7
Hamal	328° 02.7	23° 31.0
Polaris	318° 21.5	89° 18.6
Acamar	315° 19.4	-40° 15.4
Menkar	314° 16.8	4° 08.1
Mirfak	308° 42.8	49° 54.0
Aldebaran	290° 51.4	16° 31.9
Rigel	281° 13.8	-8° 11.2
Capella	280° 37.1	46° 00.4
Bellatrix	278° 34.0	6° 21.7
Elmath	278° 14.9	28° 37.0
Anilam	275° 48.2	-1° 11.6
Betelgeuse	271° 03.3	7° 24.6
Canopus	263° 56.9	-52° 41.8
Sirius	258° 35.4	-16° 43.7
Adhara	255° 14.1	-28° 59.0
Procyon	245° 01.8	5° 11.9
Pollux	243° 30.2	27° 59.9
Axiur	234° 19.1	-59° 32.4
Suhail	222° 54.2	-43° 28.4
Miaplacidus	221° 40.8	-69° 45.5
Alphard	217° 58.2	-8° 42.2
Regulus	207° 45.8	11° 54.9
Dubhe	193° 54.5	61° 41.4
Denebola	182° 35.9	14° 30.7
Gienah	175° 54.7	-17° 36.1
Acrux	173° 12.4	-63° 09.5
Gacrux	172° 03.8	-57° 10.4
Alioth	166° 22.8	55° 54.0
Spica	158° 33.6	-11° 13.0
Alkaid	153° 00.8	49° 15.6
Hadar	148° 51.4	-60° 25.6
Menkent	148° 10.3	-36° 25.4
Arcturus	145° 57.8	19° 07.7
Rigel Kent.	139° 55.1	-60° 52.8
Kochab	137° 20.7	74° 06.8
Zuben 'ubi	137° 07.9	-16° 05.2
Alphecca	126° 12.9	26° 40.9
Antares	112° 28.9	-26° 27.4
Atria	107° 32.8	-69° 03.0
Sabik	102° 14.9	-15° 44.2
Shaula	96° 24.7	-37° 06.8
Rasalhague	96° 08.4	12° 33.3
Eltanin	90° 47.2	51° 29.6
Kaus Aust.	83° 46.5	-34° 22.8
Vega	80° 40.3	38° 47.9
Nunki	76° 00.7	-26° 17.0
Altair	62° 10.1	8° 54.0
Peacock	53° 22.0	-56° 42.1
Deneb	49° 32.7	45° 19.5
Enif	33° 48.8	9° 55.7
Al Na'ir	27° 45.7	-46° 54.5
Fomalhaut	15° 25.8	-29° 33.8
Scheat	13° 55.0	28° 08.8
Markab	13° 40.0	15° 16.1

Oct 07 Thu	SHA	Mer.pass
Venus	141°33.0	13:29
Mars	137°33.8	13:47
Jupiter	2°41.5	22:43
Saturn	171°21.4	11:31

Oct 08 Fri	SHA	Mer.pass
Venus	141°32.4	13:26
Mars	136°52.4	13:46
Jupiter	2°48.2	22:39
Saturn	171°14.6	11:27

Oct 09 Sat	SHA	Mer.pass
Venus	141°34.0	13:21
Mars	136°10.8	13:45
Jupiter	2°54.8	22:34
Saturn	171°07.8	11:24

Horizontal parallax	
Venus:	0.4
Mars:	0.1

h	Sun			Moon			
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	183°00.0	S05°22.6	195°08.0	8.8'	S05°37.1	15.2'	60.9'
1	198°00.2	23.5	209°35.8	8.8'	05°52.4	15.2'	60.9'
2	213°00.4	24.5	224°03.6	8.8'	06°07.6	15.2'	60.9'
3	228°00.5	· · 25.5	238°31.4	8.8'	06°22.7	15.1'	60.9'
4	243°00.7	26.4	252°59.2	8.7'	06°37.8	15.1'	60.9'
5	258°00.9	27.4	267°26.9	8.7'	06°52.9	15.0'	60.9'
6	273°01.1	S05°28.3	281°54.6	8.7'	S07°07.9	15.0'	60.9'
7	288°01.3	29.3	296°22.3	8.6'	07°22.9	14.9'	60.8'
8	303°01.4	30.2	310°49.9	8.6'	07°37.9	14.9'	60.8'
9	318°01.6	· · 31.2	325°17.5	8.6'	07°52.7	14.8'	60.8'
10	333°01.8	32.2	339°45.1	8.5'	08°07.6	14.8'	60.8'
11	348°02.0	33.1	354°12.6	8.5'	08°22.3	14.7'	60.8'
12	3°02.2	S05°34.1	8°40.2	8.5'	S08°37.1	14.7'	60.8'
13	18°02.3	35.0	23°07.6	8.5'	08°51.7	14.6'	60.8'
14	33°02.5	36.0	37°35.1	8.4'	09°06.3	14.5'	60.8'
15	48°02.7	· · 36.9	52°02.5	8.4'	09°20.8	14.5'	60.8'
16	63°02.9	37.9	66°29.9	8.3'	09°35.3	14.4'	60.7'
17	78°03.0	38.8	80°57.2	8.3'	09°49.7	14.3'	60.7'
18	93°03.2	S05°39.8	95°24.6	8.3'	S10°04.1	14.3'	60.7'
19	108°03.4	40.8	109°51.8	8.2'	10°18.3	14.2'	60.7'
20	123°03.6	41.7	124°19.1	8.2'	10°32.5	14.1'	60.7'
21	138°03.8	· · 42.7	138°46.3	8.2'	10°46.6	14.0'	60.7'
22	153°03.9	43.6	153°13.5	8.1'	11°00.7	14.0'	60.7'
23	168°04.1	44.6	167°40.6	8.1'	11°14.7	13.9'	60.6'
SD = 16.0'			$d = 1.0'$			SD = 16.6'	

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	183°04.3	S05°45.5	182°07.7	8.1'	S11°28.6	13.8'	60.6'
1	198°04.5	46.5	196°34.8	8.0'	11°42.4	13.7'	60.6'
2	213°04.6	47.4	211°01.8	8.0'	11°56.1	13.7'	60.6'
3	228°04.8	· · 48.4	225°28.8	7.9'	12°09.8	13.6'	60.6'
4	243°05.0	49.4	239°55.7	7.9'	12°23.3	13.5'	60.5'
5	258°05.2	50.3	254°22.6	7.9'	12°36.8	13.4'	60.5'
6	273°05.3	S05°51.3	268°49.5	7.8'	S12°50.2	13.3'	60.5'
7	288°05.5	52.2	283°16.3	7.8'	13°03.5	13.2'	60.5'
8	303°05.7	53.2	297°43.1	7.8'	13°16.7	13.1'	60.5'
9	318°05.9	· · 54.1	312°09.9	7.7'	13°29.8	13.0'	60.4'
10	333°06.0	55.1	326°36.6	7.7'	13°42.9	12.9'	60.4'
11	348°06.2	56.0	341°03.3	7.6'	13°55.8	12.8'	60.4'
12	3°06.4	S05°57.0	355°29.9	7.6'	S14°08.6	12.7'	60.4'
13	18°06.6	57.9	9°56.5	7.6'	14°21.4	12.6'	60.3'
14	33°06.7	58.9	24°23.1	7.5'	14°34.0	12.5'	60.3'
15	48°06.9	05°59.8	38°49.6	7.5'	14°46.5	12.4'	60.3'
16	63°07.1	06°00.8	53°16.1	7.4'	14°58.9	12.3'	60.3'
17	78°07.3	01.7	67°42.5	7.4'	15°11.3	12.2'	60.2'
18	93°07.4	S06°02.7	82°08.9	7.4'	S15°23.5	12.1'	60.2'
19	108°07.6	03.6	96°35.3	7.3'	15°35.6	12.0'	60.2'
20	123°07.8	04.6	111°01.6	7.3'	15°47.6	11.9'	60.2'
21	138°07.9	· · 05.6	125°27.9	7.3'	15°59.5	11.8'	60.1'
22	153°08.1	06.5	139°54.2	7.2'	16°11.2	11.7'	60.1'
23	168°08.3	07.5	154°20.4	7.2'	16°22.9	11.5'	60.1'
SD = 16.0'			$d = 1.0'$			SD = 16.5'	

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	183°08.5	S06°08.4	168°46.6	7.1'	S16°34.4	11.4'	60.1'
1	198°08.6	09.4	183°12.7	7.1'	16°45.9	11.3'	60.0'
2	213°08.8	10.3	197°38.8	7.1'	16°57.2	11.2'	60.0'
3	228°09.0	· · 11.3	212°04.9	7.0'	17°08.4	11.1'	60.0'
4	243°09.2	12.2	226°30.9	7.0'	17°19.4	10.9'	59.9'
5	258°09.3	13.2	240°56.9	7.0'	17°30.4	10.8'	59.9'
6	273°09.5	S06°14.1	255°22.9	6.9'	S17°41.2	10.7'	59.9'
7	288°09.7	15.1	269°48.8	6.9'	17°51.9	10.6'	59.9'
8	303°09.8	16.0	284°14.7	6.9'	18°02.5	10.5'	59.8'
9	318°10.0	· · 17.0	298°40.6	6.8'	18°12.9	10.3'	59.8'
10	333°10.2	17.9	313°06.4	6.8'	18°23.3	10.2'	59.8'
11	348°10.3	18.9	327°32.2	6.8'	18°33.5	10.1'	59.7'
12	3°10.5	S06°19.8	341°58.0	6.7'	S18°43.5	9.9'	59.7'
13	18°10.7	20.8	356°23.7	6.7'	18°53.5	9.8'	59.7'
14	33°10.9	21.7	10°49.4	6.7'	19°03.3	9.7'	59.6'
15	48°11.0	· · 22.7	25°15.1	6.6'	19°12.9	9.5'	59.6'
16	63°11.2	23.6	39°40.7	6.6'	19°22.5	9.4'	59.6'
17	78°11.4	24.6	54°06.3	6.6'	19°31.9	9.3'	59.5'
18	93°11.5	S06°25.5	68°31.9	6.6'	S19°41.1	9.1'	59.5'
19	108°11.7	26.5	82°57.5	6.5'	19°50.2	9.0'	59.5'
20	123°11.9	27.4	97°23.0	6.5'	19°59.2	8.8'	59.4'
21	138°12.0	· · 28.4	111°48.5	6.5'	20°08.1	8.7'	59.4'
22	153°12.2	29.3	126°14.0	6.5'	20°16.8	8.6'	59.4'
23	168°12.4	30.3	140°39.5	6.4'	20°25.4	8.4'	59.3'
SD = 16.0'			$d = 1.0'$			SD = 16.4'	

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	04:23	05:42	06:50	16:43	17:51	19:09
N 70°	04:30	05:42	06:43	16:50	17:51	19:02
68°	04:37	05:42	06:37	16:56	17:52	18:56
66°	04:42	05:41	06:33	17:01	17:52	18:51
64°	04:46	05:41	06:28	17:05	17:53	18:48
62°	04:49	05:41	06:25	17:09	17:53	18:44
60°	04:52	05:40	06:22	17:12	17:54	18:42
N 58°	04:54	05:40	06:19	17:15	17:54	18:39
56°	04:56	05:39	06:17	17:17	17:55	18:38
54°	04:58	05:39	06:14	17:20	17:55	18:36
52°	05:00	05:39	06:12	17:22	17:56	18:35
50°	05:01	05:38	06:11	17:24	17:56	18:33
45°	05:03	05:37	06:06	17:28	17:57	18:31
N 40°	05:05	05:36	06:03	17:32	17:59	18:30
35°	05:05	05:35	06:00	17:35	18:00	18:29
30°	05:06	05:33	05:57	17:37	18:01	18:29
20°	05:05	05:31	05:53	17:42	18:04	18:30
N 10°	05:03	05:27	05:48	17:47	18:08	18:32
0°	04:59	05:24	05:44	17:51	18:12	18:36
S 10°	04:54	05:19	05:40	17:55	18:16	18:41
20°	04:47	05:13	05:36	18:00	18:22	18:48
30°	04:38	05:06	05:30	18:05	18:30	18:58
35°	04:31	05:02	05:27	18:09	18:34	19:04
40°	04:24	04:56	05:24	18:12	18:40	19:12
45°	04:14	04:50	05:19	18:17	18:47	19:22
S 50°	04:02	04:41	05:14	18:22	18:55	19:35
52°	03:56	04:38	05:12	18:24	18:59	19:41
54°	03:49	04:33	05:10	18:27	19:03	19:48
56°	03:41	04:28	05:07	18:30	19:08	19:56
58°	03:32	04:23	05:04	18:33	19:14	20:05
S 60°	03:22	04:17	05:00	18:37	19:20	20:16

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	06:50	09:28	■	15:27	14:45	■
N 70°	06:38	08:59	11:51	15:41	15:16	14:22
68°	06:29	08:37	10:58	15:53	15:39	15:17
66°	06:21	08:21	10:25	16:03	15:57	15:50
64°	06:14	08:07	10:02	16:11	16:12	16:15
62°	06:09	07:56	09:43	16:18	16:25	16:34
60°	06:04	07:46	09:28	16:25	16:35	16:50
N 58°	06:00	07:38	09:15	16:30	16:45	17:04
56°	05:56	07:31	09:04	16:35	16:53	17:16
54°	05:53	07:24	08:54	16:40	17:00	17:26
52°	05:50	07:18	08:45	16:44	17:07	17:35
50°	05:47	07:13	08:38	16:47	17:13	17:43
45°	05:41	07:02	08:21	16:55	17:26	18:01
N 40°	05:36	06:52	08:08	17:02	17:36	18:16
35°	05:32	06:44	07:56	17:08	17:46	18:28
30°	05:28	06:37	07:47	17:13	17:54	18:39
20°	05:22	06:26	07:30	17:22	18:08	18:57
N 10°	05:16	06:15	07:15	17:29	18:20	19:13
0°	05:11	06:06	07:01	17:37	18:32	19:29
S 10°	05:06	05:56	06:48	17:44	18:44	19:44
20°	05:01	05:46	06:34	17:52	18:56	20:00
30°	04:55	05:35	06:17	18:01	19:10	20:19
35°	04:51	05:28	06:08	18:06	19:19	20:30
40°	04:48	05:21	05:57	18:12	19:28	20:43
45°	04:43	05:12	05:45	18:19	19:40	20:58
S 50°	04:38	05:02	05:30	18:28	19:53	21:16
52°	04:35	04:57	05:23	18:32	20:00	21:25
54°	04:33	04:52	05:15	18:36	20:07	21:35
56°	04:30	04:46	05:06	18:41	20:15	21:46
58°	04:27	04:40	04:56	18:46	20:24	21:58
S 60°	04:23	04:33	04:45	18:52	20:34	22:13

Day	Sun			Moon		Age 0-2 1-2%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss	Mer. Pass hh:mm	Mer. Pass. Upper hh:mm	Lower hh:mm	
07	12:00	12:09	11:48	11:24	23:51	
08	12:17	12:26	11:48	12:19	—:—	
09	12:34	12:42	11:47	13:15	00:47	

October 10, 11, 12 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	18°29.7	160°07.5	S22°39.9	153°58.8	S17°16.9	21°30.9	S03°02.1	189°30.6	S01°30.8	Alpheratz	357°45.2 29°09.3
1	33°32.1	175°10.1	39.9	168°59.5	17.4	36°33.7	02.2	204°32.8	30.9	Ankaa	353°17.1 -42°14.7
2	48°34.6	190°12.8	40.0	184°00.2	17.9	51°36.4	02.4	219°35.0	31.0	Schedar	349°42.3 56°36.0
3	63°37.0	205°15.5	· · 40.0	199°01.0	· · 18.4	66°39.1	· · 02.5	234°37.2	· · 31.1	Diphda	348°57.5 -17°55.4
4	78°39.5	220°18.2	40.0	214°01.7	18.9	81°41.9	02.6	249°39.3	31.3	Achernar	335°27.5 -57°10.7
5	93°42.0	235°20.9	40.1	229°02.4	19.4	96°44.6	02.7	264°41.5	31.4	Hamal	328°02.7 23°31.0
6	108°44.4	250°23.6	S22°40.1	244°03.1	S17°20.0	111°47.3	S03°02.8	279°43.7	S01°31.5	Polaris	318°20.5 89°18.6
7	123°46.9	265°26.2	40.2	259°03.8	20.5	126°50.1	02.9	294°45.9	31.6	Acamar	315°19.4 -40°15.4
8	138°49.4	280°28.9	40.2	274°04.6	21.0	141°52.8	03.0	309°48.1	31.7	Menkar	314°16.8 4°08.1
9	153°51.8	295°31.7	· · 40.2	289°05.3	· · 21.5	156°55.5	· · 03.1	324°50.3	· · 31.9	Mirfak	308°42.8 49°54.0
10	168°54.3	310°34.4	40.2	304°06.0	22.0	171°58.2	03.2	339°52.4	32.0	Aldebaran	290°51.4 16°31.9
11	183°56.8	325°37.1	40.3	319°06.7	22.5	187°01.0	03.3	354°54.6	32.1	Rigel	281°13.8 -8°11.2
12	198°59.2	340°39.8	S22°40.3	334°07.5	S17°23.1	202°03.7	S03°03.5	9°56.8	S01°32.2	Capella	280°37.1 46°00.4
13	214°01.7	355°42.5	40.3	349°08.2	23.6	217°06.4	03.6	24°59.0	32.3	Bellatrix	278°33.9 6°21.7
14	229°04.2	10°45.2	40.3	4°08.9	24.1	232°09.2	03.7	40°01.2	32.4	Elnath	278°14.9 28°37.0
15	244°06.6	25°48.0	· · 40.3	19°09.6	· · 24.6	247°11.9	· · 03.8	55°03.3	· · 32.6	Alnilam	275°48.2 -1°11.6
16	259°09.1	40°50.7	40.4	34°10.3	25.1	262°14.6	03.9	70°05.5	32.7	Betelgeuse	271°03.3 7°24.6
17	274°11.5	55°53.4	40.4	49°11.0	25.6	277°17.4	04.0	85°07.7	32.8	Canopus	263°56.9 -52°41.8
18	289°14.0	70°56.2	S22°40.4	64°11.8	S17°26.2	292°20.1	S03°04.1	100°09.9	S01°32.9	Sirius	258°35.4 -16°43.7
19	304°16.5	85°58.9	40.4	79°12.5	26.7	307°22.8	04.2	115°12.1	33.0	Adhara	255°14.0 -28°59.0
20	319°18.9	101°01.7	40.4	94°13.2	27.2	322°25.6	04.3	130°14.2	33.2	Procyon	245°01.8 5°11.9
21	334°21.4	116°04.4	· · 40.4	109°13.9	· · 27.7	337°28.3	· · 04.5	145°16.4	· · 33.3	Pollux	243°30.1 27°59.9
22	349°23.9	131°07.2	40.4	124°14.6	28.2	352°31.0	04.6	160°18.6	33.4	Avior	234°19.1 -59°32.4
23	4°26.3	146°09.9	40.4	139°15.4	28.7	7°33.7	04.7	175°20.8	33.5	Suhail	222°54.1 -43°28.4
Mer.pass. 22:42		ν 2.7' d 0.0' m -4.72		ν 0.7' d 0.5' m 1.48		ν 2.7' d 0.1' m -2.90		ν 2.2' d 0.1' m 0.91		Miaplacidus	221°40.7 -69°45.5
										Alphard	217°58.2 -8°42.2
										Regulus	207°45.8 11°54.8
										Dubhe	193°54.4 61°41.4
										Denebola	182°35.9 14°30.7
										Gienah	175°54.7 -17°36.1
										Acrux	173°12.4 -63°09.5
										Gacrux	172°03.8 -57°10.4
										Alioth	166°22.8 55°54.0
										Spica	158°33.6 -11°13.0
										Alkaid	153°00.8 49°15.6
										Hadar	148°51.4 -60°25.6
										Menkent	148°10.3 -36°25.4
										Arcturus	145°57.8 19°07.6
										Rigel Kent.	139°55.1 -60°52.8
										Kochab	137°20.7 74°06.7
										Zuben'ubi	137°07.9 -16°05.2
										Alphecca	126°12.9 26°40.8
										Antares	112°28.9 -26°27.4
										Atria	107°32.8 -69°03.0
										Sabik	102°15.0 -15°44.2
										Shaula	96°24.7 -37°06.7
										Rasalhague	96°08.4 12°33.3
										Eltanin	90°47.2 51°29.6
										Kaus Aust.	83°46.5 -34°22.8
										Vega	80°40.4 38°47.9
										Nunki	76°00.7 -26°17.0
										Altair	62°10.1 8°54.0
										Peacock	53°22.0 -56°42.1
										Deneb	49°32.7 45°19.5
										Enif	33°48.9 9°55.7
										Al Na'ir	27°45.8 -46°54.5
										Fomalhaut	15°25.8 -29°33.8
										Scheat	13°55.0 28°08.8
										Markab	13°40.0 15°16.1
										Oct 10 Sun	SHA Mer.pass
										Venus	141°37.8 13:17
										Mars	135°29.1 13:43
										Jupiter	3°01.3 22:30
										Saturn	171°01.0 11:20
										Oct 11 Mon	SHA Mer.pass
										Venus	141°43.9 13:13
										Mars	134°47.3 13:42
										Jupiter	3°07.7 22:25
										Saturn	170°54.2 11:17
										Oct 12 Tue	SHA Mer.pass
										Venus	141°52.3 13:08
										Mars	134°05.2 13:41
										Jupiter	3°14.0 22:21
										Saturn	170°47.4 11:13

DUT1 = UT1-UTC = -0.0710 sec $\Delta T = TT-UT1 = +66.2550$ sec

2010 October 10 to Oct. 12 UT

h Sun Moon						
Sun	GHA	Dec	GHA	ν	Dec	d HP
0	183°12.5	S06°31.2	155°04.9	6.4'	S20°33.8	8.3' 59.3'
1	198°12.7	32.2	169°30.3	6.4'	20°42.1	8.1' 59.3'
2	213°12.9	33.1	183°55.7	6.4'	20°50.2	8.0' 59.2'
3	228°13.0	· · 34.0	198°21.1	6.4'	20°58.2	7.9' 59.2'
4	243°13.2	35.0	212°46.5	6.3'	21°06.1	7.7' 59.2'
5	258°13.4	35.9	227°11.8	6.3'	21°13.8	7.6' 59.1'
6	273°13.5	S06°36.9	241°37.1	6.3'	S21°21.4	7.4' 59.1'
7	288°13.7	37.8	256°02.4	6.3'	21°28.8	7.3' 59.0'
8	303°13.9	38.8	270°27.7	6.3'	21°36.0	7.1' 59.0'
9	318°14.0	· · 39.7	284°53.0	6.3'	21°43.2	7.0' 59.0'
10	333°14.2	40.7	299°18.3	6.3'	21°50.2	6.8' 58.9'
11	348°14.4	41.6	313°43.6	6.3'	21°57.0	6.7' 58.9'
12	3°14.5	S06°42.6	328°08.8	6.2'	S22°03.7	6.5' 58.9'
13	18°14.7	43.5	342°34.1	6.2'	22°10.2	6.4' 58.8'
14	33°14.9	44.5	356°59.3	6.2'	22°16.6	6.2' 58.8'
15	48°15.0	· · 45.4	11°24.5	6.2'	22°22.8	6.1' 58.7'
16	63°15.2	46.4	25°49.8	6.2'	22°28.9	5.9' 58.7'
17	78°15.4	47.3	40°15.0	6.2'	22°34.8	5.8' 58.7'
18	93°15.5	S06°48.2	54°40.2	6.2'	S22°40.6	5.6' 58.6'
19	108°15.7	49.2	69°05.4	6.2'	22°46.2	5.5' 58.6'
20	123°15.8	50.1	83°30.7	6.2'	22°51.7	5.3' 58.6'
21	138°16.0	· · 51.1	97°55.9	6.2'	22°57.0	5.2' 58.5'
22	153°16.2	52.0	112°21.1	6.2'	23°02.2	5.0' 58.5'
23	168°16.3	53.0	126°46.4	6.2'	23°07.2	4.9' 58.4'
SD = 16.0' $d = 0.9'$			SD = 16.2'			
Mon	GHA	Dec	GHA	ν	Dec	d HP
0	183°16.5	S06°53.9	141°11.6	6.3'	S23°12.0	4.7' 58.4'
1	198°16.7	54.9	155°36.9	6.3'	23°16.8	4.6' 58.4'
2	213°16.8	55.8	170°02.1	6.3'	23°21.3	4.4' 58.3'
3	228°17.0	· · 56.7	184°27.4	6.3'	23°25.7	4.3' 58.3'
4	243°17.1	57.7	198°52.7	6.3'	23°30.0	4.1' 58.2'
5	258°17.3	58.6	213°18.0	6.3'	23°34.1	3.9' 58.2'
6	273°17.5	S06°59.6	227°43.3	6.3'	S23°38.0	3.8' 58.2'
7	288°17.6	07°00.5	242°08.6	6.3'	23°41.8	3.6' 58.1'
8	303°17.8	01.5	256°33.9	6.4'	23°45.4	3.5' 58.1'
9	318°18.0	· · 02.4	270°59.3	6.4'	23°48.9	3.3' 58.0'
10	333°18.1	03.3	285°24.7	6.4'	23°52.3	3.2' 58.0'
11	348°18.3	04.3	299°50.1	6.4'	23°55.4	3.0' 58.0'
12	3°18.4	S07°05.2	314°15.5	6.5'	S23°58.5	2.9' 57.9'
13	18°18.6	06.2	328°41.0	6.5'	24°01.3	2.7' 57.9'
14	33°18.8	07.1	343°06.4	6.5'	24°04.1	2.6' 57.9'
15	48°18.9	· · 08.1	357°32.0	6.5'	24°06.6	2.4' 57.8'
16	63°19.1	09.0	11°57.5	6.6'	24°09.0	2.3' 57.8'
17	78°19.2	09.9	26°23.0	6.6'	24°11.3	2.1' 57.7'
18	93°19.4	S07°10.9	40°48.6	6.6'	S24°13.4	2.0' 57.7'
19	108°19.6	11.8	55°14.3	6.7'	24°15.4	1.8' 57.7'
20	123°19.7	12.8	69°39.9	6.7'	24°17.2	1.7' 57.6'
21	138°19.9	· · 13.7	84°05.6	6.7'	24°18.9	1.5' 57.6'
22	153°20.0	14.6	98°31.4	6.8'	24°20.4	1.4' 57.5'
23	168°20.2	15.6	112°57.1	6.8'	24°21.7	1.2' 57.5'
SD = 16.0' $d = 0.9'$			SD = 15.9'			
Tue	GHA	Dec	GHA	ν	Dec	d HP
0	183°20.3	S07°16.5	127°23.0	6.9'	S24°22.9	1.1' 57.5'
1	198°20.5	17.5	141°48.8	6.9'	24°24.0	0.9' 57.4'
2	213°20.7	18.4	156°14.7	6.9'	24°24.9	0.8' 57.4'
3	228°20.8	· · 19.3	170°40.6	7.0'	24°25.7	0.6' 57.3'
4	243°21.0	20.3	185°06.6	7.0'	24°26.3	0.5' 57.3'
5	258°21.1	21.2	199°32.7	7.1'	24°26.8	0.3' 57.3'
6	273°21.3	S07°22.2	213°58.7	7.1'	S24°27.1	0.2' 57.2'
7	288°21.4	23.1	228°24.9	7.2'	24°27.3	0.0' 57.2'
8	303°21.6	24.0	242°51.0	7.2'	24°27.4	-0.1' 57.2'
9	318°21.8	· · 25.0	257°17.3	7.3'	24°27.2	-0.2' 57.1'
10	333°21.9	25.9	271°43.6	7.3'	24°27.0	-0.4' 57.1'
11	348°22.1	26.9	286°09.9	7.4'	24°26.6	-0.5' 57.0'
12	3°22.2	S07°27.8	300°36.3	7.4'	S24°26.1	-0.7' 57.0'
13	18°22.4	28.7	315°02.7	7.5'	24°25.4	-0.8' 57.0'
14	33°22.5	29.7	329°29.2	7.6'	24°24.6	-1.0' 56.9'
15	48°22.7	· · 30.6	343°55.8	7.6'	24°23.6	-1.1' 56.9'
16	63°22.8	31.5	358°22.4	7.7'	24°22.5	-1.2' 56.9'
17	78°23.0	32.5	12°49.1	7.7'	24°21.3	-1.4' 56.8'
18	93°23.1	S07°33.4	27°15.8	7.8'	S24°19.9	-1.5' 56.8'
19	108°23.3	34.4	41°42.6	7.9'	24°18.4	-1.6' 56.7'
20	123°23.5	35.3	56°09.5	7.9'	24°16.8	-1.8' 56.7'
21	138°23.6	· · 36.2	70°36.4	8.0'	24°15.0	-1.9' 56.7'
22	153°23.8	37.2	85°03.4	8.1'	24°13.0	-2.1' 56.6'
23	168°23.9	38.1	99°30.4	8.1'	24°11.0	-2.2' 56.6'
SD = 16.0' $d = 0.9'$			SD = 15.7'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	04:37	05:55	07:04	16:27	17:36	18:54
N 70°	04:43	05:54	06:55	16:36	17:38	18:48
68°	04:47	05:52	06:48	16:44	17:40	18:44
66°	04:51	05:51	06:42	16:50	17:41	18:40
64°	04:54	05:49	06:37	16:55	17:43	18:37
62°	04:57	05:48	06:33	16:59	17:44	18:35
60°	04:59	05:47	06:29	17:03	17:45	18:33
N 58°	05:01	05:46	06:26	17:07	17:46	18:31
56°	05:02	05:45	06:23	17:10	17:47	18:30
54°	05:04	05:44	06:20	17:13	17:48	18:29
52°	05:05	05:44	06:17	17:15	17:49	18:28
50°	05:05	05:43	06:15	17:17	17:50	18:27
45°	05:07	05:41	06:10	17:23	17:52	18:26
N 40°	05:08	05:39	06:06	17:27	17:54	18:25
35°	05:08	05:37	06:02	17:31	17:56	18:25
30°	05:07	05:35	05:59	17:34	17:58	18:26
20°	05:06	05:31	05:54	17:40	18:02	18:27
N 10°	05:03	05:27	05:48	17:45	18:06	18:31
0°	04:58	05:23	05:43	17:50	18:11	18:35
S 10°	04:53	05:17	05:38	17:55	18:16	18:41
20°	04:45	05:11	05:33	18:01	18:23	18:49
30°	04:34	05:03	05:27	18:07	18:32	19:00
35°	04:27	04:57	05:23	18:11	18:37	19:07
40°	04:19	04:51	05:19	18:15	18:43	19:16
45°	04:08	04:44	05:14	18:20	18:51	19:27
S 50°	03:54	04:35	05:08	18:27	19:00	19:41
52°	03:48	04:30	05:05	18:29	19:04	19:47
54°	03:40	04:26	05:02	18:32	19:09	19:55
56°	03:32	04:20	04:59	18:36	19:15	20:04
58°	03:22	04:14	04:55	18:40	19:21	20:14
S 60°	03:11	04:07	04:51	18:44	19:28	20:25

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	■	■	■	■	■	■
66°	12:37	■	■	15:39	■	■
64°	11:54	13:35	14:41	16:22	16:42	17:35
62°	11:26	12:55	13:59	16:51	17:22	18:17
60°	11:04	12:28	13:30	17:13	17:50	18:45
N 58°	10:47	12:06	13:08	17:31	18:11	19:07
56°	10:32	11:49	12:50	17:46	18:29	19:25
54°	10:19	11:34	12:35	18:00	18:44	19:40
52°	10:08	11:21	12:22	18:11	18:57	19:53
50°	09:58	11:10	12:10	18:21	19:08	20:05
45°	09:37	10:46	11:46	18:43	19:32	20:28
N 40°	09:20	10:28	11:27	19:00	19:51	20:48
35°	09:06	10:12	11:11	19:15	20:07	21:03
30°	08:54	09:58	10:57	19:28	20:21	21:17
20°	08:33	09:35	10:33	19:50	20:45	21:41
N 10°	08:15	09:15	10:13	20:09	21:05	22:01
0°	07:59	08:57	09:53	20:26	21:24	22:20
S 10°	07:42	08:38	09:34	20:44	21:43	22:39
20°	07:24	08:18	09:14	21:03	22:03	22:59
30°	07:04	07:55	08:50	21:25	22:27	23:22
35°	06:52	07:42	08:36	21:38	22:41	23:36
40°	06:39	07:27	08:20	21:53	22:57	23:52
45°	06:23	07:09	08:01	22:11	23:16	· · ·
S 50°	06:04	06:46	07:37	22:33	23:39	· · ·
52°	05:54	06:35	07:26	22:43	23:51	· · ·
54°	05:44	06:23	07:13	22:55	· · ·	00:04
56°	05:33	06:09	06:58	23:09	· · ·	00:19
58°	05:19	05:53	06:40	23:25	· · ·	00:36
S 60°	05:04	05:33	06:19	23:44	· · ·	00:58

Day	Sun		Mer. Pass	Moon		Age 3-5 7-22%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
10	12:50	12:58	11:47	14:13	01:44	
11	13:06	13:14	11:47	15:10	02:41	
12	13:21	13:29	11:47	16:07	03:39	

October 13, 14, 15 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars								
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec							
0	21°27.1	163°30.0	S22°37.6	154°50.1	S17°53.6	24°47.3	S03°09.9	192°07.7	S01°39.3	Alpheratz	357°45.2 29°09.3							
1	36°29.5	178°33.0	37.5	169°50.8	54.1	39°50.0	10.0	207°09.9	39.4	Ankaa	353°17.1 -42°14.7							
2	51°32.0	193°35.9	37.4	184°51.5	54.6	54°52.7	10.1	222°12.1	39.5	Schedar	349°42.3 56°36.0							
3	66°34.5	208°38.9	•• 37.3	199°52.2	•• 55.1	69°55.4	•• 10.2	237°14.3	•• 39.6	Diphda	348°57.5 -17°55.4							
4	81°36.9	223°41.9	37.1	214°52.9	55.6	84°58.1	10.3	252°16.4	39.8	Achernar	335°27.5 -57°10.8							
5	96°39.4	238°44.9	37.0	229°53.6	56.1	100°00.9	10.4	267°18.6	39.9	Hamal	328°02.6 23°31.0							
6	111°41.9	253°47.8	S22°36.9	244°54.3	S17°56.6	115°03.6	S03°10.5	282°20.8	S01°40.0	Polaris	318°19.6 89°18.6							
7	126°44.3	268°50.8	36.8	259°55.1	57.1	130°06.3	10.7	297°23.0	40.1	Acamar	315°19.4 -40°15.4							
8	141°46.8	283°53.8	36.6	274°55.8	57.6	145°09.0	10.8	312°25.2	40.2	Menkar	314°16.8 4°08.1							
9	156°49.3	298°56.8	•• 36.5	289°56.5	•• 58.1	160°11.7	•• 10.9	327°27.4	•• 40.3	Mirfak	308°42.8 49°54.0							
10	171°51.7	313°59.8	36.4	304°57.2	58.6	175°14.5	11.0	342°29.5	40.5	Aldebaran	290°51.4 16°31.9							
11	186°54.2	329°02.8	36.2	319°57.9	59.1	190°17.2	11.1	357°31.7	40.6	Rigel	281°13.7 -8°11.2							
12	201°56.6	344°05.8	S22°36.1	334°58.6	S17°59.6	205°19.9	S03°11.2	12°33.9	S01°40.7	Capella	280°37.1 46°00.4							
13	216°59.1	359°08.8	36.0	349°59.3	18°00.1	220°22.6	11.3	27°36.1	40.8	Bellatrix	278°33.9 6°21.7							
14	232°01.6	14°11.8	35.8	5°00.0	00.6	235°25.3	11.4	42°38.3	40.9	Elnath	278°14.9 28°37.0							
15	247°04.0	29°14.8	•• 35.7	20°00.7	•• 01.1	250°28.0	•• 11.5	57°40.4	•• 41.0	Alnilam	275°48.2 -1°11.6							
16	262°06.5	44°17.8	35.5	35°01.3	01.6	265°30.8	11.6	72°42.6	41.2	Betelgeuse	271°03.3 7°24.6							
17	277°09.0	59°20.8	35.4	50°02.0	02.1	280°33.5	11.7	87°44.8	41.3	Canopus	263°56.9 -52°41.8							
18	292°11.4	74°23.9	S22°35.2	65°02.7	S18°02.6	295°36.2	S03°11.8	102°47.0	S01°41.4	Sirius	258°35.4 -16°43.7							
19	307°13.9	89°26.9	35.1	80°03.4	03.1	310°38.9	11.9	117°49.2	41.5	Adhara	255°14.0 -28°59.0							
20	322°16.4	104°29.9	34.9	95°04.1	03.6	325°41.6	12.0	132°51.4	41.6	Procyon	245°01.8 5°11.9							
21	337°18.8	119°33.0	•• 34.7	110°04.8	•• 04.1	340°44.3	•• 12.1	147°53.5	•• 41.8	Pollux	243°30.1 27°59.9							
22	352°21.3	134°36.0	34.6	125°05.5	04.6	355°47.1	12.2	162°55.7	41.9	Avior	234°19.0 -59°32.4							
23	7°23.8	149°39.0	34.4	140°06.2	05.1	10°49.8	12.3	177°57.9	42.0	Suhail	222°54.1 -43°28.4							
Mer.pass. 22:30		ν 3.0' d -0.1' m -4.63		ν 0.7' d 0.5' m 1.47		ν 2.7' d 0.1' m -2.88		ν 2.2' d 0.1' m 0.91		Alphard	217°58.2 -8°42.2							
										Regulus	207°45.8 11°54.8							
										Dubhe	193°54.4 61°41.4							
										Denebola	182°35.9 14°30.7							
										Gienah	175°54.6 -17°36.1							
										Acrux	173°12.4 -63°09.5							
										Gacrux	172°03.8 -57°10.4							
										Alioth	166°22.8 55°54.0							
										Spica	158°33.6 -11°13.0							
										Alkaid	153°00.8 49°15.6							
										Hadar	148°51.4 -60°25.6							
										Menkent	148°10.3 -36°25.4							
										Arcturus	145°57.8 19°07.6							
										Rigel Kent.	139°55.1 -60°52.8							
										Kochab	137°20.8 74°06.7							
										Zuben'ubi	137°07.9 -16°05.2							
										Alphecca	126°12.9 26°40.8							
										Antares	112°28.9 -26°27.4							
										Atria	107°32.9 -69°03.0							
										Sabik	102°15.0 -15°44.2							
										Shaula	96°24.7 -37°06.7							
										Rasalhague	96°08.4 12°33.3							
										Eltanin	90°47.2 51°29.6							
										Kaus Aust.	83°46.5 -34°22.8							
										Vega	80°40.4 38°47.9							
										Nunki	76°00.8 -26°17.0							
										Altair	62°10.1 8°54.0							
										Peacock	53°22.0 -56°42.1							
										Deneb	49°32.7 45°19.5							
										Enif	33°48.9 9°55.7							
										Al Na'ir	27°45.8 -46°54.5							
										Fomalhaut	15°25.8 -29°33.8							
										Scheat	13°55.0 28°08.8							
										Markab	13°40.0 15°16.1							
Oct 13 Wed		SHA		Mer.pass														
		Venus		142°03.0		13:03												
		Mars		133°23.1		13:40												
		Jupiter		3°20.2		22:17												
		Saturn		170°40.6		11:10												
Oct 14 Thu		SHA		Mer.pass														
		Venus		142°15.9		12:59												
		Mars		132°40.7		13:39												
		Jupiter		3°26.3		22:12												
		Saturn		170°33.9		11:06												
Oct 15 Fri		SHA		Mer.pass														
		Venus		142°31.0		12:54												
		Mars		131°58.2		13:38												
		Jupiter		3°32.2		22:08												
		Saturn		170°27.1		11:03												
											Horizontal parallax							
											Venus:	0.5						
											Mars:	0.1						

h	Sun			Moon			
Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	183°24.1	S07°39.0	113°57.6	8.2'	S24°08.8	-2.3'	56.6'
1	198°24.2	40.0	128°24.8	8.3'	24°06.5	-2.5'	56.5'
2	213°24.4	40.9	142°52.0	8.3'	24°04.0	-2.6'	56.5'
3	228°24.5	· · 41.8	157°19.4	8.4'	24°01.5	-2.7'	56.5'
4	243°24.7	42.8	171°46.8	8.5'	23°58.7	-2.8'	56.4'
5	258°24.8	43.7	186°14.2	8.5'	23°55.9	-3.0'	56.4'
6	273°25.0	S07°44.6	200°41.8	8.6'	S23°52.9	-3.1'	56.4'
7	288°25.1	45.6	215°09.4	8.7'	23°49.8	-3.2'	56.3'
8	303°25.3	46.5	229°37.1	8.8'	23°46.6	-3.3'	56.3'
9	318°25.4	· · 47.5	244°04.8	8.8'	23°43.3	-3.5'	56.3'
10	333°25.6	48.4	258°32.7	8.9'	23°39.8	-3.6'	56.2'
11	348°25.7	49.3	273°00.6	9.0'	23°36.2	-3.7'	56.2'
12	3°25.9	S07°50.3	287°28.5	9.1'	S23°32.5	-3.8'	56.1'
13	18°26.0	51.2	301°56.6	9.1'	23°28.6	-4.0'	56.1'
14	33°26.2	52.1	316°24.7	9.2'	23°24.7	-4.1'	56.1'
15	48°26.3	· · 53.1	330°52.9	9.3'	23°20.6	-4.2'	56.0'
16	63°26.5	54.0	345°21.2	9.4'	23°16.4	-4.3'	56.0'
17	78°26.6	54.9	359°49.6	9.4'	23°12.1	-4.4'	56.0'
18	93°26.8	S07°55.9	14°18.0	9.5'	S23°07.6	-4.6'	56.0'
19	108°26.9	56.8	28°46.6	9.6'	23°03.1	-4.7'	55.9'
20	123°27.1	57.7	43°15.2	9.7'	22°58.4	-4.8'	55.9'
21	138°27.2	· · 58.6	57°43.8	9.8'	22°53.6	-4.9'	55.9'
22	153°27.4	07°59.6	72°12.6	9.8'	22°48.7	-5.0'	55.8'
23	168°27.5	08°00.5	86°41.4	9.9'	22°43.7	-5.1'	55.8'
SD = 16.0'			SD = 15.4'				
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	183°27.7	S08°01.4	101°10.4	10.0'	S22°38.6	-5.2'	55.8'
1	198°27.8	02.4	115°39.4	10.1'	22°33.4	-5.3'	55.7'
2	213°28.0	03.3	130°08.4	10.2'	22°28.1	-5.4'	55.7'
3	228°28.1	· · 04.2	144°37.6	10.2'	22°22.6	-5.6'	55.7'
4	243°28.3	05.2	159°06.8	10.3'	22°17.1	-5.7'	55.6'
5	258°28.4	06.1	173°36.2	10.4'	22°11.4	-5.8'	55.6'
6	273°28.5	S08°07.0	188°05.6	10.5'	S22°05.6	-5.9'	55.6'
7	288°28.7	08.0	202°35.0	10.6'	21°59.8	-6.0'	55.6'
8	303°28.8	08.9	217°04.6	10.6'	21°53.8	-6.1'	55.5'
9	318°29.0	· · 09.8	231°34.3	10.7'	21°47.7	-6.2'	55.5'
10	333°29.1	10.7	246°04.0	10.8'	21°41.5	-6.3'	55.5'
11	348°29.3	11.7	260°33.8	10.9'	21°35.3	-6.4'	55.4'
12	3°29.4	S08°12.6	275°03.7	11.0'	S21°28.9	-6.5'	55.4'
13	18°29.6	13.5	289°33.6	11.1'	21°22.4	-6.6'	55.4'
14	33°29.7	14.5	304°03.7	11.1'	21°15.8	-6.7'	55.4'
15	48°29.8	· · 15.4	318°33.8	11.2'	21°09.2	-6.8'	55.3'
16	63°30.0	16.3	333°04.0	11.3'	21°02.4	-6.9'	55.3'
17	78°30.1	17.2	347°34.3	11.4'	20°55.6	-7.0'	55.3'
18	93°30.3	S08°18.2	2°04.7	11.5'	S20°48.6	-7.0'	55.2'
19	108°30.4	19.1	16°35.2	11.5'	20°41.6	-7.1'	55.2'
20	123°30.6	20.0	31°05.7	11.6'	20°34.4	-7.2'	55.2'
21	138°30.7	· · 21.0	45°36.3	11.7'	20°27.2	-7.3'	55.2'
22	153°30.9	21.9	60°07.0	11.8'	20°19.9	-7.4'	55.1'
23	168°31.0	22.8	74°37.8	11.9'	20°12.5	-7.5'	55.1'
SD = 16.0'			SD = 15.2'				
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	183°31.1	S08°23.7	89°08.6	11.9'	S20°05.0	-7.6'	55.1'
1	198°31.3	24.7	103°39.5	12.0'	19°57.4	-7.7'	55.1'
2	213°31.4	25.6	118°10.6	12.1'	19°49.7	-7.7'	55.0'
3	228°31.6	· · 26.5	132°41.6	12.2'	19°42.0	-7.8'	55.0'
4	243°31.7	27.4	147°12.8	12.2'	19°34.2	-7.9'	55.0'
5	258°31.8	28.4	161°44.0	12.3'	19°26.2	-8.0'	55.0'
6	273°32.0	S08°29.3	176°15.4	12.4'	S19°18.2	-8.1'	55.0'
7	288°32.1	30.2	190°46.8	12.5'	19°10.2	-8.2'	54.9'
8	303°32.3	31.1	205°18.2	12.5'	19°02.0	-8.2'	54.9'
9	318°32.4	· · 32.1	219°49.8	12.6'	18°53.8	-8.3'	54.9'
10	333°32.5	33.0	234°21.4	12.7'	18°45.4	-8.4'	54.9'
11	348°32.7	33.9	248°53.1	12.8'	18°37.0	-8.5'	54.8'
12	3°32.8	S08°34.8	263°24.9	12.8'	S18°28.6	-8.5'	54.8'
13	18°33.0	35.8	277°56.7	12.9'	18°20.0	-8.6'	54.8'
14	33°33.1	36.7	292°28.6	13.0'	18°11.4	-8.7'	54.8'
15	48°33.2	· · 37.6	307°00.6	13.1'	18°02.7	-8.8'	54.8'
16	63°33.4	38.5	321°32.7	13.1'	17°53.9	-8.8'	54.7'
17	78°33.5	39.5	336°04.8	13.2'	17°45.1	-8.9'	54.7'
18	93°33.6	S08°40.4	350°37.0	13.3'	S17°36.2	-9.0'	54.7'
19	108°33.8	41.3	5°09.3	13.3'	17°27.2	-9.1'	54.7'
20	123°33.9	42.2	19°41.6	13.4'	17°18.1	-9.1'	54.7'
21	138°34.1	· · 43.1	34°14.1	13.5'	17°09.0	-9.2'	54.6'
22	153°34.2	44.1	48°46.5	13.6'	16°59.8	-9.3'	54.6'
23	168°34.3	45.0	63°19.1	13.6'	16°50.6	-9.3'	54.6'
SD = 16.0'			SD = 15.0'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	04:50	06:08	07:19	16:12	17:21	18:39
N 70°	04:55	06:05	07:08	16:22	17:25	18:35
68°	04:58	06:02	06:59	16:31	17:28	18:32
66°	05:01	06:00	06:52	16:38	17:30	18:29
64°	05:03	05:58	06:46	16:45	17:33	18:27
62°	05:05	05:56	06:41	16:50	17:35	18:26
60°	05:06	05:54	06:36	16:54	17:36	18:24
N 58°	05:07	05:53	06:32	16:59	17:38	18:23
56°	05:08	05:51	06:29	17:02	17:40	18:23
54°	05:09	05:50	06:26	17:06	17:41	18:22
52°	05:10	05:49	06:23	17:09	17:43	18:21
50°	05:10	05:47	06:20	17:11	17:44	18:21
45°	05:11	05:45	06:14	17:17	17:47	18:21
N 40°	05:11	05:42	06:09	17:22	17:50	18:21
35°	05:10	05:39	06:05	17:27	17:52	18:21
30°	05:09	05:37	06:01	17:31	17:55	18:22
20°	05:07	05:32	05:54	17:37	18:00	18:25
N 10°	05:03	05:27	05:48	17:43	18:05	18:29
0°	04:58	05:22	05:43	17:49	18:10	18:34
S 10°	04:51	05:16	05:37	17:55	18:17	18:41
20°	04:42	05:08	05:31	18:02	18:24	18:50
30°	04:30	04:59	05:23	18:09	18:34	19:02
35°	04:23	04:53	05:19	18:14	18:39	19:10
40°	04:13	04:47	05:14	18:19	18:46	19:20
45°	04:02	04:38	05:09	18:24	18:55	19:31
S 50°	03:47	04:28	05:02	18:31	19:05	19:46
52°	03:40	04:23	04:59	18:35	19:10	19:54
54°	03:32	04:18	04:55	18:38	19:16	20:02
56°	03:23	04:12	04:51	18:42	19:22	20:11
58°	03:12	04:05	04:47	18:47	19:29	20:22
S 60°	02:59	03:57	04:42	18:52	19:37	20:36

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	■	■	17:55	■	■	19:39
N 70°	■	■	16:45	■	■	20:48
68°	■	16:39	16:08	■	19:15	21:24
66°	16:03	15:49	15:41	18:05	20:04	21:49
64°	15:08	15:17	15:21	19:00	20:35	22:09
62°	14:35	14:54	15:04	19:32	20:58	22:25
60°	14:10	14:35	14:50	19:56	21:16	22:38
N 58°	13:51	14:19	14:38	20:16	21:32	22:49
56°	13:35	14:06	14:28	20:32	21:45	22:59
54°	13:21	13:54	14:19	20:45	21:56	23:08
52°	13:09	13:44	14:10	20:57	22:06	23:16
50°	12:58	13:35	14:03	21:08	22:15	23:22
45°	12:35	13:15	13:47	21:30	22:33	23:37
N 40°	12:17	12:59	13:34	21:47	22:48	23:49
35°	12:02	12:46	13:23	22:02	23:01	· · ·
30°	11:49	12:34	13:13	22:15	23:12	· · ·
20°	11:26	12:14	12:56	22:37	23:31	· · ·
N 10°	11:06	11:56	12:41	22:56	23:48	· · ·
0°	10:48	11:40	12:28	23:13	· · ·	00:03
S 10°	10:30	11:23	12:14	23:31	· · ·	00:18
20°	10:10	11:05	11:59	23:49	· · ·	00:35
30°	09:47	10:45	11:42	· · ·	00:11	00:53
35°	09:34	10:33	11:32	· · ·	00:23	01:04
40°	09:19	10:19	11:20	· · ·	00:38	01:16
45°	09:00	10:03	11:07	00:10	00:55	01:31
S 50°	08:37	09:43	10:50	00:33	01:16	01:48
52°	08:26	09:33	10:42	00:45	01:26	01:56
54°	08:14	09:22	10:33	00:57	01:37	02:06
56°	08:00	09:10	10:24	01:12	01:49	02:16
58°	07:43	08:55	10:12	01:29	02:04	02:28
S 60°	07:22	08:38	09:59	01:49	02:21	02:41

Day	Sun		Mer. Pass hh:mm	Moon		Age 6-8 32-51%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
13	13:36	13:44	11:46	17:01	04:34	
14	13:51	13:58	11:46	17:51	05:26	
15	14:05	14:11	11:46	18:39	06:15	

October 16, 17, 18 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	24°24.5	167°12.8	S22°23.4	155°40.0	S18°29.1	28°02.6	S03°17.2	194°44.9	S01°47.7	Alpheratz	357°45.2 29°09.3
1	39°27.0	182°16.0	23.1	170°40.7	29.6	43°05.3	17.3	209°47.1	47.8	Ankaa	353°17.1 -42°14.7
2	54°29.4	197°19.3	22.8	185°41.4	30.1	58°08.0	17.4	224°49.3	47.9	Schedar	349°42.3 56°36.0
3	69°31.9	212°22.5	•• 22.5	200°42.1	•• 30.6	73°10.7	•• 17.5	239°51.4	•• 48.1	Diphda	348°57.5 -17°55.4
4	84°34.3	227°25.7	22.2	215°42.8	31.1	88°13.4	17.6	254°53.6	48.2	Achernar	335°27.5 -57°10.8
5	99°36.8	242°29.0	21.9	230°43.4	31.5	103°16.1	17.7	269°55.8	48.3	Hamal	328°02.6 23°31.0
6	114°39.3	257°32.2	S22°21.6	245°44.1	S18°32.0	118°18.8	S03°17.8	284°58.0	S01°48.4	Polaris	318°19.0 89°18.6
7	129°41.7	272°35.5	21.3	260°44.8	32.5	133°21.5	17.9	300°00.2	48.5	Acamar	315°19.3 -40°15.5
8	144°44.2	287°38.7	21.0	275°45.5	33.0	148°24.2	18.0	315°02.4	48.6	Menkar	314°16.8 4°08.1
9	159°46.7	302°42.0	•• 20.7	290°46.2	•• 33.5	163°26.9	•• 18.1	330°04.5	•• 48.7	Mirfak	308°42.7 49°54.0
10	174°49.1	317°45.3	20.4	305°46.8	34.0	178°29.6	18.2	345°06.7	48.9	Aldebaran	290°51.4 16°31.9
11	189°51.6	332°48.5	20.1	320°47.5	34.4	193°32.3	18.3	0°08.9	49.0	Rigel	281°13.7 -8°11.2
12	204°54.1	347°51.8	S22°19.8	335°48.2	S18°34.9	208°35.0	S03°18.4	15°11.1	S01°49.1	Capella	280°37.0 46°00.4
13	219°56.5	2°55.1	19.5	350°48.9	35.4	223°37.7	18.5	30°13.3	49.2	Bellatrix	278°33.9 6°21.7
14	234°59.0	17°58.4	19.2	5°49.6	35.9	238°40.4	18.6	45°15.5	49.3	Elnath	278°14.9 28°37.0
15	250°01.5	33°01.6	•• 18.9	20°50.2	•• 36.4	253°43.1	•• 18.7	60°17.7	•• 49.4	Alnilam	275°48.2 -1°11.6
16	265°03.9	48°04.9	18.5	35°50.9	36.8	268°45.9	18.8	75°19.8	49.6	Betelgeuse	271°03.2 7°24.6
17	280°06.4	63°08.2	18.2	50°51.6	37.3	283°48.6	18.9	90°22.0	49.7	Canopus	263°56.8 -52°41.8
18	295°08.8	78°11.5	S22°17.9	65°52.3	S18°37.8	298°51.3	S03°19.0	105°24.2	S01°49.8	Sirius	258°35.3 -16°43.7
19	310°11.3	93°14.8	17.6	80°52.9	38.3	313°54.0	19.0	120°26.4	49.9	Adhara	255°14.0 -28°59.0
20	325°13.8	108°18.1	17.2	95°53.6	38.8	328°56.7	19.1	135°28.6	50.0	Procyon	245°01.7 5°11.9
21	340°16.2	123°21.4	•• 16.9	110°54.3	•• 39.2	343°59.4	•• 19.2	150°30.8	•• 50.1	Pollux	243°30.1 27°59.9
22	355°18.7	138°24.7	16.6	125°55.0	39.7	359°02.1	19.3	165°32.9	50.3	Avior	234°19.0 -59°32.4
23	10°21.2	153°28.0	16.2	140°55.7	40.2	14°04.8	19.4	180°35.1	50.4	Suhail	222°54.1 -43°28.4
Mer.pass. 22:19		ν 3.2' d -0.3' m -4.51		ν 0.7' d 0.5' m 1.46		ν 2.7' d 0.1' m -2.87		ν 2.2' d 0.1' m 0.92		Miaplacidus	221°40.6 -69°45.4
										Alphard	217°58.1 -8°42.3
										Regulus	207°45.7 11°54.8
										Dubhe	193°54.4 61°41.3
										Denebola	182°35.9 14°30.7
										Gienah	175°54.6 -17°36.1
										Acrux	173°12.4 -63°09.5
										Gacrux	172°03.8 -57°10.4
										Alioth	166°22.8 55°54.0
										Spica	158°33.6 -11°13.0
										Alkaid	153°00.8 49°15.6
										Hadar	148°51.4 -60°25.5
										Menkent	148°10.3 -36°25.4
										Arcturus	145°57.8 19°07.6
										Rigel Kent.	139°55.1 -60°52.8
										Kochab	137°20.8 74°06.7
										Zuben'ubi	137°07.9 -16°05.2
										Alphecca	126°12.9 26°40.8
										Antares	112°28.9 -26°27.4
										Atria	107°32.9 -69°03.0
										Sabik	102°15.0 -15°44.2
										Shaula	96°24.8 -37°06.7
										Rasalhague	96°08.4 12°33.3
										Eltanin	90°47.3 51°29.6
										Kaus Aust.	83°46.5 -34°22.8
										Vega	80°40.4 38°47.9
										Nunki	76°00.8 -26°17.0
										Altair	62°10.1 8°54.0
										Peacock	53°22.1 -56°42.1
										Deneb	49°32.7 45°19.5
										Enif	33°48.9 9°55.7
										Al Na'ir	27°45.8 -46°54.5
										Fomalhaut	15°25.8 -29°33.8
										Scheat	13°55.0 28°08.8
										Markab	13°40.0 15°16.1
										Oct 16 Sat	SHA Mer.pass
										Venus	142°48.3 12:48
										Mars	131°15.5 13:37
										Jupiter	3°38.1 22:04
										Saturn	170°20.4 10:59
										Oct 17 Sun	SHA Mer.pass
										Venus	143°07.7 12:43
										Mars	130°32.7 13:36
										Jupiter	3°43.8 22:00
										Saturn	170°13.7 10:56
										Oct 18 Mon	SHA Mer.pass
										Venus	143°29.1 12:38
										Mars	129°49.7 13:35
										Jupiter	3°49.4 21:55
										Saturn	170°07.0 10:52

h	Sun				Moon			
Sat	GHA	Dec	GHA	ν	Dec	d	HP	
0	183°34.5	S08°45.9	77°51.7	13.7'	S16°41.3	-9.4'	54.6'	
1	198°34.6	46.8	92°24.4	13.8'	16°31.9	-9.5'	54.6'	
2	213°34.7	47.8	106°57.2	13.8'	16°22.4	-9.5'	54.5'	
3	228°34.9	· · 48.7	121°30.0	13.9'	16°12.9	-9.6'	54.5'	
4	243°35.0	49.6	136°02.9	13.9'	16°03.3	-9.6'	54.5'	
5	258°35.1	50.5	150°35.8	14.0'	15°53.7	-9.7'	54.5'	
6	273°35.3	S08°51.4	165°08.8	14.1'	S15°44.0	-9.8'	54.5'	
7	288°35.4	52.4	179°41.9	14.1'	15°34.2	-9.8'	54.5'	
8	303°35.6	53.3	194°15.1	14.2'	15°24.4	-9.9'	54.5'	
9	318°35.7	· · 54.2	208°48.3	14.3'	15°14.5	-9.9'	54.4'	
10	333°35.8	55.1	223°21.5	14.3'	15°04.6	-10.0'	54.4'	
11	348°36.0	56.0	237°54.8	14.4'	14°54.6	-10.1'	54.4'	
12	3°36.1	S08°56.9	252°28.2	14.4'	S14°44.5	-10.1'	54.4'	
13	18°36.2	57.9	267°01.7	14.5'	14°34.4	-10.2'	54.4'	
14	33°36.4	58.8	281°35.2	14.6'	14°24.3	-10.2'	54.4'	
15	48°36.5	08°59.7	296°08.7	14.6'	14°14.1	-10.3'	54.3'	
16	63°36.6	09°00.6	310°42.3	14.7'	14°03.8	-10.3'	54.3'	
17	78°36.8	01.5	325°16.0	14.7'	13°53.5	-10.4'	54.3'	
18	93°36.9	S09°02.5	339°49.7	14.8'	S13°43.1	-10.4'	54.3'	
19	108°37.0	03.4	354°23.5	14.8'	13°32.7	-10.5'	54.3'	
20	123°37.1	04.3	8°57.3	14.9'	13°22.2	-10.5'	54.3'	
21	138°37.3	· · 05.2	23°31.2	14.9'	13°11.7	-10.6'	54.3'	
22	153°37.4	06.1	38°05.1	15.0'	13°01.1	-10.6'	54.3'	
23	168°37.5	07.0	52°39.1	15.0'	12°50.5	-10.7'	54.2'	
	SD = 16.0'	d = 0.9'	SD = 14.9'					
Sun	GHA	Dec	GHA	ν	Dec	d	HP	
0	183°37.7	S09°08.0	67°13.1	15.1'	S12°39.8	-10.7'	54.2'	
1	198°37.8	08.9	81°47.2	15.1'	12°29.1	-10.8'	54.2'	
2	213°37.9	09.8	96°21.4	15.2'	12°18.3	-10.8'	54.2'	
3	228°38.1	· · 10.7	110°55.5	15.2'	12°07.5	-10.8'	54.2'	
4	243°38.2	11.6	125°29.8	15.3'	11°56.7	-10.9'	54.2'	
5	258°38.3	12.5	140°04.0	15.3'	11°45.8	-10.9'	54.2'	
6	273°38.4	S09°13.4	154°38.4	15.4'	S11°34.9	-11.0'	54.2'	
7	288°38.6	14.4	169°12.7	15.4'	11°23.9	-11.0'	54.2'	
8	303°38.7	15.3	183°47.1	15.4'	11°12.9	-11.1'	54.2'	
9	318°38.8	· · 16.2	198°21.6	15.5'	11°01.8	-11.1'	54.2'	
10	333°39.0	17.1	212°56.1	15.5'	10°50.7	-11.1'	54.1'	
11	348°39.1	18.0	227°30.6	15.6'	10°39.6	-11.2'	54.1'	
12	3°39.2	S09°18.9	242°05.2	15.6'	S10°28.4	-11.2'	54.1'	
13	18°39.3	19.8	256°39.8	15.6'	10°17.2	-11.3'	54.1'	
14	33°39.5	20.8	271°14.4	15.7'	10°06.0	-11.3'	54.1'	
15	48°39.6	· · 21.7	285°49.1	15.7'	09°54.7	-11.3'	54.1'	
16	63°39.7	22.6	300°23.8	15.8'	09°43.3	-11.4'	54.1'	
17	78°39.8	23.5	314°58.6	15.8'	09°32.0	-11.4'	54.1'	
18	93°40.0	S09°24.4	329°33.4	15.8'	S09°20.6	-11.4'	54.1'	
19	108°40.1	25.3	344°08.2	15.9'	09°09.2	-11.5'	54.1'	
20	123°40.2	26.2	358°43.1	15.9'	08°57.7	-11.5'	54.1'	
21	138°40.3	· · 27.1	13°18.0	15.9'	08°46.2	-11.5'	54.1'	
22	153°40.5	28.0	27°52.9	16.0'	08°34.7	-11.6'	54.1'	
23	168°40.6	29.0	42°27.9	16.0'	08°23.1	-11.6'	54.1'	
	SD = 16.0'	d = 0.9'	SD = 14.8'					
Mon	GHA	Dec	GHA	ν	Dec	d	HP	
0	183°40.7	S09°29.9	57°02.8	16.0'	S08°11.6	-11.6'	54.1'	
1	198°40.8	30.8	71°37.9	16.0'	07°59.9	-11.6'	54.0'	
2	213°41.0	31.7	86°12.9	16.1'	07°48.3	-11.7'	54.0'	
3	228°41.1	· · 32.6	100°48.0	16.1'	07°36.6	-11.7'	54.0'	
4	243°41.2	33.5	115°23.1	16.1'	07°24.9	-11.7'	54.0'	
5	258°41.3	34.4	129°58.2	16.1'	07°13.2	-11.8'	54.0'	
6	273°41.5	S09°35.3	144°33.3	16.2'	S07°01.4	-11.8'	54.0'	
7	288°41.6	36.2	159°08.5	16.2'	06°49.7	-11.8'	54.0'	
8	303°41.7	37.1	173°43.7	16.2'	06°37.9	-11.8'	54.0'	
9	318°41.8	· · 38.1	188°18.9	16.2'	06°26.0	-11.9'	54.0'	
10	333°42.0	39.0	202°54.1	16.3'	06°14.2	-11.9'	54.0'	
11	348°42.1	39.9	217°29.4	16.3'	06°02.3	-11.9'	54.0'	
12	3°42.2	S09°40.8	232°04.7	16.3'	S05°50.4	-11.9'	54.0'	
13	18°42.3	41.7	246°40.0	16.3'	05°38.5	-11.9'	54.0'	
14	33°42.4	42.6	261°15.3	16.3'	05°26.5	-12.0'	54.0'	
15	48°42.6	· · 43.5	275°50.6	16.3'	05°14.6	-12.0'	54.0'	
16	63°42.7	44.4	290°25.9	16.4'	05°02.6	-12.0'	54.0'	
17	78°42.8	45.3	305°01.3	16.4'	04°50.6	-12.0'	54.0'	
18	93°42.9	S09°46.2	319°36.7	16.4'	S04°38.6	-12.0'	54.0'	
19	108°43.0	47.1	334°12.1	16.4'	04°26.5	-12.1'	54.0'	
20	123°43.2	48.0	348°47.5	16.4'	04°14.5	-12.1'	54.0'	
21	138°43.3	· · 48.9	3°22.9	16.4'	04°02.4	-12.1'	54.0'	
22	153°43.4	49.8	17°58.3	16.4'	03°50.3	-12.1'	54.0'	
23	168°43.5	50.7	32°33.7	16.4'	03°38.2	-12.1'	54.0'	
	SD = 16.0'	d = 0.9'	SD = 14.7'					

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	05:03	06:22	07:33	15:56	17:07	18:25
N 70°	05:06	06:17	07:21	16:08	17:12	18:22
68°	05:08	06:13	07:11	16:19	17:16	18:20
66°	05:10	06:09	07:02	16:27	17:20	18:19
64°	05:11	06:06	06:55	16:34	17:23	18:18
62°	05:12	06:04	06:49	16:40	17:26	18:17
60°	05:13	06:01	06:44	16:46	17:28	18:16
N 58°	05:14	05:59	06:39	16:51	17:30	18:16
56°	05:14	05:57	06:35	16:55	17:32	18:15
54°	05:14	05:55	06:31	16:59	17:34	18:15
52°	05:15	05:54	06:28	17:02	17:36	18:15
50°	05:15	05:52	06:25	17:05	17:38	18:15
45°	05:14	05:48	06:18	17:12	17:42	18:16
N 40°	05:14	05:45	06:12	17:18	17:45	18:17
35°	05:12	05:42	06:07	17:23	17:48	18:18
30°	05:11	05:39	06:03	17:27	17:51	18:19
20°	05:07	05:33	05:55	17:35	17:57	18:23
N 10°	05:03	05:27	05:49	17:42	18:03	18:28
0°	04:57	05:21	05:42	17:49	18:10	18:34
S 10°	04:49	05:14	05:35	17:55	18:17	18:42
20°	04:40	05:06	05:28	18:03	18:25	18:51
30°	04:27	04:56	05:20	18:11	18:36	19:05
35°	04:18	04:49	05:15	18:16	18:42	19:13
40°	04:08	04:42	05:10	18:22	18:50	19:23
45°	03:56	04:33	05:03	18:28	18:59	19:36
S 50°	03:40	04:22	04:56	18:36	19:10	19:52
52°	03:32	04:16	04:52	18:40	19:16	20:00
54°	03:23	04:10	04:48	18:44	19:22	20:09
56°	03:13	04:04	04:44	18:49	19:29	20:20
58°	03:01	03:56	04:39	18:54	19:36	20:32
S 60°	02:47	03:48	04:33	18:59	19:45	20:46

October 19, 20, 21 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	27°21.9	171°14.3	S21°56.5	156°28.5	S19°03.4	31°16.8	S03°24.0	197°22.2	S01°56.0	Alpheratz	357°45.2 29°09.3
1	42°24.4	186°17.7	56.1	171°29.1	03.8	46°19.5	24.1	212°24.4	56.1	Ankaa	353°17.1 -42°14.7
2	57°26.8	201°21.2	55.6	186°29.8	04.3	61°22.2	24.1	227°26.6	56.3	Schedar	349°42.3 56°36.0
3	72°29.3	216°24.7	· · 55.1	201°30.4	· · 04.8	76°24.9	· · 24.2	242°28.8	· · 56.4	Diphda	348°57.5 -17°55.5
4	87°31.8	231°28.2	54.6	216°31.1	05.2	91°27.6	24.3	257°30.9	56.5	Achernar	335°27.5 -57°10.8
5	102°34.2	246°31.7	54.2	231°31.8	05.7	106°30.3	24.4	272°33.1	56.6	Hamal	328°02.6 23°31.0
6	117°36.7	261°35.1	S21°53.7	246°32.4	S19°06.2	121°32.9	S03°24.5	287°35.3	S01°56.7	Polaris	318°18.5 89°18.7
7	132°39.2	276°38.6	53.2	261°33.1	06.6	136°35.6	24.6	302°37.5	56.8	Acamar	315°19.3 -40°15.5
8	147°41.6	291°42.1	52.7	276°33.7	07.1	151°38.3	24.7	317°39.7	56.9	Menkar	314°16.8 4°08.1
9	162°44.1	306°45.6	· · 52.2	291°34.4	· · 07.5	166°41.0	· · 24.8	332°41.9	· · 57.1	Mirfak	308°42.7 49°54.0
10	177°46.5	321°49.1	51.8	306°35.1	08.0	181°43.7	24.9	347°44.1	57.2	Aldebaran	290°51.3 16°31.9
11	192°49.0	336°52.6	51.3	321°35.7	08.5	196°46.4	24.9	2°46.2	57.3	Rigel	281°13.7 -8°11.2
12	207°51.5	351°56.1	S21°50.8	336°36.4	S19°08.9	211°49.1	S03°25.0	17°48.4	S01°57.4	Capella	280°37.0 46°00.4
13	222°53.9	6°59.6	50.3	351°37.0	09.4	226°51.7	25.1	32°50.6	57.5	Bellatrix	278°33.9 6°21.7
14	237°56.4	22°03.2	49.8	6°37.7	09.9	241°54.4	25.2	47°52.8	57.6	Elnath	278°14.8 28°37.0
15	252°58.9	37°06.7	· · 49.3	21°38.4	· · 10.3	256°57.1	· · 25.3	62°55.0	· · 57.7	Alnilam	275°48.1 -1°11.6
16	268°01.3	52°10.2	48.8	36°39.0	10.8	271°59.8	25.4	77°57.2	57.9	Betelgeuse	271°03.2 7°24.6
17	283°03.8	67°13.7	48.3	51°39.7	11.3	287°02.5	25.5	92°59.4	58.0	Canopus	263°56.8 -52°41.8
18	298°06.3	82°17.2	S21°47.8	66°40.3	S19°11.7	302°05.2	S03°25.6	108°01.6	S01°58.1	Sirius	258°35.3 -16°43.7
19	313°08.7	97°20.8	47.3	81°41.0	12.2	317°07.9	25.7	123°03.7	58.2	Adhara	255°14.0 -28°59.0
20	328°11.2	112°24.3	46.8	96°41.7	12.6	332°10.5	25.7	138°05.9	58.3	Procyon	245°01.7 5°11.9
21	343°13.7	127°27.8	· · 46.2	111°42.3	· · 13.1	347°13.2	· · 25.8	153°08.1	· · 58.4	Pollux	243°30.0 27°59.9
22	358°16.1	142°31.4	45.7	126°43.0	13.6	2°15.9	25.9	168°10.3	58.5	Avior	234°19.0 -59°32.4
23	13°18.6	157°34.9	45.2	141°43.6	14.0	17°18.6	26.0	183°12.5	58.7	Suhail	222°54.1 -43°28.4
Mer.pass. 22:07		ν 3.5' d -0.5' m -4.35		ν 0.7' d 0.5' m 1.45		ν 2.7' d 0.1' m -2.85		ν 2.2' d 0.1' m 0.92		Miaplacidus	221°40.6 -69°45.4
										Alphard	217°58.1 -8°42.3
										Regulus	207°45.7 11°54.8
										Dubhe	193°54.4 61°41.3
										Denebola	182°35.9 14°30.7
										Gienah	175°54.6 -17°36.1
										Acrux	173°12.3 -63°09.5
										Gacrux	172°03.8 -57°10.4
										Alioth	166°22.8 55°54.0
										Spica	158°33.6 -11°13.0
										Alkaid	153°00.8 49°15.5
										Hadar	148°51.4 -60°25.5
										Menkent	148°10.3 -36°25.4
										Arcturus	145°57.8 19°07.6
										Rigel Kent.	139°55.1 -60°52.8
										Kochab	137°20.8 74°06.7
										Zuben'ubi	137°07.9 -16°05.2
										Alphecca	126°12.9 26°40.8
										Antares	112°28.9 -26°27.4
										Atria	107°32.9 -69°03.0
										Sabik	102°15.0 -15°44.2
										Shaula	96°24.8 -37°06.7
										Rasalhague	96°08.4 12°33.3
										Eltanin	90°47.3 51°29.5
										Kaus Aust.	83°46.5 -34°22.8
										Vega	80°40.4 38°47.9
										Nunki	76°00.8 -26°17.0
										Altair	62°10.1 8°54.0
										Peacock	53°22.1 -56°42.1
										Deneb	49°32.8 45°19.5
										Enif	33°48.9 9°55.7
										Al Na'ir	27°45.8 -46°54.5
										Fomalhaut	15°25.8 -29°33.8
										Scheat	13°55.0 28°08.8
										Markab	13°40.0 15°16.1
										Oct 19 Tue	SHA Mer.pass
										Venus	143°52.4 12:32
										Mars	129°06.6 13:34
										Jupiter	3°54.9 21:51
										Saturn	170°00.3 10:49
										Oct 20 Wed	SHA Mer.pass
										Venus	144°17.4 12:26
										Mars	128°23.2 13:32
										Jupiter	4°00.2 21:47
										Saturn	169°53.6 10:45
										Oct 21 Thu	SHA Mer.pass
										Venus	144°44.0 12:21
										Mars	127°39.8 13:31
										Jupiter	4°05.4 21:42
										Saturn	169°47.0 10:42

h Sun Moon						
Tue	GHA	Dec	GHA	ν	Dec	d HP
0	183°43.6	S09°51.6	47°09.1	16.4'	S03°26.1	-12.1' 54.0'
1	198°43.7	52.6	61°44.6	16.4'	03°14.0	-12.1' 54.0'
2	213°43.9	53.5	76°20.0	16.5'	03°01.8	-12.2' 54.0'
3	228°44.0	· · 54.4	90°55.5	16.5'	02°49.7	-12.2' 54.0'
4	243°44.1	55.3	105°30.9	16.5'	02°37.5	-12.2' 54.0'
5	258°44.2	56.2	120°06.4	16.5'	02°25.3	-12.2' 54.0'
6	273°44.3	S09°57.1	134°41.9	16.5'	S02°13.1	-12.2' 54.0'
7	288°44.4	58.0	149°17.3	16.5'	02°00.9	-12.2' 54.0'
8	303°44.6	58.9	163°52.8	16.5'	01°48.7	-12.2' 54.0'
9	318°44.7	09°59.8	178°28.3	16.5'	01°36.5	-12.2' 54.0'
10	333°44.8	10°00.7	193°03.8	16.5'	01°24.3	-12.2' 54.0'
11	348°44.9	01.6	207°39.2	16.5'	01°12.0	-12.2' 54.0'
12	3°45.0	S10°02.5	222°14.7	16.5'	S00°59.8	-12.3' 54.0'
13	18°45.1	03.4	236°50.2	16.5'	00°47.5	-12.3' 54.1'
14	33°45.3	04.3	251°25.6	16.5'	00°35.3	-12.3' 54.1'
15	48°45.4	· · 05.2	266°01.1	16.4'	00°23.0	-12.3' 54.1'
16	63°45.5	06.1	280°36.5	16.4'	S00°10.8	-12.3' 54.1'
17	78°45.6	07.0	295°12.0	16.4'	N00°01.5	12.3' 54.1'
18	93°45.7	S10°07.9	309°47.4	16.4'	N00°13.8	12.3' 54.1'
19	108°45.8	08.8	324°22.8	16.4'	00°26.1	12.3' 54.1'
20	123°45.9	09.7	338°58.2	16.4'	00°38.3	12.3' 54.1'
21	138°46.0	· · 10.6	353°33.7	16.4'	00°50.6	12.3' 54.1'
22	153°46.2	11.5	8°09.1	16.4'	01°02.9	12.3' 54.1'
23	168°46.3	12.4	22°44.4	16.4'	01°15.2	12.3' 54.1'
SD = 16.0' $d = 0.9'$ SD = 14.7'						
Wed	GHA	Dec	GHA	ν	Dec	d HP
0	183°46.4	S10°13.3	37°19.8	16.4'	N01°27.4	12.3' 54.1'
1	198°46.5	14.2	51°55.2	16.3'	01°39.7	12.3' 54.1'
2	213°46.6	15.1	66°30.5	16.3'	01°52.0	12.3' 54.1'
3	228°46.7	· · 16.0	81°05.8	16.3'	02°04.2	12.3' 54.1'
4	243°46.8	16.9	95°41.1	16.3'	02°16.5	12.3' 54.1'
5	258°46.9	17.8	110°16.4	16.3'	02°28.8	12.3' 54.1'
6	273°47.0	S10°18.7	124°51.7	16.3'	N02°41.0	12.2' 54.2'
7	288°47.2	19.6	139°27.0	16.2'	02°53.3	12.2' 54.2'
8	303°47.3	20.5	154°02.2	16.2'	03°05.5	12.2' 54.2'
9	318°47.4	· · 21.4	168°37.4	16.2'	03°17.7	12.2' 54.2'
10	333°47.5	22.2	183°12.6	16.2'	03°30.0	12.2' 54.2'
11	348°47.6	23.1	197°47.8	16.1'	03°42.2	12.2' 54.2'
12	3°47.7	S10°24.0	212°22.9	16.1'	N03°54.4	12.2' 54.2'
13	18°47.8	24.9	226°58.1	16.1'	04°06.6	12.2' 54.2'
14	33°47.9	25.8	241°33.2	16.1'	04°18.8	12.2' 54.2'
15	48°48.0	· · 26.7	256°08.2	16.0'	04°30.9	12.2' 54.2'
16	63°48.1	27.6	270°43.3	16.0'	04°43.1	12.1' 54.2'
17	78°48.2	28.5	285°18.3	16.0'	04°55.3	12.1' 54.2'
18	93°48.3	S10°29.4	299°53.3	16.0'	N05°07.4	12.1' 54.3'
19	108°48.5	30.3	314°28.3	15.9'	05°19.5	12.1' 54.3'
20	123°48.6	31.2	329°03.2	15.9'	05°31.6	12.1' 54.3'
21	138°48.7	· · 32.1	343°38.1	15.9'	05°43.7	12.1' 54.3'
22	153°48.8	33.0	358°13.0	15.8'	05°55.8	12.1' 54.3'
23	168°48.9	33.9	12°47.8	15.8'	06°07.8	12.0' 54.3'
SD = 16.1' $d = 0.9'$ SD = 14.8'						
Thu	GHA	Dec	GHA	ν	Dec	d HP
0	183°49.0	S10°34.8	27°22.6	15.8'	N06°19.8	12.0' 54.3'
1	198°49.1	35.7	41°57.4	15.7'	06°31.9	12.0' 54.3'
2	213°49.2	36.5	56°32.1	15.7'	06°43.9	12.0' 54.3'
3	228°49.3	· · 37.4	71°06.8	15.7'	06°55.8	12.0' 54.3'
4	243°49.4	38.3	85°41.5	15.6'	07°07.8	11.9' 54.4'
5	258°49.5	39.2	100°16.1	15.6'	07°19.7	11.9' 54.4'
6	273°49.6	S10°40.1	114°50.7	15.5'	N07°31.6	11.9' 54.4'
7	288°49.7	41.0	129°25.3	15.5'	07°43.5	11.9' 54.4'
8	303°49.8	41.9	143°59.8	15.5'	07°55.3	11.8' 54.4'
9	318°49.9	· · 42.8	158°34.2	15.4'	08°07.2	11.8' 54.4'
10	333°50.0	43.7	173°08.7	15.4'	08°19.0	11.8' 54.4'
11	348°50.1	44.6	187°43.0	15.3'	08°30.8	11.7' 54.4'
12	3°50.2	S10°45.4	202°17.4	15.3'	N08°42.5	11.7' 54.5'
13	18°50.3	46.3	216°51.7	15.3'	08°54.2	11.7' 54.5'
14	33°50.4	47.2	231°25.9	15.2'	09°05.9	11.7' 54.5'
15	48°50.5	· · 48.1	246°00.2	15.2'	09°17.6	11.6' 54.5'
16	63°50.6	49.0	260°34.3	15.1'	09°29.2	11.6' 54.5'
17	78°50.7	49.9	275°08.4	15.1'	09°40.8	11.6' 54.5'
18	93°50.8	S10°50.8	289°42.5	15.0'	N09°52.4	11.5' 54.5'
19	108°50.9	51.7	304°16.5	15.0'	10°03.9	11.5' 54.5'
20	123°51.0	52.5	318°50.5	14.9'	10°15.4	11.5' 54.6'
21	138°51.1	· · 53.4	333°24.4	14.9'	10°26.8	11.4' 54.6'
22	153°51.2	54.3	347°58.3	14.8'	10°38.3	11.4' 54.6'
23	168°51.3	55.2	2°32.1	14.8'	10°49.6	11.3' 54.6'
SD = 16.1' $d = 0.9'$ SD = 14.8'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	05:16	06:35	07:48	15:39	16:53	18:11
N 70°	05:18	06:29	07:34	15:54	16:59	18:10
68°	05:19	06:23	07:22	16:06	17:05	18:09
66°	05:19	06:19	07:12	16:16	17:09	18:09
64°	05:20	06:15	07:04	16:24	17:13	18:08
62°	05:20	06:12	06:57	16:31	17:17	18:08
60°	05:20	06:08	06:51	16:37	17:20	18:08
N 58°	05:20	06:06	06:46	16:43	17:23	18:08
56°	05:20	06:03	06:41	16:47	17:25	18:09
54°	05:20	06:01	06:37	16:52	17:28	18:09
52°	05:19	05:59	06:33	16:56	17:30	18:09
50°	05:19	05:57	06:30	16:59	17:32	18:10
45°	05:18	05:52	06:22	17:07	17:37	18:11
N 40°	05:17	05:48	06:16	17:14	17:41	18:12
35°	05:15	05:44	06:10	17:19	17:45	18:14
30°	05:13	05:41	06:05	17:24	17:48	18:16
20°	05:08	05:34	05:56	17:33	17:55	18:21
N 10°	05:03	05:27	05:49	17:41	18:02	18:27
0°	04:56	05:20	05:41	17:48	18:09	18:34
S 10°	04:48	05:13	05:34	17:56	18:17	18:42
20°	04:37	05:04	05:26	18:04	18:26	18:53
30°	04:23	04:52	05:17	18:13	18:38	19:07
35°	04:14	04:45	05:11	18:19	18:45	19:16
40°	04:04	04:37	05:05	18:25	18:53	19:27
45°	03:50	04:27	04:58	18:32	19:03	19:41
S 50°	03:33	04:15	04:49	18:41	19:16	19:58
52°	03:24	04:09	04:45	18:45	19:22	20:07
54°	03:15	04:03	04:41	18:50	19:28	20:17
56°	03:04	03:56	04:36	18:55	19:36	20:28
58°	02:51	03:47	04:31	19:01	19:44	20:41
S 60°	02:35	03:38	04:24	19:07	19:54	20:57

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	15:21	15:00	14:37	02:17	04:05	05:55
N 70°	15:21	15:06	14:50	02:22	04:01	05:43
68°	15:21	15:11	15:02	02:25	03:59	05:34
66°	15:20	15:15	15:11	02:28	03:57	05:27
64°	15:20	15:19	15:18	02:31	03:55	05:20
62°	15:20	15:22	15:25	02:33	03:53	05:15
60°	15:19	15:25	15:31	02:35	03:52	05:10
N 58°	15:19	15:27	15:36	02:36	03:51	05:06
56°	15:19	15:29	15:41	02:38	03:50	05:02
54°	15:19	15:31	15:45	02:39	03:49	04:59
52°	15:19	15:33	15:49	02:40	03:48	04:56
50°	15:19	15:35	15:52	02:41	03:47	04:53
45°	15:18	15:38	16:00	02:44	03:45	04:47
N 40°	15:18	15:41	16:06	02:46	03:44	04:43
35°	15:18	15:44	16:12	02:47	03:43	04:38
30°	15:18	15:46	16:16	02:49	03:41	04:35
20°	15:17	15:50	16:25	02:52	03:40	04:28
N 10°	15:17	15:54	16:32	02:54	03:38	04:23
0°	15:17	15:57	16:39	02:56	03:36	04:17
S 10°	15:17	16:01	16:46	02:58	03:35	04:12
20°	15:16	16:05	16:54	03:00	03:33	04:07
30°	15:16	16:09	17:03	03:02	03:31	04:00
35°	15:16	16:11	17:08	03:04	03:30	03:57
40°	15:16	16:14	17:13	03:05	03:29	03:53
45°	15:16	16:17	17:20	03:07	03:27	03:48
S 50°	15:15	16:21	17:28	03:09	03:25	03:42
52°	15:15	16:23	17:32	03:10	03:25	03:40
54°	15:15	16:25	17:36	03:11	03:24	03:37
56°	15:15	16:27	17:41	03:13	03:23	03:34
58°	15:15	16:30	17:46	03:14	03:22	03:30
S 60°	15:15	16:32	17:52	03:15	03:21	03:26

Day	Sun		Mer. Pass	Moon		Age 12-14 85-96%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
19	14:55	15:00	11:45	21:26	09:06	
20	15:06	15:11	11:45	22:07	09:47	
21	15:16	15:21	11:45	22:50	10:28	

October 22, 23, 24 UT (Fri., Sat., Sun.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	30°19.3	175°31.5	S21°16.8	157°15.4	S19°36.3	34°29.8	S03°30.1	199°59.7	S02°04.2	Alpheratz	357°45.2	29°09.3											
1	45°21.8	190°35.1	16.2	172°16.1	36.7	49°32.5	30.2	215°01.8	04.4	Ankaa	353°17.1	-42°14.7											
2	60°24.2	205°38.8	15.5	187°16.7	37.1	64°35.1	30.3	230°04.0	04.5	Schedar	349°42.3	56°36.1											
3	75°26.7	220°42.4	•• 14.9	202°17.4	•• 37.6	79°37.8	•• 30.4	245°06.2	•• 04.6	Diphda	348°57.5	-17°55.5											
4	90°29.2	235°46.1	14.2	217°18.0	38.0	94°40.5	30.5	260°08.4	04.7	Achernar	335°27.5	-57°10.8											
5	105°31.6	250°49.8	13.6	232°18.6	38.5	109°43.2	30.5	275°10.6	04.8	Hamal	328°02.6	23°31.0											
6	120°34.1	265°53.5	S21°12.9	247°19.3	S19°38.9	124°45.8	S03°30.6	290°12.8	S02°04.9	Polaris	318°18.0	89°18.7											
7	135°36.6	280°57.1	12.3	262°19.9	39.4	139°48.5	30.7	305°15.0	05.0	Acamar	315°19.3	-40°15.5											
8	150°39.0	296°00.8	11.6	277°20.6	39.8	154°51.2	30.8	320°17.2	05.1	Menkar	314°16.8	4°08.1											
9	165°41.5	311°04.5	•• 11.0	292°21.2	•• 40.3	169°53.8	•• 30.9	335°19.4	•• 05.3	Mirfak	308°42.7	49°54.0											
10	180°44.0	326°08.2	10.3	307°21.8	40.7	184°56.5	30.9	350°21.5	05.4	Aldebaran	290°51.3	16°31.9											
11	195°46.4	341°11.9	09.7	322°22.5	41.2	199°59.2	31.0	5°23.7	05.5	Rigel	281°13.7	-8°11.2											
12	210°48.9	356°15.5	S21°09.0	337°23.1	S19°41.6	215°01.8	S03°31.1	20°25.9	S02°05.6	Capella	280°37.0	46°00.4											
13	225°51.3	11°19.2	08.3	352°23.7	42.0	230°04.5	31.2	35°28.1	05.7	Bellatrix	278°33.9	6°21.7											
14	240°53.8	26°22.9	07.6	7°24.4	42.5	245°07.2	31.3	50°30.3	05.8	Elnath	278°14.8	28°37.0											
15	255°56.3	41°26.6	•• 07.0	22°25.0	•• 42.9	260°09.9	•• 31.3	65°32.5	•• 05.9	Alnilam	275°48.1	-1°11.6											
16	270°58.7	56°30.3	06.3	37°25.7	43.4	275°12.5	31.4	80°34.7	06.0	Betelgeuse	271°03.2	7°24.6											
17	286°01.2	71°34.0	05.6	52°26.3	43.8	290°15.2	31.5	95°36.9	06.2	Canopus	263°56.8	-52°41.8											
18	301°03.7	86°37.7	S21°04.9	67°26.9	S19°44.3	305°17.9	S03°31.6	110°39.1	S02°06.3	Sirius	258°35.3	-16°43.7											
19	316°06.1	101°41.4	04.3	82°27.6	44.7	320°20.5	31.7	125°41.2	06.4	Adhara	255°13.9	-28°59.0											
20	331°08.6	116°45.1	03.6	97°28.2	45.1	335°23.2	31.7	140°43.4	06.5	Procyon	245°01.7	5°11.9											
21	346°11.1	131°48.8	•• 02.9	112°28.8	•• 45.6	350°25.9	•• 31.8	155°45.6	•• 06.6	Pollux	243°30.0	27°59.9											
22	1°13.5	146°52.5	02.2	127°29.5	46.0	5°28.5	31.9	170°47.8	06.7	Avior	234°18.9	-59°32.4											
23	16°16.0	161°56.2	01.5	142°30.1	46.5	20°31.2	32.0	185°50.0	06.8	Suhail	222°54.0	-43°28.4											
Mer.pass. 21:55				ν 3.7' d -0.6' m -4.15				ν 0.6' d 0.4' m 1.44				ν 2.7' d 0.1' m -2.84				ν 2.2' d 0.1' m 0.92							

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	31°18.5	176°59.9	S21°00.8	157°30.8	S19°46.9	35°33.9	S03°32.1	200°52.2	S02°07.0										
1	46°20.9	192°03.6	21°00.1	172°31.4	47.3	50°36.5	32.1	215°54.4	07.1										
2	61°23.4	207°07.4	20°59.4	187°32.0	47.8	65°39.2	32.2	230°56.6	07.2										
3	76°25.8	222°11.1	•• 58.7	202°32.7	•• 48.2	80°41.8	•• 32.3	245°58.8	•• 07.3										
4	91°28.3	237°14.8	58.0	217°33.3	48.7	95°44.5	32.4	261°00.9	07.4										
5	106°30.8	252°18.5	57.3	232°33.9	49.1	110°47.2	32.4	276°03.1	07.5										
6	121°33.2	267°22.2	S20°56.6	247°34.6	S19°49.5	125°49.8	S03°32.5	291°05.3	S02°07.6										
7	136°35.7	282°26.0	55.9	262°35.2	50.0	140°52.5	32.6	306°07.5	07.7										
8	151°38.2	297°29.7	55.2	277°35.8	50.4	155°55.2	32.7	321°09.7	07.9										
9	166°40.6	312°33.4	•• 54.5	292°36.5	•• 50.9	170°57.8	•• 32.8	336°11.9	•• 08.0										
10	181°43.1	327°37.1	53.8	307°37.1	51.3	186°00.5	32.8	351°14.1	08.1										
11	196°45.6	342°40.9	53.1	322°37.7	51.7	201°03.2	32.9	6°16.3	08.2										
12	211°48.0	357°44.6	S20°52.3	337°38.4	S19°52.2	216°05.8	S03°33.0	21°18.5	S02°08.3										
13	226°50.5	12°48.3	51.6	352°39.0	52.6	231°08.5	33.1	36°20.6	08.4										
14	241°53.0	27°52.1	50.9	7°39.6	53.0	246°11.1	33.1	51°22.8	08.5										
15	256°55.4	42°55.8	•• 50.2	22°40.3	•• 53.5	261°13.8	•• 33.2	66°25.0	•• 08.6										
16	271°57.9	57°59.5	49.4	37°40.9	53.9	276°16.5	33.3	81°27.2	08.8										
17	287°00.3	73°03.3	48.7	52°41.5	54.3	291°19.1	33.4	96°29.4	08.9										
18	302°02.8	88°07.0	S20°48.0	67°42.1	S19°54.8	306°21.8	S03°33.5	111°31.6	S02°09.0										
19	317°05.3	103°10.7	47.2	82°42.8	55.2	321°24.4	33.5	126°33.8	09.1										
20	332°07.7	118°14.5	46.5	97°43.4	55.7	336°27.1	33.6	141°36.0	09.2										
21	347°10.2	133°18.2	•• 45.8	112°44.0	•• 56.1	351°29.8	•• 33.7	156°38.2	•• 09.3										
22	2°12.7	148°22.0	45.0	127°44.7	56.5	6°32.4	33.8	171°40.4	09.4										
23	17°15.1	163°25.7	44.3	142°45.3	57.0	21°35.1	33.8	186°42.5	09.5										
Mer.pass. 21:51				ν 3.7' d -0.7' m -4.09				ν 0.6' d 0.4' m 1.44				ν 2.7' d 0.1' m -2.83				ν 2.2' d 0.1' m 0.92			

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	32°17.6	178°29.5	S20°43.5	157°45.9	S19°57.4	36°37.7	S03°33.9	201°44.7	S02°09.7										
1	47°20.1	193°33.2	42.8	172°46.6	57.8	51°40.4	34.0	216°46.9	09.8										
2	62°22.5	208°37.0	42.0	187°47.2	58.3	66°43.1	34.1	231°49.1	09.9										
3	77°25.0	223°40.7	•• 41.3	202°47.8	•• 58.7	81°45.7	•• 34.1	246°51.3	•• 10.0										
4	92°27.4	238°44.5	40.5	217°48.4	59.1	96°48.4	34.2	261°53.5	10.1										
5	107°29.9	253°48.2	39.8	232°49.1	19°59.6	111°51.0	34.3	276°55.7	10.2										
6	122°32.4	268°52.0	S20°39.0	247°49.7	S20°00.0	126°53.7	S03°34.4	291°57.9	S02°10.3										
7	137°34.8	283°55.8	38.3	262°50.3	00.4	141°56.4	34.4	307°00.1	10.4										
8	152°37.3	298°59.5	37.5	277°50.9	00.8	156°59.0	34.5	322°02.3	10.5										
9	167°39.8	314°03.3	•• 36.7	292°51.6	•• 01.3	172°01.7	•• 34.6	337°04.4	•• 10.7										
10	182°42.2	329°07.0	36.0	307°52.2	01.7	187°04.3	34.7	352°06.6	10.8										
11	197°44.7	344°10.8	35.2	322°52.8	02.1	202°07.0	34.7	7°08.8	10.9										
12	212°47.2	359°14.6	S20°34.4	337°53.4	S20°02.6	217°09.6	S03°34.8	22°11.0	S02°11.0										
13	227°49.6	14°18.3	33.7	352°54.1	03.0	232°12.3	34.9	37°13.2	11.1										
14	242°52.1	29°22.1	32.9	7°54.7	03.4	247°15.0	35.0	52°15.4	11.2										
15	257°54.6	44°25.9	•• 32.1	22°55.3	•• 03.9	262°17.6	•• 35.0	67°17.6	•• 11.3										
16	272°57.0	59°29.7	31.3	37°55.9	04.3	277°20.3	35.1	82°19.8	11.4										
17	287°59.5	74°33.4	30.5	52°56.6	04.7	292°22.9	35.2	97°22.0	11.6										
18	303°01.9	89°37.2	S20°29.8	67°57.2	S20°05.1	307°25.6	S03°35.2	112°24.2	S02°11.7										
19	318°04.4	104°41.0	29.0	82°57.8	05.6	322°28.2	35.3	127°26.3	11.8										
20	333°06.9	119°44.7	28.2	97°58.4	06.0	337°30.9	35.4	142°28.5	11.9										
21	348°09.3	134°48.5	•• 27.4	112°59.1	•• 06.4	352°33.5	•• 35.5	157°30.7	•• 12.0										
22	3°11.8	149°52.3	26.6	127°59.7	06.9	7°36.2	35.5	172°32.9	12.1										
23	18°14.3	164°56.1	25.8	143°00.3	07.3	22°38.8	35.6	187°35.1	12.2										
Mer.pass. 21:47				ν 3.8' d -0.7' m -4.20				ν 0.6' d 0.4' m 1.44				ν 2.7' d 0.1' m -2.82				ν 2.2' d 0.1' m 0.92			

Oct 22 Fri	SHA	Mer.pass
Venus	145°12.1	12:15
Mars	126°56.1	13:30
Jupiter	4°10.5	21:38
Saturn	169°40.3	10:38

Oct 23 Sat	SHA	Mer.pass
Venus	145°41.5	12:09
Mars	126°12.3	13:29
Jupiter	4°15.4	21:34
Saturn	169°33.7	10:35

Oct 24 Sun	SHA	Mer.pass
Venus	146°11.9	12:03
Mars	125°28.3	13:28
Jupiter	4°20.2	21:30
Saturn	169°27.1	10:31

Horizontal parallax	
Venus:	0.5
Mars:	0.1

h	Sun			Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	183°51.4	S10°56.1	17°05.9	14.7'	N11°01.0	11.3'	54.6'
1	198°51.5	57.0	31°39.6	14.7'	11°12.3	11.3'	54.6'
2	213°51.6	57.9	46°13.3	14.6'	11°23.6	11.2'	54.6'
3	228°51.7	· · 58.7	60°46.9	14.6'	11°34.8	11.2'	54.7'
4	243°51.8	10°59.6	75°20.4	14.5'	11°46.0	11.1'	54.7'
5	258°51.9	11°00.5	89°53.9	14.4'	11°57.1	11.1'	54.7'
6	273°52.0	S11°01.4	104°27.4	14.4'	N12°08.2	11.1'	54.7'
7	288°52.1	02.3	119°00.8	14.3'	12°19.2	11.0'	54.7'
8	303°52.2	03.2	133°34.1	14.3'	12°30.2	11.0'	54.7'
9	318°52.3	· · 04.0	148°07.4	14.2'	12°41.2	10.9'	54.7'
10	333°52.4	04.9	162°40.6	14.2'	12°52.1	10.9'	54.8'
11	348°52.5	05.8	177°13.7	14.1'	13°03.0	10.8'	54.8'
12	3°52.6	S11°06.7	191°46.8	14.0'	N13°13.8	10.8'	54.8'
13	18°52.7	07.6	206°19.9	14.0'	13°24.6	10.7'	54.8'
14	33°52.8	08.4	220°52.8	13.9'	13°35.3	10.7'	54.8'
15	48°52.9	· · 09.3	235°25.7	13.9'	13°45.9	10.6'	54.8'
16	63°53.0	10.2	249°58.6	13.8'	13°56.5	10.6'	54.8'
17	78°53.0	11.1	264°31.4	13.7'	14°07.1	10.5'	54.9'
18	93°53.1	S11°12.0	279°04.1	13.7'	N14°17.6	10.4'	54.9'
19	108°53.2	12.9	293°36.8	13.6'	14°28.0	10.4'	54.9'
20	123°53.3	13.7	308°09.4	13.5'	14°38.4	10.3'	54.9'
21	138°53.4	· · 14.6	322°41.9	13.5'	14°48.7	10.3'	54.9'
22	153°53.5	15.5	337°14.4	13.4'	14°59.0	10.2'	54.9'
23	168°53.6	16.4	351°46.8	13.3'	15°09.2	10.1'	55.0'
SD = 16.1'			SD = 14.9'				
$d = 0.9'$							
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	183°53.7	S11°17.2	6°19.1	13.3'	N15°19.4	10.1'	55.0'
1	198°53.8	18.1	20°51.4	13.2'	15°29.4	10.0'	55.0'
2	213°53.9	19.0	35°23.6	13.1'	15°39.5	10.0'	55.0'
3	228°54.0	· · 19.9	49°55.7	13.1'	15°49.4	9.9'	55.0'
4	243°54.1	20.8	64°27.8	13.0'	15°59.3	9.8'	55.0'
5	258°54.1	21.6	78°59.8	12.9'	16°09.1	9.8'	55.1'
6	273°54.2	S11°22.5	93°31.7	12.9'	N16°18.9	9.7'	55.1'
7	288°54.3	23.4	108°03.6	12.8'	16°28.6	9.6'	55.1'
8	303°54.4	24.3	122°35.3	12.7'	16°38.2	9.6'	55.1'
9	318°54.5	· · 25.1	137°07.1	12.7'	16°47.8	9.5'	55.1'
10	333°54.6	26.0	151°38.7	12.6'	16°57.3	9.4'	55.1'
11	348°54.7	26.9	166°10.3	12.5'	17°06.7	9.3'	55.2'
12	3°54.8	S11°27.8	180°41.8	12.4'	N17°16.0	9.3'	55.2'
13	18°54.8	28.6	195°13.2	12.4'	17°25.3	9.2'	55.2'
14	33°54.9	29.5	209°44.6	12.3'	17°34.5	9.1'	55.2'
15	48°55.0	· · 30.4	224°15.9	12.2'	17°43.6	9.0'	55.2'
16	63°55.1	31.3	238°47.1	12.2'	17°52.6	9.0'	55.2'
17	78°55.2	32.1	253°18.3	12.1'	18°01.6	8.9'	55.3'
18	93°55.3	S11°33.0	267°49.4	12.0'	N18°10.5	8.8'	55.3'
19	108°55.4	33.9	282°20.4	11.9'	18°19.3	8.7'	55.3'
20	123°55.5	34.8	296°51.3	11.9'	18°28.0	8.6'	55.3'
21	138°55.5	· · 35.6	311°22.2	11.8'	18°36.6	8.6'	55.3'
22	153°55.6	36.5	325°52.9	11.7'	18°45.2	8.5'	55.3'
23	168°55.7	37.4	340°23.7	11.6'	18°53.6	8.4'	55.4'
SD = 16.1'			SD = 15.0'				
$d = 0.9'$							
Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	183°55.8	S11°38.2	354°54.3	11.6'	N19°02.0	8.3'	55.4'
1	198°55.9	39.1	9°24.9	11.5'	19°10.3	8.2'	55.4'
2	213°56.0	40.0	23°55.4	11.4'	19°18.5	8.1'	55.4'
3	228°56.0	· · 40.9	38°25.8	11.3'	19°26.6	8.0'	55.4'
4	243°56.1	41.7	52°56.1	11.3'	19°34.7	7.9'	55.5'
5	258°56.2	42.6	67°26.4	11.2'	19°42.6	7.9'	55.5'
6	273°56.3	S11°43.5	81°56.6	11.1'	N19°50.5	7.8'	55.5'
7	288°56.4	44.3	96°26.7	11.1'	19°58.2	7.7'	55.5'
8	303°56.5	45.2	110°56.8	11.0'	20°05.9	7.6'	55.5'
9	318°56.5	· · 46.1	125°26.8	10.9'	20°13.5	7.5'	55.6'
10	333°56.6	46.9	139°56.7	10.8'	20°20.9	7.4'	55.6'
11	348°56.7	47.8	154°26.5	10.8'	20°28.3	7.3'	55.6'
12	3°56.8	S11°48.7	168°56.3	10.7'	N20°35.6	7.2'	55.6'
13	18°56.9	49.5	183°26.0	10.6'	20°42.8	7.1'	55.6'
14	33°56.9	50.4	197°55.6	10.5'	20°49.9	7.0'	55.6'
15	48°57.0	· · 51.3	212°25.1	10.5'	20°56.9	6.9'	55.7'
16	63°57.1	52.1	226°54.6	10.4'	21°03.7	6.8'	55.7'
17	78°57.2	53.0	241°24.0	10.3'	21°10.5	6.7'	55.7'
18	93°57.2	S11°53.9	255°53.4	10.3'	N21°17.2	6.6'	55.7'
19	108°57.3	54.7	270°22.6	10.2'	21°23.8	6.5'	55.7'
20	123°57.4	55.6	284°51.8	10.1'	21°30.2	6.4'	55.8'
21	138°57.5	· · 56.5	299°20.9	10.1'	21°36.6	6.3'	55.8'
22	153°57.6	57.3	313°50.0	10.0'	21°42.8	6.1'	55.8'
23	168°57.6	58.2	328°19.0	9.9'	21°49.0	6.0'	55.8'
SD = 16.1'			SD = 15.1'				
$d = 0.9'$							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	05:29	06:48	08:04	15:23	16:39	17:58
N 70°	05:29	06:40	07:47	15:40	16:47	17:58
68°	05:29	06:34	07:34	15:54	16:53	17:58
66°	05:28	06:28	07:23	16:05	16:59	17:59
64°	05:28	06:23	07:14	16:14	17:04	17:59
62°	05:28	06:19	07:06	16:22	17:08	18:00
60°	05:27	06:16	06:59	16:29	17:12	18:00
N 58°	05:26	06:12	06:53	16:35	17:15	18:01
56°	05:26	06:09	06:48	16:40	17:19	18:02
54°	05:25	06:06	06:43	16:45	17:21	18:03
52°	05:24	06:04	06:38	16:49	17:24	18:03
50°	05:24	06:01	06:34	16:53	17:27	18:04
45°	05:22	05:56	06:26	17:02	17:32	18:06
N 40°	05:20	05:51	06:19	17:09	17:37	18:09
35°	05:17	05:47	06:13	17:16	17:41	18:11
30°	05:15	05:43	06:07	17:21	17:46	18:13
20°	05:09	05:35	05:58	17:31	17:53	18:19
N 10°	05:03	05:28	05:49	17:40	18:01	18:26
0°	04:55	05:20	05:41	17:48	18:09	18:33
S 10°	04:46	05:11	05:33	17:56	18:18	18:43
20°	04:35	05:01	05:24	18:05	18:28	18:54
30°	04:20	04:49	05:14	18:15	18:40	19:09
35°	04:10	04:42	05:08	18:21	18:48	19:19
40°	03:59	04:33	05:01	18:28	18:57	19:31
45°	03:44	04:22	04:53	18:36	19:08	19:46
S 50°	03:26	04:09	04:44	18:46	19:21	20:05
52°	03:17	04:03	04:39	18:51	19:28	20:14
54°	03:06	03:56	04:34	18:56	19:35	20:24
56°	02:54	03:48	04:29	19:01	19:43	20:37
58°	02:40	03:39	04:23	19:08	19:52	20:51
S 60°	02:23	03:28	04:16	19:15	20:03	21:09

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	14:06	13:04	□	07:56	10:33	□
N 70°	14:31	14:02	□	07:32	09:37	□
68°	14:50	14:36	14:10	07:15	09:03	11:10
66°	15:06	15:01	14:55	07:00	08:40	10:26
64°	15:19	15:20	15:25	06:49	08:21	09:56
62°	15:30	15:36	15:48	06:39	08:06	09:34
60°	15:39	15:50	16:06	06:30	07:53	09:17
N 58°	15:47	16:02	16:22	06:23	07:42	09:02
56°	15:54	16:12	16:35	06:17	07:33	08:49
54°	16:01	16:21	16:46	06:11	07:24	08:38
52°	16:07	16:29	16:56	06:06	07:17	08:28
50°	16:12	16:36	17:05	06:01	07:10	08:20
45°	16:24	16:51	17:25	05:51	06:56	08:01
N 40°	16:33	17:04	17:40	05:43	06:44	07:47
35°	16:41	17:15	17:54	05:35	06:34	07:34
30°	16:49	17:25	18:05	05:29	06:25	07:23
20°	17:01	17:41	18:25	05:18	06:10	07:04
N 10°	17:13	17:56	18:43	05:09	05:57	06:48
0°	17:23	18:09	18:59	05:00	05:45	06:33
S 10°	17:34	18:23	19:15	04:51	05:33	06:18
20°	17:45	18:38	19:33	04:42	05:20	06:02
30°	17:58	18:55	19:53	04:31	05:05	05:43
35°	18:06	19:05	20:05	04:25	04:57	05:32
40°	18:14	19:16	20:19	04:18	04:47	05:20
45°	18:24	19:30	20:36	04:10	04:36	05:06
S 50°	18:37	19:47	20:56	04:01	04:22	04:48
52°	18:43	19:54	21:06	03:56	04:16	04:40
54°	18:49	20:03	21:17	03:51	04:09	04:31
56°	18:56	20:13	21:29	03:46	04:01	04:20
58°	19:04	20:24	21:43	03:40	03:52	04:09
S 60°	19:13	20:37	22:00	03:33	03:42	03:55

Day	Sun			Moon		
	Eqn.of Time		Mer. Pass hh:mm	Mer.Pass.		Age 15-17 99-99%
	00 ^h	12 ^h		Upper	Lower	
	mm:ss	mm:ss	hh:mm	hh:mm	hh:mm	
22	15:26	15:30	11:44	23:34	11:11	
23	15:35	15:39	11:44	—:—	11:57	
24	15:43	15:47	11:44	00:21	12:46	

Aries Venus Mars

<https://pypi.org/project/sfalmanac/>

DUT1 = UT1-UTC = -0.0855 sec $\Delta T = TT-UT1 = +66.2695$ sec

2010 October 25 to Oct. 27 UT

h Sun Moon						
Mon	GHA	Dec	GHA	ν	Dec	d HP
0	183°57.7	S11°59.1	342°47.9	9.8'	N21°55.0	5.9' 55.8'
1	198°57.8	11°59.9	357°16.7	9.8'	22°00.9	5.8' 55.9'
2	213°57.9	12°00.8	11°45.5	9.7'	22°06.7	5.7' 55.9'
3	228°57.9	· · 01.6	26°14.2	9.6'	22°12.4	5.6' 55.9'
4	243°58.0	02.5	40°42.9	9.6'	22°18.0	5.5' 55.9'
5	258°58.1	03.4	55°11.4	9.5'	22°23.5	5.4' 55.9'
6	273°58.2	S12°04.2	69°39.9	9.4'	N22°28.9	5.2' 56.0'
7	288°58.2	05.1	84°08.4	9.4'	22°34.1	5.1' 56.0'
8	303°58.3	06.0	98°36.8	9.3'	22°39.2	5.0' 56.0'
9	318°58.4	· · 06.8	113°05.1	9.3'	22°44.2	4.9' 56.0'
10	333°58.5	07.7	127°33.3	9.2'	22°49.1	4.8' 56.0'
11	348°58.5	08.5	142°01.5	9.1'	22°53.9	4.7' 56.1'
12	3°58.6	S12°09.4	156°29.7	9.1'	N22°58.6	4.5' 56.1'
13	18°58.7	10.3	170°57.7	9.0'	23°03.1	4.4' 56.1'
14	33°58.8	11.1	185°25.7	8.9'	23°07.5	4.3' 56.1'
15	48°58.8	· · 12.0	199°53.7	8.9'	23°11.8	4.2' 56.1'
16	63°58.9	12.8	214°21.6	8.8'	23°16.0	4.0' 56.2'
17	78°59.0	13.7	228°49.4	8.8'	23°20.0	3.9' 56.2'
18	93°59.0	S12°14.6	243°17.2	8.7'	N23°23.9	3.8' 56.2'
19	108°59.1	15.4	257°44.9	8.7'	23°27.7	3.7' 56.2'
20	123°59.2	16.3	272°12.6	8.6'	23°31.4	3.5' 56.2'
21	138°59.3	· · 17.1	286°40.2	8.6'	23°34.9	3.4' 56.3'
22	153°59.3	18.0	301°07.7	8.5'	23°38.3	3.3' 56.3'
23	168°59.4	18.8	315°35.2	8.5'	23°41.6	3.1' 56.3'
SD = 16.1' $d = 0.9'$ SD = 15.2'						
Tue	GHA	Dec	GHA	ν	Dec	d HP
0	183°59.5	S12°19.7	330°02.7	8.4'	N23°44.7	3.0' 56.3'
1	198°59.5	20.6	344°30.1	8.4'	23°47.7	2.9' 56.3'
2	213°59.6	21.4	358°57.4	8.3'	23°50.6	2.8' 56.4'
3	228°59.7	· · 22.3	13°24.7	8.3'	23°53.4	2.6' 56.4'
4	243°59.7	23.1	27°52.0	8.2'	23°56.0	2.5' 56.4'
5	258°59.8	24.0	42°19.2	8.2'	23°58.5	2.4' 56.4'
6	273°59.9	S12°24.8	56°46.4	8.1'	N24°00.9	2.2' 56.4'
7	288°59.9	25.7	71°13.5	8.1'	24°03.1	2.1' 56.5'
8	304°00.0	26.5	85°40.5	8.0'	24°05.2	2.0' 56.5'
9	319°00.1	· · 27.4	100°07.6	8.0'	24°07.1	1.8' 56.5'
10	334°00.1	28.2	114°34.6	8.0'	24°09.0	1.7' 56.5'
11	349°00.2	29.1	129°01.5	7.9'	24°10.7	1.5' 56.5'
12	4°00.3	S12°29.9	143°28.4	7.9'	N24°12.2	1.4' 56.6'
13	19°00.3	30.8	157°55.3	7.8'	24°13.6	1.3' 56.6'
14	34°00.4	31.6	172°22.2	7.8'	24°14.9	1.1' 56.6'
15	49°00.5	· · 32.5	186°49.0	7.8'	24°16.0	1.0' 56.6'
16	64°00.5	33.3	201°15.7	7.7'	24°17.0	0.9' 56.7'
17	79°00.6	34.2	215°42.5	7.7'	24°17.9	0.7' 56.7'
18	94°00.6	S12°35.0	230°09.2	7.7'	N24°18.6	0.6' 56.7'
19	109°00.7	35.9	244°35.9	7.6'	24°19.2	0.4' 56.7'
20	124°00.8	36.7	259°02.5	7.6'	24°19.6	0.3' 56.7'
21	139°00.8	· · 37.6	273°29.1	7.6'	24°19.9	0.2' 56.8'
22	154°00.9	38.4	287°55.7	7.6'	24°20.1	0.0' 56.8'
23	169°01.0	39.3	302°22.3	7.5'	24°20.1	-0.1' 56.8'
SD = 16.1' $d = 0.9'$ SD = 15.4'						
Wed	GHA	Dec	GHA	ν	Dec	d HP
0	184°01.0	S12°40.1	316°48.9	7.5'	N24°20.0	-0.3' 56.8'
1	199°01.1	41.0	331°15.4	7.5'	24°19.7	-0.4' 56.8'
2	214°01.1	41.8	345°41.9	7.5'	24°19.3	-0.5' 56.9'
3	229°01.2	· · 42.7	0°08.4	7.5'	24°18.8	-0.7' 56.9'
4	244°01.3	43.5	14°34.8	7.4'	24°18.1	-0.8' 56.9'
5	259°01.3	44.4	29°01.3	7.4'	24°17.3	-1.0' 56.9'
6	274°01.4	S12°45.2	43°27.7	7.4'	N24°16.3	-1.1' 57.0'
7	289°01.4	46.1	57°54.1	7.4'	24°15.2	-1.3' 57.0'
8	304°01.5	46.9	72°20.5	7.4'	24°13.9	-1.4' 57.0'
9	319°01.6	· · 47.8	86°46.9	7.4'	24°12.5	-1.5' 57.0'
10	334°01.6	48.6	101°13.3	7.4'	24°11.0	-1.7' 57.0'
11	349°01.7	49.5	115°39.7	7.4'	24°09.3	-1.8' 57.1'
12	4°01.7	S12°50.3	130°06.0	7.4'	N24°07.5	-2.0' 57.1'
13	19°01.8	51.1	144°32.4	7.3'	24°05.5	-2.1' 57.1'
14	34°01.8	52.0	158°58.7	7.3'	24°03.4	-2.3' 57.1'
15	49°01.9	· · 52.8	173°25.1	7.3'	24°01.1	-2.4' 57.1'
16	64°02.0	53.7	187°51.4	7.3'	23°58.7	-2.5' 57.2'
17	79°02.0	54.5	202°17.8	7.3'	23°56.2	-2.7' 57.2'
18	94°02.1	S12°55.4	216°44.1	7.3'	N23°53.5	-2.8' 57.2'
19	109°02.1	56.2	231°10.4	7.3'	23°50.6	-3.0' 57.2'
20	124°02.2	57.0	245°36.8	7.3'	23°47.7	-3.1' 57.3'
21	139°02.2	· · 57.9	260°03.1	7.3'	23°44.5	-3.3' 57.3'
22	154°02.3	58.7	274°29.5	7.4'	23°41.3	-3.4' 57.3'
23	169°02.3	59.6	288°55.8	7.4'	23°37.9	-3.5' 57.3'
SD = 16.1' $d = 0.8'$ SD = 15.5'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	05:41	07:02	08:21	15:06	16:25	17:45
N 70°	05:40	06:52	08:01	15:26	16:34	17:46
68°	05:39	06:44	07:46	15:41	16:42	17:48
66°	05:37	06:38	07:33	15:53	16:49	17:49
64°	05:36	06:32	07:23	16:04	16:55	17:50
62°	05:35	06:27	07:14	16:13	17:00	17:52
60°	05:34	06:23	07:06	16:20	17:04	17:53
N 58°	05:33	06:19	07:00	16:27	17:08	17:54
56°	05:32	06:15	06:54	16:33	17:12	17:55
54°	05:30	06:12	06:49	16:39	17:15	17:57
52°	05:29	06:09	06:44	16:43	17:18	17:58
50°	05:28	06:06	06:39	16:48	17:21	17:59
45°	05:25	06:00	06:30	16:57	17:28	18:02
N 40°	05:23	05:54	06:22	17:05	17:33	18:05
35°	05:20	05:49	06:15	17:12	17:38	18:08
30°	05:17	05:45	06:09	17:18	17:43	18:11
20°	05:10	05:36	05:59	17:29	17:51	18:17
N 10°	05:03	05:28	05:49	17:38	18:00	18:25
0°	04:55	05:19	05:41	17:47	18:09	18:33
S 10°	04:45	05:10	05:32	17:56	18:18	18:43
20°	04:33	04:59	05:22	18:06	18:29	18:56
30°	04:17	04:46	05:11	18:18	18:42	19:12
35°	04:06	04:38	05:04	18:24	18:51	19:22
40°	03:54	04:28	04:57	18:32	19:00	19:35
45°	03:39	04:17	04:48	18:41	19:12	19:51
S 50°	03:19	04:03	04:38	18:51	19:27	20:11
52°	03:09	03:56	04:33	18:56	19:34	20:21
54°	02:58	03:48	04:28	19:02	19:41	20:32
56°	02:44	03:40	04:22	19:08	19:50	20:46
58°	02:29	03:30	04:15	19:15	20:00	21:02
S 60°	02:10	03:18	04:07	19:23	20:12	21:22

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	□	□	□	□	□	□
N 70°	□	□	□	□	□	□
68°	□	□	□	□	□	□
66°	14:48	□	15:55	12:21	□	15:03
64°	15:37	16:08	17:12	11:32	12:54	13:46
62°	16:09	16:46	17:49	11:01	12:16	13:09
60°	16:32	17:14	18:15	10:37	11:48	12:43
N 58°	16:51	17:35	18:36	10:19	11:27	12:22
56°	17:07	17:52	18:53	10:03	11:10	12:05
54°	17:21	18:07	19:07	09:50	10:55	11:50
52°	17:33	18:20	19:19	09:38	10:42	11:38
50°	17:43	18:31	19:30	09:28	10:31	11:26
45°	18:05	18:55	19:53	09:06	10:08	11:03
N 40°	18:23	19:14	20:12	08:49	09:49	10:44
35°	18:38	19:30	20:27	08:34	09:33	10:29
30°	18:51	19:43	20:41	08:22	09:20	10:15
20°	19:14	20:07	21:04	08:00	08:56	09:52
N 10°	19:33	20:27	21:23	07:41	08:36	09:32
0°	19:51	20:46	21:42	07:24	08:17	09:13
S 10°	20:10	21:05	22:00	07:06	07:59	08:54
20°	20:29	21:25	22:20	06:48	07:39	08:34
30°	20:52	21:49	22:43	06:26	07:15	08:10
35°	21:05	22:03	22:56	06:14	07:02	07:56
40°	21:21	22:19	23:12	05:59	06:46	07:40
45°	21:39	22:38	23:30	05:42	06:27	07:21
S 50°	22:02	23:02	23:53	05:21	06:04	06:57
52°	22:13	23:14	· · ·	05:11	05:53	06:46
54°	22:26	23:27	· · ·	05:00	05:40	06:33
56°	22:40	23:42	· · ·	04:47	05:25	06:18
58°	22:57	· · ·	00:00	04:33	05:08	06:00
S 60°	23:18	· · ·	00:21	04:15	04:47	05:38

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Upper hh:mm	Lower hh:mm	
25	15:51	15:54	11:44	01:11	13:37	
26	15:58	16:01	11:44	02:04	14:32	
27	16:04	16:07	11:44	02:59	15:27	

October 28, 29, 30 UT (Thu., Fri., Sat.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	36°14.2	184°33.8	S19°22.9	158°45.0	S20°37.7	40°51.8	S03°40.6	205°15.1	S02°20.3	Alpheratz	357°45.2	29°09.3											
1	51°16.6	199°37.6	22.0	173°45.6	38.1	55°54.4	40.7	220°17.3	20.4	Ankaa	353°17.1	-42°14.8											
2	66°19.1	214°41.4	21.0	188°46.2	38.5	70°57.1	40.7	235°19.5	20.5	Schedar	349°42.3	56°36.1											
3	81°21.5	229°45.2	· · 20.1	203°46.8	· · 38.9	85°59.7	· · 40.8	250°21.7	· · 20.6	Diphda	348°57.5	-17°55.5											
4	96°24.0	244°49.1	19.2	218°47.4	39.3	101°02.3	40.8	265°23.9	20.8	Achernar	335°27.5	-57°10.8											
5	111°26.5	259°52.9	18.2	233°48.0	39.7	116°05.0	40.9	280°26.1	20.9	Hamal	328°02.6	23°31.0											
6	126°28.9	274°56.7	S19°17.3	248°48.6	S20°40.1	131°07.6	S03°41.0	295°28.3	S02°21.0	Polaris	318°16.7	89°18.7											
7	141°31.4	290°00.5	16.4	263°49.2	40.5	146°10.2	41.0	310°30.5	21.1	Acamar	315°19.3	-40°15.5											
8	156°33.9	305°04.3	15.4	278°49.8	40.9	161°12.9	41.1	325°32.7	21.2	Menkar	314°16.7	4°08.1											
9	171°36.3	320°08.1	· · 14.5	293°50.4	· · 41.3	176°15.5	· · 41.2	340°34.9	· · 21.3	Mirfak	308°42.7	49°54.0											
10	186°38.8	335°11.9	13.6	308°51.0	41.7	191°18.1	41.2	355°37.1	21.4	Aldebaran	290°51.3	16°31.9											
11	201°41.3	350°15.7	12.6	323°51.6	42.1	206°20.7	41.3	10°39.3	21.5	Rigel	281°13.6	-8°11.2											
12	216°43.7	5°19.6	S19°11.7	338°52.2	S20°42.5	221°23.4	S03°41.3	25°41.5	S02°21.6	Capella	280°36.9	46°00.4											
13	231°46.2	20°23.4	10.8	353°52.8	42.9	236°26.0	41.4	40°43.7	21.7	Bellatrix	278°33.8	6°21.6											
14	246°48.6	35°27.2	09.8	8°53.3	43.3	251°28.6	41.5	55°45.9	21.9	Elnath	278°14.8	28°37.0											
15	261°51.1	50°31.0	· · 08.9	23°53.9	· · 43.7	266°31.3	· · 41.5	70°48.0	· · 22.0	Alnilam	275°48.1	-1°11.6											
16	276°53.6	65°34.8	07.9	38°54.5	44.1	281°33.9	41.6	85°50.2	22.1	Betelgeuse	271°03.2	7°24.6											
17	291°56.0	80°38.6	07.0	53°55.1	44.5	296°36.5	41.7	100°52.4	22.2	Canopus	263°56.7	-52°41.8											
18	306°58.5	95°42.4	S19°06.0	68°55.7	S20°44.9	311°39.1	S03°41.7	115°54.6	S02°22.3	Sirius	258°35.2	-16°43.7											
19	322°01.0	110°46.2	05.1	83°56.3	45.3	326°41.8	41.8	130°56.8	22.4	Adhara	255°13.9	-28°59.0											
20	337°03.4	125°50.0	04.1	98°56.9	45.7	341°44.4	41.8	145°59.0	22.5	Procyon	245°01.6	5°11.8											
21	352°05.9	140°53.8	· · 03.2	113°57.5	· · 46.1	356°47.0	· · 41.9	161°01.2	· · 22.6	Pollux	243°30.0	27°59.9											
22	7°08.4	155°57.6	02.2	128°58.1	46.5	11°49.7	42.0	176°03.4	22.7	Avior	234°18.9	-59°32.4											
23	22°10.8	171°01.4	01.3	143°58.7	46.9	26°52.3	42.0	191°05.6	22.8	Suhail	222°54.0	-43°28.4											
Mer.pass. 21:32				ν 3.8' d -0.9' m -4.20				ν 0.6' d 0.4' m 1.43				ν 2.6' d 0.1' m -2.80				ν 2.2' d 0.1' m 0.92							

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	37°13.3	186°05.3	S19°00.3	158°59.3	S20°47.3	41°54.9	S03°42.1	206°07.8	S02°22.9										
1	52°15.8	201°09.1	18°59.4	173°59.9	47.7	56°57.5	42.1	221°10.0	23.1										
2	67°18.2	216°12.9	58.4	189°00.5	48.1	72°00.2	42.2	236°12.2	23.2										
3	82°20.7	231°16.7	· · 57.5	204°01.1	· · 48.5	87°02.8	· · 42.3	251°14.4	· · 23.3										
4	97°23.1	246°20.5	56.5	219°01.7	48.9	102°05.4	42.3	266°16.6	23.4										
5	112°25.6	261°24.3	55.5	234°02.3	49.3	117°08.0	42.4	281°18.8	23.5										
6	127°28.1	276°28.1	S18°54.6	249°02.9	S20°49.7	132°10.7	S03°42.4	296°21.0	S02°23.6										
7	142°30.5	291°31.9	53.6	264°03.5	50.1	147°13.3	42.5	311°23.2	23.7										
8	157°33.0	306°35.7	52.6	279°04.1	50.5	162°15.9	42.6	326°25.4	23.8										
9	172°35.5	321°39.5	· · 51.7	294°04.6	· · 50.9	177°18.5	· · 42.6	341°27.6	· · 23.9										
10	187°37.9	336°43.3	50.7	309°05.2	51.3	192°21.2	42.7	356°29.8	24.0										
11	202°40.4	351°47.1	49.7	324°05.8	51.7	207°23.8	42.7	11°31.9	24.1										
12	217°42.9	6°50.9	S18°48.8	339°06.4	S20°52.1	222°26.4	S03°42.8	26°34.1	S02°24.3										
13	232°45.3	21°54.7	47.8	354°07.0	52.4	237°29.0	42.9	41°36.3	24.4										
14	247°47.8	36°58.5	46.8	9°07.6	52.8	252°31.7	42.9	56°38.5	24.5										
15	262°50.3	52°02.3	· · 45.9	24°08.2	· · 53.2	267°34.3	· · 43.0	71°40.7	· · 24.6										
16	277°52.7	67°06.1	44.9	39°08.8	53.6	282°36.9	43.0	86°42.9	24.7										
17	292°55.2	82°09.9	43.9	54°09.4	54.0	297°39.5	43.1	101°45.1	24.8										
18	307°57.6	97°13.6	S18°42.9	69°10.0	S20°54.4	312°42.1	S03°43.1	116°47.3	S02°24.9										
19	323°00.1	112°17.4	41.9	84°10.6	54.8	327°44.8	43.2	131°49.5	25.0										
20	338°02.6	127°21.2	41.0	99°11.1	55.2	342°47.4	43.3	146°51.7	25.1										
21	353°05.0	142°25.0	· · 40.0	114°11.7	· · 55.6	357°50.0	· · 43.3	161°53.9	· · 25.2										
22	8°07.5	157°28.8	39.0	129°12.3	56.0	12°52.6	43.4	176°56.1	25.3										
23	23°10.0	172°32.6	38.0	144°12.9	56.4	27°55.2	43.4	191°58.3	25.4										
Mer.pass. 21:28				ν 3.8' d -1.0' m -4.17				ν 0.6' d 0.4' m 1.42				ν 2.6' d 0.1' m -2.79				ν 2.2' d 0.1' m 0.92			

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	38°12.4	187°36.4	S18°37.0	159°13.5	S20°56.8	42°57.9	S03°43.5	207°00.5	S02°25.6										
1	53°14.9	202°40.2	36.1	174°14.1	57.1	58°00.5	43.5	222°02.7	25.7										
2	68°17.4	217°44.0	35.1	189°14.7	57.5	73°03.1	43.6	237°04.9	25.8										
3	83°19.8	232°47.7	· · 34.1	204°15.3	· · 57.9	88°05.7	· · 43.7	252°07.1	· · 25.9										
4	98°22.3	247°51.5	33.1	219°15.8	58.3	103°08.3	43.7	267°09.3	26.0										
5	113°24.8	262°55.3	32.1	234°16.4	58.7	118°11.0	43.8	282°11.5	26.1										
6	128°27.2	277°59.1	S18°31.1	249°17.0	S20°59.1	133°13.6	S03°43.8	297°13.7	S02°26.2										
7	143°29.7	293°02.9	30.1	264°17.6	59.5	148°16.2	43.9	312°15.9	26.3										
8	158°32.1	308°06.6	29.1	279°18.2	20°59.9	163°18.8	43.9	327°18.1	26.4										
9	173°34.6	323°10.4	· · 28.2	294°18.8	21°00.3	178°21.4	· · 44.0	342°20.3	· · 26.5										
10	188°37.1	338°14.2	27.2	309°19.4	00.6	193°24.0	44.1	357°22.5	26.6										
11	203°39.5	353°18.0	26.2	324°19.9	01.0	208°26.7	44.1	12°24.6	26.7										
12	218°42.0	8°21.8	S18°25.2	339°20.5	S21°01.4	223°29.3	S03°44.2	27°26.8	S02°26.9										
13	233°44.5	23°25.5	24.2	354°21.1	01.8	238°31.9	44.2	42°29.0	27.0										
14	248°46.9	38°29.3	23.2	9°21.7	02.2	253°34.5	44.3	57°31.2	27.1										
15	263°49.4	53°33.1	· · 22.2	24°22.3	· · 02.6	268°37.1	· · 44.3	72°33.4	· · 27.2										
16	278°51.9	68°36.8	21.2	39°22.9	03.0	283°39.7	44.4	87°35.6	27.3										
17	293°54.3	83°40.6	20.2	54°23.4	03.3	298°42.3	44.4	102°37.8	27.4										
18	308°56.8	98°44.4	S18°19.2	69°24.0	S21°03.7	313°45.0	S03°44.5	117°40.0	S02°27.5										
19	323°59.2	113°48.1	18.2	84°24.6	04.1	328°47.6	44.6	132°42.2	27.6										
20	339°01.7	128°51.9	17.2	99°25.2	04.5	343°50.2	44.6	147°44.4	27.7										
21	354°04.2	143°55.7	· · 16.2	114°25.8	· · 04.9	358°52.8	· · 44.7	162°46.6	· · 27.8										
22	9°06.6	158°59.4	15.2	129°26.4	05.3	13°55.4	44.7	177°48.8	27.9										
23	24°09.1	174°03.2	14.2	144°26.9	05.7	28°58.0	44.8	192°51.0	28.0										
Mer.pass. 21:24				ν 3.8' d -1.0' m -4.18				ν 0.6' d 0.4' m 1.42				ν 2.6' d 0.1' m -2.78				ν 2.2' d 0.1' m 0.92			

Oct 28 Thu	SHA	Mer.pass
Venus	148°19.7	11:39
Mars	122°30.8	13:24
Jupiter	4°37.6	21:13
Saturn	169°01.0	10:17

Oct 29 Fri	SHA	Mer.pass
Venus	148°52.0	11:33
Mars	121°46.0	13:24
Jupiter	4°41.6	21:09
Saturn	168°54.5	10:14

Oct 30 Sat	SHA	Mer.pass
Venus	149°24.0	11:27
Mars	121°01.1	13:23
Jupiter	4°45.4	21:04
Saturn	168°48.1	10:10

Horizontal parallax		
Venus:	0.5	
Mars:	0.1	

h	Sun			Moon			
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	184°02.4	S13°00.4	303°22.2	7.4'	N23°34.4	-3.7'	57.3'
1	199°02.4	01.2	317°48.6	7.4'	23°30.7	-3.8'	57.4'
2	214°02.5	02.1	332°14.9	7.4'	23°26.8	-4.0'	57.4'
3	229°02.6	· · 02.9	346°41.3	7.4'	23°22.9	-4.1'	57.4'
4	244°02.6	03.8	1°07.7	7.4'	23°18.8	-4.2'	57.4'
5	259°02.7	04.6	15°34.1	7.4'	23°14.5	-4.4'	57.5'
6	274°02.7	S13°05.4	30°00.5	7.4'	N23°10.1	-4.5'	57.5'
7	289°02.8	06.3	44°27.0	7.5'	23°05.6	-4.7'	57.5'
8	304°02.8	07.1	58°53.4	7.5'	23°00.9	-4.8'	57.5'
9	319°02.9	· · 07.9	73°19.9	7.5'	22°56.1	-4.9'	57.6'
10	334°02.9	08.8	87°46.4	7.5'	22°51.2	-5.1'	57.6'
11	349°03.0	09.6	102°12.9	7.5'	22°46.1	-5.2'	57.6'
12	4°03.0	S13°10.4	116°39.4	7.5'	N22°40.9	-5.4'	57.6'
13	19°03.1	11.3	131°06.0	7.6'	22°35.5	-5.5'	57.6'
14	34°03.1	12.1	145°32.5	7.6'	22°30.0	-5.6'	57.7'
15	49°03.2	· · 13.0	159°59.1	7.6'	22°24.4	-5.8'	57.7'
16	64°03.2	13.8	174°25.7	7.6'	22°18.6	-5.9'	57.7'
17	79°03.2	14.6	188°52.3	7.7'	22°12.7	-6.0'	57.7'
18	94°03.3	S13°15.5	203°19.0	7.7'	N22°06.7	-6.2'	57.8'
19	109°03.3	16.3	217°45.7	7.7'	22°00.5	-6.3'	57.8'
20	124°03.4	17.1	232°12.4	7.7'	21°54.2	-6.4'	57.8'
21	139°03.4	· · 18.0	246°39.1	7.8'	21°47.7	-6.6'	57.8'
22	154°03.5	18.8	261°05.9	7.8'	21°41.2	-6.7'	57.8'
23	169°03.5	19.6	275°32.7	7.8'	21°34.4	-6.8'	57.9'
SD = 16.1'			SD = 15.6'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	184°03.6	S13°20.5	289°59.5	7.9'	N21°27.6	-7.0'	57.9'
1	199°03.6	21.3	304°26.4	7.9'	21°20.6	-7.1'	57.9'
2	214°03.7	22.1	318°53.2	7.9'	21°13.5	-7.2'	57.9'
3	229°03.7	· · 22.9	333°20.1	7.9'	21°06.3	-7.4'	58.0'
4	244°03.8	23.8	347°47.1	8.0'	20°58.9	-7.5'	58.0'
5	259°03.8	24.6	2°14.1	8.0'	20°51.5	-7.6'	58.0'
6	274°03.8	S13°25.4	16°41.1	8.0'	N20°43.8	-7.7'	58.0'
7	289°03.9	26.3	31°08.1	8.1'	20°36.1	-7.9'	58.1'
8	304°03.9	27.1	45°35.2	8.1'	20°28.2	-8.0'	58.1'
9	319°04.0	· · 27.9	60°02.3	8.1'	20°20.2	-8.1'	58.1'
10	334°04.0	28.7	74°29.5	8.2'	20°12.1	-8.2'	58.1'
11	349°04.1	29.6	88°56.6	8.2'	20°03.9	-8.4'	58.1'
12	4°04.1	S13°30.4	103°23.9	8.3'	N19°55.5	-8.5'	58.2'
13	19°04.1	31.2	117°51.1	8.3'	19°47.0	-8.6'	58.2'
14	34°04.2	32.1	132°18.4	8.3'	19°38.4	-8.7'	58.2'
15	49°04.2	· · 32.9	146°45.7	8.4'	19°29.7	-8.8'	58.2'
16	64°04.3	33.7	161°13.1	8.4'	19°20.8	-9.0'	58.3'
17	79°04.3	34.5	175°40.5	8.4'	19°11.9	-9.1'	58.3'
18	94°04.3	S13°35.4	190°07.9	8.5'	N19°02.8	-9.2'	58.3'
19	109°04.4	36.2	204°35.4	8.5'	18°53.6	-9.3'	58.3'
20	124°04.4	37.0	219°02.9	8.6'	18°44.3	-9.4'	58.4'
21	139°04.4	· · 37.8	233°30.5	8.6'	18°34.8	-9.5'	58.4'
22	154°04.5	38.6	247°58.1	8.6'	18°25.3	-9.7'	58.4'
23	169°04.5	39.5	262°25.7	8.7'	18°15.6	-9.8'	58.4'
SD = 16.1'			SD = 15.8'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	184°04.6	S13°40.3	276°53.4	8.7'	N18°05.9	-9.9'	58.4'
1	199°04.6	41.1	291°21.1	8.7'	17°56.0	-10.0'	58.5'
2	214°04.6	41.9	305°48.8	8.8'	17°46.0	-10.1'	58.5'
3	229°04.7	· · 42.8	320°16.6	8.8'	17°35.9	-10.2'	58.5'
4	244°04.7	43.6	334°44.4	8.9'	17°25.7	-10.3'	58.5'
5	259°04.7	44.4	349°12.3	8.9'	17°15.3	-10.4'	58.6'
6	274°04.8	S13°45.2	3°40.2	8.9'	N17°04.9	-10.5'	58.6'
7	289°04.8	46.0	18°08.1	9.0'	16°54.4	-10.6'	58.6'
8	304°04.8	46.9	32°36.1	9.0'	16°43.7	-10.7'	58.6'
9	319°04.9	· · 47.7	47°04.1	9.1'	16°33.0	-10.8'	58.6'
10	334°04.9	48.5	61°32.2	9.1'	16°22.2	-10.9'	58.7'
11	349°04.9	49.3	76°00.3	9.1'	16°11.2	-11.0'	58.7'
12	4°05.0	S13°50.1	90°28.4	9.2'	N16°00.2	-11.1'	58.7'
13	19°05.0	51.0	104°56.6	9.2'	15°49.0	-11.2'	58.7'
14	34°05.0	51.8	119°24.8	9.2'	15°37.8	-11.3'	58.8'
15	49°05.1	· · 52.6	133°53.0	9.3'	15°26.5	-11.4'	58.8'
16	64°05.1	53.4	148°21.3	9.3'	15°15.0	-11.5'	58.8'
17	79°05.1	54.2	162°49.6	9.4'	15°03.5	-11.6'	58.8'
18	94°05.2	S13°55.0	177°18.0	9.4'	N14°51.9	-11.7'	58.8'
19	109°05.2	55.9	191°46.3	9.4'	14°40.2	-11.8'	58.9'
20	124°05.2	56.7	206°14.8	9.5'	14°28.3	-11.9'	58.9'
21	139°05.3	· · 57.5	220°43.2	9.5'	14°16.4	-12.0'	58.9'
22	154°05.3	58.3	235°11.7	9.5'	14°04.5	-12.1'	58.9'
23	169°05.3	59.1	249°40.2	9.6'	13°52.4	-12.2'	59.0'
SD = 16.1'			SD = 15.9'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	05:54	07:15	08:38	14:48	16:11	17:32
N 70°	05:51	07:04	08:15	15:11	16:22	17:35
68°	05:48	06:55	07:58	15:28	16:31	17:37
66°	05:46	06:47	07:44	15:42	16:39	17:40
64°	05:44	06:40	07:32	15:54	16:46	17:42
62°	05:42	06:35	07:23	16:04	16:52	17:44
60°	05:40	06:30	07:14	16:12	16:57	17:46
N 58°	05:39	06:25	07:07	16:20	17:01	17:48
56°	05:37	06:21	07:00	16:26	17:06	17:49
54°	05:36	06:17	06:54	16:32	17:09	17:51
52°	05:34	06:14	06:49	16:38	17:13	17:52
50°	05:33	06:11	06:44	16:42	17:16	17:54
45°	05:29	06:04	06:34	16:53	17:23	17:58
N 40°	05:26	05:58	06:25	17:02	17:29	18:01
35°	05:22	05:52	06:18	17:09	17:35	18:05
30°	05:19	05:47	06:11	17:16	17:40	18:08
20°	05:11	05:37	06:00	17:27	17:50	18:16
N 10°	05:03	05:28	05:50	17:37	17:59	18:24
0°	04:54	05:19	05:40	17:47	18:08	18:33
S 10°	04:44	05:09	05:31	17:57	18:19	18:44
20°	04:31	04:57	05:20	18:08	18:30	18:57
30°	04:13	04:43	05:08	18:20	18:45	19:15
35°	04:02	04:34	05:01	18:27	18:54	19:26
40°	03:49	04:24	04:53	18:35	19:04	19:39
45°	03:33	04:12	04:44	18:45	19:17	19:56
S 50°	03:12	03:57	04:32	18:56	19:32	20:17
52°	03:01	03:49	04:27	19:02	19:40	20:28
54°	02:49	03:41	04:21	19:08	19:48	20:41
56°	02:35	03:32	04:14	19:14	19:58	20:55
58°	02:17	03:21	04:07	19:22	20:09	21:13
S 60°	01:56	03:08	03:59	19:30	20:21	21:36

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	□	□	20:53	□	□	15:52
N 70°	□	18:44	21:24	□	16:07	15:19
68°	□	19:36	21:47	□	15:14	14:55
66°	18:07	20:08	22:04	14:48	14:41	14:36
64°	18:47	20:32	22:18	14:08	14:17	14:21
62°	19:14	20:50	22:30	13:40	13:58	14:08
60°	19:35	21:05	22:40	13:19	13:42	13:57
N 58°	19:52	21:18	22:49	13:01	13:28	13:47
56°	20:06	21:29	22:56	12:46	13:17	13:39
54°	20:19	21:39	23:03	12:34	13:06	13:31
52°	20:30	21:48	23:09	12:22	12:57	13:25
50°	20:40	21:55	23:15	12:12	12:49	13:19
45°	21:00	22:12	23:26	11:51	12:31	13:05
N 40°	21:17	22:25	23:36	11:34	12:17	12:54
35°	21:30	22:37	23:44	11:19	12:05	12:45
30°	21:42	22:46	23:51	11:07	11:54	12:37
20°	22:03	23:03	· ·	10:45	11:35	12:22
N 10°	22:21	23:18	· ·	10:26	11:19	12:09
0°	22:38	23:32	· ·	10:09	11:04	11:57
S 10°	22:54	23:46	· ·	09:51	10:48	11:45
20°	23:12	· ·	00:00	09:32	10:32	11:32
30°	23:32	· ·	00:17	09:10	10:13	11:17
35°	23:44	· ·	00:27	08:57	10:02	11:09
40°	23:58	· ·	00:38	08:42	09:49	10:59
45°	· ·	00:14	00:51	08:24	09:34	10:47
S 50°	· ·	00:34	01:06	08:02	09:15	10:33
52°	00:04	00:43	01:14	07:51	09:06	10:26
54°	00:16	00:53	01:22	07:39	08:56	10:18
56°	00:30	01:05	01:31	07:25	08:44	10:10
58°	00:46	01:19	01:41	07:09	08:31	10:00
S 60°	01:06	01:35	01:53	06:50	08:16	09:49

Day	Sun		Mer. Pass hh:mm	Moon		Age 21-23 76-56%
	Eqn.of Time 00 ^h mm:ss	Time 12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
28	16:10	16:12	11:44	03:55	16:23	
29	16:14	16:16	11:44	04:51	17:18	
30	16:18	16:20	11:44	05:45	18:11	

October 31, 01, 02 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	39°11.6	189°07.0	S18°13.2	159°27.5	S21°06.0	44°00.6	S03°44.8	207°53.2	S02°28.2	Alpheratz	357°45.2 29°09.3	
1	54°14.0	204°10.7	12.2	174°28.1	06.4	59°03.3	44.9	222°55.4	28.3	Ankaa	353°17.1 -42°14.8	
2	69°16.5	219°14.5	11.2	189°28.7	06.8	74°05.9	44.9	237°57.6	28.4	Schedar	349°42.3 56°36.1	
3	84°19.0	234°18.2	•• 10.2	204°29.3	•• 07.2	89°08.5	•• 45.0	252°59.8	•• 28.5	Diphda	348°57.5 -17°55.5	
4	99°21.4	249°22.0	09.1	219°29.8	07.6	104°11.1	45.0	268°02.0	28.6	Achernar	335°27.5 -57°10.8	
5	114°23.9	264°25.7	08.1	234°30.4	07.9	119°13.7	45.1	283°04.2	28.7	Hamal	328°02.6 23°31.0	
6	129°26.4	279°29.5	S18°07.1	249°31.0	S21°08.3	134°16.3	S03°45.1	298°06.4	S02°28.8	Polaris	318°16.4 89°18.7	
7	144°28.8	294°33.2	06.1	264°31.6	08.7	149°18.9	45.2	313°08.6	28.9	Acamar	315°19.3 -40°15.5	
8	159°31.3	309°37.0	05.1	279°32.2	09.1	164°21.5	45.3	328°10.8	29.0	Menkar	314°16.7 4°08.1	
9	174°33.7	324°40.7	•• 04.1	294°32.7	•• 09.5	179°24.1	•• 45.3	343°13.0	•• 29.1	Mirfak	308°42.7 49°54.0	
10	189°36.2	339°44.5	03.1	309°33.3	09.8	194°26.7	45.4	358°15.2	29.2	Aldebaran	290°51.3 16°31.9	
11	204°38.7	354°48.2	02.1	324°33.9	10.2	209°29.4	45.4	13°17.4	29.3	Rigel	281°13.6 -8°11.2	
12	219°41.1	9°52.0	S18°01.1	339°34.5	S21°10.6	224°32.0	S03°45.5	28°19.6	S02°29.4	Capella	280°36.9 46°00.4	
13	234°43.6	24°55.7	18°00.0	354°35.1	11.0	239°34.6	45.5	43°21.8	29.6	Bellatrix	278°33.8 6°21.6	
14	249°46.1	39°59.4	17°59.0	9°35.6	11.4	254°37.2	45.6	58°24.0	29.7	Elnath	278°14.8 28°37.0	
15	264°48.5	55°03.2	•• 58.0	24°36.2	•• 11.7	269°39.8	•• 45.6	73°26.2	•• 29.8	Alnilam	275°48.1 -1°11.6	
16	279°51.0	70°06.9	57.0	39°36.8	12.1	284°42.4	45.7	88°28.4	29.9	Betelgeuse	271°03.1 7°24.6	
17	294°53.5	85°10.7	56.0	54°37.4	12.5	299°45.0	45.7	103°30.6	30.0	Canopus	263°56.7 -52°41.8	
18	309°55.9	100°14.4	S17°55.0	69°37.9	S21°12.9	314°47.6	S03°45.8	118°32.8	S02°30.1	Sirius	258°35.2 -16°43.7	
19	324°58.4	115°18.1	53.9	84°38.5	13.2	329°50.2	45.8	133°35.0	30.2	Adhara	255°13.9 -28°59.0	
20	340°00.9	130°21.8	52.9	99°39.1	13.6	344°52.8	45.9	148°37.2	30.3	Procyon	245°01.6 5°11.8	
21	355°03.3	145°25.6	•• 51.9	114°39.7	•• 14.0	359°55.4	•• 45.9	163°39.4	•• 30.4	Pollux	243°29.9 27°59.9	
22	10°05.8	160°29.3	50.9	129°40.2	14.4	14°58.0	46.0	178°41.6	30.5	Avior	234°18.8 -59°32.4	
23	25°08.2	175°33.0	49.9	144°40.8	14.8	30°00.6	46.0	193°43.8	30.6	Suhail	222°54.0 -43°28.4	
Mer.pass. 21:20		ν 3.8' d -1.0' m -4.21		ν 0.6' d 0.4' m 1.42		ν 2.6' d 0.1' m -2.78		ν 2.2' d 0.1' m 0.92		Miaplacidus	221°40.4 -69°45.4	
											Alphard	217°58.0 -8°42.3
											Regulus	207°45.6 11°54.8
											Dubhe	193°54.2 61°41.3
											Denebola	182°35.8 14°30.6
											Gienah	175°54.6 -17°36.1
											Acrux	173°12.2 -63°09.4
											Gacrux	172°03.7 -57°10.3
											Alioth	166°22.7 55°53.9
											Spica	158°33.6 -11°13.0
											Alkaid	153°00.8 49°15.5
											Hadar	148°51.4 -60°25.5
											Menkent	148°10.3 -36°25.3
											Arcturus	145°57.8 19°07.6
											Rigel Kent.	139°55.1 -60°52.7
											Kochab	137°20.9 74°06.6
											Zuben'ubi	137°07.9 -16°05.2
											Alphecca	126°13.0 26°40.8
											Antares	112°28.9 -26°27.3
											Atria	107°33.0 -69°02.9
											Sabik	102°15.0 -15°44.2
											Shaula	96°24.8 -37°06.7
											Rasalhague	96°08.5 12°33.3
											Eltanin	90°47.4 51°29.5
											Kaus Aust.	83°46.6 -34°22.8
											Vega	80°40.5 38°47.9
											Nunki	76°00.8 -26°17.0
											Altair	62°10.2 8°54.0
											Peacock	53°22.2 -56°42.1
											Deneb	49°32.8 45°19.5
											Enif	33°48.9 9°55.7
											Al Na'ir	27°45.9 -46°54.6
											Fomalhaut	15°25.8 -29°33.8
											Scheat	13°55.1 28°08.8
											Markab	13°40.1 15°16.1
											Oct 31 Sun	SHA Mer.pass
											Venus	149°55.4 11:21
											Mars	120°16.0 13:22
											Jupiter	4°49.1 21:00
											Saturn	168°41.6 10:07
											Nov 01 Mon	SHA Mer.pass
											Venus	150°26.0 11:15
											Mars	119°30.7 13:21
											Jupiter	4°52.5 20:56
											Saturn	168°35.2 10:03
											Nov 02 Tue	SHA Mer.pass
											Venus	150°55.7 11:09
											Mars	118°45.2 13:20
											Jupiter	4°55.8 20:52
											Saturn	168°28.9 10:00
											Horizontal parallax	
											Venus:	0.5
											Mars:	0.1
Mer.pass. 21:12		ν 3.7' d -1.0' m -4.28		ν 0.6' d 0.4' m 1.41		ν 2.6' d 0.0' m -2.76		ν 2.2' d 0.1' m 0.92				

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	40°10.7	190°36.8	S17°48.8	159°41.4	S21°15.1	45°03.2	S03°46.1	208°46.0	S02°30.7		
1	55°13.2	205°40.5	47.8	174°42.0	15.5	60°05.9	46.1	223°48.2	30.8		
2	70°15.6	220°44.2	46.8	189°42.5	15.9	75°08.5	46.2	238°50.4	30.9		
3	85°18.1	235°47.9	•• 45.8	204°43.1	•• 16.3	90°11.1	•• 46.2	253°52.6	•• 31.1		
4	100°20.6	250°51.6	44.8	219°43.7	16.6	105°13.7	46.3	268°54.7	31.2		
5	115°23.0	265°55.3	43.7	234°44.3	17.0	120°16.3	46.3	283°56.9	31.3		
6	130°25.5	280°59.1	S17°42.7	249°44.8	S21°17.4	135°18.9	S03°46.4	298°59.1	S02°31.4		
7	145°28.0	296°02.8	41.7	264°45.4	17.7	150°21.5	46.4	314°01.3	31.5		
8	160°30.4	311°06.5	40.7	279°46.0	18.1	165°24.1	46.5	329°03.5	31.6		
9	175°32.9	326°10.2	•• 39.6	294°46.5	•• 18.5	180°26.7	•• 46.5	344°05.7	•• 31.7		
10	190°35.3	341°13.9	38.6	309°47.1	18.9	195°29.3	46.6	359°07.9	31.8		
11	205°37.8	356°17.6	37.6	324°47.7	19.2	210°31.9	46.6	14°10.1	31.9		
12	220°40.3	11°21.3	S17°36.6	339°48.3	S21°19.6	225°34.5	S03°46.7	29°12.3	S02°32.0		
13	235°42.7	26°25.0	35.4	354°48.8	20.0	240°37.1	46.7	44°14.5	32.1		
14	250°45.2	41°28.7	34.5	9°49.4	20.3	255°39.7	46.8	59°16.7	32.2		
15	265°47.7	56°32.4	•• 33.5	24°50.0	•• 20.7	270°42.3	•• 46.8	74°18.9	•• 32.3		
16	280°50.1	71°36.1	32.4	39°50.5	21.1	285°44.9	46.9	89°21.1	32.4		
17	295°52.6	86°39.8	31.4	54°51.1	21.5	300°47.5	46.9	104°23.3	32.5		
18	310°55.1	101°43.5	S17°30.4	69°51.7	S21°21.8	315°50.1	S03°47.0	119°25.5	S02°32.7		
19	325°57.5	116°47.1	29.4	84°52.2	22.2	330°52.7	47.0	134°27.7	32.8		
20	341°00.0	131°50.8	28.3	99°52.8	22.6	345°55.3	47.1	149°29.9	32.9		
21	356°02.5	146°54.5	•• 27.3	114°53.4	•• 22.9	0°57.9	•• 47.1	164°32.1	•• 33.0		
22	11°04.9	161°58.2	26.3	129°54.0	23.3	16°00.5	47.2	179°34.3	33.1		
23	26°07.4	177°01.9	25.2	144°54.5	23.7	31°03.1	47.2	194°36.5	33.2		
Mer.pass. 21:16		ν 3.7' d -1.0' m -4.25		ν 0.6' d 0.4' m 1.41		ν 2.6' d 0.1' m -2.77		ν 2.2' d 0.1' m 0.92			

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	41°09.8	192°05.6	S17°24.2	159°55.1	S21°24.0	46°05.7	S03°47.3	209°38.7	S02°33.3		
1	56°12.3	207°09.2	23.2	174°55.7	24.4	61°08.3	47.3	224°40.9	33.4		
2	71°14.8	222°12.9	22.1	189°56.2	24.8	76°10.9	47.4	239°43.1	33.5		
3	86°17.2	237°16.6	•• 21.1	204°56.8	•• 25.1	91°13.5	•• 47.4	254°45.3	•• 33.6		
4	101°19.7	252°20.2	20.1	219°57.4	25.5	106°16.1	47.4	269°47.5	33.7		
5	116°22.2	267°23.9	19.0	234°57.9	25.9	121°18.7	47.5	284°49.7	33.8		
6	131°24.6	282°27.6	S17°18.0	249°58.5	S21°26.2	136°21.3	S03°47.5	299°51.9	S02°33.9		
7	146°27.1	297°31.2	17.0	264°59.1	26.6	151°23.9	47.6	314°54.1	34.0		
8	161°29.6	312°34.9	16.0	279°59.6	27.0	166°26.5	47.6	329°56.3	34.1		
9	176°32.0	327°38.6	•• 14.9	295°00.2	•• 27.3	181°29.1	•• 47.7	344°58.5	•• 34.2		
10	191°34.5	342°42.2	13.9	310°00.8	27.7	196°31.7	47.7	0°00.7	34.4		
11	206°36.9	357°45.9	12.8	325°01.3	28.0	211°34.2	47.8	15°02.9	34.5		
12	221°39.4	12°49.5	S17°11.8	340°01.9	S21°28.4	226°36.8	S03°47.8	30°05.1	S02°34.6		
13	236°41.9	27°53.2	10.8	355°02.4	28.8	241°39.4	47.9	45°07.3	34.7		
14	251°44.3	42°56.8	09.7	10°03.0	29.1	256°42.0	47.9	60°09.5	34.8		
15	266°46.8	58°00.5	•• 08.7	25°03.6	•• 29.5	271°44.6	•• 48.0	75°11.7	•• 34.9		
16	281°49.3	73°04.1	07.7	40°04.1	29.9	286°47.2	48.0	90°13.9	35.0		
17	296°51.7	88°07.8	06.6	55°04.7	30.2	301°49.8	48.0	105°16.1	35.1		
18	311°54.2	103°11.4	S17°05.6	70°05.3	S21°30.6	316°52.4	S03°48.1	120°18.3	S02°35.2		
19	326°56.7	118°15.0	04.6	85°05.8	30.9	331°55.0	48.1	135°20.5	35.3		
20	341°59.1	133°18.7	03.5	100°06.4	31.3	346°57.6	48.2	150°22.7	35.4		
21	357°01.6	148°22.3	•• 02.5	115°06.9	•• 31.7	2°00.2	•• 48.2	165°24.9	•• 35.5		
22	12°04.1	163°25.9	01.5	130°07.5	32.0	17°02.8	48.3	180°27.1	35.6		
23	27°06.5	178°29.6	00.4	145°08.1	32.4	32°05.8	48.3	195°29.3	35.7		
Mer.pass. 21:12		ν 3.7' d -1.0' m -4.28		ν 0.6' d 0.4' m 1.41		ν 2.6' d 0.0' m -2.76		ν 2.2' d 0.1' m 0.92			

h	Sun			Moon			
Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	184°05.3	S13°59.9	264°08.8	9.6'	N13°40.2	-12.2'	59.0'
1	199°05.4	14°00.7	278°37.4	9.6'	13°28.0	-12.3'	59.0'
2	214°05.4	01.6	293°06.0	9.7'	13°15.6	-12.4'	59.0'
3	229°05.4	· · 02.4	307°34.7	9.7'	13°03.2	-12.5'	59.0'
4	244°05.5	03.2	322°03.4	9.7'	12°50.7	-12.6'	59.1'
5	259°05.5	04.0	336°32.1	9.8'	12°38.2	-12.7'	59.1'
6	274°05.5	S14°04.8	351°00.8	9.8'	N12°25.5	-12.7'	59.1'
7	289°05.5	05.6	5°29.6	9.8'	12°12.8	-12.8'	59.1'
8	304°05.6	06.4	19°58.4	9.8'	12°00.0	-12.9'	59.1'
9	319°05.6	· · 07.2	34°27.3	9.9'	11°47.1	-13.0'	59.2'
10	334°05.6	08.0	48°56.1	9.9'	11°34.1	-13.0'	59.2'
11	349°05.6	08.8	63°25.0	9.9'	11°21.1	-13.1'	59.2'
12	4°05.7	S14°09.7	77°54.0	10.0'	N11°07.9	-13.2'	59.2'
13	19°05.7	10.5	92°22.9	10.0'	10°54.8	-13.3'	59.2'
14	34°05.7	11.3	106°51.9	10.0'	10°41.5	-13.3'	59.3'
15	49°05.7	· · 12.1	121°20.9	10.0'	10°28.2	-13.4'	59.3'
16	64°05.8	12.9	135°49.9	10.1'	10°14.8	-13.5'	59.3'
17	79°05.8	13.7	150°19.0	10.1'	10°01.4	-13.5'	59.3'
18	94°05.8	S14°14.5	164°48.0	10.1'	N09°47.8	-13.6'	59.3'
19	109°05.8	15.3	179°17.1	10.1'	09°34.2	-13.6'	59.4'
20	124°05.8	16.1	193°46.3	10.1'	09°20.6	-13.7'	59.4'
21	139°05.9	· · 16.9	208°15.4	10.2'	09°06.9	-13.8'	59.4'
22	154°05.9	17.7	222°44.6	10.2'	08°53.1	-13.8'	59.4'
23	169°05.9	18.5	237°13.7	10.2'	08°39.3	-13.9'	59.4'
SD = 16.1'			SD = 16.1'				

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	184°05.9	S14°19.3	251°42.9	10.2'	N08°25.4	-13.9'	59.5'
1	199°06.0	20.1	266°12.1	10.2'	08°11.5	-14.0'	59.5'
2	214°06.0	20.9	280°41.4	10.2'	07°57.5	-14.0'	59.5'
3	229°06.0	· · 21.7	295°10.6	10.3'	07°43.4	-14.1'	59.5'
4	244°06.0	22.5	309°39.9	10.3'	07°29.4	-14.1'	59.5'
5	259°06.0	23.3	324°09.2	10.3'	07°15.2	-14.2'	59.5'
6	274°06.0	S14°24.2	338°38.4	10.3'	N07°01.0	-14.2'	59.6'
7	289°06.1	25.0	353°07.7	10.3'	06°46.8	-14.3'	59.6'
8	304°06.1	25.8	7°37.1	10.3'	06°32.5	-14.3'	59.6'
9	319°06.1	· · 26.6	22°06.4	10.3'	06°18.1	-14.4'	59.6'
10	334°06.1	27.4	36°35.7	10.3'	06°03.8	-14.4'	59.6'
11	349°06.1	28.2	51°05.1	10.4'	05°49.3	-14.5'	59.6'
12	4°06.1	S14°29.0	65°34.4	10.4'	N05°34.9	-14.5'	59.7'
13	19°06.2	29.8	80°03.8	10.4'	05°20.4	-14.5'	59.7'
14	34°06.2	30.5	94°33.1	10.4'	05°05.8	-14.6'	59.7'
15	49°06.2	· · 31.3	109°02.5	10.4'	04°51.3	-14.6'	59.7'
16	64°06.2	32.1	123°31.9	10.4'	04°36.7	-14.6'	59.7'
17	79°06.2	32.9	138°01.3	10.4'	04°22.0	-14.7'	59.7'
18	94°06.2	S14°33.7	152°30.7	10.4'	N04°07.3	-14.7'	59.8'
19	109°06.3	34.5	167°00.0	10.4'	03°52.6	-14.7'	59.8'
20	124°06.3	35.3	181°29.4	10.4'	03°37.9	-14.8'	59.8'
21	139°06.3	· · 36.1	195°58.8	10.4'	03°23.1	-14.8'	59.8'
22	154°06.3	36.9	210°28.2	10.4'	03°08.3	-14.8'	59.8'
23	169°06.3	37.7	224°57.6	10.4'	02°53.5	-14.8'	59.8'
SD = 16.1'			SD = 16.2'				

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	184°06.3	S14°38.5	239°26.9	10.4'	N02°38.7	-14.9'	59.8'
1	199°06.3	39.3	253°56.3	10.4'	02°23.8	-14.9'	59.9'
2	214°06.3	40.1	268°25.7	10.4'	02°08.9	-14.9'	59.9'
3	229°06.3	· · 40.9	282°55.1	10.4'	01°54.0	-14.9'	59.9'
4	244°06.4	41.7	297°24.4	10.3'	01°39.1	-14.9'	59.9'
5	259°06.4	42.5	311°53.8	10.3'	01°24.2	-15.0'	59.9'
6	274°06.4	S14°43.3	326°23.1	10.3'	N01°09.2	-15.0'	59.9'
7	289°06.4	44.1	340°52.4	10.3'	00°54.3	-15.0'	59.9'
8	304°06.4	44.9	355°21.8	10.3'	00°39.3	-15.0'	59.9'
9	319°06.4	· · 45.6	9°51.1	10.3'	00°24.3	-15.0'	60.0'
10	334°06.4	46.4	24°20.4	10.3'	N00°09.3	-15.0'	60.0'
11	349°06.4	47.2	38°49.6	10.3'	S00°05.7	15.0'	60.0'
12	4°06.4	S14°48.0	53°18.9	10.3'	S00°20.7	15.0'	60.0'
13	19°06.4	48.8	67°48.2	10.2'	00°35.7	15.0'	60.0'
14	34°06.4	49.6	82°17.4	10.2'	00°50.7	15.0'	60.0'
15	49°06.4	· · 50.4	96°46.6	10.2'	01°05.8	15.0'	60.0'
16	64°06.5	51.2	111°15.8	10.2'	01°20.8	15.0'	60.0'
17	79°06.5	52.0	125°45.0	10.2'	01°35.8	15.0'	60.0'
18	94°06.5	S14°52.7	140°14.2	10.1'	S01°50.8	15.0'	60.0'
19	109°06.5	53.5	154°43.3	10.1'	02°05.8	15.0'	60.0'
20	124°06.5	54.3	169°12.4	10.1'	02°20.8	15.0'	60.1'
21	139°06.5	· · 55.1	183°41.5	10.1'	02°35.8	15.0'	60.1'
22	154°06.5	55.9	198°10.6	10.0'	02°50.7	15.0'	60.1'
23	169°06.5	56.7	212°39.6	10.0'	03°05.7	15.0'	60.1'
SD = 16.1'			SD = 16.3'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	06:06	07:29	08:56	14:30	15:57	17:20
N 70°	06:02	07:16	08:30	14:56	16:10	17:24
68°	05:58	07:05	08:10	15:16	16:21	17:28
66°	05:55	06:56	07:55	15:31	16:30	17:31
64°	05:52	06:49	07:42	15:44	16:37	17:34
62°	05:50	06:42	07:31	15:55	16:44	17:36
60°	05:47	06:37	07:22	16:04	16:50	17:39
N 58°	05:45	06:32	07:14	16:13	16:55	17:41
56°	05:43	06:27	07:07	16:20	16:59	17:43
54°	05:41	06:23	07:00	16:26	17:04	17:45
52°	05:39	06:19	06:55	16:32	17:07	17:47
50°	05:37	06:15	06:49	16:37	17:11	17:49
45°	05:33	06:08	06:38	16:48	17:19	17:54
N 40°	05:29	06:01	06:29	16:58	17:26	17:58
35°	05:25	05:55	06:21	17:06	17:32	18:02
30°	05:21	05:49	06:14	17:13	17:38	18:06
20°	05:13	05:39	06:01	17:26	17:48	18:14
N 10°	05:04	05:29	05:50	17:37	17:58	18:23
0°	04:54	05:19	05:40	17:47	18:08	18:33
S 10°	04:43	05:08	05:30	17:58	18:19	18:45
20°	04:29	04:56	05:18	18:09	18:32	18:59
30°	04:10	04:40	05:06	18:22	18:47	19:17
35°	03:59	04:31	04:58	18:30	18:57	19:29
40°	03:45	04:20	04:49	18:39	19:08	19:43
45°	03:27	04:07	04:39	18:49	19:21	20:01
S 50°	03:05	03:51	04:27	19:01	19:38	20:24
52°	02:53	03:43	04:21	19:07	19:46	20:36
54°	02:40	03:34	04:15	19:14	19:55	20:49
56°	02:25	03:24	04:08	19:21	20:05	21:05
58°	02:06	03:12	04:00	19:29	20:17	21:25
S 60°	01:41	02:59	03:50	19:38	20:31	21:51

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	23:24	· · ·	01:40	15:11	14:43	14:18
N 70°	23:37	· · ·	01:43	14:55	14:36	14:19
68°	23:48	· · ·	01:46	14:42	14:31	14:21
66°	23:57	· · ·	01:48	14:31	14:26	14:22
64°	· · ·	00:04	01:49	14:22	14:22	14:22
62°	· · ·	00:10	01:51	14:14	14:19	14:23
60°	· · ·	00:16	01:52	14:08	14:16	14:24
N 58°	· · ·	00:21	01:53	14:02	14:14	14:24
56°	· · ·	00:25	01:54	13:56	14:11	14:25
54°	· · ·	00:29	01:55	13:52	14:09	14:25
52°	· · ·	00:32	01:56	13:47	14:07	14:26
50°	· · ·	00:35	01:57	13:43	14:05	14:26
45°	· · ·	00:42	01:59	13:35	14:01	14:27
N 40°	· · ·	00:48	02:00	13:27	13:58	14:28
35°	· · ·	00:52	02:01	13:21	13:55	14:28
30°	· · ·	00:57	02:02	13:16	13:53	14:29
20°	00:04	01:04	02:04	13:06	13:48	14:30
N 10°	00:15	01:10	02:06	12:58	13:44	14:31
0°	00:25	01:16	02:07	12:49	13:40	14:31
S 10°	00:35	01:22	02:09	12:41	13:37	14:32
20°	00:46	01:29	02:10	12:33	13:33	14:33
30°	00:58	01:36	02:12	12:23	13:28	14:34
35°	01:05	01:40	02:13	12:17	13:25	14:34
40°	01:13	01:45	02:15	12:10	13:22	14:35
45°	01:22	01:50	02:16	12:02	13:18	14:35
S 50°	01:33	01:57	02:18	11:53	13:14	14:36
52°	01:39	01:59	02:19	11:48	13:12	14:36
54°	01:44	02:03	02:19	11:44	13:10	14:36
56°	01:50	02:06	02:20	11:38	13:07	14:37
58°	01:57	02:10	02:21	11:32	13:04	14:37
S 60°	02:05	02:15	02:23	11:25	13:01	14:38

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time	12 ^h		Mer. Pass.		
	00 ^h	mm:ss	hh:mm	Upper	Lower	24-26
				hh:mm	hh:mm	45-23%
31	16:21	16:23	11:44	06:37	19:03	
01	16:24	16:25	11:44	07:28	19:54	
02	16:25	16:26	11:44	08:19	20:45	

November 03, 04, 05 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	42°09.0	193°33.2	S16°59.4	160°08.6	S21°32.7	47°07.9	S03°48.4	210°31.5	S02°35.8	Alpheratz	357°45.2 29°09.3
1	57°11.4	208°36.8	58.4	175°09.2	33.1	62°10.5	48.4	225°33.7	35.9	Ankaa	353°17.1 -42°14.8
2	72°13.9	223°40.4	57.3	190°09.8	33.5	77°13.1	48.4	240°35.9	36.0	Schedar	349°42.3 56°36.1
3	87°16.4	238°44.0	•• 56.3	205°10.3	•• 33.8	92°15.7	•• 48.5	255°38.1	•• 36.2	Diphda	348°57.5 -17°55.5
4	102°18.8	253°47.7	55.3	220°10.9	34.2	107°18.3	48.5	270°40.3	36.3	Achernar	335°27.5 -57°10.9
5	117°21.3	268°51.3	54.2	235°11.4	34.5	122°20.9	48.6	285°42.5	36.4	Hamal	328°02.6 23°31.0
6	132°23.8	283°54.9	S16°53.2	250°12.0	S21°34.9	137°23.5	S03°48.6	300°44.7	S02°36.5	Polaris	318°16.2 89°18.7
7	147°26.2	298°58.5	52.1	265°12.6	35.2	152°26.1	48.7	315°46.9	36.6	Acamar	315°19.3 -40°15.5
8	162°28.7	314°02.1	51.1	280°13.1	35.6	167°28.7	48.7	330°49.1	36.7	Menkar	314°16.7 4°08.1
9	177°31.2	329°05.7	•• 50.1	295°13.7	•• 36.0	182°31.3	•• 48.8	345°51.3	•• 36.8	Mirfak	308°42.6 49°54.1
10	192°33.6	344°09.3	49.0	310°14.2	36.3	197°33.8	48.8	0°53.6	36.9	Aldebaran	290°51.3 16°31.9
11	207°36.1	359°12.9	48.0	325°14.8	36.7	212°36.4	48.8	15°55.8	37.0	Rigel	281°13.6 -8°11.2
12	222°38.6	14°16.5	S16°47.0	340°15.3	S21°37.0	227°39.0	S03°48.9	30°58.0	S02°37.1	Capella	280°36.9 46°00.4
13	237°41.0	29°20.1	45.9	355°15.9	37.4	242°41.6	48.9	46°00.2	37.2	Bellatrix	278°33.8 6°21.6
14	252°43.5	44°23.7	44.9	10°16.5	37.7	257°44.2	49.0	61°02.4	37.3	Elnath	278°14.7 28°37.0
15	267°45.9	59°27.3	•• 43.9	25°17.0	•• 38.1	272°46.8	•• 49.0	76°04.6	•• 37.4	Alnilam	275°48.0 -1°11.6
16	282°48.4	74°30.8	42.8	40°17.6	38.4	287°49.4	49.0	91°06.8	37.5	Betelgeuse	271°03.1 7°24.6
17	297°50.9	89°34.4	41.8	55°18.1	38.8	302°51.9	49.1	106°09.0	37.6	Canopus	263°56.7 -52°41.8
18	312°53.3	104°38.0	S16°40.8	70°18.7	S21°39.2	317°54.5	S03°49.1	121°11.2	S02°37.7	Sirius	258°35.2 -16°43.7
19	327°55.8	119°41.6	39.7	85°19.2	39.5	332°57.1	49.2	136°13.4	37.8	Adhara	255°13.9 -28°59.0
20	342°58.3	134°45.2	38.7	100°19.8	39.9	347°59.7	49.2	151°15.6	37.9	Procyon	245°01.6 5°11.8
21	358°00.7	149°48.7	•• 37.7	115°20.4	•• 40.2	3°02.3	•• 49.3	166°17.8	•• 38.0	Pollux	243°29.9 27°59.9
22	13°03.2	164°52.3	36.6	130°20.9	40.6	18°04.9	49.3	181°20.0	38.1	Avior	234°18.8 -59°32.4
23	28°05.7	179°55.9	35.6	145°21.5	40.9	33°07.5	49.3	196°22.2	38.3	Suhail	222°53.9 -43°28.4
Mer.pass. 21:08		ν 3.6' d -1.0' m -4.26		ν 0.6' d 0.4' m 1.41		ν 2.6' d 0.0' m -2.76		ν 2.2' d 0.1' m 0.92			

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	43°08.1	194°59.4	S16°34.6	160°22.0	S21°41.3	48°10.0	S03°49.4	211°24.4	S02°38.4		
1	58°10.6	210°03.0	33.5	175°22.6	41.6	63°12.6	49.4	226°26.6	38.5		
2	73°13.0	225°06.6	32.5	190°23.1	42.0	78°15.2	49.5	241°28.8	38.6		
3	88°15.5	240°10.1	•• 31.5	205°23.7	•• 42.3	93°17.8	•• 49.5	256°31.0	•• 38.7		
4	103°18.0	255°13.7	30.4	220°24.2	42.7	108°20.4	49.5	271°33.2	38.8		
5	118°20.4	270°17.2	29.4	235°24.8	43.0	123°22.9	49.6	286°35.4	38.9		
6	133°22.9	285°20.8	S16°28.4	250°25.3	S21°43.4	138°25.5	S03°49.6	301°37.6	S02°39.0		
7	148°25.4	300°24.3	27.3	265°25.9	43.7	153°28.1	49.7	316°39.8	39.1		
8	163°27.8	315°27.9	26.3	280°26.4	44.1	168°30.7	49.7	331°42.0	39.2		
9	178°30.3	330°31.4	•• 25.3	295°27.0	•• 44.4	183°33.3	•• 49.7	346°44.2	•• 39.3		
10	193°32.8	345°34.9	24.2	310°27.5	44.8	198°35.9	49.8	1°46.4	39.4		
11	208°35.2	0°38.5	23.2	325°28.1	45.1	213°38.4	49.8	16°48.6	39.5		
12	223°37.7	15°42.0	S16°22.2	340°28.6	S21°45.4	228°41.0	S03°49.9	31°50.8	S02°39.6		
13	238°40.2	30°45.5	21.1	355°29.2	45.8	243°43.6	49.9	46°53.0	39.7		
14	253°42.6	45°49.1	20.1	10°29.7	46.1	258°46.2	49.9	61°55.2	39.8		
15	268°45.1	60°52.6	•• 19.1	25°30.3	•• 46.5	273°48.7	•• 50.0	76°57.4	•• 39.9		
16	283°47.5	75°56.1	18.0	40°30.8	46.8	288°51.3	50.0	91°59.6	40.0		
17	298°50.0	90°59.6	17.0	55°31.4	47.2	303°53.9	50.1	107°01.8	40.1		
18	313°52.5	106°03.1	S16°16.0	70°31.9	S21°47.5	318°56.5	S03°50.1	122°04.0	S02°40.2		
19	328°54.9	121°06.6	15.0	85°32.5	47.9	333°59.1	50.1	137°06.2	40.3		
20	343°57.4	136°10.1	13.9	100°33.0	48.2	349°01.6	50.2	152°08.4	40.4		
21	358°59.9	151°13.7	•• 12.9	115°33.6	•• 48.6	4°04.2	•• 50.2	167°10.6	•• 40.6		
22	14°02.3	166°17.2	11.9	130°34.1	48.9	19°06.8	50.2	182°12.8	40.7		
23	29°04.8	181°20.7	10.8	145°34.7	49.2	34°09.4	50.3	197°15.0	40.8		
Mer.pass. 21:04		ν 3.6' d -1.0' m -4.19		ν 0.6' d 0.4' m 1.40		ν 2.6' d 0.0' m -2.75		ν 2.2' d 0.1' m 0.92			

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	44°07.3	196°24.2	S16°09.8	160°35.2	S21°49.6	49°11.9	S03°50.3	212°17.3	S02°40.9		
1	59°09.7	211°27.6	08.8	175°35.8	49.9	64°14.5	50.4	227°19.5	41.0		
2	74°12.2	226°31.1	07.8	190°36.3	50.3	79°17.1	50.4	242°21.7	41.1		
3	89°14.6	241°34.6	•• 06.7	205°36.9	•• 50.6	94°19.7	•• 50.4	257°23.9	•• 41.2		
4	104°17.1	256°38.1	05.7	220°37.4	51.0	109°22.2	50.5	272°26.1	41.3		
5	119°19.6	271°41.6	04.7	235°38.0	51.3	124°24.8	50.5	287°28.3	41.4		
6	134°22.0	286°45.1	S16°03.7	250°38.5	S21°51.6	139°27.4	S03°50.5	302°30.5	S02°41.5		
7	149°24.5	301°48.6	02.6	265°39.1	52.0	154°30.0	50.6	317°32.7	41.6		
8	164°27.0	316°52.0	01.6	280°39.6	52.3	169°32.5	50.6	332°34.9	41.7		
9	179°29.4	331°55.5	16°00.6	295°40.2	•• 52.7	184°35.1	•• 50.6	347°37.1	•• 41.8		
10	194°31.9	346°59.0	15°59.6	310°40.7	53.0	199°37.7	50.7	2°39.3	41.9		
11	209°34.4	2°02.4	58.5	325°41.3	53.3	214°40.3	50.7	17°41.5	42.0		
12	224°36.8	17°05.9	S15°57.5	340°41.8	S21°53.7	229°42.8	S03°50.8	32°43.7	S02°42.1		
13	239°39.3	32°09.4	56.5	355°42.3	54.0	244°45.4	50.8	47°45.9	42.2		
14	254°41.8	47°12.8	55.5	10°42.9	54.3	259°48.0	50.8	62°48.1	42.3		
15	269°44.2	62°16.3	•• 54.5	25°43.4	•• 54.7	274°50.5	•• 50.9	77°50.3	•• 42.4		
16	284°46.7	77°19.7	53.4	40°44.0	55.0	289°53.1	50.9	92°52.5	42.5		
17	299°49.1	92°23.2	52.4	55°44.5	55.4	304°55.7	50.9	107°54.7	42.6		
18	314°51.6	107°26.6	S15°51.4	70°45.1	S21°55.7	319°58.3	S03°51.0	122°56.9	S02°42.7		
19	329°54.1	122°30.0	50.4	85°45.6	56.0	335°00.8	51.0	137°59.1	42.8		
20	344°56.5	137°33.5	49.4	100°46.1	56.4	350°03.4	51.0	153°01.3	42.9		
21	359°59.0	152°36.9	•• 48.4	115°46.7	•• 56.7	5°06.0	•• 51.1	168°03.5	•• 43.0		
22	15°01.5	167°40.3	47.3	130°47.2	57.0	20°08.5	51.1	183°05.7	43.1		
23	30°03.9	182°43.8	46.3	145°47.8	57.4	35°11.1	51.1	198°08.0	43.2		
Mer.pass. 21:00		ν 3.5' d -1.0' m -4.12		ν 0.5' d 0.3' m 1.40		ν 2.6' d 0.0' m -2.74		ν 2.2' d 0.1' m 0.92			

Nov 03 Wed	SHA	Mer.pass
Venus	151°24.2	11:03
Mars	117°59.6	13:19
Jupiter	4°59.0	20:48
Saturn	168°22.6	09:56

Nov 04 Thu	SHA	Mer.pass
Venus	151°51.3	10:57
Mars	117°13.9	13:18
Jupiter	5°01.9	20:44
Saturn	168°16.3	09:53

Nov 05 Fri	SHA	Mer.pass
Venus	152°16.9	10:52
Mars	116°28.0	13:17
Jupiter	5°04.7	20:40
Saturn	168°10.0	09:49

Horizontal parallax	
Venus:	0.5
Mars:	0.1

h	Sun			Moon			
Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	184°06.5	S14°57.5	227°08.7	10.0'	S03°20.7	14.9'	60.1'
1	199°06.5	58.2	241°37.7	10.0'	03°35.6	14.9'	60.1'
2	214°06.5	59.0	256°06.6	9.9'	03°50.5	14.9'	60.1'
3	229°06.5	14°59.8	270°35.6	9.9'	04°05.4	14.9'	60.1'
4	244°06.5	15°00.6	285°04.5	9.9'	04°20.3	14.9'	60.1'
5	259°06.5	01.4	299°33.4	9.8'	04°35.2	14.8'	60.1'
6	274°06.5	S15°02.2	314°02.2	9.8'	S04°50.0	14.8'	60.1'
7	289°06.5	02.9	328°31.0	9.8'	05°04.8	14.8'	60.1'
8	304°06.5	03.7	342°59.8	9.8'	05°19.6	14.8'	60.1'
9	319°06.5	· · 04.5	357°28.6	9.7'	05°34.3	14.7'	60.1'
10	334°06.5	05.3	11°57.3	9.7'	05°49.1	14.7'	60.1'
11	349°06.5	06.1	26°26.0	9.6'	06°03.8	14.7'	60.1'
12	4°06.5	S15°06.8	40°54.6	9.6'	S06°18.4	14.6'	60.1'
13	19°06.5	07.6	55°23.2	9.6'	06°33.1	14.6'	60.1'
14	34°06.5	08.4	69°51.8	9.5'	06°47.7	14.6'	60.1'
15	49°06.5	· · 09.2	84°20.3	9.5'	07°02.2	14.5'	60.1'
16	64°06.5	10.0	98°48.8	9.5'	07°16.7	14.5'	60.1'
17	79°06.5	10.7	113°17.3	9.4'	07°31.2	14.4'	60.1'
18	94°06.5	S15°11.5	127°45.7	9.4'	S07°45.6	14.4'	60.1'
19	109°06.5	12.3	142°14.1	9.3'	08°00.0	14.3'	60.1'
20	124°06.5	13.1	156°42.4	9.3'	08°14.4	14.3'	60.1'
21	139°06.5	· · 13.8	171°10.7	9.2'	08°28.6	14.2'	60.1'
22	154°06.5	14.6	185°38.9	9.2'	08°42.9	14.2'	60.1'
23	169°06.5	15.4	200°07.1	9.2'	08°57.1	14.1'	60.1'
SD = 16.1'			SD = 16.4'				
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	184°06.5	S15°16.2	214°35.3	9.1'	S09°11.2	14.1'	60.1'
1	199°06.5	16.9	229°03.4	9.1'	09°25.3	14.0'	60.1'
2	214°06.4	17.7	243°31.5	9.0'	09°39.3	14.0'	60.1'
3	229°06.4	· · 18.5	257°59.5	9.0'	09°53.3	13.9'	60.1'
4	244°06.4	19.2	272°27.5	8.9'	10°07.2	13.8'	60.1'
5	259°06.4	20.0	286°55.4	8.9'	10°21.1	13.8'	60.1'
6	274°06.4	S15°20.8	301°23.3	8.8'	S10°34.8	13.7'	60.1'
7	289°06.4	21.6	315°51.1	8.8'	10°48.6	13.7'	60.1'
8	304°06.4	22.3	330°18.9	8.7'	11°02.2	13.6'	60.1'
9	319°06.4	· · 23.1	344°46.6	8.7'	11°15.8	13.5'	60.1'
10	334°06.4	23.9	359°14.3	8.6'	11°29.3	13.4'	60.1'
11	349°06.4	24.6	13°41.9	8.6'	11°42.8	13.4'	60.1'
12	4°06.4	S15°25.4	28°09.5	8.5'	S11°56.1	13.3'	60.1'
13	19°06.4	26.2	42°37.0	8.5'	12°09.4	13.2'	60.1'
14	34°06.3	26.9	57°04.5	8.4'	12°22.7	13.1'	60.1'
15	49°06.3	· · 27.7	71°32.0	8.4'	12°35.8	13.1'	60.0'
16	64°06.3	28.5	85°59.3	8.3'	12°48.9	13.0'	60.0'
17	79°06.3	29.3	100°26.7	8.3'	13°01.9	12.9'	60.0'
18	94°06.3	S15°30.0	114°53.9	8.2'	S13°14.8	12.8'	60.0'
19	109°06.3	30.8	129°21.2	8.2'	13°27.6	12.7'	60.0'
20	124°06.3	31.5	143°48.3	8.1'	13°40.3	12.6'	60.0'
21	139°06.3	· · 32.3	158°15.4	8.1'	13°52.9	12.6'	60.0'
22	154°06.2	33.1	172°42.5	8.0'	14°05.5	12.5'	60.0'
23	169°06.2	33.8	187°09.5	8.0'	14°18.0	12.4'	60.0'
SD = 16.1'			SD = 16.4'				
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	184°06.2	S15°34.6	201°36.5	7.9'	S14°30.3	12.3'	60.0'
1	199°06.2	35.4	216°03.4	7.8'	14°42.6	12.2'	59.9'
2	214°06.2	36.1	230°30.2	7.8'	14°54.8	12.1'	59.9'
3	229°06.2	· · 36.9	244°57.0	7.7'	15°06.9	12.0'	59.9'
4	244°06.2	37.7	259°23.8	7.7'	15°18.9	11.9'	59.9'
5	259°06.1	38.4	273°50.4	7.6'	15°30.8	11.8'	59.9'
6	274°06.1	S15°39.2	288°17.1	7.6'	S15°42.5	11.7'	59.9'
7	289°06.1	39.9	302°43.7	7.5'	15°54.2	11.6'	59.9'
8	304°06.1	40.7	317°10.2	7.5'	16°05.8	11.5'	59.8'
9	319°06.1	· · 41.5	331°36.7	7.4'	16°17.3	11.4'	59.8'
10	334°06.1	42.2	346°03.1	7.4'	16°28.6	11.3'	59.8'
11	349°06.0	43.0	0°29.5	7.3'	16°39.9	11.1'	59.8'
12	4°06.0	S15°43.7	14°55.8	7.3'	S16°51.0	11.0'	59.8'
13	19°06.0	44.5	29°22.1	7.2'	17°02.1	10.9'	59.8'
14	34°06.0	45.3	43°48.3	7.2'	17°13.0	10.8'	59.7'
15	49°06.0	· · 46.0	58°14.5	7.1'	17°23.8	10.7'	59.7'
16	64°05.9	46.8	72°40.6	7.1'	17°34.5	10.6'	59.7'
17	79°05.9	47.5	87°06.6	7.0'	17°45.0	10.5'	59.7'
18	94°05.9	S15°48.3	101°32.7	7.0'	S17°55.5	10.3'	59.7'
19	109°05.9	49.0	115°58.6	6.9'	18°05.8	10.2'	59.7'
20	124°05.9	49.8	130°24.5	6.9'	18°16.0	10.1'	59.6'
21	139°05.8	· · 50.5	144°50.4	6.8'	18°26.1	10.0'	59.6'
22	154°05.8	51.3	159°16.2	6.8'	18°36.1	9.8'	59.6'
23	169°05.8	52.0	173°42.0	6.7'	18°45.9	9.7'	59.6'
SD = 16.1'			SD = 16.4'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	06:17	07:42	09:15	14:10	15:43	17:08
N 70°	06:12	07:28	08:45	14:41	15:58	17:14
68°	06:07	07:16	08:23	15:03	16:10	17:18
66°	06:03	07:06	08:06	15:21	16:20	17:22
64°	06:00	06:57	07:51	15:35	16:29	17:26
62°	05:57	06:50	07:40	15:47	16:36	17:29
60°	05:54	06:44	07:29	15:57	16:43	17:32
N 58°	05:51	06:38	07:21	16:06	16:48	17:35
56°	05:48	06:33	07:13	16:13	16:53	17:38
54°	05:46	06:28	07:06	16:20	16:58	17:40
52°	05:44	06:24	07:00	16:26	17:02	17:43
50°	05:42	06:20	06:54	16:32	17:06	17:45
45°	05:37	06:11	06:42	16:44	17:15	17:50
N 40°	05:32	06:04	06:32	16:54	17:23	17:55
35°	05:27	05:57	06:24	17:03	17:29	17:59
30°	05:23	05:51	06:16	17:11	17:36	18:04
20°	05:14	05:40	06:03	17:24	17:47	18:13
N 10°	05:04	05:29	05:51	17:36	17:58	18:23
0°	04:54	05:19	05:40	17:47	18:08	18:33
S 10°	04:42	05:07	05:29	17:58	18:20	18:46
20°	04:27	04:54	05:17	18:10	18:34	19:01
30°	04:07	04:38	05:03	18:24	18:50	19:20
35°	03:55	04:28	04:55	18:33	19:00	19:33
40°	03:41	04:16	04:46	18:42	19:12	19:47
45°	03:22	04:02	04:35	18:53	19:26	20:06
S 50°	02:58	03:45	04:22	19:07	19:44	20:31
52°	02:46	03:36	04:15	19:13	19:52	20:43
54°	02:32	03:27	04:09	19:20	20:02	20:58
56°	02:14	03:16	04:01	19:28	20:13	21:15
58°	01:53	03:04	03:52	19:36	20:26	21:37
S 60°	01:25	02:49	03:42	19:47	20:41	22:08

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	03:55	06:19	09:17	13:52	13:20	12:18
N 70°	03:49	06:00	08:27	14:02	13:41	13:09
68°	03:43	05:45	07:56	14:10	13:58	13:42
66°	03:39	05:33	07:33	14:17	14:12	14:06
64°	03:35	05:24	07:15	14:22	14:23	14:25
62°	03:32	05:15	07:00	14:27	14:33	14:41
60°	03:29	05:08	06:48	14:32	14:41	14:54
N 58°	03:27	05:02	06:38	14:36	14:49	15:05
56°	03:25	04:56	06:28	14:39	14:55	15:15
54°	03:23	04:51	06:20	14:42	15:01	15:24
52°	03:21	04:47	06:13	14:45	15:06	15:32
50°	03:19	04:43	06:07	14:48	15:11	15:39
45°	03:16	04:34	05:53	14:53	15:22	15:54
N 40°	03:13	04:27	05:41	14:58	15:30	16:07
35°	03:10	04:21	05:32	15:02	15:38	16:18
30°	03:08	04:15	05:23	15:06	15:45	16:27
20°	03:04	04:06	05:09	15:12	15:56	16:44
N 10°	03:01	03:58	04:56	15:18	16:06	16:58
0°	02:58	03:50	04:45	15:23	16:16	17:11
S 10°	02:55	03:43	04:33	15:28	16:26	17:25
20°	02:52	03:35	04:21	15:34	16:36	17:39
30°	02:49	03:26	04:07	15:40	16:48	17:56
35°	02:47	03:21	03:59	15:44	16:54	18:06
40°	02:44	03:16	03:50	15:48	17:02	18:17
45°	02:42	03:09	03:39	15:53	17:11	18:30
S 50°	02:39	03:01	03:27	15:59	17:22	18:46
52°	02:37	02:58	03:21	16:01	17:28	18:53
54°	02:36	02:54	03:14	16:04	17:33	19:02
56°	02:34	02:49	03:07	16:08	17:40	19:11
58°	02:32	02:44	02:59	16:11	17:47	19:22
S 60°	02:30	02:39	02:50	16:16	17:55	19:35

Day	Sun		Mer. Pass hh:mm	Moon		Age 27-29 14-2%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
03	16:26	16:26	11:44	09:10	21:37	
04	16:26	16:25	11:44	10:03	22:30	
05	16:25	16:24	11:44	10:58	23:26	

November 06, 07, 08 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	45°06.4	197°47.2	S15°45.3	160°48.3	S21°57.7	50°13.7	S03°51.2	213°10.2	S02°43.4	Alpheratz	357°45.2 29°09.3
1	60°08.9	212°50.6	44.3	175°48.9	58.0	65°16.2	51.2	228°12.4	43.5	Ankaa	353°17.1 -42°14.8
2	75°11.3	227°54.0	43.3	190°49.4	58.4	80°18.8	51.2	243°14.6	43.6	Schedar	349°42.3 56°36.1
3	90°13.8	242°57.5	•• 42.3	205°49.9	•• 58.7	95°21.4	•• 51.3	258°16.8	•• 43.7	Diphda	348°57.5 -17°55.5
4	105°16.3	258°00.9	41.3	220°50.5	59.0	110°23.9	51.3	273°19.0	43.8	Achernar	335°27.5 -57°10.9
5	120°18.7	273°04.3	40.3	235°51.0	59.4	125°26.5	51.3	288°21.2	43.9	Hamal	328°02.6 23°31.0
6	135°21.2	288°07.7	S15°39.2	250°51.6	S21°59.7	140°29.1	S03°51.4	303°23.4	S02°44.0	Polaris	318°15.8 89°18.8
7	150°23.6	303°11.1	38.2	265°52.1	22°00.0	155°31.6	51.4	318°25.6	44.1	Acamar	315°19.3 -40°15.6
8	165°26.1	318°14.5	37.2	280°52.6	00.4	170°34.2	51.4	333°27.8	44.2	Menkar	314°16.7 4°08.1
9	180°28.6	333°17.9	•• 36.2	295°53.2	•• 00.7	185°36.8	•• 51.5	348°30.0	•• 44.3	Mirfak	308°42.6 49°54.1
10	195°31.0	348°21.3	35.2	310°53.7	01.0	200°39.3	51.5	3°32.2	44.4	Aldebaran	290°51.2 16°31.9
11	210°33.5	3°24.7	34.2	325°54.3	01.4	215°41.9	51.5	18°34.4	44.5	Rigel	281°13.6 -8°11.2
12	225°36.0	18°28.0	S15°33.2	340°54.8	S22°01.7	230°44.5	S03°51.6	33°36.6	S02°44.6	Capella	280°36.9 46°00.5
13	240°38.4	33°31.4	32.2	355°55.3	02.0	245°47.0	51.6	48°38.8	44.7	Bellatrix	278°33.8 6°21.6
14	255°40.9	48°34.8	31.2	10°55.9	02.4	260°49.6	51.6	63°41.0	44.8	Elnath	278°14.7 28°37.0
15	270°43.4	63°38.2	•• 30.2	25°56.4	•• 02.7	275°52.2	•• 51.7	78°43.2	•• 44.9	Alnilam	275°48.0 -1°11.6
16	285°45.8	78°41.6	29.2	40°56.9	03.0	290°54.7	51.7	93°45.4	45.0	Betelgeuse	271°03.1 7°24.6
17	300°48.3	93°44.9	28.2	55°57.5	03.3	305°57.3	51.7	108°47.7	45.1	Canopus	263°56.6 -52°41.9
18	315°50.8	108°48.3	S15°27.2	70°58.0	S22°03.7	320°59.8	S03°51.8	123°49.9	S02°45.2	Sirius	258°35.2 -16°43.7
19	330°53.2	123°51.7	26.2	85°58.6	04.0	336°02.4	51.8	138°52.1	45.3	Adhara	255°13.8 -28°59.0
20	345°55.7	138°55.0	25.2	100°59.1	04.3	351°05.0	51.8	153°54.3	45.4	Procyon	245°01.6 5°11.8
21	0°58.1	153°58.4	•• 24.2	115°59.6	•• 04.7	6°07.5	•• 51.9	168°56.5	•• 45.5	Pollux	243°29.9 27°59.9
22	16°00.6	169°01.7	23.2	131°00.2	05.0	21°10.1	51.9	183°58.7	45.6	Avior	234°18.7 -59°32.4
23	31°03.1	184°05.1	22.2	146°00.7	05.3	36°12.7	51.9	199°00.9	45.7	Suhail	222°53.9 -43°28.4
Mer.pass. 20:56		ν 3.4' d -1.0' m -4.19		ν 0.5' d 0.3' m 1.39		ν 2.6' d 0.0' m -2.73		ν 2.2' d 0.1' m 0.92			

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	46°05.5	199°08.4	S15°21.2	161°01.2	S22°05.6	51°15.2	S03°51.9	214°03.1	S02°45.8		
1	61°08.0	214°11.8	20.2	176°01.8	06.0	66°17.8	52.0	229°05.3	45.9		
2	76°10.5	229°15.1	19.2	191°02.3	06.3	81°20.3	52.0	244°07.5	46.0		
3	91°12.9	244°18.5	•• 18.2	206°02.8	•• 06.6	96°22.9	•• 52.0	259°09.7	•• 46.1		
4	106°15.4	259°21.8	17.2	221°03.4	06.9	111°25.5	52.1	274°11.9	46.2		
5	121°17.9	274°25.1	16.2	236°03.9	07.3	126°28.0	52.1	289°14.1	46.3		
6	136°20.3	289°28.4	S15°15.2	251°04.4	S22°07.6	141°30.6	S03°52.1	304°16.3	S02°46.4		
7	151°22.8	304°31.8	14.2	266°05.0	07.9	156°33.1	52.2	319°18.6	46.5		
8	166°25.2	319°35.1	13.2	281°05.5	08.2	171°35.7	52.2	334°20.8	46.6		
9	181°27.7	334°38.4	•• 12.3	296°06.0	•• 08.5	186°38.2	•• 52.2	349°23.0	•• 46.7		
10	196°30.2	349°41.7	11.3	311°06.6	08.9	201°40.8	52.2	4°25.2	46.8		
11	211°32.6	4°45.0	10.3	326°07.1	09.2	216°43.4	52.3	19°27.4	46.9		
12	226°35.1	19°48.3	S15°09.3	341°07.6	S22°09.5	231°45.9	S03°52.3	34°29.6	S02°47.0		
13	241°37.6	34°51.6	08.3	356°08.2	09.8	246°48.5	52.3	49°31.8	47.1		
14	256°40.0	49°54.9	07.3	11°08.7	10.2	261°51.0	52.4	64°34.0	47.3		
15	271°42.5	64°58.2	•• 06.3	26°09.2	•• 10.5	276°53.6	•• 52.4	79°36.2	•• 47.4		
16	286°45.0	80°01.5	05.4	41°09.8	10.8	291°56.1	52.4	94°38.4	47.5		
17	301°47.4	95°04.8	04.4	56°10.3	11.1	306°58.7	52.4	109°40.6	47.6		
18	316°49.9	110°08.1	S15°03.4	71°10.8	S22°11.4	322°01.2	S03°52.5	124°42.8	S02°47.7		
19	331°52.4	125°11.4	02.4	86°11.3	11.8	337°03.8	52.5	139°45.0	47.8		
20	346°54.8	140°14.7	01.4	101°11.9	12.1	352°06.4	52.5	154°47.2	47.9		
21	1°57.3	155°17.9	15°00.5	116°12.4	•• 12.4	7°08.9	•• 52.6	169°49.5	•• 48.0		
22	16°59.7	170°21.2	14°59.5	131°12.9	12.7	22°11.5	52.6	184°51.7	48.1		
23	32°02.2	185°24.5	58.5	146°13.5	13.0	37°14.0	52.6	199°53.9	48.2		
Mer.pass. 20:52		ν 3.3' d -1.0' m -4.27		ν 0.5' d 0.3' m 1.39		ν 2.6' d 0.0' m -2.73		ν 2.2' d 0.1' m 0.92			

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	47°04.7	200°27.7	S14°57.5	161°14.0	S22°13.3	52°16.6	S03°52.6	214°56.1	S02°48.3		
1	62°07.1	215°31.0	56.6	176°14.5	13.7	67°19.1	52.7	229°58.3	48.4		
2	77°09.6	230°34.3	55.6	191°15.1	14.0	82°21.7	52.7	245°00.5	48.5		
3	92°12.1	245°37.5	•• 54.6	206°15.6	•• 14.3	97°24.2	•• 52.7	260°02.7	•• 48.6		
4	107°14.5	260°40.8	53.7	221°16.1	14.6	112°26.8	52.7	275°04.9	48.7		
5	122°17.0	275°44.0	52.7	236°16.6	14.9	127°29.3	52.8	290°07.1	48.8		
6	137°19.5	290°47.3	S14°51.7	251°17.2	S22°15.2	142°31.9	S03°52.8	305°09.3	S02°48.9		
7	152°21.9	305°50.5	50.8	266°17.7	15.6	157°34.4	52.8	320°11.5	49.0		
8	167°24.4	320°53.7	49.8	281°18.2	15.9	172°37.0	52.9	335°13.8	49.1		
9	182°26.9	335°57.0	•• 48.8	296°18.7	•• 16.2	187°39.5	•• 52.9	350°16.0	•• 49.2		
10	197°29.3	351°00.2	47.9	311°19.3	16.5	202°42.1	52.9	5°18.2	49.3		
11	212°31.8	6°03.4	46.9	326°19.8	16.8	217°44.6	52.9	20°20.4	49.4		
12	227°34.2	21°06.6	S14°45.9	341°20.3	S22°17.1	232°47.2	S03°53.0	35°22.6	S02°49.5		
13	242°36.7	36°09.9	45.0	356°20.8	17.4	247°49.7	53.0	50°24.8	49.6		
14	257°39.2	51°13.1	44.0	11°21.4	17.8	262°52.3	53.0	65°27.0	49.7		
15	272°41.6	66°16.3	•• 43.1	26°21.9	•• 18.1	277°54.8	•• 53.0	80°29.2	•• 49.8		
16	287°44.1	81°19.5	42.1	41°22.4	18.4	292°57.4	53.1	95°31.4	49.9		
17	302°46.6	96°22.7	41.2	56°22.9	18.7	307°59.9	53.1	110°33.6	50.0		
18	317°49.0	111°25.9	S14°40.2	71°23.5	S22°19.0	323°02.5	S03°53.1	125°35.8	S02°50.1		
19	332°51.5	126°29.1	39.3	86°24.0	19.3	338°05.0	53.1	140°38.1	50.2		
20	347°54.0	141°32.3	38.3	101°24.5	19.6	353°07.6	53.2	155°40.3	50.3		
21	2°56.4	156°35.5	•• 37.4	116°25.0	•• 19.9	8°10.1	•• 53.2	170°42.5	•• 50.4		
22	17°58.9	171°38.7	36.4	131°25.6	20.2	23°12.7	53.2	185°44.7	50.5		
23	33°01.4	186°41.9	35.5	146°26.1	20.6	38°15.2	53.2	200°46.9	50.6		
Mer.pass. 20:48		ν 3.3' d -1.0' m -4.34		ν 0.5' d 0.3' m 1.39		ν 2.6' d 0.0' m -2.72		ν 2.2' d 0.1' m 0.92			

Nov 06 Sat			SHA	Mer.pass
Venus			152°40.8	10:46
Mars			115°41.9	13:16
Jupiter			5°07.3	20:36
Saturn			168°03.8	09:46

Nov 07 Sun			SHA	Mer.pass
Venus			153°02.9	10:41
Mars			114°55.7	13:15
Jupiter			5°09.7	20:31
Saturn			167°57.6	09:42

Nov 08 Mon			SHA	Mer.pass
Venus			153°23.1	10:36
Mars			114°09.3	13:15
Jupiter			5°11.9	20:27
Saturn			167°51.4	09:39

Horizontal parallax			
Venus:			0.5
Mars:			0.1

h	Sun				Moon			
Sat	GHA	Dec	GHA	ν	Dec	d	HP	
0	184°05.8	S15°52.8	188°07.8	6.7'	S18°55.7	9.6'	59.6'	
1	199°05.8	53.6	202°33.4	6.6'	19°05.2	9.5'	59.5'	
2	214°05.7	54.3	216°59.1	6.6'	19°14.7	9.3'	59.5'	
3	229°05.7	55.1	231°24.7	6.6'	19°24.0	9.2'	59.5'	
4	244°05.7	55.8	245°50.2	6.5'	19°33.2	9.1'	59.5'	
5	259°05.7	56.6	260°15.8	6.5'	19°42.3	8.9'	59.4'	
6	274°05.6	S15°57.3	274°41.2	6.4'	S19°51.2	8.8'	59.4'	
7	289°05.6	58.1	289°06.7	6.4'	20°00.0	8.7'	59.4'	
8	304°05.6	58.8	303°32.1	6.4'	20°08.7	8.5'	59.4'	
9	319°05.6	15°59.6	317°57.4	6.3'	20°17.2	8.4'	59.3'	
10	334°05.5	16°00.3	332°22.7	6.3'	20°25.6	8.2'	59.3'	
11	349°05.5	01.1	346°48.0	6.2'	20°33.8	8.1'	59.3'	
12	4°05.5	S16°01.8	1°13.3	6.2'	S20°41.9	8.0'	59.3'	
13	19°05.4	02.5	15°38.5	6.2'	20°49.9	7.8'	59.3'	
14	34°05.4	03.3	30°03.6	6.1'	20°57.7	7.7'	59.2'	
15	49°05.4	04.0	44°28.8	6.1'	21°05.4	7.5'	59.2'	
16	64°05.4	04.8	58°53.9	6.1'	21°12.9	7.4'	59.2'	
17	79°05.3	05.5	73°19.0	6.1'	21°20.3	7.3'	59.1'	
18	94°05.3	S16°06.3	87°44.0	6.0'	S21°27.6	7.1'	59.1'	
19	109°05.3	07.0	102°09.1	6.0'	21°34.7	7.0'	59.1'	
20	124°05.2	07.8	116°34.1	6.0'	21°41.7	6.8'	59.1'	
21	139°05.2	08.5	130°59.1	6.0'	21°48.5	6.7'	59.0'	
22	154°05.2	09.2	145°24.0	5.9'	21°55.1	6.5'	59.0'	
23	169°05.1	10.0	159°49.0	5.9'	22°01.6	6.4'	59.0'	
SD = 16.1'		d = 0.8'	SD = 16.2'					
Sun	GHA	Dec	GHA	ν	Dec	d	HP	
0	184°05.1	S16°10.7	174°13.9	5.9'	S22°08.0	6.2'	59.0'	
1	199°05.1	11.5	188°38.8	5.9'	22°14.2	6.1'	58.9'	
2	214°05.1	12.2	203°03.6	5.9'	22°20.3	5.9'	58.9'	
3	229°05.0	13.0	217°28.5	5.8'	22°26.2	5.8'	58.9'	
4	244°05.0	13.7	231°53.4	5.8'	22°32.0	5.6'	58.8'	
5	259°05.0	14.4	246°18.2	5.8'	22°37.6	5.5'	58.8'	
6	274°04.9	S16°15.2	260°43.0	5.8'	S22°43.0	5.3'	58.8'	
7	289°04.9	15.9	275°07.8	5.8'	22°48.3	5.2'	58.8'	
8	304°04.9	16.6	289°32.6	5.8'	22°53.5	5.0'	58.7'	
9	319°04.8	17.4	303°57.4	5.8'	22°58.5	4.8'	58.7'	
10	334°04.8	18.1	318°22.2	5.8'	23°03.3	4.7'	58.7'	
11	349°04.7	18.9	332°47.0	5.8'	23°08.0	4.5'	58.6'	
12	4°04.7	S16°19.6	347°11.8	5.8'	S23°12.6	4.4'	58.6'	
13	19°04.7	20.3	1°36.6	5.8'	23°17.0	4.2'	58.6'	
14	34°04.6	21.1	16°01.4	5.8'	23°21.2	4.1'	58.5'	
15	49°04.6	21.8	30°26.1	5.8'	23°25.3	3.9'	58.5'	
16	64°04.6	22.5	44°50.9	5.8'	23°29.2	3.8'	58.5'	
17	79°04.5	23.3	59°15.7	5.8'	23°32.9	3.6'	58.4'	
18	94°04.5	S16°24.0	73°40.5	5.8'	S23°36.5	3.4'	58.4'	
19	109°04.4	24.7	88°05.4	5.8'	23°40.0	3.3'	58.4'	
20	124°04.4	25.5	102°30.2	5.8'	23°43.3	3.1'	58.3'	
21	139°04.4	26.2	116°55.0	5.9'	23°46.4	3.0'	58.3'	
22	154°04.3	26.9	131°19.9	5.9'	23°49.4	2.8'	58.3'	
23	169°04.3	27.7	145°44.7	5.9'	23°52.2	2.7'	58.2'	
SD = 16.1'		d = 0.7'	SD = 16.1'					
Mon	GHA	Dec	GHA	ν	Dec	d	HP	
0	184°04.3	S16°28.4	160°09.6	5.9'	S23°54.9	2.5'	58.2'	
1	199°04.2	29.1	174°34.5	5.9'	23°57.4	2.4'	58.2'	
2	214°04.2	29.8	188°59.5	6.0'	23°59.8	2.2'	58.2'	
3	229°04.1	30.6	203°24.4	6.0'	24°02.0	2.0'	58.1'	
4	244°04.1	31.3	217°49.4	6.0'	24°04.0	1.9'	58.1'	
5	259°04.0	32.0	232°14.4	6.0'	24°05.9	1.7'	58.1'	
6	274°04.0	S16°32.8	246°39.4	6.1'	S24°07.7	1.6'	58.0'	
7	289°04.0	33.5	261°04.5	6.1'	24°09.2	1.4'	58.0'	
8	304°03.9	34.2	275°29.6	6.1'	24°10.7	1.3'	57.9'	
9	319°03.9	34.9	289°54.7	6.2'	24°12.0	1.1'	57.9'	
10	334°03.8	35.7	304°19.9	6.2'	24°13.1	1.0'	57.9'	
11	349°03.8	36.4	318°45.0	6.2'	24°14.1	0.8'	57.8'	
12	4°03.7	S16°37.1	333°10.3	6.3'	S24°14.9	0.7'	57.8'	
13	19°03.7	37.8	347°35.5	6.3'	24°15.5	0.5'	57.8'	
14	34°03.7	38.6	2°00.9	6.4'	24°16.1	0.4'	57.7'	
15	49°03.6	39.3	16°26.2	6.4'	24°16.4	0.2'	57.7'	
16	64°03.6	40.0	30°51.6	6.4'	24°16.6	0.1'	57.7'	
17	79°03.5	40.7	45°17.1	6.5'	24°16.7	-0.1'	57.6'	
18	94°03.5	S16°41.5	59°42.5	6.5'	S24°16.6	-0.2'	57.6'	
19	109°03.4	42.2	74°08.1	6.6'	24°16.4	-0.4'	57.6'	
20	124°03.4	42.9	88°33.7	6.6'	24°16.0	-0.5'	57.5'	
21	139°03.3	43.6	102°59.3	6.7'	24°15.5	-0.7'	57.5'	
22	154°03.3	44.3	117°25.0	6.7'	24°14.8	-0.8'	57.5'	
23	169°03.2	45.1	131°50.7	6.8'	24°14.0	-1.0'	57.4'	
SD = 16.1'		d = 0.7'	SD = 15.9'					

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	06:29	07:56	09:37	13:49	15:30	16:57
N 70°	06:22	07:39	09:01	14:25	15:47	17:03
68°	06:17	07:26	08:36	14:50	16:00	17:09
66°	06:12	07:15	08:17	15:10	16:11	17:14
64°	06:08	07:06	08:01	15:25	16:21	17:19
62°	06:04	06:58	07:48	15:38	16:29	17:23
60°	06:00	06:51	07:37	15:49	16:36	17:26
N 58°	05:57	06:44	07:28	15:59	16:42	17:29
56°	05:54	06:39	07:19	16:07	16:48	17:33
54°	05:51	06:34	07:12	16:15	16:53	17:35
52°	05:49	06:29	07:05	16:21	16:58	17:38
50°	05:46	06:25	06:59	16:27	17:02	17:41
45°	05:40	06:15	06:46	16:40	17:11	17:46
N 40°	05:35	06:07	06:36	16:51	17:20	17:52
35°	05:30	06:00	06:26	17:00	17:27	17:57
30°	05:25	05:53	06:18	17:09	17:34	18:02
20°	05:15	05:41	06:04	17:23	17:46	18:12
N 10°	05:05	05:30	05:52	17:35	17:57	18:22
0°	04:54	05:19	05:40	17:47	18:09	18:34
S 10°	04:41	05:06	05:28	17:59	18:21	18:47
20°	04:25	04:52	05:16	18:12	18:35	19:03
30°	04:05	04:35	05:01	18:27	18:53	19:23
35°	03:52	04:25	04:52	18:36	19:03	19:36
40°	03:37	04:13	04:42	18:46	19:15	19:52
45°	03:17	03:58	04:31	18:57	19:30	20:12
S 50°	02:51	03:39	04:17	19:12	19:49	20:38
52°	02:38	03:30	04:10	19:18	19:58	20:51
54°	02:23	03:20	04:03	19:26	20:09	21:07
56°	02:04	03:09	03:55	19:34	20:20	21:26
58°	01:40	02:55	03:45	19:44	20:34	21:51
S 60°	01:06	02:39	03:35	19:55	20:51	22:27

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	10:27	■	■	13:10	■	■
66°	09:39	11:54	■	13:59	13:46	■
64°	09:08	10:56	12:22	14:30	14:44	15:21
62°	08:45	10:23	11:40	14:54	15:18	16:02
60°	08:27	09:58	11:12	15:13	15:43	16:30
N 58°	08:12	09:39	10:51	15:29	16:03	16:52
56°	07:59	09:23	10:33	15:42	16:19	17:10
54°	07:48	09:09	10:18	15:54	16:33	17:25
52°	07:38	08:57	10:05	16:04	16:46	17:38
50°	07:29	08:46	09:53	16:13	16:57	17:50
45°	07:10	08:24	09:29	16:33	17:19	18:14
N 40°	06:55	08:06	09:10	16:49	17:38	18:33
35°	06:43	07:51	08:54	17:02	17:53	18:49
30°	06:32	07:38	08:40	17:14	18:06	19:03
20°	06:13	07:16	08:17	17:35	18:29	19:26
N 10°	05:56	06:57	07:57	17:52	18:49	19:46
0°	05:41	06:39	07:38	18:09	19:07	20:05
S 10°	05:26	06:22	07:19	18:25	19:26	20:24
20°	05:10	06:03	06:59	18:43	19:46	20:45
30°	04:52	05:41	06:35	19:04	20:08	21:08
35°	04:41	05:29	06:22	19:16	20:22	21:22
40°	04:29	05:14	06:06	19:29	20:37	21:38
45°	04:15	04:57	05:47	19:46	20:56	21:56
S 50°	03:58	04:36	05:24	20:06	21:19	22:20
52°	03:49	04:26	05:13	20:16	21:30	22:31
54°	03:40	04:15	05:00	20:27	21:42	22:44
56°	03:30	04:02	04:46	20:39	21:57	22:59
58°	03:19	03:47	04:29	20:53	22:14	23:16
S 60°	03:06	03:30	04:08	21:10	22:34	23:37

November 09, 10, 11 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	48°03.8	201°45.0	S14°34.5	161°26.6	S22°20.9	53°17.7	S03°53.3	215°49.1	S02°50.7	Alpheratz	357°45.2 29°09.3
1	63°06.3	216°48.2	33.6	176°27.1	21.2	68°20.3	53.3	230°51.3	50.8	Ankaa	353°17.2 -42°14.8
2	78°08.7	231°51.4	32.6	191°27.7	21.5	83°22.8	53.3	245°53.5	50.9	Schedar	349°42.3 56°36.1
3	93°11.2	246°54.5	· · 31.7	206°28.2	· · 21.8	98°25.4	· · 53.3	260°55.7	· · 51.0	Diphda	348°57.5 -17°55.5
4	108°13.7	261°57.7	30.7	221°28.7	22.1	113°27.9	53.3	275°57.9	51.1	Achernar	335°27.5 -57°10.9
5	123°16.1	277°00.9	29.8	236°29.2	22.4	128°30.5	53.4	291°00.2	51.2	Hamal	328°02.6 23°31.0
6	138°18.6	292°04.0	S14°28.9	251°29.7	S22°22.7	143°33.0	S03°53.4	306°02.4	S02°51.3	Polaris	318°15.4 89°18.8
7	153°21.1	307°07.2	27.9	266°30.3	23.0	158°35.6	53.4	321°04.6	51.4	Acamar	315°19.3 -40°15.6
8	168°23.5	322°10.3	27.0	281°30.8	23.3	173°38.1	53.4	336°06.8	51.5	Menkar	314°16.7 4°08.1
9	183°26.0	337°13.5	· · 26.0	296°31.3	· · 23.6	188°40.6	· · 53.5	351°09.0	· · 51.6	Mirfak	308°42.6 49°54.1
10	198°28.5	352°16.6	25.1	311°31.8	23.9	203°43.2	53.5	6°11.2	51.7	Aldebaran	290°51.2 16°31.9
11	213°30.9	7°19.8	24.2	326°32.3	24.2	218°45.7	53.5	21°13.4	51.8	Rigel	281°13.6 -8°11.2
12	228°33.4	22°22.9	S14°23.2	341°32.9	S22°24.5	233°48.3	S03°53.5	36°15.6	S02°51.9	Capella	280°36.8 46°00.5
13	243°35.9	37°26.0	22.3	356°33.4	24.8	248°50.8	53.5	51°17.8	52.0	Bellatrix	278°33.7 6°21.6
14	258°38.3	52°29.2	21.4	11°33.9	25.1	263°53.3	53.6	66°20.0	52.1	Elnath	278°14.7 28°37.0
15	273°40.8	67°32.3	· · 20.5	26°34.4	· · 25.4	278°55.9	· · 53.6	81°22.3	· · 52.2	Alnilam	275°48.0 -1°11.6
16	288°43.2	82°35.4	19.5	41°34.9	25.8	293°58.4	53.6	96°24.5	52.3	Betelgeuse	271°03.1 7°24.6
17	303°45.7	97°38.5	18.6	56°35.5	26.1	309°01.0	53.6	111°26.7	52.4	Canopus	263°56.6 -52°41.9
18	318°48.2	112°41.6	S14°17.7	71°36.0	S22°26.4	324°03.5	S03°53.7	126°28.9	S02°52.5	Sirius	258°35.2 -16°43.8
19	333°50.6	127°44.7	16.8	86°36.5	26.7	339°06.0	53.7	141°31.1	52.6	Adhara	255°13.8 -28°59.0
20	348°53.1	142°47.8	15.8	101°37.0	27.0	354°08.6	53.7	156°33.3	52.7	Procyon	245°01.6 5°11.8
21	3°55.6	157°50.9	· · 14.9	116°37.5	· · 27.3	9°11.1	· · 53.7	171°35.5	· · 52.8	Pollux	243°29.9 27°59.9
22	18°58.0	172°54.0	14.0	131°38.0	27.6	24°13.7	53.7	186°37.7	52.9	Avior	234°18.7 -59°32.4
23	34°00.5	187°57.1	13.1	146°38.6	27.9	39°16.2	53.8	201°39.9	53.0	Suhail	222°53.9 -43°28.4
Mer.pass. 20:44		ν 3.2' d -0.9' m -4.40		ν 0.5' d 0.3' m 1.38		ν 2.5' d 0.0' m -2.71		ν 2.2' d 0.1' m 0.92		Miaplacidus	221°40.2 -69°45.4
										Alphard	217°58.0 -8°42.3
										Regulus	207°45.6 11°54.8
										Dubhe	193°54.1 61°41.2
										Denebola	182°35.8 14°30.6
										Gienah	175°54.5 -17°36.1
										Acrux	173°12.1 -63°09.4
										Gacrux	172°03.6 -57°10.3
										Alioth	166°22.7 55°53.8
										Spica	158°33.6 -11°13.0
										Alkaid	153°00.8 49°15.4
										Hadar	148°51.3 -60°25.4
										Menkent	148°10.2 -36°25.3
										Arcturus	145°57.8 19°07.5
										Rigel Kent.	139°55.0 -60°52.7
										Kochab	137°20.9 74°06.6
										Zuben'ubi	137°07.9 -16°05.2
										Alphecca	126°13.0 26°40.7
										Antares	112°28.9 -26°27.3
										Atria	107°33.0 -69°02.9
										Sabik	102°15.0 -15°44.2
										Shaula	96°24.8 -37°06.7
										Rasalhague	96°08.5 12°33.3
										Eltanin	90°47.4 51°29.5
										Kaus Aust.	83°46.6 -34°22.8
										Vega	80°40.5 38°47.9
										Nunki	76°00.9 -26°17.0
										Altair	62°10.2 8°54.0
										Peacock	53°22.3 -56°42.1
										Deneb	49°32.9 45°19.5
										Enif	33°48.9 9°55.7
										Al Na'ir	27°45.9 -46°54.6
										Fomalhaut	15°25.9 -29°33.8
										Scheat	13°55.1 28°08.8
										Markab	13°40.1 15°16.1
										Nov 09 Tue	SHA Mer.pass
										Venus	153°41.2 10:31
										Mars	113°22.8 13:14
										Jupiter	5°13.9 20:23
										Saturn	167°45.3 09:35
										Nov 10 Wed	SHA Mer.pass
										Venus	153°57.3 10:26
										Mars	112°36.1 13:13
										Jupiter	5°15.8 20:19
										Saturn	167°39.2 09:32
										Nov 11 Thu	SHA Mer.pass
										Venus	154°11.2 10:21
										Mars	111°49.3 13:12
										Jupiter	5°17.4 20:15
										Saturn	167°33.2 09:28

h	Sun			Moon			
Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	184°03.2	S16°45.8	146°16.5	6.9'	S24°13.0	-1.1'	57.4'
1	199°03.1	46.5	160°42.4	6.9'	24°11.9	-1.3'	57.4'
2	214°03.1	47.2	175°08.3	7.0'	24°10.6	-1.4'	57.3'
3	229°03.0	47.9	189°34.3	7.0'	24°09.2	-1.5'	57.3'
4	244°03.0	48.6	204°00.3	7.1'	24°07.7	-1.7'	57.3'
5	259°02.9	49.4	218°26.4	7.2'	24°06.0	-1.8'	57.2'
6	274°02.9	S16°50.1	232°52.6	7.2'	S24°04.1	-2.0'	57.2'
7	289°02.8	50.8	247°18.8	7.3'	24°02.2	-2.1'	57.2'
8	304°02.8	51.5	261°45.1	7.4'	24°00.1	-2.2'	57.1'
9	319°02.7	52.2	276°11.5	7.4'	23°57.8	-2.4'	57.1'
10	334°02.7	52.9	290°37.9	7.5'	23°55.4	-2.5'	57.1'
11	349°02.6	53.6	305°04.4	7.6'	23°52.9	-2.7'	57.0'
12	4°02.6	S16°54.4	319°31.0	7.6'	S23°50.3	-2.8'	57.0'
13	19°02.5	55.1	333°57.6	7.7'	23°47.5	-2.9'	57.0'
14	34°02.5	55.8	348°24.3	7.8'	23°44.5	-3.1'	56.9'
15	49°02.4	56.5	2°51.1	7.9'	23°41.5	-3.2'	56.9'
16	64°02.3	57.2	17°17.9	7.9'	23°38.3	-3.3'	56.8'
17	79°02.3	57.9	31°44.9	8.0'	23°35.0	-3.5'	56.8'
18	94°02.2	S16°58.6	46°11.9	8.1'	S23°31.5	-3.6'	56.8'
19	109°02.2	16°59.3	60°39.0	8.2'	23°27.9	-3.7'	56.7'
20	124°02.1	17°00.0	75°06.1	8.2'	23°24.2	-3.8'	56.7'
21	139°02.1	00.7	89°33.4	8.3'	23°20.4	-4.0'	56.7'
22	154°02.0	01.5	104°00.7	8.4'	23°16.4	-4.1'	56.6'
23	169°02.0	02.2	118°28.1	8.5'	23°12.3	-4.2'	56.6'
SD = 16.1'		$d = 0.7'$	SD = 15.7'				
Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	184°01.9	S17°02.9	132°55.6	8.6'	S23°08.1	-4.3'	56.6'
1	199°01.8	03.6	147°23.1	8.6'	23°03.8	-4.5'	56.5'
2	214°01.8	04.3	161°50.8	8.7'	22°59.3	-4.6'	56.5'
3	229°01.7	05.0	176°18.5	8.8'	22°54.7	-4.7'	56.5'
4	244°01.7	05.7	190°46.3	8.9'	22°50.0	-4.8'	56.4'
5	259°01.6	06.4	205°14.2	9.0'	22°45.2	-4.9'	56.4'
6	274°01.5	S17°07.1	219°42.2	9.1'	S22°40.3	-5.1'	56.4'
7	289°01.5	07.8	234°10.3	9.1'	22°35.2	-5.2'	56.3'
8	304°01.4	08.5	248°38.4	9.2'	22°30.0	-5.3'	56.3'
9	319°01.4	09.2	263°06.7	9.3'	22°24.8	-5.4'	56.3'
10	334°01.3	09.9	277°35.0	9.4'	22°19.4	-5.5'	56.3'
11	349°01.2	10.6	292°03.4	9.5'	22°13.9	-5.6'	56.2'
12	4°01.2	S17°11.3	306°31.9	9.6'	S22°08.2	-5.7'	56.2'
13	19°01.1	12.0	321°00.5	9.7'	22°02.5	-5.8'	56.2'
14	34°01.1	12.7	335°29.1	9.8'	21°56.7	-5.9'	56.1'
15	49°01.0	13.4	349°57.9	9.8'	21°50.7	-6.1'	56.1'
16	64°00.9	14.1	4°26.7	9.9'	21°44.7	-6.2'	56.1'
17	79°00.9	14.8	18°55.6	10.0'	21°38.5	-6.3'	56.0'
18	94°00.8	S17°15.5	33°24.7	10.1'	S21°32.2	-6.4'	56.0'
19	109°00.7	16.2	47°53.8	10.2'	21°25.9	-6.5'	56.0'
20	124°00.7	16.9	62°23.0	10.3'	21°19.4	-6.6'	55.9'
21	139°00.6	17.6	76°52.2	10.4'	21°12.8	-6.7'	55.9'
22	154°00.5	18.3	91°21.6	10.5'	21°06.1	-6.8'	55.9'
23	169°00.5	19.0	105°51.1	10.5'	20°59.4	-6.9'	55.8'
SD = 16.1'		$d = 0.7'$	SD = 15.4'				
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	184°00.4	S17°19.7	120°20.6	10.6'	S20°52.5	-7.0'	55.8'
1	199°00.3	20.4	134°50.2	10.7'	20°45.5	-7.1'	55.8'
2	214°00.3	21.1	149°20.0	10.8'	20°38.5	-7.2'	55.8'
3	229°00.2	21.8	163°49.8	10.9'	20°31.3	-7.3'	55.7'
4	244°00.1	22.4	178°19.7	11.0'	20°24.1	-7.3'	55.7'
5	259°00.1	23.1	192°49.7	11.1'	20°16.7	-7.4'	55.7'
6	274°00.0	S17°23.8	207°19.7	11.2'	S20°09.3	-7.5'	55.6'
7	288°59.9	24.5	221°49.9	11.2'	20°01.8	-7.6'	55.6'
8	303°59.9	25.2	236°20.1	11.3'	19°54.1	-7.7'	55.6'
9	318°59.8	25.9	250°50.5	11.4'	19°46.4	-7.8'	55.6'
10	333°59.7	26.6	265°20.9	11.5'	19°38.6	-7.9'	55.5'
11	348°59.7	27.3	279°51.4	11.6'	19°30.8	-8.0'	55.5'
12	3°59.6	S17°28.0	294°22.0	11.7'	S19°22.8	-8.0'	55.5'
13	18°59.5	28.7	308°52.7	11.8'	19°14.8	-8.1'	55.4'
14	33°59.4	29.3	323°23.4	11.8'	19°06.6	-8.2'	55.4'
15	48°59.4	30.0	337°54.3	11.9'	18°58.4	-8.3'	55.4'
16	63°59.3	30.7	352°25.2	12.0'	18°50.1	-8.4'	55.4'
17	78°59.2	31.4	6°56.2	12.1'	18°41.8	-8.5'	55.3'
18	93°59.2	S17°32.1	21°27.3	12.2'	S18°33.3	-8.5'	55.3'
19	108°59.1	32.8	35°58.5	12.3'	18°24.8	-8.6'	55.3'
20	123°59.0	33.4	50°29.8	12.3'	18°16.2	-8.7'	55.3'
21	138°58.9	34.1	65°01.1	12.4'	18°07.5	-8.8'	55.2'
22	153°58.9	34.8	79°32.6	12.5'	17°58.7	-8.8'	55.2'
23	168°58.8	35.5	94°04.1	12.6'	17°49.9	-8.9'	55.2'
SD = 16.1'		$d = 0.7'$	SD = 15.2'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	06:40	08:10	10:01	13:25	15:16	16:46
N 70°	06:33	07:51	09:18	14:09	15:35	16:54
68°	06:26	07:36	08:49	14:38	15:50	17:01
66°	06:20	07:24	08:28	14:59	16:03	17:07
64°	06:15	07:14	08:11	15:16	16:13	17:12
62°	06:11	07:05	07:57	15:30	16:22	17:16
60°	06:07	06:57	07:45	15:42	16:30	17:20
N 58°	06:03	06:51	07:35	15:52	16:36	17:24
56°	05:59	06:44	07:26	16:01	16:43	17:28
54°	05:56	06:39	07:18	16:09	16:48	17:31
52°	05:53	06:34	07:11	16:16	16:53	17:34
50°	05:50	06:29	07:04	16:23	16:58	17:37
45°	05:44	06:19	06:51	16:37	17:08	17:43
N 40°	05:38	06:10	06:39	16:48	17:17	17:49
35°	05:32	06:03	06:29	16:58	17:25	17:55
30°	05:27	05:56	06:21	17:07	17:32	18:00
20°	05:16	05:43	06:06	17:22	17:45	18:11
N 10°	05:06	05:31	05:53	17:35	17:57	18:22
0°	04:54	05:19	05:40	17:47	18:09	18:34
S 10°	04:40	05:06	05:28	18:00	18:22	18:48
20°	04:24	04:51	05:14	18:14	18:37	19:05
30°	04:02	04:33	04:59	18:29	18:55	19:26
35°	03:49	04:22	04:50	18:39	19:06	19:40
40°	03:33	04:09	04:39	18:49	19:19	19:56
45°	03:12	03:54	04:27	19:02	19:35	20:17
S 50°	02:45	03:34	04:12	19:17	19:55	20:45
52°	02:31	03:25	04:05	19:24	20:05	20:59
54°	02:14	03:14	03:57	19:32	20:16	21:16
56°	01:54	03:01	03:48	19:41	20:28	21:37
58°	01:26	02:47	03:38	19:51	20:43	22:06
S 60°	00:44	02:29	03:27	20:03	21:01	22:53

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	■	■	■	■	■	■
N 70°	■	■	15:15	■	■	18:02
68°	■	15:21	14:24	■	16:12	18:51
66°	14:20	14:00	13:53	15:21	17:32	19:22
64°	13:06	13:23	13:29	16:35	18:09	19:45
62°	12:30	12:56	13:10	17:11	18:35	20:04
60°	12:03	12:35	12:54	17:37	18:56	20:18
N 58°	11:43	12:18	12:41	17:57	19:12	20:31
56°	11:26	12:03	12:30	18:14	19:26	20:42
54°	11:11	11:51	12:20	18:28	19:38	20:52
52°	10:59	11:40	12:11	18:41	19:49	21:00
50°	10:48	11:30	12:02	18:51	19:59	21:08
45°	10:25	11:09	11:45	19:14	20:19	21:24
N 40°	10:06	10:52	11:31	19:33	20:35	21:37
35°	09:50	10:38	11:19	19:48	20:48	21:48
30°	09:37	10:26	11:09	20:01	21:00	21:58
20°	09:14	10:05	10:50	20:24	21:20	22:15
N 10°	08:54	09:46	10:35	20:43	21:38	22:29
0°	08:35	09:29	10:20	21:01	21:54	22:43
S 10°	08:16	09:12	10:05	21:19	22:10	22:56
20°	07:56	08:53	09:49	21:39	22:27	23:10
30°	07:33	08:32	09:30	22:01	22:47	23:27
35°	07:19	08:19	09:20	22:14	22:58	23:36
40°	07:04	08:05	09:07	22:29	23:11	23:47
45°	06:45	07:48	08:53	22:46	23:27	23:59
S 50°	06:22	07:27	08:35	23:08	23:46	
52°	06:11	07:16	08:26	23:19	23:55	
54°	05:58	07:05	08:17	23:31		00:04
56°	05:43	06:52	08:06	23:44		00:16
58°	05:26	06:36	07:54	□	■	00:28
S 60°	05:05	06:18	07:39		00:18	00:43

November 12, 13, 14 UT (Fri., Sat., Sun.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	51°01.2	205°24.1	S13°30.0	162°03.6	S22°42.1	56°20.1	S03°54.6	218°28.4	S02°57.9	Alpheratz	357°45.3	29°09.3											
1	66°03.7	220°27.0	29.1	177°04.1	42.4	71°22.7	54.6	233°30.6	57.9	Ankaa	353°17.2	-42°14.8											
2	81°06.2	235°29.9	28.3	192°04.6	42.7	86°25.2	54.6	248°32.8	58.0	Schedar	349°42.4	56°36.2											
3	96°08.6	250°32.8	· · 27.4	207°05.1	· · 43.0	101°27.7	· · 54.6	263°35.0	· · 58.1	Diphda	348°57.5	-17°55.5											
4	111°11.1	265°35.7	26.6	222°05.6	43.2	116°30.2	54.6	278°37.3	58.2	Achernar	335°27.5	-57°10.9											
5	126°13.6	280°38.5	25.8	237°06.1	43.5	131°32.7	54.6	293°39.5	58.3	Hamal	328°02.6	23°31.0											
6	141°16.0	295°41.4	S13°24.9	252°06.6	S22°43.8	146°35.3	S03°54.7	308°41.7	S02°58.4	Polaris	318°15.3	89°18.8											
7	156°18.5	310°44.3	24.1	267°07.1	44.1	161°37.8	54.7	323°43.9	58.5	Acamar	315°19.3	-40°15.6											
8	171°20.9	325°47.2	23.3	282°07.6	44.4	176°40.3	54.7	338°46.1	58.6	Menkar	314°16.7	4°08.1											
9	186°23.4	340°50.1	· · 22.4	297°08.1	· · 44.6	191°42.8	· · 54.7	353°48.3	· · 58.7	Mirfak	308°42.6	49°54.1											
10	201°25.9	355°52.9	21.6	312°08.6	44.9	206°45.3	54.7	8°50.6	58.8	Aldebaran	290°51.2	16°31.9											
11	216°28.3	10°55.8	20.8	327°09.1	45.2	221°47.9	54.7	23°52.8	58.9	Rigel	281°13.6	-8°11.2											
12	231°30.8	25°58.6	S13°20.0	342°09.6	S22°45.5	236°50.4	S03°54.7	38°55.0	S02°59.0	Capella	280°36.8	46°00.5											
13	246°33.3	41°01.5	19.1	357°10.1	45.8	251°52.9	54.7	53°57.2	59.1	Bellatrix	278°33.7	6°21.6											
14	261°35.7	56°04.3	18.3	12°10.6	46.0	266°55.4	54.7	68°59.4	59.2	Elnath	278°14.7	28°37.0											
15	276°38.2	71°07.2	· · 17.5	27°11.1	· · 46.3	281°57.9	· · 54.8	84°01.6	· · 59.3	Alnilam	275°48.0	-1°11.6											
16	291°40.7	86°10.0	16.7	42°11.6	46.6	297°00.4	54.8	99°03.8	59.4	Betelgeuse	271°03.1	7°24.6											
17	306°43.1	101°12.9	15.9	57°12.1	46.9	312°03.0	54.8	114°06.1	59.5	Canopus	263°56.6	-52°41.9											
18	321°45.6	116°15.7	S13°15.1	72°12.6	S22°47.1	327°05.5	S03°54.8	129°08.3	S02°59.6	Sirius	258°35.1	-16°43.8											
19	336°48.1	131°18.5	14.3	87°13.1	47.4	342°08.0	54.8	144°10.5	59.7	Adhara	255°13.8	-28°59.1											
20	351°50.5	146°21.4	13.4	102°13.6	47.7	357°10.5	54.8	159°12.7	59.8	Procyon	245°01.5	5°11.8											
21	6°53.0	161°24.2	· · 12.6	117°14.1	· · 48.0	12°13.0	· · 54.8	174°14.9	02°59.9	Pollux	243°29.8	27°59.8											
22	21°55.4	176°27.0	11.8	132°14.6	48.2	27°15.5	54.8	189°17.1	03°00.0	Avior	234°18.7	-59°32.4											
23	36°57.9	191°29.8	11.0	147°15.1	48.5	42°18.0	54.8	204°19.4	00.1	Suhail	222°53.8	-43°28.4											
Mer.pass. 20:33				ν 2.9' d -0.8' m -4.57				ν 0.5' d 0.3' m 1.37				ν 2.5' d 0.0' m -2.69				ν 2.2' d 0.1' m 0.91							

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec				
0	52°00.4	206°32.6	S13°10.2	162°15.6	S22°48.8	57°20.6	S03°54.9	219°21.6	S03°00.2	Denebola	182°35.7	14°30.6			
1	67°02.8	221°35.5	09.4	177°16.1	49.1	72°23.1	54.9	234°23.8	00.3	Gienah	175°54.5	-17°36.1			
2	82°05.3	236°38.3	08.6	192°16.6	49.3	87°25.6	54.9	249°26.0	00.4	Acrux	173°12.1	-63°09.4			
3	97°07.8	251°41.1	· · 07.8	207°17.1	· · 49.6	102°28.1	· · 54.9	264°28.2	· · 00.5	Gacrux	172°03.6	-57°10.3			
4	112°10.2	266°43.9	07.0	222°17.6	49.9	117°30.6	54.9	279°30.4	00.6	Alioth	166°22.7	55°53.8			
5	127°12.7	281°46.6	06.2	237°18.1	50.1	132°33.1	54.9	294°32.7	00.7	Spica	158°33.5	-11°13.0			
6	142°15.2	296°49.4	S13°05.5	252°18.6	S22°50.4	147°35.6	S03°54.9	309°34.9	S03°00.8	Alkaid	153°00.7	49°15.4			
7	157°17.6	311°52.2	04.7	267°19.1	50.7	162°38.1	54.9	324°37.1	00.9	Hadar	148°51.3	-60°25.4			
8	172°20.1	326°55.0	03.9	282°19.6	50.9	177°40.7	54.9	339°39.3	01.0	Menkent	148°10.2	-36°25.3			
9	187°22.6	341°57.8	· · 03.1	297°20.1	· · 51.2	192°43.2	· · 54.9	354°41.5	· · 01.1	Arcturus	145°57.7	19°07.5			
10	202°25.0	357°00.5	02.3	312°20.6	51.5	207°45.7	54.9	9°43.8	01.2	Rigel Kent.	139°55.0	-60°52.7			
11	217°27.5	12°03.3	01.5	327°21.1	51.8	222°48.2	55.0	24°46.0	01.3	Kochab	137°20.9	74°06.5			
12	232°29.9	27°06.1	S13°00.7	342°21.6	S22°52.0	237°50.7	S03°55.0	39°48.2	S03°01.4	Zuben'ubi	137°07.9	-16°05.2			
13	247°32.4	42°08.8	13°00.0	357°22.1	52.3	252°53.2	55.0	54°50.4	01.4	Alphecca	126°12.9	26°40.7			
14	262°34.9	57°11.6	12°59.2	12°22.6	52.6	267°55.7	55.0	69°52.6	01.5	Antares	112°28.9	-26°27.3			
15	277°37.3	72°14.3	· · 58.4	27°23.1	· · 52.8	282°58.2	· · 55.0	84°54.8	· · 01.6	Atria	107°33.0	-69°02.9			
16	292°39.8	87°17.1	57.6	42°23.6	53.1	298°00.7	55.0	99°57.1	01.7	Sabik	102°15.0	-15°44.2			
17	307°42.3	102°19.8	56.9	57°24.1	53.4	313°03.2	55.0	114°59.3	01.8	Shaula	96°24.8	-37°06.7			
18	322°44.7	117°22.6	S12°56.1	72°24.6	S22°53.6	328°05.8	S03°55.0	130°01.5	S03°01.9	Rasalhague	96°08.5	12°33.3			
19	337°47.2	132°25.3	55.3	87°25.1	53.9	343°08.3	55.0	145°03.7	02.0	Eltanin	90°47.4	51°29.5			
20	352°49.7	147°28.1	54.6	102°25.6	54.2	358°10.8	55.0	160°05.9	02.1	Kaus Aust.	83°46.6	-34°22.8			
21	7°52.1	162°30.8	· · 53.8	117°26.1	· · 54.4	13°13.3	· · 55.0	175°08.2	· · 02.2	Vega	80°40.5	38°47.9			
22	22°54.6	177°33.5	53.0	132°26.6	54.7	28°15.8	55.0	190°10.4	02.3	Nunki	76°00.9	-26°17.0			
23	37°57.0	192°36.2	52.3	147°27.0	55.0	43°18.3	55.0	205°12.6	02.4	Altair	62°10.2	8°54.0			
Mer.pass. 20:29				ν 2.8' d -0.8' m -4.61				ν 2.5' d 0.0' m -2.68				ν 2.2' d 0.1' m 0.91			

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	52°59.5	207°38.9	S12°51.5	162°27.5	S22°55.2	58°20.8	S03°55.0	220°14.8	S03°02.5		
1	68°02.0	222°41.7	50.8	177°28.0	55.5	73°23.3	55.1	235°17.0	02.6		
2	83°04.4	237°44.4	50.0	192°28.5	55.7	88°25.8	55.1	250°19.2	02.7		
3	98°06.9	252°47.1	· · 49.3	207°29.0	· · 56.0	103°28.3	· · 55.1	265°21.5	· · 02.8		
4	113°09.4	267°49.8	48.5	222°29.5	56.3	118°30.8	55.1	280°23.7	02.9		
5	128°11.8	282°52.5	47.7	237°30.0	56.5	133°33.3	55.1	295°25.9	03.0		
6	143°14.3	297°55.2	S12°47.0	252°30.5	S22°56.8	148°35.8	S03°55.1	310°28.1	S03°03.1		
7	158°16.8	312°57.9	46.3	267°31.0	57.0	163°38.3	55.1	325°30.3	03.2		
8	173°19.2	328°00.5	45.5	282°31.5	57.3	178°40.8	55.1	340°32.6	03.3		
9	188°21.7	343°03.2	· · 44.8	297°32.0	· · 57.6	193°43.3	· · 55.1	355°34.8	· · 03.4		
10	203°24.2	358°05.9	44.0	312°32.5	57.8	208°45.8	55.1	10°37.0	03.5		
11	218°26.6	13°08.6	43.3	327°32.9	58.1	223°48.3	55.1	25°39.2	03.6		
12	233°29.1	28°11.2	S12°42.6	342°33.4	S22°58.3	238°50.8	S03°55.1	40°41.4	S03°03.7		
13	248°31.5	43°13.9	41.8	357°33.9	58.6	253°53.3	55.1	55°43.7	03.8		
14	263°34.0	58°16.6	41.1	12°34.4	58.9	268°55.8	55.1	70°45.9	03.8		
15	278°36.5	73°19.2	· · 40.4	27°34.9	· · 59.1	283°58.3	· · 55.1	85°48.1	· · 03.9		
16	293°38.9	88°21.9	39.6	42°35.4	59.4	299°00.8	55.1	100°50.3	04.0		
17	308°41.4	103°24.5	38.9	57°35.9	59.6	314°03.3	55.1	115°52.5	04.1		
18	323°43.9	118°27.2	S12°38.2	72°36.4	S22°59.9	329°05.8	S03°55.1	130°54.8	S03°04.2		
19	338°46.3	133°29.8	37.5	87°36.9	23°00.2	344°08.3	55.1	145°57.0	04.3		
20	353°48.8	148°32.5	36.7	102°37.4	00.4	359°10.8	55.1	160°59.2	04.4		
21	8°51.3	163°35.1	· · 36.0	117°37.8	· · 00.7	14°13.3	· · 55.1	176°01.4	· · 04.5		
22	23°53.7	178°37.7	35.3	132°38.3	00.9	29°15.8	55.2	19			

h	Sun				Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP	
0	183°58.7	S17°36.2	108°35.7	12.7'	S17°41.0	-9.0'	55.2'	
1	198°58.6	36.9	123°07.3	12.8'	17°32.0	-9.0'	55.1'	
2	213°58.6	37.5	137°39.1	12.8'	17°23.0	-9.1'	55.1'	
3	228°58.5	· · 38.2	152°10.9	12.9'	17°13.9	-9.2'	55.1'	
4	243°58.4	38.9	166°42.8	13.0'	17°04.7	-9.3'	55.1'	
5	258°58.3	39.6	181°14.8	13.1'	16°55.4	-9.3'	55.0'	
6	273°58.3	S17°40.3	195°46.9	13.1'	S16°46.1	-9.4'	55.0'	
7	288°58.2	40.9	210°19.0	13.2'	16°36.7	-9.4'	55.0'	
8	303°58.1	41.6	224°51.2	13.3'	16°27.3	-9.5'	55.0'	
9	318°58.0	· · 42.3	239°23.5	13.4'	16°17.8	-9.6'	55.0'	
10	333°57.9	43.0	253°55.9	13.4'	16°08.2	-9.6'	54.9'	
11	348°57.9	43.6	268°28.3	13.5'	15°58.6	-9.7'	54.9'	
12	3°57.8	S17°44.3	283°00.8	13.6'	S15°48.9	-9.8'	54.9'	
13	18°57.7	45.0	297°33.4	13.7'	15°39.1	-9.8'	54.9'	
14	33°57.6	45.7	312°06.1	13.7'	15°29.3	-9.9'	54.8'	
15	48°57.5	· · 46.3	326°38.8	13.8'	15°19.4	-9.9'	54.8'	
16	63°57.5	47.0	341°11.6	13.9'	15°09.5	-10.0'	54.8'	
17	78°57.4	47.7	355°44.5	13.9'	14°59.5	-10.0'	54.8'	
18	93°57.3	S17°48.4	10°17.4	14.0'	S14°49.4	-10.1'	54.8'	
19	108°57.2	49.0	24°50.4	14.1'	14°39.3	-10.2'	54.7'	
20	123°57.1	49.7	39°23.5	14.1'	14°29.2	-10.2'	54.7'	
21	138°57.1	· · 50.4	53°56.6	14.2'	14°19.0	-10.3'	54.7'	
22	153°57.0	51.0	68°29.8	14.3'	14°08.7	-10.3'	54.7'	
23	168°56.9	51.7	83°03.1	14.3'	13°58.4	-10.4'	54.7'	
SD = 16.1' $d = 0.7'$ SD = 15.0'								
Sat	GHA	Dec	GHA	ν	Dec	d	HP	
0	183°56.8	S17°52.4	97°36.4	14.4'	S13°48.0	-10.4'	54.7'	
1	198°56.7	53.0	112°09.8	14.5'	13°37.6	-10.5'	54.6'	
2	213°56.6	53.7	126°43.3	14.5'	13°27.1	-10.5'	54.6'	
3	228°56.6	· · 54.4	141°16.8	14.6'	13°16.6	-10.6'	54.6'	
4	243°56.5	55.1	155°50.4	14.6'	13°06.1	-10.6'	54.6'	
5	258°56.4	55.7	170°24.0	14.7'	12°55.5	-10.7'	54.6'	
6	273°56.3	S17°56.4	184°57.7	14.8'	S12°44.8	-10.7'	54.6'	
7	288°56.2	57.0	199°31.5	14.8'	12°34.1	-10.7'	54.5'	
8	303°56.1	57.7	214°05.3	14.9'	12°23.4	-10.8'	54.5'	
9	318°56.0	· · 58.4	228°39.1	14.9'	12°12.6	-10.8'	54.5'	
10	333°55.9	59.0	243°13.1	15.0'	12°01.7	-10.9'	54.5'	
11	348°55.9	17°59.7	257°47.0	15.0'	11°50.9	-10.9'	54.5'	
12	3°55.8	S18°00.4	272°21.1	15.1'	S11°39.9	-11.0'	54.5'	
13	18°55.7	01.0	286°55.2	15.1'	11°29.0	-11.0'	54.4'	
14	33°55.6	01.7	301°29.3	15.2'	11°18.0	-11.0'	54.4'	
15	48°55.5	· · 02.3	316°03.5	15.2'	11°07.0	-11.1'	54.4'	
16	63°55.4	03.0	330°37.7	15.3'	10°55.9	-11.1'	54.4'	
17	78°55.3	03.7	345°12.0	15.3'	10°44.8	-11.1'	54.4'	
18	93°55.2	S18°04.3	359°46.3	15.4'	S10°33.6	-11.2'	54.4'	
19	108°55.1	05.0	14°20.7	15.4'	10°22.5	-11.2'	54.4'	
20	123°55.1	05.6	28°55.1	15.5'	10°11.2	-11.3'	54.4'	
21	138°55.0	· · 06.3	43°29.6	15.5'	10°00.0	-11.3'	54.3'	
22	153°54.9	07.0	58°04.1	15.6'	09°48.7	-11.3'	54.3'	
23	168°54.8	07.6	72°38.7	15.6'	09°37.4	-11.4'	54.3'	
SD = 16.2' $d = 0.7'$ SD = 14.9'								
Sun	GHA	Dec	GHA	ν	Dec	d	HP	
0	183°54.7	S18°08.3	87°13.3	15.6'	S09°26.0	-11.4'	54.3'	
1	198°54.6	08.9	101°47.9	15.7'	09°14.6	-11.4'	54.3'	
2	213°54.5	09.6	116°22.6	15.7'	09°03.2	-11.4'	54.3'	
3	228°54.4	· · 10.2	130°57.3	15.8'	08°51.8	-11.5'	54.3'	
4	243°54.3	10.9	145°32.0	15.8'	08°40.3	-11.5'	54.3'	
5	258°54.2	11.5	160°06.8	15.8'	08°28.8	-11.5'	54.3'	
6	273°54.1	S18°12.2	174°41.7	15.9'	S08°17.2	-11.6'	54.2'	
7	288°54.0	12.8	189°16.5	15.9'	08°05.7	-11.6'	54.2'	
8	303°53.9	13.5	203°51.4	15.9'	07°54.1	-11.6'	54.2'	
9	318°53.8	· · 14.1	218°26.4	16.0'	07°42.5	-11.6'	54.2'	
10	333°53.7	14.8	233°01.3	16.0'	07°30.8	-11.7'	54.2'	
11	348°53.7	15.4	247°36.3	16.0'	07°19.1	-11.7'	54.2'	
12	3°53.6	S18°16.1	262°11.4	16.1'	S07°07.5	-11.7'	54.2'	
13	18°53.5	16.7	276°46.4	16.1'	06°55.7	-11.7'	54.2'	
14	33°53.4	17.4	291°21.5	16.1'	06°44.0	-11.8'	54.2'	
15	48°53.3	· · 18.0	305°56.6	16.1'	06°32.2	-11.8'	54.2'	
16	63°53.2	18.7	320°31.8	16.2'	06°20.4	-11.8'	54.2'	
17	78°53.1	19.3	335°07.0	16.2'	06°08.6	-11.8'	54.2'	
18	93°53.0	S18°20.0	349°42.2	16.2'	S05°56.8	-11.9'	54.2'	
19	108°52.9	20.6	4°17.4	16.2'	05°44.9	-11.9'	54.2'	
20	123°52.8	21.3	18°52.6	16.3'	05°33.1	-11.9'	54.2'	
21	138°52.7	· · 21.9	33°27.9	16.3'	05°21.2	-11.9'	54.2'	
22	153°52.6	22.6	48°03.2	16.3'	05°09.3	-11.9'	54.1'	
23	168°52.5	23.2	62°38.5	16.3'	04°57.4	-11.9'	54.1'	
SD = 16.2' $d = 0.7'$ SD = 14.8'								

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	06:52	08:24	10:32	12:56	15:03	16:35
N 70°	06:42	08:03	09:36	13:52	15:24	16:45
68°	06:35	07:47	09:03	14:25	15:41	16:53
66°	06:28	07:33	08:39	14:49	15:54	16:59
64°	06:22	07:22	08:20	15:07	16:06	17:05
62°	06:17	07:12	08:05	15:22	16:15	17:10
60°	06:13	07:04	07:53	15:35	16:24	17:15
N 58°	06:09	06:57	07:42	15:46	16:31	17:19
56°	06:05	06:50	07:32	15:56	16:38	17:23
54°	06:01	06:44	07:24	16:04	16:44	17:27
52°	05:58	06:39	07:16	16:12	16:49	17:30
50°	05:55	06:34	07:09	16:19	16:54	17:33
45°	05:48	06:23	06:55	16:33	17:05	17:40
N 40°	05:41	06:14	06:43	16:46	17:14	17:47
35°	05:35	06:06	06:32	16:56	17:23	17:53
30°	05:29	05:58	06:23	17:05	17:30	17:59
20°	05:18	05:44	06:08	17:21	17:44	18:10
N 10°	05:06	05:32	05:54	17:35	17:57	18:22
0°	04:54	05:19	05:41	17:48	18:10	18:35
S 10°	04:39	05:05	05:28	18:01	18:23	18:49
20°	04:22	04:50	05:13	18:15	18:39	19:07
30°	04:00	04:31	04:57	18:32	18:58	19:29
35°	03:46	04:20	04:47	18:42	19:09	19:43
40°	03:29	04:06	04:36	18:53	19:23	20:00
45°	03:07	03:50	04:24	19:06	19:40	20:22
S 50°	02:38	03:29	04:08	19:22	20:01	20:52
52°	02:23	03:19	04:00	19:29	20:11	21:07
54°	02:05	03:08	03:52	19:38	20:23	21:26
56°	01:43	02:54	03:43	19:47	20:36	21:49
58°	01:11	02:39	03:32	19:58	20:52	22:22
S 60°	00:06	02:20	03:20	20:11	21:12	///

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	15:06	14:27	14:01	19:49	21:58	23:50
N 70°	14:31	14:08	13:51	20:22	22:14	23:57
68°	14:06	13:54	13:44	20:45	22:27	· · ·
66°	13:47	13:42	13:37	21:03	22:38	· · ·
64°	13:31	13:32	13:31	21:18	22:46	· · ·
62°	13:18	13:23	13:27	21:30	22:54	· · ·
60°	13:07	13:16	13:22	21:41	23:00	· · ·
N 58°	12:57	13:09	13:19	21:50	23:06	· · ·
56°	12:49	13:03	13:15	21:57	23:11	· · ·
54°	12:41	12:58	13:12	22:04	23:16	· · ·
52°	12:34	12:53	13:09	22:11	23:20	· · ·
50°	12:28	12:49	13:07	22:16	23:23	· · ·
45°	12:15	12:39	13:01	22:28	23:31	· · ·
N 40°	12:04	12:32	12:57	22:38	23:38	· · ·
35°	11:54	12:25	12:53	22:47	23:43	· · ·
30°	11:46	12:19	12:49	22:54	23:48	· · ·
20°	11:31	12:08	12:43	23:07	23:57	· · ·
N 10°	11:19	11:59	12:38	23:18	· · ·	00:04
0°	11:07	11:51	12:33	23:28	· · ·	00:11
S 10°	10:55	11:42	12:27	23:39	· · ·	00:18
20°	10:42	11:33	12:22	23:49	· · ·	00:25
30°	10:27	11:22	12:16	· · ·	00:02	00:33
35°	10:19	11:16	12:12	· · ·	00:09	00:38
40°	10:09	11:09	12:08	· · ·	00:17	00:43
45°	09:58	11:01	12:03	· · ·	00:26	00:50
S 50°	09:44	10:51	11:57	00:14	00:38	00:57
52°	09:37	10:47	11:55	00:22	00:43	01:00
54°	09:30	10:42	11:52	00:29	00:48	01:04
56°	09:22	10:36	11:49	00:38	00:55	01:08
58°	09:12	10:30	11:45	00:48	01:02	01:13
S 60°	09:02	10:22	11:41	00:59	01:10	01:18

Day	Sun			Moon		
	Eqn. of Time		Mer.	Mer.Pass.	Age	
	00 ^h	12 ^h	Pass	Upper	Lower	
	mm:ss	mm:ss	hh:mm	hh:mm	hh:mm	
12	15:55	15:51	11:44	17:18	04:55	
13	15:47	15:43	11:44	18:01	05:40	
14	15:39	15:34	11:44	18:42	06:22	

Aries Venus Mars

<https://pypi.org/project/sfalmanac/>

h						
Sun			Moon			
Mon	GHA	Dec	GHA	ν	Dec	HP
0	183°52.4	S18°23.8	77°13.8	16.3'	S04°45.4	-12.0' 54.1'
1	198°52.3	24.5	91°49.1	16.4'	04°33.5	-12.0' 54.1'
2	213°52.2	25.1	106°24.5	16.4'	04°21.5	-12.0' 54.1'
3	228°52.1	· · 25.8	120°59.9	16.4'	04°09.5	-12.0' 54.1'
4	243°52.0	26.4	135°35.2	16.4'	03°57.5	-12.0' 54.1'
5	258°51.9	27.0	150°10.6	16.4'	03°45.5	-12.0' 54.1'
6	273°51.8	S18°27.7	164°46.1	16.4'	S03°33.4	-12.0' 54.1'
7	288°51.7	28.3	179°21.5	16.4'	03°21.4	-12.1' 54.1'
8	303°51.5	29.0	193°56.9	16.4'	03°09.3	-12.1' 54.1'
9	318°51.4	· · 29.6	208°32.4	16.5'	02°57.2	-12.1' 54.1'
10	333°51.3	30.2	223°07.8	16.5'	02°45.2	-12.1' 54.1'
11	348°51.2	30.9	237°43.3	16.5'	02°33.1	-12.1' 54.1'
12	3°51.1	S18°31.5	252°18.8	16.5'	S02°21.0	-12.1' 54.1'
13	18°51.0	32.1	266°54.2	16.5'	02°08.9	-12.1' 54.1'
14	33°50.9	32.8	281°29.7	16.5'	01°56.7	-12.1' 54.1'
15	48°50.8	· · 33.4	296°05.2	16.5'	01°44.6	-12.1' 54.1'
16	63°50.7	34.0	310°40.7	16.5'	01°32.5	-12.1' 54.1'
17	78°50.6	34.7	325°16.2	16.5'	01°20.3	-12.1' 54.1'
18	93°50.5	S18°35.3	339°51.7	16.5'	S01°08.2	-12.2' 54.1'
19	108°50.4	35.9	354°27.2	16.5'	00°56.0	-12.2' 54.1'
20	123°50.3	36.6	9°02.7	16.5'	00°43.9	-12.2' 54.1'
21	138°50.2	· · 37.2	23°38.2	16.5'	00°31.7	-12.2' 54.1'
22	153°50.1	37.8	38°13.7	16.5'	00°19.5	-12.2' 54.1'
23	168°50.0	38.5	52°49.2	16.5'	00°07.4	-12.2' 54.1'
SD = 16.2' d = 0.6'			SD = 14.8'			
Tue	GHA	Dec	GHA	ν	Dec	HP
0	183°49.8	S18°39.1	67°24.6	16.5'	N00°04.8	12.2' 54.1'
1	198°49.7	39.7	82°00.1	16.5'	00°17.0	12.2' 54.1'
2	213°49.6	40.3	96°35.6	16.5'	00°29.2	12.2' 54.1'
3	228°49.5	· · 41.0	111°11.0	16.5'	00°41.3	12.2' 54.2'
4	243°49.4	41.6	125°46.5	16.4'	00°53.5	12.2' 54.2'
5	258°49.3	42.2	140°21.9	16.4'	01°05.7	12.2' 54.2'
6	273°49.2	S18°42.9	154°57.4	16.4'	N01°17.9	12.2' 54.2'
7	288°49.1	43.5	169°32.8	16.4'	01°30.0	12.2' 54.2'
8	303°49.0	44.1	184°08.2	16.4'	01°42.2	12.2' 54.2'
9	318°48.8	· · 44.7	198°43.6	16.4'	01°54.4	12.2' 54.2'
10	333°48.7	45.3	213°19.0	16.4'	02°06.6	12.2' 54.2'
11	348°48.6	46.0	227°54.3	16.3'	02°18.7	12.2' 54.2'
12	3°48.5	S18°46.6	242°29.7	16.3'	N02°30.9	12.2' 54.2'
13	18°48.4	47.2	257°05.0	16.3'	02°43.0	12.1' 54.2'
14	33°48.3	47.8	271°40.3	16.3'	02°55.2	12.1' 54.2'
15	48°48.2	· · 48.5	286°15.6	16.3'	03°07.3	12.1' 54.2'
16	63°48.0	49.1	300°50.9	16.3'	03°19.4	12.1' 54.2'
17	78°47.9	49.7	315°26.1	16.2'	03°31.6	12.1' 54.2'
18	93°47.8	S18°50.3	330°01.4	16.2'	N03°43.7	12.1' 54.2'
19	108°47.7	50.9	344°36.6	16.2'	03°55.8	12.1' 54.3'
20	123°47.6	51.5	359°11.8	16.2'	04°07.9	12.1' 54.3'
21	138°47.5	· · 52.2	13°46.9	16.1'	04°20.0	12.1' 54.3'
22	153°47.3	52.8	28°22.1	16.1'	04°32.0	12.1' 54.3'
23	168°47.2	53.4	42°57.2	16.1'	04°44.1	12.0' 54.3'
SD = 16.2' d = 0.6'			SD = 14.8'			
Wed	GHA	Dec	GHA	ν	Dec	HP
0	183°47.1	S18°54.0	57°32.2	16.0'	N04°56.1	12.0' 54.3'
1	198°47.0	54.6	72°07.3	16.0'	05°08.2	12.0' 54.3'
2	213°46.9	55.2	86°42.3	16.0'	05°20.2	12.0' 54.3'
3	228°46.8	· · 55.9	101°17.3	16.0'	05°32.2	12.0' 54.3'
4	243°46.6	56.5	115°52.2	15.9'	05°44.2	12.0' 54.3'
5	258°46.5	57.1	130°27.2	15.9'	05°56.2	12.0' 54.4'
6	273°46.4	S18°57.7	145°02.1	15.9'	N06°08.1	11.9' 54.4'
7	288°46.3	58.3	159°36.9	15.8'	06°20.0	11.9' 54.4'
8	303°46.2	58.9	174°11.7	15.8'	06°32.0	11.9' 54.4'
9	318°46.0	18°59.5	188°46.5	15.7'	06°43.9	11.9' 54.4'
10	333°45.9	19°00.1	203°21.3	15.7'	06°55.7	11.9' 54.4'
11	348°45.8	00.7	217°56.0	15.7'	07°07.6	11.8' 54.4'
12	3°45.7	S19°01.3	232°30.6	15.6'	N07°19.4	11.8' 54.4'
13	18°45.5	02.0	247°05.3	15.6'	07°31.3	11.8' 54.4'
14	33°45.4	02.6	261°39.8	15.5'	07°43.1	11.8' 54.5'
15	48°45.3	· · 03.2	276°14.4	15.5'	07°54.8	11.7' 54.5'
16	63°45.2	03.8	290°48.9	15.5'	08°06.6	11.7' 54.5'
17	78°45.0	04.4	305°23.4	15.4'	08°18.3	11.7' 54.5'
18	93°44.9	S19°05.0	319°57.8	15.4'	N08°30.0	11.7' 54.5'
19	108°44.8	05.6	334°32.1	15.3'	08°41.6	11.6' 54.5'
20	123°44.7	06.2	349°06.5	15.3'	08°53.3	11.6' 54.5'
21	138°44.5	· · 06.8	3°40.7	15.2'	09°04.9	11.6' 54.6'
22	153°44.4	07.4	18°15.0	15.2'	09°16.5	11.6' 54.6'
23	168°44.3	08.0	32°49.2	15.1'	09°28.0	11.5' 54.6'
SD = 16.2' d = 0.6'			SD = 14.8'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:03	08:38	11:25	12:03	14:50	16:26
N 70°	06:52	08:15	09:55	13:33	15:14	16:36
68°	06:43	07:57	09:17	14:12	15:32	16:45
66°	06:36	07:42	08:50	14:38	15:46	16:53
64°	06:29	07:30	08:30	14:59	15:59	16:59
62°	06:24	07:20	08:14	15:15	16:09	17:05
60°	06:19	07:11	08:00	15:29	16:18	17:10
N 58°	06:14	07:03	07:48	15:40	16:26	17:15
56°	06:10	06:56	07:38	15:51	16:33	17:19
54°	06:06	06:49	07:29	16:00	16:39	17:23
52°	06:02	06:44	07:21	16:08	16:45	17:26
50°	05:59	06:38	07:14	16:15	16:51	17:30
45°	05:51	06:27	06:59	16:30	17:02	17:38
N 40°	05:44	06:17	06:46	16:43	17:12	17:45
35°	05:38	06:08	06:35	16:54	17:21	17:52
30°	05:31	06:00	06:26	17:04	17:29	17:58
20°	05:19	05:46	06:09	17:20	17:43	18:10
N 10°	05:07	05:33	05:55	17:35	17:57	18:22
0°	04:54	05:19	05:41	17:48	18:10	18:36
S 10°	04:39	05:05	05:27	18:02	18:24	18:51
20°	04:21	04:49	05:13	18:17	18:41	19:09
30°	03:58	04:29	04:55	18:34	19:01	19:32
35°	03:43	04:17	04:45	18:45	19:13	19:47
40°	03:26	04:03	04:34	18:56	19:27	20:05
45°	03:03	03:46	04:20	19:10	19:44	20:28
S 50°	02:32	03:24	04:04	19:27	20:06	20:59
52°	02:16	03:14	03:56	19:35	20:17	21:15
54°	01:57	03:02	03:47	19:44	20:29	21:35
56°	01:32	02:47	03:37	19:54	20:44	22:01
58°	00:54	02:31	03:26	20:05	21:01	22:42
S 60°	////	02:10	03:12	20:19	21:22	////

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	13:39	13:19	12:57	· · ·	01:37	03:25
N 70°	13:37	13:23	13:08	· · ·	01:36	03:17
68°	13:34	13:25	13:16	00:03	01:36	03:10
66°	13:32	13:28	13:24	00:08	01:36	03:05
64°	13:31	13:30	13:30	00:12	01:35	03:00
62°	13:29	13:32	13:35	00:15	01:35	02:56
60°	13:28	13:34	13:40	00:18	01:35	02:52
N 58°	13:27	13:35	13:44	00:21	01:35	02:49
56°	13:26	13:36	13:48	00:23	01:35	02:47
54°	13:25	13:38	13:51	00:25	01:35	02:44
52°	13:24	13:39	13:54	00:27	01:34	02:42
50°	13:23	13:40	13:57	00:29	01:34	02:40
45°	13:22	13:42	14:03	00:33	01:34	02:36
N 40°	13:20	13:44	14:08	00:36	01:34	02:32
35°	13:19	13:45	14:12	00:39	01:34	02:29
30°	13:18	13:47	14:16	00:41	01:33	02:26
20°	13:16	13:49	14:23	00:45	01:33	02:21
N 10°	13:15	13:51	14:29	00:49	01:33	02:17
0°	13:13	13:54	14:35	00:52	01:33	02:13
S 10°	13:12	13:56	14:40	00:55	01:32	02:09
20°	13:10	13:58	14:47	00:59	01:32	02:05
30°	13:08	14:00	14:54	01:03	01:32	02:00
35°	13:07	14:02	14:58	01:05	01:31	01:58
40°	13:06	14:04	15:02	01:08	01:31	01:55
45°	13:04	14:06	15:08	01:11	01:31	01:51
S 50°	13:03	14:08	15:14	01:14	01:30	01:47
52°	13:02	14:09	15:17	01:16	01:30	01:45
54°	13:01	14:11	15:21	01:17	01:30	01:43
56°	13:00	14:12	15:24	01:19	01:30	01:41
58°	12:59	14:13	15:29	01:21	01:30	01:38
S 60°	12:58	14:15	15:33	01:24	01:29	01:35

Day	Sun		Mer. Pass	Moon		Age 9-11 62-79%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
15	15:29	15:25	11:45	19:23	07:03	
16	15:19	15:14	11:45	20:03	07:43	
17	15:08	15:03	11:45	20:45	08:24	

November 18, 19, 20 UT (Thu., Fri., Sat.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	56°56.1	211°41.4	S11°47.6	163°13.8	S23°18.7	62°19.8	S03°55.0	223°48.2	S03°11.5					Alpheratz	357°45.3	29°09.3	
1	71°58.5	226°43.8	47.1	178°14.3	18.9	77°22.3	55.0	238°50.4	11.6					Ankaa	353°17.2	-42°14.8	
2	87°01.0	241°46.1	46.5	193°14.8	19.2	92°24.8	55.0	253°52.6	11.7					Schedar	349°42.4	56°36.2	
3	102°03.5	256°48.4	· · 45.9	208°15.2	· · 19.4	107°27.2	· · 55.0	268°54.9	· · 11.8					Diphda	348°57.5	-17°55.5	
4	117°05.9	271°50.8	45.4	223°15.7	19.6	122°29.7	55.0	283°57.1	11.9					Achernar	335°27.5	-57°10.9	
5	132°08.4	286°53.1	44.8	238°16.2	19.8	137°32.2	54.9	298°59.3	12.0					Hamal	328°02.6	23°31.0	
6	147°10.8	301°55.4	S11°44.3	253°16.7	S23°20.0	152°34.7	S03°54.9	314°01.5	S03°12.1					Polaris	318°15.5	89°18.8	
7	162°13.3	316°57.7	43.7	268°17.1	20.3	167°37.1	54.9	329°03.8	12.2					Acamar	315°19.3	-40°15.6	
8	177°15.8	332°00.0	43.1	283°17.6	20.5	182°39.6	54.9	344°06.0	12.3					Menkar	314°16.7	4°08.1	
9	192°18.2	347°02.3	· · 42.6	298°18.1	· · 20.7	197°42.1	· · 54.9	359°08.2	· · 12.4					Mirfak	308°42.6	49°54.1	
10	207°20.7	2°04.6	42.0	313°18.5	20.9	212°44.5	54.9	14°10.4	12.5					Aldebaran	290°51.2	16°31.9	
11	222°23.2	17°06.9	41.5	328°19.0	21.2	227°47.0	54.9	29°12.7	12.5					Rigel	281°13.5	-8°11.3	
12	237°25.6	32°09.2	S11°40.9	343°19.5	S23°21.4	242°49.5	S03°54.9	44°14.9	S03°12.6					Capella	280°36.8	46°00.5	
13	252°28.1	47°11.5	40.4	358°19.9	21.6	257°52.0	54.9	59°17.1	12.7					Bellatrix	278°33.7	6°21.6	
14	267°30.6	62°13.8	39.9	13°20.4	21.8	272°54.4	54.9	74°19.4	12.8					Elnath	278°14.6	28°37.0	
15	282°33.0	77°16.1	· · 39.3	28°20.9	· · 22.0	287°56.9	· · 54.9	89°21.6	· · 12.9					Alnilam	275°48.0	-1°11.6	
16	297°35.5	92°18.4	38.8	43°21.3	22.3	302°59.4	54.8	104°23.8	13.0					Betelgeuse	271°03.0	7°24.6	
17	312°38.0	107°20.6	38.2	58°21.8	22.5	318°01.8	54.8	119°26.0	13.1					Canopus	263°56.5	-52°41.9	
18	327°40.4	122°22.9	S11°37.7	73°22.3	S23°22.7	333°04.3	S03°54.8	134°28.3	S03°13.2					Sirius	258°35.1	-16°43.8	
19	342°42.9	137°25.2	37.2	88°22.7	22.9	348°06.8	54.8	149°30.5	13.3					Adhara	255°13.7	-28°59.1	
20	357°45.3	152°27.4	36.6	103°23.2	23.1	3°09.2	54.8	164°32.7	13.4					Procyon	245°01.5	5°11.8	
21	12°47.8	167°29.7	· · 36.1	118°23.7	· · 23.3	18°11.7	· · 54.8	179°34.9	· · 13.5					Pollux	243°29.8	27°59.8	
22	27°50.3	182°32.0	35.6	133°24.1	23.6	33°14.2	54.8	194°37.2	13.6					Avior	234°18.6	-59°32.4	
23	42°52.7	197°34.2	35.1	148°24.6	23.8	48°16.6	54.8	209°39.4	13.6					Suhail	222°53.8	-43°28.4	
Mer.pass. 20:09			ν 2.3' d -0.6' m -4.78	ν 0.5' d 0.2' m 1.35	ν 2.5' d -0.0' m -2.64	ν 2.2' d 0.1' m 0.91											

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	57°55.2	212°36.5	S11°34.5	163°25.1	S23°24.0	63°19.1	S03°54.8	224°41.6	S03°13.7
1	72°57.7	227°38.7	34.0	178°25.5	24.2	78°21.6	54.8	239°43.9	13.8
2	88°00.1	242°40.9	33.5	193°26.0	24.4	93°24.0	54.7	254°46.1	13.9
3	103°02.6	257°43.2	· · 33.0	208°26.5	· · 24.6	108°26.5	· · 54.7	269°48.3	· · 14.0
4	118°05.1	272°45.4	32.5	223°26.9	24.9	123°29.0	54.7	284°50.5	14.1
5	133°07.5	287°47.6	32.0	238°27.4	25.1	138°31.4	54.7	299°52.8	14.2
6	148°10.0	302°49.9	S11°31.5	253°27.9	S23°25.3	153°33.9	S03°54.7	314°55.0	S03°14.3
7	163°12.5	317°52.1	30.9	268°28.3	25.5	168°36.4	54.7	329°57.2	14.4
8	178°14.9	332°54.3	30.4	283°28.8	25.7	183°38.8	54.7	344°59.5	14.5
9	193°17.4	347°56.5	· · 29.9	298°29.3	· · 25.9	198°41.3	· · 54.7	0°01.7	· · 14.6
10	208°19.8	2°58.7	29.4	313°29.7	26.1	213°43.7	54.6	15°03.9	14.6
11	223°22.3	18°00.9	28.9	328°30.2	26.3	228°46.2	54.6	30°06.2	14.7
12	238°24.8	33°03.1	S11°28.4	343°30.7	S23°26.6	243°48.7	S03°54.6	45°08.4	S03°14.8
13	253°27.2	48°05.3	27.9	358°31.1	26.8	258°51.1	54.6	60°10.6	14.9
14	268°29.7	63°07.5	27.5	13°31.6	27.0	273°53.6	54.6	75°12.8	15.0
15	283°32.2	78°09.7	· · 27.0	28°32.0	· · 27.2	288°56.1	· · 54.6	90°15.1	· · 15.1
16	298°34.6	93°11.9	26.5	43°32.5	27.4	303°58.5	54.6	105°17.3	15.2
17	313°37.1	108°14.1	26.0	58°33.0	27.6	319°01.0	54.6	120°19.5	15.3
18	328°39.6	123°16.3	S11°25.5	73°33.4	S23°27.8	334°03.4	S03°54.5	135°21.8	S03°15.4
19	343°42.0	138°18.5	25.0	88°33.9	28.0	349°05.9	54.5	150°24.0	15.5
20	358°44.5	153°20.6	24.5	103°34.4	28.2	4°08.4	54.5	165°26.2	15.5
21	13°46.9	168°22.8	· · 24.1	118°34.8	· · 28.4	19°10.8	· · 54.5	180°28.5	· · 15.6
22	28°49.4	183°25.0	23.6	133°35.3	28.6	34°13.3	54.5	195°30.7	15.7
23	43°51.9	198°27.1	23.1	148°35.7	28.9	49°15.7	54.5	210°32.9	15.8
Mer.pass. 20:05			ν 2.2' d -0.5' m -4.81	ν 0.5' d 0.2' m 1.34	ν 2.5' d -0.0' m -2.63	ν 2.2' d 0.1' m 0.91			

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	58°54.3	213°29.3	S11°22.6	163°36.2	S23°29.1	64°18.2	S03°54.5	225°35.1	S03°15.9
1	73°56.8	228°31.4	22.2	178°36.7	29.3	79°20.6	54.4	240°37.4	16.0
2	88°59.3	243°33.6	21.7	193°37.1	29.5	94°23.1	54.4	255°39.6	16.1
3	104°01.7	258°35.7	· · 21.2	208°37.6	· · 29.7	109°25.6	· · 54.4	270°41.8	· · 16.2
4	119°04.2	273°37.9	20.8	223°38.0	29.9	124°28.0	54.4	285°44.1	16.3
5	134°06.7	288°40.0	20.3	238°38.5	30.1	139°30.5	54.4	300°46.3	16.4
6	149°09.1	303°42.2	S11°19.8	253°39.0	S23°30.3	154°32.9	S03°54.4	315°48.5	S03°16.4
7	164°11.6	318°44.3	19.4	268°39.4	30.5	169°35.4	54.3	330°50.8	16.5
8	179°14.1	333°46.4	18.9	283°39.9	30.7	184°37.8	54.3	345°53.0	16.6
9	194°16.5	348°48.5	· · 18.5	298°40.3	· · 30.9	199°40.3	· · 54.3	0°55.2	· · 16.7
10	209°19.0	3°50.7	18.0	313°40.8	31.1	214°42.8	54.3	15°57.5	16.8
11	224°21.4	18°52.8	17.6	328°41.3	31.3	229°45.2	54.3	30°59.7	16.9
12	239°23.9	33°54.9	S11°17.1	343°41.7	S23°31.5	244°47.7	S03°54.3	46°01.9	S03°17.0
13	254°26.4	48°57.0	16.7	358°42.2	31.7	259°50.1	54.3	61°04.2	17.1
14	269°28.8	63°59.1	16.2	13°42.6	31.9	274°52.6	54.2	76°06.4	17.2
15	284°31.3	79°01.2	· · 15.8	28°43.1	· · 32.1	289°55.0	· · 54.2	91°08.6	· · 17.3
16	299°33.8	94°03.3	15.3	43°43.6	32.3	304°57.5	54.2	106°10.8	17.3
17	314°36.2	109°05.4	14.9	58°44.0	32.5	319°59.9	54.2	121°13.1	17.4
18	329°38.7	124°07.5	S11°14.5	73°44.5	S23°32.7	335°02.4	S03°54.2	136°15.3	S03°17.5
19	344°41.2	139°09.6	14.0	88°44.9	32.9	350°04.8	54.2	151°17.5	17.6
20	359°43.6	154°11.7	13.6	103°45.4	33.1	5°07.3	54.1	166°19.8	17.7
21	14°46.1	169°13.8	· · 13.2	118°45.8	· · 33.3	20°09.7	· · 54.1	181°22.0	· · 17.8
22	29°48.6	184°15.8	12.8	133°46.3	33.5	35°12.2	54.1	196°24.2	17.9
23	44°51.0	199°17.9	12.3	148°46.8	33.7	50°14.6	54.1	211°26.5	18.0
Mer.pass. 20:01			ν 2.2' d -0.5' m -4.83	ν 0.5' d 0.2' m 1.34	ν 2.5' d -0.0' m -2.62	ν 2.2' d 0.1' m 0.90			

Nov 18 Thu	SHA	Mer.pass
Venus	154°45.4	09:52
Mars	106°17.8	13:07
Jupiter	5°23.8	19:47
Saturn	166°52.1	09:03

Nov 19 Fri	SHA	Mer.pass
Venus	154°41.2	09:48
Mars	105°29.9	13:06
Jupiter	5°23.9	19:43
Saturn	166°46.4	09:00

Nov 20 Sat	SHA	Mer.pass
Venus	154°34.9	09:45
Mars	104°41.9	13:05
Jupiter	5°23.9	19:40
Saturn	166°40.8	08:56

Horizontal parallax		
Venus:	0.4	
Mars:	0.1	

h	Sun			Moon			
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	183°44.2	S19°08.6	47°23.3	15.1'	N09°39.6	11.5'	54.6'
1	198°44.0	09.2	61°57.4	15.0'	09°51.0	11.5'	54.6'
2	213°43.9	09.8	76°31.4	15.0'	10°02.5	11.4'	54.6'
3	228°43.8	· · 10.4	91°05.4	14.9'	10°13.9	11.4'	54.6'
4	243°43.7	11.0	105°39.3	14.9'	10°25.3	11.4'	54.7'
5	258°43.5	11.6	120°13.2	14.8'	10°36.7	11.3'	54.7'
6	273°43.4	S19°12.2	134°47.0	14.8'	N10°48.0	11.3'	54.7'
7	288°43.3	12.8	149°20.7	14.7'	10°59.3	11.2'	54.7'
8	303°43.1	13.4	163°54.4	14.6'	11°10.5	11.2'	54.7'
9	318°43.0	· · 14.0	178°28.1	14.6'	11°21.7	11.2'	54.7'
10	333°42.9	14.6	193°01.7	14.5'	11°32.9	11.1'	54.8'
11	348°42.8	15.2	207°35.2	14.5'	11°44.0	11.1'	54.8'
12	3°42.6	S19°15.8	222°08.7	14.4'	N11°55.1	11.0'	54.8'
13	18°42.5	16.3	236°42.1	14.3'	12°06.1	11.0'	54.8'
14	33°42.4	16.9	251°15.4	14.3'	12°17.2	11.0'	54.8'
15	48°42.2	· · 17.5	265°48.7	14.2'	12°28.1	10.9'	54.8'
16	63°42.1	18.1	280°21.9	14.2'	12°39.0	10.9'	54.9'
17	78°42.0	18.7	294°55.1	14.1'	12°49.9	10.8'	54.9'
18	93°41.8	S19°19.3	309°28.2	14.0'	N13°00.7	10.8'	54.9'
19	108°41.7	19.9	324°01.2	14.0'	13°11.5	10.7'	54.9'
20	123°41.6	20.5	338°34.2	13.9'	13°22.2	10.7'	54.9'
21	138°41.4	· · 21.1	353°07.1	13.8'	13°32.9	10.6'	55.0'
22	153°41.3	21.7	7°39.9	13.8'	13°43.5	10.6'	55.0'
23	168°41.2	22.2	22°12.7	13.7'	13°54.1	10.5'	55.0'
SD = 16.2'			SD = 14.9'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	183°41.0	S19°22.8	36°45.4	13.6'	N14°04.6	10.5'	55.0'
1	198°40.9	23.4	51°18.0	13.6'	14°15.1	10.4'	55.0'
2	213°40.8	24.0	65°50.6	13.5'	14°25.5	10.4'	55.0'
3	228°40.6	· · 24.6	80°23.1	13.4'	14°35.9	10.3'	55.1'
4	243°40.5	25.2	94°55.5	13.4'	14°46.2	10.2'	55.1'
5	258°40.3	25.8	109°27.9	13.3'	14°56.4	10.2'	55.1'
6	273°40.2	S19°26.3	124°00.1	13.2'	N15°06.6	10.1'	55.1'
7	288°40.1	26.9	138°32.3	13.1'	15°16.7	10.1'	55.1'
8	303°39.9	27.5	153°04.5	13.1'	15°26.8	10.0'	55.2'
9	318°39.8	· · 28.1	167°36.5	13.0'	15°36.8	9.9'	55.2'
10	333°39.7	28.7	182°08.5	12.9'	15°46.8	9.9'	55.2'
11	348°39.5	29.2	196°40.4	12.8'	15°56.7	9.8'	55.2'
12	3°39.4	S19°29.8	211°12.3	12.8'	N16°06.5	9.8'	55.2'
13	18°39.2	30.4	225°44.0	12.7'	16°16.2	9.7'	55.3'
14	33°39.1	31.0	240°15.7	12.6'	16°25.9	9.6'	55.3'
15	48°39.0	· · 31.6	254°47.3	12.5'	16°35.5	9.6'	55.3'
16	63°38.8	32.1	269°18.9	12.5'	16°45.1	9.5'	55.3'
17	78°38.7	32.7	283°50.3	12.4'	16°54.6	9.4'	55.4'
18	93°38.5	S19°33.3	298°21.7	12.3'	N17°04.0	9.3'	55.4'
19	108°38.4	33.9	312°53.0	12.2'	17°13.3	9.3'	55.4'
20	123°38.2	34.4	327°24.2	12.1'	17°22.6	9.2'	55.4'
21	138°38.1	· · 35.0	341°55.4	12.1'	17°31.8	9.1'	55.4'
22	153°38.0	35.6	356°26.4	12.0'	17°40.9	9.0'	55.5'
23	168°37.8	36.1	10°57.4	11.9'	17°50.0	9.0'	55.5'
SD = 16.2'			SD = 15.0'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	183°37.7	S19°36.7	25°28.3	11.8'	N17°58.9	8.9'	55.5'
1	198°37.5	37.3	39°59.1	11.7'	18°07.8	8.8'	55.5'
2	213°37.4	37.9	54°29.9	11.7'	18°16.6	8.7'	55.5'
3	228°37.2	· · 38.4	69°00.6	11.6'	18°25.4	8.6'	55.6'
4	243°37.1	39.0	83°31.1	11.5'	18°34.0	8.6'	55.6'
5	258°37.0	39.6	98°01.7	11.4'	18°42.6	8.5'	55.6'
6	273°36.8	S19°40.1	112°32.1	11.3'	N18°51.1	8.4'	55.6'
7	288°36.7	40.7	127°02.4	11.3'	18°59.5	8.3'	55.7'
8	303°36.5	41.3	141°32.7	11.2'	19°07.8	8.2'	55.7'
9	318°36.4	· · 41.8	156°02.9	11.1'	19°16.0	8.1'	55.7'
10	333°36.2	42.4	170°33.0	11.0'	19°24.1	8.0'	55.7'
11	348°36.1	43.0	185°03.0	10.9'	19°32.2	8.0'	55.7'
12	3°35.9	S19°43.5	199°32.9	10.9'	N19°40.1	7.9'	55.8'
13	18°35.8	44.1	214°02.8	10.8'	19°48.0	7.8'	55.8'
14	33°35.6	44.7	228°32.6	10.7'	19°55.8	7.7'	55.8'
15	48°35.5	· · 45.2	243°02.3	10.6'	20°03.5	7.6'	55.8'
16	63°35.3	45.8	257°31.9	10.5'	20°11.1	7.5'	55.9'
17	78°35.2	46.3	272°01.4	10.5'	20°18.5	7.4'	55.9'
18	93°35.0	S19°46.9	286°30.9	10.4'	N20°25.9	7.3'	55.9'
19	108°34.9	47.5	301°00.2	10.3'	20°33.2	7.2'	55.9'
20	123°34.7	48.0	315°29.5	10.2'	20°40.4	7.1'	55.9'
21	138°34.6	· · 48.6	329°58.7	10.1'	20°47.5	7.0'	56.0'
22	153°34.4	49.1	344°27.9	10.1'	20°54.5	6.9'	56.0'
23	168°34.3	49.7	358°56.9	10.0'	21°01.4	6.8'	56.0'
SD = 16.2'			SD = 15.1'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:13	08:53	■	■	14:37	16:16
N 70°	07:01	08:27	10:17	13:13	15:03	16:28
68°	06:52	08:07	09:31	13:58	15:23	16:38
66°	06:43	07:51	09:02	14:28	15:39	16:46
64°	06:36	07:38	08:40	14:50	15:52	16:53
62°	06:30	07:27	08:22	15:08	16:03	17:00
60°	06:25	07:17	08:07	15:23	16:13	17:05
N 58°	06:20	07:09	07:55	15:35	16:21	17:10
56°	06:15	07:01	07:44	15:46	16:29	17:15
54°	06:11	06:55	07:35	15:55	16:36	17:19
52°	06:07	06:48	07:26	16:04	16:42	17:23
50°	06:03	06:43	07:19	16:11	16:47	17:27
45°	05:55	06:31	07:03	16:28	17:00	17:36
N 40°	05:47	06:20	06:49	16:41	17:10	17:43
35°	05:40	06:11	06:38	16:52	17:19	17:50
30°	05:34	06:03	06:28	17:02	17:28	17:57
20°	05:21	05:48	06:11	17:20	17:43	18:10
N 10°	05:08	05:34	05:56	17:35	17:57	18:23
0°	04:54	05:20	05:42	17:49	18:11	18:36
S 10°	04:39	05:05	05:27	18:03	18:26	18:52
20°	04:20	04:48	05:12	18:19	18:43	19:11
30°	03:56	04:28	04:54	18:37	19:03	19:35
35°	03:41	04:15	04:44	18:48	19:16	19:50
40°	03:23	04:01	04:32	19:00	19:31	20:09
45°	02:59	03:43	04:17	19:14	19:49	20:33
S 50°	02:26	03:20	04:00	19:32	20:12	21:06
52°	02:09	03:09	03:52	19:40	20:23	21:23
54°	01:48	02:56	03:42	19:50	20:36	21:45
56°	01:20	02:41	03:32	20:00	20:52	22:15
58°	00:32	02:23	03:20	20:12	21:10	23:08
S 60°	////	02:00	03:06	20:27	21:33	////

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	12:30	11:48	□	05:20	07:36	□
N 70°	12:50	12:27	11:34	05:02	06:58	09:30
68°	13:06	12:54	12:35	04:48	06:33	08:31
66°	13:19	13:15	13:10	04:36	06:13	07:57
64°	13:30	13:31	13:35	04:27	05:57	07:32
62°	13:39	13:45	13:55	04:18	05:44	07:13
60°	13:47	13:57	14:12	04:11	05:33	06:57
N 58°	13:54	14:07	14:25	04:05	05:24	06:44
56°	14:00	14:16	14:37	04:00	05:15	06:32
54°	14:06	14:24	14:48	03:55	05:08	06:22
52°	14:11	14:31	14:57	03:51	05:01	06:13
50°	14:16	14:38	15:06	03:47	04:55	06:05
45°	14:26	14:52	15:23	03:38	04:43	05:49
N 40°	14:34	15:04	15:38	03:31	04:32	05:35
35°	14:41	15:14	15:51	03:25	04:23	05:23
30°	14:48	15:22	16:01	03:20	04:15	05:13
20°	14:59	15:37	16:20	03:11	04:02	04:56
N 10°	15:08	15:51	16:37	03:03	03:50	04:40
0°	15:18	16:03	16:52	02:55	03:39	04:26
S 10°	15:27	16:16	17:08	02:48	03:28	04:12
20°	15:37	16:29	17:24	02:40	03:17	03:57
30°	15:48	16:45	17:44	02:31	03:03	03:40
35°	15:55	16:54	17:55	02:25	02:56	03:30
40°	16:03	17:05	18:08	02:20	02:47	03:19
45°	16:11	17:17	18:23	02:13	02:37	03:06
S 50°	16:22	17:32	18:42	02:05	02:25	02:49
52°	16:27	17:39	18:51	02:01	02:19	02:42
54°	16:33	17:47	19:01	01:57	02:13	02:34
56°	16:39	17:55	19:13	01:52	02:06	02:24
58°	16:46	18:05	19:26	01:47	01:59	02:14
S 60°	16:54	18:17	19:42	01:42	01:50	02:01

Day	Sun		Mer. Pass hh:mm	Moon		Age 12-14 87-97%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
18	14:57	14:50	11:45	21:28	09:06	
19	14:44	14:38	11:45	22:15	09:51	
20	14:31	14:24	11:46	23:04	10:39	

November 21, 22, 23 UT (Sun., Mon., Tue.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Sun	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec			
0	59°53.5	214°20.0	S11°11.9	163°47.2	S23°33.9	65°17.1	S03°54.1	226°28.7	S03°18.1				Alpheratz	357°45.3	29°09.3		
1	74°55.9	229°22.0	11.5	178°47.7	34.1	80°19.5	54.1	241°30.9	18.1				Ankaa	353°17.2	-42°14.8		
2	89°58.4	244°24.1	11.1	193°48.1	34.3	95°22.0	54.0	256°33.2	18.2				Schedar	349°42.4	56°36.2		
3	105°00.9	259°26.2	•• 10.6	208°48.6	•• 34.5	110°24.4	•• 54.0	271°35.4	•• 18.3				Diphda	348°57.5	-17°55.5		
4	120°03.3	274°28.2	10.2	223°49.0	34.7	125°26.9	54.0	286°37.6	18.4				Achernar	335°27.6	-57°10.9		
5	135°05.8	289°30.3	09.8	238°49.5	34.9	140°29.3	54.0	301°39.9	18.5				Hamal	328°02.6	23°31.0		
6	150°08.3	304°32.3	S11°09.4	253°49.9	S23°35.1	155°31.8	S03°54.0	316°42.1	S03°18.6				Polaris	318°15.5	89°18.9		
7	165°10.7	319°34.4	09.0	268°50.4	35.2	170°34.2	53.9	331°44.3	18.7				Acamar	315°19.3	-40°15.6		
8	180°13.2	334°36.4	08.6	283°50.9	35.4	185°36.7	53.9	346°46.6	18.8				Menkar	314°16.7	4°08.1		
9	195°15.7	349°38.4	•• 08.2	298°51.3	•• 35.6	200°39.1	•• 53.9	1°48.8	•• 18.9				Mirfak	308°42.6	49°54.1		
10	210°18.1	4°40.5	07.8	313°51.8	35.8	215°41.6	53.9	16°51.0	18.9				Aldebaran	290°51.2	16°31.9		
11	225°20.6	19°42.5	07.4	328°52.2	36.0	230°44.0	53.9	31°53.3	19.0				Rigel	281°13.5	-8°11.3		
12	240°23.1	34°44.5	S11°07.0	343°52.7	S23°36.2	245°46.5	S03°53.8	46°55.5	S03°19.1				Capella	280°36.8	46°00.5		
13	255°25.5	49°46.6	06.6	358°53.1	36.4	260°48.9	53.8	61°57.7	19.2				Bellatrix	278°33.7	6°21.6		
14	270°28.0	64°48.6	06.2	13°53.6	36.6	275°51.4	53.8	77°00.0	19.3				Elnath	278°14.6	28°37.0		
15	285°30.4	79°50.6	•• 05.8	28°54.0	•• 36.8	290°53.8	•• 53.8	92°02.2	•• 19.4				Alnilam	275°47.9	-1°11.7		
16	300°32.9	94°52.6	05.4	43°54.5	37.0	305°56.3	53.8	107°04.4	19.5				Betelgeuse	271°03.0	7°24.6		
17	315°35.4	109°54.6	05.0	58°54.9	37.2	320°58.7	53.7	122°06.7	19.6				Canopus	263°56.5	-52°41.9		
18	330°37.8	124°56.6	S11°04.6	73°55.4	S23°37.3	336°01.1	S03°53.7	137°08.9	S03°19.6				Sirius	258°35.1	-16°43.8		
19	345°40.3	139°58.6	04.2	88°55.8	37.5	351°03.6	53.7	152°11.1	19.7				Adhara	255°13.7	-28°59.1		
20	0°42.8	155°00.6	03.9	103°56.3	37.7	6°06.0	53.7	167°13.4	19.8				Procyon	245°01.5	5°11.8		
21	15°45.2	170°02.6	•• 03.5	118°56.7	•• 37.9	21°08.5	•• 53.7	182°15.6	•• 19.9				Pollux	243°29.8	27°59.8		
22	30°47.7	185°04.6	03.1	133°57.2	38.1	36°10.9	53.6	197°17.8	20.0				Avior	234°18.6	-59°32.4		
23	45°50.2	200°06.6	02.7	148°57.6	38.3	51°13.4	53.6	212°20.1	20.1				Suhail	222°53.8	-43°28.4		
Mer.pass. 19:57			ν 2.1' d -0.4' m -4.84	ν 0.5' d 0.2' m 1.34	ν 2.5' d -0.0' m -2.62	ν 2.2' d 0.1' m 0.90											

Mon	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	60°52.6	215°08.6	S11°02.3	163°58.1	S23°38.5	66°15.8	S03°53.6	227°22.3
1	75°55.1	230°10.5	02.0	178°58.5	38.7	81°18.2	53.6	242°24.6
2	90°57.5	245°12.5	01.6	193°59.0	38.8	96°20.7	53.6	257°26.8
3	106°00.0	260°14.5	•• 01.2	208°59.5	•• 39.0	111°23.1	•• 53.5	272°29.0
4	121°02.5	275°16.5	00.9	223°59.9	39.2	126°25.6	53.5	287°31.3
5	136°04.9	290°18.4	00.5	239°00.4	39.4	141°28.0	53.5	302°33.5
6	151°07.4	305°20.4	S11°00.1	254°00.8	S23°39.6	156°30.5	S03°53.5	317°35.7
7	166°09.9	320°22.3	10°59.8	269°01.3	39.8	171°32.9	53.4	332°38.0
8	181°12.3	335°24.3	59.4	284°01.7	39.9	186°35.3	53.4	347°40.2
9	196°14.8	350°26.3	•• 59.1	299°02.2	•• 40.1	201°37.8	•• 53.4	2°42.4
10	211°17.3	5°28.2	58.7	314°02.6	40.3	216°40.2	53.4	17°44.7
11	226°19.7	20°30.1	58.4	329°03.0	40.5	231°42.6	53.4	32°46.9
12	241°22.2	35°32.1	S10°58.0	344°03.5	S23°40.7	246°45.1	S03°53.3	47°49.1
13	256°24.7	50°34.0	57.7	359°03.9	40.9	261°47.5	53.3	62°51.4
14	271°27.1	65°36.0	57.3	14°04.4	41.0	276°50.0	53.3	77°53.6
15	286°29.6	80°37.9	•• 57.0	29°04.8	•• 41.2	291°52.4	•• 53.3	92°55.9
16	301°32.0	95°39.8	56.6	44°05.3	41.4	306°54.8	53.2	107°58.1
17	316°34.5	110°41.7	56.3	59°05.7	41.6	321°57.3	53.2	123°00.3
18	331°37.0	125°43.7	S10°55.9	74°06.2	S23°41.8	336°59.7	S03°53.2	138°02.6
19	346°39.4	140°45.6	55.6	89°06.6	41.9	352°02.1	53.2	153°04.8
20	1°41.9	155°47.5	55.3	104°07.1	42.1	7°04.6	53.1	168°07.0
21	16°44.4	170°49.4	•• 54.9	119°07.5	•• 42.3	22°07.0	•• 53.1	183°09.3
22	31°46.8	185°51.3	54.6	134°08.0	42.5	37°09.5	53.1	198°11.5
23	46°49.3	200°53.2	54.3	149°08.4	42.6	52°11.9	53.1	213°13.7
Mer.pass. 19:53			ν 2.0' d -0.4' m -4.86	ν 0.5' d 0.2' m 1.33	ν 2.4' d -0.0' m -2.61	ν 2.2' d 0.1' m 0.90		

Tue	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	61°51.8	215°55.1	S10°54.0	164°08.9	S23°42.8	67°14.3	S03°53.0	228°16.0
1	76°54.2	230°57.0	53.6	179°09.3	43.0	82°16.8	53.0	243°18.2
2	91°56.7	245°58.9	53.3	194°09.8	43.2	97°19.2	53.0	258°20.5
3	106°59.2	261°00.8	•• 53.0	209°10.2	•• 43.3	112°21.6	•• 53.0	273°22.7
4	122°01.6	276°02.7	52.7	224°10.7	43.5	127°24.1	52.9	288°24.9
5	137°04.1	291°04.5	52.3	239°11.1	43.7	142°26.5	52.9	303°27.2
6	152°06.5	306°06.4	S10°52.0	254°11.5	S23°43.9	157°28.9	S03°52.9	318°29.4
7	167°09.0	321°08.3	51.7	269°12.0	44.0	172°31.4	52.9	333°31.6
8	182°11.5	336°10.2	51.4	284°12.4	44.2	187°33.8	52.8	348°33.9
9	197°13.9	351°12.0	•• 51.1	299°12.9	•• 44.4	202°36.2	•• 52.8	3°36.1
10	212°16.4	6°13.9	50.8	314°13.3	44.6	217°38.7	52.8	18°38.4
11	227°18.9	21°15.8	50.5	329°13.8	44.7	232°41.1	52.8	33°40.6
12	242°21.3	36°17.6	S10°50.2	344°14.2	S23°44.9	247°43.5	S03°52.7	48°42.8
13	257°23.8	51°19.5	49.9	359°14.7	45.1	262°45.9	52.7	63°45.1
14	272°26.3	66°21.3	49.6	14°15.1	45.2	277°48.4	52.7	78°47.3
15	287°28.7	81°23.2	•• 49.3	29°15.5	•• 45.4	292°50.8	•• 52.7	93°49.6
16	302°31.2	96°25.0	49.0	44°16.0	45.6	307°53.2	52.6	108°51.8
17	317°33.7	111°26.9	48.7	59°16.4	45.8	322°55.7	52.6	123°54.0
18	332°36.1	126°28.7	S10°48.4	74°16.9	S23°45.9	337°58.1	S03°52.6	138°56.3
19	347°38.6	141°30.5	48.1	89°17.3	46.1	353°00.5	52.5	153°58.5
20	2°41.0	156°32.4	47.8	104°17.8	46.3	8°02.9	52.5	169°00.7
21	17°43.5	171°34.2	•• 47.5	119°18.2	•• 46.4	23°05.4	•• 52.5	184°03.0
22	32°46.0	186°36.0	47.3	134°18.6	46.6	38°07.8	52.5	199°05.2
23	47°48.4	201°37.8	47.0	149°19.1	46.8	53°10.2	52.4	214°07.5
Mer.pass. 19:49			ν 1.9' d -0.3' m -4.87	ν 0.4' d 0.2' m 1.33	ν 2.4' d -0.0' m -2.60	ν 2.2' d 0.1' m 0.90		

Nov 21 Sun	SHA	Mer.pass
Venus	154°26.5	09:41
Mars	103°53.7	13:04
Jupiter	5°23.6	19:36
Saturn	166°35.2	08:53

Nov 22 Mon	SHA	Mer.pass
Venus	154°15.9	09:38
Mars	103°05.5	13:04
Jupiter	5°23.2	19:32
Saturn	166°29.7	08:49

Nov 23 Tue	SHA	Mer.pass
Venus	154°03.3	09:35
Mars	102°17.1	13:03
Jupiter	5°22.6	19:28
Saturn	166°24.2	08:46

Horizontal parallax		
Venus:	0.4	
Mars:	0.1	

Sun				Moon			
h	GHA	Dec		GHA	ν	Dec	HP
Sun	GHA	Dec		GHA	ν	Dec	HP
0	183°34.1	S19°50.3		13°25.9	9.9'	N21°08.2	6.7'
1	198°34.0	50.8		27°54.8	9.8'	21°14.9	6.6'
2	213°33.8	51.4		42°23.6	9.7'	21°21.5	6.5'
3	228°33.7	51.9		56°52.3	9.7'	21°27.9	6.4'
4	243°33.5	52.5		71°21.0	9.6'	21°34.3	6.3'
5	258°33.4	53.0		85°49.6	9.5'	21°40.5	6.1'
6	273°33.2	S19°53.6		100°18.1	9.4'	N21°46.7	6.0'
7	288°33.1	54.1		114°46.5	9.4'	21°52.7	5.9'
8	303°32.9	54.7		129°14.9	9.3'	21°58.6	5.8'
9	318°32.7	55.2		143°43.2	9.2'	22°04.5	5.7'
10	333°32.6	55.8		158°11.4	9.1'	22°10.1	5.6'
11	348°32.4	56.3		172°39.5	9.1'	22°15.7	5.5'
12	3°32.3	S19°56.9		187°07.5	9.0'	N22°21.2	5.3'
13	18°32.1	57.4		201°35.5	8.9'	22°26.5	5.2'
14	33°32.0	58.0		216°03.4	8.8'	22°31.8	5.1'
15	48°31.8	58.5		230°31.3	8.8'	22°36.9	5.0'
16	63°31.7	59.1		244°59.1	8.7'	22°41.9	4.9'
17	78°31.5	59.6		259°26.8	8.6'	22°46.7	4.7'
18	93°31.3	S20°00.2		273°54.4	8.6'	N22°51.5	4.6'
19	108°31.2	00.7		288°22.0	8.5'	22°56.1	4.5'
20	123°31.0	01.3		302°49.4	8.4'	23°00.6	4.4'
21	138°30.9	01.8		317°16.9	8.4'	23°05.0	4.2'
22	153°30.7	02.3		331°44.2	8.3'	23°09.2	4.1'
23	168°30.5	02.9		346°11.5	8.2'	23°13.3	4.0'
SD = 16.2' d = 0.6'				SD = 15.3'			
Mon	GHA	Dec		GHA	ν	Dec	HP
0	183°30.4	S20°03.4		0°38.8	8.2'	N23°17.3	3.9'
1	198°30.2	04.0		15°06.0	8.1'	23°21.2	3.7'
2	213°30.1	04.5		29°33.1	8.1'	23°24.9	3.6'
3	228°29.9	05.1		44°00.1	8.0'	23°28.5	3.5'
4	243°29.7	05.6		58°27.1	7.9'	23°32.0	3.3'
5	258°29.6	06.1		72°54.1	7.9'	23°35.3	3.2'
6	273°29.4	S20°06.7		87°20.9	7.8'	N23°38.5	3.1'
7	288°29.2	07.2		101°47.8	7.8'	23°41.6	2.9'
8	303°29.1	07.7		116°14.5	7.7'	23°44.5	2.8'
9	318°28.9	08.3		130°41.2	7.7'	23°47.4	2.7'
10	333°28.8	08.8		145°07.9	7.6'	23°50.0	2.5'
11	348°28.6	09.3		159°34.5	7.6'	23°52.6	2.4'
12	3°28.4	S20°09.9		174°01.1	7.5'	N23°55.0	2.3'
13	18°28.3	10.4		188°27.6	7.5'	23°57.2	2.1'
14	33°28.1	10.9		202°54.0	7.4'	23°59.3	2.0'
15	48°27.9	11.5		217°20.5	7.4'	24°01.3	1.8'
16	63°27.8	12.0		231°46.8	7.3'	24°03.2	1.7'
17	78°27.6	12.5		246°13.2	7.3'	24°04.9	1.6'
18	93°27.4	S20°13.1		260°39.5	7.3'	N24°06.4	1.4'
19	108°27.3	13.6		275°05.7	7.2'	24°07.8	1.3'
20	123°27.1	14.1		289°31.9	7.2'	24°09.1	1.1'
21	138°26.9	14.7		303°58.1	7.1'	24°10.3	1.0'
22	153°26.8	15.2		318°24.2	7.1'	24°11.3	0.9'
23	168°26.6	15.7		332°50.3	7.1'	24°12.1	0.7'
SD = 16.2' d = 0.5'				SD = 15.4'			
Tue	GHA	Dec		GHA	ν	Dec	HP
0	183°26.4	S20°16.2		347°16.4	7.0'	N24°12.8	0.6'
1	198°26.3	16.8		1°42.5	7.0'	24°13.4	0.4'
2	213°26.1	17.3		16°08.5	7.0'	24°13.8	0.3'
3	228°25.9	17.8		30°34.5	7.0'	24°14.1	0.1'
4	243°25.8	18.3		45°00.4	6.9'	24°14.2	-0.0'
5	258°25.6	18.9		59°26.3	6.9'	24°14.2	-0.2'
6	273°25.4	S20°19.4		73°52.2	6.9'	N24°14.0	-0.3'
7	288°25.2	19.9		88°18.1	6.9'	24°13.7	-0.5'
8	303°25.1	20.4		102°44.0	6.8'	24°13.2	-0.6'
9	318°24.9	21.0		117°09.8	6.8'	24°12.6	-0.7'
10	333°24.7	21.5		131°35.6	6.8'	24°11.9	-0.9'
11	348°24.6	22.0		146°01.5	6.8'	24°11.0	-1.0'
12	3°24.4	S20°22.5		160°27.2	6.8'	N24°10.0	-1.2'
13	18°24.2	23.0		174°53.0	6.8'	24°08.8	-1.3'
14	33°24.0	23.5		189°18.8	6.8'	24°07.4	-1.5'
15	48°23.9	24.1		203°44.6	6.7'	24°05.9	-1.6'
16	63°23.7	24.6		218°10.3	6.7'	24°04.3	-1.8'
17	78°23.5	25.1		232°36.0	6.7'	24°02.5	-1.9'
18	93°23.3	S20°25.6		247°01.8	6.7'	N24°00.6	-2.1'
19	108°23.2	26.1		261°27.5	6.7'	23°58.5	-2.2'
20	123°23.0	26.6		275°53.3	6.7'	23°56.3	-2.4'
21	138°22.8	27.1		290°19.0	6.7'	23°53.9	-2.5'
22	153°22.6	27.7		304°44.7	6.7'	23°51.4	-2.7'
23	168°22.5	28.2		319°10.4	6.7'	23°48.8	-2.8'
SD = 16.2' d = 0.5'				SD = 15.6'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:23	09:07	■	■	14:24	16:08
N 70°	07:10	08:38	10:44	12:48	14:53	16:21
68°	07:00	08:16	09:46	13:45	15:15	16:32
66°	06:51	07:59	09:13	14:18	15:32	16:41
64°	06:43	07:45	08:49	14:42	15:46	16:48
62°	06:36	07:34	08:30	15:01	15:58	16:55
60°	06:30	07:23	08:15	15:17	16:08	17:01
N 58°	06:25	07:14	08:02	15:30	16:17	17:07
56°	06:20	07:07	07:50	15:41	16:25	17:12
54°	06:15	06:59	07:40	15:51	16:32	17:16
52°	06:11	06:53	07:31	16:00	16:39	17:21
50°	06:07	06:47	07:24	16:08	16:45	17:25
45°	05:58	06:34	07:07	16:25	16:58	17:34
N 40°	05:50	06:23	06:53	16:39	17:09	17:42
35°	05:43	06:14	06:41	16:51	17:18	17:49
30°	05:36	06:05	06:31	17:01	17:27	17:56
20°	05:23	05:49	06:13	17:19	17:43	18:10
N 10°	05:09	05:35	05:57	17:35	17:57	18:23
0°	04:55	05:20	05:42	17:50	18:12	18:37
S 10°	04:39	05:05	05:28	18:05	18:27	18:54
20°	04:19	04:48	05:12	18:21	18:45	19:13
30°	03:55	04:27	04:53	18:39	19:06	19:38
35°	03:39	04:14	04:42	18:50	19:19	19:54
40°	03:20	03:58	04:30	19:03	19:34	20:13
45°	02:55	03:40	04:15	19:18	19:53	20:38
S 50°	02:21	03:16	03:56	19:37	20:17	21:13
52°	02:03	03:04	03:48	19:45	20:29	21:31
54°	01:40	02:51	03:38	19:55	20:43	21:55
56°	01:08	02:35	03:27	20:06	20:59	22:29
58°	////	02:15	03:14	20:19	21:19	////
S 60°	////	01:51	03:00	20:34	21:44	////

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	□	□	□	□	□	□
N 70°	□	□	□	□	□	□
68°	□	□	□	□	□	□
66°	13:05	12:55	□	09:48	11:51	□
64°	13:45	14:08	15:02	09:09	10:38	11:42
62°	14:13	14:44	15:40	08:41	10:02	11:04
60°	14:34	15:11	16:07	08:20	09:36	10:38
N 58°	14:52	15:31	16:27	08:03	09:16	10:17
56°	15:07	15:48	16:45	07:48	08:59	09:59
54°	15:20	16:03	16:59	07:36	08:45	09:45
52°	15:31	16:15	17:12	07:25	08:32	09:32
50°	15:41	16:26	17:23	07:15	08:21	09:21
45°	16:02	16:49	17:46	06:55	07:59	08:57
N 40°	16:19	17:08	18:05	06:38	07:40	08:38
35°	16:34	17:24	18:21	06:24	07:25	08:23
30°	16:46	17:37	18:34	06:12	07:11	08:09
20°	17:08	18:00	18:57	05:51	06:49	07:46
N 10°	17:27	18:21	19:17	05:33	06:29	07:25
0°	17:44	18:39	19:36	05:17	06:10	07:06
S 10°	18:02	18:58	19:55	05:00	05:52	06:48
20°	18:21	19:18	20:15	04:42	05:32	06:27
30°	18:43	19:42	20:38	04:22	05:10	06:04
35°	18:56	19:56	20:51	04:10	04:56	05:50
40°	19:11	20:12	21:07	03:56	04:41	05:34
45°	19:29	20:31	21:26	03:40	04:23	05:15
S 50°	19:51	20:54	21:49	03:20	04:00	04:51
52°	20:01	21:06	22:00	03:11	03:49	04:40
54°	20:13	21:19	22:13	03:00	03:37	04:27
56°	20:27	21:34	22:27	02:49	03:23	04:12
58°	20:43	21:51	22:44	02:35	03:07	03:54
S 60°	21:03	22:13	23:05	02:19	02:47	03:32

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Upper hh:mm	Lower hh:mm	
21	14:17	14:09	11:46	23:57	11:30	
22	14:02	13:54	11:46	—	12:25	
23	13:46	13:38	11:46	00:53	13:21	

November 24, 25, 26 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	62°50.9	216°39.6	S10°46.7	164°19.5	S23°46.9	68°12.7	S03°52.4	229°09.7	S03°24.3	Alpheratz	357°45.3	29°09.3
1	77°53.4	231°41.5	46.4	179°20.0	47.1	83°15.1	52.4	244°11.9	24.4	Ankaa	353°17.2	-42°14.9
2	92°55.8	246°43.3	46.2	194°20.4	47.3	98°17.5	52.4	259°14.2	24.5	Schedar	349°42.4	56°36.2
3	107°58.3	261°45.1	· · 45.9	209°20.9	· · 47.4	113°19.9	· · 52.3	274°16.4	· · 24.6	Diphda	348°57.5	-17°55.5
4	123°00.8	276°46.9	45.6	224°21.3	47.6	128°22.4	52.3	289°18.7	24.7	Achernar	335°27.6	-57°11.0
5	138°03.2	291°48.7	45.3	239°21.7	47.7	143°24.8	52.3	304°20.9	24.8	Hamal	328°02.6	23°31.0
6	153°05.7	306°50.5	S10°45.1	254°22.2	S23°47.9	158°27.2	S03°52.2	319°23.1	S03°24.9	Polaris	318°15.5	89°18.9
7	168°08.2	321°52.3	44.8	269°22.6	48.1	173°29.6	52.2	334°25.4	24.9	Acamar	315°19.3	-40°15.6
8	183°10.6	336°54.0	44.5	284°23.1	48.2	188°32.1	52.2	349°27.6	25.0	Menkar	314°16.7	4°08.1
9	198°13.1	351°55.8	· · 44.3	299°23.5	· · 48.4	203°34.5	· · 52.1	4°29.9	· · 25.1	Mirfak	308°42.6	49°54.1
10	213°15.5	6°57.6	44.0	314°23.9	48.6	218°36.9	52.1	19°32.1	25.2	Aldebaran	290°51.2	16°31.9
11	228°18.0	21°59.4	43.8	329°24.4	48.7	233°39.3	52.1	34°34.3	25.3	Rigel	281°13.5	-8°11.3
12	243°20.5	37°01.2	S10°43.5	344°24.8	S23°48.9	248°41.8	S03°52.1	49°36.6	S03°25.4	Capella	280°36.7	46°00.5
13	258°22.9	52°02.9	43.3	359°25.3	49.0	263°44.2	52.0	64°38.8	25.4	Bellatrix	278°33.7	6°21.6
14	273°25.4	67°04.7	43.0	14°25.7	49.2	278°46.6	52.0	79°41.1	25.5	Elnath	278°14.6	28°37.0
15	288°27.9	82°06.5	· · 42.7	29°26.1	· · 49.4	293°49.0	· · 52.0	94°43.3	· · 25.6	Alnilam	275°47.9	-1°11.7
16	303°30.3	97°08.2	42.5	44°26.6	49.5	308°51.4	51.9	109°45.5	25.7	Betelgeuse	271°03.0	7°24.5
17	318°32.8	112°10.0	42.3	59°27.0	49.7	323°53.9	51.9	124°47.8	25.8	Canopus	263°56.5	-52°41.9
18	333°35.3	127°11.8	S10°42.0	74°27.4	S23°49.8	338°56.3	S03°51.9	139°50.0	S03°25.9	Sirius	258°35.1	-16°43.8
19	348°37.7	142°13.5	41.8	89°27.9	50.0	353°58.7	51.8	154°52.3	26.0	Adhara	255°13.7	-28°59.1
20	3°40.2	157°15.3	41.5	104°28.3	50.2	9°01.1	51.8	169°54.5	26.0	Procyon	245°01.4	5°11.8
21	18°42.7	172°17.0	· · 41.3	119°28.8	· · 50.3	24°03.5	· · 51.8	184°56.8	· · 26.1	Pollux	243°29.7	27°59.8
22	33°45.1	187°18.8	41.1	134°29.2	50.5	39°06.0	51.8	199°59.0	26.2	Avior	234°18.5	-59°32.5
23	48°47.6	202°20.5	40.8	149°29.6	50.6	54°08.4	51.7	215°01.2	26.3	Suhail	222°53.7	-43°28.4
Mer.pass. 19:45		ν 1.8' d -0.3' m -4.88		ν 0.4' d 0.2' m 1.32		ν 2.4' d -0.0' m -2.59		ν 2.2' d 0.1' m 0.90				

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	63°50.0	217°22.2	S10°40.6	164°30.1	S23°50.8	69°10.8	S03°51.7	230°03.5	S03°26.4	Denebola	182°35.7	14°30.5
1	78°52.5	232°24.0	40.4	179°30.5	50.9	84°13.2	51.7	245°05.7	26.5	Gienah	175°54.4	-17°36.1
2	93°55.0	247°25.7	40.1	194°30.9	51.1	99°15.6	51.6	260°08.0	26.6	Acrux	173°12.0	-63°09.4
3	108°57.4	262°27.4	· · 39.9	209°31.4	· · 51.2	114°18.1	· · 51.6	275°10.2	· · 26.6	Gacrux	172°03.4	-57°10.3
4	123°59.9	277°29.1	39.7	224°31.8	51.4	129°20.5	51.6	290°12.4	26.7	Alioth	166°22.6	55°53.8
5	139°02.4	292°30.9	39.4	239°32.3	51.6	144°22.9	51.5	305°14.7	26.8	Spica	158°33.5	-11°13.1
6	154°04.8	307°32.6	S10°39.2	254°32.7	S23°51.7	159°25.3	S03°51.5	320°16.9	S03°26.9	Alkaid	153°00.7	49°15.3
7	169°07.3	322°34.3	39.0	269°33.1	51.9	174°27.7	51.5	335°19.2	27.0	Hadar	148°51.2	-60°25.4
8	184°09.8	337°36.0	38.8	284°33.6	52.0	189°30.1	51.4	350°21.4	27.1	Menkent	148°10.2	-36°25.3
9	199°12.2	352°37.7	· · 38.6	299°34.0	· · 52.2	204°32.6	· · 51.4	5°23.7	· · 27.1	Arcturus	145°57.7	19°07.5
10	214°14.7	7°39.4	38.4	314°34.4	52.3	219°35.0	51.4	20°25.9	27.2	Rigel Kent.	139°54.9	-60°52.7
11	229°17.1	22°41.1	38.2	329°34.9	52.5	234°37.4	51.3	35°28.1	27.3	Kochab	137°20.9	74°06.5
12	244°19.6	37°42.8	S10°37.9	344°35.3	S23°52.6	249°39.8	S03°51.3	50°30.4	S03°27.4	Zuben'ubi	137°07.8	-16°05.2
13	259°22.1	52°44.5	37.7	359°35.7	52.8	264°42.2	51.3	65°32.6	27.5	Alphecca	126°12.9	26°40.7
14	274°24.5	67°46.2	37.5	14°36.2	52.9	279°44.6	51.2	80°34.9	27.6	Antares	112°28.9	-26°27.3
15	289°27.0	82°47.9	· · 37.3	29°36.6	· · 53.1	294°47.0	· · 51.2	95°37.1	· · 27.6	Atria	107°33.0	-69°02.8
16	304°29.5	97°49.6	37.1	44°37.0	53.2	309°49.5	51.2	110°39.4	27.7	Sabik	102°15.0	-15°44.2
17	319°31.9	112°51.2	36.9	59°37.5	53.4	324°51.9	51.1	125°41.6	27.8	Shaula	96°24.8	-37°06.7
18	334°34.4	127°52.9	S10°36.7	74°37.9	S23°53.5	339°54.3	S03°51.1	140°43.8	S03°27.9	Rasalhague	96°08.5	12°33.2
19	349°36.9	142°54.6	36.5	89°38.3	53.7	354°56.7	51.1	155°46.1	28.0	Eltanin	90°47.5	51°29.4
20	4°39.3	157°56.3	36.3	104°38.8	53.8	9°59.1	51.0	170°48.3	28.1	Kaus Aust.	83°46.6	-34°22.8
21	19°41.8	172°57.9	· · 36.1	119°39.2	· · 54.0	25°01.5	· · 51.0	185°50.6	· · 28.1	Vega	80°40.6	38°47.8
22	34°44.3	187°59.6	35.9	134°39.6	54.1	40°03.9	51.0	200°52.8	28.2	Nunki	76°00.9	-26°17.0
23	49°46.7	203°01.3	35.8	149°40.1	54.2	55°06.3	50.9	215°55.1	28.3	Altair	62°10.3	8°54.0
Mer.pass. 19:41		ν 1.7' d -0.2' m -4.89		ν 0.4' d 0.2' m 1.32		ν 2.4' d -0.0' m -2.58		ν 2.2' d 0.1' m 0.90		Peacock	53°22.4	-56°42.1
										Deneb	49°33.0	45°19.5
										Enif	33°49.0	9°55.7
										Al Na'ir	27°46.0	-46°54.6
										Fomalhaut	15°25.9	-29°33.9
										Scheat	13°55.1	28°08.8
										Markab	13°40.1	15°16.1

Nov 24 Wed	SHA	Mer.pass
Venus	153°48.7	09:32
Mars	101°28.6	13:02
Jupiter	5°21.8	19:24
Saturn	166°18.8	08:42

Nov 25 Thu	SHA	Mer.pass
Venus	153°32.2	09:29
Mars	100°40.0	13:02
Jupiter	5°20.8	19:20
Saturn	166°13.4	08:38

Nov 26 Fri	SHA	Mer.pass
Venus	153°13.7	09:27
Mars	99°51.3	13:01
Jupiter	5°19.6	19:16
Saturn	166°08.1	08:35

Horizontal parallax		
Venus:	0.4	
Mars:	0.1	

Mer.pass. 19:37		ν 1.7' d -0.2' m -4.89		ν 0.4' d 0.1' m 1.32		ν 2.4' d -0.0' m -2.58		ν 2.2' d 0.1' m 0.89	
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Sun							Moon						
h	GHA	Dec	GHA	ν	Dec	d	HP						
Wed													
0	183°22.3	S20°28.7	333°36.2	6.7'	N23°46.0	-3.0'	57.6'						
1	198°22.1	29.2	348°01.9	6.7'	23°43.0	-3.1'	57.6'						
2	213°21.9	29.7	2°27.7	6.8'	23°39.9	-3.2'	57.6'						
3	228°21.8	· · 30.2	16°53.4	6.8'	23°36.7	-3.4'	57.7'						
4	243°21.6	30.7	31°19.2	6.8'	23°33.3	-3.5'	57.7'						
5	258°21.4	31.2	45°45.0	6.8'	23°29.7	-3.7'	57.7'						
6	273°21.2	S20°31.7	60°10.8	6.8'	N23°26.0	-3.8'	57.7'						
7	288°21.0	32.2	74°36.6	6.8'	23°22.2	-4.0'	57.7'						
8	303°20.9	32.7	89°02.4	6.8'	23°18.2	-4.1'	57.7'						
9	318°20.7	· · 33.2	103°28.2	6.9'	23°14.1	-4.3'	57.8'						
10	333°20.5	33.8	117°54.1	6.9'	23°09.9	-4.4'	57.8'						
11	348°20.3	34.3	132°20.0	6.9'	23°05.4	-4.5'	57.8'						
12	3°20.1	S20°34.8	146°45.8	6.9'	N23°00.9	-4.7'	57.8'						
13	18°20.0	35.3	161°11.8	6.9'	22°56.2	-4.8'	57.8'						
14	33°19.8	35.8	175°37.7	7.0'	22°51.4	-5.0'	57.9'						
15	48°19.6	· · 36.3	190°03.7	7.0'	22°46.4	-5.1'	57.9'						
16	63°19.4	36.8	204°29.7	7.0'	22°41.3	-5.3'	57.9'						
17	78°19.2	37.3	218°55.7	7.0'	22°36.0	-5.4'	57.9'						
18	93°19.0	S20°37.8	233°21.7	7.1'	N22°30.6	-5.5'	57.9'						
19	108°18.9	38.3	247°47.8	7.1'	22°25.1	-5.7'	57.9'						
20	123°18.7	38.8	262°13.9	7.1'	22°19.4	-5.8'	58.0'						
21	138°18.5	· · 39.3	276°40.0	7.2'	22°13.6	-6.0'	58.0'						
22	153°18.3	39.8	291°06.2	7.2'	22°07.6	-6.1'	58.0'						
23	168°18.1	40.2	305°32.4	7.2'	22°01.6	-6.2'	58.0'						
	SD = 16.2'	d = 0.5'			SD = 15.7'								
Thu	GHA	Dec	GHA	ν	Dec	d	HP						
0	183°17.9	S20°40.7	319°58.6	7.3'	N21°55.3	-6.4'	58.0'						
1	198°17.8	41.2	334°24.9	7.3'	21°49.0	-6.5'	58.0'						
2	213°17.6	41.7	348°51.2	7.3'	21°42.5	-6.6'	58.1'						
3	228°17.4	· · 42.2	3°17.5	7.4'	21°35.8	-6.8'	58.1'						
4	243°17.2	42.7	17°43.9	7.4'	21°29.1	-6.9'	58.1'						
5	258°17.0	43.2	32°10.3	7.5'	21°22.2	-7.0'	58.1'						
6	273°16.8	S20°43.7	46°36.8	7.5'	N21°15.2	-7.2'	58.1'						
7	288°16.6	44.2	61°03.3	7.5'	21°08.0	-7.3'	58.1'						
8	303°16.5	44.7	75°29.9	7.6'	21°00.7	-7.4'	58.2'						
9	318°16.3	· · 45.2	89°56.4	7.6'	20°53.3	-7.5'	58.2'						
10	333°16.1	45.7	104°23.1	7.7'	20°45.8	-7.7'	58.2'						
11	348°15.9	46.1	118°49.7	7.7'	20°38.1	-7.8'	58.2'						
12	3°15.7	S20°46.6	133°16.5	7.8'	N20°30.3	-7.9'	58.2'						
13	18°15.5	47.1	147°43.2	7.8'	20°22.4	-8.0'	58.2'						
14	33°15.3	47.6	162°10.0	7.9'	20°14.3	-8.2'	58.2'						
15	48°15.1	· · 48.1	176°36.9	7.9'	20°06.1	-8.3'	58.3'						
16	63°14.9	48.6	191°03.8	7.9'	19°57.9	-8.4'	58.3'						
17	78°14.7	49.1	205°30.7	8.0'	19°49.4	-8.5'	58.3'						
18	93°14.6	S20°49.5	219°57.7	8.0'	N19°40.9	-8.7'	58.3'						
19	108°14.4	50.0	234°24.8	8.1'	19°32.2	-8.8'	58.3'						
20	123°14.2	50.5	248°51.9	8.1'	19°23.5	-8.9'	58.3'						
21	138°14.0	· · 51.0	263°19.0	8.2'	19°14.6	-9.0'	58.4'						
22	153°13.8	51.5	277°46.2	8.2'	19°05.6	-9.1'	58.4'						
23	168°13.6	51.9	292°13.4	8.3'	18°56.4	-9.2'	58.4'						
	SD = 16.2'	d = 0.5'			SD = 15.8'								
Fri	GHA	Dec	GHA	ν	Dec	d	HP						
0	183°13.4	S20°52.4	306°40.7	8.3'	N18°47.2	-9.4'	58.4'						
1	198°13.2	52.9	321°08.1	8.4'	18°37.9	-9.5'	58.4'						
2	213°13.0	53.4	335°35.5	8.4'	18°28.4	-9.6'	58.4'						
3	228°12.8	· · 53.9	350°02.9	8.5'	18°18.8	-9.7'	58.4'						
4	243°12.6	54.3	4°30.4	8.5'	18°09.1	-9.8'	58.5'						
5	258°12.4	54.8	18°58.0	8.6'	17°59.3	-9.9'	58.5'						
6	273°12.2	S20°55.3	33°25.5	8.6'	N17°49.4	-10.0'	58.5'						
7	288°12.0	55.8	47°53.2	8.7'	17°39.4	-10.1'	58.5'						
8	303°11.9	56.2	62°20.9	8.8'	17°29.3	-10.2'	58.5'						
9	318°11.7	· · 56.7	76°48.6	8.8'	17°19.1	-10.3'	58.5'						
10	333°11.5	57.2	91°16.5	8.9'	17°08.8	-10.4'	58.5'						
11	348°11.3	57.6	105°44.3	8.9'	16°58.3	-10.5'	58.5'						
12	3°11.1	S20°58.1	120°12.2	9.0'	N16°47.8	-10.6'	58.6'						
13	18°10.9	58.6	134°40.2	9.0'	16°37.2	-10.7'	58.6'						
14	33°10.7	59.1	149°08.2	9.1'	16°26.4	-10.8'	58.6'						
15	48°10.5	20°59.5	163°36.2	9.1'	16°15.6	-10.9'	58.6'						
16	63°10.3	21°00.0	178°04.4	9.2'	16°04.7	-11.0'	58.6'						
17	78°10.1	00.5	192°32.5	9.2'	15°53.7	-11.1'	58.6'						
18	93°09.9	S21°00.9	207°00.7	9.3'	N15°42.6	-11.2'	58.6'						
19	108°09.7	01.4	221°29.0	9.3'	15°31.4	-11.3'	58.6'						
20	123°09.5	01.9	235°57.3	9.4'	15°20.1	-11.4'	58.7'						
21	138°09.3	· · 02.3	250°25.7	9.4'	15°08.7	-11.5'	58.7'						
22	153°09.1	02.8	264°54.1	9.5'	14°57.2	-11.6'	58.7'						
23	168°08.9	03.2	279°22.5	9.5'	14°45.6	-11.6'	58.7'						
	SD = 16.2'	d = 0.5'			SD = 15.9'								

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:33	09:21	■	■	14:12	16:00
N 70°	07:19	08:49	11:29	12:04	14:44	16:14
68°	07:07	08:26	10:02	13:31	15:07	16:26
66°	06:58	08:08	09:25	14:09	15:26	16:35
64°	06:49	07:53	08:58	14:35	15:41	16:44
62°	06:42	07:40	08:38	14:55	15:53	16:51
60°	06:36	07:29	08:22	15:12	16:04	16:58
N 58°	06:30	07:20	08:08	15:26	16:13	17:03
56°	06:25	07:12	07:56	15:37	16:22	17:09
54°	06:20	07:04	07:46	15:48	16:29	17:14
52°	06:15	06:57	07:36	15:57	16:36	17:18
50°	06:11	06:51	07:28	16:05	16:42	17:22
45°	06:02	06:38	07:10	16:23	16:56	17:32
N 40°	05:53	06:26	06:56	16:37	17:07	17:40
35°	05:45	06:16	06:44	16:50	17:17	17:48
30°	05:38	06:07	06:33	17:00	17:26	17:56
20°	05:24	05:51	06:15	17:19	17:43	18:10
N 10°	05:10	05:36	05:59	17:35	17:58	18:24
0°	04:56	05:21	05:43	17:51	18:13	18:38
S 10°	04:39	05:05	05:28	18:06	18:29	18:55
20°	04:19	04:47	05:11	18:23	18:47	19:15
30°	03:53	04:26	04:52	18:42	19:09	19:41
35°	03:37	04:12	04:41	18:53	19:22	19:57
40°	03:17	03:57	04:28	19:06	19:38	20:17
45°	02:52	03:37	04:13	19:22	19:57	20:43
S 50°	02:16	03:12	03:53	19:41	20:23	21:20
52°	01:56	03:00	03:44	19:50	20:35	21:39
54°	01:31	02:46	03:34	20:01	20:49	22:05
56°	00:54	02:29	03:23	20:12	21:07	22:44
58°	////	02:08	03:09	20:26	21:28	////
S 60°	////	01:41	02:54	20:42	21:55	////

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	□	□	18:18	□	□	14:19
N 70°	□	15:43	18:58	□	15:00	13:38
68°	□	17:10	19:24	□	13:32	13:09
66°	15:45	17:48	19:45	12:58	12:53	12:48
64°	16:31	18:14	20:01	12:13	12:25	12:30
62°	17:00	18:35	20:14	11:43	12:04	12:16
60°	17:23	18:52	20:26	11:20	11:47	12:04
N 58°	17:41	19:06	20:35	11:01	11:32	11:53
56°	17:56	19:18	20:44	10:46	11:20	11:44
54°	18:09	19:28	20:51	10:32	11:09	11:36
52°	18:21	19:37	20:58	10:21	10:59	11:28
50°	18:31	19:46	21:04	10:10	10:50	11:22
45°	18:52	20:03	21:17	09:48	10:32	11:07
N 40°	19:09	20:17	21:28	09:31	10:16	10:55
35°	19:23	20:30	21:37	09:16	10:03	10:45
30°	19:36	20:40	21:45	09:03	09:52	10:36

November 27, 28, 29 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	65°48.3	218°41.8	S10°31.6	164°50.8	S23°57.8	71°06.5	S03°50.0	231°51.2	S03°30.4	Alpheratz	357°45.3	29°09.3
1	80°50.8	233°43.4	31.5	179°51.3	57.9	86°08.9	50.0	246°53.4	30.5	Ankaa	353°17.2	-42°14.9
2	95°53.3	248°44.9	31.4	194°51.7	58.0	101°11.3	49.9	261°55.7	30.5	Schedar	349°42.4	56°36.2
3	110°55.7	263°46.5	· · 31.2	209°52.1	· · 58.2	116°13.7	· · 49.9	276°57.9	· · 30.6	Diphda	348°57.5	-17°55.5
4	125°58.2	278°48.1	31.1	224°52.6	58.3	131°16.1	49.9	292°00.2	30.7	Achernar	335°27.6	-57°11.0
5	141°00.6	293°49.6	30.9	239°53.0	58.4	146°18.5	49.8	307°02.4	30.8	Hamal	328°02.6	23°31.0
6	156°03.1	308°51.2	S10°30.8	254°53.4	S23°58.6	161°20.9	S03°49.8	322°04.7	S03°30.9	Polaris	318°15.8	89°18.9
7	171°05.6	323°52.8	30.7	269°53.8	58.7	176°23.3	49.7	337°06.9	31.0	Acamar	315°19.3	-40°15.6
8	186°08.0	338°54.3	30.6	284°54.3	58.8	191°25.7	49.7	352°09.2	31.0	Menkar	314°16.7	4°08.1
9	201°10.5	353°55.9	· · 30.4	299°54.7	· · 59.0	206°28.1	· · 49.7	7°11.4	· · 31.1	Mirfak	308°42.5	49°54.1
10	216°13.0	8°57.4	30.3	314°55.1	59.1	221°30.5	49.6	22°13.7	31.2	Aldebaran	290°51.2	16°31.9
11	231°15.4	23°59.0	30.2	329°55.5	59.2	236°32.9	49.6	37°15.9	31.3	Rigel	281°13.5	-8°11.3
12	246°17.9	39°00.5	S10°30.1	344°56.0	S23°59.3	251°35.3	S03°49.5	52°18.2	S03°31.4	Capella	280°36.7	46°00.5
13	261°20.4	54°02.1	29.9	359°56.4	59.5	266°37.7	49.5	67°20.4	31.4	Bellatrix	278°33.7	6°21.6
14	276°22.8	69°03.6	29.8	14°56.8	59.6	281°40.1	49.5	82°22.7	31.5	Elnath	278°14.6	28°37.0
15	291°25.3	84°05.2	· · 29.7	29°57.2	· · 59.7	296°42.5	· · 49.4	97°24.9	· · 31.6	Alnilam	275°47.9	-1°11.7
16	306°27.7	99°06.7	29.6	44°57.7	23°59.9	311°44.9	49.4	112°27.2	31.7	Betelgeuse	271°03.0	7°24.5
17	321°30.2	114°08.2	29.5	59°58.1	24°00.0	326°47.3	49.3	127°29.4	31.8	Canopus	263°56.5	-52°42.0
18	336°32.7	129°09.8	S10°29.4	74°58.5	S24°00.1	341°49.7	S03°49.3	142°31.6	S03°31.9	Sirius	258°35.0	-16°43.8
19	351°35.1	144°11.3	29.3	89°59.0	00.2	356°52.1	49.3	157°33.9	31.9	Adhara	255°13.7	-28°59.1
20	6°37.6	159°12.8	29.2	104°59.4	00.4	11°54.5	49.2	172°36.1	32.0	Procyon	245°01.4	5°11.8
21	21°40.1	174°14.3	· · 29.0	119°59.8	· · 00.5	26°56.9	· · 49.2	187°38.4	· · 32.1	Pollux	243°29.7	27°59.8
22	36°42.5	189°15.8	28.9	135°00.2	00.6	41°59.3	49.1	202°40.6	32.2	Avior	234°18.5	-59°32.5
23	51°45.0	204°17.3	28.8	150°00.6	00.7	57°01.7	49.1	217°42.9	32.3	Suhail	222°53.7	-43°28.4
Mer.pass. 19:34		ν 1.6' d -0.1' m -4.90		ν 0.4' d 0.1' m 1.31		ν 2.4' d -0.0' m -2.57		ν 2.2' d 0.1' m 0.89				

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	66°47.5	219°18.9	S10°28.7	165°01.1	S24°00.9	72°04.1	S03°49.1	232°45.1	S03°32.3	Denebola	182°35.6	14°30.5
1	81°49.9	234°20.4	28.6	180°01.5	01.0	87°06.5	49.0	247°47.4	32.4	Gienah	175°54.4	-17°36.1
2	96°52.4	249°21.9	28.5	195°01.9	01.1	102°08.9	49.0	262°49.6	32.5	Acrux	173°11.9	-63°09.4
3	111°54.9	264°23.4	· · 28.5	210°02.3	· · 01.2	117°11.3	· · 48.9	277°51.9	· · 32.6	Gacrux	172°03.4	-57°10.3
4	126°57.3	279°24.9	28.4	225°02.8	01.4	132°13.7	48.9	292°54.1	32.7	Alioth	166°22.5	55°53.7
5	141°59.8	294°26.4	28.3	240°03.2	01.5	147°16.1	48.8	307°56.4	32.7	Spica	158°33.5	-11°13.1
6	157°02.2	309°27.9	S10°28.5	255°03.6	S24°01.6	162°18.5	S03°48.8	322°58.6	S03°32.8	Alkaid	153°00.7	49°15.3
7	172°04.7	324°29.3	28.1	270°04.0	01.7	177°20.9	48.8	338°00.9	32.9	Hadar	148°51.2	-60°25.4
8	187°07.2	339°30.8	28.0	285°04.5	01.9	192°23.3	48.7	353°03.1	33.0	Menkent	148°10.1	-36°25.3
9	202°09.6	354°32.3	· · 27.9	300°04.9	· · 02.0	207°25.7	· · 48.7	8°05.4	· · 33.1	Arcturus	145°57.7	19°07.5
10	217°12.1	9°33.8	27.8	315°05.3	02.1	222°28.0	48.6	23°07.6	33.1	Rigel Kent.	139°54.9	-60°52.7
11	232°14.6	24°35.3	27.8	330°05.7	02.2	237°30.4	48.6	38°09.9	33.2	Kochab	137°20.9	74°06.4
12	247°17.0	39°36.7	S10°27.7	345°06.2	S24°02.3	252°32.8	S03°48.5	53°12.1	S03°33.3	Zuben'ubi	137°07.8	-16°05.2
13	262°19.5	54°38.2	27.6	0°06.6	02.5	267°35.2	48.5	68°14.4	33.4	Alphecca	126°12.9	26°40.7
14	277°22.0	69°39.7	27.5	15°07.0	02.6	282°37.6	48.5	83°16.6	33.5	Antares	112°28.9	-26°27.3
15	292°24.4	84°41.2	· · 27.5	30°07.4	· · 02.7	297°40.0	· · 48.4	98°18.9	· · 33.6	Atria	107°33.0	-69°02.8
16	307°26.9	99°42.6	27.4	45°07.8	02.8	312°42.4	48.4	113°21.1	33.6	Sabik	102°15.0	-15°44.2
17	322°29.4	114°44.1	27.3	60°08.3	02.9	327°44.8	48.3	128°23.4	33.7	Shaula	96°24.8	-37°06.7
18	337°31.8	129°45.5	S10°27.2	75°08.7	S24°03.0	342°47.2	S03°48.3	143°25.6	S03°33.8	Rasalhague	96°08.5	12°33.2
19	352°34.3	144°47.0	27.2	90°09.1	03.2	357°49.6	48.2	158°27.9	33.9	Eltanin	90°47.5	51°29.4
20	7°36.7	159°48.4	27.1	105°09.5	03.3	12°51.9	48.2	173°30.1	34.0	Kaus Aust.	83°46.6	-34°22.8
21	22°39.2	174°49.9	· · 27.0	120°09.9	· · 03.4	27°54.3	· · 48.1	188°32.4	· · 34.0	Vega	80°40.6	38°47.8
22	37°41.7	189°51.3	27.0	135°10.4	03.5	42°56.7	48.1	203°34.6	34.1	Nunki	76°00.9	-26°17.0
23	52°44.1	204°52.8	26.9	150°10.8	03.6	57°59.1	48.1	218°36.9	34.2	Altair	62°10.3	8°54.0
Mer.pass. 19:30		ν 1.5' d -0.1' m -4.90		ν 0.4' d 0.1' m 1.31		ν 2.4' d -0.0' m -2.56		ν 2.2' d 0.1' m 0.89				

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	67°46.6	219°54.2	S10°26.9	165°11.2	S24°03.7	73°01.5	S03°48.0	233°39.1	S03°34.3	Nov 27 Sat	SHA	Mer.pass
1	82°49.1	234°55.7	26.8	180°11.6	03.8	88°03.9	48.0	248°41.4	34.3	Venus	152°53.5	09:24
2	97°51.5	249°57.1	26.8	195°12.0	04.0	103°06.3	47.9	263°43.6	34.4	Mars	99°02.5	13:00
3	112°54.0	264°58.5	· · 26.7	210°12.5	· · 04.1	118°08.7	· · 47.9	278°45.9	· · 34.5	Jupiter	5°18.2	19:12
4	127°56.5	279°59.9	26.6	225°12.9	04.2	133°11.0	47.8	293°48.2	34.6	Saturn	166°02.9	08:31
5	142°58.9	295°01.4	26.6	240°13.3	04.3	148°13.4	47.8	308°50.4	34.7			
6	158°01.4	310°02.8	S10°26.6	255°13.7	S24°04.4	163°15.8	S03°47.7	323°52.7	S03°34.7	Nov 28 Sun	SHA	Mer.pass
7	173°03.8	325°04.2	26.5	270°14.1	04.5	178°18.2	47.7	338°54.9	34.8	Venus	152°31.4	09:22
8	188°06.3	340°05.6	26.5	285°14.6	04.6	193°20.6	47.6	353°57.2	34.9	Mars	98°13.6	13:00
9	203°08.8	355°07.0	· · 26.4	300°15.0	· · 04.7	208°23.0	· · 47.6	8°59.4	· · 35.0	Jupiter	5°16.6	19:09
10	218°11.2	10°08.5	26.4	315°15.4	04.8	223°25.4	47.6	24°01.7	35.1	Saturn	165°57.7	08:28
11	233°13.7	25°09.9	26.3	330°15.8	05.0	238°27.7	47.5	39°03.9	35.1			
12	248°16.2	40°11.3	S10°26.3	345°16.2	S24°05.1	253°30.1	S03°47.5	54°06.2	S03°35.2	Nov 29 Mon	SHA	Mer.pass
13	263°18.6	55°12.7	26.3	0°16.7	05.2	268°32.5	47.4	69°08.4	35.3	Venus	152°07.6	09:20
14	278°21.1	70°14.1	26.2	15°17.1	05.3	283°34.9	47.4	84°10.7	35.4	Mars	97°24.6	12:59
15	293°23.6	85°15.5	· · 26.2	30°17.5	· · 05.4	298°37.3	· · 47.3	99°12.9	· · 35.5	Jupiter	5°14.9	19:05
16	308°26.0	100°16.9	26.2	45°17.9	05.5	313°39.7	47.3	114°15.2	35.5	Saturn	165°52.5	08:24
17	323°28.5	115°18.3	26.1	60°18.3	05.6	328°42.0	47.2	129°17.4	35.6			
18	338°31.0	130°19.6	S10°26.1	75°18.7	S24°05.7	343°44.4	S03°47.2	144°19.7	S03°35.7	Horizontal parallax		
19	353°33.4	145°21.0	26.1	90°19.2	05.8	358°46.8	47.1	159°21.9	35.8	Venus:	0.4	
20	8°35.9	160°22.4	26.1	105°19.6	05.9	13°49.2	47.1	174°24.2	35.9	Mars:	0.1	
21	23°38.3	175°23.8	· · 26.0	120°20.0	· · 06.0	28°51.6	· · 47.0	189°26.4	· · 35.9			
22	38°40.8	190°25.2	26.0	135°20.4	06.1	43°54.0	47.0	204°28.7	36.0			
23	53°43.3	205°26.5	26.0	150°20.8	06.2	58°56.3	46.9	219°31.0	36.1			
Mer.pass. 19:26		ν 1.4' d -0.1' m -4.90		ν 0.4' d 0.1' m 1.30		ν 2.4' d -0.0' m -2.55		ν 2.3' d 0.1' m 0.89				

h		Sun			Moon		
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	183°08.7	S21°03.7	293°51.1	9.6'	N14°34.0	-11.7'	58.7'
1	198°08.5	04.2	308°19.6	9.6'	14°22.3	-11.8'	58.7'
2	213°08.3	04.6	322°48.2	9.7'	14°10.4	-11.9'	58.7'
3	228°08.1	· · 05.1	337°16.9	9.7'	13°58.5	-12.0'	58.7'
4	243°07.9	05.5	351°45.6	9.7'	13°46.6	-12.1'	58.8'
5	258°07.7	06.0	6°14.3	9.8'	13°34.5	-12.1'	58.8'
6	273°07.5	S21°06.5	20°43.1	9.8'	N13°22.4	-12.2'	58.8'
7	288°07.3	06.9	35°12.0	9.9'	13°10.1	-12.3'	58.8'
8	303°07.1	07.4	49°40.9	9.9'	12°57.8	-12.4'	58.8'
9	318°06.9	· · 07.8	64°09.8	10.0'	12°45.5	-12.4'	58.8'
10	333°06.7	08.3	78°38.8	10.0'	12°33.0	-12.5'	58.8'
11	348°06.5	08.7	93°07.8	10.1'	12°20.5	-12.6'	58.8'
12	3°06.2	S21°09.2	107°36.8	10.1'	N12°07.9	-12.7'	58.8'
13	18°06.0	09.7	122°06.0	10.1'	11°55.2	-12.7'	58.8'
14	33°05.8	10.1	136°35.1	10.2'	11°42.5	-12.8'	58.9'
15	48°05.6	· · 10.6	151°04.3	10.2'	11°29.7	-12.9'	58.9'
16	63°05.4	11.0	165°33.5	10.3'	11°16.8	-12.9'	58.9'
17	78°05.2	11.5	180°02.8	10.3'	11°03.9	-13.0'	58.9'
18	93°05.0	S21°11.9	194°32.1	10.3'	N10°50.9	-13.1'	58.9'
19	108°04.8	12.4	209°01.4	10.4'	10°37.8	-13.1'	58.9'
20	123°04.6	12.8	223°30.8	10.4'	10°24.7	-13.2'	58.9'
21	138°04.4	· · 13.3	238°00.3	10.5'	10°11.5	-13.2'	58.9'
22	153°04.2	13.7	252°29.7	10.5'	09°58.3	-13.3'	58.9'
23	168°04.0	14.2	266°59.2	10.5'	09°45.0	-13.4'	58.9'
SD = 16.2'		$d = 0.5'$	SD = 16.0'				
Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	183°03.8	S21°14.6	281°28.7	10.6'	N09°31.6	-13.4'	59.0'
1	198°03.6	15.0	295°58.3	10.6'	09°18.2	-13.5'	59.0'
2	213°03.3	15.5	310°27.9	10.6'	09°04.8	-13.5'	59.0'
3	228°03.1	· · 15.9	324°57.5	10.7'	08°51.2	-13.6'	59.0'
4	243°02.9	16.4	339°27.2	10.7'	08°37.7	-13.6'	59.0'
5	258°02.7	16.8	353°56.9	10.7'	08°24.1	-13.7'	59.0'
6	273°02.5	S21°17.3	8°26.6	10.7'	N08°10.4	-13.7'	59.0'
7	288°02.3	17.7	22°56.3	10.8'	07°56.7	-13.8'	59.0'
8	303°02.1	18.1	37°26.1	10.8'	07°42.9	-13.8'	59.0'
9	318°01.9	· · 18.6	51°55.9	10.8'	07°29.1	-13.9'	59.0'
10	333°01.7	19.0	66°25.7	10.9'	07°15.2	-13.9'	59.0'
11	348°01.5	19.5	80°55.6	10.9'	07°01.4	-13.9'	59.0'
12	3°01.2	S21°19.9	95°25.5	10.9'	N06°47.4	-14.0'	59.1'
13	18°01.0	20.3	109°55.4	10.9'	06°33.4	-14.0'	59.1'
14	33°00.8	20.8	124°25.3	10.9'	06°19.4	-14.0'	59.1'
15	48°00.6	· · 21.2	138°55.2	11.0'	06°05.4	-14.1'	59.1'
16	63°00.4	21.6	153°25.2	11.0'	05°51.3	-14.1'	59.1'
17	78°00.2	22.1	167°55.2	11.0'	05°37.2	-14.2'	59.1'
18	93°00.0	S21°22.5	182°25.2	11.0'	N05°23.0	-14.2'	59.1'
19	107°59.7	22.9	196°55.2	11.0'	05°08.8	-14.2'	59.1'
20	122°59.5	23.4	211°25.3	11.1'	04°54.6	-14.2'	59.1'
21	137°59.3	· · 23.8	225°55.3	11.1'	04°40.4	-14.3'	59.1'
22	152°59.1	24.2	240°25.4	11.1'	04°26.1	-14.3'	59.1'
23	167°58.9	24.7	254°55.5	11.1'	04°11.8	-14.3'	59.1'
SD = 16.2'		$d = 0.4'$	SD = 16.1'				
Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	182°58.7	S21°25.1	269°25.6	11.1'	N03°57.5	-14.3'	59.1'
1	197°58.5	25.5	283°55.7	11.1'	03°43.1	-14.4'	59.1'
2	212°58.2	25.9	298°25.8	11.1'	03°28.8	-14.4'	59.2'
3	227°58.0	· · 26.4	312°55.9	11.1'	03°14.4	-14.4'	59.2'
4	242°57.8	26.8	327°26.0	11.1'	03°00.0	-14.4'	59.2'
5	257°57.6	27.2	341°56.2	11.1'	02°45.5	-14.4'	59.2'
6	272°57.4	S21°27.7	356°26.3	11.2'	N02°31.1	-14.5'	59.2'
7	287°57.2	28.1	10°56.5	11.2'	02°16.6	-14.5'	59.2'
8	302°56.9	28.5	25°26.7	11.2'	02°02.1	-14.5'	59.2'
9	317°56.7	· · 28.9	39°56.8	11.2'	01°47.6	-14.5'	59.2'
10	332°56.5	29.3	54°27.0	11.2'	01°33.1	-14.5'	59.2'
11	347°56.3	29.8	68°57.1	11.2'	01°18.6	-14.5'	59.2'
12	2°56.1	S21°30.2	83°27.3	11.2'	N01°04.1	-14.5'	59.2'
13	17°55.8	30.6	97°57.5	11.2'	00°49.6	-14.5'	59.2'
14	32°55.6	31.0	112°27.6	11.2'	00°35.0	-14.5'	59.2'
15	47°55.4	· · 31.4	126°57.8	11.2'	00°20.5	-14.6'	59.2'
16	62°55.2	31.9	141°27.9	11.2'	N00°05.9	-14.6'	59.2'
17	77°55.0	32.3	155°58.1	11.1'	S00°08.6	14.6'	59.2'
18	92°54.7	S21°32.7	170°28.2	11.1'	S00°23.2	14.6'	59.2'
19	107°54.5	33.1	184°58.4	11.1'	00°37.8	14.6'	59.2'
20	122°54.3	33.5	199°28.5	11.1'	00°52.3	14.6'	59.2'
21	137°54.1	· · 33.9	213°58.6	11.1'	01°06.9	14.6'	59.2'
22	152°53.8	34.4	228°28.7	11.1'	01°21.4	14.5'	59.2'
23	167°53.6	34.8	242°58.8	11.1'	01°36.0	14.5'	59.3'
SD = 16.2'		$d = 0.4'$	SD = 16.1'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:42	09:36	■	■	13:59	15:53
N 70°	07:27	09:00	■	■	14:35	16:08
68°	07:15	08:35	10:18	13:17	15:00	16:21
66°	07:04	08:15	09:36	14:00	15:20	16:31
64°	06:55	08:00	09:07	14:28	15:36	16:40
62°	06:48	07:46	08:46	14:50	15:49	16:48
60°	06:41	07:35	08:28	15:07	16:00	16:55
N 58°	06:35	07:25	08:14	15:22	16:10	17:01
56°	06:29	07:17	08:01	15:34	16:19	17:06
54°	06:24	07:09	07:51	15:45	16:27	17:11
52°	06:19	07:02	07:41	15:54	16:34	17:16
50°	06:15	06:55	07:32	16:03	16:40	17:21
45°	06:05	06:41	07:14	16:21	16:54	17:31
N 40°	05:56	06:29	06:59	16:36	17:06	17:40
35°	05:48	06:19	06:47	16:49	17:16	17:48
30°	05:40	06:10	06:36	17:00	17:26	17:55
20°	05:26	05:53	06:17	17:19	17:43	18:10
N 10°	05:11	05:37	06:00	17:36	17:58	18:24
0°	04:56	05:22	05:44	17:52	18:14	18:40
S 10°	04:39	05:06	05:28	18:07	18:30	18:57
20°	04:19	04:47	05:11	18:25	18:49	19:17
30°	03:52	04:25	04:52	18:45	19:11	19:44
35°	03:36	04:11	04:40	18:56	19:25	20:01
40°	03:15	03:55	04:27	19:10	19:41	20:21
45°	02:49	03:35	04:11	19:26	20:02	20:48
S 50°	02:11	03:09	03:51	19:46	20:28	21:26
52°	01:50	02:56	03:42	19:55	20:41	21:47
54°	01:23	02:41	03:31	20:06	20:56	22:15
56°	00:39	02:24	03:19	20:18	21:14	23:03
58°	////	02:02	03:05	20:32	21:36	////
S 60°	////	01:32	02:48	20:49	22:06	////

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	20:55	23:11	· ·	13:30	13:01	12:37
N 70°	21:13	23:17	· ·	13:11	12:52	12:36
68°	21:26	23:22	· ·	12:56	12:45	12:35
66°	21:37	23:26	· ·	12:43	12:39	12:34
64°	21:46	23:29	· ·	12:33	12:33	12:34
62°	21:54	23:32	· ·	12:24	12:29	12:33
60°	22:00	23:34	· ·	12:16	12:25	12:33
N 58°	22:06	23:37	· ·	12:09	12:21	12:32
56°	22:11	23:39	· ·	12:03	12:18	12:32
54°	22:16	23:40	· ·	11:57	12:15	12:31
52°	22:20	23:42	· ·	11:52	12:13	12:31
50°	22:24	23:43	· ·	11:48	12:10	12:31
45°	22:32	23:47	· ·	11:38	12:05	12:30
N 40°	22:39	23:49	· ·	11:30	12:00	12:30
35°	22:44	23:52	· ·	11:23	11:57	12:29
30°	22:50	23:54	· ·	11:16	11:53	12:29
20°	22:58	23:57	· ·	11:05	11:47	12:28
N 10°	23:06	· ·	00:00	10:56	11:42	12:27
0°	23:13	· ·	00:03	10:46	11:37	12:26
S 10°	23:20	· ·	00:06	10:37	11:32	12:26
20°	23:28	· ·	00:09	10:27	11:26	12:25
30°	23:36	· ·	00:12	10:16	11:20	12:24
35°	23:41	· ·	00:14	10:09	11:16	12:23
40°	23:47	· ·	00:17	10:02	11:12	12:23
45°	23:53	· ·	00:19	09:53	11:08	12:22
S 50°	· ·	00:01	00:22	09:42	11:02	12:21
52°	· ·	00:05	00:24	09:37	10:59	12:21
54°	· ·	00:09	00:25	09:31	10:56	12:20
56°	· ·	00:13	00:27	09:25	10:53	12:20
58°	00:04	00:18	00:29	09:18	10:49	12:19
S 60°	00:13	00:23	00:31	09:10	10:45	12:19

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Upper hh:mm	Lower hh:mm	
27	12:35	12:25	11:48	04:34	17:00	
28	12:15	12:05	11:48	05:25	17:50	
29	11:55	11:44	11:48	06:15	18:39	

November 30, 01, 02 UT (Tue., Wed., Thu.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	68°45.7	220°27.9	S10°26.0	165°21.2	S24°06.3	73°58.7	S03°46.9	234°33.2	S03°36.2					Alpheratz	357°45.3	29°09.3	
1	83°48.2	235°29.3	26.0	180°21.7	06.4	89°01.1	46.8	249°35.5	36.2					Ankaa	353°17.2	-42°14.9	
2	98°50.7	250°30.6	25.9	195°22.1	06.5	104°03.5	46.8	264°37.7	36.3					Schedar	349°42.4	56°36.2	
3	113°53.1	265°32.0	· · 25.9	210°22.5	· · 06.6	119°05.8	· · 46.7	279°40.0	· · 36.4					Diphda	348°57.5	-17°55.6	
4	128°55.6	280°33.4	25.9	225°22.9	06.7	134°08.2	46.7	294°42.2	36.5					Achernar	335°27.6	-57°11.0	
5	143°58.1	295°34.7	25.9	240°23.3	06.8	149°10.6	46.6	309°44.5	36.6					Hamal	328°02.6	23°31.0	
6	159°00.5	310°36.1	S10°25.9	255°23.7	S24°06.9	164°13.0	S03°46.6	324°46.7	S03°36.6					Polaris	318°16.2	89°18.9	
7	174°03.0	325°37.4	25.9	270°24.1	07.0	179°15.4	46.5	339°49.0	36.7					Acamar	315°19.3	-40°15.7	
8	189°05.5	340°38.8	25.9	285°24.6	07.1	194°17.7	46.5	354°51.2	36.8					Menkar	314°16.7	4°08.1	
9	204°07.9	355°40.1	· · 25.9	300°25.0	· · 07.2	209°20.1	· · 46.5	9°53.5	· · 36.9					Mirfak	308°42.5	49°54.1	
10	219°10.4	10°41.5	25.9	315°25.4	07.3	224°22.5	46.4	24°55.8	37.0					Aldebaran	290°51.1	16°31.9	
11	234°12.8	25°42.8	25.9	330°25.8	07.4	239°24.9	46.4	39°58.0	37.0					Rigel	281°13.5	-8°11.3	
12	249°15.3	40°44.2	S10°25.9	345°26.2	S24°07.5	254°27.2	S03°46.3	55°00.3	S03°37.1					Capella	280°36.7	46°00.5	
13	264°17.8	55°45.5	25.9	0°26.6	07.6	269°29.6	46.3	70°02.5	37.2					Bellatrix	278°33.6	6°21.6	
14	279°20.2	70°46.8	25.9	15°27.0	07.7	284°32.0	46.2	85°04.8	37.3					Elnath	278°14.6	28°37.0	
15	294°22.7	85°48.2	· · 25.9	30°27.5	· · 07.8	299°34.4	· · 46.2	100°07.0	· · 37.3					Alnilam	275°47.9	-1°11.7	
16	309°25.2	100°49.5	25.9	45°27.9	07.9	314°36.7	46.1	115°09.3	37.4					Betelgeuse	271°03.0	7°24.5	
17	324°27.6	115°50.8	25.9	60°28.3	08.0	329°39.1	46.0	130°11.5	37.5					Canopus	263°56.5	-52°42.0	
18	339°30.1	130°52.1	S10°25.9	75°28.7	S24°08.1	344°41.5	S03°46.0	145°13.8	S03°37.6					Sirius	258°35.0	-16°43.8	
19	354°32.6	145°53.5	25.9	90°29.1	08.2	359°43.9	45.9	160°16.1	37.7					Adhara	255°13.7	-28°59.1	
20	9°35.0	160°54.8	26.0	105°29.5	08.3	14°46.2	45.9	175°18.3	37.7					Procyon	245°01.4	5°11.8	
21	24°37.5	175°56.1	· · 26.0	120°29.9	· · 08.4	29°48.6	· · 45.8	190°20.6	· · 37.8					Pollux	243°29.7	27°59.8	
22	39°39.9	190°57.4	26.0	135°30.4	08.5	44°51.0	45.8	205°22.8	37.9					Avior	234°18.5	-59°32.5	
23	54°42.4	205°58.7	26.0	150°30.8	08.6	59°53.4	45.7	220°25.1	38.0					Suhail	222°53.7	-43°28.4	
Mer.pass. 19:22			ν 1.4' d -0.0' m -4.90	ν 0.4' d 0.1' m 1.30	ν 2.4' d -0.0' m -2.54	ν 2.3' d 0.1' m 0.89											

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	69°44.9	221°00.0	S10°26.0	165°31.2	S24°08.7	74°55.7	S03°45.7	235°27.3	S03°38.0
1	84°47.3	236°01.3	26.1	180°31.6	08.8	89°58.1	45.6	250°29.6	38.1
2	99°49.8	251°02.6	26.1	195°32.0	08.9	105°00.5	45.6	265°31.8	38.2
3	114°52.3	266°03.9	· · 26.1	210°32.4	· · 09.0	120°02.8	· · 45.5	280°34.1	· · 38.3
4	129°54.7	281°05.2	26.1	225°32.8	09.1	135°05.2	45.5	295°36.4	38.3
5	144°57.2	296°06.5	26.2	240°33.2	09.1	150°07.6	45.4	310°38.6	38.4
6	159°59.7	311°07.8	S10°26.2	255°33.6	S24°09.2	165°10.0	S03°45.4	325°40.9	S03°38.5
7	175°02.1	326°09.1	26.2	270°34.1	09.3	180°12.3	45.3	340°43.1	38.6
8	190°04.6	341°10.4	26.3	285°34.5	09.4	195°14.7	45.3	355°45.4	38.7
9	205°07.1	356°11.6	· · 26.3	300°34.9	· · 09.5	210°17.1	· · 45.2	10°47.6	· · 38.7
10	220°09.5	11°12.9	26.3	315°35.3	09.6	225°19.4	45.2	25°49.9	38.8
11	235°12.0	26°14.2	26.4	330°35.7	09.7	240°21.8	45.1	40°52.2	38.9
12	250°14.4	41°15.5	S10°26.4	345°36.1	S24°09.8	255°24.2	S03°45.1	55°54.4	S03°39.0
13	265°16.9	56°16.7	26.4	0°36.5	09.9	270°26.5	45.0	70°56.7	39.0
14	280°19.4	71°18.0	26.5	15°36.9	09.9	285°28.9	45.0	85°58.9	39.1
15	295°21.8	86°19.3	· · 26.5	30°37.3	· · 10.0	300°31.3	· · 44.9	101°01.2	· · 39.2
16	310°24.3	101°20.5	26.6	45°37.8	10.1	315°33.6	44.8	116°03.5	39.3
17	325°26.8	116°21.8	26.6	60°38.2	10.2	330°36.0	44.8	131°05.7	39.3
18	340°29.2	131°23.1	S10°26.7	75°38.6	S24°10.3	345°38.4	S03°44.7	146°08.0	S03°39.4
19	355°31.7	146°24.3	26.7	90°39.0	10.4	0°40.7	44.7	161°10.2	39.5
20	10°34.2	161°25.6	26.8	105°39.4	10.4	15°43.1	44.6	176°12.5	39.6
21	25°36.6	176°26.8	· · 26.8	120°39.8	· · 10.5	30°45.5	· · 44.6	191°14.7	· · 39.7
22	40°39.1	191°28.1	26.9	135°40.2	10.6	45°47.8	44.5	206°17.0	39.7
23	55°41.6	206°29.3	26.9	150°40.6	10.7	60°50.2	44.5	221°19.3	39.8
Mer.pass. 19:18			ν 1.3' d 0.0' m -4.90	ν 0.4' d 0.1' m 1.29	ν 2.4' d -0.1' m -2.54	ν 2.3' d 0.1' m 0.88			

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	70°44.0	221°30.6	S10°27.0	165°41.0	S24°10.8	75°52.6	S03°44.4	236°21.5	S03°39.9
1	85°46.5	236°31.8	27.1	180°41.4	10.9	90°54.9	44.4	251°23.8	40.0
2	100°48.9	251°33.0	27.1	195°41.8	10.9	105°57.3	44.3	266°26.0	40.0
3	115°51.4	266°34.3	· · 27.2	210°42.3	· · 11.0	120°59.7	· · 44.2	281°28.3	· · 40.1
4	130°53.9	281°35.5	27.2	225°42.7	11.1	136°02.0	44.2	296°30.6	40.2
5	145°56.3	296°36.7	27.3	240°43.1	11.2	151°04.4	44.1	311°32.8	40.3
6	160°58.8	311°38.0	S10°27.4	255°43.5	S24°11.3	166°06.8	S03°44.1	326°35.1	S03°40.3
7	176°01.3	326°39.2	27.4	270°43.9	11.3	181°09.1	44.0	341°37.3	40.4
8	191°03.7	341°40.4	27.5	285°44.3	11.4	196°11.5	44.0	356°39.6	40.5
9	206°06.2	356°41.6	· · 27.6	300°44.7	· · 11.5	211°13.8	· · 43.9	11°41.9	· · 40.6
10	221°08.7	11°42.8	27.7	315°45.1	11.6	226°16.2	43.9	26°44.1	40.6
11	236°11.1	26°44.1	27.7	330°45.5	11.7	241°18.6	43.8	41°46.4	40.7
12	251°13.6	41°45.3	S10°27.8	345°45.9	S24°11.7	256°20.9	S03°43.7	56°48.6	S03°40.8
13	266°16.0	56°46.5	27.9	0°46.3	11.8	271°23.3	43.7	71°50.9	40.9
14	281°18.5	71°47.7	28.0	15°46.7	11.9	286°25.6	43.6	86°53.2	40.9
15	296°21.0	86°48.9	· · 28.1	30°47.1	· · 12.0	301°28.0	· · 43.6	101°55.4	· · 41.0
16	311°23.4	101°50.1	28.1	45°47.5	12.0	316°30.4	43.5	116°57.7	41.1
17	326°25.9	116°51.3	28.2	60°48.0	12.1	331°32.7	43.5	131°59.9	41.2
18	341°28.4	131°52.5	S10°28.3	75°48.4	S24°12.2	346°35.1	S03°43.4	147°02.2	S03°41.2
19	356°30.8	146°53.7	28.4	90°48.8	12.3	1°37.4	43.3	162°04.5	41.3
20	11°33.3	161°54.9	28.5	105°49.2	12.3	16°39.8	43.3	177°06.7	41.4
21	26°35.8	176°56.1	· · 28.6	120°49.6	· · 12.4	31°42.2	· · 43.2	192°09.0	· · 41.5
22	41°38.2	191°57.2	28.7	135°50.0	12.5	46°44.5	43.2	207°11.3	41.5
23	56°40.7	206°58.4	28.8	150°50.4	12.5	61°46.9	43.1	222°13.5	41.6
Mer.pass. 19:14			ν 1.2' d 0.1' m -4.90	ν 0.4' d 0.1' m 1.29	ν 2.4' d -0.1' m -2.53	ν 2.3' d 0.1' m 0.88			

Nov 30 Tue	SHA	Mer.pass
Venus	151°42.2	09:17
Mars	96°35.5	12:58
Jupiter	5°13.0	19:01
Saturn	165°47.5	08:21

Dec 01 Wed	SHA	Mer.pass
Venus	151°15.1	09:15
Mars	95°46.3	12:58
Jupiter	5°10.9	18:57
Saturn	165°42.5	08:17

Dec 02 Thu	SHA	Mer.pass
Venus	150°46.5	09:13
Mars	94°57.0	12:57
Jupiter	5°08.6	18:54
Saturn	165°37.5	08:13

Horizontal parallax		
Venus:	0.4	
Mars:	0.1	

h	Sun			Moon			
Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	182°53.4	S21°35.2	257°28.9	11.1'	S01°50.5	14.5'	59.3'
1	197°53.2	35.6	271°59.0	11.1'	02°05.0	14.5'	59.3'
2	212°52.9	36.0	286°29.1	11.0'	02°19.6	14.5'	59.3'
3	227°52.7	· · 36.4	300°59.1	11.0'	02°34.1	14.5'	59.3'
4	242°52.5	36.8	315°29.1	11.0'	02°48.6	14.5'	59.3'
5	257°52.3	37.2	329°59.1	11.0'	03°03.1	14.5'	59.3'
6	272°52.0	S21°37.6	344°29.1	11.0'	S03°17.5	14.5'	59.3'
7	287°51.8	38.0	358°59.1	10.9'	03°32.0	14.4'	59.3'
8	302°51.6	38.5	13°29.0	10.9'	03°46.4	14.4'	59.3'
9	317°51.4	· · 38.9	27°59.0	10.9'	04°00.8	14.4'	59.3'
10	332°51.1	39.3	42°28.9	10.9'	04°15.2	14.4'	59.3'
11	347°50.9	39.7	56°58.7	10.9'	04°29.6	14.4'	59.3'
12	2°50.7	S21°40.1	71°28.6	10.8'	S04°44.0	14.3'	59.3'
13	17°50.5	40.5	85°58.4	10.8'	04°58.3	14.3'	59.3'
14	32°50.2	40.9	100°28.2	10.8'	05°12.6	14.3'	59.3'
15	47°50.0	· · 41.3	114°58.0	10.7'	05°26.9	14.3'	59.3'
16	62°49.8	41.7	129°27.8	10.7'	05°41.2	14.2'	59.3'
17	77°49.6	42.1	143°57.5	10.7'	05°55.4	14.2'	59.3'
18	92°49.3	S21°42.5	158°27.2	10.7'	S06°09.6	14.2'	59.3'
19	107°49.1	42.9	172°56.8	10.6'	06°23.7	14.1'	59.3'
20	122°48.9	43.3	187°26.4	10.6'	06°37.8	14.1'	59.3'
21	137°48.6	· · 43.7	201°56.0	10.6'	06°51.9	14.1'	59.3'
22	152°48.4	44.1	216°25.6	10.5'	07°06.0	14.0'	59.3'
23	167°48.2	44.5	230°55.1	10.5'	07°20.0	14.0'	59.3'
SD = 16.2'			SD = 16.2'				

Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	182°47.9	S21°44.9	245°24.6	10.4'	S07°34.0	13.9'	59.3'
1	197°47.7	45.3	259°54.1	10.4'	07°47.9	13.9'	59.3'
2	212°47.5	45.6	274°23.5	10.4'	08°01.8	13.8'	59.3'
3	227°47.3	· · 46.0	288°52.8	10.3'	08°15.6	13.8'	59.3'
4	242°47.0	46.4	303°22.2	10.3'	08°29.4	13.8'	59.3'
5	257°46.8	46.8	317°51.5	10.3'	08°43.2	13.7'	59.3'
6	272°46.6	S21°47.2	332°20.7	10.2'	S08°56.9	13.7'	59.3'
7	287°46.3	47.6	346°49.9	10.2'	09°10.6	13.6'	59.3'
8	302°46.1	48.0	1°19.1	10.1'	09°24.2	13.6'	59.3'
9	317°45.9	· · 48.4	15°48.2	10.1'	09°37.7	13.5'	59.3'
10	332°45.6	48.8	30°17.3	10.0'	09°51.2	13.4'	59.3'
11	347°45.4	49.2	44°46.4	10.0'	10°04.7	13.4'	59.3'
12	2°45.2	S21°49.5	59°15.3	9.9'	S10°18.1	13.3'	59.3'
13	17°44.9	49.9	73°44.3	9.9'	10°31.4	13.3'	59.3'
14	32°44.7	50.3	88°13.2	9.9'	10°44.7	13.2'	59.3'
15	47°44.5	· · 50.7	102°42.0	9.8'	10°57.9	13.1'	59.2'
16	62°44.2	51.1	117°10.8	9.8'	11°11.0	13.1'	59.2'
17	77°44.0	51.5	131°39.6	9.7'	11°24.1	13.0'	59.2'
18	92°43.7	S21°51.8	146°08.3	9.7'	S11°37.1	13.0'	59.2'
19	107°43.5	52.2	160°37.0	9.6'	11°50.1	12.9'	59.2'
20	122°43.3	52.6	175°05.6	9.6'	12°03.0	12.8'	59.2'
21	137°43.0	· · 53.0	189°34.1	9.5'	12°15.8	12.7'	59.2'
22	152°42.8	53.4	204°02.6	9.4'	12°28.5	12.7'	59.2'
23	167°42.6	53.7	218°31.1	9.4'	12°41.2	12.6'	59.2'
SD = 16.2'			SD = 16.2'				

Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	182°42.3	S21°54.1	232°59.5	9.3'	S12°53.8	12.5'	59.2'
1	197°42.1	54.5	247°27.8	9.3'	13°06.3	12.4'	59.2'
2	212°41.9	54.9	261°56.1	9.2'	13°18.8	12.4'	59.2'
3	227°41.6	· · 55.2	276°24.3	9.2'	13°31.1	12.3'	59.2'
4	242°41.4	55.6	290°52.5	9.1'	13°43.4	12.2'	59.2'
5	257°41.1	56.0	305°20.6	9.1'	13°55.6	12.1'	59.2'
6	272°40.9	S21°56.4	319°48.7	9.0'	S14°07.8	12.0'	59.2'
7	287°40.7	56.7	334°16.7	9.0'	14°19.8	12.0'	59.2'
8	302°40.4	57.1	348°44.7	8.9'	14°31.8	11.9'	59.2'
9	317°40.2	· · 57.5	3°12.6	8.8'	14°43.6	11.8'	59.1'
10	332°39.9	57.9	17°40.4	8.8'	14°55.4	11.7'	59.1'
11	347°39.7	58.2	32°08.2	8.7'	15°07.1	11.6'	59.1'
12	2°39.5	S21°58.6	46°36.0	8.7'	S15°18.7	11.5'	59.1'
13	17°39.2	59.0	61°03.7	8.6'	15°30.2	11.4'	59.1'
14	32°39.0	59.3	75°31.3	8.6'	15°41.6	11.3'	59.1'
15	47°38.7	21°59.7	89°58.8	8.5'	15°53.0	11.2'	59.1'
16	62°38.5	22°00.1	104°26.3	8.5'	16°04.2	11.1'	59.1'
17	77°38.2	00.4	118°53.8	8.4'	16°15.3	11.0'	59.1'
18	92°38.0	S22°00.8	133°21.2	8.3'	S16°26.3	10.9'	59.1'
19	107°37.8	01.2	147°48.5	8.3'	16°37.3	10.8'	59.1'
20	122°37.5	01.5	162°15.8	8.2'	16°48.1	10.7'	59.0'
21	137°37.3	· · 01.9	176°43.0	8.2'	16°58.8	10.6'	59.0'
22	152°37.0	02.2	191°10.2	8.1'	17°09.4	10.5'	59.0'
23	167°36.8	02.6	205°37.3	8.1'	17°20.0	10.4'	59.0'
SD = 16.2'			SD = 16.1'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:51	09:50	■	■	13:48	15:47
N 70°	07:34	09:10	■	■	14:27	16:03
68°	07:21	08:43	10:36	13:02	14:54	16:16
66°	07:10	08:23	09:46	13:51	15:15	16:27
64°	07:01	08:06	09:16	14:22	15:31	16:36
62°	06:53	07:52	08:53	14:45	15:45	16:45
60°	06:46	07:40	08:35	15:03	15:57	16:52
N 58°	06:39	07:30	08:19	15:18	16:07	16:58
56°	06:33	07:21	08:07	15:31	16:16	17:04
54°	06:28	07:13	07:55	15:42	16:25	17:10
52°	06:23	07:06	07:45	15:52	16:32	17:15
50°	06:18	06:59	07:36	16:01	16:39	17:19
45°	06:08	06:45	07:18	16:20	16:53	17:30
N 40°	05:59	06:32	07:02	16:35	17:05	17:39
35°	05:50	06:22	06:49	16:48	17:16	17:47
30°	05:43	06:12	06:38	17:00	17:26	17:55
20°	05:28	05:55	06:19	17:19	17:43	18:10
N 10°	05:13	05:39	06:01	17:36	17:59	18:25
0°	04:57	05:23	05:45	17:53	18:15	18:41
S 10°	04:40	05:06	05:29	18:09	18:32	18:59
20°	04:19	04:47	05:12	18:27	18:51	19:20
30°	03:52	04:24	04:51	18:47	19:14	19:47
35°	03:35	04:10	04:39	18:59	19:28	20:04
40°	03:13	03:54	04:26	19:13	19:45	20:25
45°	02:46	03:33	04:09	19:29	20:06	20:53
S 50°	02:07	03:06	03:49	19:50	20:32	21:32
52°	01:45	02:53	03:39	20:00	20:46	21:55
54°	01:15	02:38	03:28	20:11	21:02	22:26
56°	00:18	02:19	03:16	20:23	21:20	23:30
58°	////	01:56	03:01	20:38	21:44	////
S 60°	////	01:23	02:44	20:56	22:18	////

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	01:22	03:36	06:08	12:13	11:45	11:03
N 70°	01:18	03:22	05:36	12:19	12:01	11:36
68°	01:15	03:11	05:13	12:25	12:14	12:01
66°	01:13	03:02	04:55	12:30	12:25	12:20
64°	01:11	02:55	04:41	12:34	12:34	12:35
62°	01:10	02:48	04:29	12:37	12:42	12:48
60°	01:08	02:43	04:19	12:40	12:49	12:59
N 58°	01:07	02:38	04:10	12:43	12:55	13:09
56°	01:06	02:34	04:03	12:45	13:00	13:18
54°	01:05	02:30	03:56	12:47	13:05	13:25
52°	01:04	02:26	03:50	12:49	13:09	13:32
50°	01:03	02:23	03:44	12:51	13:13	13:38
45°	01:01	02:17	03:33	12:55	13:22	13:52
N 40°	01:00	02:11	03:23	12:59	13:29	14:03
35°	00:59	02:06	03:15	13:01	13:35	14:12
30°	00:57	02:02	03:07	13:04	13:41	14:20
20°	00:56	01:55	02:55	13:08	13:50	14:35
N 10°	00:54	01:48	02:44	13:12	13:59	14:48
0°	00:52	01:42	02:34	13:16	14:07	14:59
S 10°	00:51	01:37	02:24	13:20	14:15	15:11
20°	00:49	01:30	02:14	13:24	14:23	15:24
30°	00:48	01:24	02:02	13:28	14:33	15:39
35°	00:47	01:20	01:55	13:31	14:39	15:47
40°	00:46	01:15	01:47	13:34	14:45	15:57
45°	00:44	01:10	01:38	13:37	14:53	16:09
S 50°	00:43	01:04	01:27	13:41	15:02	16:23
52°	00:42	01:01	01:22	13:43	15:06	16:29
54°	00:41	00:58	01:17	13:45	15:11	16:37
56°	00:41	00:55	01:11	13:47	15:16	16:45
58°	00:40	00:51	01:04	13:50	15:22	16:54
S 60°	00:39	00:47	00:56	13:53	15:28	17:05

Day	Sun		Mer. Pass hh:mm	Moon		Age 24-26 37-17%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
30	11:34	11:23	11:49	07:04	19:29	
01	11:12	11:01	11:49	07:55	20:20	
02	10:49	10:38	11:49	08:47	21:14	

December 03, 04, 05 UT (Fri., Sat., Sun.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	71°43.2	221°59.6	S10°28.8	165°50.8	S24°12.6	76°49.2	S03°43.1	237°15.8	S03°41.7					Alpheratz	357°45.3	29°09.3	
1	86°45.6	237°00.8	28.9	180°51.2	12.7	91°51.6	43.0	252°18.0	41.8					Ankaa	353°17.2	-42°14.9	
2	101°48.1	252°02.0	29.0	195°51.6	12.8	106°53.9	42.9	267°20.3	41.8					Schedar	349°42.4	56°36.2	
3	116°50.5	267°03.1	•• 29.1	210°52.0	•• 12.8	121°56.3	•• 42.9	282°22.6	•• 41.9					Diphda	348°57.5	-17°55.6	
4	131°53.0	282°04.3	29.2	225°52.4	12.9	136°58.6	42.8	297°24.8	42.0					Achernar	335°27.6	-57°11.0	
5	146°55.5	297°05.5	29.3	240°52.8	13.0	152°01.0	42.8	312°27.1	42.1					Hamal	328°02.6	23°31.0	
6	161°57.9	312°06.6	S10°29.4	255°53.2	S24°13.0	167°03.4	S03°42.7	327°29.4	S03°42.1					Polaris	318°16.5	89°18.9	
7	177°00.4	327°07.8	29.5	270°53.6	13.1	182°05.7	42.6	342°31.6	42.2					Acamar	315°19.3	-40°15.7	
8	192°02.9	342°09.0	29.6	285°54.0	13.2	197°08.1	42.6	357°33.9	42.3					Menkar	314°16.7	4°08.1	
9	207°05.3	357°10.1	•• 29.8	300°54.4	•• 13.2	212°10.4	•• 42.5	12°36.1	•• 42.4					Mirfak	308°42.5	49°54.1	
10	222°07.8	12°11.3	29.9	315°54.8	13.3	227°12.8	42.5	27°38.4	42.4					Aldebaran	290°51.1	16°31.9	
11	237°10.3	27°12.4	30.0	330°55.2	13.4	242°15.1	42.4	42°40.7	42.5					Rigel	281°13.5	-8°11.3	
12	252°12.7	42°13.6	S10°30.1	345°55.6	S24°13.4	257°17.5	S03°42.3	57°42.9	S03°42.6					Capella	280°36.7	46°00.5	
13	267°15.2	57°14.7	30.2	0°56.0	13.5	272°19.8	42.3	72°45.2	42.7					Bellatrix	278°33.6	6°21.6	
14	282°17.7	72°15.9	30.3	15°56.5	13.6	287°22.2	42.2	87°47.5	42.7					Elnath	278°14.6	28°37.0	
15	297°20.1	87°17.0	•• 30.4	30°56.9	•• 13.6	302°24.5	•• 42.2	102°49.7	•• 42.8					Alnilam	275°47.9	-1°11.7	
16	312°22.6	102°18.2	30.5	45°57.3	13.7	317°26.9	42.1	117°52.0	42.9					Betelgeuse	271°03.0	7°24.5	
17	327°25.0	117°19.3	30.7	60°57.7	13.8	332°29.2	42.0	132°54.2	43.0					Canopus	263°56.5	-52°42.0	
18	342°27.5	132°20.5	S10°30.8	75°58.1	S24°13.8	347°31.6	S03°42.0	147°56.5	S03°43.0					Sirius	258°35.0	-16°43.8	
19	357°30.0	147°21.6	30.9	90°58.5	13.9	2°33.9	41.9	162°58.8	43.1					Adhara	255°13.6	-28°59.2	
20	12°32.4	162°22.7	31.0	105°58.9	13.9	17°36.3	41.9	178°01.0	43.2					Procyon	245°01.4	5°11.8	
21	27°34.9	177°23.9	•• 31.2	120°59.3	•• 14.0	32°38.6	•• 41.8	193°03.3	•• 43.2					Pollux	243°29.7	27°59.8	
22	42°37.4	192°25.0	31.3	135°59.7	14.1	47°41.0	41.7	208°05.6	43.3					Avior	234°18.4	-59°32.5	
23	57°39.8	207°26.1	31.4	151°00.1	14.1	62°43.3	41.7	223°07.8	43.4					Suhail	222°53.6	-43°28.5	
Mer.pass. 19:10			ν 1.2' d 0.1' m -4.90			ν 0.4' d 0.1' m 1.28			ν 2.4' d -0.1' m -2.52			ν 2.3' d 0.1' m 0.88					
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec				
0	72°42.3	222°27.2	S10°31.5	166°00.5	S24°14.2	77°45.7	S03°41.6	238°10.1	S03°43.5					Denebola	182°35.6	14°30.5	
1	87°44.8	237°28.3	31.7	181°00.9	14.2	92°48.0	41.5	253°12.4	43.5					Gienah	175°54.3	-17°36.1	
2	102°47.2	252°29.5	31.8	196°01.3	14.3	107°50.4	41.5	268°14.6	43.6					Acrux	173°11.8	-63°09.4	
3	117°49.7	267°30.6	•• 31.9	211°01.7	•• 14.4	122°52.7	•• 41.4	283°16.9	•• 43.7					Gacrux	172°03.3	-57°10.3	
4	132°52.1	282°31.7	32.1	226°02.1	14.4	137°55.1	41.4	298°19.2	43.8					Alioth	166°22.5	55°53.7	
5	147°54.6	297°32.8	32.2	241°02.5	14.5	152°57.4	41.3	313°21.4	43.8					Spica	158°33.4	-11°13.1	
6	162°57.1	312°33.9	S10°32.3	256°02.9	S24°14.5	167°59.8	S03°41.2	328°23.7	S03°43.9					Alkaid	153°00.6	49°15.3	
7	177°59.5	327°35.0	32.5	271°03.3	14.6	183°02.1	41.2	343°26.0	44.0					Hadar	148°51.1	-60°25.4	
8	193°02.0	342°36.1	32.6	286°03.7	14.6	198°04.5	41.1	358°28.2	44.1					Menkent	148°10.1	-36°25.3	
9	208°04.5	357°37.2	•• 32.8	301°04.1	•• 14.7	213°06.8	•• 41.1	13°30.5	•• 44.1					Arcturus	145°57.7	19°07.4	
10	223°06.9	12°38.3	32.9	316°04.5	14.8	228°09.2	41.0	28°32.7	44.2					Rigel Kent.	139°54.9	-60°52.6	
11	238°09.4	27°39.4	33.0	331°04.9	14.8	243°11.5	40.9	43°35.0	44.3					Kochab	137°20.8	74°06.4	
12	253°11.9	42°40.5	S10°33.2	346°05.3	S24°14.9	258°13.9	S03°40.9	58°37.3	S03°44.3					Zuben'ubi	137°07.8	-16°05.2	
13	268°14.3	57°41.6	33.3	1°05.7	14.9	273°16.2	40.8	73°39.5	44.4					Alphecca	126°12.9	26°40.6	
14	283°16.8	72°42.7	33.5	16°06.1	15.0	288°18.5	40.7	88°41.8	44.5					Antares	112°28.9	-26°27.3	
15	298°19.3	87°43.8	•• 33.6	31°06.5	•• 15.0	303°20.9	•• 40.7	103°44.1	•• 44.6					Atria	107°33.0	-69°02.8	
16	313°21.7	102°44.9	33.8	46°06.9	15.1	318°23.2	40.6	118°46.3	44.6					Sabik	102°15.0	-15°44.2	
17	328°24.2	117°46.0	33.9	61°07.3	15.1	333°25.6	40.5	133°48.6	44.7					Shaula	96°24.8	-37°06.7	
18	343°26.6	132°47.0	S10°34.1	76°07.7	S24°15.2	348°27.9	S03°40.5	148°50.9	S03°44.8					Rasalhague	96°08.5	-12°33.2	
19	358°29.1	147°48.1	34.2	91°08.1	15.2	3°30.3	40.4	163°53.1	44.9					Eltanin	90°47.5	51°29.4	
20	13°31.6	162°49.2	34.4	106°08.5	15.3	18°32.6	40.4	178°55.4	44.9					Kaus Aust.	83°46.6	-34°22.7	
21	28°34.0	177°50.3	•• 34.6	121°08.9	•• 15.3	33°35.0	•• 40.3	193°57.7	•• 45.0					Vega	80°40.6	38°47.8	
22	43°36.5	192°51.3	34.7	136°09.3	15.4	48°37.3	40.2	208°59.9	45.1					Nunki	76°00.9	-26°17.0	
23	58°39.0	207°52.4	34.9	151°09.7	15.4	63°39.6	40.2	224°02.2	45.1					Altair	62°10.3	8°54.0	
Mer.pass. 19:06			ν 1.1' d 0.1' m -4.89			ν 0.4' d 0.1' m 1.28			ν 2.3' d -0.1' m -2.51			ν 2.3' d 0.1' m 0.88					
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec				
0	73°41.4	222°53.5	S10°35.0	166°10.1	S24°15.5	78°42.0	S03°40.1	239°04.5	S03°45.2					Dec 03 Fri	SHA	Mer.pass	
1	88°43.9	237°54.5	35.2	181°10.5	15.5	93°44.3	40.0	254°06.7	45.3					Venus	150°16.5	09:11	
2	103°46.4	252°55.6	35.4	196°10.9	15.6	108°46.7	40.0	269°09.0	45.4					Mars	94°07.6	12:56	
3	118°48.8	267°56.6	•• 35.5	211°11.3	•• 15.6	123°49.0	•• 39.9	284°11.3	•• 45.4					Jupiter	5°06.1	18:50	
4	133°51.3	282°57.7	35.7	226°11.7	15.7	138°51.3	39.8	299°13.5	45.5					Saturn	165°32.6	08:10	
5	148°53.8	297°58.8	35.9	241°12.1	15.7	153°53.7	39.8	314°15.8	45.6								
6	163°56.2	312°59.8	S10°36.0	256°12.5	S24°15.8	168°56.0	S03°39.7	329°18.1	S03°45.6					Dec 04 Sat	SHA	Mer.pass	
7	178°58.7	328°00.9	36.2	271°12.9	15.8	183°58.4	39.6	344°20.4	45.7					Venus	149°44.9	09:10	
8	194°01.1	343°01.9	36.4	286°13.3	15.9	199°00.7	39.6	359°22.6	45.8					Mars	93°18.2	12:56	
9	209°03.6	358°03.0	•• 36.5	301°13.7	•• 15.9	214°03.0	•• 39.5	14°24.9	•• 45.9					Jupiter	5°03.4	18:46	
10	224°06.1	13°04.0	36.7	316°14.1	16.0	229°05.4	39.4	29°27.2	45.9					Saturn	165°27.8	08:06	

h	Sun			Moon				
Fri	GHA	Dec	GHA	ν	Dec	d	HP	
0	182°36.5	S22°03.0	220°04.4	8.0'	S17°30.4	10.3'	59.0'	
1	197°36.3	03.3	234°31.3	7.9'	17°40.7	10.2'	59.0'	
2	212°36.1	03.7	248°58.3	7.9'	17°50.9	10.1'	59.0'	
3	227°35.8	· · 04.0	263°25.2	7.8'	18°00.9	10.0'	59.0'	
4	242°35.6	04.4	277°52.0	7.8'	18°10.9	9.9'	59.0'	
5	257°35.3	04.7	292°18.8	7.7'	18°20.8	9.7'	58.9'	
6	272°35.1	S22°05.1	306°45.5	7.7'	S18°30.5	9.6'	58.9'	
7	287°34.8	05.5	321°12.1	7.6'	18°40.1	9.5'	58.9'	
8	302°34.6	05.8	335°38.7	7.6'	18°49.6	9.4'	58.9'	
9	317°34.3	· · 06.2	350°05.3	7.5'	18°59.0	9.3'	58.9'	
10	332°34.1	06.5	4°31.8	7.4'	19°08.3	9.2'	58.9'	
11	347°33.8	06.9	18°58.2	7.4'	19°17.5	9.0'	58.9'	
12	2°33.6	S22°07.2	33°24.6	7.3'	S19°26.5	8.9'	58.8'	
13	17°33.4	07.6	47°51.0	7.3'	19°35.4	8.8'	58.8'	
14	32°33.1	07.9	62°17.3	7.2'	19°44.2	8.7'	58.8'	
15	47°32.9	· · 08.3	76°43.5	7.2'	19°52.8	8.5'	58.8'	
16	62°32.6	08.6	91°09.7	7.1'	20°01.4	8.4'	58.8'	
17	77°32.4	09.0	105°35.8	7.1'	20°09.8	8.3'	58.8'	
18	92°32.1	S22°09.3	120°01.9	7.0'	S20°18.0	8.1'	58.8'	
19	107°31.9	09.7	134°28.0	7.0'	20°26.2	8.0'	58.7'	
20	122°31.6	10.0	148°54.0	7.0'	20°34.2	7.9'	58.7'	
21	137°31.4	· · 10.3	163°19.9	6.9'	20°42.1	7.8'	58.7'	
22	152°31.1	10.7	177°45.8	6.9'	20°49.8	7.6'	58.7'	
23	167°30.9	11.0	192°11.7	6.8'	20°57.5	7.5'	58.7'	
SD = 16.2'			d = 0.4'	SD = 16.1'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP	
0	182°30.6	S22°11.4	206°37.5	6.8'	S21°04.9	7.3'	58.7'	
1	197°30.4	11.7	221°03.3	6.7'	21°12.3	7.2'	58.6'	
2	212°30.1	12.1	235°29.0	6.7'	21°19.5	7.1'	58.6'	
3	227°29.9	· · 12.4	249°54.7	6.7'	21°26.6	6.9'	58.6'	
4	242°29.6	12.7	264°20.4	6.6'	21°33.5	6.8'	58.6'	
5	257°29.4	13.1	278°46.0	6.6'	21°40.3	6.7'	58.6'	
6	272°29.1	S22°13.4	293°11.6	6.5'	S21°47.0	6.5'	58.6'	
7	287°28.9	13.8	307°37.2	6.5'	21°53.5	6.4'	58.5'	
8	302°28.6	14.1	322°02.7	6.5'	21°59.9	6.2'	58.5'	
9	317°28.3	· · 14.4	336°28.2	6.4'	22°06.1	6.1'	58.5'	
10	332°28.1	14.8	350°53.6	6.4'	22°12.2	6.0'	58.5'	
11	347°27.8	15.1	5°19.0	6.4'	22°18.2	5.8'	58.5'	
12	2°27.6	S22°15.4	19°44.4	6.4'	S22°24.0	5.7'	58.4'	
13	17°27.3	15.8	34°09.8	6.3'	22°29.6	5.5'	58.4'	
14	32°27.1	16.1	48°35.1	6.3'	22°35.1	5.4'	58.4'	
15	47°26.8	· · 16.4	63°00.4	6.3'	22°40.5	5.2'	58.4'	
16	62°26.6	16.7	77°25.7	6.3'	22°45.7	5.1'	58.4'	
17	77°26.3	17.1	91°51.0	6.2'	22°50.8	4.9'	58.3'	
18	92°26.1	S22°17.4	106°16.2	6.2'	S22°55.7	4.8'	58.3'	
19	107°25.8	17.7	120°41.4	6.2'	23°00.5	4.6'	58.3'	
20	122°25.5	18.1	135°06.6	6.2'	23°05.2	4.5'	58.3'	
21	137°25.3	· · 18.4	149°31.8	6.2'	23°09.6	4.3'	58.3'	
22	152°25.0	18.7	163°57.0	6.2'	23°14.0	4.2'	58.2'	
23	167°24.8	19.0	178°22.2	6.1'	23°18.2	4.0'	58.2'	
SD = 16.2'			d = 0.3'	SD = 16.0'				

Sun	GHA	Dec	GHA	ν	Dec	d	HP	
0	182°24.5	S22°19.4	192°47.3	6.1'	S23°22.2	3.9'	58.2'	
1	197°24.3	19.7	207°12.5	6.1'	23°26.1	3.7'	58.2'	
2	212°24.0	20.0	221°37.6	6.1'	23°29.8	3.6'	58.2'	
3	227°23.8	· · 20.3	236°02.7	6.1'	23°33.4	3.4'	58.1'	
4	242°23.5	20.7	250°27.8	6.1'	23°36.8	3.3'	58.1'	
5	257°23.2	21.0	264°52.9	6.1'	23°40.1	3.1'	58.1'	
6	272°23.0	S22°21.3	279°18.1	6.1'	S23°43.2	3.0'	58.1'	
7	287°22.7	21.6	293°43.2	6.1'	23°46.2	2.8'	58.0'	
8	302°22.5	21.9	308°08.3	6.1'	23°49.0	2.7'	58.0'	
9	317°22.2	· · 22.2	322°33.4	6.1'	23°51.7	2.5'	58.0'	
10	332°21.9	22.6	336°58.5	6.1'	23°54.2	2.4'	58.0'	
11	347°21.7	22.9	351°23.6	6.1'	23°56.6	2.2'	57.9'	
12	2°21.4	S22°23.2	5°48.8	6.1'	S23°58.8	2.1'	57.9'	
13	17°21.2	23.5	20°13.9	6.2'	24°00.8	1.9'	57.9'	
14	32°20.9	23.8	34°39.1	6.2'	24°02.8	1.8'	57.9'	
15	47°20.6	· · 24.1	49°04.3	6.2'	24°04.5	1.6'	57.8'	
16	62°20.4	24.4	63°29.4	6.2'	24°06.1	1.5'	57.8'	
17	77°20.1	24.8	77°54.6	6.2'	24°07.6	1.3'	57.8'	
18	92°19.9	S22°25.1	92°19.9	6.2'	S24°08.9	1.1'	57.8'	
19	107°19.6	25.4	106°45.1	6.3'	24°10.0	1.0'	57.7'	
20	122°19.3	25.7	121°10.4	6.3'	24°11.0	0.8'	57.7'	
21	137°19.1	· · 26.0	135°35.7	6.3'	24°11.9	0.7'	57.7'	
22	152°18.8	26.3	150°01.0	6.3'	24°12.6	0.5'	57.7'	
23	167°18.6	26.6	164°26.3	6.4'	24°13.1	0.4'	57.6'	
SD = 16.2'			d = 0.3'	SD = 15.9'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:59	10:03	■	■	13:36	15:41
N 70°	07:41	09:20	■	■	14:20	15:58
68°	07:27	08:51	10:55	12:45	14:49	16:12
66°	07:16	08:30	09:57	13:43	15:10	16:24
64°	07:06	08:12	09:24	14:16	15:28	16:34
62°	06:58	07:58	09:00	14:40	15:42	16:42
60°	06:50	07:45	08:41	14:59	15:54	16:50
N 58°	06:43	07:35	08:25	15:15	16:05	16:57
56°	06:37	07:25	08:11	15:29	16:15	17:03
54°	06:32	07:17	08:00	15:40	16:23	17:08
52°	06:26	07:10	07:50	15:51	16:30	17:13
50°	06:22	07:03	07:40	16:00	16:37	17:18
45°	06:11	06:48	07:21	16:19	16:52	17:29
N 40°	06:01	06:35	07:05	16:35	17:05	17:39
35°	05:53	06:24	06:52	16:48	17:16	17:47
30°	05:45	06:14	06:40	17:00	17:26	17:55
20°	05:29	05:57	06:20	17:20	17:44	18:11
N 10°	05:14	05:40	06:03	17:37	18:00	18:26
0°	04:58	05:24	05:46	17:54	18:16	18:42
S 10°	04:40	05:07	05:30	18:11	18:33	19:00
20°	04:19	04:48	05:12	18:28	18:53	19:22
30°	03:51	04:24	04:51	18:49	19:16	19:49
35°	03:34	04:10	04:39	19:02	19:31	20:07
40°	03:12	03:53	04:25	19:16	19:48	20:29
45°	02:44	03:32	04:08	19:33	20:09	20:57
S 50°	02:03	03:04	03:47	19:54	20:37	21:38
52°	01:40	02:50	03:37	20:04	20:51	22:02
54°	01:07	02:34	03:26	20:15	21:07	22:36
56°	////	02:15	03:13	20:28	21:27	////
58°	////	01:50	02:58	20:44	21:52	////
S 60°	////	01:15	02:40	21:02	22:28	////

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	■	■	■	■	■	■
N 70°	08:24	■	■	10:42	■	■
68°	07:28	■	■	11:40	■	■
66°	06:55	09:01	■	12:14	12:06	■
64°	06:30	08:19	09:56	12:39	12:48	13:12
62°	06:12	07:51	09:18	12:59	13:17	13:51
60°	05:56	07:29	08:51	13:15	13:39	14:18
N 58°	05:43	07:12	08:30	13:29	13:57	14:39
56°	05:32	06:57	08:13	13:41	14:12	14:56
54°	05:22	06:44	07:58	13:51	14:25	15:11
52°	05:13	06:33	07:46	14:00	14:37	15:24
50°	05:05	06:23	07:34	14:09	14:47	15:35
45°	04:49	06:03	07:11	14:26	15:09	15:59
N 40°	04:35	05:46	06:52	14:41	15:26	16:18
35°	04:24	05:32	06:37	14:53	15:41	16:34
30°	04:14	05:20	06:23	15:04	15:53	16:47
20°	03:57	04:59	06:00	15:23	16:15	17:11
N 10°	03:42	04:41	05:40	15:39	16:34	17:31
0°	03:28	04:24	05:22	15:55	16:52	17:50
S 10°	03:14	04:08	05:03	16:10	17:09	18:09
20°	03:00	03:50	04:44	16:26	17:28	18:29
30°	02:43	03:30	04:21	16:45	17:50	18:52
35°	02:34	03:18	04:08	16:56	18:03	19:06
40°	02:23	03:04	03:53	17:09	18:18	19:22
45°	02:10	02:48	03:35	17:24	18:36	19:40
S 50°	01:55	02:29	03:12	17:43	18:58	20:04
52°	01:48	02:20	03:01	17:51	19:08	20:15
54°	01:40	02:10	02:49	18:01	19:20	20:28
56°	01:31	01:58	02:35	18:12	19:34	20:43
58°	01:21	01:45	02:19	18:25	19:50	21:00
S 60°	01:09	01:29	02:00	18:40	20:09	21:21

Day	Sun		Mer.	Moon		Age
	Eqn. of Time	12h	Pass	Mer. Pass.	Lower	27-0
	00h	12h	hh:mm	Upper	hh:mm	9-1%
03	10:26	10:14	11:50	09:41	22:09	
04	10:02	09:50	11:50	10:38	23:07	
05	09:38	09:26	11:51	11:36	—:—	

December 06, 07, 08 UT (Mon., Tue., Wed.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	74°40.6	223°18.4	S10°39.3	166°19.6	S24°16.5	79°38.1	S03°38.5	239°58.9	S03°46.9	Alpheratz	357°45.3	29°09.3
1	89°43.0	238°19.4	39.5	181°20.0	16.6	94°40.4	38.4	255°01.2	47.0	Ankaa	353°17.3	-42°14.9
2	104°45.5	253°20.4	39.7	196°20.4	16.6	109°42.8	38.4	270°03.5	47.1	Schedar	349°42.5	56°36.2
3	119°48.0	268°21.4	· · 39.9	211°20.8	· · 16.7	124°45.1	· · 38.3	285°05.7	· · 47.1	Diphda	348°57.5	-17°55.6
4	134°50.4	283°22.4	40.1	226°21.2	16.7	139°47.4	38.2	300°08.0	47.2	Achernar	335°27.6	-57°11.0
5	149°52.9	298°23.4	40.3	241°21.6	16.7	154°49.8	38.2	315°10.3	47.3	Hamal	328°02.6	23°31.0
6	164°55.4	313°24.4	S10°40.5	256°22.0	S24°16.8	169°52.1	S03°38.1	330°12.6	S03°47.4	Polaris	318°16.7	89°18.9
7	179°57.8	328°25.4	40.7	271°22.4	16.8	184°54.4	38.0	345°14.8	47.4	Acamar	315°19.3	-40°15.7
8	195°00.3	343°26.4	40.9	286°22.8	16.8	199°56.7	37.9	0°17.1	47.5	Menkar	314°16.7	4°08.0
9	210°02.8	358°27.4	· · 41.1	301°23.2	· · 16.9	214°59.1	· · 37.9	15°19.4	· · 47.6	Mirfak	308°42.5	49°54.2
10	225°05.2	13°28.4	41.3	316°23.6	16.9	230°01.4	37.8	30°21.6	47.6	Aldebaran	290°51.1	16°31.9
11	240°07.7	28°29.3	41.5	331°24.0	16.9	245°03.7	37.7	45°23.9	47.7	Rigel	281°13.5	-8°11.3
12	255°10.1	43°30.3	S10°41.7	346°24.4	S24°17.0	260°06.1	S03°37.7	60°26.2	S03°47.8	Capella	280°36.7	46°00.5
13	270°12.6	58°31.3	41.9	1°24.7	17.0	275°08.4	37.6	75°28.5	47.8	Bellatrix	278°33.6	6°21.6
14	285°15.1	73°32.3	42.1	16°25.1	17.0	290°10.7	37.5	90°30.7	47.9	Elnath	278°14.5	28°37.0
15	300°17.5	88°33.3	· · 42.3	31°25.5	· · 17.1	305°13.1	· · 37.5	105°33.0	· · 48.0	Alnilam	275°47.9	-1°11.7
16	315°20.0	103°34.3	42.6	46°25.9	17.1	320°15.4	37.4	120°35.3	48.1	Betelgeuse	271°02.9	7°24.5
17	330°22.5	118°35.2	42.8	61°26.3	17.1	335°17.7	37.3	135°37.5	48.1	Canopus	263°56.4	-52°42.0
18	345°24.9	133°36.2	S10°43.0	76°26.7	S24°17.2	350°20.0	S03°37.2	150°39.8	S03°48.2	Sirius	258°35.0	-16°43.9
19	0°27.4	148°37.2	43.2	91°27.1	17.2	5°22.4	37.2	165°42.1	48.3	Adhara	255°13.6	-28°59.2
20	15°29.9	163°38.1	43.4	106°27.5	17.2	20°24.7	37.1	180°44.4	48.3	Procyon	245°01.4	5°11.7
21	30°32.3	178°39.1	· · 43.6	121°27.9	· · 17.2	35°27.0	· · 37.0	195°46.6	· · 48.4	Pollux	243°29.6	27°59.8
22	45°34.8	193°40.1	43.9	136°28.3	17.3	50°29.4	37.0	210°48.9	48.5	Avior	234°18.4	-59°32.5
23	60°37.3	208°41.0	44.1	151°28.7	17.3	65°31.7	36.9	225°51.2	48.5	Suhail	222°53.6	-43°28.5
Mer.pass. 18:58		ν 1.0' d 0.2' m -4.88		ν 0.4' d 0.0' m 1.27		ν 2.3' d -0.1' m -2.50		ν 2.3' d 0.1' m 0.87		Miaplacidus	221°39.7	-69°45.5
Alphard 217°57.7 -8°42.4 Regulus 207°45.4 11°54.7 Dubhe 193°53.8 61°41.1 Denebola 182°35.6 14°30.5 Gienah 175°54.3 -17°36.1 Acrux 173°11.8 -63°09.4 Gacrux 172°03.3 -57°10.3 Alioth 166°22.5 55°53.7 Spica 158°33.4 -11°13.1 Alkaid 153°00.6 49°15.3 Hadar 148°51.1 -60°25.4 Menkent 148°10.1 -36°25.3 Arcturus 145°57.6 19°07.4 Rigel Kent. 139°54.8 -60°52.6 Kochab 137°20.8 74°06.4 Zuben'ubi 137°07.8 -16°05.2 Alphecca 126°12.9 26°40.6 Antares 112°28.9 -26°27.3 Atria 107°33.0 -69°02.8 Sabik 102°15.0 -15°44.2 Shaula 96°24.8 -37°06.7 Rasalhague 96°08.5 12°33.2 Eltanin 90°47.5 51°29.4 Kaus Aust. 83°46.6 -34°22.7 Vega 80°40.6 38°47.8 Nunki 76°00.9 -26°16.9 Altair 62°10.3 8°54.0 Peacock 53°22.4 -56°42.1 Deneb 49°33.0 45°19.5 Enif 33°49.0 9°55.7 Al Na'ir 27°46.1 -46°54.6 Fomalhaut 15°26.0 -29°33.9 Scheat 13°55.2 28°08.8 Markab 13°40.2 15°16.1												
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	75°39.7	223°42.0	S10°44.3	166°29.1	S24°17.3	80°34.0	S03°36.8	240°53.4	S03°48.6			
1	90°42.2	238°42.9	44.5	181°29.5	17.4	95°36.3	36.8	255°55.7	48.7			
2	105°44.6	253°43.9	44.7	196°29.9	17.4	110°38.7	36.7	270°58.0	48.8			
3	120°47.1	268°44.8	· · 45.0	211°30.3	· · 17.4	125°41.0	· · 36.6	286°00.3	· · 48.8			
4	135°49.6	283°45.8	45.2	226°30.6	17.4	140°43.3	36.5	301°02.5	48.9			
5	150°52.0	298°46.7	45.4	241°31.0	17.5	155°45.6	36.5	316°04.8	49.0			
6	165°54.5	313°47.7	S10°45.7	256°31.4	S24°17.5	170°48.0	S03°36.4	331°07.1	S03°49.0			
7	180°57.0	328°48.6	45.9	271°31.8	17.5	185°50.3	36.3	346°09.4	49.1			
8	195°59.4	343°49.6	46.1	286°32.2	17.5	200°52.6	36.2	1°11.6	49.2			
9	211°01.9	358°50.5	· · 46.4	301°32.6	· · 17.6	215°54.9	· · 36.2	16°13.9	· · 49.2			
10	226°04.4	13°51.5	46.6	316°33.0	17.6	230°57.3	36.1	31°16.2	49.3			
11	241°06.8	28°52.4	46.8	331°33.4	17.6	245°59.6	36.0	46°18.5	49.4			
12	256°09.3	43°53.3	S10°47.1	346°33.8	S24°17.6	261°01.9	S03°36.0	61°20.7	S03°49.4			
13	271°11.8	58°54.3	47.3	1°34.2	17.6	276°04.2	35.9	76°23.0	49.5			
14	286°14.2	73°55.2	47.5	16°34.6	17.7	291°06.5	35.8	91°25.3	49.6			
15	301°16.7	88°56.1	· · 47.8	31°35.0	· · 17.7	306°08.9	· · 35.7	106°27.6	· · 49.7			
16	316°19.1	103°57.0	48.0	46°35.3	17.7	321°11.2	35.7	121°29.8	49.7			
17	331°21.6	118°58.0	48.3	61°35.7	17.7	336°13.5	35.6	136°32.1	49.8			
18	346°24.1	133°58.9	S10°48.5	76°36.1	S24°17.7	351°15.8	S03°35.5	151°34.4	S03°49.9			
19	1°26.5	148°59.8	48.8	91°36.5	17.8	6°18.2	35.4	166°36.7	49.9			
20	16°29.0	164°00.7	49.0	106°36.9	17.8	21°20.5	35.4	181°38.9	50.0			
21	31°31.5	179°01.6	· · 49.3	121°37.3	· · 17.8	36°22.8	· · 35.3	196°41.2	· · 50.1			
22	46°33.9	194°02.5	49.5	136°37.7	17.8	51°25.1	35.2	211°43.5	50.1			
23	61°36.4	209°03.5	49.8	151°38.1	17.8	66°27.4	35.1	226°45.8	50.2			
Mer.pass. 18:54		ν 1.0' d 0.2' m -4.87		ν 0.4' d 0.0' m 1.27		ν 2.3' d -0.1' m -2.49		ν 2.3' d 0.1' m 0.87				
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	76°38.9	224°04.4	S10°50.0	166°38.5	S24°17.8	81°29.8	S03°35.1	241°48.0	S03°50.3			
1	91°41.3	239°05.3	50.3	181°38.9	17.9	96°32.1	35.0	256°50.3	50.3			
2	106°43.8	254°06.2	50.5	196°39.3	17.9	111°34.4	34.9	271°52.6	50.4			
3	121°46.2	269°07.1	· · 50.8	211°39.6	· · 17.9	126°36.7	· · 34.9	286°54.9	· · 50.5			
4	136°48.7	284°08.0	51.0	226°40.0	17.9	141°39.0	34.8	301°57.1	50.5			
5	151°51.2	299°08.9	51.3	241°40.4	17.9	156°41.3	34.7	316°59.4	50.6			
6	166°53.6	314°09.8	S10°51.5	256°40.8	S24°17.9	171°43.7	S03°34.6	332°01.7	S03°50.7			
7	181°56.1	329°10.7	51.8	271°41.2	17.9	186°46.0	34.5	347°04.0	50.7			
8	196°58.6	344°11.6	52.1	286°41.6	18.0	201°48.3	34.5	2°06.3	50.8			
9	212°01.0	359°12.4	· · 52.3	301°42.0	· · 18.0	216°50.6	· · 34.4	17°08.5	· · 50.9			
10	227°03.5	14°13.3	52.6	316°42.4	18.0	231°52.9	34.3	32°10.8	50.9			
11	242°06.0	29°14.2	52.8	331°42.8	18.0	246°55.2	34.2	47°13.1	51.0			
12	257°08.4	44°15.1	S10°53.1	346°43.2	S24°18.0	261°57.6	S03°34.2	62°15.4	S03°51.1			
13	272°10.9	59°16.0	53.4	1°43.5	18.0	276°59.9	34.1	77°17.6	51.1			
14	287°13.4	74°16.9	53.6	16°43.9	18.0	292°02.2	34.0	92°19.9	51.2			
15	302°15.8	89°17.7	· · 53.9	31°44.3	· · 18.0	307°04.5	· · 33.9	107°22.2	· · 51.3			
16	317°18.3	104°18.6	54.2	46°44.7	18.0	322°06.8	33.9	122°24.5	51.4			
17	332°20.7	119°19.5	54.4	61°45.1	18.1	337°09.1	33.8	137°26.7	51.4			
18	347°23.2	134°20.4	S10°54.7	76°45.5	S24°18.1	352°11.4	S03°33.7	152°29.0	S03°51.5			
19	2°25.7	149°21.2	55.0	91°45.9	18.1	7°13.8	33.6	167°31.3	51.6			
20	17°28.1	164°22.1	55.3	106°46.3	18.1	22°16.1	33.6	182°33.6	51.6			
21	32°30.6	179°23.0	· · 55.5	121°46.6	· · 18.1	37°18.4	· · 33.5	197°35.9	· · 51.7			
22	47°33.1	194°23.8	55.8	136°47.0	18.1	52°20.7	33.4	212°38.1	51.8			
23	62°35.5	209°24.7	56.1	151°47.4	18.1	67°23.0	33.3	227°40.4	51.8			
Mer.pass. 18:50		ν 0.9' d 0.3' m -4.87		ν 0.4' d 0.0' m 1.26		ν 2.3' d -0.1' m -2.48		ν 2.3' d 0.1' m 0.87				
Dec 06 Mon SHA Mer.pass Venus 148°37.8 09:06 Mars 91°39.0 12:54 Jupiter 4°57.5 18:39 Saturn 165°18.4 07:59 Dec 07 Tue SHA Mer.pass Venus 148°02.3 09:05 Mars 90°49.4 12:54 Jupiter 4°54.3 18:35 Saturn 165°13.7 07:55 Dec 08 Wed SHA Mer.pass Venus 147°25.5 09:03 Mars 89°59.6 12:53 Jupiter 4°50.9 18:31 Saturn 165°09.2 07:52 Horizontal parallax Venus: 0.3 Mars: 0.1												

h	Sun			Moon			
Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	182°18.3	S22°26.9	178°51.7	6.4'	S24°13.5	0.2'	57.6'
1	197°18.0	27.2	193°17.1	6.4'	24°13.7	0.1'	57.6'
2	212°17.8	27.5	207°42.5	6.5'	24°13.8	-0.1'	57.6'
3	227°17.5	· · 27.8	222°08.0	6.5'	24°13.8	-0.2'	57.5'
4	242°17.2	28.1	236°33.5	6.5'	24°13.6	-0.4'	57.5'
5	257°17.0	28.4	250°59.0	6.6'	24°13.2	-0.5'	57.5'
6	272°16.7	S22°28.7	265°24.6	6.6'	S24°12.7	-0.6'	57.5'
7	287°16.5	29.0	279°50.2	6.7'	24°12.1	-0.8'	57.4'
8	302°16.2	29.3	294°15.8	6.7'	24°11.3	-0.9'	57.4'
9	317°15.9	· · 29.6	308°41.5	6.7'	24°10.3	-1.1'	57.4'
10	332°15.7	29.9	323°07.3	6.8'	24°09.2	-1.2'	57.4'
11	347°15.4	30.2	337°33.1	6.8'	24°08.0	-1.4'	57.3'
12	2°15.1	S22°30.5	351°58.9	6.9'	S24°06.6	-1.5'	57.3'
13	17°14.9	30.8	6°24.8	6.9'	24°05.1	-1.7'	57.3'
14	32°14.6	31.1	20°50.7	7.0'	24°03.4	-1.8'	57.3'
15	47°14.3	· · 31.4	35°16.7	7.0'	24°01.6	-2.0'	57.2'
16	62°14.1	31.7	49°42.8	7.1'	23°59.7	-2.1'	57.2'
17	77°13.8	32.0	64°08.9	7.2'	23°57.6	-2.2'	57.2'
18	92°13.5	S22°32.3	78°35.0	7.2'	S23°55.3	-2.4'	57.1'
19	107°13.3	32.6	93°01.2	7.3'	23°53.0	-2.5'	57.1'
20	122°13.0	32.9	107°27.5	7.3'	23°50.5	-2.6'	57.1'
21	137°12.7	· · 33.2	121°53.8	7.4'	23°47.8	-2.8'	57.1'
22	152°12.5	33.5	136°20.2	7.5'	23°45.0	-2.9'	57.0'
23	167°12.2	33.7	150°46.7	7.5'	23°42.1	-3.1'	57.0'
	SD = 16.2'	$d = 0.3'$	SD = 15.7'				
Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	182°11.9	S22°34.0	165°13.2	7.6'	S23°39.0	-3.2'	57.0'
1	197°11.7	34.3	179°39.8	7.6'	23°35.8	-3.3'	56.9'
2	212°11.4	34.6	194°06.4	7.7'	23°32.5	-3.5'	56.9'
3	227°11.1	· · 34.9	208°33.1	7.8'	23°29.1	-3.6'	56.9'
4	242°10.9	35.2	222°59.9	7.9'	23°25.5	-3.7'	56.9'
5	257°10.6	35.5	237°26.8	7.9'	23°21.7	-3.9'	56.8'
6	272°10.3	S22°35.7	251°53.7	8.0'	S23°17.9	-4.0'	56.8'
7	287°10.1	36.0	266°20.7	8.1'	23°13.9	-4.1'	56.8'
8	302°09.8	36.3	280°47.8	8.1'	23°09.8	-4.2'	56.8'
9	317°09.5	· · 36.6	295°14.9	8.2'	23°05.5	-4.4'	56.7'
10	332°09.2	36.9	309°42.1	8.3'	23°01.2	-4.5'	56.7'
11	347°09.0	37.1	324°09.4	8.4'	22°56.7	-4.6'	56.7'
12	2°08.7	S22°37.4	338°36.8	8.4'	S22°52.1	-4.7'	56.6'
13	17°08.4	37.7	353°04.2	8.5'	22°47.3	-4.9'	56.6'
14	32°08.2	38.0	7°31.8	8.6'	22°42.5	-5.0'	56.6'
15	47°07.9	· · 38.3	21°59.4	8.7'	22°37.5	-5.1'	56.6'
16	62°07.6	38.5	36°27.1	8.8'	22°32.4	-5.2'	56.5'
17	77°07.4	38.8	50°54.8	8.8'	22°27.2	-5.3'	56.5'
18	92°07.1	S22°39.1	65°22.7	8.9'	S22°21.8	-5.5'	56.5'
19	107°06.8	39.4	79°50.6	9.0'	22°16.4	-5.6'	56.4'
20	122°06.5	39.6	94°18.6	9.1'	22°10.8	-5.7'	56.4'
21	137°06.3	· · 39.9	108°46.7	9.2'	22°05.1	-5.8'	56.4'
22	152°06.0	40.2	123°14.8	9.3'	21°59.3	-5.9'	56.4'
23	167°05.7	40.4	137°43.1	9.3'	21°53.4	-6.0'	56.3'
	SD = 16.2'	$d = 0.3'$	SD = 15.5'				
Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	182°05.5	S22°40.7	152°11.4	9.4'	S21°47.4	-6.1'	56.3'
1	197°05.2	41.0	166°39.9	9.5'	21°41.2	-6.2'	56.3'
2	212°04.9	41.2	181°08.4	9.6'	21°35.0	-6.4'	56.3'
3	227°04.6	· · 41.5	195°37.0	9.7'	21°28.6	-6.5'	56.2'
4	242°04.4	41.8	210°05.6	9.8'	21°22.2	-6.6'	56.2'
5	257°04.1	42.0	224°34.4	9.8'	21°15.6	-6.7'	56.2'
6	272°03.8	S22°42.3	239°03.2	9.9'	S21°09.0	-6.8'	56.1'
7	287°03.5	42.6	253°32.2	10.0'	21°02.2	-6.9'	56.1'
8	302°03.3	42.8	268°01.2	10.1'	20°55.3	-7.0'	56.1'
9	317°03.0	· · 43.1	282°30.3	10.2'	20°48.3	-7.1'	56.1'
10	332°02.7	43.4	296°59.5	10.3'	20°41.3	-7.2'	56.0'
11	347°02.4	43.6	311°28.8	10.4'	20°34.1	-7.3'	56.0'
12	2°02.2	S22°43.9	325°58.1	10.5'	S20°26.8	-7.4'	56.0'
13	17°01.9	44.1	340°27.6	10.5'	20°19.4	-7.5'	56.0'
14	32°01.6	44.4	354°57.1	10.6'	20°12.0	-7.6'	55.9'
15	47°01.3	· · 44.7	9°26.8	10.7'	20°04.4	-7.7'	55.9'
16	62°01.1	44.9	23°56.5	10.8'	19°56.8	-7.7'	55.9'
17	77°00.8	45.2	38°26.3	10.9'	19°49.0	-7.8'	55.8'
18	92°00.5	S22°45.4	52°56.2	11.0'	S19°41.2	-7.9'	55.8'
19	107°00.2	45.7	67°26.1	11.1'	19°33.3	-8.0'	55.8'
20	122°00.0	45.9	81°56.2	11.1'	19°25.3	-8.1'	55.8'
21	136°59.7	· · 46.2	96°26.4	11.2'	19°17.2	-8.2'	55.7'
22	151°59.4	46.4	110°56.6	11.3'	19°09.0	-8.3'	55.7'
23	166°59.1	46.7	125°26.9	11.4'	19°00.7	-8.4'	55.7'
	SD = 16.2'	$d = 0.3'$	SD = 15.4'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:06	10:17	■	■	13:26	15:37
N 70°	07:48	09:29	■	■	14:13	15:55
68°	07:33	08:58	11:18	12:24	14:44	16:09
66°	07:21	08:36	10:06	13:36	15:07	16:21
64°	07:11	08:18	09:31	14:12	15:25	16:32
62°	07:02	08:03	09:06	14:37	15:40	16:41
60°	06:54	07:50	08:46	14:57	15:53	16:48
N 58°	06:47	07:39	08:30	15:13	16:04	16:55
56°	06:41	07:29	08:16	15:27	16:13	17:02
54°	06:35	07:21	08:04	15:39	16:22	17:07
52°	06:30	07:13	07:53	15:49	16:30	17:13
50°	06:25	07:06	07:44	15:59	16:37	17:18
45°	06:14	06:51	07:24	16:18	16:52	17:29
N 40°	06:04	06:38	07:08	16:35	17:05	17:39
35°	05:55	06:27	06:55	16:48	17:16	17:48
30°	05:47	06:16	06:43	17:00	17:26	17:56
20°	05:31	05:58	06:22	17:20	17:44	18:12
N 10°	05:16	05:42	06:04	17:38	18:01	18:27
0°	04:59	05:25	05:48	17:55	18:18	18:44
S 10°	04:41	05:08	05:31	18:12	18:35	19:02
20°	04:19	04:48	05:13	18:30	18:55	19:24
30°	03:51	04:24	04:51	18:52	19:19	19:52
35°	03:33	04:10	04:39	19:04	19:34	20:10
40°	03:11	03:52	04:25	19:19	19:51	20:32
45°	02:42	03:31	04:07	19:36	20:13	21:01
S 50°	02:00	03:02	03:46	19:57	20:41	21:44
52°	01:36	02:48	03:36	20:08	20:55	22:09
54°	01:00	02:32	03:24	20:19	21:12	22:45
56°	////	02:11	03:11	20:33	21:33	////
58°	////	01:45	02:55	20:49	21:59	////
S 60°	////	01:07	02:36	21:08	22:39	////

December 09, 10, 11 UT (Thu., Fri., Sat.)

Aries				Venus				Mars				Jupiter				Saturn				Stars																																																																																																																																																																																																																																																																																																																					
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec																																																																																																																																																																																																																																																																																																																								
0	77°38.0	224°25.5	S10°56.4	166°47.8	S24°18.1	82°25.3	S03°33.2	242°42.7	S03°51.9	Alpheratz	357°45.3	29°09.3	1	92°40.5	239°26.4	56.7	181°48.2	18.1	257°45.0	52.0	Ankaa	353°17.3	-42°14.9	2	107°42.9	254°27.3	56.9	196°48.6	18.1	112°29.9	33.1	272°47.3	52.0	Schedar	349°42.5	56°36.2	3	122°45.4	269°28.1	57.2	211°49.0	18.1	127°32.3	33.0	287°49.5	52.1	Diphda	348°57.5	-17°55.6	4	137°47.9	284°29.0	57.5	226°49.4	18.1	142°34.6	32.9	302°51.8	52.2	Achernar	335°27.6	-57°11.0	5	152°50.3	299°29.8	57.8	241°49.8	18.1	157°36.9	32.9	317°54.1	52.2	Hamal	328°02.6	23°31.0	6	167°52.8	314°30.7	S10°58.1	256°50.1	S24°18.1	172°39.2	S03°32.8	332°56.4	S03°52.3	Polaris	318°17.3	89°19.0	7	182°55.2	329°31.5	58.3	271°50.5	18.1	187°41.5	32.7	347°58.7	52.4	Acamar	315°19.3	-40°15.7	8	197°57.7	344°32.3	58.6	286°50.9	18.1	202°43.8	32.6	3°00.9	52.4	Menkar	314°16.7	4°08.0	9	213°00.2	359°33.2	58.9	301°51.3	18.1	217°46.1	32.5	18°03.2	52.5	Mirfak	308°42.5	49°54.2	10	228°02.6	14°34.0	59.2	316°51.7	18.1	232°48.4	32.5	33°05.5	52.6	Aldebaran	290°51.1	16°31.9	11	243°05.1	29°34.9	59.5	331°52.1	18.1	247°50.7	32.4	48°07.8	52.6	Rigel	281°13.4	-8°11.3	12	258°07.6	44°35.7	S10°59.8	346°52.5	S24°18.1	262°53.0	S03°32.3	63°10.1	S03°52.7	Capella	280°36.6	46°00.5	13	273°10.0	59°36.5	11°00.1	1°52.8	18.1	277°55.3	32.2	78°12.3	52.8	Bellatrix	278°33.6	6°21.6	14	288°12.5	74°37.4	00.4	16°53.2	18.1	292°57.7	32.1	93°14.6	52.8	Elnath	278°14.5	28°37.0	15	303°15.0	89°38.2	00.7	31°53.6	18.1	308°00.0	32.1	108°16.9	52.9	Alnilam	275°47.9	-1°11.7	16	318°17.4	104°39.0	01.0	46°54.0	18.1	323°02.3	32.0	123°19.2	52.9	Betelgeuse	271°02.9	7°24.5	17	333°19.9	119°39.8	01.3	61°54.4	18.1	338°04.6	31.9	138°21.5	53.0	Canopus	263°56.4	-52°42.0	18	348°22.4	134°40.7	S11°01.6	76°54.8	S24°18.1	353°06.9	S03°31.8	153°23.7	S03°53.1	Sirius	258°35.0	-16°43.9	19	3°24.8	149°41.5	01.9	91°55.2	18.1	8°09.2	31.8	168°26.0	53.1	Adhara	255°13.6	-28°59.2	20	18°27.3	164°42.3	02.2	106°55.6	18.1	23°11.5	31.7	183°28.3	53.2	Procyon	245°01.3	5°11.7	21	33°29.7	179°43.1	02.5	121°55.9	18.1	38°13.8	31.6	198°30.6	53.3	Pollux	243°29.6	27°59.8	22	48°32.2	194°43.9	02.8	136°56.3	18.1	53°16.1	31.5	213°32.9	53.3	Avior	234°18.4	-59°32.5	23	63°34.7	209°44.8	03.1	151°56.7	18.1	68°18.4	31.4	228°35.1	53.4	Suhail	222°53.6	-43°28.5	Mer.pass. 18:46				ν 0.9' d 0.3' m -4.86				ν 0.4' d 0.0' m 1.26				ν 2.3' d -0.1' m -2.47				ν 2.3' d 0.1' m 0.87			
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec																																																																																																																																																																																																																																																																																																																								
0	78°37.1	224°45.6	S11°03.4	166°57.1	S24°18.1	83°20.7	S03°31.3	243°37.4	S03°53.5	Denebola	182°35.5	14°30.5	1	93°39.6	239°46.4	03.7	181°57.5	18.1	258°39.7	53.5	Gienah	175°54.3	-17°36.1	2	108°42.1	254°47.2	04.0	196°57.9	18.1	113°25.3	31.2	273°42.0	53.6	Acrux	173°11.7	-63°09.4	3	123°44.5	269°48.0	04.3	211°58.3	18.1	128°27.6	31.1	288°44.3	53.7	Gacrux	172°03.2	-57°10.3	4	138°47.0	284°48.8	04.6	226°58.6	18.1	143°29.9	31.0	303°46.6	53.7	Alioth	166°22.4	55°53.7	5	153°49.5	299°49.6	04.9	241°59.0	18.1	158°32.2	30.9	318°48.8	53.8	Spica	158°33.4	-11°13.1	6	168°51.9	314°50.4	S11°05.2	256°59.4	S24°18.0	173°34.5	S03°30.9	333°51.1	S03°53.9	Alkaid	153°00.6	49°15.2	7	183°54.4	329°51.2	05.5	271°59.8	18.0	188°36.8	30.8	348°53.4	53.9	Hadar	148°51.0	-60°25.4	8	198°56.9	344°52.0	05.8	287°00.2	18.0	203°39.1	30.7	3°55.7	54.0	Menkent	148°10.0	-36°25.3	9	213°59.3	359°52.8	06.2	302°00.6	18.0	218°41.4	30.6	18°58.0	54.1	Arcturus	145°57.6	19°07.4	10	229°01.8	14°53.6	06.5	317°00.9	18.0	233°43.7	30.5	34°00.3	54.1	Rigil Kent.	139°54.8	-60°52.6	11	244°04.2	29°54.4	06.8	332°01.3	18.0	248°46.0	30.5	49°02.5	54.2	Kochab	137°20.8	74°06.4	12	259°06.7	44°55.2	S11°07.1	347°01.7	S24°18.0	263°48.3	S03°30.4	64°04.8	S03°54.3	Zuben'ubi	137°07.7	-16°05.2	13	274°09.2	59°55.9	07.4	2°02.1	18.0	278°50.6	30.3	79°07.1	54.3	Alphecca	126°12.9	26°40.6	14	289°11.6	74°56.7	07.7	17°02.5	18.0	293°52.9	30.2	94°09.4	54.4	Antares	112°28.9	-26°27.3	15	304°14.1	89°57.5	08.1	32°02.9	17.9	308°55.2	30.1	109°11.7	54.4	Atria	107°32.9	-69°02.8	16	319°16.6	104°58.3	08.4	47°03.3	17.9	323°57.5	30.0	124°14.0	54.5	Sabik	102°15.0	-15°44.3	17	334°19.0	119°59.1	08.7	62°03.6	17.9	338°59.8	30.0	139°16.2	54.6	Shaula	96°24.8	-37°06.7	18	349°21.5	134°59.8	S11°09.0	77°04.0	S24°17.9	354°02.1	S03°29.9	154°18.5	S03°54.6	Rasalhague	96°08.5	12°33.2	19	4°24.0	150°00.6	09.3	92°04.4	17.9	9°04.4	29.8	169°20.8	54.7	Eltanin	90°47.5	51°29.3	20	19°26.4	165°01.4	09.7	107°04.8	17.9	24°06.7	29.7	184°23.1	54.8	Kaus Aust.	83°46.6	-34°22.7	21	34°28.9	180°02.2	10.0	122°05.2	17.9	39°09.0	29.6	199°25.4	54.8	Vega	80°40.6	38°47.8	22	49°31.3	195°02.9	10.3	137°05.6	17.8	54°11.3	29.5	214°27.7	54.9	Nunki	76°00.9	-26°16.9	23	64°33.8	210°03.7	10.6	152°05.9	17.8	69°13.6	29.5	229°29.9	55.0	Altair	62°10.3	8°54.0	Mer.pass. 18:42				ν 0.8' d 0.3' m -4.85				ν 0.4' d -0.0' m 1.25				ν 2.3' d -0.1' m -2.47				ν 2.3' d 0.1' m 0.86			
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec																																																																																																																																																																																																																																																																																																																								
0	79°36.3	225°04.5	S11°11.0	167°06.3	S24°17.8	84°15.9	S03°29.4	244°32.2	S03°55.0	Dec 09 Thu	SHA	Mer.pass	1	94°38.7	240°05.2	11.3	182°06.7	17.8	259°34.5	55.1	Venus	146°47.5	09:02	2	109°41.2	255°06.0	11.6	197°07.1	17.8	114°20.5	29.2	274°36.8	55.2	Mars	89°09.8	12:52	3	124°43.7	270°06.8	12.0	212°07.5	17.8	129°22.8	29.1	289°39.1	55.2	Jupiter	4°47.3	18:27	4	139°46.1	285°07.5	12.3	227°07.9	17.7	144°25.1	29.0	304°41.4	55.3	Saturn	165°04.7	07:48	5	154°48.6	300°08.3	12.6	242°08.2	17.7	159°27.4	29.0	319°43.7	55.3	Dec 10 Fri	SHA	Mer.pass	6	169°51.1	315°09.0	S11°13.0	257°08.6	S24°17.7	174°29.7	S03°28.9	334°45.9	S03°55.4	Venus	146°08.4	09:00	7	184°53.5	330°09.8	13.3	272°09.0	17.7	189°32.0	28.8	349°48.2	55.5	Mars	88°20.0	12:52	8	199°56.0	345°10.5	13.6	287°09.4	17.7	204°34.3	28.7	4°50.5	55.5	Jupiter	4°43.6	18:24	9	214°58.5	0°11.3	14.0	302°09.8	17.6	219°36.6	28.6	19°52.8	55.6	Saturn	165°00.3	07:44	10	230°00.9	15°12.0	14.3	317°10.2	17.6	234°38.9	28.5	34°55.1	55.7	Dec 11 Sat	SHA	Mer.pass	11	245°03.4	30°12.8	14.7	332°10.5	17.6	249°41.2	28.4	49°57.4	55.7	Venus	145°28.2	08:59	12	260°05.8	45°13.5	S11°15.0	347°10.9	S24°17.6	264°43.5	S03°28.4	64°59.7	S03°55.8	Mars	87°30.1	12:51	13	275°08.3	60°14.3	15.3	2°11.3	17.5	279°45.8	28.3	80°01.9	55.9	Jupiter	4°39.6	18:20	14	290°10.8	75°15.0	15.7	17°11.7	17.5	294°48.1	28.2	95°04.2	55.9	Saturn	164°56.0	07:41	15	305°13.2	90°15.7	16.0	32°12.1	17.5	309°50.3	28.1	110°06.5	56.0	Horizontal parallax			16	320°15.7	105°16.5	16.4	47°12.5	17.5	324°52.6	28.0	125°08.8	56.0	Venus:	0.3	17	335°18.2	120°17.2	16.7	62°12.8	17.4	339°54.9	27.9	140°11.1	56.1	Mars:	0.1	18	350°20.6	135°17.9	S11°17.1	77°13.2	S24°17.4	354°57.2	S03°27.8	155°13.4	S03°56.2	19	5°23.1	150°18.7	17.4	92°13.6	17.4	9°59.5	27.8	170°15.7	56.2	20	20°25.6	165°19.4	17.7	107°14.0	17.4	25°01.8	27.7	185°18.0	56.3	21	35°28.0	180°20.1	18.1	122°14.4	17.3	40°04.1	27.6	200°20.2	56.4	22	50°30.5	195°20.8	18.4	137°14.8	17.3	55°06.4	27.5	215°22.5	56.4	23	65°33.0	210°21.6	18.8	152°15.1	17.3	70°08.7	27.4	230°24.8	56.5	Mer.pass. 18:39				ν 0.8' d 0.3' m -4.84				ν 0.4' d -0.0' m 1.25				ν 2.3' d -0.1' m -2.46				ν 2.3' d 0.1' m 0.86																							

h	Sun			Moon			
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	181°58.8	S22°46.9	139°57.3	11.5'	S18°52.4	-8.4'	55.7'
1	196°58.6	47.2	154°27.8	11.6'	18°43.9	-8.5'	55.6'
2	211°58.3	47.4	168°58.4	11.7'	18°35.4	-8.6'	55.6'
3	226°58.0	· · 47.7	183°29.1	11.8'	18°26.8	-8.7'	55.6'
4	241°57.7	47.9	197°59.8	11.8'	18°18.1	-8.8'	55.6'
5	256°57.5	48.2	212°30.7	11.9'	18°09.4	-8.8'	55.5'
6	271°57.2	S22°48.4	227°01.6	12.0'	S18°00.5	-8.9'	55.5'
7	286°56.9	48.7	241°32.6	12.1'	17°51.6	-9.0'	55.5'
8	301°56.6	48.9	256°03.7	12.2'	17°42.6	-9.1'	55.5'
9	316°56.3	· · 49.2	270°34.8	12.3'	17°33.6	-9.1'	55.4'
10	331°56.1	49.4	285°06.1	12.3'	17°24.4	-9.2'	55.4'
11	346°55.8	49.6	299°37.4	12.4'	17°15.2	-9.3'	55.4'
12	1°55.5	S22°49.9	314°08.8	12.5'	S17°06.0	-9.3'	55.4'
13	16°55.2	50.1	328°40.3	12.6'	16°56.6	-9.4'	55.3'
14	31°54.9	50.4	343°11.9	12.7'	16°47.2	-9.5'	55.3'
15	46°54.7	· · 50.6	357°43.6	12.7'	16°37.7	-9.5'	55.3'
16	61°54.4	50.8	12°15.3	12.8'	16°28.2	-9.6'	55.3'
17	76°54.1	51.1	26°47.1	12.9'	16°18.6	-9.7'	55.2'
18	91°53.8	S22°51.3	41°19.0	13.0'	S16°08.9	-9.7'	55.2'
19	106°53.5	51.6	55°51.0	13.1'	15°59.1	-9.8'	55.2'
20	121°53.3	51.8	70°23.0	13.1'	15°49.3	-9.9'	55.2'
21	136°53.0	· · 52.0	84°55.2	13.2'	15°39.5	-9.9'	55.2'
22	151°52.7	52.3	99°27.4	13.3'	15°29.5	-10.0'	55.1'
23	166°52.4	52.5	113°59.7	13.4'	15°19.6	-10.0'	55.1'
SD = 16.2'			SD = 15.2'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	181°52.1	S22°52.7	128°32.0	13.4'	S15°09.5	-10.1'	55.1'
1	196°51.8	53.0	143°04.4	13.5'	14°59.4	-10.2'	55.1'
2	211°51.6	53.2	157°36.9	13.6'	14°49.2	-10.2'	55.0'
3	226°51.3	· · 53.4	172°09.5	13.6'	14°39.0	-10.3'	55.0'
4	241°51.0	53.6	186°42.2	13.7'	14°28.8	-10.3'	55.0'
5	256°50.7	53.9	201°14.9	13.8'	14°18.4	-10.4'	55.0'
6	271°50.4	S22°54.1	215°47.7	13.9'	S14°08.1	-10.4'	55.0'
7	286°50.2	54.3	230°20.5	13.9'	13°57.6	-10.5'	54.9'
8	301°49.9	54.6	244°53.5	14.0'	13°47.2	-10.5'	54.9'
9	316°49.6	· · 54.8	259°26.5	14.1'	13°36.6	-10.6'	54.9'
10	331°49.3	55.0	273°59.5	14.1'	13°26.0	-10.6'	54.9'
11	346°49.0	55.2	288°32.7	14.2'	13°15.4	-10.7'	54.9'
12	1°48.7	S22°55.4	303°05.9	14.3'	S13°04.7	-10.7'	54.8'
13	16°48.5	55.7	317°39.1	14.3'	12°54.0	-10.8'	54.8'
14	31°48.2	55.9	332°12.5	14.4'	12°43.3	-10.8'	54.8'
15	46°47.9	· · 56.1	346°45.8	14.5'	12°32.4	-10.9'	54.8'
16	61°47.6	56.3	1°19.3	14.5'	12°21.6	-10.9'	54.8'
17	76°47.3	56.5	15°52.8	14.6'	12°10.7	-10.9'	54.7'
18	91°47.0	S22°56.8	30°26.4	14.6'	S11°59.8	-11.0'	54.7'
19	106°46.7	57.0	45°00.0	14.7'	11°48.8	-11.0'	54.7'
20	121°46.5	57.2	59°33.7	14.8'	11°37.7	-11.1'	54.7'
21	136°46.2	· · 57.4	74°07.5	14.8'	11°26.7	-11.1'	54.7'
22	151°45.9	57.6	88°41.3	14.9'	11°15.6	-11.1'	54.7'
23	166°45.6	57.8	103°15.2	14.9'	11°04.4	-11.2'	54.6'
SD = 16.2'			SD = 15.0'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	181°45.3	S22°58.1	117°49.1	15.0'	S10°53.3	-11.2'	54.6'
1	196°45.0	58.3	132°23.1	15.0'	10°42.1	-11.2'	54.6'
2	211°44.7	58.5	146°57.1	15.1'	10°30.8	-11.3'	54.6'
3	226°44.5	· · 58.7	161°31.2	15.1'	10°19.5	-11.3'	54.6'
4	241°44.2	58.9	176°05.3	15.2'	10°08.2	-11.3'	54.6'
5	256°43.9	59.1	190°39.5	15.2'	09°56.9	-11.4'	54.5'
6	271°43.6	S22°59.3	205°13.8	15.3'	S09°45.5	-11.4'	54.5'
7	286°43.3	59.5	219°48.1	15.3'	09°34.1	-11.4'	54.5'
8	301°43.0	59.7	234°22.4	15.4'	09°22.6	-11.5'	54.5'
9	316°42.7	22°59.9	248°56.8	15.4'	09°11.2	-11.5'	54.5'
10	331°42.4	23°00.1	263°31.2	15.5'	08°59.7	-11.5'	54.5'
11	346°42.2	00.3	278°05.7	15.5'	08°48.1	-11.6'	54.5'
12	1°41.9	S23°00.5	292°40.2	15.6'	S08°36.6	-11.6'	54.4'
13	16°41.6	00.8	307°14.8	15.6'	08°25.0	-11.6'	54.4'
14	31°41.3	01.0	321°49.4	15.7'	08°13.4	-11.6'	54.4'
15	46°41.0	· · 01.2	336°24.1	15.7'	08°01.7	-11.7'	54.4'
16	61°40.7	01.4	350°58.8	15.7'	07°50.1	-11.7'	54.4'
17	76°40.4	01.6	5°33.5	15.8'	07°38.4	-11.7'	54.4'
18	91°40.1	S23°01.8	20°08.3	15.8'	S07°26.6	-11.7'	54.4'
19	106°39.8	02.0	34°43.1	15.8'	07°14.9	-11.8'	54.4'
20	121°39.6	02.2	49°17.9	15.9'	07°03.2	-11.8'	54.4'
21	136°39.3	· · 02.3	63°52.8	15.9'	06°51.4	-11.8'	54.3'
22	151°39.0	02.5	78°27.7	16.0'	06°39.6	-11.8'	54.3'
23	166°38.7	02.7	93°02.7	16.0'	06°27.8	-11.8'	54.3'
SD = 16.2'			SD = 14.9'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:12	10:29	■	■	13:16	15:33
N 70°	07:53	09:37	■	■	14:08	15:52
68°	07:38	09:05	■	■	14:40	16:07
66°	07:26	08:41	10:15	13:30	15:04	16:20
64°	07:15	08:22	09:37	14:08	15:23	16:30
62°	07:06	08:07	09:11	14:34	15:38	16:39
60°	06:58	07:54	08:51	14:55	15:51	16:47
N 58°	06:51	07:43	08:34	15:11	16:02	16:55
56°	06:44	07:33	08:20	15:26	16:12	17:01
54°	06:38	07:24	08:07	15:38	16:21	17:07
52°	06:33	07:16	07:57	15:49	16:29	17:12
50°	06:28	07:09	07:47	15:58	16:36	17:18
45°	06:16	06:53	07:27	16:18	16:52	17:29
N 40°	06:06	06:40	07:11	16:35	17:05	17:39
35°	05:57	06:29	06:57	16:48	17:17	17:48
30°	05:49	06:19	06:45	17:01	17:27	17:57
20°	05:33	06:00	06:24	17:21	17:45	18:13
N 10°	05:17	05:43	06:06	17:39	18:02	18:28
0°	05:00	05:27	05:49	17:56	18:19	18:45
S 10°	04:42	05:09	05:32	18:14	18:37	19:04
20°	04:20	04:49	05:13	18:32	18:57	19:26
30°	03:51	04:25	04:52	18:54	19:21	19:54
35°	03:33	04:10	04:39	19:07	19:36	20:13
40°	03:11	03:52	04:25	19:21	19:54	20:35
45°	02:41	03:30	04:07	19:39	20:16	21:05
S 50°	01:58	03:01	03:45	20:01	20:45	21:48
52°	01:32	02:47	03:35	20:11	20:59	22:14
54°	00:54	02:30	03:23	20:23	21:17	22:54
56°	////	02:09	03:09	20:37	21:38	////
58°	////	01:41	02:53	20:53	22:06	////
S 60°	////	00:59	02:34	21:13	22:49	////

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	13:45	12:51	12:21	16:52	19:20	21:18
N 70°	12:55	12:27	12:09	17:40	19:41	21:28
68°	12:23	12:09	11:58	18:11	19:58	21:36
66°	12:00	11:55	11:50	18:33	20:11	21:43
64°	11:41	11:42	11:43	18:51	20:22	21:49
62°	11:26	11:32	11:36	19:05	20:32	21:54
60°	11:13	11:23	11:31	19:18	20:39	21:59
N 58°	11:02	11:15	11:26	19:28	20:46	22:03
56°	10:52	11:08	11:22	19:37	20:53	22:06
54°	10:43	11:02	11:18	19:45	20:58	22:09
52°	10:36	10:57	11:14	19:52	21:03	22:12
50°	10:29	10:52	11:11	19:59	21:07	22:14
45°	10:14	10:41	11:04	20:13	21:17	22:20
N 40°	10:01	10:31	10:58	20:24	21:25	22:24
35°	09:51	10:24	10:53	20:33	21:32	22:28
30°	09:41	10:17	10:48	20:42	21:38	22:31
20°	09:25	10:05	10:40	20:56	21:48	22:37
N 10°	09:11	09:54	10:34	21:09	21:57	22:42
0°	08:58	09:44	10:27	21:21	22:05	22:47
S 10°	08:45	09:34	10:21	21:32	22:13	22:52
20°	08:31	09:23	10:14	21:44	22:22	22:57
30°	08:14	09:11	10:06	21:58	22:32	23:02
35°	08:05	09:04	10:01	22:07	22:38	23:06
40°	07:54	08:56	09:56	22:16	22:44	23:09
45°	07:41	08:46	09:50	22:26	22:51	23:14
S 50°	07:25	08:35	09:42	22:39	23:00	23:19
52°	07:18	08:29	09:39	22:45	23:04	23:21
54°	07:10	08:23	09:35	22:52	23:09	23:23
56°	07:01	08:17	09:31	22:59	23:14	23:26
58°	06:50	08:09	09:26	23:07	23:19	23:29
S 60°	06:38	08:01	09:21	23:17	23:26	23:32

Day	Sun		Mer.	Moon		Age
	Eqn. of Time	12h	Pass	Mer. Pass.	Lower	4-6
	mm:ss	mm:ss	hh:mm	hh:mm	hh:mm	11-27%
09	07:55	07:42	11:52	15:09	02:46	
10	07:29	07:15	11:53	15:55	03:32	
11	07:01	06:47	11:53	16:37	04:16	

December 12, 13, 14 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	80°35.4	225°22.3	S11°19.1	167°15.5	S24°17.3	85°11.0	S03°27.3	245°27.1	S03°56.5	Alpheratz	357°45.3	29°09.3
1	95°37.9	240°23.0	19.5	182°15.9	17.2	100°13.3	27.2	260°29.4	56.6	Ankaa	353°17.3	-42°14.9
2	110°40.3	255°23.7	19.9	197°16.3	17.2	115°15.5	27.2	275°31.7	56.7	Schedar	349°42.5	56°36.2
3	125°42.8	270°24.4	•• 20.2	212°16.7	•• 17.2	130°17.8	•• 27.1	290°34.0	•• 56.7	Diphda	348°57.6	-17°55.6
4	140°45.3	285°25.2	20.6	227°17.0	17.1	145°20.1	27.0	305°36.3	56.8	Achernar	335°27.7	-57°11.0
5	155°47.7	300°25.9	20.9	242°17.4	17.1	160°22.4	26.9	320°38.6	56.9	Hamal	328°02.6	23°31.0
6	170°50.2	315°26.6	S11°21.3	257°17.8	S24°17.1	175°24.7	S03°26.8	335°40.8	S03°56.9	Polaris	318°18.0	89°19.0
7	185°52.7	330°27.3	21.6	272°18.2	17.0	190°27.0	26.7	350°43.1	57.0	Acamar	315°19.3	-40°15.7
8	200°55.1	345°28.0	22.0	287°18.6	17.0	205°29.3	26.6	5°45.4	57.0	Menkar	314°16.7	4°08.0
9	215°57.6	0°28.7	•• 22.3	302°18.9	•• 17.0	220°31.6	•• 26.5	20°47.7	•• 57.1	Mirfak	308°42.5	49°54.2
10	231°00.1	15°29.4	22.7	317°19.3	16.9	235°33.8	26.4	35°50.0	57.2	Aldebaran	290°51.1	16°31.9
11	246°02.5	30°30.1	23.1	332°19.7	16.9	250°36.1	26.4	50°52.3	57.2	Rigel	281°13.4	-8°11.3
12	261°05.0	45°30.8	S11°23.4	347°20.1	S24°16.9	265°38.4	S03°26.3	65°54.6	S03°57.3	Capella	280°36.6	46°00.5
13	276°07.4	60°31.5	23.8	2°20.5	16.8	280°40.7	26.2	80°56.9	57.4	Bellatrix	278°33.6	6°21.6
14	291°09.9	75°32.2	24.2	17°20.9	16.8	295°43.0	26.1	95°59.2	57.4	Elnath	278°14.5	28°37.0
15	306°12.4	90°32.9	•• 24.5	32°21.2	•• 16.8	310°45.3	•• 26.0	111°01.4	•• 57.5	Alnilam	275°47.9	-1°11.7
16	321°14.8	105°33.6	24.9	47°21.6	16.7	325°47.6	25.9	126°03.7	57.5	Betelgeuse	271°02.9	7°24.5
17	336°17.3	120°34.3	25.3	62°22.0	16.7	340°49.8	25.8	141°06.0	57.6	Canopus	263°56.4	-52°42.0
18	351°19.8	135°35.0	S11°25.6	77°22.4	S24°16.7	355°52.1	S03°25.7	156°08.3	S03°57.7	Sirius	258°35.0	-16°43.9
19	6°22.2	150°35.7	26.0	92°22.8	16.6	10°54.4	25.7	171°10.6	57.7	Adhara	255°13.6	-28°59.2
20	21°24.7	165°36.3	26.4	107°23.1	16.6	25°56.7	25.6	186°12.9	57.8	Procyon	245°01.3	5°11.7
21	36°27.2	180°37.0	•• 26.7	122°23.5	•• 16.6	40°59.0	•• 25.5	201°15.2	•• 57.8	Pollux	243°29.6	27°59.8
22	51°29.6	195°37.7	27.1	137°23.9	16.5	56°01.3	25.4	216°17.5	57.9	Avior	234°18.3	-59°32.5
23	66°32.1	210°38.4	27.5	152°24.3	16.5	71°03.5	25.3	231°19.8	58.0	Suhail	222°53.6	-43°28.5
Mer.pass. 18:35		ν 0.7' d 0.4' m -4.83		ν 0.4' d -0.0' m 1.24		ν 2.3' d -0.1' m -2.45		ν 2.3' d 0.1' m 0.86				

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	81°34.6	225°39.1	S11°27.8	167°24.7	S24°16.4	86°05.8	S03°25.2	246°22.1	S03°58.0	Denebola	182°35.5	14°30.5
1	96°37.0	240°39.7	28.2	182°25.0	16.4	101°08.1	25.1	261°24.4	58.1	Gienah	175°54.3	-17°36.1
2	111°39.5	255°40.4	28.6	197°25.4	16.4	116°10.4	25.0	276°26.6	58.1	Acrux	173°11.7	-63°09.4
3	126°41.9	270°41.1	•• 29.0	212°25.8	•• 16.3	131°12.7	•• 24.9	291°28.9	•• 58.2	Gacrux	172°03.2	-57°10.3
4	141°44.4	285°41.8	29.4	227°26.2	16.3	146°15.0	24.8	306°31.2	58.3	Alloth	166°22.4	55°53.7
5	156°46.9	300°42.4	29.7	242°26.6	16.2	161°17.2	24.8	321°33.5	58.3	Spica	158°33.3	-11°13.1
6	171°49.3	315°43.1	S11°30.1	257°26.9	S24°16.2	176°19.5	S03°24.7	336°35.8	S03°58.4	Alkaid	153°00.5	49°15.2
7	186°51.8	330°43.8	30.5	272°27.3	16.2	191°21.8	24.6	351°38.1	58.5	Hadar	148°51.0	-60°25.4
8	201°54.3	345°44.4	30.9	287°27.7	16.1	206°24.1	24.5	6°40.4	58.5	Menkent	148°10.0	-36°25.3
9	216°56.7	0°45.1	•• 31.2	302°28.1	•• 16.1	221°26.4	•• 24.4	21°42.7	•• 58.6	Arcturus	145°57.6	19°07.4
10	231°59.2	15°45.8	31.6	317°28.5	16.0	236°28.6	24.3	36°45.0	58.6	Rigel Kent.	139°54.8	-60°52.6
11	247°01.7	30°46.4	32.0	332°28.8	16.0	251°30.9	24.2	51°47.3	58.7	Kochab	137°20.7	74°06.4
12	262°04.1	45°47.1	S11°32.4	347°29.2	S24°15.9	266°33.2	S03°24.1	66°49.6	S03°58.8	Zuben'ubi	137°07.7	-16°05.2
13	277°06.6	60°47.7	32.8	2°29.6	15.9	281°35.5	24.0	81°51.9	58.8	Alphecca	126°12.9	26°40.6
14	292°09.1	75°48.4	33.2	17°30.0	15.8	296°37.8	23.9	96°54.2	58.9	Antares	112°28.9	-26°27.3
15	307°11.5	90°49.0	•• 33.5	32°30.4	•• 15.8	311°40.0	•• 23.8	111°56.4	•• 58.9	Atria	107°32.9	-69°02.8
16	322°14.0	105°49.7	33.9	47°30.7	15.7	326°42.3	23.7	126°58.7	59.0	Sabik	102°15.0	-15°44.3
17	337°16.4	120°50.3	34.3	62°31.1	15.7	341°44.6	23.7	142°01.0	59.1	Shaula	96°24.8	-37°06.7
18	352°18.9	135°51.0	S11°34.7	77°31.5	S24°15.6	356°46.9	S03°23.6	157°03.3	S03°59.1	Rasalhague	96°08.5	12°33.2
19	7°21.4	150°51.6	35.1	92°31.9	15.6	11°49.1	23.5	172°05.6	59.2	Eltanin	90°47.5	51°29.3
20	22°23.8	165°52.3	35.5	107°32.2	15.6	26°51.4	23.4	187°07.9	59.2	Kaus Aust.	83°46.6	-34°22.7
21	37°26.3	180°52.9	•• 35.9	122°32.6	•• 15.5	41°53.7	•• 23.3	202°10.2	•• 59.3	Vega	80°40.6	38°47.8
22	52°28.8	195°53.6	36.3	137°33.0	15.5	56°56.0	23.2	217°12.5	59.4	Nunki	76°00.9	-26°16.9
23	67°31.2	210°54.2	36.7	152°33.4	15.4	71°58.3	23.1	232°14.8	59.4	Altair	62°10.3	8°54.0
Mer.pass. 18:31		ν 0.7' d 0.4' m -4.82		ν 0.4' d -0.0' m 1.24		ν 2.3' d -0.1' m -2.44		ν 2.3' d 0.1' m 0.85				

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	82°33.7	225°54.8	S11°37.1	167°33.8	S24°15.4	87°00.5	S03°23.0	247°17.1	S03°59.5	Fomalhaut	15°26.0	-29°33.9
1	97°36.2	240°55.5	37.5	182°34.1	15.3	102°02.8	22.9	262°19.4	59.5	Scheat	13°55.2	28°08.8
2	112°38.6	255°56.1	37.8	197°34.5	15.2	117°05.1	22.8	277°21.7	59.6	Markab	13°40.2	15°16.1
3	127°41.1	270°56.7	•• 38.2	212°34.9	•• 15.2	132°07.4	•• 22.7	292°24.0	•• 59.7			
4	142°43.5	285°57.4	38.6	227°35.3	15.1	147°09.6	22.6	307°26.3	59.7			
5	157°46.0	300°58.0	39.0	242°35.7	15.1	162°11.9	22.5	322°28.6	59.8			
6	172°48.5	315°58.6	S11°39.4	257°36.0	S24°15.0	177°14.2	S03°22.4	337°30.9	S03°59.8			
7	187°50.9	330°59.2	39.8	272°36.4	15.0	192°16.5	22.4	352°33.2	03°59.9			
8	202°53.4	345°59.9	40.2	287°36.8	14.9	207°18.7	22.3	7°35.4	04°00.0			
9	217°55.9	1°00.5	•• 40.6	302°37.2	•• 14.9	222°21.0	•• 22.2	22°37.7	•• 00.0			
10	232°58.3	16°01.1	41.0	317°37.5	14.8	237°23.3	22.1	37°40.0	00.1			
11	248°00.8	31°01.7	41.4	332°37.9	14.8	252°25.5	22.0	52°42.3	00.1			
12	263°03.3	46°02.3	S11°41.8	347°38.3	S24°14.7	267°27.8	S03°21.9	67°44.6	S04°00.2			
13	278°05.7	61°03.0	42.2	2°38.7	14.7	282°30.1	21.8	82°46.9	00.2			
14	293°08.2	76°03.6	42.7	17°39.1	14.6	297°32.4	21.7	97°49.2	00.3			
15	308°10.7	91°04.2	•• 43.1	32°39.4	•• 14.5	312°34.6	•• 21.6	112°51.5	•• 00.4			
16	323°13.1	106°04.8	43.5	47°39.8	14.5	327°36.9	21.5	127°53.8	00.4			
17	338°15.6	121°05.4	43.9	62°40.2	14.4	342°39.2	21.4	142°56.1	00.5			
18	353°18.0	136°06.0	S11°44.3	77°40.6	S24°14.4	357°41.4	S03°21.3	157°58.4	S04°00.5			
19	8°20.5	151°06.6	44.7	92°40.9	14.3	12°43.7	21.2	173°00.7	00.6			
20	23°23.0	166°07.2	45.1	107°41.3	14.2	27°46.0	21.1	188°03.0	00.7			
21	38°25.4	181°07.8	•• 45.5	122°41.7	•• 14.2	42°48.3	•• 21.0	203°05.3	•• 00.7			
22	53°27.9	196°08.4	45.9	137°42.1	14.1	57°50.5	20.9	218°07.6	00.8			
23	68°30.4	211°09.0	46.3	152°42.5	14.1	72°52.8	20.8	233°09.9	00.8			
Mer.pass. 18:27		ν 0.6' d 0.4' m -4.81		ν 0.4' d -0.1' m 1.23		ν 2.3' d -0.1' m -2.44		ν 2.3' d 0.1' m 0.85				

Dec 12 Sun	SHA	Mer.pass
Venus	144°46.9	08:58
Mars	86°40.1	12:51
Jupiter	4°35.5	18:16
Saturn	164°51.7	07:37

Dec 13 Mon	SHA	Mer.pass
Venus	144°04.5	08:57
Mars	85°50.1	12:50
Jupiter	4°31.3	18:13
Saturn	164°47.5	07:33

Dec 14 Tue	SHA	Mer.pass
Venus	143°21.1	08:56
Mars	85°00.1	12:49
Jupiter	4°26.8	18:09
Saturn	164°43.4	07:30

Horizontal parallax		
Venus:		0.3
Mars:		0.1

h Sun Moon						
Sun	GHA	Dec	GHA	ν	Dec	d HP
0	181°38.4	S23°02.9	107°37.7	16.0'	S06°15.9	-11.9' 54.3'
1	196°38.1	03.1	122°12.7	16.1'	06°04.1	-11.9' 54.3'
2	211°37.8	03.3	136°47.7	16.1'	05°52.2	-11.9' 54.3'
3	226°37.5	· · 03.5	151°22.8	16.1'	05°40.3	-11.9' 54.3'
4	241°37.2	03.7	165°57.9	16.1'	05°28.4	-11.9' 54.3'
5	256°37.0	03.9	180°33.1	16.2'	05°16.4	-11.9' 54.3'
6	271°36.7	S23°04.1	195°08.2	16.2'	S05°04.5	-12.0' 54.3'
7	286°36.4	04.3	209°43.4	16.2'	04°52.5	-12.0' 54.2'
8	301°36.1	04.5	224°18.6	16.2'	04°40.6	-12.0' 54.2'
9	316°35.8	· · 04.6	238°53.9	16.3'	04°28.6	-12.0' 54.2'
10	331°35.5	04.8	253°29.1	16.3'	04°16.6	-12.0' 54.2'
11	346°35.2	05.0	268°04.4	16.3'	04°04.6	-12.0' 54.2'
12	1°34.9	S23°05.2	282°39.7	16.3'	S03°52.5	-12.0' 54.2'
13	16°34.6	05.4	297°15.1	16.3'	03°40.5	-12.0' 54.2'
14	31°34.3	05.6	311°50.4	16.4'	03°28.5	-12.1' 54.2'
15	46°34.0	· · 05.7	326°25.8	16.4'	03°16.4	-12.1' 54.2'
16	61°33.7	05.9	341°01.2	16.4'	03°04.3	-12.1' 54.2'
17	76°33.5	06.1	355°36.6	16.4'	02°52.3	-12.1' 54.2'
18	91°33.2	S23°06.3	10°12.0	16.4'	S02°40.2	-12.1' 54.2'
19	106°32.9	06.5	24°47.4	16.4'	02°28.1	-12.1' 54.2'
20	121°32.6	06.6	39°22.8	16.5'	02°16.0	-12.1' 54.2'
21	136°32.3	· · 06.8	53°58.3	16.5'	02°03.9	-12.1' 54.2'
22	151°32.0	07.0	68°33.8	16.5'	01°51.8	-12.1' 54.2'
23	166°31.7	07.2	83°09.2	16.5'	01°39.7	-12.1' 54.2'
SD = 16.2' d = 0.2' SD = 14.8'						
Mon	GHA	Dec	GHA	ν	Dec	d HP
0	181°31.4	S23°07.4	97°44.7	16.5'	S01°27.5	-12.1' 54.2'
1	196°31.1	07.5	112°20.2	16.5'	01°15.4	-12.1' 54.2'
2	211°30.8	07.7	126°55.7	16.5'	01°03.3	-12.1' 54.2'
3	226°30.5	· · 07.9	141°31.2	16.5'	00°51.2	-12.1' 54.2'
4	241°30.2	08.0	156°06.7	16.5'	00°39.0	-12.1' 54.2'
5	256°29.9	08.2	170°42.2	16.5'	00°26.9	-12.1' 54.2'
6	271°29.6	S23°08.4	185°17.7	16.5'	S00°14.8	-12.1' 54.1'
7	286°29.3	08.6	199°53.3	16.5'	S00°02.6	-12.1' 54.1'
8	301°29.1	08.7	214°28.8	16.5'	N00°09.5	12.1' 54.1'
9	316°28.8	· · 08.9	229°04.3	16.5'	00°21.7	12.1' 54.1'
10	331°28.5	09.1	243°39.8	16.5'	00°33.8	12.1' 54.1'
11	346°28.2	09.2	258°15.3	16.5'	00°45.9	12.1' 54.1'
12	1°27.9	S23°09.4	272°50.9	16.5'	N00°58.0	12.1' 54.2'
13	16°27.6	09.6	287°26.4	16.5'	01°10.2	12.1' 54.2'
14	31°27.3	09.7	302°01.9	16.5'	01°22.3	12.1' 54.2'
15	46°27.0	· · 09.9	316°37.4	16.5'	01°34.4	12.1' 54.2'
16	61°26.7	10.0	331°12.8	16.5'	01°46.5	12.1' 54.2'
17	76°26.4	10.2	345°48.3	16.5'	01°58.6	12.1' 54.2'
18	91°26.1	S23°10.4	0°23.8	16.5'	N02°10.7	12.1' 54.2'
19	106°25.8	10.5	14°59.3	16.5'	02°22.8	12.1' 54.2'
20	121°25.5	10.7	29°34.7	16.4'	02°34.9	12.1' 54.2'
21	136°25.2	· · 10.8	44°10.1	16.4'	02°47.0	12.1' 54.2'
22	151°24.9	11.0	58°45.6	16.4'	02°59.1	12.1' 54.2'
23	166°24.6	11.2	73°21.0	16.4'	03°11.1	12.1' 54.2'
SD = 16.2' d = 0.2' SD = 14.8'						
Tue	GHA	Dec	GHA	ν	Dec	d HP
0	181°24.3	S23°11.3	87°56.4	16.4'	N03°23.2	12.0' 54.2'
1	196°24.0	11.5	102°31.8	16.4'	03°35.2	12.0' 54.2'
2	211°23.7	11.6	117°07.1	16.3'	03°47.2	12.0' 54.2'
3	226°23.4	· · 11.8	131°42.5	16.3'	03°59.3	12.0' 54.2'
4	241°23.1	11.9	146°17.8	16.3'	04°11.3	12.0' 54.2'
5	256°22.8	12.1	160°53.1	16.3'	04°23.3	12.0' 54.2'
6	271°22.5	S23°12.2	175°28.4	16.3'	N04°35.2	12.0' 54.2'
7	286°22.2	12.4	190°03.6	16.2'	04°47.2	12.0' 54.2'
8	301°22.0	12.5	204°38.9	16.2'	04°59.2	11.9' 54.2'
9	316°21.7	· · 12.7	219°14.1	16.2'	05°11.1	11.9' 54.2'
10	331°21.4	12.8	233°49.3	16.2'	05°23.0	11.9' 54.2'
11	346°21.1	13.0	248°24.4	16.1'	05°34.9	11.9' 54.3'
12	1°20.8	S23°13.1	262°59.5	16.1'	N05°46.8	11.9' 54.3'
13	16°20.5	13.3	277°34.7	16.1'	05°58.7	11.9' 54.3'
14	31°20.2	13.4	292°09.7	16.0'	06°10.5	11.8' 54.3'
15	46°19.9	· · 13.6	306°44.8	16.0'	06°22.4	11.8' 54.3'
16	61°19.6	13.7	321°19.8	16.0'	06°34.2	11.8' 54.3'
17	76°19.3	13.8	335°54.8	15.9'	06°46.0	11.8' 54.3'
18	91°19.0	S23°14.0	350°29.7	15.9'	N06°57.7	11.7' 54.3'
19	106°18.7	14.1	5°04.6	15.9'	07°09.5	11.7' 54.3'
20	121°18.4	14.3	19°39.5	15.8'	07°21.2	11.7' 54.3'
21	136°18.1	· · 14.4	34°14.3	15.8'	07°32.9	11.7' 54.4'
22	151°17.8	14.5	48°49.1	15.8'	07°44.6	11.7' 54.4'
23	166°17.5	14.7	63°23.9	15.7'	07°56.3	11.6' 54.4'
SD = 16.2' d = 0.2' SD = 14.8'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:17	10:40	■	■	13:08	15:31
N 70°	07:58	09:44	■	■	14:04	15:50
68°	07:42	09:10	■	■	14:38	16:06
66°	07:29	08:46	10:23	13:25	15:02	16:19
64°	07:19	08:27	09:43	14:05	15:22	16:29
62°	07:09	08:11	09:16	14:32	15:37	16:39
60°	07:01	07:58	08:55	14:53	15:51	16:47
N 58°	06:54	07:46	08:38	15:11	16:02	16:54
56°	06:47	07:36	08:23	15:25	16:12	17:01
54°	06:41	07:27	08:11	15:37	16:21	17:07
52°	06:35	07:19	08:00	15:48	16:29	17:13
50°	06:30	07:12	07:50	15:58	16:36	17:18
45°	06:19	06:56	07:30	16:19	16:52	17:29
N 40°	06:09	06:43	07:13	16:35	17:06	17:40
35°	05:59	06:31	06:59	16:49	17:17	17:49
30°	05:51	06:21	06:47	17:01	17:28	17:57
20°	05:35	06:02	06:26	17:22	17:46	18:14
N 10°	05:19	05:45	06:08	17:41	18:03	18:30
0°	05:02	05:28	05:50	17:58	18:20	18:47
S 10°	04:43	05:10	05:33	18:15	18:38	19:05
20°	04:21	04:50	05:14	18:34	18:59	19:28
30°	03:52	04:25	04:53	18:56	19:23	19:56
35°	03:34	04:10	04:40	19:09	19:38	20:15
40°	03:11	03:52	04:25	19:24	19:56	20:38
45°	02:41	03:30	04:07	19:41	20:19	21:08
S 50°	01:56	03:01	03:45	20:04	20:48	21:53
52°	01:30	02:46	03:34	20:14	21:03	22:19
54°	00:48	02:28	03:22	20:26	21:20	23:02
56°	////	02:07	03:08	20:40	21:42	////
58°	////	01:38	02:52	20:57	22:11	////
S 60°	////	00:53	02:32	21:17	22:57	////

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	11:59	11:38	11:17	23:06	· · ·	00:53
N 70°	11:53	11:39	11:25	23:09	· · ·	00:48
68°	11:49	11:40	11:31	23:11	· · ·	00:44
66°	11:45	11:41	11:36	23:12	· · ·	00:40
64°	11:42	11:41	11:41	23:14	· · ·	00:37
62°	11:39	11:42	11:45	23:15	· · ·	00:35
60°	11:37	11:43	11:48	23:16	· · ·	00:33
N 58°	11:35	11:43	11:51	23:17	· · ·	00:31
56°	11:33	11:43	11:54	23:18	· · ·	00:29
54°	11:31	11:44	11:57	23:18	· · ·	00:28
52°	11:30	11:44	11:59	23:19	· · ·	00:26
50°	11:28	11:44	12:01	23:20	· · ·	00:25
45°	11:25	11:45	12:05	23:21	· · ·	00:22
N 40°	11:22	11:46	12:09	23:22	· · ·	00:20
35°	11:20	11:46	12:12	23:23	· · ·	00:18
30°	11:18	11:47	12:15	23:24	· · ·	00:16
20°	11:14	11:47	12:20	23:25	· · ·	00:13
N 10°	11:11	11:48	12:25	23:27	· · ·	00:11
0°	11:08	11:49	12:29	23:28	· · ·	00:08
S 10°	11:05	11:49	12:34	23:29	· · ·	00:06
20°	11:02	11:50	12:38	23:30	· · ·	00:03
30°	10:59	11:51	12:43	23:31	· · ·	00:00
35°	10:57	11:51	12:46	23:32	23:58	· · ·
40°	10:54	11:52	12:50	23:33	23:56	· · ·
45°	10:52	11:53	12:54	23:34	23:54	· · ·
S 50°	10:48	11:53	12:59	23:35	23:51	· · ·
52°	10:47	11:54	13:01	23:36	23:50	· · ·
54°	10:45	11:54	13:04	23:36	23:49	· · ·
56°	10:43	11:55	13:06	23:37	23:47	23:58
58°	10:41	11:55	13:09	23:38	23:46	23:54
S 60°	10:39	11:56	13:13	23:38	23:44	23:50

Day	Sun		Mer. Pass	Moon		Age 7-9 35-54%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
12	06:34	06:20	11:54	17:18	04:58	
13	06:06	05:51	11:54	17:58	05:38	
14	05:37	05:23	11:55	18:39	06:19	

December 15, 16, 17 UT (Wed., Thu., Fri.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	83°32.8	226°09.6	S11°46.7	167°42.8	S24°14.0	87°55.1	S03°20.7	248°12.2	S04°00.9					Alpheratz	357°45.4 29°09.3		
1	98°35.3	241°10.2	47.2	182°43.2	13.9	102°57.3	20.6	263°14.5	00.9					Ankaa	353°17.3 -42°14.9		
2	113°37.8	256°10.8	47.6	197°43.6	13.9	117°59.6	20.6	278°16.8	01.0					Schedar	349°42.5 56°36.2		
3	128°40.2	271°11.4	· · 48.0	212°44.0	· · 13.8	133°01.9	· · 20.5	293°19.1	· · 01.1					Diphda	348°57.6 -17°55.6		
4	143°42.7	286°12.0	48.4	227°44.3	13.7	148°04.1	20.4	308°21.4	01.1					Achernar	335°27.7 -57°11.0		
5	158°45.1	301°12.6	48.8	242°44.7	13.7	163°06.4	20.3	323°23.7	01.2					Hamal	328°02.6 23°31.0		
6	173°47.6	316°13.2	S11°49.2	257°45.1	S24°13.6	178°08.7	S03°20.2	338°26.0	S04°01.2					Polaris	318°18.8 89°19.0		
7	188°50.1	331°13.7	49.7	272°45.5	13.5	193°10.9	20.1	353°28.3	01.3					Acamar	315°19.3 -40°15.7		
8	203°52.5	346°14.3	50.1	287°45.8	13.5	208°13.2	20.0	8°30.6	01.4					Menkar	314°16.7 4°08.0		
9	218°55.0	1°14.9	· · 50.5	302°46.2	· · 13.4	223°15.5	· · 19.9	23°32.9	· · 01.4					Mirfak	308°42.5 49°54.2		
10	233°57.5	16°15.5	50.9	317°46.6	13.3	238°17.7	19.8	38°35.2	01.5					Aldebaran	290°51.1 16°31.9		
11	248°59.9	31°16.1	51.3	332°47.0	13.3	253°20.0	19.7	53°37.5	01.5					Rigel	281°13.4 -8°11.4		
12	264°02.4	46°16.6	S11°51.8	347°47.4	S24°13.2	268°22.3	S03°19.6	68°39.8	S04°01.6					Capella	280°36.6 46°00.5		
13	279°04.9	61°17.2	52.2	2°47.7	13.1	283°24.5	19.5	83°42.1	01.6					Bellatrix	278°33.6 6°21.6		
14	294°07.3	76°17.8	52.6	17°48.1	13.1	298°26.8	19.4	98°44.4	01.7					Elnath	278°14.5 28°37.0		
15	309°09.8	91°18.4	· · 53.0	32°48.5	· · 13.0	313°29.1	· · 19.3	113°46.7	· · 01.8					Alnilam	275°47.9 -1°11.7		
16	324°12.3	106°18.9	53.4	47°48.9	12.9	328°31.3	19.2	128°49.0	01.8					Betelgeuse	271°02.9 7°24.5		
17	339°14.7	121°19.5	53.9	62°49.2	12.9	343°33.6	19.1	143°51.3	01.9					Canopus	263°56.4 -52°42.1		
18	354°17.2	136°20.1	S11°54.3	77°49.6	S24°12.8	358°35.8	S03°19.0	158°53.6	S04°01.9					Sirius	258°35.0 -16°43.9		
19	9°19.6	151°20.6	54.7	92°50.0	12.7	13°38.1	18.9	173°55.9	02.0					Adhara	255°13.6 -28°59.2		
20	24°22.1	166°21.2	55.2	107°50.4	12.7	28°40.4	18.8	188°58.2	02.0					Procyon	245°01.3 5°11.7		
21	39°24.6	181°21.8	· · 55.6	122°50.7	· · 12.6	43°42.6	· · 18.7	204°00.5	· · 02.1					Pollux	243°29.6 27°59.8		
22	54°27.0	196°22.3	56.0	137°51.1	12.5	58°44.9	18.6	219°02.8	02.2					Avior	234°18.3 -59°32.6		
23	69°29.5	211°22.9	56.4	152°51.5	12.4	73°47.2	18.5	234°05.1	02.2					Suhail	222°53.5 -43°28.5		
Mer.pass. 18:23			ν 0.6' d 0.4' m -4.80			ν 0.4' d -0.1' m 1.23			ν 2.3' d -0.1' m -2.43			ν 2.3' d 0.1' m 0.85					

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	84°32.0	226°23.4	S11°56.9	167°51.9	S24°12.4	88°49.4	S03°18.4	249°07.4	S04°02.3					Denebola	182°35.5 14°30.5		
1	99°34.4	241°24.0	57.3	182°52.2	12.3	103°51.7	18.3	264°09.7	02.3					Gienah	175°54.2 -17°36.2		
2	114°36.9	256°24.5	57.7	197°52.6	12.2	118°53.9	18.2	279°12.0	02.4					Acrux	173°11.7 -63°09.4		
3	129°39.4	271°25.1	· · 58.2	212°53.0	· · 12.1	133°56.2	· · 18.1	294°14.3	· · 02.4					Gacrux	172°03.2 -57°10.3		
4	144°41.8	286°25.6	58.6	227°53.4	12.1	148°58.5	18.0	309°16.6	02.5					Alioth	166°22.4 55°53.7		
5	159°44.3	301°26.2	59.0	242°53.8	12.0	164°00.7	17.9	324°18.9	02.6					Spica	158°33.3 -11°13.1		
6	174°46.8	316°26.7	S11°59.5	257°54.1	S24°11.9	179°03.0	S03°17.8	339°21.2	S04°02.6					Alkaid	153°00.5 49°15.2		
7	189°49.2	331°27.3	11°59.9	272°54.5	11.8	194°05.2	17.7	354°23.5	02.7					Hadar	148°51.0 -60°25.4		
8	204°51.7	346°27.8	12°00.3	287°54.9	11.8	209°07.5	17.6	9°25.8	02.7					Menkent	148°10.0 -36°25.3		
9	219°54.1	1°28.4	· · 00.8	302°55.3	· · 11.7	224°09.8	· · 17.5	24°28.1	· · 02.8					Arcturus	145°57.6 19°07.4		
10	234°56.6	16°28.9	01.2	317°55.6	11.6	239°12.0	17.4	39°30.4	02.8					Rigel Kent.	139°54.7 -60°52.6		
11	249°59.1	31°29.5	01.6	332°56.0	11.5	254°14.3	17.3	54°32.7	02.9					Kochab	137°20.7 74°06.3		
12	265°01.5	46°30.0	S12°02.1	347°56.4	S24°11.5	269°16.5	S03°17.2	69°35.0	S04°02.9					Zuben'ubi	137°07.7 -16°05.2		
13	280°04.0	61°30.5	02.5	2°56.8	11.4	284°18.8	17.1	84°37.3	03.0					Alphecca	126°12.8 26°40.6		
14	295°06.5	76°31.1	03.0	17°57.1	11.3	299°21.1	17.0	99°39.6	03.1					Antares	112°28.9 -26°27.3		
15	310°08.9	91°31.6	· · 03.4	32°57.5	· · 11.2	314°23.3	· · 16.9	114°41.9	· · 03.1					Atria	107°32.9 -69°02.7		
16	325°11.4	106°32.1	03.9	47°57.9	11.1	329°25.6	16.8	129°44.2	03.2					Sabik	102°15.0 -15°44.3		
17	340°13.9	121°32.7	04.3	62°58.3	11.0	344°27.8	16.7	144°46.5	03.2					Shaula	96°24.8 -37°06.6		
18	355°16.3	136°33.2	S12°04.7	77°58.6	S24°11.0	359°30.1	S03°16.6	159°48.8	S04°03.3					Rasalhague	96°08.5 12°33.2		
19	10°18.8	151°33.7	05.2	92°59.0	10.9	14°32.3	16.5	174°51.1	03.3					Eltanin	90°47.5 51°29.3		
20	25°21.2	166°34.2	05.6	107°59.4	10.8	29°34.6	16.4	189°53.4	03.4					Kaus Aust.	83°46.6 -34°22.7		
21	40°23.7	181°34.8	· · 06.1	122°59.8	· · 10.7	44°36.9	· · 16.3	204°55.7	· · 03.4					Vega	80°40.6 38°47.7		
22	55°26.2	196°35.3	06.5	138°00.1	10.6	59°39.1	16.2	219°58.0	03.5					Nunki	76°00.9 -26°16.9		
23	70°28.6	211°35.8	07.0	153°00.5	10.6	74°41.4	16.1	235°00.3	03.6					Altair	62°10.3 8°54.0		
Mer.pass. 18:19			ν 0.6' d 0.4' m -4.79			ν 0.4' d -0.1' m 1.23			ν 2.3' d -0.1' m -2.42			ν 2.3' d 0.1' m 0.85					

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	85°31.1	226°36.3	S12°07.4	168°00.9	S24°10.5	89°43.6	S03°16.0	250°02.6	S04°03.6					Fomalhaut	15°26.0 -29°33.9
1	100°33.6	241°36.8	07.9	183°01.3	10.4	104°45.9	15.9	265°04.9	03.7					Scheat	13°55.2 28°08.8
2	115°36.0	256°37.4	08.3	198°01.6	10.3	119°48.1	15.8	280°07.2	03.7					Markab	13°40.2 15°16.1
3	130°38.5	271°37.9	· · 08.8	213°02.0	· · 10.2	134°50.4	· · 15.7	295°09.5	· · 03.8					Dec 15 Wed	
4	145°41.0	286°38.4	09.2	228°02.4	10.1	149°52.6	15.6	310°11.8	03.8					Venus	142°36.8 08:55
5	160°43.4	301°38.9	09.7	243°02.8	10.0	164°54.9	15.5	325°14.2	03.9					Mars	84°10.0 12:49
6	175°45.9	316°39.4	S12°10.1	258°03.1	S24°09.9	179°57.1	S03°15.4	340°16.5	S04°03.9					Jupiter	4°22.2 18:06
7	190°48.4	331°39.9	10.6	273°03.5	09.9	194°59.4	15.3	355°18.8	04.0					Saturn	164°39.4 07:26
8	205°50.8	346°40.4	11.0	288°03.9	09.8	210°01.6	15.2	10°21.1	04.1					Dec 16 Thu	
9	220°53.3	1°40.9	· · 11.5	303°04.3	· · 09.7	225°03.9	· · 15.1	25°23.4	· · 04.1					Venus	141°51.5 08:54
10	235°55.7	16°41.4	11.9	318°04.6	09.6	240°06.2	15.0	40°25.7	04.2					Mars	83°19.9 12:48
11	250°58.2	31°41.9	12.4	333°05.0	09.5	255°08.4	14.9	55°28.0	04.2					Jupiter	4°17.5 18:02
12	266°00.7	46°42.4	S12°12.8	348°05.4	S24°09.4	270°10.7	S03°14.8	70°30.3	S04°04.3					Saturn	164°35.4 07:22
13	281°03.1	61°42.9	13.3	3°05.8	09.3	285°12.9	14.7	85°32.6	04.3					Dec 17 Fri	
14	296°05.6	76°43.4	13.7	18°06.1	09.2	300°15.2	14.6	100°34.9	04.4					Venus	141°05.2 08:53
15	311°08.1	91°43.9	· · 14.2	33°06.5	· · 09.1	315°17.4	· · 14.5	115°37.2	· · 04.4					Mars	82°29.8 12:48
16	326°10.5	106°44.4	14.6	48°06.9	09.1	330°19.7	14.4								

h Sun Moon							
Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	181°17.2	S23°14.8	77°58.6	15.7'	N08°07.9	11.6'	54.4'
1	196°16.9	14.9	92°33.3	15.6'	08°19.5	11.6'	54.4'
2	211°16.6	15.1	107°07.9	15.6'	08°31.1	11.6'	54.4'
3	226°16.3	· · 15.2	121°42.5	15.6'	08°42.6	11.5'	54.4'
4	241°16.0	15.4	136°17.1	15.5'	08°54.1	11.5'	54.4'
5	256°15.7	15.5	150°51.6	15.5'	09°05.6	11.5'	54.5'
6	271°15.4	S23°15.6	165°26.1	15.4'	N09°17.1	11.4'	54.5'
7	286°15.1	15.7	180°00.5	15.4'	09°28.5	11.4'	54.5'
8	301°14.8	15.9	194°34.9	15.3'	09°39.9	11.4'	54.5'
9	316°14.5	· · 16.0	209°09.2	15.3'	09°51.3	11.3'	54.5'
10	331°14.2	16.1	223°43.5	15.2'	10°02.7	11.3'	54.5'
11	346°13.9	16.3	238°17.7	15.2'	10°14.0	11.3'	54.5'
12	1°13.6	S23°16.4	252°51.9	15.1'	N10°25.2	11.2'	54.6'
13	16°13.3	16.5	267°26.0	15.1'	10°36.5	11.2'	54.6'
14	31°13.0	16.6	282°00.1	15.0'	10°47.7	11.2'	54.6'
15	46°12.7	· · 16.8	296°34.1	15.0'	10°58.9	11.1'	54.6'
16	61°12.4	16.9	311°08.1	14.9'	11°10.0	11.1'	54.6'
17	76°12.1	17.0	325°42.0	14.9'	11°21.1	11.1'	54.6'
18	91°11.8	S23°17.1	340°15.8	14.8'	N11°32.1	11.0'	54.6'
19	106°11.5	17.3	354°49.6	14.7'	11°43.2	11.0'	54.7'
20	121°11.2	17.4	9°23.3	14.7'	11°54.1	10.9'	54.7'
21	136°10.9	· · 17.5	23°57.0	14.6'	12°05.1	10.9'	54.7'
22	151°10.6	17.6	38°30.6	14.6'	12°16.0	10.9'	54.7'
23	166°10.3	17.7	53°04.2	14.5'	12°26.8	10.8'	54.7'
SD = 16.2' d = 0.1' SD = 14.8'							
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	181°10.0	S23°17.8	67°37.7	14.4'	N12°37.6	10.8'	54.8'
1	196°09.7	18.0	82°11.1	14.4'	12°48.4	10.7'	54.8'
2	211°09.4	18.1	96°44.5	14.3'	12°59.1	10.7'	54.8'
3	226°09.1	· · 18.2	111°17.8	14.2'	13°09.8	10.6'	54.8'
4	241°08.8	18.3	125°51.0	14.2'	13°20.4	10.6'	54.8'
5	256°08.5	18.4	140°24.2	14.1'	13°31.0	10.5'	54.8'
6	271°08.2	S23°18.5	154°57.3	14.0'	N13°41.5	10.5'	54.9'
7	286°07.9	18.6	169°30.4	14.0'	13°52.0	10.4'	54.9'
8	301°07.6	18.8	184°03.3	13.9'	14°02.4	10.4'	54.9'
9	316°07.2	· · 18.9	198°36.2	13.8'	14°12.8	10.3'	54.9'
10	331°06.9	19.0	213°09.1	13.8'	14°23.1	10.3'	55.0'
11	346°06.6	19.1	227°41.8	13.7'	14°33.4	10.2'	55.0'
12	1°06.3	S23°19.2	242°14.5	13.6'	N14°43.6	10.2'	55.0'
13	16°06.0	19.3	256°47.1	13.5'	14°53.8	10.1'	55.0'
14	31°05.7	19.4	271°19.7	13.5'	15°03.9	10.1'	55.0'
15	46°05.4	· · 19.5	285°52.2	13.4'	15°14.0	10.0'	55.1'
16	61°05.1	19.6	300°24.6	13.3'	15°24.0	9.9'	55.1'
17	76°04.8	19.7	314°56.9	13.2'	15°33.9	9.9'	55.1'
18	91°04.5	S23°19.8	329°29.1	13.2'	N15°43.8	9.8'	55.1'
19	106°04.2	19.9	344°01.3	13.1'	15°53.6	9.8'	55.1'
20	121°03.9	20.0	358°33.4	13.0'	16°03.4	9.7'	55.2'
21	136°03.6	· · 20.1	13°05.4	12.9'	16°13.0	9.6'	55.2'
22	151°03.3	20.2	27°37.4	12.9'	16°22.7	9.6'	55.2'
23	166°03.0	20.3	42°09.2	12.8'	16°32.2	9.5'	55.2'
SD = 16.2' d = 0.1' SD = 14.9'							
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	181°02.7	S23°20.4	56°41.0	12.7'	N16°41.7	9.4'	55.3'
1	196°02.4	20.5	71°12.7	12.6'	16°51.2	9.4'	55.3'
2	211°02.1	20.6	85°44.3	12.5'	17°00.5	9.3'	55.3'
3	226°01.8	· · 20.7	100°15.9	12.5'	17°09.8	9.2'	55.3'
4	241°01.5	20.8	114°47.3	12.4'	17°19.1	9.2'	55.4'
5	256°01.2	20.9	129°18.7	12.3'	17°28.2	9.1'	55.4'
6	271°00.9	S23°21.0	143°50.0	12.2'	N17°37.3	9.0'	55.4'
7	286°00.6	21.1	158°21.2	12.1'	17°46.3	8.9'	55.4'
8	301°00.3	21.2	172°52.3	12.0'	17°55.3	8.9'	55.5'
9	316°00.0	· · 21.3	187°23.4	12.0'	18°04.1	8.8'	55.5'
10	330°59.7	21.4	201°54.3	11.9'	18°12.9	8.7'	55.5'
11	345°59.4	21.4	216°25.2	11.8'	18°21.6	8.6'	55.5'
12	0°59.0	S23°21.5	230°56.0	11.7'	N18°30.2	8.6'	55.6'
13	15°58.7	21.6	245°26.7	11.6'	18°38.8	8.5'	55.6'
14	30°58.4	21.7	259°57.3	11.5'	18°47.3	8.4'	55.6'
15	45°58.1	· · 21.8	274°27.9	11.4'	18°55.7	8.3'	55.6'
16	60°57.8	21.9	288°58.3	11.4'	19°04.0	8.2'	55.7'
17	75°57.5	22.0	303°28.7	11.3'	19°12.2	8.1'	55.7'
18	90°57.2	S23°22.0	317°58.9	11.2'	N19°20.3	8.1'	55.7'
19	105°56.9	22.1	332°29.1	11.1'	19°28.4	8.0'	55.8'
20	120°56.6	22.2	346°59.2	11.0'	19°36.3	7.9'	55.8'
21	135°56.3	· · 22.3	1°29.2	10.9'	19°44.2	7.8'	55.8'
22	150°56.0	22.4	15°59.1	10.8'	19°52.0	7.7'	55.8'
23	165°55.7	22.4	30°29.0	10.7'	19°59.7	7.6'	55.9'
SD = 16.2' d = 0.1' SD = 15.1'							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:21	10:49	■	■	13:02	15:30
N 70°	08:01	09:49	■	■	14:02	15:50
68°	07:46	09:14	■	■	14:37	16:05
66°	07:33	08:49	10:29	13:22	15:02	16:18
64°	07:22	08:30	09:47	14:04	15:21	16:29
62°	07:12	08:14	09:19	14:32	15:37	16:39
60°	07:04	08:01	08:58	14:53	15:51	16:47
N 58°	06:56	07:49	08:41	15:10	16:02	16:55
56°	06:50	07:39	08:26	15:25	16:12	17:01
54°	06:43	07:30	08:13	15:38	16:21	17:08
52°	06:38	07:22	08:02	15:49	16:30	17:13
50°	06:33	07:14	07:53	15:58	16:37	17:18
45°	06:21	06:58	07:32	16:19	16:53	17:30
N 40°	06:11	06:45	07:15	16:36	17:06	17:40
35°	06:01	06:33	07:01	16:50	17:18	17:50
30°	05:53	06:22	06:49	17:02	17:29	17:59
20°	05:36	06:04	06:28	17:23	17:47	18:15
N 10°	05:20	05:46	06:09	17:42	18:05	18:31
0°	05:03	05:29	05:52	17:59	18:22	18:48
S 10°	04:44	05:11	05:34	18:17	18:40	19:07
20°	04:22	04:51	05:15	18:36	19:00	19:30
30°	03:53	04:26	04:54	18:58	19:25	19:59
35°	03:34	04:11	04:41	19:11	19:40	20:17
40°	03:11	03:53	04:26	19:26	19:59	20:40
45°	02:41	03:30	04:08	19:44	20:21	21:11
S 50°	01:56	03:01	03:45	20:06	20:51	21:56
52°	01:28	02:46	03:34	20:17	21:06	22:23
54°	00:44	02:28	03:22	20:29	21:24	23:09
56°	////	02:06	03:08	20:43	21:46	////
58°	////	01:37	02:51	21:00	22:15	////
S 60°	////	00:48	02:31	21:20	23:04	////

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	10:53	10:20	08:50	02:43	04:46	07:53
N 70°	11:09	10:49	10:16	02:29	04:19	06:28
68°	11:22	11:11	10:56	02:19	03:59	05:49
66°	11:32	11:28	11:23	02:10	03:43	05:23
64°	11:41	11:42	11:44	02:02	03:30	05:02
62°	11:48	11:53	12:01	01:56	03:19	04:46
60°	11:55	12:04	12:16	01:50	03:10	04:32
N 58°	12:01	12:12	12:28	01:46	03:02	04:21
56°	12:06	12:20	12:39	01:41	02:55	04:11
54°	12:11	12:27	12:48	01:37	02:49	04:02
52°	12:15	12:34	12:57	01:34	02:43	03:54
50°	12:19	12:39	13:04	01:31	02:38	03:47
45°	12:27	12:52	13:20	01:24	02:27	03:32
N 40°	12:34	13:02	13:34	01:18	02:18	03:19
35°	12:40	13:11	13:45	01:13	02:10	03:09
30°	12:46	13:18	13:55	01:09	02:03	03:00
20°	12:55	13:32	14:13	01:02	01:52	02:44
N 10°	13:03	13:44	14:28	00:55	01:41	02:30
0°	13:11	13:55	14:42	00:49	01:32	02:17
S 10°	13:19	14:06	14:56	00:43	01:22	02:04
20°	13:27	14:18	15:12	00:37	01:12	01:51
30°	13:37	14:32	15:30	00:29	01:01	01:35
35°	13:42	14:40	15:40	00:25	00:54	01:26
40°	13:49	14:49	15:52	00:20	00:46	01:16
45°	13:56	15:00	16:06	00:15	00:38	01:04
S 50°	14:05	15:14	16:23	00:08	00:27	00:50
52°	14:10	15:20	16:31	00:05	00:22	00:43
54°	14:14	15:27	16:41	00:02	00:17	00:35
56°	14:19	15:34	16:51	· · ·	00:11	00:27
58°	14:25	15:43	17:03	· · ·	00:05	00:18
S 60°	14:32	15:53	17:17	23:57	· · ·	00:07

Day	Sun		Mer. Pass	Moon		Age 10-12 63-81%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
15	05:09	04:54	11:55	19:21	07:00	
16	04:40	04:25	11:56	20:06	07:43	
17	04:11	03:56	11:56	20:54	08:29	

December 18, 19, 20 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	86°30.2	226°48.3	S12°18.3	168°09.9	S24°08.3	90°37.7	S03°13.5	250°58.0	S04°04.9	Alpheratz	357°45.4	29°09.3
1	101°32.7	241°48.8	18.8	183°10.3	08.2	105°39.9	13.4	266°00.3	05.0	Ankaa	353°17.3	-42°14.9
2	116°35.2	256°49.3	19.3	198°10.6	08.1	120°42.1	13.3	281°02.6	05.0	Schedar	349°42.5	56°36.2
3	131°37.6	271°49.8	· · 19.7	213°11.0	· · 08.0	135°44.4	· · 13.2	296°04.9	· · 05.1	Diphda	348°57.6	-17°55.6
4	146°40.1	286°50.2	20.2	228°11.4	07.9	150°46.6	13.1	311°07.2	05.1	Achernar	335°27.7	-57°11.0
5	161°42.6	301°50.7	20.7	243°11.8	07.8	165°48.9	13.0	326°09.5	05.2	Hamal	328°02.6	23°31.0
6	176°45.0	316°51.2	S12°21.1	258°12.1	S24°07.7	180°51.1	S03°12.9	341°11.8	S04°05.2	Polaris	318°19.4	89°19.0
7	191°47.5	331°51.7	21.6	273°12.5	07.6	195°53.4	12.8	356°14.1	05.3	Acamar	315°19.3	-40°15.7
8	206°50.0	346°52.1	22.1	288°12.9	07.5	210°55.6	12.7	11°16.4	05.4	Menkar	314°16.7	4°08.0
9	221°52.4	1°52.6	· · 22.5	303°13.2	· · 07.4	225°57.9	· · 12.6	26°18.7	· · 05.4	Mirfak	308°42.5	49°54.2
10	236°54.9	16°53.1	23.0	318°13.6	07.3	241°00.1	12.5	41°21.0	05.5	Aldebaran	290°51.1	16°31.9
11	251°57.4	31°53.5	23.5	333°14.0	07.2	256°02.4	12.4	56°23.4	05.5	Rigel	281°13.4	-8°11.4
12	266°59.8	46°54.0	S12°23.9	348°14.4	S24°07.1	271°04.6	S03°12.3	71°25.7	S04°05.6	Capella	280°36.6	46°00.5
13	282°02.3	61°54.4	24.4	3°14.7	07.0	286°06.9	12.2	86°28.0	05.6	Bellatrix	278°33.6	6°21.6
14	297°04.7	76°54.9	24.9	18°15.1	06.9	301°09.1	12.0	101°30.3	05.7	Elnath	278°14.5	28°37.0
15	312°07.2	91°55.4	· · 25.3	33°15.5	· · 06.8	316°11.3	· · 11.9	116°32.6	· · 05.7	Alnilam	275°47.8	-1°11.7
16	327°09.7	106°55.8	25.8	48°15.9	06.7	331°13.6	11.8	131°34.9	05.8	Betelgeuse	271°02.9	7°24.5
17	342°12.1	121°56.3	26.3	63°16.2	06.6	346°15.8	11.7	146°37.2	05.8	Canopus	263°56.4	-52°42.1
18	357°14.6	136°56.7	S12°26.8	78°16.6	S24°06.5	1°18.1	S03°11.6	161°39.5	S04°05.9	Sirius	258°34.9	-16°43.9
19	12°17.1	151°57.2	27.2	93°17.0	06.4	16°20.3	11.5	176°41.8	05.9	Adhara	255°13.6	-28°59.2
20	27°19.5	166°57.6	27.7	108°17.4	06.3	31°22.6	11.4	191°44.1	06.0	Procyon	245°01.3	5°11.7
21	42°22.0	181°58.1	· · 28.2	123°17.7	· · 06.2	46°24.8	· · 11.3	206°46.5	· · 06.0	Pollux	243°29.6	27°59.8
22	57°24.5	196°58.5	28.7	138°18.1	06.1	61°27.0	11.2	221°48.8	06.1	Avior	234°18.3	-59°32.6
23	72°26.9	211°59.0	29.1	153°18.5	06.0	76°29.3	11.1	236°51.1	06.1	Suhail	222°53.5	-43°28.5
Mer.pass. 18:11		ν 0.5' d 0.5' m -4.77		ν 0.4' d -0.1' m 1.22		ν 2.2' d -0.1' m -2.41		ν 2.3' d 0.1' m 0.84				

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	87°29.4	226°59.4	S12°29.6	168°18.9	S24°05.9	91°31.5	S03°11.0	251°53.4	S04°06.2
1	102°31.8	241°59.9	30.1	183°19.2	05.7	106°33.8	10.9	266°55.7	06.3
2	117°34.3	257°00.3	30.6	198°19.6	05.6	121°36.0	10.8	281°58.0	06.3
3	132°36.8	272°00.8	· · 31.0	213°20.0	· · 05.5	136°38.2	· · 10.7	297°00.3	· · 06.4
4	147°39.2	287°01.2	31.5	228°20.3	05.4	151°40.5	10.5	312°02.6	06.4
5	162°41.7	302°01.6	32.0	243°20.7	05.3	166°42.7	10.4	327°04.9	06.5
6	177°44.2	317°02.1	S12°32.5	258°21.1	S24°05.2	181°45.0	S03°10.3	342°07.3	S04°06.5
7	192°46.6	332°02.5	33.0	273°21.5	05.1	196°47.2	10.2	357°09.6	06.6
8	207°49.1	347°02.9	33.4	288°21.8	05.0	211°49.4	10.1	12°11.9	06.6
9	222°51.6	2°03.4	· · 33.9	303°22.2	· · 04.9	226°51.7	· · 10.0	27°14.2	· · 06.7
10	237°54.0	17°03.8	34.4	318°22.6	04.8	241°53.9	09.9	42°16.5	06.7
11	252°56.5	32°04.2	34.9	333°23.0	04.6	256°56.2	09.8	57°18.8	06.8
12	267°59.0	47°04.7	S12°35.4	348°23.3	S24°04.5	271°58.4	S03°09.7	72°21.1	S04°06.8
13	283°01.4	62°05.1	35.9	3°23.7	04.4	287°00.6	09.6	87°23.4	06.9
14	298°03.9	77°05.5	36.3	18°24.1	04.3	302°02.9	09.5	102°25.7	06.9
15	313°06.3	92°05.9	· · 36.8	33°24.5	· · 04.2	317°05.1	· · 09.4	117°28.1	· · 07.0
16	328°08.8	107°06.4	37.3	48°24.8	04.1	332°07.3	09.2	132°30.4	07.0
17	343°11.3	122°06.8	37.8	63°25.2	04.0	347°09.6	09.1	147°32.7	07.1
18	358°13.7	137°07.2	S12°38.3	78°25.6	S24°03.8	2°11.8	S03°09.0	162°35.0	S04°07.1
19	13°16.2	152°07.6	38.8	93°25.9	03.7	17°14.1	08.9	177°37.3	07.2
20	28°18.7	167°08.0	39.3	108°26.3	03.6	32°16.3	08.8	192°39.6	07.2
21	43°21.1	182°08.5	· · 39.8	123°26.7	· · 03.5	47°18.5	· · 08.7	207°41.9	· · 07.3
22	58°23.6	197°08.9	40.2	138°27.1	03.4	62°20.8	08.6	222°44.3	07.3
23	73°26.1	212°09.3	40.7	153°27.4	03.3	77°23.0	08.5	237°46.6	07.4
Mer.pass. 18:07		ν 0.4' d 0.5' m -4.76		ν 0.4' d -0.1' m 1.21		ν 2.2' d -0.1' m -2.40		ν 2.3' d 0.1' m 0.84	

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	88°28.5	227°09.7	S12°41.2	168°27.8	S24°03.1	92°25.2	S03°08.4	252°48.9	S04°07.4
1	103°31.0	242°10.1	41.7	183°28.2	03.0	107°27.5	08.3	267°51.2	07.5
2	118°33.5	257°10.5	42.2	198°28.6	02.9	122°29.7	08.1	282°53.5	07.5
3	133°35.9	272°10.9	· · 42.7	213°28.9	· · 02.8	137°31.9	· · 08.0	297°55.8	· · 07.6
4	148°38.4	287°11.3	43.2	228°29.3	02.7	152°34.2	07.9	312°58.1	07.6
5	163°40.8	302°11.7	43.7	243°29.7	02.5	167°36.4	07.8	328°00.5	07.7
6	178°43.3	317°12.1	S12°44.2	258°30.1	S24°02.4	182°38.6	S03°07.7	343°02.8	S04°07.7
7	193°45.8	332°12.5	44.7	273°30.4	02.3	197°40.9	07.6	358°05.1	07.8
8	208°48.2	347°12.9	45.2	288°30.8	02.2	212°43.1	07.5	13°07.4	07.8
9	223°50.7	2°13.3	· · 45.7	303°31.2	· · 02.1	227°45.3	· · 07.4	28°09.7	· · 07.9
10	238°53.2	17°13.7	46.2	318°31.5	01.9	242°47.6	07.3	43°12.0	07.9
11	253°55.6	32°14.1	46.6	333°31.9	01.8	257°49.8	07.1	58°14.3	08.0
12	268°58.1	47°14.5	S12°47.1	348°32.3	S24°01.7	272°52.0	S03°07.0	73°16.7	S04°08.0
13	284°00.6	62°14.9	47.6	3°32.7	01.6	287°54.3	06.9	88°19.0	08.1
14	299°03.0	77°15.3	48.1	18°33.0	01.4	302°56.5	06.8	103°21.3	08.1
15	314°05.5	92°15.7	· · 48.6	33°33.4	· · 01.3	317°58.7	· · 06.7	118°23.6	· · 08.2
16	329°08.0	107°16.1	49.1	48°33.8	01.2	333°01.0	06.6	133°25.9	08.2
17	344°10.4	122°16.5	49.6	63°34.2	01.1	348°03.2	06.5	148°28.2	08.3
18	359°12.9	137°16.8	S12°50.1	78°34.5	S24°00.9	3°05.4	S03°06.4	163°30.6	S04°08.3
19	14°15.3	152°17.2	50.6	93°34.9	00.8	18°07.6	06.2	178°32.9	08.4
20	29°17.8	167°17.6	51.1	108°35.3	00.7	33°09.9	06.1	193°35.2	08.4
21	44°20.3	182°18.0	· · 51.6	123°35.6	· · 00.5	48°12.1	· · 06.0	208°37.5	· · 08.5
22	59°22.7	197°18.4	52.1	138°36.0	00.4	63°14.3	05.9	223°39.8	08.5
23	74°25.2	212°18.7	52.6	153°36.4	00.3	78°16.6	05.8	238°42.1	08.6
Mer.pass. 18:03		ν 0.4' d 0.5' m -4.75		ν 0.4' d -0.1' m 1.21		ν 2.2' d -0.1' m -2.40		ν 2.3' d 0.1' m 0.83	

h	Sun			Moon			
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	180°55.4	S23°22.5	44°58.7	10.7'	N20°07.3	7.5'	55.9'
1	195°55.1	22.6	59°28.4	10.6'	20°14.8	7.4'	55.9'
2	210°54.8	22.7	73°57.9	10.5'	20°22.2	7.3'	55.9'
3	225°54.5	· · 22.8	88°27.4	10.4'	20°29.6	7.2'	56.0'
4	240°54.2	22.8	102°56.8	10.3'	20°36.8	7.1'	56.0'
5	255°53.9	22.9	117°26.1	10.2'	20°43.9	7.0'	56.0'
6	270°53.6	S23°23.0	131°55.3	10.1'	N20°51.0	6.9'	56.1'
7	285°53.2	23.1	146°24.5	10.0'	20°57.9	6.8'	56.1'
8	300°52.9	23.1	160°53.5	10.0'	21°04.7	6.7'	56.1'
9	315°52.6	· · 23.2	175°22.5	9.9'	21°11.5	6.6'	56.1'
10	330°52.3	23.3	189°51.3	9.8'	21°18.1	6.5'	56.2'
11	345°52.0	23.3	204°20.1	9.7'	21°24.6	6.4'	56.2'
12	0°51.7	S23°23.4	218°48.8	9.6'	N21°31.0	6.3'	56.2'
13	15°51.4	23.5	233°17.4	9.5'	21°37.3	6.2'	56.3'
14	30°51.1	23.5	247°45.9	9.4'	21°43.5	6.1'	56.3'
15	45°50.8	· · 23.6	262°14.3	9.3'	21°49.6	6.0'	56.3'
16	60°50.5	23.7	276°42.7	9.3'	21°55.6	5.9'	56.3'
17	75°50.2	23.7	291°11.0	9.2'	22°01.5	5.8'	56.4'
18	90°49.9	S23°23.8	305°39.1	9.1'	N22°07.2	5.6'	56.4'
19	105°49.6	23.9	320°07.2	9.0'	22°12.9	5.5'	56.4'
20	120°49.3	23.9	334°35.2	8.9'	22°18.4	5.4'	56.5'
21	135°49.0	· · 24.0	349°03.2	8.8'	22°23.8	5.3'	56.5'
22	150°48.6	24.0	3°31.0	8.8'	22°29.1	5.2'	56.5'
23	165°48.3	24.1	17°58.8	8.7'	22°34.3	5.1'	56.5'
SD = 16.2'			SD = 15.2'				

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	180°48.0	S23°24.2	32°26.4	8.6'	N22°39.4	4.9'	56.6'
1	195°47.7	24.2	46°54.0	8.5'	22°44.3	4.8'	56.6'
2	210°47.4	24.3	61°21.5	8.4'	22°49.2	4.7'	56.6'
3	225°47.1	· · 24.3	75°49.0	8.4'	22°53.9	4.6'	56.7'
4	240°46.8	24.4	90°16.3	8.3'	22°58.4	4.5'	56.7'
5	255°46.5	24.4	104°43.6	8.2'	23°02.9	4.3'	56.7'
6	270°46.2	S23°24.5	119°10.8	8.1'	N23°07.2	4.2'	56.8'
7	285°45.9	24.6	133°37.9	8.0'	23°11.4	4.1'	56.8'
8	300°45.6	24.6	148°05.0	8.0'	23°15.5	3.9'	56.8'
9	315°45.3	· · 24.7	162°32.0	7.9'	23°19.5	3.8'	56.8'
10	330°45.0	24.7	176°58.9	7.8'	23°23.3	3.7'	56.9'
11	345°44.7	24.8	191°25.7	7.8'	23°27.0	3.6'	56.9'
12	0°44.3	S23°24.8	205°52.4	7.7'	N23°30.5	3.4'	56.9'
13	15°44.0	24.9	220°19.1	7.6'	23°33.9	3.3'	57.0'
14	30°43.7	24.9	234°45.7	7.5'	23°37.2	3.2'	57.0'
15	45°43.4	· · 24.9	249°12.3	7.5'	23°40.4	3.0'	57.0'
16	60°43.1	25.0	263°38.8	7.4'	23°43.4	2.9'	57.1'
17	75°42.8	25.0	278°05.2	7.3'	23°46.3	2.7'	57.1'
18	90°42.5	S23°25.1	292°31.5	7.3'	N23°49.0	2.6'	57.1'
19	105°42.2	25.1	306°57.8	7.2'	23°51.7	2.5'	57.1'
20	120°41.9	25.2	321°24.0	7.2'	23°54.1	2.3'	57.2'
21	135°41.6	· · 25.2	335°50.2	7.1'	23°56.5	2.2'	57.2'
22	150°41.3	25.3	350°16.3	7.0'	23°58.7	2.1'	57.2'
23	165°41.0	25.3	4°42.3	7.0'	24°00.7	1.9'	57.3'
SD = 16.2'			SD = 15.4'				

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	180°40.6	S23°25.3	19°08.3	6.9'	N24°02.6	1.8'	57.3'
1	195°40.3	25.4	33°34.3	6.9'	24°04.4	1.6'	57.3'
2	210°40.0	25.4	48°00.1	6.8'	24°06.0	1.5'	57.3'
3	225°39.7	· · 25.4	62°25.9	6.8'	24°07.5	1.3'	57.4'
4	240°39.4	25.5	76°51.7	6.7'	24°08.8	1.2'	57.4'
5	255°39.1	25.5	91°17.4	6.7'	24°10.0	1.0'	57.4'
6	270°38.8	S23°25.6	105°43.1	6.6'	N24°11.1	0.9'	57.5'
7	285°38.5	25.6	120°08.7	6.6'	24°12.0	0.8'	57.5'
8	300°38.2	25.6	134°34.3	6.5'	24°12.7	0.6'	57.5'
9	315°37.9	· · 25.7	148°59.8	6.5'	24°13.4	0.5'	57.5'
10	330°37.6	25.7	163°25.3	6.4'	24°13.8	0.3'	57.6'
11	345°37.3	25.7	177°50.8	6.4'	24°14.1	0.2'	57.6'
12	0°36.9	S23°25.7	192°16.2	6.4'	N24°14.3	0.0'	57.6'
13	15°36.6	25.8	206°41.5	6.3'	24°14.3	-0.1'	57.7'
14	30°36.3	25.8	221°06.9	6.3'	24°14.2	-0.3'	57.7'
15	45°36.0	· · 25.8	235°32.2	6.3'	24°13.9	-0.4'	57.7'
16	60°35.7	25.9	249°57.5	6.2'	24°13.4	-0.6'	57.7'
17	75°35.4	25.9	264°22.7	6.2'	24°12.8	-0.7'	57.8'
18	90°35.1	S23°25.9	278°47.9	6.2'	N24°12.1	-0.9'	57.8'
19	105°34.8	25.9	293°13.1	6.2'	24°11.2	-1.0'	57.8'
20	120°34.5	25.9	307°38.2	6.1'	24°10.1	-1.2'	57.8'
21	135°34.2	· · 26.0	322°03.4	6.1'	24°08.9	-1.4'	57.9'
22	150°33.9	26.0	336°28.5	6.1'	24°07.6	-1.5'	57.9'
23	165°33.5	26.0	350°53.5	6.1'	24°06.1	-1.7'	57.9'
SD = 16.2'			SD = 15.6'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:24	10:55	■	■	12:59	15:30
N 70°	08:04	09:53	■	■	14:01	15:50
68°	07:48	09:18	■	■	14:36	16:06
66°	07:35	08:52	10:33	13:21	15:02	16:19
64°	07:24	08:33	09:51	14:03	15:21	16:30
62°	07:14	08:16	09:22	14:32	15:38	16:40
60°	07:06	08:03	09:01	14:53	15:51	16:48
N 58°	06:58	07:51	08:43	15:11	16:03	16:56
56°	06:52	07:41	08:28	15:26	16:13	17:02
54°	06:46	07:32	08:16	15:38	16:22	17:08
52°	06:40	07:24	08:05	15:49	16:30	17:14
50°	06:35	07:16	07:55	15:59	16:38	17:19
45°	06:23	07:00	07:34	16:20	16:54	17:31
N 40°	06:12	06:47	07:17	16:37	17:08	17:42
35°	06:03	06:35	07:03	16:51	17:19	17:51
30°	05:54	06:24	06:51	17:04	17:30	18:00
20°	05:38	06:05	06:29	17:25	17:49	18:16
N 10°	05:22	05:48	06:11	17:43	18:06	18:33
0°	05:05	05:31	05:53	18:01	18:23	18:50
S 10°	04:46	05:13	05:36	18:18	18:41	19:09
20°	04:23	04:52	05:17	18:37	19:02	19:31
30°	03:54	04:27	04:55	18:59	19:27	20:00
35°	03:35	04:12	04:42	19:12	19:42	20:19
40°	03:12	03:54	04:27	19:28	20:00	20:42
45°	02:41	03:31	04:09	19:46	20:23	21:13
S 50°	01:56	03:01	03:46	20:08	20:53	21:58
52°	01:28	02:46	03:35	20:19	21:08	22:26
54°	00:41	02:28	03:23	20:31	21:26	23:13
56°	////	02:06	03:09	20:46	21:48	////
58°	////	01:36	02:52	21:02	22:18	////
S 60°	////	00:46	02:31	21:23	23:09	////

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	□	□	□	□	□	□
N 70°	□	□	□	□	□	□
68°	10:25	□	□	08:03	□	□
66°	11:18	11:12	□	07:10	09:08	□
64°	11:51	12:07	12:47	06:38	08:13	09:31
62°	12:15	12:40	13:25	06:15	07:40	08:53
60°	12:34	13:04	13:52	05:56	07:16	08:26
N 58°	12:50	13:24	14:13	05:40	06:57	08:04
56°	13:04	13:40	14:31	05:27	06:41	07:47
54°	13:16	13:54	14:45	05:16	06:27	07:32
52°	13:26	14:06	14:58	05:06	06:15	07:20
50°	13:36	14:17	15:10	04:57	06:05	07:08
45°	13:56	14:39	15:33	04:38	05:43	06:45
N 40°	14:12	14:58	15:52	04:22	05:25	06:26
35°	14:26	15:13	16:08	04:09	05:10	06:10
30°	14:37	15:26	16:21	03:58	04:57	05:56
20°	14:58	15:49	16:45	03:38	04:35	05:33
N 10°	15:16	16:09	17:05	03:22	04:16	05:13
0°	15:33	16:27	17:24	03:06	03:58	04:54
S 10°	15:50	16:45	17:43	02:50	03:41	04:35
20°	16:08	17:05	18:03	02:34	03:22	04:15
30°	16:29	17:28	18:27	02:14	03:00	03:52
35°	16:41	17:42	18:40	02:03	02:47	03:38
40°	16:55	17:58	18:56	01:51	02:32	03:22
45°	17:12	18:16	19:15	01:36	02:15	03:03
S 50°	17:33	18:40	19:39	01:17	01:53	02:39
52°	17:43	18:51	19:51	01:09	01:43	02:28
54°	17:54	19:03	20:03	00:59	01:31	02:15
56°	18:07	19:18	20:18	00:48	01:18	02:01
58°	18:22	19:35	20:36	00:36	01:03	01:43
S 60°	18:40	19:57	20:57	00:22	00:44	01:22

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Upper hh:mm	Lower hh:mm	
18	03:42	03:27	11:57	21:45	09:19	
19	03:12	02:57	11:57	22:40	10:13	
20	02:43	02:28	11:58	23:38	11:09	

December 21, 22, 23 UT (Tue., Wed., Thu.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	89°27.7	227°19.1	S12°53.1	168°36.8	S24°00.2	93°18.8	S03°05.7	253°44.5	S04°08.6	Alpheratz	357°45.4	29°09.3					
1	104°30.1	242°19.5	53.6	183°37.1	24°00.0	108°21.0	05.6	268°46.8	08.7	Ankaa	353°17.3	-42°14.9					
2	119°32.6	257°19.9	54.1	198°37.5	23°59.9	123°23.2	05.5	283°49.1	08.7	Schedar	349°42.5	56°36.2					
3	134°35.1	272°20.2	· · 54.6	213°37.9	· · 59.8	138°25.5	· · 05.3	298°51.4	· · 08.8	Diphda	348°57.6	-17°55.6					
4	149°37.5	287°20.6	55.1	228°38.3	59.6	153°27.7	05.2	313°53.7	08.8	Achernar	335°27.7	-57°11.0					
5	164°40.0	302°21.0	55.7	243°38.6	59.5	168°29.9	05.1	328°56.1	08.9	Hamal	328°02.6	23°31.0					
6	179°42.5	317°21.3	S12°56.2	258°39.0	S23°59.4	183°32.1	S03°05.0	343°58.4	S04°08.9	Polaris	318°20.0	89°19.0					
7	194°44.9	332°21.7	56.7	273°39.4	59.2	198°34.4	04.9	359°00.7	09.0	Acamar	315°19.3	-40°15.7					
8	209°47.4	347°22.1	57.2	288°39.7	59.1	213°36.6	04.8	14°03.0	09.0	Menkar	314°16.7	4°08.0					
9	224°49.8	2°22.4	· · 57.7	303°40.1	· · 59.0	228°38.8	· · 04.7	29°05.3	· · 09.1	Mirfak	308°42.5	49°54.2					
10	239°52.3	17°22.8	58.2	318°40.5	58.8	243°41.1	04.5	44°07.6	09.1	Aldebaran	290°51.1	16°31.9					
11	254°54.8	32°23.2	58.7	333°40.9	58.7	258°43.3	04.4	59°10.0	09.2	Rigel	281°13.4	-8°11.4					
12	269°57.2	47°23.5	S12°59.2	348°41.2	S23°58.6	273°45.5	S03°04.3	74°12.3	S04°09.2	Capella	280°36.6	46°00.5					
13	284°59.7	62°23.9	12°59.7	3°41.6	58.4	288°47.7	04.2	89°14.6	09.3	Bellatrix	278°33.6	6°21.6					
14	300°02.2	77°24.2	13°00.2	18°42.0	58.3	303°50.0	04.1	104°16.9	09.3	Elnath	278°14.5	28°37.0					
15	315°04.6	92°24.6	· · 00.7	33°42.4	· · 58.2	318°52.2	· · 04.0	119°19.2	· · 09.4	Alnilam	275°47.8	-1°11.7					
16	330°07.1	107°24.9	01.2	48°42.7	58.0	333°54.4	03.9	134°21.6	09.4	Betelgeuse	271°02.9	7°24.5					
17	345°09.6	122°25.3	01.7	63°43.1	57.9	348°56.6	03.7	149°23.9	09.5	Canopus	263°56.4	-52°42.1					
18	0°12.0	137°25.7	S13°02.2	78°43.5	S23°57.7	3°58.8	S03°03.6	164°26.2	S04°09.5	Sirius	258°34.9	-16°43.9					
19	15°14.5	152°26.0	02.8	93°43.8	57.6	19°01.1	03.5	179°28.5	09.6	Adhara	255°13.6	-28°59.2					
20	30°17.0	167°26.3	03.3	108°44.2	57.5	34°03.3	03.4	194°30.8	09.6	Procyon	245°01.3	5°11.7					
21	45°19.4	182°26.7	· · 03.8	123°44.6	· · 57.3	49°05.5	· · 03.3	209°33.2	· · 09.7	Pollux	243°29.5	27°59.8					
22	60°21.9	197°27.0	04.3	138°45.0	57.2	64°07.7	03.2	224°35.5	09.7	Avior	234°18.3	-59°32.6					
23	75°24.3	212°27.4	04.8	153°45.3	57.0	79°10.0	03.1	239°37.8	09.8	Suhail	222°53.5	-43°28.5					
Mer.pass. 17:59			ν 0.4' d 0.5' m -4.74			ν 0.4' d -0.1' m 1.20			ν 2.2' d -0.1' m -2.39			ν 2.3' d 0.0' m 0.83					

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	90°26.8	227°27.7	S13°05.3	168°45.7	S23°56.9	94°12.2	S03°02.9	254°40.1	S04°09.8					
1	105°29.3	242°28.1	05.8	183°46.1	56.8	109°14.4	02.8	269°42.4	09.9					
2	120°31.7	257°28.4	06.3	198°46.4	56.6	124°16.6	02.7	284°44.8	09.9					
3	135°34.2	272°28.8	· · 06.9	213°46.8	· · 56.5	139°18.8	· · 02.6	299°47.1	· · 09.9					
4	150°36.7	287°29.1	07.4	228°47.2	56.3	154°21.1	02.5	314°49.4	10.0					
5	165°39.1	302°29.4	07.9	243°47.6	56.2	169°23.3	02.4	329°51.7	10.0					
6	180°41.6	317°29.8	S13°08.4	258°47.9	S23°56.1	184°25.5	S03°02.2	344°54.1	S04°10.1					
7	195°44.1	332°30.1	08.9	273°48.3	55.9	199°27.7	02.1	359°56.4	10.1					
8	210°46.5	347°30.4	09.4	288°48.7	55.8	214°29.9	02.0	14°58.7	10.2					
9	225°49.0	2°30.8	· · 10.0	303°49.1	· · 55.6	229°32.2	· · 01.9	30°01.0	· · 10.2					
10	240°51.5	17°31.1	10.5	318°49.4	55.5	244°34.4	01.8	45°03.3	10.3					
11	255°53.9	32°31.4	11.0	333°49.8	55.3	259°36.6	01.7	60°05.7	10.3					
12	270°56.4	47°31.7	S13°11.5	348°50.2	S23°55.2	274°38.8	S03°01.5	75°08.0	S04°10.4					
13	285°58.8	62°32.1	12.0	3°50.5	55.0	289°41.0	01.4	90°10.3	10.4					
14	301°01.3	77°32.4	12.5	18°50.9	54.9	304°43.3	01.3	105°12.6	10.5					
15	316°03.8	92°32.7	· · 13.1	33°51.3	· · 54.7	319°45.5	· · 01.2	120°15.0	· · 10.5					
16	331°06.2	107°33.0	13.6	48°51.7	54.6	334°47.7	01.1	135°17.3	10.6					
17	346°08.7	122°33.3	14.1	63°52.0	54.4	349°49.9	01.0	150°19.6	10.6					
18	1°11.2	137°33.7	S13°14.6	78°52.4	S23°54.3	4°52.1	S03°00.8	165°21.9	S04°10.7					
19	16°13.6	152°34.0	15.1	93°52.8	54.1	19°54.3	00.7	180°24.2	10.7					
20	31°16.1	167°34.3	15.7	108°53.2	54.0	34°56.6	00.6	195°26.6	10.7					
21	46°18.6	182°34.6	· · 16.2	123°53.5	· · 53.8	49°58.8	· · 00.5	210°28.9	· · 10.8					
22	61°21.0	197°34.9	16.7	138°53.9	53.7	65°01.0	00.4	225°31.2	10.8					
23	76°23.5	212°35.2	17.2	153°54.3	53.5	80°03.2	00.2	240°33.5	10.9					
Mer.pass. 17:55			ν 0.3' d 0.5' m -4.73			ν 0.4' d -0.1' m 1.20			ν 2.2' d -0.1' m -2.38			ν 2.3' d 0.0' m 0.83		

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	91°26.0	227°35.5	S13°17.8	168°54.6	S23°53.4	95°05.4	S03°00.1	255°35.9	S04°10.9					
1	106°28.4	242°35.9	18.3	183°55.0	53.2	110°07.6	03°00.0	270°38.2	11.0					
2	121°30.9	257°36.2	18.8	198°55.4	53.1	125°09.9	02°59.9	285°40.5	11.0					
3	136°33.3	272°36.5	· · 19.3	213°55.8	· · 52.9	140°12.1	· · 59.8	300°42.8	· · 11.1					
4	151°35.8	287°36.8	19.8	228°56.1	52.8	155°14.3	59.7	315°45.2	11.1					
5	166°38.3	302°37.1	20.4	243°56.5	52.6	170°16.5	59.5	330°47.5	11.2					
6	181°40.7	317°37.4	S13°20.9	258°56.9	S23°52.5	185°18.7	S02°59.4	345°49.8	S04°11.2					
7	196°43.2	332°37.7	21.4	273°57.2	52.3	200°20.9	59.3	0°52.1	11.3					
8	211°45.7	347°38.0	21.9	288°57.6	52.2	215°23.1	59.2	15°54.5	11.3					
9	226°48.1	2°38.3	· · 22.5	303°58.0	· · 52.0	230°25.3	· · 59.1	30°56.8	· · 11.3					
10	241°50.6	17°38.6	23.0	318°58.4	51.8	245°27.6	58.9	45°59.1	11.4					
11	256°53.1	32°38.9	23.5	333°58.7	51.7	260°29.8	58.8	61°01.4	11.4					
12	271°55.5	47°39.2	S13°24.1	348°59.1	S23°51.5	275°32.0	S02°58.7	76°03.8	S04°11.5					
13	286°58.0	62°39.5	24.6	3°59.5	51.4	290°34.2	58.6	91°06.1	11.5					
14	302°00.4	77°39.8	25.1	18°59.9	51.2	305°36.4	58.5	106°08.4	11.6					
15	317°02.9	92°40.0	· · 25.6	34°00.2	· · 51.0	320°38.6	· · 58.3	121°10.7	· · 11.6					
16	332°05.4	107°40.3	26.2	49°00.6	50.9	335°40.8	58.2	136°13.1	11.7					
17	347°07.8	122°40.6	26.7	64°01.0	50.7	350°43.0	58.1	151°15.4	11.7					
18	2°10.3	137°40.9	S13°27.2	79°01.3	S23°50.6	5°45.3	S02°58.0	166°17.7	S04°11.8					
19	17°12.8	152°41.2	27.7	94°01.7	50.4	20°47.5	57.9	181°20.1	11.8					
20	32°15.2	167°41.5	28.3	109°02.1	50.2	35°49.7	57.7	196°22.4	11.8					
21	47°17.7	182°41.8	· · 28.8	124°02.5	· · 50.1	50°51.9	· · 57.6	211°24.7	· · 11.9					
22	62°20.2	197°42.0	29.3	139°02.8	49.9	65°54.1	57.5	226°27.0	11.9					
23	77°22.6	212°42.3	29.9	154°03.2	49.8	80°56.3	57.4	241°29.4	12.0					
Mer.pass. 17:51			ν 0.3' d 0.5' m -4.72			ν 0.4' d -0.2' m 1.19			ν 2.2' d -0.1' m -2.38			ν 2.3' d 0.0' m 0.82		

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h	Sun			Moon			
Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	180°33.2	S23°26.0	5°18.6	6.0'	N24°04.4	-1.8'	58.0'
1	195°32.9	26.1	19°43.7	6.0'	24°02.6	-2.0'	58.0'
2	210°32.6	26.1	34°08.7	6.0'	24°00.7	-2.1'	58.0'
3	225°32.3	· · 26.1	48°33.7	6.0'	23°58.5	-2.3'	58.0'
4	240°32.0	26.1	62°58.7	6.0'	23°56.3	-2.4'	58.1'
5	255°31.7	26.1	77°23.7	6.0'	23°53.8	-2.6'	58.1'
6	270°31.4	S23°26.1	91°48.7	6.0'	N23°51.3	-2.7'	58.1'
7	285°31.1	26.1	106°13.7	6.0'	23°48.5	-2.9'	58.1'
8	300°30.8	26.2	120°38.7	6.0'	23°45.6	-3.0'	58.2'
9	315°30.5	· · 26.2	135°03.6	6.0'	23°42.6	-3.2'	58.2'
10	330°30.1	26.2	149°28.6	6.0'	23°39.4	-3.3'	58.2'
11	345°29.8	26.2	163°53.6	6.0'	23°36.1	-3.5'	58.2'
12	0°29.5	S23°26.2	178°18.5	6.0'	N23°32.6	-3.7'	58.3'
13	15°29.2	26.2	192°43.5	6.0'	23°28.9	-3.8'	58.3'
14	30°28.9	26.2	207°08.5	6.0'	23°25.1	-4.0'	58.3'
15	45°28.6	· · 26.2	221°33.5	6.0'	23°21.2	-4.1'	58.3'
16	60°28.3	26.2	235°58.5	6.0'	23°17.0	-4.3'	58.4'
17	75°28.0	26.2	250°23.5	6.0'	23°12.8	-4.4'	58.4'
18	90°27.7	S23°26.2	264°48.5	6.0'	N23°08.4	-4.6'	58.4'
19	105°27.4	26.3	279°13.5	6.0'	23°03.8	-4.7'	58.4'
20	120°27.1	26.3	293°38.5	6.1'	22°59.1	-4.9'	58.4'
21	135°26.7	· · 26.3	308°03.6	6.1'	22°54.2	-5.0'	58.5'
22	150°26.4	26.3	322°28.7	6.1'	22°49.2	-5.2'	58.5'
23	165°26.1	26.3	336°53.8	6.1'	22°44.1	-5.3'	58.5'
SD = 16.3'			SD = 15.8'				

Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	180°25.8	S23°26.3	351°18.9	6.1'	N22°38.8	-5.5'	58.5'
1	195°25.5	26.3	5°44.0	6.2'	22°33.3	-5.6'	58.6'
2	210°25.2	26.3	20°09.2	6.2'	22°27.7	-5.8'	58.6'
3	225°24.9	· · 26.3	34°34.4	6.2'	22°21.9	-5.9'	58.6'
4	240°24.6	26.3	48°59.6	6.2'	22°16.1	-6.0'	58.6'
5	255°24.3	26.3	63°24.8	6.3'	22°10.0	-6.2'	58.6'
6	270°24.0	S23°26.2	77°50.1	6.3'	N22°03.8	-6.3'	58.7'
7	285°23.6	26.2	92°15.4	6.3'	21°57.5	-6.5'	58.7'
8	300°23.3	26.2	106°40.7	6.4'	21°51.0	-6.6'	58.7'
9	315°23.0	· · 26.2	121°06.1	6.4'	21°44.4	-6.8'	58.7'
10	330°22.7	26.2	135°31.5	6.4'	21°37.6	-6.9'	58.7'
11	345°22.4	26.2	149°56.9	6.5'	21°30.7	-7.0'	58.8'
12	0°22.1	S23°26.2	164°22.4	6.5'	N21°23.7	-7.2'	58.8'
13	15°21.8	26.2	178°47.9	6.5'	21°16.5	-7.3'	58.8'
14	30°21.5	26.2	193°13.4	6.6'	21°09.2	-7.5'	58.8'
15	45°21.2	· · 26.2	207°39.0	6.6'	21°01.7	-7.6'	58.8'
16	60°20.9	26.2	222°04.6	6.7'	20°54.1	-7.7'	58.8'
17	75°20.5	26.1	236°30.3	6.7'	20°46.4	-7.9'	58.9'
18	90°20.2	S23°26.1	250°56.0	6.8'	N20°38.6	-8.0'	58.9'
19	105°19.9	26.1	265°21.8	6.8'	20°30.6	-8.1'	58.9'
20	120°19.6	26.1	279°47.6	6.8'	20°22.4	-8.3'	58.9'
21	135°19.3	· · 26.1	294°13.4	6.9'	20°14.2	-8.4'	58.9'
22	150°19.0	26.1	308°39.3	6.9'	20°05.8	-8.5'	58.9'
23	165°18.7	26.0	323°05.3	7.0'	19°57.3	-8.7'	59.0'
SD = 16.3'			SD = 16.0'				

Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	180°18.4	S23°26.0	337°31.3	7.0'	N19°48.6	-8.8'	59.0'
1	195°18.1	26.0	351°57.3	7.1'	19°39.8	-8.9'	59.0'
2	210°17.8	26.0	6°23.4	7.1'	19°30.9	-9.0'	59.0'
3	225°17.4	· · 26.0	20°49.5	7.2'	19°21.9	-9.2'	59.0'
4	240°17.1	25.9	35°15.7	7.2'	19°12.7	-9.3'	59.0'
5	255°16.8	25.9	49°42.0	7.3'	19°03.5	-9.4'	59.1'
6	270°16.5	S23°25.9	64°08.3	7.3'	N18°54.1	-9.5'	59.1'
7	285°16.2	25.9	78°34.6	7.4'	18°44.5	-9.6'	59.1'
8	300°15.9	25.8	93°01.0	7.5'	18°34.9	-9.8'	59.1'
9	315°15.6	· · 25.8	107°27.5	7.5'	18°25.2	-9.9'	59.1'
10	330°15.3	25.8	121°54.0	7.6'	18°15.3	-10.0'	59.1'
11	345°15.0	25.8	136°20.6	7.6'	18°05.3	-10.1'	59.1'
12	0°14.7	S23°25.7	150°47.2	7.7'	N17°55.2	-10.2'	59.1'
13	15°14.3	25.7	165°13.9	7.7'	17°45.0	-10.3'	59.2'
14	30°14.0	25.7	179°40.6	7.8'	17°34.7	-10.4'	59.2'
15	45°13.7	· · 25.6	194°07.4	7.8'	17°24.2	-10.5'	59.2'
16	60°13.4	25.6	208°34.2	7.9'	17°13.7	-10.7'	59.2'
17	75°13.1	25.6	223°01.1	8.0'	17°03.0	-10.8'	59.2'
18	90°12.8	S23°25.5	237°28.1	8.0'	N16°52.3	-10.9'	59.2'
19	105°12.5	25.5	251°55.1	8.1'	16°41.4	-11.0'	59.2'
20	120°12.2	25.5	266°22.2	8.1'	16°30.4	-11.1'	59.2'
21	135°11.9	· · 25.4	280°49.3	8.2'	16°19.4	-11.2'	59.2'
22	150°11.6	25.4	295°16.5	8.2'	16°08.2	-11.3'	59.3'
23	165°11.2	25.4	309°43.8	8.3'	15°56.9	-11.4'	59.3'
SD = 16.3'			SD = 16.1'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:26	10:58	■	■	12:59	15:31
N 70°	08:06	09:55	■	■	14:02	15:51
68°	07:50	09:20	■	■	14:38	16:07
66°	07:37	08:54	10:35	13:22	15:03	16:20
64°	07:26	08:34	09:53	14:05	15:23	16:31
62°	07:16	08:18	09:24	14:33	15:39	16:41
60°	07:08	08:05	09:02	14:55	15:52	16:49
N 58°	07:00	07:53	08:45	15:12	16:04	16:57
56°	06:53	07:43	08:30	15:27	16:14	17:04
54°	06:47	07:33	08:17	15:40	16:24	17:10
52°	06:41	07:25	08:06	15:51	16:32	17:16
50°	06:36	07:18	07:56	16:01	16:39	17:21
45°	06:24	07:02	07:36	16:22	16:55	17:33
N 40°	06:14	06:48	07:19	16:38	17:09	17:43
35°	06:05	06:36	07:04	16:53	17:21	17:52
30°	05:56	06:26	06:52	17:05	17:31	18:01
20°	05:39	06:07	06:31	17:26	17:50	18:18
N 10°	05:23	05:49	06:12	17:45	18:08	18:34
0°	05:06	05:32	05:55	18:02	18:25	18:51
S 10°	04:47	05:14	05:37	18:20	18:43	19:10
20°	04:24	04:54	05:18	18:39	19:03	19:33
30°	03:55	04:29	04:56	19:01	19:28	20:02
35°	03:37	04:13	04:43	19:14	19:44	20:21
40°	03:13	03:55	04:28	19:29	20:02	20:44
45°	02:43	03:32	04:10	19:47	20:25	21:14
S 50°	01:57	03:03	03:47	20:10	20:54	22:00
52°	01:29	02:47	03:36	20:21	21:10	22:28
54°	00:42	02:29	03:24	20:33	21:28	23:15
56°	////	02:07	03:10	20:47	21:50	////
58°	////	01:37	02:53	21:04	22:20	////
S 60°	////	00:46	02:32	21:25	23:11	////

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	□	□	15:14	□	□	13:08
N 70°	□	□	16:17	□	□	12:04
68°	□	14:22	16:52	□	12:00	11:27
66°	13:07	15:15	17:17	11:13	11:06	11:01
64°	14:04	15:47	17:36	10:15	10:33	10:41
62°	14:38	16:11	17:52	09:41	10:09	10:24
60°	15:02	16:29	18:05	09:16	09:50	10:10
N 58°	15:22	16:45	18:16	08:57	09:33	09:58
56°	15:38	16:58	18:26	08:40	09:20	09:48
54°	15:52	17:10	18:35	08:26	09:08	09:39
52°	16:04	17:20	18:42	08:14	08:57	09:31
50°	16:14	17:29	18:49	08:03	08:48	09:23
45°	16:37	17:48	19:04	07:40	08:28	09:07
N 40°	16:55	18:04	19:16	07:22	08:11	08:54
35°	17:10	18:17	19:26	07:07	07:58	08:43
30°	17:23	18:28	19:35	06:53	07:46	08:33
20°	17:45	18:47	19:50	06:30	07:25	08:16
N 10°	18:04	19:04	20:03	06:10	07:07	08:01
0°	18:22	19:20	20:16	05:52	06:50	07:47
S 10°	18:40	19:36	20:28	05:33	06:33	07:33
20°	18:59	19:52	20:41	05:13	06:15	07:17
30°	19:22	20:11	20:56	04:50	05:54	07:00
35°	19:34	20:23	21:05	04:36	05:41	06:49
40°	19:49	20:35	21:15	04:21	05:27	06:38
45°	20:07	20:50	21:26	04:02	05:10	06:24
S 50°	20:29	21:09	21:40	03:38	04:48	06:06
52°	20:39	21:17	21:47	03:27	04:38	05:58
54°	20:51	21:27	21:54	03:14	04:27	05:49
56°	21:04	21:38	22:02	02:59	04:14	05:39
58°	21:20	21:50	22:11	02:42	03:59	05:27
S 60°	21:39	22:04	22:21	02:21	03:40	05:13

Day	Sun		Mer. Pass	Moon		Age 16-18 100-97%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Upper hh:mm	Lower hh:mm	
21	02:13	01:58	11:58	—:—	12:07	
22	01:43	01:28	11:59	00:36	13:05	
23	01:14	00:59	11:59	01:33	14:01	

December 24, 25, 26 UT (Fri., Sat., Sun.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	92°25.1	227°42.6	S13°30.4	169°03.6	S23°49.6	95°58.5	S02°57.2	256°31.7	S04°12.0	Alpheratz	357°45.4	29°09.3					
1	107°27.6	242°42.9	30.9	184°04.0	49.4	111°00.7	57.1	271°34.0	12.1	Ankaa	353°17.3	-42°14.9					
2	122°30.0	257°43.1	31.5	199°04.3	49.3	126°02.9	57.0	286°36.3	12.1	Schedar	349°42.6	56°36.2					
3	137°32.5	272°43.4	•• 32.0	214°04.7	•• 49.1	141°05.1	•• 56.9	301°38.7	•• 12.2	Diphda	348°57.6	-17°55.6					
4	152°34.9	287°43.7	32.5	229°05.1	48.9	156°07.3	56.8	316°41.0	12.2	Achernar	335°27.7	-57°11.0					
5	167°37.4	302°44.0	33.1	244°05.4	48.8	171°09.6	56.6	331°43.3	12.2	Hamal	328°02.6	23°31.0					
6	182°39.9	317°44.2	S13°33.6	259°05.8	S23°48.6	186°11.8	S02°56.5	346°45.7	S04°12.3	Polaris	318°20.9	89°19.0					
7	197°42.3	332°44.5	34.1	274°06.2	48.4	201°14.0	56.4	1°48.0	12.3	Acamar	315°19.3	-40°15.8					
8	212°44.8	347°44.8	34.7	289°06.6	48.3	216°16.2	56.3	16°50.3	12.4	Menkar	314°16.7	4°08.0					
9	227°47.3	2°45.0	•• 35.2	304°06.9	•• 48.1	231°18.4	•• 56.2	31°52.7	•• 12.4	Mirfak	308°42.5	49°54.2					
10	242°49.7	17°45.3	35.7	319°07.3	47.9	246°20.6	56.0	46°55.0	12.5	Aldebaran	290°51.1	16°31.9					
11	257°52.2	32°45.6	36.3	334°07.7	47.8	261°22.8	55.9	61°57.3	12.5	Rigel	281°13.4	-8°11.4					
12	272°54.7	47°45.8	S13°36.8	349°08.0	S23°47.6	276°25.0	S02°55.8	76°59.6	S04°12.6	Capella	280°36.6	46°00.6					
13	287°57.1	62°46.1	37.3	4°08.4	47.4	291°27.2	55.7	92°02.0	12.6	Bellatrix	278°33.6	6°21.5					
14	302°59.6	77°46.4	37.9	19°08.8	47.3	306°29.4	55.5	107°04.3	12.6	Elnath	278°14.5	28°37.0					
15	318°02.1	92°46.6	•• 38.4	34°09.2	•• 47.1	321°31.6	•• 55.4	122°06.6	•• 12.7	Alnilam	275°47.8	-1°11.7					
16	333°04.5	107°46.9	38.9	49°09.5	46.9	336°33.8	55.3	137°09.0	12.7	Betelgeuse	271°02.9	7°24.5					
17	348°07.0	122°47.1	39.5	64°09.9	46.7	351°36.0	55.2	152°11.3	12.8	Canopus	263°56.4	-52°42.1					
18	3°09.4	137°47.4	S13°40.0	79°10.3	S23°46.6	6°38.2	S02°55.0	167°13.6	S04°12.8	Sirius	258°34.9	-16°43.9					
19	18°11.9	152°47.6	40.6	94°10.7	46.4	21°40.4	54.9	182°15.9	12.9	Adhara	255°13.5	-28°59.3					
20	33°14.4	167°47.9	41.1	109°11.0	46.2	36°42.6	54.8	197°18.3	12.9	Procyon	245°01.3	5°11.7					
21	48°16.8	182°48.1	•• 41.6	124°11.4	•• 46.0	51°44.8	•• 54.7	212°20.6	•• 12.9	Pollux	243°29.5	27°59.8					
22	63°19.3	197°48.4	42.2	139°11.8	45.9	66°47.0	54.6	227°22.9	13.0	Avior	234°18.2	-59°32.6					
23	78°21.8	212°48.6	42.7	154°12.1	45.7	81°49.2	54.4	242°25.3	13.0	Suhail	222°53.5	-43°28.6					
Mer.pass. 17:47			ν 0.3' d 0.5' m -4.71			ν 0.4' d -0.2' m 1.19			ν 2.2' d -0.1' m -2.37			ν 2.3' d 0.0' m 0.82					

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	93°24.2	227°48.9	S13°43.2	169°12.5	S23°45.5	96°51.4	S02°54.3	257°27.6	S04°13.1	Denebola	182°35.4	14°30.4		
1	108°26.7	242°49.1	43.8	184°12.9	45.4	111°53.6	54.2	272°29.9	13.1	Gienah	175°54.2	-17°36.2		
2	123°29.2	257°49.4	44.3	199°13.3	45.2	126°55.8	54.1	287°32.3	13.2	Acrux	173°11.5	-63°09.4		
3	138°31.6	272°49.6	•• 44.9	214°13.6	•• 45.0	141°58.1	•• 53.9	302°34.6	•• 13.2	Gacrux	172°03.0	-57°10.3		
4	153°34.1	287°49.9	45.4	229°14.0	44.8	157°00.3	53.8	317°36.9	13.3	Alioth	166°22.2	55°53.6		
5	168°36.6	302°50.1	45.9	244°14.4	44.6	172°02.5	53.7	332°39.3	13.3	Spica	158°33.2	-11°13.2		
6	183°39.0	317°50.3	S13°46.5	259°14.8	S23°44.5	187°04.7	S02°53.6	347°41.6	S04°13.3	Alkaid	153°00.4	49°15.2		
7	198°41.5	332°50.6	47.0	274°15.1	44.3	202°06.9	53.4	2°43.9	13.4	Hadar	148°50.8	-60°25.4		
8	213°43.9	347°50.8	47.6	289°15.5	44.1	217°09.1	53.3	17°46.3	13.4	Menkent	148°09.9	-36°25.3		
9	228°46.4	2°51.1	•• 48.1	304°15.9	•• 43.9	232°11.3	•• 53.2	32°48.6	•• 13.5	Arcturus	145°57.5	19°07.3		
10	243°48.9	17°51.3	48.7	319°16.2	43.8	247°13.5	53.1	47°50.9	13.5	Rigel Kent.	139°54.6	-60°52.6		
11	258°51.3	32°51.5	49.2	334°16.6	43.6	262°15.7	52.9	62°53.3	13.5	Kochab	137°20.6	74°06.3		
12	273°53.8	47°51.8	S13°49.7	349°17.0	S23°43.4	277°17.9	S02°52.8	77°55.6	S04°13.6	Zuben'ubi	137°07.6	-16°05.2		
13	288°56.3	62°52.0	50.3	4°17.4	43.2	292°20.1	52.7	92°57.9	13.6	Alphecca	126°12.8	26°40.5		
14	303°58.7	77°52.2	50.8	19°17.7	43.0	307°22.3	52.6	108°00.3	13.7	Antares	112°28.8	-26°27.3		
15	319°01.2	92°52.4	•• 51.4	34°18.1	•• 42.9	322°24.5	•• 52.4	123°02.6	•• 13.7	Atria	107°32.8	-69°02.7		
16	334°03.7	107°52.7	51.9	49°18.5	42.7	337°26.7	52.3	138°04.9	13.8	Sabik	102°14.9	-15°44.3		
17	349°06.1	122°52.9	52.5	64°18.9	42.5	352°28.9	52.2	153°07.3	13.8	Shaula	96°24.7	-37°06.6		
18	4°08.6	137°53.1	S13°53.0	79°19.2	S23°42.3	7°31.0	S02°52.1	168°09.6	S04°13.8	Rasalhague	96°08.4	12°33.1		
19	19°11.0	152°53.3	53.5	94°19.6	42.1	22°33.2	51.9	183°11.9	13.9	Eltanin	90°47.5	51°29.3		
20	34°13.5	167°53.6	54.1	109°20.0	41.9	37°35.4	51.8	198°14.3	13.9	Kaus Aust.	83°46.6	-34°22.7		
21	49°16.0	182°53.8	•• 54.6	124°20.4	•• 41.7	52°37.6	•• 51.7	213°16.6	•• 14.0	Vega	80°40.6	38°47.7		
22	64°18.4	197°54.0	55.2	139°20.7	41.6	67°39.8	51.6	228°18.9	14.0	Nunki	76°00.9	-26°16.9		
23	79°20.9	212°54.2	55.7	154°21.1	41.4	82°42.0	51.4	243°21.3	14.1	Altair	62°10.3	8°53.9		
Mer.pass. 17:43			ν 0.2' d 0.5' m -4.69			ν 0.4' d -0.2' m 1.18			ν 2.2' d -0.1' m -2.36			ν 2.3' d 0.0' m 0.82		

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	94°23.4	227°54.4	S13°56.3	169°21.5	S23°41.2	97°44.2	S02°51.3	258°23.6	S04°14.1	Dec 24 Fri	SHA	Mer.pass
1	109°25.8	242°54.7	56.8	184°21.8	41.0	112°46.4	51.2	273°25.9	14.1	Venus	135°17.5	08:49
2	124°28.3	257°54.9	57.4	199°22.2	40.8	127°48.6	51.1	288°28.3	14.2	Mars	76°38.5	12:43
3	139°30.8	272°55.1	•• 57.9	214°22.6	•• 40.6	142°50.8	•• 50.9	303°30.6	•• 14.2	Jupiter	3°33.4	17:34
4	154°33.2	287°55.3	58.4	229°23.0	40.4	157°53.0	50.8	318°32.9	14.3	Saturn	164°06.6	06:53
5	169°35.7	302°55.5	59.0	244°23.3	40.3	172°55.2	50.7	333°35.3	14.3			
6	184°38.2	317°55.7	S13°59.5	259°23.7	S23°40.1	187°57.4	S02°50.5	348°37.6	S04°14.3	Dec 25 Sat	SHA	Mer.pass
7	199°40.6	332°55.9	14°00.1	274°24.1	39.9	202°59.6	50.4	3°40.0	14.4	Venus	134°24.7	08:49
8	214°43.1	347°56.1	00.6	289°24.5	39.7	218°01.8	50.3	18°42.3	14.4	Mars	75°48.3	12:43
9	229°45.5	2°56.3	•• 01.2	304°24.8	•• 39.5	233°04.0	•• 50.2	33°44.6	•• 14.5	Jupiter	3°27.2	17:30
10	244°48.0	17°56.5	01.7	319°25.2	39.3	248°06.2	50.0	48°47.0	14.5	Saturn	164°03.4	06:49
11	259°50.5	32°56.7	02.3	334°25.6	39.1	263°08.4	49.9	63°49.3	14.5			
12	274°52.9	47°56.9	S14°02.8	349°25.9	S23°38.9	278°10.6	S02°49.8	78°51.6	S04°14.6	Dec 26 Sun	SHA	Mer.pass
13	289°55.4	62°57.1	03.4	4°26.3	38.7	293°12.8	49.7	93°54.0	14.6	Venus	133°31.1	08:48
14	304°57.9	77°57.3	03.9	19°26.7	38.5	308°15.0	49.5	108°56.3	14.7	Mars	74°58.1	12:42
15	320°00.3	92°57.5	•• 04.5	34°27.1	•• 38.3	323°17.1	•• 49.4	123°58.6	•• 14.7	Jupiter	3°20.9	17:27
16	335°02.8	107°57.7	05.0	49°27.4	38.2	338°19.3	49.3	139°01.0	14.8	Saturn	164°00.2	06:45
17	35											

h	Sun			Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	180°10.9	S23°25.3	324°11.1	8.4'	N15°45.6	-11.5'	59.3'
1	195°10.6	25.3	338°38.4	8.4'	15°34.1	-11.6'	59.3'
2	210°10.3	25.2	353°05.9	8.5'	15°22.5	-11.7'	59.3'
3	225°10.0	· · 25.2	7°33.3	8.5'	15°10.9	-11.7'	59.3'
4	240°09.7	25.1	22°00.9	8.6'	14°59.1	-11.8'	59.3'
5	255°09.4	25.1	36°28.5	8.7'	14°47.3	-11.9'	59.3'
6	270°09.1	S23°25.1	50°56.1	8.7'	N14°35.4	-12.0'	59.3'
7	285°08.8	25.0	65°23.8	8.8'	14°23.4	-12.1'	59.3'
8	300°08.5	25.0	79°51.6	8.8'	14°11.3	-12.2'	59.3'
9	315°08.2	· · 24.9	94°19.4	8.9'	13°59.1	-12.3'	59.3'
10	330°07.8	24.9	108°47.3	8.9'	13°46.8	-12.3'	59.4'
11	345°07.5	24.8	123°15.2	9.0'	13°34.5	-12.4'	59.4'
12	0°07.2	S23°24.8	137°43.2	9.0'	N13°22.0	-12.5'	59.4'
13	15°06.9	24.7	152°11.3	9.1'	13°09.5	-12.6'	59.4'
14	30°06.6	24.7	166°39.4	9.2'	12°57.0	-12.7'	59.4'
15	45°06.3	· · 24.6	181°07.5	9.2'	12°44.3	-12.7'	59.4'
16	60°06.0	24.6	195°35.7	9.3'	12°31.6	-12.8'	59.4'
17	75°05.7	24.5	210°04.0	9.3'	12°18.7	-12.9'	59.4'
18	90°05.4	S23°24.5	224°32.3	9.4'	N12°05.9	-13.0'	59.4'
19	105°05.1	24.4	239°00.6	9.4'	11°52.9	-13.0'	59.4'
20	120°04.7	24.4	253°29.1	9.5'	11°39.9	-13.1'	59.4'
21	135°04.4	· · 24.3	267°57.5	9.5'	11°26.8	-13.2'	59.4'
22	150°04.1	24.2	282°26.1	9.6'	11°13.6	-13.2'	59.4'
23	165°03.8	24.2	296°54.6	9.6'	11°00.4	-13.3'	59.4'
SD = 16.3' $d = -0.0'$ SD = 16.2'							
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	180°03.5	S23°24.1	311°23.2	9.7'	N10°47.1	-13.3'	59.4'
1	195°03.2	24.1	325°51.9	9.7'	10°33.8	-13.4'	59.4'
2	210°02.9	24.0	340°20.6	9.8'	10°20.4	-13.5'	59.4'
3	225°02.6	· · 24.0	354°49.4	9.8'	10°06.9	-13.5'	59.4'
4	240°02.3	23.9	9°18.2	9.9'	09°53.4	-13.6'	59.4'
5	255°02.0	23.8	23°47.1	9.9'	09°39.8	-13.6'	59.4'
6	270°01.7	S23°23.8	38°16.0	10.0'	N09°26.2	-13.7'	59.4'
7	285°01.3	23.7	52°44.9	10.0'	09°12.5	-13.7'	59.4'
8	300°01.0	23.6	67°13.9	10.0'	08°58.8	-13.8'	59.4'
9	315°00.7	· · 23.6	81°43.0	10.1'	08°45.0	-13.8'	59.4'
10	330°00.4	23.5	96°12.0	10.1'	08°31.2	-13.9'	59.4'
11	345°00.1	23.4	110°41.2	10.2'	08°17.3	-13.9'	59.4'
12	359°59.8	S23°23.4	125°10.3	10.2'	N08°03.4	-14.0'	59.4'
13	14°59.5	23.3	139°39.5	10.2'	07°49.4	-14.0'	59.4'
14	29°59.2	23.2	154°08.8	10.3'	07°35.4	-14.0'	59.4'
15	44°58.9	· · 23.2	168°38.1	10.3'	07°21.4	-14.1'	59.4'
16	59°58.6	23.1	183°07.4	10.4'	07°07.3	-14.1'	59.4'
17	74°58.2	23.0	197°36.8	10.4'	06°53.1	-14.2'	59.4'
18	89°57.9	S23°22.9	212°06.2	10.4'	N06°39.0	-14.2'	59.4'
19	104°57.6	22.9	226°35.6	10.5'	06°24.8	-14.2'	59.4'
20	119°57.3	22.8	241°05.1	10.5'	06°10.5	-14.3'	59.4'
21	134°57.0	· · 22.7	255°34.6	10.5'	05°56.3	-14.3'	59.4'
22	149°56.7	22.6	270°04.1	10.6'	05°42.0	-14.3'	59.4'
23	164°56.4	22.6	284°33.7	10.6'	05°27.7	-14.4'	59.4'
SD = 16.3' $d = -0.1'$ SD = 16.2'							
Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	179°56.1	S23°22.5	299°03.2	10.6'	N05°13.3	-14.4'	59.4'
1	194°55.8	22.4	313°32.9	10.7'	04°58.9	-14.4'	59.4'
2	209°55.5	22.3	328°02.5	10.7'	04°44.5	-14.4'	59.4'
3	224°55.2	· · 22.2	342°32.2	10.7'	04°30.1	-14.4'	59.4'
4	239°54.9	22.2	357°01.9	10.7'	04°15.7	-14.5'	59.4'
5	254°54.5	22.1	11°31.7	10.8'	04°01.2	-14.5'	59.4'
6	269°54.2	S23°22.0	26°01.4	10.8'	N03°46.7	-14.5'	59.4'
7	284°53.9	21.9	40°31.2	10.8'	03°32.2	-14.5'	59.4'
8	299°53.6	21.8	55°01.0	10.8'	03°17.7	-14.5'	59.4'
9	314°53.3	· · 21.7	69°30.8	10.9'	03°03.2	-14.5'	59.4'
10	329°53.0	21.7	84°00.7	10.9'	02°48.6	-14.6'	59.4'
11	344°52.7	21.6	98°30.6	10.9'	02°34.0	-14.6'	59.4'
12	359°52.4	S23°21.5	113°00.5	10.9'	N02°19.5	-14.6'	59.4'
13	14°52.1	21.4	127°30.4	10.9'	02°04.9	-14.6'	59.4'
14	29°51.8	21.3	142°00.3	10.9'	01°50.3	-14.6'	59.4'
15	44°51.5	· · 21.2	156°30.3	11.0'	01°35.7	-14.6'	59.4'
16	59°51.2	21.1	171°00.2	11.0'	01°21.1	-14.6'	59.4'
17	74°50.8	21.0	185°30.2	11.0'	01°06.5	-14.6'	59.4'
18	89°50.5	S23°20.9	200°00.2	11.0'	N00°51.9	-14.6'	59.3'
19	104°50.2	20.8	214°30.2	11.0'	00°37.3	-14.6'	59.3'
20	119°49.9	20.7	229°00.2	11.0'	00°22.7	-14.6'	59.3'
21	134°49.6	· · 20.7	243°30.2	11.0'	N00°08.1	-14.6'	59.3'
22	149°49.3	20.6	258°00.2	11.0'	S00°06.5	14.6'	59.3'
23	164°49.0	20.5	272°30.3	11.0'	00°21.1	14.6'	59.3'
SD = 16.3' $d = -0.1'$ SD = 16.2'							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:27	10:57	■	■	13:03	15:33
N 70°	08:07	09:55	■	■	14:05	15:53
68°	07:51	09:20	■	■	14:40	16:09
66°	07:38	08:55	10:35	13:25	15:05	16:22
64°	07:27	08:35	09:53	14:07	15:25	16:33
62°	07:17	08:19	09:25	14:35	15:41	16:43
60°	07:09	08:06	09:03	14:57	15:54	16:51
N 58°	07:01	07:54	08:46	15:14	16:06	16:59
56°	06:55	07:44	08:31	15:29	16:16	17:06
54°	06:48	07:35	08:19	15:42	16:25	17:12
52°	06:43	07:26	08:07	15:53	16:34	17:17
50°	06:37	07:19	07:58	16:03	16:41	17:23
45°	06:26	07:03	07:37	16:23	16:57	17:34
N 40°	06:15	06:49	07:20	16:40	17:11	17:45
35°	06:06	06:38	07:06	16:54	17:22	17:54
30°	05:57	06:27	06:53	17:07	17:33	18:03
20°	05:41	06:08	06:32	17:28	17:52	18:19
N 10°	05:25	05:51	06:14	17:46	18:09	18:36
0°	05:08	05:34	05:56	18:04	18:26	18:53
S 10°	04:49	05:16	05:39	18:21	18:44	19:11
20°	04:26	04:55	05:20	18:40	19:05	19:34
30°	03:57	04:30	04:58	19:02	19:30	20:03
35°	03:38	04:15	04:45	19:15	19:45	20:22
40°	03:15	03:57	04:30	19:30	20:03	20:45
45°	02:44	03:34	04:12	19:48	20:26	21:15
S 50°	01:59	03:04	03:49	20:11	20:55	22:01
52°	01:31	02:49	03:38	20:22	21:11	22:29
54°	00:45	02:31	03:26	20:34	21:28	23:15
56°	////	02:09	03:12	20:48	21:51	////
58°	////	01:39	02:55	21:05	22:20	////
S 60°	////	00:49	02:34	21:25	23:10	////

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	18:21	20:44	22:56	11:54	11:21	10:56
N 70°	18:43	20:53	22:55	11:30	11:10	10:53
68°	19:00	21:00	22:55	11:12	11:00	10:50
66°	19:14	21:06	22:54	10:57	10:52	10:48
64°	19:25	21:11	22:54	10:44	10:45	10:46
62°	19:34	21:15	22:54	10:33	10:39	10:44
60°	19:42	21:19	22:53	10:24	10:34	10:42
N 58°	19:49	21:22	22:53	10:16	10:30	10:41
56°	19:56	21:25	22:53	10:09	10:26	10:40
54°	20:01	21:27	22:53	10:03	10:22	10:39
52°	20:06	21:30	22:52	09:57	10:19	10:38
50°	20:11	21:32	22:52	09:52	10:16	10:37
45°	20:20	21:37	22:52	09:40	10:09	10:35
N 40°	20:28	21:40	22:52	09:31	10:03	10:33
35°	20:35	21:44	22:51	09:23	09:58	10:32
30°	20:41	21:47	22:51	09:15	09:54	10:30
20°	20:52	21:52	22:51	09:03	09:47	10:28
N 10°	21:01	21:56	22:51	08:52	09:40	10:26
0°	21:09	22:00	22:50	08:41	09:34	10:24
S 10°	21:18	22:05	22:50	08:31	09:27	10:22
20°	21:27	22:09	22:50	08:20	09:20	10:20
30°	21:37	22:14	22:50	08:07	09:13	10:17
35°	21:43	22:17	22:50	07:59	09:08	10:16
40°	21:49	22:20	22:49	07:50	09:03	10:14
45°	21:57	22:24	22:49	07:40	08:57	10:12
S 50°	22:06	22:28	22:49	07:28	08:49	10:10
52°	22:10	22:30	22:49	07:22	08:46	10:09
54°	22:15	22:33	22:49	07:15	08:42	10:08
56°	22:20	22:35	22:49	07:08	08:38	10:07
58°	22:26	22:38	22:49	07:00	08:33	10:05
S 60°	22:32	22:41	22:49	06:51	08:28	10:03

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time	12 ^h		Mer. Pass.		
	00 ^h	mm:ss	hh:mm	Upper	Lower	19-21
				hh:mm	hh:mm	91-74%
24	00:44	00:29	12:00	02:29	14:55	
25	00:14	00:01	12:00	03:21	15:47	
26	00:16	00:30	12:01	04:12	16:37	

December 27, 28, 29 UT (Mon., Tue., Wed.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	95°22.5	227°59.3	S14°09.4	169°30.4	S23°36.6	98°36.9	S02°48.2	259°19.7	S04°15.1	Alpheratz	357°45.4 29°09.3
1	110°25.0	242°59.5	10.0	184°30.8	36.4	113°39.1	48.1	274°22.0	15.1	Ankaa	353°17.4 -42°14.9
2	125°27.4	257°59.6	10.5	199°31.2	36.2	128°41.2	48.0	289°24.4	15.2	Schedar	349°42.6 56°36.2
3	140°29.9	272°59.8	· · 11.1	214°31.5	· · 36.0	143°43.4	· · 47.9	304°26.7	· · 15.2	Diphda	348°57.6 -17°55.6
4	155°32.4	288°00.0	11.6	229°31.9	35.8	158°45.6	47.7	319°29.0	15.2	Achernar	335°27.8 -57°11.1
5	170°34.8	303°00.2	12.2	244°32.3	35.6	173°47.8	47.6	334°31.4	15.3	Hamal	328°02.6 23°31.0
6	185°37.3	318°00.4	S14°12.7	259°32.7	S23°35.4	188°50.0	S02°47.5	349°33.7	S04°15.3	Polaris	318°21.9 89°19.0
7	200°39.8	333°00.5	13.3	274°33.0	35.2	203°52.2	47.3	4°36.1	15.4	Acamar	315°19.3 -40°15.8
8	215°42.2	348°00.7	13.8	289°33.4	35.0	218°54.4	47.2	19°38.4	15.4	Menkar	314°16.7 4°08.0
9	230°44.7	3°00.9	· · 14.4	304°33.8	· · 34.8	233°56.6	· · 47.1	34°40.7	· · 15.4	Mirfak	308°42.6 49°54.2
10	245°47.1	18°01.1	14.9	319°34.2	34.6	248°58.8	46.9	49°43.1	15.5	Aldebaran	290°51.1 16°31.9
11	260°49.6	33°01.2	15.5	334°34.5	34.4	264°00.9	46.8	64°45.4	15.5	Rigel	281°13.4 -8°11.4
12	275°52.1	48°01.4	S14°16.0	349°34.9	S23°34.2	279°03.1	S02°46.7	79°47.8	S04°15.5	Capella	280°36.6 46°00.6
13	290°54.5	63°01.6	16.6	4°35.3	34.0	294°05.3	46.6	94°50.1	15.6	Bellatrix	278°33.6 6°21.5
14	305°57.0	78°01.8	17.1	19°35.7	33.8	309°07.5	46.4	109°52.4	15.6	Elnath	278°14.5 28°37.0
15	320°59.5	93°01.9	· · 17.7	34°36.0	· · 33.6	324°09.7	· · 46.3	124°54.8	· · 15.7	Alnilam	275°47.8 -1°11.7
16	336°01.9	108°02.1	18.3	49°36.4	33.4	339°11.9	46.2	139°57.1	15.7	Betelgeuse	271°02.9 7°24.5
17	351°04.4	123°02.3	18.8	64°36.8	33.2	354°14.1	46.0	154°59.5	15.7	Canopus	263°56.4 -52°42.1
18	6°06.9	138°02.4	S14°19.4	79°37.2	S23°33.0	9°16.2	S02°45.9	170°01.8	S04°15.8	Sirius	258°34.9 -16°43.9
19	21°09.3	153°02.6	19.9	94°37.5	32.8	24°18.4	45.8	185°04.1	15.8	Adhara	255°13.5 -28°59.3
20	36°11.8	168°02.8	20.5	109°37.9	32.5	39°20.6	45.6	200°06.5	15.9	Procyon	245°01.2 5°11.7
21	51°14.3	183°02.9	· · 21.0	124°38.3	· · 32.3	54°22.8	· · 45.5	215°08.8	· · 15.9	Pollux	243°29.5 27°59.8
22	66°16.7	198°03.1	21.6	139°38.7	32.1	69°25.0	45.4	230°11.2	15.9	Avior	234°18.2 -59°32.6
23	81°19.2	213°03.2	22.1	154°39.0	31.9	84°27.2	45.2	245°13.5	16.0	Suhail	222°53.4 -43°28.6
Mer.pass. 17:36		ν 0.2' d 0.6' m -4.67		ν 0.4' d -0.2' m 1.17		ν 2.2' d -0.1' m -2.35		ν 2.3' d 0.0' m 0.81		Miaplacidus	221°39.5 -69°45.6
Alphard 217°57.6 -8°42.5 Regulus 207°45.2 11°54.6 Dubhe 193°53.5 61°41.1 Denebola 182°35.4 14°30.4 Gienah 175°54.1 -17°36.2 Acrux 173°11.5 -63°09.4 Gacrux 172°03.0 -57°10.3 Alioth 166°22.2 55°53.6 Spica 158°33.2 -11°13.2 Alkaid 153°00.4 49°15.2 Hadar 148°50.8 -60°25.4 Menkent 148°09.9 -36°25.3 Arcturus 145°57.5 19°07.3 Rigel Kent. 139°54.6 -60°52.6 Kochab 137°20.5 74°06.3 Zuben'ubi 137°07.6 -16°05.2 Alphecca 126°12.8 26°40.5 Antares 112°28.8 -26°27.3 Atria 107°32.8 -69°02.7 Sabik 102°14.9 -15°44.3 Shaula 96°24.7 -37°06.6 Rasalhague 96°08.4 12°33.1 Eltanin 90°47.5 51°29.2 Kaus Aust. 83°46.6 -34°22.7 Vega 80°40.6 38°47.7 Nunki 76°00.9 -26°16.9 Altair 62°10.3 8°53.9 Peacock 53°22.5 -56°42.0 Deneb 49°33.1 45°19.4 Enif 33°49.1 9°55.7 Al Na'ir 27°46.1 -46°54.6 Fomalhaut 15°26.0 -29°33.9 Scheat 13°55.3 28°08.8 Markab 13°40.2 15°16.1											
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	96°21.6	228°03.4	S14°22.7	169°39.4	S23°31.7	99°29.4	S02°45.1	260°15.9	S04°16.0	Alpheratz	357°45.4 29°09.3
1	111°24.1	243°03.6	23.2	184°39.8	31.5	114°31.5	45.0	275°18.2	16.0	Ankaa	353°17.4 -42°14.9
2	126°26.6	258°03.7	23.8	199°40.2	31.3	129°33.7	44.8	290°20.5	16.1	Schedar	349°42.6 56°36.2
3	141°29.0	273°03.9	· · 24.4	214°40.5	· · 31.1	144°35.9	· · 44.7	305°22.9	· · 16.1	Diphda	348°57.6 -17°55.6
4	156°31.5	288°04.0	24.9	229°40.9	30.9	159°38.1	44.6	320°25.2	16.2	Achernar	335°27.8 -57°11.1
5	171°34.0	303°04.2	25.5	244°41.3	30.7	174°40.3	44.4	335°27.6	16.2	Hamal	328°02.6 23°31.0
6	186°36.4	318°04.3	S14°26.0	259°41.6	S23°30.5	189°42.5	S02°44.3	350°29.9	S04°16.2	Polaris	318°21.9 89°19.0
7	201°38.9	333°04.5	26.6	274°42.0	30.2	204°44.6	44.2	5°32.3	16.3	Acamar	315°19.3 -40°15.8
8	216°41.4	348°04.6	27.1	289°42.4	30.0	219°46.8	44.1	20°34.6	16.3	Menkar	314°16.7 4°08.0
9	231°43.8	3°04.8	· · 27.7	304°42.8	· · 29.8	234°49.0	· · 43.9	35°36.9	· · 16.4	Mirfak	308°42.6 49°54.2
10	246°46.3	18°04.9	28.3	319°43.1	29.6	249°51.2	43.8	50°39.3	16.4	Aldebaran	290°51.1 16°31.9
11	261°48.8	33°05.1	28.8	334°43.5	29.4	264°53.4	43.7	65°41.6	16.4	Rigel	281°13.4 -8°11.4
12	276°51.2	48°05.2	S14°29.4	349°43.9	S23°29.2	279°55.5	S02°43.5	80°44.0	S04°16.5	Capella	280°36.6 46°00.6
13	291°53.7	63°05.3	29.9	4°44.3	29.0	294°57.7	43.4	95°46.3	16.5	Bellatrix	278°33.6 6°21.5
14	306°56.1	78°05.5	30.5	19°44.6	28.8	309°59.9	43.3	110°48.7	16.5	Elnath	278°14.5 28°37.0
15	321°58.6	93°05.6	· · 31.0	34°45.0	· · 28.5	325°02.1	· · 43.1	125°51.0	· · 16.6	Alnilam	275°47.8 -1°11.7
16	337°01.1	108°05.8	31.6	49°45.4	28.3	340°04.3	43.0	140°53.4	16.6	Betelgeuse	271°02.9 7°24.5
17	352°03.5	123°05.9	32.2	64°45.8	28.1	355°06.4	42.9	155°55.7	16.7	Canopus	263°56.4 -52°42.1
18	7°06.0	138°06.0	S14°32.7	79°46.1	S23°27.9	10°08.6	S02°42.7	170°58.0	S04°16.7	Sirius	258°34.9 -16°43.9
19	22°05.3	153°06.2	33.3	94°46.5	27.7	25°10.8	42.6	186°00.4	16.7	Adhara	255°13.5 -28°59.3
20	37°10.9	168°06.3	33.8	109°46.9	27.5	40°13.0	42.5	201°02.7	16.8	Procyon	245°01.2 5°11.7
21	52°13.4	183°06.5	· · 34.4	124°47.3	· · 27.2	55°15.2	· · 42.3	216°05.1	· · 16.8	Pollux	243°29.5 27°59.8
22	67°15.9	198°06.6	34.9	139°47.6	27.0	70°17.3	42.2	231°07.4	16.8	Avior	234°18.2 -59°32.6
23	82°18.3	213°06.7	35.5	154°48.0	26.8	85°19.5	42.1	246°09.8	16.9	Suhail	222°53.4 -43°28.6
Mer.pass. 17:32		ν 0.2' d 0.6' m -4.66		ν 0.4' d -0.2' m 1.17		ν 2.2' d -0.1' m -2.34		ν 2.3' d 0.0' m 0.81		Miaplacidus	221°39.5 -69°45.6
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	97°20.8	228°06.8	S14°36.1	169°48.4	S23°26.6	100°21.7	S02°41.9	261°12.1	S04°16.9	Alpheratz	357°45.4 29°09.3
1	112°23.2	243°07.0	36.6	184°48.8	26.4	115°23.9	41.8	276°14.5	16.9	Ankaa	353°17.4 -42°14.9
2	127°25.7	258°07.1	37.2	199°49.1	26.1	130°26.1	41.7	291°16.8	17.0	Schedar	349°42.6 56°36.2
3	142°28.2	273°07.2	· · 37.7	214°49.5	· · 25.9	145°28.2	· · 41.5	306°19.2	· · 17.0	Diphda	348°57.6 -17°55.6
4	157°30.6	288°07.4	38.3	229°49.9	25.7	160°30.4	41.4	321°21.5	17.1	Achernar	335°27.8 -57°11.1
5	172°33.1	303°07.5	38.9	244°50.3	25.5	175°32.6	41.2	336°23.8	17.1	Hamal	328°02.6 23°31.0
6	187°35.6	318°07.6	S14°39.4	259°50.6	S23°25.2	190°34.8	S02°41.1	351°26.2	S04°17.1	Polaris	318°21.9 89°19.0
7	202°38.0	333°07.7	40.0	274°51.0	25.0	205°36.9	41.0	6°28.5	17.2	Acamar	315°19.3 -40°15.8
8	217°40.5	348°07.8	40.5	289°51.4	24.8	220°39.1	40.8	21°30.9	17.2	Menkar	314°16.7 4°08.0
9	232°43.0	3°08.0	· · 41.1	304°51.8	· · 24.6	235°41.3	· · 40.7	36°33.2	· · 17.2	Mirfak	308°42.6 49°54.2
10	247°45.4	18°08.1	41.7	319°52.1	24.4	250°43.5	40.6	51°35.6	17.3	Aldebaran	290°51.1 16°31.9
11	262°47.9	33°08.2	42.2	334°52.5	24.1	265°45.6	40.4	66°37.9	17.3	Rigel	281°13.4 -8°11.4
12	277°50.4	48°08.3	S14°42.8	349°52.9	S23°23.9	280°47.8	S02°40.3	81°40.3	S04°17.3	Capella	280°36.6 46°00.6
13	292°52.8	63°08.4	43.3	4°53.3	23.7	295°50.0	40.2	96°42.6	17.4	Bellatrix	278°33.6 6°21.5
14	307°55.3	78°08.5	43.9	19°53.6	23.5	310°52.2	40.0	111°45.0	17.4	Elnath	278°14.5 28°37.0
15	322°57.7	93°08.7	· · 44.5	34°54.0	· · 23.2	325°54.3	· · 39.9	126°47.3	· · 17.5	Alnilam	275°47.8 -1°11.7
16	338°00.2	108°08.8	45.0	49°54.4	23.0	340°56.5	39.8	141°49.7	17.5	Betelgeuse	271°02.9 7°24.5
17	353°02.7	123°08.9	45.6	64°54.8	22.8	355°58.7	39.6	156°52.0	17.5	Canopus	263°56.4 -52°42.1
18	8°05.1	138°09.0	S14°46.1	79°55.1	S23°22.5	11°00.9	S02°39.5	171°54.4	S04°17.6	Sirius	258°34.9 -16°43.9
19	23°07.6	153°09.1	46.7	94°55.5	22.3	26°03.0	39.3	186°56.7	17.6	Adhara	255°13.5 -28°59.3
20	38°10.1	168°09.2	47.3	109°55.9	22.1	41°05.2	39.2	201°59.1	17.6	Procyon	245°01.2 5°11.7
21	53°12.5	183°09.3	· · 47.8	124°56.3	· · 21.9	56°07.4	· · 39.1	217°01.4	· · 17.7	Pollux	243°29.5 27°59.8
22	68°15.0	198°09.4	48.4	139°56.7	21.6	71°09.6	38.9	232°03.8	17.7	Avior	234°18.2 -59°32.6
23	83°17.5	213°09.5	48.9	154°57.0	21.4	86°11.7	38.8	247°06.1	17.7	Suhail	222°53.4 -43°28.6
Mer.pass. 17:28		ν 0.1' d 0.6' m -4.65		ν 0.4' d -0.2' m 1.16		ν 2.2' d -0.1' m -2.34		ν 2.3' d 0.0' m 0.80		Miaplacidus	221°39.5 -69°45.6

Dec 27 Mon	SHA	Mer.pass
Venus	132°36.8	08:48
Mars	74°07.9	12:42
Jupiter	3°14.4	17:23
Saturn	163°57.2	06:42

Dec 28 Tue	SHA	Mer.pass
Venus	131°41.8	08:48
Mars	73°17.8	12:41
Jupiter	3°07.7	17:20
Saturn	163°54.2	06:38

Dec 29 Wed	SHA	Mer.pass
Venus	130°46.1	08:47
Mars	72°27.6	12:40
Jupiter	3°00.9	17:16
Saturn	163°51.3	06:34

Horizontal parallax		
Venus:	0.2	
Mars:	0.1	

h Sun Moon						
Mon	GHA	Dec	GHA	ν	Dec	d HP
0	179°48.7	S23°20.4	287°00.3	11.0'	S00°35.7	14.6' 59.3'
1	194°48.4	20.3	301°30.4	11.1'	00°50.3	14.6' 59.3'
2	209°48.1	20.2	316°00.4	11.1'	01°04.9	14.6' 59.3'
3	224°47.8	· · 20.1	330°30.5	11.1'	01°19.5	14.6' 59.3'
4	239°47.5	20.0	345°00.5	11.1'	01°34.0	14.5' 59.3'
5	254°47.1	19.9	359°30.6	11.1'	01°48.6	14.5' 59.3'
6	269°46.8	S23°19.8	14°00.7	11.1'	S02°03.1	14.5' 59.3'
7	284°46.5	19.7	28°30.7	11.1'	02°17.6	14.5' 59.3'
8	299°46.2	19.5	43°00.8	11.1'	02°32.1	14.5' 59.3'
9	314°45.9	· · 19.4	57°30.8	11.1'	02°46.6	14.5' 59.2'
10	329°45.6	19.3	72°00.9	11.1'	03°01.1	14.4' 59.2'
11	344°45.3	19.2	86°30.9	11.0'	03°15.5	14.4' 59.2'
12	359°45.0	S23°19.1	101°01.0	11.0'	S03°29.9	14.4' 59.2'
13	14°44.7	19.0	115°31.0	11.0'	03°44.3	14.4' 59.2'
14	29°44.4	18.9	130°01.1	11.0'	03°58.7	14.3' 59.2'
15	44°44.1	· · 18.8	144°31.1	11.0'	04°13.0	14.3' 59.2'
16	59°43.8	18.7	159°01.1	11.0'	04°27.3	14.3' 59.2'
17	74°43.5	18.6	173°31.1	11.0'	04°41.6	14.3' 59.2'
18	89°43.2	S23°18.5	188°01.1	11.0'	S04°55.9	14.2' 59.2'
19	104°42.8	18.3	202°31.1	11.0'	05°10.1	14.2' 59.2'
20	119°42.5	18.2	217°01.0	11.0'	05°24.3	14.2' 59.1'
21	134°42.2	· · 18.1	231°31.0	10.9'	05°38.5	14.1' 59.1'
22	149°41.9	18.0	246°00.9	10.9'	05°52.6	14.1' 59.1'
23	164°41.6	17.9	260°30.8	10.9'	06°06.7	14.1' 59.1'
SD = 16.3' d = -0.1' SD = 16.2'						
Tue	GHA	Dec	GHA	ν	Dec	d HP
0	179°41.3	S23°17.8	275°00.7	10.9'	S06°20.8	14.0' 59.1'
1	194°41.0	17.7	289°30.6	10.9'	06°34.8	14.0' 59.1'
2	209°40.7	17.5	304°00.5	10.8'	06°48.8	13.9' 59.1'
3	224°40.4	· · 17.4	318°30.3	10.8'	07°02.7	13.9' 59.1'
4	239°40.1	17.3	333°00.2	10.8'	07°16.6	13.8' 59.1'
5	254°39.8	17.2	347°30.0	10.8'	07°30.4	13.8' 59.1'
6	269°39.5	S23°17.0	1°59.7	10.8'	S07°44.2	13.8' 59.0'
7	284°39.2	16.9	16°29.5	10.7'	07°58.0	13.7' 59.0'
8	299°38.9	16.8	30°59.2	10.7'	08°11.7	13.7' 59.0'
9	314°38.6	· · 16.7	45°28.9	10.7'	08°25.4	13.6' 59.0'
10	329°38.3	16.6	59°58.6	10.7'	08°39.0	13.6' 59.0'
11	344°37.9	16.4	74°28.3	10.6'	08°52.5	13.5' 59.0'
12	359°37.6	S23°16.3	88°57.9	10.6'	S09°06.0	13.5' 59.0'
13	14°37.3	16.2	103°27.5	10.6'	09°19.5	13.4' 59.0'
14	29°37.0	16.0	117°57.0	10.5'	09°32.9	13.3' 59.0'
15	44°36.7	· · 15.9	132°26.6	10.5'	09°46.2	13.3' 58.9'
16	59°36.4	15.8	146°56.1	10.5'	09°59.5	13.2' 58.9'
17	74°36.1	15.6	161°25.5	10.4'	10°12.7	13.2' 58.9'
18	89°35.8	S23°15.5	175°55.0	10.4'	S10°25.9	13.1' 58.9'
19	104°35.5	15.4	190°24.4	10.4'	10°39.0	13.0' 58.9'
20	119°35.2	15.3	204°53.7	10.3'	10°52.0	13.0' 58.9'
21	134°34.9	· · 15.1	219°23.1	10.3'	11°05.0	12.9' 58.9'
22	149°34.6	15.0	233°52.4	10.3'	11°17.9	12.8' 58.9'
23	164°34.3	14.8	248°21.6	10.2'	11°30.8	12.8' 58.9'
SD = 16.3' d = -0.1' SD = 16.1'						
Wed	GHA	Dec	GHA	ν	Dec	d HP
0	179°34.0	S23°14.7	262°50.9	10.2'	S11°43.5	12.7' 58.8'
1	194°33.7	14.6	277°20.0	10.1'	11°56.3	12.6' 58.8'
2	209°33.4	14.4	291°49.2	10.1'	12°08.9	12.6' 58.8'
3	224°33.1	· · 14.3	306°18.3	10.1'	12°21.4	12.5' 58.8'
4	239°32.8	14.2	320°47.4	10.0'	12°33.9	12.4' 58.8'
5	254°32.5	14.0	335°16.4	10.0'	12°46.4	12.3' 58.8'
6	269°32.2	S23°13.9	349°45.4	9.9'	S12°58.7	12.3' 58.8'
7	284°31.8	13.7	4°14.3	9.9'	13°11.0	12.2' 58.7'
8	299°31.5	13.6	18°43.2	9.9'	13°23.2	12.1' 58.7'
9	314°31.2	· · 13.4	33°12.1	9.8'	13°35.3	12.0' 58.7'
10	329°30.9	13.3	47°40.9	9.8'	13°47.3	11.9' 58.7'
11	344°30.6	13.2	62°09.6	9.7'	13°59.2	11.9' 58.7'
12	359°30.3	S23°13.0	76°38.4	9.7'	S14°11.1	11.8' 58.7'
13	14°30.0	12.9	91°07.1	9.6'	14°22.9	11.7' 58.7'
14	29°29.7	12.7	105°35.7	9.6'	14°34.6	11.6' 58.7'
15	44°29.4	· · 12.6	120°04.3	9.5'	14°46.2	11.5' 58.6'
16	59°29.1	12.4	134°32.8	9.5'	14°57.7	11.4' 58.6'
17	74°28.8	12.3	149°01.3	9.5'	15°09.2	11.3' 58.6'
18	89°28.5	S23°12.1	163°29.8	9.4'	S15°20.5	11.3' 58.6'
19	104°28.2	12.0	177°58.2	9.4'	15°31.8	11.2' 58.6'
20	119°27.9	11.8	192°26.5	9.3'	15°42.9	11.1' 58.6'
21	134°27.6	· · 11.6	206°54.8	9.3'	15°54.0	11.0' 58.6'
22	149°27.3	11.5	221°23.1	9.2'	16°05.0	10.9' 58.5'
23	164°27.0	11.3	235°51.3	9.2'	16°15.8	10.8' 58.5'
SD = 16.3' d = -0.1' SD = 16.0'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:27	10:53	■	■	13:10	15:37
N 70°	08:07	09:54	■	■	14:09	15:56
68°	07:51	09:20	■	■	14:43	16:12
66°	07:38	08:55	10:34	13:30	15:08	16:25
64°	07:27	08:35	09:53	14:11	15:28	16:36
62°	07:18	08:20	09:25	14:38	15:44	16:45
60°	07:09	08:06	09:04	15:00	15:57	16:54
N 58°	07:02	07:55	08:46	15:17	16:09	17:01
56°	06:55	07:44	08:32	15:31	16:19	17:08
54°	06:49	07:35	08:19	15:44	16:28	17:14
52°	06:44	07:27	08:08	15:55	16:36	17:19
50°	06:38	07:20	07:58	16:05	16:43	17:25
48°	06:27	07:04	07:38	16:25	16:59	17:36
N 40°	06:16	06:50	07:21	16:42	17:13	17:47
35°	06:07	06:39	07:07	16:56	17:24	17:56
30°	05:58	06:28	06:55	17:08	17:35	18:05
20°	05:42	06:10	06:34	17:29	17:53	18:21
N 10°	05:26	05:52	06:15	17:48	18:11	18:37
0°	05:09	05:35	05:58	18:05	18:28	18:54
S 10°	04:50	05:17	05:40	18:23	18:46	19:13
20°	04:28	04:57	05:21	18:42	19:06	19:35
30°	03:59	04:32	04:59	19:03	19:31	20:04
35°	03:40	04:17	04:47	19:16	19:46	20:23
40°	03:17	03:59	04:32	19:31	20:04	20:46
45°	02:47	03:36	04:14	19:49	20:26	21:16
S 50°	02:02	03:07	03:51	20:11	20:56	22:01
52°	01:34	02:52	03:40	20:22	21:11	22:28
54°	00:50	02:34	03:28	20:34	21:29	23:12
56°	////	02:12	03:14	20:48	21:50	////
58°	////	01:43	02:58	21:05	22:19	////
S 60°	////	00:54	02:37	21:25	23:07	////

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	· ·	01:08	03:30	10:32	10:06	09:31
N 70°	· ·	00:58	03:05	10:36	10:19	09:57
68°	· ·	00:49	02:47	10:40	10:30	10:18
66°	· ·	00:42	02:32	10:43	10:38	10:34
64°	· ·	00:37	02:20	10:46	10:46	10:47
62°	· ·	00:32	02:10	10:48	10:52	10:58
60°	· ·	00:27	02:02	10:50	10:58	11:08
N 58°	· ·	00:24	01:54	10:52	11:03	11:16
56°	· ·	00:20	01:48	10:53	11:08	11:24
54°	· ·	00:17	01:42	10:55	11:12	11:31
52°	· ·	00:14	01:36	10:56	11:15	11:37
50°	· ·	00:12	01:32	10:57	11:19	11:42
45°	· ·	00:07	01:21	11:00	11:26	11:54
N 40°	· ·	00:02	01:13	11:02	11:32	12:04
35°	23:58	· ·	01:06	11:04	11:37	12:12
30°	23:55	· ·	00:59	11:06	11:42	12:20
20°	23:49	· ·	00:49	11:09	11:50	12:33
N 10°	23:44	· ·	00:39	11:11	11:57	12:44
0°	23:40	· ·	00:30	11:14	12:04	12:55
S 10°	23:35	· ·	00:22	11:16	12:10	13:05
20°	23:30	· ·	00:12	11:19	12:18	13:17
30°	23:25	· ·	00:02	11:22	12:26	13:30
35°	23:22	23:56	· ·	11:23	12:31	13:38
40°	23:18	23:49	· ·	11:25	12:36	13:47
45°	23:14	23:41	· ·	11:27	12:42	13:57
S 50°	23:10	23:32	23:57	11:30	12:50	14:09
52°	23:08	23:28	23:51	11:31	12:53	14:15
54°	23:05	23:23	23:44	11:33	12:57	14:22
56°	23:03	23:18	23:36	11:34	13:02	14:29
58°	23:00	23:12	23:27	11:36	13:06	14:37
S 60°	22:56	23:05	23:17	11:38	13:12	14:46

Day	Sun		Mer. Pass	Moon		Age 22-24 63-41%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
27	00:45	01:00	12:01	05:02	17:27	
28	01:15	01:29	12:01	05:52	18:17	
29	01:44	01:59	12:02	06:42	19:08	

December 30, 31, 01 UT (Thu., Fri., Sat.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	98°19.9	228°09.6	S14°49.5	169°57.4	S23°21.2	101°13.9	S02°38.7	262°08.5	S04°17.8	Alpheratz	357°45.4	29°09.3											
1	113°22.4	243°09.7	50.1	184°57.8	20.9	116°16.1	38.5	277°10.8	17.8	Ankaa	353°17.4	-42°14.9											
2	128°24.9	258°09.8	50.6	199°58.2	20.7	131°18.2	38.4	292°13.2	17.8	Schedar	349°42.6	56°36.2											
3	143°27.3	273°09.9	· · 51.2	214°58.5	· · 20.5	146°20.4	· · 38.3	307°15.5	· · 17.9	Diphda	348°57.6	-17°55.6											
4	158°29.8	288°10.0	51.7	229°58.9	20.2	161°22.6	38.1	322°17.9	17.9	Achernar	335°27.8	-57°11.1											
5	173°32.2	303°10.1	52.3	244°59.3	20.0	176°24.8	38.0	337°20.2	17.9	Hamal	328°02.6	23°31.0											
6	188°34.7	318°10.2	S14°52.9	259°59.7	S23°19.8	191°26.9	S02°37.8	352°22.6	S04°18.0	Polaris	318°22.8	89°19.1											
7	203°37.2	333°10.3	53.4	275°00.0	19.5	206°29.1	37.7	7°24.9	18.0	Acamar	315°19.4	-40°15.8											
8	218°39.6	348°10.4	54.0	290°00.4	19.3	221°31.3	37.6	22°27.3	18.1	Menkar	314°16.7	4°08.0											
9	233°42.1	3°10.5	· · 54.6	305°00.8	· · 19.1	236°33.4	· · 37.4	37°29.6	· · 18.1	Mirfak	308°42.6	49°54.2											
10	248°44.6	18°10.6	55.1	320°01.2	18.8	251°35.6	37.3	52°32.0	18.1	Aldebaran	290°51.1	16°31.9											
11	263°47.0	33°10.7	55.7	335°01.5	18.6	266°37.8	37.2	67°34.3	18.2	Rigel	281°13.4	-8°11.4											
12	278°49.5	48°10.8	S14°56.2	350°01.9	S23°18.4	281°39.9	S02°37.0	82°36.7	S04°18.2	Capella	280°36.6	46°00.6											
13	293°52.0	63°10.9	56.8	5°02.3	18.1	296°42.1	36.9	97°39.0	18.2	Bellatrix	278°33.6	6°21.5											
14	308°54.4	78°10.9	57.4	20°02.7	17.9	311°44.3	36.7	112°41.4	18.3	Elnath	278°14.5	28°37.0											
15	323°56.9	93°11.0	· · 57.9	35°03.0	· · 17.7	326°46.5	· · 36.6	127°43.7	· · 18.3	Alnilam	275°47.8	-1°11.8											
16	338°59.3	108°11.1	58.5	50°03.4	17.4	341°48.6	36.5	142°46.1	18.3	Betelgeuse	271°02.9	7°24.5											
17	354°01.8	123°11.2	59.0	65°03.8	17.2	356°50.8	36.3	157°48.4	18.4	Canopus	263°56.4	-52°42.2											
18	9°04.3	138°11.3	S14°59.6	80°04.2	S23°16.9	11°53.0	S02°36.2	172°50.8	S04°18.4	Sirius	258°34.9	-16°44.0											
19	24°06.7	153°11.4	15°00.2	95°04.6	16.7	26°55.1	36.0	187°53.1	18.4	Adhara	255°13.5	-28°59.3											
20	39°09.2	168°11.4	00.7	110°04.9	16.5	41°57.3	35.9	202°55.5	18.5	Procyon	245°01.2	5°11.7											
21	54°11.7	183°11.5	· · 01.3	125°05.3	· · 16.2	56°59.5	· · 35.8	217°57.8	· · 18.5	Pollux	243°29.5	27°59.8											
22	69°14.1	198°11.6	01.9	140°05.7	16.0	72°01.6	35.6	233°00.2	18.5	Avior	234°18.2	-59°32.6											
23	84°16.6	213°11.7	02.4	155°06.1	15.7	87°03.8	35.5	248°02.5	18.6	Suhail	222°53.4	-43°28.6											
Mer.pass. 17:24				ν 0.1' d 0.6' m -4.64				ν 0.4' d -0.2' m 1.16				ν 2.2' d -0.1' m -2.33				ν 2.3' d 0.0' m 0.80							

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	99°19.1	228°11.7	S15°03.0	170°06.4	S23°15.5	102°06.0	S02°35.4	263°04.9	S04°18.6										
1	114°21.5	243°11.8	03.5	185°06.8	15.3	117°08.1	35.2	278°07.3	18.6										
2	129°24.0	258°11.9	04.1	200°07.2	15.0	132°10.3	35.1	293°09.6	18.7										
3	144°26.5	273°12.0	· · 04.7	215°07.6	· · 14.8	147°12.5	· · 34.9	308°12.0	· · 18.7										
4	159°28.9	288°12.0	05.2	230°08.0	14.5	162°14.6	34.8	323°14.3	18.7										
5	174°31.4	303°12.1	05.8	245°08.3	14.3	177°16.8	34.7	338°16.7	18.8										
6	189°33.8	318°12.2	S15°06.4	260°08.7	S23°14.0	192°18.9	S02°34.5	353°19.0	S04°18.8										
7	204°36.3	333°12.2	06.9	275°09.1	13.8	207°21.1	34.4	8°21.4	18.8										
8	219°38.8	348°12.3	07.5	290°09.5	13.5	222°23.3	34.2	23°23.7	18.9										
9	234°41.2	3°12.4	· · 08.1	305°09.8	· · 13.3	237°25.4	· · 34.1	38°26.1	· · 18.9										
10	249°43.7	18°12.4	08.6	320°10.2	13.0	252°27.6	34.0	53°28.4	18.9										
11	264°46.2	33°12.5	09.2	335°10.6	12.8	267°29.8	33.8	68°30.8	19.0										
12	279°48.6	48°12.6	S15°09.7	350°11.0	S23°12.6	282°31.9	S02°33.7	83°33.2	S04°19.0										
13	294°51.1	63°12.6	10.3	5°11.3	12.3	297°34.1	33.5	98°35.5	19.0										
14	309°53.6	78°12.7	10.9	20°11.7	12.1	312°36.3	33.4	113°37.9	19.1										
15	324°56.0	93°12.8	· · 11.4	35°12.1	· · 11.8	327°38.4	· · 33.3	128°40.2	· · 19.1										
16	339°58.5	108°12.8	12.0	50°12.5	11.6	342°40.6	33.1	143°42.6	19.1										
17	355°01.0	123°12.9	12.6	65°12.9	11.3	357°42.7	33.0	158°44.9	19.2										
18	10°03.4	138°12.9	S15°13.1	80°13.2	S23°11.1	12°44.9	S02°32.8	173°47.3	S04°19.2										
19	25°05.9	153°13.0	13.7	95°13.6	10.8	27°47.1	32.7	188°49.6	19.2										
20	40°08.3	168°13.0	14.2	110°14.0	10.6	42°49.2	32.5	203°52.0	19.3										
21	55°10.8	183°13.1	· · 14.8	125°14.4	· · 10.3	57°51.4	· · 32.4	218°54.4	· · 19.3										
22	70°13.3	198°13.1	15.4	140°14.8	10.1	72°53.5	32.3	233°56.7	19.3										
23	85°15.7	213°13.2	15.9	155°15.1	09.8	87°55.7	32.1	248°59.1	19.3										
Mer.pass. 17:20				ν 0.1' d 0.6' m -4.62				ν 0.4' d -0.2' m 1.15				ν 2.2' d -0.1' m -2.32				ν 2.4' d 0.0' m 0.79			

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	100°18.2	228°13.2	S15°16.5	170°15.5	S23°09.6	102°57.9	S02°32.0	264°01.4	S04°19.4										
1	115°20.7	243°13.3	17.1	185°15.9	09.3	118°00.0	31.8	279°03.8	19.4										
2	130°23.1	258°13.3	17.6	200°16.3	09.0	133°02.2	31.7	294°06.1	19.4										
3	145°25.6	273°13.4	· · 18.2	215°16.6	· · 08.8	148°04.4	· · 31.6	309°08.5	· · 19.5										
4	160°28.1	288°13.4	18.7	230°17.0	08.5	163°06.5	31.4	324°10.9	19.5										
5	175°30.5	303°13.5	19.3	245°17.4	08.3	178°08.7	31.3	339°13.2	19.5										
6	190°33.0	318°13.5	S15°19.9	260°17.8	S23°08.0	193°10.8	S02°31.1	354°15.6	S04°19.6										
7	205°35.5	333°13.6	20.4	275°18.2	07.8	208°13.0	31.0	9°17.9	19.6										
8	220°37.9	348°13.6	21.0	290°18.5	07.5	223°15.1	30.8	24°20.3	19.6										
9	235°40.4	3°13.6	· · 21.6	305°18.9	· · 07.3	238°17.3	· · 30.7	39°22.7	· · 19.7										
10	250°42.8	18°13.7	22.1	320°19.3	07.0	253°19.5	30.6	54°25.0	19.7										
11	265°45.3	33°13.7	22.7	335°19.7	06.7	268°21.6	30.4	69°27.4	19.7										
12	280°47.8	48°13.7	S15°23.3	350°20.1	S23°06.5	283°23.8	S02°30.3	84°29.7	S04°19.8										
13	295°50.2	63°13.8	23.8	5°20.4	06.2	298°25.9	30.1	99°32.1	19.8										
14	310°52.7	78°13.8	24.4	20°20.8	06.0	313°28.1	30.0	114°34.4	19.8										
15	325°55.2	93°13.8	· · 24.9	35°21.2	· · 05.7	328°30.2	· · 29.8	129°36.8	· · 19.9										
16	340°57.6	108°13.9	25.5	50°21.6	05.4	343°32.4	29.7	144°39.2	19.9										
17	356°00.1	123°13.9	26.1	65°21.9	05.2	358°34.6	29.6	159°41.5	19.9										
18	11°02.6	138°13.9	S15°26.6	80°22.3	S23°04.9	13°36.7	S02°29.4	174°43.9	S04°19.9										
19	26°05.0	153°14.0	27.2	95°22.7	04.7	28°38.9	29.3	189°46.2	20.0										
20	41°07.5	168°14.0	27.8	110°23.1	04.4	43°41.0	29.1	204°48.6	20.0										
21	56°10.0	183°14.0	· · 28.3	125°23.5	· · 04.1	58°43.2	· · 29.0	219°51.0	· · 20.0										
22	71°12.4	198°14.1	28.9	140°23.8	03.9	73°45.3	28.8	234°53.3	20.1										
23	86°14.9	213°14.1	29.4	155°24.2	03.6	88°47.5	28.7	249°55.7	20.1										
Mer.pass. 17:16				ν 0.0' d 0.6' m -4.61				ν 0.4' d -0.3' m 1.15				ν 2.2' d -0.1' m -2.32				ν 2.4' d 0.0' m 0.79			

Dec 30 Thu	SHA	Mer.pass
Venus	129°49.7	08:47
Mars	71°37.5	12:40
Jupiter	2°54.0	17:13
Saturn	163°48.5	06:30

Dec 31 Fri	SHA	Mer.pass
Venus	128°52.7	08:47
Mars	70°47.4	12:39
Jupiter	2°46.9	17:09
Saturn	163°45.8	06:27

Jan 01 Sat	SHA	Mer.pass
Venus	127°55.0	08:47
Mars	69°57.3	12:39
Jupiter	2°39.7	17:06
Saturn	163°43.2	06:23

Horizontal parallax		
Venus:	0.2	
Mars:	0.1	

h	Sun						Moon					
Thu	GHA	Dec	GHA	ν	Dec	d	HP	GHA	ν	Dec	d	HP
0	179°26.7	S23°11.2	250°19.5	9.1'	S16°26.6	10.7'	58.5'	250°19.5	9.1'	S16°26.6	10.7'	58.5'
1	194°26.4	11.0	264°47.6	9.1'	16°37.3	10.6'	58.5'	279°15.7	9.0'	16°47.9	10.5'	58.5'
2	209°26.1	10.9	279°15.7	9.0'	16°47.9	10.5'	58.5'	293°43.7	9.0'	16°58.4	10.4'	58.5'
3	224°25.8	· · 10.7	293°43.7	9.0'	16°58.4	10.4'	58.5'	308°11.7	8.9'	17°08.8	10.3'	58.5'
4	239°25.5	10.5	308°11.7	8.9'	17°08.8	10.3'	58.5'	322°39.6	8.9'	17°19.1	10.2'	58.4'
5	254°25.2	10.4	322°39.6	8.9'	17°19.1	10.2'	58.4'	337°07.5	8.8'	S17°29.2	10.1'	58.4'
6	269°24.9	S23°10.2	337°07.5	8.8'	S17°29.2	10.1'	58.4'	351°35.3	8.8'	17°39.3	10.0'	58.4'
7	284°24.6	10.1	351°35.3	8.8'	17°39.3	10.0'	58.4'	6°03.1	8.7'	17°49.3	9.9'	58.4'
8	299°24.3	09.9	6°03.1	8.7'	17°49.3	9.9'	58.4'	20°30.8	8.7'	17°59.1	9.8'	58.4'
9	314°24.0	· · 09.7	20°30.8	8.7'	17°59.1	9.8'	58.4'	34°58.5	8.6'	18°08.9	9.6'	58.4'
10	329°23.7	09.6	34°58.5	8.6'	18°08.9	9.6'	58.4'	49°26.1	8.6'	18°18.5	9.5'	58.4'
11	344°23.4	09.4	49°26.1	8.6'	18°18.5	9.5'	58.4'	63°53.7	8.5'	S18°28.1	9.4'	58.3'
12	359°23.1	S23°09.2	63°53.7	8.5'	S18°28.1	9.4'	58.3'	78°21.2	8.5'	18°37.5	9.3'	58.3'
13	14°22.8	09.1	78°21.2	8.5'	18°37.5	9.3'	58.3'	92°48.7	8.4'	18°46.8	9.2'	58.3'
14	29°22.5	08.9	92°48.7	8.4'	18°46.8	9.2'	58.3'	107°16.1	8.4'	18°56.0	9.1'	58.3'
15	44°22.2	· · 08.7	107°16.1	8.4'	18°56.0	9.1'	58.3'	121°43.5	8.3'	19°05.1	9.0'	58.3'
16	59°21.9	08.6	121°43.5	8.3'	19°05.1	9.0'	58.3'	136°10.9	8.3'	19°14.1	8.9'	58.3'
17	74°21.6	08.4	136°10.9	8.3'	19°14.1	8.9'	58.3'	150°38.2	8.2'	S19°23.0	8.7'	58.2'
18	89°21.3	S23°08.2	150°38.2	8.2'	S19°23.0	8.7'	58.2'	165°05.4	8.2'	19°31.7	8.6'	58.2'
19	104°21.0	08.1	165°05.4	8.2'	19°31.7	8.6'	58.2'	179°32.6	8.2'	19°40.3	8.5'	58.2'
20	119°20.7	07.9	179°32.6	8.2'	19°40.3	8.5'	58.2'	193°59.8	8.1'	19°48.8	8.4'	58.2'
21	134°20.4	· · 07.7	193°59.8	8.1'	19°48.8	8.4'	58.2'	208°26.9	8.1'	19°57.2	8.3'	58.2'
22	149°20.1	07.5	208°26.9	8.1'	19°57.2	8.3'	58.2'	222°53.9	8.0'	20°05.5	8.1'	58.2'
23	164°19.8	07.4	222°53.9	8.0'	20°05.5	8.1'	58.2'	SD = 16.0'				
SD = 16.3' d = -0.2'												
Fri	GHA	Dec	GHA	ν	Dec	d	HP	GHA	ν	Dec	d	HP
0	179°19.5	S23°07.2	237°21.0	8.0'	S20°13.6	8.0'	58.1'	237°21.0	8.0'	S20°13.6	8.0'	58.1'
1	194°19.2	07.0	251°47.9	7.9'	20°21.6	7.9'	58.1'	251°47.9	7.9'	20°21.6	7.9'	58.1'
2	209°18.9	06.8	266°14.9	7.9'	20°29.5	7.8'	58.1'	266°14.9	7.9'	20°29.5	7.8'	58.1'
3	224°18.6	· · 06.7	280°41.7	7.8'	20°37.3	7.6'	58.1'	280°41.7	7.8'	20°37.3	7.6'	58.1'
4	239°18.3	06.5	295°08.6	7.8'	20°44.9	7.5'	58.1'	295°08.6	7.8'	20°44.9	7.5'	58.1'
5	254°18.0	06.3	309°35.4	7.8'	20°52.5	7.4'	58.1'	309°35.4	7.8'	20°52.5	7.4'	58.1'
6	269°17.7	S23°06.1	324°02.2	7.7'	S20°59.9	7.3'	58.0'	324°02.2	7.7'	S20°59.9	7.3'	58.0'
7	284°17.4	05.9	338°28.9	7.7'	21°07.1	7.1'	58.0'	338°28.9	7.7'	21°07.1	7.1'	58.0'
8	299°17.1	05.8	352°55.6	7.6'	21°14.3	7.0'	58.0'	352°55.6	7.6'	21°14.3	7.0'	58.0'
9	314°16.8	· · 05.6	7°22.2	7.6'	21°21.3	6.9'	58.0'	7°22.2	7.6'	21°21.3	6.9'	58.0'
10	329°16.5	05.4	21°48.8	7.6'	21°28.2	6.7'	58.0'	21°48.8	7.6'	21°28.2	6.7'	58.0'
11	344°16.2	05.2	36°15.4	7.5'	21°34.9	6.6'	58.0'	36°15.4	7.5'	21°34.9	6.6'	58.0'
12	359°15.9	S23°05.0	50°41.9	7.5'	S21°41.5	6.5'	57.9'	50°41.9	7.5'	S21°41.5	6.5'	57.9'
13	14°15.6	04.8	65°08.4	7.5'	21°48.0	6.3'	57.9'	65°08.4	7.5'	21°48.0	6.3'	57.9'
14	29°15.3	04.6	79°34.8	7.4'	21°54.3	6.2'	57.9'	79°34.8	7.4'	21°54.3	6.2'	57.9'
15	44°15.0	· · 04.5	94°01.2	7.4'	22°00.6	6.1'	57.9'	94°01.2	7.4'	22°00.6	6.1'	57.9'
16	59°14.7	04.3	108°27.6	7.4'	22°06.6	5.9'	57.9'	108°27.6	7.4'	22°06.6	5.9'	57.9'
17	74°14.4	04.1	122°54.0	7.3'	22°12.6	5.8'	57.9'	122°54.0	7.3'	22°12.6	5.8'	57.9'
18	89°14.1	S23°03.9	137°20.3	7.3'	S22°18.4	5.7'	57.8'	137°20.3	7.3'	S22°18.4	5.7'	57.8'
19	104°13.8	03.7	151°46.6	7.3'	22°24.1	5.5'	57.8'	151°46.6	7.3'	22°24.1	5.5'	57.8'
20	119°13.5	03.5	166°12.9	7.2'	22°29.6	5.4'	57.8'	166°12.9	7.2'	22°29.6	5.4'	57.8'
21	134°13.2	· · 03.3	180°39.1	7.2'	22°35.0	5.3'	57.8'	180°39.1	7.2'	22°35.0	5.3'	57.8'
22	149°12.9	03.1	195°05.3	7.2'	22°40.3	5.1'	57.8'	195°05.3	7.2'	22°40.3	5.1'	57.8'
23	164°12.6	02.9	209°31.5	7.2'	22°45.4	5.0'	57.8'	209°31.5	7.2'	22°45.4	5.0'	57.8'
SD = 16.3' d = -0.2'												
SD = 15.9'												
Sat	GHA	Dec	GHA	ν	Dec	d	HP	GHA	ν	Dec	d	HP
0	179°12.3	S23°02.7	223°57.6	7.1'	S22°50.4	4.8'	57.7'	223°57.6	7.1'	S22°50.4	4.8'	57.7'
1	194°12.0	02.5	238°23.8	7.1'	22°55.2	4.7'	57.7'	238°23.8	7.1'	22°55.2	4.7'	57.7'
2	209°11.7	02.3	252°49.9	7.1'	22°59.9	4.6'	57.7'	252°49.9	7.1'	22°59.9	4.6'	57.7'
3	224°11.4	· · 02.1	267°16.0	7.1'	23°04.5	4.4'	57.7'	267°16.0	7.1'	23°04.5	4.4'	57.7'
4	239°11.1	01.9	281°42.0	7.0'	23°08.9	4.3'	57.7'	281°42.0	7.0'	23°08.9	4.3'	57.7'
5	254°10.8	01.7	296°08.1	7.0'	23°13.2	4.1'	57.6'	296°08.1	7.0'	23°13.2	4.1'	57.6'
6	269°10.5	S23°01.5	310°34.1	7.0'	S23°17.3	4.0'	57.6'	310°34.1	7.0'	S23°17.3	4.0'	57.6'
7	284°10.2	01.3	325°00.1	7.0'	23°21.3	3.9'	57.6'	325°00.1	7.0'	23°21.3	3.9'	57.6'
8	299°09.9	01.1	339°26.1	7.0'	23°25.2	3.7'	57.6'	339°26.1	7.0'	23°25.2	3.7'	57.6'
9	314°09.6	· · 00.9	353°52.1	7.0'	23°28.9	3.6'	57.6'	353°52.1	7.0'	23°28.9	3.6'	57.6'
10	329°09.3	00.7	8°18.1	7.0'	23°32.4	3.4'	57.6'	8°18.1	7.0'	23°32.4	3.4'	57.6'
11	344°09.0	00.5	22°44.0	6.9'	23°35.8	3.3'	57.5'	22°44.0	6.9'	23°35.8	3.3'	57.5'
12	359°08.7	S23°00.3	37°10.0	6.9'	S23°39.1	3.1'	57.5'	37°10.0	6.9'	S23°39.1	3.1'	57.5'
13	14°08.4	23°00.1	51°35.9	6.9'	23°42.3	3.0'	57.5'	51°35.9	6.9'	23°42.3	3.0'	57.5'
14	29°08.1	22°59.9	66°01.8	6.9'	23°45.2	2.8'	57.5'	66°01.8	6.9'	23°45.2	2.8'	57.5'
15	44°07.8	· · 59.7	80°27.8	6.9'	23°48.1	2.7'	57.5'	80°27.8	6.9'	23°48.1	2.7'	57.5'
16	59°07.5	59.5	94°53.7	6.9'	23°50.8	2.6'	57.4'	94°53.7	6.9'	23°50.8	2.6'	57.4'
17	74°07.2	59.3	109°19.6	6.9'	23°53.3	2.4'	57.4'	109°19.6	6.9'	23°53.3	2.4'	57.4'
18	89°06.9	S22°59.1	123°45.5	6.9'	S23°55.7	2.3'	57.4'	123°45.5	6.9'	S23°55.7	2.3'	57.4'
19	104°06.6	58.9	138°11.4	6.9'	23°58.0	2.1'	57.4'	138°11.4	6.9'	23°58.0	2.1'	57.4'
20	119°06.4	58.7	152°37.4	6.9'	24°00.1	2.0'	57.4'	152°37.4	6.9'	24°00.1	2.0'	57.4'
21	134°06.1	· · 58.5	167°03.3	6.9'	24°02.1	1.8'	57.3'	167°03.3	6.9'	24°02.1	1.8'	57.3'
22	149°05.8	58.2	181°29.2	6.9'	24°03.9	1.7'	57.3'	181°29.2	6.9'	24°03.9	1.7'	57.3'
23	164°05.5	58.0	195°55.1	6.9'	24°05.6	1.5'	57.3'	195°55.1	6.9'	24°05.6	1.5'	57.3'
SD = 16.3' d = -0.2'												
SD = 15.7'												

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:25	10:46	■	■	13:20	15:41
N 70°	08:06	09:51	■	■	14:15	16:00
68°	07:50	09:18	■	■	14:48	16:16
66°	07:38	08:54	10:30	13:36	15:12	16:28
64°	07:27	08:35	09:51	14:15	15:31	16:39
62°	07:18	08:19	09:24	14:42	15:47	16:48
60°	07:10	08:06	09:03	15:03	16:00	16:57
N 58°	07:02	07:55	08:46	15:20	16:12	17:04
56°	06:56	07:45	08:32	15:35	16:22	17:10
54°	06:50	07:36	08:19	15:47	16:30	17:16
52°	06:44	07:28	08:08	15:58	16:38	17:22
50°	06:39	07:20	07:59	16:07	16:46	17:27
45°	06:27	07:05	07:38	16:28	17:02	17:39
N 40°	06:17	06:51	07:22	16:44	17:15	17:49
35°	06:08	06:40	07:08	16:58	17:26	17:58
30°	05:59	06:29	06:56	17:10	17:37	18:07
20°	05:43	06:11	06:35	17:31	17:55	18:23
N 10°	05:27	05:54	06:16	17:50	18:12	18:39
0°	05:11	05:37	05:59	18:07	18:29	18:55
S 10°	04:52	05:19	05:42	18:24	18:47	19:14
20°	04:30	04:59	05:23	18:43	19:07	19:36
30°	04:01	04:34	05:01	19:04	19:32	20:05
35°	03:42	04:19	04:49	19:17	19:47	20:23
40°	03:20	04:01	04:34	19:32	20:05	20:46
45°	02:50	03:39	04:16	19:50	20:27	21:16
S 50°	02:05	03:10	03:54	20:12	20:56	22:00
52°	01:39	02:55	03:43	20:22	21:10	22:26
54°	00:57	02:37	03:31	20:34	21:28	23:07
56°	///	02:16	03:17	20:48	21:49	///
58°	///	01:47	03:01	21:04	22:17	///
S 60°	///	01:02	02:41	21:24	23:02	///

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	06:57	■	■	07:54	■	■
N 70°	05:32	■	■	09:21	■	■
68°	04:53	07:27	■	10:01	09:21	■
66°	04:27	06:27	08:39	10:28	10:21	10:07
64°	04:06	05:53	07:34	10:50	10:56	11:12
62°	03:50	05:29	06:59	11:07	11:21	11:47
60°	03:36	05:09	06:34	11:21	11:41	12:13
N 58°	03:25	04:53	06:14	11:34	11:58	12:33
56°	03:15	04:40	05:58	11:44	12:12	12:50
54°	03:06	04:28	05:43	11:54	12:24	13:04
52°	02:58	04:17	05:31	12:02	12:35	13:16
50°	02:51	04:08	05:20	12:10	12:44	13:27
45°	02:36	03:49	04:58	12:26	13:04	13:50
N 40°	02:24	03:33	04:40	12:40	13:21	14:09
35°	02:13	03:20	04:25	12:51	13:35	14:24
30°	02:04	03:08	04:12	13:01	13:47	14:38
20°	01:48	02:49	03:49	13:18	14:08	15:01
N 10°	01:35	02:32	03:30	13:33	14:26	15:20
0°	01:22	02:16	03:12	13:48	14:43	15:39
S 10°	01:10	02:01	02:54	14:02	15:00	15:58
20°	00:56	01:44	02:35	14:17	15:18	16:18
30°	00:41	01:25	02:13	14:35	15:39	16:41
35°	00:33	01:14	02:00	14:45	15:51	16:54
40°	00:23	01:01	01:46	14:57	16:05	17:10
45°	00:11	00:46	01:28	15:11	16:22	17:28
S 50°		00:28	01:07	15:28	16:43	17:51
52°		00:20	00:56	15:36	16:53	18:02
54°		00:10	00:45	15:45	17:04	18:15
56°	23:59		00:32	15:55	17:17	18:29
58°	23:47		00:17	16:07	17:32	18:47
S 60°	23:33	23:58		16:20	17:50	19:07